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KAPLAN & SADOCK'S SYNOPSIS OF PSYCHIATRY

TWELFTH EDITION

Robert Boland, M.D.

Marcia L. Verduin, M.D.

CONSULTING EDITOR

Pedro Ruiz, M.D.

Wolters Kluwer

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To my wife, Susan, who contributed to this work—and everything else—in more ways than she will ever know. Also to my sons, Rich and Greg—perhaps I helped encourage you once but now you are both my inspiration.

—RB

To my husband, Jonathan, and my children, Max and Sarah—for their encouragement, support, and inspiration in all that I do.

—MV



Preface

"No one can replace him, Sir; I am only his successor."

Thomas Jefferson would routinely say this when asked if he was replacing Benjamin Franklin as Minister to France. Although not quite the same historical scale, this aptly describes our feelings about taking over as editors for *Kaplan & Sadock's Synopsis of Psychiatry*.

The Synopsis was one of the first psychiatry books we encountered and was our "Bible" during our respective training years. As we became educators, it was the book that we recommended to our students and residents. In our roles as editors of the Psychiatry Resident-In-Training Examination®, administered by the American College of Psychiatrists to virtually all trainees in the United States, the *Synopsis* was the book we would most often turn to when deciding whether a given question on the test was within the accepted body of knowledge for all psychiatrists.

As we approach the book again from this new perspective, the time we have spent editing and revising the book has only given renewed respect for this fantastic work. The previous authors have done an incredible job of creating a book that is comprehensive, approachable, and frankly enjoyable to read.

Much of our approach to this new edition has been careful, and we have preserved much of the previous editions, often verbatim. Our edits have been either to incorporate discoveries in our rapidly growing field or to reorganize the book in a way we think practical for a new generation of psychiatrists. We have emphasized the clinical material upfront. The underlying science and theory of the field have been moved later in the book. This reorganization does not reflect any value or order of importance; it is meant to be practical: one can imagine a busy trainee grabbing the book during the

day to answer a crucial question about the diagnosis or management of a patient, and later that day, with, hopefully, more time for reflection, sitting down and reading further to understand their patient more deeply.

Whenever we could, we grouped disorders syndromally. Thus, we first discuss the various depressive disorders' phenomenology as a group before discussing diagnosis and, eventually, treatment. This approach was a strategic choice as much modern research no longer groups patients by clinical diagnoses, and many incorporate larger groupings: depressed patients or psychotic patients, for example. That said, in deference to Emerson, we did not let our desire for organization and consistency become so rigid as to be foolish, and in cases where the disorders within a chapter are fundamentally different, we did create separate sections for each.

We realize that some may worry that our "practical" approach might take away some of the artistry of this book. Indeed, we no longer include many paintings, creative photos, and other artistic touches that were a defining part of previous volumes. These are personal choices reflecting on our personal approaches. We certainly do not wish to wring out all of the personality from the book, and we hope that our preservation of much of the prose, case histories, and historical references keep some of what made the previous volumes special.

The other philosophy we bring to this work is that mental health is part of physical health, and one cannot discuss psychiatric disease except within the context of all medicines. Even the term "mind—body" seems obsolete and implying a divide that does not exist. We were frequently reminded of this as we edited during the COVID-19 pandemic and witnessed in our patients, faculty, and trainees how a simple virus can wreak such havoc on the psyche.

HISTORY

This textbook evolved as a condensation of the *Comprehensive Textbook of Psychiatry*. That book is an extensive, two-volume work with many contributions by experts in the field. It helps those who require an exhaustive, detailed, and encyclopedic survey of the entire field. That work aims at all levels of expertise in the field who desire an up-to-date work that is as comprehensive as possible. This textbook is aimed more at the trainee, who needs a brief and more condensed statement of the field of psychiatry.

Although, in essence, this book derives from the *Comprehensive Textbook of Psychiatry*, different purposes for each work result in a divergence. However, although we have reworked some sections of this book and written new content, we still owe an outstanding debt of gratitude to the more than 2,000 contributors to the current and previous editions of the *Comprehensive Textbook of Psychiatry*, all of whom have allowed us to synopsize their work. At the same time, we must accept responsibility for the modifications and changes in the new work.

COMPREHENSIVE TEACHING SYSTEM

This textbook forms one part of a comprehensive system developed to facilitate the teaching of psychiatry and the behavioral sciences. At the head of the system is the *Comprehensive Textbook of Psychiatry*, which is global in depth and scope; it is designed for and used by psychiatrists, behavioral scientists, and all workers in the mental health field. *Synopsis of Psychiatry* is a briefer, highly modified, and current version useful for medical students, psychiatric residents, practicing psychiatrists, and mental health professionals. Two special editions derived from the *Synopsis*, *Concise Textbook of Clinical Psychiatry* and *Concise Textbook of Child and Adolescent Psychiatry*, contain descriptions of all psychiatric disorders, including their diagnosis and treatment in adults and children, respectively. They will be useful for clinical clerks and psychiatric residents who need a succinct overview of the management of clinical problems. Another part of the system, *Study Guide and Self-Examination Review of Psychiatry*, consists of multiple-choice questions and answers; it is designed for students of psychiatry and for clinical psychiatrists who require a review of the behavioral sciences and general psychiatry in preparation for a variety of examinations. The questions are consistent with the format used by most standardized examinations. Other parts of the system are the various editions of the pocket handbooks: *Pocket Handbook of Clinical Psychiatry*, *Pocket Handbook of Psychiatric Drug Treatment*, *Pocket Handbook of Emergency Psychiatric Medicine*, and *Pocket Handbook of Primary Care Psychiatry*. Those books cover the diagnosis and treatment of psychiatric disorders, psychopharmacology, psychiatric emergencies, and primary care psychiatry, respectively, and are designed and written to be carried by clinical clerks and

practicing physicians, whatever their specialty, for a quick reference. Finally, *Comprehensive Glossary of Psychiatry and Psychology* provides written definitions for psychiatrists and other physicians, psychologists, students, other mental health professionals, and the public. Together, these books create multiple approaches to the teaching, study, and learning of psychiatry.

CLASSIFICATION OF DISORDERS

DSM-5

The American Psychiatric Association published the fifth edition of the *American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders* in 2013; it is commonly called the DSM-5. It contains the official nomenclature used by psychiatrists and other mental health professionals in the United States, and many other countries also use the manual. The psychiatric disorders discussed in the textbook are consistent with and follow that nosology. We have chosen not to directly reproduce the DSM-5 criteria in either prose or table form. We expect that most psychiatrists and trainees already have access to the DSM-5; if not, we do highly recommend it. We include some tables outlining the primary symptoms, time courses, and other criteria associated with the major diagnoses to aid the reader with diagnostic reasoning.

ICD-10

Readers also should be aware of a parallel classification system developed by the World Health Organization (WHO) called the *International Statistical Classification of Diseases and Related Health Problems* (ICD-10). There are textual differences between DSM and ICD, but according to treaties between the United States and the WHO, the diagnostic code numbers are identical to ensure uniform national and international psychiatric statistics. ICD diagnoses and numerical codes are accepted by Medicare, Medicaid, and private insurance companies for reimbursement purposes. ICD-10 is a comprehensive list of all medical diagnoses, not just psychiatric. It tends to be descriptive and allows more judgment than the DSM-5, which contains fixed numbers of criteria. ICD-10 is also older than DSM-5 and contains diagnoses no longer included in the newer manual. ICD-11 is currently

available in draft form and is online at the WHO's website. The ICD-11 is more aligned with DSM-5. The WHO plans for this new edition of the ICD to go into effect in 2022. As much of the world still uses ICD-10 (including the United States, especially for billing purposes), we include it in our tables devoted to diagnostic reading. The reader will note the many similarities between DSM-5 and ICD-10 in those tables, as well as some interesting differences.

CASE HISTORIES

Case histories are an integral part of *Synopsis.* They are used extensively throughout the text to add clarity and bring life to the clinical disorders described. Cases come from various sources. Many are from the Sadocks' hospital colleagues or the Sadocks' own experience at Bellevue Hospital in New York. We have preserved most of these case studies as we believe that they expertly illustrate the sometimes abstract concepts described, and we thank the many contributors of those cases for allowing the Synopsis to use these cases in past and future editions.

REFERENCES

Each section in *Synopsis* ends with several citations that include reviews of the literature and references in addition to relevant chapters in our more extensive textbook, *Comprehensive Textbook of Psychiatry*. References are limited in number; in part, this was to conserve space, but more importantly, we are mindful that modern-day readers consult Internet databases such as *PubMed* and *Google Scholar* to stay abreast of the most current literature, and we encourage that trend.

ACKNOWLEDGMENTS

We sincerely appreciate the work of our distinguished Editorial Assistant, Arya Shah, M.D., who helped us to update much of the work, including epidemiology and much other research. Dr. Shah also took on the challenge of reviewing DSM-5 and ICD-10 to create the diagnostic comparative tables, which are new to this edition.

We also wish to thank our colleagues and friends at our respective

departments. This impressive group includes the members of the Department of Psychiatry at the Brigham and Women's Hospital. A special thanks to David Silbersweig, M.D., the Chair of Psychiatry at the Brigham and Women's, who offered invaluable advice and mentorship throughout the creation of this work, and to Deborah German, M.D., the Vice President for Health Affairs and Dean of the College of Medicine at the University of Central Florida, for her support, encouragement, and mentorship.

We would also thank our many friends in psychiatry and medicine who live throughout the country. One of them, Josepha Cheong, M.D., at the University of Florida, has dubbed us an "Invisible College," and this is an apt description of the loose network of dear friends, many of them leaders in their fields, who are available at any time to offer advice, counsel, critique, and perspective, all delivered with an infectious sense of humor.

We want to take this opportunity to acknowledge those who have translated this and other *Kaplan & Sadock* books into foreign languages, including Chinese, Croatian, French, German, Greek, Indonesian, Italian, Japanese, Polish, Portuguese, Romanian, Russian, Spanish, and Turkish, in addition to a special Asian and international student edition.

A special thanks to the editorial team at Wolters Kluwer, who were patient with us as we took on this work. We put their patience to the test during the pandemic, and the editorial team was both supportive and encouraging as we helped manage the ramifications of the disease on our respective home fronts while we worked on this edition. Particular thanks go to Chris Teja, our Acquisitions Editor; Anne Malcolm, the Associate Director for Content Development; Ariel Winter, our Development Editor; Ashley Pfeiffer, our Editorial Coordinator; Karan Rana, our Project Manager and Bridgett Dougherty, the Vendor Manager.

Our most tremendous thanks go to Benjamin and Virginia Sadock. Sadly, the pandemic has required that we only know them through their reputations and electronic communication. We do thank them for trusting us with their brainchild. We cannot replace them, but we hope we can still do them proud.

INTERVIEW

Our Meeting with the Sadocks

In planning for this book, we were hoping to visit Benjamin and Virginia Sadock. However, like so many things, the pandemic made that impossible. We settled for a Zoom meeting. We wished to hear some of the book's history and get some advice from the experts. We also wanted to understand the goals and values that motivated them to take on this monumental task. Most of all, we just wanted to meet the people who were responsible for this fantastic work. So we joined the video call with great anticipation, not knowing what to expect.

Meeting people you have only known by reputation can play out in various ways. In this case, the experience was a lovely one. Ben and Virginia turned out to be a warm and friendly couple who were very gracious. They were a smart and energetic pair who played well off each other, at times reflective, other times witty, and always surprisingly humble. They were also happy to tell us about their life's work. They explained that the history of the *Synopsis* began with the *Comprehensive Textbook*, first published in 1967. It was initially about 1,000 pages, with about 200 contributors, edited by Harold Kaplan and Benjamin. Now, of course, it is closer to 5,000 pages with thousands of contributors. Still, even then, it was a massive book, so eventually, they had the idea to try to condense it into something that could be used by medical students and residents. They explained that they did a "cut and paste" of the more extensive book for the first edition of the *Synopsis*. We asked them to clarify and found that they meant a *literal* cutting up of portions of the *Comprehensive Textbook* with an X-ACTO knife, which they then pasted together to form the first edition, calling it the *Modern Synopsis of the Comprehensive Textbook of Psychiatry*. Several editions later, they shortened the title. They explained that as they continued to write new editions of the *Synopsis*, the authors felt that simply extracting bits from the *Comprehensive Textbook* read poorly. They gradually translated the grander work into their words while keeping some of the original book's prose until it became the hybrid between a multiple-contributor and consistently authored work that exists today.

They recalled that, sadly, Harold Kaplan died in 1998 while working on the seventh edition of the *Synopsis*, which is when Virginia became an official author. Benjamin clarified "official," as she had long been unofficially contributing to the book. Virginia noted that, among other things, she helped to add a female perspective to the book—a critical perspective:

she was, she noted, only one of eight women in her class at New York Medical College.

They felt that an important goal for them in writing the *Synopsis* was to address their growing concern with the rise of technology and corporate medicine and their dehumanizing effect on medical education. Medical students, they felt, were trained to be "robotic physicians." This dilemma is a problem all academic physicians still worry about as we heap more checklists, protocols, and electronic record-keeping on physicians. They hoped that their book could help to restore some humanism in psychiatric education.

Thus, although there has always been much science and groundbreaking research in the book, they have always kept the practicing physician in mind, which was easy for them as they continued to treat patients as they wrote the book. Indeed, the book is enormously popular among practicing psychiatrists. However, they related their surprise when they found that many nonpsychiatrists were also buying the book, including psychologists, various therapists, and many other professionals involved in treating patients with mental illnesses.

The Sadocks noted that there had been specific challenges, such as keeping up to date with the many revisions of the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders* (DSM). Although they acknowledge the occasional competition between these two popular books, they see the DSM and the *Synopsis* as complementary, with the DSM defining and their book explaining.

Much of their advice involved the aesthetics of the work. We were impressed with Virginia and Ben's careful thought to details such as the layout, the color scheme, and even the quality of paper used. Their advice helped us appreciate the importance of the work as a whole, including the book's look and feel.

In discussing the book's more than 50-year history, it was clear that the Sadocks rightly took a great deal of pride from their masterpiece. They described being somewhat critical about their first edition, and each time working to make the book better. They felt that their most recent 11th edition was as near to perfect as they could make it. Clearly, the book has grown from the "cut-and-paste" work of the first edition into a thoughtful work with an engaging style and a humanistic approach that incorporates art, history,

philosophy, and many case studies. It also grew from a work that, as they related, was initially difficult to interest a publisher in, to become one of the most popular books in psychiatry. Most importantly, they rightfully took great pride in achieving their original goals of creating a work that moves beyond the checklist and outline style that defines many references, to a work that reflects both the art and science of medicine, emphasizing the whole patient.

The meeting ended with their blessing and good wishes. For us, it was a revelation and a great chance to gain some insight into the passion that goes into creating great work. We can only hope that we can learn from their advice and work and try our best to continue the book in the rich tradition that they have built. We also hope that they remain in the excellent health and spirits we encountered when we virtually met and that we can meet in person sometime soon.

Bob Boland and Marcy Verduin. 10/16/2020.



Contents

A Clinical Psychiatry

1. Examination and Diagnosis of the Psychiatric Patient
   - 1.1 The Adult Patient
   - 1.2 Children and Adolescents
   - 1.3 Geriatric Patients
2. Neurodevelopmental Disorders and Other Childhood Disorders
   - 2.1 Intellectual Disability
   - 2.2 Communication Disorders
   - 2.3 Autism Spectrum Disorder
   - 2.4 Attention-Deficit/Hyperactivity Disorder
   - 2.5 Specific Learning Disorder
   - 2.6 Motor Disorders
   - 2.7 Feeding and Eating Disorders of Infancy or Early Childhood
   - 2.8 Trauma- and Stressor-Related Disorders in Children
   - 2.9 Depressive Disorders and Suicide in Children and Adolescents
   - 2.10 Early-Onset Bipolar Disorder
   - 2.11 Disruptive Mood Dysregulation Disorder
   - 2.12 Disruptive Behaviors of Childhood
   - 2.13 Anxiety Disorders of Infancy, Childhood, and Adolescence: Separation Anxiety Disorder, Generalized Anxiety Disorder, and Social Anxiety Disorder (Social Phobia)
   - 2.14 Selective Mutism
   - 2.15 Obsessive-Compulsive Disorder in Childhood and Adolescence
   - 2.16 Early-Onset Schizophrenia

- 2.17 Adolescent Substance Use Disorders
3. Neurocognitive Disorders
   - 3.1 Delirium
   - 3.2 Dementia (Major Neurocognitive Disorder)
   - 3.3 Major or Minor Neurocognitive Disorder due to Another Medical Condition (Amnestic Disorders)
   - 3.4 Neurocognitive and Other Psychiatric Disorders due to a General Medical Condition
   - 3.5 Mild Cognitive Impairment
4. Substance Use and Addictive Disorders
   - 4.1 General Features of the Substance-Related Disorders
   - 4.2 Alcohol-Related Disorders
   - 4.3 Cannabis-Related Disorders
   - 4.4 Opioid-Related Disorders
   - 4.5 Sedative-, Hypnotic-, or Anxiolytic-Related Disorders
   - 4.6 Stimulant-Related Disorders
   - 4.7 Tobacco-Related Disorders
   - 4.8 Caffeine-Related Disorders
   - 4.9 Hallucinogen-Related Disorders
   - 4.10 Inhalant-Related Disorders
   - 4.11 Anabolic-Androgenic Steroid Use
   - 4.12 Gambling Disorder
   - 4.13 Other Disorders Related to the Addictive Disorders
5. Schizophrenia Spectrum and Other Psychotic Disorders
6. Bipolar Disorders
7. Depressive Disorders
8. Anxiety Disorders
9. Obsessive-Compulsive and Related Disorders
10. Trauma- and Stressor-Related Disorders
11. Dissociative Disorders
12. Somatic Symptom and Related Disorders
13. Feeding and Eating Disorders
14. Elimination Disorders

15. Sleep–Wake Disorders
16. Human Sexuality and Sexual Dysfunctions
17. Gender Dysphoria, Gender Identity, and Related Conditions
18. Disruptive, Impulse-Control, and Conduct Disorders
19. Personality Disorders
20. Other Conditions that May be a Focus of Clinical Attention

B Treatment Across the Lifespan

21. Psychopharmacology
    21.1 Antipsychotics
    21.2 Antidepressants
    21.3 Mood Stabilizers
    21.4 Anxiolytics
    21.5 Drugs Used to Treat Sleep Disorders
    21.6 Stimulants
    21.7 Drugs Used to Treat Substance Use Disorders
    21.8 Drugs Used for Cognitive Enhancement
    21.9 Drugs Used to Treat Sexual Disorders
    21.10 Drugs Used to Treat the Side Effects of Psychotropic Drugs
    21.11 Nutritional Supplements and Related
22. Other Somatic Therapies
23. Psychotherapy
24. Psychiatric Rehabilitation and Other Interventions
25. Consultation to Other Disciplines
    25.1 Consultation-Liaison Psychiatry
    25.2 Geriatric Psychiatry
    25.3 Psychiatric Emergencies
26. Level of Care

C Other Issues Relevant to Psychiatry

27. Ethics and Professionalism
28. Forensic and Legal Issues

29. End-of-Life Issues and Palliative Care
29.1 Death, Dying, and Bereavement
29.2 Palliative Care
29.3 Euthanasia and Physician-Assisted Suicide
30. Community Psychiatry
31. Global and Cultural Issues in Psychiatry

D Contributions from the Sciences and Social Sciences to Psychiatry

32. Normal Development and Aging
32.1 Infant, Child, and Adolescent Development
32.2 Adulthood
32.3 Old Age
33. Contributions from the Neurosciences
34. Contributions from the Behavioral and Social Sciences
34.1 Contributions from the Psychosocial Sciences
34.2 Normality and Mental Health
34.3 Contributions from the Sociocultural Sciences
34.4 Theories of Personality and Psychopathology
35. A Brief History of Psychiatry

Glossary of Terms Relating to Signs and Symptoms
Index

A Clinical Psychiatry



1 Examination and Diagnosis of the Psychiatric Patient

1.1 The Adult Patient

PSYCHIATRIC INTERVIEW, HISTORY, AND MENTAL STATUS EXAMINATION

The psychiatric interview is the most crucial element in the evaluation and care of persons with mental illness. A significant purpose of the initial psychiatric interview is to obtain information that will establish a criteria-based diagnosis. This process, helpful in the prediction of the course of the illness and the prognosis, leads to treatment decisions. A well-conducted psychiatric interview provides a multidimensional understanding of the disorder's biopsychosocial elements. It provides the information necessary for the psychiatrist, in collaboration with the patient, to develop a person-centered treatment plan.

Equally important, the interview itself is often an essential part of the treatment process. From the very first moments of the encounter, the interview shapes the nature of the patient–physician relationship, which can have a profound influence on the outcome of treatment. The settings in which the psychiatric interview takes place include psychiatric inpatient units, medical nonpsychiatric inpatient units, emergency rooms, outpatient offices, nursing homes, other residential programs, and correctional facilities. The length of time for the interview, and its focus, will vary depending on the setting, the specific purpose of the interview, and other factors (including

concurrent competing demands for professional services).

Nevertheless, some basic principles and techniques are essential for all psychiatric interviews, and these will be the focus of this section. There are particular issues in the evaluation of children that will not be addressed. This section focuses on the psychiatric interview of adult patients.

General Principles

Agreement as to Process. At the beginning of the interview, the psychiatrist should introduce themself and, depending on the circumstances, may need to identify why he or she is speaking with the patient. Unless implicit (the patient coming to the office), consent to proceed with the interview should be obtained, and the nature of the interaction and the approximate (or specific) amount of time for the interview should be stated. The patient should be encouraged to identify any elements of the process that he or she wishes to alter or add.

A crucial issue is whether the patient is, directly or indirectly, seeking the evaluation voluntarily or has been brought involuntarily for the assessment. This issue should be established before the interview begins, and this information will guide the interviewer, especially in the early stages of the process.

Privacy and Confidentiality. Issues concerning confidentiality are crucial in the evaluation/treatment process and may need to be discussed on multiple occasions. Health Insurance Portability and Accountability Act (HIPAA) regulations must be carefully followed, and the appropriate paperwork must be presented to the patient.

Confidentiality is an essential component of the patient–doctor relationship. The interviewer should make every attempt to ensure that others cannot overhear the content of the interview. Sometimes, in a hospital unit or other institutional setting, this may be difficult. If the patient is sharing a room with others, an attempt should be made to use a different place for the interview. If this is not feasible, the interviewer may need to avoid specific topics or indicate that these issues can be discussed later when privacy can be ensured. Generally, in the beginning, the interviewer should suggest that the content of the session(s) will remain confidential except for what needs to be

shared with the referring physician or treatment team. Some evaluations, including forensic and disability evaluations, are less confidential, and what is discussed may be shared with others. In those cases, the interviewer should be explicit in stating that the session is not confidential and identify who will receive a report of the evaluation. This information should be carefully and thoroughly documented in the patient's record.

A special issue concerning confidentiality is when the patient indicates that he or she intends to harm another person. When the psychiatrist's evaluation suggests that this might indeed happen, the psychiatrist may have a legal obligation to warn the potential victim. (The law concerning notification of a potential victim varies by state.) Psychiatrists should also consider their ethical obligations. Part of this obligation may be met by appropriate clinical measures such as increasing the dose of antipsychotic medication or hospitalizing the patient.

Often members of the patient's family, including spouse, adult children, or parents, come with the patient to the first session or are present in the hospital or other institutional setting when the psychiatrist first sees the patient. If a family member wishes to talk to the psychiatrist, it is generally preferable to meet with the family member(s) and the patient together after the session and after the patient's consent has been obtained. The psychiatrist should not bring up material the patient has shared but listen to the input from family members and discuss items that the patient introduces during the joint session. Occasionally, when family members have not asked to be seen, the psychiatrist may feel that including a family member or caregiver might be helpful and raise this subject with the patient. This may be the case when the patient is not able to communicate effectively. As always, the patient must give consent except if the psychiatrist determines that the patient is a danger to himself or herself or others. Sometimes family members might telephone the psychiatrist. Except in an emergency, consent should be obtained from the patient before the psychiatrist speaks to the relative. As indicated above, the psychiatrist should not bring up material that the patient has shared but listen to the input from the family member. The patient should be told when a family member has contacted the psychiatrist even if the patient has given consent for this to occur.

In educational and, occasionally, forensic settings, there may be occasions when the session is recorded. The patient must be fully informed about the

recording and how the recording will be used. The length of time the recording will be kept and how access to it will be restricted must be discussed. Occasionally in educational settings, one-way mirrors may be used as a tool to allow trainees to benefit from the observation of an interview. The patient should be informed of the use of the one-way mirror and the category of the observers and be reassured that the rules of confidentiality also bound the observers. The patient's consent for proceeding with the recording or use of the one-way mirror must be obtained, and it should be made clear that the patient's receiving care will not be determined by whether he or she agrees to its use. These devices will have an impact on the interview that the psychiatrist should be open to discussing as the session unfolds.

Respect and Consideration. As should happen in all clinical settings, the patient must be treated with respect, and the interviewer should be considerate of the circumstances of the patient's condition. The patient is often experiencing considerable pain or other distress and frequently is feeling vulnerable and uncertain of what may happen. Because of the stigma of mental illness and misconceptions about psychiatry, the patient may be especially concerned, or even frightened, about seeing a psychiatrist. The skilled psychiatrist is aware of these potential issues and interacts in a manner to decrease, or at least not increase the distress. The success of the initial interview will often depend on the physician's ability to alleviate excessive anxiety.

Rapport/Empathy. Respect for and consideration of the patient will contribute to the development of rapport. In the clinical setting, rapport can be defined as the harmonious responsiveness of the physician to the patient and the patient to the physician. It is crucial that patients increasingly feel that the evaluation is a joint effort and that the psychiatrist is genuinely interested in their story. Empathic interventions ("*That must have been very difficult for you*" or "*I'm beginning to understand how awful that felt*") further increase the rapport. Frequently a nonverbal response (raised eyebrows or leaning toward the patient) or a very brief response ("*Wow*") will be similarly effective. Empathy is understanding what the patient is thinking and feeling. It occurs when the psychiatrist can put himself or herself in the patient's place while at the same time maintaining objectivity. For the psychiatrist to

truly understand what the patient is thinking and feeling requires an appreciation of many issues in the patient's life. As the interview progresses, the patient's story unfolds, and patterns of behaviors become evident, and it becomes clearer what the patient may have experienced. Early in the interview, the psychiatrist may not be as fully confident of where the patient is or was (although the patient's nonverbal cues can be very beneficial). If the psychiatrist is uncertain about the patient's experience, it is often best not to guess but to encourage the patient to continue. Head nodding, putting down one's pen, leaning toward the patient, or a brief comment, "*I see,*" can accomplish this objective and simultaneously indicate that this is important material. The large majority of empathic responses in an interview are nonverbal.

An essential ingredient in empathy is retaining objectivity. Maintaining objectivity is crucial in a therapeutic relationship, and it differentiates empathy from identification. With identification, psychiatrists not only understand the emotion but also experience it to the extent that they lose the ability to be objective. This blurring of boundaries between the patient and psychiatrist can be confusing and distressing to many patients, especially to those who, as part of their illness, already have significant boundary problems (e.g., individuals with borderline personality disorder). Identification can also be draining to the psychiatrist and lead to disengagement and, ultimately, burnout.

Patient–Physician Relationship. The patient–physician relationship is the core of the practice of medicine. (For many years, the term used was "physician–patient" or "doctor–patient," but the order is sometimes reversed to reinforce that the treatment should always be patient-centered.) Although the relationship between any single patient and physician will vary depending on each of their personalities and past experiences as well as the setting and purpose of the encounter, there are general principles that, when followed, help to ensure that the relationship established is helpful.

The patient comes to the interview seeking help. Even in those instances when the patient arrives on the insistence of others (i.e., spouse, family, courts), assistance may be sought by the patient in dealing with the person requesting or requiring the evaluation or treatment. This desire for help motivates the patient to share with a stranger information and feelings that are

distressing, personal, and often private. The patient is willing, to various degrees, to do so because of a belief that the doctor has the expertise, by training and experience, to be of help. Right from the very first encounter (sometimes the initial phone call), the patient's willingness to share is increased or decreased depending on the verbal and often the nonverbal interventions of the physician and other staff. As the physician's behaviors demonstrate respect and consideration, rapport begins to develop. This rapport is increased as the patient feels safe and comfortable. If the patient feels secure that what is said in the interview remains confidential, he or she will be more open to sharing.

The sharing is reinforced by the nonjudgmental attitude and behavior of the physician. The patient may have been exposed to considerable negative responses, actual or feared, to their symptoms or behaviors, including criticism, disdain, belittlement, anger, or violence. Being able to share thoughts and feelings with a nonjudgmental listener is generally a positive experience.

There are two additional essential ingredients in a helpful patient–physician relationship. One is the demonstration by physicians that they understand what the patient is stating and emoting. It is not enough that the physician understands what the patient is relating, thinking, and feeling; this understanding must be conveyed to the patient if it is to nurture the therapeutic relationship. The interview is not just an intellectual exercise to arrive at a supportable diagnosis. The other essential ingredient in a helpful patient–physician relationship is the recognition by the patient that the physician cares. As the patient becomes aware that the physician not only understands but also cares, trust increases, and the therapeutic alliance becomes stronger.

The genuineness of the physician reinforces the patient–physician relationship. Being able to laugh in response to a humorous comment, admit a mistake, or apologize for an error that inconvenienced the patient (e.g., being late for or missing an appointment) strengthens the therapeutic alliance. It is also essential to be flexible in the interview and responsive to patient initiatives. If the patient brings in an item, for example, a photo that he or she wants to show the psychiatrist, it is good to look at it, ask questions, and thank the patient for sharing it. Much can be learned about the family history and dynamics from such a seemingly sidebar moment. Also, the therapeutic

alliance is strengthened. The psychiatrist should be mindful of the reality that there are no irrelevant moments in the interview room.

At times patients will ask questions about the psychiatrist. A good rule of thumb is that questions about the physician's qualifications and position should generally be answered directly (e.g., board certification, hospital privileges). On occasion, such a question might be a sarcastic comment ("Did you really go to medical school?"). In this case, it would be better to address the issue that provoked the comment rather than respond concretely. There is no easy answer to the question of how the psychiatrist should respond to personal questions ("Are you married?" "Do you have children?" "Do you watch football?"). Advice on how to respond will vary depending on several issues, including the type of psychotherapy being used or considered, the context in which the question is asked, and the wishes of the psychiatrist. Often, especially if the patient is being, or might be, seen for insight-oriented psychotherapy, it is useful to explore why the question is being asked. The question about children may be precipitated by the patient wondering if the psychiatrist has had personal experience in raising children, or more generally does the psychiatrist have the skills and expertise necessary to meet the patient's needs. In this instance, part of the psychiatrist's response may be that he or she has had considerable experiences in helping people deal with issues of parenting. For patients being seen for supportive psychotherapy or medication management, answering the question, especially if it is not very personal, such as "Do you watch football?" is entirely appropriate. A significant reason for not directly answering personal questions is that the interview may become psychiatrist-centered rather than patient-centered.

Occasionally, again depending on the nature of the treatment, it can be helpful for the psychiatrist to share some personal information even if it is not asked directly by the patient. The purpose of the self-revelation should always be to strengthen the therapeutic alliance to be helpful to the patient. Personal information should not be shared to meet the psychiatrist's needs.

Conscious/Unconscious. Unconscious processes must be considered to understand more fully the patient–physician relationship. The reality is that the majority of mental activity remains outside of conscious awareness. In the interview, unconscious processes may be suggested by tangential references to an issue, slips of the tongue or mannerisms of speech, what is not said or

avoided, and other defense mechanisms. For example, phrases such as “to tell you the truth” or “to speak frankly” suggest that the speaker does not usually tell the truth or speak frankly. In the initial interview, it is best to note such mannerisms or slips but not to explore them. It may or may not be helpful to pursue them in subsequent sessions. In the interview, transference and countertransference are very significant expressions of unconscious processes. *Transference* is the process of the patient unconsciously and inappropriately displacing onto individuals in his or her current life those patterns of behavior and emotional reactions that originated with significant figures from earlier in life, often childhood. In the clinical situation, the displacement is onto the psychiatrist, who is usually an authority figure or a parent surrogate. The psychiatrist must recognize that the transference may be driving the behaviors of the patient, and the interactions with the psychiatrist may be based on distortions that have their origins much earlier in life. The patient may be angry, hostile, demanding, or obsequious not because of the reality of the relationship with the psychiatrist but because of former relationships and patterns of behaviors. Failure to recognize this process can lead to the psychiatrist inappropriately reacting to the patient’s behavior as if it were a personal attack on the psychiatrist.

Similarly, *countertransference* is the process where the physician unconsciously displaces onto the patient patterns of behaviors or emotional reactions as if he or she were a significant figure from earlier in the physician’s life. Psychiatrists should be alert to signs of countertransference issues (missed appointments by the psychiatrist, boredom, or sleepiness in a session). Supervision or consultations can be helpful as can personal therapy in helping the psychiatrist recognize and deal with these issues.

Although the patient comes for help, there may be forces that impede the movement to health. Resistances are the processes, conscious or unconscious, that interfere with the therapeutic objectives of treatment. The patient is generally unaware of the impact of these feelings, thinking, or behaviors, which take many different forms, including exaggerated emotional responses, intellectualization, generalization, missed appointments or acting out behaviors. Resistance may be fueled by repression, which is an unconscious process that keeps issues or feelings out of awareness. Because of repression, patients may not be aware of the conflicts that may be central to their illness. In insight-oriented psychotherapy, interpretations are interventions that undo

the process of repression and allow the unconscious thoughts and feelings to come to awareness so that they can be handled. As a result of these interventions, the primary gain of the symptom, the unconscious purpose that it serves, may become apparent. In the initial session, interpretations are generally avoided. The psychiatrist should make a note of potential areas for exploration in subsequent sessions.

Person-Centered and Disorder-Based Interviews. A psychiatric interview should be person- (patient-) centered. That is, the focus should be on understanding the patient and enabling the patient to tell his or her story. The individuality of the patient's experience is a central theme, and the patient's life history is elicited, subject to the constraints of time, the patient's willingness to share some of this material, and the skill of the interviewer. Adolf Meyer's "life-charts" were graphic representations of the material collected in this endeavor and were a core component of the "psychobiological" understanding of the illness. The patient's early life experiences, family, education, occupation(s), religious beliefs and practices, hobbies, talents, relationships, and losses are some of the areas that, in concert with genetic and biologic variables, contribute to the development of the personality. An appreciation of these experiences and their impact on the person is necessary for forming an understanding of the patient. It is not only the history that should be person-centered. The resulting treatment plan must be based on the patient's goals, not the psychiatrist's. Numerous studies have demonstrated that often the patient's goals for treatment (e.g., safe housing) are not the same as the psychiatrist's (e.g., decrease in hallucinations). This dichotomy can often be traced to the interview where the focus was not sufficiently person-centered but instead was exclusively or mostly symptom-based. Even when the interviewer asks explicitly about the patient's goals and aspirations, the patient, having been exposed on numerous occasions to what a professional is interested in hearing about, may attempt to focus on "acceptable" or "expected" goals rather than his or her own goals. The patient should be explicitly encouraged to identify his or her goals and aspirations in his or her own words.

Traditionally, medicine has focused on illness and deficits rather than strengths and assets. A person-centered approach focuses on strengths and assets as well as deficits. During the assessment, it is often helpful to ask the

patient, "*Tell me about some of the things you do best,*" or, "*What do you consider your greatest asset?*" A more open-ended question, such as "*Tell me about yourself,*" may elicit information that focuses more on either strengths or deficits depending on several factors, including the patient's mood and self-image.

Safety and Comfort. Both the patient and the interviewer must feel safe. This includes physical safety. On occasion, especially in hospital or emergency room settings, this may require that other staff be present or that the door to the room where the interview is conducted be left ajar. In emergency room settings, it is generally advisable for the interviewer to have a clear, unencumbered exit path. Patients, especially if psychotic or confused, may feel threatened and need to be reassured that they are safe, and the staff will do everything possible to ensure their safety. Sometimes it is useful to explicitly state and sometimes demonstrate that there is sufficient staff to prevent a situation from spiraling out of control. For some, often psychotic patients who are fearful of losing control, this can be reassuring. The interview may need to be shortened or quickly terminated if the patient becomes more agitated and threatening. Once issues of safety have been assessed (and for many outpatients, this may be accomplished within a few seconds), the interviewer should inquire about the patient's comfort and continue to be alert to the patient's comfort throughout the interview. A direct question may be helpful in not only making the patient feel more comfortable but also in enhancing the patient–doctor relationship. This might include, "Are you warm enough?" or "Is that chair comfortable for you?" As the interview progresses, if the patient desires tissues or water, it should be provided.

Time and Number of Sessions. For an initial interview, 45 to 90 minutes are generally allotted. For inpatients on a medical unit or at times for confused patients, in considerable distress, or psychotic, the length of time that can be tolerated in one sitting may be 20 to 30 minutes or less. In those instances, several brief sessions may be necessary. Even for patients who can tolerate longer sessions, more than one session may be required to complete an evaluation. The clinician must accept the reality that the history obtained is never complete or entirely accurate. An interview is dynamic, and some

aspects of the assessment are ongoing, such as how a patient responds to exploration and consideration of new material that emerges. If the patient is coming for treatment, as the initial interview progresses, the psychiatrist makes decisions about what can be continued in subsequent sessions.

Process of the Interview

Before the Interview. For outpatients, the first contact with the psychiatrist's office is often a telephone call. Whoever is receiving the call must understand how to respond if the patient is acutely distressed, confused, or expresses suicidal or homicidal intent. If the receiver of the call is not a mental health professional, the call should be transferred to the psychiatrist or other mental health professional, if available. If not available, the caller should be directed to a psychiatric emergency center or an emergency hotline. The receiver of the call should obtain the name and phone number of the caller and offer to initiate the call to the hotline if the caller prefers that.

Most calls are not of such an urgent nature. The receptionist (or whoever receives the call) should obtain the information that the setting has deemed relevant for the first contact. Although the requested information varies considerably, it generally includes the name, age, address, and telephone number(s) of the patient, who referred the patient, the reason for the referral, and insurance information. The patient is given relevant information about the office, including the length of time for the initial session, fees, and whom to call if there are additional questions. In many practices, the psychiatrist will contact the patient to discuss the reason for the appointment and to determine if indeed an appointment appears warranted. The timing of the appointment should reflect the apparent urgency of the problem. Asking the patient to bring information about past psychiatric and medical treatments as well as a list of medications (or preferably the medications themselves) can be very helpful. Frequently a patient is referred to the psychiatrist or a psychiatric facility. If possible, reviewing records that precede the patient can be quite beneficial. Some psychiatrists prefer not to read records before the initial interview so that prior evaluations will not unduly influence their initial view of the patient's problems. Whether or not records are reviewed, the reason for the referral must be understood as clearly as possible. This is especially important for forensic evaluations where the reason for the referral

and the question(s) posed will help to shape the assessment. Often, especially in the outpatient setting, a patient is referred to the psychiatrist by a primary care physician or other health care provider. Although not always feasible, communicating with the referring professional before the evaluation can be very helpful. It is critical to determine whether the patient is referred for only an evaluation with the ongoing treatment to be provided by the primary care physician or mental health provider (e.g., social worker) or if the patient is being referred for evaluation and treatment by the psychiatrist.

If the court refers the patient or a lawyer or some other non–treatment-oriented agency such as an insurance company, the goals of the interview may be different from diagnosis and treatment recommendations. These goals can include determination of disability, questions of competence or capacity, or determining, if possible, the cause or contributors of the psychiatric illness. In these particular circumstances, the patient and clinician are not entering a treatment relationship, and often the usual rules of confidentiality do not apply. This limited confidentiality must be explicitly established with the patient and must include a discussion of who will be receiving the information gathered during the interview.

The Waiting Room. When the patient arrives for the initial appointment, he or she is often given forms to complete. These generally include demographic and insurance information. Also, the patient receives information about the practice (including contact information for evenings and weekends) and HIPAA-mandated information that must be read and signed. Many practices also ask for a list of medications, the name and address of the primary care physician, and the identification of major medical problems and allergies. Sometimes the patient is asked what his or her primary reason is for coming to the office. Increasingly, some psychiatrists ask the patient to fill out a questionnaire or a rating scale that identifies significant symptoms. Such scales include the Patient Health Questionnaire 9 (PHQ-9) or the Quick Inventory of Depression Symptomatology Self Report (QIDS-SR), which are scales of depressive symptoms based on the *Diagnostic and Statistical Manual of Mental Diseases* (DSM).

The Interview Room. The interview room itself should be relatively soundproof. The decor should be pleasant and not distracting. If feasible, it is

a good idea to give the patient the choice of a soft chair or a hard-back chair. Sometimes the selection of the chair or how the chair is chosen can reveal characteristics of the patient. Many psychiatrists suggest that the interviewer's chair and the patient's chair be of relatively equal height so that the interviewer does not tower over the patient (or vice versa). It is generally agreed that the patient and the psychiatrist should be seated approximately 4 to 6 ft apart. The psychiatrist should not be sitting behind a desk. The psychiatrist should dress professionally and be well-groomed. Distractions should be kept to a minimum. Unless there is an urgent matter, there should be no telephone or beeper interruptions during the interview. The patient should feel that the time has been set aside just for him or her and that for this designated time, he or she is the exclusive focus of the psychiatrist's attention.

Initiation of the Interview. The patient is greeted in the waiting room by the psychiatrist who, with a friendly face, introduces themself, extends a hand, and, if the patient reciprocates, gives a firm handshake. If the patient does not extend their hand, it is probably best not to comment at that point but warmly indicate the way to the interview room. The refusal to shake hands is probably an important issue, and the psychiatrist can keep this in mind for a potential inquiry if it is not brought up subsequently by the patient. Upon entering the interview room, if the patient has a coat, the psychiatrist can offer to take the coat and hang it up. The psychiatrist then indicates where the patient can sit. A brief pause can be helpful as there may be something the patient wants to say immediately. If not, the psychiatrist can inquire if the patient prefers to be called Mr. Smith, Thomas, or Tom. If this question is not asked, it is best to use the last name as some patients will find it presumptive to be called by their first name, especially if the interviewer is many years younger. These first few minutes of the encounter, even before the formal interview begins, can be crucial to the success of the interview and the development of a helpful patient–doctor relationship. The patient, who is often anxious, forms an initial impression of the psychiatrist and begins to make decisions as to how much can be shared with this doctor. Psychiatrists can convey interest and support by exhibiting a warm, friendly face and other nonverbal communications such as leaning forward in their chair. It is generally useful for the psychiatrist to indicate how much time is available

for the interview. The patient may have some questions about what will happen during this time, confidentiality, and other issues, and these questions should be answered directly by the psychiatrist. The psychiatrist can then continue with an open-ended inquiry, "*Why don't we start by you telling me what has led to your being here,*" or simply, "*What has led to your being here?*" Often the response to this question will establish whether or not the patient has been referred. When a referral has been made, it is important to elicit from the patient his or her understanding of why he or she has been referred. Not uncommonly, the patient may be uncertain as to why he or she has been referred or may even feel angry at the referrer, often a primary care physician.

Open-Ended Questions. As the patient responds to these initial questions, the psychiatrist must interact in a manner that allows the patient to tell his or her story. This is the primary goal of the data collection part of the interview to elicit the patient's story of his or her health and illness. Open-ended questions are a necessity to accomplish this objective. Open-ended questions identify an area but provide minimal structure as to how to respond. A typical open-ended question is, "*Tell me about your pain.*" This is in contrast to closed-ended questions that provide much structure and narrow the field from which a response may be chosen. "*Is your pain sharp?*" The ultimate closed-ended question leads to a "yes" or "no" answer. In the initial portion of the interview, questions should be primarily open-ended. As the patient responds, the psychiatrist reinforces the patient continuing by nodding or other supportive interventions. As the patient continues to share his or her story about an aspect of his or her health or illness, the psychiatrist may ask some increasingly closed-ended questions to understand some of the specifics of the history. Then, when that area is understood, the psychiatrist may make a transition to another area again using open-ended questions and eventually closed-ended questions until that area is well described. Hence, the interview should not be a single funnel of open-ended questions in the beginning and closed-ended questions at the end of the interview but rather a series of funnels, each of which begins with open-ended questions.

Elements of the Initial Psychiatric Interview

The interview is now well launched into the present illness. Table 1-1 lists the sections or parts of the initial psychiatric interview. Although not necessarily obtained during the interview in precisely this order, these are the categories that conventionally have been used to organize and record the elements of the evaluation.

The two overarching elements of the psychiatric interview are patient history and the mental status examination. The patient history is based on the subjective report of the patient and, in some cases, the account of collaterals, including other health care providers, family, and other caregivers. The mental status examination, on the other hand, is the interviewer's objective tool similar to the physical examination in other areas of medicine. The physical examination, although not part of the interview itself, is included because of its potential relevance in the psychiatric diagnosis and also because it usually is included as part of the psychiatric evaluation, especially in the inpatient setting. (Also, much relevant information can be verbally obtained by the physician as parts of the physical examination are performed.) Similarly, the formulation, diagnosis, and treatment plan are included because they are products of the interview and also influence the course of the interview dynamically as the interview moves back and forth pursuing, for example, whether specific diagnostic criteria are met or whether potential elements of the treatment plan are realistic. Details of the psychiatric interview are discussed below.



Table 1-1
Parts of the Initial Psychiatric Interview

I. Identifying data
II. Source and reliability
III. Chief complaint
IV. Present illness
V. Past psychiatric history
VI. Substance use/abuse
VII. Past medical history
VIII. Family history
IX. Developmental and social history

X. Review of systems
XI. Mental status examination
XII. Physical examination
XIII. Formulation
XIV. DSM-5 diagnoses
XV. Treatment plan

I. Identifying Data. This section is brief, one or two sentences, and typically includes the patient's name, age, sex, marital status (or significant other relationship), race or ethnicity, and occupation. Often the referral source is also included.

II. Source and Reliability. It is important to clarify where the information has come from, primarily if others have provided information or records reviewed, and the interviewer's assessment of how reliable the data are.

III. Chief Complaint. This should be the patient's presenting complaint, ideally in his or her own words. Examples include, "I'm depressed" or "I have a lot of anxiety."

A 64-year-old man presented in a psychiatric emergency room with a chief complaint, "I'm melting away like a snowball." He had become increasingly depressed over 3 months. Four weeks before the emergency room visit, he had seen his primary care physician who had increased his antidepressant medication (imipramine) from 25 to 75 mg and also added hydrochlorothiazide (50 mg) because of mild hypertension and slight pedal edema. Over the ensuing 4 weeks, the patient's condition deteriorated. In the emergency department, he was noted to have a depressed mood, hopelessness, weakness, significant weight loss, and psychomotor retardation and was described as appearing "depleted." He also seemed to be dehydrated, and blood work indicated he was hypokalemic. Examination of his medication revealed that the medication bottles had been mislabeled; he was taking 25 mg of imipramine (generally a nontherapeutic dose) and 150 mg of hydrochlorothiazide. He was indeed, "melting away like a snowball." Fluid and potassium

replacement and a therapeutic dose of an antidepressant resulted in significant improvement.

IV. History of Present Illness. The present illness is a chronologic description of the evolution of the symptoms of the current episode. Also, the account should include any other changes that have occurred during this same period in the patient's interests, interpersonal relationships, behaviors, personal habits, and physical health. As noted above, the patient may provide much of the essential information for this section in response to an open-ended question such as, "*Can you tell me in your own words what brings you here today?*" Other times the clinician may have to lead the patient through parts of the presenting problem. Details that should be gathered include the length of time that the current symptoms have been present and whether there have been fluctuations in the nature or severity of those symptoms over time. ("I have been depressed for the past 2 weeks" vs. "I've had depression all my life"). The presence or absence of stressors should be established, and these may include situations at home, work, school, legal issues, medical comorbidities, and interpersonal difficulties. Also important are factors that alleviate or exacerbate symptoms such as medications, support, coping skills, or time of day. The essential questions to be answered in the history of the present illness include what (symptoms), how much (severity), how long, and associated factors. It is also essential to identify why the patient is seeking help now and what are the "triggering" factors ("I'm here now because my girlfriend told me if I don't get help with this nervousness she is going to leave me."). Identifying the setting in which the illness began can be revealing and helpful in understanding the etiology of or significant contributors to the condition. If any treatment has been received for the current episode, it should be defined in terms of who saw the patient and how often, what was done (e.g., psychotherapy or medication), and the specifics of the modality used. Also, is that treatment continuing and, if not, why not? The psychiatrist should be alert for any hints of abuse by former therapists as this experience, unless addressed, can be a significant impediment to a healthy and helpful therapeutic alliance.

Table 1-2

Psychiatric Review of Systems

1. Mood
   A. Depression: Sadness, tearfulness, sleep, appetite, energy, concentration, sexual function, guilt, psychomotor agitation or slowing, interest. A common pneumonic used to remember the symptoms of major depression is SIGECAPS (**S**leep, **I**nterest, **G**uilt, **E**nergy, **C**oncentration, **A**ppetite, **P**sychomotor agitation or slowing, **S**uicidality).
   B. Mania: Impulsivity, grandiosity, recklessness, excessive energy, decreased need for sleep, increased spending beyond means, talkativeness, racing thoughts, hypersexuality.
   C. Mixed/Other: Irritability, liability.
2. Anxiety
   A. Generalized anxiety symptoms: Where, when, who, how long, how frequent.
   B. Panic disorder symptoms: How long until peak, somatic symptoms including racing heart, sweating, shortness of breath, trouble swallowing, sense of doom, fear of recurrence, agoraphobia.
   C. Obsessive-compulsive symptoms: Checking, cleaning, organizing, rituals, hang-ups, obsessive thinking, counting, rational vs. irrational beliefs.
   D. Posttraumatic stress disorder: Nightmares, flashbacks, startle response, avoidance.
   E. Social anxiety symptoms.
   F. Simple phobias, for example, heights, planes, spiders, etc.
3. Psychosis
   A. Hallucinations: Auditory, visual, olfactory, tactile.
   B. Paranoia.
   C. Delusions: TV, radio, thought broadcasting, mind control, referential thinking.
   D. Patient’s perception: Spiritual or cultural context of symptoms, reality testing.

4. Other
    A. Attention-deficit/hyperactivity disorder symptoms.
    B. Eating disorder symptoms: Binging, purging, excessive exercising.

Often it can be helpful to include a psychiatric review of systems in conjunction with the history of the present illness to help rule in or out psychiatric diagnoses with pertinent positives and negatives. This may help to identify whether some comorbid disorders or disorders are actually more bothersome to the patient but are not initially identified for a variety of reasons. This review can be split into four major categories of mood, anxiety, psychosis, and other (Table 1-2). The clinician will want to ensure that these areas are covered in the comprehensive psychiatric interview.

V. Past Psychiatric History. In the past psychiatric history, the clinician should obtain information about all psychiatric illnesses and their course over the patient's lifetime, including symptoms and treatment. Because comorbidity is the rule rather than the exception, also to prior episodes of the same illness (e.g., past episodes of depression in an individual who has a major depressive disorder), the psychiatrist should also be alert for the signs and symptoms of other psychiatric disorders. Description of past symptoms should include when they occurred, how long they lasted, and the frequency and severity of episodes.

Past treatment episodes should be reviewed in detail. These include outpatient treatment such as psychotherapy (individual, group, couple, or family), day treatment or partial hospitalization, inpatient treatment, including voluntary or involuntary, and what precipitated the need for the higher level of care, support groups, or other forms of treatment such as vocational training. Medications and other modalities such as electroconvulsive therapy (ECT), light therapy, or alternative treatments should be carefully reviewed. One should explore what was tried (may have to offer lists of names to patients), how long and at what doses they were used (to establish the adequacy of the trials), and why they were stopped. Essential questions include what the response to the medication or modality and whether there were side effects was. It is also helpful to establish whether there was reasonable compliance with the recommended treatment. The

psychiatrist should also inquire whether a diagnosis was made, what it was, and who made the diagnosis. Although a diagnosis made by another clinician should not be automatically accepted as valid, it is essential information that can be used by the psychiatrist in forming his or her opinion.

Special consideration should be given to establishing a lethality history that is important in the assessment of current risk. Past suicidal ideation, intent, plan, and attempts should be reviewed, including the nature of attempts, perceived lethality of the attempts, save potential, suicide notes, giving away things, or other death preparations. Violence and homicidality history will include any violent actions or intent. Specific questions about domestic violence, legal complications, and the outcome of the victim may help define this history more clearly. History of nonsuicidal self-injurious behavior should also be covered, including any history of cutting, burning, banging head, and biting oneself. The feelings, including relief of distress that accompany or follow the behavior, should also be explored as well as the degree to which the patient has gone to hide the evidence of these behaviors.

VI. Substance Use, Abuse, and Addictions. A careful review of substance use, abuse, and addictions are essential to the psychiatric interview. The clinician should keep in mind that this information may be difficult for the patient to discuss, and a nonjudgmental style will elicit more accurate information. If the patient seems reluctant to share such information, specific questions may be helpful (e.g., "*Have you ever used marijuana?*" or "*Do you typically drink alcohol every day?*"). History of use should include which substances have been used, including alcohol, drugs, medications (prescribed or not prescribed to the patient), and routes of use (oral, snorting, or intravenous). The frequency and amount of use should be determined, keeping in mind the tendency for patients to minimize or deny use that may be perceived as socially unacceptable. Also, there are many misconceptions about alcohol that can lead to erroneous data. The definition of alcohol may be misunderstood, for example, "No, I don't use alcohol," yet later in the same interview, "I drink a fair amount of beer." Also, the amount of alcohol can be confused with the volume of the drink: "I'm not worried about my alcohol use. I mix my drinks, and I add a lot of water." in response to a follow-up question, "*How much bourbon? Probably three or four shots?*" Tolerance, the need for increasing amounts of use, and any withdrawal

symptoms should be established to help determine abuse versus dependence. The impact of use on social interactions, work, school, legal consequences, and driving while intoxicated (DWI) should be covered. Some psychiatrists use a brief standardized questionnaire, the CAGE or RAPS4, to identify alcohol abuse or dependence.

CAGE includes four questions: Have you ever **C**ut down on your drinking? Have people **A**nnoyed you by criticizing your drinking? Have you ever felt bad or **G**uilty about your drinking? Have you ever had a drink the first thing in the morning, as an **E**ye-opener, to steady your nerves or get rid of a hangover? The Rapid Alcohol Problem Screen 4 (RAPS4) also consists of four questions: Have you ever felt guilty after drinking (**R**emorse), could not remember things said or did after drinking (**A**mnesia), failed to do what was normally expected after drinking (**P**erform), or had a morning drink (**S**tarter)?

Any periods of sobriety should be noted, including the length of time and setting such as in jail, legally mandated, and so forth. A history of treatment episodes should be explored, including inpatient detoxification or rehabilitation, outpatient treatment, group therapy, or other settings including self-help groups, Alcoholics Anonymous (AA) or Narcotics Anonymous (NA), halfway houses, or group homes.

Current substance abuse or dependence can have a significant impact on psychiatric symptoms and treatment course. The patient's readiness for change should be determined, including whether they are in the precontemplative, contemplative, or action phase. Referral to the appropriate treatment setting should be considered.

Other vital substances and addictions that should be covered in this section include tobacco and caffeine use, gambling, eating behaviors, and Internet use. Exploration of tobacco use is especially crucial because persons abusing substances are more likely to die as a result of tobacco use than because of the identified abused substance. Gambling history should include casino visits, horse racing, lottery, and scratch cards, and sports betting. Addictive type eating may include binge eating disorder. Overeaters Anonymous (OA) and Gamblers Anonymous (GA) are 12-step programs, similar to AA, for patients with addictive eating behaviors and gambling addictions.

VII. Past Medical History. The past medical history includes an account of

major medical illnesses and conditions as well as treatments, both past, and present. Any past surgeries should also be reviewed. It is essential to understand the patient's reaction to these illnesses and the coping skills employed. The past medical history is an important consideration when determining potential causes of mental illness as well as comorbid or confounding factors and may dictate possible treatment options or limitations. Medical illnesses can precipitate a psychiatric disorder (e.g., anxiety disorder in an individual recently diagnosed with cancer), mimic a psychiatric disorder (hyperthyroidism resembling an anxiety disorder), be precipitated by a psychiatric disorder or its treatment (a metabolic syndrome in a patient on a second-generation antipsychotic medication), or influence the choice of treatment of a psychiatric disorder (renal disorder and the use of lithium carbonate). It is vital to pay special attention to neurologic issues, including seizures, head injury, and pain disorder. Any known history of prenatal or birthing problems or issues with developmental milestones should be noted. In women, a reproductive and menstrual history is essential as well as a careful assessment of the potential for current or future pregnancy. ("*How do you know you are not pregnant?*" may be answered with "Because I have had my tubes tied" or "I just hope I'm not.")

A careful review of all current medications is critical. This should include all current psychiatric medications with attention to how long they have been used, compliance with schedules, the effect of the medications, and any side effects. It is often helpful to be very specific in determining compliance and side effects, including asking questions such as, "*How many days of the week are you able to actually take this medication?*" or "*Have you noticed any change in your sexual function since starting this medication?*" as the patient may not spontaneously offer this information, which may be embarrassing or perceived to be treatment interfering.

Nonpsychiatric medications, over-the-counter medications, sleep aids, herbal, and alternative medications should also be reviewed. These can all potentially have psychiatric implications, including side effects or produce symptoms, as well as potential medication interactions dictating treatment options. Optimally the patient should be asked to bring all medications currently being taken, prescribed or not, over-the-counter preparations, vitamins, and herbs to the interview.

Allergies to medications must be covered, including which medication and

the nature of the extent of, and the treatment of the allergic response. Psychiatric patients should be encouraged to have adequate and regular medical care. The sharing of appropriate information among the primary care physicians, other medical specialists, and the psychiatrist can be beneficial for optimal patient care. The initial interview is an opportunity to reinforce that concept with the patient. At times a patient may not want the information to be shared with his or her primary care physician. This wish should be respected, although it may be useful to explore if there is some information that can be shared. Often patients want to restrict certain social or family information (e.g., an extramarital affair) but are comfortable with other details (medication prescribed) being shared.

VIII. Family History. Because many psychiatric illnesses are familial and a significant number of those have a genetic predisposition, if not cause, a careful review of family history is an essential part of the psychiatric assessment. Furthermore, an accurate family history helps not only in defining a patient's potential risk factors for specific illnesses but also the formative psychosocial background of the patient. Psychiatric diagnoses, medications, hospitalizations, substance use disorders, and lethality history should all be covered. The importance of these issues is highlighted, for example, by the evidence that, at times, there appears to be a familial response to medications, and a family history of suicide is a significant risk factor for suicidal behaviors in the patient. The interviewer must keep in mind that the diagnosis ascribed to a family member may or may not be accurate, and some data about the presentation and treatment of that illness may be helpful. Medical illnesses present in family histories may also be significant in both the diagnosis and the treatment of the patient.

An example is a family history of diabetes or hyperlipidemia affecting the choice of antipsychotic medication that may carry a risk for the development of these illnesses in the patient. Family traditions, beliefs, and expectations may also play a significant role in the development, expression, or course of the illness. Also, the family history is essential in identifying potential support as well as stresses for the patient and, depending on the degree of disability of the patient, the availability and adequacy of likely caregivers.

IX. Developmental and Social History. The developmental and social

history reviews the stages of the patient's life. It is an essential tool in determining the context of psychiatric symptoms and illnesses and may identify some of the significant factors in the evolution of the disorder. Frequently, current psychosocial stressors will be revealed in the course of obtaining a social history. It can often be helpful to review the social history chronologically to ensure all information is covered.

Any available information concerning prenatal or birthing history and developmental milestones should be noted. For the large majority of adult patients, such information is not readily available, and when it is, it may not be entirely accurate. Childhood history will include childhood home environments, including members of the family and social environment, including the number and quality of friendships. A detailed school history including how far the patient went in school and how old he or she was at that level, any special education circumstances or learning disorders, behavioral problems at school, academic performance, and extracurricular activities should be obtained. Childhood physical and sexual abuse should be carefully queried.

Work history will include types of jobs, performance at jobs, reasons for changing jobs, and current work status. The nature of the patient's relationships with supervisors and coworkers should be reviewed. The patient's income, financial issues, and insurance coverage, including pharmacy benefits, are often significant issues.

Military history, where applicable, should be noted, including rank achieved, combat exposure, disciplinary actions, and discharge status. Marriage and relationship history, including sexual preferences and current family structure, should be explored. This should include the patient's capacity to develop and maintain stable and mutually satisfying relationships as well as issues of intimacy and sexual behaviors. Current relationships with parents, grandparents, children, and grandchildren are an essential part of social history. Legal history is also relevant, especially any pending charges or lawsuits. The social history also includes hobbies, interests, pets, and leisure time activities and how this has fluctuated over time. It is important to identify cultural and religious influences on the patient's life and current religious beliefs and practices. A brief overview of the sexual history is given in Table 1-3.

Table 1-3
Sexual History

1. Screening questions
   a. Are you sexually active?
   b. Have you noticed any changes or problems with sex recently?
2. Developmental
   a. Acquisition of sexual knowledge
   b. Onset of puberty/menarche
   c. Development of sexual identity and orientation
   d. First sexual experiences
   e. Sex in romantic relationship
   f. Changing experiences or preferences over time
   g. Sex and advancing age
3. Clarification of sexual problems
   a. Desire phase
      Presence of sexual thoughts or fantasies:
      When do they occur and what is their object?
      Who initiates sex and how?
   b. Excitement phase
      Difficulty in sexual arousal (achieving or maintaining erections, lubrication), during foreplay and preceding orgasm
   c. Orgasm phase
      Does orgasm occur?
      Does it occur too soon or too late?
      How often and under what circumstances does orgasm occur?
      If orgasm does not occur, is it because of not being excited or lack of orgasm despite being aroused?
   d. Resolution phase
      What happens after sex is over (e.g., contentment, frustration, continued arousal)?

X. Review of Systems. The review of systems attempts to capture any current physical or psychological signs and symptoms not already identified

in the present illness. Particular attention is paid to neurologic and systemic symptoms (e.g., fatigue or weakness). Illnesses that might contribute to the presenting complaints or influence the choice of therapeutic agents should be carefully considered (e.g., endocrine, hepatic, or renal disorders). Generally, the review of systems is organized by the major systems of the body.

XI. Mental Status Examination. The mental status examination (MSE) is the psychiatric equivalent of the physical examination in the rest of medicine. The MSE explores all the areas of mental functioning and denotes evidence of signs and symptoms of mental illnesses. Data are gathered for the mental status examination throughout the interview from the initial moments of the interaction, including what the patient is wearing and their general presentation. Most of the information does not require direct questioning, and the information gathered from observation may give the clinician a different dataset than patient responses. Direct questioning augments and rounds out the MSE. The MSE provides the clinician with a snapshot of the patient's mental status at the time of the interview and is useful for subsequent visits to compare and monitor changes over time. The psychiatric MSE includes cognitive screening most often in the form of the Mini-Mental Status Examination (MMSE), but the MMSE is not to be confused with the MSE overall. The components of the MSE are presented in this section in the order one might include them in the written note for organizational purposes. Still, as noted above, the data are gathered throughout the interview.

APPEARANCE AND BEHAVIOR. This section consists of a general description of how the patient looks and acts during the interview. Does the patient appear to be his or her stated age, younger or older? Is this related to the patient's style of dress, physical features, or style of interaction? Items to be noted include what the patient is wearing, including body jewelry, and whether it is appropriate for the context. For example, a patient in a hospital gown would be appropriate in the emergency room or inpatient unit but not in an outpatient clinic. Distinguishing features, including disfigurations, scars, and tattoos, are noted. Grooming and hygiene also are included in the overall appearance and can be clues to the patient's level of functioning.

The description of a patient's behavior includes a general statement about whether he or she is exhibiting acute distress and then a more specific

statement about the patient's approach to the interview. The patient may be described as cooperative, agitated, disinhibited, disinterested, and so forth. Once again, appropriateness is an essential factor to consider in the interpretation of the observation. If a patient is brought involuntarily for examination, it may be appropriate, certainly understandable, that he or she is somewhat uncooperative, especially at the beginning of the interview.

MOTOR ACTIVITY. Motor activity may be described as normal, slowed (bradykinesia), or agitated (hyperkinesia). This can give clues to diagnoses (e.g., depression vs. mania) as well as confounding neurologic or medical issues. Gait, freedom of movement, any unusual or sustained postures, pacing, and hand wringing are described. The presence or absence of any tics should be noted, as should be jitteriness, tremor, apparent restlessness, lip-smacking, and tongue protrusions. These can be clues to adverse reactions or side effects of medications such as tardive dyskinesia, akathisia, or parkinsonian features from antipsychotic medications or suggestion of symptoms of illnesses such as attention-deficit/hyperactivity disorder (ADHD).

SPEECH. Evaluation of speech is an integral part of the MSE. Elements considered include fluency, amount, rate, tone, and volume. Fluency can refer to whether the patient has full command of the English language as well as potentially more subtle fluency issues such as stuttering, word-finding difficulties, or paraphasic errors. (A Spanish-speaking patient with an interpreter would be considered not fluent in English, but an attempt should be made to establish whether he or she is fluent in Spanish.) The evaluation of the amount of speech refers to whether it is normal, increased, or decreased. Decreased amounts of speech may suggest several different things ranging from anxiety or disinterest to thought blocking or psychosis. Increased amounts of speech often (but not always) are suggestive of mania or hypomania. A related element is the speed or rate of speech. Is it slowed or rapid (pressured)? Finally, speech can be evaluated for its tone and volume. Descriptive terms for these elements include irritable, anxious, dysphoric, loud, quiet, timid, angry, or childlike.

MOOD. The terms *mood* and *affect* vary in their definition, and several authors

have recommended combining the two elements into a new label, “emotional expression.” Traditionally, *mood* is defined as the patient’s internal and sustained emotional state. Its experience is subjective, and hence it is best to use the patient’s own words in describing his or her mood. Terms such as “sad,” “angry,” “guilty,” or “anxious” are common descriptions of mood.

AFFECT. *Affect* differs from mood in that it is the expression of mood or what the patient’s mood appears to be to the clinician. Affect is often described with the following elements: quality, quantity, range, appropriateness, and congruence. Terms used to describe the quality (or tone) of a patient’s affect include dysphoric, happy, euthymic, irritable, angry, agitated, tearful, sobbing, and flat. Speech is often an essential clue to the assessment of affect, but it is not exclusive. Quantity of affect is a measure of its intensity. Two patients, both described as having depressed affect can be very different if one is described as mildly depressed and the other as severely depressed. The range can be restricted, normal, or labile. *Flat* is a term that has been used for a severely restricted range of affect that is described in some patients with schizophrenia. Appropriateness of affect refers to how the affect correlates to the setting. A patient who is laughing at a solemn moment of a funeral service is described as having an inappropriate affect. Affect can also be congruent or incongruent with the patient’s described mood or thought content. A patient may report feeling depressed or describe a depressive theme but do so with laughter, smiling, and no suggestion of sadness.

THOUGHT CONTENT. Essentially, thought content is what thoughts are occurring to the patient. This is inferred by what the patient spontaneously expresses, as well as responses to specific questions aimed at eliciting particular pathology. Some patients may perseverate or ruminate on specific content or thoughts. They may focus on material that is considered obsessive or compulsive. *Obsessional thoughts* are unwelcome and repetitive thoughts that intrude into the patient’s consciousness. They are generally ego alien and resisted by the patient. *Compulsions* are repetitive, ritualized behaviors that patients feel compelled to perform to avoid an increase in anxiety or some dreaded outcome. Another broad category of thought content pathology is delusions. *Delusions* are false, fixed ideas that are not shared by others and can be divided into bizarre and nonbizarre (nonbizarre delusions refer to

thought content that is not true but is not out of the realm of possibility). Common delusions include grandiose, erotomanic, jealous, somatic, and persecutory. It is often helpful to suggest delusional content to patients who may have learned not to spontaneously discuss them. Questions that can be helpful include, "*Do you ever feel like someone is following you or out to get you*?" and "*Do you feel like the TV or radio has a special message for you?*" An affirmative answer to the latter question indicates an "idea of reference." Paranoia can be closely related to delusional material and can range from "soft" paranoia, such as general suspiciousness, to more severe forms that impact daily functioning. Questions that elicit paranoia can include asking about the patient worrying about cameras, microphones, or the government.

Suicidality and homicidality fall under the category of thought content, but here are discussed separately because of their particular importance in being addressed in every initial psychiatric interview. Simply asking if someone is suicidal or homicidal is not adequate. One must get a sense of ideation, intent, plan, and preparation. Although completed suicide is hugely challenging to predict accurately, there are identified risk factors, and these can be used in conjunction with an evaluation of the patient's intent and plan for acting on thoughts of suicide.

THOUGHT PROCESS. Thought process differs from thought content in that it does not describe what the person is thinking but rather how the thoughts are formulated, organized, and expressed. A patient can have normal thought process with significantly delusional thought content. Conversely, there may be generally normal thought content but significantly impaired thought process. Normal thought process is typically described as linear, organized, and goal-directed. With flight of ideas, the patient rapidly moves from one thought to another, at a pace that is difficult for the listener to keep up with, but all of the ideas are logically connected. The circumstantial patient overincludes details and material that is not directly relevant to the subject or an answer to the question but does eventually return to address the subject or answer the question. Typically the examiner can follow a circumstantial train of thought, seeing connections between the sequential statements. Tangential thought process may at first appear similar, but the patient never returns to the original point or question. The tangential thoughts are seen as irrelevant and related in a minor, insignificant manner. Loose thoughts or associations

differ from circumstantial and tangential thoughts in that with loose thoughts it is difficult or impossible to see the connections between the sequential content. Perseveration is the tendency to focus on a specific idea or content without the ability to move on to other topics. The perseverative patient will repeatedly come back to the same topic despite the interviewer's attempts to change the subject. Thought blocking refers to a disordered thought process in which the patient appears to be unable to complete a thought. The patient may stop midsentence or midthought and leave the interviewer waiting for the completion. When asked about this, patients will often remark that they don't know what happened and may not remember what was being discussed. Neologisms refer to a new word or condensed combination of several words that is not a true word and is not readily understandable. However, sometimes the intended meaning or partial meaning may be apparent. Word salad is speech characterized by confused, and often repetitious, language with no apparent meaning or relationship attached to it. A description of formal thought disorders is given in Table 1-4.

PERCEPTUAL DISTURBANCES. Perceptual disturbances include hallucinations, illusions, depersonalization, and derealization. Hallucinations are perceptions in the absence of stimuli to account for them. Auditory hallucinations are the hallucinations most frequently encountered in the psychiatric setting. Other hallucinations can include visual, tactile, olfactory, and gustatory (taste). In the North American culture, nonauditory hallucinations are often clues that there is a neurologic, medical, or substance withdrawal issue rather than a primary psychiatric issue. In other cultures, visual hallucinations have been reported to be the most common form of hallucinations in schizophrenia. The interviewer should make a distinction between a real hallucination and a misperception of stimuli (illusion). Hearing the wind rustle through the trees outside one's bedroom and thinking a name is being called is an illusion. Hypnagogic hallucinations (at the interface of wakefulness and sleep) may be normal phenomena. At times patients without psychosis may hear their name called or see flashes or shadows out of the corners of their eyes. In describing hallucinations, the interviewer should include what the patient is experiencing, when it occurs, how often it occurs, and whether or not it is uncomfortable (ego-dystonic). In the case of auditory hallucinations, it can be useful to learn if the patient hears words, commands, or conversations and

whether the voice is recognizable to the patient.

Depersonalization is a feeling that one is not oneself or that something has changed. Derealization is a feeling that one's environment has changed in some strange way that is difficult to describe.



Table 1-4
Formal Thought Disorders

Circumstantiality. Overinclusion of trivial or irrelevant details that impede the sense of getting to the point.

Clang associations. Thoughts are associated by the sound of words rather than by their meaning (e.g., through rhyming or assonance).

Derailment. (Synonymous with loose associations.) A breakdown in both the logical connection between ideas and the overall sense of goal directedness. The words make sentences, but the sentences do not make sense.

Flight of ideas. A succession of multiple associations so that thoughts seem to move abruptly from idea to idea; often (but not invariably) expressed through rapid, pressured speech.

Neologism. The invention of new words or phrases or the use of conventional words in idiosyncratic ways.

Perseveration. Repetition of out of context words, phrases, or ideas.

Tangentiality. In response to a question, the patient gives a reply that is appropriate to the general topic without actually answering the question. Example:

Doctor: "Have you had any trouble sleeping lately?"
Patient: "I usually sleep in my bed, but now I'm sleeping on the sofa."

Thought blocking. A sudden disruption of thought or a break in the flow of ideas.

COGNITION. The elements of cognitive functioning that should be assessed are alertness, orientation, concentration, memory (both short and long term), calculation, fund of knowledge, abstract reasoning, insight, and judgment.

Note should be made of the patient's level of alertness. The amount of

detail in assessing cognitive function will depend on the purpose of the examination and also what has already been learned in the interview about the patient's level of functioning, performance at work, handling daily chores, balancing one's checkbook, among others. Also, the psychiatrist will have already elicited data concerning the patient's memory for both remote and recent past. A general sense of intellectual level and how much schooling the patient has had can help distinguish intelligence and educational issues versus cognitive impairment that might be seen in delirium or dementia. Table 1-5 presents an overview of the questions used to test cognitive function in the mental status examination.

Table 1-5
Questions Used to Test Cognitive Functions in the Sensorium Section of the Mental Status Examination

1. Alertness	(Observation)
2. Orientation	What is your name? Who am I? What place is this? Where is it located? What city are we in?
3. Concentration	Starting at 100, count backward by 7 (or 3). Say the letters of the alphabet backward starting with Z. Name the months of the year backward starting with December.
4. Memory: Immediate	Repeat these numbers after me: 1, 4, 9, 2, 5.
Recent	What did you have for breakfast? What were you doing before we started talking this morning? I want you to remember these three things: a yellow pencil, a cocker spaniel, and Cincinnati. After a few minutes I'll ask you to repeat them.
Long term	What was your address when you were in the

	third grade? Who was your teacher? What did you do during the summer between high school and college?
5. Calculations	If you buy something that costs $3.75 and you pay with a $5 bill, how much change should you get? What is the cost of three oranges if a dozen oranges cost $4.00?
6. Fund of knowledge	What is the distance between New York and Los Angeles? What body of water lies between South America and Africa?
7. Abstract reasoning	Which one does not belong in this group: a pair of scissors, a canary, and a spider? Why? How are an apple and an orange alike?

ABSTRACT REASONING. Abstract reasoning is the ability to shift back and forth between general concepts and specific examples. Having the patient identify similarities between like objects or concepts (apple and pear, bus and airplane, or a poem and a painting) as well as interpreting proverbs can be useful in assessing one's ability to abstract. Cultural and educational factors and limitations should be kept in mind when assessing the ability to abstract. Occasionally, the inability to abstract or the idiosyncratic manner of grouping items can be dramatic.

INSIGHT. Insight, in the psychiatric evaluation, refers to the patient's understanding of how he or she is feeling, presenting, and functioning as well as the potential causes of his or her psychiatric presentation. The patient may have no insight, partial insight, or full insight. A component of insight often is reality testing in the case of a patient with psychosis. An example of intact reality testing would be, "I know that there are not really little men talking to me when I am alone, but I feel like I can see them and hear their voices." As indicated by this example, the amount of insight is not an indicator of the severity of the illness. A person with psychosis may have good insight, while

a person with a mild anxiety disorder may have little or no insight.

JUDGMENT. Judgment refers to the person's capacity to make good decisions and act on them. The level of judgment may or may not correlate to the level of insight. A patient may have no insight into his or her illness but have good judgment. It has been traditional to use hypothetical examples to test judgment, for example, "*What would you do if you found a stamped envelope on the sidewalk?*" It is better to use real situations from the patient's own experience to test judgment. The important issues in assessing judgment include whether a patient is doing things that are dangerous or going to get him or her into trouble and whether the patient can effectively participate in his or her own care. Significantly impaired judgment can be cause for considering a higher level of care or more restrictive setting such as inpatient hospitalization. Table 1-6 lists some common questions for the psychiatric history and mental status.

XII. Physical Examination. The inclusion and extent of physical examination will depend on the nature and setting of the psychiatric interview. In the outpatient setting, little or no physical examination may be routinely performed, while in the emergency room or inpatient setting, a more complete physical examination is warranted. Vital signs, weight, waist circumference, body mass index, and height may be important measurements to follow particularly given the potential effects of psychiatric medications or illnesses on these parameters. The Abnormal Involuntary Movement Scale (AIMS) is an important screening test to be followed when using antipsychotic medication to monitor for potential side effects such as tardive dyskinesia. A focused neurologic evaluation is an important part of the psychiatric assessment.

In those instances where a physical examination is not performed the psychiatrist should ask the patient when the last physical examination was performed and by whom. As part of the communication with that physician, the psychiatrist should inquire about any abnormal findings.

XIII. Formulation. The culmination of the data-gathering aspect of the psychiatric interview is developing a formulation and diagnosis (diagnoses) as well as recommendations and treatment planning. In this part of the

evaluation process, the data gathering is supplanted by data processing where the various themes contribute to a biopsychosocial understanding of the patient's illness. Although the formulation is placed near the end of the reported or written evaluation, it is developed as part of a dynamic process throughout the interview as new hypotheses are created and tested by further data that are elicited. The formulation should include a brief summary of the patient's history, presentation, and current status. It should include discussion of biologic factors (medical, family, and medication history) as well as psychological factors such as childhood circumstances, upbringing, and past interpersonal interactions and social factors including stressors, and contextual circumstances such as finances, school, work, home, and interpersonal relationships. These elements should lead to a differential diagnosis of the patient's illness (if any) as well as a provisional diagnosis. Finally, the formulation should include a summary of the safety assessment, which contributes to the determination of the level of care recommended or required.

XIV. Treatment Planning. The assessment and formulation will appear in the written note correlating to the psychiatric interview. Still, the discussion with the patient may only be a summary of this assessment geared toward the patient's ability to understand and interpret the information. Treatment planning and recommendations, in contrast, are integral parts of the psychiatric interview and should be explicitly discussed with the patient in detail.



Table 1-6
Common Questions for Psychiatric History and Mental Status

Topic	Questions	Comments and Clinical Hints
Identifying data	Be direct in obtaining identifying data. Request specific answers.	If patient cannot cooperate, get information from family member or friend; if referred by a

		physician, obtain medical record.
Chief complaint (CC)	Why are you going to see a psychiatrist? What brought you to the hospital? What seems to be the problem?	Records answers verbatim; a bizarre complaint points to psychotic process.
History of present illness (HPI)	When did you first notice something happening to you? Were you upset about anything when symptoms began? Did they begin suddenly or gradually?	Record in patient's own words as much as possible. Get history of previous hospitalizations and treatment. Sudden onset of symptoms may indicate drug-induced disorder.
Previous psychiatric and medical disorders	Did you ever lose consciousness? Have a seizure?	Ascertain extent of illness, treatment, medications, outcomes, hospitals, doctors. Determine whether illness serves some additional purpose (secondary gain).
Personal history	Do you know anything about your birth? If so, from whom? How old was your mother when you were born? Your father?	Older mothers (>35) have high risk for Down syndrome babies; older fathers (>45) may contribute damaged sperm, producing deficits including schizophrenia.

Childhood	Toilet training? Bed-wetting? Sex play with peers? What is your first childhood memory?	Separation anxiety and school phobia are associated with adult depression; enuresis is associated with fire setting. Childhood memories before the age of 3 are usually imagined, not real.
Adolescence	Adolescents may refuse to answer questions, but they should be asked. Adults may distort memories of emotionally charged experiences. Sexual molestation?	Poor school performance is a sensitive indicator of emotional disorder. Schizophrenia begins in late adolescence.
Adulthood	Open-ended questions are preferable. Tell me about your marriage. Be nonjudgmental; What role does religion play in your life, if any? What is your sexual preference in a partner?	Depending on the chief complaint, some areas require more detailed inquiry. Manic patients frequently go into debt or are promiscuous. Overvalued religious ideas are associated with paranoid personality disorder.
Sexual history	Are there or have there been any problems or concerns about your sex life? How did you learn about sex? Has	Be nonjudgmental. Asking *when* masturbation began is a better approach than asking *do you* or

	there been any change in your sex drive?	*did you ever* masturbate.
Family history	Have any members in your family been depressed? Alcoholic? In a mental hospital? Describe your living conditions. Did you have your own room?	Genetic loading in anxiety, depression, schizophrenia. Get medication history of family (medications effective in family members for similar disorders may be effective in patient).
Mental Status		
General appearance	Introduce yourself and direct patient to take a seat. In the hospital, bring your chair to bedside; do not sit on the bed.	Unkempt and disheveled in cognitive disorder, pinpoint pupils in narcotic addiction, withdrawal and stooped posture in depression.
Motoric behavior	Have you been more active than usual? Less active? You may ask about obvious mannerisms, such as, “I notice that your hand still shakes, can you tell me about that?” Stay aware of smells, such as alcoholism/ketoacidosis.	Fixed posturing, odd behavior in schizophrenia. Hyperactive with stimulant (cocaine) abuse and in mania. Psychomotor retardation in depression; tremors with anxiety or medication side effect (lithium). Eye contact is normally made during the

		interview. Minimal eye contact in schizophrenia. Scanning of environment in paranoid states.
Attitude during interview	You may comment about attitude: “You seem irritated about something; is that an accurate observation?”	Suspiciousness in paranoia; seductive in hysteria; apathetic in conversion disorder (*la belle indifference*); punning (*witzelsucht*) in frontal lobe syndromes.
Mood	How do you feel? How are your spirits? Do you have thoughts that life is not worth living or that you want to harm yourself? Do you have plans to take your own life? Do you want to die? Has there been a change in your sleep habits?	Suicidal ideas in 25% of depressives; elation in mania. Early morning awakening in depression; decreased need for sleep in mania.
Affect	Observe nonverbal signs of emotion, body movements, facies, rhythm of voice (prosody). Laughing when talking about sad subjects, such as death, is inappropriate.	Changes in affect usual with schizophrenia: loss of prosody in cognitive disorder, catatonia. Do not confuse medication adverse effect with flat affect.

Speech	Ask patient to say “Methodist Episcopalian” to test for dysarthria.	Manic patients show pressured speech; paucity of speech in depression; uneven or slurred speech in cognitive disorders.
Perceptual disorders	Do you ever see things or hear voices? Do you have strange experiences as you fall asleep or upon awakening? Has the world changed in any way? Do you have strange smells?	Visual hallucinations suggest schizophrenia. Tactile hallucinations suggest cocainism, delirium tremens (DTs). Olfactory hallucinations common in temporal lobe epilepsy.
Thought content	Do you feel people want to harm you? Do you have special powers? Is anyone trying to influence you? Do you have strange body sensations? Are there thoughts that you can’t get out of your mind? Do you think about the end of the world? Can people read your mind? Do you ever feel the TV is talking to you? Ask about fantasies and dreams.	Are delusions congruent with mood (grandiose delusions with elated mood) or incongruent? Mood-incongruent delusions point to schizophrenia. Illusions are common in delirium. Thought insertion is characteristic of schizophrenia.
Thought process	Ask meaning of proverbs to test abstraction, such	Loose associations point to

	as, “People in glass houses should not throw stones.” Concrete answer is, “Glass breaks.” Abstract answers deal with universal themes or moral issues. Ask similarity between bird and butterfly (both alive), bread and cake (both food).	schizophrenia; flight of ideas to mania; inability to abstract to schizophrenia, brain damage.
Sensorium	What place is this? What is today’s date? Do you know who I am?	Delirium or dementia shows clouded or wandering sensorium. Orientation to person remains intact longer than orientation to time or place.
Remote memory (long-term memory)	Where were you born? Where did you go to school? Date of marriage? Birthdays of children? What were last week’s newspaper headlines?	Patients with dementia of the Alzheimer type retain remote memory longer than recent memory. Gaps in memory may be localized or filled in with confabulatory details. Hypermnesia is seen in paranoid personality.
Immediate memory (very short-term	Ask patient to repeat six digits forward, then backward (normal	Loss of memory occurs with cognitive, dissociative, or

memory)	responses). Ask patient to try to remember three nonrelated items; test patient after 5 min.	conversion disorder. Anxiety can impair immediate retention and recent memory. Anterograde memory loss (amnesia) occurs after taking certain drugs, such as benzodiazepines. Retrograde memory loss occurs after head trauma.
Concentration and calculation	Ask patient to count from 1 to 20 rapidly; do simple calculations (2 × 4, 4 × 9); do serial 7 test (i.e., subtract 7 from 100 and keep subtracting 7). How many nickels in $1.35?	Rule out medical cause for any defects vs. anxiety or depression (pseudodementia). Make tests congruent with educational level of patient.
Information and intelligence	Distance from New York City to Los Angeles. Name some vegetables. What is the largest river in the United States?	Check educational level to results. Rule out mental retardation, borderline intellectual functioning.
Judgment	What is the thing to do if you find an envelope in the street that is sealed, stamped, and addressed?	Impaired in brain disease, schizophrenia, borderline intellectual functioning, intoxication.
Insight level	Do you think you have a problem? Do you need	Impaired in delirium, dementia, frontal lobe

	treatment? What are your plans for the future?	syndrome, psychosis, borderline intellectual functioning.

From Sadock BJ, Sadock V. *Kaplan and Sadock's Pocket Handbook of Clinical Psychiatry*. Philadelphia, PA: Lippincott Williams & Wilkins, 2010, with permission.

The first part of treatment planning involves determining whether a treatment relationship is to be established between the interviewer and patient. Cases, where this may not be the case, include if the interview was done in consultation, for a legal matter or as a third-party review, or in the emergency room or other acute setting. If a treatment relationship is not being started, then the patient should be informed as to what the recommended treatment is (if any). In some instances, this may not be voluntary (as in the case of an involuntary hospitalization). In most cases, there should be a discussion of the options available so that the patient can participate in the decisions about the next steps. If a treatment relationship is being initiated, then the structure of that treatment should be discussed. Will the primary focus be on medication management, psychotherapy, or both? What will the frequency of visits be? How will the clinician be paid for service, and what are the expectations for the patient to be considered engaged in treatment?

Medication recommendations should include a discussion of possible therapeutic medications, the risks and benefits of no medication treatment, and alternative treatment options. The prescriber must obtain informed consent from the patient for any medications (or other treatments) initiated.

Other clinical treatment recommendations may include referral for psychotherapy, group therapy, chemical dependency evaluation or treatment, or medical assessment. There also may be recommended psychosocial interventions, including case management, group home or assisted living, social clubs, support groups such as a mental health alliance, the National Alliance for the Mentally Ill, and AA.

Collaboration with primary care doctors, specialists, or other clinicians should always be a goal, and proper patient consent must be obtained for this. Similarly, family involvement in a patient's care can often be a useful and integral part of treatment and requires proper patient consent.

A thorough discussion of safety planning and contact information should occur during the psychiatric interview. The clinician's contact information, as

well as after-hours coverage scheme, should be reviewed. The patient needs to be informed of what he or she should do in the case of an emergency, including using the emergency room or calling 911 or available crisis hotlines.

Techniques

General principles of the psychiatric interview, such as the patient–doctor relationship, open-ended interviewing, and confidentiality, are described above. Also, to the general principles, several specific techniques can be useful in obtaining information in a manner consistent with the general principles. These helpful techniques can be described as facilitating interventions and expanding interventions. Some interventions are generally counterproductive and interfere with the goals of helping the patient tell his or her story and reinforcing the therapeutic alliance.

Facilitating Interventions. These are some of the interventions that are effective in enabling the patient to continue sharing his or her story and also help promote a positive patient–doctor relationship. At times some of these techniques may be combined in a single intervention.

REINFORCEMENT. Reinforcement interventions, although seemingly simplistic, are very important in the patient sharing material about himself or herself and other influential individuals and events in the patient's life. Without these reinforcements, often, the interview will become less productive. A brief phrase such as "I see," "Go on," "Yes," "Tell me more," "Hmm," or "Uh-huh" all convey the interviewer's interest in the patient continuing. These phrases must fit naturally into the dialogue.

REFLECTION. By using the patient's words, the psychiatrist indicates that he or she has heard what the patient is saying and conveys an interest in hearing more.

This response is not a question. A question, with a slight inflection at the end, calls for some clarification. It should also not be said with a tone that is challenging or disbelieving but rather as a statement of fact. The fact is that this is the patient's experience that the psychiatrist hears. Sometimes it is helpful to paraphrase the patient's statement, so it doesn't sound like it is

coming from anautomation.

SUMMARIZING. Periodically during the interview, it is helpful to summarize what has been identified about a certain topic. This provides the opportunity for the patient to clarify or modify the psychiatrist's understanding and possibly add new material. When new material is introduced, the psychiatrist may decide to continue with a further exploration of the previous discussion and return to the new information at a later point.

EDUCATION. At times in the interview, it is helpful for the psychiatrist to educate the patient about the interview process.

REASSURANCE. It is often appropriate and helpful to provide reassurance to the patient. For example, accurate information about the usual course of an illness can decrease anxiety, encourage the patient to continue to discuss his or her illness, and strengthen his or her resolve to continue in treatment. It is generally inappropriate for psychiatrists to reassure patients when the psychiatrist does not know what the outcome will be. In these cases, psychiatrists can assure patients they will continue to be available and will help in whatever way they can.

ENCOURAGEMENT. It is difficult for many patients to come for a psychiatric evaluation. Often they are uncertain as to what will happen, and receiving encouragement can facilitate their engagement. Psychiatrists should be careful not to overstate the patient's progress in the interview. The psychiatrist may provide the patient with feedback about his or her efforts, but the secondary message should be that there is more work to be done.

ACKNOWLEDGMENT OF EMOTION. The interviewer needs to acknowledge the expression of emotion by the patient. This frequently leads to the patient sharing more feelings and being relieved that he or she can do so. Sometimes a nonverbal action, such as moving a tissue box closer, can suffice or be used adjunctly. If the display of the emotion is clear (e.g., patient openly crying), then it is not helpful to comment directly on the expression of the emotion. It is better to comment on the associated feelings.

HUMOR. At times the patient may make a humorous comment or tell a brief joke. It can be beneficial if the psychiatrist smiles, laughs, or even, when appropriate, add another punch line. This sharing of humor can decrease tension and anxiety and reinforce the interviewer's genuineness. It is crucial to be certain that the patient's comment was indeed meant to be humorous and that the psychiatrist clearly conveys that he or she is laughing with the patient, not at the patient.

SILENCE. Careful use of silence can facilitate the progression of the interview. The patient may need time to think about what has been said or to experience a feeling that has arisen in the interview. The psychiatrist, whose own anxiety results in any silence quickly being terminated, can retard the development of insight or the expression of feeling by the patient. On the other hand, extended or repeated silences can deaden an interview and become a struggle as to who can outwait the other. If the patient is looking at his or her watch or looking about the room, then it might be helpful to comment, "*It looks like there are other things on your mind.*" If the patient has become silent and looks like he or she is thinking about the subject, then the psychiatrist might ask, "*What thoughts do you have about that*?"

NONVERBAL COMMUNICATION. In many good interviews, the most common facilitating interventions are nonverbal. Nodding of the head, body posture including leaning toward the patient, body positioning becoming more open, moving the chair closer to the patient, putting down the pen and folder, and facial expressions including arching of eyebrows all indicate that the psychiatrist is concerned, listening attentively, and engaged in the interview. Although these interventions can be constructive, they can also be overdone, especially if the same action is repeated too frequently or done exaggeratedly. The interviewer does not want to reinforce the popular caricature of a psychiatrist nodding his or her head repeatedly regardless of the content of what is being said or the emotion being expressed.

EXPANDING INTERVENTIONS. Several interventions can be used to expand the focus of the interview. These techniques are helpful when the line of discussion has been sufficiently mined, at least for the time being, and the interviewer wants to encourage the patient to talk about other issues. These

interventions are most successful when a degree of trust has been established in the interview, and the patient feels that the psychiatrist is nonjudgmental about what is being shared.

CLARIFYING. At times carefully clarifying what the patient has said can lead to unrecognized issues or psychopathology.

A 62-year-old widow describes how it feels since her husband died 14 months ago. She repeatedly comments that “everything is empty inside.” The resident interprets this as meaning her world feels empty without her spouse and makes this interpretation on a few occasions. The patient’s nonverbal cues suggest that she is not on the same wavelength. The supervisor asks the patient to clarify what she means by “empty inside.” After some avoidance, the patient states that she is indeed empty inside; all her organs are missing—they have “disappeared.”

The resident’s interpretation may have been psychodynamically accurate, but a somatic delusion was not identified. The correct identification of what the patient was saying led to an exploration of other thoughts and other delusions were uncovered. This vignette of “missing” the delusion is an example of the interviewer “normalizing” what the patient is saying. The interviewer was using secondary process thinking in understanding the words of the patient, while the patient was using primary process thinking.

ASSOCIATIONS. As the patient describes his or her symptoms, other areas are related to a symptom that should be explored. For example, the symptom of nausea leads to questions about appetite, bowel habits, weight loss, and eating habits. Also, experiences that are temporally related may be investigated. When a patient is talking about his or her sleeping pattern, it can be a good opportunity to ask about dreams.

LEADING. Often, continuing the story can be facilitated by asking a “what,” “when,” “where,” or “who” question. Sometimes the psychiatrist may suggest or ask about something that has not been introduced by the patient but that the psychiatrist surmises may be relevant.

PROBING. The interview may point toward an area of conflict, but the patient may minimize or deny any difficulties. Gently encouraging the patient to talk more about this issue may be quite productive.

TRANSITIONS. Sometimes transitions occur very smoothly. The patient is talking about her primary education major in college and the psychiatrist asks, "*Did that lead to your work after college?*" On other occasions, the transition means moving to a different area of the interview, and a bridge statement is useful.

REDIRECTING. A difficult technique for unseasoned interviewers is redirecting the focus of the patient. If the interviewer is concentrating on reinforcing the patient's telling of his or her story, it can be especially difficult to move the interview in a different direction. However, this is often crucial to a successful interview because of the time constraints and the necessity to obtain a broad overview of the patient's life as well as the current problems. Also, the patient may, for conscious or often unconscious reasons, avoid certain important areas, and need guidance in approaching these subjects. Redirection can be used when the patient changes the topic or when the patient continues to focus on a nonproductive or well-covered area.

Obstructive Interventions. Although supportive and expanding techniques facilitate the gathering of information and the development of a positive patient–doctor relationship, several other interventions are not helpful for either task. Some of these activities are in the same categories as the more useful interventions but are unclear, unconnected, poorly timed, and not responsive to the patient's issues or concerns.

CLOSED-ENDED QUESTIONS. A series of closed-ended questions early in the interview can retard the natural flow of the patient's story and reinforces the patient giving one word or brief answers with little or no elaboration.

A patient can be a partner in the interview unless the psychiatrist blocks them. Many patients, some of whom have previous experiences in therapy, come prepared to talk about even painful matters. Over time, psychiatrists, especially if they have had the benefit of supervision, learn from patients, and refine their interviewing skills.

COMPOUND QUESTIONS. Some questions are difficult for patients to respond to because more than one answer is being sought.

WHY QUESTIONS. Especially early in the psychiatric interview, "why" questions are often nonproductive. Very often, the answer to that question is one of the reasons that the patient has sought help.

JUDGMENTAL QUESTIONS OR STATEMENTS. Judgmental interventions are generally nonproductive for the issue at hand and also inhibit the patient from sharing even more private or sensitive material. Instead of telling a patient that particular behavior was right or wrong, it would be better for the psychiatrist to help the patient reflect on how successful that behavior was.

MINIMIZING PATIENT'S CONCERNS. In an attempt to reassure patients, psychiatrists sometimes make the error of minimizing a concern. This can be counterproductive in that rather than being reassured, and the patient may feel that the psychiatrist does not understand what he or she is trying to express. It is much more productive to explore the concern; there is likely much more material that has not yet been shared.

PREMATURE ADVICE. Advice given too early is often bad advice because the interviewer does not yet know all of the variables. Also, it can preempt the patient from arriving at a plan for himself or herself.

PREMATURE INTERPRETATION. Even if it is accurate, a premature interpretation can be counterproductive as the patient may respond defensively and feel misunderstood.

TRANSITIONS. Some transitions are too abrupt and may interrupt essential issues that the patient is discussing.

NONVERBAL COMMUNICATION. The psychiatrist that repeatedly looks at a watch turns away from the patient, yawns, or refreshes the computer screen conveys boredom, disinterest, or annoyance. Just as reinforcing nonverbal communications can be powerful facilitators of a good interview, these obstructive actions can quickly shatter an interview and undermine the

patient–doctor relationship.

Closing of Interview. The last 5 to 10 minutes of the interview are crucial and are often not given sufficient attention by inexperienced interviewers. It is essential to alert the patient to the remaining time: "*We have to stop in about 10 minutes.*" Not infrequently, a patient will have kept an important issue or question until the end of the interview, and having at least a brief time to identify the issue is helpful. If there is to be another session, then the psychiatrist can indicate that this issue will be addressed at the beginning of the next session or ask the patient to bring it up at that time. If the patient repeatedly brings up important information at the end of sessions, then this should be explored as to its meaning. If the patient spontaneously brings up no such item, then it can be useful to ask the patient if any other issues have not been covered that the patient wanted to share. If such an issue can be dealt with in short order, then it should be; if not, then it can be put on the agenda for the next session. It can also be useful to allow the patient to ask a question: "*I've asked you a lot of questions today. Are there any other questions you'd like to ask me at this point?*"

If this interview was to be a single evaluative session, then a summary of the diagnosis and options for treatment should generally be shared with the patient (exceptions may be a disability or forensic evaluation for which it was established at the outset that a report would be made to the referring entity). If a primary care physician referred the patient, then the psychiatrist also indicates that he or she will communicate with the primary care physician and share the findings and recommendations. If this was not to be a single session and the patient will be seen again, then the psychiatrist may indicate that he or she and the patient can work further on the treatment plan in the next session. A mutually agreed upon time is arrived at, and the patient is escorted to the door.

Motivational Interviewing. Motivational interviewing is a technique used to motivate the patient to change his or her maladaptive behavior. The therapist relies on empathy to convey understanding, provides support by noting the patient's strengths, and explores the ambivalence and conflicting thoughts or feelings the patient may have about change. Guidance is provided in the interview by imparting information about issues (e.g., alcoholism,

diabetes) while at the same time getting the patient to talk about resistances to altering behavior. It has been used effectively in persons with substance use disorders to get them to join AA, to help change lifestyles, or to enter psychotherapy. It has the potential to combine diagnosis and therapy in a single interview with the patient and can be applied to a wide range of mental disorders.

Medical Record

Most psychiatrists take notes throughout the interview. Generally, these are not verbatim recordings, except for the chief complaint or other key statements. Many psychiatrists use a form that covers the basic elements in the psychiatric evaluation. Occasionally, patients may have questions or concerns about note-taking. These concerns, which often have to do with confidentiality, should be discussed (and during this discussion, notes should not be taken). After the discussion, it is rare for a patient to insist that notes not be taken. It is much more common for patients to feel comfortable about the note-taking, feeling reassured that their experiences and feelings are important enough to be written. However, too much attention to the record can be distracting. Eye contact must be maintained as much as possible during the note-taking. Otherwise, patients will feel that the record is more important than what they are saying. Also, the interviewer may miss nonverbal communications that can be more important than the words being recorded.

Increasingly, the electronic health record (EHR) is now being used throughout medicine. There are great advantages of computerized records, including rapid retrieval of information, appropriately sharing data among various members of the health care team, access to important data in an emergency, decreasing errors, and as a tool for research and quality improvement activities. Evidence-based practice guidelines can also be integrated with EHRs so that information or recommendations can be provided at the point of service. However, the use of computers can also present significant challenges to the developing patient–physician relationship. Frequently, physicians using computers during an interview will turn away from the patient to enter data. Especially in a psychiatric interview, this can be very disruptive to a smooth and dynamic interaction. As improved

technology becomes more widespread (e.g., the use of notepads held in the lap) and psychiatrists become more accustomed to using the equipment, some of these disruptions can be minimized.

Cultural Issues

Culture can be defined as a common heritage, a set of beliefs, and values that set expectations for behaviors, thoughts, and even feelings. Several culture-bound syndromes that are unique to a particular population have been described (see Chapter 34). Culture can influence the presentation of illness, the decision when and where to seek care, the decision as to what to share with the physician, and the acceptance of and participation in treatment planning. Often, individuals from a minority population may be reluctant to seek help from a physician who is from the majority group, especially for emotional difficulties. Some minority groups have strong beliefs in faith healers, and in some areas of the US "root doctors" carry significant influence. These beliefs may not be apparent in the interview as the patient may have learned to be quite guarded about such matters. A patient may only report that he or she is "frightened" and not discuss the reality that this fear began when he or she realized someone was working "roots" on him or her. The psychiatrist needs to be alert to the possibility that the patient's thoughts about what has happened may be unusual from a traditional Western medical perspective and at the same time recognize that these culturally shared beliefs are not indications of psychosis. By being humble, open, and respectful the psychiatrist increases the possibility of developing a trusting working relationship with the patient and learning more about the patient's actual experiences.

The psychiatrist clearly understanding what the patient is saying, and the patient clearly understanding what the psychiatrist is saying are crucial for an effective interview. It is not just both being fluent in the language of the interview, but the psychiatrist should also be aware of common slang words and phrases that the patient, depending on their cultural background, may use. If the psychiatrist does not understand a particular phrase or comment, then he or she should ask for clarification. If the patient and psychiatrist are not both fluent in the same language, then an interpreter is necessary.

Interviewing with an Interpreter. When translation is needed, it should be provided by a non–family-member professional interpreter. Translation by family members is to be avoided because (1) a patient, with a family member as an interpreter, may justifiably be very reluctant to discuss sensitive issues, including suicidal ideation or drug use, and (2) family members may be hesitant to portray a patient's deficits accurately. Both of these issues make accurate assessment very difficult.

It is helpful to speak with the interpreter before the interview to clarify the goals of the exam. If the interpreter does not primarily work with psychiatric patients, then it is important to highlight the need for verbatim translation even if the responses are disorganized or tangential. If the translator is not aware of this issue, then the psychiatrist may have difficulty diagnosing thought disorders or cognitive deficits. Occasionally, the patient will say several sentences in response to a question, and the interpreter will remark, "He said it's okay." The interpreter should again be reminded that the psychiatrist wants to hear everything that the patient is saying.

It is helpful to place the chairs in a triangle so that the psychiatrist and patient can maintain eye contact. The psychiatrist should continue to refer to the patient directly to maintain the therapeutic connection rather than speaking to the interpreter. The examiner may need to take a more directive approach and interrupt the patient's responses more frequently to allow for accurate and timely translation.

Once the interview is concluded, it may be helpful to again meet briefly with the interpreter. If the interpreter is especially knowledgeable about the patient's cultural background, they may be able to provide helpful insights regarding cultural norms.

Interviewing the Difficult Patient

Patients with Psychosis. Patients with psychotic illnesses are often frightened and guarded. They may have difficulty with reasoning and thinking clearly. Also, they may be actively hallucinating during the interview, causing them to be inattentive and distracted. They may have suspicions regarding the purpose of the interview. All of these possibilities are reasons that the interviewer may need to alter the usual format and adapt the interview to match the capacity and tolerance of the patient.

Auditory hallucinations are the most common hallucinations in psychiatric illnesses in North America. Many patients will not interpret their experiences as hallucinations, and it is useful to begin with a more general question: "*Do you ever hear someone talking to you when no one else is there?*" The patient should be asked about the content of the hallucinations, the clarity, and the situations in which they occur. Often it is helpful to ask the patient about a specific instance and if he or she can repeat verbatim the content of the hallucination. It is important to specifically ask if the patient has ever experienced command hallucinations, hallucinations in which a patient is ordered to perform a specific act. If so, the nature of the commands should be clarified, specifically if the commands have ever included orders to harm himself or herself or others, and if the patient has ever felt compelled to follow the commands.

The validity of the patient's perception should not be dismissed, but it is helpful to test the strength of the belief in the hallucinations: "*Does it seem that the voices are coming from inside your head? Who do you think is speaking to you?*"

Other perceptual disturbances should be explored, including visual, olfactory, and tactile hallucinations. These disturbances are less common in psychiatric illness and may suggest a primary medical etiology to the psychosis.

The psychiatrist should be alert for cues that psychotic processes may be part of the patient's experience during the interview. It is usually best to ask directly about such behaviors or comments.

By definition, patients with delusions have fixed false beliefs. With delusions, as with hallucinations, it is important to explore the specific details. Patients are often very reluctant to discuss their beliefs, as many have had their beliefs dismissed or ridiculed. They may ask the interviewer directly if the interviewer believes the delusion. Although an interviewer should not directly endorse the false belief, it is rarely helpful to challenge the delusion, particularly in the initial examination directly. It can be useful to shift the attention back to the patient's rather than the examiner's beliefs and acknowledge the need for more information: "*I believe that what you are experiencing is frightening and I would like to know more about your experiences.*"

For patients with paranoid thoughts and behaviors, it is important to

maintain a respectful distance. Their suspiciousness may be increased by an overly warm interview. It may be helpful to avoid sustained direct eye contact, as this may be perceived as threatening. Harry Stack Sullivan recommended that rather than sitting face to face with the patient who is paranoid, the psychiatrist might sit more side by side, "looking out" with the patient. Interviewers should keep in mind that they themselves may become incorporated into the paranoid delusions, and it is helpful to ask directly about such fears: "*Are you concerned that I am involved?*" The psychiatrist should also ask whether there is a specific target related to paranoid thinking. When asked regarding thoughts about hurting others, the patient may not disclose plans for violence. Exploration of the patient's plan on how to manage his or her fears may elicit information regarding violence risk: "*Do you feel you need to protect yourself in any way? How do you plan to do so?*" If there is some expression of possible violence toward others, the psychiatrist then needs to do further risk assessment. This is further discussed in the section below on hostile, agitated, and violent patients.

Depressed and Potentially Suicidal Patients. The depressed patient may have particular difficulty during the interview as he or she may have cognitive deficits as a result of the depressive symptoms. The patients may have impaired motivation and may not spontaneously report their symptoms. Feelings of hopelessness may contribute to a lack of engagement. Depending on the severity of symptoms, patients may need more direct questioning rather than an open-ended format.

A suicide assessment should be performed for all patients, including prior history, family history of suicide attempts and completed suicides, and current ideation, plan, and intent. An open-ended approach is often helpful: "*Have you ever had thoughts that life wasn't worth living?*" It is important to detail prior attempts. The lethality risk of prior attempts and any potential triggers for the attempt should be clarified. This can help with assessing the current risk.

The patient should be asked about any current thoughts of suicide, and if thoughts are present, what is the patient's intent. Some patients will describe having thoughts of suicide but do not intend to act on these thoughts or wish to be dead. They report that although the thoughts are present, they have no intent to act on the thoughts. This is typically referred to as passive suicidal

ideation. Other patients will express their determination to end their life and are at higher risk. The presence of psychotic symptoms should be assessed. Some patients may have hallucinations compelling them to hurt themselves even though they do not have a desire to die.

If the patient reports suicidal ideation, they should be asked if they have a plan to end his or her life. The specificity of the plan should be determined and whether the patient has access to the means to complete the plan. The interviewer should pursue this line of questioning in detail if the patient has taken any preparatory steps to move forward with the plan. (A patient who has purchased a gun and has given away important items would be at high risk.)

If the patient has not acted upon these urges, then it is helpful to ask what has prevented him or her from acting on these thoughts: "*What do you think has kept you from hurting yourself?*" The patient may disclose information that may decrease their acute risk, such as religious beliefs that prohibit suicide or awareness of the impact of suicide on family members. This information is essential to keep in mind during treatment especially if these preventative factors change. (A patient who states he or she could never abandon a beloved pet may be at increased risk if the pet dies.)

Although the psychiatric interview intends to build rapport and gather information for treatment and diagnosis, the patient's safety must be the priority. If the patient is viewed to be at imminent risk, then an interview may need to be terminated, and the interviewer must take action to secure the safety of the patient.

Hostile, Agitated, and Potentially Violent Patients. Safety for the patient and the psychiatrist is the priority when interviewing agitated patients. Hostile patients are often interviewed in emergency settings, but angry and agitated patients can present in any setting. If interviewing in an unfamiliar setting, then the psychiatrist should familiarize himself or herself with the office setup, paying particular attention to the chair placement. The chairs should ideally be placed in a way in which both the interviewer and patient could exit if necessary and not be obstructed. The psychiatrist should be aware of any available safety features (emergency buttons or number for security) and should be familiar with the facility's security plan. If the psychiatrist is aware in advance that the patient is agitated, then he or she can

take additional preparatory steps such as having security closely available if necessary.

As increased stimulation can be agitating for a hostile patient, care should be taken to decrease excess stimulation as much as feasible. The psychiatrist should be aware of their own body position and avoid postures that could be seen as threatening, including clenched hands or hands behind the back.

The psychiatrist should approach the interview in a calm, direct manner and take care not to bargain or promise to elicit cooperation in the interview: "*Once we finish here, you will be able to go home.*" These tactics may only escalate agitation.

As stated above, the priority must be safety. An intimidated psychiatrist who is fearful regarding his or her physical safety will be unable to perform an adequate assessment. Similarly, a patient who feels threatened will be unable to focus on the interview and may begin to escalate thinking that he or she needs to defend himself or herself. An interview may need to be terminated early if the patient's agitation escalates. Generally, unpremeditated violence is preceded by a period of gradually escalating psychomotor agitation such as pacing, loud speech, and threatening comments. At this point, the psychiatrist should consider whether other measures are necessary, including assistance from security personnel or need for medication or restraint.

If the patient makes threats or gives some indication that he or she may become violent outside the interview setting, then further assessment is necessary. Because a past history of violence is the best predictor of future violence, past episodes of violence should be explored as to setting, what precipitated the episode, and what was the outcome or potential outcome (if the act was interrupted). Also, what has helped in the past in preventing violent episodes (medication, timeout, physical activity, or talking to a particular person) should be explored. Is there an identified victim and is there a plan for the violent behavior? Has the patient taken steps to fulfill the plan? Depending on the answers to these questions the psychiatrist may decide to prescribe or increase antipsychotic medication, recommend hospitalization, and perhaps, depending on the jurisdiction, notify the threatened victim.

Deceptive Patients. Psychiatrists are trained to diagnose and treat

psychiatric illness. Although psychiatrists are well trained in eliciting information and maintaining awareness for deception, these abilities are not foolproof. Patients lie or deceive their psychiatrists for many different reasons. Some are motivated by secondary gain (e.g., for financial resources, absence from work, or for a supply of medication). Some patients may deceive, not for an external advantage, but for the psychological benefits of assuming a sick role. As noted above, unconscious processes may result in events or feelings being outside the patient's awareness.

There are no current biologic markers to validate a patient's symptoms definitively. Psychiatrists are dependent on the patient's self-report. Given these limitations, it may be useful, especially when there is a question about the patient's reliability (possibly related to inconsistencies in the patient's report), to gather collateral information regarding the patient. This allows the psychiatrist to have a more broad understanding of the patient outside the interview setting, and discrepancies in symptom severity between self-report and collateral information may suggest deception. Some psychological tests can help in further evaluating the reliability of the patient.

PHYSICAL EXAMINATION OF THE PSYCHIATRIC PATIENT

Confronted with a patient who has a mental disorder, the psychiatrist must decide whether a medical, surgical, or neurologic condition may be the cause. Once satisfied that no disease process can be held accountable, then the diagnosis of mental disorder not attributable to a medical illness can be made. Although psychiatrists do not perform routine physical examinations of their patients, knowledge, and understanding of physical signs and symptoms is part of their training, which enables them to recognize signs and symptoms that may indicate possible medical or surgical illness. For example, palpitations can be associated with mitral valve prolapse, which is diagnosed by cardiac auscultation. Psychiatrists are also able to recognize and treat the adverse effects of psychotropic medications, which are used by an increasing number of patients seen by psychiatrists and nonpsychiatric physicians.

Some psychiatrists insist that every patient has a complete medical workup; others may not. Whatever their policy, psychiatrists should consider patients' medical status at the outset of a psychiatric evaluation. Psychiatrists

must often decide whether a patient needs a medical examination and, if so, what it should include—most commonly, a thorough medical history, including a review of systems, a physical examination, and relevant diagnostic laboratory studies. A recent study of 1,000 medical patients found that in 75 percent of cases, no cause of symptoms (i.e., subjective complaints) could be found, and a psychological basis was assumed in 10 percent of those cases.

History of Medical Illness

In the course of conducting a psychiatric evaluation, the information should be gathered about known bodily diseases or dysfunctions, hospitalizations and operative procedures, medications are taken recently or at present, personal habits and occupational history, family history of illnesses, and specific physical complaints. Information about medical illnesses should be gathered from the patient, the referring physician, and the family, if necessary.

Information about previous episodes of illness may provide valuable clues about the nature of the present disorder. For example, a distinctly delusional disorder in a patient with a history of several similar episodes that responded promptly to diverse forms of treatment strongly suggests the possibility of substance-induced psychotic disorder. To pursue this lead, the psychiatrist should order a drug screen. The history of a surgical procedure may also be useful; for instance, a thyroidectomy suggests hypothyroidism as the cause of depression.

Depression is an adverse effect of several medications prescribed for hypertension. Medication taken in a therapeutic dosage occasionally reaches high concentrations in the blood. Digitalis intoxication, for example, can occur under such circumstances and result in impaired mental functioning. Proprietary drugs can cause or contribute to an anticholinergic delirium. The psychiatrist, therefore, must inquire about over-the-counter remedies as well as prescribed medications. A history of herbal intake and alternative therapy is essential, given their increased use.

An occupational history may also provide essential information. Exposure to mercury can result in complaints suggesting a psychosis, and exposure to lead, as in smelting, can produce a cognitive disorder. The latter clinical

picture can also result from imbibing moonshine whiskey with high lead content.

In eliciting information about specific symptoms, the psychiatrist brings medical and psychological knowledge into full play. For example, the psychiatrist should elicit sufficient information from the patient complaining of headaches to predict whether the pain results from an intracranial disease that requires neurologic testing. Also, the psychiatrist should be able to recognize that the pain in the right shoulder of a hypochondriacal patient with abdominal discomfort may be the classic referred pain of gallbladder disease.

Review of Systems

An inventory by systems should follow the open-ended inquiry. The review can be organized according to organ systems (e.g., liver, pancreas), functional systems (e.g., gastrointestinal), or a combination of the two, as in the outline presented in the following subsections. In all cases, the review should be comprehensive and thorough. Even if a psychiatric component is suspected, a complete workup is still indicated.

Head. Many patients give a history of headache; its duration, frequency, character, location, and severity should be ascertained. Headaches often result from substance abuse, including alcohol, nicotine, and caffeine. Vascular (migraine) headaches are precipitated by stress. Temporal arteritis causes unilateral throbbing headaches and can lead to blindness. Brain tumors are associated with headaches as a result of increased intracranial pressure, but some may be silent, the first signs being a change in personality or cognition.

> A 63-year-old woman in treatment for depression began to complain of difficulties in concentration. The psychiatrist attributed the complaint to the depressive disorder; however, when the patient began to complain of balance difficulties, magnetic resonance imaging was obtained, which revealed the presence of meningioma.

A head injury can result in subdural hematoma and, in boxers, can cause progressive dementia with extrapyramidal symptoms. The headache of

subarachnoid hemorrhage is sudden, severe, and associated with changes in the sensorium. Normal-pressure hydrocephalus can follow a head injury or encephalitis and be associated with dementia, shuffling gait, and urinary incontinence. Dizziness occurs in up to 30 percent of persons, and determining its cause is challenging and often complicated. A change in the size or shape of the head may be indicative of Paget disease.

Eye, Ear, Nose, and Throat. Visual acuity, diplopia, hearing problems, tinnitus, glossitis, and bad taste are covered in this area. A patient taking antipsychotics who gives a history of twitching about the mouth or disturbing movements of the tongue may be in the early and potentially reversible stage of tardive dyskinesia. Impaired vision can occur with thioridazine (Mellaril) in high doses (over 800 mg a day). A history of glaucoma contraindicates drugs with anticholinergic effects. Complaints of bad odors may be a symptom of temporal lobe epilepsy rather than schizophrenia. Aphonia may be hysterical. The late stage of cocaine abuse can result in perforations of the nasal septum and difficulty breathing. A transitory episode of diplopia may herald multiple sclerosis. Delusional disorder is more common in hard of hearing persons than in those with normal hearing. Transient blue-tinged vision can occur when using sildenafil (Viagra) or similar drugs.

Respiratory System. Cough, asthma, pleurisy, hemoptysis, dyspnea, and orthopnea are considered in this subsection. Hyperventilation is suggested if the patient's symptoms include all or a few of the following: onset at rest, sighing respirations, apprehension, anxiety, depersonalization, palpitations, inability to swallow, numbness of the feet and hands, and carpopedal spasm. Dyspnea and breathlessness can occur in depression. In pulmonary or obstructive airway disease, the onset of symptoms is usually insidious, whereas, in depression, it is sudden. In depression, breathlessness is experienced at rest, shows little change with exertion, and can fluctuate within a matter of minutes; the onset of breathlessness coincides with the onset of a mood disorder and is often accompanied by attacks of dizziness, sweating, palpitations, and paresthesias.

In obstructive airway disease, patients with the most advanced respiratory incapacity experience breathlessness at rest. Most striking and of greatest assistance in making a differential diagnosis is an emphasis placed on the

difficulty in inspiration experienced by patients with depression and on the difficulty in expiration experienced by patients with pulmonary disease. Bronchial asthma has sometimes been associated with a childhood history of extreme dependence on the mother. Patients with bronchospasm should not receive propranolol (Inderal) because it can block catecholamine-induced bronchodilation; propranolol is specifically contraindicated for patients with bronchial asthma because epinephrine given to such patients in an emergency will not be effective. Patients taking angiotensin-converting enzyme (ACE) inhibitors can develop a dry cough as an adverse effect of the drug.

Cardiovascular System. Tachycardia, palpitations, and cardiac arrhythmia are among the most common signs of anxiety about which the patient may complain. Pheochromocytoma usually produces symptoms that mimic anxiety disorders, such as rapid heartbeat, tremors, and pallor. Increased urinary catecholamines are diagnostic of pheochromocytoma. Patients taking guanethidine (Ismelin) for hypertension should not receive tricyclic drugs, which reduce or eliminate the antihypertensive effect of guanethidine. A history of hypertension can preclude the use of MAOIs because of the risk of a hypertensive crisis if such patients with hypertension inadvertently ingest foods high in tyramine. Patients with suspected cardiac disease should have an electrocardiogram before tricyclics or lithium (Eskalith) is prescribed. A history of substernal pain should be evaluated, and the clinician should keep in mind that psychological stress can precipitate angina-type chest pain in the presence of normal coronary arteries. Patients taking opioids should never receive MAOIs; the combination can cause cardiovascular collapse.

Gastrointestinal System. Such topics as appetite, distress before or after meals, food preferences, diarrhea, vomiting, constipation, laxative use, and abdominal pain relate to the gastrointestinal system. A history of weight loss is common in depressive disorders, but depression can accompany the weight loss caused by ulcerative colitis, regional enteritis, and cancer. Atypical depression is accompanied by hyperphagia and weight gain. Anorexia nervosa is accompanied by severe weight loss in the presence of normal appetite. Avoidance of certain foods may be a phobic phenomenon or part of an obsessive ritual. Laxative abuse and induced vomiting are common in

bulimia nervosa. Constipation can be caused by opioid dependence and psychotropic drugs with anticholinergic side effects. Cocaine or amphetamine abuse causes a loss of appetite and weight loss. Weight gain can occur under stress or in association with atypical depression. Polyphagia, polyuria, and polydipsia are the triad of diabetes mellitus. Polyuria, polydipsia, and diarrhea are signs of lithium toxicity. Some patients take enemas routinely as part of paraphilic behavior, and anal fissures or recurrent hemorrhoids may indicate anal penetration by foreign objects. Some patients may ingest foreign objects that produce symptoms that can be diagnosed only by x-ray (Fig. 1-1).

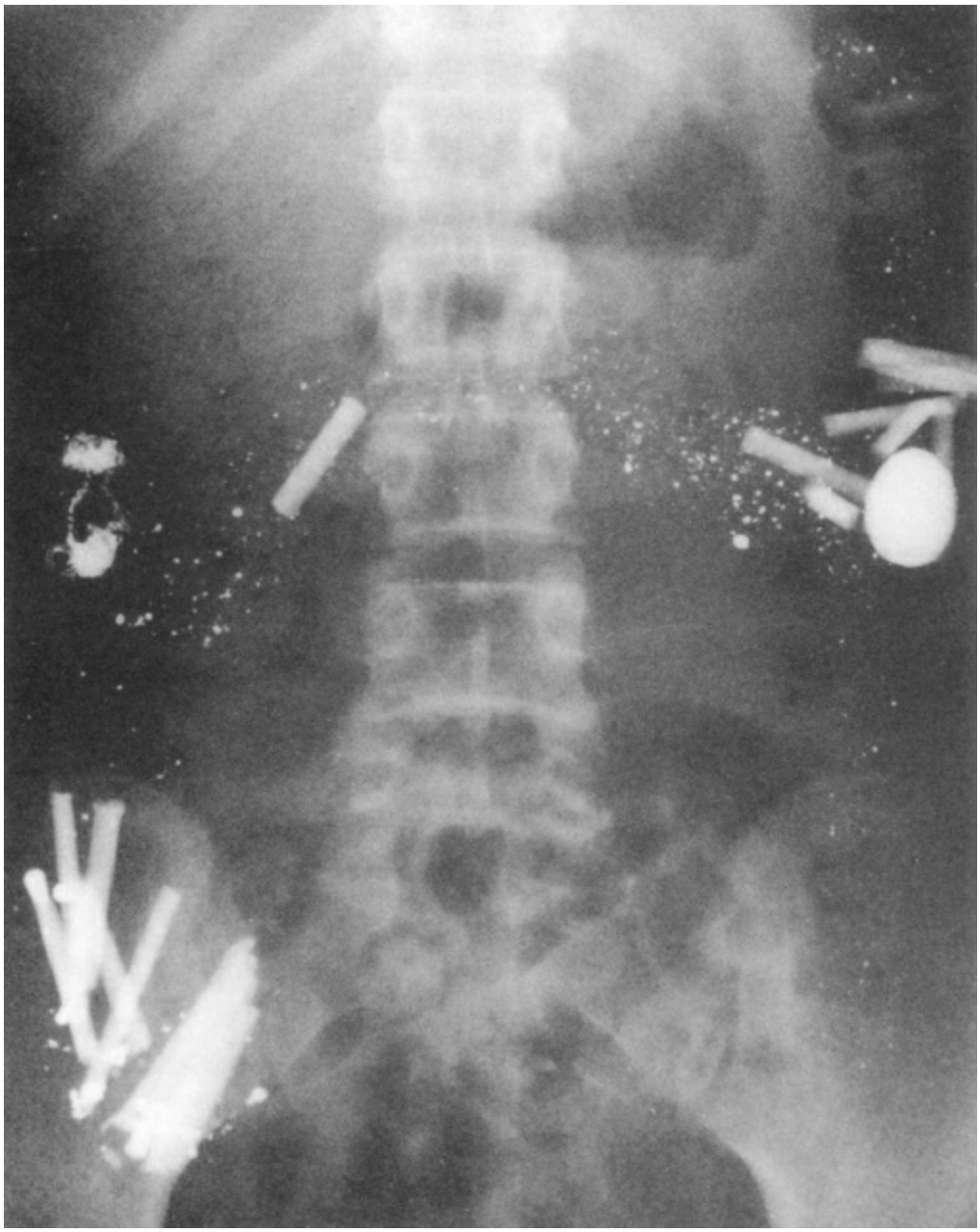


FIGURE 1-1
A mentally ill patient who is a habitual swallower of foreign objects. Included in his colonic lumen are 13 thermometers and 8 pennies. The dense, round, almost punctate densities are globules of liberated liquid mercury. (Courtesy of Stephen R. Baker, M.D. and Kyunghee C. Cho, M.D.)

Genitourinary System. Urinary frequency, nocturia, pain or burning on urination, and changes in the size and the force of the stream are some of the signs and symptoms emanating from the genitourinary system. Anticholinergic adverse effects associated with antipsychotics and tricyclic

drugs can cause urinary retention in men with prostate hypertrophy. Erectile difficulty and retarded ejaculation are also common adverse effects of these drugs, and retrograde ejaculation occurs with thioridazine. A baseline level of sexual responsiveness before using pharmacologic agents should be obtained. A history of STDs—for example, gonorrheal discharge, chancre, herpes, and pubic lice—may indicate sexual promiscuity or unsafe sexual practices.

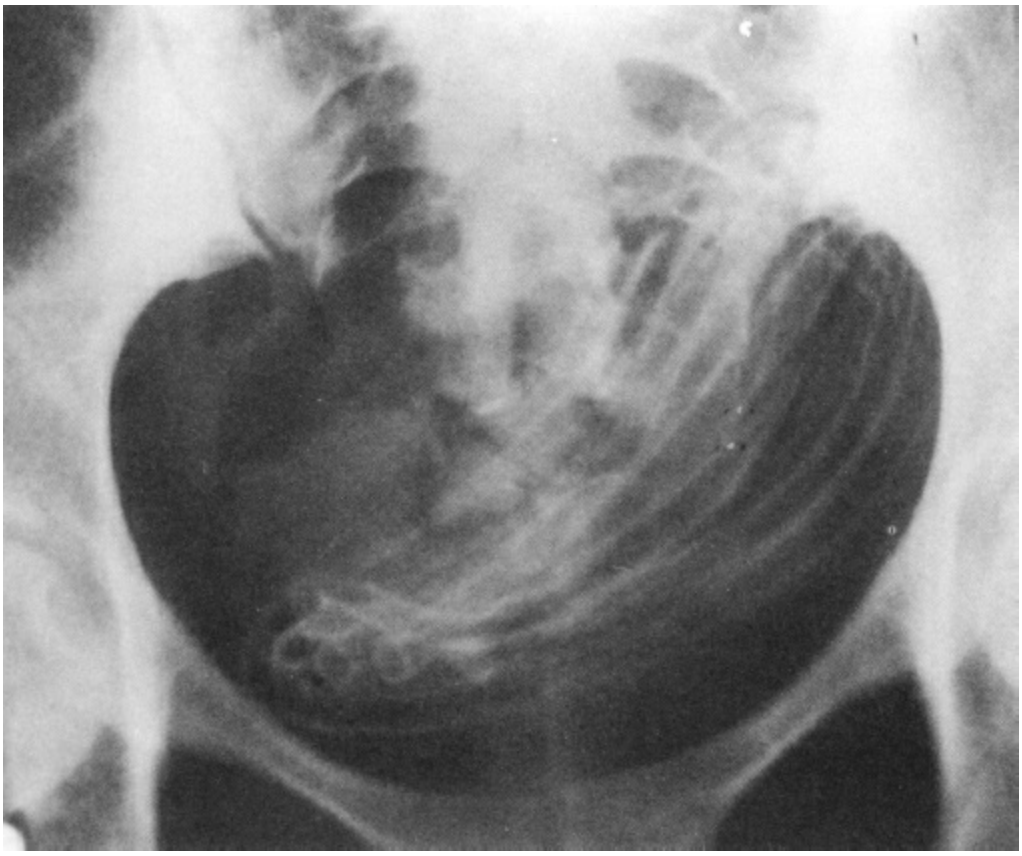


FIGURE 1-2
A patient brought to the emergency room with lower abdominal pain. X-ray shows a nasogastric tube folded into the bladder. The patient would insert the tube into his urethra as part of a masturbatory ritual (urethral eroticism). (Courtesy of Stephen R. Baker, M.D. and Kyunghee C. Cho, M.D.)

In some cases, the first symptom of acquired immune deficiency syndrome (AIDS) is the gradual onset of mental confusion leading to dementia. Incontinence should be evaluated carefully, and if it persists, further investigation for more extensive disease should include a workup for HIV infection. Drugs with anticholinergic adverse effects should be avoided in men with prostatism. Urethral eroticism, in which catheters or other objects

are inserted into the urethra, can cause infection or laceration (Fig. 1-2).

Orgasm causes prostatic contractions, which may artificially raise prostate-specific antigen (PSA) and give a false-positive result for prostatic cancer. Men scheduled to have a PSA test should avoid masturbation or coitus for 7 to 10 days before the test.

Menstrual History. A menstrual history should include the age of the onset of menarche (and menopause, if applicable); the interval, regularity, duration, and amount of flow of periods; irregular bleeding; dysmenorrhea; and abortions. Amenorrhea is characteristic of anorexia nervosa and also occurs in women who are psychologically stressed. Women who are afraid of becoming pregnant or who have a wish to be pregnant may have delayed periods. *Pseudocyesis* is false pregnancy with complete cessation of the menses. Perimenstrual mood changes (e.g., irritability, depression, and dysphoria) should be noted. Painful menstruation can result from uterine disease (e.g., myomata), from psychological conflicts about the menses, or from a combination of the two. Some women report a perimenstrual increase in sexual desire. The emotional reaction associated with abortion should be explored because it can be mild or severe.

General Observation

An important part of the medical examination is subsumed under the broad heading of general observation—visual, auditory, and olfactory. Such nonverbal clues as posture, facial expression, and mannerisms should also be noted.

Visual Inspection. Scrutiny of the patient begins at the first encounter. When the patient goes from the waiting room to the interview room, the psychiatrist should observe the patient's gait. Is the patient unsteady? Ataxia suggests diffuse brain disease, alcohol or other substance intoxication, chorea, spinocerebellar degeneration, weakness based on a debilitating process, and an underlying disorder, such as myotonic dystrophy. Does the patient walk without the usual associated arm movements and turn in a rigid fashion, such as a toy soldier, as is seen in early Parkinson disease? Does the patient have asymmetry of gait, such as turning one foot outward, dragging a leg, or not swinging one arm, suggesting a focal brain lesion?

As soon as the patient is seated, the psychiatrist should direct attention to grooming. Is the patient's hair combed, are the nails clean, and are the teeth brushed? Has clothing been chosen with care and is it appropriate? Although inattention to dress and hygiene is common in mental disorders—in particular, depressive disorders—it is also a hallmark of cognitive disorders. Lapses, such as mismatching socks, stockings, or shoes, may suggest a cognitive disorder.

The patient's posture and automatic movements or the lack of them should be noted. A stooped, flexed posture with a paucity of automatic movements may be caused by Parkinson disease or diffuse cerebral hemispheric disease or be an adverse effect of antipsychotics. An unusual tilt of the head may be adopted to avoid eye contact, but it can also result from diplopia, a visual field defect, or focal cerebellar dysfunction. Frequent quick, purposeless movements are characteristic of anxiety disorders, but they are equally characteristic of chorea and hyperthyroidism. Tremors, although commonly seen in anxiety disorders, may point to Parkinson disease, essential tremor, or adverse effects of psychotropic medication. Patients with essential tremor sometimes seek psychiatric treatment because they believe the tremor must be caused by unrecognized fear or anxiety, as others often suggest. Unilateral paucity or excess of movement suggests focal brain disease.

The patient's appearance is then scrutinized to assess general health. Does the patient appear to be robust or is there a sense of ill health? Does looseness of clothing indicate recent weight loss? Is the patient short of breath or coughing? Does the patient's general physiognomy suggest a specific disease? Men with Klinefelter syndrome have a feminine fat distribution and lack the development of secondary male sex characteristics. Acromegaly is usually immediately recognizable by the large head and jaw.

What is the patient's nutritional status? Recent weight loss, although often seen in depressive disorders and schizophrenia, may be caused by gastrointestinal disease, diffuse carcinomatosis, Addison disease, hyperthyroidism, and many other somatic disorders. Obesity can result from either emotional distress or organic disease. Moon facies, truncal obesity, and buffalo hump are striking findings in Cushing syndrome. The puffy, bloated appearance seen in hypothyroidism and the massive obesity and periodic respiration seen in Pickwickian syndrome are easily recognized in patients referred for psychiatric help. Hyperthyroidism is indicated by exophthalmos.

The skin frequently provides valuable information. The yellow discoloration of hepatic dysfunction and the pallor of anemia are reasonably distinctive. Intense reddening may be caused by carbon monoxide poisoning or by photosensitivity resulting from porphyria or phenothiazines. Eruptions can be manifestations of such disorders as SLE (e.g., the butterfly on the face), tuberous sclerosis with adenoma sebaceum, and sensitivity to drugs. A dusky purplish cast to the face, plus telangiectasia, is almost pathognomonic of alcohol abuse.

Careful observation may reveal clues that lead to the correct diagnosis in patients who create their skin lesions. For example, the location and shape of the lesions and the time of their appearance may be characteristic of dermatitis factitia.

The patient's face and head should be scanned for evidence of disease. Premature whitening of the hair occurs in pernicious anemia, and thinning and coarseness of the hair occur in myxedema. In alopecia areata, patches of hair are lost, leaving bald spots; hair-pulling disorder (trichotillomania) presents a similar picture. Pupillary changes are produced by various drugs—constriction by opioids and dilation by anticholinergic agents and hallucinogens. The combination of dilated and fixed pupils and dry skin and mucous membranes should immediately suggest the likelihood of atropine use or atropine-like toxicity. The diffusion of the conjunctiva suggests alcohol abuse, cannabis abuse, or obstruction of the superior vena cava. Flattening of the nasolabial fold on one side or weakness of one side of the face—as manifested in speaking, smiling, and grimacing—may be the result of focal dysfunction of the contralateral cerebral hemisphere or Bell palsy. A drooping eyelid may be an early sign of myasthenia gravis.

The patient's state of alertness and responsiveness should be evaluated carefully. A psychological problem may cause drowsiness and inattentiveness, but they are more likely to result from organic brain dysfunction, whether secondary to an intrinsic brain disease or an exogenous factor, such as substance intoxication.

Listening. Listening intently is just as important as looking intently for evidence of somatic disorders. Slowed speech is characteristic not only of depression but also of diffuse brain dysfunction and subcortical dysfunction; unusually rapid speech is characteristic of manic episodes and anxiety

disorders and also of hyperthyroidism. A weak voice with a monotonous tone may be a clue to Parkinson disease in patients who complain mainly of depression. A slow, low-pitched, hoarse voice should suggest the possibility of hypothyroidism; this voice quality has been described as sounding like a drowsy, slightly intoxicated person with a bad cold and a plum in the mouth. A soft or tremulous voice accompanies anxiety.

Difficulty initiating speech may be owing to anxiety or stuttering or may indicate Parkinson disease or aphasia. Easy fatigability of speech is sometimes a manifestation of an emotional problem, but it is also characteristic of myasthenia gravis. Patients with these complaints are likely to be seen by a psychiatrist before the correct diagnosis is made.

Word production, as well as the quality of speech, is important. Mispronounced or incorrectly used words suggest a possibility of aphasia caused by a lesion of the dominant hemisphere. The same possibility exists when the patient perseverates, has trouble finding a name or a word, or describes an object or an event in an indirect fashion (paraphasia). When not consonant with patients' socioeconomic and educational levels, coarseness, profanity, or inappropriate disclosures may indicate loss of inhibition caused by dementia.

Smell. Smell may also provide useful information. The unpleasant odor of a patient who fails to bathe suggests a cognitive or a depressive disorder. The odor of alcohol or substances used to hide it is revealing in a patient who attempts to conceal a drinking problem. Occasionally, a uriniferous odor calls attention to bladder dysfunction secondary to a nervous system disease. Characteristic odors are also noted in patients with diabetic acidosis, flatulence, uremia, and hepatic coma. Precocious puberty can be associated with the smell of adult sweat produced by mature apocrine glands.

A 23-year-old woman was referred to a psychiatrist for a second opinion. She had been diagnosed 6 months earlier with schizophrenia after complaining of smelling foul odors that were considered to be hallucinatory. She had been placed on antipsychotic medication (perphenazine) and was compliant despite the side effects of tremor and lethargy. Although there was some improvement in her symptoms, they

did not remit entirely. The consulting psychiatrist obtained an electroencephalogram, which showed abnormal waveforms consistent with a diagnosis of temporal lobe epilepsy. The antipsychotic medication was replaced with an anticonvulsant (phenytoin), after which she no longer experienced an olfactory hallucination, nor did she have to endure the unpleasant side effects of the previous medication.

Physical Examination

Patient Selection. The nature of the patient's complaints is critical in determining whether a complete physical examination is required. Complaints fall into the three categories of body, mind, and social interactions. Bodily symptoms (e.g., headaches and palpitations) call for a thorough medical examination to determine what part if any, somatic processes play in causing the distress. The same can be said for mental symptoms such as depression, anxiety, hallucinations, and persecutory delusions, which can be expressions of somatic processes. If the problem is limited to the social sphere (e.g., long-standing difficulties in interactions with teachers, employers, parents, or a spouse), there may be no particular indication for a physical examination. Personality changes, however, can result from a medical disorder (e.g., early Alzheimer disease) and cause interpersonal conflicts.

Psychological Factors. Even a routine physical examination may evoke adverse reactions; instruments, procedures, and the examining room may be frightening. A simple running account of what is being done can prevent much needless anxiety. Moreover, if the patient is consistently forewarned of what will be done, the dread of being suddenly and painfully surprised recedes. Comments such as "There's nothing to this" and "You don't have to be afraid because this won't hurt" leave the patient in the dark and are much less reassuring than a few words about what actually will be done.

Although the physical examination is likely to engender or intensify a reaction of anxiety, it can also stir up sexual feelings. Some women with fears or fantasies of being seduced may misinterpret an ordinary movement in the physical examination as a sexual advance. Similarly, a delusional man

with homophobic fears may perceive a rectal examination as a sexual attack. Lingering over the examination of a particular organ because an unusual but normal variation has aroused the physician's scientific curiosity is likely to raise concern in the patient that a severe pathologic process has been discovered. Such a reaction may be profound in an anxious or hypochondriacal patient.

The physical examination occasionally serves a psychotherapeutic function. Anxious patients may be relieved to learn that, despite troublesome symptoms, no evidence is found of the serious illness that they fear. The young person who complains of chest pain and is confident that the pain heralds a heart attack can usually be reassured by the report of normal findings after a physical examination and electrocardiogram. The reassurance relieves only the worry occasioned by the immediate episode, however. Unless psychiatric treatment succeeds in dealing with the determinants of the reaction, recurrent episodes are likely.

Sending a patient who has a deeply rooted fear of malignancy for still another test that is intended to be reassuring is usually unrewarding. Some patients may have a false fixed belief that a disorder is present.

During the performance of the physical examination, an observant physician may note indications of emotional distress. For instance, during genital examinations, a patient's behavior may reveal information about sexual attitudes and problems, and these reactions can be used later to open this area for exploration.

Timing of the Physical Examination. Circumstances occasionally make it desirable or necessary to defer a complete medical assessment. For example, a delusional or manic patient may be combative, resistive, or both. In this instance, a medical history should be elicited from a family member, if possible, but unless a pressing reason exists to proceed with the examination, it should be deferred until the patient is tractable.

For psychological reasons, it may be ill-advised to recommend a medical assessment at the time of an initial office visit. Because of today's increased sensitivity and openness about sexual matters and a tendency to turn quickly to psychiatric help, young men may complain about their failure to consummate their first coital attempt. After taking a detailed history, the psychiatrist may conclude that the failure was because of situational anxiety.

If so, neither a physical examination nor psychotherapy should be recommended; they would have the undesirable effect of reinforcing the notion of pathology. Should the problem be recurrent, further evaluation would be warranted.

Neurologic Examination. If the psychiatrist suspects that the patient has an underlying somatic disorder, such as diabetes mellitus or Cushing syndrome, a referral is usually made for diagnosis and treatment. The situation is different when a cognitive disorder is suspected. The psychiatrist often chooses to assume responsibility in these cases. At some point, however, a thorough neurologic evaluation may be indicated.

During the history-taking process in such cases, the patient's level of awareness, attentiveness to the details of the examination, understanding, facial expression, speech, posture, and gait are noted. It is also assumed that a thorough mental status examination will be performed. The neurologic examination is carried out with two objectives in mind: to elicit (1) signs pointing to focal, circumscribed cerebral dysfunction, and (2) signs suggesting diffuse, bilateral cerebral disease. The first objective is met by the routine neurologic examination, which is designed primarily to reveal asymmetries in the motor, perceptual, and reflex functions of the two sides of the body, caused by focal hemispheric disease. The second objective is met by seeking to elicit signs that have been attributed to diffuse brain dysfunction and to frontal lobe disease. These signs include the sucking, snout, palmomental, and grasp reflexes and the persistence of the glabella tap response. Regrettably, except for the grasp reflex, such signs do not correlate strongly with the presence of underlying brain pathology.

Other Findings. Psychiatrists should be able to evaluate the significance of findings uncovered by consultants. With a patient who complains of a lump in the throat (globus hystericus) and who is found on examination to have hypertrophied lymphoid tissue, it is tempting to wonder about a causal relation. How can a clinician be sure that the finding is not incidental? Has the patient been known to have hypertrophied lymphoid tissue at a time when no complaint was made? Do many persons with hypertrophied lymphoid tissue never experience the sensation of a lump in the throat?

With a patient with multiple sclerosis who complains of an inability to

walk but, on neurologic examination, has only mild spasticity and a unilateral Babinski sign, it is tempting to ascribe the symptom to the neurologic disorder. However, the complaint may be aggravated by emotional distress. The same holds for a patient with profound dementia in whom a small frontal meningioma is seen on CT scan. Dementia is not always correlated with the findings. Significant brain atrophy could cause very mild dementia, and minimal brain atrophy could cause significant dementia.

A lesion is often found that can account for a symptom, but the psychiatrist should make every effort to separate an incidental finding from a causative one and to distinguish a lesion merely found in the area of the symptom from a lesion producing the symptom.

Patients Undergoing Psychiatric Treatment

While patients are being treated for psychiatric disorders, psychiatrists should be alert to the possibility of intercurrent illnesses that call for diagnostic studies. Patients in psychotherapy, particularly those in psychoanalysis, may be all too willing to ascribe their new symptoms to emotional causes. Attention should be given to the possible use of denial, especially if the symptoms seem to be unrelated to the conflicts currently in focus.

Not only may patients in psychotherapy be likely to attribute new symptoms to emotional causes, but sometimes their therapists do so as well. The danger of providing psychodynamic explanations for physical symptoms is ever-present.

Symptoms such as drowsiness and dizziness and signs such as a skin eruption and a gait disturbance, common adverse effects of psychotropic medication, call for a medical reevaluation if the patient fails to respond in a reasonable time to changes in the dose or the kind of medication prescribed. If patients who are receiving tricyclic or antipsychotic drugs complain of blurred vision (usually an anticholinergic adverse effect) and the condition does not recede with a reduction in dose or a change in medication, they should be evaluated to rule out other causes. In one case, the diagnosis proved to be toxoplasma chorioretinitis. The absence of other anticholinergic adverse effects, such as a dry mouth and constipation, is an additional clue alerting the psychiatrist to the possibility of a concomitant medical illness.

Early in the illness, there may be few, if any, positive physical or

laboratory results. In such instances, especially if the evidence of psychic trauma or emotional conflicts is glaring, all symptoms are likely to be regarded as psychosocial in origin, and new symptoms are also seen in this light. Indications for repeating portions of the medical workup may be missed unless the psychiatrist is alert to clues suggesting that some symptoms do not fit the original diagnosis and, instead, point to a medical illness. Occasionally, a patient with an acute illness, such as encephalitis, is hospitalized with the diagnosis of schizophrenia, or a patient with a subacute illness, such as carcinoma of the pancreas, is treated in a private office or clinic with the diagnosis of a depressive disorder. Although it may not be possible to make the correct diagnosis at the time of the initial psychiatric evaluation, continued surveillance, and attention to clinical details usually provide clues leading to the recognition of the cause.

The likelihood of intercurrent illness is higher with some psychiatric disorders than with others. Substance abusers, for example, because of their life patterns, are susceptible to infection and are likely to suffer from the adverse effects of trauma, dietary deficiencies, and poor hygiene. Depression decreases the immune response.

When somatic and psychological dysfunctions are known to coexist, the psychiatrist should be thoroughly conversant with the patient's medical status. In cases of cardiac decompensation, peripheral neuropathy, and other disabling disorders, the nature and degree of impairment that can be attributed to the physical disorder should be assessed. It is important to answer the question: Does the patient exploit a disability, or is it ignored or denied with resultant overexertion? To answer this question, the psychiatrist must assess the patient's capabilities and limitations, rather than make sweeping judgments based on a diagnostic label.

Special vigilance about medical status is required for some patients in treatment for somatoform and eating disorders. Such is the case for patients with ulcerative colitis who are bleeding profusely and for patients with anorexia nervosa who are losing appreciable weight. These disorders can become life-threatening.

Importance of Medical Screening. Numerous articles have called attention to the need for thorough medical screening of patients seen in psychiatric inpatient services and clinics. (A similar need has been

demonstrated for the psychiatric evaluation of patients seen in medical inpatient services and clinics.) The concept of *medical clearance* remains ambiguous and has meaning in the context of psychiatric admission or clearance for transfers from different settings or institutions. It implies that no medical condition exists to account for the patient's condition.

Among identified psychiatric patients, from 24 to 60 percent have been shown to suffer from associated physical disorders. In a survey of 2,090 psychiatric clinic patients, 43 percent were found to have associated physical disorders; of these, almost half the physical disorders had not been diagnosed by the referring sources. (In this study, 69 patients were found to have diabetes mellitus, but only 12 of these cases had been diagnosed before referral.)

Expecting psychiatrists to be experts in internal medicine is unrealistic, but they should be able to recognize or have a high suspicion of physical disorders when they are present. Moreover, they should make appropriate referrals and collaborate in treating patients who have both physical and mental disorders.

Psychiatric symptoms are nonspecific; they can herald medical as well as psychiatric illness. They often precede the appearance of definitive medical symptoms. Some psychiatric symptoms (e.g., visual hallucinations, distortions, and illusions) should evoke a high level of suspicion of medication toxicity.

The medical literature abounds with case reports of patients whose disorders were initially considered emotional but ultimately proved to be secondary to medical conditions. The data in most of the reports revealed features pointing toward organicity. Diagnostic errors arose because such features were accorded too little weight.

THE PSYCHIATRIC REPORT AND MEDICAL RECORD

Psychiatric Report

This section complements the previous section, "Psychiatric Interview, History, and Mental Status Examination," in that it provides a comprehensive outline on how to write the psychiatric report (see Table 1-7). The need to follow some sort of outline in gathering data about a person to make a

psychiatric diagnosis is universally recognized. The one that follows calls for including a tremendous amount of potential information about the patient, not all of which need be obtained, depending on the circumstances in the case. Beginning clinicians are advised to get as much information as possible; more experienced clinicians can pick and choose among the series of questions they might ask. In all cases, however, the person is best understood within the context of his or her life events.



Table 1-7
Psychiatric Report

I. Psychiatric History
   A. Identification: Name, age, marital status, sex, occupation, language if other than English, race, nationality, and religion if pertinent; previous admissions to a hospital for the same or a different condition; with whom the patient lives
   B. Chief complaint: Exactly why the patient came to the psychiatrist, preferably in the patient's own words; if that information does not come from the patient, note who supplied it
   C. History of present illness: Chronologic background and development of the symptoms or behavioral changes that culminated in the patient's seeking assistance; patient's life circumstances at the time of onset; personality when well; how illness has affected life activities and personal relations—changes in personality, interests, mood, attitudes toward others, dress, habits, level of tenseness, irritability, activity, attention, concentration, memory, speech; psychophysiological symptoms—nature and details of dysfunction; pain—location, intensity, fluctuation; level of anxiety—generalized and nonspecific (free floating) or specifically related to particular situations, activities, or objects; how anxieties are handled—avoidance, repetition of feared situation, use of drugs or other activities for alleviation
   D. Past psychiatric and medical history: (1) Emotional or mental disturbances—extent of incapacity, type of treatment, names

of hospitals, length of illness, effect of treatment; (2) psychosomatic disorders: hay fever, arthritis, colitis, rheumatoid arthritis, recurrent colds, skin conditions; (3) medical conditions: follow customary review of systems—sexually transmitted diseases, alcohol or other substance abuse, at risk for acquired immunodeficiency syndrome (AIDS); (4) neurologic disorders: headache, craniocerebral trauma, loss of consciousness, seizures, or tumors

E. Family history: Elicited from patient and from someone else, because quite different descriptions may be given of the same persons and events; ethnic, national, and religious traditions; other persons in the home, descriptions of them—personality and intelligence—and what has become of them since patient's childhood; descriptions of different households lived in; present relationships between patient and those who were in family; role of illness in the family; family history of mental illness; where does patient live—neighborhood and particular residence of the patient; is home crowded; privacy of family members from each other and from other families; sources of family income and difficulties in obtaining it; public assistance (if any) and attitude about it; will patient lose job or apartment by remaining in the hospital; who is caring for children

F. Personal history (anamnesis): History of the patient's life from infancy to the present to the extent it can be recalled; gaps in history as spontaneously related by the patient; emotions associated with different life periods (painful, stressful, conflictual) or with phases of life cycle

1. Early childhood (Birth through age 3)
    a. Prenatal history and mother's pregnancy and delivery: Length of pregnancy, spontaneity and normality of delivery, birth trauma, whether patient was planned and wanted, birth defects
    b. Feeding habits: Breast-fed or bottle-fed, eating problems
    c. Early development: Maternal deprivation, language

development, motor development, signs of unmet needs, sleep pattern, object constancy, stranger anxiety, separation anxiety

   d. Toilet training: Age, attitude of parents, feelings about it
   e. Symptoms of behavior problems: Thumb sucking, temper tantrums, tics, head bumping, rocking, night terrors, fears, bed-wetting or bed soiling, nail biting, masturbation
   f. Personality and temperament as a child: Shy, restless, overactive, withdrawn, studious, outgoing, timid, athletic, friendly patterns of play, reactions to siblings

2. Middle childhood (ages 3–11): Early school history—feelings about going to school, early adjustment, sex identification, conscience development, punishment; social relationships, attitudes toward siblings and playmates
3. Later childhood (prepuberty through adolescence)
   a. Peer relationships: Number and closeness of friends, leader or follower, social popularity, participation in group or gang activities, idealized figures; patterns of aggression, passivity, anxiety, antisocial behavior
   b. School history: How far the patient went, adjustment to school, relationships with teachers—teacher's pet or rebellious—favorite studies or interests, particular abilities or assets, extracurricular activities, sports, hobbies, relationships of problems or symptoms to any school period
   c. Cognitive and motor development: Learning to read and other intellectual and motor skills, minimal cerebral dysfunction, learning disabilities—their management and effects on the child
   d. Particular adolescent emotional or physical problems: Nightmares, phobias, masturbation, bed-wetting, running away, delinquency, smoking, drug or alcohol use, weight problems, feeling of inferiority
   e. Psychosexual history

i. Early curiosity, infantile masturbation, sex play
ii. Acquiring of sexual knowledge, attitude of parents toward sex, sexual abuse
iii. Onset of puberty, feelings about it, kind of preparation, feelings about menstruation, development of secondary sexual characteristics
iv. Adolescent sexual activity: Crushes, parties, dating, petting, masturbation, wet dreams and attitudes toward them
v. Attitudes toward same and opposite sex: Timid, shy, aggressive, need to impress, seductive, sexual conquests, anxiety
vi. Sexual practices: Sexual problems, homosexual and heterosexual experiences, paraphilias, promiscuity

f. Religious background: Strict, liberal, mixed (possible conflicts), relation of background to current religious practices

4. Adulthood
   a. Occupational history: Choice of occupation, training, ambitions, conflicts; relations with authority, peers, and subordinates; number of jobs and duration; changes in job status; current job and feelings about it
   b. Social activity: Whether patient has friends or not; is patient withdrawn or socializing well; social, intellectual, and physical interests; relationships with same sex and opposite sex; depth, duration, and quality of human relations
   c. Adult sexuality
      i. Premarital sexual relationships, age of first coitus, sexual orientation
      ii. Marital history: Common-law marriages, legal marriages, description of courtship and role played by each partner, age at marriage, family planning and contraception, names and ages of children, attitudes toward raising children, problems of any

family members, housing difficulties if important to the marriage, sexual adjustment, extramarital affairs, areas of agreement and disagreement, management of money, role of in-laws

iii. Sexual symptoms: Anorgasmia, impotence, premature ejaculation, lack of desire

iv. Attitudes toward pregnancy and having children; contraceptive practices and feelings about them

v. Sexual practices: Paraphilias such as sadism, fetishes, voyeurism; attitude toward fellation, cunnilingus; coital techniques, frequency

d. Military history: General adjustment, combat, injuries, referral to psychiatrists, type of discharge, veteran status

e. Value systems: Whether children are seen as a burden or a joy; whether work is seen as a necessary evil, an avoidable chore, or an opportunity; current attitude about religion; belief in heaven and hell

G. Summation of the examiner's observations and impressions derived from the initial interview

II. Mental Status

A. Appearance

1. Personal identification: May include a brief nontechnical description of the patient's appearance and behavior as a novelist might write it; attitude toward examiner can be described here—cooperative, attentive, interested, frank, seductive, defensive, hostile, playful, ingratiating, evasive, guarded

2. Behavior and psychomotor activity: Gait, mannerisms, tics, gestures, twitches, stereotypes, picking, touching examiner, echopraxia, clumsy, agile, limp, rigid, retarded, hyperactive, agitated, combative, waxy

3. General description: Posture, bearing, clothes, grooming, hair, nails; healthy, sickly, angry, frightened, apathetic, perplexed, contemptuous, ill at ease, poised, old looking, young looking, effeminate, masculine; signs of anxiety—

moist hands, perspiring forehead, restlessness, tense posture, strained voice, wide eyes; shifts in level of anxiety during interview or with particular topic

B. Speech: Rapid, slow, pressured, hesitant, emotional, monotonous, loud, whispered, slurred, mumbled, stuttering, echolalia, intensity, pitch, ease, spontaneity, productivity, manner, reaction time, vocabulary, prosody

C. Mood and affect
   1. Mood (a pervasive and sustained emotion that colors the person's perception of the world): How does patient say he or she feels; depth, intensity, duration, and fluctuations of mood—depressed, despairing, irritable, anxious, terrified, angry, expansive, euphoric, empty, guilty, awed, futile, self-contemptuous, anhedonic, alexithymic
   2. Affect (the outward expression of the patient's inner experiences): How examiner evaluates patient's affects—broad, restricted, blunted or flat, shallow, amount and range of expression; difficulty in initiating, sustaining, or terminating an emotional response; is the emotional expression appropriate to the thought content, culture, and setting of the examination; give examples if emotional expression is not appropriate

D. Thinking and perception
   1. Form of thinking
      a. Productivity: Overabundance of ideas, paucity of ideas, flight of ideas, rapid thinking, slow thinking, hesitant thinking; does patient speak spontaneously or only when questions are asked, stream of thought, quotations from patient
      b. Continuity of thought: Whether patient's replies really answer questions and are goal directed, relevant, or irrelevant; loose associations; lack of causal relations in patient's explanations; illogic, tangential, circumstantial, rambling, evasive, perseverative statements, blocking or distractibility
      c. Language impairments: Impairments that reflect

disordered mentation, such as incoherent or incomprehensible speech (word salad), clang associations, neologisms

2. Content of thinking
    a. Preoccupations: About the illness, environmental problems; obsessions, compulsions, phobias; obsessions or plans about suicide, homicide; hypochondriacal symptoms, specific antisocial urges or impulses
3. Thought disturbances
    a. Delusions: Content of any delusional system, its organization, the patient's convictions as to its validity, how it affects his or her life: persecutory delusions—isolated or associated with pervasive suspiciousness; mood congruent or mood incongruent
    b. Ideas of reference and ideas of influence: How ideas began, their content, and the meaning the patient attributes to them
4. Perceptual disturbances
    a. Hallucinations and illusions: Whether patient hears voices or sees visions; content, sensory system involvement, circumstances of the occurrence; hypnagogic or hypnopompic hallucinations; thought broadcasting
    b. Depersonalization and derealization: Extreme feelings of detachment from self or from the environment
5. Dreams and fantasies
    a. Dreams: Prominent ones, if patient will tell them; nightmares
    b. Fantasies: Recurrent, favorite, or unshakable daydreams

E. Sensorium
1. Alertness: Awareness of environment, attention span, clouding of consciousness, fluctuations in levels of awareness, somnolence, stupor, lethargy, fugue state, coma

2. Orientation
    a. Time: Whether patient identifies the day correctly; or approximate date, time of day; if in a hospital, knows how long he or she has been there; behaves as though oriented to the present
    b. Place: Whether patient knows where he or she is
    c. Person: Whether patient knows who the examiner is and the roles or names of the persons with whom in contact
3. Concentration and calculation: Subtracting 7 from 100 and keep subtracting 7s; if patient cannot subtract 7s, can easier tasks be accomplished—4 × 9; 5 × 4; how many nickels are in $1.35; whether anxiety or some disturbance of mood or concentration seems to be responsible for difficulty
4. Memory: Impairment, efforts made to cope with impairment—denial, confabulation, catastrophic reaction, circumstantiality used to conceal deficit: whether the process of registration, retention, or recollection of material is involved
    a. Remote memory: Childhood data, important events known to have occurred when the patient was younger or free of illness, personal matters, neutral material
    b. Recent past memory: Past few months
    c. Recent memory: Past few days, what did patient do yesterday, the day before, have for breakfast, lunch, dinner
    d. Immediate retention and recall: Ability to repeat six figures after examiner dictates them—first forward, then backward, then after a few minutes' interruption; other test questions; did same questions, if repeated, call forth different answers at different times
    e. Effect of defect on patient: Mechanisms patient has developed to cope with defect
5. Fund of knowledge: Level of formal education and self-education; estimate of the patient's intellectual capability

and whether capable of functioning at the level of his or her basic endowment; counting, calculation, general knowledge; questions should have relevance to the patient's educational and cultural background

6. Abstract thinking: Disturbances in concept formation; manner in which the patient conceptualizes or handles his or her ideas; similarities (e.g., between apples and pears), differences, absurdities; meanings of simple proverbs (e.g., "A rolling stone gathers no moss") answers may be concrete (giving specific examples to illustrate the meaning) or overly abstract (giving generalized explanation); appropriateness of answers

F. Insight: Degree of personal awareness and understanding of illness
   1. Complete denial of illness
   2. Slight awareness of being sick and needing help but denying it at the same time
   3. Awareness of being sick but blaming it on others, on external factors, on medical or unknown organic factors
   4. Intellectual insight: Admission of illness and recognition that symptoms or failures in social adjustment are due to irrational feelings or disturbances, without applying that knowledge to future experiences
   5. True emotional insight: Emotional awareness of the motives and feelings within, of the underlying meaning of symptoms; does the awareness lead to changes in personality and future behavior; openness to new ideas and concepts about self and the important persons in his or her life

G. Judgment
   1. Social judgment: Subtle manifestations of behavior that are harmful to the patient and contrary to acceptable behavior in the culture; does the patient understand the likely outcome of personal behavior and is patient influenced by that understanding; examples of impairment
   2. Test judgment: Patient's prediction of what he or she

would do in imaginary situations (e.g., what patient would do with a stamped addressed letter found in the street)

III. Further Diagnostic Studies
   A. Physical examination
   B. Neurologic examination
   C. Additional psychiatric diagnostic studies
   D. Interviews with family members, friends, or neighbors by a social worker
   E. Psychological, neurologic, or laboratory tests as indicated: Electroencephalogram, computed tomography scan, magnetic resonance imaging, tests of other medical conditions, reading comprehension and writing tests, test for aphasia, projective or objective psychological tests, dexamethasone-suppression test, 24-hour urine test for heavy metal intoxication, urine screen for drugs of abuse

IV. Summary of Findings

Summarize mental symptoms, medical and laboratory findings, and psychological and neurologic test results, if available; include medications patient has been taking, dosage, duration. Clarity of thinking is reflected in clarity of writing. When summarizing the mental status (e.g., the phrase "Patient denies hallucinations and delusions" is not as precise as "Patient denies hearing voices or thinking that he is being followed."). The latter indicates the specific question asked and the specific response given. Similarly, in the conclusion of the report one would write "Hallucinations and delusions were not elicited."

V. Diagnosis

Diagnostic classification is made according to DSM-5. The diagnostic numerical code should be used from DSM-5 or ICD-10. It might be prudent to use both codes to cover current and future regulatory guidelines.

VI. Prognosis

Opinion about the probable future course, extent, and outcome of the disorder; good and bad prognostic factors; specific goals of therapy

VII. Psychodynamic Formulation

Causes of the patient's psychodynamic breakdown—influences in the patient's life that contributed to present disorder; environmental, genetic, and personality factors relevant to determining patient's symptoms; primary and secondary gains; outline of the major defense mechanism used by the patient

VIII. Comprehensive Treatment Plan

Modalities of treatment recommended, role of medication, inpatient or outpatient treatment, frequency of sessions, probable duration of therapy; type of psychotherapy; individual, group, or family therapy; symptoms or problems to be treated. Initially, treatment must be directed toward any life-threatening situations such as suicidal risk or risk of danger to others that require psychiatric hospitalization. Danger to self or others is an acceptable reason (both legally and medically) for involuntary hospitalization. In the absence of the need for confinement, a variety of outpatient treatment alternatives are available: day hospitals, supervised residences, outpatient psychotherapy or pharmacotherapy, among others. In some cases, treatment planning must attend to vocational and psychosocial skills training and even legal or forensic issues.

Comprehensive treatment planning requires a therapeutic team approach using the skills of psychologists, social workers, nurses, activity and occupational therapists, and a variety of other mental health professionals, with referral to self-help groups (e.g., Alcoholics Anonymous [AA]) if needed. If either the patient or family members are unwilling to accept the recommendations of treatment and the clinician thinks that the refusal of the recommendations may have serious consequences, the patient, parent, or guardian should sign a statement to the effect that the recommended treatment was refused.

The psychiatric report covers both the psychiatric history and the mental status. The history, or *anamnesis* (from the Greek meaning “to remember”), describes life events within the framework of the life cycle, from infancy to old age, and the clinician should attempt to elicit the emotional reaction to each event as remembered by the patient. The mental status examination covers what the patient is thinking and feeling at the moment and how he or she responds to specific questions from the examiner. Sometimes it may be necessary to report, in detail, the questions asked and the answers received; but this should be kept to a minimum, so that the report does not read like a verbatim transcript. Nevertheless, the clinician should try to use the patient’s own words as much as possible, especially when describing certain symptoms such as hallucinations or delusions.

Finally, the psychiatric report includes more than the psychiatric history and mental status. It also includes a summary of positive and negative findings and an interpretation of the data. It has more than descriptive value; it has meaning that helps provide an understanding of the case. The examiner addresses critical questions in the report: Are future diagnostic studies needed, and, if so, which ones? Is a consultant needed? Is a comprehensive neurologic workup, including an electroencephalogram (EEG) or computed tomography (CT) scan, needed? Are psychological tests indicated? Are psychodynamic factors relevant? Has the cultural context of the patient’s illness been considered? The report includes a diagnosis made according to the 5th edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5). Prognosis is also discussed in the report, with good and bad prognostic factors listed. The report concludes with a discussion of a treatment plan and makes firm recommendations about the management of the case.

Medical Record

The psychiatric report is a part of the medical record; however, the medical record is more than the psychiatric report. It is a narrative that documents all events that occur during treatment, most often referring to the patient’s stay in the hospital. Progress notes record every interaction between doctor and patient; reports of all special studies, including laboratory tests; and prescriptions and orders for all medications. Nurses’ notes help describe the

patient's course: Is the patient beginning to respond to treatment? Are there times during the day or night when symptoms get worse or remit? Are there adverse effects or complaints by the patient about prescribed medication? Are there signs of agitation, violence, or mention of suicide? If the patient requires restraints or seclusion, are the proper supervisory procedures being followed? The medical record tells what happened to the patient since first making contact with the health care system. It concludes with a discharge summary that provides a concise overview of the patient's course with recommendations for future treatment, if necessary. Evidence of contact with a referral agency should be documented in the medical record to establish continuity of care if further intervention is necessary.

Use of the Record. The medical record is not only used by physicians. It is also used by regulatory agencies and managed care companies to determine the length of stay, quality of care, and reimbursement to doctors and hospitals. In theory, the inpatient medical record is accessible to authorized persons only and is safeguarded for confidentiality. In practice, however, absolute confidentiality cannot be guaranteed. Guidelines for what material needs to be incorporated into the medical record are provided in Table 1-8.



Table 1-8
Medical Record

There shall be an individual record for each person admitted to the psychiatric inpatient unit. Patient records shall be safeguarded for confidentiality and should be accessible only to authorized persons. Each case record shall include:

- Legal admission documents
- Identifying information on the individual and family
- Source of referral, date of commencement of service, and name of staff member carrying overall responsibility for treatment and care
- Initial, intercurrent, and final diagnoses, including psychiatric or mental retardation diagnoses in official terminology
- Reports of all diagnostic examinations and evaluations, including findings and conclusions

Reports of all special studies performed, including x-rays, clinical laboratory tests, clinical psychological testing, electroencephalograms, and psychometric tests
The individual written plan of care, treatment, and rehabilitation
Progress notes written and signed by all staff members having significant participation in the program of treatment and care
Summaries of case conferences and special consultations
Dated and signed prescriptions or orders for all medications, with notation of termination dates
Closing summary of the course of treatment and care
Documentation of any referrals to another agency

Adapted from the 1995 guidelines of the New York State Office of Mental Health.

The medical record is also crucial in malpractice litigation. Robert I. Simon summarized the liability issues as follows:

Properly kept medical records can be the psychiatrist's best ally in malpractice litigation. If no record is kept, numerous questions will be raised regarding the psychiatrist's competence and credibility. This failure to keep medical records may also violate state statutes or licensing provisions. Failure to keep medical records may arise out of the psychiatrist's concern that patient treatment information be totally protected. Although this is an admirable ideal, in real life, the psychiatrist may be legally compelled under certain circumstances to testify directly about confidential treatment matters.

Outpatient records are also subject to scrutiny by third parties under certain circumstances, and psychiatrists in private practice are under the same obligation to maintain a record of the patient in treatment as the hospital psychiatrist. Table 1-9 lists documentation issues of concern to third-party payers.

Personal Notes and Observations. According to laws relating to access to medical records, some jurisdictions (such as in the Public Health Law of New York State) have a provision that applies to a physician's personal notes and observations. *Personal notes* are defined as "a practitioner's speculations, impressions (other than tentative or actual diagnosis) and reminders." The data are maintained only by the clinician and cannot be disclosed to any other

person, including the patient. Psychiatrists concerned about material that may prove damaging or otherwise hurtful to the patient if released to a third party may consider using this provision to maintain doctor–patient confidentiality.

Psychotherapy Notes. Psychotherapy notes include details of transference, fantasies, dreams, personal information about persons with whom the patient interacts, and other intimate details of the patient's life. They may also include the psychiatrist's comments on his or her countertransference and feelings toward the patient. Psychotherapy notes should be kept separate from the rest of the medical records.



Table 1-9
Documentation Issues

Are patient's areas of dysfunction described? From the biologic, psychological, and social points of view?
Is alcohol or substance abuse addressed?
Do clinical activities happen at the expected time? If too late or never, why?
Are issues identified in the treatment plan and followed in progress notes?
When there is a variance in the patient's outcome, is there a note in the progress notes to that effect? Is there also a note in the progress notes reflecting the clinical strategies recommended to overcome the impediments to the patient's improvement?
If new clinical strategies are implemented, how is their impact evaluated? When?
Is there a sense of multidisciplinary input and coordination of treatment in the progress notes?
Do progress notes indicate the patient's functioning in the therapeutic community and its relationship to their discharge criteria?
Can one extrapolate from the patient's behavior in the therapeutic community how he or she will function in the community at large?
Are there notes depicting the patient's understanding of his or her discharge planning? Family participation in discharge planning

must be entered in the progress notes with their reaction to the plan.

Do attending progress notes bridge the differences in thinking of other disciplines?

Are the patient's needs addressed in the treatment plan?

Are the patient's family's needs evaluated and implemented?

Is patient and family satisfaction evaluated in any way?

Is alcohol and substance abuse addressed as a possible contributor to readmission?

If the patient was readmitted, are there indications that previous records were reviewed, and, if the patient is on medication other than that prescribed on discharge, is there a rationale for this change?

Do the progress notes identify the type of medication used and the rationale for increase, decrease, discontinuation, or augmentation of medication?

Are medication effects documented, including dosages, response, and adverse or other side effects?

Note: Documentation issues are of concern to third-party payers, such as insurance companies and health maintenance organizations who examine patients' charts to see if the areas listed above are covered. In many cases, however, the review is conducted by persons with little or no background in psychiatry or psychology who do not recognize the complexities of psychiatric diagnosis and treatment. Payments to hospitals, doctors, and patients are often denied because of what such reviewers consider to be the so-called inadequate documentation.

Patient Access to Records. Patients have a legal right to access their medical records. This right represents society's belief that the responsibility for medical care has become a collaborative process between doctor and patient. Patients see many different physicians, and they can be more effective historians and coordinators of their care with such information.

Psychiatrists must be careful in releasing their records to the patient if, in their judgment, the patient can be harmed emotionally as a result. Under these circumstances, the psychiatrist may choose to prepare a summary of the patient's course of treatment, holding back material that might be hurtful—especially if it were to get into the hands of third parties. In malpractice cases,

however, it may not be possible to do so. When litigation occurs, the entire medical record is subject to discovery. Psychotherapy notes are usually protected, but not always. If psychotherapy notes are ordered to be produced, the judge would probably review them privately and select what is relevant to the case in question.

Blogs. Blogs or weblogs are used by persons who wish to record their day-to-day experiences or to express their thoughts and feelings about events. Physicians should be especially cautious about such activities because they are subject to discovery in lawsuits. Pseudonyms and aliases offer no protection because they can be traced. Writing about patients on blogs is a breach of confidentiality. In one case, a doctor detailed his thoughts about a lawsuit that included hostile comments about the plaintiff and his attorney. His blog was discovered inadvertently and was used against him in court. Physicians are advised not to use blogs to vent emotions and to write nothing that they would not write for attribution even if their identity were discovered.

E-Mail. Physicians are increasingly using e-mail as a quick and efficient way to communicate not only with patients but also with other doctors about their patients; however, it is a public document and should be treated as such. The dictum of not diagnosing or prescribing medication over the telephone to a patient one has not examined should also apply to e-mail. It is not only dangerous but also unethical. All e-mail messages should be printed to include with the paper chart unless electronic archives are regularly backed up and secure.

Ethical Issues and the Medical Record. Psychiatrists continually make judgments about what is appropriate material to include in the psychiatric report, the medical record, the case report, and other written communications about a patient. Such judgments often involve ethical issues. In a case report, for example, the patient should not be identifiable, a position made clear in the American Psychiatric Association's (APA's) *Principles of Medical Ethics with Annotations Especially Applicable to Psychiatry*, which states that published case reports must be suitably disguised to safeguard patient confidentiality without altering material to provide a less-than-complete

portrayal of the patient's actual condition. In some instances, obtaining a written release from the patient that allows the psychiatrist to publish the case may also be advisable, even if the patient is appropriately disguised.

Psychiatrists sometimes include material in the medical record that is specifically directed toward warding off future culpability if liability issues are ever raised. This may include having advised the patient about specific adverse effects of medication to be prescribed.

Health Insurance Portability and Accountability Act (HIPAA). The HIPAA was passed in 1996 to address the medical delivery system's mounting complexity and its rising dependence on electronic communication. The act orders that the federal Department of Health and Human Services (HHS) develop rules protecting the transmission and confidentiality of patient information, and all units under HIPAA must comply with such rules.

Two rules were finalized in February 2003: the Transaction Rule and the Privacy Rule (see Tables 1-10 and 1-11). The Transaction Rule facilitates transferring health information effectively and efficiently using regulations created by the HHS that established a uniform set of formats, code sets, and data requirements. The Privacy Rule, administered by the Office of Civil Rights (OCR) at HHS, protects the confidentiality of patient information. This means that a patient's medical information belongs to the patient and that the patient has the right to access it, except for psychotherapy notes, which are deemed as property of the psychotherapist who wrote them.



Table 1-10
Transaction Rule Code Sets

Health care information: The Transactions Rule defines standards and establishes code sets and forms to be used for electronic transaction that involve the following health care information: Claims or equivalent encounter information Eligibility inquiries Referral certification and authorization Claims status inquiries Enrollment and disenrollment information Payment and remittance advice

Health plan premium payments
Coordination of benefits

Code sets: Under the Transaction Rule, the following code sets are required for filing claims with Medicare:
Procedure codes
American Medical Association Current Procedural Terminology codes
Healthcare Common Procedure Coding System codes
Diagnosis codes
International Classification of Disease, 10th edition, clinical modification, codes
Drugs and biologicals
National Drug Codes
Dental codes
Code on dental procedures
Nomenclature for dental services

Adapted from Jaffe E. HIPAA basics for psychiatrists. *Psych Pract Manage Care*. 2002;8:15.

In 2003, the Privacy Rule was executed. Under the Privacy Rule, there are certain guidelines by which every practice must abide:

1. Every practice must establish written privacy procedures. These include administrative, physical, and technical safeguards that establish who has access to the patient's information, how this information is used within the facility, and when the information will and will not be disclosed to others.
2. Every practice must take steps to make sure that its business associates protect the privacy of medical records and other health information.
3. Every practice must train employees to comply with the rule.
4. Every practice must have a designated person to serve as a privacy officer. If it is an individual practice or private practice, this person can be the physician.
5. Every practice must establish complaint procedures for patients who wish to ask or to complain about the privacy of their records.

The OCR at HHS is responsible for making sure that Privacy Rule is enforced; however, it is not clear as to how it will be done. One method expressed by the government is a complaint-driven system in which the OCR will respond to complaints made by patients concerning confidentiality violations or denied access to records, all of which are covered under HIPAA. In such cases, OCR may follow up and audit compliance.

The APA's Committee on Confidentiality, along with legal experts, has developed a set of sample forms. They are part of the APA's HIPAA educational packet, which can be obtained on the APA website (www.psych.org/). On the website, there are also recommendations for enabling physicians to comply with HIPAA.



Table 1-11
Patient's Rights under the Privacy Rule

Physician must give the patient a written notice of his or her privacy rights, the privacy policies of the practice, and how patient information is used, kept, and disclosed. A written acknowledgment should be taken from the patient verifying that he or she seen such notice.
Patients should be able to obtain copies of their medical records and to request revisions to those records within a stated amount of time (usually 30 days). Patients do not have the right to see psychotherapy notes.
Physicians must provide the patient with a history of most disclosures of his or her medical history on request. There are some exceptions. The APA Committee on Confidentiality has developed a model document for this requirement.
Physicians must obtain authorization from the patient for disclosure of information other than for treatment, payment, and health care operations (these three are considered to be routine uses, for which consent is not required). The APA Committee on Confidentiality has developed a model document for this requirement.
Patients may request another means of communication of their

protected information (i.e., request that the physician contact them at a specific phone number or address).

Physicians cannot generally limit treatment to obtaining patient authorization for disclosure of the patient's information for nonroutine uses.

Patients have the right to complain about Privacy Rule violations to the physician, their health plan, or to the secretary of HHS.

APA, American Psychiatric Association; HHS, Department of Health and Human Services.
Adapted from Jaffe E. HIPAA basics for psychiatrists. *Psych Pract Manage Care*. 2002;8:15.

PSYCHIATRIC RATING SCALES

The term *psychiatric rating scales* encompasses a variety of questionnaires, interviews, checklists, outcome assessments, and other instruments that are available to inform psychiatric practice, research, and administration. Psychiatrists must keep up with significant developments in rating scales for several reasons. Most critically, many such scales are useful in psychiatric practice for monitoring patients over time or for providing information that is more comprehensive than what is generally obtained in a routine clinical interview. Also, health care administrators and payors are increasingly requiring standardized assessments to justify the need for services or to assess the quality of care. Lastly, but equally important, rating scales are used in research that informs the practice of psychiatry, so familiarity with them provides a deeper understanding of the results of that research and the degree to which it applies to psychiatric practice.

Potential Benefits and Limitations of Rating Scales in Psychiatry

The key role of rating scales in psychiatry and elsewhere is to standardize the information collected across time and by various observers. This standardization ensures a consistent, comprehensive evaluation that may aid treatment planning by establishing a diagnosis, ensuring a thorough description of symptoms, identifying comorbid conditions, and characterizing other factors affecting treatment response. Also, the use of a rating scale can establish a baseline for follow-up of the progression of an illness over time or in response to specific interventions. This is particularly useful when more than one clinician is involved—for instance, in a group practice or in the

conduct of psychiatric research.

Also to standardization, most rating scales also offer the user the advantages of a formal evaluation of the measure's performance characteristics. This allows the clinician to know to what extent a given scale produces reproducible results (reliability) and how it compares to more definitive or established ways of measuring the same thing (validity).

Types of Scales and What They Measure

Scales are used in psychiatric research and practice to achieve a variety of goals. They also cover a broad range of areas and use a broad range of procedures and formats.

Measurement Goals. Most psychiatric rating scales in common use fall into one or more of the following categories: making a diagnosis; measuring severity and tracking change in specific symptoms, in general functioning, or overall outcome; and screening for conditions that may or may not be present.

Constructs Assessed. Psychiatric practitioners and investigators assess a broad range of areas, referred to as *constructs*, to underscore the fact that they are not simple, direct observations of nature. These include diagnoses, signs and symptoms, severity, functional impairment, quality of life, and many others. Some of these constructs are relatively complex and are divided into two or more domains (e.g., positive and negative symptoms in schizophrenia or mood and neurovegetative symptoms in major depression).

CATEGORICAL VERSUS CONTINUOUS CLASSIFICATION. Some constructs are viewed as *categorical* or classifying, whereas others are seen as *continuous* or measuring. Categorical constructs describe the presence or absence of a given attribute (e.g., competency to stand trial) or the category best suited to a given individual among a finite set of options (e.g., assigning a diagnosis). Continuous measures provide a quantitative assessment along a continuum of intensity, frequency, or severity. Also, to symptom severity and functional status, multidimensional personality traits, cognitive status, social support, and many other attributes are generally measured continuously.

The distinction between categorical and continuous measures is by no

means absolute. *Ordinal* classification, which uses a finite, ordered set of categories (e.g., unaffected, mild, moderate, or severe) stands between the two.

Measurement Procedures. Rating scales differ in measurement methods. Issues to be considered include format, raters, and sources of information.

FORMAT. Rating scales are available in a variety of formats. Some are simply checklists or guides to observation that help the clinician achieve a standardized rating. Others are self-administered questionnaires or tests. Still, others are formal interviews that may be *fully structured* (i.e., specifying the exact wording of questions to be asked) or *partly structured* (i.e., providing only some specific wording, along with suggestions for additional questions or probes).

RATERS. Some instruments are designed to be administered by doctoral-level clinicians only, whereas others may be administered by psychiatric nurses or social workers with more limited clinical experience. Other instruments are designed primarily for use by lay raters with little or no experience with psychopathology.

SOURCE OF INFORMATION. Instruments also vary in the source of information used to make the ratings. Information may be obtained solely from the patient, who generally knows the most about his or her condition. In some instruments, some or all of the information may be obtained from a knowledgeable informant. When the construct involves limited insight (e.g., cognitive disorders or mania) or significant social undesirability (e.g., antisocial personality or substance abuse), other informants may be preferable. Informants may also be helpful when the subject has limited ability to recall or report symptoms (e.g., delirium, dementia, or any disorder in young children). Some rating scales also allow or require information to be included from medical records or patient observation.

Assessment of Rating Scales

In clinical research, rating scales are mandatory to ensure interpretable and potentially generalizable results. They are selected based on coverage of the

relevant constructs, expense (based on the nature of the raters, purchase price if any, and necessary training), length and administration time, comprehensibility to the intended audience, and quality of the ratings provided. In clinical practice, one considers these factors and, also, whether a scale would provide more or better information than what would be obtained in ordinary clinical practice or would contribute to the efficiency of obtaining that information. In either case, the assessment of quality is based on *psychometric,* or mind-measuring, properties.

Psychometric Properties. The two principal psychometric properties of a measure are *reliability* and *validity.* Although these words are used almost interchangeably in everyday speech, they are distinct in the context of evaluating rating scales. Scales should be *reliable,* or consistent and repeatable even if performed by different raters at different times or under different conditions, and they should be *valid* or accurate in representing the actual state of nature.

RELIABILITY. *Reliability* refers to the consistency or repeatability of ratings and is mainly empirical. An instrument is more likely to be reliable if the instructions and questions are clearly and simply worded, and the format is easy to understand and score. There are three standard ways to assess reliability: *internal consistency*, *interrater*, and *test–retest.*

INTERNAL CONSISTENCY. Internal consistency assesses agreement among the individual items in a measure. This provides information about reliability because each item is viewed as a single measurement of the underlying construct. Thus, the coherence of the items suggests that each is measuring the same thing.

INTERRATER AND TEST–RETEST RELIABILITY. Interrater (also called *interjudge* or *joint*) reliability is a measure of agreement between two or more observers evaluating the same subjects using the same information. Estimates may vary with assessment conditions—for instance, estimates of interrater reliability based on videotaped interviews tend to be higher than those based on interviews conducted by one of the raters. Test–retest evaluations measure reliability only to the extent that the subject's true condition remains stable in

the time interval.

ISSUES IN INTERPRETING RELIABILITY DATA. When interpreting reliability data, it is important to bear in mind that reliability estimates published in the literature may not generalize to other settings. Factors to consider are the nature of the sample, the training and experience of the raters, and the test conditions. Issues regarding the sample are especially critical. In particular, reliability tends to be higher in samples with high variability in which it is easier to discriminate among individuals.

VALIDITY. *Validity* refers to conformity with truth, or a gold standard that can stand for truth. In the categorical context, it refers to whether an instrument can make correct classifications. In the continuous context, it refers to accuracy, or whether the score assigned can be said to represent the true state of nature. Although reliability is an empirical question, validity is partly theoretical—for many constructs measured in psychiatry, there is no absolute underlying truth. Even so, some measures yield more useful and meaningful data than others do. Validity assessment is generally divided into face and content validity, criterion validity, and construct validity.

Face and Content Validity. Face validity refers to whether the items appear to assess the construct in question. Although a rating scale may purport to measure a construct of interest, a review of the items may reveal that it embodies a very different conceptualization of the construct. For instance, an insight scale may define *insight* in either psychoanalytic or neurologic terms. However, items with a transparent relationship to the construct may be a disadvantage when measuring socially undesirable traits, such as substance abuse or malingering. *Content validity* is similar to face validity but describes whether the measure provides good balanced coverage of the construct and is less focused on whether the items give the appearance of validity. Content validity is often assessed with formal procedures such as expert consensus or factor analysis.

Criterion Validity. Criterion validity (sometimes called *predictive* or *concurrent validity*) refers to whether or not the measure agrees with a gold standard or criterion of accuracy. Suitable gold standards include the long

form of an established instrument for a new, shorter version, a clinician-rated measure for a self-report form, and blood or urine tests for measures of drug use. For diagnostic interviews, the generally accepted gold standard is the **L**ongitudinal, **E**xpert, **A**ll **D**ata (LEAD) standard, which incorporates expert clinical evaluation, longitudinal data, medical records, family history, and any other sources of information.

Construct Validity. When an adequate gold standard is not available—a frequent state of affairs in psychiatry—or when additional validity data are desired, construct validity must be assessed. To accomplish this, one can compare the measure to *external validators. These attributes bear a well-characterized relationship to the construct under study but are not measured directly by the instrument. External validators used to validate psychiatric diagnostic criteria and the diagnostic instruments that aim to operationalize them include the course of illness, family history, and treatment response. For example, when compared with schizophrenia measures, mania measures are expected to identify more individuals with a remitting course, a family history of major mood disorders, and a good response to lithium.*

Selection of Psychiatric Rating Scales

The scales discussed below cover various areas such as diagnosis, functioning, and symptom severity, among others. Selections were made based on coverage of major areas and common use in clinical research or current (or potential) use in clinical practice. Only a few of the many scales available in each category are discussed here.

Disability Assessment. One of the most widely used scales to measure disability was developed by the World Health Association (WHO), known as the WHO Disability Assessment Schedule, now in its second iteration (WHODAS 2.0). It is self-administered and measures disability along several parameters such as cognition, interpersonal relations, work and social impairment, among many others. It can be taken at intervals along the course of a person's illness and is reliable in tracking changes that indicate a positive or negative response to therapeutic interventions or course of illness (Table 1-12).

Several assessment scales were developed for inclusion in the 5th edition

of the *Diagnostic and Statistical Manual of Mental Disorders* of the American Psychiatric Association, (DSM-5); however, they were developed by and intended for use by research psychiatrists and are not as well tested as the WHO scales. It is expected that, in time, they will eventually be better adapted for clinical use. Some clinicians may wish to use the scales known as *Cross-Cutting Symptom Measure Scales,* but at this time, the WHO scale is recommended for general use.

Psychiatric Diagnosis. Instruments assessing psychiatric diagnoses are central to psychiatric research and may be useful in clinical practice as well. However, they tend to be rather long, especially with individuals reporting many symptoms, potentially requiring many follow-up questions. When such instruments are evaluated, it is essential to ensure they implement the current diagnostic criteria and cover the diagnostic areas of interest.

STRUCTURED CLINICAL INTERVIEW FOR DSM (SCID). The SCID begins with a section on demographic information and clinical background. Then there are seven diagnostic modules focused on different diagnostic groups: mood, psychotic, substance abuse, anxiety, somatic, eating, and adjustment disorders; the modules can be administered separately. Both required and optional probes are provided and skip outs are suggested where no further questioning is warranted. All available information, including that from hospital records, informants, and patient observation, should be used to rate the SCID. The SCID is designed to be administered by experienced clinicians and is generally not recommended for use by lay interviewers. Also, formal training in the SCID is required, and training books and videos are available to facilitate this. Although the primary focus is research with psychiatric patients, a nonpatient version (with no reference to a chief complaint) and a more clinical version (without as much detailed subtyping) are also available. Reliability data on the SCID suggest that it performs better on more severe disorders (e.g., bipolar disorder or alcohol dependence) than on milder ones (e.g., dysthymia). Validity data are limited, as the SCID is more often used as the gold standard to evaluate other instruments. It is considered the standard interview to verify the diagnosis in clinical trials and is extensively used in other forms of psychiatric research. Although its length precludes its use in routine clinical practice, the SCID can sometimes be useful to ensure a

systematic evaluation in psychiatric patients—for instance, on admission to an inpatient unit or at intake into an outpatient clinic. It is also used in forensic practice to ensure a formal and reproducible examination.

Psychotic Disorders. A variety of instruments are used for patients with psychotic disorders. Those discussed here are symptom severity measures. A developing consensus suggests that the distinction between positive and negative symptoms in schizophrenia is worthwhile, and more recently developed instruments implement this distinction.



Table 1-12
WHODAS 2.0

World Health Organization Disability Assessment Schedule 2.0

36-item version, self-administered

Patient Name:_____ **Age:**_____ **Sex:** Male Female **Date:** _____

This questionnaire asks about difficulties due to health/mental health conditions. Health conditions include **diseases or illnesses, other health problems that may be short or long lasting, injuries, mental or emotional problems, and problems with alcohol or drugs.** Think back over the **past 30 days** and answer these questions thinking about how much difficulty you had doing the following activities. For each question, please circle only **one** response.

							Clinician Use Only		
Numeric Scores Assigned to Each of the Items:		**1**	**2**	**3**	**4**	**5**	**Raw Item Score**	**Raw Domain Score**	**Average Domain Score**
In the last 30 days, how much difficulty did you have in:									
Understanding and Communicating									
D1.1	Concentrating on doing something for ten minutes?	None	Mild	Moderate	Severe	Extreme or cannot do			
D1.2	Remembering to do important things?	None	Mild	Moderate	Severe	Extreme or cannot do			
D1.3	Analyzing and finding solutions to problems in day-to-day life?	None	Mild	Moderate	Severe	Extreme or cannot do		_____ 30	_____ 5
D1.4	Learning a new task for example, learning how to get to a new place?	None	Mild	Moderate	Severe	Extreme or cannot do			
D1.5	Generally understanding what people say?	None	Mild	Moderate	Severe	Extreme or cannot do			
D1.6	Starting and maintaining a conversation?	None	Mild	Moderate	Severe	Extreme or cannot do			
Getting Around									
D2.1	Standing for long periods, such as 30 minutes?	None	Mild	Moderate	Severe	Extreme or cannot do			
D2.2	Standing up from sitting down?	None	Mild	Moderate	Severe	Extreme or cannot do			
D2.3	Moving around inside your home?	None	Mild	Moderate	Severe	Extreme or cannot do		_____ 25	_____ 5
D2.4	Getting out of your home?	None	Mild	Moderate	Severe	Extreme or cannot do			
D2.5	Walking a long distance, such as a kilometer (or equivalent)?	None	Mild	Moderate	Severe	Extreme or cannot do			
Self-care									
D3.1	Washing your whole body?	None	Mild	Moderate	Severe	Extreme or cannot do			
D3.2	Getting dressed?	None	Mild	Moderate	Severe	Extreme or cannot do			
D3.3	Eating?	None	Mild	Moderate	Severe	Extreme or cannot do		_____ 20	_____ 5
D3.4	Staying by yourself for a few days?	None	Mild	Moderate	Severe	Extreme or cannot do			
Getting Along with People									
D4.1	Dealing with people you do not know?	None	Mild	Moderate	Severe	Extreme or cannot do			
D4.2	Maintaining a friendship?	None	Mild	Moderate	Severe	Extreme or cannot do			
D4.3	Getting along with people who are close to you?	None	Mild	Moderate	Severe	Extreme or cannot do		_____ 25	_____ 5
D4.4	Making new friends?	None	Mild	Moderate	Severe	Extreme or cannot do			
D4.5	Sexual activities?	None	Mild	Moderate	Severe	Extreme or cannot do			

Numeric Scores Assigned to Each of the Items:		1	2	3	4	5	Clinician Use Only: Raw Item Score	Raw Domain Score	Average Domain Score
Life Activities–Household									
D5.1	Taking care of your household responsibilities?	None	Mild	Moderate	Severe	Extreme or cannot do		20	5
D5.2	Doing most important household tasks well?	None	Mild	Moderate	Severe	Extreme or cannot do			
D5.3	Getting all of the household work done that you needed to do?	None	Mild	Moderate	Severe	Extreme or cannot do			
D5.4	Getting your household work done as quickly as needed?	None	Mild	Moderate	Severe	Extreme or cannot do			
Life Activities–School/Work									
D5.5	Your day-to-day work/school?	None	Mild	Moderate	Severe	Extreme or cannot do		20	5
D5.6	Doing your most important work/ school tasks well?	None	Mild	Moderate	Severe	Extreme or cannot do			
D5.7	Getting all of the work done that you need to do?	None	Mild	Moderate	Severe	Extreme or cannot do			
D5.8	Getting your work done as quickly as needed?	None	Mild	Moderate	Severe	Extreme or cannot do			
Participation in Society									
In the past 30 days:									
D6.1	How much of a problem did you have in joining in community activities (for example, festivities, religious, or other activities) in the same way as anyone else can?	None	Mild	Moderate	Severe	Extreme or cannot do		40	5
D6.2	How much of a problem did you have because of barriers or hindrances around you?	None	Mild	Moderate	Severe	Extreme or cannot do			
D6.3	How much of a problem did you have living with dignity because of the attitudes and actions of others?	None	Mild	Moderate	Severe	Extreme or cannot do			
D6.4	How much time did you spend on your health condition or its consequences?	None	Mild	Moderate	Severe	Extreme or cannot do			
D6.5	How much have you been emotionally affected by your health condition?	None	Mild	Moderate	Severe	Extreme or cannot do			
D6.6	How much has your health been a drain on the financial resources of you or your family?	None	Mild	Moderate	Severe	Extreme or cannot do			
D6.7	How much of a problem did your family have because of your health problems?	None	Mild	Moderate	Severe	Extreme or cannot do			
D6.8	How much of a problem did you have in doing things by yourself for relaxation or pleasure?	None	Mild	Moderate	Severe	Extreme or cannot do			
						General Disability Score (Total):		180	5

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Table 1-13
Brief Psychiatric Rating Scale

DEPARTMENT OF HEALTH AND HUMAN SERVICES PUBLIC HEALTH SERVICE	PATIENT NUMBER – – – –	DATA GROUP bprs	EVALUATION DATE M D Y
Alcohol, Drug Abuse, and Mental Health Administration NIMH Treatment Strategies in Schizophrenia Society	PATIENT NAME		
Brief Psychiatric Rating Scale – Anchored Overall and Gorham	RATER NUMBER		

RATER NUMBER – – –	EVALUATION TYPE (*Circle*)			
	1. Baseline	4. Major evaluation	7. During open meds	10. Study completion
	2. 4-wk minor	5. Other	8. Stop open minds	
	3. Start double-blind	6. Start open meds	9. Early termination	

Introduce all questions with "During the past week have you…"

*1. **Somatic Concern:** Degree of concern over present bodily health. Rate the degree to which physical health is perceived as a problem by the patient, whether complaints have a realistic basis or not. Do not rate mere reporting of somatic symptoms. Rate only concern for (or worrying about) physical problems (real or imagined). **Rate on the basis of reported (i.e., subjective) information pertaining to the past week.**
   1 = Not reported
   2 = Very Mild: occasionally is somewhat concerned about body, symptoms, or physical illness
   3 = Mild: occasionally is moderately concerned, or often is somewhat concerned
   4 = Moderate: occasionally is very concerned, or often is moderately concerned
   5 = Moderately Severe: often is very concerned
   6 = Severe: is very concerned most of the time
   7 = Very Severe: is very concerned nearly all of the time
   8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

2. **Anxiety:** Worry, fear, or overconcern for present or future: Rate solely on the basis of verbal report of patient's own subjective experiences pertaining to the past week.
   Do not infer anxiety from physical signs or from neurotic defense mechanisms. Do not rate if restricted to somatic concern.
   1 = Not reported
   2 = Very Mild: occasionally feels somewhat anxious
   3 = Mild: occasionally feels moderately anxious, or often feels somewhat anxious
   4 = Moderate: occasionally feels very anxious, or often feels moderately anxious
   5 = Moderately Severe: often feels very anxious
   6 = Severe: feels very anxious most of the time
   7 = Very Severe: feels very anxious nearly all of the time
   8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

3. **Emotional Withdrawal:** Deficiency in relating to the interviewer and to the interview situation. Overt manifestations of this deficiency include poor/absence of eye contact, failure to orient oneself physically toward the interviewer, and a general lack of involvement or engagement in the interview. Distinguish from BLUNTED AFFECT, in which deficits in facial expression, body gesture, and voice pattern are scored.
   1 = Not observed
   2 = Very Mild: e.g., occasionally exhibits poor eye contact
   3 = Mild: e.g., as above, but more frequent
   4 = Moderate: e.g., exhibits little eye contact, but still seems engaged in the interview and is appropriately responsive to all questions
   5 = Moderately Severe: e.g., stares at floor or orients self away from interviewer but still seems moderately engaged
   6 = Severe: e.g., as above, but more persistent or pervasive
   7 = Very Severe: e.g., appears "spacey" or "out of it" (total absence of emotional relatedness) and is disproportionately uninvolved or unengaged in the interview (DO NOT SCORE IF EXPLAINED BY DISORIENTATION)
   8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

4. **Conceptual Disorganization:** Degree of speech incomprehensibility. Include any type of formal thought disorder (e.g., loose associations, incoherence, flight of ideas, neologisms). DO NOT include mere circumstantiality or pressured speech, even if marked. DO NOT rate on the basis of the patient's subjective impressions (e.g., "my thoughts are racing. I can't hold a thought," "My thinking gets all mixed up"). Rate ONLY on the basis of observations made during the interview.
   1 = Not observed
   2 = Very Mild: e.g., somewhat vague, but of doubtful clinical significance
   3 = Mild: e.g., frequently vague, but the interview is able to progress smoothly; occasional loosening of associations
   4 = Moderate: e.g., occasional irrelevant statements, infrequent use of neologisms, or moderate loosening of associations
   5 = Moderately Severe: as above, but more frequent
   6 = Severe: formal thought disorder is present for most of the interview, and the interview is severely strained
   7 = Very Severe: e.g., appears "spacey" or "out of it" (total absence of emotional relatedness) and is disproportionately uninvolved or unengaged in the interview (DO NOT SCORE IF EXPLAINED BY DISORIENTATION)

DEPARTMENT OF HEALTH AND HUMAN SERVICES PUBLIC HEALTH SERVICE	PATIENT NUMBER – – – –	DATA GROUP bprs	EVALUATION DATE M D Y
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Brief Psychiatric Rating Scale – Anchored Overall and Gorham	RATER NUMBER		

RATER NUMBER – – –	EVALUATION TYPE (*Circle*)			
	1. Baseline	4. Major evaluation	7. During open meds	10. Study completion
	2. 4-wk minor	5. Other	8. Stop open minds	
	3. Start double-blind	6. Start open meds	9. Early termination	

5. **Guilt Feelings:** Overconcern or remorse for past behavior. Rate on the basis of the patient's subjective experiences of guilt as evidenced by verbal report pertaining to the past week. Do not infer guilt feelings from depression, anxiety, or neurotic defenses.
   1 = Not reported
   2 = Very Mild: occasionally feels somewhat guilty
   3 = Mild: occasionally feels moderately guilty, or often feels somewhat guilty
   4 = Moderate: occasionally feels very guilty, or often feels moderately guilty
   5 = Moderately Severe: often feels very guilty
   6 = Severe: feels very guilty most of the time, or encapsulated delusion of guilt
   7 = Very Severe: agonizing constant feelings of guilt, or pervasive delusion(s) of guilt
   8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

6. **Tension:** Rate motor restlessness (agitation) observed during the interview.
   DO NOT rate on the basis of subjective experiences reported by the patient. Disregard suspected pathogenesis (e.g., tardive dyskinesia).
   1 = Not reported
   2 = Very Mild: e.g., occasionally fidgets
   3 = Mild: e.g., frequently fidgets
   4 = Moderate: e.g., constantly fidgets, or frequently fidgets, wrings hands and pulls clothing
   5 = Moderately Severe: e.g., constantly fidgets, wrings hands and pulls clothing
   6 = Severe: e.g., cannot remain seated (i.e., must pace)
   7 = Very Severe: e.g., paces in a frantic manner

7. **Mannerisms and Posturing:** Unusual and unnatural motor behavior.
   Rate only abnormality of movements. Do not rate simple heightened motor activity here. Consider frequency, duration, and degree of bizarreness. Disregard suspected pathogenesis.
   1 = Not observed
   2 = Very Mild: odd behavior but of doubtful clinical significance, e.g., occasional unprompted smiling, infrequent lip movements
   3 = Mild: strange behavior but not obviously bizarre, e.g., infrequent head-tilting (side to side) in a rhythmic fashion, intermittent abnormal finger movements
   4 = Moderate: e.g., assumes unnatural position for a brief period of time, infrequent tongue protrusions, rocking, facial grimacing
   5 = Moderately Severe: e.g., assumes and maintains unnatural position throughout interview, unusual movements in several body areas
   6 = Severe: as above, but more frequent, intense, or pervasive
   7 = Very Severe: e.g., bizarre posturing throughout most of the interview, continuous abnormal movements in several body areas

[a]8. **Grandiosity:** Inflated self-esteem (self-confidence), or inflated appraisal of one's talents, powers, abilities, accomplishments, knowledge, importance, or identity. Do not score mere grandiose quality of claims (e.g., "I'm the worst sinner in the world," "The entire country is trying to kill me") unless the guilt/persecution is related to some special, exaggerated attributes of the individual. Also, the patient must claim exaggerated attributes: e.g., if patient denies talents, powers, etc., even if he or she states that others indicate that he or she has these attributes, this item should not be scored. Rate on the basis of reported (i.e., subjective) information pertaining to the past week.
   1 = Not reported
   2 = Very Mild: e.g., is more confident than most people, but of only possible clinical significance
   3 = Mild: e.g., definitely inflated self-esteem or exaggerates talents somewhat out of proportion to the circumstances
   4 = Moderate: e.g., inflated self-esteem clearly out of proportion to the circumstances or suspected grandiose delusion(s)
   5 = Moderately Severe: e.g., a single (definite) encapsulated grandiose delusion, or multiple (definite) encapsulated grandiose delusion, or multiple (definite) fragmentary grandiose delusions
   6 = Severe: e.g., a single (definite) grandiose delusion/delusional system, or multiple (definite) grandiose delusions that the patient seems preoccupied with
   7 = Very Severe: e.g., as above, but nearly all conversation is directed toward the patient's grandiose delusion(s)
   8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

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Alcohol, Drug Abuse, and Mental Health Administration NIMH Treatment Strategies in Schizophrenia Society	PATIENT NAME		
Brief Psychiatric Rating Scale – Anchored Overall and Gorham	RATER NUMBER		

RATER NUMBER – – –	EVALUATION TYPE (*Circle*)			
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	2. 4-wk minor	5. Other	8. Stop open minds	
	3. Start double-blind	6. Start open meds	9. Early termination	

[a]9. **Depressive Mood:** Subjective report of feeling depressed, blue, "down in the dumps," etc. Rate only degree of reported depression. Do not rate on the basis of inferences concerning depression based upon general retardation and somatic complaints. Rate on the basis of report (i.e., subjective) information pertaining to the past week.
1 = Not reported
2 = Very Mild: occasionally feels somewhat depressed
3 = Mild: occasionally feels moderately depressed, or often feels somewhat depressed
4 = Moderate: occasionally feels very depressed, or often feels moderately depressed
5 = Moderately Severe: often feels very depressed
6 = Severe: feels very depressed most of the time
7 = Very Severe: feels very depressed nearly all of the time
8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

[a]10. **Hostility:** Animosity, contempt, belligerence, disdain for other people outside the interview situation. Rate solely on the basis of the verbal report of feelings and actions of the patient toward others during the past week. Do not infer hostility from neurotic defenses, anxiety, somatic complaints.
1 = Not reported
2 = Very Mild: occasionally feels somewhat angry
3 = Mild: often feels somewhat angry, or occasionally feels moderately angry
4 = Moderate: occasionally feels very angry, or often feels moderately angry
5 = Moderately Severe: often feels very angry
6 = Severe: has acted on his anger by becoming verbally or physically abusive on one or two occasions
7 = Very Severe: has acted on his anger on several occasions
8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not Assessed

[a]11. **Suspiciousness:** Belief (delusional or otherwise) that others have now, or have had in the past, malicious or discriminatory intent toward the patient. On the basis of verbal report, rate only those suspicions which are currently held whether they concern past or present circumstances. Rate on the basis of reported (i.e., subjective) information pertaining to the past week.
1 = Not reported
2 = Very Mild: rare instances of distrustfulness which may or may not be warranted by the situation
3 = Mild: occasional instances of suspiciousness that are definitely not warranted by the situation
4 = Moderate: more frequent suspiciousness, or transient ideas of reference
5 = Moderately Severe: pervasive suspiciousness, frequent ideas of reference, or an encapsulated delusion
6 = Severe: definite, delusion(s) of reference or persecution that is (are) not wholly pervasive (e.g., an encapsulated delusion)
7 = Very Severe: as above, but more widespread, frequent, or intense
8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

[a]12. **Hallucinatory Behavior:** Perceptions (in any sensory modality) in the absence of an identifiable external stimulus. Rate only those experiences that have occurred during the last week. DO NOT rate "voices in my head," or "visions in my mind" unless the patient can differentiate between these experiences and his or her thoughts.
1 = Not reported
2 = Very Mild: suspected hallucinations only
3 = Mild: definite hallucinations, but insignificant, infrequent, or transient (e.g., occasional formless visual hallucinations, a voice calling the patient's name)
4 = Moderate: as above, but more frequent or extensive (e.g., frequently sees the devil's face, two voices carry on lengthy conversations)
5 = Moderately Severe: hallucinations are experienced nearly every day, or are a source of extreme distress
6 = Severe: as above, and has had a moderate impact on the patient's behavior (e.g., concentration difficulties leading to impaired work functioning)
7 = Very Severe: as above, and has had a severe impact (e.g., attempts suicide in response to command hallucinations)
8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

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	3. Start double-blind	6. Start open meds	9. Early termination	

13. **Motor Retardation:** Reduction in energy level evidenced in slowed movements.
Rate on the basis of observed behavior of the patient only. Do not rate on the basis of the patient's subjective impression of his or her own energy level.
1 = Not observed
2 = Very Mild: Very Mild and of doubtful clinical significance
3 = Mild: e.g., conversation is somewhat retarded, movements somewhat slowed
4 = Moderate: e.g., conversation is noticeably retarded but not strained
5 = Moderately Severe: e.g., conversation is strained, moves very slowly
6 = Severe: e.g., conversation is difficult to maintain, hardly moves at all
7 = Very Severe: e.g., conversation is almost impossible, does not move at all throughout interview

14. **Uncooperativeness:** Evidence of resistance, unfriendliness, resentment, and lack of readiness to cooperate with the interviewer. Rate only on the basis of the patient's attitude and responses to the interviewer and the interview situation. Do not rate on the basis of reported resentment or uncooperativeness outside the interview situation.
1 = Not observed
2 = Very Mild: e.g., does not seem motivated
3 = Mild: e.g., seems evasive in certain areas
4 = Moderate: e.g., monosyllabic, fails to elaborate spontaneously, somewhat unfriendly
5 = Moderately Severe: e.g., expresses resentment and is unfriendly throughout the interview
6 = Severe: e.g., refuses to answer a number of questions
7 = Very Severe: e.g., refuses to answer most questions

15. **Unusual Thought Content:** Severity of delusions of any type—consider conviction, and effect on actions. Assume full conviction if patient has acted on his or her beliefs. Rate on the basis of reported (i.e., subjective) information pertaining to past week.
1 = Not reported
2 = Very Mild: delusion(s) suspected or likely
3 = Mild: at times, patient questions his or her belief(s) (partial delusion)
4 = Moderate: full delusional conviction, but delusion(s) has little or no influence on behavior
5 = Moderately Severe: full delusional conviction, but delusion(s) has only occasional impact on behavior
6 = Severe: delusion(s) has significant effect, e.g., neglects responsibilities because of preoccupation with belief that he/she is God
7 = Very Severe: delusion(s) has major impact, e.g., stops eating because believes food is poisoned

16. **Blunted Affect:** Diminished affective responsivity, as characterized by deficits in facial expression, body gesture, and voice pattern. Distinguish from EMOTIONAL WITHDRAWAL, in which the focus is on interpersonal impairment rather than affect. Consider degree and consistency of impairment. Rate based on observations made during interview.
1 = Not reported
2 = Very Mild: e.g., occasionally seems indifferent to material that is usually accompanied by some show of emotion
3 = Mild: e.g., somewhat diminished facial expression, or somewhat monotonous voice or somewhat restricted gestures
4 = Moderate: e.g., as above, but more intense, prolonged, or frequent
5 = Moderately Severe: e.g., flattening of affect, including at least two of the three features: severe lack of facial expression, monotonous voice or restricted body gestures
6 = Severe: e.g., profound flattening of affect
7 = Very Severe: e.g., totally monotonous voice, and total lack of expressive gestures throughout the evaluation

17. **Excitement:** Heightened emotional tone, including irritability and expansiveness (hypomanic affect). Do not infer affect from statements of grandiose delusions. Rate based on observations made during interview.
1 = Not reported
2 = Very Mild and of doubtful clinical significance
3 = Mild: e.g., irritable or expansive at times
4 = Moderate: e.g., frequently irritable or expansive
5 = Moderately Severe: e.g., constantly irritable or expansive; or, at times, enraged or euphoric
6 = Severe: e.g., enraged or euphoric throughout most of the interview
7 = Very Severe: e.g., as above, but to such a degree that the interview must be terminated prematurely

DEPARTMENT OF HEALTH AND HUMAN SERVICES PUBLIC HEALTH SERVICE	PATIENT NUMBER – – – –	DATA GROUP bprs	EVALUATION DATE M D Y
Alcohol, Drug Abuse, and Mental Health Administration NIMH Treatment Strategies in Schizophrenia Society	PATIENT NAME		
Brief Psychiatric Rating Scale – Anchored Overall and Gorham	RATER NUMBER		

RATER NUMBER – – –	EVALUATION TYPE (*Circle*)			
	1. Baseline	4. Major evaluation	7. During open meds	10. Study completion
	2. 4-wk minor	5. Other	8. Stop open minds	
	3. Start double-blind	6. Start open meds	9. Early termination	

18. **Disorientation:** Confusion or lack of proper association for person, place or time.
Rate based on observations made during interview.
1 = Not reported
2 = Very Mild: e.g., seems somewhat confused
3 = Mild: e.g., indicated 1982 when, in fact, it is 1983
4 = Moderate: e.g., indicates 1978
5 = Moderately Severe: e.g., is unsure where he/she is
6 = Severe: e.g., has no idea where he/she is
7 = Very Severe: e.g., does not know who he/she is
8 = Cannot be assessed adequately because of severe formal thought disorder, uncooperativeness, or marked evasiveness/guardedness; or Not assessed

19. **Severity of Illness:** Considering your total clinical experience with this patient population, how mentally ill is the patient at this time?
1 = Normal, not at all ill
2 = Borderline mentally ill
3 = Mildly ill
4 = Moderately ill
5 = Markedly ill
6 = Severely ill
7 = Among the most severely ill patients

20. **Global Improvement:** Rate total improvement whether or not, in your judgment, it is due to treatment. At baseline assessment, mark "Not assessed" for item 20. For assessments up to the start of double-blind medication, rate Global Improvement compared to baseline. For assessments following the start of double-blind medication, rate Global Improvement compared to the start of double-blind.
1 = Very much improved
2 = Much improved
3 = Minimally improved
4 = No change
5 = Minimally worse
6 = Much worse
7 = Very much worse
8 = Not assessed

[a]Ratings based primarily on verbal report.
From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:1043, with permission.

BRIEF PSYCHIATRIC RATING SCALE (BPRS). The BPRS (Table 1-13) was developed in the late 1960s as a short scale for measuring the severity of psychiatric symptomatology. It was designed primarily to assess change in psychotic inpatients and covers a broad range of areas, including thought disturbance, emotional withdrawal and retardation, anxiety and depression, and hostility and suspiciousness. The reliability of the BPRS is good to excellent when raters are experienced, but this is difficult to achieve without substantial training; a semistructured interview has been developed to increase reliability. Validity is also good as measured by correlations with other measures of symptom severity, especially those assessing schizophrenia symptomatology. The BPRS has been used extensively for decades as an outcome measure in treatment studies of schizophrenia; it functions well as a measure of the change in this context and offers the advantage of

comparability with earlier trials. However, it has been largely supplanted in more recent clinical trials by the newer measures described below. Also, given its focus on psychosis and associated symptoms, it is only suitable for patients with fairly significant impairment. Its use in clinical practice is less well supported, in part because considerable training is required to achieve the necessary reliability.

POSITIVE AND NEGATIVE SYNDROME SCALE (PANSS). The PANSS was developed in the late 1980s to remedy perceived deficits in the BPRS in the assessment of positive and negative symptoms of schizophrenia and other psychotic disorders by adding additional items and providing careful anchors for each. The PANSS requires a clinician rater because it requires considerable probing and clinical judgment. A semistructured interview guide is available. Reliability for each scale is reasonably high, with excellent internal consistency and interrater reliability. Validity also appears good based on correlation with other symptom severity measures and factor analytic validation of the subscales. The PANSS has become the standard tool for assessing clinical outcome in treatment studies of schizophrenia and other psychotic disorders and is easy to administer reliably and sensitive to change with treatment. Its high reliability and good coverage of both positive and negative symptoms make it excellent for this purpose. It may also be useful for tracking severity in clinical practice, and its clear anchors make it easy to use in this setting.

SCALE FOR THE ASSESSMENT OF POSITIVE SYMPTOMS (SAPS) AND SCALE FOR THE ASSESSMENT OF NEGATIVE SYMPTOMS (SANS). The SAPS and SANS (Tables 1-14 and 1-15) were designed to provide a detailed assessment of positive and negative symptoms of schizophrenia and may be used separately or in tandem. SAPS assesses hallucinations, delusions, bizarre behavior, and thought disorder, and SANS assesses affective flattening, poverty of speech, apathy, anhedonia, and inattentiveness. The SAPS and SANS are mainly used to monitor treatment effects in clinical research.

Mood Disorders. The domain of mood disorders includes both unipolar and bipolar disorder, and the instruments described here assess depression and mania. For mania, the issues are similar to those for psychotic disorders

in that limited insight and agitation may hinder accurate symptom reporting, so clinician ratings, including observational data, are generally required. Rating depression, on the other hand, depends, to a substantial extent, on subjective assessment of mood states, so interviews and self-report instruments are both common. Because depression is common in the general population and involves significant morbidity and even mortality, screening instruments—especially those using a self-report format—are potentially quite useful in primary care and community settings.

HAMILTON RATING SCALE FOR DEPRESSION (HAM-D). The HAM-D was developed in the early 1960s to monitor the severity of major depression, with a focus on somatic symptomatology. The 17-item version is the most commonly used version, although versions with different numbers of items, including the 24-item version in Table 1-16, have been used in many studies as well. The 17-item version does not include some of the symptoms for depression in DSM-III and its successors, most notably the so-called reverse neurovegetative signs (increased sleep, increased appetite, and psychomotor retardation). The HAM-D was designed for clinician raters but has been used by trained lay administrators as well. Ratings are completed by the examiner based on the patient interview and observations. A structured interview guide has been developed to improve reliability. The ratings can be completed in 15 to 20 minutes. Reliability is good to excellent, particularly when the structured interview version is used. Validity appears good based on correlation with other depression symptom measures. The HAM-D has been used extensively to evaluate a change in response to pharmacologic and other interventions and, thus, offers the advantage of comparability across a broad range of treatment trials. It is more problematic in the elderly and the medically ill, in whom the presence of somatic symptoms may not be indicative of major depression.

BECK DEPRESSION INVENTORY (BDI). The BDI was developed in the early 1960s to rate depression severity, with a focus on behavioral and cognitive dimensions of depression. The current version, the Beck-II, has added more coverage of somatic symptoms and covers the most recent 2 weeks. Earlier versions are focused on the past week or even shorter intervals, which may be preferable for monitoring treatment response. The scale can be completed in

5 to 10 minutes. Internal consistency has been high in numerous studies. Test–retest reliability is not consistently high, but this may reflect changes in underlying symptoms. Validity is supported by correlation with other depression measures. The principal use of the BDI is as an outcome measure in clinical trials of interventions for major depression, including psychotherapeutic interventions. Because it is a self-report instrument, it is sometimes used to screen for major depression.

Anxiety Disorders. The anxiety disorders addressed by the measures below include panic disorder, generalized anxiety disorder, posttraumatic stress disorder (PTSD), and obsessive-compulsive disorder (OCD). When anxiety measures are examined, it is important to be aware that there have been significant changes over time in how anxiety disorders are defined. Both panic and OCD are relatively recently recognized, and the conceptualization of generalized anxiety disorder has shifted over time. Thus, older measures have somewhat less relevance for diagnostic purposes, although they may identify symptoms causing considerable distress. Whether reported during an interview or on a self-report rating scale, virtually all measures in this domain, like the measures of depression discussed above, depend on subjective descriptions of inner states.

HAMILTON ANXIETY RATING SCALE (HAM-A). The HAM-A (Table 1-17) was developed in the late 1950s to assess anxiety symptoms, both somatic and cognitive. Because the conceptualization of anxiety has changed considerably, the HAM-A provides limited coverage of the "worry" required for a diagnosis of generalized anxiety disorder and does not include the episodic anxiety found in panic disorder. A score of 14 has been suggested as the threshold for clinically significant anxiety, but scores of 5 or less are typical in individuals in the community. The scale is designed to be administered by a clinician, and formal training or the use of a structured interview guide is required to achieve high reliability. A computer-administered version is also available. Reliability is fairly good based on internal consistency, interrater, and test–retest studies. However, given the lack of specific anchors, reliability should not be assumed to be high across different users in the absence of formal training. Validity appears good based on correlation with other anxiety scales but is limited by the relative lack of

coverage of domains critical to the modern understanding of anxiety disorders. Even so, the HAM-A has been used extensively to monitor treatment response in clinical trials of generalized anxiety disorder and may also be useful for this purpose in clinical settings.

PANIC DISORDER SEVERITY SCALE (PDSS). The PDSS was developed in the 1990s as a brief rating scale for the severity of panic disorder. It was based on the Yale-Brown Obsessive-Compulsive Scale and has seven items, each of which is rated on an item-specific, 5-point Likert scale. The seven items address frequency of attacks, distress associated with attacks, anticipatory anxiety, phobic avoidance, and impairment. Reliability is excellent based on interrater studies, but, in keeping with the small number of items and multiple dimensions, internal consistency is limited. Validity is supported by correlations with other anxiety measures, both at the total and item levels; lack of correlation with the HAM-D; and, more recently, by brain imaging studies. Growing experience with the PDSS suggests that it is sensitive to change with treatment and is useful as a change measure in clinical trials or other outcome studies for panic disorder, as well as for monitoring panic disorder in clinical practice.



Table 1-14
Scale for the Assessment of Positive Symptoms (SAPS)

0 = None	1 = Questionable	2 = Mild	3 = Moderate	4 = Marked	5 = Severe	
Hallucinations						
1	*Auditory hallucinations* The patient reports voices, noises, or other sounds that no one else hears.					0 1 2 3 4 5
2	*Voices commenting* The patient reports a voice which makes a running commentary on his [her] behavior or thoughts.					0 1 2 3 4 5
3	*Voices conversing* The patient reports hearing two or more voices conversing.					0 1 2 3 4 5
4	*Somatic or tactile hallucinations* The patient reports experiencing peculiar physical sensations in the body.					0 1 2 3 4 5
5	*Olfactory hallucinations* The patient reports experiencing unusual smells which no one else notices.					0 1 2 3 4 5
6	*Visual hallucinations* The patient sees shapes or people that are not actually present.					0 1 2 3 4 5
7	*Global rating of hallucinations* This rating should be based on the duration and severity of the hallucinations and their effects on the patient's life.					0 1 2 3 4 5
Delusions						
8	*Persecutory delusions* The patient believes he [she] is being conspired against or persecuted in some way.					0 1 2 3 4 5
9	*Delusions of jealousy* The patient believes his [her] spouse is having an affair with someone.					0 1 2 3 4 5
10	*Delusions of guilt or sin* The patient believes that he [she] has committed some terrible sin or done something unforgivable.					0 1 2 3 4 5
11	*Grandiose delusions* The patient believes he [she] has special powers or abilities.					0 1 2 3 4 5
12	*Religious delusions* The patient is preoccupied with false beliefs of a religious nature.					0 1 2 3 4 5
13	*Somatic delusions* The patient believes that somehow his [her] body is diseased, abnormal, or changed.					0 1 2 3 4 5
14	*Delusions of reference* The patient believes that insignificant remarks or events refer to him [her] or have some special meaning.					0 1 2 3 4 5
15	*Delusions of being controlled* The patient feels that his [her] feelings or actions are controlled by some outside force.					0 1 2 3 4 5
16	*Delusions of mind reading* The patient feels that people can read his [her] mind or know his [her] thoughts.					0 1 2 3 4 5
17	*Thought broadcasting* The patient believes that his [her] thoughts are broadcast so that he himself [she herself] or others can hear them.					0 1 2 3 4 5
18	*Thought insertion* The patient believes that thoughts that are not his [her] own have been inserted into his [her] mind.					0 1 2 3 4 5
19	*Thought withdrawal* The patient believes that thoughts have been taken away from his [her] mind.					0 1 2 3 4 5
20	*Global rating of delusions* This rating should be based on the duration and persistence of the delusions and their effect on the patient's life.					0 1 2 3 4 5
Bizarre Behavior						
21	*Clothing and appearance* The patient dresses in an unusual manner or does other strange things to alter his [her] appearance.					0 1 2 3 4 5
22	*Social and sexual behavior* The patient may do things considered inappropriate according to usual social norms (e.g., masturbating in public).					0 1 2 3 4 5
23	*Aggressive and agitated behavior* The patient may behave in an aggressive, agitated manner, often unpredictably.					0 1 2 3 4 5
24	*Repetitive or stereotyped behavior* The patient develops a set of repetitive actions or rituals that he [she] must perform over and over.					0 1 2 3 4 5
25	*Global rating of bizarre behavior* This rating should reflect the type of behavior and the extent to which it deviates from social norms.					0 1 2 3 4 5
Positive Formal Thought Disorder						
26	*Derailment* A pattern of speech in which ideas slip off track onto ideas obliquely related or unrelated.					0 1 2 3 4 5
27	*Tangentiality* Replying to a question in an oblique or irrelevant manner.					0 1 2 3 4 5
28	*Incoherence* A pattern of speech which is essentially incomprehensible at times.					0 1 2 3 4 5
29	*Illogicality* A pattern of speech in which conclusions are reached which do not follow logically.					0 1 2 3 4 5
30	*Circumstantiality* A pattern of speech which is very indirect and delayed in reaching its goal idea.					0 1 2 3 4 5
31	*Pressure of speech* The patient's speech is rapid and difficult to interrupt; the amount of speech produced is greater than that considered normal.					0 1 2 3 4 5
32	*Distractible speech* The patient is distracted by nearby stimuli which interrupt his [her] flow of speech.					0 1 2 3 4 5
33	*Clanging* A pattern of speech in which sounds rather than meaningful relationships govern word choice.					0 1 2 3 4 5
34	*Global rating of positive formal thought disorder* This rating should reflect the frequency of abnormality and degree to which it affects the patient's ability to communicate.					0 1 2 3 4 5
Inappropriate Affect						
35	*Inappropriate affect* The patient's affect is inappropriate or incongruous, not simply flat or blunted.					0 1 2 3 4 5



Table 1-15
Scale for the Assessment of Negative Symptoms (SANS)

0 = None	1 = Questionable	2 = Mild	3 = Moderate	4 = Marked	5 = Severe
Affective Flattening or Blunting					
1 *Unchanging facial expression* The patient's face appears wooden, changes less than expected as emotional content of discourse changes.					0 1 2 3 4 5
2 *Decreased spontaneous movements* The patient shows few or no spontaneous movements, does not shift position, move extremities, etc.					0 1 2 3 4 5
3 *Paucity of expressive gestures* The patient does not use hand gestures, body position, etc., as an aid to expressing his ideas.					0 1 2 3 4 5
4 *Poor eye contact* The patient avoids eye contact or "stares through" interviewer even when speaking.					0 1 2 3 4 5
5 *Affective nonresponsivity* The patient fails to smile or laugh when prompted.					0 1 2 3 4 5
6 *Lack of vocal inflections* The patient fails to show normal vocal emphasis patterns, is often monotonic.					0 1 2 3 4 5
7 *Global rating of affective flattening* This rating should focus on overall severity of symptoms, especially unresponsiveness, eye contact, facial expression, and vocal inflections.					0 1 2 3 4 5
Alogia					
8 *Poverty of speech* The patient's replies to questions are restricted in *amount* tend to be brief, concrete, and unelaborated.					0 1 2 3 4 5
9 *Poverty of content of speech* The patient's replies are adequate in amount but tend to be vague, overconcrete, or overgeneralized, and convey little information.					0 1 2 3 4 5
10 *Blocking* The patient indicates, either spontaneously or with prompting, that his [her] train of thought was interrupted.					0 1 2 3 4 5
11 *Increased latency of response* The patient takes a long time to reply to questions; prompting indicates the patient is aware of the question.					0 1 2 3 4 5
12 *Global rating of alogia* The core features of alogia are poverty of speech and poverty of content.					0 1 2 3 4 5
Avolition-apathy					
13 *Grooming and hygiene* The patient's clothes may be sloppy or soiled, and he [she] may have greasy hair, body odor, etc.					0 1 2 3 4 5
14 *Impersistence at work or school* The patient has difficulty seeking or maintaining employment, completing school work, keeping house, etc. If an inpatient, cannot persist at ward activities, such as occupational therapy, playing cards, etc.					0 1 2 3 4 5
15 *Physical anergia* The patient tends to be physically inert. He [she] may sit for hours and does not initiate spontaneous activity.					0 1 2 3 4 5
16 *Global rating of avolition-apathy* Strong weight may be given to one or two prominent symptoms if particularly striking.					0 1 2 3 4 5
Anhedonia-asociality					
17 *Recreational interests and activities* The patient may have few or no interests. Both the quality and quantity of interests should be taken into account.					0 1 2 3 4 5
18 *Sexual activity* The patient may show a decrease in sexual interest and activity, or enjoyment when active.					0 1 2 3 4 5
19 *Ability to feel intimacy and closeness* The patient may display an inability to form close or intimate relationships, especially with the opposite sex and family.					0 1 2 3 4 5
20 *Relationships with friends and peers* The patient may have few or no friends and may prefer to spend all of his [her] time isolated.					0 1 2 3 4 5
21 *Global rating of anhedonia-asociality* This rating should reflect overall severity, taking into account the patient's age, family status, etc.					0 1 2 3 4 5
Attention					
22 *Social inattentiveness* The patient appears uninvolved or unengaged. He [she] may seem spacey.					0 1 2 3 4 5
23 *Inattentiveness during mental status testing* Tests of "serial 7s" (at least five subtractions) and spelling *world* backward: Score: 2 = 1 error; 3 = 2 errors; 4 = 3 errors.					0 1 2 3 4 5
24 *Global rating of attention* This rating should assess the patient's overall concentration, clinically and on tests.					0 1 2 3 4 5

From Nancy C. Andreasen, M.D., Ph.D., Department of Psychiatry, College of Medicine, The University of Iowa, Iowa City, IA 52242, with permission.

CLINICIAN-ADMINISTERED PTSD SCALE (CAPS). The CAPS includes 17 items required to make the diagnosis, covering all four criteria: (1) the event itself, (2) reexperiencing of the event, (3) avoidance, and (4) increased arousal. The diagnosis requires evidence of a traumatic event, one symptom of reexperiencing, three of avoidance, and two of arousal (typically, an item is counted if the frequency is rated at least 1 and intensity is at least 2). The items can also be used to generate a total PTSD severity score obtained by summing the frequency and intensity scales for each item. The CAPS also

includes several global rating scales for the impact of PTSD symptomatology on social and occupational functioning, for general severity, for recent changes, and the validity of the patient's report. The CAPS must be administered by a trained clinician and requires 45 to 60 minutes to complete, with follow-up examinations somewhat briefer. It has demonstrated reliability and validity in multiple settings and multiple languages, although it has had more limited testing in the setting of sexual and criminal assault. It performs well in the research setting for diagnosis and severity assessment but is generally too long for use in clinical practice.



Table 1-16
Hamilton Rating Scale for Depression

For each item select the "cue" which best characterizes the patient.

1: Depressed mood (Sadness, hopeless, helpless, worthless)
- 0 Absent
- 1 These feeling states indicated only on questioning
- 2 These feeling states spontaneously reported verbally
- 3 Communicates feeling states nonverbally—i.e., through facial expression, posture, voice, and tendency to weep
- 4 Patient reports VIRTUALLY ONLY these feeling states in his spontaneous verbal and nonverbal communication

2: Feelings of guilt
- 0 Absent
- 1 Self-reproach, feels he has let people down
- 2 Ideas of guilt or rumination over past errors or sinful deeds
- 3 Present illness is a punishment. Delusions of guilt.
- 4 Hears accusatory or denunciatory voices and/or experiences threatening visual hallucinations

3: Suicide
- 0 Absent
- 1 Feels life is not worth living
- 2 Wishes he were dead or any thoughts of possible death to self
- 3 Suicide ideas or gesture
- 4 Attempts at suicide (any serious attempt rates 4)

4: Insomnia early
- 0 No difficulty falling asleep
- 1 Complains of occasional difficulty falling asleep—i.e., more than ¼ hour
- 2 Complains of nightly difficulty falling asleep

5: Insomnia middle
- 0 No difficulty
- 1 Patient complains of being restless and disturbed during the night
- 2 Waking during the night—any getting out of bed rates 2 (except for purpose of voiding)

6: Insomnia late
- 0 No difficulty
- 1 Waking in early hours of the morning but goes back to sleep
- 2 Unable to fall asleep again if gets out of bed

7: Work and activities
- 0 No difficulty
- 1 Thoughts and feelings of incapacity, fatigue, or weakness related to activities, work, or hobbies
- 2 Loss of interest in activity, hobbies, or work—either directly reported by patient, or indirect in listlessness, indecision, and vacillation (feels he has to push self to work or activities)
- 3 Decrease in actual time spent in activities or decrease in productivity. In hospital, rate 3 if patient does not spend at least 3 hours a day in activities (hospital job or hobbies) exclusive of ward chores
- 4 Stopped working because of present illness. In hospital, rate 4 if patient engages in no activities except ward chores, or if patient fails to perform ward chores unassisted

8: Retardation (Slowness of thought and speech; impaired ability to concentrate; decreased motor activity)
- 0 Normal speech and thought
- 1 Slight retardation at interview
- 2 Obvious retardation at interview
- 3 Interview difficult
- 4 Complete stupor

9: Agitation
- 0 None
- 1 "Playing with" hands, hair, etc.
- 2 Hand-wringing, nail biting, hair pulling, biting of lips

10: Anxiety psychic
- 0 No difficulty
- 1 Subjective tension and irritability
- 2 Worrying about minor matters
- 3 Apprehensive attitude apparent in face or speech
- 4 Fears expressed without questioning

11: Anxiety somatic

0 Absent	Physiologic concomitants of anxiety, such as:
1 Mild	Gastrointestinal—dry mouth, wind, indigestion, diarrhea, cramps, belching
2 Moderate	Cardiovascular—palpitations, headaches
3 Severe	Respiratory—hyperventilation, sighing
4 Incapacitating	Urinary frequency
	Sweating

12: Somatic symptoms gastrointestinal
- 0 None
- 1 Loss of appetite but eating without staff encouragement Heavy feelings in abdomen
- 2 Difficulty eating without staff urging; requests or requires laxatives or medication for bowels or medication for GI symptoms

13: Somatic symptoms general
- 0 None
- 1 Heaviness in limbs, back, or head. Backaches, headache, muscle aches. Loss of energy and fatigability
- 2 Any clear-cut symptom rates 2

14: Genital symptoms

0 Absent	Symptoms such as:
1 Mild	Loss of libido
2 Severe	Menstrual disturbances

15: Hypochondriasis
- 0 Not present
- 1 Self-absorption (bodily)
- 2 Preoccupation with health
- 3 Frequent complaints, requests for help, etc.
- 4 Hypochondriacal delusions

16: Loss of weight

When rating by history
- 0 No weight loss
- 1 Probable weight loss associated with present illness
- 2 Definite (according to patient) weight loss

On weekly ratings by ward psychiatrist, when actual weight changes are measured
- 0 Less than 1 lb weight loss in week
- 1 Greater than 1 lb weight loss in week
- 2 Greater than 2 lb weight loss in week

17: Insight
- 0 Acknowledges being depressed and ill
- 1 Acknowledges illness but attributes cause to bad food, climate, overwork, virus, need for rest, etc.
- 2 Denies being ill at all

18: Diurnal variation

AM	PM		
0	0	Absent	If symptoms are worse in the morning or
1	1	Mild	evening, note which it is and rate
2	2	Severe	severity of variation

19: Depersonalization and derealization
- 0 Absent
- 1 Mild — Such as:
- 2 Moderate — Feeling of unreality
- 3 Severe — Nihilistic ideas
- 4 Incapacitating

20: Paranoid symptoms
- 0 None
- 1 Suspiciousness
- 2
- 3 Ideas of reference
- 4 Delusions of reference and persecution

21: Obsessional and compulsive symptoms
- 0 Absent
- 1 Mild
- 2 Severe

22: Helplessness
- 0 Not present
- 1 Subjective feelings which are elicited only by inquiry
- 2 Patient volunteers his helpless feelings
- 3 Requires urging, guidance, and reassurance to accomplish ward chores or personal hygiene
- 4 Requires physical assistance for dress, grooming, eating, bedside tasks, or personal hygiene

23: Hopelessness
- 0 Not present
- 1 Intermittently doubts that "things will improve" but can be reassured
- 2 Consistently feels "hopeless" but accepts reassurances
- 3 Expresses feelings of discouragement, despair, pessimism about future, which cannot be dispelled
- 4 Spontaneously and inappropriately perseverates: "I'll never get well" or its equivalent

24: Worthlessness (Ranges from mild loss of esteem, feelings of inferiority, self-deprecation to delusional notions of worthlessness)
- 0 Not present
- 1 Indicates feelings of worthlessness (loss of self-esteem) only on questioning
- 2 Spontaneously indicates feelings of worthlessness (loss of self-esteem)
- 3 Different from 2 by degree. Patient volunteers that he is "no good," "inferior," etc.
- 4 Delusional notions of worthlessness—i.e., "I am a heap of garbage" or its equivalent

From Hamilton M. A rating scale for depression. *J Neurol Neurosurg Psychiatry*. 1960;23:56, with permission.



Table 1-17
Hamilton Anxiety Rating Scale

Instructions: This checklist is to assist the physician or psychiatrist in evaluating each patient as to his degree of anxiety and pathologic condition. Please fill in the appropriate rating:

NONE = 0	MILD = 1	MODERATE = 2	SEVERE = 3	SEVERE, GROSSLY DISABLING = 4

Item		Rating
Anxious	Worries, anticipation of the worst, fearful anticipation, irritability	
Tension	Feelings of tension, fatigability, startle response, moved to tears easily, trembling, feelings of restlessness, inability to relax	
Fears	Of dark, of strangers, of being left alone, of animals, of traffic, of crowds	
Insomnia	Difficulty in falling asleep, broken sleep, unsatisfying sleep and fatigue on waking, dreams, nightmares, night-terrors	
Intellectual (cognitive)	Difficulty in concentration, poor memory	
Depressed mood	Loss of interest, lack of pleasure in hobbies, depression, early waking, diurnal swing	
Somatic (muscular)	Pains and aches, twitching, stiffness, myoclonic jerks, grinding of teeth, unsteady voice, increased muscular tone	
Somatic (sensory)	Tinnitus, blurring of vision, hot and cold flushes, feelings of weakness, picking sensation	
Cardiovascular symptoms	Tachycardia, palpitations, pain in chest, throbbing of vessels, fainting feelings, missing beat	
Respiratory symptoms	Pressure or constriction in chest, choking feelings, sighing, dyspnea	
Gastrointestinal symptoms	Difficulty in swallowing, wind, abdominal pain, burning sensations, abdominal fullness, nausea, vomiting, borborygmi, looseness of bowels, loss of weight, constipation	
Genitourinary symptoms	Frequency of micturition, urgency of micturition, amenorrhea, menorrhagia, development of frigidity, premature ejaculation, loss of libido, impotence	
Autonomic symptoms	Dry mouth, flushing, pallor, tendency to sweat, giddiness, tension headache, raising of hair	
Behavior at interview	Fidgeting, restlessness or pacing, tremor of hands, furrowed brow, strained face, sighing or rapid respiration, facial pallor, swallowing, belching, brisk tendon jerks, dilated pupils, exophthalmos	

ADDITIONAL COMMENTS
Investigator's signature:

From Hamilton M. The assessment of anxiety states by rating. *Br J Psychiatry*. 1959;32:50, with permission.

YALE-BROWN OBSESSIVE-COMPULSIVE SCALE (YBOCS). The YBOCS was

developed in the late 1980s to measure the severity of symptoms in OCD. It has 10 items rated based on a semistructured interview. The first five items concern obsessions: the amount of time that they consume, the degree to which they interfere with normal functioning, the distress that they cause, the patient's attempts to resist them, and the patient's ability to control them. The remaining five items ask parallel questions about compulsions. The semistructured interview and ratings can be completed in 15 minutes or less. A self-administered version has recently been developed and can be completed in 10 to 15 minutes. Computerized and telephone use have also been found to provide acceptable ratings. Reliability studies of the YBOCS show good internal consistency, interrater reliability, and test–retest reliability over a 1-week interval. Validity appears good, although data are fairly limited in this developing field. The YBOCS has become the standard instrument for assessing OCD severity and is used in virtually every drug trial. It may also be used clinically to monitor treatment response.

Substance Use Disorders. Substance use disorders include abuse and dependence on both alcohol and drugs. These disorders, particularly those involving alcohol, are common and debilitating in the general population, so screening instruments are particularly helpful. Because these behaviors are socially undesirable, underreporting of symptoms is a significant problem; thus, the validity of all substance use measures is limited by the honesty of the patient. Validation against drug tests or other measures is of great value, particularly when working with patients who have known substance abuse.

CAGE. The CAGE was developed in the mid-1970s to serve as a very brief screen for significant alcohol problems in a variety of settings, which could then be followed up by clinical inquiry. CAGE is an acronym for the four questions that comprise the instrument: (1) Have you ever felt you should **C**ut down on your drinking? (2) Have people **A**nnoyed you by criticizing your drinking? (3) Have you ever felt bad or **G**uilty about your drinking? (4) Have you ever had a drink first thing in the morning to steady your nerves or to get rid of a hangover (**E**ye-opener)? Each "yes" answer is scored as 1, and these are summed to generate a total score. Scores of 1 or more warrant follow-up, and scores of 2 or more strongly suggest significant alcohol problems. The instrument can be administered in a minute or less, either

orally or on paper. Reliability has not been formally assessed. Validity has been assessed against a clinical diagnosis of alcohol abuse or dependence, and these four questions perform surprisingly well. Using a threshold score of 1, the CAGE achieves excellent sensitivity and fair to good specificity. A threshold of 2 provides still greater specificity but at the cost of a drop in sensitivity. The CAGE performs well as an extremely brief screening instrument for use in primary care or in psychiatric practice focused on problems unrelated to alcohol. However, it has limited ability to pick up early indicators of problem drinking that might be the focus of preventive efforts.

ADDICTION SEVERITY INDEX (ASI). The ASI was developed in the early 1980s to serve as a quantitative measure of symptoms and functional impairment due to alcohol or drug disorders. It covers demographics, alcohol use, drug use, psychiatric status, medical status, employment, legal status, and family and social issues. Frequency, duration, and severity are assessed. It includes both subjective and objective items reported by the patient and observations made by the interviewer.

Eating Disorders. Eating disorders include anorexia nervosa, bulimia, and binge-eating disorder. A wide variety of instruments, particularly self-report scales, are available. Because of the secrecy that may surround dieting, bingeing, purging, and other symptoms, validation against other indicators (e.g., body weight for anorexia or dental examination for bulimia) may be very helpful. Such validation is particularly critical for patients with anorexia, who may lack insight into their difficulties.

EATING DISORDERS EXAMINATION (EDE). The EDE was developed in 1987 as the first interviewer-based comprehensive assessment of eating disorders, including diagnosis, severity, and an assessment of subthreshold symptoms. A self-report version (the EDE-Q), as well as an interview for children, have since been developed. The EDE focuses on symptoms during the preceding 4 weeks, although longer-term questions are included to assess diagnostic criteria for eating disorders. Each item on the EDE has a required probe with suggested follow-up questions to judge severity, frequency, or both, which are then rated on a 7-point Likert scale. For the self-report version, subjects are asked to make similar ratings of frequency or severity. The instrument

provides both global severity ratings and ratings on four subscales: restraint, eating concern, weight concern, and shape concern. The interview, which must be administered by a trained clinician, requires 30 to 60 minutes to complete, whereas the self-report version can be completed more quickly. Reliability and validity data for both the EDE and EDE-Q are excellent, although the EDE-Q may have greater sensitivity for binge-eating disorder. The EDE performs well in both the diagnosis and the detailed assessment of eating disorders in the research context. It also has the sensitivity to change as is required for use in clinical trials or monitoring of individual therapy. Even in the research setting, however, the EDE is relatively lengthy for repeated use, and the EDE-Q may be preferable for some purposes. Although the EDE is too lengthy for routine clinical practice, the EDE or EDE-Q might help provide a comprehensive assessment of a patient with a suspected eating disorder, particularly during an evaluation visit or on entry into an inpatient facility.

BULIMIA TEST-REVISED (BULIT-R). The BULIT-R was developed in the mid-1980s to provide both a categorical and a continuous assessment of bulimia. Patients with bulimia typically score above 110, whereas patients without disordered eating typically score below 60. The instrument can be completed in approximately 10 minutes. The BULIT-R shows high reliability based on studies of internal consistency and test–retest reliability in multiple studies. High correlations support validity with other bulimia assessments. The recommended cutoff of 104 suggested to identify probable cases of bulimia shows high sensitivity and specificity for a clinical diagnosis of bulimia nervosa. With cutoffs between 98 and 104, the BULIT-R has been used successfully to screen for cases of bulimia nervosa. As with any screening procedure, follow-up by clinical examination is indicated for individuals scoring positive; clinical follow-up is particularly critical because the BULIT-R does not distinguish clearly between different types of eating disorders. The BULIT-R may also be useful in clinical and research practice to track symptoms over time or in response to treatment. However, more detailed measures of the frequency and severity of bingeing and purging may be preferable in research settings.

Cognitive Disorders. A wide variety of measures of dementia are

available. Most involve cognitive testing and provide objective, quantifiable data. However, scores vary by educational level in subjects without dementia, so these instruments tend to be most useful when the patient's baseline scores are known. Other measures focus on functional status, which can be assessed based on a comparison with a description of the subject's baseline function; these types of measures generally require a knowledgeable informant and, thus, may be more cumbersome to administer but tend to be less subject to educational biases. A third type of measure focuses on the associated behavioral symptoms that are frequently seen in demented patients.

MINI-MENTAL STATE EXAMINATION (MMSE). The MMSE is a 30-point cognitive test developed in the mid-1970s to provide a bedside assessment of a broad array of cognitive functions, including orientation, attention, memory, construction, and language. It can be administered in less than 10 minutes by a busy doctor or a technician and scored rapidly by hand. The MMSE has been extensively studied and shows excellent reliability when raters refer to consistent scoring rules. Validity appears good based on correlations with a wide variety of more comprehensive measures of mental functioning and clinicopathologic correlations.

Since its development in 1975, the MMSE was widely distributed in textbooks, pocket guides, and on websites and has been used at the bedside. In 2001 the authors granted a worldwide exclusive license to Psychological Assessment Resources (PAR) to publish, distribute, and manage all intellectual property rights to the test. A licensed version of the MMSE, officially called the MMSE-2, must now be purchased from PAR per test. The MMSE form is gradually disappearing from textbooks, websites, and clinical tool kits.

In an article in the *New England Journal of Medicine* (2011;365:2447–2449) John C. Newman and Robin Feldman concluded: "The restrictions on the MMSE's use present clinicians with difficult choices: increase practice costs and complexity, risk copyright infringement, or sacrifice 30 years of practical experience and validation to adopt new cognitive assessment tools."

NEUROPSYCHIATRIC INVENTORY (NPI). The NPI was developed in the mid-1990s to assess a wide range of behavioral symptoms that are often seen in Alzheimer disease and other dementing disorders. The current version rates

12 areas: delusions, hallucinations, dysphoria, anxiety, agitation/aggression, euphoria, disinhibition, irritability/lability, apathy, aberrant motor behavior, nocturnal disturbances, appetite, and eating. The standard NPI is an interview with a caregiver or other informant that can be performed by a clinician or trained lay interviewer and requires 15 to 20 minutes to complete. There is also a nursing home interview version, the NPI-NH, and a self-report questionnaire, the NPI-Q. For each area, the NPI asks whether a symptom is present and, if so, assesses the frequency, severity, and associated caregiver distress. The instrument has demonstrated reliability and validity and is useful to screen for problem behaviors in both clinical and research settings. Because of the detailed frequency and severity ratings, it is also helpful to monitor change with treatment.

SCORED GENERAL INTELLIGENCE TEST (SGIT). This test was developed and validated by N. D. C. Lewis at the New York State Psychiatric Institute in the 1930s. It is one of the few tests that attempts to measure general intelligence that can be administered by the clinician during the psychiatric interview. A decline in general intelligence will be seen in cognitive disorders, and the SGIT can alert the clinician to begin a workup for disease states that interfere with cognition. This test deserves more widespread use (Table 1-18).

Personality Disorders and Personality Traits. Personality may be conceptualized categorically as personality disorders or dimensionally as personality traits, which may be viewed as normal or pathologic. The focus here is on personality disorders and the maladaptive traits generally viewed as their milder forms. Ten personality disorders are divided into three clusters. Patients tend not to fall neatly into DSM personality categories; instead, most patients who meet the criteria for one personality disorder also meet the criteria for at least one other, particularly within the same cluster. This and other limitations in the validity of the constructs themselves make it difficult to achieve validity in personality measures. Personality measures include both interviews and self-report instruments. Self-report measures are appealing in that they require less time and may appear less threatening to the patient. However, they tend to overdiagnose personality disorders. Because many of the symptoms suggesting personality problems are socially undesirable and because patients' insight tends to be limited, clinician-

administered instruments, which allow for probing and patient observation, may provide more accurate data.

PERSONALITY DISORDER QUESTIONNAIRE (PDQ). The PDQ was developed in the late 1980s as a simple self-report questionnaire designed to provide a categorical and dimensional assessment of personality disorders. The PDQ includes 85 yes–no items intended primarily to assess the diagnostic criteria for personality disorders. Within the 85 items, two validity scales are embedded to identify underreporting, lying, and inattention. There is also a brief clinician-administered Clinical Significance Scale to address the impact of any personality disorder identified by the self-report PDQ. The PDQ can provide categorical diagnoses, a scaled score for each, or an overall index of personality disturbance based on the sum of all of the diagnostic criteria. Overall scores range from 0 to 79; normal controls tend to score below 20, personality disordered patients generally score above 30, and psychotherapy outpatients without such disorders tend to score in the 20 to 30 range.

Childhood Disorders. A wide variety of instruments are available to assess mental disorders in children. Despite this rich array of instruments, however, the evaluation of children remains difficult for several reasons. First, the child psychiatric nosology is at an earlier stage of development, and construct validity is often problematic. Second, because children change markedly with age, it is virtually impossible to design a measure that covers children of all ages. Lastly, because children, particularly young children, have limited ability to report their symptoms, other informants are necessary. This often creates problems because there are frequent disagreements among child, parent, and teacher reports of symptoms, and the optimal way to combine information is unclear.

CHILD BEHAVIOR CHECKLIST (CBCL). The CBCL is a family of self-rated instruments that survey a broad range of difficulties encountered in children from preschool through adolescence. One version of the CBCL is designed for completion by parents of children between 4 and 18 years of age. Another version is available for parents of children between 2 and 3 years of age. Children between 11 and 18 years of age complete the Youth Self-Report and the Teacher Report Form is completed by teachers of school-age children.

The scale includes not only problem behaviors but also academic and social strengths. Each version includes approximately 100 items scored on a 3-point Likert scale. Scoring can be done by hand or computer, and normative data are available for each of the three subscales: problem behaviors, academic functioning, and adaptive behaviors. A computerized version is also available. The CBCL does not generate diagnoses but, instead, suggests cutoff scores for problems in the "clinical range." Parent, teacher, and child versions all show high reliability on the problem subscale, but the three informants frequently do not agree with one another. The CBCL may be useful in clinical settings as an adjunct to clinical evaluation, as it provides a good overall view of symptomatology and may also be used to track change over time. It is used frequently for similar purposes in research involving children and, thus, can be compared with clinical experience. The instrument does not, however, provide diagnostic information, and its length limits its efficiency for tracking purposes.



Table 1-18
Scored General Intelligence Test (SGIT)

Indications: When a cognitive disorder is suspected because of apparent intellectual defects, impairment in the ability to make generalizations, the ability to maintain a trend of thought, or to show good judgment, a scored test can be of value.
Directions: Ask the following questions as part of the mental status examination. A conversational manner should be used and the questions may be adapted to cultural differences.
Scoring: If the patient obtains a score of 25 or under (out of a maximum of 40), it is indicative of a cognitive problem and further examination should follow.
Questions: There are 13 questions that follow.

1. What are houses made of? (Any material you can think of)	1–4
One point for each item, up to four.	
2. What is sand used for?	1, 2, or 4
Four points for manufacture of glass. Two points for mixing with concrete, road building, or other constructive use. One point for play or sandboxes. Credit not cumulative.	
3. If the flag floats to the south, from what direction is the wind?	3
Three points for north, no partial credits. It is permissible to say: "Which way is the wind coming from?"	
4. Tell me the names of some fish.	1–4
One point for each, up to four. If the subject stops with one, encourage him or her to go on.	
5. At what time of day is your shadow shortest?	3
Noon, three points. If correct response is suspected of being a guess, inquire why.	
6. Give the names of some large cities.	1–4
One point for each, up to four. When any state is named as a city, no credit, that is, New York unless specified as New York City. No credit for hometown, except when it is an outstanding city.	
7. Why does the moon look larger than the stars?	2, 3, or 4
Make it clear that the question refers to any particular star, and give assurance that the moon is actually smaller than any star. Encourage the subject to guess. Two points for "Moon is lower down." Three points for nearer or closer. Four points for generalized statement that nearer objects look larger than more distant objects.	
8. What metal is attracted by a magnet?	2 or 4
Four points for iron, two for steel.	
9. If your shadow points to the northeast, where is the sun?	4
Four points for southwest, no partial credits.	
10. How many stripes are in the American flag?	2
Thirteen, two points. A subject who responds 50 may be permitted to correct the mistake. Explain, if necessary, that the white stripes are included as well as the red ones.	
11. What does ice become when it melts?	1
Water, one point.	
12. How many minutes in an hour?	1
60, one point.	
13. Why is it colder at night than in the daytime?	1–2
Two points for "sun goes down," or any recognition of direct rays of sun as source of heat. Question may be reversed: "What makes it warmer in the daytime than at night?" Only one point for answer to reverse question.	

This test was developed and validated by N. D. C. Lewis, M.D.
From Sadock BJ, Sadock VA. *Pocket Handbook of Clinical Psychiatry*. 5th ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2010, with permission.

DIAGNOSTIC INTERVIEW SCHEDULE FOR CHILDREN (DISC). The current DISC, the DISC-IV, covers a broad range of DSM diagnoses, both current and lifetime. It has nearly 3,000 questions but is structured with a series of stem questions that serve as gateways to each diagnostic area, with the remainder of each section skipped if the subject answers no. Subjects who enter each section have very few skips, so complete diagnostic and symptom scale information can be obtained. Child, parent, and teacher versions are available. Computer programs are available to implement diagnostic criteria and generate severity scales based on each version or to combine parent and child information. A typical DISC interview may take more than 1 hour for a child, plus an additional hour for a parent. However, because of the stem question structure, the actual time varies widely with the number of symptoms endorsed. The DISC was designed for lay interviewers. It is relatively complicated to administer, and formal training programs are highly recommended. The

reliability of the DISC is only fair to good and generally better for the combined child and parent interview. Validity judged against a clinical interview by a child psychiatrist is also fair to good—better for some diagnoses and better for the combined interview. The DISC is well tolerated by parents and children and can be used to supplement a clinical interview to ensure comprehensive diagnostic coverage. Because of its inflexibility, some clinicians find it uncomfortable to use, and its length makes it less than optimal for use in clinical practice. However, it is frequently used in a variety of research settings.

CONNERS RATING SCALES. The Conners Rating Scales are a family of instruments designed to measure a range of childhood and adolescent psychopathology. They are most commonly used in the assessment of ADHD. The main uses of the Conners Rating Scales are in screening for ADHD in school or clinic populations and following changes in symptom severity over time; sensitivity to change in response to specific therapies has been demonstrated for most versions of the Conners Rating Scales. There are teacher, parent, and self-report (for adolescents) versions and both short (as few as 10 items) and long (as many as 80 items, with multiple subscales) forms. Reliability data are excellent for the Conners Rating Scales. However, the teacher and parent versions tend to show poor agreement. Validity data suggest that the Conners Rating Scales are excellent at discriminating between ADHD patients and normal controls.

AUTISM DIAGNOSTIC INTERVIEW-REVISED (ADI-R). The Autism Diagnostic Interview (ADI) was developed in 1989 as a clinical assessment of autism and related disorders. The ADI-R was developed in 2003 to provide a shorter instrument with a better ability to discriminate autism from other developmental disorders. The instrument has 93 items, is designed for individuals with a mental age greater than 18 months, and covers three broad areas, consistent with the diagnostic criteria for autism: language and communication; reciprocal social interactions; and restricted, repetitive, and stereotyped behaviors and interests. There are three versions: one for lifetime diagnosis, one for current diagnosis, and one for patients under age 4 focused on an initial diagnosis. It must be administered by a clinician trained in its use and takes about 90 minutes to complete. When clinicians are properly trained,

it has good to excellent reliability and validity but performs poorly in the setting of severe developmental disabilities. It is generally intended for the research setting when a thorough assessment of autism is required but may have use in clinical practice as well.

CLINICAL NEUROPSYCHOLOGY AND INTELLECTUAL ASSESSMENT OF ADULTS

Clinical neuropsychology is a specialty in psychology that examines the relationship between behavior and brain functioning in the realms of cognitive, motor, sensory, and emotional functioning. The clinical neuropsychologist integrates the medical and psychosocial history with the reported complaints and the pattern of performance on neuropsychological procedures to determine whether results are consistent with a particular area of brain damage or a specific diagnosis.

Neuroanatomical Correlates

The early history of neuropsychology was driven in large part by the goal of linking behavioral deficits to specific neuroanatomical areas of dysfunction or damage. Although this historic assessment method helped to validate neuropsychological tests that are commonly used today, the localizing function of neuropsychological assessment is now considered less important in light of recent advances in neuroimaging techniques. Increasing knowledge in the neurosciences has also led to a more sophisticated view of brain–behavior relationships, in which complex cognitive, perceptual, and motor activities are controlled by neural circuits rather than single structures within the brain. An understanding of these brain–behavior relationships is particularly helpful when evaluating patients with focal damage. It is crucial to ensure that the neuropsychological evaluation adequately assesses relevant behavior that is likely to be associated with that area and its interconnecting pathways.



Table 1-19
Selected Neuropsychological Deficits Associated with Left or Right Hemisphere Damage

Left Hemisphere	Right Hemisphere
Aphasia	Visuospatial deficits
Right–left disorientation	Impaired visual perception

Finger agnosia	Neglect
Dysgraphia (aphasic)	Dysgraphia (spatial, neglect)
Dyscalculia (number alexia)	Dyscalculia (spatial)
Constructional apraxia (details)	Constructional apraxia (Gestalt)
Limb apraxia	Dressing apraxia
	Anosognosia

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.

Hemispheric Dominance and Intrahemispheric Localization. Many functions are mediated by both the right and left hemispheres. However, critical qualitative differences between the two hemispheres can be demonstrated in the presence of lateralized brain injury. Various cognitive skills that have been linked to the left or right hemisphere in right-handed individuals are listed in Table 1-19. Although language is the most apparent function that is mostly controlled by the left hemisphere, especially among right-handed individuals, the left hemisphere is also generally considered to be dominant for limb praxis (i.e., performing complex movements, such as brushing teeth, to command, or imitation). It has been associated with the cluster of deficits identified as Gerstmann syndrome (i.e., finger agnosia, dyscalculia, dysgraphia, and right–left disorientation). In contrast, the right hemisphere is thought to play a more critical role in controlling visuospatial abilities and hemispatial attention, which are associated with the clinical presentations of constructional apraxia and neglect, respectively.

Although lateralized deficits such as these are typically characterized in terms of *damage* to the right or left hemisphere, it is essential to keep in mind that the patient's performance can also be described in terms of *preserved* brain functions. In other words, it is the remaining intact brain tissue that drives many behavioral responses following an injury to the brain and not only the absence of critical brain tissue.

LANGUAGE DISORDERS. Appreciation for the special role of the left hemisphere in the control of language functions in most right-handed

individuals has been validated in many studies. These include the results of sodium amytal testing in epilepsy surgery patients, as well as the incidence of aphasia following unilateral stroke to the left versus right hemisphere. Although it is rare for right-handed individuals to be right hemisphere dominant for language, it does occur in about 1 percent of the cases. Hemispheric dominance for language in left-handed individuals varies. About two-thirds of left-handed individuals are left-hemisphere dominant for language, while about 20 percent each are right hemisphere dominant or bilaterally dominant.

Several classification systems have been developed over the years for describing various patterns of language breakdown. A standard method takes into account the presence or absence of three key features: (1) fluency, (2) comprehension, and (3) repetition (i.e., intact ability to repeat verbally presented words or phrases).

Broca Aphasia. Broca Aphasia (also called *nonfluent* or *expressive aphasia*) has traditionally been characterized by nonfluent speech but intact auditory comprehension and somewhat impaired repetition. It has long been thought to be associated with damage to Broca area (i.e., left inferior frontal convolution) or Brodmann area 44 (Fig. 1-3). However, more recent neuroimaging data in stroke patients have shown that the full syndrome of Broca aphasia, including *agrammatism* (telegraphic speech), is found only in the presence of more extensive damage, which encompasses the suprasylvian area from Broca area to the posterior extent of the Sylvian fissure.

Wernicke Aphasia. Wernicke aphasia (also called *fluent* or *receptive aphasia*) is characterized by fluent speech, impaired comprehension, and somewhat impaired repetition. It has been associated with damage to Wernicke area in the region of the superior temporal gyrus. The impaired ability to comprehend language directly affects the individual's ability to self-monitor language output and may be related to a breakdown of the syntactic structure of language. Unlike patients with Broca aphasia, who are usually painfully aware of their communication difficulty, patients with Wernicke aphasia are typically not aware of their communication problems because Wernicke area is critical for comprehending their speech as well as the language of others. This lack of insight is similar to the condition of *anosognosia,* in which

patients fail to appreciate their deficits, and presents an incredibly frustrating condition for many family members and caregivers.

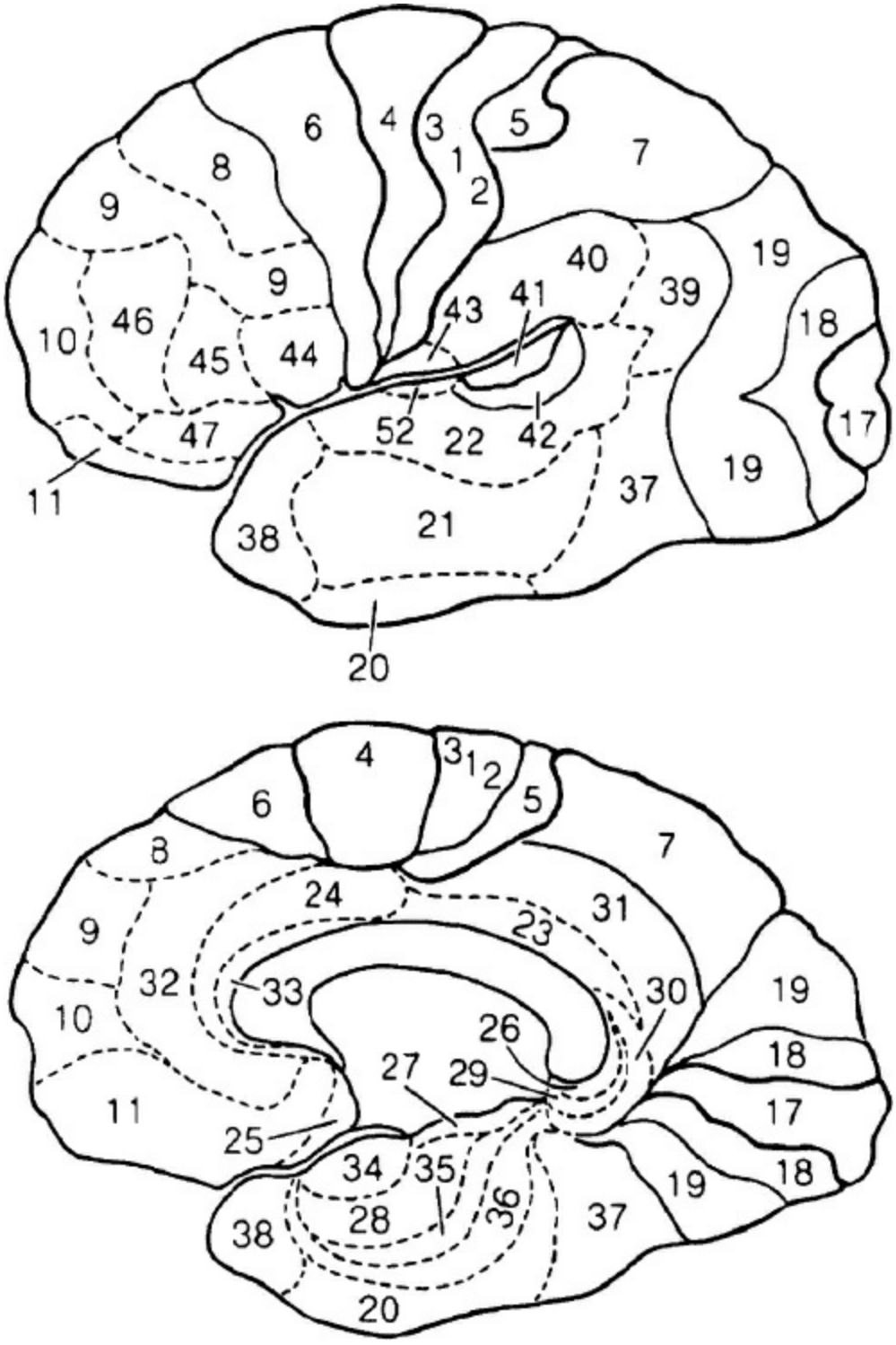


FIGURE 1-3
Brodmann areas of the human cortex, showing convex surface (*top*) and medial

surface (*bottom*). (From Elliott HC. *Textbook of Neuroanatomy*. Philadelphia, PA: Lippincott; 1969, with permission.)

Conduction Aphasia. Patients with *conduction aphasia demonstrate relatively intact auditory comprehension and spontaneous speech, due to the preservation of Wernicke and Broca areas. However, the ability to repeat words and phrases is specifically impaired and has traditionally been attributed to damage to the arcuate fasciculus, which interconnects Wernicke and Broca areas. This type of aphasia is much more subtle and tends to have a less negative impact on daily functioning.*

Global Aphasia. Another common classification, *global aphasia, is characterized by impairment in all three dimensions of fluency, comprehension, and repetition due to damage to the core language areas on the lateral surface of the left hemisphere. In reality, many aphasic patients cannot be neatly classified within a specific system because the pattern of deficits does not precisely fit clear, descriptive categories. Detailed language assessment of most aphasic patients typically demonstrates deficits in all three areas, although the degree of deficit among the three areas varies.*

LIMB APRAXIA. Limb apraxia and other cognitive-motor skills deficits are more commonly seen with left than with right hemisphere damage. However, Kathleen Haaland and Deborah Harrington reviewed data showing that the difference in the incidence of limb apraxia after left or right hemisphere damage is not as great as with language, suggesting that left hemisphere dominance for disorders of complex movement is not as strong as that for language. Although limb apraxia has not traditionally been considered to be of substantial functional importance, more recently, new information suggests it may significantly affect rehabilitation outcome. *Conceptual apraxia* might result in using the wrong object to perform a movement, such as attempting to use a toothbrush to eat. Finally, sequencing errors and ideational errors can lead to disrupted activities, such as trying to light a candle before striking the match.

ARITHMETIC. Arithmetic skills can be impaired after either left or right hemisphere damage. Left hemisphere damage, especially of the parietal lobe,

produces difficulty in reading and appreciating the symbolic meaning of numbers (*number dyslexia*). Left hemisphere damage also can be associated with an impaired conceptual understanding of the arithmetic problem (*anarithmetria*). In contrast, the deficits in arithmetic computation that can accompany right hemisphere damage are more likely to be observed in written problems. These emerge as problems with the spatial aspects of arithmetic, such as errors resulting from hemispatial visual neglect, poor alignment of columns, or visual misperceptions and rotations that can confuse signs for addition and multiplication.

SPATIAL DISORDERS. Right hemisphere damage in right-handed individuals is frequently associated with deficits in visuospatial skills. Common assessment techniques include drawings and constructional or spatial assembly tasks.

Visuospatial Impairment. Distinctive qualitative errors in constructing block designs and in drawing a complex geometric configuration (e.g., Rey–Osterrieth Complex Figure test) can be seen with either right or left hemisphere damage. In the presence of lateralized damage to the right hemisphere, impaired performance often reflects the patient's inability to appreciate the "Gestalt" or global features of a design. In the example shown in Figure 1-4, this is seen in the patient's failure to maintain the 2 × 2 matrix of blocks and instead of converting this matrix into a column of four blocks. In contrast, damage to the left hemisphere commonly results in inaccurate reproduction of internal details of the design, including improper orientation of individual blocks. However, the 2 × 2 matrix (i.e., the Gestalt) is more likely to be preserved. Many neuropsychologists emphasize that a neuropsychological understanding of the impairment depends not just on a set of test scores but also on a qualitative description of the type of error. This often allows the impairment to be linked to a specific neuroanatomical region as well as enabling a better understanding of the mechanisms of the deficit for rehabilitation purposes. This qualitative focus on the type of error is similar to the *pathognomonic* approach that is often used by behavioral neurologists.

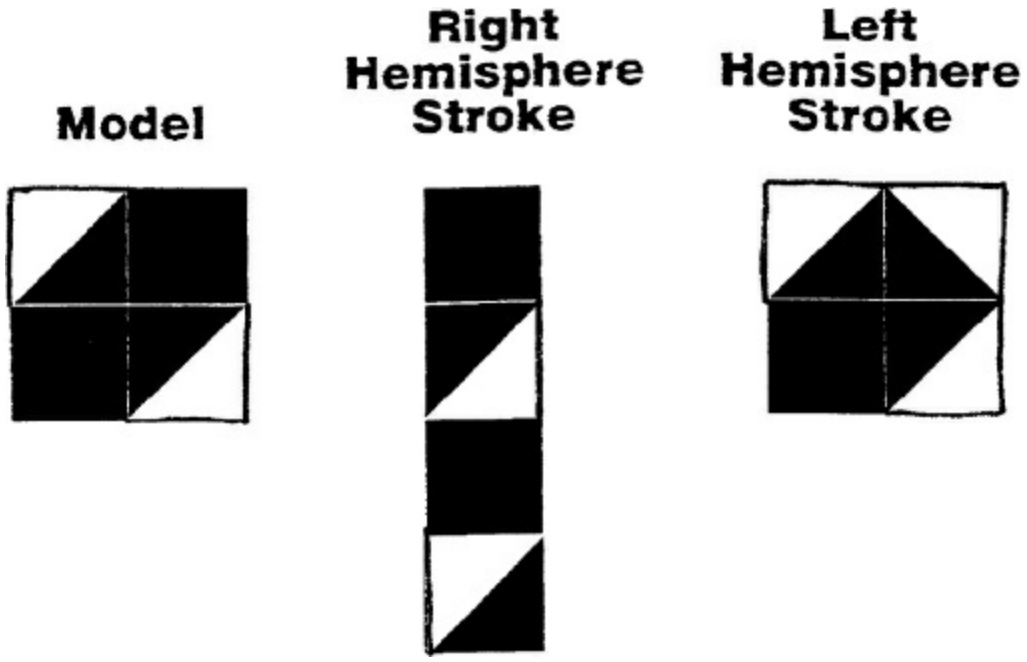


FIGURE 1-4
Examples of block design construction seen in a right hemisphere stroke patient and a left hemisphere stroke patient. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.)

In another example, damage to the right hemisphere tends to be associated with a decreased appreciation of global features of visual stimuli. In contrast, left hemisphere damage tends to be associated with the decreased analysis of local features and detail. This notion is illustrated in Figure 1-5, where a patient with left hemisphere damage focuses on the larger Gestalt of the triangle or letter M with no regard for the internal characters that make up the designs. In contrast, the "local" approach of a patient with right hemisphere damage emphasizes the internal details (small rectangles or letter Z) without an appreciation of the Gestalt that is formed by the internal details. This example also illustrates the important point that behavioral responses (including errors) are driven as much by preserved regions of intact brain functioning as by the loss of other regions of brain functioning.

Neglect. Neglect syndromes are characterized by failure to detect visual or tactile stimuli or to move the limb in the contralateral hemispace. They are most commonly associated with right hemisphere damage in the parietal region, but damage to other areas within the cerebral cortex and subcortical

areas can also produce this problem. Although neglect syndromes have a similar incidence and may co-occur with visual field cuts or somatosensory deficits, the neglect syndrome is distinct and not explained by any motor or sensory problems that may be present. Visual neglect can be assessed with line cancellation and line bisection tasks, in which the paper is placed at the patient's midline, and the patient is asked to either cross out all of the lines on the page or to bisect the single line presented. The method of double simultaneous stimulation or visual extinction is another standard procedure for demonstrating the deficit. Neglect syndromes can have devastating functional effects on safety and the ability to live independently and should be taken into account as a standard consideration in the evaluation process.

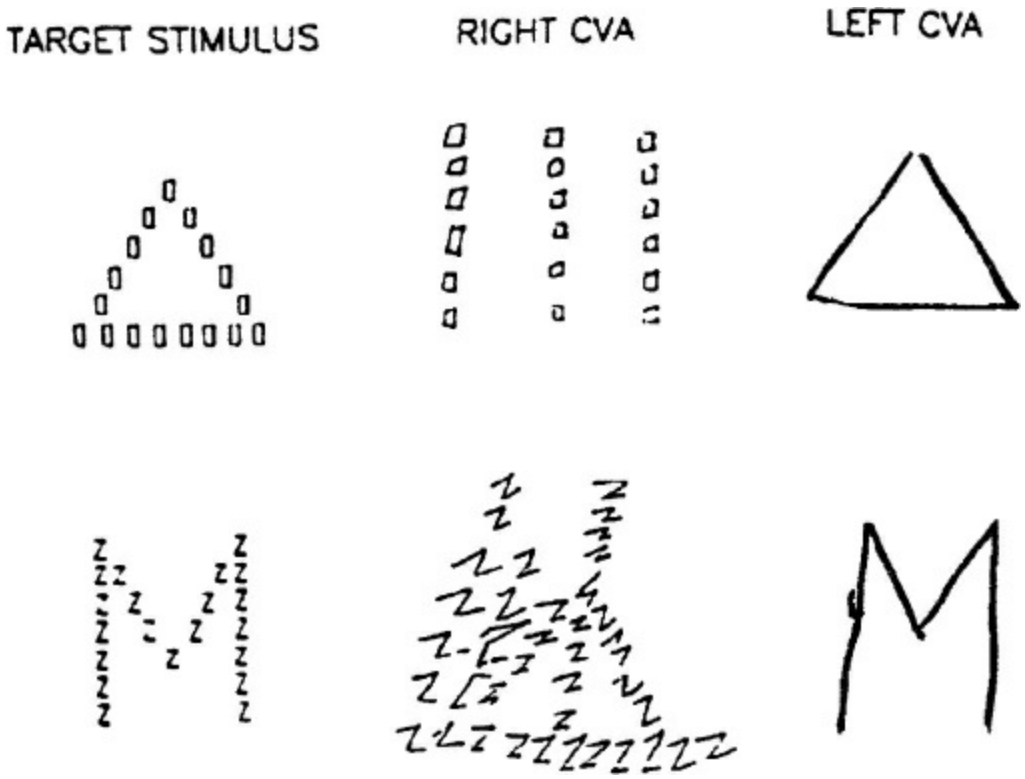


FIGURE 1-5
Global local target stimuli with drawings from memory by a patient with right hemisphere cerebrovascular accident (CVA) and by a patient with left hemisphere CVA. (From Robertson LC, Lamb MR. Neuropsychological contributions to theories of part/whole organization. *Cogn Psychol*. 1991;23:325, with permission from Elsevier Science.)

Dressing Apraxia. The syndrome of *dressing apraxia tends to arise in association with spatial deficits following right hemisphere damage. The*

resulting difficulty in coordinating the spatial and tactual demands of dressing can be seen in the patient's difficulty in identifying the top or bottom of a garment, as well as right–left confusion in inserting his or her limbs into the garment. As a result, dressing time can be painfully protracted, and the patient may present with a greater level of functional dependence than might otherwise be expected from the assessment of simple motor or spatial skills alone.

MEMORY DISORDERS. Memory complaints constitute the most common referral to neuropsychology. A thorough neuropsychological examination of memory considers the modality (e.g., verbal vs. spatial) in which the material is presented, as well as presentation formats that systematically assess different aspects of the information-processing and storage system that forms the basis for memory. Accumulated research indicates that specialized processing of verbal and spatial memory material tends to be differentially mediated by the left and right hemispheres, respectively. Also, to interhemispheric differences in functional localization, specific memory problems can be associated with a breakdown at any stage in the information-processing model of memory. These stages include (1) registration of the material through *attention*, (2) initial processing and encoding of the material within *short-term memory*, also known as *working memory*, (3) consolidation and storage of material in *long-term memory*, and (4) *retrieval* processes, in which material moves from long-term memory storage back into consciousness. A great advantage of neuropsychological assessment is that these various types of memory problems can be readily isolated and described in the course of the examination procedures. Once identified, the specific nature of the deficit can then have important implications for diagnosis, treatment, and prognosis.

Encoding. The initial encoding of new material can be influenced by a variety of factors, including deficits in attention, language, and spatial processing abilities. It is usually measured by the immediate recall of newly learned information (e.g., narrative stories or designs) or by demonstrating the ability to learn new material that has been presented across multiple "learning trials" (e.g., word lists). Attention itself is a relatively fragile cognitive function that can be affected by many factors, including neurologically based disorders

(e.g., head injury or acute confusional state) and psychiatric disorders (e.g., depression or anxiety), so it is a crucial aspect of a proper assessment of memory.

Storage and Retrieval. Deficits in recall can be associated with impaired storage of information, or it can be due to impaired retrieval, in which case the material is still present but not readily accessible. The best way of differentiating these problems is to examine *recognition memory, in which a patient is typically asked to choose from a set of multiple-choice alternatives or to discriminate target words from false-positive foils. If the patient demonstrates accurate recognition but poor recall, then the problem most likely lies in poor retrieval. However, if recognition is impaired, then the problem is more likely to be related to impaired storage of new information. This distinction is important because different neuroanatomical structures subserve the functions of retrieval and storage. Impaired storage is more often associated with dysfunction of the medial temporal lobe (MTL)-diencephalic systems, whereas impaired retrieval can be associated with a variety of structures, including the frontal lobes.*

EXECUTIVE FUNCTION. The prefrontal lobes and their interconnections to the rest of the brain are known to play an important role in *executive functions*, which are essential in planning and organizing, self-monitoring, and controlling complex problem-solving responses. Damage to the frontal lobes also has been associated with significant personality changes. This was historically exemplified by the famous 19th-century case of Phineas Gage, who became irresponsible, socially inappropriate, and unable to carry out plans after a tamping iron was blown through his frontal lobes. As conceptualized by Muriel Lezak, the executive functions include volition (i.e., formulation of a goal, motivation to achieve the goal, and awareness of one's own ability to achieve the goal), planning, purposive action (response selection and initiation, maintenance, switching, and stopping), and execution, which involves self-monitoring and self-correction as well as control of the spatiotemporal aspects of the response. Hemispheric differences in the control of executive functions by the frontal lobes have not been as well documented as in the parietal and temporal lobes.

MOTOR SKILLS. The neuropsychological evaluation commonly includes formal tests of motor skills, such as measures of finger tapping speed, grip strength, and fine motor dexterity. These tests, which have demonstrated validity and reliability, are useful for assessing lateralized motor impairment and have implications for functioning in daily life as well as vocational planning.

General Referral Issues

Referents turn to neuropsychology for many reasons that include differential diagnosis, baseline measurement, and treatment planning, as well as opinions regarding causality and decisional capacity. Because many referents have limited experience and knowledge of the scope of neuropsychology, it is both reasonable and essential for the neuropsychologist to take an active role in refining the specific questions that are asked and providing realistic information about the limitations of the consultation.

Level of Functioning. A common referral issue involves documentation of the level of functioning for a variety of purposes, including assessment of change or capacity to make decisions, especially in the presence of diagnoses such as dementia, stroke, and head injury.

Differential Diagnosis. Like any other diagnostic procedure, the results of a neuropsychological examination must be interpreted in light of all available information, including the history and any associated medical factors that are documented or reported for the individual. Many neurologic and psychiatric disorders have similar clusters of symptoms in common, with complaints of concentration or memory problems being among the most frequently reported problems.

AGE- OR STRESS-RELATED COGNITIVE CHANGE. Many middle-aged and older adults have concerns about everyday concentration and memory failures. With heightened public awareness about conditions such as Alzheimer disease, an increasing number of these individuals seek evaluations for these concerns. Neuropsychological testing provides a detailed, objective picture of different aspects of memory and attention, which can be helpful in reassuring healthy persons about their abilities. It also offers an opportunity for assessing undetected mood or anxiety disorders that may be reflected in

cognitive concerns and for offering suggestions about mnemonic strategies that can sharpen everyday function.

A 77-year-old, left-handed man with a high school education was referred for neuropsychological assessment by his primary care physician after the patient mentioned a recent episode of getting turned around while driving. Results of the neuropsychological assessment indicated variable performance on tests of attention and concentration. His performance was excellent on tests of memory, language, and executive problem-solving abilities, but visual-spatial and constructional abilities were moderately impaired.

MILD TRAUMATIC BRAIN INJURY. Traumatic brain injury (TBI) is usually classified as mild, moderate, or severe. However, the vast majority of TBI cases referred for neuropsychological consultation involve mild TBI. A significant proportion of persons who have suffered a mild TBI complain of problems with attention and inefficient information processing, memory, and mood, also to headache or other forms of pain, for many months after the injury. Neuropsychological testing plays a crucial role in determining the extent of objective cognitive deficit and examining the possible role of psychological factors in perpetuating cognitive problems.

The neuropsychologist should bear in mind that many patients with mild TBI are involved in litigation, which can complicate the neuropsychologist's ability to identify the causes for impairment. Although outright malingering is probably relatively infrequent, subtle presentations of chronic illness behavior should be a prominent consideration when potential legal settlements or disability benefits are in question. This is a particularly important factor in the case of mild head injury, when subjective complaints may be disproportionate to the objectively reported circumstances of the injury, especially because most follow-up studies of mild head injury indicate a return to neuropsychological baseline with no objective evidence of significant cognitive sequelae after 3 to 12 months following injury.

POSTSTROKE SYNDROMES. After the acute phase of recovery from stroke, patients may be left with residual deficits, which can affect memory,

language, sensory/motor skills, reasoning, or mood. Neuropsychological testing can help to identify areas of strength, which can be used in planning additional rehabilitation and can provide feedback on the functional implications of residual deficits for work or complex activities of daily living. Assessment of functional skills can also be helpful to a psychiatrist who is managing mood and behavioral symptoms or dealing with family caregivers.

DETECTING EARLY DEMENTIA. Conditions that particularly warrant neuropsychological assessment for early detection and potential treatment include HIV-related cognitive deficits and normal pressure hydrocephalus. When concerns about a person's memory functioning are expressed by relatives instead of the patient, there is a higher probability of a neurologic basis for the functional problems. Neuropsychological testing, combined with a good clinical history and other medical screening tests, can be highly effective in distinguishing early dementia from the mild changes in memory and executive functioning that can be seen with normal aging. Neuropsychological evaluation is particularly helpful in documenting cognitive deterioration and differentiating among different forms of dementia. An additional incentive for early diagnosis of dementia now lies in the fact that a portion of patients with early dementia may be candidates for memory-enhancing therapies (e.g., acetylcholinesterase inhibitors), and testing can provide an objective means of monitoring treatment efficacy.

DISTINGUISHING DEMENTIA AND DEPRESSION. A substantial minority of patients with severe depression exhibit serious generalized impairment of cognitive functioning. Also, to problems with attention and slowing of thought and action, there may be significant forgetfulness and problems with reasoning. By examining the pattern of cognitive impairment, neuropsychological testing can help to identify a dementia syndrome that is associated with depression, usually known as *depression-related cognitive dysfunction.* Mixed presentations are also common, in which symptoms of depression coexist with various forms of cognitive decline and exacerbate the effects of cognitive dysfunction beyond what would be expected from the neurologic impairment alone. Neuropsychological testing, in this case, can be beneficial by providing a baseline for measuring the effect of antidepressants or other therapy in alleviating cognitive and mood symptoms.

A 75-year-old man with a Ph.D. in the social sciences sought neuropsychological reexamination for ongoing memory complaints, stating that “several of my friends have Alzheimer’s.” In an initial examination 1 year prior, he had performed in the expected range (above average) for most procedures, despite variable performance on measures of attention and concentration. Results of the follow-up examination again clustered in the expected above-average range with uneven performance on measures of attention. On list learning tests of memory, his initial learning of a word list was lower than expected. Still, delayed retention of the material was above average, with excellent discrimination of target items on a recognition subtest. He also endorsed a large number of symptoms of depression on a self-report inventory.

Change in Functioning Over Time. Because many neurologic diagnoses carry clear expectations regarding normal rates of recovery and decline over time, it is frequently important to reexamine a given patient with follow-up neuropsychological assessment after 6 months to a year. For example, it might be essential to monitor declines in independent functioning that could be associated with progressive dementia or to identify improvement following a stroke or tumor resection. Follow-up examinations also provide an opportunity to objectively examine complaints of long-standing or worsening cognitive sequelae following mild head trauma, even though the current literature indicates that the most significant proportion of recovery of function is likely to occur over the initial 6 months to 1-year postinjury. Although continuing subtle signs of recovery can continue after that period, failure to improve following the injury—or a worsening of complaints—would suggest the possibility of contributing psychological factors or the existence of a preexisting or coexisting condition, such as substance abuse, dementia, or outright malingering.

Assessment of Decision-Making Capacity. Neuropsychologists are often asked to assist in determining an individual’s capacity to make decisions or to manage personal affairs. Neuropsychological testing can be useful in these cases by documenting areas of significant impairment and by identifying areas of strength and well-preserved skills. Opinions about

decision-making capacity are seldom based on test findings alone. Usually, they rely heavily on information gleaned from the clinical interview, collateral interviews with family or caregivers, and direct observations (e.g., in-home assessment) of everyday function. Appraisal of an individual's level of insight and capacity to appreciate his or her limitations is typically the most critical aspect of the assessment. State statutes generally define standards for decision-making capacity, and, of course, the ultimate determination of competence rests in the authority of the presiding judge. However, the neuropsychologist or other health care professional can play a significant role in shaping the judge's ruling by providing a professional opinion that is supported by compelling behavioral data that have strong face validity. As a general rule of thumb, consideration of decision-making capacity is usually best approached in the narrowest possible sense to infringe as little as possible on the individual's freedom to represent his or her interests. Therefore, consultation requests for assessment of decision-making capacity should identify specific areas of decision making and behavior that are of concern. Frequent concerns having to do with decision-making capacity involve the areas of (1) financial and legal matters, (2) health care and medical treatment, and (3) ability to live independently. Some capacity issues involve higher standards, such as the ability to drive, the ability to work, or practice in a given profession (e.g., air traffic controller, surgeon, or financial advisor). In such cases, the neuropsychologist needs to rely on normative expectations that are appropriate for the type of activity, as well as the patient's demographics.

Forensic Evaluation. Neuropsychological evaluation of individuals in matters of criminal or civil law usually requires specialized knowledge beyond expertise in neuropsychology. Neuropsychologists are frequently called upon as experts in matters involving head injury, especially in the case of mild head injury associated with a motor vehicle accident. As a distinct subspecialty, this area of practice requires the integration of knowledge of statutes, laws, precedents, and legal procedures as well as expertise in identifying and describing the impact of an injury or event on cognitive, emotional, and behavioral functioning.

Approaches to Neuropsychological Assessment

The neuropsychological examination systematically assesses functioning in the realms of attention and concentration, memory, language, spatial skills, sensory and motor abilities, as well as executive functioning and emotional status. Because deficits in cognitive performance can only be interpreted in comparison to a person's long-standing or *premorbid level of functioning*, overall intellectual abilities are typically examined to measure the current level of overall functioning and to identify any changes in intellectual functioning. Psychological contributions to performance are also considered concerning personality and coping style, emotional lability, presence of thought disorder, developmental history, and significant past or current stressors. The expertise of the neuropsychologist lies in integrating findings that are obtained from many diverse sources, including the history, clinical presentation, and several dozen discrete performance scores that make up the neuropsychological data.

Battery Approach. The battery approach, exemplified by the Halstead–Reitan Neuropsychological Test Battery (HRNTB) or the Neuropsychological Assessment Battery (NAB), grew directly out of the psychometric tradition in psychology. This approach typically includes a large variety of tests that measure most cognitive domains as well as sensory and motor skills. Traditionally, all parts of the test battery are administered regardless of the patient's presenting problem, although the NAB has a screening examination that covers all appropriate domains. The battery approach has the advantage of identifying issues that the patient might not have mentioned and that the medical history may not necessarily predict. However, it has the disadvantage of being very time-consuming (i.e., 6- to 8-hour examination for the HRNTB).

Hypothesis Testing Approach. The qualitative hypothesis testing approach is historically best exemplified by the work of Alexander Luria and more recently developed as the Boston Process Approach by Edith Kaplan and her colleagues. It is characterized by a detailed evaluation of areas of functioning that are related to the patient's complaints and predicted areas of impairment, with relatively less emphasis on aspects of functioning that are less likely to be impaired. The hypothesis testing approach has been particularly helpful in illuminating the differential roles of the two

hemispheres, as discussed above. This approach has the advantage of efficiently honing in on areas of impairment and producing a detailed description of the deficits from a cognitive processing standpoint. Still, it has the shortcoming of potentially overlooking unexpected areas of deficits.

Screening Approaches. Many practitioners have moved away from strict battery or hypothesis testing approaches since the 1990s and developed more flexible and efficient screening approaches. In this model, the neuropsychologist utilizes a core set of screening procedures as a first step in determining whether a diagnosis can be made with less information or whether additional testing is necessary to identify more subtle problems. Therefore, a screening protocol that efficiently assesses the major areas of neuropsychological functioning may or may not be followed by more detailed testing in selected areas that might provide a better understanding of the reasons for the deficits demonstrated on the screening evaluation.

Mental Status Examinations. In some cases, usually involving very acute or severe cognitive impairment, it is simply not feasible to administer extensive cognitive examination procedures, so the neuropsychologist might appropriately rely on bedside mental status examination or very brief cognitive screening procedures to address the referral issues. However, research has shown that, even with brief screening procedures, the systematic use of a structured examination format can significantly increase the accuracy of detecting cognitive impairment.

One of the most widely used screening instruments for documenting gross changes in mental status is the Mini-Mental State Examination (MMSE). However, it is essential to note that the MMSE does have distinct limitations. Other than serial seven countings, the MMSE does not assess executive functions, which are often impaired in dementing patients. Also, the MMSE is likely to underestimate the prevalence of cognitive deficits in well-educated older persons with early Alzheimer disease or in younger adults with focal brain injury. Still, it is more likely to overestimate the presence of cognitive deficits in persons with little education. Therefore, cutoff scores should be adjusted for age and education before concluding that impairment is present. Although mental status examinations can be beneficial in screening for gross signs of cognitive impairment, they do not provide a

sufficient foundation for diagnosing specific etiologies of cognitive impairment, and they are not interchangeable with neuropsychological testing.

Domains of Formal Neuropsychological Assessment

The past decade has seen a virtual explosion in the growth of more sophisticated and better-standardized tests and procedures for neuropsychological evaluation. A list of examples of standard neuropsychological tests and techniques is provided in Table 1-20.

Interview. The clinical interview provides the single best opportunity for identifying the patient's concerns and questions, eliciting a direct description of current complaints from the patient, and understanding the context of the patient's history and current circumstances. Although the patient typically serves as the primary interview source, it is crucial to seek corroborating information for the patient's account from interviews with caregivers or family members as well as a thorough review of relevant records, such as medical and mental health treatment, educational, and employment experiences.

Intellectual Functioning. Assessment of intellectual functioning serves as the cornerstone of the neuropsychological examination. The Wechsler Intelligence Scales have represented the traditional gold standard in intellectual assessment for many years, based on carefully developed normative standards. The scope and variety of subtests on which the summary IQ values are based also provide useful benchmarks against which to compare performance on other tests of specific abilities. The latest revision of this instrument, the Wechsler Adult Intelligence Scale III (WAIS-III), offers the additional advantage of greatly extended age norms (ages 16 to 89) that are directly related to normative performances on the Wechsler Memory Scale III (WMS-III). The Wechsler Intelligence Scales utilize a broad set of complex verbal and visuospatial tasks that have traditionally been summarized as a verbal IQ, a performance IQ, and full-scale IQ. In the context of a neuropsychological examination, the patient's performance across the procedures provides useful information regarding long-standing abilities as well as current functioning. Most neuropsychologists recognize

that the summary IQ values provide only a ballpark range for characterizing an individual's general level of functioning. Therefore, it is usually more appropriate and meaningful to characterize an individual's intellectual functioning in terms of the range of functioning (e.g., borderline, low average, average, high average, or superior) that is represented by the IQ value rather than the specific value itself.



Table 1-20
Selected Tests of Neuropsychological Functioning

Area of Function	Comment
Intellectual Functioning	
Wechsler Intelligence Scales	Age-stratified normative references; appropriate for adults up to age 89, adolescents, and young children
Shipley Scale	Scale Brief (20-min) paper-and-pencil measure of multiple-choice vocabulary and open-ended verbal abstraction
Attention and Concentration	
Digit Span	Auditory–verbal measure of simple span of attention (*digits forward*) and cognitive manipulation of increasingly longer strings of digits (*digits backward*)
Visual Memory Span	Visual–spatial measure of ability to reproduce a spatial sequence in forward and reverse order
Paced Auditory Serial Addition Test	Requires double tracking to add pairs of digits at increasing rates; particularly sensitive to subtle simultaneous processing deficits, especially in head injury

(PASAT)	
Digit Vigilance Test	Timed measure of speed and accuracy in cancelling a specific digit on a page of random digits; directly examines an individual's tendency to sacrifice either speed or accuracy in favor of the other
Memory	
Wechsler Memory Scale III	Comprehensive set of subtests measuring attention and encoding, retrieval, and recognition of various types of verbal and visual material with both immediate recall and delayed retention; excellent age-stratified normative comparisons for adults up to age 89 with intellectual data for direct comparison
California Verbal Learning Test II	Documents encoding, recognition, and both immediate and 30-min recall; affords examination of possible learning strategies as well as susceptibility to semantic interference with alternate and short forms available
Fuld Object Memory Evaluation	Selective reminding format requires patient to identify objects tactually, then assesses consistency of retrieval and storage as well as ability to benefit from cues; normative reference group is designed for use with older individuals
Benton Visual Retention Test	Assesses memory for 10 geometric designs after 10-s exposures; requires graphomotor response
Brief Visuospatial Memory Test—Revised	Serial learning approach used to assess recall and recognition memory for an array of six geometric figures; six alternate forms
Language	

Boston Diagnostic Aphasia Examination	Comprehensive assessment of expressive and receptive language functions
Boston Naming Test—Revised	Documents word finding difficulty in a visual confrontation format
Verbal Fluency	Measures ability to fluently generate words within semantic categories (e.g., animals) or phonetic categories (e.g., words beginning with “S”)
Token Test	Systematically assesses comprehension of complex commands using standard token stimuli that vary in size, shape, and color
Visuospatial-Constructional	
Judgment of Line Orientation	Ability to judge angles of lines on a page presented in a match-to-sample format
Facial Recognition	Assesses matching and discrimination of unfamiliar faces
Clock Drawing	Useful screening technique is sensitive to organization and planning as well as constructional ability
Rey–Osterrieth Complex Figure test	Ability to draw and later recall a complex geometric configuration; sensitive visual memory as well as executive deficits in development of strategies and planning
Motor	
Finger	Standard measure of simple motor speed;

Tapping	particularly useful for documenting lateralized motor impairment
Grooved Pegboard	Ability to rapidly place notched pegs in slotted holes; measures fine finger dexterity as well as eye–hand coordination
Grip Strength	Standard measure of lateralizing differences in strength
Executive Functions	
Wisconsin Card Sorting Test	Measure of problem-solving efficiency is particularly sensitive to executive deficits of perseveration and impaired ability to flexibly generate alternative strategies in response to feedback
Category Test	This measure of problem-solving ability also examines ability to benefit from feedback while flexibly generating alternative response strategies; regarded as one of the most sensitive measures of general brain dysfunction in the Halstead–Reitan Battery
Trail-Making Test	Requires rapid and efficient integration of attention, visual scanning, and cognitive sequencing
Delis–Kaplan Executive Function System (D-KEFS)	Battery of measures that are sensitive to executive functions
Psychological Factors	
Beck Depression Inventory	Brief (5–10 min) self-report measure that is sensitive to symptoms of depression; best for screening depression in adults up to late middle age, who can be expected to frankly report symptoms; available in standard (21 four-choice items) or

	short (13-item) form
Geriatric Depression Scale	30-item self-report screen for symptoms of depression; the yes–no format is less cognitively demanding than other scales
Minnesota Multiphasic Personality Inventory 2	This psychometrically developed self-report instrument remains highly useful for documenting quantitative levels of self-reported symptoms that can be objectively compared with known populations; drawbacks include administration time (567 true–false questions, requires about 1–1.5 hr or more) for frail individuals, and the emphasis on pathologic features for persons who are generally psychologically healthy; advantages include well-developed validity scales and availability of many symptom-specific subscales that have been identified over the years

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.

Careful examination of the individual's performance across the various verbal and performance subtests can provide information regarding the patient's pattern of strengths and weaknesses as well as the degree to which these performance characteristics are consistent with the history and performance on other aspects of the neuropsychological examination. Tests of long-standing knowledge, such as for vocabulary or general information, provide a basis for estimating an individual's long-standing (or premorbid) level of intellectual abilities, which in turn can help to gauge the degree to which an individual may have deteriorated.

The verbal IQ and performance IQ (VIQ and PIQ) have historically been reported to be associated with left and right hemisphere functioning, respectively. However, more recent research indicates that, also to language and spatial skills, the subtests of the Wechsler Intelligence Scales reflect other contributions such as speed, sustained concentration, and novel experience. Therefore, experienced neuropsychologists do not merely assume

that a discrepancy between VIQ and PIQ is due to unilateral hemispheric damage. Important clues to the nature of the contributing problem can often be gleaned by considering the pattern of performance across other aspects of the examination and by carefully analyzing the types of errors that are observed.

Attention. Attention underlies performance in virtually all other areas of functioning. It should always be considered a potential contributor to impairment on any tests that require sustained concentration and vigilance or rapid integration of new information. Measures of attention and concentration have traditionally been included in the Wechsler Intelligence and Wechsler Memory Scales to assess orientation and "freedom from distractibility." These procedures also provide a useful basis for "previewing" the individual's ability to comprehend, process information, and otherwise engage in the assessment process. *Digit span* requires patients to repeat longer strings of digits increasingly as a way of assessing the ability to process relatively simple information. In contrast, *digit span backward* reflects more complex simultaneous processing and cognitive manipulation demands or working memory.

Memory. Complaints of memory problems constitute one of the most common reasons for referral to neuropsychology. As described above, the neuropsychologist utilizes an information-processing approach to assess memory problems that might involve difficulty with encoding, retrieval, or storage of new information. The WMS-III is the latest revision of a widely used battery of subtests that utilizes several measures of attention, memory, and new learning ability.

Language. The assessment of language examines both expressive abilities and comprehension. However, most neuropsychologists screen for language impairment rather than administer an extensive formal language assessment battery, such as the Boston Diagnostic Aphasia Examination. Expressive language is commonly assessed by measures of *verbal fluency*, which require the patient to rapidly generate words within semantic (e.g., names of animals) and phonetic categories (e.g., words beginning with specified letters of the alphabet).

Visuospatial Functions. Complex visuospatial abilities can be assessed through procedures that were developed in Arthur Benton's laboratory, such as *facial recognition* and *judgment of line orientation.* Measures of visual constructional ability examine the person's ability to draw spatial designs or assemble two- or three-dimensional figures (see Fig. 1-5). Also, to the significant visuospatial component, these tasks reflect the contributions of executive planning and organizational abilities. More impaired individuals can be asked to copy simple geometric forms, such as a Greek cross or intersecting pentagons, to examine visuospatial abilities that are less influenced by planning and organization.

The widely used technique of *clock drawing* provides a surprisingly sensitive measure of planning and organization, especially for older individuals who are at risk for dementia. Although problems involving poor organization, perseveration, and possible neglect are apparent in the drawing that is illustrated in Figure 1-6, more subtle difficulties can also be detected, especially when a patient's performance is evaluated in light of premorbid expectations.

Sensory and Motor Functions. *Double simultaneous stimulation* in the visual, tactile, and auditory modalities is a standard component of the HRNTB. It can be useful for assessing the integrity of basic sensory functions as well as neglect if deficits are present on one side only on the bilateral simultaneous trials and not when stimulation is unilateral. *Grip strength* and rapid *finger tapping* are commonly used measures of motor strength and speed that are sensitive to lateralized brain dysfunction.

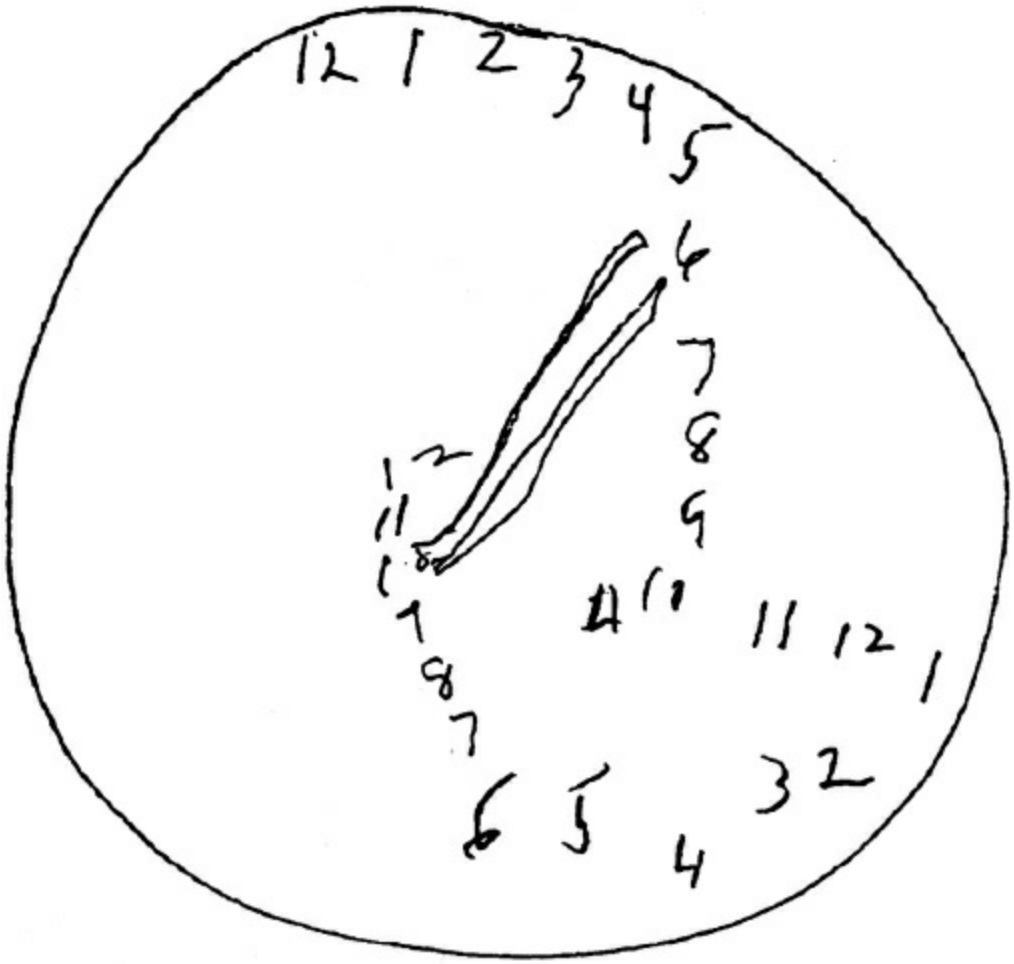


FIGURE 1-6
Clock drawing by a patient with vascular dementia, showing poor planning and organization, perseveration, and possible neglect. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.)

Executive Functions. One of the most critical aspects of the neuropsychological examination lies in the assessment of higher *executive* functions, which play an essential role in the planning and initiation of independent activities, self-monitoring of performance, inhibition of inappropriate responses, switching between tasks, and planning and control of complex motor and problem-solving responses. Although the prefrontal lobes have long been regarded as a vital component in mediating these functions, more recent developments in the neurosciences have also led to an increased appreciation for the essential role that is played by extensive cerebral interconnections between subcortical and cortical regions of the brain.

Psychological Factors. A key component of any neuropsychological examination involves consideration of the degree to which long-standing personality or other psychological factors (including current stressors) might contribute to the patient's presentation. Common techniques for assessing personality and psychological factors include the Minnesota Multiphasic Personality Inventory 2 (MMPI-2) and paper and pencil techniques, such as the Beck Depression Inventory II.

Assessment of Effort and Motivation. Because the results of neuropsychological examinations may eventually be introduced as evidence in litigation or other forensic proceedings or be used for determining disability compensation, the neuropsychologist must address any possible concerns about effort and motivation as a routine matter. Several instruments have been developed recently that directly assess a patient's level of effort and motivation to perform at his or her best. Normative research indicates that patients with histories of bona fide brain injury or even dementia perform close to perfect levels on many such instruments, so poor performance suggests poor effort or tendencies to exaggerate symptoms. Many other indicators of effort are based on the pattern of an individual's performance on standard procedures in a neuropsychological examination.

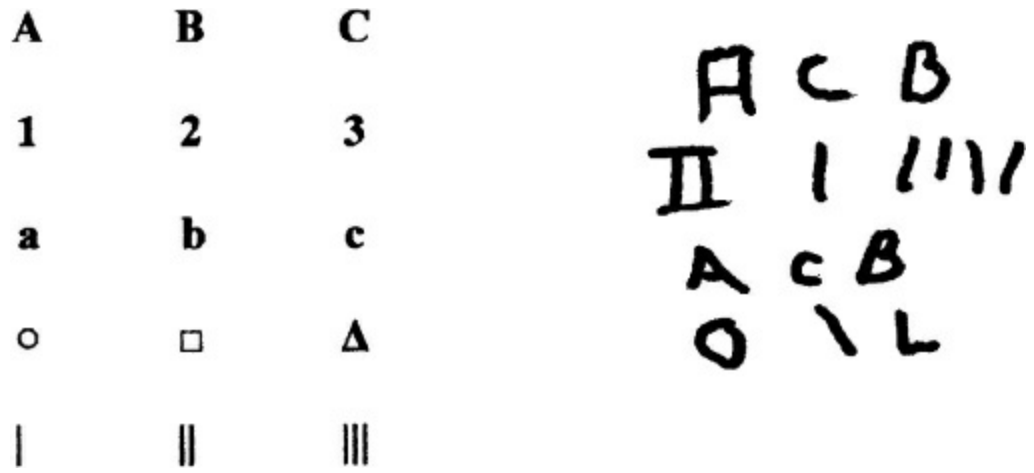


FIGURE 1-7
Rey's Memory Test with example of a response that is typical of exaggerated "memory" problems. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.)

A 32-year-old woman with 13 years' education was seen for disability evaluation, claiming current "trouble remembering things." Her account of personal history was vague, and she "forgot" information, such as her birth date and mother's maiden name. Response latencies were extremely long, even for highly familiar information (e.g., count from 1 to 20), she could not repeat more than three digits forward consistently, and on a word list learning procedure, she was not able to correctly recognize more items (only five) than she could freely recall (also five). Despite otherwise fluent language, she was only able to generate five examples of animals in 1 minute. When asked to recall 15 items on a procedure (Rey's Memory Test) that is presented as a challenging task, but in reality is fairly simple, her performance demonstrated exaggerated errors of commission (Fig. 1-7). The evaluation concluded that current levels of cognitive functioning could not be conclusively established due to overt symptoms exaggeration.

Therapeutic Discussion of Results

A critical component of the neuropsychological examination process is found in the opportunity to discuss the results of the examination with the patient and family or other caregivers. This meeting can represent a potential therapeutic opportunity to educate and clarify individual and relationship issues, which can impact the identified patient's functioning. If the patient's active cooperation in the initial examination has been appropriately enlisted, then the patient will be prepared to invest value and confidence in the findings of the examination. At the time of the discussion of the results, it is useful to review the goals of the examination with the patient and supportive family or caregivers and to clarify the expectations of those who are present. Typically, these sessions will include information about the patient's diagnosis, with emphasis on the natural course and prognosis as well as compensation and coping strategies for the patient and family. Given the impact of chronic neurologic disease on the family system as well as the patient, explicit discussion of these issues is critical in maximizing adjustment to brain injury. It is equally important to relate the impact of the results on the patient's current living circumstances, future goals, and course

of adjustment. It is not unusual for strong emotions and underlying tensions within family relationships to come to light in the context of honest discussion, so the results discussion can represent a significant therapeutic opportunity to model effective communication and problem-solving techniques.

PERSONALITY ASSESSMENT: ADULTS AND CHILDREN

Personality is defined as an individual's enduring and pervasive motivations, emotions, interpersonal styles, attitudes, and traits. Personality assessment is the systematic measurement of these personality characteristics. Personality tests measure such difficult-to-define concepts as depression, anger, and anxiety. Even more challenging personality concepts such as somatization, ability to delay gratification or suicide potential can be quantified using personality assessment. Personality assessment can be of utmost importance in the scientific study of psychology and psychiatry.

Purposes of Psychological Testing

Personality testing can be an expensive undertaking. A considerable amount of time is required to administer, score, and interpret psychological test results. Personality testing should not be routinely obtained from all psychiatric patients. Personality testing can be helpful with selective patients from both a clinical and a cost–benefit analysis perspective.

Assisting in Differential Diagnosis. Psychiatric diagnosis can be difficult and, at times, confusing exercise. However, knowing a patient's diagnosis is essential to treatment, as a proper diagnosis can assist in understanding the etiology of the presenting psychiatric problem and the prognosis of the disorder.

A 49-year-old man had abruptly resigned his position as an accountant and decided he was going to start an oil exploration business. He had never worked in the oil business and knew nothing about the profession. The patient had received a revelation from an unknown entity through an auditory hallucination. This voice told him he would become quite

wealthy in the business if he would simply follow the directions given to him. Around this time, the patient had a marked change in personality. Although his grooming was formally very neat and appropriate, he became disheveled. He began sleeping about 3 hours a night. He became somewhat agitated and talked loudly to those around him.

The differential diagnosis, in this case, includes schizophrenia and bipolar disorder. Psychological testing might help assist in this differential diagnosis, as well as in the formulation of a treatment plan.

Aiding in Psychotherapy. Psychological tests can be useful in psychotherapy. The usefulness of these tests can be even more important for short-term, problem-centered therapy, where understanding the patient and his or her problem must be accomplished quickly. Psychological assessment can be used in pretreatment planning, assessing progress once therapy begins, and in evaluating the effectiveness of therapy. Patients need to have objective information about themselves at the time of therapy if they are to go about changing themselves productively. Personality tests, particularly objective tests, allow patients to compare themselves to objective norms and evaluate the extent and magnitude of their problem. Testing also can reveal areas of the patient's life that may be problematic but for which the patient may not have a full appreciation. Information about patients' willingness to disclose information about themselves can also be helpful. Psychological tests may reveal considerable information concerning the patient's inner life, feelings, and images, which may make therapy progress faster. Psychological testing can provide baseline information at the beginning of therapy, and repeat testing can then be used to assess the change that occurred during therapy.

Providing Narrow-Band Assessment. *Narrow-band personality tests* measure a single personality characteristic or a few related characteristics. *Broad-bank personality tests,* on the other hand, are designed to measure a wide spectrum of personality characteristics. A psychiatrist may need answers to specific questions, such as those that arise when assessing the degree of clinical depression, measuring the intensity of the state or trait anxiety, or, possibly, quantifying the amount of a patient's anger. Such quantification can help measure severity or in providing a baseline for future

assessment.

Psychometric Properties of Personality Assessment Instruments

The quality of personality tests varies widely. On the one hand, there are well-constructed, empirically validated instruments, and, on the other hand, there are "psychological tests" that one can find in the Sunday supplement of the newspaper or on the Internet. Evaluating the usefulness of particular psychological instruments can be challenging, even to the well informed.

Normative Sample. To construct a personality test, a representative sample of subjects (normative sample) should be administered the test to establish expected performance. Fundamental issues, such as the size and representativeness of the sample used to construct the test, must be evaluated. To illustrate this point, the Minnesota Multiphasic Personality Inventory 2 (MMPI-2), a well-constructed instrument, initially tested approximately 2,900 subjects. However, about 300 subjects were eliminated because of test invalidity or incompleteness of needed information.

Test Characteristics. Any psychological test must be completed, in its entirety, by the intended test taker to be useful. If the questions are offensive or are difficult to understand, then the individual taking the test may not complete all items. These omissions can create problems, especially when normative tables are used to interpret results.

Validity Issues. Perhaps the most crucial characteristic in evaluating the scientific merit of a given personality test is the validity of the instrument. Does the test measure what it purports to measure? If a test is designed to measure depression, does it indeed measure depression? Although validity may seem like a simple issue to address, it can become complex, especially when attempting to measure such characteristics as self-esteem, assertiveness, hostility, or self-control.

FACE VALIDITY. *Face validity* refers to the content of the test items themselves. In other words, do the items appear to measure what they purport to measure? One problem with face validity is that professionals differ in their subjective appraisal of individual items.

CRITERIA AND CONSTRUCT VALIDITY. Although face validity refers to the degree that test items appear on the surface to measure what the instrument, as a whole, purports to measure, *criterion validity* uses data outside the test itself to measure validity. For example, if a test were designed to measure hypochondria, one would expect that a patient with high scores would have more visits to the physician's office, complain of more physical symptoms, and use prescribed and over-the-counter medications more extensively.

CONCURRENT AND PREDICTIVE VALIDITY. To determine test *concurrent validity*, external measures are obtained at the same time that the test is given to the sample of subjects. Thus, the concurrent validity of the test reveals that, at a given point of time, high scorers on a test may be more likely than low scorers on a test to manifest the behavior reflected in the criteria (e.g., more physician visits or more medication for a hypochondriac patient). Occasionally, however, a test developer is interested in predicting future events. The *discriminant validity* of a test tells whether the test can discriminate between known groups of patients at a given time. Is a measure of depression able to statistically differentiate among mild, moderate, and severe major depressive disorder?

FACTOR VALIDITY. Factor validity utilizes a multivariate statistical technique known as *factor analysis* to determine if certain significant groups of items on a given test empirically cluster together. For example, on a personality test measuring depression, do items concerning vegetative symptoms tend to covary together?

Reliability. *Reliability* refers to the degree that a test measures what it purports to measure, consistently. The key word here is *consistently*. There are several means of checking reliability, including test–retest reliability, internal consistency reliability, and parallel form reliability.

TEST–RETEST RELIABILITY. *Test–retest reliability* is obtained by simply administering the same test on two occasions to a group of subjects and statistically correlating the results. The correlation coefficient should be at least 0.80 if the two tests were administered within 2 weeks of each other, and if the trait in question is stable.

INTERNAL CONSISTENCY RELIABILITY. Another approach to determining *internal consistency reliability* is to divide a given test into two equal parts and statistically correlate the two halves for the test with each other. This technique determines the *split-half reliability* of a test. The first half of the test should be highly correlated with the second half of the test if the test is consistently measuring what it purportedly measures. Alternatively, the odd-numbered items could be correlated with the even-numbered items (*odd–even consistent reliability*). A reliability coefficient of 0.80 to 0.85 is needed to demonstrate usefulness in most circumstances. However, the higher the reliability as measured by the correlation coefficient, the better the test instrument.

PARALLEL FORM RELIABILITY. Sometimes, two separate forms of the same test are needed. For example, if the process of taking a test at one point in time would by itself influence a patient's score the second time he or she took the same test, then parallel forms of the tests are needed. *Parallel forms* of a test measure the same construct but use different items to do so. To ensure that the test does measure the same construct, the correlation coefficient between the two parallel forms of the same test is computed. Such parallel form reliability should be at least 0.90 or higher.

USE OF STANDARD ERROR OF MEASUREMENT TO ASSESS RELIABILITY. Another way to assess the usefulness of a given test is to examine the test's standard error of measurement (SEM), which should be included in the test's manual. The SEM is a single statistic that is used to estimate what the score of a given patient would be on the test if the patient retook the same test within a short time.

Adult Psychological Tests

Objective Personality Tests. *Objective personality tests* are rather straightforward in approach. Patients are usually asked specific and standard questions in a structured written or oral format. Each patient is typically asked the same question. The data obtained from a given patient are compared to similar data obtained from the normative group. The degree to which the patient deviates from the norm is noted and is used in the

interpretive process. The patient's responses are scored according to certain agreed-upon criteria. The obtained scores are then compared with normative tables and often converted to standardized scores or percentiles, or both. The MMPI-2 is an example of an objective personality test. Table 1-21 lists a sample of objective personality test along with a brief description and brief list of strengths and weaknesses.

MINNESOTA MULTIPHASIC PERSONALITY INVENTORY. The MMPI-2 is relatively easy to administer and score and takes approximately 1.5 hours for most patients to complete. It consists of 567 true or false questions concerning a wide variety of issues and requires only eighth-grade reading comprehension. Scoring of the MMPI-2 involves adding up the number of responses on numerous scales and comparing the results to certain normative information. Interpretation of the MMPI-2 is more straightforward than with many other tests.

When a patient takes the MMPI-2, questions are not grouped in any particular order to aid in interpretation. Various items in the MMPI-2 can be selected, sorted, and analyzed according to various criteria.

A new version of the MMPI-2 was developed in 2008, the MMPI-2 Restructured Form (MMPI-2 RF). It contains 338 questions and allows less time to administer. The MMPI-2 RF is meant to be an alternative to the MMPI-2, not a replacement.

PERSONALITY ASSESSMENT INVENTORY (PAI). Another increasingly popular objective personality test is the Personality Assessment Inventory (PAI). This test consists of 344 items that are written at a fourth-grade reading level. This reading level ensures that most patients can complete it without experiencing any reading problems. The PAI takes about 45 to 50 minutes to complete for most patients. The PAI was normed on 1,000 community-dwelling individuals stratified according to sex, race, and age. There are no separate norms for male and female as there are in the MMPI. Also, data were gathered on 1,246 clinical subjects and 1,051 college students in the normative process. The clinical subjects were drawn from a variety of different clinical settings, including inpatient psychiatric facilities (25 percent), outpatient psychiatric facilities (35 percent), correctional institutions (12 percent), medical settings (2 percent), and substance abuse treatment

programs (15 percent).

The PAI has 11 clinical scales. These main clinical scales are similar to the MMPI-2 clinical scales and measure such personality issues as somatic concerns, depression, paranoia, borderline features, or alcohol or drug problems. The PAI also has five treatment-related scales that are designed to address such issues as treatment rejection, suicide ideation, or aggression.

Projective Personality Test. Projective personality tests, in contrast to objective personality instruments, are more indirect and unstructured. Unlike objective tests in which the patient may simply mark true or false to given questions, the variety of responses to projective personality tests are almost unlimited. Instructions are usually very general, allowing the patient's fantasies to be expressed. The patient generally does not know how his or her responses will be scored or analyzed. Consequently, trying to fake the test becomes difficult. Projective tests typically do not measure one particular personality characteristic, such as "type A personality" (e.g., narrow-band measurement). Instead, they are designed to assess one's personality as a whole (e.g., broad-band measurement).



Table 1-21
Objective Measures of Personality

Name	Description	Strengths	Weaknesses
Minnesota Multiphasic Personality Inventory-2 (MMPI-2)	567 Items; true–false; self-report format; 20 primary scales	Current revision of MMPI that has updated the response booklet; revised scaling methods and new validity scores; new normative data	Preliminary data indicate that t MMPI-2 and the MMPI can provide discrepant results; normative sample biased toward upper socioeconomi status; no

			normative dat for adolescen
Million Clinical Multiaxial Inventory (MCMI)	175 Items; true–false; self-report format; 20 primary scales	Brief administration time; corresponds well with diagnostic classifications	In need of more validation research; no information or disorder severity; need revision for DSM-5
Million Clinical Multiaxial Inventory-II (MCMI-II)	175 Items; true–false; self-report format; 25 primary scales	Brief administration time	High degree of item overlap i various scales no information on disorder or trait severity
16 Personality Factor Questionnaire (16 PF)	True–false; self-report format; 16 personality dimensions	Sophisticated psychometric instrument with considerable research conducted on nonclinical populations	Limited usefulness wi clinical populations
Personality Assessment Inventory (PAI)	344 Items; Likert-type format; self-report; 22 scales	Includes measures of psychopathology, personality dimensions, validity scales, and specific concerns to psychotherapeutic treatment	The inventory is new and has not yet generated a supportive research base

California Personality Inventory (CPI)	True–false; self-report format; 17 scales	Well-accepted method of assessing patients who do not present with major psychopathology	Limited usefulness wi clinical populations
Jackson Personality Inventory (JPI)	True–false; self-report format; 15 personality scales	Constructed in accord with sophisticated psychometric techniques; controls for response sets	Unproved usefulness in clinical setting
Edwards Personal Preference Schedule (EPPS)	Forced choice; self-report format	Follows Murray's theory of personology; accounts for social desirability	Not widely used clinically because of restricted nature of information obtained
Psychological Screening Inventory (PSI)	103 Items; true–false; self-report format	Yields four scores, which can be used as screening measures on the possibility of a need for psychological help	The scales are short and hav correspondin low reliability
Eysenck Personality Questionnaire (EPQ)	True–false; self-report format	Useful as a screening device; test has a theoretical basis	Scales are shor and items are transparent a to purpose; n

		with research support	recommende for other than screening device
Adjective Checklist (ACL)	True–false; self-report or informant report	Can be used for self-rating or other rating	Scores rarely correlate high with conventional personality inventories
Comrey Personality Scales (CPS)	True–false; self-report format; eight scales	Factor analytic techniques used with a high degree of sophistication in test constructed	Not widely used factor analytic interpretation problems
Tennessee Self-Concept Scale (TSCS)	100 Items; true–false; self-report format; 14 scales	Brief administration time yields considerable information	Brevity is also a disadvantage lowering reliability and validity; usefu as a screenin device only

Courtesy of Robert W. Butler, Ph.D. and Paul Satz, Ph.D.

Projective tests often focus on “latent” or unconscious aspects of personality. Psychologists and others differ in the degree to which they rely on “unconscious” information. In many projective techniques, the patient is simply shown a picture of something and asked to tell what the picture reminds him or her of. An underlying assumption of projective techniques (projective hypothesis) is that, when presented with an ambiguous stimulus, such as an inkblot, for which there are an almost unlimited number of responses, the patient’s responses will reflect fundamental aspects of his or her personality. The ambiguous stimulus is a sort of screen on which the

individual projects his or her own needs, thoughts, or conflicts. Different persons have different thoughts, needs, and conflicts and, hence, have widely different responses. A schizophrenic's responses often reflect a rather bizarre, idiosyncratic view of the world.

Table 1-22 lists the common projective tests together with a description and strengths and weaknesses for each test.

RORSCHACH TEST. Herman Rorschach, a Swiss psychiatrist, developed the first significant use of projective techniques around 1910. The Rorschach test is the most frequently used projective personality instrument (Fig. 1-8). The test consists of 10 ambiguous symmetrical inkblots. The inkblot card appears as if a blot of ink were poured onto a piece of paper and folded over—hence, the symmetrical appearance.



Table 1-22
Projective Measures of Personality

Name	Description	Strengths	Weaknesses
Rorschach test	10 Stimulus cards of inkblots, some colored, others achromatic	Most widely used projective device and certainly the best researched; considerable interpretative data available	Some Rorschach interpretive systems have unimproved validity
Thematic Apperception Test (TAT)	20 Stimulus cards depicting a number of scenes of varying ambiguity	A widely used method that, in the hands of a well-trained person, provides	No generally accepted scoring system results in poor consistence in interpretation; time-

		valuable information	consuming administration
Sentence completion test	A number of different devices available, all sharing the same format with more similarities than differences	Brief administration time; can be a useful adjunct to clinical interviews if supplied beforehand	Stimuli are obvious as to intent and subject to easy falsification
Holtzman Inkblot Technique (HIT)	Two parallel forms of inkblot cards with 45 cards per form	Only one response is allowed per card, making research less troublesome	Not widely accepted and rarely used; not directly comparable to Rorschach interpretive strategies
Figure drawing	Typically human forms but can involve houses or other forms	Quick administration	Interpretive strategies have typically been unsupported by research
Make-a-Picture Story (MAPS)	Similar to TAT; however, stimuli can be manipulated by the patient	Provides idiographic personality information through thematic analysis	Minimal research support; not widely used

Courtesy of Robert W. Butler, Ph.D. and Paul Satz, Ph.D.

Minimal interaction between the examiner and the patient occurs while the Rorschach is administered, which ensures standardization procedures are upheld. The examiner writes down verbatim what the patient says during the above-described “free association” or “response proper” phase. If the patient rotates the card during his or her response, then the examiner makes the appropriate notation on the test protocol. After the patient has given answers to all 10 cards, an inquiry phase of administration begins. The examiner asks the patient to go through the cards again and help the examiner see the responses he or she gave. The examiner reads the patient’s initial response and asks the patient to point out what he or she saw and explain what made it look like that to him or her. An almost unlimited range of responses is possible with the Rorschach test and most projective tests.



FIGURE 1-8
Card I of the Rorschach test. (From Hermann Rorschach, Rorschach®-Test. Copyright © Verlag Hans Hubar AG, Bern, Switzerland, 1921, 1948, 1994, with permission.)

THEMATIC APPERCEPTION TEST. Although the Rorschach test is the most frequently used projective personality test, the Thematic Apperception Test (TAT) is probably in second place. Many clinicians will include both the TAT and the Rorschach test in a battery of tests for personality assessment. The TAT consists of a series of 10 black-and-white pictures that depict individuals of both sexes and of different age groups, who are involved in a

variety of different activities. An example of a TAT card is presented in Figure 1-9.



FIGURE 1-9
Card 12F of the Thematic Apperception Test. (Reprinted from Henry A. Murray, Thematic Apperception Test, Harvard University Press, Cambridge, MA. Copyright © 1943 President and Fellows of Harvard College, © 1971 Henry A. Murray, with permission.)

Henry Murray developed the TAT in 1943 at the Harvard Psychological Clinic. The stories that the patient makes up concerning the pictures, according to the projective hypothesis, reflect the patient's own needs, thoughts, feelings, stresses, wishes, desires, and views of the future. According to the theory underlying the test, a patient identifies with a particular individual in the picture. This individual is called the *hero.* The hero is usually close to the age of the patient and frequently of the same sex, although not necessarily so. Theoretically, the patient would attribute his or

her own needs, thoughts, and feelings to this hero. The forces present in the hero's environment represent the *press* of the story, and the *outcome* is the resolution of the interaction between the hero's needs and desires and the press of the environment.

SENTENCE COMPLETION TEST. Although a projective instrument, the sentence completion test is much more direct in soliciting responses from the patient. He or she is simply presented with a series of incomplete sentences and is asked to complete the sentence with the first response that comes to mind. The following are examples of possible incomplete sentences:

My father seldom…
Most people don't know that I'm afraid of…
When I was a child, I…
When encountering frustration, I usually…

The purpose of the test is to elicit, in a somewhat indirect manner, information about the patient that cannot be elicited from other measures. Because the patient responds in writing, the examiner's time is limited. The length of time it takes to complete the sentence completion varies greatly depending on the number of incomplete sentences. Tests can range from less than 10 sentences to greater than 75.

Behavioral Assessment. Behavioral assessment involves the direct measurement of a given behavior. Rather than focus primarily on human characteristics, such as repression, ego strength, or self-esteem (vague terms to a behaviorist), strict behavioral measurement concentrates on the direct measurement that can be observed, such as several temper tantrums per unit of time, duration, and intensity and number of hyperventilation episodes, or the number of cigarettes smoked per 24-hour period.

Although early strict behaviorists would count only observable behaviors, a broader definition of behavior has emerged, under which just about anything people do—whether it is overt such as crying, swearing, or hand-washing or covert such as feeling and thinking—is considered behavior.

DIRECT COUNTING OF BEHAVIOR. Measuring overt behavior is direct and can be

done by the patient himself or herself, a family member, or an impartial observer.

Cognitive behavior therapists use these measurements to establish baselines of a given undesirable behavior (i.e., violent thoughts that the patient may wish to reduce). Similarly, therapists can measure behavior that the patient wants to increase (time studying, time out of bed, or distance walked on a treadmill). Follow-up measures of the same behavior monitor progress and quantify improvement.

MEDICAL ASSESSMENT AND LABORATORY TESTING IN PSYCHIATRY

Two recent issues have pushed medical assessment and laboratory testing in psychiatric patients to the forefront of attention for most clinicians: the widespread recognition of the pervasive problem of metabolic syndrome in clinical psychiatry and the shorter life expectancy of psychiatric patients compared with that of the general population. Factors that may contribute to medical comorbidity include abuse of tobacco, alcohol, and drugs, poor dietary habits, and obesity. Further, many psychotropic medications are associated with health risks that include obesity, metabolic syndrome, and hyperprolactinemia. Consequently, monitoring the physical health of psychiatric patients has become a more prominent issue.

A logical and systematic approach to the use of medical assessment and laboratory testing by the psychiatrist is vital to achieving the goals of arriving at accurate diagnoses, identifying medical comorbidities, implementing appropriate treatment, and delivering cost-effective care. Concerning the diagnosis or management of medical disease, it is crucial to consult with colleagues in other specialties. Good clinicians recognize the limits of their expertise and the need for consultation with their nonpsychiatric colleagues.

Physical Health Monitoring

Monitoring the physical health of psychiatric patients has two goals: to provide appropriate care for existing illnesses and to protect the patient’s current health from possible future impairment. Disease prevention should begin with a clear concept of the condition to be avoided. Ideally, in psychiatry, this would be a focus on commonly found conditions that could

be a significant source of morbidity or mortality. It is clear that in psychiatry, a small number of clinical problems underlie a significant number of impairments and premature deaths.

Role of History and Physical Examination

A thorough history, including a review of systems, is the basis for a comprehensive patient assessment. The history guides the clinician in the selection of laboratory studies that are relevant for a specific patient. Many psychiatric patients, owing to their illnesses, are not capable of providing sufficiently detailed information. Collateral sources of information, including family members and prior clinicians and their medical records, may be particularly helpful in the assessment of such patients.

The patient's medical history is an important component of the history. It should include notation of prior injuries and, in particular, head injuries that resulted in loss of consciousness and other causes of unconsciousness. The patient's medical history also should note pain conditions, ongoing medical problems, prior hospitalizations, prior surgeries, and a list of the patient's current medications. Toxic exposures are another important component of the medical history. Such exposures are often workplace-related.

The social history contains many of the details relevant to the assessment of character pathology, including risk factors for personality disorders as well as information relevant to the assessment of major disorders. Commonly, the social history includes a legal history, information about family and other significant relationships, and an occupational history.

In evaluating patients who appear demented, the role of the physical examination is to elucidate possible causative factors such as the cogwheel rigidity and tremor associated with Parkinson disease or neurologic deficits suggestive of prior strokes. Standard laboratory studies commonly assessed in dementia patients include a complete blood count (CBC), serum electrolytes, liver function tests (LFTs), blood urea nitrogen (BUN), creatinine (Cr), thyroid function tests, serum $B_{12,}$ and folate levels, Venereal Disease Research Laboratory (VDRL) test, and a urinalysis. Currently, there is no clear clinical indication for testing for the apolipoprotein E epsilon 4 allele. Often, a CT scan is performed if there are focal neurologic findings, and an electroencephalogram (EEG) may be performed if there is delirium.

When patients are delirious, the neurologic examination may be complicated by inattention due to altered levels of consciousness. Delirium workup often includes the same laboratory workup described above for dementia. Urine or blood cultures, chest radiograph, neuroimaging studies, or EEG also may be appropriate.

Imaging of the Central Nervous System

Imaging of the central nervous system (CNS) can be broadly divided into two domains: structural and functional. Structural imaging provides detailed, noninvasive visualization of the morphology of the brain. Functional imaging provides a visualization of the spatial distribution of specific biochemical processes. Structural imaging includes x-ray, CT, and magnetic resonance imaging (MRI). Functional imaging includes PET, single-photon emission computed tomography (SPECT), fMRI, and magnetic resonance spectroscopy (MRS). With the limited exception of PET scanning, functional imaging techniques are still considered research tools that are not yet ready for routine clinical use.

Magnetic Resonance Imaging. MRI scans are used to distinguish structural brain abnormalities that may be associated with a patient's behavioral changes. These studies provide the clinician with images of anatomical structures viewed from cross-sectional, coronal, or oblique perspectives. MRI scans can detect a large variety of structural abnormalities. The MRI is particularly useful in examining the temporal lobes, the cerebellum, and the deep subcortical structures. It is unique in its ability to identify periventricular white matter hyperintensities. MRI scans are useful in examining the patient for particular diseases, such as nonmeningeal neoplasms, vascular malformations, seizure foci, demyelinating disorders, neurodegenerative disorders, and infarctions. The advantages of MRI include the absence of ionizing radiation and the absence of iodine-based contrast agents. MRI scans are contraindicated when the patient has a pacemaker, aneurysm clips, or ferromagnetic foreign bodies.

Computed Tomography. CT scans are used to identify structural brain abnormalities that may contribute to a patient's behavioral abnormalities. These studies provide the clinician with cross-sectional x-ray images of the

brain. CT scans can detect a large variety of structural abnormalities in the cortical and subcortical regions of the brain. CT scans are useful when a clinician is looking for evidence of a stroke, subdural hematoma, tumor, or abscess. These studies also permit visualization of skull fractures. CT scans are the preferred modality when there is suspicion of a meningeal tumor, calcified lesions, acute subarachnoid or parenchymal hemorrhage, or acute parenchymal infarction.

CT scans may be performed with or without contrast. The purpose of contrast is to enhance the visualization of diseases that alter the blood–brain barrier, such as tumors, strokes, abscesses, and other infections.

Positron Emission Tomography. PET scans are performed predominately at university medical centers. PET scans require a positron emission tomograph (the scanner) and a cyclotron to create the relevant isotopes. This type of scan involves the detection and measurement of emitted positron radiation after the injection of a compound that has been tagged with a positron-emitting isotope. Typically, PET scans use fluorodeoxyglucose (FDG) to measure regional brain glucose metabolism. Glucose is the principal energy source for the brain. These scans can provide information about the relative activation of brain regions because regional glucose metabolism is directly proportionate to neuronal activity. Brain FDG scans are useful in the differential diagnosis of dementing disease. The most consistent finding in the PET literature is the pattern of temporal-parietal glucose hypometabolism in patients with Alzheimer type dementia.

PET scanning using FDDNP (2-(1-{6-[(2-[fluorine-18]fluoroethyl) (methyl)amino]-2-naphthyl}-ethylidene) malononitrile) can differentiate between normal aging, mild cognitive impairment, and Alzheimer disease by determining regional cerebral patterns of plaques and tangles associated with Alzheimer disease. FDDNP binds to the amyloid senile plaques and tau neurofibrillary tangles. FDDNP appears to be superior to FDG PET in differentiating Alzheimer's patients from those with mild cognitive impairment and subjects with normal aging and no cognitive impairment.

Single-Photon Emission Computed Tomography. SPECT is available in most hospitals but is rarely used to study the brain. SPECT is more commonly used to study other organs, such as the heart, liver, and

spleen. Some recent work, however, attempts to correlate SPECT brain imaging with mental disorders.

Functional Magnetic Resonance Imaging. fMRI is a research scan used to measure regional cerebral blood flow. Often, fMRI data are superimposed on conventional MRI images, resulting in detailed brain maps of brain structure and function. The measurement of blood flow involves the use of the heme molecule as an endogenous contrast agent. The rate of flow of heme molecules can be measured, resulting in an assessment of regional cerebral metabolism.

Magnetic Resonance Spectroscopy. MRS is another research method to measure regional brain metabolism. MRS scans are performed on conventional MRI devices that have had specific upgrades to their hardware and software. The upgrades permit the signal from protons to be suppressed and other compounds to be measured. (Conventional MRI images are, in reality, a map of the spatial distribution of protons found in water and fat.)

Magnetic Resonance Angiography. Magnetic resonance angiography (MRA) is a method for creating three-dimensional maps of cerebral blood flow. Neurologists and neurosurgeons more commonly use this test. Psychiatrists rarely use it.

Toxicology Studies

Urine drugs of abuse screens are immunoassays that detect barbiturates, benzodiazepines, cocaine metabolites, opiates, phencyclidine, tetrahydrocannabinol, and tricyclic antidepressants. These rapid tests provide results within an hour. However, they are screening tests; additional testing is required to confirm the results of this screening.

Testing to determine blood concentrations of certain psychotropic medications enables the clinician to ascertain whether blood levels of medications are at therapeutic, subtherapeutic, or toxic levels. Psychiatric symptoms are not uncommon when prescribed medications are at toxic levels. In the debilitated and the elderly, pathologic symptoms may occur at therapeutic concentrations. The normal reference range varies between laboratories. It is essential to check with the laboratory performing the test to

obtain the normal reference range for that laboratory.

Testing for drugs of abuse is usually performed on urine specimens. It also may be performed on specimens of blood, breath (alcohol), hair, saliva, and sweat. Urine screens provide information about the recent use of frequently abused drugs such as alcohol, amphetamines, cocaine, marijuana, opioids, and phencyclidine along with 3,4-methylenedioxymethamphetamine (MDMA) (ecstasy). Many substances may produce false positives with urine drug screening tests. When a false positive is suspected, a confirmatory test may be requested.

Comprehensive qualitative toxicology screening is usually performed by liquid and gas chromatography. This may require many hours to perform and is rarely done in routine clinical situations. It is usually performed in patients with unexplained toxicity and an atypical clinical picture.

Qualitative toxicology assessments may be useful in managing patients who have overdosed when combined with clinical assessment and knowledge of when the ingestion occurred.

Drug Abuse. Patients are frequently unreliable when reporting their drug abuse history. Drug-induced mental disorders often resemble primary psychiatric disorders. Furthermore, substance abuse can exacerbate preexisting mental illness. Indications for ordering a drug abuse screen include unexplained behavioral symptoms, a history of illicit drug use or dependence in a new patient evaluation, or a high-risk background (e.g., criminal record, adolescents, and prostitutes). A drug abuse screen is also frequently used to monitor patient abstinence during treatment of substance abuse. Such tests can be ordered on a scheduled or random basis. Many clinicians believe random testing may be more accurate in the assessment of abstinence. The tests also may help to motivate the patient.

Other laboratory data may suggest a problem with a substance use disorder. An increase in the mean corpuscular volume is associated with alcohol use disorder. Liver enzymes may be increased with alcohol use disorder or from hepatitis B or C acquired from intravenous (IV) drug abuse. Serologic testing for hepatitis B or C can confirm that diagnosis. IV drug abusers are at risk for bacterial endocarditis. If bacterial endocarditis is suspected, a further medical workup is indicated.

TESTED SUBSTANCES. Routine tests are available for phencyclidine (PCP), cocaine, tetrahydrocannabinol (THC; also known as *marijuana*), benzodiazepines, methamphetamine, and its metabolite amphetamine, morphine (Duramorph), codeine, methadone (Dolophine), propoxyphene (Darvon), barbiturates, lysergic acid diethylamide (LSD), and MDMA.

Drug screening tests may have high false-positive rates. This is often due to the interaction of prescribed medication with the test, resulting in false-positive results and a lack of confirmatory testing. False-negative tests are common as well. False-negative results may be due to problems with specimen collection and storage.

Testing is most commonly performed on urine, although serum testing is also possible for most agents. Hair and saliva testing are also available in some laboratories. Alcohol can also be detected in the breath (breathalyzer). Except for alcohol, drug levels are not usually determined. Instead, only the presence or absence of the drug is determined. There is usually not a meaningful or useful correlation between the level of the drug and clinical behavior. The length of time that a substance can be detected in the urine is listed in Table 1-23.



Table 1-23
Drugs of Abuse that Can Be Detected in Urine

Drug	Length of Time Detected in Urine
Alcohol	7–12 hr
Amphetamine	48–72 hr
Barbiturate	24 hr (short acting); 3 wk (long acting)
Benzodiazepine	3 days
Cocaine	6–8 hr (metabolites 2–4 days)
Codeine	48 hr
Heroin	36–72 hr
Marijuana	2–7 days

Methadone	3 days
Methaqualone	7 days
Morphine	48–72 hr

Alcohol. There is no single test or finding on physical examination that is diagnostic for alcohol use disorder. The history of the pattern of alcohol ingestion is most important in making the diagnosis. Laboratory test results and findings on physical examination may help to confirm the diagnosis. In patients with acute alcohol intoxication, a blood alcohol level (BAL) may be useful. A high BAL in a patient who clinically does not show significant intoxication is consistent with tolerance. Significant clinical evidence of intoxication with a low BAL should suggest intoxication with additional agents. Intoxication is commonly found with levels between 100 and 300 mg/dL. The degree of alcohol intoxication can also be assessed using the concentration of alcohol in expired respirations (breathalyzer). Chronic alcohol use is commonly associated with other laboratory abnormalities, including elevation in liver enzymes, such as aspartate aminotransferase (AST), which is usually greater than serum alanine aminotransferase (ALT). Bilirubin also is often elevated. Total protein and albumin may be low, and prothrombin time (PT) may be increased. A macrocytic anemia may be present.

Alcohol use disorder may be associated with rhinophyma, telangiectasias, hepatomegaly, and evidence of trauma on physical examination. In withdrawal, patients may have hypertension, tremulousness, and tachycardia.

Laboratory studies in patients who abuse alcohol may reveal macrocytosis. This occurs in most patients who consume four or more drinks per day. Elevations in AST and ALT characterize alcoholic liver disease, typically in a ratio of AST to ALT of 2:1 or greater. The γ-glutamyl transpeptidase (GGT) level may be elevated. Carbohydrate-deficient transferrin (CDT) may be helpful in the identification of chronic heavy alcohol use. It has a sensitivity of 60 to 70 percent and a specificity of 80 to 90 percent.

BAL is used to legally define intoxication in the determination of whether an individual is driving under the influence. The legal limit in many states is 80 mg/dL. However, clinical manifestations of intoxication vary with an

individual's degree of alcohol tolerance. At the same BAL, an individual who chronically abuses alcohol may exhibit less impairment than an alcohol-naive individual. Generally, a BAL in the range of 50 to 100 mg/dL is associated with impaired judgment and coordination and levels greater than 100 mg/dL produce ataxia.

Environmental Toxins. Specific toxins are associated with a variety of behavioral abnormalities. Exposure to toxins commonly occurs through occupation or hobbies.

Aluminum intoxication can cause a dementia-like condition. Aluminum can be detected in the urine or blood.

Arsenic intoxication may cause fatigue, loss of consciousness, anemia, and hair loss. Arsenic can be detected in urine, blood, and hair.

Manganese intoxication may present with delirium, confusion, and a parkinsonian syndrome. Manganese may be detected in urine, blood, and hair.

Symptoms of mercury intoxication include apathy, poor memory, lability, headache, and fatigue. Mercury can be detected in urine, blood, and hair.

Manifestations of lead intoxication include encephalopathy, irritability, apathy, and anorexia. Lead can be detected in blood or urine. Lead levels typically are assessed by collecting a 24-hour urine sample. The free erythrocyte protoporphyrin test is a screening test for chronic lead intoxication. This test is commonly coupled with a blood lead level. The Centers for Disease Control and Prevention specify that a lead level greater than 5 μg/dL requires further evaluation, and above 20 μg/dL requires a thorough investigation of the home and other possible sources of lead exposure. The incidence of lead toxicity in children has been falling recently.

Significant exposure to organic compounds, such as insecticides, may produce behavioral abnormalities. Many insecticides have strong anticholinergic effects. There are no readily available laboratory tests to detect these compounds. Poison control centers may assist in the identification of appropriate testing facilities.

Volatile Solvent Inhalation. Volatile substances produce vapors that are inhaled for their psychoactive effect. The most commonly abused volatile solvents include gasoline, glue, paint thinner, and correction fluid (white-

out). The aerosol propellants from cleaning sprays, deodorant sprays, and whipped cream containers may be abused. Nitrites, such as amyl nitrite ("poppers") and butyl nitrite vials ("rush"), and anesthetic gases, such as chloroform, ether, and nitrous oxide, are also abused.

Chronic abuse of volatile solvents is associated with damage to the brain, liver, kidneys, lung, heart, bone marrow, and blood. Abuse may produce hypoxia or anoxia. Signs of abuse include short-term memory loss, cognitive impairment, slurred and "scanning" speech, and tremor. Cardiac arrhythmias may occur. Exposure to toluene, which is present in many cleaning solutions, paints, and glues, has been associated with loss of clear gray–white matter differentiation and with brain atrophy on MRI scans. Methemoglobinemia has occurred with butyl nitrite abuse. Chronic use of volatile solvents is associated with the production of panic attacks and an organic personality disorder. Chronic use may also produce impairment in working memory and executive cognitive function.

Serum Medication Concentrations

Serum concentrations of psychotropic medications are assessed to minimize the risk of toxicity to patients receiving these medications and to ensure the administration of amounts sufficient to produce a therapeutic response. This is particularly true for medications with therapeutic blood levels. Medication levels are often influenced by hepatic metabolism. This metabolism occurs via the action of enzymes in the liver.

Acetaminophen. Acetaminophen may produce hepatic necrosis, which in some cases may be fatal. Acetaminophen is one of the most frequently used agents in intentional drug overdoses and is a common cause of overdose-related deaths. Toxicity is associated with levels greater than 140 mcg/mL at 4 hours from ingestion in patients without preexisting liver disease. Chronic abusers of alcohol are particularly vulnerable to the effects of an overdose. Acetylcysteine (Mucomyst) treatment must occur promptly after an overdose to prevent hepatotoxicity.

Salicylate Toxicity. Aspirin is frequently ingested in overdose. Consequently, serum salicylate levels often are obtained in overdose cases. Some rheumatic patients may chronically ingest large amounts of salicylate

for therapeutic reasons. Ingestion of 500 mg/kg of aspirin may be fatal. Most patients will develop symptoms of toxicity when salicylate levels are greater than 40 mg/dL. Common symptoms of toxicity include acid–base abnormalities, tachypnea, tinnitus, nausea, and vomiting. In cases of severe toxicity, symptoms may include hyperthermia, altered mental status, pulmonary edema, and death.

Antipsychotic Agents

CLOZAPINE. Clozapine (Clozaril) levels are trough levels determined in the morning before administration of the morning dose of medication. A therapeutic range for clozapine has not been established; however, a level of 100 mg/mL is widely considered to be the minimum therapeutic threshold. At least 350 mg/mL of clozapine is considered to be necessary to achieve therapeutic response in patients with refractory schizophrenia. The likelihood of seizures and other side effects increase with clozapine levels greater than 1,200 mg/mL or doses greater than 600 mg per day or both. Clozapine is a common cause of leukopenia in psychiatry. When moderate to severe leucopenia develops, clozapine treatment must be interrupted, but patients may be retreated with clozapine in the future.

Mood Stabilizers

CARBAMAZEPINE. Carbamazepine (Tegretol) may produce changes in the levels of white blood cells, platelets, and, under rare circumstances, red blood cells. Anemia, aplastic anemia, leucopenia, and thrombocytopenia may all occur but are rare. Pretreatment evaluations typically include CBC.

Carbamazepine may produce hyponatremia. This hyponatremia is usually mild and does not produce clinical symptoms. However, carbamazepine may cause the syndrome of inappropriate secretion of antidiuretic hormone (SIADH). Carbamazepine may produce a variety of congenital abnormalities, including spina bifida and anomalies of the fingers. Manifestations of toxicity may include nausea, vomiting, urinary retention, ataxia, confusion, drowsiness, agitation, or nystagmus. At very high levels, symptoms may also include cardiac dysrhythmias, seizures, and respiratory depression.

LITHIUM. Lithium (Eskalith) has a narrow therapeutic index. Consequently, blood levels of lithium must be monitored to achieve therapeutic dosing and avoid toxicity. Side effects are dose-dependent. Symptoms of toxicity include tremors, sedation, and confusion. At higher levels, delirium, seizures, and coma may occur. Symptoms of toxicity may begin to manifest with serum levels of greater than 1.2 mEq/L and are common with levels greater than 1.4 mEq/L. Elderly or debilitated patients may show signs of toxicity with levels of less than 1.2 mEq/L.

VALPROATE. Because of the risk of hepatotoxicity, ranging from mild dysfunction to hepatic necrosis, pretreatment LFTs are usually obtained. More commonly valproate (valproic acid [Depakene] and divalproex [Depakote]) may cause a sustained elevation in liver transaminase levels of as much as three times the upper limit of normal.

Valproate may increase the risk of congenital disabilities. A pretreatment urine pregnancy test is usually obtained in women of childbearing years. Women should be cautioned to use adequate contraception.

Hematologic abnormalities are also possible and include leucopenia and thrombocytopenia. Treatment with valproate may increase serum ammonia levels. It is prudent to obtain an ammonia level in a patient undergoing valproate treatment who presents with altered mental status or lethargy. Acute pancreatitis may also occur.

Antidepressants

MONOAMINE OXIDASE INHIBITORS. Treatment with monoamine oxidase inhibitors (MAOIs) can cause orthostasis and, rarely, hypertensive crisis. Baseline blood pressure measurement should be obtained before the initiation of treatment, and blood pressure should be monitored during treatment.

There are no meaningful blood levels for MAOIs, and direct monitoring of MAOI blood levels is not clinically indicated. Treatment with MAOIs is occasionally associated with hepatotoxicity. For this reason, LFTs usually are obtained at the initiation of treatment and periodically after.

TRICYCLIC AND TETRACYCLIC ANTIDEPRESSANTS. Routine laboratory studies obtained before initiation of tricyclic or tetracyclic antidepressants (TCAs)

typically include CBC, serum electrolytes, and LFTs. Because TCAs affect cardiac conduction, clinicians also may obtain an electrocardiogram (ECG) to assess for the presence of abnormal cardiac rhythms and prolonged PR, QRS, and QTc complexes before initiation of these medications.

Neuroleptic Malignant Syndrome

Neuroleptic malignant syndrome (NMS) is a rare, potentially fatal consequence of neuroleptic administration. The syndrome consists of autonomic instability, hyperpyrexia, severe extrapyramidal symptoms (i.e., rigidity), and delirium. Sustained muscle contraction results in peripheral heat generation and muscle breakdown. Muscle breakdown contributes to elevated levels of creatine kinase (CK). Peripheral heat generation with impaired central mechanisms of thermoregulation results in hyperpyrexia. Myoglobinuria and leukocytosis are common. Hepatic and renal failure may occur. Liver enzymes become elevated with liver failure. Patients may die from hyperpyrexia, aspiration pneumonia, renal failure, hepatic failure, respiratory arrest, or cardiovascular collapse. Treatment includes discontinuation of the neuroleptic, hydration, administration of muscle relaxants, and general supportive nursing care.

A typical laboratory workup for NMS includes a CBC, serum electrolytes, BUN, Cr, and CK. A urinalysis, including an assessment of urine myoglobin, is also usually performed. As part of the differential diagnosis, blood and urine cultures are performed as part of a fever workup. Pronounced elevations in the white blood cell (WBC) count may occur in NMS. White blood cell counts are typically in the range from 10,000 to 40,000 per mm^3.

Muscle Injury. Serum CK levels may rise in response to repeated intramuscular (IM) injections, prolonged or agitated periods in restraint, or NMS. Dystonic reactions from neuroleptic administration may also result in elevated levels of CK.

Electroconvulsive Therapy

ECT is usually reserved for patients with the most treatment-resistant depression. Typical laboratory tests obtained before the administration of ECT include a CBC, serum electrolytes, urinalysis, and LFTs. However, no

specific laboratory tests are required in the pre-ECT evaluation. Usually, an ECG is also obtained. A spinal x-ray series is no longer considered routinely indicated because of the low risk of spinal injury associated with modern administration techniques that use paralyzing agents. A comprehensive medical history and physical examination are useful screening tools to identify possible conditions that could complicate treatment.

Endocrine Evaluations

Endocrine disease is of great relevance to psychiatry. Management of psychiatric illness is complicated by comorbid endocrine disease. Endocrine illness frequently has psychiatric manifestations. For these reasons, screening for endocrine disease is often of relevance to the psychiatrist.

Adrenal Disease. Adrenal disease may have psychiatric manifestations, including depression, anxiety, mania, dementia, psychosis, and delirium. However, patients with adrenal disease rarely come to the attention of psychiatrists. Assessment and management of these patients are best done in conjunction with specialists.

Low plasma levels of cortisol are found in Addison disease. These patients may have symptoms that are also common in psychiatric conditions, including fatigue, anorexia, weight loss, and malaise. Patients may also have memory impairment, confusion, or delirium. Depression or psychosis with hallucinations and delusions may occur.

Elevated levels of cortisol are seen in Cushing syndrome. About half of all patients with Cushing syndrome develop psychiatric symptoms. These symptoms may include lability, irritability, anxiety, panic attacks, depressed mood, euphoria, mania, or paranoia. Cognitive dysfunctions may include cognitive slowing and poor short-term memory. Symptoms usually improve when cortisol normalizes. If not, or if symptoms are severe, psychiatric treatment may be necessary.

Cortisol levels are not useful in the assessment or management of primary psychiatric disease. In particular, the dexamethasone-suppression test (DST) remains a research tool in psychiatry that is not used in routine clinical care.

Anabolic Steroid Use. The use of anabolic steroids has been associated with irritability, aggression, depression, and psychosis. Athletes and

bodybuilders are common abusers of anabolic steroids. Urine specimens can be used to screen for these agents. Because so many compounds have been synthesized, a variety of tests may be required to confirm the diagnosis, depending on the compound that has been used. Consultation with a specialist is advised. Generally, androgens other than testosterone can be detected by gas chromatography and mass spectroscopy.

Antidiuretic Hormone. Arginine vasopressin (AVP), also called antidiuretic hormone (ADH), is decreased in central diabetes insipidus (DI). DI may be central (due to the pituitary or hypothalamus) or nephrogenic. Nephrogenic DI may be acquired or due to an inherited X-linked condition. Lithium-induced DI is an example of an acquired form of DI. Lithium has been shown to decrease the sensitivity of renal tubules to AVP. Patients with central DI respond to the administration of vasopressin with a decrease in urine output. Secondary central DI may develop in response to head trauma that produces damage in the pituitary or hypothalamus.

About one-fifth of patients taking lithium develop polyuria, and a more considerable amount may have some degree of impairment in concentrating urine. Chronic treatment with lithium is a common cause of nephrogenic DI. However, there are other causes of polyuria in lithium-treated patients also to nephrogenic DI. Primary polydipsia is common and is often associated with the dry mouth associated with many psychiatric medications. Central diabetes has also been associated with lithium treatment.

Excessive secretion of AVP results in increased retention of fluid in the body. This condition is called SIADH. Water retention in SIADH causes hyponatremia. SIADH may develop in response to injury to the brain or from medication administration (including phenothiazines, butyrophenones, carbamazepine, and oxcarbazepine). The hyponatremia associated with this condition may produce delirium.

Human Chorionic Gonadotropin. Human chorionic gonadotropin (hCG) can be assessed in the urine and blood. The urine test for hCG is the basis for the commonly used urine pregnancy test. This immunometric test can detect pregnancy approximately 2 weeks after an expected menstrual period has passed. Routine tests are most accurate when performed 1 to 2 weeks after a missed period and are not reliably accurate until the 2-week period has

passed. However, there are ultrasensitive urine hCG tests that can accurately detect pregnancy 7 days after fertilization. Pregnancy tests often are obtained before initiating certain psychotropic medications, such as lithium, carbamazepine, and valproate, which are associated with congenital anomalies.

Parathormone. Parathormone (parathyroid hormone) modulates serum concentrations of calcium and phosphorus. Dysregulation in this hormone and the resulting production of abnormalities in calcium and phosphorus may produce depression or delirium.

Prolactin. Prolactin levels may become elevated in response to the administration of antipsychotic agents. Elevations in serum prolactin result from the blockade of dopamine receptors in the pituitary. This blockade produces an increase in prolactin synthesis and release.

Cerebral MRI is not usually performed if the patient is taking an antipsychotic drug known to cause hyperprolactinemia, and the magnitude of the prolactin elevation is consistent with drug-induced causes.

Prolactin levels may briefly rise after a seizure. For this reason, prompt measurement of a prolactin level after possible seizure activity may assist in differentiating a seizure from a pseudoseizure.

Thyroid Hormone. The disease of the thyroid is associated with many psychiatric manifestations. Thyroid disease is most commonly associated with depression and anxiety but may also give rise to symptoms of panic, dementia, and psychosis. Thyroid disease may mimic depression. It is challenging to achieve euthymia if a patient is not euthyroid.

Systemic Lupus Erythematosus. Systemic lupus erythematosus (SLE) is an autoimmune disorder. Tests for SLE are based on the detection of antibodies formed as part of the disease. Antinuclear antibodies are found in virtually all patients with SLE. Antibody levels also are used to monitor the severity of the illness. A fluorescent test is used to detect antinuclear antibodies. This test can be positive in a variety of rheumatic diseases.

For this reason, a positive result usually is followed by additional tests, including a test to detect anti-deoxyribonucleic acid (DNA) antibodies. Anti-

DNA antibodies, when associated with antinuclear antibodies, are strongly suggestive of a diagnosis of lupus. Anti-DNA antibodies are followed to monitor the response to treatment.

Psychiatric manifestations of lupus include depression, dementia, delirium, mania, and psychosis. About 5 percent of patients with lupus present with symptoms of psychosis, including hallucinations and delusions.

Pancreatic Function. Measurement of serum amylase is used to monitor pancreatic function. Elevations in amylase levels may occur in alcohol-abusing patients who develop pancreatitis. Serum amylase levels also may be fractionated into salivary and pancreatic components.

Clinical Chemistry

Serum Electrolytes. Serum electrolyte levels may be useful in the initial evaluation of a psychiatric patient. Levels of serum electrolytes often are abnormal in patients with delirium. Abnormalities also may occur in response to the administration of psychotropic medications. Low serum chloride levels may occur in eating disorder patients who purge by self-induced vomiting. Serum bicarbonate levels may be elevated in patients who purge or who abuse laxatives. Bicarbonate levels are commonly low in patients who hyperventilate in response to anxiety.

Hypokalemia may be present in eating disorder patients who purge or abuse laxatives and in psychogenic vomiting. Diuretic abuse by eating disorder patients also may produce hypokalemia. Low levels of potassium are associated with weakness and fatigue. Characteristic ECG changes occur with hypokalemia and consist of cardiac arrhythmias, U waves, flattened T waves, and ST-segment depression.

Eating disorder patients with anorexia nervosa or bulimia nervosa usually receive a relatively standard set of laboratory studies, including serum electrolytes (particularly potassium and phosphorus), blood glucose, thyroid function tests, liver enzymes, total protein, serum albumin, BUN, Cr, CBC, and ECG. Serum amylase is often assessed in bulimic patients.

Magnesium levels may be low in alcohol-abusing patients. Low magnesium levels are associated with agitation, confusion, and delirium. If untreated, convulsions and coma may follow.

Low levels of serum phosphorus may be present in eating disorder patients with purging behavior. Phosphorus levels may also be low in anxiety patients who hyperventilate. Hyperparathyroidism may produce low serum phosphorus levels. Elevated serum phosphorus levels are seen in hypoparathyroidism.

Hyponatremia is seen in psychogenic polydipsia and SIADH and in response to certain medications, such as carbamazepine. Low sodium levels are associated with delirium.

Serum calcium abnormalities are associated with a variety of behavioral abnormalities. Low serum calcium levels are associated with depression, delirium, and irritability. Elevated levels are associated with depression, psychosis, and weakness. Laxative abuse, common in eating disorder patients, can be associated with hypocalcemia. Hypocalcemia secondary to hypoparathyroidism may occur in patients who have undergone surgery for thyroid disease.

Serum copper levels are low in Wilson disease, a rare abnormality in copper metabolism. Copper is deposited in the brain and liver, resulting in decreased intellectual functioning, personality changes, psychosis, and a movement disorder. Symptoms are usually present in the second and third decades of life. Laboratory assessment for Wilson disease includes the measurement of serum ceruloplasmin, the transport protein for copper, which is low, and urine copper, measured in a 24-hour specimen, which is elevated.

Renal Function. Tests of renal function include BUN and Cr. Other relevant laboratory studies include routine urinalysis and Cr clearance. An elevated BUN often results in lethargy or delirium. BUN is commonly elevated with dehydration. Elevations in BUN often are associated with impaired clearance of lithium. A less sensitive index of renal function is Cr. Elevations in Cr may indicate extensive renal impairment. Elevated levels occur when approximately 50 percent of the nephrons are damaged.

Cr clearance is often assessed in patients taking lithium. It is a sensitive measurement of renal function. The test is performed in a well-hydrated patient by collecting all of the patient’s urine for 24 hours. During the midpoint of the 24-hour collection period, a serum Cr level also is obtained. The resulting data are used to calculate the patient’s Cr clearance. Usually, the laboratory performs the calculation.

Elevated levels of porphobilinogen are found in the urine of symptomatic patients with acute intermittent porphyria. Symptoms of this disease include psychosis, apathy, or depression, along with intermittent abdominal pain, neuropathy, and autonomic dysfunction. If urine porphobilinogen levels are elevated when the patient is symptomatic, collection of a 24-hour urine specimen for quantitative assessment of porphobilinogen and aminolevulinic acid is indicated.

Liver Function. LFTs commonly include the serum aminotransferases, alkaline phosphatase, γ-glutamyl transpeptidase, and tests of synthetic function, usually the serum albumin concentration and prothrombin time, and the serum bilirubin, which reflects hepatic transport capability.

Elevations in AST may occur with diseases of the liver, heart, lungs, kidneys, and skeletal muscle. In patients with alcohol-induced liver disease, AST typically is more elevated than ALT. In viral- and drug-induced liver disease, ALT is often elevated. Serum GGT is elevated in hepatobiliary disease, including alcohol-induced liver disease and cirrhosis.

Alkaline phosphatase elevations occur in many diseases, including diseases of the liver, bone, kidney, and thyroid. Levels of alkaline phosphatase may be elevated in response to some psychiatric medications, most notably the phenothiazines.

Serum ammonia levels are often elevated in patients with hepatic encephalopathy. High levels are associated with the delirium of hepatic encephalopathy. Serum ammonia levels also may be elevated in patients undergoing treatment with valproate.

Serum bilirubin is an index of hepatic and bile duct function. Prehepatic, unconjugated, or indirect bilirubin and posthepatic, conjugated, or direct bilirubin are often assessed to help elucidate the origin of the elevation in bilirubin.

Lactate dehydrogenase (LDH) may be elevated in diseases of the liver, skeletal muscle, heart, and kidney. It is also elevated in pernicious anemia.

Vitamins

FOLATE AND B_{12}. Folate and B_{12} deficiencies are common in patients who

abuse alcohol. Folate and B_{12} deficiencies are associated with dementia; delirium; psychosis, including paranoia; fatigue; and personality change. Folate and B_{12} can be directly measured. Low folate levels may be found in patients who use contraceptive pills or other forms of estrogen, who drink alcohol, or who take phenytoin.

Infectious Disease Testing

Testing for sexually transmitted diseases (STDs) has become common, given the current frequency of these diseases. Some psychiatric illnesses, such as mania and substance abuse, are associated with a higher risk of contracting STDs. STDs include herpes simplex virus types 1 and 2, chlamydia, hepatitis viruses, gonorrhea, syphilis, and human immunodeficiency virus (HIV). Risk factors for STD include contact with sex workers, drug abuse, prior history of STDs, meeting partners on the Internet, multiple sex partners, a new sex partner, and being young or unmarried. Other diseases to think about are Epstein–Barr virus and gonorrhea.

Intravenous Drug Use. The IV route is used for many substances of abuse. Most commonly, heroin, amphetamines, and cocaine are used alone or in combination via the IV route. Because needles often are contaminated, IV drug users are at risk for bacterial endocarditis, hepatitis B and C, HIV infection, and acquired immunodeficiency syndrome (AIDS) from HIV infection. It has been estimated that over 60 percent of new cases of hepatitis C occur in individuals with a history of injecting illicit drugs.

CBC AND SERUM BLOOD CULTURES. The use of contaminated needles or nonsterile injection sites places IV drug users at risk for bacterial infections, including abscesses, bacteremia, and bacterial endocarditis. Findings on physical examination suggestive of endocarditis, possible bacteremia, or abscess necessitate obtaining a CBC to rule out an elevated WBC count. Blood cultures should be obtained from at least two different sites if the patient is febrile or if findings are suggestive of bacteremia or endocarditis, and internal medicine consultation should be obtained.

Syphilis. The fluorescent treponemal antibody absorption (FTA-ABS) test

detects antibodies against *Treponema pallidum* spirochetes and is more sensitive and specific than nontreponemal tests for syphilis. The test is used to confirm positive screening tests for syphilis, such as the rapid plasma reagin (RPR) test and the VDRL test. The FTA-ABS test is also used when neurosyphilis is suspected. Once positive, a patient usually remains so for life. False-positive results may occur in patients with SLE.

Viral Hepatitis. Several types of viruses can cause viral hepatitis. Viral hepatitis produces abnormalities in LFTs, including elevation of liver enzymes, especially ALT. Symptoms range from mild flu-like manifestations to rapidly progressive and fatal liver failure. Psychiatric manifestations include depression, anxiety, weakness, and psychosis. Viral hepatitis can also impair the metabolism of psychotropic medications that are metabolized by the liver. Impaired liver metabolism requires an adjustment of the dose of medications metabolized by the liver or consideration of agents that are less affected by alterations in liver metabolism. Viruses causing hepatitis include hepatitis A virus (HAV), hepatitis B virus (HBV), hepatitis C virus (HCV), and hepatitis D virus (HDV) (delta agent).

The WBC is normal to low in patients with hepatitis, especially in the preicteric phase. Large atypical lymphocytes occasionally are present. Rarely, aplastic anemia follows an episode of acute hepatitis not caused by any of the known hepatitis viruses. Mild proteinuria is common, and bilirubinuria often precedes the appearance of jaundice. Acholic stools frequently are present during the icteric phase. Strikingly elevated AST or ALT occurs early, followed by elevations of bilirubin and alkaline phosphatase. In a minority of patients, elevations of bilirubin and alkaline phosphatase persist after aminotransferase levels have normalized. Cholestasis may be substantial in acute hepatitis A. Marked prolongation of the PT in severe hepatitis correlates with increased mortality.

Chronic hepatitis, characterized by elevated aminotransferase levels for more than 6 months, develops in 1 to 2 percent of immunocompetent adults with acute hepatitis B. More than 80 percent of all persons with acute hepatitis C develop chronic hepatitis, which, in many cases, progresses slowly. Ultimately, cirrhosis develops in as many as 30 percent of those with chronic hepatitis C and 40 percent of those with chronic hepatitis B; the risk of cirrhosis is even higher in patients coinfected with both viruses or with

HIV. Patients with cirrhosis are at risk, with a rate of 3 to 5 percent per year, of hepatocellular carcinoma. Even in the absence of cirrhosis, patients with chronic hepatitis B—particularly those with active viral replication—are at an increased risk.

Electroencephalogram

The EEG assesses regional cerebral cortical electrical activity. Clinical neuroscience has a long history of using EEG. The EEG can be used in different ways to study specific brain states or activities by modifications to the technique of data collection or the data themselves. EEG data can be displayed on paper tracings in the manner of conventional EEG recordings. Alternatively, the data can be digitized, and the digitized data can be transformed, often using a Fourier transformation, to yield color-coded topographic brain maps of regional activity. The collection periods can be prolonged, and the data can be electronically displayed along with video monitoring of the patient to provide telemetry assessments of patients with epilepsy. Telemetry assessments are typically performed to correlate behavioral abnormalities with brain electrical activity as part of the workup of seizure disorders. Prolonged periods of EEG recording during sleep, when coupled with a recording of a limited lead ECG and facial muscle activity, result in the sleep EEG or polysomnography. Many clinicians also use the EEG to monitor ECT administration.

Clinicians use the EEG to localize seizure foci and to evaluate delirium. The EEG and its topographical descendants have not found an exact role in the diagnostic assessment of psychiatric patients. The EEG is usually used in psychiatry to rule out nonpsychiatric diseases, such as seizure disorders or delirium, as a cause of psychiatric symptoms. When the differential diagnosis includes strokes, tumors, subdural hematomas, or dementia, the yield is usually higher with imaging tests. Not surprisingly, the yield is the highest in patients with a history of a seizure disorder or a clinical history that is strongly suggestive of a recent seizure or other organic illness. Such clinical features would include a history of altered consciousness, atypical hallucinations (e.g., olfactory), head injury, and automatism. Also, the EEG is commonly obtained when there is an abnormal CT or MRI. It is important to remember that seizures are a clinical diagnosis; a normal EEG does not rule

out the possibility of a seizure disorder.

Evoked Potential. Evoked potential (EP) testing is the measurement of the EEG response to specific sensory stimulation. The stimulation may be visual, auditory, or somatosensory. During visual EPs, the patient is exposed to flashing lights or a checkerboard pattern. With auditory EP, the patient hears a specific tone. In somatosensory EP, the patient experiences an electrical stimulation to an extremity. These stimuli repeatedly occur while the patient undergoes a routine EEG. Using a computer, the responses to these stimuli are recorded and averaged. The time frame is measured in milliseconds. These tests are useful in neurology and neurosurgery. For example, they assist in the assessment of demyelinating disorders such as multiple sclerosis (MS). In psychiatry, EP testing may help in the differentiation of organic from functional impairments. A classical example is the use of EP testing to evaluate possible hysterical blindness. The usefulness of these tests in psychiatry is still under investigation.

Polysomnography. Polysomnography is used to assess disorders of sleep by concurrently assessing the EEG, ECG, blood oxygen saturation, respiration, body temperature, electromyogram, and electrooculogram. Polysomnography has demonstrated an increase in the overall amount of rapid eye movement (REM) sleep and a shortened period before the onset of REM sleep (decreased REM latency) in patients with major depression. These studies may assist in differentiating depression from other conditions that mimic depression. For example, patients who appear depressed from dementia do not have decreased REM latency or an increase in the amount of REM sleep.

Electrocardiogram

The ECG is a graphical representation of the electrical activity of the heart. Abnormalities in this activity correlate with cardiac pathology. The ECG is most commonly used in psychiatry to assess the side effects of psychotropic medications.

Ziprasidone has been associated with a dose-related prolongation of the QTc interval. There is a known association of fatal arrhythmias (e.g., torsades de pointes) with QTc prolongation from some other medications. For this

reason, clinicians usually obtain an ECG before initiation of treatment with ziprasidone. Ziprasidone is contraindicated in patients with a known history of QTc prolongation (including congenital long QT syndrome), with recent acute myocardial infarction, or with uncompensated heart failure. Bradycardia, hypokalemia or hypomagnesemia, or the concurrent use of other drugs that prolong the QTc interval all increase the risk of severe arrhythmias. Ziprasidone should be discontinued in patients who have persistent QTc measurements greater than 500 milliseconds.

Like ziprasidone, thioridazine has been associated with prolongation of the QTc interval in a dose-related manner. Prolongation of the QTc interval has been associated with torsades de pointes arrhythmias and sudden death. An ECG should be obtained before initiating treatment with thioridazine to rule out QTc prolongation. Many other antipsychotics can along prolong the QTc interval particularly in patients with preexisting arrhythmia.

TCAs are, at times, associated with ECG changes. Anticholinergic effects may increase heart rate. Prolongation of the PR, QT, and QRS intervals, along with ST-segment and T-wave abnormalities, may occur. The TCAs can cause or increase preexisting atrioventricular or bundle branch block. When the QTc exceeds 0.440 seconds, a patient is at an increased risk for sudden death due to cardiac arrhythmias. Many clinicians obtain an ECG before beginning a TCA in a patient older than 40 years of age and any patient with known cardiovascular disease.

Lithium therapy can cause benign reversible T-wave changes, which can impair sinoatrial (SA) node function and can cause heart block. ECGs are often obtained before initiation of treatment with lithium and in cases of lithium toxicity or overdose.

Psychiatrists, when treating patients with specific psychiatric diagnoses, also use the ECG. Eating disorder patients commonly have low potassium levels that may result in abnormal ECG recordings. As the serum potassium drops below normal, T waves become flat (or inverted), and U waves may appear.

Holter Monitoring. Holter monitoring is the continuous recording of a patient’s ECG activity for a sustained period (e.g., 24 hours). Patients are ambulatory during this time. It is useful for the evaluation of dizziness, palpitations, and syncope. It is commonly used in the evaluation of patients

with panic disorder who manifest cardiac symptoms.

Cardiac Ultrasound. Cardiac ultrasound is the visualization of cardiac anatomy by the use of computer-transformed echoes of ultrasound. It is commonly used in the evaluation of mitral valve prolapse. There is an unclear association between mitral valve prolapse and panic attacks and anxiety disorders.

NEUROIMAGING

The primary observation of structural and functional brain imaging in neuropsychiatric disorders such as dementia, movement disorders, demyelinating disorders, and epilepsy has contributed to a greater understanding of the pathophysiology of neurologic and psychiatric illnesses and helps practicing clinicians in difficult diagnostic situations.

Neuroimaging methodologies allow measurement of the structure, function, and chemistry of the living human brain. Over the past decade, studies using these methods have provided new information about the pathophysiology of psychiatric disorders that may prove to be useful for diagnosing illness and for developing new treatments. Computed tomographic (CT) scanners, the first widely used neuroimaging devices, allowed assessment of structural brain lesions such as tumors or strokes. MRI scans, developed next, distinguish gray and white matter better than CT scans do and allow visualizations of smaller brain lesions as well as white matter abnormalities. Also, to structural neuroimaging with CT and MRI, a revolution in functional neuroimaging has enabled clinical scientists to obtain unprecedented insight into the diseased human brain. The foremost techniques for functional neuroimaging include PET and SPECT.

Uses of Neuroimaging

Indications for Ordering Neuroimaging in Clinical Practice

NEUROLOGIC DEFICITS. In a neurologic examination, any change that can be localized to the brain or spinal cord requires neuroimaging. Neurologic examination includes mental status, cranial nerves, motor system,

coordination, sensory system, and reflex components. The mental status examination assesses arousal, attention, and motivation; memory; language; visuospatial function; complex cognition; and mood and affect. Consultant psychiatrists should consider a workup, including neuroimaging for patients with new-onset psychosis and acute changes in mental status. The clinical examination always assumes priority, and neuroimaging is ordered based on clinical suspicion of a CNS disorder.

DEMENTIA. Loss of memory and cognitive abilities affects more than 10 million persons in the United States and will affect an increasing number as the population ages. Reduced mortality from cancer and heart disease has increased life expectancy and has allowed persons to survive to the age of onset of degenerative brain disorders, which have proved more challenging to treat. Depression, anxiety, and psychosis are common in patients with dementia. The most common cause of dementia is Alzheimer disease, which does not have a characteristic appearance on routine neuroimaging but, instead, is associated with diffuse loss of brain volume.

One treatable cause of dementia that requires neuroimaging for diagnosis is *normal pressure hydrocephalus*, a disorder of the drainage of cerebrospinal fluid (CSF). This condition does not progress to the point of acutely increased intracranial pressure but stabilizes at a pressure at the upper end of the normal range. The dilated ventricles, which may be readily visualized with CT or MRI, exert pressure on the frontal lobes. A gait disorder is almost uniformly present; dementia, which may be indistinguishable from Alzheimer disease, appears less consistently. Relief of the increased CSF pressure may completely restore gait and mental function.

Infarction of the cortical or subcortical areas, or stroke, can produce focal neurologic deficits, including cognitive and emotional changes. Strokes are easily seen on MRI scans. Depression is common among stroke patients, either because of direct damage to the emotional centers of the brain or because of the patient’s reaction to the disability. Depression, in turn, can cause pseudodementia. Also, to major strokes, extensive atherosclerosis in brain capillaries can cause countless tiny infarctions of brain tissue; patients with this phenomenon may develop dementia as fewer and fewer neural pathways participate in cognition. This state, called *vascular dementia* (or Vascular Neurocognitive Disorder in DSM-5), is characterized on MRI scans

by patches of increased signal in the white matter.

Certain degenerative disorders of basal ganglia structures, associated with dementia, may have a characteristic appearance on MRI scans. Huntington disease typically produces atrophy of the caudate nucleus; thalamic degeneration can interrupt the neural links to the cortex (Fig. 1-10).

Space-occupying lesions can cause dementia. Chronic subdural hematomas and cerebral contusions, caused by head trauma, can produce focal neurologic deficits or may only produce dementia. Brain tumors can affect cognition in several ways. Skull-based meningiomas can compress the underlying cortex and impair its processing. Infiltrative glial cell tumors, such as astrocytoma or glioblastoma multiforme, can cut off communication between brain centers by interrupting white matter tracts. Tumors located near the ventricular system can obstruct the flow of CSF and gradually increase the intracranial pressure.

Chronic infections, including neurosyphilis, cryptococcosis, tuberculosis, and Lyme disease, can cause symptoms of dementia and may produce a characteristic enhancement of the meninges, especially at the base of the brain. Serologic studies are needed to complete the diagnosis. HIV infection can cause dementia directly, in which case is seen as a diffuse loss of brain volume, or it can allow a proliferation of Creutzfeldt–Jakob virus to yield progressive multifocal leukoencephalopathy, which affects white matter tracts and appears as increased white matter signal on MRI scans.

Chronic demyelinating diseases, such as multiple sclerosis, can affect cognition because of white matter disruption. Multiple sclerosis plaques are easily seen on MRI scans as periventricular patches of increased signal intensity.

Any evaluation of dementia should consider medication effects, metabolic derangements, infections, and nutritional causes that may not produce abnormalities on neuroimaging.

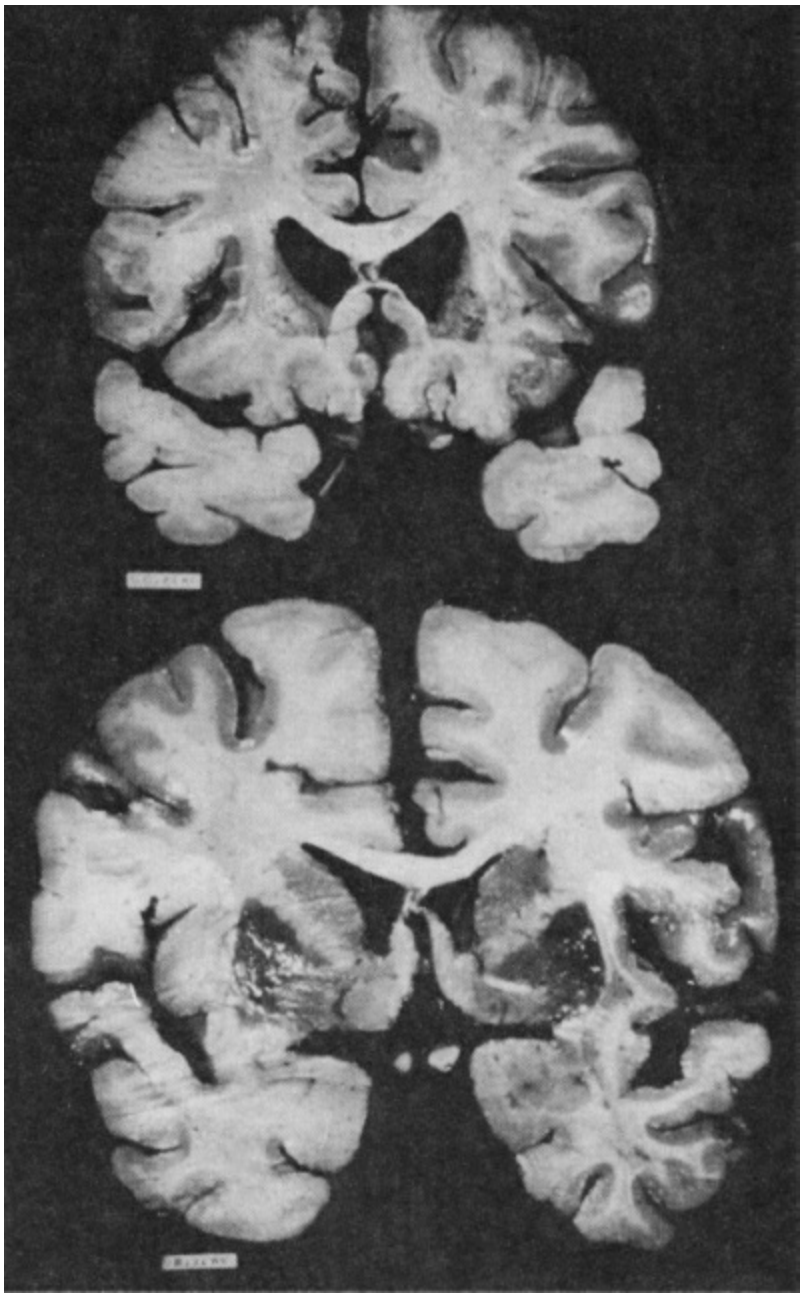


FIGURE 1-10
Brain slices. **Top:** Huntington disease. Atrophy of caudate nucleus and lentiform nuclei with dilatation of lateral ventricle. **Bottom:** Normal brain. (From Fahn S. Huntington disease. In: Rowland LP, ed. *Merritt's Textbook of Neurology*. 10th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2000:659, with permission.)

Indications for Neuroimaging in Clinical Research

ANALYSIS OF CLINICALLY DEFINED GROUPS OF PATIENTS. Psychiatric research aims to categorize patients with psychiatric disorders to facilitate the

discovery of neuroanatomical and neurochemical bases of mental illness. Researchers have used functional neuroimaging to study groups of patients with such psychiatric conditions as schizophrenia, affective disorders, and anxiety disorders, among others. In schizophrenia, for example, neuropathologic volumetric analyses have suggested a loss of brain weight, specifically of gray matter. A paucity of axons and dendrites appears present in the cortex, and CT and MRI may show compensatory enlargement of the lateral and third ventricles. Specifically, the temporal lobes of persons with schizophrenia appear to lose the most volume relative to healthy persons. Recent studies have found that the left temporal lobe is generally more affected than the right. The frontal lobe may also have abnormalities, not in the volume of the lobe, but in the level of activity detected by functional neuroimaging. Persons with schizophrenia consistently exhibit decreased metabolic activity in the frontal lobes, especially during tasks that require the prefrontal cortex. As a group, patients with schizophrenia are also more likely to have an increase in ventricular size than are healthy controls.

Disorders of mood and affect can also be associated with loss of brain volume and decreased metabolic activity in the frontal lobes. Inactivation of the left prefrontal cortex appears to depress mood; inactivation of the right prefrontal cortex elevates it. Among anxiety disorders, studies of OCD with conventional CT and MRI have shown either no specific abnormalities or a smaller caudate nucleus. Functional PET and SPECT studies suggest abnormalities in the corticolimbic, basal ganglia, and thalamic structures in the disorder. When patients are experiencing OCD symptoms, the orbital prefrontal cortex shows abnormal activity. A partial normalization of caudate glucose metabolism appears in patients taking medications such as fluoxetine or clomipramine or undergoing behavior modification.

Functional neuroimaging studies of persons with ADHD either have shown no abnormalities or have shown a decreased volume of the right prefrontal cortex and the right globus pallidus. Also, whereas usually the right caudate nucleus is larger than the left caudate nucleus, persons with ADHD may have caudate nuclei of equal size. These findings suggest dysfunction of the right prefrontal-striatal pathway for control of attention.

ANALYSIS OF BRAIN ACTIVITY DURING PERFORMANCE OF SPECIFIC TASKS. Many original conceptions of different brain region functions emerged from

observing deficits caused by local injuries, tumors, or strokes. Functional neuroimaging allows researchers to review and reassess classic teachings in the intact brain. Most work, to date, has been aimed at language and vision. Although many technical peculiarities and limitations of SPECT, PET, and fMRI have been overcome, none of these techniques has demonstrated clear superiority. Studies require carefully controlled conditions, which subjects may find arduous. Nonetheless, functional neuroimaging has contributed major conceptual advances, and the methods are now limited mainly by the creativity of the investigative protocols.

Studies have been designed to reveal the functional neuroanatomy of all sensory modalities, gross and fine motor skills, language, memory, calculations, learning, and disorders of thought, mood, and anxiety. Unconscious sensations transmitted by the autonomic nervous system have been localized to specific brain regions. These analyses provide a basis for comparison with results of studies of clinically defined patient groups and may lead to improved therapies for mental illnesses.

Specific Techniques

CT Scans. In 1972, CT scanning revolutionized diagnostic neuroradiology by allowing imaging of the brain tissue in live patients. CT scanners are currently the most widely available and convenient imaging tools available in clinical practice; practically, every hospital emergency room has immediate access to a CT scanner at all times. CT scanners effectively take a series of head x-ray pictures from all vantage points, 360 degrees around a patient's head. The amount of radiation that passes through, or is not absorbed from, each angle is digitized and entered into a computer. The computer uses matrix algebra calculations to assign a specific density to each point within the head and displays these data as a set of two-dimensional images. When viewed in sequence, the images allow mental reconstruction of the shape of the brain.

The CT image is determined only by the degree to which tissues absorb x-irradiation. The bony structures absorb high amounts of irradiation and tend to obscure details of neighboring structures, an especially troublesome problem in the brainstem, which is surrounded by a thick skull base. Within the brain itself, there is relatively little difference in the attenuation between

gray matter and white matter in x-ray images. Although the gray–white border is usually distinguishable, details of the gyral pattern may be difficult to appreciate in CT scans. Certain tumors may be invisible on CT because they absorb as much irradiation as the surrounding normal brain.

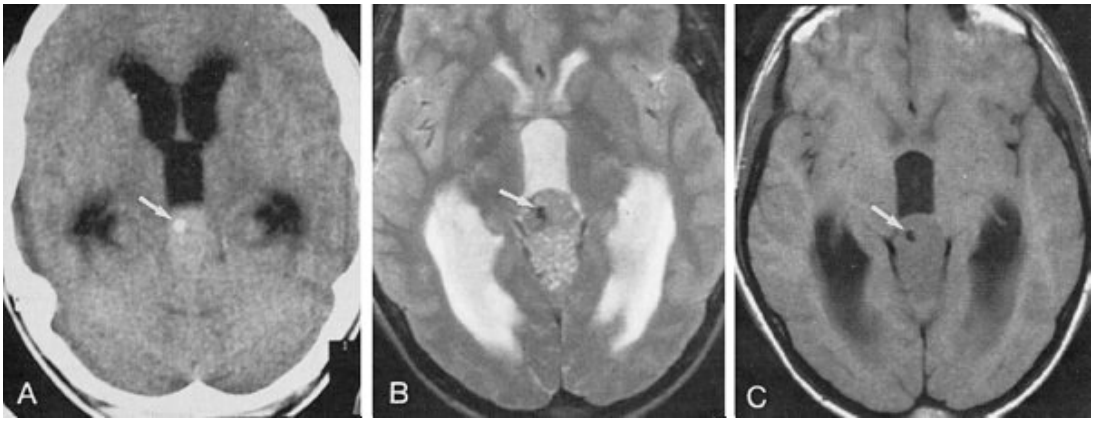


FIGURE 1-11
Comparison of computed tomography (CT) and magnetic resonance imaging (MRI). **A:** CT scan in the axial plane at the level of the third ventricle. The cerebrospinal fluid (CSF) within the ventricles appears black, the brain tissue appears gray, and the skull appears white. There is very poor discrimination between the gray and white matter of the brain. The *arrow* indicates a small calcified lesion in a tumor of the pineal gland. Detection of calcification is one role in which CT is superior to MRI. **B:** T2-weighted image of the same patient at roughly the same level. With T2, the CSF appears white, the gray matter appears gray, the white matter is clearly distinguished from the gray matter; the skull and indicated calcification appear black. Much more detail of the brain is visible than with CT. **C:** T1-weighted image of the same patient at roughly the same level. With T1, the CSF appears dark, the brain appears more uniformly gray, and the skull and indicated calcification appear black. T1 MRI images are the most similar to CT images. (Reprinted from Grossman CB. *Magnetic Resonance Imaging and Computed Tomography of the Head and Spine*. 2nd ed. Baltimore, MD: Williams & Wilkins; 1996:101, with permission.)

Appreciation of tumors and areas of inflammation, which can cause changes in behavior, can be increased by intravenous infusion of iodine-containing contrast agents. Iodinated compounds, which absorb much more irradiation than the brain, appear white. The intact brain is separated from the bloodstream by the blood–brain barrier, which normally prevents the passage of the highly charged contrast agents. The blood–brain barrier, however, breaks down in the presence of inflammation or fails to form within tumors and thus allows accumulation of contrast agents. These sites appear whiter than the surrounding brain. Iodinated contrast agents must be used with

caution in patients who are allergic to these agents or shellfish.

With the introduction of MRI scanning, CT scans have been supplanted as the nonemergency neuroimaging study of choice (Fig. 1-11). The increased resolution and delineation of detail afforded by MRI scanning is often required for diagnosis in psychiatry. Also, performing the most detailed study available inspires the most confidence in the analysis. The only component of the brain better seen on CT scanning is calcification, which may be invisible on MRI.

MRI Scans. MRI scanning entered clinical practice in 1982 and soon became the test of choice for clinical psychiatrists and neurologists. The technique does not rely on the absorption of x-rays but is based on nuclear magnetic resonance (NMR). The principle of NMR is that the nuclei of all atoms are thought to spin about an axis, which is randomly oriented in space. When atoms are placed in a magnetic field, the axes of all odd-numbered nuclei align with the magnetic field. The axis of a nucleus deviates away from the magnetic field when exposed to a pulse of radiofrequency electromagnetic radiation oriented at 90 or 180 degrees to the magnetic field. When the pulse terminates, the axis of the spinning nucleus realigns itself with the magnetic field, and during this realignment, it emits its own radiofrequency signal. MRI scanners collect the emissions of individuals, realigning nuclei, and use computer analysis to generate a series of two-dimensional images that represent the brain. The images can be in the axial, coronal, or sagittal planes.

By far, the most abundant odd-numbered nucleus in the brain belongs to hydrogen. The rate of realignment of the hydrogen axis is determined by its immediate environment, a combination of both the nature of the molecule of which it is a part and the degree to which it is surrounded by water. Hydrogen nuclei within fat realign rapidly, and hydrogen nuclei within water realign slowly. Hydrogen nuclei in proteins and carbohydrates realign at intermediate rates.

Routine MRI studies use three different radiofrequency pulse sequences. The two varied parameters are the duration of the radiofrequency excitation pulse and the length of the time that data are collected from the realigning nuclei. Because T1 pulses are brief and data collection is brief, hydrogen nuclei in hydrophobic environments are emphasized. Thus, fat is bright on

T1, and CSF is dark. The T1 image most closely resembles that of CT scans and is most useful for assessing overall brain structure. T1 is also the only sequence that allows contrast enhancement with the contrast agent gadolinium-diethylenetriamine pentaacetic acid (gadolinium-DTPA). As with the iodinated contrast agents used in CT scanning, gadolinium remains excluded from the brain by the blood–brain barrier, except in areas where this barrier breaks down, such as inflammation or tumor. On T1 images, gadolinium-enhanced structures appear white.

T2 pulses last four times as long as T1 pulses, and the collection times are also extended to emphasize the signal from hydrogen nuclei surrounded by water. Thus, brain tissue is dark, and CSF is white on T2 images. Areas within the brain tissue that have abnormally high water content, such as tumors, inflammation, or strokes, appear brighter on T2 images. T2 images reveal brain pathology most clearly. The third routine pulse sequence is the proton density or balanced sequence. In this sequence, a short radio pulse is followed by a prolonged period of data collection, which equalizes the density of the CSF and the brain and allows the distinction of tissue changes immediately adjacent to the ventricles.

An additional technique, sometimes used in clinical practice for specific indications, is fluid-attenuated inversion recovery (FLAIR). In this method, the T1 image is inverted and added to the T2 image to double the contrast between gray matter and white matter. Inversion recovery imaging is useful for detecting sclerosis of the hippocampus caused by temporal lobe epilepsy and for localizing areas of abnormal metabolism in degenerative neurologic disorders.

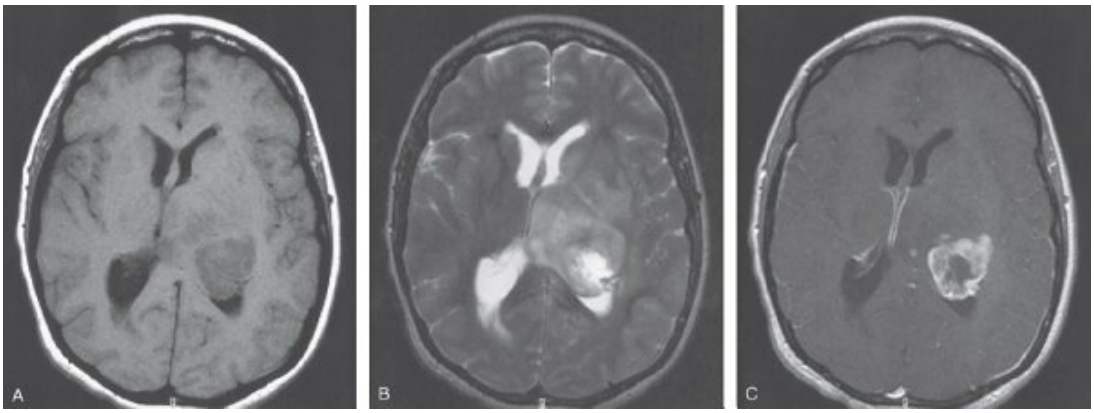


FIGURE 1-12
Three axial images from a 46-year-old woman who was hospitalized for the first

time for depression and suicidality following the end of a long-standing relationship. A malignant neoplasm extending into the posterior aspect of the left lateral ventricle is clearly seen in all three images. Images **A** and **B** are T1 and T2 weighted, respectively. Image **C** demonstrates the effects of postcontrast enhancement. (Courtesy of Craig N. Carson, M.D. and Perry F. Renshaw, M.D.)

MRI magnets are rated in teslas (T), units of magnetic field strength. MRI scanners in clinical use range from 1.5 to 3.0 T. Higher field–strength scanners produce images of markedly higher resolution. In research settings for humans, magnets as powerful as 7.0 T are used (and a few centers have used magnetic fields greater than 10 T in humans); for animals, magnets greater than 20 T are used. Unlike the well-known hazards of x-irradiation, exposure to electromagnetic fields of the strength used in MRI machines has not been shown to damage biologic tissues.

MRI scans cannot be used for patients with pacemakers or implants of ferromagnetic metals. MRI involves enclosing a patient in a narrow tube, in which the patient must remain motionless for up to 20 minutes. The radiofrequency pulses create a loud banging noise that may be obscured by music played in headphones. A significant number of patients cannot tolerate the claustrophobic conditions of routine MRI scanners and may need an open MRI scanner, which has less power and thus produces images of lower resolution. The resolution of brain tissue of even the lowest power MRI scan, however, exceeds that of CT scanning. Figure 1-12 reveals that a brain tumor is the cause of a patient's depression.

MRI APPLICATIONS TO DEMENTIA. Several MRI changes, including an increased number of subcortical hyperintensities, generalized atrophy, and ventricular enlargement, are associated with normal aging. However, it is well established that some changes seem more specific to the diagnosis of Alzheimer disease and may be clinically useful in formulating the diagnosis and prognosis of the disorder. MRI evidence of MTL atrophy appears to be most closely associated with the disorder. One approach that may help to improve the clinical utility of MRI in the diagnosis and prognosis of Alzheimer disease and other forms of dementia is to follow the rate of change in brain structure over time. Longitudinal follow-up studies have shown the rates of volume loss to be significantly greater in subjects with prodromal Alzheimer disease (up to 5 percent brain volume per year) compared with

those experiencing normal age-related reductions (0.1 percent brain volume per year).

MRI APPLICATIONS TO ALCOHOL DEPENDENCE. MRI studies have been the principal tool to describe in vivo the many sources of neurotoxicity associated with alcoholism, including (1) the direct neurotoxic and gliotoxic effect of ethanol, (2) the neurotoxic effects of poor nutrition that often accompany the abuse of alcohol, (3) the excitotoxicity associated with the ethanol withdrawal state, and (4) the possible disruption in adult-neurogenesis-associated ethanol intoxication and withdrawal. These studies documented a striking age dependence of the overall neurotoxicity associated with alcoholism.

Magnetic Resonance Spectroscopy. Whereas routine MRI detects hydrogen nuclei to determine brain structure, MRS can detect several odd-numbered nuclei (Table 1-24). The ability of MRS to detect a wide range of biologically important nuclei allows the use of the technique to study many metabolic processes. Although the resolution and sensitivity of MRS machines are poor compared with those of currently available PET and SPECT devices, the use of stronger magnetic fields will improve this feature to some extent in the future. MRS can image nuclei with an odd number of protons and neutrons. The unpaired protons and neutrons (nucleons) appear naturally and are nonradioactive. As in MRI, the nuclei align themselves in the strong magnetic field produced by an MRS device. A radiofrequency pulse causes the nuclei of interest to absorb and then emit energy. The readout of an MRS device is usually in the form of a spectrum, such as those for phosphorus-31 and hydrogen-1 nuclei, although the spectrum can also be converted into a pictorial image of the brain. The multiple peaks for each nucleus reflect that the same nucleus is exposed to different electron environments (electron clouds) in different molecules. The hydrogen-1 nuclei in a molecule of creatine, therefore, have a different chemical shift (position in the spectrum) than the hydrogen-1 nuclei in a choline molecule, for example. Thus, the position in the spectrum (the chemical shift) indicates the identity of the molecule in which the nuclei are present. The height of the peak for a reference standard of the molecule indicates the amount of the molecule present.

The MRS of the hydrogen-1 nuclei is best at measuring *N*-acetylaspartate (NAA), creatine, and choline-containing molecules, but MRS can also detect glutamate, glutamine, lactate, and myoinositol. Although glutamate and γ-aminobutyric acid (GABA), the major amino acid neurotransmitters, can be detected by MRS, the biogenic amine neurotransmitters (e.g., dopamine) are present in concentrations too low to be detected with the technique. MRS of phosphorus-31 can be used to determine the pH of brain regions and the concentrations of phosphorus-containing compounds (e.g., adenosine triphosphate [ATP] and guanosine triphosphate [GTP]), which are important in the energy metabolism of the brain.



Table 1-24
Nuclei Available for In Vivo Magnetic Resonance Spectroscopy[a]

Nucleus	Natural Abundance	Sensitivity	Relative Potential Clinical Uses
^{1}H	99.99	1.00	Magnetic resonance imaging (MRI) Analysis of metabolism Identification of unusual metabolites Characterization of hypoxia
^{19}F	100.00	0.83	Measurement of pO_2 Analysis of glucose metabolism Measurement of pH Noninvasive pharmacokinetics
^{7}Li	92.58	0.27	Pharmacokinetics
^{23}Na	100.00	0.09	MRI
^{31}P	100.00	0.07	Analysis of bioenergetics Identification of unusual metabolites Characterization of hypoxia

			Measurement of pH
^{14}N	93.08	0.001	Measurement of glutamate, urea, ammonia
^{39}K	93.08	0.0005	Similar to sodium (^{23}Na) but less specific, hence rarely used
^{13}C	1.11	0.0002	Analysis of metabolite turnover rate Pharmacokinetics of labeled drugs
^{17}O	0.04	0.00001	Measurement of metabolic rate
^{2}H	0.02	0.000002	Measurement of perfusion

[a]Natural abundance is given as percentage abundance of the isotope of interest. Nuclei are tabulated in order of decreasing relative sensitivity; relative sensitivity is calculated by multiplying the relative sensitivity for equal numbers of nuclei (at a given field strength) by the natural abundance of that nucleus. A considerable gain in relative sensitivity can be obtained by isotopic enrichment of the nucleus of choice or by the use of novel pulse sequences.

Reprinted from Dager SR, Steen RG with minor adaptations. Applications of magnetic resonance spectroscopy to the investigation of neuropsychiatric disorders. *Neuropsychopharmacology*. 1992;6:249, with permission.

MRS has revealed decreased concentrations of NAA in the temporal lobes and increased concentrations of inositol in the occipital lobes of persons with dementia of the Alzheimer type. In a series of subjects with schizophrenia, decreased NAA concentrations were found in the temporal and frontal lobes. MRS has been used to trace the levels of ethanol in various brain regions. In panic disorder, MRS has been used to record the levels of lactate, whose intravenous infusion can precipitate panic episodes in about three-fourths of patients with either panic disorder or major depression. Brain lactate concentrations were found to be elevated during panic attacks, even without provocative infusion.

Additional indications include the use of MRS to measure concentrations of psychotherapeutic drugs in the brain. One study used MRS to measure lithium (Eskalith) concentrations in the brains of patients with bipolar disorder and found that lithium concentrations in the brain were half those in the plasma during depressed and euthymic periods but exceeded those in the

plasma during manic episodes. Some compounds, such as fluoxetine and trifluoperazine (Stelazine), contain fluorine-19, which can also be detected in the brain and measured by MRS. For example, MRS has demonstrated that it takes 6 months of steady use for fluoxetine to reach maximal concentrations in the brain, which equilibrate at about 20 times the serum concentrations.

MRS IN DEMENTIA. MRS presents the opportunity to noninvasively obtain measures of several neurochemicals related to neurotransmission, energy metabolism, and cellular function. Studies using MRS have shown a trend for a general reduction in NAA measures with increasing age in MTL and frontal cortical brain regions. The studies in MCI and Alzheimer disease report patients with these disorders have decreased levels of NAA and increased levels of myoinositol (a form of inositol normally found in the brain that contributes to osmotic regulation) compared with those of age-matched comparison subjects.

MRS IN SCHIZOPHRENIA. MRS has been applied widely in studies of cortical chemistry in schizophrenia. These studies documented reductions in NAA levels in many cortical and limbic brain regions in schizophrenic individuals and smaller reductions in family members of people diagnosed with schizophrenia. Other metabolites have been measured in MRS studies of schizophrenic patients. The most interesting finding may be the description of normal or low levels of glutamate and increased levels of glutamine in medication-free patients with schizophrenia. One preliminary study suggested that glutamine elevations were not present in medication-free patients who were receiving benzodiazepines, drugs that would be predicted to suppress excitatory neurotransmission.

MRS IN ALCOHOL DEPENDENCE. MRS studies evaluating NAA and choline have provided neurochemical evidence that complements the MRI findings related to the emergence and recovery from alcohol-related neurotoxicity. MRS studies of GABA have provided insights into alterations in cortical inhibitory neurotransmissions associated with the recovery from alcohol dependence. During acute withdrawal, cortical GABA levels appear to be normal. With recovery from alcohol dependence, cortical GABA levels appear to decline and may be significantly below the level seen in healthy

subjects with extended sobriety.

Functional MRI. Recent advances in data collection and computer data processing have reduced the acquisition time for an MRI image to less than 1 second. A new sequence of particular interest to psychiatrists is the T2 or blood oxygen level-dependent (BOLD) sequence, which detects levels of oxygenated hemoglobin in the blood. Neuronal activity within the brain causes a local increase in blood flow, which in turn increases the local hemoglobin concentration. Although neuronal metabolism extracts more oxygen in active areas of the brain, the net effect of neuronal activity is to increase the local amount of oxygenated hemoglobin. This change can be detected nearly in real-time with the T2 sequence, which thus detects the functionally active brain regions. This process is the basis for the technique of fMRI.

What fMRI detects is not brain activity per se, but blood flow. The volume of the brain in which blood flow increases exceeds the volume of activated neurons by about 1 to 2 cm and limits the resolution of the technique. Sensitivity and resolution can be improved with the use of nontoxic, ultrasmall iron oxide particles. Thus, two tasks that activate clusters of neurons 5 mm apart, such as recognizing two different faces, yield overlapping signals on fMRI and so are usually indistinguishable by this technique. fMRI is useful to localize neuronal activity to a particular lobe or subcortical nucleus and has even been able to localize activity to a single gyrus. The method detects tissue perfusion, not neuronal metabolism. In contrast, PET scanning may give information specifically about neuronal metabolism.

No radioactive isotopes are administered in fMRI, a great advantage over PET and SPECT. A subject can perform a variety of tasks, both experimental and control, in the same imaging session. First, a routine T1 MRI image is obtained; then, the T2 images are superimposed to allow more precise localization. Acquisition of sufficient images for study can require 20 minutes to 3 hours, during which time the subject’s head must remain in exactly the same position. Several methods, including a frame around the head and a special mouthpiece, have been used. Although realignments of images can correct for some head movement, small changes in head position may lead to erroneous interpretations of brain activation.

fMRI has recently revealed unexpected details about the organization of language within the brain. Using a series of language tasks requiring semantic, phonemic, and rhyming discrimination, one study found that rhyming (but not other types of language processing) produced a different pattern of activation in men and women. Rhyming activated the inferior frontal gyrus bilaterally in women, but only on the left in men. In another study, fMRI revealed a previously suspected, but unproved, neural circuit for lexical categories, interpolated between the representations for concepts and those for phonemes. This novel circuit was located in the left anterior temporal lobe. Data from patients with dyslexia (reading disorder) doing simple rhyming tasks demonstrated a failure to activate Wernicke area and the insula, which were active in normal subjects doing the same task.

Sensory functions have also been mapped in detail with fMRI. The activation of the visual and auditory cortices has been visualized in real-time. In a recent intriguing study, the areas that were activated while a subject with schizophrenia listened to speech were also activated during auditory hallucinations. These areas included the primary auditory cortex as well as higher-order auditory processing regions. fMRI is the imaging technique most widely used to study brain abnormality related to cognitive dysfunction.

FMRI OF DEMENTIA. fMRI methods provide information that can potentially be used in the study, diagnosis, and prognosis of Alzheimer disease and other forms of dementia as well as proving insights into normal age-related changes in cognitive processing. Evidence that aging is associated with weaker and more diffused activations as well as decreased hemispheric lateralization, suggests either compensation for lost regional intensity or dedifferentiation of processing. The weaker activation, especially prefrontal, suggests potential encoding-stage dysfunctions associated with aging. fMRI studies have consistently demonstrated that patients with Alzheimer disease have decreased fMRI activation in the hippocampus and related structures within the MTL during the encoding of new memories compared with cognitively intact older subjects. More recently, fMRI studies of subjects at risk for Alzheimer disease, by their genetics or evidence of minimal cognitive impairment, have yielded variable results with some studies suggesting there may be a phase of paradoxically increased activation early in the course of prodromal Alzheimer disease.

FMRI OF ALCOHOL DEPENDENCE. fMRI studies have provided insights into the functional consequences of alcoholism-related neurotoxicity. Studies suggest that recovering alcohol-dependent patients show abnormal activation patterns in the frontal cortex, thalamus, striatum, cerebellum, and hippocampus related to impairments in attention, learning and memory, motor coordination, and inhibitory control of behavior. Studies have begun to explore pharmacologic modulation of resting circuit activity to probe mechanisms underlying circuit dysfunction in alcoholism, illustrated by blunted responses to benzodiazepines.

SPECT Scanning. Manufactured radioactive compounds are used in SPECT to study regional differences in cerebral blood flow within the brain. This high-resolution imaging technique records the pattern of photon emission from the bloodstream according to the level of perfusion in different regions of the brain. As with fMRI, it provides information on the cerebral blood flow, which is highly correlated with the rate of glucose metabolism, but does not measure neuronal metabolism directly.

SPECT uses compounds labeled with single photon-emitting isotopes: iodine-123, technetium-99m, and xenon-133. Xenon-133 is a noble gas that is inhaled directly. The xenon quickly enters the blood and is distributed to areas of the brain as a function of regional blood flow. Xenon-SPECT is thus referred to as the *regional cerebral blood flow* (rCBF) *technique.* For technical reasons, xenon-SPECT can measure blood flow only on the surface of the brain, which is an important limitation. Many mental tasks require communication between the cortex and subcortical structures, and this activity is poorly measured by xenon-SPECT.

Assessment of blood flow over the whole brain with SPECT requires the injectable tracers, technetium-99m-D,L-hexamethylpropyleneamine oxime (HMPAO [Ceretec]) or iodoamphetamine (Spectamine). These isotopes are attached to molecules that are highly lipophilic and rapidly cross the blood–brain barrier and enter cells. Once inside the cell, the ligands are enzymatically converted to charged ions, which remain trapped in the cell. Thus, over time, the tracers are concentrated in areas of relatively higher blood flow. Although blood flow is usually assumed to be the major variable tested in HMPAO SPECT, local variations in the permeability of the blood–brain barrier and in the enzymatic conversion of the ligands within cells also

contribute to regional differences in signal levels.

Also, to these compounds used for measuring blood flow, iodine-123–labeled ligands for the muscarinic, dopaminergic, and serotonergic receptors, for example, can be used to study these receptors by SPECT technology. Once photon-emitting compounds reach the brain, detectors surrounding the patient's head pick up their light emissions. This information is relayed to a computer, which constructs a two-dimensional image of the isotope's distribution within a slice of the brain. A critical difference between SPECT and PET is that in SPECT a single particle is emitted, whereas in PET two particles are emitted; the latter reaction gives a more precise location for the event and better resolution of the image. Increasingly, for both SPECT and PET studies, investigators are performing prestudy MRI or CT studies, then superimposing the SPECT or PET image on the MRI or CT image to obtain a more accurate anatomical location for the functional information. SPECT is useful in diagnosing decreased or blocked cerebral blood flow in stroke victims. Some have described abnormal flow patterns in the early stages of Alzheimer disease that may aid in early diagnosis.

PET Scanning. The isotopes used in PET decay by emitting positrons, antimatter particles that bind with and annihilate electrons, thereby giving off photons that travel in 180-degree opposite directions. Because detectors have twice as much signal from which to generate an image as SPECT scanners have, the resolution of the PET image is higher than with SPECT. A wide range of compounds can be used in PET studies, and the resolution of PET continues to be refined closer to its theoretical minimum of 3 mm, which is the distance positrons move before colliding with an electron. Relatively few PET scanners are available because they require an onsite cyclotron to make the isotopes.

The most commonly used isotopes in PET are fluorine-18 (^{18}F), nitrogen-13, and oxygen-15. These isotopes are usually linked to another molecule, except in the case of oxygen-15 (^{15}O). The most commonly reported ligand has been [^{18}F]fluorodeoxyglucose (FDG), an analog of glucose that the brain cannot metabolize. Thus, the brain regions with the highest metabolic rate and the highest blood flow take up the most FDG but cannot metabolize and excrete the usual metabolic products. The concentration of ^{18}F builds up in these neurons and is detected by the PET camera. Water-15 ($H_2{}^{15}O$) and

nitrogen-13 are used to measure blood flow, and ^{15}O can be used to determine the metabolic rate. Glucose is by far the predominant energy source available to brain cells, and its use is thus a highly sensitive indicator of the rate of brain metabolism. ^{18}F-labeled 3,4-dihydroxyphenylalanine (DOPA), the fluorinated precursor to dopamine, has been used to localize dopaminergic neurons.

PET has been used increasingly to study normal brain development and function as well as to study neuropsychiatric disorders. Concerning brain development, PET studies have found that glucose use is greatest in the sensorimotor cortex, thalamus, brainstem, and cerebellar vermis when an infant is 5 weeks of age or younger. By 3 months of age, most areas of the cortex show increased use, except for the frontal and association cortices, which do not begin to exhibit an increase until the infant is 8 months of age. An adult pattern of glucose metabolism is achieved by the age of 1 year, but use in the cortex continues to rise above adult levels until the child is about 9 years of age when use in the cortex begins to decrease and reaches its final adult level in the late teen years.

In another study, subjects listened to a rapidly presented list of thematically related words. When asked to recall words in the thematic category that may or may not have been on the list, some subjects falsely recalled that they had heard words that were not on the list. By PET scanning, the hippocampus was active during both true and false recollections, whereas the auditory cortex was only active during recollection of words that were heard. When pressed to determine whether memories were true or false, subjects activated the frontal lobes. FDG studies have also investigated pathology in neurologic disorders and psychiatric disorders. Two other types of studies used precursor molecules and receptor ligands. The dopamine precursor dopa has been used to visualize pathology in patients with Parkinson disease, and radiolabeled ligands for receptors have been useful in determining the occupancy of receptors by specific psychotherapeutic drugs. Neurochemical findings from PET radiotracer scan are listed in Table 1-25.

For example, dopamine receptor antagonists such as haloperidol block almost 100 percent of D_2 receptors. The atypical antipsychotic drugs block serotonin 5-HT_2 receptors also to D_2 receptors; hence, they are referred to as *serotonin-dopamine receptor antagonists.* The case study presented

illustrates the potential diagnostic value of three-dimensional PET imaging.

Patient A is a 70-year-old man who had gotten more forgetful, to the point that his family was worried about him. The patient's family was interested in getting a diagnostic workup to evaluate the possible causes of his memory disorder. His PET scan showed that he had functional parietotemporal decrease, which corroborated other neurologic evaluations, suggesting that he had Alzheimer disease. The patient was treated with tacrine (Cognex) and benefited from some stabilization of his symptoms. (Courtesy of Joseph C. Wu, M.D., Daniel G. Amen, M.D., and H. Stefan Bracha, M.D.)

Pharmacologic and Neuropsychological Probes. With both PET and SPECT and eventually, with MRS, more studies and possibly more diagnostic procedures will use pharmacologic and neuropsychological probes. The purpose of such probes is to stimulate particular regions of brain activity, so that, when compared with a baseline, workers can reach conclusions about the functional correspondence to particular brain regions. One example of the approach is the use of PET to detect regions of the brain involved in the processing of shape, color, and velocity in the visual system. Another example is the use of cognitive activation tasks (e.g., the Wisconsin Card Sorting Test) to study frontal blood flow in patients with schizophrenia. A key consideration in the evaluation of reports that measure blood flow is the establishment of actual baseline value in the study design. Typically, the reports use an awake, resting state, but there is variability in whether the patients have their eyes closed or their ears blocked; both conditions can affect brain function. There is also variability in such baseline brain function factors as sex, age, anxiety about the test, nonpsychiatric drug treatment, vasoactive medications, and time of day.

1.2 Children and Adolescents

A comprehensive evaluation of a child is composed of interviews with the

parents, the child, and other family members, gathering information regarding the child's current school functioning; and often, a standardized assessment of the child's intellectual level and academic achievement. In some cases, standardized measures of developmental level and neuropsychological assessments are useful. The child rarely initiates psychiatric evaluations, so we must obtain information from the family and the school to understand the reasons for the evaluation. In some cases, the court or a child protective service agency may initiate a psychiatric evaluation. Children can be excellent informants about symptoms related to mood and inner experiences, such as psychotic phenomena, sadness, fears, and anxiety. However, they often have difficulty with the chronology of symptoms and are sometimes reticent about reporting behaviors that have gotten them into trouble. Very young children often cannot articulate their experiences verbally and do better, showing their feelings and preoccupations in a play situation. Assessment of a child or adolescent includes identifying the reasons for referral; assessing the nature and extent of the child's psychological and behavioral difficulties; and determining family, school, social, and developmental factors that may be influencing the child's emotional well-being.



Table 1-25
Neurochemical Findings from Positron Emission Tomography Radiotracer Scans

Dopamine	Decreased uptake of dopamine in striatum in parkinsonian patients
	Dopamine release higher in patients with schizophrenia than in controls
	High dopamine release associated with positive symptoms in schizophrenia
Receptors	
• D_1 receptor	Lower D_1 receptor binding in prefrontal cortex of patients with schizophrenia compared with controls; correlates with negative symptoms

• D_2 receptor	Schizophrenia associated with small elevations of binding at D_2 receptor
• Serotonin type 1A (5-HT_{1A})	Reduction in receptor binding in patients with unipolar major depression
Transporters	
• Dopamine	Amphetamine and cocaine cause an increase in dopamine
	Tourette disorder shows increase in dopamine transporter system (may account for success of dopamine blocking therapies)
• Serotonin	Serotonin binding is low in depression, alcoholism, cocainism, binge eating, and impulse control
Metabolism	
• Nicotine	Cigarette smoking inhibits monamine oxidase activity in brain
• Amyloid-β Deposits	Can be visualized in vivo with positron emission tomography
Pharmacology	Plasma levels of cocaine peak at 2 min
	D_2 receptor occupancy lasts for several weeks after discontinuation of antipsychotic medication
	D_2 receptor occupancy is lower for atypical antipsychotics than typical antipsychotics (may account for decrease in extrapyramidal side effects)
	Low doses (10–20 mg) of selective serotonin reuptake inhibitors cause occupancy of up to 90% of serotonin receptors

The first step in the comprehensive evaluation of a child or adolescent is to obtain a full description of the current concerns and a history of the child's previous psychiatric and medical problems. This evaluation is often done with the parents for school-age children, whereas we may choose to see adolescents first, to get their perception of the situation. A direct interview and observation of the child are usually next, followed by psychological testing when indicated.

Clinical interviews offer the most flexibility in understanding the evolution of problems and in establishing the role of environmental factors and life events, but they may not systematically cover all psychiatric diagnostic categories. To increase the breadth of information generated, we may use semistructured interviews such as the *Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children* (K-SADS); structured interviews such as the *National Institute for Mental Health Diagnostic Interview Schedule for Children Version IV* (NIMH DISC-IV); and rating scales, such as the *Child Behavior Checklist* and *Connors Parent or Teacher Rating Scale for ADHD.*

It is not uncommon for interviews from different sources, such as parents, teachers, and school counselors, to reflect different or even contradictory information about a given child. When faced with conflicting information, we must determine whether apparent contradictions reflect an accurate picture of the child in different settings. Once obtaining a complete history from the parents, we should examine the child and assess the child's current functioning at home and school. Once we complete the evaluation and psychological testing, we can use all the available information to make a best-estimate diagnosis and can then make recommendations.

Once we obtain clinical information about a given child or adolescent, it is our task to determine whether that child meets the criteria for one or more psychiatric disorders according to the Fifth Edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5). Whereas clinical situations requiring intervention do not always fall within the context of a given psychiatric disorder, the importance of identifying psychiatric disorders when they arise is to facilitate meaningful investigation of childhood psychopathology.

CLINICAL INTERVIEWS

To conduct a useful child interview, we must be familiar with typical development to place the child's responses in the proper perspective. For example, a young child's discomfort on separation from a parent and a school-age child's lack of clarity about the purpose of the interview is perfectly normal. Furthermore, behavior that is normal in a child at one age, such as temper tantrums in a 2-year-old, takes on a different meaning, for example, in a teenager. Chapter 32 presents a detailed discussion of healthy childhood development.

The interviewer's first task is to engage the child and develop a rapport so that the child is comfortable. The interviewer should inquire about the child's concept of the purpose of the interview and should ask what the parents have told the child. If the child appears to be confused about the reason for the interview, the examiner may opt to summarize the parents' concerns in a developmentally appropriate and supportive manner. During the interview with the child, we seek to learn about the child's relationships with family members and peers, academic achievement, and peer relationships in school, and the child's pleasurable activities. An estimate of the child's cognitive functioning is a part of the mental status examination.

The extent of confidentiality in child assessment correlates with the age of the child. In most cases, almost all specific information can appropriately be shared with the parents of a very young child, whereas the privacy and permission of an older child or adolescent are necessary before we can share any information with parents. We should inform school-age and older children that if we become concerned that any child is dangerous to himself or herself or others, we must share this information with parents and, at times, additional adults. As part of our assessment, we must determine whether that child is safe in his or her environment and must develop an index of suspicion about whether the child is a victim of abuse or neglect. Whenever there is a suspicion of child maltreatment, we are required to inform the local child protective service agency.

Toward the end of the interview, the child may be asked in an open-ended manner whether he or she would like to bring up anything else. We should complement each child for his or her cooperation and thank them for participating in the interview, and the interview should end on a positive

note.

Infants and Young Children

Assessments of infants usually begin with the parents present, because very young children may be frightened by the interview situation; the interview with the parents present also allows us to assess the parent–infant interaction. Infants may be referred for a variety of reasons, including high levels of irritability, difficulty being consoled, eating disturbances, poor weight gain, sleep disturbances, withdrawn behavior, lack of engagement in play, and developmental delay. We should assess areas of functioning that include motor development, activity level, verbal communication, ability to engage in play, problem-solving skills, adaptation to daily routines, relationships, and social responsiveness.

We should use our observations during the interview and well as standardized developmental measures to determine a child's developmental level of functioning. Observations of play reveal a child's developmental level and reflect the child's emotional state and preoccupations. The examiner can interact with an infant age 18 months or younger in a playful manner by using such games as peek-a-boo. We may observe children between the ages of 18 months and 3 years in a playroom. Children ages 2 years or older may exhibit symbolic play with toys, revealing more in this mode than through conversation. The use of puppets and dolls with children younger than 6 years of age is often an effective way to elicit information, particularly if questions are directed to the dolls, rather than to the child.

School-Age Children

Some school-age children are at ease when conversing with an adult; others may show fear, anxiety, poor verbal skills, or oppositional behavior. School-age children can usually tolerate a 45-minute session. The room should be sufficiently spacious for the child to move around, but not so large as to reduce intimate contact between the examiner and the child. We may reserve part of the interview for unstructured play, and various toys can be made available to capture the child's interest and to elicit themes and feelings. Children in lower grades may be more interested in the toys in the room, whereas by the sixth grade, children may be more comfortable with the

interview process and less likely to show spontaneous play.

The first part of the interview explores the child's understanding of the reasons for the meeting. We should reassure the child that they are not here because they are "in trouble." Techniques that can facilitate disclosure of feelings include asking the child to draw peers, family members, a house, or anything else that comes to mind. We can then question the child about the drawings. We may ask children to reveal three wishes, to describe the best and worst events of their lives, and to name a favorite person to be stranded with on a desert island. Games such as Donald W. Winnicott's "squiggle," in which the examiner draws a curved line and then the child and the examiner take turns continuing the drawing, may facilitate conversation.

Questions that are partially open-ended with some multiple choices may elicit the best answers from school-age children. Simple, closed (yes or no) questions may not elicit sufficient information, and entirely open-ended questions can overwhelm a school-age child who cannot construct a chronologic narrative. These techniques often result in a shoulder shrug from the child. The use of indirect commentaries—such as, "I once knew a child who felt very sad when he moved away from all his friends"—is helpful, although we must be careful not to lead the child into confirming what the child thinks we want to hear. School-age children respond well to clinicians who help them compare moods or feelings by asking them to rate feelings on a scale of 1 to 10.

Adolescents

Adolescents usually have distinct ideas about why the evaluation was initiated, and can usually give a chronologic account of the recent events leading to the evaluation, although some may disagree with the need for the evaluation. We should communicate the value of hearing the story from an adolescent's point of view and must be careful to reserve judgment and not assign blame. Adolescents may be concerned about confidentiality, and clinicians can assure them that permission will be requested from them before we share any specific information with parents, except in situations involving danger to the adolescent or others, in which case we must sacrifice confidentiality. We can approach adolescents in an open-ended manner; however, when silences occur during the interview, we should attempt to

reengage the patient. We can explore what the adolescent believes the outcome of the evaluation will be (change of school, hospitalization, removal from the home, removal of privileges).

Some adolescents approach the interview with apprehension or hostility but open up when it becomes evident that we are neither punitive nor judgmental. We must be aware of our responses to adolescents' behavior (countertransference) and stay focused on the therapeutic process even in the face of defiant, angry, or difficult teenagers. We should set appropriate limits and should postpone or discontinue an interview if they feel threatened or if they become destructive or self-injurious. Every interview should include an exploration of suicidal thoughts, assaultive behavior, psychotic symptoms, substance use, and knowledge of safe sexual practices along with a sexual history. Once we establish rapport, many adolescents appreciate the opportunity to tell their side of the story and may reveal things that they have not disclosed to anyone else.

Family Interview

An interview with parents and the patient may take place first or may occur later in the evaluation. Sometimes, an interview with the entire family, including siblings, can be enlightening. During it, we can observe the interactions between the parents and children. Our job is to maintain a nonthreatening atmosphere in which each member of the family can speak freely without feeling that we are taking sides with any particular member. Although child psychiatrists generally function as advocates for the child, we must validate each family member's feelings in this setting, because lack of communication often contributes to the patient's problems.

Parents

The interview with the patient's parents or caretakers is necessary to get a chronologic picture of the child's growth and development. We should obtain a thorough developmental history and details of any stressors or significant events that have influenced the child's development. We should also elicit the parents' view of the family dynamics, their marital history, and their emotional adjustment. The family's psychiatric history and the upbringing of the parents are pertinent. Parents are usually the best informants about the

child's early development and previous psychiatric and medical illnesses. They may be better able to provide an accurate chronology of past evaluations and treatment. In some cases, especially with older children and adolescents, the parents may be unaware of significant current symptoms or social difficulties of the child. Clinicians elicit the parents' formulation of the causes and nature of their child's problems and ask about their expectations for the current assessment.

DIAGNOSTIC INSTRUMENTS

The two main types of diagnostic instruments used by clinicians are diagnostic interviews and questionnaires. We administer diagnostic interviews to either children or their parents, which should elicit sufficient information on various aspects of functioning in order to determine whether the child meets DSM-5 criteria.

Semistructured interviews, or "interviewer-based" interviews, such as K-SADS and the *Child and Adolescent Psychiatric Assessment* (CAPA), serve as guides. They help us clarify answers to questions about symptoms. Structured interviews, or "respondent-based" interviews, such as NIMH DISC-IV, the *Children's Interview for Psychiatric Syndromes* (ChIPS), and the *Diagnostic Interview for Children and Adolescents* (DICA), provide a script for the interviewer without interpretation of the patient responses during the interview process. Two other diagnostic instruments, the *Dominic-R* and the *Pictorial Instrument for Children and Adolescents* (PICA-III-R), use pictures as cues along with an accompanying question to elicit information about symptoms, which can be especially useful for young children as well as for adolescents.

Diagnostic instruments systematically aid the collection of information. Diagnostic instruments, even the most comprehensive, however, cannot replace clinical interviews because clinical interviews are superior in understanding the chronology of symptoms, the interplay between environmental stressors and emotional responses, and developmental issues. Clinicians often find it helpful to combine data from diagnostic instruments with clinical material gathered in a comprehensive evaluation.

Questionnaires can cover a broad range of symptom areas, such as the *Achenbach Child Behavior Checklist*, or they can focus on a particular type

of symptomatology, such as the *Connors Parent Rating Scale for ADHD.*

Semistructured Diagnostic Interviews

Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS). The K-SADS is intended for children and adolescents from 6 to 18 years of age. It contains multiple items with some space for further clarification of symptoms. It elicits information on current diagnosis and symptoms present in the previous year. Another version can also ascertain lifetime diagnoses. This instrument has been used extensively, especially in the evaluation of mood disorders, and includes measures of impairment caused by symptoms. There is also a form for parents as well as the child. The schedule takes about 1 to 1.5 hours to administer. The interviewer should have some training in the field of child psychiatry, but need not be a psychiatrist.

Child and Adolescent Psychiatric Assessment (CAPA). The CAPA is an "interviewer-based" instrument that can be used for children from 9 to 17 years of age. It comes in modular form so that we can focus on particular diagnostic entities without having to give the entire interview. It covers disruptive behavior disorders, mood disorders, anxiety disorders, eating disorders, sleep disorders, elimination disorders, substance use disorders, tic disorders, schizophrenia, PTSD, and somatization symptoms. It focuses on the 3 months before the interview, called the "primary period." In general, it takes about 1 hour to administer. It has a glossary to help clarify symptoms, and it provides separate ratings for the presence and severity of symptoms. The CAPA can be used to obtain information that applies to DSM-5 diagnoses. Training is necessary to administer this interview, and the interviewer must be prepared to use some clinical judgment in interpreting elicited symptoms.

Structured Diagnostic Interviews

National Institute of Mental Health Interview Schedule for Children Version IV (NIMH DISC-IV). The NIMH DISC-IV is a highly structured interview designed to assess more than 30 DSM-IV diagnostic

entities administered by trained “laypersons.” Although developed for DSM-IV, it can still be helpful diagnoses in DSM-5. It is available in parallel child and parent forms. The parent form can be used for children from 6 to 17 years of age, and the direct child form of the instrument was designed for children from 9 to 17 years of age. A computer scoring algorithm is available. This instrument assesses for diagnoses present within the last 4 weeks, and also within the last year. Because it is a fully structured interview, the instructions serve as a complete guide for the questions, and the examiner need not have any knowledge of child psychiatry to administer the interview correctly.

Children’s Interview for Psychiatric Syndromes (ChIPS). The ChIPS is a highly structured interview designed for use by trained interviewers with children from 6 to 18 years of age. It is composed of 15 sections, and it elicits information on psychiatric symptoms as well as psychosocial stressors targeting 20 psychiatric disorders, according to DSM-IV criteria; however, it also applies to diagnoses in DSM-5. There are parent and child forms. It takes approximately 40 minutes to administer the ChIPS. Diagnoses covered include depression, mania, ADHD, separation disorder, OCD, conduct disorder, substance use disorder, anorexia, and bulimia. The ChIPS is a screening instrument for clinicians and a diagnostic instrument for clinical and epidemiologic research.

Diagnostic Interview for Children and Adolescents (DICA). The current version of the DICA was developed in 1997 to assess information resulting in diagnoses according to either DSM-IV or DSM-III-R. This instrument is relevant to DSM-5 as well. Although designed as a highly structured interview, it can be used in a semistructured format. This means that, although interviewers are allowed to use additional questions and probes to clarify elicited information, the method of probing is standardized so that all interviewers will follow a specific pattern. When using the interview with younger children, more flexibility is built-in, allowing interviewers to deviate from written questions to ensure that the child understands the question. Parent and child interviews are expected to be used. The DICA is designed for use with children 6 to 17 years of age and generally takes 1 to 2 hours to administer. It covers externalizing behavior disorders, anxiety disorders,

depressive disorders, and substance abuse disorders, among others.

Pictorial Diagnostic Instruments

Dominic-R. The Dominic-R is a pictorial, fully structured interview designed to elicit psychiatric symptoms from children 6 to 11 years of age. The pictures illustrate abstract emotional and behavioral content of diagnostic entities according to the DSM-III-R; however, information gleaned from this instrument also applies in conjunction with clinical information to the DSM-5. The instrument uses a picture of a child called "Dominic," who is experiencing the symptom in question. Some symptoms have more than one picture, with a brief story. Each picture includes a question about the situation depicted and whether the child has experienced something similar. Diagnostic entities covered by the Dominic-R include separation anxiety, generalized anxiety, depression and dysthymia, ADHD, oppositional defiant disorder, conduct disorder, and specific phobia. The symptom list is comprehensive; however, the interview does not explicitly ask about the frequency, duration, or age of onset of the symptom. The paper version of this interview takes about 20 minutes, and the computerized version of this instrument takes about 15 minutes. Trained lay-interviewers can administer this interview. Computerized versions of this interview are available with pictures of a child who is white, black, Latino, or Asian.

Pictorial Instrument for Children and Adolescents (PICA-III-R). PICA-III-R is composed of 137 pictures organized in modules and designed to cover five diagnostic categories, including disorders of anxiety, mood, psychosis, disruptive disorders, and substance use disorder. It is designed to be administered by clinicians and can be used for children and adolescents ranging from 6 to 16 years of age. The PICA-III_R provides a categorical (diagnosis present or absent) and a dimensional (range of severity) assessment. This instrument presents pictures of a child experiencing emotional, behavioral, and cognitive symptoms. The clinician asks the child, "How much are you like him/her?" and a five-point rating scale with pictures of a person with open arms in increasing degrees is shown to the child to help the child identify the severity of the symptoms. It takes about 40 minutes to 1 hour to administer the interview. This instrument was designed for DSM-III-

R but is useful for DSM-5 diagnoses. This assessment can be used to aid in clinical interviews and research diagnostic protocols.

QUESTIONNAIRES AND RATING SCALES

Achenbach Child Behavior Checklist

The parent and teacher versions of the *Achenbach Child Behavior Checklist* cover a broad range of symptoms and several positive attributes related to academic and social competence. The checklist presents items related to mood, frustration tolerance, hyperactivity, oppositional behavior, anxiety, and various other behaviors. The parent version consists of 118 items to be rated 0 (not true), 1 (sometimes true), or 2 (very true). The teacher version is similar, but without the items that apply only to home life. Profiles were developed based on normal children of three different age groups (4 to 5, 6 to 11, and 12 to 16).

Such a checklist identifies specific problem areas that we might otherwise overlook, and it may point out areas in which the child's behavior deviates from that of normal children of the same age group. The checklist does not make diagnoses.

Revised Achenbach Behavior Problem Checklist

Consisting of 150 items that cover a variety of childhood behavioral and emotional symptoms, the *Revised Achenbach Behavior Problem Checklist* discriminates between clinic-referred and nonreferred children. Separate subscales correlate in the appropriate direction with other measures of intelligence, academic achievement, clinical observations, and peer popularity. As with the other broad rating scales, this instrument can help elicit a comprehensive view of a multitude of behavioral areas, but it does not make psychiatric diagnoses.

Connors Abbreviated Parent–Teacher Rating Scale for ADHD

In its original form, the *Connors Abbreviated Parent–Teacher Rating Scale for ADHD* consisted of 93 items rated on a 0 to 3 scale, subgrouped into 25 clusters, including problems with restlessness, temper, school, stealing, eating, and sleeping. Over the years, multiple versions of this scale were

developed and used to aid in the systematic identification of children with ADHD. A highly abbreviated form of this rating scale, the *Connors Abbreviated Parent–Teacher Questionnaire*, was developed in 1973. It consists of 10 items that assess both hyperactivity and inattention.

Brief Impairment Scale

A newly validated 23-item instrument suitable to obtain information on children ranging from 4 to 17 years, the *Brief Impairment Scale* (BIS) evaluates three domains of functioning: interpersonal relations, school/work functioning, and care/self-fulfillment. This scale is administered to an adult informant about his or her child, does not take long to administer, and provides a global measure of impairment along the above three dimensions. This scale does not make clinical decisions on individual patients, but it can provide information on the degree of impairment that a given child is experiencing in a particular area.

COMPONENTS OF THE CHILD PSYCHIATRIC EVALUATION

Psychiatric evaluation of a child includes a description of the reason for the referral, the child's past and present functioning, and any test results. Table 1-26 gives an outline of the evaluation.

Identifying Data

Identifying data for a child includes the child's gender, age, as well as the family constellation surrounding the child.



Table 1-26
Child Psychiatric Evaluation

Identifying data
Identified patient and family members
Source of referral
Informants
History

Chief complaint
History of present illness
Developmental history and milestones
Psychiatric history
Medical history, including immunizations
Family social history and parents' marital status
Educational history and current school functioning
Peer relationship history
Current family functioning
Family psychiatric and medical histories
Current physical examination
Mental status examination
Neuropsychiatric examination (when applicable)
Developmental, psychological, and educational testing
Formulation and summary
DSM-5 diagnosis
Recommendations and treatment plan

History

A comprehensive history contains information about the child's current and past functioning from the child's report, from clinical and structured interviews with the parents, and information from teachers and previous treating clinicians. We usually obtain the chief complaint and the history of the present illness from both the child and the parents. Naturally, the child will articulate the situation according to his or her developmental level. The parents are usually the best source for determining the child's developmental history. Past clinicians can help augment the psychiatric and medical histories. The child's report is critical for understanding the current situation regarding peer relationships and adjustment to school. Adolescents are the best informants regarding knowledge of safe sexual practices, drug or alcohol use, and suicidal ideation. The parents are usually the best source to understand the family's psychiatric and social histories and family functioning.

Mental Status Examination

We should obtain a detailed description of the child's current mental functioning through observation and specific questioning. Table 1-27 presents an outline of the mental status examination, and Table 1-28 lists components of a comprehensive neuropsychiatry mental status.

Physical Appearance. The examiner should document the child's size, grooming, nutritional state, bruising, head circumference, physical signs of anxiety, facial expressions, and mannerisms.

Parent–Child Interaction. Before the interview, the examiner can observe the interactions between parents and the child in the waiting area. How the parents and child interact, both verbally and emotionally, are pertinent.



Table 1-27
Mental Status Examination for Children

1. Physical appearance
2. Parent–child interaction
3. Separation and reunion
4. Orientation to time, place, and person
5. Speech and language
6. Mood
7. Affect
8. Thought process and content
9. Social relatedness
10. Motor behavior
11. Cognition
12. Memory
13. Judgment and insight

Separation and Reunion. The examiner should note how the child responds to separation from a parent, and how they react to the reunion. Either lack of affect at separation and reunion or severe distress on separation or reunion can indicate problems in the parent–child relationship or other psychiatric disturbances.

Orientation to Time, Place, and Person. Impairments in orientation can reflect neurologic damage, low intelligence, or a thought disorder. The age of the child must be kept in mind, however, because very young children may not know the date, other chronologic information, or the name of the interview site.

Speech and Language. The examiner should evaluate the child's speech and language acquisition. Is it appropriate for the child's age? A disparity between expressive language usage and receptive language is notable. The examiner should also note the child's rate of speech, rhythm, latency to answer, the spontaneity of speech, intonation, articulation of words, and prosody. Echolalia, repetitive stereotypical phrases and unusual syntax are significant psychiatric findings. Children who do not use words by age 18 months or who do not use phrases by ages 2.5 to 3 years, but who have a history of typical babbling and responding appropriately to nonverbal cues, are probably developing typically. The examiner should consider the possibility that a hearing loss is contributing to a speech and language deficit.

Mood. A child's sad expression, lack of appropriate smiling, tearfulness, anxiety, euphoria, and anger are valid indicators of mood, as are verbal admissions of feelings. Persistent themes in play and fantasy also reflect the child's mood.

Affect. The examiner should note the child's range of emotional expressivity, appropriateness of affect to thought content, ability to move smoothly from one affect to another, and sudden labile emotional shifts.

Thought Process and Content. In evaluating a thought disorder in a child, we must always consider what is typical for the child's age and what is deviant for any age group. The evaluation of thought-form considers loosening of associations, excessive magical thinking, perseveration, echolalia, the ability to distinguish fantasy from reality, sentence coherence, and the ability to reason logically. The evaluation of thought content considers delusions, obsessions, themes, fears, wishes, preoccupations, and interests.

Table 1-28
Neuropsychiatric Mental Status Examination[a]

A. General Description
   1. General appearance and dress
   2. Level of consciousness and arousal
   3. Attention to environment
   4. Posture (standing and seated)
   5. Gait
   6. Movements of limbs, trunk, and face (spontaneous, resting, and after instruction)
   7. General demeanor (including evidence of responses to internal stimuli)
   8. Response to examiner (eye contact, cooperation, ability to focus on interview process)
   9. Native or primary language

B. Language and Speech
   1. Comprehension (words, sentences, simple and complex commands, and concepts)
   2. Output (spontaneity, rate, fluency, melody or prosody, volume, coherence, vocabulary, paraphasic errors, complexity of usage)
   3. Repetition
   4. Other aspects
      a. Object naming
      b. Color naming
      c. Body part identification
      d. Ideomotor praxis to command

C. Thought
   1. Form (coherence and connectedness)
   2. Content
      a. Ideational (preoccupations, overvalued ideas, delusions)
      b. Perceptual (hallucinations)

D. Mood and Affect
   1. Internal mood state (spontaneous and elicited; sense of

humor)
2. Future outlook
3. Suicidal ideas and plans
4. Demonstrated emotional status (congruence with mood)

E. Insight and Judgment
1. Insight
   a. Self-appraisal and self-esteem
   b. Understanding of current circumstances
   c. Ability to describe personal psychological and physical status
2. Judgment
   a. Appraisal of major social relationships
   b. Understanding of personal roles and responsibilities

F. Cognition
1. Memory
   a. Spontaneous (as evidenced during interview)
   b. Tested (incidental, immediate repetition, delayed recall, cued recall, recognition; verbal, nonverbal; explicit, implicit)
2. Visuospatial skills
3. Constructional ability
4. Mathematics
5. Reading
6. Writing
7. Fine sensory function (stereognosis, graphesthesia, two-point discrimination)
8. Finger gnosis
9. Right–left orientation
10. “Executive functions”
11. Abstraction

[a]Questions should be adapted to the age of the child.

Courtesy of Eric D. Caine, M.D. and Jeffrey M. Lyness, M.D.

Suicidal ideation is always a part of the mental status examination for children who are sufficiently verbal to understand the questions and old

enough to understand the concept. Generally, children of average intelligence who are older than 4 years of age have some understanding of what is real and what is make-believe and may be asked about suicidal ideation, although a firm concept of the permanence of death may not be present until several years later.

We should also assess for aggressive thoughts and homicidal ideation. Perceptual disturbances, such as hallucinations, are also assessed. Very young children are expected to have short attention spans and may change the topic and conversation abruptly without exhibiting a symptomatic flight of ideas. Transient visual and auditory hallucinations in very young children do not necessarily represent major psychotic illnesses, but they do deserve further investigation.

Social Relatedness. We should assess the appropriateness of the child's response to the interviewer, general level of social skills, eye contact, and degree of familiarity or withdrawal in the interview process. Overly friendly or familiar behavior may be as troublesome as extremely retiring and withdrawn responses. The examiner assesses the child's self-esteem, general and specific areas of confidence, and success with family and peer relationships.

Motor Behavior. The motor behavior part of the mental status examination includes observations of the child's coordination and activity level and ability to pay attention and carry out developmentally appropriate tasks. It also involves involuntary movements, tremors, motor hyperactivity, and any unusual focal asymmetries of muscle movement.

Cognition. We also should assess the child's intellectual functioning and problem-solving abilities. We can estimate the relative level of intelligence from a child's general information, vocabulary, and comprehension. For a specific assessment of the child's cognitive abilities, we can use a standardized test.

Memory. School-age children should be able to remember three objects after 5 minutes and to repeat five digits forward and three digits backward. Anxiety can interfere with the child's performance, but an apparent inability

to repeat digits or to add simple numbers may reflect brain damage, mental retardation, or learning disabilities.

Judgment and Insight. The child's view of the problems, reactions to them, and suggested solutions may give us a good idea of the child's judgment and insight. Also, the child's understanding of what he or she can realistically do to help and what we can do adds to the assessment of the child's judgment.

Neuropsychiatric Assessment

A neuropsychiatric assessment is appropriate for children suspected of having a psychiatric disorder that coexists with neuropsychiatric impairment, or psychiatric symptoms that may be caused by neuropsychiatric dysfunction, or a neurologic disorder. Although a neuropsychiatric assessment is not sufficient in most cases to make a psychiatric diagnosis, neuropsychological profiles have been, in some cases, correlated with particular psychiatric symptoms and syndromes. For example, neuropsychological differences in executive function, language, and memory functions, as well as measures of mood and anxiety, have been found between youth with histories of childhood maltreatment and those without it. The neuropsychiatric evaluation combines information from neurologic, neuropsychological testing, and mental status examinations. The neurologic examination can identify asymmetrical abnormal signs (hard signs) that may indicate lesions in the brain. A physical examination can evaluate the presence of physical stigmata of particular syndromes in which neuropsychiatric symptoms or developmental aberrations play a role (e.g., fetal alcohol syndrome, Down syndrome).

A neuropsychiatric examination also includes neurologic soft signs and minor physical anomalies. Loretta Bender coined the term *neurological soft signs* in the 1940s to describe the nondiagnostic abnormalities in the neurologic examinations of children with schizophrenia. Soft signs do not indicate focal neurologic disorders, but they are associated with a wide variety of developmental disabilities and frequently occur in children with low intelligence, learning disabilities, and behavioral disturbances. Soft signs may refer to both behavioral symptoms (which are sometimes associated with

brain damage, such as severe impulsivity and hyperactivity), physical findings (including contralateral overflow movements), and a variety of nonfocal signs (e.g., mild choreiform movements, poor balance, mild incoordination, asymmetry of gait, nystagmus, and the persistence of infantile reflexes). Soft signs include those that are normal in a young child, but become abnormal when they persist in an older child and those that are abnormal at any age. The *Physical and Neurological Examination for Soft Signs* (PANESS) is an instrument used with children up to the age of 15 years. It consists of 15 questions about the general physical status and medical history and 43 physical tasks (e.g., touch your finger to your nose, hop on one foot to the end of the line, tap quickly with your finger). Neurologic soft signs are useful to note, but they are not useful in making a specific psychiatric diagnosis.

Minor physical anomalies or dysmorphic features occur with a higher than usual frequency in children with developmental disabilities, learning disabilities, speech and language disorders, and hyperactivity. As with soft signs, the documentation of minor physical anomalies is part of the neuropsychiatric assessment, but it is rarely helpful in the diagnostic process and does not imply a good or bad prognosis. Minor physical anomalies include a high-arched palate, epicanthal folds, hypertelorism, low-set ears, transverse palmar creases, multiple hair whorls, a large head, a furrowed tongue, and partial syndactyl of several toes.

When considering a seizure disorder in the differential diagnosis or a structural abnormality in the brain, electroencephalography (EEG), CT, or MRI may be indicated.

Developmental, Psychological, and Educational Testing

Psychological testing, structured developmental assessments, and achievement testing are valuable in evaluating a child's developmental level, intellectual functioning, and academic difficulties. A measure of adaptive functioning (including the child's competence in communication, daily living skills, socialization, and motor skills) is the most definitive way to determine the level of intellectual disability in a child. Table 1-29 outlines the general categories of psychological tests.

Development Tests for Infants and Preschoolers. The *Gesell Infant Scale*, the *Cattell Infant Intelligence Scale*, *Bayley Scales of Infant Development*, and the *Denver Developmental Screening Test* include developmental assessments of infants as young as 2 months of age. When used with very young infants, the tests focus on sensorimotor and social responses to a variety of objects and interactions. When using these instruments with older infants and preschoolers, the emphasis is on language acquisition. The *Gesell Infant Scale* measures development in four areas: motor, adaptive functioning, language, and social.

An infant's score on one of these developmental assessments is not a reliable way to predict a child's future IQ in most cases. Infant assessments are valuable, however, in detecting developmental deviation and mental retardation and in raising suspicions of a developmental disorder. Whereas infant assessments rely heavily on sensorimotor functions, intelligence testing in older children and adolescents includes later-developing functions, including verbal, social, and abstract cognitive abilities.

Intelligence Tests for School-Age Children and Adolescents. The most widely used test of intelligence for school-age children and adolescents is the third edition of the *Wechsler Intelligence Scale for Children* (WISC-III-R). It can be given to children from 6 to 17 years of age and yields a verbal IQ, a performance IQ, and a combined full-scale IQ. The verbal subtests consist of vocabulary, information, arithmetic, similarities, comprehension, and digit span (supplemental) categories. The performance subtests include block design, picture completion, picture arrangement, object assembly, coding, mazes (supplemental), and symbol search (supplemental). The IQ computation does not include the scores of the supplemental subtests.

Each subcategory is scored from 1 to 19, with 10 being the average score. An average full-scale IQ is 100; 70 to 80 represents borderline intellectual function; 80 to 90 is in the low average range; 90 to 109 is average; 110 to 119 is a high average, and above 120 is in the superior or very superior range. The multiple breakdowns of the performance and verbal subscales allow great flexibility in identifying specific areas of deficit and scatter in intellectual abilities. Because a large part of intelligence testing measures abilities used in academic settings, the breakdown of the WISC-III-R can also point out skills in which a child is weak and may benefit from remedial

education.

The *Stanford-Binet Intelligence Scale* covers an age range from 2 to 24 years. It relies on pictures, drawings, and objects for very young children and verbal performance for older children and adolescents. This intelligence scale, the earliest version of an intelligence test of its kind, leads to a mental age score as well as an IQ.

The *McCarthy Scales of Children's Abilities* and the *Kaufman Assessment Battery for Children* are two other intelligence tests that are available for preschool and school-age children. They do not cover adolescents.

LONG-TERM STABILITY OF INTELLIGENCE. Although a child's intelligence is relatively stable throughout the school-age years and adolescence, some factors can influence intelligence and a child's score on an intelligence test. The intellectual functions of children with severe mental illnesses and of those from deprived and neglectful environments may decrease over time, whereas the IQs of children with intensively enriched environments may increase over time. Factors that influence a child's score on a given test of intellectual functioning and, thus, affect the accuracy of the test are motivation, emotional state, anxiety, and cultural milieu. The interactions between cognitive ability and anxiety and depression and psychosis are complex. One study of 4,405 youth from the Canadian National Longitudinal Study of Children and Youth (NLSCY), by Weeks and colleagues (2013) found that more exceptional cognitive ability was associated with less risk for anxiety and depressive symptoms in youth from 12 to 13 years of age, however, by ages 14 to 15 years, cognitive ability did not affect the odds of anxiety or depression.



Table 1-29
Commonly Used Child and Adolescent Psychological Assessment Instruments

Test	Age/Grades	Data Generated and Comments
Intellectual Ability		
Wechsler Intelligence	6–16	Standard scores: verbal,

Scale for Children—Third Edition (WISC-III-R)		performance and full-scale IQ; scaled subtest scores permitting specific skill assessment.
Wechsler Adult Intelligence Scale—(WAIS-III)	16–adult	Same as WISC-III-R.
Wechsler Preschool and Primary Scale of Intelligence—Revised (WPPSI-R)	3–7	Same as WISC-III-R.
Kaufman Assessment Battery for Children (K-ABC)	2.6–12.6	Well-grounded in theories of cognitive psychology and neuropsychology. Allows immediate comparison of intellectual capacity with acquired knowledge. Scores: Mental Processing Composite (IQ equivalent); sequential and simultaneous processing and achievement standard scores; scaled mental processing and achievement subtest scores; age equivalents; percentiles.
Kaufman Adolescent and Adult Intelligence Test (KAIT)	11–85+	Composed of separate Crystallized and Fluid scales. Scores: Composite Intelligence Scale; Crystallized and Fluid IQ; scaled subtest scores; percentiles.

Stanford-Binet, 4th Edition (SB:FE)	2–23	Scores: IQ; verbal, abstract/visual, and quantitative reasoning; short-term memory; standard age.
Peabody Picture Vocabulary Test—III (PPVT-III)	4–adult	Measures receptive vocabulary acquisition; standard scores, percentiles, age equivalents.
Achievement		
Woodcock–Johnson Psycho-Educational Battery—Revised (W-J)	K–12	Scores: reading and mathematics (mechanics and comprehension), written language, other academic achievement; grade and age scores, standard scores, percentiles.
Wide Range Achievement Test—3, Levels 1 and 2 (WRAT-3)	Level 1: 1–5 Level 2: 12–75	Permits screening for deficits in reading, spelling, and arithmetic; grade levels, percentiles, stanines, standard scores.
Kaufman Test of Educational Achievement, Brief and Comprehensive Forms (K-TEA)	1–12	Standard scores: reading, mathematics, and spelling; grade and age equivalents, percentiles, stanines. Brief Form is sufficient for most clinical applications; Comprehensive Form

		allows error analysis and more detailed curriculum planning.
Wechsler Individual Achievement Test (WIAT)	K–12	Standard scores: basic reading, mathematics reasoning, spelling, reading comprehension, numerical operations, listening comprehension, oral expression, written expression. Co-normal with WISC-III-R.
Adaptive Behavior		
Vineland Adaptive Behavior Scales	Normal: 0–19 Retarded: All ages	Standard scores: adaptive behavior composite and communication, daily living skills, socialization, and motor domains; percentiles, age equivalents, developmental age scores. Separate standardization groups for normal, visually handicapped, hearing impaired, emotionally disturbed, and retarded.
Scales of Independent Behavior—Revised	Newborn–adult	Standard scores: four adaptive (motor, social interaction, communication, personal living, community living) and three maladaptive (internalized, asocial, and externalized) areas;

		General Maladaptive Index and Broad Independence cluster.
Attentional Capacity		
Trail Making Test	8–adult	Standard scores, standard deviations, ranges; corrections for age and education.
Wisconsin Card Sorting Test	6.6–adult	Standard scores, standard deviations, T-scores, percentiles, developmental norms for number of categories achieved, perseverative errors, and failures to maintain set; computer measures.
Behavior Assessment System for Children (BASC)	4–18	Teacher and parent rating scales and child self-report of personality permitting multireporter assessment across a variety of domains in home, school, and community. Provides validity, clinical, and adaptive scales. ADHD component avails.
Home Situations Questionnaire—Revised (HSQ-R)	6–12	Permits parents to rate child's specific problems with attention or concentration. Scores for number of problem settings, mean severity, and factor scores for

		compliance and leisure situations.
ADHD Rating Scale	6–12	Score for number of symptoms keyed to DSM cutoff for diagnosis of ADHD; standard scores permit derivation of clinical significance for total score and two factors (Inattentive-Hyperactive and Impulsive-Hyperactive).
School Situations Questionnaire (SSQ-R)	6–12	Permits teachers to rate a child's specific problems with attention or concentration. Scores for number of problem settings and mean severity.
Child Attention Profile (CAP)	6–12	Brief measure allowing teachers' weekly ratings of presence and degree of child's inattention and overactivity. Normative scores for inattention, overactivity, and total score.
Projective Tests		
Rorschach Inkblots	3–adult	Special scoring systems. Most recently developed and increasingly universally accepted is

		John Exner's Comprehensive System (1974). Assesses perceptual accuracy, integration of affective and intellectual functioning, reality testing, and other psychological processes.
Thematic Apperception Test (TAT)	6–adult	Generates stories that are analyzed qualitatively. Assumed to provide especially rich data regarding interpersonal functioning.
Machover Draw-A-Person Test (DAP)	3–adult	Qualitative analysis and hypothesis generation, especially regarding subject's feelings about self and significant others.
Kinetic Family Drawing (KFD)	3–adult	Qualitative analysis and hypothesis generation regarding an individual's perception of family structure and sentient environment. Some objective scoring systems in existence.
Rotter Incomplete Sentences Blank	Child, adolescent, and adult forms	Primarily qualitative analysis, although some objective scoring systems have been developed.
Personality Tests		

Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A)	14–18	1992 version of widely used personality measure, developed specifically for use with adolescents. Standard scores: 3 validity scales, 14 clinical scales, additional content and supplementary scales.
Millon Adolescent Personality Inventory (MAPI)	13–18	Standard scores for 20 scales grouped into three categories: Personality styles; expressed concerns; behavioral correlates. Normed on adolescent population. Focuses on broad functional spectrum, not just problem areas. Measures 14 primary personality traits, including emotional stability, self-concept level, excitability, and self-assurance.
Children's Personality Questionnaire	8–12	Generates combined broad trait patterns including extraversion and anxiety.
Neuropsychological Screening Tests and Test Batteries		
Developmental Test of Visual-Motor Integration (VMI)	2–16	Screening instrument for visual motor deficits. Standard scores, age equivalents, percentiles.

Benton Visual Retention Test	6–adult	Assesses presence of deficits in visual-figure memory. Mean scores by age.
Benton Visual Motor Gestalt Test	5–adult	Assesses visual-motor deficits and visual-figural retention. Age equivalents.
Reitan-Indiana Neuropsychological Test Battery for Children	5–8	Cognitive and perceptual-motor tests for children with suspected brain damage.
Halstead–Reitan Neuropsychological Test Battery for Older Children	9–14	Same as Reitan-Indiana.
Luria-Nebraska Neuropsychological Battery: Children's Revision LNNB:C	8–12	Sensory-motor, perceptual, cognitive tests measuring 11 clinical and 2 additional domains of neuropsychological functioning. Provides standard scores.
Developmental Status		
Bayley Scales of Infant Development-Second Edition	16 days–42 mo	Mental, motor, and behavior scales measuring infant, development. Provides standard scores.
Mullen Scales of Early Learning	Newborn–5 yr	Language and visual scales for receptive and expressive ability. Yields age scores and T scores.

Adapted from Racusin G, Moss N. Psychological assessment of children and adolescents. In: Lewis M, ed. *Child and Adolescent Psychiatry: A Comprehensive Textbook*. Philadelphia, PA: Williams Wilkins; 1991, with permission.

Perceptual and Perceptual Motor Tests. The *Bender Visual-Motor Gestalt Test* can be given to children between the ages of 4 and 12 years. The test consists of a set of spatially related figures that the child copies. The number of errors determines the score. Although not a diagnostic test, it is useful in identifying developmentally age-inappropriate perceptual performances.

Personality Tests. Personality tests are not of much use in making diagnoses, and they are less satisfactory than intelligence tests concerning norms, reliability, and validity, but they can help elicit themes and fantasies.

The Rorschach test is a projective technique in which ambiguous stimuli—a set of bilaterally symmetrical inkblots—are shown to a child, who then describes what he or she sees in each. The hypothesis is that the child's interpretation of the vague stimuli reflects the basic characteristics of their personality. The examiner notes the themes and patterns.

A more structured projective test is the *Children's Apperception Test* (CAT), which is an adaptation of the *Thematic Apperception Test* (TAT). The CAT consists of cards with pictures of animals in scenes that are somewhat ambiguous but are related to parent–child and sibling issues, caretaking, and other relationships. The child describes what is happening and tells a story about the scene. The choice to use animals for the pictures is because some believe that children might respond more readily to animal images than to human figures.

Drawings, toys, and play are also applications of projective techniques used to evaluate children. Dollhouses, dolls, and puppets have been especially helpful in allowing a child a nonconversational mode in which to express a variety of attitudes and feelings. Play materials that reflect household situations are likely to elicit a child's fears, hopes, and conflicts about the family.

Projective techniques have not fared well as standardized instruments. Rather than being considered tests, projective techniques are better used as additional clinical modalities.

Educational Tests. Achievement tests measure the attainment of knowledge and skills in a particular academic curriculum. The *Wide-Range Achievement Test-Revised* (WRAT-R) consists of tests of knowledge and skills and timed performances of reading, spelling, and mathematics. It is used with children from 5 years of age to adulthood. The test yields a score that we can compare to the average expected score for the child's chronologic age and grade level.

The *Peabody Individual Achievement Test* (PIAT) includes word identification, spelling, mathematics, and reading comprehension.

The *Kaufman Test of Educational Achievement,* the *Gray Oral Reading Test-Revised* (GORT-R), and the *Sequential Tests of Educational Progress* (STEP) are achievement tests that determine whether a child has achieved the educational level expected for his or her grade level. Often, children whose achievement is significantly lower than expected for their grade level in one or more subjects exhibit a specific learning disorder.

Biopsychosocial Formulation. Our task is to integrate all of the information obtained into a formulation that takes into account the biologic predisposition, psychodynamic factors, environmental stressors, and life events that have led to the child's current level of functioning. We must consider psychiatric disorders and any specific physical, neuromotor, or developmental abnormalities in the formulation of etiologic factors for current impairment. Our conclusions are an integration of clinical information along with data from standardized psychological and developmental assessments. The psychiatric formulation includes an assessment of family function as well as the appropriateness of the child's educational setting. We should then determine the child's overall safety in his or her current situation, and report any suspected maltreatment to the local child protective service agency. We then consider the child's overall well-being regarding growth, development, and academic and play activities.

Diagnosis

Structured and semistructured (evidence-based) assessment tools often enhance a clinician's ability to make the most accurate diagnoses. These instruments described earlier include the K-SADS, the CAPA, and the NIMH

DISC-IV interviews. The advantages of including an evidence-based instrument in the diagnostic process include decreasing potential clinician bias to make a diagnosis without all of the necessary symptoms information and serving as guides for us to consider each symptom that could contribute to a given diagnosis. These data can enable us to optimize his expertise to make challenging judgments regarding child and adolescent disorders, which may possess overlapping symptoms. Our ultimate task includes making all appropriate diagnoses according to the DSM-5. Some clinical situations do not fulfill the criteria for DSM-5 diagnoses, but cause impairment and require psychiatric attention and intervention. Clinicians who evaluate children are frequently in the position of determining the impact of the behavior of family members on the child's well-being. In many cases, a child's level of impairment is related to factors extending beyond a psychiatric diagnosis, such as the child's adjustment to his or her family life, peer relationships, and educational placement.

RECOMMENDATIONS AND TREATMENT PLAN

To recommend treatment, we should use our formulation, integrating all of the data gained during our assessments. Optimal treatment often flows from this formulation. That is, identification of a biologic predisposition to a particular psychiatric disorder may be clinically relevant to inform a psychopharmacologic recommendation. An understanding of the psychodynamic interactions between family members may lead a clinician to recommend treatment that includes a family component. Educational and academic problems addressed in the formulation may lead to a recommendation to seek a more effective academic placement. We should also take into account the overall social situation of the child or adolescent when recommending treatment. Of utmost important—and at the top of our recommendation list—is the physical and emotional safety of a child or adolescent.

The child or adolescent's family, school life, peer interactions, and social activities often have a direct impact on the child's success in overcoming his or her difficulties. The psychological education and cooperation of a child or adolescent's family are essential ingredients in the successful application of treatment recommendations. Patients and families often see communications

that balance the observed positive qualities of the child and family with the weak areas as more helpful than a focus only on the problem areas. Finally, the most successful treatment plans are those developed cooperatively between us, the child, and the family.

INTELLECTUAL DISABILITY PSYCHIATRIC INTERVIEW AND TESTS

Psychiatric Interview

A psychiatric interview of a child or adolescent with intellectual disability requires a high level of sensitivity in order to elicit information at the appropriate intellectual level while remaining respectful of the patient's age and emotional development. The patient's verbal abilities, including receptive and expressive language, can be initially screened by observing the communication between the caretakers and the patient. If the patient communicates mostly through gestures or sign language, the parents may serve as interpreters. Patients with milder forms of intellectual disability are often well aware of their differences from others and their failures and may be anxious and ashamed during the interview. Approaching patients with a clear, supportive, concrete explanation of the diagnostic process, particularly patients with sufficiently receptive language ability, may allay anxiety and fears. Providing support and praise in language appropriate to the patient's age and understanding is beneficial. Subtle direction, structure, and reinforcement may be necessary to keep patients focused on the task or topic.

In general, the psychiatric examination of an intellectually disabled child or adolescent should reveal how the patient has coped with stages of development. Frustration tolerance, impulse control, and over-aggressive motor and sexual behavior are essential areas of attention in the interview. It is equally crucial to elicit the patient's self-image, areas of self-confidence, and an assessment of tenacity, persistence, curiosity, and willingness to explore the environment.

Structured Instruments, Rating Scales, and Psychological Assessment

In children and adolescents who have acquired language, we usually use one of several standardized instruments that include numerous domains of cognitive function. For children ages 6 to 16 years, the Wechsler Intelligence Test for Children is most common, and for children ages 3 to 6 years, the most commonly used is the Wechsler Preschool and Primary Scale of

Intelligence-Revised. The Stanford-Binet Intelligence Scale, Fourth Edition, has the advantage that it can be administered to children even younger, starting at the age of 2 years. The Kaufman Assessment Battery for Children is appropriate for children ages 2½ to 12½ years, whereas the Kaufman Adolescent and Adult Intelligence Test apply to a wide range of ages, from 11 to 85 years. All of these standardized instruments evaluate cognitive abilities across multiple domains, including verbal, performance, memory, and problem-solving. Standardized instruments measuring adaptive function (functions of "everyday" life) assume that adaptive skills increase with age, and that adaptation may vary across different settings such as school, peer relationships, and family life. The Vineland Adaptive Behavior Scales can be used in infants through youth 18 years of age and includes four basic domains including *Communication* (Receptive, Expressive, and Written); *Daily Living Skills (*Personal, Domestic, and Community); *Socialization (*Interpersonal Relations, Play and Leisure, and Coping Skills); *Motor Skills* (Fine and Gross).

Several behavioral rating scales exist for the population with intellectual disabilities. General behavioral rating scales include the Aberrant Behavior Checklist (ABC) and the Developmental Behavior Checklist (DBC). The Behavior Problem Inventory (BPI) is a useful screening instrument for self-injurious, aggressive, and stereotyped behaviors. The Psychopathology Inventory for Mentally Retarded Adults (PIMRA) helps identify the presence of comorbid psychiatric symptoms and disorders.

Examining clinicians can use several screening instruments for developmental and intellectual delay or disability in infants and toddlers. However, the controversy over the predictive value of infant psychological tests is heated. Some report the correlation of abnormalities during infancy with later abnormal functioning as very low, and others report it to be very high. The correlation rises in direct proportion to the age of the child at the time of the developmental examination. Some exercises such as copying geometric figures, the *Goodenough Draw-a-Person Test,* the *Kohs Block Test,* and geometric puzzles all are useful quick screening tests of visual-motor coordination. For infants, the *Gesell* and *Bayley scales* and the *Cattell Infant Intelligence Scale* are useful.

The *Peabody Vocabulary Test* is the most widely used picture-based vocabulary test. Other tests often found useful in detecting intellectual

disability are the *Bender Gestalt Test* and the *Benton Visual Retention Test.* The psychological evaluation should assess perceptual, motor, linguistic, and cognitive abilities.

Physical Examination

Various parts of the body may demonstrate identifying characteristics of specific perinatal and prenatal events or conditions associated with intellectual disabilities. For example, the configuration and the size of the head may offer clues to a variety of conditions, such as microcephaly, hydrocephalus, or Down syndrome. A patient's facial characteristics, for example, hypertelorism, a flat nasal bridge, prominent eyebrows, epicanthal folds, may provide clues to a recognizable syndrome such as FAS. Additional facial characteristics including corneal opacities, retinal changes, low-set, and small or misshapen ears, a protruding tongue, and disturbance in dentition may be stigmata of a variety of known syndromes. The facial expression, color, and texture of the skin and hair, a high-arched palate, the size of the thyroid gland, and the proportions of a child's trunk and extremities may offer clues for particular syndromes. We should measure the circumference of the head as part of the clinical investigation. Dermatoglyphics (studying skin markings) may offer another diagnostic tool because unique ridge patterns and flexion creases on the hand occur in persons who are intellectually disabled, including chromosomal disorders and rubella. Table 1-30 lists syndromes with intellectual disability and their behavioral phenotypes.

Neurologic Examination

Sensory impairments frequently occur among persons with intellectual disabilities. For example, hearing impairment occurs in 10 percent of persons with intellectual disability, a rate that is four times that of the general population. Visual disturbances can range from blindness to disturbances of spatial concepts, design recognition, and concepts of body image. Seizure disorders occur in about 10 percent of intellectually disabled populations and one-third of those with severe intellectual disabilities. Neurologic abnormalities increase in incidence and severity in direct proportion to the degree of intellectual disability. Disturbances in motor areas cause

abnormalities of muscle tone (spasticity or hypotonia), reflexes (hyperreflexia), and involuntary movements (choreoathetosis). Less disability may also be associated with clumsiness and poor coordination.



Table 1-30
Syndromes with Intellectual Disability and Behavioral Phenotypes

Disorder	Pathophysiology	Clinical Features and Behavioral Phenotype
Down syndrome	Trisomy 21, 95% nondisjunction, approx. 4% translocation; 1/1,000 live births: 1:2,500 in women less than 30 yr old, 1:80 over 40 yr old, 1:32 at 45 yr old; possible overproduction of β-amyloid due to defect at 21q21.1	Hypotonia, upward-slanted palpebral fissures, midface depression, flat wide nasal bridge, simian crease, short stature, increased incidence of thyroid abnormalities and congenital heart disease Passive, affable, hyperactivity in childhood, stubborn; verbal > auditory processing, increased risk of depression, and dementia of the Alzheimer type in adulthood
Fragile X syndrome	Inactivation of *FMR-1* gene at X q27.3 due to CGG base repeats, methylation; recessive;	Long face, large ears, midface hypoplasia, high arched palate, short stature, macroorchidism, mitral valve prolapse, joint

	1:1,000 male births, 1:3,000 female; accounts for 10–12% of intellectual disability in males	laxity, strabismus Hyperactivity, inattention, anxiety, stereotypies, speech and language delays, IQ decline, gaze aversion, social avoidance, shyness, irritability, learning disorder in some females; mild intellectual disability in affected females, moderate to severe in males; verbal IQ > performance IQ
Prader–Willi syndrome	Deletion in 15q12 (15q11–15q13) of paternal origin; some cases of maternal uniparental disomy; dominant 1/10,000 live births; 90% sporadic; candidate gene: small nuclear ribonucleoprotein polypeptide (SNRPN)	Hypotonia, failure to thrive in infancy, obesity, small hands and feet microorchidism, cryptorchidism, short stature, almond-shaped eyes, fair hair and light skin, flat face, scoliosis, orthopedic problems, prominent forehead and bitemporal narrowing Compulsive behavior, hyperphagia, hoarding, impulsivity, borderline to moderate intellectual disability, emotional lability, tantrums, excess daytime

		sleepiness, skin picking, anxiety, aggression
Angelman syndrome	Deletion in 15q12 (15q11–15q13) of maternal origin; dominant; frequent deletion of γ-aminobutyric acid (GABA) B-3 receptor subunit, prevalence unknown but rare, estimated 1/20,000–1/30,000	Fair hair and blue eyes (66%); dysmorphic faces including wide smiling mouth, thin upper lip, and pointed chin; epilepsy (90%) with characteristic EEG; ataxia; small head circumference, 25% microcephalic Happy disposition, paroxysmal laughter, hand flapping, clapping; profound intellectual disability; sleep disturbance with nighttime waking; possible increased incidence of autistic features; anecdotal love of water and music
Cornelia de Lange syndrome	Lack of pregnancy-associated plasma protein A (PAPPA) linked to chromosome 9q33; similar phenotype associated with trisomy 5p, ring chromosome 3;	Continuous eyebrows, thin downturning upper lip, microcephaly, short stature, small hands and feet, small upturned nose, anteverted nostrils, malformed upper limbs, failure to thrive Self-injury, limited speech

	rare (1/40,000–1/100,000 live births); possible association with 3q26.3	in severe cases, language delays, avoidance of being held, stereotypic movements, twirling, severe to profound intellectual disability
Williams syndrome	1/20,000 births; hemizygous deletion that includes elastin locus chromosome 7q11–23; autosomal dominant	Short stature, unusual facial features including broad forehead, depressed nasal bridge, stellate pattern of the iris, widely spaced teeth, and full lips; elfinlike facies; renal and cardiovascular abnormalities; thyroid abnormalities; hypercalcemia Anxiety, hyperactivity, fears, outgoing, sociable, verbal skills > visual spatial skills
Cri-du-chat syndrome	Partial deletion 5p; 1/50,000; region may be 5p15.2	Round face with hypertelorism, epicanthal folds, slanting palpebral fissures, broad flat nose, low-set ears, micrognathia; prenatal growth retardation; respiratory and ear infections; congenital

		heart disease; gastrointestinal abnormalities Severe intellectual disability, infantile catlike cry, hyperactivity, stereotypies, self-injury
Smith–Magenis syndrome	Incidence unknown, estimated 1/25,000 live births; complete or partial deletion of 17p11.2	Broad face; flat midface; short, broad hands; small toes; hoarse, deep voice Severe intellectual disability; hyperactivity; severe self-injury including hand biting, head banging, and pulling out finger- and toenails; stereotyped self-hugging; attention seeking; aggression; sleep disturbance (decreased REM)
Rubinstein–Taybi syndrome	1/250,000, approx. male = female; sporadic; likely autosomal dominant; documented microdeletions in some cases at 16p13.3	Short stature and microcephaly, broad thumb and big toes, prominent nose, broad nasal bridge, hypertelorism, ptosis, frequent fractures, feeding difficulties in infancy, congenital heart disease, EEG abnormalities, seizures

		Poor concentration, distractible, expressive language difficulties, performance IQ > verbal IQ; anecdotally happy, loving, sociable, responsive to music, self-stimulating behavior; older patients have mood lability and temper tantrums
Tuberous sclerosis complex 1 and 2	Benign tumors (hamartomas) and malformations (hamartias) of central nervous system (CNS), skin, kidney, heart; dominant; 1/10,000 births; 50% TSC 1, 9q34; 50% TSC 2, 16p13	Epilepsy, autism, hyperactivity, impulsivity, aggression; spectrum of intellectual disability from none (30%) to profound; self-injurious behaviors, sleep disturbances
Neurofibromatosis type 1 (NF1)	1/2,500–1/4,000; male = female; autosomal dominant; 50% new mutations; more than 90% paternal NF1 allele mutated; *NFl* gene 17q11.2; gene product is	Variable manifestations; café-au-lait spots, cutaneous neurofibromas, Lisch nodules; short stature and macrocephaly in 30–45 Half with speech and language difficulties; 10% with moderate to

	neurofibromin thought to be tumor suppressor gene	profound intellectual disability; verbal IQ > performance IQ; distractible, impulsive, hyperactive, anxious; possibly associated with increased incidence of mood and anxiety disorders
Lesch–Nyhan syndrome	Defect in hypoxanthine guanine phosphoribosyl-transferase with accumulation of uric acid; Xq26–27; recessive; rare (1/10,000–1/38,000)	Ataxia, chorea, kidney failure, gout Often severe self-biting behavior; aggression; anxiety; mild to moderate intellectual disability
Galactosemia	Defect in galactose-1-phosphate uridyltransferase or galactokinase or empiramase; autosomal recessive; 1/62,000 births in the United States	Vomiting in early infancy, jaundice, hepatosplenomegaly; later cataracts, weight loss, food refusal, increased intracranial pressure and increased risk for sepsis, ovarian failure, failure to thrive, renal tubular damage Possible intellectual disability even with treatment, visuospatial deficits, language disorders, reports of

		increased behavioral problems, anxiety, social withdrawal, and shyness
Phenylketonuria	Defect in phenylalanine hydroxylase (PAH) or cofactor (biopterin) with accumulation of phenylalanine; approximately 1/11,500 births; varies with geographical location; gene for PAH, 12q22–24.1; autosomal recessive	Symptoms absent neonatally, later development of seizures (25% generalized), fair skin, blue eyes, blond hair, rash Untreated: mild to profound intellectual disability, language delay, destructiveness, self-injury, hyperactivity
Hurler syndrome	1/100,000; deficiency in α-L-iduronidase activity; autosomal recessive	Early onset; short stature, hepatosplenomegaly; hirsutism, corneal clouding, death before age 10 yr, dwarfism, coarse facial features, recurrent respiratory infections Moderate to severe intellectual disability, anxious, fearful, rarely aggressive
Hunter syndrome	1/100,000, X-linked recessive;	Normal infancy; symptom onset at age 2–4 yr;

	iduronate sulfatase deficiency; X q28	typical coarse faces with flat nasal bridge, flaring nostrils; hearing loss, ataxia, hernia common; enlarged liver and spleen, joint stiffness, recurrent infections, growth retardation, cardiovascular abnormality
		Hyperactivity, intellectual disability by 2 yr; speech delay; loss of speech at 8–10 yr; restless, aggressive, inattentive, sleep abnormalities; apathetic, sedentary with disease progression
Fetal alcohol syndrome	Maternal alcohol consumption (trimester III > II > I); 1/3,000 live births in Western countries; 1/300 with fetal alcohol effects	Microcephaly, short stature, midface hypoplasia, short palpebral fissure, thin upper lip, retrognathia in infancy, micrognathia in adolescence, hypoplastic long or smooth philtrum
		Mild to moderate intellectual disability, irritability, inattention, memory impairment

Table by B. H. King, M.D., R. M. Hodapp, Ph.D., and E. M. Dykens, Ph.D.

1.3 Geriatric Patients

PSYCHIATRIC EXAMINATION OF THE OLDER PATIENT

Psychiatric history taking and the mental status examination of older adults follow the same format as for younger adults; however, because of the high prevalence of cognitive disorders in older persons, psychiatrists must determine whether a patient understands the nature and purpose of the examination. When a patient is cognitively impaired, an independent history should be obtained from a family member or caretaker. The patient still should be seen alone—even in cases of clear evidence of impairment—to preserve the privacy of the doctor–patient relationship and to elicit any suicidal thoughts or paranoid ideation, which may not be voiced in the presence of a relative or nurse.

When approaching the examination of the older patient, it is essential to remember that older adults differ markedly from one another. The approach to examining the older patient must take into account whether the person is a healthy 75-year-old who recently retired from a second career or a frail 96-year-old who just lost the only surviving relative with the death of the 75-year-old care-giving daughter.

Psychiatric History

A complete psychiatric history includes preliminary identification (name, age, sex, marital status), chief complaint, history of the present illness, history of previous illnesses, personal history, and family history. A review of medications (including over-the-counter medications), both current and recent, is also essential.

Patients older than age 65 often have subjective complaints of minor memory impairments, such as forgetting persons' names and misplacing objects. Minor cognitive problems also can occur because of anxiety in the interview situation. These age-associated memory impairments are of no significance; the term *benign senescent forgetfulness* has been used to

describe them.

A patient's childhood and adolescent history can provide information about personality organization and give important clues about coping strategies and defense mechanisms used under stress. A history of learning disability or minimal cerebral dysfunction is significant. The psychiatrist should inquire about friends, sports, hobbies, social activity, and work. The occupational history should include the patient's feelings about work, relationships with peers, problems with authority, and attitudes toward retirement. The patient also should be questioned about plans for the future. What are the patient's hopes and fears?

The family history should include a patient's description of parents' attitudes and adaptation to their old age and, if applicable, information about the causes of their deaths. Alzheimer disease is transmitted as an autosomal-dominant trait in 10 to 30 percent of the offspring of parents with Alzheimer disease; depression and alcohol dependence also run in families. The patient's current social situation should be evaluated. Who cares for the patient? Does the patient have children? What are the characteristics of the patient's parent–child relationships? A financial history helps the psychiatrist evaluate the role of economic hardship in the patient's illness and to make realistic treatment recommendations.

The marital history includes a description of the spouse and the characteristics of the relationship. If the patient is a widow or a widower, the psychiatrist should explore how grieving was handled. If the loss of the spouse occurred within the past year, the patient is at high risk for an adverse physical or psychological event.

The patient's sexual history includes sexual activity, orientation, libido, masturbation, extramarital affairs, and sexual symptoms (e.g., impotence and anorgasmia). Young clinicians may have to overcome their own biases about taking a sexual history: Sexuality is an area of concern for many geriatric patients, who welcome the chance to talk about their sexual feelings and attitudes.

Mental Status Examination

The mental status examination offers a cross-sectional view of how a patient thinks, feels, and behaves during the examination. With older adults, a

psychiatrist may not be able to rely on a single examination to answer all of the diagnostic questions. Repeat mental status examinations may be needed because of fluctuating changes in the patient's family.

General Description. A general description of the patient includes appearance, psychomotor activity, attitude toward the examiner, and speech activity.

Motor disturbances (e.g., shuffling gait, stooped posture, "pill-rolling" movements of the fingers, tremors, and body asymmetry) should be noted. Involuntary movements of the mouth or tongue may be adverse effects of phenothiazine medication. Many depressed patients seem to be slow in speech and movement. Mask-like facies occurs in Parkinson disease.

The patient's speech may be pressured in agitated, manic, and anxious states. Tearfulness and overt crying occur in depressive and cognitive disorders, especially if the patient feels frustrated about being unable to answer one of the examiner's questions. The presence of a hearing aid or another indication that the patient has a hearing problem (e.g., requesting repetition of questions) should be noted.

The patient's attitude toward the examiner—cooperative, suspicious, guarded, ingratiating—can give clues about possible transference reactions. Because of transference, older adults can react to younger physicians as if the physicians were parent figures, despite the age difference.

Functional Assessment. Patients older than 65 years of age should be evaluated for their capacity to maintain independence and to perform the activities of daily life, which include toileting, preparing meals, dressing, grooming, and eating. The degree of functional competence in their everyday behaviors is an essential consideration in formulating a treatment plan for these patients.

Mood, Feelings, and Affect. Suicide is a leading cause of death of older persons, and an evaluation of a patient's suicidal ideation is essential. Loneliness is the most common reason cited by older adults who consider suicide. Feelings of loneliness, worthlessness, helplessness, and hopelessness are symptoms of depression, which carries a high risk of suicide. Nearly 75 percent of all suicide victims suffer from depression, alcohol abuse, or both.

The examiner should specifically ask the patient about any thoughts of suicide: Does the patient feel life is no longer worth living? Does the patient think he or she would be better off dead or, when dead, would no longer be a burden to others? Such thoughts—significantly when associated with alcohol abuse, living alone, the recent death of a spouse, physical illness, and somatic pain—indicate a high suicidal risk.

Disturbances in mood states, most notably depression and anxiety, can interfere with memory functioning. An expansive or euphoric mood may indicate a manic episode or may signal a dementing disorder. Frontal lobe dysfunction often produces *witzelsucht,* which is the tendency to make puns and jokes and then laugh aloud at them.

The patient's affect may be flat, blunted, constricted, shallow, or inappropriate, all of which can indicate a depressive disorder, schizophrenia, or brain dysfunction. Such affects are significant abnormal findings, although they are not pathognomonic of a specific disorder. Dominant lobe dysfunction causes *dysprosody,* an inability to express emotional feelings through speech intonation.

Perceptual Disturbances. Hallucinations and illusions by older adults can be transitory phenomena resulting from decreased sensory acuity. The examiner should note whether the patient is confused about time or place during the hallucinatory episode; confusion points to an organic condition. It is particularly important to ask the patient about distorted body perceptions. Because brain tumors and other focal pathology can cause hallucinations, a diagnostic workup may be indicated. Brain diseases cause perceptive impairments; agnosia, the inability to recognize and interpret the significance of sensory impressions, is associated with organic brain diseases. The examiner should note the type of agnosia. Types of agnosia include denial of illness (anosognosia), denial of a body part (atopognosia), and inability to recognize objects (visual agnosia) or faces (prosopagnosia).

Language Output. The language output category of the geriatric mental status examination covers the aphasias, which are disorders of language output related to organic lesions of the brain. The best described are nonfluent or Broca aphasia, fluent or Wernicke aphasia, and global aphasia, a combination of fluent and nonfluent aphasias. In nonfluent or Broca aphasia,

the patient's understanding remains intact, but the ability to speak is impaired. The patient cannot pronounce "Methodist Episcopalian." Words are generally mispronounced, and speech may be telegraphic. A simple test for Wernicke aphasia is to point to some everyday objects—such as a pen or a pencil, a doorknob, and a light switch—and ask the patient to name them. The patient also may be unable to demonstrate the use of simple objects, such as a key and a match (ideomotor apraxia).

Visuospatial Functioning. Some decline in visuospatial capability is expected with aging. Asking a patient to copy figures or a drawing may help assess the function. A neuropsychological assessment should be performed when visuospatial functioning is impaired.

Thought. Disturbances in thinking include neologisms, word salad, circumstantiality, tangentiality, loosening of associations, flight of ideas, clang associations, and blocking. The loss of the ability to appreciate nuances of meaning (abstract thinking) may be an early sign of dementia. Thinking is then described as concrete or literal.

Thought content should be examined for phobias, obsessions, somatic preoccupations, and compulsions. Ideas about suicide or homicide should be discussed. The examiner should determine whether delusions are present and how such delusions affect the patient's life. Delusions may be present in nursing home patients and may have been a reason for admission. Ideas of reference or influence should be described. Patients who are hard of hearing can be classified mistakenly as paranoid or suspicious.

Sensorium and Cognition. *Sensorium* concerns the functioning of the special senses; *cognition* concerns information processing and intellect. The survey of both areas, known as the neuropsychiatric examination, consists of the clinician's assessment and a comprehensive battery of psychological tests.

CONSCIOUSNESS. A sensitive indicator of brain dysfunction is an altered state of consciousness in which the patient does not seem to be alert, shows fluctuations in levels of awareness, or seems to be lethargic. In severe cases, the patient is somnolescent or stuporous.

ORIENTATION. Impairment in orientation to time, place, and person is associated with cognitive disorders. Cognitive impairment often is observed in mood disorders, anxiety disorders, factitious disorders, conversion disorder, and personality disorders, especially during periods of severe physical or environmental stress. The examiner should test for orientation to place by asking the patient to describe his or her present location. Orientation to a person may be approached in two ways: Does the patient know their name, and are nurses and doctors identified as such? Time is tested by asking the patient the date, the year, the month, and the day of the week. The patient also should be asked about the length of time spent in a hospital, during what season of the year, and how the patient knows these facts. Greater significance is given to difficulties concerning person than to difficulties of time and place, and more significance is given to orientation to place than to orientation to time.

MEMORY. Memory usually is evaluated in terms of immediate, recent, and remote memory. Immediate retention and recall are tested by giving the patient six digits to repeat forward and backward. The examiner should record the result of the patient's capacity to remember. Persons with unimpaired memory usually can recall six digits forward and five or six digits backward. The clinician should be aware that the ability to do well on digit-span tests is impaired in very anxious patients. Remote memory can be tested by asking for the patient's place and date of birth, the patient's mother's name before she was married, and the names and birthdays of the patient's children.

In cognitive disorders, recent memory deteriorates first. Recent memory assessment can be approached in several ways. Some examiners give the patient the names of three items early in the interview and ask for recall later. Others prefer to tell a brief story and ask the patient to repeat it verbatim. We can test for recent memory by asking for the patient's place of residence, including the street number, the method of transportation to the hospital, and some current events. If the patient has a memory deficit, such as amnesia, careful testing should be performed to determine whether it is retrograde amnesia (loss of memory before an event) or anterograde amnesia (loss of memory after the event). Retention and recall also can be tested by having the patient retell a simple story. Patients who confabulate make up new material

in retelling the story.

INTELLECTUAL TASKS, INFORMATION, AND INTELLIGENCE. Various intellectual tasks can be presented to estimate the patient's fund of general knowledge and intellectual functioning. Counting and calculation can be tested by asking the patient to subtract 7 from 100 and to continue subtracting 7 from the result until the number 2 is reached. The examiner records the responses as a baseline for future testing. The examiner can also ask the patient to count backward from 20 to 1 and can record the time necessary to complete the exercise.

The patient's fund of general knowledge is related to intelligence. The patient can be asked to name the president of the United States, to name the three largest cities in the United States, to give the population of the United States, and to give the distance from New York to Paris. The examiner must take into account the patient's educational level, socioeconomic status, and general life experience in assessing the results of some of these tests.

READING AND WRITING. It may be necessary for the clinician to examine the patient's reading and writing and to determine whether the patient has a specific speech deficit. The examiner may have the patient read a simple story aloud or write a short sentence to test for a reading or writing disorder. Whether the patient is right-handed or left-handed should be noted.

JUDGMENT. *Judgment* is the capacity to act appropriately in various situations. Does the patient show impaired judgment? What would the patient do on finding a stamped, sealed, addressed envelope in the street? What would the patient do if he or she smelled smoke in a theater? Can the patient discriminate? What is the difference between a dwarf and a boy? Why are couples required to get a marriage license?

Neuropsychological Evaluation

A thorough neuropsychological examination includes a comprehensive battery of tests that can be replicated by various examiners and can be repeated over time to assess the course of a specific illness. The most widely used test of current cognitive functioning is the Mini-Mental State Examination (MMSE), which assesses orientation, attention, calculation,

immediate and short-term recall, language, and the ability to follow simple commands. The MMSE is used to detect impairments, follow the course of an illness, and monitor the patient's treatment responses. It is not used to make a formal diagnosis. The maximal MMSE score is 30. Age and educational level influence cognitive performance.

The assessment of intellectual abilities is performed with the Wechsler Adult Intelligence Scale-Revised (WAIS-R), which gives verbal, performance, and full-scale IQ scores. Some test results, such as those of vocabulary tests, hold up as aging progresses; results of other tests, such as tests of similarities and digit-symbol substitution, do not. The performance part of the WAIS-R is a more sensitive indicator of brain damage than the verbal part.

Visuospatial functions are sensitive to the normal aging process. The Bender Gestalt Test is one of a large number of instruments used to test visuospatial functions; another is the Halstead–Reitan Battery, which is the most complex battery of tests covering the entire spectrum of information processing and cognition. Depression, even in the absence of dementia, often impairs psychomotor performance, especially visuospatial functioning and timed motor performance. The Geriatric Depression Scale is a useful screening instrument that excludes somatic complaints from its list of items. The presence of somatic complaints on a rating scale tends to confound the diagnosis of a depressive disorder.

Medical History. Elderly patients have more concomitant, chronic, and multiple medical problems and take more medications than younger adults; many of these medications can influence their mental status. The medical history includes all significant illnesses, trauma, hospitalizations, and treatment interventions. The psychiatrist should also be alert to underlying medical illness. Psychiatric symptoms may first manifest infections, metabolic and electrolyte disturbances, and myocardial infarction and stroke. Depressed mood, delusions, and hallucinations may precede other symptoms of Parkinson disease by many months. On the other hand, a psychiatric disorder can also cause such somatic symptoms as weight loss, malnutrition, and inanition of severe depression.

A careful review of medications (including over-the-counter medications, laxatives, vitamins, tonics, and lotions) and even substances recently

discontinued is critical. Drug effects can be long-lasting and may induce depression (e.g., antihypertensives), cognitive impairment (e.g., sedatives), delirium (e.g., anticholinergics), and seizures (e.g., neuroleptics). The review of medications must include sufficient detail to identify misuse (overdose, underuse) and relate medication use to special diets. A dietary history is also essential; deficiencies and excesses (e.g., protein, vitamins) can influence physiologic function and mental status.

EARLY DETECTION AND PREVENTION STRATEGIES

Many age-related illnesses develop insidiously and gradually progress over the years. The most common cause of late-life cognitive impairment, Alzheimer disease, is characterized neuropathologically by a gradual accumulation of neuritic plaques and neurofibrillary tangles in the brain. Clinically, a progression of cognitive decline is seen, which begins with mild memory loss and ends with severe cognitive and behavioral deterioration.

Because it will likely be simpler to prevent neural damage than to repair it once it occurs, investigators are developing strategies for early detection and prevention of age-related illnesses, such as Alzheimer disease. Considerable progress has been made in the detection component of this strategy, using brain imaging technologies, such as PET and fMRI, in combination with genetic risk measures. With these approaches, subtle brain changes can now be detected that progress and can be followed over time. Such surrogate markers allow clinical scientists to track disease progression and to test novel treatments designed to decelerate brain aging. Clinical trials of cholinesterase inhibitor drugs, anticholesterol drugs, anti-inflammatory drugs, and others (e.g., vitamin E) are in progress to determine if such treatments delay the onset of Alzheimer disease or the progression of brain metabolic or cognitive decline.

Novel approaches to measuring the physical evidence of Alzheimer disease, the plaques, and tangles in the cerebral cortex, have been successful in initial studies and will likely facilitate the testing of innovative treatments designed to rid the brain of these pathognomonic lesions. Scientists may not be able to cure Alzheimer disease in its advanced stages, but they may be able to delay its onset effectively, thus helping patients live longer without the debilitating manifestations of the disease, including cognitive decline.

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2

Neurodevelopmental Disorders and Other Childhood Disorders

This section contains disorders typically found in children. Some disorders are disorders of development and invariably present during childhood or adolescence. Others are more common disorders, such as depression or anxiety, that may have unique presentations or treatment issues in children.

2.1 Intellectual Disability

Intellectual disability (ID), formerly known as *mental retardation*, can be caused by a range of environmental and genetic factors that lead to a combination of cognitive and social impairments. The American Association on Intellectual and Developmental Disability (AAIDD) defines intellectual disability as a disability characterized by significant limitations in both intellectual functioning (reasoning, learning, and problem-solving) and in adaptive behavior (conceptual, social, and practical skills) that emerges before the age of 18 years.

The widespread acceptance of this definition has led to the international consensus that an assessment of both social adaptation and intelligence quotient (IQ) is necessary to determine the level of intellectual disability. Measures of adaptive function assess competency in social functioning, understanding of societal norms, and performance of everyday tasks, whereas measures of intellectual function focus on cognitive abilities. Although individuals with a given intellectual level do not all have identical levels of adaptive function, epidemiologic data suggest that a person's intellectual

level and level of adaptive function largely determine the prevalence of intellectual disability and typically correspond closely with cognitive ability.

In the *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition (DSM-5), various levels of severity of intellectual disability are determined based on adaptive functioning, not on IQ scores. DSM-5 has adopted this change in emphasis from prior diagnostic manuals because adaptive functioning determines the level of support that is required. Furthermore, IQ scores are less valid in the lower portions of the IQ range. Deciding the severity level of intellectual disability, according to DSM-5, includes an assessment of functioning in a conceptual domain (e.g., academic skills), a social domain (e.g., relationships), and a practical domain (e.g., personal hygiene).

As it is older, the International Classification of Diseases, 10th Edition (ICD-10) continues to rely on IQ as the main determiner of severity; this is being revised for the next edition.

Societal approaches to children with intellectual disabilities have shifted significantly over time. Historically, in the mid-1800s, many children with intellectual disabilities were placed in residential, educational facilities based on the belief that with sufficient intensive training, these children would be able to return to their families and function in society at a higher level. However, many residential programs increased in size, and eventually, the focus began to shift from intensive education to custodial care. Residential settings for children with an intellectual disability received their maximal use in the mid-1900s until public awareness of the crowded, unsanitary, and, in some cases, abusive conditions sparked the movement toward "deinstitutionalization." A vital force in the deinstitutionalization of children with intellectual disability was the philosophy of "normalization" in living situations, and "inclusion" in educational settings. Since the late 1960s, placing children with intellectual disabilities into institutions is rare, and the concepts of normalization and inclusion remain prominent among advocacy groups and parents.

The passage of Public Law 94–142 (the Education for all Handicapped Children Act) in 1975 mandates that the public school system provides appropriate educational services to all children with disabilities. The Individuals with Disabilities Act in 1990 extended and modified the above legislation. Currently, the provision of public education for all children,

including those with disabilities, “within the least restrictive environment” is mandated by law.

In addition to the educational system, advocacy groups, including the Council for Exceptional Children (CEC) and the Arc of the United States (The Arc), are well-known parental lobbying organizations for children with intellectual disability and were instrumental in advocating for Public Law 94–142. The American Association on Intellectual and Developmental Disabilities (AAIDD) is the most prominent advocacy organization in this field. It has been very influential in educating the public about and supporting research and legislation relating to intellectual disability.

The AAIDD promotes a view of intellectual disability as a functional interaction between an individual and the environment, rather than a static designation of a person’s limitations. Within this conceptual framework, a child or adolescent with an intellectual disability is determined to need intermittent, limited, extensive, or pervasive “environmental support” concerning a specific set of adaptive function domains. These include communication, self-care, home living, social or interpersonal skills, use of community resources, self-direction, functional academic skills, work, leisure, health, and safety.

The United Nations Convention on the Rights of Persons with Disabilities (2006) has created a forum to promote the full social inclusion of people with intellectual disabilities. Through its recognition and focus on social barriers, this international forum aims to provide protections for individuals with intellectual disabilities and to seek more inclusion of those with intellectual disabilities in social, civic, and educational activities.



Table 2-1
Intellectual Disability

	DSM-5	ICD-10
Diagnostic name	Intellectual Disability (Intellectual Developmental Disorder)	Mental Retardation
Duration	The symptoms arise during development	Occurs during the developmental

		period
Symptoms	Deficits in: • Reasoning • Abstraction • Judgment • Learning • Adaptive functioning	Incomplete mental development Affects skills related to intelligence: • Cognitive • Language • Motor • Social abilities
Psychosocial consequences of symptoms	Symptoms delay developmental milestones, achieving independence and social functioning	
Severity specifiers	Profound Severe Moderate Mild	Estimated by IQ: Mild: 50–69 Moderate: 35–49 Severe: 20–34 Profound: <20

NOMENCLATURE

The accurate definition of intellectual disability has been a challenge for clinicians over the centuries. All current classification systems underscore that intellectual disability is more than a cognitive deficit. That is, it also includes impaired social adaptive function. According to DSM-5, we should only make a diagnosis of intellectual disability when there are deficits in intellectual functioning and deficits in adaptive functioning (Table 2-1). Once intellectual disability is recognized, we determine the level of severity by the level of adaptive functional impairment.

Table 2-2
Developmental Characteristics of Intellectual Disability

Level of Intellectual Disability	Preschool Age (0–5 yr) Maturation and Development	School Age (6–20 yr) Training and Education	Adult (21 yr and Above) Social and Vocational Adequacy
Profound	Gross disability; minimal capacity for functioning in sensorimotor areas; needs nursing care; constant aid and supervision required	Some motor development present; may respond to minimal or limited training in self-help	Some motor and speech development; may achieve very limited self-care; needs nursing care
Severe	Poor motor development; speech minimal; generally unable to profit from training in self-help; little or no communication skills	Can talk or learn to communicate; can be trained in elemental health habits; profits from systematic habit training; unable to profit from vocational training	May contribute partially to self-maintenance under complete supervision; can develop self-protection skills to a minimal useful level in controlled environment
Moderate	Can talk or learn to communicate; poor social awareness;	Can profit from training in social and occupational skills; unlikely	May achieve self-maintenance in unskilled or semiskilled work under

	fair motor development; profits from training in self-help; can be managed with moderate supervision	to progress beyond second-grade level in academic subjects; may learn to travel alone in familiar places	sheltered conditions; needs supervision and guidance when under mild social or economic stress
Mild	Can develop social and communication skills; minimal retardation in sensorimotor areas; often not distinguished from normal until later age	Can learn academic skills up to approximately sixth-grade level by late teens; can be guided toward social conformity	Can usually achieve social and vocational skills adequate for minimal self-support, but may need guidance and assistance when under unusual social or economic stress

Adapted from *Mental Retarded Activities of the US Department of Health, Education and Welfare*. Washington, DC: US Government Printing Office; 1989:2, with permission.

CLASSIFICATION

DSM-5 criteria for intellectual disability include significantly subaverage general intellectual functioning associated with concurrent impairment in adaptive behavior manifested before the age of 18. The diagnosis is made independent of coexisting physical or mental disorders. Table 2-2 presents an overview of developmental levels in communication, academic functioning, and vocational skills expected of persons with various degrees of intellectual disability.

If we choose to use a standardized test of intelligence—which is still a common practice—the term *significantly subaverage* is defined as an IQ of

approximately 70 or below or two standard deviations below the mean for the particular test. We can measure adaptive functioning by using a standardized scale, such as the *Vineland Adaptive Behavior Scale.* This scale scores communications, daily living skills, socialization, and motor skills (up to 4 years, 11 months) and generates an adaptive behavior composite that correlates with the expected skills at a given age.

Approximately 85 percent of individuals who have an intellectual disability fall within the DSM-5 mild intellectual disability category. This category requires an IQ between 50 and 70 and an adaptive function severity in the mild range. Adaptive function includes skills such as communication, self-care, social skills, work, leisure, and understanding of safety. Genetic, environmental, and psychosocial factors all influence intellectual disability. A host of subtle environmental and developmental factors, including subclinical lead intoxication and prenatal exposure to drugs, alcohol, and other toxins, can contribute to intellectual disability. Specific genetic syndromes associated with intellectual disability such as fragile X syndrome, Down syndrome, and Prader–Willi syndrome have characteristic patterns of social, linguistic, and cognitive development and typical behavioral manifestations.

DEGREES OF SEVERITY OF INTELLECTUAL DISABILITY

DSM-5 classifies the severity levels of intellectual disability as mild, moderate, severe, and profound. “Borderline intellectual functioning,” a term previously used to describe individuals with an IQ in the range of 70 to 80, is no longer described as a diagnosis in DSM-5, and is instead listed as a condition that may be the focus of clinical attention; although DSM-5 does not offer any criteria.

Mild intellectual disability represents approximately 85 percent of persons with intellectual disabilities. Children with mild intellectual disabilities often are not identified until the first or second grade, when academic demands increase. By late adolescence, they often acquire academic skills at approximately a sixth-grade level. Specific causes for intellectual disability are often unidentified in this group. Many adults with mild intellectual disabilities can live independently with appropriate support and raise their own families. IQ for this level of adaptive function may typically range from 50 to 70.

Moderate intellectual disability represents about 10 percent of persons with intellectual disabilities. Most children with a moderate intellectual disability acquire language and can communicate adequately during early childhood. They are challenged academically and often are not able to achieve above a second- to third-grade level. During adolescence, socialization difficulties often set these persons apart, and a great deal of social and vocational support is beneficial. As adults, individuals with moderate intellectual disability may be able to perform semiskilled work under appropriate supervision. IQ for this level of adaptive function may typically range from 35 to 50.

Severe intellectual disability represents about 4 percent of individuals with intellectual disabilities. They may be able to develop communication skills in childhood and often can learn to count as well as recognize words that are critical to functioning. In this group, the cause for the intellectual disability is more likely to be identified than in milder forms of intellectual disability. In adulthood, persons with severe intellectual disability may adapt well to supervised living situations, such as group homes, and may be able to perform work-related tasks under supervision. IQ in individuals with this level of adaptive function may typically range from 20 to 35.

Profound intellectual disability constitutes approximately 1 to 2 percent of individuals with intellectual disabilities. Most individuals with profound intellectual disabilities have identifiable causes for their condition. Children with profound intellectual disabilities may be taught some self-care skills and learn to communicate their needs, given the appropriate training. IQ in individuals with this level of adaptive function may typically be less than 20.

The DSM-5 also includes a disorder called “Unspecified Intellectual Disability” (Intellectual Developmental Disorder), reserved for individuals over the age of 5 years who are difficult to evaluate but likely have an intellectual disability. Individuals with this diagnosis may have sensory or physical impairments such as blindness or deafness, or concurrent mental disorders, making it challenging to administer standard assessment tools.

CLINICAL FEATURES

Mild intellectual disability may not be recognized or diagnosed in a child until school challenges the child’s social and communication skills. Cognitive deficits include reduced ability to abstract and egocentric thinking, both of

which become more readily evident as a child reaches middle childhood. Children with milder intellectual disabilities may function academically at the high elementary level and may acquire vocational skills sufficient to support themselves in some cases; however, social assimilation may be problematic. Communication deficits, poor self-esteem, and dependence may further contribute to a relative lack of social spontaneity.

Moderate levels of intellectual disability are significantly more likely to be observed at a younger age since communication skills develop more slowly, and social isolation may ensue in the elementary school years. Academic achievement is usually limited to the middle-elementary level. Children with moderate intellectual deficits benefit from individual attention focused on the development of self-help skills. However, these children are aware of their deficits and often feel alienated from their peers and frustrated by their limitations. They continue to require a relatively high level of supervision but can become competent at occupational tasks in supportive settings.

Severe intellectual disability is typically evident in the preschool years; affected children have minimal speech and impaired motor development. Some language development may occur in the school-age years. By adolescence, if the language has not improved significantly, nonverbal forms of communication may have evolved. Behavioral approaches are useful means to promote some self-care, although those with severe intellectual disabilities generally need extensive supervision.

Children with profound intellectual disabilities require constant supervision and are severely limited in both communication and motor skills. By adulthood, some speech development may be present, and simple self-help skills may be acquired. Clinical features frequently observed in populations with intellectual disability, either in isolation or as part of a mental disorder, include hyperactivity, low frustration tolerance, aggression, affective instability, repetitive and stereotypic motor behaviors, and self-injurious behaviors. Self-injurious behaviors occur more frequently and with higher intensity in more severe intellectual disability.

Dylan was a full-term infant, the second child born to his 42-year-old mother, a medical technician, and a 48-year-old father, a high school basketball coach. The pregnancy was unremarkable, and Dylan’s sister,

who was 2 years older, was healthy and developing typically. The family lived in a rural town in the Midwest.

Dylan was an extremely fussy, active newborn with extended periods of crying that the pediatrician labeled classic colic. As a newborn, Dylan seemed to have large ears and strabismus, which the pediatrician said would probably resolve spontaneously. At 2 months of age, at a regular pediatric visit, a systolic heart murmur was heard, and electrocardiography (ECG) revealed a mitral valve prolapse. Because Dylan was not cyanotic and had no other cardiac symptoms, his pediatrician monitored him without specific treatment. Although Dylan became less fussy over time, he remained very active, did not sleep through the night, and was a picky eater, refusing solid foods.

Milestones were slightly delayed, with Dylan sitting unassisted at 10 months and walking at 18 months. He had delayed language, and, although his first words appeared at 20 months, Dylan had always made his wants and needs known. Dylan's parents were concerned about his activity level and his developmental delays compared with his sister; however, the pediatrician reassured them that boys often develop more slowly than girls in the first 2 years.

When Dylan was 3 years of age, his preschool teacher noted that he was unable to pay attention, and he was hyperactive compared to his classmates, prompting his parents to obtain a developmental evaluation. Results showed modest delays in cognitive, linguistic, and motor functioning, with a developmental quotient (DQ) of 74. Dylan was inattentive, shy, and anxious, and he had poor eye contact. Enrolled in a special kindergarten, Dylan remained in a combination of special education and mainstreamed classes throughout his academic life.

At 7 years of age, the school psychologist evaluated Dylan, and results indicated that he met the criteria for a "learning disability" profile. Dylan had an overall IQ of 66, with close to normal functioning in short-term memory and pronounced deficits in long-term memory, expressive language, and visual–spatial functioning. Dylan struggled with writing tasks and arithmetic but loved science. Due to his significant problems with inattention and hyperactivity, Dylan was placed on Concerta, which was beneficial, and titrated up to 54 mg/day. He displayed transient, intense interests in unusual items, such as vacuum cleaners. When Dylan

reached the older elementary grades, he began to have more difficulty socially, and some peers bullied him about being in special education and teased for his long head and big ears.

As he entered adolescence, Dylan became increasingly anxious, so much so that he occasionally rubbed his hands or rocked, and he “fretted” about day-to-day issues and what would happen next. His long-term sensitivities to loud sounds seemed to wane slightly, but he developed fears of storm clouds and dogs and refused to ride on elevators. Dylan became tearful and upset after his older sister left for a party, and worried that she might have a car accident. Dylan was very shy and would occasionally pace with worry and complain of stomachaches, but he attended school and had a small group of acquaintances in the Special Olympics bowling league. He enjoyed activities that did not involve much talking or sustained attention.

When Dylan was 17 years of age, his parents happened to watch a television documentary on genetic causes of intellectual disability. They were overwhelmed by the similarities between Dylan and some of the people described in the program. They later described the experience as a “jolt.” They had always accepted Dylan, quirks and all, and had stopped pushing their doctors for reasons “why” when Dylan was a preschooler. Nevertheless, they immediately called the informational number offered in the show, and within 2 months, genetic tests confirmed a diagnosis of fragile X syndrome.

Although Dylan’s day-to-day life did not change dramatically after the diagnosis, his parents reported a big difference in their approach to his shyness, restricted interests, and inattention. Dylan later received a selective serotonin reuptake inhibitor (SSRI) for anxiety, which decreased his social anxiety and facilitated activities with a few peers. Dylan’s parents reported a mixture of feelings at having such a late diagnosis—disappointment in their doctors, relief in finally knowing, and twinges of guilt. They were energized by Dylan’s positive responses to treatments for his attentional and anxiety symptoms and were pleased with Dylan’s recent increased interest in sharing activities with classmates and peers.

A negative self-image and poor self-esteem are standard features of mildly and moderately intellectually disabled persons who are aware of their social and academic differences from others. Given their experience of repeated failure and disappointment in being unable to meet their parents' and society's expectations, they may also fall progressively behind younger siblings. Communication difficulties further increase their vulnerability to feelings of ineptness and frustration. Inappropriate behaviors, such as withdrawal, are common. The perpetual sense of isolation and inadequacy is related to feelings of anxiety, anger, dysphoria, and depression.

DIAGNOSIS

The diagnosis of intellectual disability occurs after obtaining the history, using information from a standardized intellectual assessment, and a standardized measure of adaptive function indicating that a child is significantly below the expected level in both areas. The severity of the intellectual disability is determined based on the level of adaptive function. A history and psychiatric interview are useful in obtaining a longitudinal picture of the child's development and functioning. Examination of physical signs, neurologic abnormalities, and in some cases, laboratory tests can help ascertain the cause and prognosis.

History

When we take the history, which may elucidate pathways to intellectual disability, we should pay particular attention to the mother's pregnancy, labor, and delivery; the presence of a family history of intellectual disability; consanguinity of the parents; and known familial hereditary disorders.

Chapter 1 details some approaches to interviewing and examining children with intellectual disabilities.

Laboratory Examination

Laboratory tests that may elucidate the causes of intellectual disability include chromosomal analysis, urine and blood testing for metabolic disorders, and neuroimaging. Chromosomal abnormalities are the most common known cause of intellectual disability.

Chromosome Studies. When multiple physical anomalies, developmental delays, and intellectual disability present together, we should order chromosome analysis. Current techniques can identify chromosomal regions with specific fluorescent in situ hybridization (FISH) markers, enabling the identification of microscopic deletions in up to 7 percent of persons with moderate to severe intellectual disability. A history of growth retardation, the presence of microcephaly, a family history of intellectual disability, short stature, hypertelorism, and other facial abnormalities increase the risk of finding subtelomeric defects.

Amniocentesis, in which clinicians remove a small amount of amniotic fluid from the amniotic cavity transabdominally at about 15 weeks of gestation, has been useful for diagnosing prenatal chromosomal abnormalities. Its use is considered when an increased fetal risk exists, such as with increased maternal age. Amniotic fluid cells, mostly fetal in origin, are cultured for cytogenetic and biochemical studies.

Chorionic villi sampling (CVS) is a screening technique to determine fetal chromosomal abnormalities. It is done at 8 to 10 weeks of gestation, 6 weeks earlier than amniocentesis. The results are available in a short time (hours or days) and, if the result is abnormal, it is possible to decide to terminate the pregnancy within the first trimester. The procedure has a miscarriage risk between 2 and 5 percent; the risk in amniocentesis is lower (1 in 200). A noninvasive blood test called MaterniT21 is a proprietary prenatal test that detects abnormalities of chromosomes 21, 18, 13, X, and Y. It is highly specific for Down syndrome. There is a slight risk of miscarriage.

Urine and Blood Analysis. Lesch–Nyhan syndrome, galactosemia, PKU, Hurler syndrome (Fig. 2-1), and Hunter syndrome (Fig. 2-2) are examples of disorders characterized by an intellectual disability that we can identify through assays of the appropriate enzyme or organic or amino acids. Enzymatic abnormalities in chromosomal disorders, such as Down syndrome, promise to become useful diagnostic tools.

Electroencephalography. Whenever considering a seizure disorder, we should request electroencephalography (EEG). “Nonspecific” EEG changes, characterized by slow frequencies with bursts of spikes and sharp or blunt wave complexes, are found with higher frequency among populations with

intellectual disability than in the general population; however, these findings do not elucidate specific diagnoses.



FIGURE 2-1
A 6-year-old girl with Hurler syndrome. Her care has involved a class for seriously multihandicapped children, attention to cardiac problems, and special counseling for patients. (Courtesy of L.S. Syzmanski, M.D. and A.C. Crocker, M.D.)



FIGURE 2-2
Two brothers, age 6 and 8 years, with Hunter syndrome, shown with their developmentally normal older sister. They have had siganificant developmental delay, trouble with recurrent respiratory infection, and behavioral abnormalities. (Courtesy of L.S. Syzmanski, M.D. and A.C. Crocker, M.D.)

Neuroimaging. Neuroimaging studies with populations of intellectually disabled patients using either computerized tomography (CT) or magnetic resonance imaging (MRI) have found high rates of abnormalities in those patients with microcephaly, significant delay, cerebral palsy, and profound disability. Among patients with intellectual disability, seizures, microcephaly or macrocephaly, loss of previously acquired skills, or neurologic signs such as dystonia, spasticity, or altered reflexes all are indications for neuroimaging.

Although clinically not diagnostic, neuroimaging studies are currently also utilized to gather data that may eventually uncover biologic mechanisms contributing to intellectual disability. Structural MRI, functional MRI (fMRI), and diffusion tensor imaging (DTI) are useful in current research. For example, current data suggest that individuals with fragile X syndrome and concurrent attentional deficits are also more likely to show aberrant

frontal–striatal pathways on MRI than those patients without attentional problems. MRI is also useful to elucidate myelination patterns. MRI studies can also provide a baseline for comparison of a later, potentially degenerative process in the brain.

Hearing and Speech Evaluations. We should evaluate a patient's hearing and speech routinely. Speech development may be the most reliable criterion in investigating intellectual disability. Various hearing impairments often occur in persons who are intellectually disabled, but in some instances, hearing impairments can simulate intellectual disability. The commonly used methods of hearing and speech evaluation, however, require the patient's cooperation and, thus, are often unreliable in severely disabled persons.

DIFFERENTIAL DIAGNOSIS

By definition, an intellectual disability must begin before the age of 18. In some cases, severe child maltreatment in the form of neglect or abuse may contribute to delays in development, which can appear to be an intellectual disability. However, these damages are partially reversible when a child grows up in a corrective, enriched, and stimulating environment. Sensory disabilities, especially deafness and blindness, can be mistaken for intellectual disability when a lack of awareness of the sensory deficit leads to inappropriate testing. Expressive and receptive speech disorders may give the impression of intellectual disability in a child of average intelligence, and cerebral palsy may be mistaken for intellectual disability. Chronic, debilitating medical diseases may depress and delay a child's functioning and achievement despite average intelligence. Seizure disorders, especially uncontrolled, may contribute to persisting intellectual disability. Specific syndromes leading to isolated handicaps such as failure to read (alexia), failure to write (agraphia), or failure to communicate (aphasia), may occur in a child of average and even superior intelligence. Children with learning disorders (which can coexist with intellectual disability) experience a delay or failure of development in a specific area, such as reading or mathematics, but they usually develop in other areas. In contrast, children with intellectual disabilities show delays in most areas of development.

Intellectual disability and autism spectrum disorder (ASD) often coexist;

70 to 75 percent of those with ASD have an IQ below 70. Also, epidemiologic data indicate that ASD occurs in approximately 19.8 percent of persons with intellectual disabilities. Children with ASD have relatively more severe impairment in social relatedness and language than other children with the same level of intellectual disability.

A child younger than the age of 18 years with significant adaptive functional impairment, with an IQ less than 70, who also meets diagnostic criteria for dementia, will receive both a diagnosis of dementia and intellectual disability. However, a child whose IQ drops below 70 after the age of 18 years with newly acquired cognitive impairment will receive only the diagnosis of dementia.

COMORBIDITY

Many psychiatric disorders are common with intellectual disabilities, including depressive disorders, attention-deficit disorders, anxiety and neurologic disorders.

Attention-Deficit/Hyperactivity Disorder

Estimates of attention-deficit/hyperactivity disorder (ADHD) and ADHD-like symptoms among children with subaverage intelligence, genetic disorders, and developmental delay are estimated to be significantly higher than rates in the community.

Neurologic Disorders

Seizure disorders occur more frequently in individuals with intellectual disabilities than in the general population, and prevalence rates for seizures increase proportionally to the severity level of intellectual disability. A review of psychiatric disorders in children and adolescents with intellectual disability and epilepsy found that approximately one-third had comorbid autism spectrum disorder. The combination of intellectual disability, epilepsy, and autism spectrum disorder may occur in 0.07 percent of the general population.

COURSE AND PROGNOSIS

Although the underlying intellectual impairment does not improve, in most cases of intellectual disability, the level of adaptation increases with age and can be influenced positively by an enriched and supportive environment. In general, persons with mild and moderate intellectual disabilities have the most flexibility in adapting to various environmental conditions. Comorbid psychiatric disorders negatively impact overall prognosis. When psychiatric disorders co-occur with intellectual disabilities, standard treatments for comorbid mental disorders are often beneficial; however, less robust responses and increased vulnerability to side effects of psychopharmacologic agents are common.

TREATMENT

Interventions for children and adolescents with intellectual disabilities incorporate an assessment of social, educational, psychiatric, and environmental needs. Intellectual disability is associated with a variety of comorbid psychiatric disorders that often require specific treatment, in addition to psychosocial support. Of course, when preventive measures are available, the optimal approach includes primary, secondary, and tertiary interventions.

Prevention

Primary Prevention. Primary prevention comprises actions taken to eliminate or reduce the conditions that lead to the development of intellectual disability, as well as associated disorders. For example, screening babies for PKU, and administrating a low phenylalanine diet when PKU is present, significantly alters the emergence of intellectual disability in those affected children. Additional primary prevention steps include education of the general public about strategies to prevent intellectual disability, such as abstinence from alcohol during pregnancy, continuing efforts of health professionals to ensure and upgrade public health policies, and legislation to provide optimal maternal and child health care. Family and genetic counseling help reduce the incidence of intellectual disability in a family with a history of a genetic disorder.

Secondary and Tertiary Prevention. Prompt attention to medical and psychiatric complications of intellectual disability can diminish their course (secondary prevention) and minimize the sequelae or consequent disabilities (tertiary prevention). Hereditary metabolic and endocrine disorders, such as PKU and hypothyroidism, can be treated effectively in an early stage by dietary control or hormone replacement therapy.

Psychosocial Interventions

Educational Interventions. Educational settings for children with intellectual disability should include a comprehensive program that addresses academics and training in adaptive skills, social skills, and vocational skills. Particular attention should focus on communication and efforts to improve the quality of life.

Behavioral and Cognitive-Behavioral Interventions. The difficulties in adaptation among the intellectual disability populations are widespread and so varied that several interventions alone or in combination may be beneficial. Behavior therapy has been used for many years to shape and enhance social behaviors and to control and minimize aggressive and destructive behaviors. Positive reinforcement for desired behaviors and benign punishment (e.g., loss of privileges) for objectionable behaviors has been helpful. Cognitive therapy, such as dispelling false beliefs and relaxation exercises with self-instruction, has also been recommended for intellectually disabled persons who can follow the instructions. Psychodynamic therapy has been used with patients and their families to decrease conflicts about expectations that result in persistent anxiety, rage, and depression. Psychiatric treatment modalities require modifications that take into consideration the patient's level of intelligence.

Family Education. One of the most critical areas that we can address is educating the family of a child or adolescent with intellectual disability about ways to enhance competence and self-esteem while maintaining realistic expectations for the patient. The family often finds it challenging to balance the fostering of independence and the providing of a nurturing and supportive environment for an intellectually disabled child, who is likely to experience

some rejection and failure outside the family context. The parents may benefit from continuous counseling or family therapy and should be allowed opportunities to express their feelings of guilt, despair, anguish, recurring denial, and anger about their child's disorder and future. We should give the parents all the necessary and current medical information regarding causes, treatment, and other pertinent areas (e.g., specialized training and the correction of sensory defects).

Social Intervention. One of the most prevalent problems among persons with an intellectual disability is a sense of social isolation and social skills deficits. Thus, improving the quantity and quality of social competence is a critical part of their care. Special Olympics International is the most extensive recreational sports program geared for this population. In addition to providing a forum to develop physical fitness, Special Olympics also enhances social interactions, friendships, and (hopefully) general self-esteem.

Psychopharmacologic Interventions

Pharmacologic approaches to the treatment of behavioral and psychological symptoms in children with an intellectual disability follow the paradigms of the evidence-based literature on treatment for all children with psychiatric disorders. However, given the paucity of randomized trials in the childhood intellectual disability population, an empirical approach must also be taken.

Treating Comorbid Disorders and Symptoms

AGGRESSION, IRRITABILITY, AND SELF-INJURIOUS BEHAVIOR. Antipsychotic medicine may be useful for reducing challenging behaviors. Among the antipsychotics, both aripiprazole and risperidone had reasonable effect sizes in multiple studies. However, we should view these findings cautiously as many of the studies had significant methodologic limitations. Overall, the consensus seems to be that antipsychotics are helpful in the short term. Among them, aripiprazole and risperidone have the majority of studies and show effect sizes in the moderate to large range. There is little evidence on the long-term use of these medications. Given the significant side effects (weight gain, sedation, cardiovascular risks, extrapyramidal side effects, and increased prolactin levels, which are of particular concern in children), we

should be very cautious when choosing to use these medications beyond an acute crisis.

Other medications, such as anticonvulsants and antioxidants, may be helpful; however, the data are not conclusive. GABA analogs, such as piracetam, may have promise, but the supportive data are limited.

ATTENTION-DEFICIT/HYPERACTIVITY DISORDER. The existing data for the treatment of ADHD and ADHD-like symptoms in youth with sub-average intelligence and developmental disorders suggest that agents, particularly stimulants used to treat ADHD in typically developing children, provide some degree of benefit to children with intellectual disability and ADHD. However, these children experience more side effects than ADHD children without an intellectual disability. Thus, recommendations regarding the treatment of ADHD in children and adolescents with comorbid intellectual disability include close monitoring for side effects. Studies of methylphenidate treatment in those mildly intellectually disabled with ADHD show significant improvement in the ability to maintain attention and to stay focused on tasks. Methylphenidate treatment studies have not shown evidence of long-term improvement in social skills or learning. Risperidone may be beneficial in reducing symptoms of ADHD in this population; however, as with all antipsychotics, it has substantial side effects. Thus stimulant medications are the preferred choice.

In addition to stimulants and antipsychotics, clonidine has been used clinically in this population, especially to ameliorate hyperactivity and impulsivity. Although there are scant data, clinical ratings by parents and clinicians suggest its efficacy. Atomoxetine may also help for this population.

DEPRESSIVE DISORDERS. The identification of depressive disorders among individuals with intellectual disabilities requires careful evaluation since they are easy to miss when behavioral problems are prominent. There have been anecdotal reports of disinhibition in response to SSRIs in intellectually disabled individuals with ASD. Given the relative safety of SSRI antidepressants, it is reasonable to try them when a child with an intellectual disability is depressed.

STEREOTYPICAL MOTOR MOVEMENTS. Antipsychotic medications are used in

the treatment of repetitive self-stimulatory behaviors in children with intellectual disability when these behaviors are either harmful to the child or disruptive. Anecdotal reports indicate that these agents may diminish self-stimulatory behaviors; however, there is no improvement seen in adaptive behavior. Obsessive-compulsive symptoms often overlap with the repetitive stereotypical behaviors seen in children and adolescents with intellectual disability, particularly in those with comorbid ASD. SSRIs such as fluoxetine, fluvoxamine, paroxetine, and sertraline have efficacy for treating obsessive-compulsive symptoms in children and adolescents and may have some efficacy for stereotyped motor movements.

EXPLOSIVE RAGE BEHAVIOR. Antipsychotic medications, particularly risperidone, are typically used for the treatment of explosive rage. As with the treatment of aggression, there is a need for more data investigating this use. β-Adrenergic receptor antagonists (β-blockers), such as propranolol, have been anecdotally reported to result in fewer explosive rages in some children with intellectual disability and ASD.

Services and Support for Children with Intellectual Disability

Early Intervention. Early intervention programs serve individuals for the first 3 years of life. Such services are generally provided by the state and begin with a specialist visiting the home for several hours per week. Since the passage of Public Law 99–457, the Education of the Handicapped Amendments of 1986, the emphasis is on early intervention services for the entire family. Agencies are required to develop an Individualized Family Service Plan (IFSP) for each family, which identifies specific interventions to best help the family and child.

School. From ages 3 to 21 years, school is responsible by law to provide appropriate educational services to children and adolescents with intellectual disability in the United States. Two laws codified these mandates, Public Law 94–142, the Education for all Handicapped Children Act of 1975, and the Individuals with Disabilities Act (IDEA) of 1990. Through these laws, public schools must develop and provide an individualized educational program for each student with intellectual disability, determined at a meeting designated

as the Individualized Education Plan (IEP) with school personnel and the family. Schools should educate the children in the "least restrictive environment" that will allow the child to learn.

Supports. A wide variety of organized groups and services are available for children with intellectual disability and their families. These include short-term respite care, which allows families a break and is generally set up by state agencies. Other programs include the Special Olympics, which allows children with intellectual disabilities to participate in team sports and sports competitions. Many organizations also exist for families who wish to connect with others who have children with intellectual disabilities.

EPIDEMIOLOGY

The majority of population-based prevalence estimates for intellectual disability in developing countries range from 10 to 15 per 1,000 children. The prevalence of intellectual disability at any one time is estimated to range from 1 to 3 percent of the population in Western societies. The incidence of intellectual disability is difficult to calculate because mild disabilities may be unrecognized until middle childhood. In some cases, even when the intellectual function is limited, adaptive social skills may not be challenged until late childhood or early adolescence. The highest incidence of intellectual disability is reported in school-age children, with a peak at ages 10 to 14 years. Intellectual disability is about 1.5 times more common among males than females.

Prevalence

Epidemiologic surveys indicate that up to two-thirds of children and adults with intellectual disabilities have comorbid psychiatric disorders, and this rate is several times higher than that in community samples without intellectual disability. The prevalence of psychopathology appears to correlate with the severity of the intellectual disability; the more severe the intellectual disability, the higher the risk for coexisting psychiatric disorders. An epidemiologic study found that 40.7 percent of intellectually disabled children between 4 and 18 years of age met the criteria for at least one additional psychiatric disorder.

The severity of intellectual disability influences the risk of particular comorbid psychiatric disorders. Disruptive and conduct-disorder behaviors occur more frequently in those diagnosed with mild intellectual disability. In contrast, those with more severe intellectual disabilities are more likely to meet criteria for autism spectrum disorder and exhibit symptoms such as self-stimulation and self-mutilation. Comorbidity of psychiatric disorders with intellectual disability in children do not appear to correlate with age or gender. Children diagnosed with profound intellectual disabilities are less likely to exhibit comorbid psychiatric disorders.

Psychiatric disorders among persons with intellectual disabilities vary and include mood disorders, schizophrenia, ADHD, and conduct disorder. Children diagnosed with severe intellectual disabilities have an unusually high rate of comorbid autism spectrum disorder. Approximately 2 to 3 percent of those with an intellectual disability meet diagnostic criteria for schizophrenia, which is several times higher than the rate for the general population. Up to 50 percent of children and adults with intellectual disabilities meet the criteria for a mood disorder as measured on an appropriate depression scale. However, a limitation of these studies is that these instruments are not standardized within intellectual disability populations. Frequent psychiatric symptoms that occur in children with intellectual disability, outside the context of a full psychiatric disorder, include hyperactivity and short attention span, self-injurious behaviors (e.g., head-banging and self-biting), and repetitive stereotypical behaviors (hand-flapping and toe-walking). In children and adults with milder forms of intellectual disability, negative self-image, low self-esteem, poor frustration tolerance, interpersonal dependence, and a rigid problem-solving style are frequent.

ETIOLOGY

Etiologic factors in intellectual disability can be genetic, developmental, environmental, or a combination. Genetic causes include chromosomal and inherited conditions; developmental and environmental factors include prenatal exposure to infections and toxins; and environmental or acquired factors include prenatal trauma (e.g., prematurity) and sociocultural factors. The severity of intellectual disability may relate to the timing and duration of

a given trauma as well as to the degree of exposure to the central nervous system (CNS). In about three-fourths of persons diagnosed with severe intellectual disability, there is an identifiable cause, whereas the etiology is apparent in only about half of those diagnosed with mild intellectual disability. No cause is known for three-fourths of persons with IQ ranging from 70 to 80 and variable adaptive functioning. Among chromosomal disorders, Down syndrome and fragile X syndrome are the most common disorders that usually produce at least moderate intellectual disability. A prototype of a metabolic disorder associated with an intellectual disability is phenylketonuria (PKU). Deprivation of nutrition, nurturance, and social stimulation can potentially contribute to the development of at least mild forms of intellectual disability. Current knowledge suggests that genetic, environmental, biologic, and psychosocial factors work additively to the emergence of intellectual disability.

Genetic Etiologic Factors in Intellectual Disability

Single-Gene Causes. One of the most well-known single-gene causes of intellectual disability is fragile X syndrome, a mutation of the *FMR1* gene. It is the most common and first X-linked gene to be identified as a direct cause of intellectual disability. Abnormalities in autosomal chromosomes are frequently associated with intellectual disability. In contrast, aberrations in sex chromosomes can result in characteristic physical syndromes that do not include intellectual disability (e.g., Turner syndrome with XO and Klinefelter syndrome with XXY, XXXY, and XXYY variations). Some children with Turner syndrome have average to superior intelligence. An agreement exists on a few predisposing factors for chromosomal disorders—among them, advanced maternal age, increased age of the father, and x-ray radiation.

Visible and Submicroscopic Chromosomal Causes of Intellectual Disability. Trisomy 21 (Down syndrome) is a prototype of a cytogenetically visible abnormality that accounts for about two-thirds of the 15 percent of intellectual disability attributable to visible abnormal cytogenetics. Other microscopically visible chromosomal abnormalities associated with intellectual disability include deletions, translocations, and supernumerary marker chromosomes. Typically, microscopic chromosome analysis can

identify abnormalities of 5 to 10 million base pairs or higher.

Submicroscopic identification requires the use of microarrays that can identify losses of chromosomal segments too small to be picked up by light microscopy. The altered copy number variants (CNVs) in submicroscopic segments of the chromosome are associated with up to 13 to 20 percent of cases of intellectual disability. That is, the genes associated with a particular developmental abnormality are in the critical regions of the pathogenic CNVs.

Genetic Intellectual Disability and Behavioral Phenotype

Specific and predictable behaviors are associated with certain genetically based cases of intellectual disability. These behavioral phenotypes are a syndrome of observable behaviors that occur with a significantly higher probability than expected among those individuals with a specific genetic abnormality.

Examples of behavioral phenotypes occur in genetically determined syndromes such as fragile X syndrome, Prader–Willi syndrome, and Down syndrome in which we can expect specific behavioral manifestations. Persons with fragile X syndrome have incredibly high rates (up to three-fourths of those studied) of ADHD. High rates of aberrant interpersonal behavior and language function often meet the criteria for autistic disorder and avoidant personality disorder. Prader–Willi syndrome is almost always associated with compulsive eating disturbances, hyperphagia, and obesity. Socialization is an area of weakness, especially in coping skills. Externalizing behavior problems—such as temper tantrums, irritability, and arguing—seem to be heightened in adolescence.

Down Syndrome. The etiology of Down syndrome, known to be caused by an extra copy of the entire chromosome 21, makes it one of the more complicated disorders. The original description of Down syndrome, first made by the English physician Langdon Down in 1866, was based on physical characteristics associated with subnormal mental functioning. Since then, Down syndrome has been the most investigated, and most discussed, syndrome in intellectual disability. Recent data have suggested that Down syndrome may be more amenable to postnatal interventions to address the

cognitive deficits that it produces than was previously thought. Although still in the early stages of animal research, data from experiments with one mouse model, the Ts65Dn, indicate that pharmacologic interventions may influence learning and memory deficits known to occur in Down syndrome.

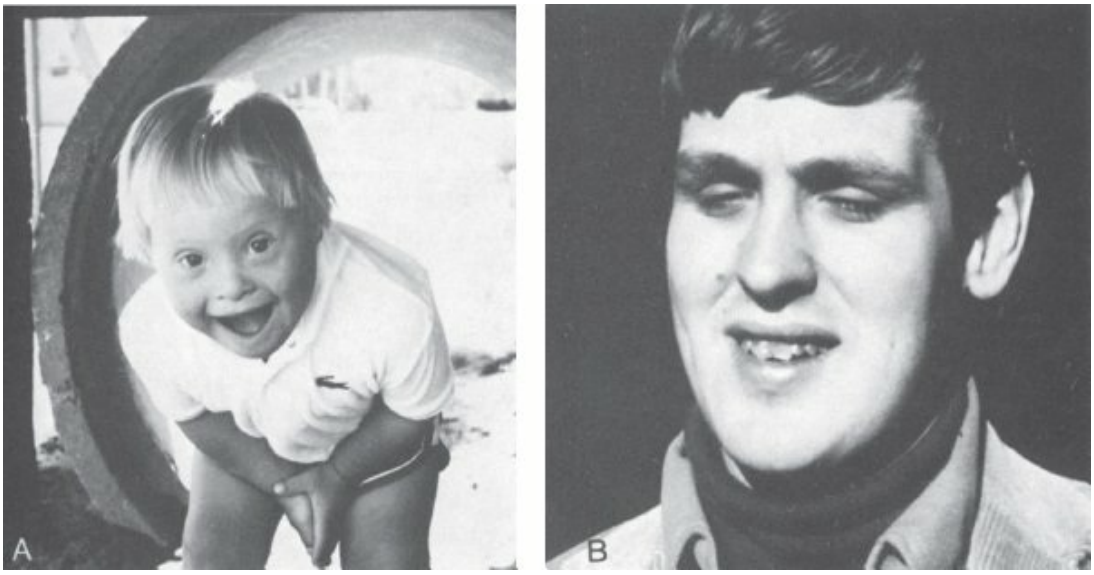


FIGURE 2-3
Down Syndrome and Fragile X Syndrome. **A**. An young child with Down syndrome. **B**. A young adult with fragile X syndrome. (Courtesy of L.S. Syzmanski, M.D. and A.C. Crocker, M.D.)

Phenotypically, children with Down syndrome have characteristic physical attributes, including slanted eyes, epicanthal folds, and a flat nose. Figure 2-3 illustrates the typical features.

There are three types of chromosomal aberrations in Down syndrome, which complicate the diagnosis. They are:

1. Patients with trisomy 21 (three chromosomes 21, instead of the usual two) represent the overwhelming majority; they have 47 chromosomes, with an extra chromosome 21. The mothers' karyotypes are normal. A nondisjunction during meiosis, occurring for unknown reasons, is held responsible for the disorder.
2. Nondisjunction occurring after fertilization in any cell division results in mosaicism, a condition in which there are both normal and trisomic cells in various tissues.
3. In translocation, a fusion occurs of two chromosomes, usually 21 and

15, resulting in a total of 46 chromosomes, despite the presence of an extra chromosome 21. The disorder, unlike trisomy 21, is usually inherited, and the translocated chromosome may occur in unaffected parents and siblings. The asymptomatic carriers have only 45 chromosomes.

Approximately 6,000 babies are affected with Down syndrome in the United States annually, which makes the incidence of Down syndrome 1 in every 700 births, or 15 per 10,000 live births. For women older than 32 years of age, the risk of having a child with Down syndrome (trisomy 21) is about 1 in 100 births, but when translocation is present, the risk is about 1 in 3. Most children with Down syndrome are mildly to moderately intellectually disabled, with a minority having an IQ above 50. Cognitive development appears to progress normally from birth to 6 months of age; IQ scores gradually decrease from near normal at 1 year of age to about 30 to 50 as development proceeds. The decline in intellectual function may not be readily apparent. Infant tests may not reveal the full extent of the deficits. According to anecdotal clinical reports, children with Down syndrome are typically placid, cheerful, and cooperative and adapt quickly at home. With adolescence, the picture changes: youth with Down syndrome may experience more social and emotional difficulties and behavior disorders, and there is an increased risk for psychotic disorders.

In Down syndrome, language function is a relative weakness, whereas sociability and social skills, such as interpersonal cooperation and conformity with social conventions, are relative strengths. Children with Down syndrome typically manifest deficits in scanning the environment; they are more likely to focus on a single stimulus, leading to difficulty noticing environmental changes. A variety of comorbid psychiatric disorders emerge in persons with Down syndrome; however, the rates appear to be lower than in children with intellectual disability and an autism spectrum disorder.

The diagnosis of Down syndrome is relatively simple in an older child, but it is often tricky in newborn infants. The most important signs in a newborn include general hypotonia; oblique palpebral fissures; abundant neck skin; a small, flattened skull; high cheekbones; and a protruding tongue. The hands are broad and thick, with a single palmar transversal crease, and the little fingers are short and curved inward. Moro reflex is weak or absent. More

than 100 signs or stigmata may occur in Down syndrome, but rarely all in one person. Commonly occurring physical problems in Down syndrome include cardiac defects, thyroid abnormalities, and gastrointestinal problems. Life expectancy was once drastically limited to about the age of 40. However, it is now significantly increased, although still shorter than those without intellectual disability.

Down syndrome includes deterioration in language, memory, self-care skills, and problem-solving by the third decade of life. Postmortem studies of individuals with Down syndrome older than age 40 have shown a high incidence of senile plaques and neurofibrillary tangles, similar to those seen in Alzheimer disease. Neurofibrillary tangles are known to occur in a variety of degenerative diseases, whereas senile plaques seem to be found most often in Alzheimer disease and Down syndrome.

Fragile X Syndrome. Fragile X syndrome is the second most common single cause of intellectual disability. The syndrome results from a mutation on the X chromosome at what is known as the fragile site (Xq27.3). The fragile site occurs in only some cells, and it may be absent in asymptomatic males and female carriers. Much variability is present in both genetic and phenotypic expression. Fragile X syndrome occurs in about 1 of every 1,000 males and 1 of every 2,000 females. The typical phenotype includes a large, long head and ears, short stature, hyperextensible joints, and postpubertal macroorchidism (see Figure 2-3). Associated intellectual disability ranges from mild to severe. The behavioral profile of persons with the syndrome includes a high rate of ADHD, learning disorders, and autism spectrum disorder. Deficits in language function include rapid perseverative speech with abnormalities in combining words into phrases and sentences. Persons with fragile X syndrome seem to have relatively strong skills in communication and socialization; their intellectual functions seem to decline in the pubertal period. Often, female carriers are less impaired than males with fragile X syndrome, but females can also manifest the typical physical characteristics and may have a mild intellectual disability.

Prader–Willi Syndrome. Prader–Willi syndrome likely results from a small deletion involving chromosome 15, occurring sporadically. Its prevalence is less than 1 in 10,000. Persons with the syndrome exhibit

compulsive eating behavior and often obesity, intellectual disability, hypogonadism, small stature, hypotonia, and small hands and feet.

Cat's Cry (Cri-du-Chat) Syndrome. Children with cat's cry syndrome have a deletion in chromosome 5. They are typically severely intellectually disabled and show many signs often associated with chromosomal aberrations, such as microcephaly, low-set ears, oblique palpebral fissures, hypertelorism, and micrognathia. The characteristic cat-like cry that gave the syndrome its name is caused by laryngeal abnormalities that gradually change and disappear with increasing age.

Phenylketonuria (PKU). Ivar Asbjörn Fölling first described PKU in 1934 as an inborn error of metabolism. PKU is transmitted as a recessive autosomal Mendelian trait and occurs in about 1 of every 10,000 to 15,000 live births. For parents who have already had a child with PKU, the chance of having another child with PKU is 20 to 25 percent of successive pregnancies. PKU is reported predominantly in persons of North European origin; a few cases may occur in African Americans, Yemenite Jews, and Asians. The underlying metabolic defect in PKU is an inability to convert phenylalanine, an essential amino acid, to paratyrosine because of the absence or inactivity of the liver enzyme phenylalanine hydroxylase, which catalyzes the conversion. Therefore, PKU is mostly preventable with a screening for it, which, if positive, should be followed with a low phenylalanine diet. Other types of hyperphenylalaninemia exist, but are rare and often fatal.

Most patients with PKU are severely intellectually disabled, but some have borderline or average intelligence. Eczema, vomiting, and convulsions occur in about one-third of all patients. Although the clinical picture varies, typically, children with PKU are reported to be hyperactive and irritable. They frequently exhibit temper tantrums and often display bizarre movements of their bodies and upper extremities, including twisting hand mannerisms. Verbal and nonverbal communication is commonly severely impaired or nonexistent. The child has reduced coordination, and they have many perceptual difficulties.

Currently, the Guthrie inhibition assay is a widely applied screening test using a bacteriologic procedure to detect phenylalanine in the blood. In the United States, screening newborn infants for PKU is routine. Early diagnosis

is essential, because a low-phenylalanine diet, in use since 1955, significantly improves both behavior and developmental progress. The best results seem to be obtained with early diagnosis and the start of dietary treatment before the child is 6 months of age. Dietary treatment, however, is not without risk. Phenylalanine is an essential amino acid, and its omission from the diet can lead to severe complications such as anemia, hypoglycemia, or edema. The dietary treatment of PKU continues indefinitely. Children who receive a diagnosis before the age of 3 months and have an optimal dietary regimen may have average intelligence. A low-phenylalanine diet does not reverse intellectual disability in untreated older children and adolescents with PKU, but the diet does decrease irritability and abnormal EEG changes and does increase social responsiveness and attention span. The parents of children with PKU and some of the children's unaffected siblings are heterozygous carriers.

Rett Syndrome. Rett syndrome, now diagnosed in the DSM-5 as a form of autism spectrum disorder, is believed to be caused by a dominant X-linked gene. It is degenerative and affects only females. In 1966, Andreas Rett reported on 22 girls with a severe progressive neurologic disability. Deterioration in communications skills, motor behavior, and social functioning starts at about 1 year of age. Symptoms include ataxia, facial grimacing, teeth-grinding, and loss of speech. Intermittent hyperventilation and a disorganized breathing pattern are characteristic while the child is awake. Stereotypical hand movements, including hand-wringing, are typical. Progressive gait disturbance, scoliosis, and seizures occur. Severe spasticity is usually present in middle childhood. Cerebral atrophy occurs with decreased pigmentation of the substantia nigra, which suggests abnormalities of the dopaminergic nigrostriatal system.

Associated features include seizures in up to 75 percent of affected children and disorganized EEG findings with some epileptiform discharges in almost all young children with Rett syndrome, even in the absence of clinical seizures. An additional associated feature is irregular respiration, with episodes of hyperventilation, apnea, and breath-holding. The disorganized breathing occurs in most patients while they are awake; during sleep, the breathing usually normalizes. Many patients with Rett syndrome also have scoliosis. As the disorder progresses, muscle tone seems to change from an

initial hypotonic condition to spasticity to rigidity.

Although children with Rett syndrome may live for well over a decade after the onset of the disorder, after 10 years, many patients are wheelchair-bound, with muscle wasting, rigidity, and virtually no language ability. Long-term receptive and expressive communication and socialization abilities remain at a developmental level of less than 1 year.

Dana was born as a full-term and healthy baby after an uncomplicated pregnancy. An amniocentesis had been obtained because of advanced maternal age of 40 years, and findings were normal. At birth, Dana received good Apgar scores and her weight, height, and head circumference were all near the 50th percentile. Her development during the first months of life was unremarkable. At approximately 8 months of age, her development seemed to wane, and her interest in the environment, including the social environment, declined. Dana's developmental milestones failed to progress, and she became markedly delayed; she was just starting to walk at her second birthday and had no spoken language. Evaluation at that time revealed that head growth had decelerated. Self-stimulatory behaviors emerged, and in addition, marked cognitive and communicative delays were noted on formal testing. Dana began to lose purposeful hand movements and developed unusual stereotypical hand-washing behaviors. By age 6, her EEG was abnormal and abnormal hand movements were prominent. Subsequently, Dana developed truncal ataxia and breath-holding spells, and motor skills further deteriorated. (Adapted from Fred Volkmar, M.D.)

Neurofibromatosis. Also called *von Recklinghausen's* disease, neurofibromatosis is the most common of the neurocutaneous syndromes caused by a single dominant gene, which may be inherited or occur as a new mutation. The disorder occurs in about 1 of 5,000 births and is characterized by café-au-lait spots on the skin and by neurofibromas, including optic gliomas and acoustic neuromas, caused by abnormal cell migration. Mild intellectual disability occurs in up to one-third of those with the disease.

Tuberous Sclerosis. Tuberous sclerosis is the second most common of the

neurocutaneous syndromes; a progressive intellectual disability occurs in up to two-thirds of all affected persons. It occurs in about 1 of 15,000 persons who inherit it through autosomal dominant transmission. Seizures are present in all those with intellectual disability, and in two-thirds of those without. Infantile spasms may occur as early as 6 months of age. The phenotypic presentation includes adenoma sebaceum and ash-leaf spots, identifiable with a slit lamp.

Childhood Disintegrative Disorder. The previous diagnosis of childhood disintegrative disorder, now included in autism spectrum disorder, is characterized by marked regression in several areas of functioning after at least 2 years of apparently normal development. Childhood disintegrative disorder, also called *Heller syndrome* and *disintegrative psychosis,* was described in 1908 as a deterioration over several months of intellectual, social, and language function occurring in 3- and 4-year-olds with previously normal function. After the deterioration, the children closely resembled children with autistic disorder.

Ron's early history was within normal limits. By age 2, he was speaking in sentences, and his development appeared to be proceeding appropriately. At 3½ years of age, he abruptly exhibited a period of marked behavioral regression shortly after the birth of a sibling. Ron lost previously acquired skills in communication and was no longer toilet trained. Ron became more withdrawn and less interested in social interaction, exhibiting various self-stimulatory behaviors repeatedly. Comprehensive medical examination failed to reveal any conditions that might account for this developmental regression. Behaviorally, Ron exhibited features of autism spectrum disorder. At follow-up at age 12, he spoke only an occasional single word and had severe mental retardation. (Adapted from Fred Volkmar, M.D.)

Lesch–Nyhan Syndrome. Lesch–Nyhan syndrome is a rare disorder caused by a deficiency of an enzyme involved in purine metabolism. The disorder is X-linked; patients have intellectual disability, microcephaly, seizures, choreoathetosis, and spasticity. The syndrome is also associated

with severe compulsive self-mutilation by biting the mouth and fingers. Lesch–Nyhan syndrome is another example of a genetically determined syndrome with a specific, predictable behavioral pattern.

Adrenoleukodystrophy. The most common of several disorders of sudanophilic cerebral sclerosis, adrenoleukodystrophy is characterized by diffuse demyelination of the cerebral white matter resulting in visual and intellectual impairment, seizures, spasticity, and progression to death. Adrenocortical insufficiency accompanies the cerebral degeneration. A sex-linked gene located on the distal end of the long arm of the X chromosome transmits the disorder. The clinical onset is generally between 5 and 8 years of age, with early seizures, disturbances in gait, and mild intellectual impairment. Abnormal pigmentation reflecting adrenal insufficiency sometimes precedes the neurologic symptoms, and attacks of crying are frequent. Spastic contractures, ataxia, and swallowing disturbances are also frequent. Although the course is often rapidly progressive, some patients may have a relapsing and remitting course.

Maple Syrup Urine Disease. The clinical symptoms of maple syrup urine disease appear during the first week of life. The infant deteriorates rapidly and has decerebrate rigidity, seizures, respiratory irregularity, and hypoglycemia. If untreated, maple syrup urine disease is usually fatal in the first months of life, and the survivors have a severe intellectual disability. Some variants have transient ataxia and only mild intellectual disability. Treatment follows the general principles established for PKU and consists of a diet deficient in the three involved amino acids—leucine, isoleucine, and valine.

Other Enzyme Deficiency Disorders. We have identified several enzyme deficiency disorders associated with intellectual disability, and are adding more as discoveries are made, including Hartnup disease, galactosemia, and glycogen-storage disease. Table 2-3 lists 30 significant disorders with inborn errors of metabolism, hereditary transmission patterns, defective enzymes, clinical signs, and relation to intellectual disability.

Acquired and Developmental Factors

Prenatal Period. Essential prerequisites for the overall development of the fetus include the mother's physical, psychological, and nutritional health during pregnancy. Many illnesses and conditions can affect fetal brain development, including uncontrolled diabetes, anemia, emphysema, hypertension, and long-term use of alcohol and narcotic substances. Maternal infections during pregnancy, especially viral infections, have been known to cause fetal damage and intellectual disability. The extent of fetal damage depends on several factors. These factors include the type and severity of the viral infection and the gestational age of the fetus.

Rubella (German Measles). Rubella has replaced syphilis as the primary cause of congenital malformations and intellectual disability caused by maternal infection. The children of affected mothers may show several abnormalities, including congenital heart disease, intellectual disability, cataracts, deafness, microcephaly, and microphthalmia. Timing is crucial because the extent and frequency of the complications are inversely related to the duration of the pregnancy at the time of maternal infection. When mothers become infected in the first trimester of pregnancy, 10 to 15 percent of the children are affected. This incidence rises to almost 50 percent when the infection occurs in the first month of pregnancy. Undetected subclinical forms of maternal infection can complicate the situation. Immunization can prevent maternal rubella.

Cytomegalic Inclusion Disease. In many cases, cytomegalic inclusion disease remains dormant in the mother. Some children are stillborn, and others have jaundice, microcephaly, hepatosplenomegaly, and radiographic findings of intracerebral calcification. Children with intellectual disability from the disease frequently have cerebral calcification, microcephaly, or hydrocephalus. Positive findings of the virus in throat and urine cultures and the recovery of inclusion-bearing cells in the urine all can confirm the diagnosis.

Syphilis. Syphilis in pregnant women was once the leading cause of various neuropathologic changes in their offspring, including intellectual disability. Today, the incidence of syphilitic complications of pregnancy fluctuates with the incidence of syphilis in the general population. Some recent alarming

statistics from several major cities in the United States indicate that there is still no room for complacency.

Toxoplasmosis. The mother can transmit toxoplasmosis to the fetus. It causes mild or severe intellectual disability and, in severe cases, hydrocephalus, seizures, microcephaly, and chorioretinitis.

Herpes Simplex. The herpes simplex virus can be transmitted transplacentally, although the most common mode of infection is during birth. Microcephaly, intellectual disability, intracranial calcification, and ocular abnormalities may result.

Human Immunodeficiency Virus (HIV). Cognitive impairments are well known to be associated with the transmission of HIV from mothers to their babies. HIV may have both direct and indirect influences on the developing brain. A subset of infants born infected with HIV may develop progressive encephalopathy, intellectual disabilities, and seizures within the first year of life. Fortunately, over the last two decades, there has been a dramatic decrease in perinatal HIV transmission due to a combination of antiviral agents provided to mothers during pregnancy and delivery, obstetric interventions that reduce risk, and administration of zidovudine (ZDV) as prophylaxis for 6 weeks to newborns exposed to HIV. In the United States, the annual number of HIV infections through perinatal transmission had declined about 40 percent over the last 5 years and is dramatically less than the 1990s when the transmission was at its peak. In 2015 the rates were less than 1 percent in white and Hispanic/Latino HIV-infected mothers. However, that rate is about five times greater in African American mothers with HIV. Vertical transmission of HIV from mother to child around the world, especially in Africa, is also considerable.

Fetal Alcohol Syndrome. Fetal alcohol syndrome (FAS) results from prenatal alcohol exposure and can lead to a wide range of problems in the newborn. According to the Centers for Disease Control and Prevention, FAS in the United States occurs at a rate ranging from 0.2 to 1.5 per 1,000 live births. FAS is one of the leading preventable causes of intellectual disability and physical disabilities. Table 2-4 lists the typical characteristics of FAS,

and Figure 2-4 demonstrates examples of the typical defects.

The entire syndrome occurs in up to 15 percent of babies born to women who regularly ingest large amounts of alcohol. Babies born to women who consume alcohol regularly during pregnancy have a high incidence of ADHD, learning disorders, and intellectual disability without facial dysmorphism.

Prenatal Drug Exposure. Prenatal exposure to opioids, such as heroin, often results in infants who are small for their gestational age, with head circumference below the tenth percentile and withdrawal symptoms that appear within the first 2 days of life. The withdrawal symptoms of infants include irritability, hypertonia, tremor, vomiting, a high-pitched cry, and an abnormal sleep pattern. Seizures are unusual, but the withdrawal syndrome can be life-threatening to infants if it is untreated. Diazepam, phenobarbital, chlorpromazine, and paregoric have been used to treat neonatal opioid withdrawal. The long-term sequelae of prenatal opioid exposure are not fully known; the children's developmental milestones and intellectual functions may be within the normal range, but they have an increased risk for impulsivity and behavioral problems. Infants prenatally exposed to cocaine are at high risk for low birth weight and premature delivery. In the early neonatal period, they may have transient neurologic and behavioral abnormalities, including abnormal results on EEG studies, tachycardia, poor feeding patterns, irritability, and excessive drowsiness. The physiologic and behavioral abnormalities are from cocaine intoxication, not withdrawal, as cocaine excretion may take up to a week after.



Table 2-3
Impairment in Disorders with Inborn Errors of Metabolism

Disorder	Hereditary Transmission[a]	Enzyme Defect	Prenatal Diagnosis	Intellectual Disability	Clinical Signs
I. LIPID METABOLISM					
Niemann-Pick disease					
Group A, infantile		Unknown			Hepatomegaly
Group B, adult	A.R.	Sphingomyelinase	+	±	Hepatosplenomegaly
Groups C and D, intermediate		Unknown	–	+	Pulmonary infiltration
Infantile Gaucher disease	A.R.	β-Glucosidase	+	±	Hepatosplenomegaly, pseudobulbar palsy
Tay–Sachs disease	A.R.	Hexosaminidase A	+	+	Macular changes, seizures, spasticity
Generalized gangliosidosis	A.R.	β-Galactosidase	+	+	Hepatosplenomegaly, bone changes
Krabbe disease	A.R.	Galactocerebroside β-Galactosidase	+	+	Stiffness, seizures
Metachromatic leukodystrophy	A.R.	Cerebroside sulfatase	+	+	Stiffness, developmental failure
Wolman disease	A.R.	Acid lipase	+	–	Hepatosplenomegaly, adrenal calcification, vomiting, diarrhea
Farber lipogranulomatosis	A.R.	Acid ceramidase	+	+	Hoarseness, arthropathy, subcutaneous nodules
Fabry disease	X.R.	β-Galactosidase	+	–	Angiokeratomas, renal failure
II. MUCOPOLYSACCHARIDE METABOLISM					
Hurler syndrome MPS I	A.R.	Iduronidase	+	+	?
Hurler disease II	X.R.	Iduronate sulfatase	+	+	?
Sanfilippo syndrome III	A.R.	Various sulfatases (types A–D)	+	+	Varying degrees of bone changes, hepatosplenomegaly, joint restriction, etc.
Morquio disease IV	A.R.	*N*-Acetylgalactosamine-6-sulfate sulfatase	+	–	?
Maroteaux–Lamy syndrome VI	A.R.	Arylsulfatase B	+	±	?
III. OLIGOSACCHARIDE AND GLYCOPROTEIN METABOLISM					
I-cell disease	A.R.	Glycoprotein *N*-acetylglucosaminyl-phospho transferase	+	+	Hepatomegaly, bone changes, swollen gingivae
Mannosidosis	A.R.	Mannosidase	+	+	Hepatomegaly, bone changes, facial coarsening
Fucosidosis	A.R.	Fucosidase	+	+	Same as above
IV. AMINO ACID METABOLISM					
Phenylketonuria	A.R.	Phenylalanine hydroxylase	–	+	Eczema, blonde hair, musty odor
Hemocystinuria	A.R.	Cystathionine β-synthetase	+	+	Ectopia lentis, Marfan-like phenotype, cardiovascular anomalies
Tyrosinosis	A.R.	Tyrosine amine transaminase	–	+	Hyperkeratotic skin lesions, conjunctivitis
Maple syrup urine disease	A.R.	Branched-chain ketoacid decarboxylase	+	+	Recurrent ketoacidosis
Methylmalonic acidemia	A.R.	Methylmalonyl-CoA mutase	+	+	Recurrent ketoacidosis, hepatomegaly, growth retardation
Propionic acidemia	A.R.	Propionyl-CoA carboxylase	+	+	Same as above
Nonketotic hyperglycinemia	A.R.	Glycine cleavage enzyme	+	+	Seizures
Urea cycle disorders	Mostly A.R.	Urea cycle enzymes	+	+	Recurrent acute encephalopathy, vomiting
Hartnup disease	A.R.	Renal transport disorder	–	–	None consistent
V. OTHERS					
Galactosemia	A.R.	Galactose-1-phosphate uridyltransferase	+	+	Hepatomegaly, cataracts, ovarian failure
Wilson's hepatolenticular degeneration	A.R.	Unknown factor in copper metabolism	–	±	Liver disease, Kayser-Fleischer ring, neurologic problems
Menkes kinky-hair disease	X.R.	Same as above	+	–	Abnormal hair, cerebral degeneration
Lesch–Nyhan syndrome	X.R.	Hypoxanthine guanine phosphoribosyltransferase	+	+	Behavioral abnormalities

[a]A.R., autosomal recessive transmission; X.R., X-linked recessive transmission.

Adapted from Leroy JC. Hereditary, development, and behavior. In: Levine MD, Carey WB, Crocker AC, eds. *Developmental-Behavioral Pediatrics*. Philadelphia, PA: WB Saunders; 1983:315, with permission.



Table 2-4
Characteristics of Fetal Alcohol Syndrome

Growth retardation of prenatal origin (height, weight)
Facial dysmorphism
- Microcephaly (head circumference below the third percentile)
- Hypertelorism (large distance between eyes)
- Microphthalmia (small eyeballs)
- Short palpebral fissures
- Inner epicanthal folds
- Midface hypoplasia (underdevelopment)
- Smooth or short philtrum
- Thin upper lip
- Short, turned up nose

Cardiac Defects
Central nervous system (CNS) manifestations
- Delayed development
- Hyperactivity
- Attention deficits
- Learning disabilities
- Intellectual deficits
- Seizures

Complications of Pregnancy. Toxemia of pregnancy and uncontrolled maternal diabetes present hazards to the fetus and can potentially result in intellectual disability. Maternal malnutrition during pregnancy often results in prematurity and other obstetrical complications. Vaginal hemorrhage, placenta previa, premature separation of the placenta, and prolapse of the cord can damage the fetal brain by causing anoxia.

Perinatal Period. Some evidence indicates that premature infants and infants with low birth weight are at high risk for neurologic and subtle intellectual impairments that may not be apparent until their school years. Infants who sustain intracranial hemorrhages or show evidence of cerebral ischemia are especially vulnerable to cognitive abnormalities. The degree of neurodevelopmental impairment generally correlates with the severity of the intracranial hemorrhage. Recent studies have documented that, among

children with low birth weight (less than 1 kg), 20 percent had significant disabilities, including cerebral palsy, intellectual disability, autism, and low intelligence with severe learning problems. Very premature children and those who had intrauterine growth retardation were at high risk for developing both social problems and academic difficulties. Socioeconomic deprivation can also affect the adaptive function of these vulnerable infants. Early intervention may improve their cognitive, language, and perceptual abilities.

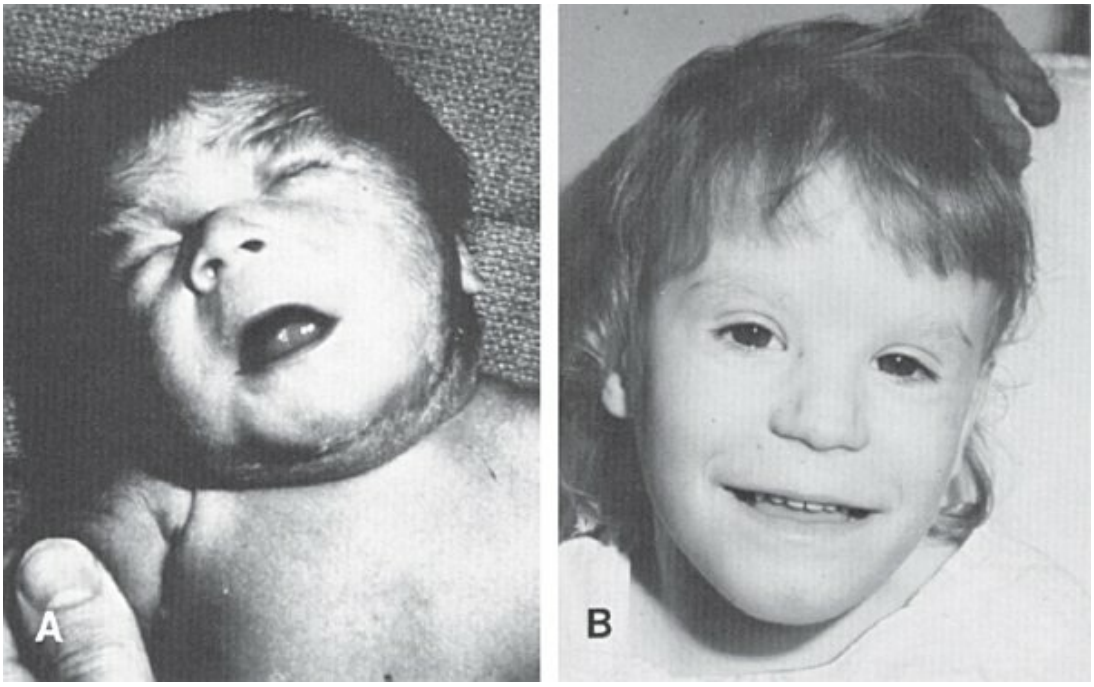


FIGURE 2-4
Fetal alcohol syndrome. Photographs of children with "fetal-alcohol syndrome." **A**. Severe case. **B**. Slightly affected child. Note in both children the short palpebral fissures and hypoplasia of the maxilla. Usually, the defect includes other craniofacial abnormalities. Cardiovascular defects and limb deformities are also common symptoms of the fetal alcohol syndrome. (From Langman J. *Medical Embryology*. 7th ed. Philadelphia, PA: Williams & Wilkins; 1995:108, with permission.)

Acquired Childhood Disorders

Infection. The most severe infections affecting cerebral integrity are encephalitis and meningitis. The universal use of the measles vaccine had virtually eliminated measles encephalitis. However, a recent refusal by some individuals and communities to vaccinate their children has increased the risk

of this as well as subacute sclerosing panencephalitis, a rare and usually fatal degenerative disease that can occur years after the infection. Antibacterial agents have reduced the incidence of other bacterial infections, at least in the developed world. Viruses now cause most episodes of encephalitis. Sometimes a clinician must retrospectively consider a probable encephalitic component in a previously unknown illness with a high fever. A delayed diagnosis of meningitis, even when followed by antibiotic treatment, can seriously affect a child's cognitive development. Thrombotic and purulent intracranial phenomena secondary to septicemia are rarely seen today except in small infants.

Head Trauma. The best-known causes of head injury in children that produce developmental handicaps, including seizures, are motor vehicle accidents (household accidents are the most common cause of head injuries, such as falls from tables, open windows, and on stairways). Child maltreatment is a frequent cause of head traumas or intracranial trauma such as bleeding due to "shaken baby" syndrome.

Asphyxia. Brain damage due to asphyxia associated with near-drowning is not an uncommon cause of intellectual disability.

Long-Term Exposures. Long-term exposure to lead is a well-established cause of compromised intelligence and learning skills. Intracranial tumors of various types and origins, surgery, and chemotherapy can also adversely affect brain function.

Environmental and Sociocultural Factors

Mild intellectual disability is associated with significant deprivation of nutrition and nurturance. Children who have endured these conditions are at risk for a host of psychiatric disorders, including mood disorders, posttraumatic stress disorder (PTSD), and attentional and anxiety disorders. A prenatal environment compromised by poor medical care and poor maternal nutrition may be contributing factors in the development of mild intellectual disability. Teenage pregnancies are at risk for mild intellectual disability in the baby due to the increased risk of obstetrical complications, prematurity, and low birth weight. Poor postnatal medical care, malnutrition,

exposure to toxic substances such as lead, and potential physical trauma are additional risk factors for mild intellectual disabilities. Child neglect and inadequate caretaking may deprive an infant of both physical and emotional nurturances, leading to failure to thrive syndromes.

2.2 Communication Disorders

Communication disorders range from mild delays in acquiring language to expressive or mixed receptive–expressive disorders, phonologic disorders, and stuttering, which may remit spontaneously or persist into adolescence or even adulthood. Language delay is one of the most common very early childhood developmental delays, affecting up to approximately 7 percent of 5-year-olds. The rates of language disorders are understandably higher in preschoolers than in school-age children; rates were close to 20 percent of 4-year-olds in the Early Language in Victoria Study (ELVS). To communicate effectively, children must have a mastery of multiple aspects of language—that is, the ability to understand and express ideas—using words and speech to express themselves in vernacular language. In DSM-5, Language Disorder includes both expressive and mixed receptive–expressive problems. DSM-5 speech disorders include Speech Sound Disorder (formerly known as Phonologic Disorder) and Childhood-Onset Fluency Disorder (Stuttering). Children with expressive language deficits have difficulties expressing their thoughts with words and sentences at a level of sophistication expected for their age and developmental level in other areas. These children may struggle with a limited vocabulary, speak in sentences that are short or ungrammatical, and often present descriptions of situations that are disorganized, confusing, and infantile. They may be slow to develop an understanding and a memory of words compared with others their age. Children with language disorders are at higher risk for developing reading difficulties. Current expert consensus considers reading comprehension impairment a form of language impairment, distinct from other reading deficits such as dyslexia.

Language and speech are pragmatically intertwined, despite the distinct categories of language disorders and speech disorders in DSM-5. Language competence spans four domains: phonology, grammar, semantics, and

pragmatics. *Phonology* refers to the ability to produce sounds that constitute words in a given language and the skills to discriminate the various phonemes (sounds that are made by a letter or group of letters in a language). A child must be able to produce the sounds of a word to imitate them. *Grammar* designates the organization of words and the rules for placing words in an order that makes sense in that language. *Semantics* refers to the organization of concepts and the acquisition of words themselves. A child draws from a mental list of words to produce sentences. Children with language impairments exhibit a wide range of difficulties with semantics that includes acquiring new words, storage and organization of known words, and word retrieval. Speech and language evaluations that are sufficiently broad to test all of the other skill levels will be more accurate in evaluating a child's remedial needs. *Pragmatics* is a branch of linguistics concentrating on the skill of using language. It involves the ability to understand the context of one's speech and how to interact and converse. It requires not only understanding the literal meaning of a sentence but also the speaker's intention. By age 2 years, toddlers without speech or language delay may know a few words or up to 200 words, and by age 3 years, most children understand the basic rules of language and can converse effectively. Table 2-5 provides an overview of typical milestones in language and nonverbal development.

Over the last decade, there have been an increasing number of investigative studies of speech and language interventions with positive outcomes identified in numerous areas of language. These include improvements in expressive vocabulary, syntax usage, and overall phonologic development. Most interventions are targeted strategies for the child's particular deficit and delivered by speech and language therapists.

LANGUAGE DISORDER

Language disorder consists of difficulties in the acquisition and use of language across many modalities, including spoken and written, due to deficits in comprehension or production based on both expressive and receptive skills. These deficits include reduced vocabulary, limited abilities in forming sentences using the rules of grammar, and impairments in conversing based on difficulties using vocabulary to connect sentences in descriptive

ways. Table 2-6 lists the approaches to diagnosing language disorders.

Expressive Language Deficits

Expressive language deficits are present when a child demonstrates a selective deficit in expressive language development relative to receptive language skills and nonverbal intellectual function. Infants and young children with typically developing expressive language will laugh and coo by about 6 months of age, babble and verbalize syllables such as *mama* or *dada* by about 9 months, and by 1 year, babies imitate vocalizations and can often speak at least one word. Expressive speech and language generally continue to develop in a stepwise fashion, so that at a year and a half, children typically can say a handful of words, and by 2 years, children generally are combining words into simple sentences. By the age of 2½ years, children can name actions in a picture and can make themselves understood through their verbalizations about half of the time. By 3 years, most children can speak understandably and can name a color and describe what they see with several adjectives. At 4 years, children typically can name at least four colors and can converse understandably. In the early years, before entering preschool, the development of proficiency in vocabulary and language usage is highly variable and influenced by the amount and quality of verbal interactions with family members. After beginning school, the level of verbal engagement in school can significantly influence a child's language skills. A child with expressive language deficits may demonstrate a verbal intellectual level that appears to be depressed compared with the child's overall IQ. A child with expressive language problems is likely to function below the expected levels of acquired vocabulary, correct tense usage, complex sentence constructions, and word recall. Children with expressive language deficits often present verbally as younger than their age. Language disability can be acquired during childhood (e.g., secondary to a trauma or a neurologic disorder), although less frequently, or it can be developmental; it is usually congenital, without an apparent cause. Most childhood language disorders fall into the developmental category. In either case, deficits in receptive skills (language comprehension) or expressive skills (ability to use language) can occur. Expressive language disturbance often appears in the absence of comprehension difficulties, whereas receptive dysfunction generally

diminishes proficiency in the expression of language. Children with expressive language disturbance alone have better prognoses, and less interference with learning, than children with mixed receptive–expressive language disturbances.



Table 2-5
Language Development

Age and Stage of Development	Mastery of Comprehension	Mastery of Expression
0–6 mo	Shows startle response to loud or sudden sounds; Attempts to localize sounds, turning eyes or head; Appears to listen to speakers, may respond with smile; Recognizes warning, angry, and friendly voices; Responds to hearing own name	Has vocalizations other than crying; Has differential cries for hunger, pain; Makes vocalizations to show pleasure; Plays at making sounds; Babbles (a repeated series of sounds)
7–11 mo Attending-to-Language	Shows listening selectivity (voluntary control over responses to sounds); Listens to music or singing with interest; Recognizes “no,” “hot,” own name; Looks at pictures being named for	Responds to own name with vocalizations; Imitates the melody of utterances; Uses jargon (own language); Has gestures (shakes head for no); Has exclamation (“oh-oh”); Plays language games (pat-a-cake, peekaboo)

	up to 1 min; Listens to speech without being distracted by other sounds	
12–18 mo Single-Word	Shows gross discriminations between dissimilar sounds (bells vs. dog vs. horn vs. mother's or father's voice); Understands basic body parts, names of common objects; Acquires understanding of some new words each week; Can identify simple objects (baby, ball, etc.) from a group of objects or pictures; Understands up to 150 words by age 18 mo	Uses single words (mean age of first word is 11 mo; by age 18 mo, child is using up to 20 words); "Talks" to toys, self, or others using long patterns of jargon and occasional words; Approximately 25% of utterances are intelligible; All vowels articulated correctly; Initial and final consonants often omitted
12–24 mo Two-Word Messages	Responds to simple directions ("Give me the ball") Responds to action commands ("Come here," "Sit down") Understands	Uses two-word utterances ("Mommy sock," "all gone," "ball here"); Imitates environmental sounds in play ("moo," "mmm, mmm," etc.); Refers to self by name, begins to use pronouns;

	pronouns (me, him, her, you) Begins to understand complex sentences (“When we go to the store, I’ll buy you some candy”)	Echoes two or more last words of sentences; Begins to use three-word telegraphic utterances (“all gone ball,” “me go now”); Utterances 26–50% intelligible; Uses language to ask for needs
24–36 mo Grammar Formation	Understands small body parts (elbow, chin, eyebrow); Understands family name categories (grandma, baby) Understands size (little one, big one) Understands most adjectives Understands functions (why do we eat, why do we sleep)	Uses real sentences with grammatical function words (can, will, the, a); Usually announces intentions before acting; “Conversations” with other children, usually just monologues; Jargon and echolalia gradually drop from speech; Increased vocabulary (up to 270 words at 2 yr, 895 words at 3 yr); Speech 50–80% intelligible; P, b, m articulated correctly; Speech may show rhythmic disturbances
36–54 mo Grammar Development	Understands prepositions (under, behind, between) Understands many words (up to 3,500	Correct articulation of *n, w, ng, h, t, d, k, g*; Uses language to relate incidents from the past; Uses wide range of grammatical forms: plurals, past tense,

	at 3 yr, 5,500 at 4 yr) Understands cause and effect (What do you do when you're hungry? cold?) Understands analogies (Food is to eat, milk is to ____)	negatives, questions; Plays with language: rhymes, exaggerates; Speech 90% intelligible, occasional errors in the ordering of sounds within words; Able to define words; Egocentric use of language rare; Can repeat a 12-syllable sentence correctly; Some grammatical errors still occur
55 mo on True Communication	Understands concepts of number, speed, time, space; Understands left and right; Understands abstract terms; Is able to categorize items into semantic classes	Uses language to tell stories, share ideas, and discuss alternatives; Increasing use of varied grammar; Spontaneous self-correction of grammatical errors; Stabilizing of articulation f, v, s, z, l, r, th, and consonant clusters; Speech 100% intelligible

Reprinted from Rutter M, Hersov L, eds. *Child and Adolescent Psychiatry*. London: Blackwell; 1985, with permission.



Table 2-6
Language Disorders

	DSM-5	ICD-10
Diagnostic name	Language Disorder	Specific Developmental Disorders of Speech and Language

Duration	Onset during early development	Onset during early development
Symptoms	Difficulties with acquiring language skills, symptoms include decreased/limited: • Vocabulary • Sentence structure • Discourse Symptoms beginning during school-age years (but may become obvious during later part of academic career)	**Specific speech articulation disorder:** speech sounds **Expressive language disorder:** spoken language **Receptive language disorder:** understanding of language **Acquired aphasia with epilepsy (Landau–Kleffner):** sudden loss of expressive and receptive skills after onset of paroxysmal EEG abnormalities **Other development disorders of speech and language:** e.g., lisping
Exclusions (not result of):	Age appropriate language development Sensory impairment Motor dysfunction Intellectual disability A neurologic disorder Another medical disorder	Mechanical speech abnormality Sensory impairment Mental retardation Neurologic disorder (aphasias, etc.) Environmental factors

Although language use depends on both expressive and receptive skills, the degree of deficits in a given individual may be severe in one area, and hardly impaired at all in the other. Thus, language disorders can exist in children with expressive language disturbance in the absence of receptive language problems, or both receptive and expressive language syndromes can be

present. In general, when receptive skills are sufficiently impaired to warrant a diagnosis, expressive skills are also impaired. In DSM-5, a language disorder is not limited to developmental language disabilities; it also includes acquired forms of language disturbances. To meet the DSM-5 criteria for language disorder, patients must have scores on standardized measures of expressive or receptive language markedly below those of standardized nonverbal IQ subtests and standardized tests.

Clinical Features. Children with expressive language deficits are vague when telling a story and use many filler words such as “stuff” and “things” instead of naming specific objects.

The essential feature of expressive deficits in language disorder is marked impairment in the development of age-appropriate expressive language, which results in the use of verbal or sign language markedly below the expected level given a child’s nonverbal intellectual capacity. Language understanding (decoding) skills remain relatively intact. When severe, the disorder becomes recognizable by about the age of 18 months, when a child fails to utter spontaneously or even echo single words or sounds. Even simple words, such as “Mama” and “Dada,” are absent from the child’s active vocabulary, and the child points or uses gestures to indicate desires. The child seems to want to communicate, maintains eye contact, relates well to the mother, and enjoys games such as pat-a-cake and peek-a-boo. The child’s vocabulary is severely limited. At 18 months, the child may only point at common objects when they are named.

When a child with expressive language deficits begins to speak, language impairment gradually becomes apparent. Articulation is often immature; numerous articulation errors occur but are inconsistent, particularly with such sounds as *th, r, s, z, y,* and *l,* which the child either omits or substitutes with other sounds.

By the age of 4 years, most children with expressive language disturbance can speak in short phrases but may have difficulty retaining new words. After beginning to speak, they acquire language more slowly than do most children. Their use of various grammatical structures is also markedly below the age-expected level, and they may have delayed milestones. Emotional problems involving poor self-image, frustration, and depression may develop in school-age children.

Diagnosis. We should diagnose a language disorder of the expressive disturbance type when a child has a selective deficit in language skills and is functioning well in nonverbal areas. Markedly below-age-level verbal or sign language, accompanied by a low score on standardized expressive verbal tests, is diagnostic of expressive deficits in language disorder. Although expressive language deficits often occur in children with autism spectrum disorders, these disturbances also frequently occur in the absence of autism spectrum disorder and are characterized by the following features: limited vocabulary, simple grammar, and variable articulation. “Inner language” or the appropriate use of toys and household objects is present. One assessment tool, the *Carter Neurocognitive Assessment,* itemizes and quantifies skills in areas of social awareness, visual attention, auditory comprehension, and vocal communication even when there are compromised expressive language and motor skills in very young children—up to 2 years of age. Also, we should administer standardized expressive language and nonverbal intelligence tests. Observations of children’s verbal and sign language patterns in various settings (e.g., schoolyard, classroom, home, and playroom) and during interactions with other children help ascertain the severity and specific areas of a child’s impairment and aid in early detection of behavioral and emotional complications. Family history should include the presence or absence of expressive language disorder among relatives.

Damien was a friendly, alert, and hyperactive 2-year-old, whose expressive vocabulary was limited to only two words (*mama, daddy*). He used these words one at a time in inappropriate situations. He supplemented his infrequent verbal communications with pointing and other simple gestures to request desired objects or actions. He was unable to communicate for other purposes (e.g., commenting or protesting). Damien appeared to be developing typically in other areas, especially in gross motor skills, although his fine motor skills were also poor. Damien sat, stood, and walked, and played happily with other children, enjoying activities and toys that were appropriate for 2-year-olds. Although he had a history of frequent ear infections, a recent hearing test revealed normal hearing. Despite his expressive limitations, Damien exhibited age-appropriate comprehension for the names of familiar objects and actions

and simple verbal instructions (e.g., “Put that down.” “Get your shirt.” “Clap your hands.”). However, due to his hyperactivity and impulsivity, he often required multiple directions to complete a simple task.

Despite Damien’s slow start in language development, his pediatrician had reassured his parents that most of the time, toddlers like Damien spontaneously overcome their initial slow start in language development. Fortunately, Damien’s language delay spontaneously remitted by the time he entered preschool at 3½ years of age, although he showed symptoms of an attention-deficit/hyperactivity disorder.

Jessica was a sociable, active 5-year-old, who presented with a language disorder. She was well-liked in kindergarten despite her language deficits and played with many of her classmates. During an activity in which each student recounted the story of Little Red Riding Hood to her doll, Jessica’s classmate’s story began: “Little Red Riding Hood was taking a basket of food to her grandmother who was sick. A bad wolf stopped Red Riding Hood in the forest. He tried to get the basket away from her, but she would not give it to him.”

Jessica tried to avoid participating, but when called on, her version sounded quite different. Jessica struggled and came up with: “Riding Hood going to grandma house. Her taking food. Bad wolf in a bed. Riding Hood say, what big ears, and grandma? Hear you, dear. What big eyes, grandma? See you, dear. What big mouth, grandma? Eat you all up!”

Jessica’s story was characteristic of expressive language deficits at her age: including short, incomplete sentences; simple sentence structures; omission of grammatical function words (e.g., *is* and *the*) and inflectional endings (e.g., possessives and present tense verbs); problems in question formation; and incorrect use of pronouns (e.g., *her* for *she*). Jessica, however, understood the story as well as her classmates. Jessica also demonstrated adequate comprehension skills in her kindergarten classroom, where she readily followed the teacher’s complex, multistep verbal instructions (e.g., “After you write your name in the top left corner of your paper, get your crayons and scissors, put your library books under

your chair, and line up at the back of the room.").

Ramon was a quiet, sullen 8-year-old boy whose expressive language problems had improved over time and were no longer evident in play with peers. His speech now rarely contained the incomplete sentences and grammatical errors that were so evident when he was younger. Ramon's expressive problems, however, were still impairing him in tasks involving abstract use of language, and he was struggling in his third-grade academic work. An example was Ramon's explanation of a recent science experiment: "The teacher had stuff in some jars. He poured it, and it got pink. The other thing made it white." Although each sentence was grammatical, his explanation was difficult to follow, because he vaguely explained critical ideas and details. Ramon also showed problems in word-finding, and he relied on vague and nonspecific terms, such as thing, *stuff,* and *got.*

In the first and second grades, Ramon had struggled to keep up with his classmates in reading, writing, and other academic skills. By third grade, however, the increasing demands for written work were beyond his abilities. Ramon's written work was disorganized and lacked specificity. Also, classmates began to tease him about his difficulties, and he was ashamed of his disability and reacted quite aggressively, often leading to physical fighting. Nonetheless, Ramon continued to show relatively good comprehension of spoken language, including classroom teaching concerning abstract concepts. He also comprehended sentences that were grammatically and conceptually complex (e.g., "The car the truck hit had hubcaps that were stolen. Had it been possible, she would have notified us by mail or by phone.")

PATHOLOGY AND LABORATORY EXAMINATION. Children with speech and language disorders should have an audiogram to rule out hearing loss.

Course and Prognosis. The prognosis for expressive language disturbance worsens the longer it persists in a child; prognosis is also dependent on the severity of the disorder. Studies of infants and toddlers who

are “late talkers” concur that 50 to 80 percent of these children master language skills that are within the expected level during the preschool years. Most children with language delays later catch up in their preschool years. Other comorbid disorders may influence the outcome of expressive language deficits. If children do not develop mood disorders or disruptive behavior problems, the prognosis is better. The rapidity and extent of recovery depend on the severity of the disorder, the child’s motivation to participate in speech and language therapy, and the timely initiation of therapeutic interventions. The presence or absence of hearing loss, or intellectual disability, impedes remediation and leads to a worse prognosis. Up to 50 percent of children with mild expressive language disorder recover spontaneously without any sign of language impairment, but those children with a severe expressive speech disorder may persist in exhibiting some symptoms into middle childhood or later.

Current literature shows that children who demonstrate poor comprehension, poor articulation, or poor academic performance tend to continue to have problems in these areas at follow-up 7 years later. There is also an association between particular language impairment profiles and persistent mood and behavior problems. Children with poor comprehension associated with expressive difficulties seem to be more socially isolated and impaired concerning peer relationships.

Differential Diagnosis. Language disorders are associated with various psychiatric disorders, including other learning disorders and ADHD, and in some cases, the language disorder is challenging to separate from another dysfunction. In mixed receptive–expressive language disorder, language comprehension (decoding) is markedly below the expected age-appropriate level, whereas in expressive language disorder, language comprehension remains within normal limits.

In autism spectrum disorders, children often have impaired language, symbolic and imagery play, appropriate use of gesture, or capacity to form typical social relationships. In contrast, children with expressive language disorder become very frustrated with their disorder and are usually highly motivated to make friends despite their disability.

Children with acquired aphasia or dysphasia have a history of early normal language development; the disordered language had its onset after head

trauma or other neurologic disorder (e.g., a seizure disorder). Children with selective mutism have normal language development. Often these children will speak only in front of family members (e.g., mother, father, and siblings). Children affected by selective mutism are socially anxious and withdrawn outside the family. Table 2-7 lists the differential diagnosis for language disorders.



Table 2-7
Differential Diagnosis of Language Disorder

	Hearing Impairment	Intellectual Disability	Autism Spectrum Disorder	Expressive Language Deficits Disturbance	Receptive–Expressive Language Deficits Disturbance	Selective Mutism	Speech Sound Disorder
Language comprehension	–	–	–	+	–	+	+
Expressive language	–	–	–	–	–	Variable	+
Audiogram	–	+	+	+	Variable	+	+
Articulation	–	–	– (Variable)	– (Variable)	– (Variable)	+	–
Inner language	+	+ (Limited)	–	+	+ (Slightly limited)	+	+
Uses gestures	+	+ (Limited)	–	+	+	+ (Variable)	+
Echoes	–	+	+ (Inappropriate)	+	+	+	+
Attends to sounds	Loud or low frequency only	+	–	+	Variable	+	+
Watches faces	+	+	–	+	+	+	+
Performance	+	–	+	+	+	+	+

+, normal; –, abnormal.
Adapted from Dennis Cantwell, M.D. and Lorian Baker, Ph.D, 1991.

Comorbidity. Children with language disorders have above-average rates of comorbid psychiatric disorders. In one large study of children with speech and language disorders, the most common comorbid disorders were ADHD (19 percent), anxiety disorders (10 percent), oppositional defiant disorder, and conduct disorder (7 percent combined). Children with expressive language disorder are also at higher risk for a speech disorder, receptive difficulties, and other learning disorders. Many disorders—such as reading disorder, developmental coordination disorder, and other communication disorders—are associated with expressive language disturbance. Children with expressive language disturbance often have some receptive impairment, although not always sufficiently significant for the diagnosis of language

disorder on this basis. Speech sound disorder, formerly known as phonologic disorder, is commonly found in young children with a language disorder, and neurologic abnormalities exist in some children, including soft neurologic signs, depressed vestibular responses, and EEG abnormalities.

Treatment. The primary goals for early childhood speech and language treatment are to guide children and their parents toward greater production of meaningful language. There are more data to support improvements through speech and language interventions for expressive language deficit in young school-age children with primary deficits than in preschool children. A study investigating Parent–Child Interaction Therapy (PCIT) for school-age children with expressive language impairment found that PCIT was particularly efficacious in improving a child's verbal initiation, mean length of utterances, and the proportion of child-to-parent utterances. A large-scale randomized trial of a yearlong intervention targeting preschoolers with language delay in Australia found that a community-based program did not affect language acquisition in 2- and 3-year-olds. Given the high rate of spontaneous remission of language deficits in preschoolers, and less than robust effects of interventions for children that are young, treatment for expressive language disorder is generally not initiated unless it persists after the preschool years. Various techniques may help a child improve their use of such parts of speech as pronouns, correct tenses, and question forms. Direct interventions use a speech and language pathologist who works directly with the child. Mediated interventions, in which a speech and language professional teaches a child's teacher or parent how to promote therapeutic language techniques, have also been efficacious. Language therapy works on using words to improve communication strategies and social interactions, as well. Such therapy consists of behaviorally reinforced exercises and training with phonemes (sound units), vocabulary, and sentence construction. The goal is to increase the number of phrases by using block-building methods and conventional speech therapies.

Etiology. The specific causes of the expressive components of language disorder are likely to be multifactorial. Scant data are available on the specific brain structure of children with a language disorder. However, limited MRI studies suggest that language disorders are associated with

diminished left-right brain asymmetry in the perisylvian and planum temporale regions. Results of one small MRI study suggested possible inversion of brain asymmetry (right > left). Left-handedness or ambilaterality appears to be associated with expressive language problems with more frequency than righthandedness. Evidence shows that language disorders occur more frequently within some families, and several studies of twins show significant concordance for monozygotic twins for language disorders. Environmental and educational factors also likely contribute to developmental language disorders.

Mixed Receptive and Expressive Deficits

Children with both receptive and expressive language impairment may have impaired ability in sound discrimination, deficits in auditory processing, or poor memory for sound sequences. Children with mixed receptive–expressive disturbance have impaired skills in the expression and reception (understanding and comprehension) of spoken language. The expressive difficulties in these children may be similar to those of children with only expressive language disturbance, which is characterized by limited vocabulary, use of simple sentences, and short sentence usage. Children with receptive language difficulties may be experiencing additional deficits in basic auditory processing skills, such as discriminating between sounds, rapid sound changes, the association of sounds and symbols, and the memory of sound sequences. These deficits may lead to a whole host of communication barriers for a child, including a lack of understanding of questions or directives from others, or inability to follow the conversations of peers or family members. Teachers or parents may initially misattribute communication difficulties as behavioral problems, delaying the recognition of the actual diagnosis.

We can measure the essential features of mixed receptive–expressive language disturbance on standardized tests; both receptive (comprehension) and expressive language development scores fall substantially below those obtained from standardized measures of nonverbal intellectual capacity. Language difficulties must be sufficiently severe to impair academic achievement or daily social communication.

Clinical Features. The essential clinical feature of this language disturbance is a significant impairment in both language comprehension and language expression. In the mixed type, expressive impairments are similar to those of expressive language disturbance but can be more severe. The clinical features of the receptive component of the disorder typically appear before the age of 4 years. Severe forms are apparent by the age of 2 years; mild forms may not become evident until age 7 (second grade) or older when language becomes complex. Children with language disorder characterized by mixed receptive–expressive disturbance show markedly delayed and below-normal ability to comprehend (decode) verbal or sign language, although they have age-appropriate nonverbal intellectual capacity. Patients with receptive dysfunction also are impaired in verbal or sign expression (encoding) of language. The clinical features of mixed receptive–expressive language disturbance in children between the ages of 18 and 24 months result from a child's failure to utter a single phoneme spontaneously or to mimic another person's words.

Many children with mixed receptive–expressive language deficits have auditory sensory difficulties and compromised ability to process visual symbols, such as explaining the meaning of a picture. They have deficits in integrating both auditory and visual symbols—for example, recognizing the common attributes of a toy truck and a toy passenger car. Whereas at 18 months, a child with expressive language deficits only comprehends simple commands and can point to familiar household objects when told to do so, a child of the same age with mixed receptive–expressive language disturbance typically cannot either point to everyday objects or obey simple commands. A child with mixed receptive–expressive language deficits may appear to be deaf. He or she responds appropriately to sounds from the environment, but not to spoken language. If the child later starts to speak, the speech contains numerous articulation errors, such as omissions, distortions, and substitutions of phonemes. Language acquisition is much slower for children with mixed receptive–expressive language disturbance than for other children of the same age.

Children with mixed receptive–expressive language disturbance have difficulty recalling early visual and auditory memories and recognizing and reproducing symbols in a proper sequence. Some children with mixed receptive–expressive language deficits have a partial hearing defect for true

tones, an increased threshold of auditory arousal, and an inability to localize sound sources. Seizure disorders and reading disorders are more common among the relatives of children with mixed receptive–expressive problems than in the general population.

Diagnosis. Children with mixed receptive–expressive language deficits develop language more slowly than their peers and have trouble understanding conversations that peers can follow. In mixed receptive–expressive language disorder, receptive dysfunction coexists with expressive dysfunction. Therefore, we should give standardized tests for both receptive and expressive language abilities to anyone suspected of having a language disorder with mixed receptive–expressive disturbance.

A markedly below-expected level of comprehension of verbal or sign language with intact age-appropriate nonverbal intellectual capacity, confirmation of language difficulties by standardized receptive language tests, and the absence of autism spectrum disorder, confirm the diagnosis of mixed receptive–expressive language deficits; however, in DSM-5, these deficits are included in the diagnosis of language disorder.

PATHOLOGY AND LABORATORY EXAMINATION. All children thought to have mixed receptive–expressive language disturbance should have an audiogram to rule out or confirm the presence of deafness or auditory deficits. A history of the child and family and observation of the child in various settings help to clarify the diagnosis.

Jenna was a pleasant 2-year-old who did not yet use any spoken words and did not respond to simple commands without gestures. She made her needs known with vocalizations and simple gestures (e.g., showing or pointing) such as those typically used by younger children. She seemed to understand the names for only a few familiar people and objects (e.g., *mommy, daddy*, *cat, bottle,* and *cookie*). Compared with other children her age, she had a small comprehension vocabulary and showed limited understanding of simple verbal directions (e.g., “Get your doll.” “Close your eyes.”). Nonetheless, her hearing was reasonable, and her motor and play skills were developing as expected for her age. She showed interest

in her environment and the activities of the other children at her daycare.

Lena was a shy, reserved 5-year-old who grew up in a bilingual home. Lena's parents and older siblings spoke English and Cantonese proficiently. Her grandparents, who lived in the same home, spoke only Cantonese. Lena began to understand and speak both languages much later than her older siblings had. Throughout her preschool years, Lena continued to develop slowly in comprehension and production. At the start of kindergarten, Lena understood fewer English words for objects, actions, and relations than her classmates did. Lena was unable to follow complicated classroom instructions, particularly those that involved words for concepts of time (e.g., *tomorrow*, *before*, or *day*) and space (e.g., *behind*, *next to*, or *under*). It was also hard for Lena to match one of several pictures to a syntactically complex sentence that she had heard (e.g., "It was not the train she was waiting for." "Because he had already completed his work, he was not kept after school."). Lena played with other children but only rarely tried to speak with them, which led to her classmates ostracizing her. Lena's attempts at conversations usually broke down because she misinterpreted what others said or could not express her thoughts clearly.

Consequently, her classmates generally ignored her, preferring instead to play with more verbally competent peers. Lena's infrequent interactions further limited her opportunities to learn and to practice her already weak language skills. Lena also showed limited receptive and expressive skills in Cantonese, as revealed by an assessment conducted with the assistance of a Cantonese interpreter. Nonetheless, her nonverbal cognitive and motor skills were within the normal range for her age. Lena was quite proficient in solving spatial and numerical problems, provided they were presented on paper and were not word problems.

Mark received a diagnosis of Language Disorder, based on mixed receptive–expressive deficits when he was a preschooler. By 7 years of age, he had also received the comorbid diagnoses of reading disorder and

ADHD. This combination of language, reading, and attention problems made it virtually impossible for Mark to succeed in school, although he was able to engage his peers during free play. His comprehension and attention difficulties limited his ability to understand and to learn valuable information or to follow classroom instructions or discussions. Mark fell further and further behind his classmates. He was also disadvantaged because he could read only a few familiar words. This disadvantage meant that Mark was neither motivated nor able to learn academic information outside of the classroom by reading. Mark received tutoring and speech and language interventions, and despite some improvements, he continued to lag behind his classmates academically. Despite his academic problems, however, Mark made friends during sports activities in which he excelled, and continued to show nonverbal intellectual skills within the average range.

Differential Diagnosis. Children with language disorder characterized by mixed receptive–expressive deficits have a deficit in language comprehension as well as in language production. The receptive deficit may be overlooked at first, because the expressive language deficit may be more prominent. In expressive language disturbance alone, comprehension of spoken language (decoding) remains within age norms. Children with speech sound disorder and child-onset fluency disorder (stuttering) have normal expressive and receptive language competence, despite the speech impairments.

Most children with mixed receptive–expressive language disturbance have a history of variable and inconsistent responses to sounds; they respond more often to environmental sounds than to speech sounds (Table 2-7). We should also rule out intellectual disability, selective mutism, acquired aphasia, and autism spectrum disorder.

Comorbidity. Children with mixed receptive–expressive deficits are at high risk for additional speech and language disorders, learning disorders, and additional psychiatric disorders. About half of children with these deficits also have pronunciation difficulties leading to speech sound disorder, and about half also have a reading disorder. These rates are significantly higher than the comorbidity found in children with only expressive language

problems. ADHD is present in at least one-third of children with mixed receptive–expressive language disturbances.

Course and Prognosis. The overall prognosis for language disorder with mixed receptive–expressive disturbance is less favorable than that for expressive language disturbance alone. When the mixed disorder is identified in a young child, it is usually severe, and the short-term prognosis is poor. Language develops at a rapid rate in early childhood, and young children with the disorder may appear to be falling behind. Given the likelihood of comorbid learning disorders and other mental disorders, there is a guarded prognosis. Young children with severe mixed receptive–expressive language deficits are likely to have learning disorders in the future. In children with mild versions, it may take several years to identify the mixed disorder, and the disruption in everyday life may be less overwhelming than that in severe forms of the disorder. Over the long run, some children with mixed receptive–expressive language disturbance achieve close to normal language functions. The prognosis for children who have mixed receptive–expressive language disturbances varies widely and depends on the nature and severity of the damage.

Treatment. Children with mixed receptive–expressive language disturbance should have a comprehensive speech and language evaluation, given the complexities of having both deficits. Some controversy exists as to whether remediation of receptive deficits before expressive deficits improves the overall efficacy of treatment. A review of the literature indicates that it is not more beneficial to address receptive deficits before expressive, and, in some cases, remediation of expressive language may reduce or eliminate the need for receptive language remediation. Thus, current recommendations are either to address both simultaneously, or to provide interventions for the expressive component first, and then address the receptive language. Preschoolers with mixed receptive–expressive language problems optimally receive interventions designed to promote social communication and literacy as well as oral language. For children at the kindergarten level, the optimal intervention includes direct teaching of key pre-reading skills as well as social skills training. An important early goal of interventions for young children with mixed receptive–expressive language disturbance is the

achievement of rudimentary reading skills, in that these skills are protective against the academic and psychosocial ramifications of falling behind early on in reading. Some language therapists favor a low-stimuli setting, in which they give children individual linguistic instruction. Others recommend integrating speech and language instruction into a varied setting with several children whom they teach several language structures simultaneously. Often, a child with receptive and expressive language deficits will benefit from a small, special-educational setting that allows more individualized learning.

Psychotherapy may be helpful for children with mixed language disorder who have associated emotional and behavioral problems. We should pay particular attention to evaluating the child's self-image and social skills. Family counseling in which parents and children can develop more effective, less frustrating means of communicating may be beneficial.

Epidemiology. Mixed receptive–expressive language deficits occur less frequently than expressive deficits; however, epidemiologic data are scant regarding specific prevalence rates. The language disturbance is believed to occur in about 5 percent of preschoolers and persists in about 3 percent of school-age children. It is less common than expressive language disturbance. Mixed receptive–expressive language disorder is likely twice as prevalent in boys as in girls.

Etiology. Language disorders most likely have multiple determinants, including genetic factors, developmental brain abnormalities, environmental influences, neurodevelopmental immaturity, and auditory processing features in the brain. As with expressive language disturbance alone, there is evidence of familial aggregation of mixed receptive–expressive language deficits. Twin studies implicate a genetic contribution to this disorder, but there is no proven mode of genetic transmission. Some studies of children with various speech and language disorders have also shown cognitive deficits, notably slower processing of tasks involving naming objects, as well as fine motor tasks. Slower myelinization of neural pathways may account for the slow processing found in children with developmental language disorders. Several studies suggest an underlying impairment of auditory discrimination because most children with the disorder are more responsive to environmental sounds than to speech sounds.

SPEECH SOUND DISORDER

Children with speech sound disorder have difficulty pronouncing speech sounds correctly due to omissions of sounds, distortions of sounds, or atypical pronunciation. Formerly called phonologic disorder, typical speech disturbances in speech sound disorder include omitting the last sounds of the word (e.g., saying *mou* for *mouse* or *drin* for *drink*), or substituting one sound for another (saying *bwu* instead of *blue* or *tup* for *cup*). Distortions in sounds can occur when children allow too much air to escape from the side of their mouths while saying sounds like *sh* or producing sounds like *s* or *z* with their tongue protruded. Speech sound errors can also occur in patterns because a child has an interrupted airflow instead of a steady airflow preventing their pronouncing words (e.g., *pat* for *pass* or *bacuum* for *vacuum*). Children with a speech sound disorder can be mistaken for younger children because of their difficulties in producing speech sounds correctly. We make a diagnosis of a speech sound disorder by comparing the skills of a given child with the expected skill level of others of the same age. The disorder results in errors in whole words because of incorrect pronunciation of consonants, substitution of one sound for another, omission of entire phonemes, and, in some cases, dysarthria (slurred speech because of incoordination of speech muscles) or dyspraxia (difficulty planning and executing speech). Speech sound development likely depends on both linguistic and motor development that we must integrate to produce sounds.

Speech sound disturbances such as dysarthria and dyspraxia are not diagnosed as speech sound disorder if they are known to have a neurologic basis, according to DSM-5. Thus, speech sound abnormalities accounted for by cerebral palsy, cleft palate, deafness or hearing loss, traumatic brain injury, or neurologic conditions are not diagnosed as speech sound disorder. Articulation difficulties not associated with a neurologic condition are the most common components of speech sound disorder in children. Articulation deficits include poor articulation, sound substitution, and speech sound omission, and give the impression of “baby talk.” Typically, these deficits are not caused by anatomical, structural, physiologic, auditory, or neurologic abnormalities. They vary from mild to severe and result in speech that ranges from entirely intelligible to unintelligible.

Clinical Features

Children with speech sound disorder have a delay in, or are incapable of, producing accurate speech sounds typical for their age, intelligence, and dialect. The sounds are often substitutions—for example, the use of *t* instead of *k*—and omissions, such as leaving off the final consonants of words. We can usually recognize speech sound disorder in early childhood. In severe cases, we can recognize it between 2 and 3 years of age. In less severe cases, the disorder may not be apparent until the age of 6 years. A child's articulation is disordered when it is significantly behind that of most children at the same age level, intellectual level, and educational level.

In very mild cases, a single speech sound (i.e., phoneme) may be affected. When a single phoneme is affected, it is usually one that is acquired late in normal language acquisition. The speech sounds most frequently misarticulated are also those acquired late in the developmental sequence, including *r, sh, th, f, z, l,* and *ch.* In severe cases and young children, sounds such as *b, m, t, d, n,* and *h* may be mispronounced. One or many speech sounds may be affected, but vowel sounds are not among them.

Children with speech sound disorder cannot articulate specific phonemes correctly and may distort, substitute, or even omit the affected phonemes. With omissions, the phonemes are absent entirely—for example, bu for blue, ca for car, or whaa? For what's that? With substitutions, children may replace difficult phonemes with incorrect ones—for example, wabbit for rabbit, fum for thumb, or whath dat? For what's that? With distortions, the child may approximate the correct phoneme but articulate it incorrectly. Rarely, additions (usually of the vowel uh) occur—for example, puhretty for pretty, what's uh that uh? For what's that?

Omissions are the most severe type of misarticulating, with substitutions the next most serious, and distortions the least severe type. Omissions, which are most frequent in the speech of young children, usually occur at the ends of words or in clusters of consonants (ka for car, scisso for scissors). Distortions, which are found mainly in the speech of older children, result in a sound that is not part of the speaker's dialect. Distortions may be the last type of misarticulating remaining in the speech of children whose articulation problems have mostly remitted. The most common types of distortions are the lateral slip—in which a child pronounces *s* sounds with the airstream

going across the tongue, producing a whistling effect—and the palatal or lisp—in which the *s* sound, formed with the tongue too close to the palate, produces a *ssh* sound effect.

The misarticulating of children with speech sound disorder is often inconsistent and random. A child may pronounce a phoneme correctly one time and incorrectly another time. Misarticulating is most common at the ends of words, in long and syntactically complex sentences, and during rapid speech.

Omissions, distortions, and substitutions also usually occur in the speech of young children learning to talk. However, whereas young, typically speaking children soon replace their misarticulating, children with speech sound disorder do not. Even as children with articulation problems grow and finally acquire the correct phoneme, they may use it only in newly acquired words and may not correct the words learned earlier that they have been mispronouncing for some time.

Most children eventually outgrow speech sound disorder, usually by the third grade. After the fourth grade, however, spontaneous recovery is unlikely, and so it is essential to try to remediate the disorder before the development of complications. Often, beginning kindergarten or school precipitates the improvement when recovery from speech sound disorder is spontaneous. Children who have not shown spontaneous improvement by the third or fourth grade should have speech therapy. Speech therapy should be initiated at an early age for children whose articulation is significantly unintelligible and who are troubled by their inability to speak clearly.

Children with speech sound disorder may have various concomitant social, emotional, and behavioral problems, mainly when comorbid expressive language problems are present. Children with chronic expressive language deficits and severe articulation impairment are the ones most likely to suffer from psychiatric problems.

Martin was a talkative, likable 3-year-old with virtually unintelligible speech, despite excellent receptive language skills and normal hearing. Martin’s level of expressive language development was difficult to quantify due to his very poor pronunciation. The rhythm and melody of his speech, however, suggested that he was trying to produce multiword

utterances, as would be expected at his age. Martin produced only a few vowels (*/ee*/, */ah*/, and */oo*/), some early developing consonants (*/m/, /n/, /d/, /t/, /p/, /b/, /h/,* and */w/*), and limited syllables. This reduced sound repertoire made many of his spoken words indistinguishable from one another (e.g., he said *bahbah* for *bottle, baby*, and *bubble,* and he used *nee* for *knee, need,* and *Anita* [his sister]). Moreover, he consistently omitted consonant sounds at the end of words and in consonant cluster sequences (e.g., */tr-*/, */st-*/, */-nt*/, and */-mp*/). Understandably, on occasion, Martin reacted with frustration and tantrums to his difficulties in making his needs understood.

Brad was a pleasant, cooperative 5-year-old, who was recognized as early as preschool to have articulation problems, and these persisted into kindergarten. His language comprehension skills and hearing were within normal limits. He showed some mild expressive language problems, however, in the use of certain grammatical features (e.g., pronouns, auxiliary verbs, and past-tense word endings) and in the formulation of complex sentences. He correctly produced all vowel sounds and most of the early developing consonants, but he was inconsistent in his attempts to produce later-developing consonants (e.g., */r/, /l/, /s/, /z/, /sh/, /th/,* and */ch/*). Sometimes, he omitted them; sometimes, he substituted other sounds for them (e.g., */w*/ for */r*/ or */f/* for */th*/); occasionally, he even produced them correctly. Brad had particular problems in correctly producing consonant cluster sequences and multisyllabic words. Cluster sequences had omitted or incorrect sounds (e.g., *blue* might be produced as *bue* or *bwue,* and *hearts* as *hots* or *hars*). Multisyllabic words had syllables omitted (e.g., *efant* for *elephant* and *getti* for *spaghetti*) and sounds mispronounced or even transposed (e.g., *aminal* for *animal* and *lemon* for *melon*). Strangers were unable to understand approximately 80 percent of Brad's speech. Brad often spoke more slowly and clearly than usual, however, when asked to repeat something, as he often was.

Jane was a hyperactive 8-year-old, with a history of significant speech

delay. During her preschool and early school years, she had overcome many of her earlier speech errors. A few late-developing sounds (*/r/,/l/,* and */th/*), however, continued to pose a challenge for her. Jane often substituted */f/* or */d/* for*/th/*and produced */w/* for */r/* and */l/*. Overall, her speech was easily understood, despite these minor errors. Nonetheless, she became somewhat aggressive with her peers because of the teasing she received from her classmates about her speech.

Diagnosis

The essential feature of speech sound disorder is a child's delay or failure to produce developmentally expected speech sounds, especially consonants, resulting in sound omissions, substitutions, and distortions of phonemes. A rough guideline for clinical assessment of children's articulation is that normal 3-year-olds correctly articulate *m, n, ng, b, p, h, t, k, q,* and *d;* ordinary 4-year-olds correctly articulate *f, y, ch, sh,* and *z;* and typical 5-year-olds correctly articulate *th, s, and r.*

Speech sound disorder cannot be accounted for by structural or neurologic abnormalities, and typically, normal language development accompanies it. Table 2-8 lists the diagnostic approaches to speech sound disorder.

Differential Diagnosis

The differential diagnosis of speech sound disorder includes a careful determination of symptoms, severity, and possible medical conditions that might be producing the symptoms. First, we must determine that the misarticulating is sufficiently severe to impair the child, rather than a normative developmental process of learning to speak. Second, we must determine that no physical abnormalities account for the articulation errors and must rule out neurologic disorders that may cause dysarthria, hearing impairment, mental retardation, and pervasive developmental disorders. Third, we must obtain an evaluation of receptive and expressive language to determine that the speech difficulty is not solely attributable to the disorders, as mentioned earlier.

Table 2-8

Speech Sound Disorder

	DSM-5	ICD-10
Diagnostic name	Speech Sound Disorder	Specific Speech Articulation Disorder
Duration	Onset during early development	Onset during early development
Symptoms	Difficulty producing spoken sounds Difficult to understand Limits social/academic/occupational achievement	Disorder of speech sounds. Examples: • Developmental disorder: • phonologic disorder • speech articulation disorder • Dyslalia • Functional speech articulation disorder • Lalling
Exclusions (not result of):	Mechanical disorders of speech or hearing (e.g., deafness, cleft palate) A neurologic disorder Another medical disorder	Aphasia NOS Apraxia Hearing loss Mental retardation With language developmental disorder: • Expressive • Receptive

Comments	ICD groups this with the language disorder as per Table 2-6



Table 2-9
Differential Diagnosis of Speech Sound Disorder

Criteria	Speech Sound Dysfunction due to Structural or Neurologic Abnormalities (Dysarthria)	Speech Sound Dysfunction due to Hearing Impairment	Speech Sound Disorder	Speech Sound Dysfunction Associated with Intellectual Disability, Autism Spectrum Disorder, Developmental Dysphasia, Acquired Aphasia, or Deafness
Language development	Within normal limits	Within normal limits unless hearing impairment is serious	Within normal limits	Not within normal limits
Examination	Possible abnormalities of lips, tongue, or palate; muscular weakness, incoordination, or disturbance of vegetative functions, such as sucking or chewing	Hearing impairment shown on audiometric testing	Normal	
Rate of speech	Slow; marked deterioration of articulation with increased rate	Normal	Normal; possible deterioration of articulation with increased rate	
Phonemes affected	Any phonemes, even vowels	*f, th, sh,* and *s*	*r, sh, th, ch, dg, j, f, v, s,* and *z* are most commonly affected	

Adapted from Dennis Cantwell, M.D. and Lorian Baker, Ph.D., 1991.

Neurologic, oral structural, and audiometric examinations may be necessary to rule out physical factors that cause certain types of articulation abnormalities. Children with dysarthria, a disorder caused by structural or neurologic abnormalities, differ from children with speech sound disorder in that dysarthria is less likely to remit spontaneously and may be more difficult to remediate. Drooling, slow, or uncoordinated motor behavior, abnormal chewing or swallowing, and awkward or slow protrusion and retraction of the tongue indicate dysarthria. A slow rate of speech also indicates dysarthria (Table 2-9).

Comorbidity

More than half of children with speech sound disorder have some difficulty with language. Disorders most commonly present with speech sound disorders are language disorder, reading disorder, and developmental

coordination disorder. Enuresis may also accompany the disorder. A delay in reaching speech milestones (e.g., first word and first sentence) occurs in some children with speech sound disorder, but most children with the disorder begin speaking at the appropriate age. Children with both speech sound and language disorders are at the highest risk for attentional problems and specific learning disorders. Children with speech sound disorder in the absence of language disorder have a lower risk of comorbid psychiatric disorders and behavioral problems.

Course and Prognosis

Spontaneous remission of symptoms is common in children whose misarticulating involves only a few phonemes. Children who persist in exhibiting articulation problems after the age of 5 years may be experiencing a myriad of other speech and language impairments so that it is appropriate to perform a comprehensive evaluation at that time. Children older than age 5 with articulation problems are at higher risk for auditory perceptual problems. Spontaneous recovery is rare after the age of 8 years. Some debate exists regarding the relationship between articulation problems and reading disorder or dyslexia. A recent study comparing children with phonologic problems only, with children who had dyslexia only, and those with both phonologic difficulties and dyslexia concluded that children with both disorders have somewhat distinct profiles and are comorbid disorders rather than one mixed disorder.

Treatment

Two main approaches can successfully improve speech sound difficulties. The first one, the *phonologic approach*, is usually chosen for children with extensive patterns of multiple speech sound errors that may include final consonant deletion or consonant cluster reduction. Exercises in this approach to treatment focus on the guided practice of specific sounds, such as final consonants, and when mastering that skill, extending the practice to use it in meaningful words and sentences. The other approach, the *traditional approach*, is used for children who produce substitution or distortion errors in just a few sounds. In this approach, the child practices the production of the problem sound while we provide immediate feedback and cues concerning

the correct placement of the tongue and mouth for improved articulation. Children who have errors in articulation because of abnormal swallowing resulting in tongue thrust and lisps should receive exercises to improve their swallowing patterns and, in turn, improve their speech. A speech-language pathologist typically provides speech therapy, yet parents can learn to provide adjunctive help by practicing techniques used in the treatment. Early intervention can be helpful, because, for many children with mild articulation difficulties, even several months of intervention may be helpful in early elementary school. In general, when a child's articulation and intelligibility is noticeably different from peers by 8 years of age, speech deficits often lead to problems with peers, learning, and self-image, especially when the disorder is so severe that many consonants are misarticulated, and when errors involve omissions and substitutions of phonemes, rather than distortions.

Children with persistent articulation problems are likely to be teased or ostracized by peers and may become isolated and demoralized. Therefore, it is crucial to give support to children with phonologic disorders and, whenever possible, to support prosocial activities and social interactions with peers. Parental counseling and monitoring of child–peer relationships and school behavior can help minimize social impairment in children with speech sound and language disorder.

Epidemiology

Epidemiologic studies suggest that the prevalence of speech sound disorder is at least 3 percent in preschoolers, 2 percent in children 6 to 7 years of age, and 0.5 percent in 17-year-old adolescents. Approximately 7 to 8 percent of 5-year-old children in one large community sample had speech sound production problems of developmental, structural, or neurologic origins. Another study found that up to 7.5 percent of children between the ages of 7 and 11 years had speech sound disorders. Of those, 2.5 percent had speech delay (deletion and substitution errors past the age of 4 years), and 5 percent had residual articulation errors beyond the age of 8 years. Speech sound disorders occur much more frequently than disorders with a known structural or neurologic origin. Speech sound disorder is approximately two to three times more common in boys than in girls. It is also more common among first-degree relatives of patients with the disorder than in the general

population. Although speech sound mistakes are quite common in children younger than 3 years of age, these mistakes are usually self-corrected by age 7 years. Misarticulating after the age of 7 years is likely to represent a speech sound disorder. The prevalence of speech sound disorders reportedly falls to 0.5 percent by mid to late adolescence.

Etiology

Contributing factors leading to speech disturbance may include perinatal problems, genetic factors, and auditory processing problems. Given the high rates of spontaneous remission in very young children, there may be a maturational delay in the developmental brain process underlying speech in some cases. The likelihood of neuronal cause is supported by the observation that children with speech sound disorder are also more likely to manifest "soft neurologic signs" as well as language disorder and a higher-than-expected rate of reading disorder. Data from twin studies implicate genetic factors, as they show concordance rates for monozygotic twins that are higher than chance.

Articulation disorders caused by structural or mechanical problems are rare. Articulation problems not caused by speech sound disorder could represent a neurologic impairment, such as dysarthria and apraxia or dyspraxia. Dysarthria results from an impairment in the neural mechanisms regulating the muscular control of speech. This impairment can occur in congenital conditions, such as cerebral palsy, muscular dystrophy, or head injury, or because of infectious processes. Apraxia or dyspraxia is characterized by difficulty in the execution of speech, even when no apparent paralysis or weakness of the muscles used in speech exists.

Environmental factors may play a role in speech sound disorder, but constitutional factors seem to make the most significant contribution. The high proportion of speech sound disorder in individual families implies a genetic component in the development of this disorder. Developmental coordination disorder and coordination in the mouth, such as in chewing and blowing the nose, may be associated.

CHILD-ONSET FLUENCY DISORDER (STUTTERING)

Child-onset fluency disorder (stuttering) usually begins during the first years

of life and is characterized by disruptions in the normal flow of speech by involuntary speech motor events. Stuttering can include a variety of specific disruptions of fluency, including sound or syllable repetitions, sound prolongations, dysrhythmic phonations, and complete blocking or unusual pauses between sounds and syllables of words. In severe cases, the stuttering may include accessory or secondary attempts to compensate, such as respiratory, abnormal voice phonations, or tongue clicks. Associated behaviors, such as eye blinks, facial grimacing, head jerks, and abnormal body movements, may be observed before or during the disrupted speech.

Early intervention is necessary because children who receive early intervention can be more than seven times more likely to have full resolution of their stuttering. In severe and some untreated cases, stuttering can become an entrenched pattern that is more challenging to remediate later in life and is associated with significant psychological and social distress. When stuttering becomes chronic, persisting into adulthood, the rate of concurrent social anxiety disorder is between 40 and 60 percent.

Diagnosis

The diagnosis of childhood-onset fluency disorder (stuttering) is not difficult when the clinical features are apparent and well developed, and each of the following four phases (described in the next section) are readily recognized. Diagnostic difficulties can arise when evaluating for stuttering in young children because some preschool children experience transient dysfluency. It may not be clear whether the nonfluent pattern is part of normal speech and language development or whether it represents the initial stage in the development of stuttering. The child should be referred to a speech therapist if incipient stuttering is suspected. Table 2-10 compares the approaches to diagnosing stuttering.

Clinical Features

Stuttering usually appears between the ages of 18 months and 9 years, with two sharp peaks of onset between the ages of 2 to 3.5 years and 5 to 7 years. Some, but not all, stutterers have other speech and language problems, such as phonologic disorder and expressive language disorder. Stuttering does not begin suddenly; it typically develops over weeks or months with a repetition

of initial consonants, whole words that are usually the first words of a phrase, or long words. As the disorder progresses, the repetitions become more frequent, with consistent stuttering on the most important words or phrases. Even after it develops, stuttering may be absent during oral readings, singing, and talking to pets or inanimate objects.



Table 2-10
Child Onset Fluency Disorder (Stuttering)

	DSM-5	ICD-10
Diagnostic name	Childhood-Onset Fluency Disorder (Stuttering)	Stuttering (stammering)
Duration	Onset of symptoms during the developmental period	
Symptoms	Fluency/speech abnormalities: • Repetition of sounds/ syllables • Prolonging sounds or consonants • Using broken words • Blocking/pausing • Frequent word substitution (circumlocution) • Excess physical tension during word production • Repeating monosyllabic words	Speech characterized by: • Repetition or prolongation of sounds or words • Frequent pauses
Psychosocial consequences of symptoms	Causing severe anxiety and/or impairment in areas of functioning	Resulting in disruption of normal flow or rhythm of speech
Exclusions (not	Speech-motor or sensory	Tic disorders

result of):	issues Neurologic injury Another medical condition Another mental disorder Note: later-onset cases after the developmental period are coded as “adult-onset fluency disorder”

There are four gradually evolving phases in the development of stuttering:

- **Phase 1** occurs during the preschool period. Initially, the difficulty tends to be episodic and appears for weeks or months between long interludes of normal speech. A high percentage of recovery from these periods of stuttering occurs. During this phase, children stutter most often when excited or upset, when they seem to have a great deal to say, and under other conditions of communicative pressure.
- **Phase 2** usually occurs in the elementary school years. The disorder is chronic, with few, if any, intervals of normal speech. Affected children become aware of their speech difficulties and regard themselves as stutterers. In phase 2, the stuttering occurs mainly with the significant parts of speech—nouns, verbs, adjectives, and adverbs.
- **Phase 3** usually appears after the age of 8 years and up to adulthood, most often in late childhood and early adolescence. During phase 3, stuttering comes and goes mainly in response to specific situations, such as reciting in class, speaking to strangers, making purchases in stores, and using the telephone. Some words and sounds are more difficult than others.
- **Phase 4** typically appears in late adolescence and adulthood.

Stutterers show vivid, fearful anticipation of stuttering. Words, sounds, and situations can all be intimidating. Word substitutions and circumlocutions are common. Stutterers avoid situations requiring speech and show other evidence of fear and embarrassment.

Stutterers may have associated clinical features: vivid, fearful anticipation of stuttering, with avoidance of particular words, sounds, or situations in

which stuttering is anticipated; and eye blinks, tics, and tremors of the lips or jaw. Frustration, anxiety, and depression are common among those with chronic stuttering.

Differential Diagnosis

Normal speech dysfluency in preschool years is challenging to differentiate from incipient stuttering. In stuttering occurs more nonfluencies, part-word repetitions, sound prolongations, and disruptions in voice airflow through the vocal tract. Children who stutter appear to be tense and uncomfortable with their speech pattern, in contrast to young children who are nonfluent in their speech but seem to be at ease. Spastic dysphonia is a stuttering-like speech disorder distinguished from stuttering by the presence of an abnormal breathing pattern.

Cluttering is a speech disorder characterized by erratic and dysrhythmic speech patterns of rapid and jerky spurts of words and phrases. In cluttering, those affected are usually unaware of the disturbance, whereas after the initial phase of the disorder, stutterers are aware of their speech difficulties. Cluttering is often an associated feature of expressive language disturbance.

Comorbidity

Very young children who stutter typically show some delay in the development of language and articulation without additional disorders of speech and language. Preschoolers and school-age children who stutter exhibit an increased incidence of social anxiety, school refusal, and other anxiety symptoms. Older children who stutter also do not necessarily have comorbid speech and language disorders, but often manifest anxiety symptoms and disorders. When stuttering persists into adolescence, social isolation occurs at higher rates than in the general adolescent population. Stuttering is also associated with a variety of abnormal motor movements, upper body tics, and facial grimaces. Other disorders that coexist with stuttering include phonologic disorder, expressive language disorder, mixed receptive–expressive language disorder, and ADHD.

Course and Prognosis

The course of stuttering is often long term, with periods of partial remission

lasting for weeks or months and exacerbations occurring most frequently when a child is under pressure to communicate. In children with mild cases, 50 to 80 percent recover spontaneously. School-age children who stutter chronically may have impaired peer relationships as a result of teasing and social rejection. These children may face academic difficulties, especially if they persistently avoid speaking in class. Stuttering is associated with anxiety disorders in chronic cases, and approximately half of individuals with persistent stuttering have social anxiety disorder.

Treatment

There are several evidence-based treatments for stuttering. One such treatment is the Lidcombe Program, which uses an operant conditioning model in which parents use praise for periods in which the child does not stutter, and intervene when the child does stutter to request the child to self-correct the stuttered word. This treatment program is primarily administered at home by parents, under the supervision of a speech and language therapist. A second treatment program in clinical trials is a family-based, PCIT that identifies stressors possibly associated with increased stuttering and aims to diminish these stressors. A third treatment under investigation in clinical trials relies on the knowledge that speaking each syllable in time to a particular rhythm has led to diminished stuttering in adults. This treatment program appears to be promising when administered early on, to preschoolers.

Distinct forms of interventions have historically helped stuttering. The first approach, direct speech therapy, targets modification of the stuttering response to fluent-sounding speech by systematic steps and rules of speech mechanics that the person can practice. The other form of therapy for stuttering targets diminishing tension and anxiety during speech. These treatments may utilize breathing exercises and relaxation techniques to help children slow the rate of speaking and modulate speech volume. Relaxation techniques use the premise that it is nearly impossible to be relaxed and stutter in the usual manner at the same time—current interventions for stuttering use individualized combinations of behavioral distraction, relaxation techniques, and directed speech modification.

Stutterers who have a poor self-image, comorbid anxiety disorders, or

depressive disorders are likely to require additional treatments with cognitive-behavioral therapy (CBT) or pharmacologic agents such as one of the selective serotonin reuptake inhibitor (SSRI) antidepressants.

An approach to stuttering proposed by the Speech Foundation of America is labeled self-therapy, based on the premise that stuttering is not a symptom, but a behavior that can be modified. Stutterers can learn to control their difficulty partly by modifying their feelings about stuttering and attitudes toward it and partly by modifying the deviant behaviors associated with their stuttering blocks. The approach includes desensitizing, reducing the emotional reaction to and fears of, stuttering, and substituting positive action to control the moment of stuttering.

Epidemiology

An epidemiologic survey of 3- to 17-year-olds derived from the United States National Health Interview Surveys reports that the prevalence of stuttering is approximately 1.6 percent. Stuttering tends to be most common in young children and has often resolved spontaneously as the child ages. The typical age of onset is 2 to 7 years of age, with 90 percent of children exhibiting symptoms by age 7 years. Approximately 65 to 80 percent of young children who stutter are likely to have a spontaneous remission over time. According to the DSM-5, the rate dips to 0.8 percent by adolescence. Stuttering affects about three to four males for every one female. The disorder is significantly more common among family members of affected children than in the general population. Reports suggest that for male persons who stutter, 20 percent of their male children and 10 percent of their female children will also stutter.

Etiology

Converging evidence indicates that the cause of stuttering is multifactorial, including genetic, neurophysiologic, and psychological factors that predispose a child to have poor speech fluency. Although research evidence does not indicate that anxiety or conflicts cause stuttering or that persons who stutter have more psychiatric disturbances than those with other forms of speech and language disorders, stressful situations can exacerbate stuttering.

Other theories about the cause of stuttering include organic models and

learning models. Organic models include those that focus on incomplete lateralization or abnormal cerebral dominance. Several studies using EEG found that stuttering males had right hemispheric α-suppression across stimulus words and tasks; nonstutterers had left hemispheric suppression. Some studies of stutterers have noted an overrepresentation of lefthandedness and ambidexterity. Twin studies and striking gender differences in stuttering indicate that stuttering has some genetic basis.

Learning theories about the cause of stuttering include the semantogenic theory, in which stuttering is a learned response to normative early childhood disfluencies. Another learning model focuses on classical conditioning, in which the stuttering becomes conditioned to environmental factors. In the cybernetic model, speech is a process that depends on appropriate feedback for regulation; stuttering is hypothesized to occur because of a breakdown in the feedback loop. The observations that white noise reduces stuttering and delayed auditory feedback produces stuttering in regular speakers lend support to the feedback theory.

The motor functioning of some children who stutter appears to be delayed or slightly abnormal. The observation of difficulties in speech planning exhibited by some children who stutter suggests that higher-level cognitive dysfunction may contribute to stuttering. Although children who stutter do not routinely exhibit other speech and language disorders, family members of these children often exhibit an increased incidence of a variety of speech and language disorders. Stuttering is most likely to be caused by a set of interacting variables that include both genetic and environmental factors.

SOCIAL (PRAGMATIC) COMMUNICATION DISORDER

Social (pragmatic) communication disorder is a newly added diagnosis to DSM-5 characterized by persistent deficits in using verbal and nonverbal communication for social purposes in the absence of restricted and repetitive interests and behaviors. Examples of observable deficits include difficulty in understanding and following social rules of language, gesture, and social context. This deficit may limit a child’s ability to communicate effectively with peers, in academic settings, and family activities. For successful social and pragmatic communication, a child or adolescent must integrate gestures, language, and social context of a given interaction to correctly infer its

meaning. Thus, the child or adolescent must understand another speaker's "intention" of the communication with verbal and nonverbal cues. This understanding requires that they understand the environmental and social context of the interaction. One of the reasons DSM-5 introduced the diagnosis of social (pragmatic) communication disorder was to include those children with social communication impairment who do not exhibit restrictive and repetitive interests and behaviors and therefore do not fulfill the criteria for autism spectrum disorders. Pragmatic communication encompasses the ability to infer meaning in a given communication by not only understanding the words used but also integrating the phrases into their prior understanding of the social environment. Social (pragmatic) communication disorder is a new disorder; however, the concept of children with social communication deficits without repetitive and restrictive interests and behaviors has been identified for many years and is often associated with delayed language acquisition and language disorder.



Table 2-11
Social (Pragmatic) Communication Disorder

	DSM-5	ICD-10
Diagnostic name	Social (Pragmatic) Communication Disorder	Developmental disorder of speech and language, unspecified
Duration	Begins in the early developmental period	
Symptoms	Difficulties with verbal/nonverbal communication in social situations. Examples include difficulties with: • Greetings • Sharing information	(No criteria listed)

	• Matching conversational tone • Matching conversation context • Following normal rules for social conversations • Understanding implied meanings
Psychosocial consequences of symptoms	Causing functional limitation
Exclusions (not result of):	• Another medical condition • Another neurologic condition • Another mental disorder • Autism spectrum disorder • Intellectual disability • Global developmental delay

Clinical Features

Social (pragmatic) communication disorder is an impaired ability to use verbal and nonverbal communication for social purposes effectively and occurs in the absence of restricted and repetitive interests and behaviors. Although the preceding deficits begin in the early developmental period, we would rarely make the diagnosis in a child younger than 4 years of age. In milder cases, the difficulties may not become apparent until adolescence, with its associated increased demands for language and social understanding. The deficits in social communication lead to impairment in function in social situations, in developing relationships, and in family and academic settings.

Diagnosis

According to the DSM-5, all of the following features must be present in order to meet diagnostic criteria: (1) Deficits in using appropriate communication such as greeting or sharing information in a social situation

or context. (2) Impaired ability to modulate the tone, level, or vocabulary used in social communication to match the listener and the situation, such as inability to simplify communication when speaking to a young child. (3) Impaired ability in following the rules for conversations such as taking turns or rephrasing a statement for clarification and failure to recognize and respond socially appropriately to verbal and nonverbal feedback. (4) Difficulty understanding things that are not explicitly stated, impaired ability to make inferences, understand humor, or interpret socially ambiguous stimuli.

The diagnosis of social (pragmatic) communication disorder can be difficult to distinguish from mild variants of autism spectrum disorder in which repetitive and restricted interests and behaviors are minimal. There are discrepant data regarding how many children previously diagnosed with autism would be excluded from the DSM-5 criteria, which now focus on only two symptom domains: social communication deficits and restricted, repetitive interests and behaviors. In one study, only 60.6 percent of children who had previously met the criteria for autistic spectrum disorder in the previous edition of the DSM met DSM-5 criteria for autistic spectrum disorder. However, in another study, up to 91 percent of patients with autism continued to meet the same DSM-5 criteria.

The essential features of social (pragmatic) communication disorder are persistently impaired social pragmatic communication resulting in limited effective communication, compromised social relationships, and difficulties with academic or occupational achievement.

Table 2-11 lists the diagnostic approach to social (pragmatic) communication disorder.

Differential Diagnosis

The primary diagnostic consideration in social (pragmatic) communication disorder is autism spectrum disorder. The two disorders are most easily distinguished when the prominence of restricted and repetitive interests and behaviors characteristic of autistic spectrum disorder is present. However, in many cases of autism, restrictive interests and repetitive behaviors manifest more prominently in the early developmental period and are not apparent in older childhood. However, even when these features are not observable, if we

can obtain them in the patient's history, we would diagnose autism rather than social (pragmatic) communication disorder. We should consider social (pragmatic) communication disorder only when the restricted interests and repetitive behaviors have never been present. ADHD may overlap with a social (pragmatic) communication disorder in social communication disturbance; however, the core features of ADHD are not likely to be confused with autism spectrum disorder.

In some cases, however, the two disorders may coexist. Another childhood disorder with socially impairing symptoms that may overlap with social (pragmatic) communication disorder is social anxiety disorder. In social anxiety disorder, however, social communication skills are present, but not manifested in feared social situations. In social (pragmatic) communication disorder, appropriate social communication skills are not present in any setting. Both social anxiety disorder and social (pragmatic) communication disorder may occur comorbidly, however, and children with social (pragmatic) communication disorder may be at higher risk for social anxiety disorder. Finally, intellectual disability may be confused with social (pragmatic) communication disorder, in that social communication skills may be deficits in children with intellectual disability. We should only make a diagnosis of social (pragmatic) communication disorder when social communication skills are more severe than intellectual disability.

Comorbidity

Social (pragmatic) communication disorder is commonly associated with language disorder, consisting of diminished vocabulary for expected age, deficits in receptive skills, as well as impaired ability to use expressive language. ADHD is often concurrent with social (pragmatic) communication disorder. Specific learning disorders with impairments in reading and writing are also commonly comorbid disorders with social (pragmatic) communication disorder. Although some symptoms of social anxiety disorder may overlap with social (pragmatic) communication disorder, the full disorder of social anxiety disorder may emerge comorbidly with social (pragmatic) communication disorder.

Course and Prognosis

The course and outcome of social (pragmatic) communication disorder are highly variable and dependent on both the severity of the disorder and potential interventions administered. By age 5 years, most children demonstrate enough speech and language to be able to discern the presence of deficits in social communication. However, in the milder forms of the disorder, social communication deficits may not be identified until adolescence, when language and social interactions are sufficiently complex that deficits stand out. Many children have significant improvement over time; however, even so, some early pragmatic deficits may cause lasting impairment in social relationships and academic progress. There is a newly growing body of investigations on therapeutic interventions that may affect future outcomes of social (pragmatic) communication disorder.

Treatment

There are few data to date to inform an evidence-based treatment for social (pragmatic) communication disorder or to sufficiently distinguish it from other disorders with overlapping symptoms such as autism spectrum disorder, ADHD, and social anxiety disorder. A randomized controlled trial (RCT) of a social communication intervention explicitly directed at children with social (pragmatic) communication disorder aimed at three areas of communication: (1) social understanding and social interaction; (2) verbal and nonverbal pragmatic skills, including conversation; and (3) language processing, involving making inferences, and learning new words. Although the primary outcome measure in this study did not show significant differences for the intervention group versus the “treatment as usual” group, there were several ratings by parents and teachers that demonstrate potential improvements in social communication skills after a 20-session intensive intervention for social (pragmatic) communication disorder. Continued investigation is necessary to both validate the preceding results and to promote evidence-based treatments for children with social (pragmatic) communication disorder.

Epidemiology

It is difficult to estimate the prevalence of social (pragmatic) communication disorder. Nevertheless, a body of literature has documented a profile of

children who present with these persistent difficulties in pragmatic language, who do not meet the criteria for autism spectrum disorder.

Etiology

A family history of communication disorders, autism spectrum disorder, or specific learning disorder all appear to increase the risk for social (pragmatic) communication disorder. This increased risk suggests that genetic influences are contributing factors in the development of this disorder. The etiology of social (pragmatic) communication disorder, however, is likely to be multifactorial, and given its frequent comorbidity with both language disorder and ADHD, developmental and environmental influences are also likely to play a role.

UNSPECIFIED COMMUNICATION DISORDER

Disorders that do not meet the diagnostic criteria for any specific communication disorder fall into the category of unspecified communication disorder. An example is voice disorder, in which the patient has an abnormality in pitch, loudness, quality, tone, or resonance. To be coded as a disorder, the voice abnormality must be sufficiently severe to impair academic achievement or social communication. Operationally, speech production includes five interacting subsystems, including respiration (airflow from the lungs), phonation (sound generation in the larynx), resonance (shaping of the sound quality in the pharynx and nasal cavity), articulation (modulation of the sound stream into consonant and vowel sounds with the tongue, jaw, and lips), and suprasegmentalia (speech rhythm, loudness, and intonation). These systems work together to convey information, and voice quality conveys information about the speaker's emotional, psychological, and physical status. Thus, voice abnormalities can cover a broad area of communication as well as indicate many different types of abnormalities.

Cluttering is not a disorder in the DSM-5, but it is an associated speech abnormality in which a disturbed rate and rhythm of speech impair intelligibility. ICD-10 does include it as a behavioral disorder usually occurring in childhood or adolescence. Speech is erratic and dysrhythmic and consists of rapid, jerky spurts that are inconsistent with standard phrasing

patterns. The disorder usually occurs in children between 2 and 8 years of age; in two-thirds of cases, the patient recovers spontaneously by early adolescence. Cluttering is associated with learning disorders and other communication disorders.

2.3 Autism Spectrum Disorder

Autism spectrum disorder (previously known as pervasive developmental disorders), describes a wide range of impairments in social communication and restricted and repetitive behaviors. It is a phenotypically heterogeneous group of neurodevelopmental syndromes, with polygenic heritability. Before DSM-5, five overlapping disorders captured the spectrum: *autistic disorder, Asperger disorder, childhood disintegrative disorder, Rett syndrome,* and *pervasive developmental disorder not otherwise specified.* These differed by the level of severity, specific syndrome, and in some cases underlying pathology. ICD-10 continues to follow this approach to a degree. However, the recent clinical consensus has shifted the conceptualization of autism spectrum disorder toward a continuum model in which heterogeneity of symptoms is inherent in the disorder. DSM-5 collapses the core diagnostic impairments into two domains: deficits in social communication, and restricted and repetitive behaviors. Aberrant language development and usage are no longer considered a core feature of autism spectrum disorder. This diagnostic change, is based, in part, on recent studies in siblings with diagnoses of autistic disorder, suggesting that symptom domains may be transmitted separately and that aberrant language development and usage is not a defining feature, but an associated feature in some individuals with autism spectrum disorder. Autism spectrum disorder is typically evident during the second year of life, and in severe cases, a lack of developmentally appropriate interest in social interactions may be noted even in the first year. Some studies suggest that a decline in social interaction may ensue between the first and second years of life. However, in milder cases, core impairments in autism spectrum disorder may not be identified for several more years. Although language impairment is not a core diagnostic criterion in autism spectrum disorder, clinicians and parents share concerns about a child who,

by 12 to 18 months, has not developed any language, and delayed language accompanied by diminished social behavior are frequently the heralding symptoms in autism spectrum disorder. In up to 25 percent of cases of autism spectrum disorder, some language develops and is subsequently lost. Autism spectrum disorder in children with normal intellectual function and mild impairment in language function may not be identified until middle childhood when both academic and social demands increase. Children with autism spectrum disorder often exhibit intense, idiosyncratic interest in a narrow range of activities, resist change, and typically do not respond to their social environment per their peers.

According to the DSM-5, the diagnostic criteria for autism spectrum disorder include deficits in social communication and restricted interests, which present in the early developmental period. However, when subtle, the child's caregivers may not identify the symptoms until several years later. Approximately one-third of children who meet the DSM-5 criteria for autism spectrum disorder show intellectual disability.

Rett syndrome was in DSM-IV and remains in ICD-10. This disorder appeared to occur exclusively in females and is characterized by normal development for at least 6 months, followed by stereotyped hand movements, a loss of purposeful motions, diminishing social engagement, poor coordination, and decreasing language use. In the formerly labeled *childhood disintegrative disorder,* development progresses normally for approximately 2 years, after which the child shows a loss of previously acquired skills in two or more of the following areas: language use, social responsiveness, play, motor skills, and bladder or bowel control. The former *Asperger disorder* is an impairment in social relatedness and repetitive and stereotyped patterns of behavior without delay or marked aberrant language development and usage. In Asperger disorder, cognitive abilities and significant adaptive skills are age-appropriate, although social communication is impaired. A survey of children undertaken with the former autism spectrum disorders revealed that the average age of diagnosis was 3.1 years for children with autistic disorder, 3.9 years for children diagnosed with pervasive developmental disorder not otherwise specified, and 7.2 years for those youth with Asperger disorder. Children with autism spectrum disorder who exhibited severe language deficits received an autism spectrum disorder diagnosis, on average, a year earlier than children without impairment in language. Children with autism

spectrum disorder who exhibited repetitive behaviors such as hand-flapping, toe-walking, and odd play were diagnosed with autism spectrum disorders at a younger age than those who did not exhibit such behaviors. The current DSM-5 autism spectrum disorder criteria provide specifiers for the severity of the main domains of impairment and also specifiers for the presence or absence of language impairment and intellectual impairment.

DIAGNOSIS AND CLINICAL FEATURES

Table 2-12 describes the diagnostic approaches to autism spectrum disorder.

Core Symptoms of Autism Spectrum Disorder

Persistent Deficits in Social Communication and Interaction. Children with autism spectrum disorder characteristically do not conform to the expected level of reciprocal social skills and spontaneous nonverbal social interactions. Infants with autism spectrum disorder may not develop a social smile, and as older babies may lack the anticipatory posture for being picked up by a caretaker. Less frequent and poor eye contact is common during childhood and adolescence compared to other children. The social development of children with autism spectrum disorder is characterized by atypical, but not absent, attachment behavior. Children with autism spectrum disorder may not explicitly acknowledge or differentiate the most influential persons in their lives—parents, siblings, and teachers—and on the other hand, may not react as strongly to being left with a stranger compared to others their age. Children with autism spectrum disorder often feel and display extreme anxiety when someone disrupts their usual routine. By the time children with this disorder reach school age, their social skills may have increased, and social withdrawal may be less obvious, particularly in higher-functioning children. An observable deficit, however, often remains in spontaneous play with peers and in subtle social abilities that promote developing friendships. The social behavior of children with autism spectrum disorder is often awkward and may be inappropriate. In older school-age children, social impairments may be manifested in a lack of conventional back and forth conversation, fewer shared interests, and less body and facial gestures during conversations. Cognitively, children with autism spectrum

disorder are frequently more skilled in visual–spatial tasks than in tasks requiring skill in verbal reasoning.

Table 2-12
Autism Spectrum Disorder

	DSM-5	ICD-10
Diagnostic name	Autism Spectrum Disorder	Childhood A
Duration	Begins in the early developmental period	Occurs durir developm before ag
Symptoms	Deficits in social interaction. Example deficits: • Emotional reciprocity/appropriate response to social interactions • Nonverbal communication/eye contact/ body language/gesturing and/or facial expression • Forming/developing/maintaining relationships and friendships Restricted/repetitive behaviors or interests. Examples: • Stereotyped motor movements • Rigid about daily routines/rituals • Restricted/fixed interests • Hyper- or hyporeactivity to sensations	Abnormal: • Social fu • Commu • Behavio (repetiti
Required number of symptoms	Symptoms from each category above	
Psychosocial consequences of symptoms	Functional impairment	

Exclusions (not result of):	Intellectual disability Global developmental delay	Atypical auti symptoms all diagnos for childho Asperger Sy difficulties interaction that found addition to interests; deficits in cognition a
Symptom Specifiers	***With or without accompanying intellectual impairment*** ***With or without accompanying language impairment*** ***Associated with another neurodevelopmental, mental or behavioral disorder*** ***With catatonia***	
Severity specifiers	For social impairment: **mild, moderate, severe** For behavioral patterns: **mild, moderate, severe**	

One observation of the cognitive style of children with autism spectrum disorder is an impaired ability to infer the feelings or emotional state of others around them. That is, individuals with autism spectrum disorder have difficulty with making attributions about the motivation or intentions of others (also termed “theory of mind”) and thus have difficulty developing empathy. The lack of a “theory of mind” produces difficulties interpreting the social behavior of others and leads to a lack of social reciprocation.

Individuals with autism spectrum disorder generally desire friendships, and higher functioning children may be aware that their lack of spontaneity and

poor skills in responding to the emotions and feelings of their peers are significant obstacles in developing friendships. Children with autism spectrum disorder are often avoided or shunned by peers who expect them to conform to their mainstream activities and experience their behavior as awkward and alienating. Adolescents and adults with autism spectrum disorder often desire romantic relationships, and for some, their increase in social competence and skills over time enables them to develop long-term relationships.

Restricted, Repetitive Patterns of Behavior, Interests, and Activities. From the first years of life, in a child with autism spectrum disorder, developmentally expected exploratory play is restricted and muted. They may not use toys and objects in a typical manner. Instead, they often manipulate the toys in a ritualistic manner, with fewer symbolic features. Children with autism spectrum disorder generally do not show the level of imitative play or abstract pantomime that other children of their age exhibit spontaneously. The activities and play of children with autism spectrum disorder may appear more rigid, repetitive, and monotonous than their peers. Ritualistic and compulsive behaviors are common in early and middle childhood. Children with autism spectrum disorder often seem to enjoy spinning, banging, and watching water flow. Frank compulsive behaviors are not uncommon among children with autism spectrum disorder, such as lining up objects, and not infrequently a child with autism spectrum disorder may exhibit a strong attachment to a particular inanimate object. Children with autism spectrum disorder who are severely intellectually disabled have increased rates of self-stimulatory and self-injurious behaviors. Stereotypies, mannerisms, and grimacing emerge most frequently when a child with autism spectrum disorder is in a less-structured situation. Children with autism spectrum disorder often find transitions and change intimidating. Moving to a new house, rearranging furniture in a room, or even a change such as eating a meal before a bath when the reverse was the routine, may evoke panic, fear, or temper tantrums in a child with autism spectrum disorder.

Associated Physical Characteristics. At first glance, children with autism spectrum disorder do not show any physical signs indicating the disorder. Children with autism spectrum disorder, overall, do exhibit higher

rates of minor physical anomalies, such as ear malformations, and others that may reflect abnormalities in fetal development of those organs along with parts of the brain.

A higher-than-expected number of children with autism spectrum disorder do not show early handedness and lateralization and remain ambidextrous at an age when most children establish cerebral dominance. Children with autism spectrum disorder may have a higher incidence of abnormal dermatoglyphics (e.g., fingerprints) than those in the general population. This finding may suggest a disturbance in neuroectodermal development.

Associated Behavioral Symptoms that May Occur in Autism Spectrum Disorder

Disturbances in Language Development and Usage. Deficits in language development and difficulty using language to communicate ideas are not among the core criteria for diagnosing autism spectrum disorder; however, they occur in a subset of those individuals with autism spectrum disorder. Some children with autism spectrum disorder are not merely reluctant to speak, and their speech abnormalities do not result from a lack of motivation. Language deviance, as much as language delay, is characteristic of more severe subtypes of autism spectrum disorder. Children with severe autism spectrum disorder have significant difficulty putting meaningful sentences together, even when they have extensive vocabularies. When children with autism spectrum disorder with delayed language learn to converse fluently, their conversations may impart information without typical prosody or inflection.

In the first year of life, a typical pattern of babbling may be minimal or absent. Some children with autism spectrum disorder vocalize noises—clicks, screeches, or nonsense syllables—in a stereotyped fashion, without a seeming intent of communication. Unlike most young children who generally have better receptive language skills than expressive ones, children with autism spectrum disorder may express more than they understand. Words and even entire sentences may drop in and out of a child’s vocabulary. It is not atypical for a child with autism spectrum disorder to use a word once and then not use it again for a week, a month, or years. Children with autism spectrum disorder may exhibit speech that contains echolalia, both immediate and

delayed, or stereotyped phrases that seem out of context. These language patterns are frequently associated with pronoun reversals. A child with autistic disorder might say, “You want the toy” when she means that she wants it. Difficulties in articulation are also common. Many children with autistic disorder use peculiar voice quality and rhythm. About 50 percent of autistic children never develop useful speech. Some of the brightest children show a particular fascination with letters and numbers. Children with autism spectrum disorder sometimes excel in specific tasks or have special abilities; for example, a child may learn to read fluently at preschool age (hyperlexia), often astonishingly well. Very young children with autism spectrum disorder who can read many words, however, have little comprehension of the words read.

Intellectual Disability. About 30 percent of children with autism spectrum disorder are intellectually disabled. Of those, about 30 percent of children function in the mild to moderate range, and about 45 to 50 percent are severe to profoundly intellectually disabled. The IQ scores of autism spectrum disorder children with intellectual impairments tend to reflect the most severe problems with verbal sequencing and abstraction skills, with relative strengths in visuospatial or rote memory skills. This finding suggests the importance of defects in language-related functions.

Irritability. Broadly defined, irritability includes aggression, self-injurious behaviors, and severe temper tantrums. These phenomena are common in children and adolescents with autism spectrum disorder. Severe temper tantrums may be difficult to subdue, and self-injurious behaviors are often problematic to control. Everyday situations can cause these symptoms in which one would expect a youth to transition from one activity to another, sit in a classroom setting, or remain still when they desire to run around. In children with autism spectrum disorder who are lower functioning and have intellectual deficits, aggression may emerge unexpectedly without an obvious trigger or purpose, and self-injurious behaviors such as head banging, skin picking, and biting oneself may also occur.

Instability of Mood and Affect. Some children with autism spectrum disorder exhibit sudden mood changes, with bursts of laughing or crying

without an apparent reason. It is difficult to learn more about these episodes if the child cannot express the thoughts related to the affect.

Response to Sensory Stimuli. Children with autism spectrum disorder may overrespond to some stimuli and underrespond to other sensory stimuli (e.g., to sound and pain). It is not uncommon for a child with autism spectrum disorder to appear deaf, at times showing little response to a normal speaking voice; on the other hand, the same child may show intent interest in the sound of a wristwatch. Some children have a heightened pain threshold or an altered response to pain. Indeed, some children with autism spectrum disorder do not respond to an injury by crying or seeking comfort. Some youth with autism spectrum disorder perseverate on a sensory experience; for example, they frequently hum a tune or sing a song or commercial jingle before saying words or using speech. Some particularly enjoy vestibular stimulation—spinning, swinging, and up-and-down movements.

Hyperactivity and Inattention. Hyperactivity and inattention are both common behaviors in young children with autism spectrum disorder. Lower than average activity level is less frequent; when present, it often alternates with hyperactivity. Short attention span, poor ability to focus on a task, may also interfere with daily functioning.

Precocious Skills. Some individuals with autism spectrum disorder have precocious or splinter skills of high proficiency, such as prodigious rote memories or calculating abilities, usually beyond the capabilities of their conventional peers. Other potential precocious abilities in some children with autism spectrum disorder include hyperlexia, an early ability to read well (even though they cannot understand what they read), memorizing and reciting, and musical abilities (singing or playing tunes or recognizing musical pieces).

Insomnia. Insomnia is a frequent sleep problem among children and adolescents with autism spectrum disorder, estimated to occur in 44 to 83 percent of school-age children.

Minor Infections and Gastrointestinal Symptoms. Young children

with autism spectrum disorder can have a higher-than-expected incidence of upper respiratory infections and other minor infections. They may also have febrile seizures. Some do not become febrile when they have a minor infection, and they may not have the typical malaise. The children can also have a great many gastrointestinal symptoms. These include such symptoms as excessive burping, constipation, and loose bowel movements. In other children, behavior problems and relatedness seem to improve noticeably during a minor illness, and in some, such changes are a clue to physical illness.

Assessment Tools

A standardized instrument that can be very helpful in eliciting comprehensive information regarding autism spectrum disorder is the *Autism Diagnostic Observation Schedule-Generic* (ADOS-G).

Brett was the first of two children born to middle-class parents both in their early 40s after a difficult pregnancy, with induced labor at 36 weeks due to fetal distress. As an infant, Brett was undemanding and relatively placid; he did not have colic, and motor development proceeded appropriately, but he had delayed language development. Brett's parents first became concerned about his development when he was 18 months of age and still not speaking; however, upon questioning, they noted that, in comparison to other toddlers in his playgroup, Brett had seemed less interested in social interaction and the social games with toddlers and adults. Stranger anxiety became marked at 18 months, much later compared to the other toddlers in his daycare program. Brett would become extremely upset if his usual daycare worker was not present and would tantrum until his mother took him home. Brett's pediatrician initially reassured his parents that he was a "late talker"; however, when Brett was 24 months old, he was referred for a developmental evaluation. At 24 months, motor skills were age-appropriate. His language and social development, however, were severely delayed, and he was resistant to changes in routine and unusually sensitive to aspects of the inanimate environment. Brett's play skills were quite limited, and he played with toys in repetitive and idiosyncratic ways. His younger sister, now 12

months, was beginning to say a few words, and the family history was negative for language and developmental disorders. A comprehensive medical evaluation revealed a normal EEG and CT scan; genetic screening and chromosome analysis were routine as well.

Brett was diagnosed with autism spectrum disorder, and he started a special education program where he gradually began to speak. His speech was extremely literal and characterized by monotonic voice quality and an occasional pronoun reversal. Brett often spoke and was able to make his needs known; however, his language was odd, and the other toddlers did not play with him. Brett pursued mainly solo activities and remained quite isolated. By age 5 years, Brett was quite attached to his mother and often became separation anxious and upset when she went out, exhibiting severe tantrums. Brett also had developed several self-stimulatory behaviors, such as waving his fingers in front of his eyes. His extreme sensitivity to change continued over the next few years. Intelligence testing revealed a full-scale IQ in the average range with relative weakness in the verbal subtests compared to the performance subtests. In the fourth grade, Brett began to have severe behavioral problems at school and at home. Brett was unable to complete his classwork, would wander around the classroom, and would begin to tantrum when the teacher insisted that he sit in his seat. Brett would sometimes begin screaming so loudly that he had to leave the classroom. He would then become upset and throw all of his books off his desk in a rage, sometimes inadvertently hitting other students. It took him up to 2 hours to calm down. At home, Brett would fly into a tantrum if anyone touched his things, and he would become stubborn and belligerent when asked to do anything that he was not expecting. Brett's tantrum behavior continued into middle school, and by the eighth grade, when he was 13 years old, these behaviors became so severe that the school warned his parents that he was becoming unmanageable.

Brett was evaluated by a child and adolescent psychiatrist who recommended a social skills group for him and prescribed risperidone. Brett's tantrums became less frequent and less severe. Brett seemed calmer and did not become physically out of control during tantrums. Brett continued in middle school in a combination of special education classes and regular classes. Brett's social skills group was helpful in

terms of teaching him how to approach peers in ways that would lead to less rejection. Brett had made some acquaintances, and by the time he started high school, he had acquired two friends who would come to his home and play video games with him. Brett knew that he was different from the other students, but he had trouble articulating what was different about him. Brett continued in high school with a combination of special and regular education and had plans to attend a community college and live at home for the first year. (Adapted from a case by Fred Volkmar, M.D.)

DIFFERENTIAL DIAGNOSIS

Disorders to consider in the differential diagnosis of autism spectrum disorder include social (pragmatic) communication disorder, the newly described DSM-5 communication disorder; schizophrenia with childhood-onset; congenital deafness or severe hearing disorder; and psychosocial deprivation. It is also challenging to make the diagnosis of autism spectrum disorder because of its potentially overlapping symptoms with childhood schizophrenia, intellectual disability syndromes with behavioral symptoms, and language disorders. Given the many concurrent problems often encountered in autism spectrum disorder, Michael Rutter and Lionel Hersov suggested a stepwise approach to the differential diagnosis.

Social (Pragmatic) Communication Disorder

Patients with this disorder have difficulty with conforming to typical storytelling, understanding the rules of social communication through language, exemplified by a lack of conventional greeting of others, taking turns in a conversation, and responding to verbal and nonverbal cues of a listener. Other forms of language impairment may accompany social communication disorder such as delay in learning language or expressive and receptive difficulties. Social communication disorder occurs with higher frequency in relatives of individuals with autism spectrum disorder, which increases the difficulty in discriminating this disorder from autism spectrum disorder. Although relationships may be negatively affected by social communication disorder, this disorder does not include restricted or repetitive

behaviors and interests, as autism spectrum disorder does.

Childhood-Onset Schizophrenia

Schizophrenia is rare in children younger than 12 years and almost nonexistent before the age of 5 years. Characterized by hallucinations or delusions, childhood-onset schizophrenia has a lower incidence of seizures and intellectual disability and poor social skills. Table 2-13 compares autism spectrum disorder and schizophrenia with childhood-onset.

Intellectual Disability with Behavioral Symptoms

Children with intellectual disability may exhibit behavioral symptoms that overlap with some autism spectrum disorder features. The main differentiating features between autism spectrum disorder and intellectual disability are that children with intellectual disability syndromes generally display global impairments in both verbal and nonverbal areas. In contrast, children with autism spectrum disorder are relatively weak in social interactions compared to other areas of performance. Children with intellectual disability generally relate verbally and socially to adults and peers per their mental age, and they exhibit a relatively even profile of limitations.



Table 2-13
Autism Spectrum Disorder versus Childhood-Onset Schizophrenia

Criteria	Autism Spectrum Disorder	Schizophrenia (with Onset before Puberty)
Age of onset	Early developmental period	Rarely under the age of 5
Incidence	1%	<1 in 10,000
Sex ratio (M:F)	4:1	1.67:1 (slight preponderance of males)

Family history of schizophrenia	Not increased	Likely Increased
Prenatal and perinatal complications	Increased	Not increased
Behavioral characteristics	Poor social relatedness; may have aberrant language, speech or echolalia; stereotyped phrases; may have stereotypies, repetitive behaviors	Hallucinations and delusions; thought disorder
Adaptive functioning	Impaired	Deterioration in functioning
Level of intelligence	Wide range, may be intellectually disabled (30%)	Usually within normal range, may be low average normal
Pattern of IQ	Typical higher performance than verbal	More even
Grand mal seizures	4–32%	Low incidence

Adapted from Magda Campbell, M.D. and Wayne Green, M.D.

Language Disorder

Some children with language disorders also have autism spectrum disorder features, which may present a diagnostic challenge. Table 2-14 summarizes the significant differences between autism spectrum disorder and language disorders.

Table 2-14
Autism Spectrum Disorder versus Language Disorder

Criteria	Autism Spectrum Disorder	Language Disorder
Incidence	1%	5 of 10,000
Sex ratio (M:F)	4:1	Equal or almost equal sex ratio
Family history of speech delay or language problems	<25% cases	<25% cases
Associated deafness	Very infrequent	Not infrequent
Nonverbal communication (e.g., gestures)	Impaired	Actively utilized
Language abnormalities (e.g., echolalia, stereotyped phrases out of context)	Present in a subset	Uncommon
Articulation problems	Infrequent	Frequent
Intellectual level	Impaired in a subset (about 30%)	Uncommon, less frequently severe
Patterns of intelligence quotient (IQ) tests	Typically lower on verbal scores than performance scores	Often verbal scores lower than performance scores
Impaired social communication, restricted and repetitive behaviors	Present	Absent or, if present, mild
Imaginative play	Often impaired	Usually intact

Adapted from Magda Campbell, M.D. and Wayne Green, M.D.

Congenital Deafness or Hearing Impairment

Because children with autism spectrum disorder may appear mute or lack language development, we should rule out hearing impairment or congenital deafness. Differentiating factors include the following: infants with autism spectrum disorder may babble only infrequently, whereas deaf infants often have a history of relatively normal babbling that then gradually tapers off and may stop at 6 months to 1 year of age. Deaf children generally respond only to loud sounds, whereas children with autism spectrum disorder may ignore loud or normal sounds and respond to soft or low sounds. Most importantly, an audiogram or auditory-evoked potentials indicate a significant hearing loss in deaf children. Deaf children usually seek out nonverbal social communication with regularity and seek social interactions with peers and family members more consistently than children with autism spectrum disorder.

Psychosocial Deprivation

Severe neglect, maltreatment, and lack of parental care can lead children to appear apathetic, withdrawn, and alienated. They may have delayed language and motor skills. Children with these signs generally improve when placed in a favorable and enriched psychosocial environment, but such improvement is not the case with children with autism spectrum disorder.

COURSE AND PROGNOSIS

Autism spectrum disorder is typically a lifelong, albeit heterogeneous disorder with highly variable severity and prognosis. Children with autism spectrum disorder and IQs above 70 with average adaptive skills, who develop communicative language by ages 5 to 7 years, have the best prognoses. A longitudinal study comparing symptoms in children with high-IQ autism spectrum disorder at the age of 5 years, with their symptoms at age 13 through young adulthood, found that a small proportion no longer met criteria for autism spectrum disorder. Most of these youth demonstrated positive changes in communication and social domains over time. Early intensive behavioral interventions can provide a profound positive impact on many children with the disorder, and in some cases, lead to recovery and

function in the average range.

The symptoms that do not seem to improve substantively over time with early behavioral interventions are related to ritualistic and repetitive behaviors. However, currently, evidence-based behavioral interventions specifically targeting repetitive behaviors may ameliorate them. The prognosis of a given child with autism spectrum disorder is generally improved if the home environment is supportive.

TREATMENT

The goals of treatment for children with autism spectrum disorder are to target core behaviors to improve social interactions, communication, broaden strategies to integrate into schools, develop meaningful peer relationships, and increase long-term skills in independent living. Psychosocial treatment interventions aim to help children with autism spectrum disorder to develop skills in social conventions, increase socially acceptable and prosocial behavior with peers, and to decrease odd behavioral symptoms. In many cases, a child may require language and academic remediation. Also, treatment goals generally include the reduction of irritable and disruptive behaviors that may emerge in school and at home and may exacerbate during transitions. Children with intellectual disability require developmentally appropriate behavioral interventions to reinforce socially acceptable behaviors and encourage self-care skills. Also, parents of children with autism spectrum disorder often benefit from psychoeducation, support, and counseling in order to optimize their relationships and effectiveness with their children. Comprehensive treatment for autism spectrum disorder, including intensive behavioral programs, parent training and participation, and academic/educational interventions, have provided the most promising results. Components of these comprehensive treatments include expanding social skills, communication, and language, often through practicing imitation, joint attention, social reciprocity, and play in a directed but child-centered manner. Five RCTs of early intensive, comprehensive behavioral interventions targeting core features of autism spectrum disorder in children ranging in age from 2 to 5 years of age have shown increases in language acquisition, social interactions, and educational achievement at the end of the study period compared to control groups. The study periods ranged from 12

weeks to several years, and the settings were at home, in a clinic, or at school. The next section describes some comprehensive treatment models or adapted versions based on these studies.

Psychosocial Interventions

Early Intensive Behavioral and Developmental Interventions

UCLA/LOVAAS-BASED MODEL. This intensive and manualized intervention primarily utilizes techniques derived from applied behavior analysis, given on a one-to-one basis for many hours per week. A therapist and a child will work on practicing specific social skills, language usage, and other target play skills, with reinforcement and rewards provided for accomplishments and mastery of skills.

EARLY START DENVER MODEL (ESDM). Interventions occur in naturalistic settings such as in daycare, at home, and during play with other children. Parents learn to be co-therapists and provide the training at home while educational settings also provide the interventions. The focus of the interventions is on developing basic play skills and relationship skills and integrating applied behavior analysis into the interventions. This approach is focused on training for very young children and occurs within the context of the child's daily routine.

PARENT TRAINING APPROACHES. This includes Pivotal Response Training, in which parents learn to facilitate social and communication development within the home and during activities by targeting gateway or pivotal social behaviors for mastery by the child, with the expectation that once they master these central social skills, a natural generalizing of social behaviors would follow. The approach integrates extensive parent and family components into this type of intervention. Once parents learn the interventions, they gradually increase the frequency until they occur throughout the day with the child. Another example of a parent training approach is the Hanen More Than Words Program.

Social Skills Approaches

SOCIAL SKILLS TRAINING. Typically provided by therapeutic leaders to children of various ages in a group setting with peers, children are given guided practice in initiating social conversation, greetings, initiating games, and joint attention. The approach includes emotion identification and regulation in practice with recognizing and learning how to label emotions in given social situations, learning to attribute appropriate emotional reactions in others, and social problem-solving techniques. The goals are that with practice in the group setting, the child will be able to use the techniques in less-structured settings and internalize strategies to interact positively with peers.

Behavioral Interventions (BIs) and Cognitive-Behavioral Therapy (CBT) for Repetitive Behaviors and Associated Symptoms

BEHAVIORAL THERAPY. Applied behavioral analysis is somewhat effective in reducing some repetitive behaviors in children and adolescents with autism spectrum disorder. Early intervention helps for repetitive self-injurious behaviors; behavioral interventions may need to be combined with pharmacologic treatments to manage the symptoms adequately.

COGNITIVE-BEHAVIORAL THERAPY. There is a significant evidence base from RCTs for the efficacy of CBT for symptoms of anxiety, depression, and obsessive-compulsive disorders (OCDs) in children. There are fewer controlled trials of this treatment in children with autism spectrum disorder, although there are at least two published studies in which CBT was used to treat repetitive behavior in individuals with autism spectrum disorder.

Interventions for Comorbid Symptoms in Autism Spectrum Disorder

NEUROFEEDBACK. This modality can influence symptoms of ADHD, anxiety, and increased social interaction by providing computer games or other games which reinforce the desired behavior. At the same time, the child wears electrodes that monitor electrical activity in the brain. The aim is to influence brainwave activity to prolong or produce electrical activity present during the desired behaviors. This modality is still under investigation in the treatment of symptoms in autism spectrum disorder.

INSOMNIA. Insomnia is a prevalent concern among children and adolescents with autism spectrum disorder, and both behavioral and pharmacologic interventions may improve this condition. The most common behavioral intervention for insomnia in autism spectrum disorder is based on changing the parents' behavior first toward the child at bedtime and throughout the night, such that there is a removal of reinforcement and attention for being awake, leading to gradual extinction of the "staying awake" behavior. Several studies using massage therapy before bedtime in children with autism spectrum disorder between the ages of 2 years and 13 years provided an improvement in falling asleep and a sense of relaxation.

Educational Interventions for Children with Autism Spectrum Disorder

TREATMENT AND EDUCATION OF AUTISTIC AND COMMUNICATION-RELATED HANDICAPPED CHILDREN (TEACCH). Originally developed at the University of North Carolina at Chapel Hill in the 1970s, TEACCH involves structured teaching based on the notion that children with autism spectrum disorder have difficulty with perception. This teaching method incorporates many visual supports and a picture schedule to aid in teaching academic subjects as well as socially appropriate responses. Caregivers arrange the physical environment to support visual learning, and the day is structured to promote autonomy and social relatedness.

BROAD-BASED APPROACHES. These educational plans include a blend of teaching strategies that use behavioral analysis and also focus on language remediation. Behavioral reinforcement is provided for socially acceptable behaviors while teaching academic subjects. TEACCH can be part of a broader special educational program for autism spectrum disorder.

COMPUTER-BASED APPROACHES AND VIRTUAL REALITY. Computer-based and virtual reality approaches use computer programs to teach language acquisition and reading skills. This approach provides the child with a sense of mastery. It also delivers a behaviorally based instruction in a modality that is appealing for the child. The Let's Face It! program is a computerized game that helps to teach children with autism spectrum disorder to recognize faces.

It consists of seven interactive computer games that target changes in facial expression, attention to the eye region of the face, holistic face recognition, and identifying emotional expression. An RCT of this program with children with autism spectrum disorder provided evidence that after 20 hours of face training with Let's Face It!, compared to the control group, the trained children demonstrated improvement in their ability to focus on the eye region of a face and improved their analytic and holistic face-processing skills. Several studies using virtual reality environments to teach children with autism spectrum disorder social skills and interaction have provided evidence of their value. In one study, a virtual café for children with autism spectrum disorder allowed the children to practice ordering and paying for drinks and food by navigation with the use of a computer mouse.

Psychopharmacological Interventions

Psychopharmacological interventions in autism spectrum disorder help ameliorate behavioral symptoms rather than core features of autism spectrum disorder. Target symptoms include irritability, broadly including aggression, temper tantrums, and self-injurious behaviors, hyperactivity, impulsivity, and inattention.

Irritability. Two second-generation antipsychotics, risperidone, and aripiprazole, have been approved by the Food and Drug Administration (FDA) in the United States for treatment of irritability in individuals with autism spectrum disorder. Risperidone, a high-potency antipsychotic with combined dopamine (D_2) and serotonin ($5\text{-}HT_2$) receptor antagonist properties, has been shown to subdue aggressive or self-injurious behaviors in children with and without autism spectrum disorder. There have been seven RCTs, three reanalysis studies, and two add-on studies, which have converged to confirm risperidone as an efficacious pharmacologic treatment for irritability in children and adolescents with autism spectrum disorder. Typical doses range from 0.5 to 1.5 mg. Some of the preschoolers in this study were also receiving intensive behavioral treatments. Risperidone is considered the first-line of medication treatment for children and adolescents with autism spectrum disorder who exhibit severe irritability. Despite its efficacy, risperidone's main side effects of weight gain and increased

appetite, metabolic side effects such as hyperglycemia, prolactin elevation, and dyslipidemia, along with other common adverse effects such as fatigue, drowsiness, dizziness, and drooling, have limited its use in some individuals. Risperidone should be used with caution in individuals with underlying cardiac abnormalities or hypotension, since risperidone may contribute to orthostatic hypotension. In further continuation studies of risperidone in the treatment of irritability in autism spectrum disorder, persistent efficacy and tolerability were found over 6 months, with a rapid return of symptoms in good responders when discontinuing the risperidone. Other drugs studied in the treatment of irritability in autism spectrum disorder include aripiprazole and olanzapine.

Two extensive studies utilizing aripiprazole in the treatment of tantrums, aggression, and self-injury in children and adolescents with autism spectrum disorder found that aripiprazole was both efficacious and safe. Doses ranged from 5 to 15 mg/day. The main side effects included sedation, dizziness, insomnia, akathisia, nausea, and vomiting. Although weight gain was not as pronounced as with risperidone, it was still considered a moderate adverse event, with approximately 1.3 to 1.5 kg gained during an 8-week study period. The weight gain was similar at the lower and higher doses. Olanzapine, which blocks 5-HT_{2A} and D_2 receptors and also blocks muscarinic receptors, has been studied in children and adolescents with autism spectrum disorder for the treatment of irritability with a trend toward a positive response; however, significant weight gain of approximately 3.5 kg occurred. The main side effect was sedation.

Hyperactivity, Impulsivity, and Inattention. There have been several randomized placebo-controlled trials of methylphenidate for the treatment of hyperactivity, impulsivity, and inattention in children and adolescents with autism spectrum disorder. The Research Units of Pediatric Psychopharmacology found methylphenidate to be at least moderately efficacious at doses of 0.25 to 0.5 mg/kg for youth with autism spectrum disorder and ADHD symptoms. The efficacy of methylphenidate in this population was less effective than in children with ADHD without autism spectrum disorder, and children with autism spectrum disorder developed more frequent side effects, including increased irritability, compared to ADHD children. A study of methylphenidate in the treatment of hyperactivity

and inattention in preschoolers with autism spectrum disorder found the stimulant safe and relatively efficacious; half of the preschoolers developed side effects including increased stereotypies, gastrointestinal upset, sleep problems, and emotional lability. Among nonstimulants, one double-blind placebo-controlled study of hyperactivity, impulsivity, and inattention using atomoxetine in children with autism spectrum disorder found that it was significantly more effective than placebo. Side effects included sedation, irritability, constipation, and nausea. Clonidine, an α-agonist, has also been studied in children with autism spectrum disorder for the treatment of hyperactivity with mixed results. Guanfacine may also help in some cases.

Repetitive and Stereotypic Behavior. Antidepressants and mood stabilizers have been studied for the treatment of these core symptoms of autism spectrum disorder. One study with fluoxetine found the medication group only slightly better and not significantly better than the placebo group regarding the target symptoms, and another trial with escitalopram found no difference between groups. Risperidone, however, was found to be effective in targeting irritability, and restrictive and repetitive behaviors were improved. One study using valproate in a 12-week trial with 55 children with a mean age of 9½ years with autism spectrum disorder found that those who were considered responders to irritability also spent less time engaging in repetitive behaviors.

Agents Administered for Behavioral Impairment in Autism Spectrum Disorder Based On Open Trials. Quetiapine is an antipsychotic with more potent 5-HT_{2A} than D_2 receptor blocking properties. Although there are only open-label trials with this agent, some experts use this when risperidone and olanzapine fail or are intolerable. Doses range from 50 to 200 mg/day. Adverse effects include drowsiness, tachycardia, agitation, and weight gain.

Clozapine has a heterocyclic chemical structure that is related to certain first-generation antipsychotics, such as loxapine, although clozapine carries a lower risk of extrapyramidal symptoms. It is not generally used in the treatment of aggression and self-injurious behavior unless those behaviors coexist with psychotic symptoms. The most severe adverse effect is agranulocytosis, which necessitates monitoring white blood cell count weekly

during clozapine use. Its use is generally limited to treatment-resistant psychotic patients.

Ziprasidone has receptor-blocking properties at the 5-HT_{2A} and D_2 receptor sites and carries little risk of extrapyramidal and antihistaminic effects. No guidelines exist for its use in autistic children with aggressive and self-injurious behaviors; clinicians sometimes use it to treat these behaviors in treatment-resistant children. In studies of its use in adults with schizophrenia, dose ranges of 40 to 160 mg were effective. Adverse effects include sedation, dizziness, and lightheadedness. Before using this medication, the clinician should check an electrocardiogram.

Lithium is efficacious in children with aggression without autism spectrum disorder, and it is used clinically in the treatment of aggressive or self-injurious behaviors when antipsychotic medications are not helpful.

Agents Used for Behavioral Impairment in Autism Spectrum Disorder without Evidence of Efficacy. A double-blind study investigated the efficacy of amantadine, which blocks *N*-methyl-D-aspartate (NMDA) receptors, in the treatment of behavioral disturbance, such as irritability, aggression, and hyperactivity, in children with autism. Some researchers have suggested that abnormalities of the glutamatergic system may contribute to the emergence of autism spectrum disorders. High glutamate levels occur in children with the formerly labeled Rett syndrome. In the amantadine study, 47 percent of children on amantadine were "improved" per their parents, and 37 percent of children on placebo were rated "improved" by parents in irritability and hyperactivity, although this difference was not statistically significant. Investigators rated the children on amantadine "significantly improved" for hyperactivity. A double-blind, placebo-controlled study of the efficacy of the anticonvulsant lamotrigine on hyperactivity in children with autism showed high rates of placebo improvement in ratings of hyperactivity, which were similar to response on the medication.

Clomipramine is sometimes used but lacks RCTs to provide evidence of positive results. Fenfluramine, which reduces blood serotonin levels, has also been used but appears to be ineffective. Improvement does not seem to be associated with a reduction in blood serotonin levels. Naltrexone, an opioid

receptor antagonist, has been investigated without much success, based on the notion that blocking endogenous opioids would reduce autistic symptoms.

Tetrahydrobiopterin, a coenzyme that enhances the action of enzymes, was studied. However, the results were not significant. Post hoc analysis of the three core symptoms of autism—social interaction, communication, and stereotyped behaviors—revealed a significant improvement in social interaction score after 6 months of active treatment. There was a positive correlation between social response and IQ. These results suggest that there is a possible effect of tetrahydrobiopterin on the social functioning of children with autism.

A case report suggested that low-dose venlafaxine was efficacious in three adolescents and young adults with autistic disorder with self-injurious behavior and hyperactivity. The dose of venlafaxine used was 18.75 mg/day, and the efficacy continued over 6 months.

Complementary and Alternative Medicine (CAM) Approaches to Autism Spectrum Disorder

Complementary and alternative medicine (CAM) is a group of nontraditional treatments used in conjunction with conventional treatments. Safe interventions that target both core and associated behavioral features of autism spectrum disorder with unknown efficacy include the following: music therapy, to promote communication and expression, and yoga, to promote attention and decrease activity level. A biologically based practice that appears to be safe and efficacious is melatonin, which reduces sleep-onset latency in children. Other biologic practices that appear to be safe but with unknown efficacy include vitamin C, multivitamins, essential fatty acids, and the amino acids carnosine and carnitine. Secretin is ineffective in RCTs in the treatment of autism spectrum disorder.

EPIDEMIOLOGY

Prevalence

The diagnostic rate for autism spectrum disorders has increased over the last two decades, and currently 1 in 54 children is diagnosed with autism spectrum disorder in the United States. Autistic disorder, based on DSM-IV-

TR criteria, is believed to occur at a rate of about 8 cases per 10,000 children (0.08 percent). By definition, the onset of autism spectrum disorder is in the early developmental period; however, some children go undetected until they are much older. Because of this delay between onset and diagnosis, the prevalence rates increase with age in young children.

Sex Distribution

Autism spectrum disorder is diagnosed four times more often in boys than in girls. In clinical samples, girls with autism spectrum disorder more often exhibit intellectual disability than boys. One potential explanation for this is that girls with autism spectrum disorder without an intellectual disability may be less likely to be identified, referred clinically, and diagnosed.

ETIOLOGY AND PATHOGENESIS

Genetic Factors

Family and twin studies suggest that autism spectrum disorder has a significant heritable contribution; however, it does not appear to be fully penetrant. Although up to 15 percent of cases of autism spectrum disorder appear to be associated with a known genetic mutation, in most cases, its expression is dependent on multiple genes. Family studies have demonstrated increased rates of autism spectrum disorder in siblings of an index child, as high as 50 percent in some families with two or more children with autism spectrum disorder. Siblings of a child with autism spectrum disorder are also at increased risk for a variety of developmental impairments in communication and social skills, even when they do not meet the criteria for autism spectrum disorder.

The concordance rate of autistic disorder in two extensive twin studies was 36 percent in monozygotic pairs versus 0 percent in dizygotic pairs in one study and about 96 percent in monozygotic pairs versus about 27 percent in dizygotic pairs in the second study. High rates of cognitive impairments, in the nonautistic twin in monozygotic twins with perinatal complications, suggest that contributions of perinatal environmental factors interact with genetic vulnerability differentially in autism spectrum disorder.

The heterogeneity in the expression of symptoms in families with autism

spectrum disorder suggests that there are multiple patterns of genetic transmission. Studies indicate that both an increase and decrease in specific genetic patterns may be risk factors for autism spectrum disorder. In addition to specific genetic factors, gender plays a substantial role in the expression of autism spectrum disorder. Genetic studies have identified two biologic systems involved in autism spectrum disorder: the consistent finding of elevated platelet serotonin (5-HT) and the mTOR, that is, mammalian target of rapamycin–linked synaptic plasticity mechanisms, which is disrupted in autism spectrum disorder. These will be discussed further in the next section.

Several genetic disorders cause autism spectrum disorder symptoms as part of a broader phenotype. The most common of these inherited disorders is fragile X syndrome, an X-linked recessive disorder that is present in 2 to 3 percent of individuals with autism spectrum disorder. Fragile X syndrome exhibits a nucleotide repeat in the 5’ untranslated region of the *FMNR1* gene, resulting in symptoms of autism spectrum disorder. Children with fragile X syndrome characteristically exhibit intellectual disability, gross and fine motor impairments, an unusual facies, macroorchidism, and significantly diminished expressive language ability. Tuberous sclerosis, another genetic disorder characterized by multiple benign tumors, inherited by autosomal dominant transmission, is found with higher frequency among children with autism spectrum disorder. Up to 2 percent of children with autism spectrum disorder also have tuberous sclerosis.

Researchers who screened the DNA of more than 150 pairs of siblings with autism spectrum disorder found evidence of two regions on chromosomes 2 and 7 containing genes that may contribute to autism spectrum disorder. Additional genes hypothesized to be involved in autism spectrum disorder are on chromosomes 16 and 17.

Biomarkers in Autism Spectrum Disorder

Autism spectrum disorder is associated with several biomarkers, potentially resulting from interactions of genes and environmental factors, which then influence neuronal function, dendrite development, and contribute to altered neuronal information processing. Researchers have identified several biomarkers of abnormal signaling in the 5-HT system, the mTOR-linked synaptic plasticity mechanisms, and alterations of the γ-aminobutyric acid

(GABA) inhibitory system.

The first biomarker identified in autism spectrum disorder was elevated serotonin in whole blood, almost exclusively in the platelets. Platelets acquire 5-HT through the process of SERT (serotonin transporter), known to be hereditary, as they pass through the intestinal circulation. The genes that mediate SERT (*SLC64A*), and the 5-HT receptor 5-HT_{2A} gene (*HTR2A*) are known to be more heritable than autism spectrum disorder, and encode the same protein in the platelets and the brain. Because 5-HT is known to be involved in brain development, the changes in 5-HT regulation may lead to alterations in neuronal migration and growth in the brain.

Both structural and functional neuroimaging studies have suggested specific biomarkers associated with autism spectrum disorder. Several studies found increased total brain volume in children younger than 4 years of age with autism spectrum disorder, whose neonatal head circumferences were within normal limits or slightly below. By about age 5 years, however, 15 to 20 percent of children with autism spectrum disorder developed macrocephaly. Additional studies found confirmatory data in samples of infants who later had autism spectrum disorder, who exhibited normal head circumferences at birth; by 4 years, 90 percent had larger brain volumes than controls, with 37 percent of the disordered group meeting criteria for macrocephaly. In contrast, structural magnetic resonance imaging (sMRI) studies of children with autism spectrum disorder ranging from 5 to 16 years did not find mean values of total brain volume increased. One study followed the size of the amygdala in youth with autism spectrum disorder in the first few years of life, and similarly, found an increased size in the first few years of life, followed by a decrease in size over time. Several studies have found enlarged striatum in young children with autism spectrum disorder, with a positive correlation of striatal size with the frequency of repetitive behaviors. The dynamic process of the atypical and changing total brain volume observed in children with autism spectrum disorder lends support for the overarching hypothesis that there are sensitive periods or “critical periods” within the brain’s plasticity, which when disrupted, may contribute to the emergence of autism spectrum disorder.

Functional MRI (fMRI) studies have focused on identifying biomarkers; that is, the functional brain correlates of various observed core symptoms in

autism spectrum disorder. fMRI studies of children, adolescents, and adults with autism spectrum disorder have employed tasks including face perception, neutral face tasks, “theory of mind” deficits, language and communication impairments, working memory, and repetitive behaviors. fMRI studies have provided evidence that individuals with autism spectrum disorder tend to scan faces differently than controls, in that they focus more on the mouth region of the face rather than on the eye region and rather than scanning the entire face multiple times, individuals with autism spectrum disorder focus more on individual features of the face. In response to socially relevant stimuli, researchers have concluded that individuals with autism spectrum disorder have greater amygdala hyperarousal. In terms of “theory of mind,” that is, the ability to attribute emotional states to others, and to oneself, fMRI studies find differences in activation in brain regions such as the right temporal lobe and other areas of the brain known to become activated in controls during tasks involving the theory of mind. This difference has been hypothesized by some researchers to represent dysfunction of the mirror neuron system (MNS). Atypical patterns of frontal lobe activation have been found in multiple studies of autism spectrum disorder during face processing tasks, suggesting that this area of the brain may be critical in social perception and emotional reasoning. Decreased activation in individuals with autism spectrum disorder in the left frontal regions of the brain during memory and language-based tasks led researchers to hypothesize that individuals with autism spectrum disorder utilized more visual strategies during language processing than controls did.

Both sMRI and fMRI research have contributed to demonstrating brain correlates of core impairments observed in individuals with autism spectrum disorder.

Immunologic Factors

Several reports have suggested that immunologic incompatibility (i.e., maternal antibodies directed at the fetus) may contribute to autistic disorder. The lymphocytes of some autistic children react with maternal antibodies, which raise the possibility that embryonic neural tissues damage during gestation. These reports usually reflect single cases rather than controlled studies, and this hypothesis is still under investigation.

Prenatal and Perinatal Factors

A higher-than-expected incidence of prenatal and perinatal complications seems to occur in infants who later have autism spectrum disorder. The most significant prenatal factors associated with autism spectrum disorder in the offspring are advanced maternal and paternal age at birth, maternal gestational bleeding, gestational diabetes, and first-born baby. Perinatal risk factors for autism spectrum disorder include umbilical cord complications, birth trauma, fetal distress, small for gestational age, low birth weight, low 5-minute Apgar score, congenital malformation, ABO blood group system or Rh factor incompatibility and hyperbilirubinemia. Many of the obstetrical complications that are associated with risk for autism spectrum disorder are also risk factors for hypoxia, which may be an underlying risk factor itself. There is no sufficient evidence to implicate any one single perinatal or prenatal factor in autism spectrum disorder etiology, and a genetic predisposition to autism spectrum disorder may be interacting with perinatal factors.

Comorbid Neurologic Disorders

EEG abnormalities and seizure disorders occur with higher-than-expected frequency in individuals with autism spectrum disorder. Four percent to 32 percent of individuals with autism spectrum disorder have grand mal seizures at some time, and about 20 to 25 percent show ventricular enlargement on computed tomography (CT) scans. Various EEG abnormalities occur in 10 to 83 percent of children with the previously defined autistic disorder, and although no EEG finding is specific to autistic disorder, there is some indication of failed cerebral lateralization. The current consensus is that autism spectrum disorder is a set of behavioral syndromes caused by a multitude of factors acting on the CNS.

Psychosocial Theories

Studies comparing parents of children with autism spectrum disorder with parents of typical children have shown no significant differences in child-rearing skills. Researchers have disproved earlier speculations that parental emotional factors contribute to the development of autism spectrum disorder.

▲ 2.4 Attention-Deficit/Hyperactivity Disorder

ADHD is a neuropsychiatric condition affecting preschoolers, children, adolescents, and adults around the world, characterized by a pattern of diminished sustained attention, and increased impulsivity or hyperactivity. Based on family history, genotyping, and neuroimaging studies, there is clear evidence to support a biologic basis for ADHD. Although multiple regions of the brain and several neurotransmitters contribute to the emergence of symptoms, dopamine continues to be a focus of investigation regarding ADHD symptoms. The prefrontal cortex of the brain is a focus of interest because of its high utilization of dopamine and its reciprocal connections with other brain regions involved in attention, inhibition, decision-making, response inhibition, working memory, and vigilance. ADHD affects up to 5 to 8 percent of school-age children, with 60 to 85 percent of those diagnosed as children continuing to meet criteria for the disorder in adolescence, and up to 60 percent continuing to be symptomatic into adulthood. Children, adolescents, and adults with ADHD often have significant impairment in academic functioning as well as in social and interpersonal situations. ADHD is frequently associated with comorbid disorders, including learning disorders, anxiety disorders, mood disorders, and disruptive behavior disorders.

The DSM-5 has made several changes to the diagnostic criteria of ADHD in youth and adults. Whereas in the past, ADHD symptoms had to be present by age 7 years, in DSM-5, “several inattentive or hyperactive-impulsive symptoms” must be present by age 12 years. Previously, there were two subtypes: Inattentive and Hyperactive/Impulsive type. In DSM-5, however, subtypes have been replaced by the following three specifiers, which largely denote the same groups: (1) combined presentation, (2) predominantly inattentive presentation, and (3) predominantly hyperactive/impulsive presentation. Additional changes in DSM-5 include permitting a comorbid ADHD and autism spectrum diagnosis. Finally, in DSM-5, for adolescents 17 years and older and adults, only five symptoms, rather than six symptoms of

either inattention or hyperactivity and impulsivity, are required. Also, DSM-5 added symptoms to its list to reflect different presentations across the life span. To confirm a diagnosis of ADHD, impairment from inattention or hyperactivity and impulsivity must be present in at least two settings and interfere with developmentally appropriate social or academic functioning.

ADHD has historically been described in the literature using different terminology. In the early 1900s, impulsive, disinhibited, and hyperactive children—many of whom also had neurologic damage due to encephalitis—were grouped under the label *hyperactive syndrome*. In the 1960s, a heterogeneous group of children with poor coordination, learning disabilities, and emotional lability, but without specific neurologic disorders, were described as having "minimal brain damage"; however, over time, it became clear that this was an inappropriate term. There are many explanations for ADHD symptoms, including theories of abnormal arousal and reduced ability to modulate emotions. This theory was initially supported by the observation that stimulant medications increased sustained attention and improved focus. ADHD is one of the most well-researched childhood psychiatric disorders with several evidence-based treatments.

CLINICAL FEATURES

ADHD can have its onset in infancy, although it is rarely recognized until a child is at least toddler age. More commonly, infants with ADHD are active in the crib, sleep little, and cry a great deal.

In school, children with ADHD may attack a test rapidly but may answer only the first two questions. They may be unable to wait to be called on in school and may respond before everyone else. At home, caregivers cannot put them off for even a minute. Impulsiveness and an inability to delay gratification are characteristic. Children with ADHD are often susceptible to accidents.

The most cited characteristics of children with ADHD, in order of frequency, are hyperactivity, attention deficit (short attention span, distractibility, perseveration, failure to finish tasks, inattention, poor concentration), impulsivity (action before thought, abrupt shifts in activity, lack of organization, jumping up in class), memory and thinking deficits, specific learning disabilities, and speech and hearing deficits. Associated

features often include perceptual-motor impairment, emotional lability, and developmental coordination disorder. A significant percentage of children with ADHD show behavioral symptoms of aggression and defiance. School difficulties, both learning and behavioral, commonly exist with ADHD. Comorbid communication disorders or learning disorders that hamper the acquisition, retention, and display of knowledge complicate the course of ADHD.

A psychiatrist evaluated Justin, a 9-year-old boy, after his teacher informed his adoptive parents that she was unable to manage Justin's impulsive and aggressive behaviors in the classroom. Justin was attending public school and was in a regular classroom with two resource room periods per day to help him with reading and math. Justin also received speech therapy once a week. Justin's adoptive parents knew very little about his biologic family other than that his biologic mother was known to be a polydrug abuser. Currently, she was incarcerated. His parents adopted Justin as an infant, and the pediatrician had told his adoptive parents that Justin was entirely healthy at birth. However, ever since kindergarten, Justin's teachers had complained that Justin did "not seem to listen," had "poor concentration," and was unable to stay in his seat. Because Justin was an engaging and cute child, his teachers in kindergarten and first grade made accommodations for him in their classrooms despite their complaints. When Justin entered the second grade, however, it became clear that he was struggling with reading and writing, and the school initiated an individualized educational program (IEP). Justin was provided with resource room periods for remediation during the school day but continued to have additional problems getting along with his peers during lunch, and even at recess. Justin was often arguing or fighting with other children, who said that he did not know the rules of their games. Justin became angry when his peers criticized him, and he would often push his classmates. At home, Justin's adoptive parents were becoming more and more frustrated with Justin because he seemed to take hours to do a few math problems and was unable to write a paragraph without much help. Justin would become easily annoyed when frustrated with himself and then run around the house in a silly and

disruptive manner. Justin was a good-hearted child who seemed to get along best with children who were younger than he was. Justin did not seem to make any close friends among his classmates, and the teachers indicated that Justin's peers sometimes avoided him because he was too rough during play, and he did not follow the rules of their games. Justin had a difficult time waiting for his turn, and when reprimanded, he became easily provoked. Consequently, Justin became alienated and often bullied by his classmates. Justin was aware that he was not able to keep up with the classwork, and he told his adoptive parents that he was just "stupid." Although Justin acted in a rambunctious and impulsive manner, he also appeared sad, and one day after a fight with several peers, he told his adoptive parents that he was going to "kill" himself. At this point, Justin's parents became worried and decided that Justin's teacher was right, and they would seek a psychiatric evaluation for Justin. During the initial evaluation with a child and adolescent psychiatrist, Justin was a well-developed, cute, and active child, who appeared distracted and fidgety and somewhat sad. When asked about it, Justin said that he wanted to do "better" in school but that nobody liked him, he was failing his classes, and that he did not like doing homework. He denied suicidal thoughts and reported that he had only said that to his parents because he was angry at his peers. Justin admitted that it was tough for him to understand his school work and impossible to complete his assignments. The evaluation included several parent and teacher rating scales. These included The Child Behavior Checklist and the SNAP Rating Scale. Justin's teacher and parents endorsed similar symptoms, including poor organization, inability to follow directions, being forgetful in daily activities, impulsivity, with several episodes of running into the street without looking, blurting things out in the classroom without raising his hand, and recurrent fights with peers. Justin looked dejected in school when he was excluded from play activities by peers, and sullen or angry at home when his parents asked him to read or do homework. Based on the clinical history, the rating scales, and the teacher's report, a diagnosis of attention-deficit/hyperactivity disorder, with the DSM-5 specifier of combined presentation, was made. Also, Justin had a mood disorder with depressed mood, which did not qualify for major depression. The suggested treatment plan included a behavioral

plan allowing Justin to receive rewards for effort on his homework along with a trial of stimulant medication. An extensive medical history and recent physical examination did not reveal any systemic illnesses. After obtaining a normal ECG, the psychiatrist started Justin on a trial dose of a short-acting stimulant, methylphenidate at 10 mg, to determine if he could tolerate a stimulant without any unexpected sensitivities. Justin had no adverse effects, and soon the doctor switched him to a long-acting formulation (Concerta), 36 mg, which would last between 10 and 12 hours. Justin became more vigilant in class and seemed to be less restless and more focused, and his teacher reported that he was not getting out of his seat as often, although he continued to blurt out in class inappropriately, and he continued to have difficulty following directions and forgetting things. Because Justin was not experiencing any adverse effects and was still displaying some ADHD symptoms, his psychiatrist increased his Concerta to 54 mg/day. At this dose, Justin was better at sitting and finishing his classwork and homework. However, he began to have significant problems with insomnia and was becoming fatigued from not being able to fall asleep until about 2 AM on a nightly basis. The child and adolescent psychiatrist and Justin's parents discussed two options to address insomnia. One was to add a dose of short-acting clonidine in the evenings to cause a calming effect along with some sedative properties. The other was to initiate a methylphenidate transdermal patch, which could be applied to deliver a similar dose of methylphenidate throughout the day, and the patch could be removed at approximately 4 PM or 5 PM to determine which produced the desired effect for the target symptoms for the most optimal amount of time. Because the patch may deliver medication for an hour or so after its removal, Justin would need to try several different removal times to find the optimal treatment time. Justin's family and his child psychiatrist determined that it would be the best next step for Justin to try the transdermal patch rather than add medication to treat his insomnia. Justin was tried on the transdermal 20-mg patch and found that if it was removed by 5 PM, he was able to fall asleep within 30 to 45 minutes after getting into bed. Despite some mild erythema around the site of the patch, Justin experienced no other side effects and was glad that he did not have

to take pills each morning. It was determined by Justin's parents, teachers, and child and adolescent psychiatrist that Justin's ADHD symptoms were now under much-improved control. Justin began to receive better grades, and his self-esteem improved. However, Justin still had difficulties with peers and felt that he was not making as many friends as he wanted. Justin's child psychiatrist suggested that Justin receive weekly social skills group therapy that was led by a psychologist who had experience with group interventions for children with ADHD. At first, Justin did not want to attend these sessions, but after a few sessions, during which the group praised Justin for his appropriate interactions, he decided he liked the group. Over time, he even invited a few of his peers from the group to his home to play. The combination of the medication and the social skills group resulted in a significant improvement in Justin's ADHD symptoms. It also improved the quality of his relationships with peers and even his family. (Adapted from Greenhill LL, Hechtman LI. Attention-deficit/hyperactivity disorder. In: Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Vol. 2. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:3571.)

DIAGNOSIS

We can elicit the principal signs of inattention, impulsivity, and hyperactivity from a detailed history of a child's early developmental patterns along with direct observation of the child, especially in situations that require sustained attention. Hyperactivity may be more severe in some situations (e.g., school) and less marked in others (e.g., one-on-one interviews), and may be less evident in pleasant structured activities (sports). The diagnosis of ADHD requires persistent, impairing symptoms of either hyperactivity/impulsivity or inattention in at least two different settings. For example, most children with ADHD have symptoms in school and at home. Table 2-15 outlines the diagnostic criteria for ADHD.

Distinguishing features of ADHD are short attention span and high levels of distractibility for chronologic age and developmental level. In school, children with ADHD often exhibit difficulties following instructions and

require increased individualized attention from teachers. At home, children with ADHD frequently have difficulty complying with their parents' directions and may need to be asked multiple times to complete relatively simple tasks. Children with ADHD typically act impulsively, are emotionally labile, explosive, lack focus, and are irritable.

Children for whom hyperactivity is a predominant feature are more likely to be referred for treatment earlier than are children whose primary symptoms are attention deficit. Children with the combined inattentive and hyperactive-impulsive symptoms of ADHD, or predominantly hyperactive-impulsive symptoms of ADHD, are more apt to have a stable diagnosis over time and to exhibit comorbid conduct disorder than those children with inattentive ADHD. Specific learning disorders in the areas of reading, arithmetic, language, and writing frequently occur in association with ADHD. We should assess global development to rule out other sources of inattention.

School history and teachers' reports are critical in evaluating whether a child's difficulties in learning and school behavior are caused primarily by inattention or compromised understanding of the academic material. In addition to intellectual limitations, poor performance in school may result from maturational problems, social rejection, mood disorders, anxiety, or poor self-esteem due to learning disorders. Assessment of social relationships with siblings, peers, and adults, and engagement in free and structured activities may yield valuable diagnostic clues to the presence of ADHD.

The mental status examination in a given child with ADHD who is aware of his or her impairment may reflect a demoralized or depressed mood; however, we would not expect to see a thought disorder or impaired reality testing. A child with ADHD may exhibit distractibility and perseveration and signs of visual-perceptual, auditory-perceptual, or language-based learning disorders. A neurologic examination may reveal visual, motor, perceptual, or auditory discriminatory immaturity or impairments without overt signs of visual or auditory disorders. Children with ADHD often have problems with motor coordination and difficulty copying age-appropriate figures, rapid alternating movements, right-left discrimination, ambidexterity, reflex asymmetries, and a variety of subtle nonfocal neurologic signs (soft signs).

If there are indications of possible absence spells, clinicians should obtain a neurologic consultation and an EEG to rule out seizure disorders. A child with an unrecognized temporal lobe seizure focus may have behavior

disturbances which can resemble those of ADHD.

Pathology and Laboratory Examination

Evaluation for ADHD includes a comprehensive psychiatric and medical history. This history includes prenatal, perinatal, and toddler information. Complications during the mother's pregnancy are also informative. Medical problems that may produce symptoms overlapping with ADHD include petit mal epilepsy, hearing and visual impairments, thyroid abnormalities, and hypoglycemia. A thorough cardiac history should be taken, including an investigation of the life history of syncope, family history of sudden death, and a cardiac examination of the child. Although it is reasonable to obtain electrocardiography (ECG) study before treatment, if any cardiac risk factors are present, a cardiology consultation and examination are warranted. No specific laboratory measures are pathognomonic of ADHD.

A continuous performance task, a computerized task in which a child presses a button each time a particular sequence of letters or numbers flashes on a screen, is not specifically a useful diagnostic tool for ADHD; however, it may be useful in comparing a child's performance before and after medication treatment, particularly at different doses. Children with poor attention tend to make errors of omission—that is, they fail to press the button when the sequence has flashed. Impulsivity causes errors of commission, in which an impulsive child cannot resist pushing the button, even when the desired sequence has not yet appeared on the screen.



Table 2-15
Attention-Deficit/Hyperactivity Disorder

	DSM-5	ICD-10
Diagnostic name	Attention-Deficit/Hyperactivity Disorder	Hyperkinetic Disorders Disturbance of activity and attention
Duration	≥6 mo Symptoms present before the age of 12	Usually occurs before the age of 5

Symptoms	Inattention: • Poor attention to detail/frequent mistakes • Hard to maintain attention/focus • When spoken to, appears not to listen • Poor follow through on tasks • Poor organizational skills • Procrastinates from tasks requiring attention • Loses things • Distractible • Forgetfulness Hyperactivity/impulsivity: • Fidgety/restless • Cannot stay seated • Runs/climbs inappropriately • Cannot do things quietly (e.g., play) • Cannot keep still • Inappropriately talkative • Inappropriately blurts out answers • Cannot wait his turn • Interrupts others	Cannot engage in activities requiring prolonged attention Disorganized about completing tasks Excess activity May have: • Impulsivity • Recklessness • Social disinhibition
Required	≥6 of each category (≥5	

number of symptoms	for age ≥17) Symptoms occur in >1 setting	
Psychosocial consequences of symptoms	Functional impairment	
Exclusions (not result of):	Another mental illness	Hyperkinetic conduct disorder—referring to symptoms present in the context of conduct disorder
Symptom Specifiers	**Combined presentation** **Predominantly inattentive presentation** **Predominantly hyperactive/impulsive presentation**	
Course specifiers	**In partial remission:** last 6 mo, symptoms continue but less than required for diagnosis	
Severity specifiers	**Mild:** minimal symptoms required for diagnosis; Minor functional impairment **Moderate:** Intermediate symptoms and impairment **Severe:** Severe symptoms, marked impairment	

DIFFERENTIAL DIAGNOSIS

A temperamental constellation of high activity level and short attention span, in the normal range for the child's age, and without impairment, should be ruled out. Differentiating these temperamental characteristics from the cardinal symptoms of ADHD before the age of 3 years is difficult, mainly because of the overlapping features of an ordinarily immature nervous system and the emerging signs of visual-motor-perceptual impairments frequently seen in ADHD. It is critical to evaluate for anxiety. Anxiety can accompany ADHD as a symptom or comorbid disorder, and anxiety can manifest with overactivity and easy distractibility.

It is not uncommon for a child with ADHD to become demoralized or, in some cases, to develop depressive symptoms in reaction to persistent frustration with academic difficulties and resulting in low self-esteem. Mania and ADHD share many core features, such as excessive verbalization, motoric hyperactivity, and high levels of distractibility. Also, in children with mania, irritability seems to be more common than euphoria. Although mania and ADHD can coexist, children with bipolar I disorder exhibit more waxing and waning of symptoms than those with ADHD. Recent follow-up data for children who met the criteria for ADHD and subsequently developed bipolar disorder suggest that certain clinical features occurring during ADHD predict future mania. Children with ADHD who had developed bipolar I disorder at 4-year follow-up had a higher co-occurrence of additional disorders and a higher family history of bipolar disorders and other mood disorders than children without bipolar disorder.

Frequently, oppositional defiant disorder, or conduct disorder and ADHD, may coexist, and when that occurs, we should diagnose both disorders. We must also distinguish specific learning disorders from ADHD; a child may be unable to read or do mathematics because of a learning disorder, rather than because of inattention. ADHD often coexists with one or more learning problems, including deficits in reading, mathematics, or written expression.

The DSM-5 includes Unspecified ADHD as a category for disturbances of inattention or hyperactivity that cause impairment, but do not meet the full criteria for ADHD.

COURSE AND PROGNOSIS

The course of ADHD is variable. Symptoms persist into adolescence in 60 to 85 percent of cases, and into adult life in approximately 60 percent of cases. The remaining 40 percent of cases may remit at puberty or in early adulthood. In some cases, the hyperactivity may disappear, but the decreased attention span and impulse-control problems persist. Overactivity is usually the first symptom to remit, and distractibility is the last. ADHD does not usually remit during middle childhood. A family history of the disorder predicts persistence, as do adverse life events, and comorbidity with conduct symptoms, depression, and anxiety disorders. When remission occurs, it is usually between the ages of 12 and 20. After remission, the child can go on to have a productive adolescence and adult life, satisfying interpersonal relationships, and few significant sequelae. Most patients with the disorder, however, undergo partial remission and are vulnerable to antisocial behavior, substance use disorders, and mood disorders. Learning problems often continue throughout life.

In about 60 percent of cases, some symptoms persist into adulthood. Those who persist with the disorder may show diminished hyperactivity but remain impulsive and accident-prone. Although the educational attainments of people with ADHD as a group are lower than those of people without ADHD, early employment histories do not differ from those of people with similar educations.

Children with ADHD whose symptoms persist into adolescence are at higher risk for developing conduct disorder. Children with both ADHD and conduct disorder are also at risk for developing substance use disorders. The development of substance use disorders among ADHD youth in adolescence appears to be more related to the presence of conduct disorder rather than to ADHD.

Most children with ADHD have some social difficulties. Socially dysfunctional children with ADHD have significantly higher rates of comorbid psychiatric disorders and experience more problems with behavior in school as well as with peers and family members. Overall, the outcome of ADHD in childhood seems to be related to the degree of persistent comorbid psychopathology, primarily conduct disorder, social disability, and chaotic family factors. For optimal outcomes, it is essential to ameliorate children’s social functioning, diminish aggression, and improve family situations as early as possible.

TREATMENT

Pharmacotherapy

Pharmacologic treatment is considered the first line of treatment for ADHD. CNS stimulants are the first choice of agents in that they have the highest efficacy with generally mild tolerable side effects. Children, adolescents, and adults with known cardiac risks and abnormalities should not use stimulants. In healthy youths, both short- and sustained-release preparations have excellent safety records. The newer preparations aim to maximize the target effects and minimize the adverse effects in individuals with ADHD who obtain a partial response from methylphenidate or whose dose was limited by side effects.

Current strategies favor once a day sustained-release stimulant preparations for their convenience and diminished rebound side effects. Advantages of the sustained-release preparations are that a single dose will sustain the effects all day, and the medication is sustained at an approximately even level in the body throughout the day. In general the immediate-release preparations are expected to last 1 to 4 hours and extended-release preparations are expected to last up to 8 hours, with exceptions for the transdermal patch, which lasts an hour after it is removed.

Nonstimulant medications approved by the FDA in the treatment of ADHD include atomoxetine, a norepinephrine uptake inhibitor. Unlike the stimulants, atomoxetine carries with it a black-box warning for potential increases in suicidal thoughts or behaviors and requires children with ADHD to be monitored for these symptoms, similarly to children taking antidepressants. α-agonists, including clonidine and guanfacine, are also useful in treating ADHD. The FDA has approved the extended-release form of clonidine and the extended-release form of guanfacine for the treatment of ADHD in children 6 years and older. Antidepressants, such as bupropion, have been used with variable success in the treatment of ADHD. Table 2-16 indicates FDA-approved ages for ADHD medications.)

Stimulant Medications. Methylphenidate and amphetamine preparations are dopamine agonists; however, the precise mechanism of the stimulant's

central action remains unknown. Methylphenidate preparations are highly effective in up to three-fourths of children with ADHD, with relatively few adverse effects. Concerta, the 10- to 12-hour extended-release OROS (osmotic controlled-release extended delivery system) form of methylphenidate, is administered once daily in the morning and is useful during school hours as well as after school during the afternoon and early evening. Both shorter forms of methylphenidate and Concerta have similar common adverse effects, including headaches, stomachaches, nausea, and insomnia. Some children experience a rebound effect, in which they become mildly irritable and appear to be slightly hyperactive for a brief period when the medication wears off. In children with a history of motor tics, we should observe them as, in some cases, methylphenidate can exacerbate the tics, whereas, in other children, the tics are unaffected or even improved. Because tics wax and wane, it is essential to observe their patterns over some time. Another common concern about the use of methylphenidate preparations over long periods is potential growth suppression. During periods of use, methylphenidate may cause slightly decreased rates of growth, and if used over many years continuously without any drug holidays, experts have noted a growth suppression of about several centimeters. When given “drug holidays” on weekends or summers, children tend to eat more and also make up the growth. The methylphenidate products can improve ADHD children’s scores on tasks of vigilance, such as on math calculation tests, the continuous performance task, and paired associations. Transdermal methylphenidate is available for children and adolescents. Advantages of this preparation include an alternative for children who have difficulties swallowing pills, and that the patch can individualize how many hours per day a given child with ADHD receives the medication. This option is useful because a child with ADHD who needs the medication in the late afternoons to do homework but develops insomnia if the medication is still present after dinner can remove the patch at the desired time. Thus, individualized delivery time may be provided for each child by how many hours they wear the patch.

In contrast, oral sustained-release forms of methylphenidate are those in which the release time continues for 12 hours after swallowing the pill. A double-blind, randomized study in children with ADHD who wore the methylphenidate patch for 12 hours at a time, showed the efficacy of the patch preparation doses ranging from patches delivering 0.45 to 1.8 mg/hr of

methylphenidate. A delay in the onset of the transdermal medication effect was approximately an hour. Side effects were similar to oral preparations of methylphenidate. Approximately half of the children exhibited at least minor erythematous reactions to the patch; however, these side effects are usually well tolerated by children on the patch. Dextroamphetamine and dextroamphetamine/amphetamine salt combinations are usually the second drugs of choice when methylphenidate fails.



Table 2-16
FDA Approval for ADHD Medications

Medication	Generic Name	FDA Approva Age (yr)
Methylphenidate		
Concerta	Methylphenidate (OROS long acting)	6 and older
Ritalin	Methylphenidate	6 and older
Ritalin LA	Methylphenidate (long acting)	6 and older
Metadate ER	Methylphenidate (extended release)	6 and older
Metadate CD	Methylphenidate (extended release)	6 and older
Methylin	Methylphenidate (oral solution and chewable tablet)	6 and older
Daytrana	Methylphenidate (patch)	6 and older
Adhansia XR	Methylphenidate (extended release)	6 and older

Aptensio XR	Methylphenidate (extended release)	6 and older
Cotempla XR-ODT	Methylphenidate (extended-release orally disintegrating tablet)	6–17
Jornay PM	Methylphenidate (extended release)	6 and older
Quillichew	Methylphenidate (extended-release chewable)	6 and older
Quillivant ER	Methylphenidate (extended-release suspension)	6 and older
Dexmethylphenidate		
Focalin	Dexmethylphenidate	6 and older
Focalin XR	Dexmethylphenidate (extended release)	6 and older
Dextroamphetamine		
Dexedrine	Dextroamphetamine	3 and older
Amphetamine		
Adzenys ER	Amphetamine (extended-release suspension)	6–12
Adzenys XR-ODT	Amphetamine (extended-release orally disintegrating tablet)	6–17
Dynavel XR	Amphetamine (extended-release suspension)	6 and older
Evekeo	Amphetamine	3 and older
Evekeo ODT	Amphetamine (orally disintegrating	6–17

	tablet)	
Dextroamphetamine/amphetamine		
Adderall	Dextroamphetamine/amphetamine	3 and older
Adderall XR	Dextroamphetamine/amphetamine (extended release)	6 and older
Mydayis	Dextroamphetamine/amphetamine (extended release)	13 and older
Lisdexamfetamine		
Vyvanse	Lisdexamfetamine	6 and older
Nonstimulants		
Strattera	Atomoxetine	6 and older
Alpha Agonists		
Kapvay	Clonidine (extended release)	6–17
Intuniv	Guanfacine (extended release)	6–17

CNS STIMULANT SIDE EFFECTS. CNS stimulants are generally well tolerated, and the current consensus is that once a day dosing is preferable for convenience and to minimize rebound side effects. Long-term tolerability of once-daily mixed amphetamine salts has shown mild side effects, most commonly decreased appetite, insomnia, and headache. There are a variety of strategies for children or adolescents with ADHD who respond favorably to methylphenidate, but for whom insomnia has become a significant problem. Clinical strategies to manage insomnia include the use of diphenhydramine (25 to 75 mg), a low dose of trazodone (25 to 50 mg), or the addition of an α-adrenergic agent, such as guanfacine. In some cases, insomnia may attenuate on its own after several months of treatment.

Nonstimulant Medications. Atomoxetine HCl is a norepinephrine uptake inhibitor approved by the FDA for the treatment of ADHD in children age 6 years and older. The mechanism of action is not well understood, but likely involves selective inhibition of the presynaptic norepinephrine transporter. Atomoxetine absorbs well in the gastrointestinal tract, and it reaches maximal plasma levels in 1 to 2 hours after ingestion. It is useful for inattention as well as impulsivity in children and adults with ADHD. Its half-life is approximately 5 hours and is given twice daily in most cases. The most common side effects include diminished appetite, abdominal discomfort, dizziness, and irritability. In some cases, there were increases in blood pressure and heart rate. Atomoxetine is metabolized by the cytochrome P450 (CYP) 2D6 hepatic enzyme system. A small fraction of the population are poor metabolizers of CYP 2D6–metabolized drugs, and, for those individuals, plasma concentrations of the drug may increase as much as fivefold for a given dose of medication. Drugs that inhibit CYP 2D6, including fluoxetine, paroxetine, and quinidine, may lead to increased plasma levels of this medication. Despite its short half-life, research suggests that atomoxetine can reduce symptoms of ADHD in children during the school day when administered once daily. Another recent study of a combination of atomoxetine alone and combined with fluoxetine in the treatment of 127 children with ADHD and symptoms of anxiety or depression suggested that atomoxetine alone can lead to improvements in mood and anxiety. Children who received combined atomoxetine and fluoxetine experienced more significant increases in blood pressure and pulse than those taking atomoxetine only.

α-Agonists, both short-acting and the extended-release forms of clonidine hydrochloride and guanfacine, are FDA approved for the treatment of ADHD in children and adolescents from 6 to 7 years of age. Clonidine, a centrally acting α_2-adrenergic receptor agonist, likely exerts its effect on the prefrontal cortex, although the mechanism of action is unknown. It is available in 0.1-mg and 0.2-mg tablets, and is generally used twice daily, once in the morning and once at night, to provide an around-the-clock effect. Clonidine is initiated at 0.1 mg at bedtime, with incremental weekly increases of 0.1 mg. The maximum dose recommended is 0.2 mg twice daily. The extended-release formulation is not interchangeable with the short-acting clonidine. Because it

is also an antihypertensive agent, it causes a decrease in blood pressure and heart rate. We should monitor vital signs in patients, especially during initiation and titration of the dose. Common side effects include somnolence, headache, upper abdominal pain, and fatigue. When tapering the drug, the rate should be no more than 0.1 mg every 3 to 7 days.

Extended-release guanfacine is a once-a-day medication for children between 6 and 17 years of age, available in 1-mg, 2-mg, 3-mg, and 4-mg tabs. It is swallowed whole with liquids, and the patient should not take it with a high-fat meal. It is initiated as a 1-mg tab daily and titrated by 1 mg/day at 1-week intervals. The maximum dose approved is 4 mg/day. As a monotherapy, improvement in ADHD symptoms occurs at 0.05 to 0.08 mg/kg once daily. As an adjunctive treatment, optimal doses are reported to range from 0.05 to 0.12 mg/kg/day. Common side effects include somnolence, sedation, fatigue, nausea, hypotension, insomnia, and dizziness. We should monitor the heart rate and blood pressure. When discontinuing the drug, one should use a gradual taper, decreasing by 1 mg every 3 to 7 days.

α-Adrenergic agents, including the short- and extended-release preparations of guanfacine and clonidine, are sometimes preferred treatments in children with ADHD and comorbid tic disorders when stimulants exacerbated the tics. Bupropion is somewhat useful for some children and adolescents in the treatment of ADHD. One multisite, double-blind, placebo-controlled study found a positive result regarding the efficacy of bupropion. No further studies have compared bupropion with other stimulants. There is a higher risk of seizures at doses of 400 mg/day or more.

Few data confirm the efficacy of SSRIs in the treatment of ADHD, but due to the frequency of comorbid depression and anxiety with ADHD, in cases of comorbidity, the SSRIs are likely to be considered at least in conjunction with a stimulant.

We should not use tricyclic drugs for ADHD due to potential cardiac arrhythmia effects. The reports of sudden death in at least four children with ADHD using desipramine have made the tricyclic antidepressants an unlikely choice. Antipsychotics may treat refractory severely hyperactive children and adolescents who are significantly dysfunctional. Antipsychotics are generally not chosen in the treatment of ADHD due to the risks of tardive dyskinesia, withdrawal dyskinesia, neuroleptic malignant syndrome, and weight gain.

Modafinil, another type of CNS stimulant, a narcolepsy treatment, may

help treat adults with ADHD. Only one randomized, double-blind, placebo-controlled study of the efficacy and safety of modafinil film-coated tablets in approximately 250 adolescents with ADHD showed that 48 percent of those on active treatment were rated as “much” or “very much” improved compared with 17 percent of patients receiving placebo. The dosage range was from 170 to 425 mg administered once daily, titrated to optimal doses based on efficacy and tolerability. Modafinil failed to receive FDA approval based on a Stevens–Johnson skin rash that occurred in a patient during the trial. The most common side effects included insomnia, headache, and decreased appetite.

Some clinicians use venlafaxine, especially for children and adolescents, with combinations of ADHD and depression or anxiety features. No clear empirical evidence supports the use of venlafaxine in the treatment of ADHD.

One open-label report of reboxetine, a selective norepinephrine reuptake inhibitor that is not available in the US, in 31 children and adolescents with ADHD who were resistant to methylphenidate treatment, suggested that this agent may have efficacy. In this open trial, reboxetine was initiated and maintained at 4 mg/day. The most common side effects included drowsiness, sedation, and gastrointestinal symptoms. Reboxetine and other new agents in this class await controlled studies.

Monitoring Pharmacologic Treatment

STIMULANTS. Stimulant medications have adrenergic effects and cause moderate increases in blood pressure and pulse rate. At baseline, the most recent American Academy of Child and Adolescent Psychiatry (AACAP) practice parameters recommend the following workup before starting the use of stimulant medications: physical examination, blood pressure, pulse, weight, and height. Screening electrocardiograms are recommended prior to initiating stimulant medications, particularly if there is a family history or risk factors for cardiac disease.

Children and adolescents using stimulants should have their height, weight, blood pressure, and pulse checked quarterly and have a physical examination annually. Monitoring starts with the initiation of medication. Because school performance is most markedly affected, we should give individual attention

and effort to establishing and maintaining a close collaborative working relationship with a child's school personnel. In most patients, stimulants reduce overactivity, distractibility, impulsiveness, explosiveness, and irritability. No evidence indicates that medications directly improve any existing impairments in learning, although when the attention deficits diminish, children can learn more effectively. Also, medication can improve self-esteem when children are no longer constantly reprimanded for their behavior. Children treated with medications should be taught the purpose of the medication and encouraged to describe any side effects that they may be experiencing.

Psychosocial Interventions

Psychosocial interventions for children with ADHD include psychoeducation, academic organization skills remediation, parent training, behavior modification in the classroom and at home, CBT, and social skills training. There are various studies of social skills groups, behavioral training for parents of children with ADHD, and behavioral interventions at school and home, alone and in combination with medication management for ADHD. Evaluation and treatment of coexisting learning disorders or additional psychiatric disorders are essential.

When we help children structure their environment, their anxiety diminishes. Parents and teachers should work together to develop concrete expectations for the child, and a system of rewards when the child meets these expectations.

A common goal of therapy is to help parents of children with ADHD recognize and promote the notion that, although the child may not "voluntarily" exhibit symptoms of ADHD, he or she is still capable of being responsible for meeting reasonable expectations. We should also help parents recognize that, despite their child's difficulties, every child faces the usual tasks of maturation, including the significant building of self-esteem when they develop a sense of mastery. Therefore, children with ADHD do not benefit from being exempted from the requirements, expectations, and planning applicable to other children. Parental training is an integral part of the psychotherapeutic interventions for ADHD. Most parental training helps parents develop usable behavioral interventions with positive reinforcement

that target both social and academic behaviors.

Group therapy aimed at both refining social skills and increasing self-esteem and a sense of success may be beneficial for children with ADHD who have great difficulty functioning in group settings, especially in school. A recent year-long group therapy intervention in a clinical setting for boys with the disorder described the goals as helping the boys improve skills in game-playing and feeling a sense of mastery with peers. The researchers asked the boys to do a fun task, in pairs, and then they were gradually asked to do projects in a group. The researchers directed them on how to follow instructions, wait, and pay attention, and praised them for successful cooperation.

Multimodal Treatment Study of Children with ADHD (MTA Study)

The National Institute of Mental Health (NIMH)–supported Multimodal Treatment Study of Children with ADHD (The MTA Cooperative Group, 1999) was a 14-month–long randomized clinical trial involving six clinical sites comparing four treatment strategies. More than 500 children diagnosed with DSM-IV ADHD, combined type, were randomly assigned to (1) systematic medication management utilizing an initial placebo-controlled titration and t.i.d. dosing 7 days per week and monthly 30-minute clinic visits, (2) behavior therapy consisting of 27 sessions of group parent training, eight individual parent sessions, an 8-week summer treatment program, 12 weeks of classroom administered behavior therapy with a half-time aide, and 10 teacher consultation sessions, (3) a combination of medication and behavior therapy, or (4) usual community care. All groups showed improvement over baseline; however, a combination of medication management and behavior therapy led to a greater reduction in symptoms in children with ADHD alone or ADHD and oppositional defiant disorder than behavior therapy alone or community care. The combination treatment had significantly better outcomes for those children with ADHD and anxiety or mood disorders compared to behavioral treatment and community care. Combined treatment but not medication management was superior for improvement in oppositional and aggressive symptoms, anxiety and mood symptoms, teacher-rated social skills, parent–child relationships, and reading achievement. Furthermore, the mean dose of medication per day was less in

the combination group than in the medication-only management group.

A follow-up of the MTA sample at 6 and 8 years revealed that the clinical presentation of the disorder, including the severity of ADHD, comorbid conduct disturbance, and intellect, were more reliable predictors of later functioning than the type of treatment received in childhood during the 14-month study period. The children maintained the improvements as long as they continued treatment, but 3 years after treatment, there was no difference between groups.

Overall, the evidence suggests that medication and psychosocial interventions for the combined type of ADHD in childhood provide the broadest benefit in functioning for this population. This recommendation is especially pertinent given the comorbidity of learning disorders, anxiety, mood disorders, and other disruptive behavior disorders that occur in children with ADHD.

EPIDEMIOLOGY

Rates of ADHD are 7 to 8 percent in prepubertal elementary school children. Epidemiologic studies suggest that ADHD occurs in about 5 percent of youth, including children and adolescents, and about 2.5 percent of adults. The rate of ADHD in parents and siblings of children with ADHD is two to eight times greater than in the general population. ADHD is more prevalent in boys than in girls, with the ratio ranging from 2:1 to as high as 9:1. First-degree biologic relatives (e.g., siblings of probands with ADHD) are at high risk for developing ADHD as well as other psychiatric disorders, including disruptive behavior disorders, anxiety disorders, and depressive disorders. Siblings of children with ADHD are also at higher risk than the general population for learning disorders and academic difficulties. The parents of children with ADHD show an increased incidence of substance use disorders. Symptoms of ADHD are often present by age 3 years, but unless they are very severe, the diagnosis is frequently not made until the child is in kindergarten or elementary school when teacher information is available comparing the index child peers of the same age.

ETIOLOGY

Data suggest that the etiology of ADHD is mainly genetic, with a heritability of approximately 75 percent. ADHD symptoms are the product of complex interactions of neuroanatomical and neurochemical systems evidenced by data from twin and adoption family genetic studies, dopamine transport gene studies, neuroimaging studies, and neurotransmitter data. Most children with ADHD have no evidence of gross structural damage in the CNS. In some cases, contributory factors for ADHD may include prenatal toxic exposures, prematurity, and prenatal mechanical insult to the fetal nervous system. Some have suggested that food additives, colorings, preservatives, and sugar are possible contributing causes of hyperactive behavior; however, studies have not confirmed these theories. No research has established artificial food coloring nor sugar as causes of ADHD. There is no clear evidence that omega-3 fatty acids are beneficial in the treatment of ADHD.

Genetic Factors

Evidence for a significant genetic contribution to ADHD has emerged from family studies, which reveal an increased concordance in monozygotic compared to dizygotic twins, as well as a marked increased risk of two to eight times for siblings as well as parents of an ADHD child, compared to the general population. Clinically, one sibling may have impulsivity/hyperactivity symptoms predominantly, and others may have predominantly inattention symptoms. Up to 70 percent of children with ADHD meet criteria for a comorbid psychiatric disorder, including learning disorders, anxiety disorders, mood disorder, conduct disorders, and substance use disorders. Several hypotheses of the mode of transmission of ADHD have been proposed, including a sex-linked hypothesis, which would explain the significantly increased rates of ADHD in males. Other theories have focused on a model of interaction of multiple genes that produces the various symptoms of ADHD. Numerous investigations continue to identify specific genes involved in ADHD. Cook and colleagues have found an association of the dopamine transporter gene (DAT1) with ADHD, although data from other research groups have not confirmed that result. Family studies and population-based studies have found an association between the dopamine four receptor seven-repeat allele (DRD4) gene and ADHD. Most molecular research on ADHD has focused on genes that influence the metabolism or

action of dopamine. Continued investigation is necessary to clarify the complex relationships between multiple interactive genes and the emergence of ADHD.

Neurochemical Factors

Many neurotransmitters are likely associated with ADHD symptoms; however, dopamine is a primary focus of clinical investigation, and the prefrontal cortex has been implicated based on its role in attention and regulation of impulse control. Animal studies have shown that other brain regions such as locus ceruleus, which consists predominantly of noradrenergic neurons, also play a significant role in attention. The noradrenergic system includes the central system (originating in the locus ceruleus) and the peripheral sympathetic system. Dysfunction in peripheral epinephrine, which causes the hormone to accumulate peripherally, may potentially feedback to the central system and “reset” the locus ceruleus to a lower level. In part, hypotheses regarding the neurochemistry of ADHD have arisen from the predictable effect of medications. Stimulants, known to be the most effective medications in the treatment of ADHD, affect both dopamine and norepinephrine, leading to neurotransmitter hypotheses that may include dysfunction in both the adrenergic and dopaminergic systems. Stimulants increase catecholamine concentrations by promoting their release and blocking their uptake.

Neurophysiologic Factors

EEG studies in ADHD children and adolescents over the last several decades have found evidence of increased theta activity, especially in the frontal regions. Further studies of youth with ADHD have provided data showing elevated beta activity in their EEG studies. Clarke and colleagues, studying EEG findings in children and adolescents over the last two decades, found that those ADHD children with the combined type of ADHD were the ones who showed significantly elevated beta activity on EEG, and further studies indicate that these youth also tend to show increased mood lability and temper tantrums. The current investigation of EEG in youth with ADHD has identified behavioral symptom clusters among children with similar EEG profiles.

Neuroanatomical Aspects

Researchers have hypothesized networks within the brain for promoting components of attention, including focusing, sustaining attention, and shifting attention. They describe neuroanatomical correlations for the superior and temporal cortices with focusing attention; external parietal and corpus striatal regions with motor executive functions; the hippocampus with encoding of memory traces; and the prefrontal cortex with shifting from one stimulus to another. Further hypotheses suggest that the brainstem, which contains the reticular thalamic nuclei function, is involved in sustained attention. A review of MRI, positron emission tomography (PET), and single-photon emission computerized tomography (SPECT) suggests that populations of children with ADHD show evidence of both decreased volume and decreased activity in prefrontal regions, anterior cingulate, globus pallidus, caudate, thalamus, and cerebellum. PET scans have also shown that female adolescents with ADHD have globally lower glucose metabolism than both control female and male adolescents without ADHD. One theory postulates that the frontal lobes in children with ADHD do not adequately inhibit lower brain structures, an effect leading to disinhibition.

Developmental Factors

Higher rates of ADHD are present in children who were born prematurely and whose mothers had infections during pregnancy. Perinatal insult to the brain during early infancy caused by infection, inflammation, and trauma may, in some cases, be contributing factors in the emergence of ADHD symptoms. Children with ADHD exhibit nonfocal (soft) neurologic signs at higher rates than those in the general population. Reports in the literature indicate that September is a peak month for births of children with ADHD with and without comorbid learning disorders. The implication is that prenatal exposure to winter infections during the first trimester may contribute to the emergence of ADHD symptoms in some susceptible children.

Psychosocial Factors

Severe chronic abuse, maltreatment, and neglect are associated with specific behavioral symptoms that overlap with ADHD, including poor attention and

poor impulse control. Predisposing factors may include the child's temperament and genetic–familial factors.

ADULT MANIFESTATIONS OF ADHD

We usually think of ADHD as a childhood condition resulting in delayed development of impulse control. Historically, many experts thought that, by adolescence, patients would grow out of the disorder. However, in the last few decades, many more adults with ADHD have been identified, diagnosed, and successfully treated. Longitudinal follow-up has shown that up to 60 percent of children with ADHD have persistent impairment from symptoms into adulthood. Genetic studies, brain imaging, and neurocognitive and pharmacologic studies in adults with ADHD have replicated findings demonstrated in children with ADHD. Increased public awareness and treatment studies within the last decade have led to widespread acceptance of the need for diagnosis and treatment of adults with ADHD.

Diagnosis and Clinical Features

The clinical phenomenology of ADHD features inattention and manifestations of impulsivity prevailing as the core of this disorder. A leading figure in the development of criteria for adult manifestations of ADHD is Paul Wender, from the University of Utah, who began his work on adult ADHD in the 1970s. Wender developed criteria for adults (Table 2-17). They included a retrospective diagnosis of ADHD in childhood and evidence of current impairment from ADHD symptoms in adulthood. Furthermore, evidence exists of several additional symptoms that are typical of adult behavior as opposed to childhood behaviors.

In adults, residual signs of the disorder include impulsivity and attention deficit (e.g., difficulty in organizing and completing work, inability to concentrate, increased distractibility, and sudden decision-making without thought of the consequences). Many people with the disorder have a secondary depressive disorder associated with low self-esteem related to their impaired performance and which affects both occupational and social functioning.

Brett was a 26-year-old man convinced by his new wife to seek an evaluation for his distractibility, forgetfulness, and “not listening” after a minor traffic accident. After consulting his mother, Brett reported that in grade school, he was often “in trouble” for talking out of turn, and his mother recalled teachers’ reports that Brett often made careless mistakes on tests, forgot his assignments, and had great difficulty sitting still. Although as a young child people considered him to be intellectually gifted, when Brett got to the third grade, his grades were only average, and he seemed more interested in getting his work done quickly than correctly. Brett was talkative and loud and enjoyed sports, although he was not particularly talented at them. Nevertheless, Brett had acquaintances and superficial friends because he was likable, funny, and even entertaining. Brett had no idea what he wanted to do when he grew up, and during his senior year in high school, he neglected to finish any of his college applications on time and ended up attending a community college part-time. During the 2 years after high school, Brett held down a series of jobs only briefly, including a construction job, a waiter position in a restaurant, and a Fed-Ex driver, and then decided that he wanted to become an actor. Brett went on a series of auditions but found that he would become distracted and did poorly remembering his lines and even spaced out during readings. Despite that, he acted in one commercial.

Brett reported that he had never had problems with the abuse of drugs or alcohol, and he occasionally drank beer socially. During his evaluation with a child and adolescent psychiatrist, Brett disclosed that his most significant difficulties were with tasks that seemed boring to him. He had difficulty maintaining his attention, was easily distracted, felt restless most of the time, and became frustrated when he had to sit still for long periods. Brett endorsed six inattentive and five hyperactive/impulsive symptoms on a DSM ADHD Checklist of current symptoms. Brett met the diagnostic criteria for adult ADHD, combined presentation, with a probable onset in childhood. Brett’s medical history was negative for all major illnesses, and neither he nor his parents had a history of cardiac abnormalities. He took no prescribed medications. After discussing the situation with his psychiatrist and his wife, Brett decided that he would like to try a stimulant medication. A trial of a once-a-day extended-

release formulation of stimulant medication was selected: Adderall XR 10 mg.

At his first follow-up visit, a week later, Brett reported that he felt a slight effect from this medication, but it was not enough to improve his functioning, so Brett and his psychiatrist agreed that he would increase his dose to 20 mg/day. At his next follow-up appointment, Brett reported that he had noticed a significant improvement in his ability to focus, concentrate, and remember his lines in auditions. He had just received a small part in an upcoming movie. Brett and his wife were both thrilled with the results, and Brett continued to return for monthly follow-up visits. (Adapted from McGough J. Adult manifestations of attention-deficit/hyperactivity disorder. In: Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:3577.)



Table 2-17
Utah Criteria for Adult Attention-Deficit/Hyperactivity Disorder (ADHD)

I. Retrospective childhood ADHD diagnosis
   A. Narrow criterion: met DSM-IV criteria in childhood by parent interview[a]
   B. Broad criterion: both (1) and (2) are met as reported by patient[b]
      1. Childhood hyperactivity
      2. Childhood attention deficits

II. Adult characteristics: five additional symptoms, including ongoing difficulties with inattentiveness and hyperactivity and at least three other symptoms:
   A. Inattentiveness
   B. Hyperactivity
   C. Mood lability
   D. Irritability and hot temper
   E. Impaired stress tolerance
   F. Disorganization

G. Impulsivity

III. Exclusions: not diagnosed in presence of severe depression, psychosis, or severe personality disorder

[a]Parent report aided with 10-item *Parent Rating Scale of Childhood Behavior.*

[b]Patient self-report of retrospective childhood symptoms aided by *Wender Utah Rating Scale.*

Differential Diagnosis

A diagnosis of ADHD is likely when adults describe symptoms of inattention and impulsivity as a life-long problem, not as episodic events. The overlap of ADHD and hypomania, bipolar II disorder, and cyclothymia is controversial and challenging to sort out retrospectively. Clear-cut histories of discrete episodes of hypomania and mania, with or without periods of depression, are suggestive of a mood disorder rather than a clinical picture of ADHD; however, ADHD may have predated the emergence of a mood disorder in some individuals. In such a case, ADHD and bipolar disorder may be comorbid. Adults with an early history of chronic school difficulties related to paying attention, activity level, and impulsive behavior are generally diagnosed with ADHD, even when a mood disorder occurs later in life. Anxiety disorders can coexist with ADHD and are less complicated than hypomania to distinguish from it.

Course and Prognosis

The prevalence of ADHD diminishes over time, although at least half of children and adolescents may have the disorder into adulthood. Many children, initially diagnosed with ADHD, combined presentation, exhibit fewer impulsive-hyperactive symptoms as they get older and, by the time they are adults, will meet criteria for ADHD, predominantly inattentive presentation. As with children, adults with ADHD demonstrate higher rates of learning disorders, anxiety disorders, mood disorders, and substance use disorders compared with the general population.

Treatment

The treatment of ADHD in adults targets pharmacotherapy, mainly long-acting stimulants, similar to that used with children and adolescents with ADHD. In adults, only the long-acting stimulants are FDA approved in the

treatment of ADHD. Signs of a positive response are an increased attention span, decreased impulsiveness, and improved mood. Psychopharmacological therapy may be needed indefinitely. Clinicians should use standard ways to monitor drug response and patient compliance.

Epidemiology

Among adults, evidence suggests an approximate 4 percent prevalence of ADHD in the population. ADHD in adulthood is generally diagnosed by self-report, given the lack of school information and observer information available; therefore, it is more challenging to make an accurate diagnosis.

2.5 Specific Learning Disorder

Specific learning disorder in youth is a neurodevelopmental disorder produced by the interactions of heritable and environmental factors that influence the brain's ability to perceive or process verbal and nonverbal information efficiently. Children with the disorder have persistent difficulty learning academic skills in reading, written expression, or mathematics, beginning in early childhood, which is inconsistent with the overall intellectual ability of a child. Children with specific learning disorder often find it challenging to keep up with their peers in certain academic subjects, whereas they may excel in others. Several academic skills may be compromised in specific learning disorder include reading single words and sentences fluently, written expression and spelling, and calculation and solving mathematical problems. Specific learning disorder results in underachievement that is unexpected based on the child's potential as well as the opportunity to have learned more. Specific learning disorder in reading, spelling, and mathematics appears to aggregate in families. There is an increased risk of four to eight times in first-degree relatives for reading deficits, and about five to ten times for mathematics deficits, compared to the general population. Specific learning disorder occurs two to three times more often in males than in females. Learning problems in a child or adolescent identified in this manner can establish eligibility for academic services through the public school system.

The American Psychiatric Association's DSM-5, combines the DSM-IV diagnoses of reading disorder, mathematics disorder, and disorder of written expression and learning disorder not otherwise specified into a single diagnosis: Specific learning disorder. Learning deficits in reading, written expression, and mathematics in the DSM-5 are designated using specifiers. ICD-10 continues to separate the disorders. DSM-5 notes that the term *dyslexia* is an equivalent term describing a pattern of learning difficulties, including deficits in accurate or fluent word recognition, poor decoding, and poor spelling skills. *Dyscalculia* is an alternative term referring to a pattern of deficits related to learning arithmetic facts, processing numerical information, and performing accurate calculations. Table 2-18 lists the approaches to diagnosis of specific learning disorder.

Specific learning disorder of all types affects approximately 10 percent of youth. This disorder represents approximately half of all public school children who receive special education services in the United States. In 1975, Public Law 94–142 (the Education for All Handicapped Children Act, now known as the Individual with Disabilities Education Act [IDEA]) mandated all states to provide free and appropriate educational services to all children. Since that time, the number of children identified with learning disorders has increased, and a variety of definitions of learning disabilities have arisen. For this diagnosis, a child's achievement must be significantly lower than expected in one or more of the following: reading skills, comprehension, spelling, written expression, calculation, mathematical reasoning, or the learning problems interfere with academic achievement or activities of daily living. It is common for specific learning disorder to include more than one area of skill deficits.

Children with specific learning disorder in the area of reading can be identified by poor word recognition, slow reading rate, and impaired comprehension compared with most children of the same age. Current data suggest that most children with reading difficulties have deficits in speech sound processing skills, regardless of their IQ. In DSM-5, there is no longer a diagnostic criterion for specific learning disorder comparing the specific deficit to overall IQ. The current consensus is that children with reading impairment have trouble with word recognition and "sounding out" words because they cannot efficiently process and use phonemes (the smaller bits of words that are associated with particular sounds). An epidemiologic study

found four profiles, including (1) weak reading, (2) weak language, (3) weak math, or (4) combined weak math and reading, accounting for 70 percent of children with specific learning impairments. Low scores in short-term memory for speech sounds characterized the profile with weak language, whereas, low speech sound awareness was associated with the weak reading group, but not the weak language group. Finally, in another study, it was found that the weak math group did not show speech sound deficits.

Severe specific learning disorder may make it agonizing for a child to succeed in school, often leading to demoralization, low self-esteem, chronic frustration, and compromised peer relationships. Specific learning disorder is associated with an increased risk of comorbid disorders, including ADHD, communication disorders, conduct disorders, and depressive disorders. Adolescents with specific learning disorder are at least 1.5 times more likely to drop out of school, approximating rates of 40 percent. Adults with specific learning disorder are at increased risk for difficulties in employment and social adjustment. Specific learning disorder often extends to skills deficits in multiple areas such as reading, writing, and mathematics.

Moderate to high heritability contributes to specific learning disorder, and it appears that many cognitive traits are polygenic. Also, there is pleiotropy; that is, the same genes may affect the skills necessary for diverse learning tasks. Factors such as perinatal injury and specific neurologic conditions may contribute to the development of specific learning disorder. Conditions such as lead poisoning, FAS, and in utero drug exposure are also associated with increased rates of specific learning disorder.



Table 2-18
Specific Learning Disorder

	DSM-5	ICD-10
Diagnostic name	Specific Learning Disorder	Specific Developmental Disorders of Scholastic Skills • Specific reading disorder • Specific spelling

		disorder • Specific disorder of arithmetical skills • Mixed disorder of scholastic skills
Duration	≥6 mo Symptoms beginning during school-age years, may not be apparent until later	
Symptoms	Difficulties with learning or academics. Example difficulties: • Reading (slow/inaccurate) • Reading comprehension • Spelling • Written expression • Numbers/calculation • Math reasoning	Deficits in the specific domain (reading, spelling, arithmetic) relevant to the above listed disorder
Exclusions (not result of):	Intellectual disability Visual or auditory impairment Another mental disorder Another medical disorder Psychosocial adversity Inadequate education	Intellectual disability Mental age Visual impairment Inadequate education
Symptom specifiers	***With impairment in reading*** ***With impairment in***	

	written expression ***With impairment in mathematics***
Severity specifiers	**Mild:** minimal accommodations or support needed in school **Moderate:** some accommodations needed **Severe:** intensive individualized and specialized teaching most of the time in school

SPECIFIC LEARNING DISORDER WITH IMPAIRMENT IN READING

Reading impairment is present in up to 75 percent of children and adolescents with a specific learning disorder. Students who have learning problems in other academic areas most commonly experience difficulties with reading as well.

Reading impairment is characterized by difficulty in recognizing words, slow and inaccurate reading, poor comprehension, and difficulties with spelling. Reading impairment is often comorbid with other disorders in children, particularly ADHD. The historical term *developmental alexia* defines a developmental deficit in recognizing printed symbols. The term was simplified to *dyslexia* in the 1960s. Dyslexia was used extensively for many years to describe a reading disability syndrome that often included speech and language deficits and right-left confusion. Reading impairment is frequently accompanied by disabilities in other academic skills, and the term dyslexia remains as an alternate term for a pattern of reading and spelling difficulties.

Clinical Features

We can usually identify children with reading disabilities by the age of 7 years (second grade). Reading difficulty may be apparent among students in classrooms that expect reading skills earlier. Children can sometimes compensate for reading disorders in the early elementary grades by the use of memory and inference, particularly in children with high intelligence. In such instances, the disorder may not be apparent until age 9 (fourth grade) or later. Children with reading impairment make many errors in their oral reading. The errors include omissions, additions, and distortions of words. Such children have difficulty in distinguishing between printed letter characters and sizes, especially those that differ only in spatial orientation and length of a line. The problems in managing printed or written language can pertain to individual letters, sentences, and even a page. The child's reading speed is slow, often with minimal comprehension. Most children with reading disability have an age-appropriate ability to copy from a written or printed text, but nearly all spell poorly.

Associated problems include language difficulties: discrimination, and difficulty in sequencing words properly. A patient might start with a word that occurs midway or at the end of a written sentence. Most children with reading disorder dislike and avoid reading and writing. They become anxious when confronted with demands that involve printed language. Many children with specific learning disorders who do not receive remedial education have a sense of shame and humiliation because of their continuing failure and subsequent frustration. The intensity of these feelings grows over time. Older children tend to be angry and depressed and exhibit poor self-esteem.

Jackson, a 10-year-old boy, was referred for evaluation of failing to complete in-class assignments and homework and failing tests in reading, spelling, and arithmetic. For the past 2 years (grades 5 and 6), he had been attending a particular education class every morning in the local community school, based on an assessment from the second grade. A subsequent psychoeducational assessment by a clinical psychologist confirmed reading problems. Jackson was eligible for a full-day special education class, after which he started attending a program with eight other students ranging from 6 to 12 years of age.

Clinical interview with his parents revealed a healthy pregnancy and neonatal period, and a history of language delay. In preschool and kindergarten, Jackson had difficulty with rhyming games and showed a lack of interest in books and preferred to play with construction toys. In the first grade, Jackson had more difficulty learning to read than other boys in his class and continued to have problems pronouncing multisyllabic words (e.g., he said “aminals” for “animals” and “sblanation” for “explanation”). Family history was positive for reading deficits and ADHD. Jackson’s father disclosed a history of personal reading problems, and Jackson’s older brother, 15 years of age, had ADHD, for which he took stimulant medication. Jackson’s parents were concerned about his poor focus in school and wondered whether he had ADHD. In the clinical interview with Jackson, he rarely made eye contact, mumbled a lot, and struggled to find the right words (e.g., manifested many false starts, hesitations, and nonspecific terms, such as “the thing that you draw … um … pencil—no … um … lines with”). He admitted to disliking school, adding, “Reading is boring and stupid—I’d rather be skateboarding.” Jackson complained about how much reading he was given—even in math—and commented, “Reading takes so much time. By the time I figure out a word, I can’t remember what I just read, and so have to read the stuff again.”

Psychoeducational assessment included the Wechsler Intelligence Scale for Children-IV, Clinical Evaluation of Language Fundamentals-IV (CELF-IV), the Wechsler Individual Achievement Test-II, and self-ratings of anxiety, depression, and self-esteem. Results indicated low-average verbal and above-average performance IQ, poor word attack and word identification skills (below 12th percentile), poor comprehension (below ninth percentile), poor spelling (below sixth percentile), weak comprehension of oral language (below 16th percentile), elevated but subthreshold scores on the Children’s Depression Inventory, and low self-esteem. Although Jackson manifested symptoms of inattention, restlessness, and oppositional behavior (mainly at school), he did not meet the criteria for ADHD. Jackson met DSM-5 criteria for a specific learning disorder, with deficits in reading and written expression. Recommendations included continuation in special education plus attendance at a summer camp specializing in children with reading

disorder, as well as ongoing monitoring of self-esteem and depressive traits.

At 1-year follow-up, Jackson and his parents reported striking improvements in his reading, overall school performance, mood, and self-esteem. Both Jackson and his family felt that the specialized instruction provided during the summer camp was beneficial. The program had provided one-on-one focused and explicit instruction for 1 hour a day for a total of 70 hours. Jackson explained that he had been taught "like a game plan" to read, and challenged the clinician to give him a "really tough long word to read." He demonstrated strategies that he had learned to read the word "unconditionally" and also explained what it meant. To boost his fluency in reading and comprehension, he was provided with assignments to read along with audio-taped versions of books, use of graphic organizers to facilitate reading comprehension, and continued participation in the summer camp reading program. (Adapted from Rosemary Tannock, Ph.D.)

Diagnosis

Reading impairment is diagnosed when a child's reading achievement is significantly below that expected of a child of the same age. Characteristic diagnostic features include difficulty recalling, evoking, and sequencing printed letters and words, processing sophisticated grammatical constructions, and making inferences. School failure and ensuing poor self-esteem can exacerbate the problems as a child becomes more consumed with a sense of failure and spends less time focusing on academic work. Students with reading impairment are entitled to an educational evaluation through the school district to determine eligibility for special education services. Special education classification, however, is not uniform across states or regions, and students with identical reading difficulties may be eligible for services in one region, but ineligible in another.

Pathology and Laboratory Examination. No specific physical signs or laboratory measures are helpful in the diagnosis of reading deficits. Psychoeducational testing, however, is critical in determining these deficits. The diagnostic battery generally includes a standardized spelling test, written

composition, processing, and using oral language, design copying, and judgment of the adequacy of pencil use. The reading subtests of the *Woodcock-Johnson Psycho-Educational Battery-Revised* and the *Peabody Individual Achievement Test-Revised* are useful in identifying reading disability. A screening projective battery may include human-figure drawings, picture-story tests, and sentence completion. The evaluation should also include systematic observation of behavioral variables.

Differential Diagnosis

Reading deficits are often accompanied by comorbid disorders, such as language disorder, disability in written expression, and ADHD. Data indicate that children with reading disability consistently present difficulties with linguistic skills, whereas children with ADHD only do not. Children with reading disability without ADHD, however, may have some overlapping deficits in cognitive inhibition. For example, they perform impulsively on continuous performance tasks. Deficits in expressive language and speech discrimination, along with reading disorder, may lead to a comorbid diagnosis of language disorder. We should differentiate reading impairment from intellectual disability syndromes in which reading, along with most other skills, are below the achievement expected for a child's chronologic age. Intellectual testing helps to differentiate global deficits from more specific reading difficulties.

We can detect poor reading skills resulting from inadequate schooling by comparing a given child's achievement with classmates on reading performance on standardized reading tests. We should rule out hearing and visual impairments with screening tests.

Comorbidity

Children with reading difficulties are at high risk for additional learning deficits, including mathematics and written expression. The DSM-5 Language disorder, also known as specific language impairment, has traditionally been viewed as distinct from dyslexia and dyscalculia. Children with language disorder have poor word knowledge, limited abilities to form accurate sentence structure, and impairments in the ability to put words together to produce clear explanations. Children with language disorder may

have delayed development of language acquisition, and difficulties with grammar and syntactical knowledge. Specific learning disorder in the areas of reading and mathematics frequently occur comorbidly with language disorder. In one study, 19 percent to 63 percent of reading disorder patients also had language impairment.

Conversely, reading impairment occurs in 12.5 to 85 percent of individuals with language disorder. In twin studies, reading impairments are significantly higher in those children with specific learning impairment and family members of children with the disorder. There are also high rates of comorbidity between reading impairment and mathematics impairment; in some studies, the comorbidity is as high as 60 percent. It appears that children with both reading and math impairment may perform more poorly in mathematics; however, the reading skills of the comorbid children were no different from children who had only reading disorder and not math disorder.

Comorbid psychiatric disorders are also frequent, such as ADHD, oppositional defiant disorder, conduct disorders, and depressive disorders, especially in adolescents. Data suggest that up to 25 percent of children with reading impairment may have comorbid ADHD. Alternately, between 15 and 30 percent of children diagnosed with ADHD have specific learning disorder. Family studies suggest that ADHD and reading impairment may share some degree of heritability. That is, some genetic factors contribute to both reading impairment and attentional syndromes. Youth with reading impairments have higher than average rates of depression on self-report measures and experience higher levels of anxiety symptoms than children without specific learning disorder. Furthermore, children with reading impairment are at increased risk for poor peer relationships and exhibit less skill in responding to subtle social cues.

Course and Prognosis

Children with a reading disability may gain knowledge of printed language during their first 2 years in grade school without remedial assistance. By the end of the first grade, many children with reading problems have learned how to read a few words; however, by the third grade, keeping up with classmates is exceedingly hard without remedial educational intervention. When remediation is instituted early, in milder cases, it may not be necessary after

the first or second grade. In severe cases and depending on the pattern of deficits and strengths, remediation may continue into the middle and high school years.

Treatment

Remediation strategies for children with reading impairments focus on direct instruction that leads a child's attention to the connections between speech sounds and spelling. Effective remediation programs begin by teaching the child to make accurate associations between letters and sounds. This approach relies on the theory that the core deficits in reading impairments are related to difficulty recognizing and remembering the associations between letters and sounds. After mastering individual letter-sound associations, remediation can target more significant components of reading, such as syllables and words. We can determine the exact focus of any reading program only after an accurate assessment of a child's specific deficits and weaknesses. Positive strategies include small, structured reading groups that offer individual attention and make it easier for a child to ask for help.

Children and adolescents with reading difficulties are entitled to an individual education program (IEP) provided by the public school system. However, for high school students with persistent reading disorders and ongoing difficulties with decoding and word identification, IEP services may not be sufficient to remediate their problems. A study of students with reading disorders in 54 schools indicated that, at the high school level, specific goals are not adequately met solely through school remediation. High schoolers with persisting reading difficulties may likely have more benefit from individualized reading remediation.

Reading instruction programs such as the Orton Gillingham and Direct Instructional System for Teaching and Remediation (DISTAR) approaches begin by concentrating on individual letters and sounds, advance to the mastery of simple phonetic units, and then blend these units into words and sentences. Thus, if children learn to cope with graphemes, they will learn to read. Other reading remediation programs, such as the Merrill program, and the *Science Research Associates, Inc. (SRA) Basic Reading Program*, begin by introducing whole words first and then teach children how to break them down and recognize the sounds of the syllables and the individual letters in

the word. Another approach teaches children with reading disorders to recognize whole words through the use of visual aids and bypasses the sounding-out process. One such program is called the *Bridge Reading Program.* The Fernald method uses a multisensory approach that combines teaching whole words with a tracing technique so that the child has kinesthetic stimulation while learning to read the words.

Epidemiology

An estimated 4 to 8 percent of youth in the United States have been identified with dyslexia, encompassing a variety of reading, spelling, and comprehension deficits. Three to four times as many boys as girls have reading impairments in clinically referred samples. In epidemiologic samples, however, rates of reading impairments are much closer among boys and girls. Boys with reading impairment are referred for psychiatric evaluation more often than girls due to comorbid ADHD and disruptive behavior problems. There is no apparent gender differential among adults who report reading difficulties.

Etiology

Data from cognitive, neuroimaging and genetic studies suggest that reading impairment is a neurobiologic disorder with a significant genetic contribution. It reflects a deficiency in processing sounds of speech sounds, and thus, spoken language. Children who struggle with reading most likely also have a deficit in speech sound processing skills. Children with this deficit cannot adequately identify the parts of words that denote specific sounds, leading to difficulty in recognizing and “sounding out” words. Youths with reading impairment are slower than peers in naming letters and numbers. The core deficits for children with reading impairment include poor processing of speech sounds and deficits in comprehension, spelling, and sounding out words.

Because reading impairment typically includes a language deficit, we believe that the left brain is the anatomical site of this dysfunction. Several studies using MRI have suggested that the planum temporale in the left brain shows less asymmetry than the same site in the right brain in children with both language disorders and specific learning disorder. PET studies have led

some researchers to conclude that left temporal blood flow patterns during language tasks differ between children with and without learning disorders. Cell analysis studies suggest that, in reading impaired individuals, the visual magnocellular system (which typically contains large cells) contains more disorganized and smaller cell bodies than expected. Studies indicate that 35 to 40 percent of first-degree relatives of children with reading deficits also have reading disability. Several studies have suggested that chromosome 6 maps to phonologic awareness (i.e., the ability to decode sounds and sound out words).

Furthermore, the ability to identify single words maps to chromosome 15. Impairment in reading and spelling likely links to susceptibility loci on multiple chromosomes. Although a recent research study identified a locus on chromosome 18 as a strong influence on single-word reading and phoneme awareness, generalist genes are likely responsible for learning disorders. Many genes believed to be associated with specific learning disorder may also influence normal variation in learning abilities. Also, genes that affect abilities in reading, for example, are also hypothesized to affect written expression and potentially mathematics skills.

Several historical hypotheses about the origin of reading deficits are now known to be untrue. The first myth is that visual-motor problems cause reading impairments, sometimes called *scotopic sensitivity syndrome.* There is no evidence that children with reading impairment have visual problems or difficulties with their visual-motor system. The second false theory is that allergies can cause, or contribute to, reading disability. Finally, unsubstantiated theories have implicated the cerebellar–vestibular system as the source of reading disabilities.

Research in cognitive neuroscience and neuropsychology supports the hypothesis that encoding processes and working memory, rather than attention or long-term memory, are areas of weakness for children with reading impairment. One study found an association between dyslexia and birth in May, June, and July, suggesting that prenatal exposure to maternal infectious illness, such as influenza, in the winter months may contribute to reading disabilities. Complications during pregnancy and prenatal and perinatal difficulties are common in the histories of children with reading disabilities. Meager birth weight and severely premature children are at higher risk for specific learning disorder. Children born very preterm are at

increased risk for minor motor, behavioral, and specific learning disorder.

An increased incidence of reading impairment occurs in intellectually average children with cerebral palsy and epilepsy. Children with postnatal brain lesions in the left occipital lobe, resulting in right visual-field blindness, experience reading impairments. Similarly, youths with lesions in the splenium of the corpus callosum that blocks the transmission of visual information from the intact right hemisphere to the language areas of the left hemisphere experience reading impairments.

Children malnourished for long periods during early childhood are at increased risk of compromised performance cognition, including reading.

SPECIFIC LEARNING DISORDER WITH IMPAIRMENT IN MATHEMATICS

Children with mathematics difficulties have difficulty learning and remembering numerals, cannot remember basic facts about numbers, and are slow and inaccurate in computation. There are four groups of skills for which children with this disorder have poor achievement: linguistic skills (those related to understanding mathematical terms and converting written problems into mathematical symbols), perceptual skills (the ability to recognize and understand symbols and order clusters of numbers), mathematical skills (basic addition, subtraction, multiplication, division, and following the sequencing of basic operations), and attentional skills (copying figures correctly and observing operational symbols correctly). There are a variety of terms used over the years to denote various difficulties with math skills, including *dyscalculia*, *congenital arithmetic disorder*, *acalculia*, *Gerstmann syndrome*, and *developmental arithmetic disorder;* these terms have been used to denote the difficulties present in mathematics disorder. Core deficits in dyscalculia are in processing numbers, and good language abilities are needed for accurate counting, calculating, and understanding mathematical principles.

Mathematics deficits can, however, occur in isolation or conjunction with language and reading impairments. According to the DSM-5, the diagnosis of specific learning disorder with impairment in mathematics consists of deficits in arithmetic counting and calculations, difficulty remembering mathematics facts, and potentially counting on fingers instead. Additional deficits include

difficulty with mathematical concepts and reasoning, leading to difficulties in applying procedures to solve quantitative problems. These deficits lead to skills that are substantially below what is typical for the child's age and cause significant interference in academic success, as documented by standardized academic achievement testing.

Clinical Features

Common features of mathematics deficit include difficulty learning number names, remembering the signs for addition and subtraction, learning multiplication tables, translating word problems into computations, and performing calculations at the expected pace. We can detect most children with mathematics deficits during the second and third grades in elementary school. A child with poor mathematics abilities typically has problems with concepts, such as counting and adding even one-digit numbers, compared with classmates of the same age. During the first 2 or 3 years of elementary school, a child with poor mathematics skill may just get by in mathematics by relying on rote memory. However, soon, as mathematics problems require discrimination and manipulation of spatial and numerical relations, a child with mathematics difficulties is overwhelmed.

Some investigators have classified mathematics deficiencies into the following categories: difficulty learning to count meaningfully, difficulty mastering cardinal and ordinal systems, difficulty performing arithmetic operations, and difficulty envisioning clusters of objects as groups. Children with mathematics difficulty have trouble associating auditory and visual symbols, understanding the conservation of quantity, remembering sequences of arithmetic steps, and choosing principles for problem-solving activities. Children with these problems are presumed to have good auditory and verbal abilities; however, in many cases, the mathematics deficits may occur in conjunction with reading, writing, and language problems. In these cases, the other deficiencies may compound the impairment of poor mathematics skill.

Mathematics difficulty, in fact, often coexists with other disorders affecting reading, expressive writing, coordination, and language. Spelling problems, deficits in memory or attention, and emotional or behavioral problems may be present. Young grade-school children may exhibit specific learning problems in reading and writing, and we should evaluate these

children for mathematics deficits. The exact relationship between mathematics deficits and the deficits in language and dyslexia is not clear. Although children with language disorder do not necessarily experience mathematics deficiencies, these conditions often coexist, and both are associated with impairments in decoding and encoding processes.

Lena, an 8-year-old girl, was referred for evaluation of impairing problems in attention and academic achievement, which were first noted in kindergarten but were now causing difficulty at home and school. Lena attended a regular third-grade class in a local public school, which she had been attending since midway through kindergarten.

Lena's history included a mild delay in speech acquisition (e.g., first words at approximately 18 months of age and short sentences at approximately 3 years of age). However, otherwise, she had no major developmental problems until kindergarten, when her teacher had raised concerns about inattentiveness, difficulty following instructions, and her difficulty in mastering basic number concepts (e.g., inaccurate counting of sets of objects). A speech, language, and hearing assessment completed at the end of kindergarten revealed mild language problems that did not warrant specific intervention. School reports from grades 1 and 2 noted ongoing concerns about inattention, poor reading skills, and difficulty mastering simple arithmetic facts, and "making careless mistakes in copying numbers from the board and in doing addition and subtraction." These problems continued through grade 2, despite some in-school accommodations (e.g., moving Lena's seat closer to the teacher) and modifications (e.g., providing her with printed sheets of arithmetic problems, so she did not need to copy them herself). Lena's parents reported a 3-year history of losing things, fidgeting at the dinner table, and difficulty concentrating on games and homework and forgetting to bring notes to and from school. The psychological assessment included the Wechsler Intelligence Scale for Children-III, Clinical Evaluation of Language Fundamentals-IV, Comprehensive Test of Phonological Processing, and the Woodcock-Johnson Psycho-Educational Battery–III. Results indicated average intelligence, with relatively weaker performance on tests of perceptual organization, weak phonologic

(speech sound) awareness, mild deficits in receptive and expressive language, and reading and arithmetic abilities that were well below grade level. Parent and teacher ratings on a standardized behavior questionnaire (Conners' Rating Scales-Long Form) were above the clinical threshold for ADHD.

Lena was given a diagnosis of ADHD, predominantly inattentive type, and specific learning disorder with impairment in reading, based on the history, school achievement, and standardized assessment. She did not meet the criteria for communication disorder, and her doctors thought that her mathematics problems did not cause impairment like her reading disorder and ADHD did. Recommendations included the following: family psychoeducation clarifying the ADHD and specific learning disorder, remedial interventions for reading, and treatment of her ADHD with a long-acting stimulant agent.

At 1-year follow-up, Lena and her parents reported noticeable improvement with inattention, but ongoing problems with reading and more significant deficits in mathematics. Her clinicians added mathematics remediation to her weekly schedule. Two years later, when Lena was 11 years of age, her parents called for an "urgent reevaluation" due to a sudden worsening of her difficulties at home and school. Clinical evaluation revealed adequate stimulant treatment response of her ADHD, more marked deficits in reading speed accuracy compared to others her age, and significant deficits in mathematics. Lena's parents reported that she had started lying about having mathematics homework or refused to do it, was suspended from mathematics class twice in the past 3 months because of oppositional behavior, and had failed sixth-grade mathematics. Lena acknowledged disliking and worrying about math: "whenever the teacher starts asking questions and looks in my direction, my mind just goes blank, and I feel sort of shaky—it's so bad in tests that I have to leave class to get myself together." At this point, her team noted an additional component of anxiety to be contributing to her school impairments. Added recommendations included increased specific educational remediation for mathematics. At follow-up, Lena reported that the resource teacher had taught her some helpful strategies to address her mathematics anxiety, as well as ways of classifying word problems and differentiating critical information from irrelevant information. She

continued to be a robust responder to long-acting stimulants for her ADHD and had only minimal difficulties concentrating on homework after school. (Adapted from case material by Rosemary Tannock, Ph.D.)

Diagnosis

We make a diagnosis of specific learning disorder in mathematics when a child's skill in mathematical reasoning, or calculation, remains significantly below that expected for that child's age, for at least 6 months, even when administering remedial interventions. Many different skills contribute to mathematics proficiency. These include linguistic skills, conceptual skills, and computational skills. Linguistic skills involve being able to understand mathematical terms, understand word problems, and translate them into the proper mathematical process. Conceptual skills involve the recognition of mathematical symbols and being able to use mathematical signs correctly. Computational skills include the ability to line up numbers correctly and to follow the "rules" of the mathematical operation.

Pathology and Laboratory Examination. No physical signs or symptoms indicate mathematics disorder, but educational testing and standardized measurement of intellectual function are necessary to make this diagnosis. The *Keymath Diagnostic Arithmetic Test* measures several areas of mathematics, including knowledge of mathematical content, function, and computation. It assesses ability in mathematics of children in grades 1 to 6.

Course and Prognosis

A child with a specific learning disorder in mathematics is usually identifiable by the age of 8 years (third grade). In some children, the disorder is apparent as early as 6 years (first grade); in others, it may not be apparent until age 10 (fifth grade) or later. Too few data are currently available from longitudinal studies to predict patterns of developmental and academic progress of children classified as having mathematics disorder in early school grades. On the other hand, children with a moderate mathematics disorder who do not receive intervention may have complications, including continuing academic difficulties, shame, poor self-concept, frustration, and depression. These complications can lead to reluctance to attend school and

demoralization about academic success.

Differential Diagnosis

We should differentiate mathematics deficits from global causes of impaired functioning such as intellectual disability. Inadequate schooling can affect a child's arithmetic performance. Conduct disorder or ADHD can occur comorbidly with specific learning disorder in mathematics, and, in these cases, we would make both diagnoses.

Comorbidity

Mathematics deficits are comorbid with deficits in both reading and written expression. Children with mathematics difficulties may also be at higher risk for expressive language problems and developmental coordination disorder.

Treatment

It is best to treat mathematics difficulties for children with early interventions that lead to improved skills in basic computation. The presence of specific learning disorder in reading, along with mathematics difficulties, can impede progress; however, children are quite responsive to remediation in early grade school. Children with indications of mathematics disorder as early as in kindergarten require help in understanding which digit in a pair is larger, counting abilities, identification of numbers, and remembering sequences of numbers. Flashcards, workbooks, and computer games can be a viable part of this treatment. One study indicated that mathematics instruction is most helpful when the focus is on problem-solving activities, including word problems, rather than only computation. *Project MATH*, a multimedia self-instructional or group-instructional in-service training program, has been successful for some children with mathematics disorder. Computer programs can be helpful and can increase compliance with remediation efforts.

Social skills deficits can contribute to a child's hesitation in asking for help, so a child identified with a mathematics disorder may benefit from gaining positive problem-solving skills in the social arena as well as in mathematics.

Epidemiology

Mathematics disability alone is estimated to occur in about 1 percent of school-age children, that is about one of every five children with specific learning disorder. Epidemiologic studies have indicated that up to 6 percent of school-age children have some difficulty with mathematics, with a prevalence of 3.5 to 6.5 for impairing forms of dyscalculia. Although specific learning disorder overall occurs two to three times more often in males, mathematics deficits may be relatively more frequent in girls than reading deficits. Many studies of learning disorders in children have grouped reading, writing, and mathematics disability, which makes it more difficult to ascertain the precise prevalence of mathematics disability.

Etiology

Mathematics deficiency, as with other areas of specific learning disorder, has a significant genetic contribution. Comorbidity with reading deficits is common and in the range of 17 percent up to 60 percent. One theory proposed a neurologic deficit in the right cerebral hemisphere, particularly in the occipital lobe areas. These regions are responsible for processing visual–spatial stimuli that, in turn, are responsible for mathematical skills. This theory, however, has received little support in subsequent neuropsychiatric studies.

The causes of deficits in mathematics are multifactorial, including genetic, maturational, cognitive, emotional, educational, and socioeconomic factors. Prematurity and very low birth weight are also risk factors for specific learning disorder, including mathematics. Compared with reading abilities, arithmetic abilities seem to depend more on the amount and quality of instruction.

SPECIFIC LEARNING DISORDER WITH IMPAIRMENT IN WRITTEN EXPRESSION

Written expression is the most complex skill acquired to convey an understanding of language and to express thoughts and ideas. Writing skills are highly correlated with reading for most children; however, for some youth, reading comprehension may far surpass their ability to express complex thoughts. Written expression, in some cases, is a sensitive index of more subtle deficits in language usage that typically are not detected by

standardized reading and language tests.

Deficits in written expression include writing skills that are significantly below the expected level for a child's age and education. Such deficits impair the child's academic performance and writing in everyday life. Components of writing disorder include poor spelling, errors in grammar and punctuation, and poor handwriting. Spelling errors are among the most common difficulties for a child with a writing disorder. Spelling mistakes are most often phonetic errors; that is, an erroneous spelling that sounds like the correct spelling. Examples of common spelling errors are: fone for phone or beleeve for believe.

Historically, experts considered dysgraphia (i.e., poor writing skills) a reading disorder; however, it is now clear that impairment in written expression can occur on its own. Terms once used to describe writing disability include *spelling disorder* and *spelling dyslexia.* Writing disabilities are often associated with other forms of specific learning disorders; however, impaired writing ability may be identified later than other forms because it is generally acquired later than verbal language and reading.

In contrast with the DSM-5, which includes specific learning disorder in written expression, ICD-10 includes a separate specific spelling disorder.

Clinical Features

Youth with impairments in written expression struggle early in grade school with spelling words and expressing their thoughts according to age-appropriate grammatical norms. Their spoken and written sentences contain an unusually large number of grammatical errors and poor paragraph organization. Affected children commonly make simple grammatical errors, even when writing a short sentence. For example, despite constant reminders, affected youth frequently fail to capitalize the first letter of the first word in a sentence, and fail to end the sentence with a period. Typical features of impaired written expression include spelling errors, grammatical errors, punctuation errors, poor paragraph organization, and poor handwriting.

In higher grades in school, affected youth's written sentences become more conspicuously primitive, odd, and inaccurate compared to what is typical for students at their grade level. For youth with impaired written expression, word choices are often erroneous and inappropriate, paragraphs are

disorganized and not in proper sequence, and spelling accuracy becomes increasingly difficult as their vocabulary becomes more extensive and more abstract. Associated features of writing impairments may include reluctance to go to school, refusal to do assigned written homework, and concurrent academic difficulties in other areas.

Many children with impaired written expression understandably become frustrated and angry, and harbor feelings of shame and inadequacy regarding poor academic achievement. In some cases, depressive disorders can result from a growing sense of isolation, estrangement, and despair. Young adults with impaired written expression who do not receive remedial intervention continue to have writing skills deficits and a persistent sense of incompetence and inferiority.

Brett, an 11-year-old boy, was referred for evaluation of increasing problems in school over 2 years, including failure to complete assigned schoolwork and homework, inattention and oppositional behavior, and deteriorating grades and test scores. At the time of assessment, he was in a regular fifth-grade class in a public school, which he had been attending since grade 1.

Clinical interview with parents revealed that Brett had a twin brother (monozygotic) with a history of language problems for which he had received speech-language therapy in the preschool years and remedial reading in the primary grades. Brett, however, had not exhibited difficulty in speech or language development, according to parental reports and scores on standardized tests of oral language administered in the preschool years. His current and previous school reports indicate that Brett participated well in class discussions and had no difficulty in reading or mathematics; however, his written work was far below grade level. In each of the last 2 years, his teachers had expressed increasing concerns about Brett's refusal to complete written work, failure to hand in homework, daydreaming and fidgeting in class, and withdrawal from class activities. Brett admitted to an increasing dislike of school and especially writing assignments. He explained, "It's writing, writing all day long—even in math and science. I know how to do the problems and the experiments, but I hate having to write it all down—my mind just

goes blank." Brett complained, "My teacher is always on me, telling me that I'm lazy and haven't done enough and that my writing is atrocious. He tells me I've got a bad attitude—so why would I want to go to school?" Brett and his parents reported that, over the past year, he has been down, increasingly frustrated with school, and has refused to do homework. They all agree that Brett has had a few brief episodes of depressed mood.

Testing by a clinical psychologist revealed average to high-average scores on the verbal and performance scales of the Wechsler Intelligence Scale for Children-III and average scores on the reading and arithmetic subtests of the Wide Range Achievement Test-3 (WRAT-3). However, scores on the WRAT-3 spelling subtest were below the 9th percentile, which was significantly below expectations for age and ability. Examination of his spelling errors revealed that his spelling was typically phonologically accurate (i.e., could plausibly be pronounced to sound like the target word). However, he used letter sequences that did not resemble English, regardless of pronunciation (e.g., "houses" was written as "howssis," "phones" was written as "fones," and "exact" was "egszakt"). Moreover, his performance was well below age and grade on standardized tests of written expression (TOWL-3), as well as on a brief (5-minute) informal assessment of expository text generation on a favorite topic (e.g., newspaper article on recent sports event). During the 5-minute writing activity, he frequently stared out the window, shifted positions, and chewed on his pencil. He would often get up to sharpen his pencil and sigh when he put pencil to paper, writing slowly and laboriously. At the end of 5 minutes, he had produced three short sentences without any punctuation or capitalization that were barely legible, containing several misspellings and grammatical errors, and that were not linked semantically. By contrast, later in the assessment, he described the sporting event with detail and enthusiasm. A speech-language evaluation revealed average scores on standard tests of oral language (Clinical Evaluation of Language Fundamentals-IV). However, he omitted sounds or syllables in a multisyllabic word in a nonword repetition test, which was sensitive to mild residual language impairments and written language impairments.

The clinical team formulated a diagnosis of specific learning disorder

with impairment in written expression, based on Brett's inability to compose written text, poor spelling, and grammatical errors, without problems in reading or mathematics or a history of language impairments. He did not meet full diagnostic criteria for any other DSM-5 disorder, including oppositional defiant disorder, ADHD, or mood disorder. Recommendations included the following: psychoeducation, the need for educational accommodations (e.g., provision of additional time for test-taking and written assignments, specific educational intervention to facilitate written expression and to teach note-taking, and use of specific computer software to support written composition and spelling), and counseling should his depressed mood continue or worsen. (Adapted from case material from Rosemary Tannock, Ph.D.)

Diagnosis

The DSM-5 diagnosis of specific learning disorder with impairment in written expression depends on a child's poor ability to use punctuation and grammar accurately in sentences, inability to organize paragraphs or to articulate ideas in writing. Poor performance on composing written text may also include poor handwriting and impaired ability to spell and to place words sequentially in coherent sentences, compared to others of the same age. In addition to spelling mistakes, youth with impaired written expression make grammatical mistakes, such as using incorrect tenses, forgetting words in sentences, and placing words in the wrong order. Punctuation may be incorrect, and the child may have a reduced ability to remember which words begin with capital letters. Additional symptoms of impaired written expression include the formation of letters that are not legible, inverted letters, and mixtures of capital and lowercase letters in a given word. Other features of writing disorders include poor organization of written stories, which lack critical elements such as "where," "when," and "who" or clear expression of the plot.

Pathology and Laboratory Examination. Whereas no physical signs of a writing disorder exist, educational testing helps to make a diagnosis of writing disorder. Diagnosis depends on a child's writing performance being markedly below expected production for his age, as confirmed by an

individually administered standardized expressive writing test. Currently available tests of written language include the Test of Written Language (TOWL), the Diagnostic Evaluation of Writing Skills (DEWS), and the Test of Early Written Language (TEWL). We should also evaluate for impaired vision and hearing.

When there are impairments in written expression, we should administer a standardized intelligence test, such as WISC-R to determine the child's overall intellectual capacity.

Course and Prognosis

Specific learning disorder with impairment in writing, reading, and mathematics often coexist, and additional language disorder may be present as well. A child with all of the above disabilities will likely be diagnosed with language disorder first and impaired written expression last. In severe cases, an impaired written expression is apparent by age 7 (second grade); in less severe cases, the disorder may not be apparent until age 10 (fifth grade) or later. Youth with mild and moderate impairment in written expression fare well if they receive timely remedial education early in grade school. Severely impaired written expression requires continual, extensive remedial treatment through the late part of high school and even into college.

The prognosis depends on the severity of the disorder, the age or grade when starting the remedial intervention, the length and continuity of treatment, and the presence or absence of associated or secondary emotional or behavioral problems.

Differential Diagnosis

It is crucial to determine whether disorders such as ADHD or major depression are interfering with a child's focus and thereby preventing the production of adequate writing in the absence of a specific writing impairment. If true, treatment for the other disorder should improve a child's writing performance. Commonly comorbid disorders with writing disability are language disorder, mathematics disorder, developmental coordination disorder, disruptive behavior disorders, and ADHD.

Comorbidity

Children with impaired writing ability are significantly more likely to have language disorder and impairments in reading and mathematics compared to the general population of youth. ADHD occurs with higher frequency in children with writing disability than in the general population. Youth with specific learning disorder, including writing disability, are at higher risk for social skills difficulties, and some develop poor self-esteem and depressive symptoms.

Treatment

Remedial treatment for writing disability includes direct practice in spelling and sentence writing as well as a review of grammatical rules. Intensive and continuous administration of individually tailored, one-on-one expressive and creative writing therapy appears to influence a favorable outcome. Teachers in some special schools devote as much as 2 hours a day to such writing instruction. This intervention largely depends on the relationship between the child and the writing specialist. Success or failure in sustaining the patient's motivation affects the treatment's long-term efficacy. Associated secondary emotional and behavioral problems should be given prompt attention, with appropriate psychiatric treatment and parental counseling.

Epidemiology

The prevalence of specific learning disorder with impairment in written expression occurs in the range of 5 to 15 percent of school-age children. Over time, specific learning disorder remits in many youths, leading to a persistent rate of specific learning disorder of 4 percent in adults. The gender ratio in writing deficits is two to three to one in boys compared with girls. Impaired written expression often occurs along with deficits in reading, but not always.

Etiology

Causes of writing disability are likely similar to those of reading disorder, that is, underlying deficits in using the components of language related to letter sounds. Genetic factors are a significant factor in the development of writing disability. Writing difficulties often accompany language disorder, leading an affected child to have trouble with understanding grammatical rules, finding words, and expressing ideas clearly. According to one

hypothesis, impairment in written expression may result from combined effects of language disorder and reading disorder. Most youths with impaired written expression have first-degree relatives with similar difficulties. Children with limited attention spans and high levels of distractibility may find writing an arduous task.

2.6 Motor Disorders

DEVELOPMENTAL COORDINATION DISORDER

Developmental coordination disorder is a neurodevelopmental disorder in which a child's fine or gross motor coordination is slower, less accurate, and more variable than in peers of the same age. Affecting about 5 to 6 percent of school-age children, 50 percent of children with developmental coordination disorder also have comorbid ADHD or dyslexia. A meta-analysis of recent research on developmental coordination disorder concluded that three general areas of deficits contribute to the disorder: (1) Poor predictive control of motor movements; (2) deficits in rhythmic coordination and timing; and (3) deficits in executive functions, including working memory, inhibition, and attention.

Children with developmental coordination disorder struggle to perform the motor activities of daily life, such as jumping, hopping, running, or catching a ball. Children with coordination problems may also agonize over using utensils, tying their shoelaces, or writing. A child with developmental coordination disorder may exhibit delays in achieving motor milestones, such as sitting, crawling, and walking, because of clumsiness, and yet excel at verbal skills.

Developmental coordination disorder, thus, may be characterized by either clumsy gross or fine motor skills, resulting in poor performance in sports and even in academic achievement because of poor writing skills. A child with developmental coordination disorder may bump into things more often than siblings or drop things. In the 1930s, the term *clumsy child syndrome* began to be used in the literature to denote a condition of awkward motor behaviors not due to any specific neurologic disorder or damage. This term continues to

be used to identify imprecise or delayed gross and fine motor behavior in children, resulting in subtle motor inabilities but often significant social rejection. Gross and fine motor impairment in developmental coordination disorder is not from a medical condition, such as cerebral palsy, muscular dystrophy, or a neuromuscular disorder. Currently, specific indications are that perinatal problems, such as prematurity, low birth weight, and hypoxia may contribute to the emergence of developmental coordination disorders. Children with developmental coordination disorder are at higher risk for language and learning disorders. A strong association exists between speech and language problems and coordination problems, as well as an association of coordination difficulties with hyperactivity, impulsivity, and reduced attention span.

Children with developmental coordination disorder may resemble younger children because of their inability to master motor activities typical for their age group. For example, children with developmental coordination disorder in elementary school may not be adept at bicycle riding, skateboarding, running, skipping, or hopping. In the middle school years, children with this disorder may have trouble in team sports, such as soccer, baseball, or basketball. Fine motor skill manifestations of developmental coordination disorder typically include clumsiness using utensils and difficulty with buttons and zippers in the preschool-age group. In older children, using scissors and more complex grooming skills, such as styling hair or putting on makeup, is difficult. Peers often ostracize children with developmental coordination disorder because of their poor skills in many sports, and they often have longstanding difficulties with peer relationships. DSM-5 categorizes developmental coordination disorder as a Motor Disorder, along with stereotypic movement disorder and tic disorders.

Clinical Features

The clinical signs suggesting the existence of developmental coordination disorder are evident as early as infancy in some cases, when a child begins to attempt tasks requiring motor coordination. The essential clinical feature is significantly impaired performance in motor coordination. The difficulties in motor coordination may vary with a child’s age and developmental stage (Table 2-19).

In infancy and early childhood, the disorder may manifest in delays in developmental motor milestones, such as turning over, crawling, sitting, standing, walking, buttoning shirts, and zipping up pants. Between the ages of 2 and 4 years, clumsiness appears in almost all activities requiring motor coordination. Affected children cannot hold objects and drop them easily, their gait may be unsteady, they often trip over their own feet, and they may bump into other children while attempting to go around them. Older children may display impaired motor coordination in table games, such as putting together puzzles or building blocks, and in any type of ball game. Although no specific features are pathognomonic of developmental coordination disorder, there is a delay in developmental milestones. Many children with the disorder also have speech and language difficulties. Older children may have secondary problems, including academic difficulties, as well as poor peer relationships based on social rejection. Children with motor coordination problems are more likely to have problems understanding subtle social cues, and their peers often reject them. A recent study indicated that children with motor difficulties perform more poorly on scales that measure the recognition of static and changing facial expressions of emotion. This finding is likely to be correlated to the clinical observations that children with motor coordination have difficulties in social behavior and peer relationships.



Table 2-19
Manifestations of Developmental Coordination Disorder

Gross Motor Manifestations
Preschool age
Delays in reaching motor milestones, such as sitting, crawling, and walking
Balance problems: falling, getting bruised frequently, and poor toddling
Abnormal gait
Knocking over objects, bumping into things, and destructiveness
Primary-school age
Difficulty with riding bikes, skipping, hopping, running, jumping, and doing somersaults

Awkward or abnormal gait

Older

Poor at sports, throwing, catching, kicking, and hitting a ball

Fine Motor Manifestations

Preschool age

Difficulty learning dressing skills (tying, fastening, zipping, and buttoning)

Difficulty learning feeding skills (handling knife, fork, or spoon)

Primary-school age

Difficulty assembling jigsaw pieces, using scissors, building with blocks, drawing, or tracing

Older

Difficulty with grooming (putting on makeup, blow-drying hair, and doing nails)

Messy or illegible writing

Difficulty using hand tools, sewing, and playing piano

Billy was brought for evaluation of suicidal ideation at 8 years of age, after complaining to his parents that peers were bullying him for being "bad" in sports and that nobody liked him. He only had one friend who also laughed at him sometimes, because he always dropped the ball and he looked "funny" while running. He was so upset about being rejected by peers when he tried to play sports that he refused to go to physical education class. Instead, he voluntarily went to the school counselor's office and stayed there until the period was over. Billy was already irritated with a diagnosis of ADHD for which he took medication.

On top of that, he had difficulty with reading. Billy became so distraught that one day he told his school counselor that he wanted to kill himself. A developmental history revealed that he was delayed for sitting, which he finally did at 10 months of age, and he could not walk without falling over until 30 months of age. Billy's parents were aware that he was very clumsy, but they believed that he would outgrow that. Even at 8 years of age, Billy's parents reported that, during meals, Billy often spilled his drinks and was quite awkward when he used a fork. Some of

his food typically fell off of his fork or spoon before it reached his mouth, and he had great difficulty using a knife and a fork.

A comprehensive assessment of fine and gross motor skills demonstrated the following: Billy was able to hop, but he could not skip without briefly stopping after each step. Billy could stand with both feet together but was unable to stand on tiptoe. Although Billy could catch a ball, he held a ball bounced to himself at chest level, and was unable to catch a ball bounced to him on the ground from a distance of 15 ft. The Bruininks–Oseretsky Test of Motor Development, a test of agility and coordination, revealed functioning levels commensurate with those of an average 6-year-old child.

Billy was referred to a neurologist for a comprehensive evaluation because he appeared to be generally weak, and his muscles seemed floppy. Neurologic evaluation was negative for diagnosable neurologic disorders, and his muscle strength was normal, despite his appearance. Based on the negative neurologic examination and the finding of the Bruininks-Oseretsky Test of Motor Development, Billy had a diagnosis of developmental coordination disorder. Billy's symptoms included mild hypotonia and fine motor clumsiness.

After the diagnosis of developmental coordination disorder was made, in addition to his already diagnosed ADHD and reading disorder, his treatment plan included private sessions with an occupational therapist who used perceptual-motor exercises to improve Billy's fine motor skills, targeting writing and use of utensils. His clinician requested an Individualized Educational Plan (IEP) evaluation from the school to obtain an adaptive physical education program. Also, the clinical team requested that he have a reading tutor and that he sit close to the front of the classroom. Billy was enrolled in a treatment program using motor imagery training to reduce his clumsiness and improve coordination.

Billy was relieved to be receiving help, especially for his reading and for sports activities, and no longer felt suicidal. Over 3 months of treatment, Billy showed a noticeable improvement in his reading. His mood improved further, mainly because he was receiving praise from his teachers and parents. Billy's classmates were not picking on him the way they used to. As Billy began to feel better about himself, he began to play sports informally with his peers, although not competitively. Billy was

granted an adaptive physical education program in school, and he was not required to play on teams. Instead, he practiced throwing and catching a ball and playing basketball with a staff member.

Billy continued to show some degree of clumsiness, especially in his fine motor skills over the next few years, yet he was cooperative, with the occupational therapy interventions, his mood was bright, and he demonstrated continual improvement. (Courtesy of Caroly Pataki, M.D. and Sarah Spence, M.D.)

Diagnosis

The diagnosis of developmental coordination disorder depends on poor performance in activities requiring coordination for a child's age and intellectual level. Diagnosis is based on a history of the child's delay in achieving early motor milestones, as well as on direct observation of current deficits in coordination. An informal screen for developmental coordination disorder involves asking the child to perform tasks involving gross motor coordination (e.g., hopping, jumping, and standing on one foot); fine motor coordination (e.g., finger-tapping and shoelace tying); and hand–eye coordination (e.g., catching a ball and copying letters). We should consider what is expected for a child's age to judge possible poor performance. A mildly clumsy child, but whose functioning is not impaired, does not qualify for a diagnosis of developmental coordination disorder. Table 2-20 compares the diagnostic approaches to developmental coordination disorder.

The diagnosis may be associated with below-normal scores on performance subtests of standardized intelligence tests and by normal or above-normal scores on verbal subtests. Specialized tests of motor coordination can be useful, such as the *Bender Visual Motor Gestalt Test*, the *Frostig Movement Skills Test Battery*, and the *Bruininks-Oseretsky Test of Motor Development.* We should take into account the child's age, and ensure that there is not a better explanation, such as a neurologic or neuromuscular condition. Examination, however, may occasionally reveal slight reflex abnormalities and other soft neurologic signs.



Table 2-20

Developmental Coordination Disorder

	DSM-5	ICD-10
Diagnostic name	Developmental Coordination Disorder	Specific Developmental Disorder of Motor Function
Duration	Occurs during early developmental period	
Symptoms	Delayed development of motor coordination	Significant impairment in motor coordination
Psychosocial consequences of symptoms	Functional impairment (self-care, play, other functions)	
Exclusions (not result of):	Another neurologic condition Intellectual disability Visual impairment	General intellectual retardation A specific congenital or neurologic disorder

Differential Diagnosis

The differential diagnosis includes medical conditions that produce coordination difficulties (e.g., cerebral palsy and muscular dystrophy). In autism spectrum disorder and intellectual disability, coordination usually does not stand out as a significant deficit compared with other skills. Children with neuromuscular disorders may exhibit more global muscle impairment rather than clumsiness and delayed motor milestones. Neurologic examination and workup usually reveal more extensive deficits in neurologic conditions than in developmental coordination disorder. Extremely hyperactive and impulsive children may be physically careless because of their high levels of motor activity. Clumsy gross and fine motor behavior and ADHD, as well as reading difficulties, are highly associated.

Comorbidity

Developmental coordination disorder is strongly associated with ADHD, specific learning disorder, particularly in reading, as well as language disorder. Children with coordination difficulties have higher-than-expected rates of language disorder, and studies of children with language disorder report very high rates of “clumsiness.” Developmental coordination disorder is also associated, but less strongly, with specific learning disorder with impairment in mathematics, and in written expression. A study of children with developmental coordination disorder reported that, although motor coordination is critical for accuracy in tasks that require speed, poor motor coordination does not correlate with the degree of inattention. Thus, in children comorbid for ADHD and developmental coordination disorder, children with the most severe ADHD do not necessarily have the worst developmental coordination disorder. Functional neuroimaging, pharmacologic, and neuroanatomical studies suggest that motor coordination depends on the integration of sensory input and an action response, not purely through sensorimotor function and higher-level thinking. Investigations of comorbid developmental coordination disorder and ADHD are trying to ascertain whether this comorbidity is due to overlapping genetic factors.

Peer relationship problems are common among children with developmental coordination disorders because of rejection that often occurs along with their poor performance in sports and games that require good motor skill. Adolescents with coordination problems often exhibit poor self-esteem and academic difficulties. Recent studies underscore the importance of attention to both victimization of children and adolescents with developmental motor coordination by peers and the potential resulting damage to self-worth. Children and adolescents with developmental coordination disorder often are victims of bullying and have higher rates of poor self-esteem that often deserves clinical attention.

Course and Prognosis

Historically, experts believed that developmental coordination spontaneously improved over time. However, longitudinal studies have shown that motor coordination problems can persist into adolescence and adulthood. When mild to moderate clumsiness is persistent, some children can compensate by developing interests in other skills. Some studies suggest a more favorable

outcome for children who have average or above-average intellectual capacity, in that they come up with strategies to develop friendships that do not depend on physical activities. Clumsiness typically persists into adolescence and adult life. One study following a group of children with developmental coordination problems over a decade found that the clumsy children remained less dexterous, showed poor balance and continued to be physically awkward. The affected children were also more likely to have both academic problems and poor self-esteem. Children with developmental coordination disorder have also been shown to be at higher risk for obesity, have difficulties with running, and are at greater risk of future cardiovascular diseases.

Treatment

Interventions for children with developmental coordination disorder utilize multiple modalities, including visual, auditory, and tactile materials targeting perceptual-motor training for specific motor tasks. Two broad categories of interventions are the following: (1) deficit-oriented approaches, including sensory integration therapy, sensorimotor-oriented treatment, and process-oriented treatment; and (2) task-specific interventions, including neuromotor task training and cognitive orientation to daily occupational performance (CO-OP). More recently, therapists incorporate motor imagery into treatment. These approaches involve visual imagery exercises using a computer; they have a broad range of foci, including predictive timing for motor tasks, relaxation and mental preparation, visual modeling of fundamental motor skills, and mental rehearsal of various tasks. This type of intervention derives from the notion that improved internal representation of a movement task will improve a child’s actual motor behavior.

The treatment of developmental coordination disorder generally includes versions of sensory integration programs and modified physical education. Sensory integration programs, usually administered by occupational therapists, consist of physical activities that increase awareness of motor and sensory function. For example, a child who bumps into objects often might be given the task of trying to balance on a scooter, under supervision, to improve balance and body awareness. Therapists often give children with difficulty writing letters tasks to increase awareness of hand movements.

School-based occupational therapies for motor coordination problems in writing include utilizing mechanisms that provide resistance or vibration during writing exercises, to improve grip, and practicing vertical writing on a chalkboard to increase arm strength and stability while writing. These programs have been shown to improve the legibility of student's writing, but not necessarily speed, because students learn to write with greater accuracy and deliberate letter formation. Currently, many schools also allow and may even encourage children with coordination difficulties that affect writing to use computers to aid in writing reports and long papers.

Adaptive physical education programs help children enjoy exercise and physical activities without the pressures of team sports. These programs generally incorporate certain sports actions, such as kicking a soccer ball or throwing a basketball. Children with coordination disorder may also benefit from social skills groups and other prosocial interventions. The Montessori technique may promote motor skill development, especially with preschool children, because this educational program emphasizes the development of motor skills. Small studies have suggested that exercise in rhythmic coordination, practicing motor movements, and learning to use word processing keyboards may be beneficial. Parental counseling may help reduce parents' anxiety and guilt about their child's impairment, increase their awareness, and facilitate their confidence to cope with the child.

An investigation of children with developmental coordination disorder showed positive results using a computer game designed to improve the ability to catch a ball. These children were able to improve their game scores by practicing virtual catching without specific instructions on how to utilize the visual cues. This has implications for treatment, in that certain types of motor task coordination can be positively influenced through the practice of specific motor tasks, even without overt instructions.

Epidemiology

The prevalence of developmental coordination disorder is about 5 to 6 percent of school-age children. The male-to-female ratio in referred populations tends to show increased rates of the disorder in males, but schools refer boys more often for testing and special education evaluations. About two males for every one female are affected.

Etiology

The causes of developmental coordination disorder are multifactorial and likely include both genetic and developmental factors. Risk factors postulated to contribute to this disorder include prematurity, hypoxia, perinatal malnutrition, and low birth weight. Prenatal exposure to alcohol, cocaine, and nicotine likely contribute to both low birth weight and cognitive and behavioral abnormalities. Children born prematurely have developmental coordination disorder rates of up to 50 percent. Researchers have proposed that the cerebellum may be the neurologic substrate for comorbid cases of developmental coordination disorder and ADHD. Neurochemical abnormalities and parietal lobe lesions may contribute to coordination deficits. Studies of postural control, that is, the ability to regain balance after being in motion, indicate that children with developmental coordination disorder who have adequate balance when standing still are unable to accurately correct for movement, resulting in impaired balance, compared with other children. A study concluded that, in children with developmental coordination disorder, neural signals from the brain to particular muscles involved in balance are neither being optimally sent or received. These findings have also implicated the cerebellum as a potential anatomical site for the dysfunction of developmental coordination disorder. There are several theories regarding the mechanisms of developmental coordination disorder. One example, called the automatization deficit hypothesis, suggests that similar to dyslexia, children with developmental coordination disorder have difficulty developing automatic motor skills. Another popular suggestion is called the internal modeling deficit hypothesis, which suggests that children with developmental coordination disorder are unable to perform the typical internal cognitive models that predict the sensory consequences of motor commands. In both scenarios, the cerebellum plays a vital role in motor coordination and developmental coordination disorder.

STEREOTYPIC MOVEMENT DISORDER

Stereotypic movements include a diverse range of repetitive behaviors that usually emerge in the early developmental period, appear to lack a clear function, and sometimes cause an interruption in daily life. These movements are typically rhythmic, such as hand flapping, body rocking, hand waving,

hair-twirling, lip-licking, skin picking, or self-hitting. Stereotypic movements often appear to be self-soothing or self-stimulating; however, they can result in self-injury in some cases. Stereotypic movements appear to be involuntary; however, they frequently can be suppressed with a concentrated effort. Stereotypic movement disorder occurs with increased frequency in children with autism spectrum disorder and intellectual disability, but they also exist in typically developing children. Stereotypic movements, such as head-banging, face slapping, eye-poking, or hand-biting, can cause significant self-harm. Nail-biting, thumb-sucking, and nose-picking are often not included as symptoms of stereotypic movement disorder because they rarely cause impairment. When impairment occurs, however, they can be included in stereotypic movement disorder. Stereotypic movements share several features with tics, including the repetitive, seemingly involuntary, and characteristically identical nature of the movements each time they are displayed. However, distinguishing features of stereotypical movements compared to tics include a younger age of onset, lack of changing anatomical locations, lack of premonitory "urge," and decreased response to medication management.

According to DSM-5, stereotypic movement disorder includes repetitive, seemingly driven, and purposeless motor behavior that interferes with social, academic, or other activities and may result in self-harm.

Diagnosis and Clinical Features

The presence of multiple repetitive stereotyped symptoms tends to occur frequently among children with autism spectrum disorder and intellectually disability, particularly when the intellectual disability is severe. Patients with multiple stereotyped movements frequently have other significant mental disorders, including disruptive behavior disorders or neurologic conditions. In extreme cases, severe mutilation and life-threatening injuries can result from self-inflicted trauma. Table 2-21 compares the diagnostic approaches to diagnosing stereotypic movement disorder.



Table 2-21
Stereotypic Movement Disorder

DSM-5	ICD-10

Diagnostic name	Stereotypic Movement Disorder	Stereotyped Movement Disorders
Duration	Onset during the developmental period	
Symptoms	Repetitive/seemingly purposeless movements	Movements that are: Voluntary Repetitive Stereotyped Seemingly purposeless Movements outside of any other psychiatric or neurologic condition; may include such things as repeatedly rocking, hand-flapping, or biting
Psychosocial consequences of symptoms	Marked interference in activities and functioning	
Exclusions (not result of):	Substance use Another medical condition Another mental disorder Another neurodevelopmental disorder	Tic disorder Trichotillomania Nail biting Nose-picking Another neurologic disorder (i.e., movement disorders) Another mental disorder
Symptom specifiers	With self-injurious behavior Without self-injurious behavior Associated with known	

	medical or genetic condition, neurodevelopmental disorder or environmental factor
Severity specifiers	**Mild:** symptoms easily suppressed by sensory stimulus or distraction **Moderate:** significant behavioral modification needed to control symptoms **Severe:** continuous measures taken to prevent harm or injury

Head-Banging. Head-banging exemplifies a stereotypic movement disorder that can result in functional impairment. Typically, head-banging begins during infancy, between 6 and 12 months of age. Infants strike their heads with a definite rhythmic and monotonous continuity against the crib or another hard surface. They seem to be absorbed in the activity, which can persist until they become exhausted and fall asleep. The head-banging is often transitory but sometimes persists into middle childhood. Head-banging that is a component of temper tantrums differs from stereotypic head-banging and ceases after the tantrums, and controlling their secondary gains.

Nail-Biting. Nail-biting begins as early as 1 year of age and increases in incidence until age 12. Most cases are not sufficiently severe to meet the DSM-5 diagnostic criteria for stereotypic movement disorder. In rare cases, children cause physical damage to the fingers themselves, usually by associated biting of the cuticles, which leads to secondary infections of the fingers and nail beds. Nail-biting seems to occur or increase in intensity when a child is either anxious or stressed. Some of the most severe nail-biting

occurs in children with severe or profound intellectual disability. However, many nail-biters have no obvious emotional disturbance.

Tim, a 14-year-old with autism spectrum disorder (ASD) and severe intellectual disability, was evaluated when he transferred to a new private school for children with ASD. Observed in his classroom, he was a small boy who appeared younger than his age. He held his hands in his pockets and spun around in place. When offered a toy, he took it and manipulated it for a while. When prompting him to engage in various tasks that required that he take his hands out of his pockets, he began hitting his head with his hands. If the teacher held his hands, he hit his head with his knees. He was adept in contorting himself so that he could hit or kick himself in almost any position, even while walking. Soon, his face and forehead were covered with bruises.

He had delayed development in all spheres, and he never developed language. He lived at home and attended a special educational program. His self-injurious behaviors developed early in life, and, when his parents tried to stop him, he became aggressive. Gradually, he became too difficult to manage in public school, and, at 5 years of age, his parents placed him in a special school. The self-abusive and self-restraining (i.e., holding his hands in his pockets) behavior was present throughout his stay there, and, virtually all of the time, he tried several second-generation antipsychotics with only minimal improvement. Although the psychiatrist’s notes mentioned some improvement in his self-injurious behavior, it was continuing and fluctuating. The school system transferred him to a new school because of lack of progress and difficulties in managing him as he became bigger and stronger. His intellectual functioning was within the 34 to 40 intelligence quotient (IQ) range. His adaptive skills were poor. He required full assistance in self-care, could not provide even for his own simple needs and required constant supervision for his safety.

In a few months, Tim settled into the routine in his new school. His self-injurious behavior fluctuated. He could reduce or stop the behavior by restraining himself. For example, he learned to hold his hands in his pockets or inside his shirt. He also might manipulate an object with his

hands. If left to himself, he could contort himself, while holding his hands inside his shirt. Because the stereotypic self-injurious and self-restraining behavior interfered with his daily activities and education, it became a primary focus of a behavior modification program. For a few months, he did well, especially when he developed a good relationship with a new teacher, who was firm, consistent, and nurturing. With him, Tim could successfully engage in some school tasks. When the teacher left, Tim regressed. To prevent injuries, the staff started blocking his self-hitting with a pillow. They offered him activities that he liked and in which he could engage without resorting to self-injury. After several months, his antipsychotic medication was slowly discontinued, over 11 months, without any behavioral deterioration. (Adapted from case material from Bhavik Shah, M.D.)

Pathology and Laboratory Examination. No specific laboratory measures are helpful in the diagnosis of stereotypic movement disorder.

Differential Diagnosis

The differential diagnosis of stereotypic movement disorder includes OCD and tic disorders, both of which are exclusionary criteria in DSM-5. Although stereotypic movements can often be voluntarily suppressed, and are not spasmodic, it is difficult to differentiate these features from tics in all cases. A study of stereotyped movements compared with tics found that stereotyped movements tended to be longer in duration, and displayed more rhythmic qualities than tics. Tics seemed to occur more when a child was in an “alone” condition, rather than when the child was in a play condition, whereas stereotypic movements occurred with the same frequency in these two different conditions. Stereotypic movements seem to be self-soothing, whereas tics are often associated with distress.

Differentiating dyskinetic movements from stereotypic movements can be difficult. Because antipsychotic medications can sometimes suppress stereotypic movements, clinicians should note any stereotypic movements before initiating treatment with an antipsychotic agent. Stereotypic movement disorder may be diagnosed concurrently with substance-related disorders (e.g., amphetamine use disorders), severe sensory impairments, CNS and

degenerative disorders (e.g., Lesch–Nyhan syndrome), and severe schizophrenia.

Course and Prognosis

The duration and course of stereotypic movement disorder vary, and the symptoms may wax and wane. Up to 60 to 80 percent of normal toddlers show transient rhythmic activities that seem purposeful and comforting and tend to disappear by 4 years of age. When stereotypic movements emerge more severely later in childhood, they typically range from brief episodes occurring under stress to an ongoing pattern in the context of a chronic condition, such as ASD or intellectual disability. Even in chronic conditions, stereotypic behaviors may come and go. In many cases, stereotypic movements are prominent in early childhood and diminish as the child ages.

The severity of the dysfunction caused by stereotypic movements varies with the frequency, amount, and degree of associated self-injury. Children who exhibit frequent, severe, self-injurious stereotypic behaviors have the poorest prognosis. Repetitive episodes of head-banging, self-biting, and eye-poking can be difficult to control without physical restraints. Most nail-biting is benign and often does not meet the diagnostic criteria for stereotypic movement disorder. In severe cases in which the child repetitively damages his nail beds, bacterial and fungal infections can occur. Although chronic stereotypic movement disorders can severely impair daily functioning, several treatments help control the symptoms.

Treatment

When stereotypic movements occur in the absence of any other symptoms or disorders, they may not warrant pharmacologic treatment. Treatment modalities yielding the most promising effects include behavioral techniques, such as habit reversal and differential reinforcement of other behavior, as well as pharmacologic interventions. A recent report on utilizing both habit reversal (in which the therapist trains the child to replace the undesired repetitive behavior with a more acceptable behavior) and reinforcement for reducing the unwanted behavior, indicated that these treatments had efficacy among 12 typically developing children between 6 and 14 years. One case report detailed a successful habit reversal treatment of a 3-year-old with

severe stereotypic movements, which was largely implemented at home by her parents.

Pharmacologic interventions have been used in clinical practice to minimize self-injury in children whose stereotyped movements caused significant harm to their bodies. Small open-label studies have reported some benefit from atypical antipsychotics, and case reports have indicated the use of SSRIs in the management of self-injurious stereotypies. The dopamine receptor antagonists have been tried most often for treating stereotypic movements and self-injurious behavior. The SSRI agents may be influential in diminishing stereotypies; however, this is still under investigation. Open trials suggest that both clomipramine and fluoxetine may decrease self-injurious behaviors and other stereotypic movements in some patients.

Epidemiology

Repetitive movements are common in infants and young children, with greater than 60 percent of parents of children between the ages of 2 and 4 years reporting transient emergence of these behaviors. The most frequent age of onset is in the second year of life. Epidemiologic surveys estimate that up to 7 percent of otherwise typically developing children exhibit stereotypic behaviors. A prevalence of about 15 to 20 percent in children younger than the age of 6 years has been reported for displaying stereotypic behavior, with diminishing rates over time. The prevalence of self-injurious behaviors, however, is in the range of 2 to 3 percent among children and adolescents with intellectual disability. Stereotypic movements appear to occur in about twice as many boys as girls. Determining which cases are sufficiently severe to confirm a diagnosis of stereotypic movement disorder may be difficult. Stereotypic behaviors occur in 10 to 20 percent of children with intellectual disability, with increased rates being proportional to the level of severity. Self-injurious behaviors frequently occur in genetic syndromes, such as Lesch–Nyhan syndrome and children with sensory impairments, such as blindness and deafness.

Etiology

The etiology of stereotypic movement disorder includes environmental, genetic, and neurobiologic factors. Although the neurobiologic mechanisms

of stereotypic movement disorder have yet to be proven, given their similarity to other involuntary movements, stereotypic movement disorder is hypothesized to originate from the basal ganglia. Dopamine and serotonin are likely to be involved in their emergence. Dopamine agonists tend to induce or increase stereotypic behaviors, whereas dopamine antagonists sometimes decrease them. One study found that 17 percent of typically developing children with stereotypic movement disorder had a first-degree relative with the disorder, and 25 percent had a first- or second-degree relative with stereotypic movement disorder. Transient stereotypic behaviors in very young children can be considered a normal developmental phenomenon. Genetic factors likely play a role in some stereotypic movements, such as the X-linked recessive deficiency of enzymes leading to Lesch–Nyhan syndrome, which has predictable features including intellectual disability, hyperuricemia, spasticity, and self-injurious behaviors. Other minimal stereotypic movements that do not usually cause impairment (e.g., nail-biting) appear to run in families as well. Some stereotypic behaviors seem to emerge or become exaggerated in situations of neglect or deprivation; such behaviors as head-banging correlate with psychosocial deprivation.

TOURETTES DISORDER

Tics are neuropsychiatric events characterized by brief, rapid motor movements or vocalizations in response to irresistible premonitory urges. Although frequently rapid, tics may include more complex patterns of movements and longer vocalizations. Converging evidence from many lines of research suggests that the production of tics involves dysfunction in the basal ganglia region of the brain, particularly of dopaminergic transmission in the cortico-striatothalamic circuits. Because tic disorders are significantly more common in children than in adults, the postulated alterations in dopamine circuitry in many affected children appear to improve over time. Tics may be transient or chronic, with a waxing and waning course. Tics typically emerge at age 5 to 6 years of age and tend to reach their highest severity between 10 and 12 years. About one half to two-thirds of children with tic disorders will be much improved or in remission by adolescence or early adulthood. We distinguish tic disorder by the type of tics, their frequency, and the pattern in which they emerge over time. Motor tics most

commonly affect the muscles of the face and neck, such as eye-blinking, head-jerking, mouth-grimacing, or head-shaking. Typical vocal tics include throat-clearing, grunting, snorting, and coughing. Tics are repetitive muscle contractions resulting in movements or vocalizations that are involuntary. However, patients can sometimes voluntarily suppress them. Children and adolescents may exhibit tic behaviors that occur after a stimulus or in response to a premonitory internal urge.

The most widely studied and most severe tic disorder is Gilles de la Tourette syndrome, also known as Tourette disorder. Georges Gilles de la Tourette (1857–1904) first described a patient with a syndrome, which became known as Tourette disorder in 1885, while he was studying with Jean-Martin Charcot in France. De la Tourette noted a syndrome in several patients that included multiple motor tics, coprolalia, and echolalia. Tics often consist of motions we might use in volitional movements. One half to two-thirds of children with Tourette disorder exhibit a reduction in or complete remission of tic symptoms during adolescence. There are many common comorbid psychiatric disorders and behavioral problems likely to emerge along with Tourette disorder. For example, the relationship between Tourette disorder, ADHD, and OCD is not clear. Epidemiologic surveys indicate that more than half of children with Tourette disorder also meet criteria for ADHD. There appears to be a bidirectional relationship between Tourette disorder and OCD, with 20 to 40 percent of Tourette disorder patients meeting full criteria for OCD. First-degree relatives of patients with OCD have higher rates of tic disorders compared to the general population.

There are a few small reports suggesting that the obsessive-compulsive symptoms most likely to occur in Tourette disorder are characteristically related to ordering and symmetry, counting, and repetitive touching. In contrast, OCD symptoms in the absence of tic disorders usually stem from fears of contamination and fear of harming. Motor and vocal tics can be simple or complex. *Simple motor tics* are those composed of repetitive, rapid contractions of functionally similar muscle groups—for example, eye-blinking, neck-jerking, shoulder-shrugging, and facial-grimacing. Common *simple vocal tics* include coughing, throat-clearing, grunting, sniffing, snorting, and barking. *Complex motor tics* appear to be more purposeful and ritualistic than simple tics. Common *complex motor tics* include grooming behaviors, the smelling of objects, jumping, touching behaviors, echopraxia

(imitation of observed behavior), and copropraxia (display of obscene gestures). *Complex vocal tics* include repeating words or phrases out of context, coprolalia (use of obscene words or phrases), palilalia (a person repeating their own words), and echolalia (repetition of the last-heard words of others).

Although older children and adolescents with tic disorders may be able to suppress their tics for minutes or hours, young children are often not cognizant of their tics or experience their urges to perform their tics as irresistible. Sleep, relaxation, or absorption in activity can all attenuate tics. Tics often disappear during sleep.

Diagnosis and Clinical Features

A diagnosis of Tourette disorder depends on a history of multiple motor tics that generally emerge over months or years and the emergence of at least one vocal tic at some point. According to the DSM-5, tics may wax and wane in frequency but must have persisted for more than a year since the first tic emerged to meet the diagnosis. The average age of onset of tics is between 4 and 6 years of age, although in some cases, tics may occur as early as 2 years of age. The peak age for the severity of tics is between 10 and 12 years of age. For a diagnosis of Tourette disorder, the onset must occur before the age of 18 years. Table 2-22 compares the diagnostic approaches to Tourette disorder.

In Tourette disorder, typically, the initial tics are in the face and neck. Over time, the tics tend to occur in a downward progression. The most commonly described tics are those affecting the face and head, the arms and hands, the body and lower extremities, and the respiratory and alimentary systems. In these areas, the tics take the form of grimacing; forehead puckering; eyebrow-raising; eyelid-blinking; winking; nose-wrinkling; nostril-trembling; mouth-twitching; displaying the teeth; biting the lips and other parts; tongue-extruding; protracting the lower jaw; nodding, jerking, or shaking the head; twisting the neck; looking sideways; head-rolling; hand-jerking; arm-jerking; plucking fingers; writhing fingers; fist-clenching; shoulder-shrugging; foot, knee, or toe shaking; walking peculiarly; body writhing; jumping; hiccupping; sighing; yawning; snuffing; blowing through the nostrils; whistling; belching; sucking or smacking sounds; and clearing the throat.

Several assessment instruments are currently available that are useful in making diagnoses of tic disorders, including comprehensive self-report assessment tools, such as the *Tic Symptom Self Report* and the *Yale Global Tic Severity Scale*, administered by a clinician (Table 2-23).



Table 2-22
Tourette Disorder

	DSM-5	ICD-10
Diagnostic name	Tourette Disorder	Combined Vocal and Multiple Motor Tic Disorder (de la Tourette)
Duration	>1 yr duration (may include waxing or waning) Onset before the age of 18	Presents in childhood/adolescence Usually persists into adulthood
Symptoms	Motor/vocal tics	Motor tics ≥1 Vocal tic
Exclusions (not result of):	Substance use Another medical illness	
	If symptoms of ONLY motor OR vocal tics, diagnosis would be of "Persistent (Chronic) Motor or Vocal Tic	NOTE: for tics that don't meet above criteria, consider: Tic Disorder—Presence of a tic, defined as an involuntary, fast, repeated and nonrhythmic motor movement or vocal production that is sudden and seemingly purposeless. Such tics are experienced as urges that are strong, but that can be

	Disorder”	suppressed variably. Transient tic disorder (lasting <12 mo) Chronic motor or vocal tic disorder (lasting longer than a year)

Because Tourette disorder is frequently comorbid with attentional, obsessional, and oppositional behaviors, these symptoms often emerge before the tics. In some studies, more than 25 percent of children with Tourette disorder received stimulants for a diagnosis of ADHD before receiving a diagnosis of Tourette disorder. The most frequent initial symptom is an eye-blink tic, followed by a head tic or a facial grimace. Most complex motor and vocal symptoms emerge several years after the initial symptoms. Coprolalia, a very unusual symptom involving shouting or speaking socially unacceptable or obscene words, occurs in less than 10 percent of patients and rarely in the absence of comorbid psychiatric disturbance. Mental coprolalia —in which a patient experiences a sudden, intrusive, socially unacceptable thought or obscene word—occurs more often than coprolalia. In severe cases, physical self-injury has occurred due to tic behaviors.

Jake, age 10 years, came to the Tourette Disorder Clinic for an evaluation of motor tics in the head and neck, occasional coughing and grunting, and a new symptom of throat-clearing many times per day. Jake had a history of ADHD, which included significant hyperactivity and impulsive and oppositional behavior. He is a fifth-grade student in a regular class at the local public school. Before the consultation, parent and teacher ratings, including the *Child Behavior Checklist* (CBCL), *Swanson, Nolan, and Pelham-IV* (SNAP-IV), *Conners’ Parent and Teacher Questionnaires*, *Tic Symptom Self-Report* (TSSR), and medical history survey, were sent to his family. His mother and the classroom teacher rated him well above the norm for hyperactivity, inattention, and impulsiveness. He was failing several subjects in school, often argued with adults, was occasionally aggressive, and had few friends. His tics were moderate.

Jake’s mother recalled difficulties with overactivity, oppositional and defiant behaviors, and behavior since preschool. At age 5, due to his

activity level and argumentative and aggressive behavior, his kindergarten teacher encouraged the family to obtain a psychiatric consultation. Jake's pediatrician made a diagnosis of ADHD and recommended a trial of Concerta (methylphenidate extended-release tablets) at 36 mg/day, which Jake began in the first grade. Within a week of starting medication, Jake's overly active and impulsive behavior showed a dramatic improvement; however, he remained argumentative and oppositional. However, when on his Concerta, Jake was able to stay in his seat and complete his work and was better able to wait his turn on the playground. The next few months went well; however, by early spring, Jake seemed to be returning to some of his old ways. He disrupted the class, talking out of turn and getting out of his seat. After an increase in Concerta to 54 mg/day, in the spring of his first-grade year, however, he began showing motor and phonic tics consisting of head-jerking, facial movements, coughing, and grunting. He discontinued the Concerta to see if this made a difference and, although the tics transiently decreased, they came back in full force within a month. In hindsight, Jake's mother recalled that Jake had exhibited eye blinking and grunting before starting the Concerta, but she had dismissed these events as unimportant, and they did not seem to disrupt Jake's daily life.

While Jake was off Concerta during a period when he began middle school in the sixth grade, he was disruptive to his classes, and he began to be severely teased by several classmates for his impulsivity, frequent motor tics, and loud grunting and throat clearing. Jake became despondent and began to refuse to go to school. At this point, his parents and clinicians decided to place Jake in a special education class. However, after several months of this placement, Jake felt worse about himself, despised school, and begged to be returned to regular classes. At this point, Jake's pediatrician made the referral to a child and adolescent psychiatrist at a local university Tourette Disorder Clinic.

During his evaluation at the Tourette Disorder Clinic, Jake was a healthy child who was the product of an uncomplicated pregnancy, labor, and delivery, and whose developmental milestones were appropriate. Intellectual testing completed by the school psychologist revealed a full-scale IQ of 105. Jake's mother noted that Jake has had longstanding trouble falling asleep but sleeps through the night. Jake has always been

described as argumentative and easily frustrated with frequent outbursts of temper; however, when he is not having a tantrum, his mood is generally upbeat.

Jake was noted by the child and adolescent psychiatrist to be of average height and weight with no dysmorphic features. His speech was rapid in tempo but normal in tone and volume. His speech is coherent and developmentally appropriate, without evidence of thought disorder; however, he had vocal tics, including grunting, coughing, and noticeable throat clearing. Jake denied depressed mood or suicidal ideation, although he reported distress about everyday issues such as being teased by peers, not having enough friends, and his poor school performance. Jake also denied recurring worries about contamination or harm coming to him or family members, or fears of acting on unwanted impulses. Other than mild touching habits involving the need to touch objects with each hand three times or in combinations of three, Jake denies repetitive rituals. Several motor tics occurred during the evaluation session, including blinking, head-jerking, and shoulder tics. Jake was restless and easily distracted throughout the session and often needed assistance with entertaining himself when not directly involved in the conversation.

Given the history of enduring motor and phonic tics, confirmed by direct observation, the diagnosis of Tourette disorder and ADHD, as well as oppositional defiant disorder, were confirmed.

Jake and his family attended several sessions with the child and adolescent psychiatrist to learn about the waxing and waning nature of tic symptoms and the natural history of Tourette disorder, as well as ADHD. Jake and his family were heartened to hear that, in general, tics tend to be at their maximum around his age, and it was somewhat likely that Jake’s tics would lessen over time or possibly fully remit. Jake began seeing a behavioral psychologist specializing in habit reversal training. In this treatment, Jake learned to engage in a behavior physically incompatible with his tic (a competing response) each time he experienced the urge to perform this tic. The competing response for Jake’s shoulder tic, which consisted of raising his shoulders as far as he could, was to gently press his shoulders down and extend his neck each time he felt the urge to engage in this tic. With repeated practice of his competing response, Jake’s urge to engage in this tic greatly diminished to the point where he

was able to manage the urge without performing the tic. The treatment team referred Jake to a child and adolescent psychiatrist who decided to re-start the Concerta at 36 mg/day and titrated it back up to 54 mg/day without worsening of the tics. Jake responded well to his behavioral therapy, and over 8 weeks, he had learned how to become aware of the urges that occurred before his tics and to replace his usual tics with less-distressing and less-disruptive behaviors.

However, when Jake entered the seventh grade, he had an exacerbation of his motor and vocal tics and was also touching objects repeatedly throughout the day. Jake again became despondent, not wanting to go to school. It was decided by his psychologist to add relaxation training to his behavioral treatment, and his child and adolescent psychiatrist added another medication to his pharmacologic regimen. His psychiatrist prescribed risperidone, 0.5 mg/day, which was titrated up to 1 mg twice daily. Jake became stabilized within a month and was able to continue in his school and even went to some parties. Jake and his parents understood the waxing and waning nature of his tics and were hopeful that they would begin to see some decrease in his tic symptoms within the next few years. At follow-up, when Jake was 15 years of age, Jake had minimal tic symptoms; an occasional eye blink and rare throat clearing was all that was observable. Jake was not currently in behavioral treatment. However, over the years, he had, on a few occasions, received some booster therapy sessions to brush up on his habit reversal training when he had a minor exacerbation of tics. Jake had been taken off his risperidone 2 years before without exacerbation of tics. Jake continued well controlled on Concerta, did well in school, and became more popular after joining the soccer team. (Adapted from L. Scahill M.S.N., Ph.D. and J.F. Leckman, M.D.)



Table 2-23
Clinical Assessment Tools in Tic Disorders

Domain	Type	Reliability and Validity	Sensitive to Change

Tics			
Tic Symptom Self-Report	Parent/self	Good	Yes
Yale Global Tic Severity Scale	Clinician	Excellent	Yes
Attention-Deficit/Hyperactivity Disorder			
Swanson, Nolan, and Pelham-IV	Parent/teacher	Excellent	Yes
Abbreviated Conners' Questionnaire	Parent/teacher	Excellent	Yes
Obsessive-Compulsive Disorder			
Yale-Brown Obsessive Compulsive Scale and Children's Yale-Brown Obsessive Compulsive Scale	Clinician	Excellent	Yes
National Institute of Mental Health Global Obsessive Compulsive Rating Scale	Clinician	Excellent	Yes
General			
Child Behavior Checklist	Parent/teacher	Excellent	No

Pathology and Laboratory Examination. No specific laboratory diagnostic test exists for Tourette disorder, but many patients with Tourette disorder have nonspecific abnormal EEG findings. CT and MRI scans have revealed no specific structural lesions, although about 10 percent of all patients with Tourette disorder show some nonspecific abnormality on CT scans.

Differential Diagnosis

We should differentiate tics from other movements and movement disorders

(e.g., dystonic, choreiform, athetoid, myoclonic, and hemiballismic movements) and the neurologic diseases that they may characterize (e.g., Huntington disease, parkinsonism, Sydenham chorea, and Wilson disease), as listed in Table 2-24. We should also distinguish tremors, mannerisms, and stereotypic movement disorder (e.g., head-banging or body-rocking) from tic disorders. Stereotypic movement disorders, including movements such as rocking, hand-gazing, and other self-stimulatory behaviors, seem to be voluntary and often produce a sense of comfort, in contrast to tic disorders. Although tics in children and adolescents may or may not feel controllable, they rarely produce a sense of well-being. Compulsions are sometimes difficult to distinguish from complex tics and may be on the same continuum biologically. Tic disorders may also occur comorbidly with mood disturbances. In a recent survey, the greater the severity of tics, the higher the probability of both aggressive and depressive symptoms in children. When a child experiences an exacerbation of tic symptoms, behavior and mood also seem to deteriorate.

Course and Prognosis

Tourette disorder is a childhood-onset neuropsychiatric disorder characterized by both motor and vocal tics, which usually emerge in early childhood, with a natural history leading to reduction or complete resolution of tics symptoms in most cases by adolescence or early adulthood. During childhood, individual tic symptoms may decrease, persist, or increase, and new symptoms may replace old ones. Severely afflicted persons may have serious emotional problems, including major depressive disorder. Impairment may also be associated with the motor and vocal tic symptoms of Tourette disorder; however, in many cases, interference in function is exacerbated by comorbid ADHD and OCD, both of which frequently coexist with the disorder. When the above three disorders are comorbid, severe social, academic, and occupational problems may ensue. Although most children with Tourette disorder will experience a decline in the frequency and severity of tic symptoms during adolescence, at present, no clinical measures exist to predict which children may have persistent symptoms into adulthood. Children with mild forms of Tourette disorder often have satisfactory peer relationships, function well in school, and develop adequate self-esteem, and

may not require treatment.



Table 2-24
Differential Diagnosis of Tic Disorders

Disease or Syndrome	Age at Onset	Associated Features	Course	Predominant Type of Movement
Hallervorden-Spatz	Childhood–adolescence	May be associated with optic atrophy, club feet, retinitis pigmentosa, dysarthria, dementia, ataxia, emotional lability, spasticity, autosomal recessive inheritance	Progressive to death in 5–20 yr	Choreic, athetoid, myoclonic
Dystonia musculorum deformans	Childhood–adolescence	Autosomal recessive inheritance commonly, primarily among Ashkenazi Jews; a more benign autosomal dominant form also occurs	Variable course, often progressive but with rare remissions	Dystonia
Sydenham chorea	Childhood, usually 5–15 yr	More common in females, usually associated with rheumatic fever (carditis elevated ASO titers)	Usually self-limited	Choreiform
Huntington disease	Usually 30–50 yr, but childhood forms are known	Autosomal dominant inheritance, dementia, caudate atrophy on CT scan	Progressive to death in 10–15 yr after onset	Choreiform
Wilson disease (hepatolenticular degeneration)	Usually 10–25 yr	Kayser–Fleischer rings, liver dysfunction, inborn error of copper metabolism; autosomal recessive inheritance	Progressive to death without chelating therapy	Wing-beating tremor, dystonia
Hyperreflexias (including latah, myriachit, Jumping Frenchman of Maine)	Generally in childhood (dominant inheritance)	Familial; may have generalized rigidity and autosomal inheritance	Nonprogressive	Excessive startle response; may have echolalia, coprolalia, and forced obedience
Myoclonic disorders	Any age	Numerous causes, some familial, usually no vocalizations	Variable, depending on cause	Myoclonus
Myoclonic dystonia	5–47 yr	Nonfamilial, no vocalizations	Nonprogressive	Torsion dystonia with myoclonic jerks
Paroxysmal myoclonic dystonia with vocalization	Childhood	Attention, hyperactive, and learning disorders; movements interfere with ongoing activity	Nonprogressive	Bursts of regular, repetitive clonic (less tonic) movements and vocalizations
Tardive Tourette syndrome	Variable (after antipsychotic medication use)	Reported to be precipitated by discontinuation or reduction of medication	May terminate after increase or decrease of dosage	Orofacial dyskinesias, choreoathetosis, tics, vocalization
Neuroacanthocytosis	Third or fourth decade	Acanthocytosis, muscle wasting, parkinsonism, autosomal recessive inheritance	Variable	Orofacial dyskinesia and limb chorea, tics, vocalization
Encephalitis lethargica	Variable	Shouting fits, bizarre behavior, psychosis, Parkinson disease	Variable	Simple and complex motor and vocal tics, coprolalia, echolalia, echopraxia, palilalia
Gasoline inhalation	Variable	Abnormal EEG; symmetrical theta and theta bursts frontocentrally	Variable	Simple motor and vocal tics
Postangiographic complications	Variable	Emotional lability, amnestic syndrome	Variable	Simple motor and complex vocal tics, palilalia
Postinfectious	Variable	EEG: occasional asymmetrical theta bursts before movements, elevated ASO titers	Variable	Simple motor and vocal tics, echopraxia
Posttraumatic	Variable	Asymmetrical tic distribution	Variable	Complex motor tics
Carbon monoxide poisoning	Variable	Inappropriate sexual behavior	Variable	Simple and complex motor and vocal tics, coprolalia, echolalia, palilalia
XYY genetic disorder	Infancy	Aggressive behavior	Static	Simple motor and vocal tics

Disease or Syndrome	Age at Onset	Associated Features	Course	Predominant Type of Movement
XXY and 9p mosaicism	Infancy	Multiple physical anomalies, mental retardation	Static	Simple motor and vocal tics
Duchenne muscular dystrophy (X-linked recessive)	Childhood	Mild intellectual disability	Progressive	Motor and vocal tics
Fragile X syndrome	Childhood	Intellectual disability, facial dysmorphism, seizures, autistic features	Static	Simple motor and vocal tics, coprolalia
Developmental and perinatal disorders	Infancy, childhood	Seizures, EEG and CT abnormalities, psychosis, aggressivity, hyperactivity, Ganser syndrome, compulsivity, torticollis	Variable	Motor and vocal tics, echolalia

ASLO, Antistreptolysin O; CT, computed tomography; EEG, electroencephalogram.

Treatment

Psychoeducation is a useful intervention in order for families to gain an understanding of the variability of tics, the natural history of the disorder, and ways to support stress reduction. Families need to be well-informed advocates for their children since it is easy to misinterpret tics as a child's purposeful misbehavior, rather than a response to an irresistible urge. The subjective distress of the child and the functional disruptions caused by the disorder argue for active treatment. In mild cases, children with tic disorders who are functioning well socially and academically may not seek, nor require treatment. In more severe cases, peers may ostracize children with tic disorders, and the tics can disrupt academic work. We should consider a variety of interventions, including psychosocial, pharmacologic, and school-based. A scale to measure tic severity, the *Premonitory Urge for Tics Scale* (PUTS), is internally consistent and correlates with overall tic severity in youths over 10 years of age.

The European clinical guidelines for Tourette syndrome and other tic disorders summarized and reviewed the evidence-based treatments for Tourette disorder and developed a consensus for psychosocial and pharmacologic treatments. This guideline recommends that both behavioral and pharmacologic interventions be considered in more severe cases, with behavioral interventions typically the first line of treatment. Indications for treatment include, but are not limited to, the following clinical presentations. Tics require treatment when they cause social and emotional problems, depression, or isolation. Children who are prone to severe persistent complex motor tics or loud vocal tics may be the objects of bullying and social rejection. In these cases, depressive symptoms commonly result. Tic

reduction and psychoeducation to the school may help preserve healthy social relationships, and to diminish depressive and anxiety symptoms. Tics may also lead to impairment in academic achievement. School difficulties in children with Tourette disorder are not uncommon, and reduction in tics may support increased academic success. Tics may also lead to physical discomfort, based on the repetitive musculoskeletal exertion, especially concerning head and neck tics. In some children with Tourette disorder, tics can worsen headaches and migraines. Behavioral and pharmacologic interventions can both target tic reductions, which can lead to improved quality of life.

Evidence-Based Behavioral and Psychosocial Treatment. The Canadian guidelines for the evidence-based treatment of tic disorders: behavioral therapy, deep brain stimulation and transcranial magnetic stimulation, and a large multi-site RCT of "Comprehensive Behavioral Intervention for Tics," (CBIT) both found converging evidence supporting *habit-reversal training* and *exposure and response prevention* as efficacious treatments for tic reduction. In an RCT of CBIT, 61 children received habit reversal training as their main component of treatment, and they also received relaxation treatment and a functional intervention to identify situations that worsened or sustained tics and strategies to decrease exposure to these situations. The control group of 65 children received supportive psychotherapy and psychoeducation. After 10 weeks of treatment, the intervention group had a significantly reduced Yale Global Tic Severity Scale Total Tic score compared with the control group.

HABIT REVERSAL. The primary components of habit reversal are awareness training, in which the child uses self-monitoring to enhance awareness of tic behaviors and the premonitory urges or sensations, indicating that a tic is about to occur. In competing-response training, the patient learns to voluntarily perform a behavior that is physically incompatible with the tic, contingent on the onset of the premonitory urge or the tic itself, blocking expression of the tic. The competing-response strategy relies on the self-reported observations of patients that tics occur in response to irresistible premonitory urges in order to diminish the urge. Performing the tic satisfies or reduces the premonitory urges, thus reinforcing the tics, and over time they

become repeated entrenched behaviors. Competing-response training is different from voluntary tic suppression in that the patient initiates a voluntary behavior to manage the premonitory urge and thus disrupts the reinforcement of the tic, rather than merely trying to suppress the tic. Successful competing-response training significantly reduces the premonitory urge and can decrease or eliminate the urge. For motor tics, the patient may choose a less noticeable behavior, whereas for vocal tics, slow rhythmic breathing is the most common voluntary competing response. Patients can perform competing responses without disrupting usual activities.

EXPOSURE AND RESPONSE PREVENTION. The rationale for this treatment relies on the notion that tics occur as a conditioned response to unpleasant premonitory urges, and since the tics reduce the urge, they become associated with the premonitory urge. Each time the tic reduces the urge, this strengthens the association. Rather than using competing responses, as in habit-reversal training, exposure and response prevention asks the patient to suppress tics for increasingly prolonged periods in order to break the association between the urges and the tics. Theoretically, if a patient learns to resist performing the tic in response to the urge for long enough periods, the urge may become more tolerable, or attenuate, and the need to perform the tic may diminish.

Many other behavioral interventions, such as relaxation training, self-monitoring, bio (neuro) feedback, and cognitive-behavioral treatment (CBT), are not useful for reducing tics on their own. However, it may help to include some of these strategies in comprehensive treatment programs for children with tic disorders who are receiving habit-reversal training. Habit reversal has been the most extensively researched behavioral treatment for tic disorders; it is highly effective and is currently the first-line behavioral treatment for tic disorders.

Evidence-Based Pharmacotherapy. Several reviews of pharmacologic treatments for tics suggest that the following classes of pharmacologic agents have an evidence base for treating tics: typical and atypical antipsychotics; noradrenergic agents; and alternative treatments such as tetrabenazine, topiramate, and tetrahydrocannabinol.

ATYPICAL AND TYPICAL ANTIPSYCHOTIC AGENTS. Risperidone, with its high

affinity for dopamine D_2 and serotonin 5-HT_2 receptors, is the most well-studied atypical antipsychotic in the treatment of tics. There is considerable evidence for its efficacy. Multiple randomized, controlled studies in children and adolescents have shown favorable results compared to placebo as well as in head-to-head studies with the typical antipsychotics haloperidol and pimozide. Risperidone was associated with fewer adverse events compared to typical antipsychotics; however, it can cause weight gain, metabolic side effects, and hyperprolactinemia. In a randomized, double-blind, parallel-group study of Tourette disorder comparing risperidone to pimozide, risperidone showed superiority in reducing comorbid obsessive-compulsive symptoms as well as reducing tics. In other randomized clinical trials, risperidone reduced tics in children, adolescents, and adults with mean daily doses of 2.5 mg with a range of 1 to 6 mg.

Haloperidol and pimozide are the two most well-investigated and FDA-approved antipsychotic agents in the treatment of Tourette disorder. However, atypical antipsychotics such as risperidone are often first-line agents due to their safer side-effect profiles. Both haloperidol and pimozide are efficacious in multiple randomized clinical trials in the treatment of Tourette disorder. Both haloperidol and pimozide present significant risks for extrapyramidal side effects; in a long-term naturalistic follow-up study, haloperidol produces more significant acute dyskinesia and dystonia compared to pimozide.

A third typical antipsychotic, fluphenazine, has been used in the United States for many years in the treatment of tic disorders in the absence of robust data supporting its efficacy. A small controlled study of fluphenazine, trifluphenazine, and haloperidol found similar reductions in tics; however, haloperidol has more extrapyramidal side effects and more sedation. The frequency of sedation, dystonia, and akathisia of typical antipsychotics, probably due to their predominant dopaminergic blockade in the nigrostriatal pathways, limits their use and increases the appeal of the atypical antipsychotics. Risperidone and pimozide had equal efficacy in one study of children, adolescents, and adults with Tourette disorder.

Aripiprazole has become a pharmacologic agent of interest in the treatment of tic disorders due to its mode of action; in addition to its D_2 receptor antagonistic actions, aripiprazole is also a partial D_2 and 5-HT_{1A} receptor

agonist and a 5-HT_{2A} antagonist. A multisite double-blind controlled study of aripiprazole in children with Tourette disorder in China found a reduction in tic behaviors in about 60 percent of the aripiprazole group compared to about 64 percent reduction in a group treated with tiapride, a benzamide with selective D_2 receptor antagonism. There was no significant difference between the two groups. Although sedation and sleep disturbance are common side effects with aripiprazole, weight gain is less pronounced than with risperidone.

Olanzapine and ziprasidone are also efficacious in the treatment of tic disorders in at least one RCT. Sedation and weight gain were prominent side effects with olanzapine, and potential QT prolongation was an issue with ziprasidone. Quetiapine is a potentially useful agent in the treatment of tics, with its higher affinity for 5-HT_2 receptors than for D_2 receptors. However, randomized clinical trials are needed. Clozapine, contrary to many other atypical antipsychotics, is not useful in the treatment of tics.

NORADRENERGIC AGENTS. Noradrenergic agents, including clonidine and guanfacine, as well as atomoxetine, are frequently used in children as primary treatments or adjunctive treatments for comorbid ADHD and tics. Several studies have provided some evidence for the efficacy of clonidine, an α_2-adrenergic agent, in the treatment of tics in children, adolescents, and adults with tic disorders. The largest randomized trial with oral clonidine compared to placebo found a modest reduction in tics with clonidine. A multisite randomized, double-blind placebo-controlled trial using the clonidine patch in the treatment of tic disorders in children found a significant improvement in tic symptoms (about 69 percent) compared to about 47 percent of the children in the control group. The usual dose range for clonidine is from 0.05 mg orally three times daily to 0.1 mg four times daily, and for guanfacine, it is from 1 to 4 mg/day. When used in these dosage ranges, adverse effects of the α-adrenergic agents may include drowsiness, headache, irritability, and occasional hypotension.

Guanfacine can treat children with ADHD successfully, although its efficacy regarding reducing tics is controversial. In one randomized clinical trial treating 34 children with ADHD and tics, guanfacine was superior to placebo in the reduction of tics. In another double-blind placebo-controlled

trial of 24 children with Tourette disorder, guanfacine was not superior to placebo.

Atomoxetine, a selective norepinephrine reuptake inhibitor, was found to reduce both tics and ADHD symptoms in a multicenter industry trial of 148 children. Atomoxetine also reduced both tics and ADHD in a subgroup of patients in this study who had Tourette disorder. Additional studies are needed to confirm the safety and efficacy of atomoxetine in the treatment of children with Tourette disorder.

Given the frequent comorbidity of tic behaviors and obsessive-compulsive symptoms or disorders, the SSRIs have been used alone or in combination with antipsychotics in the treatment of Tourette disorder. They seem to be useful, however, there have not been controlled trials yet to determine the effect of SSRIs on tic reduction.

Although clinicians must weigh the risks and benefits of using stimulants in cases of severe hyperactivity and comorbid tics, data suggest that methylphenidate does not increase the rate or intensity of motor or vocal tics in most children with hyperactivity and tic disorders.

ALTERNATIVE AGENTS: TETRABENAZINE, TOPIRAMATE, AND TETRAHYDROCANNABINOL

Tetrabenazine. A vesicular monoamine transporter type 2 inhibitor, tetrabenazine depletes presynaptic dopamine and serotonin and blocks postsynaptic dopamine receptors. There are no randomized clinical trials of this agent in the treatment of Tourette disorder in children; however, clinical experience suggests that this agent may have benefit in tic reduction. In a follow-up of 2 years of treatment in 77 children and adolescents, one study reports tic reduction improvement in 80 percent of subjects. Side effects of this agent include sedation, parkinsonism, depression, insomnia, anxiety, and akathisia.

Topiramate. A GABAergic drug, used primarily as an anticonvulsant, topiramate, was effective for reducing tics in a small randomized clinical trial of children and adults with Tourette disorder. Side effects were minimal. Although this does not confirm its efficacy, GABA-modulating agents deserve further study in the treatment of tic disorders.

Tetrahydrocannabinol. A small randomized trial suggested that tetrahydrocannabinol (THC) may be safe and efficacious in the treatment of tics without neuropsychological impairment. In this trial, reported adverse effects included dizziness, fatigue, and dry mouth. Potential additional side-effects include anxiety, depressive symptoms, tremor, and insomnia. This small trial does not confirm efficacy for this agent in the treatment of tics. Instead, it raises questions about the potential improvements in treatment-resistant tic disorders using this agent.

In summary, the most significant evidence for the safe and efficacious pharmacologic treatment of Tourette disorder seems to be associated with the atypical antipsychotics, in particular, risperidone. Pharmacologic treatment may be combined with and enhanced by a variety of behavioral interventions such as habit reversal and school interventions that may diminish stressful situations in the school environment.

Epidemiology

The estimated prevalence of Tourette disorder ranges from 3 to 8 per 1,000 school-age children. Males are affected between two and four times more often than females. The unique features of Tourette disorder, in which tics wax and wane and may change in character, frequency, and severity over relatively short periods, have made ascertainment of its prevalence challenging. Furthermore, remission of tics is particularly age-dependent in that tics tend to emerge and increase from ages 5 to 10 years of age, and in many cases, decrease in frequency and severity after the age of 10 to 12 years. At age 13 years, however, using stringent criteria, the prevalence rate for Tourette disorder drops to 0.3 percent. The lifetime prevalence of Tourette disorder is estimated to be approximately 1 percent.

Etiology

Genetic Factors. Twin studies, adoption studies, and segregation analysis studies all support a genetic basis, albeit a complex one, for Tourette disorder. Twin studies indicate that concordance for the disorder in monozygotic twins is significantly higher than that in dizygotic twins. Tourette disorder and chronic motor or vocal tic disorder are likely to occur

in the same families; this lends support to the view that the disorders are part of a genetically determined spectrum. The sons of mothers with Tourette disorder seem to be at the highest risk for the disorder. Evidence in some families suggests an autosomal dominant transmission for Tourette disorder. Studies of an extended family pedigree suggest that transmission of Tourette disorder is bilinear; that is, Tourette disorder appears to be inherited through an autosomal pattern in some families, intermediate between dominant and recessive. A study of 174 unrelated probands with Tourette disorder identified a higher than chance occurrence of a rare sequence variant in SLITRK1, believed to be a candidate gene on chromosome 13q31.

Up to half of all patients with Tourette disorder also have ADHD, and up to 40 percent of those with Tourette disorder also have OCD. These frequent comorbidities with Tourette disorder can lead to a plethora of overlapping symptoms. Family studies have provided compelling evidence for the association between tic disorders and OCD. First-degree relatives of persons with Tourette disorder are at high risk for the development of Tourette disorder, chronic motor or vocal tic disorder, and OCD. The current understanding of the genetic bases of Tourette disorder implicates multiple vulnerability genes that may serve to mediate the type and severity of tics. Candidate genes associated with Tourette disorder include dopamine receptor genes, dopamine transporter genes, several noradrenergic genes, and serotonergic genes.

Neuroimaging Studies. A functional magnetic resonance imaging (fMRI) study of brain activity 2 seconds before and after a tic found that paralimbic and sensory association areas were involved. Furthermore, evidence suggests that voluntary tic suppression involves deactivation of the putamen and globus pallidus, along with partial activation of regions of the prefrontal cortex and caudate nucleus. Compelling, but indirect, evidence of dopamine system involvement in tic disorders includes the observations that pharmacologic agents that antagonize dopamine (haloperidol, pimozide, and fluphenazine) suppress tics and that agents that increase central dopaminergic activity (methylphenidate, amphetamines, and cocaine) tend to exacerbate tics. The relation of tics to neurotransmitter systems is complex and not yet well understood; for example, in some cases, antipsychotic medications, such as haloperidol, are not effective in reducing tics, and the effect of stimulants

on tic disorders reportedly varies. In some cases, Tourette disorder has emerged during treatment with antipsychotic medications.

More direct analyses of the neurochemistry of Tourette disorder have been possible utilizing brain proton magnetic resonance spectroscopy (MRS). Neuroimaging studies using cerebral blood flow in PET and SPECT suggest that alterations of activity may occur in various brain regions in patients with Tourette disorder compared to controls, including the frontal and orbital cortex, striatum, and putamen. An investigation examining the cellular neurochemistry of patients with Tourette disorder utilizing MRS of the frontal cortex, caudate nucleus, putamen, and thalamus demonstrated that these patients had a reduced amount of choline and *N*-acetylaspartate in the left putamen along with reduced levels bilaterally in the putamen. In the frontal cortex, patients with Tourette disorder were found to have lower concentrations of *N*-acetylaspartate bilaterally, lower levels of creatine on the right side, and reduced myoinositol on the left side. These results suggest that deficits in the density of neuronal and nonneuronal cells are present in patients with the disorder. Abnormalities in the noradrenergic system have been implicated in some cases by the reduction of tics with clonidine. This adrenergic agonist reduces the release of norepinephrine in the CNS and, thus, may reduce activity in the dopaminergic system. Abnormalities in the basal ganglia are known to result in various movement disorders, such as Huntington disease, and are likely sites of disturbance in Tourette disorder.

Immunologic Factors and Postinfection. An autoimmune process and, in particular, one that is secondary to group A β-hemolytic streptococcal infection, is a potential mechanism for the development of tics and obsessive-compulsive symptoms in some cases. Data have been conflicting and controversial, and this mechanism appears to be unlikely as an etiology of Tourette disorder in most cases. One case-control study found little evidence of the development or exacerbation of tics, or obsessions or compulsions, in children with well-documented and treated group A β-hemolytic streptococcal infections.

PERSISTENT (CHRONIC) MOTOR OR VOCAL TIC DISORDER

Chronic motor or vocal tic disorder is defined as the presence of either motor tics or vocal tics, but not both. Tics may wax and wane but must have persisted for more than 1 year since the first tic onset to meet the diagnosis for persistent (chronic) motor or vocal tic disorder. According to DSM-5, this disorder must have its onset before the age of 18 years. We would not diagnose chronic motor or vocal tic disorder if the patient already has Tourette disorder.

Diagnosis and Clinical Features

The onset of chronic motor or vocal tic disorder typically occurs in early childhood. Chronic vocal tics are considerably rarer than chronic motor tics. Chronic vocal tics, in the absence of motor tics, are typically less conspicuous than the vocal tics in Tourette disorder. The vocal tics are usually not loud or intense and are not mainly from the vocal cords; they consist of grunts or other noises caused by thoracic, abdominal, or diaphragmatic contractions.

Differential Diagnosis

We should differentiate chronic motor tics from a variety of other motor movements, including choreiform movements, myoclonus, restless legs syndrome, akathisia, and dystonias. Involuntary vocal utterances can occur in certain neurologic disorders, such as Huntington disease and Parkinson disease.

Course and Prognosis

Children whose tics emerge between the ages of 6 and 8 years seem to have the best outcomes. Symptoms often last for 4 to 6 years and remit in early adolescence. Children whose tics involve the limbs or trunk may have less prompt remission than those with only facial tics.

Treatment

The treatment of chronic motor or vocal tic disorder depends on several factors, including the severity and frequency of the tics; the patient’s subjective distress; the effects of the tics on school or work, job performance, and socialization; and the presence of any other concomitant mental disorder.

Psychotherapy may help minimize the secondary social difficulties caused by severe tics. Behavioral techniques, particularly habit reversal treatments, are effective in treating chronic motor or vocal tic disorder. When severe, atypical antipsychotics such as risperidone may reduce the tics. If not effective, typical antipsychotics such as pimozide or haloperidol may be helpful. Behavioral interventions are the first line of treatment.

Epidemiology

The rate of chronic motor or vocal tic disorder is 100 to 1,000 times greater than that of Tourette disorder in school-age children. School-age boys are at the highest risk. The estimated prevalence of chronic motor or vocal tic disorder is from 1 to 2 percent.

Etiology

Chronic motor or vocal tic disorder, as well as Tourette disorder, tend to aggregate in the same families. Twin studies have found a high concordance for either Tourette disorder or chronic motor tics in monozygotic twins. This finding supports the importance of hereditary factors in the transmission of tic disorders.

▲ 2.7 Feeding and Eating Disorders of Infancy or Early Childhood

Feeding and eating disorders of infancy and childhood are persistent disturbances in eating or eating-related behaviors that can lead to significant impairments in physical health and psychosocial functioning. The DSM-5 category *Feeding and Eating Disorders* includes three disorders that are often, but not always, associated with infancy and early childhood: pica, rumination disorder, and avoidant/restrictive food intake disorder (formerly known as feeding disorder of infancy or early childhood). As these are specialized diagnoses that are usually made in collaboration between a pediatrician, a child psychiatrist, and often a pediatric gastroenterologist, we discuss them only briefly. Anorexia nervosa, bulimia nervosa, and binge-eating disorder are also part of that DSM category, however they are more often associated with young adulthood and we discuss them in Chapter 13.

PICA

Pica is persistent eating of nonnutritive substances. Typically, no specific biologic abnormalities account for pica. In many cases, we only discover the disorder when medical problems such as intestinal obstruction, intestinal infections, or poisonings arise, such as lead poisoning due to the ingestion of lead-containing paint chips. Pica is more frequent in the context of autism spectrum disorder or intellectual disability; we only diagnose pica when it is of sufficient severity and persistence to warrant clinical attention. Pica can emerge in young children, adolescents, or adults; however, a minimum of 2 years of age is suggested by DSM-5 in the diagnosis of pica, in order to exclude developmentally appropriate mouthing of objects by infants that may accidentally result in ingestion. Pica occurs in both males and females, and in rare cases, a cultural belief in the spiritual or medicinal benefit of ingesting nonfood substances may be the cause. In this context, we would not diagnose pica. Among adults, certain forms of pica, including geophagia (clay eating) and amylophagia (starch eating), have been reported in pregnant women.

Diagnosis and Clinical Features

Eating nonedible substances repeatedly after 18 months of age is not typical; however, DSM-5 suggests a minimum age of 2 years when making a diagnosis of pica. Pica behaviors, however, may begin in infants 12 to 24 months of age. Specific substances ingested vary with their accessibility, and they increase with a child's mastery of locomotion, and the resultant increased independence and decreased parental supervision. Typically, infants may ingest paint, plaster, string, hair, and cloth, whereas older toddlers and young children with pica may ingest dirt, animal feces, small stones, and paper. The clinical implications can be benign or life-threatening, depending on the objects ingested. Among the most severe complications are lead poisoning (usually from lead-based paint), intestinal parasites after ingestion of soil or feces, anemia and zinc deficiency after ingestion of clay, severe iron deficiency after ingestion of large quantities of starch, and intestinal obstruction from the ingestion of hairballs, stones, or gravel. Except in autism spectrum disorder and intellectual disability, pica often remits by adolescence. Pica associated with pregnancy is usually limited to the pregnancy itself.

Chantal was 2½ years of age when her mother urgently brought her to her pediatrician due to severe abdominal pain and lack of appetite. Chantal's mother complained that she still put everything in her mouth but refused to eat regular food. The pediatrician observed Chantal to be pale, thin, and withdrawn. She sucked her thumb and quietly looked down while her mother reported that Chantal often chewed on newspapers and put plaster in her mouth.

The medical examination revealed that Chantal was anemic and suffered from lead poisoning. She was hospitalized, and a child psychiatric evaluation revealed that Chantal's mother was overwhelmed, caring for five young children, and had little affection for Chantal. Chantal's mother was a single mother, living with her five children and four other family members in a three-bedroom apartment in an old housing project. Her 7-year-old daughter had behavior problems, and her 6- and 4-year-old sons were impulsive and hyperactive and required constant supervision. Chantal's 18-month-old sister was an engaging and

active little girl, whereas Chantal was withdrawn, and would sit quietly, rocking herself, sucking her thumb, or chewing on newspaper.

The treatment plan included the involvement of social services and protective services to remove any lead paint from the walls in their current apartment, seek better living arrangements for the family, and provide a safe environment for the children. Chantal's mother received guidance in enrolling Chantal in a preschool program, and her older sister and two brothers in an after-school program that provided structure and stimulation, and some respite time for her mother. Chantal, her mother, and her younger sister started family therapy to help their mother's understanding of her children's needs and to increase her positive interactions with Chantal. Once Chantal's mother felt more supported and less overwhelmed, she was able to become more empathic and warm toward Chantal. When Chantal began chewing on paper, her mother learned to instead engage her in a play activity rather than screaming at her and grabbing her mouth. Chantal and her mother continued in therapy for a year, during which their relationship gradually became more interactive and warm, while Chantal's chewing behaviors decreased, and even her thumb sucking abated.

Pathology and Laboratory Examination. No single laboratory test confirms or rules out a diagnosis of pica, but several laboratory tests are useful because abnormal levels of lead are sometimes associated. Levels of iron and zinc in serum should be determined and corrected if low. In rare cases, when this is the etiology, giving the patient oral iron and zinc may ameliorate the pica. A hemoglobin level should be determined to rule out anemia.

Differential Diagnosis

The differential diagnosis of pica includes avoidance of food, anorexia, or rarely iron and zinc deficiencies. Pica may occur in conjunction with failure to thrive and be comorbid with schizophrenia, autism spectrum disorder, and Kleine–Levin syndrome. In psychosocial dwarfism, a dramatic but reversible endocrinologic and behavioral form of failure to thrive, children often show bizarre behaviors, including ingesting toilet water, garbage, and other

nonnutritive substances. Lead intoxication may be associated with pica. In children who exhibit pica that warrants clinical intervention, along with a known medical disorder, we should diagnose both disorders.

In some areas of the world and among certain cultures, such as the Australian aborigines, rates of pica in pregnant women are reportedly high. According to DSM-5, however, if such practices are culturally accepted, the diagnostic criteria for pica are not met.

Course and Prognosis

The prognosis for pica is usually good, and typically in children with normal intellectual function, pica generally remits spontaneously within several months. In childhood, pica usually resolves with increasing age; in pregnant women, pica resolves after delivery. In some adults with pica, particularly those who also have autism spectrum disorder and intellectual disability, pica can continue for years. Follow-up data on these populations are too limited to permit conclusions.

Treatment

The first step in determining the appropriate treatment of pica is to investigate the specific situation whenever possible. When pica occurs in the context of child neglect or maltreatment, those circumstances must be immediately corrected. Exposure to toxic substances, such as lead, must also be eliminated. No definitive treatment exists for pica per se beyond education and behavior modification. Treatments emphasize psychosocial, environmental, behavioral, and family guidance approaches. It is essential to address significant psychosocial stressors. When lead is present in the surroundings, it must be eliminated or rendered inaccessible, or the child and their family should move.

When pica persists in the absence of any toxic manifestations, we should use behavioral techniques. Techniques include positive reinforcement, modeling, behavioral shaping, and overcorrection treatment. Increasing parental attention, stimulation, and emotional nurturance may yield positive results. A study found that pica occurred most frequently in impoverished environments, and in some patients, correcting an iron or zinc deficiency has eliminated pica. Medical complications (e.g., lead poisoning) that develop

secondarily to the pica also require treatment.

Epidemiology

The prevalence of pica is unclear. A survey of a large clinic population reported that 75 percent of 12-month-old infants and 15 percent of 2- to 3-year-old toddlers placed nonnutritive substances in their mouth; however, this behavior is developmentally appropriate and typically does not result in ingestion. Pica is common among children and adolescents with autism spectrum disorder and intellectual disability. Up to 15 percent of persons with severe intellectual disability have engaged in pica. Pica appears to affect both sexes equally.

Etiology

Pica is most often a transient disorder that typically lasts for several months and then remits. In younger children, those with developmental speech and social developmental delays are more likely to have it. Among adolescents with pica, a substantial number of them exhibited depressive symptoms and using substances. Nutritional deficiencies in minerals such as zinc or iron (found in, e.g., dirt or ice) may rarely be a cause. In some cases of pica, we also find severe child maltreatment in the form of parental neglect and deprivation. Lack of supervision, as well as in adequate feeding of infants and toddlers, may increase the risk of pica.

RUMINATION DISORDER

Rumination is an effortless and painless regurgitation of partially digested food into the mouth soon after a meal, which is either swallowed or spit out. We can observe rumination in developmentally normal infants who put their thumb or hand in the mouth, suck their tongue rhythmically, and arch their back to initiate regurgitation. We may observe this pattern in infants who receive inadequate emotional interaction and have learned to soothe and may stimulate themselves through rumination. However, rumination syndromes also occur in children and adolescents, in which case rumination is a functional gastrointestinal disorder. The pathophysiology of rumination is not well understood; however, it often involves a rise in intragastric pressure

generated by either voluntary or unintentional contraction of the abdominal wall muscles causing movement of gastric contents back up into the esophagus. The onset of the disorder can occur in infancy, childhood, or adolescence. In infants, it typically occurs between 3 and 12 months of age, and once the regurgitation occurs, the food may be swallowed or spit out. Infants who ruminate strain with their backs arched and head back to bring the food back into their mouths and appear to find the experience pleasurable. Infants who are "experienced" ruminators can bring up the food through tongue movements and may not spit out the food at all, but hold it in their mouths and re-swallow it. The disorder is not common in older children, adolescents, and adults. It varies in severity and is sometimes associated with medical conditions, such as hiatal hernia, leading to esophageal reflux. In its most severe form, the disorder can cause malnutrition and be fatal.

We can diagnose rumination disorder even if an infant is a healthy weight. Failure to thrive, therefore, is not a necessary criterion of this disorder, but it is sometimes a sequela. According to DSM-5, the disorder must be present for at least 1 month after a period of normal functioning, and not better accounted for by gastrointestinal illness, or psychiatric or medical conditions.

Doctors have recognized rumination for hundreds of years. An awareness of the disorder is essential to avoid misdiagnosis and unwarranted tests and treatment. *Rumination* comes from the Latin *ruminare*, which means, "to chew the cud." The Greek equivalent is *merycism,* the act of regurgitating food from the stomach into the mouth, re-chewing the food, and re-swallowing it.

Diagnosis and Clinical Features

The DSM-5 notes that the essential feature of the disorder is repeated regurgitation and re-chewing of food for at least 1 month after a period of normal functioning. Patients bring partially digested food into their mouth without nausea, retching, or disgust; on the contrary, it may appear to be pleasurable. This activity may be distinguished from vomiting by painless and purposeful movements observable in some infants who induce it. The food is then ejected from the mouth or swallowed. A characteristic position of straining and arching of the back, with the head held back, is observed. The infant makes sucking movements with the tongue and gives the

impression of gaining considerable satisfaction from the activity. Usually, the infant is irritable and hungry between episodes of rumination.

Initially, rumination may be challenging to distinguish from the regurgitation that frequently occurs in healthy infants. In infants with persistent and frequent rumination behaviors, however, the differences are apparent. Although spontaneous remissions are common, secondary complications can develop, such as progressive malnutrition, dehydration, and lowered resistance to disease. Failure to thrive, with the absence of growth and developmental delays in all areas, can occur in the most severe cases. Additional complications may occur if the mother of a given infant with rumination becomes discouraged by the persistent symptoms, viewing it as her feeding failure, as this may lead to more tension and more rumination after feedings.

Luca was 9 months old when his pediatrician referred him to a gastroenterologist, and by his gastroenterologist for psychiatric evaluation due to persistent and frequent rumination. Luca was born full-term and had developed typically until 6 weeks of age when he began to regurgitate large amounts of milk just after feedings. He was evaluated and diagnosed with gastroesophageal reflux, and the gastroenterologist had his mother thicken his feedings. Luca responded well to the treatment; his regurgitation diminished, and he gained weight adequately. Luca continued to do well, and his mother decided to go back to work when Luca was 8 months old. Luca's mother transitioned his care to a young nanny who cared for Luca while she worked. Luca and the nanny seemed to have a warm relationship; however, he started again to regurgitate his meals soon after his mother left the house. The regurgitation seemed to increase in frequency and intensity within 2 weeks of the mother's return to work. At this point, Luca regurgitated after almost every meal, and he was losing weight. A gastroenterologist evaluated Luca, and during the barium swallow, his doctor noted that Luca put his hand in his mouth, which seemed to induce the regurgitation. Luca was administered some medication for gastroesophageal reflux; however, he continued to induce regurgitation after meals with increasing frequency, prompting the psychiatric

consultation.

Observation of mother and infant during feeding at home revealed that as soon as Luca finished feeding, he purposefully placed his hand in his mouth and induced the regurgitation. When his mother restricted his hand, Luca moved his tongue back and forth in a rhythmic manner until he regurgitated again. Luca engaged in this rhythmic tongue movement repeatedly, even when he could not bring up any more milk, and appeared to be enjoying this behavior.

Due to Luca's poor nutritional state and moderate dehydration, the doctor hospitalized him and inserted a nasojejunal tube for feedings. When Luca was awake during feedings, a special duty nurse or his parents played with him and distracted him during attempts to put his hand in his mouth or thrust his tongue rhythmically. Luca became increasingly engaged in this playful activity, and his ruminative activity decreased accordingly. After 1 week in the hospital, they started small feedings; however, Luca again successfully was able to bring up his food by his rumination activity, and the treatment team had to stop oral feedings. At this point, Luca's mother decided to stop working and take Luca home to continue an intensive behavioral "distracting" intervention in order to interrupt his rumination during meals. Luca's mother started small feedings while playing with him during and after feedings, and was able to interest him in other activities so that he would not ruminate. After 4 weeks of slow increments in his feedings, Luca was able to take all his feedings by mouth without ruminating, and they removed his nasojejunal tube. Luca and his mother continued to use stimulating and distracting activities during and just after meals, which over time, became more attractive to Luca than his previous ruminating behavior.

Pathology and Laboratory Examination. No specific laboratory examination is pathognomonic of rumination disorder; however, rumination disorder is not uncommonly associated with gastrointestinal abnormalities. Clinicians should evaluate other physical causes of vomiting, such as pyloric stenosis and hiatal hernia, before making the diagnosis of rumination disorder. Rumination disorder can lead to states of malnutrition and dehydration. In very severe cases, laboratory measures of endocrinologic

function, serum electrolytes, and a hematologic workup may determine the need for medical intervention.

Differential Diagnosis

To make the diagnosis of rumination disorder, clinicians must rule out primary gastrointestinal congenital anomalies, infections, and other medical illnesses that could account for frequent regurgitation. Pyloric stenosis is usually associated with projectile vomiting and is generally evident before 3 months of age when rumination has its onset. Rumination occurs with both autism spectrum disorder and intellectual disability in which stereotypic behaviors and eating disturbances are not uncommon. Rumination behavior may occur comorbidly in youth with severe anxiety disorders as well. Rumination disorder may also occur in patients with other eating disorders, such as anorexia nervosa and bulimia nervosa.

Course and Prognosis

Rumination disorder has a high rate of spontaneous remission. Indeed, many cases of rumination disorder may develop and remit without ever being diagnosed. Limited data are available about the prognosis of rumination disorder in adolescents and adults. Behavioral interventions using habit-reversal techniques may significantly lead to improved prognosis.

Treatment

The treatment of rumination disorder is often a combination of education and behavioral techniques. Sometimes, an evaluation of the mother–child relationship reveals deficits that can be influenced by offering guidance to the mother. Behavioral interventions, such as habit-reversal, can reinforce an alternate behavior that becomes more compelling than the behaviors leading to regurgitation. Aversive behavioral interventions, such as squirting lemon juice into the infant's mouth whenever rumination occurs, have been used in the past to diminish rumination behavior. Although aversive behavioral interventions have anecdotal support, current recommendations support the use of habit-reversal techniques.

When features of child maltreatment or neglect may have contributed to rumination behaviors in an infant, treatments include improvement of the

child’s psychosocial environment, increased tender loving care from the mother or caretakers, and psychotherapy for the mother or both parents. Anatomical abnormalities, such as hiatal hernia, are not uncommon and must be evaluated, in some cases leading to surgical repair. In severe cases in which malnutrition and weight loss have occurred, treatment may include jejunal tube placement before implementing other treatments.

Medication is not a standard part of the treatment of rumination. Case reports, however, cite a variety of medications, including metoclopramide, cimetidine, and even antipsychotics such as haloperidol, as helpful. The treatment of adolescents with rumination disorder is often complicated and includes a multidisciplinary approach consisting of individual psychotherapy, nutritional intervention, and pharmacologic treatment for frequent comorbid anxiety and depressive symptoms.

Epidemiology

Rumination is a rare disorder. It seems to be more common among male infants and emerges between 3 months and 1 year of age. It persists more frequently among children, adolescents, and adults with intellectual disability. Adults with rumination usually are normal weight.

Etiology

Rumination is associated with high intragastric pressure and the ability to contract the abdominal wall to cause retrograde movement of the gastric contents into the esophagus. Several studies have elucidated other gastrointestinal symptoms such as gastroesophageal reflux that may accompany rumination.

In a study of 2,163 children in Sri Lanka between the ages of 10 and 16 years, it was found that rumination behaviors were present in 5.1 percent of boys and 5.0 percent of girls. In 94.5 percent of youth who ruminated, the regurgitation occurred in the first hour after the meal, and 73.6 percent reported re-swallowing of the regurgitated food, whereas the rest spit it out. Only 8.2 percent of this sample reported daily episodes of regurgitation, whereas 62.7 percent experienced weekly symptoms. Associated gastrointestinal symptoms reported in this sample included abdominal pain, bloating, and weight loss. Approximately 20 percent of youth with

rumination in this sample also experienced other gastrointestinal symptoms. Another survey of 147 patients from 5 to 20 years of age found the mean age of onset of rumination was 15 years, and these patients were symptomatic after each meal; 16 percent of this sample met criteria for a psychiatric disorder, 3.4 percent had anorexia or bulimia nervosa, and 11 percent had a surgical procedure. Additional gastrointestinal symptoms in this sample included abdominal pain in 38 percent, constipation in 21 percent, nausea in 17 percent, and diarrhea in 8 percent. In some cases, vomiting secondary to gastroesophageal reflux or an acute illness precedes a pattern of rumination that lasts for several months. In many cases, children classified as ruminators have gastroesophageal reflux or hiatal hernia.

It appears, for some infants, that the rumination behavior is self-soothing or produces a sense of relief, leading to a continuation of behaviors to bring it about. In youth with autism spectrum disorder or intellectual disability, rumination may serve as a self-stimulatory behavior. Overstimulation and tension is likely a contributing factor in rumination. Behaviorists attribute persistent rumination to the positive reinforcement of pleasurable self-stimulation and to the attention a baby receives from others as a consequence of the disorder.

AVOIDANT/RESTRICTIVE FOOD INTAKE DISORDER

Avoidant/restrictive food intake disorder, formerly known as feeding disorder of infancy or early childhood, is characterized by a lack of interest in food, or its avoidance based on the sensory features of the food or the perceived consequences of eating. This newly included DSM-5 disorder adds more detail about the nature of the eating problems and includes adolescents and adults. The disorder is manifested by a persistent failure to meet nutritional or energy needs, as evidenced by one or more of the following: significant weight loss or failure to achieve expected weight, nutritional deficiency, dependence on enteral feedings or nutritional supplements, or marked interference with psychosocial functioning. It may take the form of outright food refusal, food selectivity, eating too little, food avoidance, and delayed self-feeding. We should not make the diagnosis in the context of anorexia nervosa or bulimia nervosa, or if caused by a medical condition, by another mental disorder, or by a genuine lack of available food.

Infants and children with the disorder may be withdrawn, irritable, apathetic, or anxious. Because of the avoidant behavior during feeding, there is less touching and holding between mothers and infants during the feeding process compared with other children. Some reports suggest that food avoidance or restriction may be relatively longstanding; however, in many cases, normal adult functioning is eventually achieved.

Differential Diagnosis

We should differentiate the disorder from structural problems with the infants' gastrointestinal tract that may be contributing to discomfort during the feeding process. Because feeding disorders and organic causes of swallowing difficulties often coexist, it is essential to rule out medical reasons for feeding difficulties. A study of videofluoroscopic evaluation of children with feeding and swallowing problems revealed that clinical evaluation was 92 percent accurate in identifying those children at increased risk of aspiration. This type of evaluation is necessary before psychotherapeutic interventions in cases where a medical contribution to feeding problems is suspected.

Course and Prognosis

Most infants identified with feeding disorder within the first year of life and who receive treatment do not go on to develop malnutrition, growth delay, or failure to thrive. When feeding disorders have their onset later, in children 2 to 3 years of age, growth and development can be affected when the disorder lasts for several months. In older children or adolescents, the feeding disorder typically interferes with social functioning until treated. About 70 percent of infants who persistently refuse food in the first year of life continue to have some eating problems during childhood.

Jennifer was 6 months old when she saw a child psychiatrist for feeding difficulties, irritability, and poor weight gain since birth. She was small and slight, but she did not appear to be lethargic or malnourished. Her parents were college-educated, and both had pursued their professional careers until Jennifer was born. Although Jennifer was full-term and weighed 7 pounds at birth, she had been unable to breastfeed due to

turning away and not ingesting enough milk. When she was 4 weeks old, Jennifer's mother had reluctantly switched her to bottle feedings because Jennifer was losing weight. Although her intake improved somewhat on bottle feedings, she gained weight very slowly and was still less than 8 lb at 3 months of age. Since then, she had gained a minimal amount each month to maintain a low but adequate weight. Jennifer's mother appeared tired and described that Jennifer would drink only up to about 6 oz at a time, or two bites of baby food, and then wiggle and cry; and refuse to continue with the feeding. However, after a few hours, she might cry again as if she were hungry. However, she could not settle her into a good rhythm of feeding and continued attempts to feed her would lead her to cry inconsolably. Jennifer's mother described approximately 10 to 15 attempts at feeding her both liquids and solids in 24 hours. Jennifer was reported to be an irritable and fussy infant, who cried multiple times during the day and at night, and woke her family often during the night with her crying. Jennifer's developmental milestones, such as sitting up, tracking, and making sounds, were normal.

The observation of mother–infant interactions during feeding and play revealed that Jennifer was a very alert and wiggly baby who had difficulty sitting still. While drinking from the bottle, she would kick her feet and move around, and if the bottle slipped out of her mouth, she did not try to recapture it. When eating baby foods, she was not interested, and her mother had to coax her to open her mouth. This upset Jennifer, and she would start crying. Jennifer's mother reported that she was always anxious during meals, and would try to convince Jennifer to take spoonfuls of baby food while sitting in her high chair. After repeated unsuccessful attempts of adequate feeding, Jennifer and her mother both appeared exhausted and took a break.

The history and examination revealed that Jennifer was a very active and excitable baby who had difficulty keeping calm during feedings. After reviewing the videotape with the mother, the therapist explored ways in which the mother could better facilitate calming Jennifer before and during meals. Using a quiet corner in the house and singing to Jennifer before meals resulted in Jennifer remaining calmer during meals, and she was able to drink more substantial amounts of milk, eat more solid foods, and waited longer between meals. This improvement, in turn,

relieved her mother's anxiety and helped both to have calmer interactions. (Adapted by Caroly Pataki, M.D.)

Treatment

Most interventions for feeding disorders aim to optimize the interaction between the mother and infant during feedings and identifying any factors that can be changed to promote better ingestion. The psychiatrist helps the mother to become more aware of the infant's stamina for the length of individual feedings, the infant's biologic regulation patterns, and the infant's fatigue level to increase the level of engagement between mother and infant during feeding.

Some experts have proposed a transactional model of intervention for infants who exhibit the "difficult" temperamental traits of emotional intensity, stubbornness, lack of hunger cues, and irregular eating and sleeping patterns. The treatment includes education for the parents regarding the temperamental traits of the infant, exploration of the parents' anxieties about the infant's nutrition, and training for the parents regarding changing their behaviors to promote internal regulation of eating in the infant. Parents are encouraged to feed the infant regularly at 3- to 4-hour intervals and offer only water between meals. The parents learn to deliver praise to the infant for any self-feeding efforts, regardless of the amount of food ingested. Furthermore, parents learn to limit any distracting stimulation during meals and give attention and praise to positive eating behaviors rather than intense negative attention to inappropriate behavior during meals. This training process for parents is intense and is brief. Many parents can facilitate improved eating patterns in the infant as a result. If the mother or caregiver is unable to participate in the intervention, it may be necessary to include additional caregivers to contribute to feeding the infant. In rare cases, an infant may require hospitalization until they receive adequate nutrition. If an infant tires before ingesting an adequate amount of nutrition, it may be necessary to begin treatment with the placement of a nasogastric tube for supplemental oral feedings.

For older children with failure-to-thrive syndromes, hospitalization and nutritional supplementation may be necessary. Medication is not a standard component of treatment for feeding disorders; however, there are anecdotal

reports of preadolescents with failure-to-thrive and feeding disorders who were comorbid for anxiety and mood symptoms and who received enteral nutritional interventions in addition to risperidone, and who were observed to have an increase in oral intake and accelerated weight gain.

Epidemiology

Between 15 and 35 percent of infants and young children have transient feeding difficulties. A study of restrictive eating difficulties in Swedish 9- and 12-year-olds found that restrictive eating problems were present in 0.6 percent of their sample. However, another study of avoidant eating patterns in young children in Germany found that some degree of avoidance was present in up to 53 percent of children. Thus, we should separate avoidant eating behaviors without impairment of nutritional state or psychosocial functioning from restricted eating disturbances leading to significant functional impairment. A survey of feeding problems in nursery school children revealed a prevalence of 4.8 percent with equal gender distribution. In that study, children with feeding problems exhibited more somatic complaints, and mothers of affected infants exhibited an increased risk of anxiety symptoms. Data from community samples estimate a prevalence of failure to thrive syndromes in approximately 3 percent of infants, with approximately half of those infants exhibiting feeding disorders.

▲ 2.8 Trauma- and Stressor-Related Disorders in Children

This section includes disorders in which a traumatic or significantly stressful event is a necessary diagnostic criterion, according to the DSM-5. Included here are reactive attachment disorder, disinhibited social engagement disorder, and PTSD. ICD-10 also includes a category of disorders of social functioning with onset specific to childhood and adolescence. Although we cover PTSD in Chapter 10, it can present differently in children and has important implications for child development. Hence we discuss that in this chapter.

DSM-IV divided reactive attachment disorder into two subtypes: emotionally withdrawn/inhibited and indiscriminately social/disinhibited. In DSM-5, however, the preceding two subtypes have been defined as two distinct disorders, with the DSM-5 reactive attachment disorder equivalent to the previous emotionally withdrawn/inhibited subtype, and disinhibited social engagement disorder representing the previous social disinhibited subtype.

REACTIVE ATTACHMENT DISORDER AND DISINHIBITED SOCIAL ENGAGEMENT DISORDER

Reactive attachment disorder and disinhibited social engagement disorder are clinical disorders characterized by aberrant social behaviors in a young child that reflect grossly negligent parenting and maltreatment that disrupted the development of normal attachment behavior. A diagnosis of either reactive attachment disorder or disinhibited social engagement disorder rests on the presumption that the cause is caregiving deprivation. DSM-III was the first to define the disorder in 1980. It derives from attachment theory, which describes the quality of a child's affective relationship with primary caregivers, usually parents. This fundamental relationship is the product of a young child's need for protection, nurturance, and comfort, and the interaction of the parents and child in fulfilling these needs.

Based on observations of a young child and parents during a brief separation and reunion, designated the "strange situation procedure," pioneered by Mary Ainsworth and colleagues, researchers have designated a child's basic pattern of attachment to be characterized as secure, insecure, or disorganized. Children who exhibit secure attachment behavior experience their caregivers as emotionally available and appear to be more exploratory and well adjusted than children who exhibit insecure or disorganized attachment behavior. Insecure attachment results from a young child's perception that the caregiver is not consistently available. In contrast, disorganized attachment behavior in a child results from experiencing both the need for proximity to the caregiver and apprehension in approaching the caregiver. These early patterns of attachment influence a child's future capacities for affect regulation, self-soothing, and relationship building. According to the DSM-5, reactive attachment disorder is a consistent pattern of emotionally withdrawn responses toward adult caregivers, limited positive

affect, sadness, and minimal social responsiveness to others, and concomitant neglect, deprivation, and lack of appropriate nurturance from caregivers. Reactive attachment disorder is, presumably, due to grossly pathologic caregiving received by the child. The pattern of care may exhibit a disregard for a child's emotional or physical needs or repeated changes of caregivers, as when a child is frequently relocated during foster care. The symptoms are not due to autism spectrum disorder, and the child must have a developmental age of at least 9 months.

Pathologic caretaking can result in two distinct disorders: reactive attachment disorder, in which the disturbance takes the form of the child's constantly failing to initiate and respond to most social interactions in a developmentally normal way; and disinhibited social engagement disorder, in which the disturbance takes the form of undifferentiated, unselective, and inappropriate social relatedness, with familiar and unfamiliar adults.

In disinhibited social engagement disorder, according to DSM-5, a child actively approaches and interacts with unfamiliar adults in an overly familiar way, either verbally or physically. They check for or seek out their caregiver less often and are willing to go with unfamiliar adults without hesitation. These behaviors in disinhibited social engagement disorder are not accounted for by impulsivity, although socially disinhibited behavior is predominant. These patterns of disinhibited, developmentally inappropriate behaviors are presumed to be caused by pathogenic caregiving. Thus, for both reactive attachment disorder and disinhibited social engagement disorder, aberrant caretaking is presumed to be the predominant cause of the child's inappropriate behaviors. However, there have been cases of less severe disturbances in parenting that may also be associated with young children who exhibit some characteristics of reactive attachment disorder or disinhibited social engagement disorder.

These disorders may also result in a picture of failure to thrive, in which an infant shows physical signs of malnourishment and does not exhibit the expected developmental motor and verbal milestones.

Diagnosis and Clinical Features

Children with reactive attachment disorder and disinhibited social engagement disorder may initially be identified by a preschool teacher or by a

pediatrician based on direct observation of the child's inappropriate social responses. These diagnoses rely at least partially on documented evidence of pervasive disturbance of attachment leading to inappropriate social behaviors present before the age of 5 years. The clinical picture varies greatly, depending on a child's chronologic and mental ages, but expected social interaction and liveliness are not present. Often, the child is not progressing developmentally or is frankly malnourished. Perhaps the most common clinical picture of an infant with reactive attachment disorder is the nonorganic failure to thrive. Such infants usually exhibit hypokinesis, dullness, listlessness, and apathy, with poverty of spontaneous activity. Infants look sad, joyless, and miserable. Some infants also appear frightened and watchful, with a radar-like gaze.



Table 2-25
Reactive Attachment Disorder

	DSM-5	ICD-10
Diagnostic name	Reactive Attachment Disorder	Reactive Attachment Disorder of Childhood
Duration	Begins < age 5	Begins < age 5
Symptoms	Emotional inhibition, rarely seeking or responding to comfort when distressed Social and emotional disturbance: • ↓ emotional/social responsiveness • ↓ affect • Fear, sadness or irritability in non-threatening interactions with adults History of insufficient care: • Neglect	Abnormal social relationships Emotional disturbances that are reactive to environmental change *May be present:* Fear Aggression Hypervigilance Failure to thrive History of neglect or abuse

	• Deprivation • Repeated changes in caregivers • Institutional care Child has developmental age of ≥9 mo	
Required number of symptoms	Symptoms in each category ≥2 of the social/emotional disturbance symptoms	
Exclusions (not result of):	Autism spectrum disorder	Asperger syndrome Disinhibited attachment disorder of childhood
Course specifiers	***Persistent:*** >12 mo	
Severity specifiers	***Severe:*** all symptoms, high severity	

Table 2-25 compares the diagnostic approaches to reactive attachment disorder, and Table 2-26 compares them for disinhibited social engagement disorder.

Infants may exhibit delayed responsiveness to a stimulus that would elicit fright or withdrawal from a healthy infant. Infants with failure to thrive and reactive attachment disorder appear significantly malnourished, and many have protruding abdomens. Occasionally, they have foul-smelling, celiac-like stools. In unusually severe cases, a clinical picture of marasmus appears.

The infant's weight is often below the third percentile and markedly below the appropriate weight for his or her height. If serial weights are available, the weight percentiles may have decreased progressively because of an actual weight loss or a failure to gain weight as height increases. Head circumference is usually average for the infant's age. Muscle tone may be weak. The skin may be colder and paler or more mottled than the skin of a healthy child. Laboratory findings may indicate coincident malnutrition, dehydration, or concurrent illness. Bone age is usually retarded. Growth

hormone levels are usually normal or elevated, a finding suggesting that growth failure in these children is secondary to caloric deprivation and malnutrition. Cortisol secretion in children with reactive attachment disorder or disinhibited social engagement disorder is lower than in typically developing children. For children with failure to thrive, improvement physically and weight gain generally rapidly occurs after hospitalization.

Socially, the infants with reactive attachment disorder usually show little spontaneous activity and a marked diminution of both initiative toward others and reciprocity in response to the caregiving adult or examiner. Both mother and infant may be indifferent to separation on hospitalization or to termination of subsequent hospital visits. The infants frequently show none of the normal upset, fretting, or protest about hospitalization. Older infants usually show little interest in their environment. They may not play with toys, even if encouraged; however, they rapidly or gradually take an interest in and relate to their caregivers in the hospital.



Table 2-26
Disinhibited Social Engagement Disorder

	DSM-5	ICD-10
Diagnostic name	Disinhibited Social Engagement Disorder	Disinhibited Attachment Disorder of Childhood
Duration		Begins during the first 5 yr of age
Symptoms	Disinhibited social behavior: • ↓ hesitation around adult strangers • ↑ familiarity with adult strangers • ↓ checking back with caregivers in unfamiliar situations	Abnormal pattern of social functioning • Nonselective attachment behavior • Attention seeking • Overly friendly behaviors

	• Willing/less hesitant to go off with an adult stranger History of insufficient care: • Neglect • Deprivation • Repeated changes in caregivers • Institutional care	
Required number of symptoms	Each category ≥2 symptoms of disinhibited social behavior All of the above, with child having developmental age of ≥9 mo	
Exclusions (not result of):	ADHD	Asperger syndrome Hyperkinetic disorder Reactive attachment disorder of childhood
Course specifiers	***Persistent:*** >12 mo	
Severity specifiers	***Severe:*** if all symptom criteria met at relatively high level	

Children with disinhibited social engagement disorder appear to be overly friendly and familiar with little fear.

His adoptive parents brought a 7-year-old boy for evaluation because of hyperactivity and inappropriate social behavior at school. They adopted him at 4 years of age, after living most of his life in a Chinese orphanage in which he received care from a rotating shift of caregivers. Although he had been below the 5th percentile for height and weight on arrival, he

quickly approached the 15th percentile in his new home. However, his adoptive parents were frustrated by his inability to bond with them. They had initially worried about an intellectual problem, although testing and his capacity to engage almost any adult and many children verbally suggested otherwise. He appeared to be too friendly, talking to anyone and often following strangers willingly. He showed little empathy when others were hurt, and yet he would sit on the laps of teachers and students without asking. He was frequently injured because of seemingly reckless behavior, although he had an extremely high pain tolerance. His parents focused on problem behaviors at home to decrease his impulsive behavior, which improved with much prompting; however, he remained oddly overfriendly at home and in school. The psychiatrist diagnosed the child with disinhibited social engagement disorder. (Adapted from Neil W. Boris, M.D. and Charles H. Zeanah, Jr., M.D.)

Pathology and Laboratory Examination. Although no single specific laboratory test can make a diagnosis, many children with reactive attachment disorder have disturbances of growth and development. Thus, establishing a growth curve and examining the progression of developmental milestones may help determine whether associated phenomena, such as failure to thrive, are present.

Differential Diagnosis

The differential diagnosis of reactive attachment disorder and disinhibited social engagement disorder must take into account that many other psychiatric disorders may arise in conjunction with maltreatment, including depressive disorders, anxiety disorders, and PTSD. Psychiatric disorders to consider in the differential diagnosis include language disorders, autism spectrum disorder, intellectual disability, and metabolic syndromes. Children with autism spectrum disorder are typically well nourished and of age-appropriate size and weight, and are generally alert and active, despite their impairments in reciprocal social interactions. Significant intellectual disability is often present in children with an autism spectrum disorder, whereas when intellectual disability occurs with reactive attachment disorder or disinhibited social engagement disorder, it is generally relatively mild.

Children with disinhibited social engagement disorder often show comorbid ADHD, PTSD, and language disorder or delay. Furthermore, children with disinhibited social engagement disorder symptoms may have complex neuropsychiatric problems.

Course and Prognosis

Most of the data available on the natural course of children with reactive attachment disorder and disinhibited social engagement disorder come from follow-up studies of children in residential facilities with histories of severe neglect. Findings from these studies suggest that children with reactive attachment disorder, who are later adopted into caring environments, improve in their attachment behaviors and may normalize over time. Children with disinhibited social engagement disorder, however, appear to have more difficulty developing attachments to new caregivers. Children with disinhibited social engagement disorder who exhibit indiscriminate social behavior also tend to have poor peer relationships. The duration and severity of the neglect influence the prognosis for both disorders. The degree of resulting impairment does as well. Constitutional and nutritional factors interact in children, who may either respond resiliently to treatment or continue to fail to thrive. The amount of treatment and rehabilitation that the family receives affects the child. Children who have multiple problems stemming from pathogenic caregiving may recover physically faster and more completely than they do emotionally.

Treatment

The first consideration in treating reactive attachment disorder or disinhibited social engagement disorder is a child's safety. Thus, the management of these disorders must begin with a comprehensive assessment of the current level of safety and adequate caregiving. When there is suspicion of maltreatment persisting in the home, usually, the first decision is whether to hospitalize the child or to attempt treatment while the child remains in the home. If we suspect neglect or emotional, physical, or sexual abuse, we must report it to the appropriate law enforcement and child protective services in the area. The therapeutic strategy depends on the child's physical and emotional state and the level of pathologic caregiving. We should determine the nutritional status

of the child and the presence of ongoing physical abuse or threat. Hospitalization is necessary for children with malnourishment.

Along with an assessment of the child's physical well-being, an evaluation of the child's emotional condition is essential. Immediate intervention must address the parents' awareness and capacity to participate in altering the injurious patterns that have heretofore ensued. The treatment team must begin to improve the unsatisfactory relationship between caregiver and child. This intervention usually requires extensive and intensive therapy and education with the mother or with both parents when possible.

In one study, parents of 120 children between 11.7 and 31.9 months, identified as being at risk for neglect, were randomly assigned to an intervention for at-risk parents called Attachment and Biobehavioral Catch-up (ABC) or to a control intervention. The purpose of the ABC intervention was to decrease frightening behavior toward the infant by parents and to increase sensitive and nurturing interactions between parents and infants. The investigators manualized the intervention to guide the parents in how to provide those interactions with their infants. They evaluated the children after 10 sessions, and the 60 children who received the ABC intervention showed significantly lower rates of disorganized attachment (32 percent), and higher rates of secure attachment (52 percent) compared to those who received the control intervention (disorganized attachment 57 percent; secure attachment 33 percent). The authors concluded that we could enhance parental nurturance and sensitivity through a comprehensive and explicit intervention such as the ABC intervention, and can measure significant improvements in attachment behaviors in young children after 10 sessions.

The caregiver–child relationship is the basis of the assessment of reactive attachment disorder and disinhibited social engagement disorder symptoms and the substrate from which to modify attachment behaviors. Structured observations allow a clinician to determine the range of attachment behaviors established with various family members. We may work closely with the caregiver and the child to facilitate higher sensitivity in their interactions. Three primary psychotherapeutic modalities help promote bonds between children and caregivers. First, a clinician can target the caregiver to promote positive interaction with a child who does not yet have the repertoire to respond positively. Second, a clinician can work with the child and the caregiver together as a dyad to advocate for practicing appropriate positive

reinforcement for each other. Through the use of videotapes, therapists can view parent–child interactions and suggest modifications to increase positive engagement. The third modality for clinical intervention is through individual work with the child. Working with the child and caregiver together is often more effective in producing more emotionally meaningful exchanges than working with the parent or child individually.

Psychosocial interventions for families in which a child has reactive attachment disorder or disinhibited social engagement disorder include (1) psychosocial support services, including hiring a homemaker, improving the physical condition of the apartment, or obtaining adequate housing, improving the family's financial status, and decreasing the family's isolation; (2) psychotherapeutic interventions, including individual psychotherapy, psychotropic medications, and family or marital therapy; (3) educational counseling services, including mother–infant or mother–toddler groups, and counseling to increase awareness and understanding of the child's needs and to develop parenting skills; and (4) provisions for close monitoring of the progression of the patient's emotional and physical well-being. Sometimes, separating a child from the stressful home environment temporarily, as in hospitalization, allows the child to break out of the accustomed pattern. A neutral setting, such as the hospital, is the best place to start with families who are genuinely available emotionally and physically for intervention. If interventions are not feasible or inadequate or if they fail, we must consider placing the child with relatives or in foster care, adoption, or in a group home or residential treatment facility.

Epidemiology

Few data exist on the prevalence, sex ratio, or familial pattern of reactive attachment disorder and disinhibited social engagement disorder. Both disorders probably occur in less than 1 percent of the population. A study of 1,646 children aged 6 to 8 years old, living in a deprived sector of urban United Kingdom, found that the prevalence of reactive attachment disorder in this population was 1.4 percent. However, other studies of selected high-risk populations have estimated that about 10 percent of young children with documented neglectful and grossly pathologic caregiving exhibit reactive attachment disorder, and up to 20 percent of children in this situation exhibit

disinhibited social engagement disorder. In a retrospective report of children in one county of the United States who were removed from their homes because of neglect or abuse before the age of 4 years, 38 percent exhibited signs of either reactive attachment disorder or disinhibited social engagement disorder. Another study established the reliability of the diagnosis by reviewing videotaped assessments of at-risk children interacting with caregivers, along with a structured interview with the caregivers. Given that pathogenic care, including maltreatment, occurs more frequently in the presence of general psychosocial risk factors, such as poverty, disrupted families, and mental illness among caregivers, these circumstances are likely to increase the risk of reactive attachment disorder and disinhibited social engagement disorder.

Etiology

The core features of reactive attachment disorder and disinhibited social engagement disorder are disturbances of normal attachment behaviors. The inability of a young child to develop normative social interactions that culminate in aberrant attachment behaviors in reactive attachment disorder is inherent in the disorder's definition. Reactive attachment disorder and disinhibited social engagement disorder are presumably due to maltreatment of the child, including emotional neglect, physical abuse, or both. Grossly pathogenic care of an infant or young child by the caregiver presumably causes the markedly disturbed social relatedness that is evident. The emphasis is on the unidirectional cause; that is, the caregiver does something inimical or neglects to do something essential for the infant or child. In evaluating a patient for whom such a diagnosis is appropriate, however, clinicians should consider the contributions of each member of the caregiver–child dyad and their interactions. Clinicians should weigh such things as infant or child temperament, deficient or defective bonding, a developmentally disabled child, and a particular caregiver–child mismatch. The likelihood of neglect increases with a parental psychiatric disorder, substance abuse, intellectual disability, the parent's harsh upbringing, social isolation, deprivation, and premature parenthood (i.e., adolescent). These factors compromise parental ability to attend to the needs of the child, as the parents focus primarily on their existence. Frequent changes of the primary

caregivers, for example, from multiple foster care placements or repeated lengthy hospitalizations, may also lead to impaired attachment. In the general population, a study of 1,600 children found that those children with reactive attachment disorder/disinhibited social engagement disorder showed a constellation of symptoms characterized by early symptomatic syndromes eliciting neurodevelopmental clinical examinations (ESSENCE). Some of the associated symptoms in children with reactive attachment disorder/disinhibited social engagement disorder include a higher risk of failure to gain weight as neonates, feeding difficulty, and poor impulse control. These traits are likely to emerge because of both genetic and environmental factors. The authors found that children with reactive attachment disorder/disinhibited social engagement disorder were more likely to have multiple psychiatric comorbidities, lower IQs compared to the general population, and more behavioral problems. Thus, a broad assessment may be necessary to identify symptoms and disorders associated with reactive attachment disorder/disinhibited social engagement disorder.

POSTTRAUMATIC STRESS DISORDER OF INFANCY, CHILDHOOD, AND ADOLESCENCE

Posttraumatic stress disorder (PTSD), formerly grouped with anxiety disorders, now has its own chapter in DSM-5 along with other disorders caused by trauma or stress. Chapter 10 describes this disorder in detail. The purpose here is to focus on how the disorder presents in children.

In the United States, the rates of children and adolescents exposed to violence and traumatic events are incredibly high. In a nationally representative sample of children and adolescents, exposure to a traumatic event was 60 percent, with a lifetime rate ranging from 80 to 90 percent. Traumatic events can include physical or sexual abuse, domestic violence, motor vehicle accidents, severe medical illnesses, or natural or human-created disasters. A significant number of children and adolescents who experience traumatic events will develop PTSD. In children younger than the age of 6 years, spontaneous and intrusive memories may be expressed in their play, or occur in frightening dreams; it may be challenging to connect these intrusive thoughts to a traumatic event. The child may also display inexplicable agitation, fear, or disorganization.

Diagnosis and Clinical Features

For PTSD to ensue, exposure to a traumatic event consisting of either direct personal experience or witnessing an event involving the threat of death, serious injury, or severe harm must occur. The most common traumatic exposures for children and adolescents include physical or sexual abuse; domestic, school, or community violence; being kidnapped; terrorist attacks; motor vehicle or household accidents; or disasters, such as floods, hurricanes, tornadoes, fires, explosions, or airline crashes. A child with PTSD experiences either intrusive memories of the event, recurrent frightening dreams, dissociative reactions including flashbacks in which the child feels as if the traumatic event is recurring, or intense psychological distress when exposed to reminders of the trauma.

Symptoms of PTSD include *reexperiencing* the traumatic event in at least one of the following ways. Children may have intrusive thoughts, memories, or images that spontaneously recur, or body sensations that remind them of the event. In very young children, it is common to observe play that includes elements of the traumatic event, or behaviors, such as inappropriate sexual behaviors. Children may experience periods during which they either act or feel as though the event is taking place presently; this is a dissociative event usually described by adults as “flashbacks.”

Another critical symptom cluster of PTSD is *avoidance*, in which children may make active physical efforts to avoid the places, people, or situations that would present traumatic reminders of the event. The third cluster of diagnostic criteria for PTSD is negative alterations in cognition and mood following the trauma. In children 6 years or younger, according to DSM-5, negative alterations in cognition may take the form of socially withdrawn behavior, reduction of expressing positive emotions, diminished interest in play, and feelings of shame, fear, and confusion. In children older than 6 years of age, these may take the form of an inability to remember parts of a traumatic event, that is, *psychological amnesia* or persistent negative feelings about oneself, including horror, anger, guilt, or shame. After a traumatic event, children may experience a sense of detachment from their usual play activities (“psychological numbing”) or a diminished capacity to feel emotions. Older adolescents may express a fear that they expect to die young (sense of foreshortened future).

Other typical responses to traumatic events include symptoms of hyperarousal that were not present before the traumatic exposure, such as difficulty falling asleep or staying asleep, hypervigilance regarding safety, and increased checking that doors are locked or exaggerated startle reaction. In some children, hyperarousal can present as a generalized inability to relax with increased irritability, outbursts, and impaired ability to concentrate.

To meet the diagnostic criteria for PTSD, according to the DSM-5, the symptoms must be present for at least 1 month and cause distress and impairment in critical functional areas of life. When all of the diagnostic symptoms of PTSD are met following the traumatic event, persist for at least 3 days, but resolve within 1 month, we diagnose it as acute stress disorder. When the full syndrome of PTSD persists beyond 3 months, it is chronic PTSD. In some cases, the PTSD symptoms increase over time, and it is not until more than 6 months have elapsed after the exposure to the trauma that the whole syndrome emerges; in that case, the diagnosis is PTSD, delayed onset.

It is not uncommon for children and adolescents with PTSD to experience feelings of guilt, mainly if they have survived the trauma and others in the situation did not. They may blame themselves for the demise of the others and may go on to develop a comorbid depressive episode. Childhood PTSD is also associated with increased rates of other anxiety disorders, depressive episodes, substance use disorders, and attentional difficulties. DSM-5 includes a specifier *With dissociative symptoms,* which can present as either *Depersonalization,* in which there are recurrent experiences of feeling detached, as if outside of one's own body, or *Derealization*, in which the world feels unreal, dreamlike, and distant. A final specifier, *With delayed expression,* indicates that the full diagnostic criteria were not met until 6 months after the traumatic event, although some symptoms may present earlier.

Pathology and Laboratory Examination. Although reports indicate some alterations in both neurophysiologic and neuroimaging studies of children and adolescents with PTSD, no current laboratory tests can help in making this diagnosis.

Differential Diagnosis

Several overlapping symptoms exist between childhood PTSD and presentations of childhood anxiety disorders, such as separation anxiety disorder, OCD, or social anxiety disorder, in which recurrent intrusive thoughts or avoidant behaviors occur. Children with depressive disorders often exhibit withdrawal and a sense of isolation from peers as well as guilt about life events over which they have no control. Irritability, poor concentration, sleep disturbance, and decreased interest in usual activities occur in both PTSD and major depressive disorder.

Children who have lost a loved one in a traumatic event may go on to experience both PTSD and a major depressive disorder when bereavement persists beyond its expected course. Children with PTSD may also be confused with children who have disruptive behavior disorders because they often show poor concentration, inattention, and irritability. It is critical to elicit a history of traumatic exposure and evaluate the chronology of the trauma and the onset of the symptoms to make an accurate diagnosis of PTSD.

Course and Prognosis

For some children and adolescents with milder forms of PTSD, symptoms may persist for 1 to 2 years, after which they diminish and attenuate. In more severe circumstances, however, PTSD syndromes persist for many years or decades in children and adolescents, with spontaneous remission in only a portion of them.

The prognosis of untreated PTSD has become an issue of growing concern for researchers and clinicians who have documented a variety of severe comorbidities and psychobiological abnormalities associated with PTSD. In one study, children and adolescents with severe PTSD were at risk for decreased intracranial volume, diminished corpus callosum area, and lower IQs compared to children without PTSD. Children and adolescents with histories of physical and sexual abuse have higher rates of depression and suicidality themselves and in their offspring as well. This risk highlights the importance of early recognition and treatment of PTSD that may significantly improve the long-term outcome among youth.

Treatment

Trauma-Focused Cognitive-Behavioral Therapy. Randomized clinical trials have provided evidence for the efficacy of trauma-focused CBT in the treatment of PTSD in children and adolescents. We would administer this treatment over 10 to 16 treatment sessions. The treatment includes nine components itemized in the acronym PRACTICE. Trauma-focused CBT entails the inclusion of gradual exposure to feared stimuli as a critical element. Such stimuli encompass places, people, sounds, and situations. The first component of trauma-focused CBT is *Psychoeducation* regarding the nature of typical emotional and physiologic reactions to traumatic events and PTSD. Next, *Parenting Skills* involve sessions focused on guiding parents on providing praise, administering a time out, contingency reinforcement programs, and troubleshooting for specific symptoms in a given child. Component 3 is *Relaxation,* in which children learn to utilize muscle relaxation, focused breathing, affective modulation, thought-stopping, and other cognitive techniques to diminish feelings of helplessness and distress. Component 4 is *Affective Expression and Modulation,* geared to help children and their parents to identify their feelings, interrupt disturbing thoughts with positive imagery, and teach positive self-talk and social skills building. Component 5 is *Cognitive coping and processing,* which deals specifically with reviewing the Cognitive Triangle, in which the therapist and child explore the relationship between thoughts, feelings, and behaviors. The child would learn to challenge unhelpful thoughts with practice. In Component 6, *Trauma narrative,* the story of the traumatic event and its sequelae are developed over time by the child, with the therapist's support, using a depiction of words, art, or another creative form. Eventually, they share this with the parent. Component 7, *In Vivo Exposure and Mastery of Trauma Reminders,* is a session that reviews with the child how to deal with situations that are a reminder of the trauma and how to maintain control over distressing feelings associated with it. Component 8 is *Conjoint Child-Parent Sessions*; this component may involve several sessions in which the child and parent share their understanding of the process of the therapy and the gains that they have made. Finally, Component 10, *Enhancing future safety,* involves sessions that focus on the changes made in the family to ensure the safety of the child. These final sessions also promote healthy communication between the child and the parents.

A variant of trauma-focused CBT for PTSD is called *eye movement*

desensitization and reprocessing (EMDR), in which the therapist combines exposure and cognitive reprocessing interventions with directed eye movements. This technique is not as well accepted as the more extensive trauma-focused CBT detailed above.

Cognitive-Behavioral Intervention for Trauma in Schools (CBITS). CBITS is an intervention that administers treatment in the school setting for children who screen positive for PTSD and whose parents agree to treatment in school. It consists of ten weekly group sessions, one to three individual imaginal exposure sessions, two to four optional sessions with parents, and one parent education session. Similar to trauma-focused CBT (TF-CBT), CBITS incorporates psychoeducation, relaxation training, cognitive coping skills, gradual exposure to traumatic memories through a narrative, in vivo exposure, and affect modulation, cognitive restructuring, and social problem-solving. In one RCT, 86 percent of students in the CBITS group reported significantly decreased PTSD symptoms compared to the waitlist controls. Students who received CBITS also reported lower depression scores. Among parents whose children received CBITS treatment, 78 percent reported decreased psychosocial problems in their children. After CBITS treatment, the children sustained improvements in both the PTSD and depression symptoms at 6 months.

Structured Psychotherapy for Adolescents Responding to Chronic Stress (SPARCS). SPARCS consists of a group intervention, generally administered in 16 sessions, with a focus on the needs of adolescents between the ages of 12 and 19 years who have lived with chronic trauma and may also carry a diagnosis of PTSD. Investigators have tested SPARCS in a trial of multicultural teens and young adults with moderate or severe trauma exposure. Most of the participants were female and comprised multiple ethnic groups: 67 percent African American; 12 percent Latino; 21 percent Caucasian. SPARCS demonstrated efficacy in reducing traumatic stress symptoms, mainly in the largest group, the African American group. SPARCS utilizes cognitive-behavioral techniques and also incorporates many of the components of TF-CBT. Also, SPARCS includes mindfulness techniques and relaxation.

Trauma Affect Regulation: Guide for Education and Therapy (TARGET). TARGET, an affect regulation therapy, combines CBT components, such as cognitive processing, with affect modulation. We would usually consider this for adolescents between the ages of 13 and 19 who experience maltreatment or chronic traumatic exposure to such things as community violence or domestic violence. It usually requires 12 sessions, which focus on past or current situations. As with SPARCS treatment, gradual exposure may occur in the context of recounting past trauma but is not a core component of the treatment. A randomized trial with 59 delinquent girls aged 13 to 17 years who met full or partial criteria for PTSD found that TARGET reduced anxiety, anger, depression, and PTSD cognitions. TARGET is a promising treatment for girls with histories of delinquency, especially to reduce anger and to enhance optimism and self-efficacy.

Psychopharmacological Treatment. Clinicians have used several pharmacologic agents to treat children and adolescents with PTSD, often focused on diminishing intrusive thoughts, hyperarousal, and avoidance, with some success and mixed results. Given the frequent comorbidity of depressive disorder, anxiety disorders, and behavioral problems associated with PTSD, we might use a multitude of psychopharmacological agents.

Psychiatrists often use antidepressant agents as adjuncts to psychosocial treatments in youth with PTSD. Although sertraline and paroxetine are approved by the FDA in the treatment of PTSD in adults, there is scant evidence to support their use for the core symptoms of PTSD in youth. An RCT of TF-CBT plus sertraline compared to TF-CBT plus placebo in 24 children with PTSD found that both groups had a significant reduction in PTSD symptoms, with no significant difference between the groups. One multicenter placebo-controlled study of 131 children aged 6 to 17 years with PTSD tested sertraline over 10 weeks. Results showed sertraline to be a safe treatment; however, it was not effective when compared to placebo. An RCT using citalopram did not show the superiority of citalopram over placebo in the treatment of core PTSD symptoms. There is, however, evidence suggesting that the use of SSRIs in traumatized children with burns may be preventive regarding the development of PTSD. Published literature demonstrates that up to 50 percent of children with moderate to severe burns develop PTSD. Thus preventive strategies are essential. A randomized

controlled study of sertraline to prevent PTSD found that children who received sertraline, flexibly dosed between 25 and 150 mg/day, had a decrease in parent-reported symptoms of PTSD over 8 weeks compared to a placebo group. Among the child-reported symptoms, however, there was no significant difference between the two groups.

Psychiatrists sometimes use antiadrenergic agents to treat dysregulation of the noradrenergic system in adults and youth with PTSD. α_2-Agonists such as clonidine and guanfacine, for example, have been used to decrease norepinephrine release, whereas centrally acting β-antagonists such as propranolol, and α_1-antagonists such as prazosin, are hypothesized to improve hyperarousal and intrusive thoughts through attenuation of norepinephrine postsynaptically. Although there are some data in adults with PTSD to support the use of these agents, there are mainly case reports for youths. There is a suggestion that guanfacine may reduce nightmares in children with PTSD and that clonidine may diminish symptoms of reenactment of traumatic events in children. One report of propranolol treatment in 11 pediatric patients with PTSD from sexual or physical abuse with a mean age of 8.5 years, who exhibited agitation and hyperarousal, indicated some decrease in symptoms in 8 of the 11 children studied. Another open study of transdermal clonidine treatment of preschoolers with PTSD suggests that clonidine may be efficacious in this population in decreasing activation and hyperarousal. An additional open trial of oral clonidine with dosage ranges of 0.05 to 0.1 mg twice daily similarly suggests that this medication may provide some relief for the symptoms of hyperarousal, impulsivity, and agitation in young children with PTSD.

Second-generation antipsychotics such as risperidone, olanzapine, quetiapine, ziprasidone, and aripiprazole have been studied in adults with PTSD with mixed results. Risperidone and aripiprazole both have FDA approval for use in children and adolescents with aggression, severe behavioral dyscontrol, and severe psychiatric disorders; however, there are no controlled trials with children with PTSD. For example, there is one case series of three preschool-age children who exhibited symptoms of acute stress disorder and who had severe thermal burns improved after being treated with risperidone.

In children and adolescents with PTSD, there is one open-label trial of

carbamazepine and one trial of sodium valproate. In the carbamazepine trial, all 28 improved at blood levels of 10 to 11.5 micrograms/mL. In the sodium valproate trial, there was some improvement at higher doses of the drug.

Clinicians also use benzodiazepines to treat anxiety symptoms in patients with PTSD, although there are no controlled trials to support their use in youth with PTSD.

Given that many children and adolescents with PTSD have comorbid depressive and anxiety disorders, we would recommend SSRIs as a first-line pharmacologic option.

Epidemiology

Epidemiologic studies of children 9 to 17 years of age have found 3-month prevalence rates of PTSD ranging from 0.5 to 4 percent. An epidemiologic survey of preschoolers aged 4 to 5 years found a rate of 1.3 percent of PTSD.

Children exposed chronically to trauma, such as child abuse, or traumas resulting in a broader disruption of entire communities, such as war, have the highest risk of developing PTSD. In addition to the staggering rate of the full-blown disorder of PTSD among youth, several studies indicate that most children exposed to severe or chronic trauma develop PTSD symptoms sufficiently severe to disrupt functioning, even in the absence of the full diagnosis.

Etiology

Biologic Factors. Risk factors in children for developing PTSD include preexisting anxiety disorders and depressive disorders. A prospective study found that among children exposed to traumatic events, those with anxiety disorders and teacher ratings of externalizing behavior problems by the age of 6 years were at increased risk for PTSD. Furthermore, children with an IQ higher than 115 at age 6 years were at lower risk for developing PTSD. Also, among children exposed to trauma, those who developed PTSD were at higher risk of developing comorbid disorders such as depression. This comorbidity suggests that a genetic predisposition for anxiety disorders, as well as a family history indicating an increased risk of depressive disorders, may predispose a trauma-exposed child to develop PTSD. Children with

PTSD exhibit increased excretion of adrenergic and dopaminergic metabolites, smaller intracranial volume and corpus callosum, memory deficits, and lower IQs compared with age-matched controls. Adults with PTSD have been found to have an overactive amygdala and decreased hippocampal volume. Whether the above findings are sequelae of PTSD or markers of vulnerability to the disorder remains a focus of investigation.

Psychological Factors. Although the exposure to trauma is the initial etiologic factor in the development of PTSD, the enduring symptoms typical of PTSD, such as avoidance of the place where the trauma occurred, can be conceptualized, in part, as the result of both classical and operant conditioning. Extreme physiologic responses may accompany the fear of a given traumatic event, such as an adolescent who was terrorized by an attack by a group of students near a school, who then develops an extreme adverse physiologic reaction each time he or she is near the school. This reaction is an example of classical conditioning in that a neutral cue (the school) has become paired with an intensely fearful past event. Operant conditioning occurs when a child learns to avoid traumatic reminders to prevent distressing feelings from arising. For example, if a child was in a motor vehicle accident, the child may then refuse to ride in cars altogether to prevent adverse physiologic reactions and fear from occurring.

Another mechanism in developing and maintaining symptoms of PTSD is through modeling, which is a form of learning. For example, when parents and children experience traumatic events, such as natural disasters, children may emulate parental responses, such as avoidance, withdrawal, or extreme expressions of fear, and “learn” to respond to their memories of the traumatic event in the same manner.

Social Factors. Family support and reactions to traumatic events in children may play a significant role in the development of PTSD in that adverse parental emotional reactions to a child’s abuse may increase that child’s risk of developing PTSD. Lack of parental support and psychopathology among parents—especially maternal depression—have been identified as risk factors in the development of PTSD after a child experiences a traumatic event.

▲ 2.9 Depressive Disorders and Suicide in Children and Adolescents

Chapter 7 discusses depressive disorders in detail. This section focuses on the presentation of this disorder in youths.

Depressive disorders in youth represent a significant public health concern, in that they are prevalent and result in long-term adverse effects on the individual's cognitive, social, and psychological development. These disorders affect approximately 2 to 3 percent of children and up to 8 percent of adolescents, so the need for early identification and access to evidence-based interventions such as CBTs and antidepressant agents, is essential. Although significant depression runs in families, with the highest risk in children whose parents experienced early-onset depression, twin studies have demonstrated that major depression is only moderately heritable, approximately 40 to 50 percent, highlighting environmental stressors and adverse events as significant contributors to major depressive disorder in youth. The core features of major depression in children, adolescents, and adults bear a striking resemblance; however, the developmental level of the child or adolescent influences the clinical presentation. The DSM-5 uses the same criteria for major depressive disorder in youth as in adults, except that for children and adolescents, *irritable mood* may replace a *depressed mood* in the diagnostic criteria.

Most children and adolescents with depressive disorders neither attempt nor complete suicide; however, severely depressed youth often have suicidal ideation, and suicide remains the most severe risk of major depression. Nevertheless, many depressed youths do not ever have suicidal ideation, and many children and adolescents who engage in suicidal behavior do not have a depressive disorder. There is epidemiologic evidence to suggest that depressed youth with recurrent active suicidal ideation, including a plan, and who have made prior attempts, are at higher risk to complete suicide, compared to youth who express only passive suicidal ideation.

Mood disorders in children and adolescents have been studied increasingly over the last two decades, culminating in large sample multisite RCTs such as the Treatment of Adolescent Depression Study (TADS), which provides

evidence of the efficacy of both CBT as well as SSRIs. Furthermore, when the preceding modalities are combined, the highest efficacy is achieved. Increased recognition of depressive disorders in preschool populations has sparked clinicians and researchers to develop psychosocial interventions such as the Parent–Child Interaction Therapy Emotion Development (PCIT-ED), which target treatment specifically for this age group. The expression of disturbed and depressed mood appears to vary with the developmental stage. Very young children with major depression are sad, listless, or apathetic, even though they may not articulate these feelings verbally. Perhaps surprisingly, mood-congruent auditory hallucinations occur in young children with major depression. Somatic complaints such as headaches and stomachaches, withdrawn and sad appearance, and poor self-esteem are more universal symptoms. Patients in late adolescence with more severe forms of depression often display pervasive anhedonia, severe psychomotor retardation, delusions, and a sense of hopelessness. Symptoms that appear with the same frequency, regardless of age and developmental status, include suicidal ideation, depressed or irritable mood, insomnia, and diminished ability to concentrate.

Developmental issues, however, influence the expression of depressive symptoms. For example, unhappy young children who exhibit recurrent suicidal ideation are rarely able to propose a realistic suicide plan or to carry out such a plan. Children’s moods are especially vulnerable to the influences of severe social stressors, such as chronic family discord, abuse and neglect, and academic failure. Many young children with major depressive disorder have histories of abuse, neglect, and families with significant psychosocial burdens such as parental mental illness, substance abuse, or poverty. Children who develop depressive disorders amid acute toxic family stressors may have remission of depressive symptoms when the stressors diminish or when in a more nurturing family environment. Depressive disorders are generally episodic, albeit typically lasting close to a year; however, their onset may be insidious and remain unidentified until significant impairment in peer relationships, deterioration in academic function, or withdrawal from activities emerges. ADHD, oppositional defiant disorder, and conduct disorder are not infrequently comorbid with a major depressive episode. In some cases, conduct disturbances or disorders occur in the context of a major depressive episode and resolve with the resolution of the depressive episode.

Clinicians must clarify the chronology of the symptoms to determine whether a given behavior (e.g., poor concentration, defiance, or temper tantrums) was present before the depressive episode and is unrelated to it or whether the behavior is occurring for the first time and is related to the depressive episode.

DIAGNOSIS AND CLINICAL FEATURES

Major Depressive Disorder

It is easiest to diagnose major depressive disorder in children when the disorder is acute and occurs in a child without previous psychiatric symptoms. Often, however, the onset is insidious, and the disorder occurs in a child who has had several years of difficulties with hyperactivity, separation anxiety disorder, or intermittent depressive symptoms.

A major depressive episode in a prepubertal child is likely to be manifest by somatic complaints, psychomotor agitation, and mood-congruent hallucinations. Anhedonia is also frequent, but anhedonia, as well as hopelessness, psychomotor retardation, and delusions, are more common in adolescent and adult major depressive episodes than in those of young children. Adults have more problems than depressed children and adolescents with sleep and appetite. In adolescence, negativistic or frankly antisocial behavior and the use of alcohol or illicit substances can occur and may justify the additional diagnoses of oppositional defiant disorder, conduct disorder, and substance use disorder. Feelings of restlessness, irritability, aggression, reluctance to cooperate in family ventures, withdrawal from social activities, and isolation from peers often occur in adolescents. School difficulties are likely. Depressed adolescents may become less attentive to personal appearance and show increased sensitivity to rejection by peers and in romantic relationships.

Children can be reliable reporters about their emotions, relationships, and difficulties in psychosocial functions. They may, however, refer to depressive feelings in terms of anger, or feeling “mad” rather than sad. Clinicians should assess the duration and periodicity of the depressive mood to differentiate relatively universal, short-lived, and sometimes frequent periods of sadness, usually after a frustrating event, from a valid, persistent depressive mood.

The younger the child, the more imprecise his or her time estimates are likely to be.

Mood disorders tend to be chronic if they begin early. Childhood-onset may be the most severe form of mood disorder and tends to appear in families with a high incidence of mood disorders and alcohol use disorder. The children are likely to have such secondary complications as conduct disorder, alcohol and other substance use disorders, and antisocial behavior. Functional impairment associated with a depressive disorder in childhood extends to practically all areas of a child's psychosocial world; school performance and behavior, peer relationships, and family relationships all suffer. Only highly intelligent and academically oriented children with no more than a moderate depression can compensate for their difficulties in learning by substantially increasing their time and effort. Otherwise, school performance is invariably affected by a combination of difficulty concentrating, slowed thinking, lack of interest and motivation, fatigue, sleepiness, depressive ruminations, and preoccupations. Sometimes a clinician may misdiagnose a learning disorder as major depression. Learning problems secondary to depression, even when longstanding, are corrected rapidly after a child's recovery from the depressive episode.

Children and adolescents with severe forms of major depressive disorder may have hallucinations or delusions. Usually, these psychotic symptoms are thematically consistent with the depressed mood, occur with the depressive episode (usually at its worst), and do not include certain types of hallucinations (such as conversing voices and a commenting voice, which are specific to schizophrenia). Depressive hallucinations usually consist of a single voice speaking to the person from outside his or her head, with derogatory or suicidal content. Depressive delusions center on themes of guilt, physical disease, death, nihilism, deserved punishment, personal inadequacy, and (sometimes) persecution. These delusions are rare in prepuberty, probably because of cognitive immaturity, but are present in about half of psychotically depressed adolescents.

Alcohol or drug use can complicate adolescent depression. One study found that up to 17 percent of adolescents with depressive disorder received an initial evaluation due to substance abuse.

Persistent Depressive Disorder (Dysthymia)

In children and adolescents, persistent depressive disorder consists of a depressed or irritable mood for most of the day, for more days than not, for at least 1 year. DSM-5 notes that in children and adolescents, irritable mood can replace the depressed mood criterion for adults and that the duration criterion is not 2 years but 1 year for children and adolescents. According to the DSM-5 diagnostic criteria, two or more of the following symptoms must accompany the depressed or irritable mood: low self-esteem, hopelessness, poor appetite or overeating, insomnia or hypersomnia, low energy or fatigue, or poor concentration or difficulty making decisions. During the year of the disturbance, these symptoms do not resolve for more than 2 months at a time. Also, the diagnostic criteria for dysthymic disorder specify that during the first year, no major depressive episode emerges. To meet the DSM-5 diagnostic criteria for persistent depressive disorder, a child must not have a history of a manic or hypomanic episode. Persistent depressive disorder is also not diagnosed if the symptoms occur exclusively during a chronic psychotic disorder or if they are the direct effects of a substance or a general medical condition. DSM-5 provides specifiers for early-onset (before 21 years of age) or late-onset (after 21 years of age).

A child or adolescent with persistent depressive disorder may have had a major depressive episode before developing persistent depressive disorder; however, it is much more common for a child with persistent depressive disorder for more than 1 year to develop a concurrent episode of major depressive disorder. In this case, both depressive diagnoses apply (double depression). Persistent depressive disorder in youth is known to have an average age of onset that is several years earlier than the typical onset of major depressive disorder. Occasionally, youth fulfill the criteria for persistent depressive disorder, except that their episode does not last for a whole year, or they experience remission from symptoms for more than 2 months. These mood presentations in youth may predict additional mood disorder episodes in the future. Current knowledge suggests that the longer, more recurrent, and less directly related to social stress these episodes are, the higher the likelihood of future severe mood disorder. When minor depressive episodes follow a significant stressful life event by less than 3 months, it is more likely to be an adjustment disorder.

Bereavement

Bereavement is a state of grief related to the death of a loved one, which presents with an overlap of symptoms characteristic of a major depressive episode. Typical depressive symptoms associated with bereavement include feelings of sadness, insomnia, diminished appetite, and, in some cases, weight loss. Grieving children may become withdrawn and appear sad and avoid even their favorite activities.

In DSM-5, bereavement is not a mental disorder; however, uncomplicated bereavement is a condition that may be a focus of clinical attention. Children during a typical bereavement period may also meet the criteria for major depressive disorder. Symptoms indicating major depressive disorder exceeding typical bereavement include intense guilt related to issues beyond those surrounding the death of the loved one, preoccupation with death other than thoughts about being dead to be with the deceased person, morbid preoccupation with worthlessness, marked psychomotor retardation, prolonged severe functional impairment, and hallucinations other than transient perceptions of the voice of the deceased person.

The duration of bereavement varies; in children, the duration may depend partly on the support system in place. For example, a child may feel devastated and abandoned if they have to leave their home after the death of an only parent. Children who lose loved ones may feel a sense of guilt that death may have occurred because they were “bad” or did not perform as expected.

Ryan was a 12-year-old seventh grader in middle school who was brought to the emergency room in handcuffs by police after walking into oncoming traffic right after school. Ryan walked in front of a city bus; the driver began honking at the boy who kept ambling into the traffic. Two police stationed in their car across the street from the school heard the bus honking and noticed Ryan and confronted him. The police were about to issue the boy a citation for crossing against the red light; however, when they inquired as to why he had crossed against the traffic light, he informed them that he was trying to kill himself. The police handcuffed Ryan, placed him in the police car without a struggle, and brought him to the local hospital’s emergency room. Ryan’s mother was contacted and met her son in the emergency room. Ryan was physically

intact, without injury, and a team of child psychiatrists evaluated him. When asked what had happened, Ryan became tearful. He said he walked in front of the bus in the hope of being hit and killed. Ryan reported that numerous peers had bullied him over the last 2 years because he is short and overweight. Ryan reported that on this day, a girl in his class had pushed him down and started hitting him and laughing at him. Ryan reported that peer teased and assaulted him and called him stupid and fat. Ryan has some friends who usually defend him, but on this day, his friends were not close by, and he became desperate. Ryan disclosed, however, that even before this day, he has been sad for the last 2 years and thought about suicide recurrently over the last year.

Ryan was a relatively good student, earning good grades, especially in math, although he was failing history. Upon a separate interview with Ryan's mother, she did not know of any of Ryan's problems. She felt the team was mistaken and that she should take Ryan home.

Ryan had previously seen a counselor in school a few times last year, but the issues of depression or suicide did not come up. Ryan has an older brother and a younger brother who are well adjusted. When interviewing Ryan and his mother together, Ryan was able with some encouragement to let his mother know how depressed, hopeless, and suicidal he felt, and why. Ryan's mother burst into tears, and Ryan tried to comfort his mother.

Ryan was placed on a 72-hour hold and referred to a children's psychiatric inpatient unit for further evaluation and treatment. There the team treated him with an SSRI antidepressant as well as psychoeducation and family sessions. He and his family continued treatment after his hospitalization.

Pathology and Laboratory Examination

No laboratory test is useful for diagnosis. If a child or adolescent also complains of symptoms of hypothyroidism, that is, dry skin, coldness, lethargy, for example, then a screening test for thyroid function may be indicated.

Rating scales for depressive symptoms administered by us to the child and parent may be helpful in the evaluation. The Children's Depression Rating

Scale-Revised (CDRS-R) is a 17-item instrument administered by the clinician separately to the parent and child or adolescent. The clinician scores a rating for each item using the information from both the parent and the child. The scale assesses affective, somatic, cognitive, and psychomotor symptoms. A cumulative score of 40 is a marker for moderate depression and a score of 45 or higher for significant depression.

DIFFERENTIAL DIAGNOSIS

A substance-induced mood disorder may be challenging to differentiate from other mood disorders until detoxification occurs. Anxiety symptoms and disorders often coexist with depressive disorders. Of particular importance in the differential diagnosis is the distinction between agitated depressive or manic episodes and ADHD, in which the persistent excessive activity and restlessness are confusing. Prepubertal children generally do not show classic forms of agitated depression, such as hand-wringing and pacing. Instead, an inability to sit still, irritability, and frequent temper tantrums are the most common symptoms. Sometimes, the correct diagnosis becomes evident only after remission of the depressive episode.

COURSE AND PROGNOSIS

The course and prognosis of major depression in youths depend on the severity of illness. Also important is the rapidity of interventions, and the degree of response to the interventions. In general, 90 percent of youth recover from a first episode of moderate to severe major depressive disorder within 1 to 2 years. The age of onset, episode severity, and the presence of comorbid disorders also influence course and prognosis. In general, the younger the age of onset, the higher the recurrence of multiple episodes, and the presence of comorbid disorders predict a poorer prognosis. The mean length of an untreated episode of major depression in children and adolescents is about 8 to 12 months; the cumulative probability of recurrence is 20 to 60 percent within 2 years and 70 percent by 5 years. The most significant risk for relapse is in the 6 months to 1 year after discontinuing treatment. Depressed children who live in families with high levels of chronic conflict are more likely to have relapses. The relapse rate for childhood

depression into adulthood is also high. In a community sample, 45 percent of adolescents with a history of major depression developed another episode of major depression in early adulthood.

Youth with major depression are at higher risk for the development of a future bipolar disorder, compared to adults. Overall estimates of children with an episode of major depression developing bipolar disorder are about 20 to 40 percent. Clinical characteristics of a depressive episode in youth suggesting the highest risk of developing bipolar I disorder include hallucinations and delusions, psychomotor retardation, and a family history of bipolar illness. In a longitudinal study of prepubertal children with major depression, 33 percent developed bipolar I disorder, whereas 48 percent went on to develop bipolar II or bipolar disorder not otherwise specified by early adulthood.

Depressive disorders are associated with short- and long-term peer relationship difficulties and complications, compromised academic achievement, and persistently low self-esteem. Persistent depressive disorder has an even more protracted recovery than major depressive disorder; the mean episode length is about 4 years. Early-onset persistent depressive disorder is associated with significant risks of comorbidity with major depressive disorder (70 percent), bipolar disorder (13 percent), and future substance abuse (15 percent). The risk of suicide, which accounts for about 12 percent of adolescent mortalities, is significant among adolescents with depressive disorders.

TREATMENT

The AACAP practice parameters, as well as a consensus of experts who developed the Texas Children's Medication Algorithm Project (TMAP), made evidence-based recommendations for the treatment of children and adolescents with depressive disorders. These include psychoeducation and supportive interventions for youth with mild forms of depression. For youth with moderate to severe depression or recurrent episodes of major depression with significant impairment and with active suicidal thoughts or behaviors or psychosis, the optimal intervention includes both pharmacotherapy and CBT. CBT or interpersonal therapy (IPT) alone may be sufficient for moderate depression, especially when continuing treatment for 6 months or longer.

Psychiatric Hospitalization

We should assess for suicidal thoughts, behaviors, and a history of suicidal behavior in every child or adolescent with major depression. Safety is the most immediate consideration in assessing depression in youth, that is, a determination as to whether immediate psychiatric hospitalization is necessary. Depressed children and adolescents who express suicidal thoughts or behaviors most often require some extended evaluation in the safety of the psychiatric hospital to provide maximal protection from self-destructive impulses and behaviors.

Evidence-Based Treatment Studies

The Treatment for Adolescents with Depression Study (TADS) divided 439 adolescents between 12 and 17 years of age into three treatment groups of 12 weeks, composed of either fluoxetine alone (10 to 40 mg/day), fluoxetine with the same dose range in combination with CBT, or CBT alone. Based on ratings of the *Children's Depression Rating Scale-Revised* (CDRS-R), combination treatment had significantly superior response rates compared with either treatment alone. Based on CGI scores at 12 weeks, rates of much or very much improved were 71 percent for the combined treatment group, 60.6 percent for the fluoxetine group, 43.2 percent for the CBT alone group, and 34.4 percent for the placebo group. At 12 weeks, combination treatment was the optimal strategy in the treatment of adolescent depression. By the end of 9 months of treatment, however, response rates for each group had converged, so that response for the combination group was 86 percent, fluoxetine group response was about 81 percent, and CBT-alone group response rate was 81 percent. The long-term effectiveness of treatments for adolescent depression demonstrates that for moderately ill adolescents, fluoxetine, CBT, or the combination is efficacious. However, the addition of CBT to fluoxetine decreased persistent suicidal ideation and potential treatment-related emergence of suicidal ideation.

A second large multicenter randomized placebo-controlled trial, Treatment of SSRI-Resistant Depression in Adolescents (TORDIA), included adolescents with major depression who had not responded to a 2-month trial with an SSRI antidepressant. The investigators in this study randomly assigned 334 adolescents between 12 and 18 years of age to a different SSRI

agent (either citalopram, paroxetine, citalopram, or another antidepressant class, venlafaxine) with or without concurrent CBT. The SSRI plus CBT group and the venlafaxine plus CBT group had higher response rates of improvement (54.8 percent) than the group on medications alone (40.5 percent). There were no differences found in the response rates between antidepressant agents.

Psychosocial Interventions

CBT is an efficacious intervention for the treatment of moderately severe depression in children and adolescents. CBT aims to challenge maladaptive beliefs and enhance problem-solving abilities and social competence. A review of controlled cognitive-behavioral studies in children and adolescents revealed that, as with adults, both children and adolescents showed consistent improvement with these methods. Other "active" treatments, including relaxation techniques, were also shown to be helpful as an adjunctive treatment for mild to moderate depression. Findings from one large controlled study comparing cognitive-behavioral interventions with nondirective supportive psychotherapy and systemic behavioral family therapy showed that 70 percent of adolescents had some improvement with each of the interventions; cognitive-behavioral intervention had the most rapid effect. Another controlled study comparing a brief course of CBT with relaxation therapy favored the cognitive-behavioral intervention. At a 3- to 6-month follow-up, however, no significant differences existed between the two treatment groups. This effect resulted from relapse in the cognitive-behavioral group, along with continued recovery in some patients in the relaxation group. Factors that seem to interfere with treatment responsiveness include the presence of comorbid anxiety disorder that probably was present before the depressive episode. Longer-term CBT is efficacious in the treatment of depression and has the advantage of mitigating suicidal ideation.

Interpersonal psychotherapy (IPT) focuses on improving depression through a focus on ways in which depression interferes with interpersonal relationships and overcoming these challenges. The four main areas of focus with interpersonal psychotherapy include loss, interpersonal disputes, role transition, and interpersonal deficits. A modification of interpersonal therapy to more specifically address depression for adolescents (IPT-A) includes a

focus on separation from parents, authority figures, peer pressures, and dyadic relationships. IPT-A has been studied on an outpatient basis as well as in a school-based clinic setting. A 12-week study of 48 adolescents with major depression randomly assigned to IPT-A or clinical monitoring found that the group receiving IPT-A showed decreased depressive symptoms, increased social functioning, and improved problem solving compared to the other group. In the school-based health clinic, depressed adolescents were randomly assigned to IPT-A or treatment as usual for 16 weeks. Trained clinic staff administered the treatment. At the end of 16 weeks, those adolescents receiving IPT-A had more significant symptom reduction and improved overall functioning, especially older and more severely depressed adolescents seemed to benefit most significantly.

PCIT-ED for preschool depression, a modification of PCIT historically used in the treatment of disruptive disorders for children, was piloted in an RCT for 54 depressed preschoolers. Fifty-four depressed young children from ages 3 to 7 years received either PCIT-ED or psychoeducation with their caregivers. PCIT-ED was manualized and consisted of three modules conducted over 14 sessions in 12 weeks. The core modules of PCIT—Child-Directed Interaction (CDI) and Parent-Directed Interaction (PDI)—were utilized and limited to four sessions each. The focus of these modules is to strengthen the parent–child relationship by coaching parents in positive play techniques, giving effective directives to the child, and responding to disruptive behavior in firm but not punitive ways. The novel portion of the treatment targeting the preschool depression consisted of a 6-week Emotion Development (ED) module, which focused on helping the parent to be a more effective emotion guide and affect regulator for the child. As part of the ED module, the parent learns to accurately recognize their emotions as well as the child’s and serves to help regulate the child’s emotions. The team developed a psychoeducation control condition, Developmental Education and Parenting Intervention (DEPI), for parents using small group sessions. The DEPI condition was designed to educate parents about child development and emphasized emotional and social development without individual coaching or practice with behavioral techniques as provided in the PCIT-ED group. Primary outcome measures included parent’s report of the child’s symptoms of depression using a structured instrument, the Preschool Age Psychiatric Assessment (PAPA), and the investigators measured

depression severity pretreatment and posttreatment using parent ratings on the Preschool Reelings Checklist-Scale Version (PFC-S), a 20-item checklist. Results revealed that both groups showed significant improvement, with particular improvement in the PCIT-ED group concerning emotion recognition, child executive functioning, and parenting stress. This pilot study indicates that PCIT-ED is a promising novel intervention for preschool depression that deserves further investigation.

Pharmacotherapy

Fluoxetine and escitalopram have FDA approval in the treatment of major depression in adolescents. Three RCTs of fluoxetine in depressed children and adolescents demonstrate its efficacy. The common side effects are similar as for adults: headache, gastrointestinal symptoms, sedation, and insomnia.

Short-term randomized clinical trials have demonstrated the efficacy of citalopram and sertraline compared with placebo in the treatment of major depression in children and adolescents. Sertraline was efficacious in two multicenter, double-blind, placebo-controlled trials of 376 children and adolescents. The doses of sertraline ranged from 50 to 200 mg a day or placebo. Nearly 70 percent of the subjects had a greater than 40 percent decrease in depression rating scale scores (compared with 56 percent in the placebo group). The most common side effects are anorexia, vomiting, diarrhea, and agitation.

Citalopram has been demonstrated in one RCT in the United States to be efficacious in 174 children and adolescents treated with citalopram at doses of 20 to 40 mg a day or placebo for 8 weeks. Significantly more of the group on citalopram showed improvement compared with placebo on the depression rating scale (CDRS-R). A significantly increased response rate (response defined as less than 28 on CDRS-R) of 35 percent occurred in the citalopram group, compared with 24 percent of the placebo group. Common side effects that emerged included headache, nausea, insomnia, rhinitis, abdominal pain, dizziness, fatigue, and flu-like symptoms.

RCTs to date that have not shown efficacy on primary outcome measures include those using mirtazapine and tricyclic antidepressants. A meta-analysis of SSRI trials in depressed children and adolescents found the efficacy of SSRIs compared to placebo with an average response rate of 60

percent for the SSRI compared to 49 percent for placebo.

Starting doses of SSRIs for prepubertal children are lower than doses recommended for adults, but adolescents usually use the same doses as adults.

A potential side effect of SSRIs in depressed children is behavioral activation or induction of hypomanic symptoms. We should discontinue the medication in those cases to see if the symptoms resolve. Activation due to SSRIs, however, does not necessarily predict a diagnosis of bipolar disorder.

Venlafaxine was effective in the TORDIA study; however, adverse effects, including increased blood pressure, have made this agent a second-line choice compared to the SSRIs.

Tricyclic antidepressants lack good data and have significant cardiac risks, including arrhythmias, and we do not recommend them.

FDA Warning and Suicidality. In September 2004, the FDA received information from their Psychopharmacologic Drug and Pediatric Advisory Committee indicating, based on their review of reported suicidal thoughts and behavior among depressed children and adolescents who participated in randomized clinical trials with nine different antidepressants, an increased risk of suicidality in those children who were on active antidepressant medications. Although there were no suicides, the rates of suicidal thinking and behaviors were 2 percent for patients on placebo, versus 4 percent among patients on antidepressant medications. The FDA, following the recommendation of their advisory committees, instituted a “black-box” warning to the health professional label of all antidepressant medication indicating the increased risk of suicidal thoughts and behaviors in children and adolescents, and the need for close monitoring for these symptoms. Several reviews since 2004, however, concluded that the data do not indicate a significant increase in the risk of suicide or serious suicide attempts after starting treatment with antidepressant drugs.

Duration of Treatment. Based on available longitudinal data and the natural history of major depression in children and adolescents, current recommendations include maintaining antidepressant treatment for 1 year in a depressed child who has achieved a good response, and to then discontinue the medication at a time of relatively low stress for a medication-free period.

Pharmacologic Treatment Strategies for Resistant Depression. Pharmacologic recommendations, per an expert consensus panel that developed the TMAP, as well as the Treatment of SSRI-resistant Adolescents with Depression Study (TORDIA), in the treatment of children or adolescents who have not responded to treatment with an SSRI agent, is to change to another SSRI medication. If a child is not responsive to the second SSRI medication, then either a combination of antidepressants or augmentation strategies may be reasonable choices as well as an antidepressant from another class of medications.

Electroconvulsive Therapy

ECT is rarely used for adolescents, although published case reports indicate its efficacy in adolescents with depression and mania. Currently, case reports suggest that ECT may be a relatively safe and useful treatment for adolescents who have persistent severe affective disorders, particularly with psychotic features, catatonic symptoms, or persistent suicidality.

EPIDEMIOLOGY

Depressive disorders increase in frequency with increasing age in the general population. Mood disorders among preschool-age children are estimated to occur in about 0.3 percent of community samples, and 0.9 percent in clinic settings. The prevalence of major depression in school-age children is 2 to 3 percent. Depression in referred samples of school-age children occurs at the same frequency in boys as in girls, with some surveys indicating a slightly increased rate among boys. In adolescents, the prevalence rate of major depression is from 4 to 8 percent and two to three times more likely in females than males. By age 18 years, the cumulative incidence of major depression is 20 percent. Children with a family history of major depression in a first-degree relative are about three times more likely to develop the disorder than in those without family histories of affective disorders. The prevalence of persistent depressive disorder in children ranges from 0.6 to 4.6 percent and in adolescence increases to 1.6 to 8 percent. Children and adolescents with persistent depressive disorder have a high likelihood of developing major depressive disorder at some point after 1 year of persistent depressive disorder. The rate of developing a major depression on top of

persistent depressive disorder (double depression) within 6 months of persistent depressive disorder is estimated to be about 9.9 percent.

Among psychiatrically hospitalized children and adolescents, the rates of major depressive disorder are close to 20 percent for children and 40 percent for adolescents.

ETIOLOGY

Considerable evidence indicates that major depression in youth is the same fundamental disorder experienced by adults and that its neurobiology is likely to be an interaction of genetic vulnerability and environmental stressors. The etiology of the depressive disorders is discussed in Chapter 7.

SUICIDE

In the United States, suicide is the third leading cause of death among adolescents, after accidental death and homicide. Throughout the world, suicide rarely occurs in children who have not reached puberty. In the last 15 years, the rates of both completed suicide and suicidal ideation have decreased among adolescents. This decrease appears to coincide with the increase in SSRI medications prescribed to adolescents with mood and behavioral disturbance.

Suicidal Ideation and Behavior

Suicidal ideation, gestures, and attempts are frequently, but not always, associated with depressive disorders. Reports indicate that as many as half of suicidal individuals express suicidal intentions to a friend or a relative within 24 hours before enacting suicidal behavior.

Suicidal ideation occurs in all age groups and with the highest frequency in children and adolescents with severe mood disorders. More than 12,000 children and adolescents are hospitalized in the United States each year because of suicidal threats or behavior, but completed suicide is rare in children younger than 12 years of age. A young child is hardly capable of designing and carrying out a realistic suicide plan. Cognitive immaturity seems to play a protective role in preventing even children who wish they were dead from committing suicide. Completed suicide occurs about five

times more often in adolescent boys than in girls, although the rate of suicide attempts is at least three times higher among adolescent girls than among boys. Suicidal ideation is not a static phenomenon; it can wax and wane with time. The decision to engage in suicidal behavior may be made impulsively without much forethought, or the decision may be the culmination of prolonged rumination.

The method of the suicide attempt influences the morbidity and completion rates, independent of the severity of the intent to die at the time of the suicidal behavior. The most common method of completed suicide in children and adolescents is the use of firearms, which accounts for about two-thirds of all suicides in boys and almost one-half of suicides in girls. The second most common method of suicide in boys, occurring in about one-fourth of all cases, is hanging; in girls, about one-fourth commit suicide through the ingestion of toxic substances. Carbon monoxide poisoning is the next most common method of suicide in boys, but it occurs in less than 10 percent; suicide by hanging and carbon monoxide poisoning are equally frequent among girls and account for about 10 percent each. Additional risk factors in suicide include a family history of suicidal behavior, exposure to family violence, impulsivity, substance abuse, and availability of lethal methods. Gender differences in nonfatal suicidal behavior among ninth-grade adolescents in a survey of students in 100 high schools found that 19.8 percent of female students had serious suicidal thoughts, and 10.8 percent of females attempted suicide. In male students, 9.3 percent had a history of suicidal thoughts, and 4.9 percent attempted suicide. In this study, female students showed evidence of higher levels of mood and anxiety problems, whereas males had a slightly higher level of disruptive behavior problems. Female students reported higher levels of depression, anxiety, somatic complaints, and increased levels of emotional and behavioral problems than males. In young adolescents, even without meeting full criteria for psychiatric disorders, females report more psychopathology along with a higher likelihood of nonfatal suicidal behavior.

Diagnosis and Clinical Features

The characteristics of adolescents who attempt suicide and those who complete suicide are similar, and up to 40 percent of suicidal persons have

made a previous attempt. Direct questioning of children and adolescents about suicidal thoughts is necessary because studies have consistently shown that caregivers are frequently unaware of these ideas in their children. Suicidal thoughts (i.e., children talking about wanting to harm themselves) and suicidal threats (e.g., children stating that they want to jump in front of a car) are more common than suicide completion.

Most older adolescents with suicidal behavior meet the criteria for one or more psychiatric disorders, often including major depressive disorder, bipolar disorder, and psychotic disorders. Youth with mood disorders in combination with substance abuse and a history of aggressive behavior are at particularly high risk for suicide. The most common precipitating factors in younger adolescent suicide completers appear to be impending disciplinary actions, impulsive behavioral histories, and access to loaded guns, particularly in the home. Adolescents without mood disorders with histories of disruptive and violent, aggressive, and impulsive behavior may be susceptible to suicide during family or peer conflicts. High levels of hopelessness, poor problem-solving skills, and a history of aggressive behavior are risk factors for suicide. A less common profile of an adolescent who completes suicide is one of high achievement and perfectionistic character traits facing a perceived failure, such an academically proficient adolescent humiliated by a poor grade on an examination.

Findings from a World Health Organization mental health survey reveals that a range of psychiatric disorders increases the risk of suicidal ideation across the lifespan. Youth with psychiatric disorders characterized by severe anxiety and poor impulse control are at higher risk to act on suicidal ideation. In psychiatrically disturbed and vulnerable adolescents, suicide behavior may represent impulsive responses to recent stressors. Typical precipitants of suicidal behavior include conflicts and arguments with family members and boyfriends or girlfriends. Alcohol and other substance use can further predispose an already vulnerable adolescent to suicidal behavior. In other cases, an adolescent attempts suicide in anticipation of punishment after being caught by the police or other authority figures for a forbidden behavior.

About 40 percent of youth who complete suicide had previous psychiatric treatment, and about 40 percent had made a previous suicide attempt. A child who has lost a parent by any means before age 13 is at higher risk for mood disorders and suicide. The precipitating factors include loss of face with

peers, a broken romance, school difficulties, unemployment, bereavement, separation, and rejection. There are reports of clusters of suicides that occur among adolescents who know one another and go to the same school. Suicidal behavior can precipitate other such attempts within a peer group through identification—so-called copycat suicides. Some studies have found a transient increase in adolescent suicides after television programs in which the central theme was the suicide of a teenager.

The tendency of disturbed young persons to imitate highly publicized suicides has been referred to as *Werther syndrome*, after the protagonist in Johann Wolfgang von Goethe's novel, *The Sorrows of Young Werther*. The novel, in which the hero kills himself, was banned in some European countries after its publication more than 200 years ago because of a rash of suicides by young men who read it; some dressed like Werther before killing themselves or left the book open at the passage describing his death. In general, although imitation may play a role in the timing of suicide attempts by vulnerable adolescents, the overall suicide rate does not seem to increase when media exposure increases. In contrast, direct exposure to peer suicide increases the risk of depression and PTSD rather than suicide.

Treatment

The prognostic significance of suicidal ideation and behaviors in adolescents ranges from relatively low lethality to high risk for completion. One of the challenges in addressing suicide is to identify children and adolescents with suicidal ideation, and particularly to treat those who have untreated psychiatric disorders, as the risk of completed suicide increases with age, as does the onset of an untreated psychiatric disorder. We should always evaluate adolescents who come to medical attention because of suicidal attempts to determine whether hospitalization is necessary. Pediatric patients who present to the emergency room with suicidal ideation benefit from an intervention that occurs in the emergency room to ensure that the patient is transitioned to outpatient care when hospitalization is not necessary. We should hospitalize youths who fall into high-risk groups until the acute suicidality is no longer present. Adolescents at higher risk include those who have made previous suicide attempts, especially with a lethal method, males older than 12 years of age with histories of aggressive behavior or substance

abuse, use of a lethal method, and severe major depressive disorder with social withdrawal, hopelessness, and persistent suicidal ideation.

Relatively few adolescents evaluated for suicidal behavior in a hospital emergency room subsequently receive ongoing psychiatric treatment. Factors that may increase the probability of psychiatric treatment include psychoeducation for the family in the emergency room, diffusing acute family conflict, and setting up an outpatient follow-up during the emergency room visit. Emergency room discharge plans often include providing an alternative if suicidal ideation reoccurs, and a telephone hotline number provided to the adolescent and the family in case suicidal ideation reappears.

Scant data exist to evaluate the efficacy of various interventions in reducing suicidal behavior among adolescents. CBT alone and in combination with SSRIs decreased suicidal ideation in depressed adolescents over time in the TADS, a large multisite study; however, these interventions do not work immediately, so we should take safety precautions for high-risk situations. Dialectical behavior therapy (DBT), a long-term behavioral intervention useful for individuals or groups of patients, can reduce suicidal behavior in adults, but there are no significant studies in adolescents. Components of DBT include mindfulness training to improve self-acceptance, assertiveness training, instruction on avoiding situations that may trigger self-destructive behavior, and increasing the ability to tolerate psychological distress. This approach warrants investigation among adolescents.

Given the reduction in completed suicide among adolescents over the last decade, during the same period in which SSRI treatment in the adolescent population has markedly risen, SSRIs may have been instrumental in this effect. Given the risk of the increased rate of suicidal thoughts and behaviors among depressed children and adolescents (indicated in randomized clinical trials with antidepressant medications and leading to the “black-box” warning for all antidepressants for depressed youth), we must closely monitor youths for increased suicidality while taking antidepressants.

Epidemiology

In a study of 9- to 16-year-olds in 3 months, passive suicidal thoughts were approximately 1 percent, suicidal ideation with a plan was 0.3 percent, and

suicide attempt was 0.25 percent. In adolescents 14 to 18 years, the current rate of suicidal ideation was 2.7 percent, and the annual incidence was 4.3 percent. Among this population of adolescents, the lifetime prevalence of suicide attempts was 7.1 percent, with a much higher rate of suicidal behavior for girls than for boys: 10.1 percent compared to 3.8 percent. Completed suicide rates in youth are much less common in children and younger teens 10 to 14 years, with a slighter lower rate of 0.95 per 100,000 for females compared to 1.71 per 100,000 for males. In older adolescents 15 to 19 years of age, completed suicide is considerably lower for females, 3.52 per 100,000 compared to males, 12.65 per 100,000 in the United States in 2004.

Etiology

Universal features in adolescents who resort to suicidal behaviors are the inability to synthesize viable solutions to ongoing problems and the lack of coping strategies to deal with immediate crises. Therefore, a narrow view of the options available to deal with recurrent family discord, rejection, or failure contributes to a decision to commit suicide.

Genetic Factors. Completed suicide and suicidal behavior are two to four times more likely to occur in individuals with a first-degree family member with similar behavior. Family suicide risk studies support a genetic contribution to suicidal behavior. There is also a higher concordance for suicide among monozygotic twins compared to dizygotic twins. Recent studies have investigated the possible contributions of the short allele of the serotonin transporter promoter polymorphism (5-HTTLPR) to suicidal behaviors, although, to date, the evidence has not been consistent. Current studies are seeking to investigate correlations between genetic vulnerability and environment and timing interactions as multiple variables that may interact to increase the risk of suicidal behavior.

Biologic Factors. Investigators have found a relationship between altered central serotonin with suicide as well as impulsive aggression. This relationship is also true in adults. Studies document a reduction in the density of serotonin transporter receptors in the prefrontal cortex and serotonin receptors among individuals with suicidal behaviors. Postmortem studies in adolescents who have completed suicide show the most significant alterations

in the prefrontal cortex and hippocampus, brain regions that are also associated with emotion regulation and problem-solving. These studies have found altered serotonin metabolites, alteration in 5-HT_{2A} binding, and decreased activity of protein kinase A and C. Decreased levels of the serotonin metabolite 5-hydroxyindoleacetic acid (5-HIAA) are found in the cerebrospinal fluid (CSF) of depressed adults who attempted suicide by violent methods. Meta-analyses suggest an association between the short S-allele of the serotonin transporter promoter gene and depression as well as suicidal behavior, particularly when combined with adverse life events.

Psychosocial Factors. Although severe major depressive illness is the most significant risk factor for suicide, increasing its risk by 20 percent, many severely depressed individuals are not suicidal. A sense of hopelessness, impulsivity, recurrent substance use, and a history of aggressive behavior have been associated with an increased risk of suicide. Exposure to violent and abusive homes results in a wide range of psychopathological symptoms. Aggressive, self-destructive, and suicidal behaviors seem to occur with the greatest frequency among youth who have endured chronically stressful family lives. The most significant family risk factor for suicidal behavior is maltreatment, including physical and sexual abuse and neglect. The single largest association is between sexual abuse and suicidal behavior. Large community studies have provided data suggesting that youth at risk for suicidal behavior include those who feel disconnected, isolated, or alienated from peers. Sexual orientation is a risk factor, with increased rates of suicidal behavior of two to six times among youth who identify themselves as gay, lesbian, or bisexual. Protective factors mitigating the risk of suicidal behavior are youth who have a secure connection to school and peers, even in the face of other risk factors.

▲ 2.10 Early-Onset Bipolar Disorder

Early-onset bipolar disorder has been recognized in children as a rare disorder with greater continuity with its adult counterpart when it occurs in adolescents than in prepubertal children. Over the last decade, there has been

a significant increase in the diagnosis of bipolar I disorder made in youth referred to psychiatric outpatient clinics and inpatient units. Questions have arisen regarding the phenotype of bipolar disorder in youth, mainly because of the continuous irritability and mood dysregulation and lack of discrete mood episodes in most prepubertal children who have received the diagnosis. The "atypical" bipolar symptoms among prepubertal children often include extreme mood dysregulation, severe temper tantrums, intermittent aggressive or explosive behavior, and high levels of distractibility and inattention. This constellation of mood and behavior disturbance in the majority of prepubertal children with a current diagnosis of bipolar disorder is nonepisodic, although some fluctuation in mood may occur. The high frequency of the above symptoms in combination with chronic irritability has led to the inclusion of a new mood disorder in youth in DSM-5 called *Disruptive Mood Dysregulation Disorder*, and we discuss that in the next section. Many children with nonepisodic mood disorders often have past histories of severe ADHD, making the diagnosis of bipolar disorder even more complicated. Family studies of children with ADHD have not revealed an increased rate of bipolar I disorder. Children with "atypical" bipolar disorders, however, are frequently severely impaired, are challenging to manage in school and at home, and often require psychiatric hospitalization. Longitudinal follow-up studies are underway with groups of children diagnosed with subthreshold bipolar disorders and nonepisodic mood disorders, to determine how many will develop classic bipolar disorder. In one recent study of 140 children with bipolar disorder not otherwise specified (i.e., the presence of distinct manic symptoms but subthreshold for manic episodes), 45 percent developed bipolar I or bipolar II illness over a follow-up period of 5 years. In another study, investigators followed 84 children with "severe mood dysregulation" who also had at least three manic symptoms plus distractibility for approximately 2 years. The investigators found that only one child experienced a hypomanic or mixed episode. Although childhood severe mood dysregulation is common in community samples—one study reported a lifetime prevalence of 3.3 percent in youth 9 to 19 years of age—its relationship to future bipolar disorder remains questionable. A longitudinal community-based study that followed children and adolescents with nonepisodic irritability over 20 years found that these children were at higher risk for depressive disorders and generalized anxiety disorder than bipolar

disorders.

Among adults and older adolescents with bipolar disorder who present with classic manic episodes, a major depressive episode typically precedes a manic episode. A classic manic episode in an adolescent, similar to in a young adult, may emerge as a distinct departure from a preexisting state often characterized by grandiose and paranoid delusions and hallucinatory phenomena. According to DSM-5, the criteria for a manic episode in children or adults are the same. We discuss these in Chapter 6.

When mania appears in an adolescent, there is a high incidence of psychotic features, including delusions and hallucinations, which most typically involve grandiose notions about their power, worth, and relationships. Persecutory delusions and flight of ideas are also common. Overall, gross impairment of reality testing is common in adolescent manic episodes. In adolescents with major depressive disorder destined for bipolar I disorder, those at highest risk have family histories of bipolar I disorder and exhibit acute, severe depressive episodes with psychosis, hypersomnia, and psychomotor retardation.

DIAGNOSIS AND CLINICAL FEATURES

Early-onset bipolar disorder often includes extreme irritability that is severe and persistent and may include aggressive outbursts and violent behavior. In between outbursts, children with the broad diagnosis may continue to be angry or dysphoric. It is rare for a prepubertal child to exhibit grandiose thoughts or euphoric mood; for the most part, children diagnosed with early-onset bipolar disorder are intensely emotional with a fluctuating but overriding negative mood. Current diagnostic criteria for bipolar disorders in children and adolescents in DSM-5 are the same as those used in adults (see Chapter 6, Tables 6-2 and 6-3). The clinical picture of early-onset bipolar disorder, however, is complicated by the prevalence of comorbid psychiatric disorders.

Pathology and Laboratory Examination

No specific laboratory indices are currently helpful in making the diagnosis of bipolar disorders among children and adolescents.

DIFFERENTIAL DIAGNOSIS

The most critical clinical entities to distinguish from early-onset bipolar disorder are also the disorders with which it is most frequently comorbid. Included are ADHD, oppositional defiant disorder, conduct disorder, anxiety disorders, and depressive disorders.

Although childhood ADHD tends to have its onset earlier than pediatric mania, current evidence from family studies supports the presence of ADHD and bipolar disorders as highly comorbid in children, and the concurrence is not because of the overlapping symptoms that the two disorders share. In a recent study of more than 300 children and adolescents who attended a psychopharmacology clinic and received a diagnosis of ADHD, bipolar disorder was also evident in almost one-third of those children with ADHD who had combined–type and hyperactive-types. It was less frequent (i.e., in less than 10 percent) in children with ADHD, inattentive-type.

COMORBIDITY

ADHD is the most common comorbid condition among youth with early-onset bipolar disorder and occurs in up to 90 percent of prepubertal children and up to 50 percent of adolescents diagnosed with bipolar disorder. Comorbid ADHD creates a significant source of diagnostic confusion in children with early-onset bipolar disorder since the two disorders share many diagnostic criteria, including distractibility, hyperactivity, and talkativeness. Even when removing the overlapping symptoms from the diagnostic count, a significant percentage of children with bipolar disorder continued to meet the full criteria for ADHD. This implies that both disorders, with their distinct features, are present in many cases.

Children and adolescents with bipolar disorder have higher-than-expected rates of panic and other anxiety disorders. In youth with the narrow phenotype of bipolar disorders, up to 77 percent have an anxiety disorder. The lifetime prevalence of panic disorder was 21 percent among subjects with the broader phenotype of bipolar disorder compared with 0.8 percent in those without mood disorders. Patients diagnosed with bipolar disorder who have comorbid high levels of anxiety symptoms are reported as adults to have higher risks of alcohol abuse and suicidal behavior. On the other hand,

children who exhibit the broader phenotype of bipolar disorder are at higher risk of going on to have anxiety disorders as well as depressive disorders.

Jeanie is a 13-year-old adopted teen who was hospitalized after assaulting her adoptive mother, causing bruises on her arms and legs from Jeanie's kicks and punches. Jeanie had a long history of excessively severe tantrums, which include assaultive and self-injurious behavior, since before her parents adopted her when she was three. Jeanie had always been a child who was irritable and explosive, with a short fuse, who could blow up with minimal provocation, even when things were going her way. Jeanie had become increasingly hard to manage at home, refused to go to school, yelled and screamed for hours on a daily basis, and often hit and kicked her adoptive parents by the time she was 10 years old. Jeannie was in residential treatment from age 11 to almost 13, where the clinicians diagnosed her with bipolar disorder and started lithium and citalopram. She was doing so well there that Jeanie's adoptive mother decided to take her home. After a few weeks at home, however, Jeanie began to decompensate, having daily explosive tantrums, becoming aggressive and out of control. On multiple occasions, she had hurt herself and her adoptive mother and father. Upon arriving at the hospital, Jeanie was calm; however, her adoptive mother refused to consider taking her home until undergoing a full psychiatric evaluation and treatment for unsafe behaviors. The child psychiatrist initially evaluated Jeanie, after which she went to a pediatric inpatient unit to await a psychiatric bed. The psychiatrist learned that Jeanie had been born prematurely to a teenage mother and placed in multiple foster homes until she was adopted. Jeanie was a small girl who appeared younger than her stated age, although her demeanor was bossy and pedantic. Jeanie's biologic family history was unknown, and although she had at least one stigmata of fetal alcohol syndrome, her IQ was in the average range, and there was no other evidence to corroborate this possibility. On mental status examination in the hospital, Jeanie reported that things were fine, that she was not depressed, and that she did not get along with kids her age but that she had a few friends. Jeanie admitted that she had a bad temper and that she did not remember what she did after she was in a

rage. Jeanie's affect was odd, and she seemed to enjoy having the psychiatrist as her audience. Jeanie denied suicidal ideation or past attempts and denied having been a danger to herself or her adoptive parents. Jeanie seemed annoyed when she was asked about the reasons for her placement in a residential facility, and she became irritable when questioned about the reasons for her current admission. The team then transferred Jeanie to an adolescent psychiatric inpatient unit to start a trial of an atypical antipsychotic. The team recommended that her psychiatrist reconsider a more structured school program, either a day program or residential facility. The diagnosis of bipolar disorder remained in question, as she did not meet the narrow phenotype for this disorder.

COURSE AND PROGNOSIS

There are several pathways regarding the course and prognosis of children diagnosed with early-onset bipolar disorder. Those who present with severe mood dysregulation at an early age, without discrete mood cycles, are most likely to develop anxiety and depressive disorders as they mature. Youths who present in adolescence with a recognizable manic episode are most likely to continue to meet the criteria for bipolar I disorder in adulthood. In both cases, long-term impairment is considerable.

A longitudinal study of 263 child and adolescent inpatients and outpatients with bipolar disorder followed for an average of 2 years found that approximately 70 percent recovered from their index episode within that period. Half of these patients had at least one recurrence of a mood disorder during this time, more frequently a depressive episode than a mania. There were no differences in the rates of recovery for children and adolescents whose diagnosis was bipolar I disorder, bipolar II disorder, or bipolar disorder not otherwise specified. However, those whose diagnosis was bipolar disorder not otherwise specified had a significantly longer duration of illness before recovery, with less frequent recurrences once they recovered. About 19 percent of patients changed polarity once per year or less, 61 percent shifted five or more times per year, about half cycled more than ten times per year, and about one-third cycled more than 20 times per year. Predictors of more rapid cycling included lower socioeconomic status (SES),

presence of lifetime psychosis, and bipolar disorder not otherwise specified diagnosis. Over the follow-up period, about 20 percent of subjects with bipolar II disorder converted to bipolar I disorder, and 25 percent of the bipolar disorder not otherwise specified subjects developed bipolar I disorder or bipolar II disorder during the follow-up period.

Similar to the natural history of bipolar disorders in adults, children have a wide range of symptom severity in manic and depressed episodes. The more frequent diagnostic conversions from bipolar II disorder to bipolar I disorder among children and adolescents, compared with adults, highlight the lack of stability of the bipolar II disorder diagnosis in youth. This is also the case concerning conversion from bipolar disorder not otherwise specified to other bipolar disorders. When bipolar disorder occurs in young children, recovery rates are lower. Also, there is a higher likelihood mixed states and rapid cycling, and higher rates of polarity changes compared with those who develop bipolar disorders in late adolescence or early adulthood.

TREATMENT

Treatment of early-onset bipolar disorder incorporates multimodal interventions including pharmacotherapy, psychoeducation, psychosocial intervention with the family and the child, and school interventions to optimize a child's school adjustment and achievement.

Pharmacotherapy

Two classes of medications—atypical antipsychotics and mood-stabilizing agents—are the most well-studied agents that provide efficacy in the treatment of early-onset bipolar disorders. Eight RCTs have shown the efficacy of atypical antipsychotic agents in the treatment of bipolar disorder in youth between the ages of 10 and 17 years. These studies compared an atypical antipsychotic to placebo, or compared an atypical antipsychotic to a mood stabilizer, or added an antipsychotic to a mood-stabilizing agent. The atypical antipsychotics included olanzapine, quetiapine, risperidone, aripiprazole, and ziprasidone. All five of the atypical antipsychotic studies demonstrated significant efficacy in the treatment of early-onset bipolar manic or mixed states. A recent trial comparing quetiapine and valproate found that both were efficacious, but the quetiapine was superior in the speed

of its effect. In another trial comparing risperidone and sodium valproate treatment for bipolar disorder in youth, risperidone had more rapid improvement and a higher final reduction in manic symptoms compared to sodium valproate.

Mood-stabilizing agents have been used in open trials and anecdotally with early-onset bipolar illness with little evidence of efficacy at this time. In trials using lithium or sodium valproate for the treatment of early-onset bipolar disorder, responses were less robust compared to results with atypical antipsychotics. Controlled trials have provided some evidence suggesting that lithium is efficacious in the management of aggression behavior disorders. Although lithium is approved for adolescent mania, more research is needed to know if lithium is useful for more classic forms of mania in adolescents. The Collaborative Lithium Trials (CoLT) established a set of protocols to establish the safety and potential efficacy of lithium in youth, and to develop studies to provide evidence-based dosing of lithium for youth. A group of researchers studied the first-dose pharmacokinetics of lithium carbonate in youth and found that clearance and volume correlate with total body weight in youth, and particularly with fat-free mass. The difference in body size was consistent with the pharmacokinetics of lithium metabolism in children and adults. An open-label trial of lamotrigine in the treatment of bipolar depression among youth provides possible support for its use in children and adolescents.

Current evidence suggests a faster response and a more robust effect with atypical antipsychotics compared to mood-stabilizing agents in the treatment of early-onset bipolar disorder. However, given the severity and impairment of bipolar disorder in youth, when there is only partial recovery, we should consider adding another agent.

Psychosocial Treatment

Psychosocial treatment interventions for early-onset bipolar illness have included a family-focused treatment. This treatment consists of several sessions of psychoeducation, then sessions focusing on current stressors and a mood management plan, and then several sessions of communication enhancement training and problem-solving skills training. The use of this type of intervention for youths diagnosed with bipolar disorder or at risk (by

family history or subthreshold symptoms) is of value. Adjunctive family-focused psychoeducational treatment modified for children and adolescents reduces the relapse rate. Children and adolescents treated with mood-stabilizing agents in addition to a psychosocial intervention showed improvement in depressive symptoms, manic symptoms, and behavioral disturbance over 1 year.

A year-long trial of a modified Family Focused Treatment-High Risk in youth with bipolar disorder showed significant improvement in mood disturbance, especially depressive mood and hypomania, and improved psychosocial functioning. Family-focused treatment for high-risk youth is a promising intervention that deserves further investigation as a longitudinal follow-up to determine the course of youth at risk to develop bipolar disorder.

EPIDEMIOLOGY

The prevalence rates of bipolar disorder among youth vary depending on the age group studied, and on whether the diagnostic criteria are applied narrowly, restricting it to discrete mood episodes or more broadly, to include nonepisodic mood and behavioral states. In younger children, bipolar disorder is infrequent, with no cases of bipolar I disorder identified in children between the ages of 9 and 13 years by the Great Smokey Mountain Study. However, severe mood dysregulation, often a prominent feature in prepubertal children receiving a diagnosis of bipolar disorder, was found in 3.3 percent of an epidemiologic sample. In adolescents, bipolar disorder is more frequent, found to range from 0.06 to 0.1 percent of the general population of 16-year-olds in studies using a narrow definition of bipolar I disorder. The prevalence of subthreshold symptoms of bipolar illness was 5.7 percent in one study to at least 10 percent in another. Follow-up studies into adulthood revealed that the subthreshold manic symptoms predicted high levels of impairment with progression to depression and anxiety disorders, not bipolar I or II disorders.

Community use of the diagnosis of bipolar disorder in youth has increased markedly over the last 15 years in both outpatient and inpatient psychiatric settings. One survey suggested there was a 40-fold increase in the diagnosis of early-onset bipolar disorder at outpatient clinics from the 1990s to the 2000s. Furthermore, from 2000 through 2006, the rate of youth hospitalized

with a primary diagnosis of bipolar disorder nearly doubled.

ETIOLOGY

Genetic Factors

High rates of bipolar disorder occur in the relatives of the narrow phenotype of early-onset bipolar disorder compared to young adult-onset of bipolar disorder. The high rates of comorbid ADHD among children with early-onset bipolar disorder have led to questions regarding the co-transmission of these disorders in family members. However, children with the broader phenotype of bipolar disorder, that is, severe mood dysregulation without episodes of mania, have not been found to have higher rates of bipolar disorder in family members, which suggests that the narrow and broad phenotypes of bipolar disorder may be distinct and separate entities. Nearly 25 percent of adolescent offspring of families with probands with bipolar disorder experienced a mood disorder by the age of 17 years old, compared to 4 percent of controls, with approximately 8 percent representing bipolar I, bipolar II, or bipolar disorder not otherwise specified. Most of the risk in the offspring, therefore, is for unipolar major depressive disorder. Disruptive behavior disorders did not increase, in a longitudinal study, in the offspring of families with a bipolar proband, compared to controls. The combination of ADHD and bipolar disorder is not found as frequently in relatives of children with ADHD alone compared with first-degree relatives of children with the combination.

Although bipolar disorder appears to have a significant heritable component, its mode of inheritance remains unknown. Several research groups have concluded that early-onset bipolar disorder is a more severe form of the illness, characterized by more mixed episodes, greater psychiatric comorbidity, more lifetime psychotic symptoms, poorer response to prophylactic lithium treatment, and a greater heritability. The European collaborative study of early-onset bipolar disorder (France, Germany, Ireland, Scotland, Switzerland, England, and Slovenia) carried out a genome-wide linkage analysis of both the narrow and the broad phenotype of early-onset bipolar disorder. This group concluded that a genetic factor located in the 2q14 region is either specifically involved in the etiology of early-onset bipolar disorder, or that a gene in this region exerts influence as a modifier of

other genes in the development of bipolar disorder in this age group. Other linkage regions that were found by this collaborative did not pertain only to the early-onset group of bipolar disorder, suggesting that there may be some genetic factors common to early-onset and adult-onset bipolar disorder. This conclusion is consistent with the increased incidence of adult-onset bipolar disorder among siblings of early-onset disease. Further genome-wide studies are needed to elucidate the genetic etiology of early-onset bipolar disorder.

Neurobiologic Factors

Converging data suggest that early-onset bipolar disorder is associated with both structural and functional brain alterations in prefrontal cortical and subcortical regions associated with the processing and regulation of emotional stimuli. MRI studies suggest that altered development of white matter and a decreased amygdalar volume are found more frequently in this population than in the general population.

Functional MRI (fMRI) studies are essential in that they can identify altered brain function in vulnerable populations such as youth with early-onset bipolar disorder at baseline. They can also elucidate functional changes toward normalization in brain functioning after various treatments, and potentially identify pretreatment neural predictors of good response to various treatments. An fMRI study of pediatric bipolar patients documented pretreatment brain activity and posttreatment effects of a trial of risperidone versus sodium valproate. This double-blind study included 24 unmedicated manic patients with a mean age of 13 years, randomized to either risperidone or sodium valproate treatment, and 14 healthy controls examined over 6 weeks. Before treatment, the patient group showed increased amygdala activity compared to healthy controls, which was poorly controlled by the higher ventrolateral prefrontal cortex (VLPFC) and the dorsolateral prefrontal cortex (DLPFC), which exert influence on the amygdala to control emotional regulation and processing. Increased amygdala activity at baseline predicted poorer treatment response to both risperidone and sodium valproate in the patient group. Patients were given an affective color-matching word task involving matching positive words (i.e., happiness, achievement, success), negative words (i.e., disappointment, depression, or rejection), or neutral words, with one of two colored circles displayed on a screen while

administering the fMRI. Higher pretreatment right amygdala activity during a word task with positive and negative words in the risperidone group and higher pretreatment left amygdala activity with a positive word task in the sodium valproate group, predicted a poor response on the Young Mania Rating Scale. Increased amygdala activity in early-onset bipolar patients might be a potential biomarker predicting resistance and poor treatment response to both risperidone and sodium valproate.

Neuropsychological Studies

Impairments in verbal memory, processing speed, executive function, working memory, and attention are common in early-onset bipolar disorder. Data suggest that on tasks of working memory, processing speed, and attention, children and adolescents with comorbid bipolar disorder and ADHD demonstrate more pronounced impairments compared with those without ADHD. Other studies found that children with bipolar disorder make a higher number of emotion recognition errors compared with controls. The children more frequently identified faces as “angry” when presented with adult faces; however, these errors did not occur when they looked at children’s faces. Impaired perception of facial expression also occurs in studies of adults with bipolar disorder.

▲ 2.11 Disruptive Mood Dysregulation Disorder

Disruptive mood dysregulation disorder, a new inclusion in the DSM-5, describes severe, developmentally inappropriate, and recurrent temper outbursts at least three times per week, along with a persistently irritable or angry mood between temper outbursts. To meet diagnostic criteria, the symptoms must be present for at least a year, and the onset of symptoms must be present by the age of 10 years old. Clinicians usually diagnose these children with bipolar disorder or a combination of oppositional defiant disorder, ADHD, and intermittent explosive disorder. Recent longitudinal data suggest, however, that these children do not typically develop classic

bipolar disorder in late adolescence or early adulthood. Instead, studies suggest that youth with chronic irritability and severe mood dysregulation are at higher risk for future unipolar depressive disorders and anxiety disorders. Some experts include hyperarousal as a symptom, but DSM-5 does not.

DIAGNOSIS AND CLINICAL FEATURES

The DSM-5 diagnostic criteria for disruptive mood dysregulation disorder require outbursts that are grossly out of proportion to the situation. These temper outbursts present with verbal rages or physical aggression toward people or property and are inappropriate for the child's developmental level. Temper outbursts occur, on average, three or more times per week, with variations in mood between outbursts. Symptoms must exhibit before age 10 years, be present for at least 12 months, and be present within at least two settings (i.e., home and school). We would not diagnose this in children younger than 6 years or older than 18 years. In between temper outbursts, the child's mood is persistently irritable and angry, and this mood is observable by others, such as parents, teachers, or peers. A diagnosis of bipolar disorder would take precedence over this one, and the symptoms cannot occur only during a major depressive episode. Table 2-27 compares the diagnostic approaches.



Table 2-27
Disruptive Mood Dysregulation Disorder

	DSM-5	ICD-10
Diagnostic name	Disruptive Mood Dysregulation Disorder	Unspecified Behavioral and Emotional Disorders with onset usually occurring in childhood and adolescence
Duration	Outbursts occurring ≥3/wk Occurs for ≥12 mo No more than 3 mo during that time	

	when criteria not met Onset between ages 6 and 10 yr
Symptoms	• Recurrent inappropriate outbursts (verbal/physical) • Developmentally inappropriate • Irritable/angry most days between outbursts • Irritable/mad most days
Required number of symptoms	All the above
Exclusions (not result of):	• Another mental disorder • Substance use • Another neurologic disorder Cannot co-occur with: • Oppositional defiant disorder • Intermittent explosive disorder • Bipolar disorder

Daniel, a 12-year-old seventh-grade boy, was brought to his pediatrician by his mother. His rages and inappropriate tantrums were exasperating her. Daniel was on the floor in the waiting room, pounding his hands on

the floor, yelling at his mother, “Get me out of here!” and crying. His mother had bruises on both legs from Dylan’s kicks, and she appeared distressed. Daniel’s mother walked into the office, leaving Daniel on the floor in the waiting room and burst into tears. “I cannot deal with him anymore.” She recounted the problems that Daniel had been having for the last 2 years: Severe recurrent tantrums four to five times/week. “He tantrums like a 6-year-old, and even when he is not having a tantrum, he is perpetually angry and irritable.” She reported that Daniel had lost all of his friends due to his short fuse and frequent verbal and physical outbursts. He was almost always irritable, even on his birthday. Daniel’s mother wonders whether there is anything physically wrong with him, but physical examination and routine blood tests reveal no abnormalities. Daniel’s tantrums had lessened somewhat last summer during the 2-month summer vacation; however, as soon as school resumed, he was back to consistent irritability. After an interview with Daniel, his pediatrician determined that he was not acutely suicidal; however, he required urgent psychotherapeutic intervention. Daniel started cognitive-behavioral therapy with a clinical psychologist and also saw a child and adolescent psychiatrist for possible medication. Daniel resisted psychotherapy; however, after several sessions, Daniel’s parents felt more hopeful than they had in a long time, and learned that Daniel’s problems were not “all their fault.” Daniel agreed to begin a trial of fluoxetine, titrated to 30 mg over several weeks, and after about a month, it became clear that his irritability had diminished noticeably. Daniel still had many problems with peers, and he still had one or two tantrums per week; however, the tantrums were becoming less prolonged and less intense. Daniel seemed genuinely happy when a classmate invited him to a birthday party, and he was able to interact successfully with his peers during the party without any conflicts. Daniel continues to benefit from CBT, and he remains on fluoxetine 40 mg a day. Daniel is still a “temperamental” boy, but he is doing well in school, has rekindled several friendships, and can participate in family gatherings without a major tantrum.

DIFFERENTIAL DIAGNOSIS

Bipolar Disorder

Disruptive mood dysregulation disorder closely resembles the broad phenotype of bipolar disorder. Although not episodic, it has been theorized by some clinicians and researchers that the chronic and persistent symptoms of mood disturbance and irritability may be an early developmental presentation of bipolar disorder. Disruptive mood dysregulation, however, does not meet formal diagnostic criteria for mania in bipolar disorder, because irritability in disruptive mood dysregulation disorder is chronic and nonepisodic.

Oppositional Defiant Disorder

Disruptive mood dysregulation disorder is similar to oppositional defiant disorder in that they both include irritability, temper outbursts, and anger. Oppositional defiant disorder includes symptoms of annoyance and defiance, which disruptive mood dysregulation disorder does not have. Disruptive mood dysregulation disorder requires that irritable outbursts be present in at least two settings, whereas oppositional defiant disorder requires that they be present in only one setting.

COMORBIDITY

Disruptive mood dysregulation disorder often co-occurs with other psychiatric disorders. The most common comorbidities are ADHD (94 percent), oppositional defiant disorder (84 percent), anxiety disorders (47 percent), and major depressive disorder (20 percent). The relationship of severe mood dysregulation and disruptive mood dysregulation disorder to bipolar disorder has been a topic of clinical investigation. Some experts consider youths with severe mood dysregulation and hyperarousal symptoms to be part of the broad phenotype of pediatric bipolar disorder. However, the term “severe mood dysregulation” was utilized by researchers for these youth because it remains unclear whether these youth go on to meet the criteria for a bipolar disorder. Disruptive mood dysregulation disorder is not considered episodic and may coexist with ADHD. Current evidence does not support its continuity with an emerging bipolar disorder.

COURSE AND PROGNOSIS

Disruptive mood dysregulation disorder is a chronic disorder. Longitudinal studies thus far have shown that patients with disruptive mood dysregulation disorder in childhood have a high risk of progressing to major depressive disorder, dysthymic disorder, and anxiety disorders over time.

TREATMENT

The current treatment of disruptive mood dysregulation focuses on symptomatic interventions because we do not understand its etiology. If disruptive mood dysregulation disorder is confirmed to resemble unipolar depression and anxiety disorders in its pathophysiology, and it is often comorbid with ADHD, then SSRIs and stimulants would likely be the pharmacologic agents of first choice. However, if the pathophysiology of disruptive mood dysregulation disorder is like that of bipolar disorder, then first-line treatments for youth would include atypical antipsychotic agents and mood stabilizers. There are scant treatment studies of disruptive mood dysregulation disorder in the literature. One controlled trial of youths with symptoms of severe mood dysregulation and ADHD symptoms who did not respond to stimulants responded to sodium valproate combined with behavioral psychotherapy compared to placebo and behavioral psychotherapy. There are treatment studies underway of youth who exhibit symptoms of severe mood dysregulation utilizing an SSRI plus a stimulant compared to a stimulant and placebo.

Psychosocial interventions such as cognitive-behavioral psychotherapy are likely to be an essential component of treatment for youth with disruptive dysregulation disorder, and psychosocial interventions targeting children diagnosed with bipolar disorder may be beneficial.

EPIDEMIOLOGY

Most of the epidemiologic data applied to disruptive mood dysregulation disorder come from children and adolescents with severe mood dysregulation, which includes hyperarousal symptoms. Because disruptive mood dysregulation disorder differs from severe mood dysregulation disorder only in the absence of hyperarousal symptoms, the epidemiologic data from

the severe mood dysregulation disorder studies is a useful proxy for disruptive mood dysregulation disorder. Severe mood dysregulation has a lifetime prevalence of 3 percent in children aged 9 to 19 years. Within that percentage, males (78 percent) are more prevalent than females (22 percent). The mean age of onset is 5 to 11 years of age.

2.12 Disruptive Behaviors of Childhood

Disruptive behaviors, especially oppositional patterns and aggressive behaviors, are among the most frequent reasons for child and adolescent psychiatric evaluation. Demonstration of impulsive and oppositional behaviors are developmentally normative in young children; many youths who continue to display excessive patterns in middle childhood will find other forms of expression as they mature and will no longer demonstrate these behaviors in adolescence or adulthood. The origin of stable patterns of oppositional defiant behavior is widely accepted as a convergence of multiple contributing factors, including biologic, temperamental, learned, and psychological conditions. Risk factors for the development of aggressive behavior in youth include childhood maltreatment such as physical or sexual abuse, neglect, emotional abuse, and overly harsh and punitive parenting.

OPPOSITIONAL DEFIANT DISORDER

Oppositional defiant disorder describes enduring patterns of negativistic, disobedient, and hostile behavior toward authority figures, as well as an inability to take responsibility for mistakes, leading to placing blame on others. Children with oppositional defiant disorder frequently argue with adults and become easily annoyed by others, leading to a state of anger and resentment. Children with oppositional defiant disorder may have difficulty in the classroom and with peer relationships, but generally do not resort to physical aggression or significantly destructive behavior.

In contrast, children with conduct disorder engage in severe, repeated acts of aggression that can cause physical harm to themselves and others and frequently violate the rights of others.

In oppositional defiant disorder, a child’s temper outbursts, active refusal

to comply with rules, and annoying behaviors exceed expectations for these behaviors for children of the same age. The disorder is an enduring pattern of negativistic, hostile, and defiant behaviors in the absence of significant violations of the rights of others.

Diagnosis and Clinical Features

Children with oppositional defiant disorder often argue with adults, lose their temper, and are angry, resentful, and easily annoyed by others at a level and frequency that is outside of the expected range for their age and developmental level. Frequently, youth with oppositional defiant disorder actively defy adults' requests or rules and deliberately annoy other persons. They tend to blame others for their own mistakes and misbehavior, more often than is appropriate for their developmental age. Manifestations of the disorder are almost invariably present in the home, but they may not be present at school or with other adults or peers. In some cases, the child has symptoms outside the home from the start; in others, it starts at home and later moves beyond that. Typically, symptoms of the disorder are most evident in interactions with adults or peers whom the child knows well. Thus, a child with oppositional defiant disorder may not show signs of the disorder when examined clinically. Although children with oppositional defiant disorder may be aware that others disapprove of their behavior, they may still justify it as a response to unfair or unreasonable circumstances. The disorder appears to cause more distress to those around the child than to the child. Table 2-28 compares the diagnostic approaches to oppositional defiant disorder.



Table 2-28
Oppositional Defiant Disorder

	DSM-5	ICD-10
Diagnostic name	Oppositional Defiant Disorder	Oppositional Defiant Disorder
Duration	≥6 mo For age <5: occurs more	

	days than not For age ≥5: occurs at least 1/wk	
Symptoms	• Losing temper • Sensitive/easily annoyed • Angry/resentful • Arguing with authority figures • Refusing request from authorities or rules • Deliberately annoys others • Blames others for mistakes/behaviors • Spiteful/Vindictiveness (at least 2× in 6 mo)	Defined as a conduct disorder in younger children Predominant symptoms: • Disobedience • Defiance • Disruptive behavior
Required number of symptoms	≥4 symptoms	
Psychosocial consequences of symptoms	Marked distress and/or impairment	
Exclusions (not result of):	Substance use Another mental illness	No delinquent, severely aggressive or dissocial behaviors
Severity specifiers	**Mild:** symptoms in 1 setting **Moderate:** symptoms in 2 settings **Severe:** symptoms in ≥3 settings	

The DSM-5 has divided oppositional defiant disorder into three types: angry/irritable mood, argumentative/defiant behavior, and vindictiveness. A child may meet diagnostic criteria for oppositional defiant disorder with a 6-month pattern of at least four symptoms from the three types above. Angry/Irritable children with oppositional defiant disorder often lose their tempers, are easily annoyed, and feel irritable much of the time. Argumentative/Defiant children display a pattern of arguing with authority figures and adults such as parents, teachers, and relatives. Children with this type of oppositional defiant disorder actively refuse to comply with requests, deliberately break the rules, and purposely annoy others. These children often do not take responsibility for their actions, and often blame others for their misbehavior. Children with the Vindictive type of oppositional defiant disorder are spiteful and have shown vindictive or spiteful actions at least twice in 6 months to meet the diagnostic criteria.

Chronic oppositional defiant disorder or irritability almost always interferes with interpersonal relationships and school performance. Peers often reject these children, and they may become isolated and lonely. Despite adequate intelligence, they may do poorly or fail in school due to their lack of cooperation, reduced participation, and inability to accept help. Secondary to these difficulties are low self-esteem, poor frustration tolerance, depressed mood, and temper outbursts. Adolescents who are ostracized may turn to alcohol and illegal substances as a modality to fit in with peers. Chronically irritable children often develop mood disorders in adolescence or adulthood.

Pathology and Laboratory Examination. No specific laboratory tests or pathologic findings help diagnose oppositional defiant disorder. Because some children with oppositional defiant disorder become physically aggressive and violate the rights of others as they age, they may share some characteristics with people with high levels of aggression, such as low CNS serotonin.

Differential Diagnosis

Oppositional behaviors are both healthy and adaptive within an expected range at specific developmental stages. We should distinguish periods of normative negativism from oppositional defiant disorder. Developmentally

appropriate oppositional behavior is neither considerably more frequent nor more intense than that seen in other children of the same mental age. According to the DSM-5, we should not diagnose oppositional defiant disorder in the presence of disruptive mood dysregulation disorder. (See Section 2.11 for a further discussion of disruptive mood dysregulation disorder.)

Oppositional defiant behavior occurring temporarily in reaction to a stressor should be diagnosed as an adjustment disorder. When features of oppositional defiant disorder appear during conduct disorder, schizophrenia, or a mood disorder, we should not diagnose oppositional defiant disorder. Oppositional and negativistic behaviors can also be present in ADHD, cognitive disorders, and mental retardation. Whether we diagnose oppositional defiant disorder along with ADHD depends on the severity, pervasiveness, and duration of such behavior. Some young children who receive a diagnosis of oppositional defiant disorder go on in several years to meet the criteria for conduct disorder. Some investigators believe that the two disorders may be developmental variants of each other, with conduct disorder being the natural progression of oppositional defiant behavior when a child matures. Most children with oppositional defiant disorder, however, do not later meet the criteria for conduct disorder, and up to one-fourth of children with oppositional defiant disorder may not meet the diagnosis several years later.

The subtype of oppositional defiant disorder that tends to progress to conduct disorder is one in which aggression is prominent, for example, the Angry/Irritable type and the Vindictive type. Many children who have ADHD and oppositional defiant disorder develop conduct disorder before the age of 12 years. Many children who develop conduct disorder have a history of oppositional defiant disorder. Overall, the current consensus is that two subtypes of oppositional defiant disorder may exist. One type, which is likely to progress to conduct disorder, includes some symptoms of conduct disorder (e.g., fighting, bullying). The other type, which is characterized by less aggression and fewer antisocial traits, does not progress to conduct disorder. However, in either case, when both oppositional defiant disorder and conduct disorder are present, according to DSM-5, they may be diagnosed concurrently.

Jackson, age 8 years, was brought to the clinic for evaluation of irritability, negativity, and defiant behavior by his mother. She complained that he had frequent, prolonged tantrums, triggered by not "getting his way." Jackson's mother described the tantrums as consisting of shouting, cursing, crying, slamming doors, and sometimes throwing books or objects on the floor. Jackson had been having troubles in school as well, and his teacher had reported to the family that he seemed to have a habit of provoking other students as well as the teacher by making noises, rocking in his seat, and whistling in class. Recently, at home, Jackson was kicking his foot against his mother's chair, and she asked him to stop. He looked at her and continued to kick her chair until she became angry and sent him to his room. He then started yelling and stated that he wasn't doing anything and that his mother was just picking on him. Jackson's mother reported that she has given up on asking him to help with chores because it inevitably results in an argument. Jackson appeared sullen and irritable during the interview. He insisted that his problems are all his mother's fault, and she is always nagging him unfairly. During the interview with his mother, Jackson interrupted her several times to say that she was lying, and contradicted her story. Despite Jackson's behavioral problem, he has been able to succeed academically and scores highly on standardized tests. His mother reports that Jackson used to have some friends in kindergarten, but as he aged, he has lost almost all of his friends because he has difficulty sharing his things and tends to be bossy. Jackson's mother reports that ever since his sister was born when he was 2 years old, he has been aggressive and rivalrous toward her. Jackson's parents separated and divorced when he was 3. He has had no contact with his father since then. Jackson's mother was depressed for a year after the divorce until she sought treatment. She has always felt guilty that his father is not in his life, and Jackson blames her for not having his father around. She believes his behaviors have become worse since she recently started dating again.

Course and Prognosis

The course of oppositional defiant disorder depends on the severity of the

symptoms and the ability of the child to develop more adaptive responses to authority. The stability of oppositional defiant disorder varies over time, with approximately 25 percent of children with the disorder no longer meeting diagnostic criteria. Persistence of oppositional defiant symptoms poses an increased risk of additional disorders, such as mood disorders, conduct disorder, and substance use disorders. Positive outcomes are more likely for intact families who can modify their expression of demands and give less attention to the child's argumentative behaviors.

An association exists between oppositional defiant disorder and ADHD, as well as with mood disorders. In children who have a long history of aggression and oppositional defiant disorder, there is a higher risk of the development of conduct disorder and later substance use disorders. Parental psychopathology, such as antisocial personality disorder and substance abuse, appears to be more common in families with children who have oppositional defiant disorder than in the general population, which creates additional risks for chaotic and troubled home environments. The prognosis for oppositional defiant disorder in a child depends somewhat on family functioning and the development of comorbid psychopathology.

Treatment

The primary treatment of oppositional defiant disorder is family intervention using both direct training of the parents in child management skills and careful assessment of family interactions. The goals of this intervention are to reinforce more prosocial behaviors and to diminish undesired behaviors at the same time. Cognitive-behavioral therapists emphasize teaching parents how to alter their behavior to discourage the child's oppositional behavior by diminishing attention to it, and Parent Child Interaction therapy focuses on selectively reinforcing and praising appropriate behavior and ignoring or not reinforcing undesired behavior.

Children with oppositional defiant behavior may also benefit from individual psychotherapy in which they role-play and "practice" more adaptive responses. In the therapeutic relationship, the child can learn new strategies to develop a sense of mastery and success in social situations with peers and families. In the safety of a more "neutral" relationship, children may discover that they are capable of less provocative behavior. Often, we

must help restore the child's self-esteem must before they can make more positive responses to external control. Parent–child conflict strongly predicts conduct problems; patterns of harsh physical and verbal punishment, particularly evoke the emergence of aggression in children. Replacing harsh, punitive parenting and increasing positive parent–child interactions may positively influence the course of oppositional and defiant behaviors.

Epidemiology

Oppositional and negativistic behavior, in moderation, is developmentally normal in early childhood and adolescence. Epidemiologic studies of negativistic traits in nonclinical populations found such behavior in 16 to 22 percent of school-age children. Although oppositional defiant disorder can begin as early as 3 years of age, it typically is noted by 8 years of age and usually not later than early adolescence. Oppositional defiant disorder occurs at rates ranging from 2 to 16 percent with increased rates reported in boys before puberty, and an equal sex ratio reported after puberty. The prevalence of oppositional defiant behavior in males and females diminishes in youth older than 12 years of age.

Etiology

The most dramatic example of typical oppositional behavior peaks between 18 and 24 months, the "terrible twos," when toddlers are dramatically expressing their growing autonomy. Pathology begins when this developmental phase persists abnormally, authority figures overreact, or oppositional behavior recurs considerably more frequently than in most children of the same mental age. Among the criteria included in oppositional defiant disorder, irritability appears to be the one most predictive of later psychiatric disorders, whereas the other elements may be considered components of temperament.

Children exhibit a range of temperamental predispositions to strong will, stable preferences, or high assertiveness. Parents who model more extreme ways of expressing and enforcing their own will may contribute to the development of chronic struggles with their children, which the children then reenact with other authority figures. What begins for an infant as an effort to establish self-determination may become transformed into an exaggerated

behavioral pattern. In late childhood, environmental trauma, illness, or chronic incapacity, such as mental retardation, can trigger oppositionality as a defense against helplessness, anxiety, and loss of self-esteem. Another normative oppositional stage occurs in adolescence as an expression of the need to separate from the parents and to establish an autonomous identity.

Classic psychoanalytic theory implicates unresolved conflicts as fueling defiant behaviors targeting authority figures. Behaviorists have observed that in children, oppositionality may be a reinforced, learned behavior through which a child exerts control over authority figures; for example, if having a temper tantrum when making a request of the child coerces the parents to withdraw their request, then tantrum behavior becomes strongly reinforced. Also, increased parental attention during a tantrum can reinforce the behavior.

CONDUCT DISORDER

Aggressive patterns of behavior are among the most frequent reasons psychiatrists see children and adolescents. Although the demonstration of impulsive behaviors is developmentally normative in children, many youths who continue to display excessive patterns of aggression in middle childhood generally require intervention. Children who develop enduring patterns of aggressive behaviors that begin in early childhood and violate the fundamental rights of peers and family members, however, may be destined for an entrenched pattern of conduct disordered behaviors over time. Controversy remains as to whether a set of “voluntary” behaviors can constitute a valid psychiatric disorder or whether it is more accurate to see them as maladaptive responses to adverse events, harsh or punitive parenting, or a threatening environment. Longitudinal studies have demonstrated that, for some youth, early patterns of disruptive behavior may become a lifelong pervasive repertoire culminating in adult antisocial personality disorder. The etiology of enduring patterns of aggressive behavior is widely accepted as a convergence of multiple contributing factors, including biologic, temperamental, learned, and psychological conditions. Risk factors for the development of aggressive behavior in youth include childhood maltreatment such as physical or sexual abuse, neglect, emotional abuse, and overly harsh and punitive parenting. Chronic exposure to violence in the media, including

television, video games, and music videos, may promote lower levels of empathy in children, which may add a risk factor for the development of aggressive behavior.

Conduct disorder is an enduring set of behaviors in a child or adolescent that evolves, usually characterized by aggression and violation of the rights of others. Youth with conduct disorder often demonstrate behaviors in the following four categories: physical aggression or threats of harm to people, destruction of their property or that of others, theft or acts of deceit, and frequent violation of age-appropriate rules. Conduct disorder is associated with many other psychiatric disorders, including ADHD, depression, and learning disorders. It is also associated with certain psychosocial factors, including childhood maltreatment, harsh or punitive parenting, family discord, lack of appropriate parental supervision, lack of social competence, and low socioeconomic level. The DSM-5 criteria require three persistent specific behaviors of 15 conduct disorder symptoms listed, over the past 12 months, with at least one of them present in the past 6 months. Conduct disorder symptoms include bullying, threatening, or intimidating others, and staying out at night despite parental prohibition. DSM-5 also specifies that when truancy from school is a symptom, it begins before 13 years of age. The disorder may be diagnosed in a person older than 18 years only if the person does not meet the criteria for antisocial personality disorder.

DSM-5 includes specifiers denoting the severity of the disorder, including “mild” in which there are few conduct problems above those needed to make the diagnosis and behaviors cause only minor harm to others. In “moderate” cases, symptoms exceed the minimum; however, there is less confrontation that may cause harm to individuals than in “severe” cases. According to DSM-5, the “severe” level shows many conduct problems over the minimal diagnostic criteria or conduct problems that cause considerable harm to others. DSM-5 has also added the following specifier: “With limited prosocial emotions.” For this, the individual must show a persistent interpersonal and emotional pattern that can be characterized by at least two of the following: (1) Lack of remorse or guilt, (2) callous lack of empathy, (3) unconcerned about performance, (4) shallow or deficient affect. Individuals with conduct disorder who qualify for this specifier are more likely to have a childhood-onset type and meet the criteria for a “severe” disorder. Children with conduct disorder engage in severe, repeated acts of

aggression that can cause physical harm to themselves and others and frequently violate the rights of others. Children with conduct disorder usually have behaviors characterized by aggression to persons or animals, destruction of property, deceitfulness or theft, and multiple violations of rules, such as truancy from school. These behavioral patterns cause distinct difficulties in school life as well as in peer relationships. Conduct disorder has three subtypes based on the age of onset of the disorder. Childhood-onset subtype, in which at least one symptom has emerged repeatedly before age 10 years; adolescent-onset type, in which no characteristic persistent symptoms occur until after age 10 years; and unspecified-onset, in which age of onset is unknown. Although some young children show persistent patterns of behavior consistent with violating the rights of others or destroying property, the diagnosis of conduct disorder in children appears to increase with age. Epidemiologic surveys indicate that geographic locations representing a broad range of different cultures are not associated with significant variability in prevalence rates of either oppositional defiant disorder or conduct disorder. A longitudinal study of population density and antisocial behaviors in youth found no relationship in children 4 to 13 years of age between conduct problems and density of the living area. However, higher rates of conduct problems were self-reported by youths 10 to 17 years who lived in higher-density communities.



Table 2-29
Conduct Disorder

	DSM-5	ICD-10
Diagnostic name	Conduct Disorder	Conduct Disorder
Duration	≥1 criteria for ≥6 mo Other symptoms occur over ≥12 mo	≥6 mo
Symptoms	• Bullying others • Initiating physical fights • Using weapon to inflict	Repeated and pervasive pattern of antisocial behavior

	serious harm • Physical cruelty to others • Physical cruelty to animals • Confronting and robbing someone • Forcing sex • Fire setting • Property damage • Breaking into properties • Lying for personal gain • Stealing valuable items • Staying out late against parent's will < age 13 • Running away from home ≥2 times (or once with prolonged absence) • Frequent truancy <age 13	• Aggression • Defiance May be • Angry • Cruel • Bullying • Destructive • Lying • Truant/running away from home
Required number of symptoms	≥3	
Exclusions (not result of):		Age appropriate behavior
Symptom specifiers	**With limited prosocial emotions:** demonstrating at least two of the following over 12 mo in multiple contexts: • *Lack of remorse or guilt*	Conduct disorder confined to the family context (within nuclear family) Unsocialized conduct disorder (predominant

	• *Callous-–lack of empathy* • *Unconcerned about performance* • *Shallow or deficient affect*	aggression, often toward other children) Socialized conduct disorder (despite aggression, there is good integration into their peer group)
Course specifiers	**Childhood onset type:** < age 10 **Adolescent onset type:** ≥ age 10 **Unspecified onset**	
Severity specifiers	**Mild:** minimal symptoms, minor harm/consequences **Moderate:** intermediate symptoms and consequences **Severe:** many symptoms, considerable harm caused	

Diagnosis and Clinical Features

Conduct disorder does not develop overnight. Instead, many symptoms evolve until a consistent pattern develops that involves violating the rights of others. Very young children are unlikely to meet the criteria for the disorder because they are not developmentally able to exhibit the symptoms typical of older children with conduct disorder. A 3-year-old does not break into someone's home, steal with confrontation, force someone into sexual activity, or deliberately use a weapon that can cause serious harm. School-age children, however, can become bullies, initiate physical fights, destroy property, or set fires. Table 2-29 compares the diagnostic approaches to conduct disorder.

The average age of onset of conduct disorder is younger in boys than in girls. Boys most commonly meet the diagnostic criteria by 10 to 12 years of

age, whereas girls met them closer to 14 to 16 years of age.

Children who meet the criteria for conduct disorder express their overt aggressive behavior in various forms. Aggressive antisocial behavior can take the form of bullying, physical aggression, and cruel behavior toward peers. Children may be hostile, verbally abusive, impudent, defiant, and negativistic toward adults. Persistent lying, frequent truancy, and vandalism are common. In severe cases, destructiveness, stealing, and physical violence often occur. Some adolescents make little effort to conceal their antisocial behavior. Sexual behavior and regular use of tobacco, liquor, or illicit psychoactive substances begin unusually early for such children and adolescents. Suicidal thoughts, gestures, and acts are frequent in children and adolescents with conduct disorder who are in conflict with peers, family members, or the law and are unable to problem-solve their difficulties.

Some children with aggressive behavioral patterns have impaired social attachments, as evidenced by their difficulties with peer relationships. Some may befriend a much older or younger person or have superficial relationships with other antisocial youngsters. Many children with conduct problems have poor self-esteem, although they may project an image of toughness. They may lack the skills to communicate in socially acceptable ways and appear to have little regard for the feelings, wishes, and welfare of others. Children and adolescents with conduct disorders often feel guilt or remorse for some of their behaviors, but try to blame others to avoid punishment.

Many children and adolescents with conduct disorder suffer from the deprivation of having few of their dependency needs met and may have had either overly harsh parenting or a lack of appropriate supervision. The deficient socialization of many children and adolescents with conduct disorder can be expressed in physical violation of others and, for some, in sexual violation of others. Severe punishments for behavior in children with conduct disorder almost invariably increase their maladaptive expression of rage and frustration rather than ameliorating the problem.

In evaluation interviews, children with aggressive conduct disorders are typically uncooperative, hostile, and provocative. Some have a superficial charm and compliance until they discuss their problem behaviors. Then, they often deny any problems. If the interviewer persists, the child may attempt to justify misbehavior or become suspicious and angry about the source of the

examiner’s information and perhaps bolt from the room. Most often, the child becomes angry with the examiner and expresses resentment of the examination with open belligerence or sullen withdrawal. Their hostility is not limited to adult authority figures, but they also direct it to their peers and younger children. They often bully those who are smaller and weaker. By boasting, lying, and expressing little interest in a listener’s responses, such children reveal their lack of trust in adults to understand their position.

Evaluation of the family situation often reveals severe marital disharmony, which initially may center on disagreements about the management of the child. Because of a tendency toward family instability, parent surrogates are often in the picture. Children with conduct disorder are more likely to be unplanned or unwanted babies. The parents of children with conduct disorder, especially the father, have higher rates of antisocial personality disorder or alcohol use disorder. Aggressive children and their families show a stereotyped pattern of impulsive and unpredictable verbal and physical hostility. A child’s aggressive behavior rarely seems directed toward any definable goal and offers little pleasure, success, or even sustained advantages with peers or authority figures.

In other cases, conduct disorder includes repeated truancy, vandalism, and physical severe aggression or assault against others by a gang, such as mugging, gang fighting, and beating. Usually, children who become part of a gang have the skills for age-appropriate friendships. They are likely to show concern for the welfare of their friends or their gang members and are unlikely to blame them or inform on them. In most cases, gang members have a history of adequate or even excessive conformity during early childhood that ended when the youngster became a member of the delinquent peer group, usually in preadolescence or during adolescence. Also present in history is some evidence of early problems, such as marginal or poor school performance, mild behavior problems, anxiety, and depressive symptoms. Some family social or psychological pathology is usually evident. Patterns of paternal discipline are rarely ideal and can vary from harshness and excessive strictness to inconsistency or relative absence of supervision and control. The mother has often protected the child from the consequences of early mild misbehavior but does not seem to encourage delinquency actively. Delinquency, also called juvenile delinquency, is most often associated with conduct disorder but can also result from other psychological or neurologic

disorders.

Pathology and Laboratory Examination. No specific laboratory test or neurologic pathology helps make the diagnosis of conduct disorder. Some evidence indicates that amounts of certain neurotransmitters, such as serotonin in the CNS, are low in some persons with a history of violent or aggressive behavior toward others or themselves. Whether this association is related to the cause or is the effect of violence or is unrelated to the violence is not clear.

Differential Diagnosis

Disturbances of conduct, including impulsivity and aggression, may occur in many childhood psychiatric disorders, ranging from ADHD to oppositional defiant disorder, to disruptive mood dysregulation disorder, to major depression, to bipolar disorder, specific learning disorders, and psychotic disorders. Therefore, clinicians must obtain a comprehensive history of the chronology of the symptoms to determine whether the conduct disturbance is a transient or an enduring pattern. Isolated acts of aggressive behavior do not justify a diagnosis of conduct disorder; an entrenched pattern must be present. The relationship of conduct disorder to oppositional defiant disorder is still under debate. Typically, we think of oppositional defiant disorder as a mild precursor of conduct disorder, without the violation of rights, likely to be diagnosed in younger children who may be at risk for conduct disorder. Children who progress from oppositional defiant disorder to conduct disorder over time maintain their oppositional characteristics, and some evidence indicates that the two disorders are independent. Currently, in the DSM-5, oppositional defiant disorder and conduct disorder are considered distinct, but they can coexist. Many children with oppositional defiant disorder do not develop conduct disorder, and conduct disorder emerging in adolescence is not necessarily preceded by oppositional defiant disorder. The main distinguishing clinical feature between these two disorders is that in conduct disorder, the youths violate the fundamental rights of others, whereas, in oppositional defiant disorder, hostility and negativism fall short of seriously violating the rights of others.

Mood disorders are often present in children who exhibit irritability and

aggressive behavior. We should rule out both major depressive disorder and bipolar disorder, but the full syndrome of conduct disorder can occur during the onset of a mood disorder. Substantial comorbidity exists between conduct disorder and depressive disorders. A recent report concludes that the high correlation between the two disorders arises from shared risk factors for both disorders rather than a causal relation. Thus, a series of factors, including family conflict, adverse life events, an early history of conduct disturbance, level of parental involvement, and affiliation with delinquent peers, contribute to the development of affective disorders and conduct disorder. This comorbidity is not the case with oppositional defiant disorder, which we cannot diagnose if it occurs exclusively during a mood disorder.

ADHD and learning disorders are commonly associated with conduct disorder. Usually, the symptoms of these disorders predate the diagnosis of conduct disorder. Substance use disorders are also more common in adolescents with conduct disorder than in the general population. Evidence indicates an association between fighting behaviors as a child and substance use as an adolescent. Once there is a pattern of drug use, this pattern may interfere with the development of positive mediators, such as social skills and problem-solving, which could enhance the remission of the conduct disorder. Thus, once substance abuse develops, it may promote the continuation of the conduct disorder. OCD also frequently seems to coexist with disruptive behavior disorders. All the disorders described here should be noted when they co-occur. Children with ADHD often exhibit impulsive and aggressive behaviors that may not meet the full criteria for conduct disorder.

Damien, age 12 years, was referred for psychiatric evaluation after being picked up by police for truancy and running away from home. Damien explained that he just wanted to get out of his house and see his friends. He does not like to be at home because his mother tries to tell him what to do. Damien’s mother says that he left and stayed out overnight multiple times in the past year, but that he usually returns the next morning. She complains that he is always in trouble. He has shoplifted on several occasions that she knows of, the first time at age 8 years. She suspects that he also steals from neighbors or school. The police have been involved on many occasions, including for truancy, staying out all

night, stealing from a neighborhood store, and smoking marijuana. Damien has a quick temper, and his mother knows he was involved in several fights over the past year in the neighborhood. Damien is particularly cruel to his younger brother, continually taunting and teasing him.

Damien's mother stated that he continually lies, sometimes for no apparent reason. When he was 6 years of age, he was fascinated with fire and set several small fires at home, fortunately with no serious injury or damage. Damien's mother was tearful when she disclosed that Damien is just like his no-good father and that she wished she never had him. Damien initially refused to answer questions and turned away scowling, but gradually began to talk. Damien presented a tough image with an indifferent attitude toward the interviewer. Damien denied any abuse at home, saying that he ran off because he was bored. However, upon further questioning, Damien admitted that his mother's previous boyfriend, who was in the home when Damien was between 6 and 8 years of age, used to hit him with a belt when he got out of line. Damien justified his behaviors as just having fun. He said others provoked the fights. Damien also denied using weapons, although he bragged about breaking the nose of another youth.

Damien's school records indicate that he required an Individualized Educational Plan (IEP) when he was in the second grade and was evaluated for symptoms of ADHD when he was in first grade. His psychiatrist prescribed methylphenidate; however, the family did not continue with treatment, and he is currently on no medication. Damien is currently in sixth-grade special education classes, having failed and repeated fifth grade. Damien's grades are failing, and he may have to repeat sixth grade. Damien admits to truancy on several occasions this year in addition to his problems with completing schoolwork. His previous evaluation indicates that child protective services evaluated the family for possible neglect when he was 5 years of age after he and his brother were found barefoot on the street late one evening without his mother. Damien's family was referred for counseling and never attended. Both Damien's parents have a history of drug and alcohol abuse. Damien's birth was unplanned, and his mother used drugs during pregnancy. His parents separated soon after his birth, and his mother

returned to live with her parents briefly. Damien and his mother moved to live with her boyfriend when Damien was 1 year of age after she became pregnant with his younger brother. Damien's mother's relationship ended within a year, and only Damien, his mother, and his brother live in their apartment. Damien's mother has worked several different jobs, and Damien wonders if she has a drinking problem.

Comorbidity

ADHD and conduct disorder often coexist, with ADHD often predating the development of conduct disorder, and not infrequently substance abuse. CNS injury, dysfunction, or damage predispose a child to impulsivity and behavioral disturbances, which sometimes evolve into conduct disorder.

Course and Prognosis

Negative prognostic signs for conduct disorder include young age, high number of symptoms, and severe symptoms. This finding is true partly because those with severe conduct disorder seem to be most vulnerable to comorbid disorders later in life, such as mood disorders and substance use disorders. A longitudinal study found that, although assaultive behavior in childhood and parental criminality predict a high risk for incarceration later in life, the diagnosis of conduct disorder does not correlate with imprisonment. Mild symptoms, no coexisting psychopathology, and normal intellectual functioning are good prognostic signs.

Treatment

Psychosocial Interventions. Early sustained preventive interventions (e.g., at a kindergarten age) can significantly alter the course and prognosis of aggressive behavior. A screening program used with kindergarteners predicted lifetime disruptive behavior disorder by age 18 years, with the highest risk group demonstrating an 82 percent chance of a disruptive behavior diagnosis without intervention. One prevention program, *the Fast Track Preventive Intervention,* randomized 891 kindergarteners to either a 10-year prevention program or a control condition. The 10-year intervention

included parent behavior management, child social cognitive skills, reading, home visiting, mentoring, and classroom curricula. The children in the Fast Track Intervention were less likely to develop conduct disorder during those 10 years and for 2 years after that.

A meta-analysis of controlled trials of CBT programs indicates that CBT can result in significant reductions in conduct-disordered symptoms in children and adolescents. CBT treatment interventions that are proven to be efficacious include the following:

Kazdin's Problem-Solving Skills Training (PSST) in which a 12-week sequential program helps children develop problem-solving solutions when faced with conflictual situations. Assignments called "supersolvers" provide vignette situations in which children can practice these techniques. Therapists can add a companion program, Parent Management Training (PMT), to the intervention, but PSST can be useful even without the parent component. Another CBT-based intervention, the Incredible Years (IY), targeting young children from 3 to 8 years, is administered over 22 weeks and delivers sessions to the child and has a parent training component and a teacher training. Another CBT-based intervention is the Anger Coping Program, an 18-session intervention for school-age children in grades 4 to 6 focused on a child's increased development of emotion recognition and regulation, and managing anger. Anger coping strategies include distraction, self-talk, perspective taking, goal setting, and problem-solving.

Overall, treatment programs have been more successful in decreasing overt symptoms of conduct, such as aggression, than the covert symptoms, such as lying or stealing. Treatment strategies for young children that focus on increasing social behavior and social competence reduce aggressive behavior. In one study, 548 third graders were administered a school-based intervention instead of a regular health curriculum in several public schools in North Carolina, called *Making Choices: Social Problem Solving Skills for Children* (MC), along with supplemental teacher and parent components. Compared with third graders receiving the routine health curriculum, children exposed to the MC program were rated lower on the posttest social and overt aggression, and higher on social competence. Also, they scored higher on information-processing skills. These findings support the notion that school-based prevention programs have the potential to strengthen social and emotional skills and diminish aggressive behavior among normal populations

of school-age children. School settings can also use behavioral techniques to promote socially acceptable behavior toward peers and to discourage covert antisocial incidents.

Psychopharmacologic Interventions. The efficacy of psychopharmacologic interventions includes several placebo-controlled studies of risperidone for aggression in youth associated with disruptive behavior disorders or mental retardation. Also, risperidone is superior to placebo in reducing aggressive behavior in a large 6-month placebo-substitution study. One randomized, double-blind placebo-controlled trial with quetiapine also showed efficacy for aggressive behavior. Early studies of antipsychotics, most notably haloperidol, reported decreased aggressive and assaultive behaviors in children with a variety of psychiatric disorders. Atypical antipsychotics risperidone, olanzapine, quetiapine, ziprasidone, and aripiprazole have generally replaced the older antipsychotics in clinical practice due to their comparable efficacy and improved side effect profiles. Side effects of second-generation antipsychotics include sedation, increased prolactin levels (with risperidone use), and extrapyramidal symptoms, including akathisia. In general, however, the atypical antipsychotics appear to be well tolerated. A study of sodium valproate in youth with conduct disorder showed that those who responded most robustly exhibited aggression characterized by agitation, dysphoria, and distress. Although early trials suggested that carbamazepine was useful to control aggression, a double-blind, placebo-controlled study did not show the superiority of carbamazepine over placebo in decreasing aggression. A pilot study found that clonidine may decrease aggression. The SSRIs, including fluoxetine, sertraline, paroxetine, and citalopram, are used clinically to target symptoms of impulsivity, irritability, and mood lability, which frequently accompany conduct disorder. Conduct disorder often coexists with ADHD, learning disorders, and, over time, mood disorders, and substance-related disorders; thus, we also need to treat concurrent disorders.

Epidemiology

In the United States, conduct disorder ranges from 6 to 16 percent for males, and from 2 to 9 percent for females. The ratio of conduct disorder in males

compared to females ranges from 4:1 to as much as 12:1. Conduct disorder occurs with higher frequency in the children of parents with antisocial personality disorder and alcohol use disorder than in the general population. The prevalence of conduct disorder and antisocial behavior is associated with socioeconomic factors, as well as parental psychopathology.

Etiology

A meta-analysis of longitudinal studies indicates that the most critical risk factors that predict conduct disorder include impulsivity, physical or sexual abuse or neglect, poor parental supervision and harsh and punitive parental discipline, low IQ, and poor school achievement.

Parental Factors. Harsh, punitive parenting characterized by severe physical and verbal aggression is associated with the development of children's maladaptive aggressive behaviors. Chaotic home conditions are associated with conduct disorder and delinquency. Divorce itself is not necessarily a risk factor, but the persistence of hostility, resentment, and bitterness between divorced parents may be the more significant contributor to maladaptive behavior. Parental psychopathology, child abuse, and negligence often contribute to conduct disorder. Sociopathy, alcohol use disorder, and substance use disorder in the parents are associated with conduct disorder in their children. Parents may be so negligent that a child's care is shared by relatives or assumed by foster parents. Many such parents were scarred by their upbringing and tended to be abusive, negligent, or engrossed in getting their personal needs met.

Studies indicate that parents of children with conduct disorder have high rates of severe psychopathology, including psychotic disorders. Data show that children who exhibit a pattern of aggressive behavior often experience physically or emotionally harsh parenting.

Genetic Factors. A study of more than 6,000 male, female, and opposite-sex twins found that genetic and environmental factors accounted for proportionally the same amount of variance in males and females. Genetic or shared environmental factors exert different effects on males and females in childhood conduct disorder, but by adulthood, the gender-specific influences on antisocial behavior are no longer apparent. The sex-specific effects on

antisocial behavior in youth along with the replicated finding of a potential role for the X-linked monoamine oxidase A gene in the etiology of antisocial behavior leads to the need for further genetic investigation of conduct disorder on the X chromosome and for analyses of these behaviors to be done separately by gender.

Sociocultural Factors. Youth living in population-dense areas have higher rates of aggression and delinquency. Unemployed parents, lack of a supportive social network, and lack of positive participation in community activities seem to predict conduct disorder. Associated findings that may influence the development of conduct disorder in urban areas are exposure to and prevalence of substance use. A survey of alcohol use and mental health in adolescents found that weekly alcohol use among adolescents is associated with increased delinquent and aggressive behavior. Significant interactions between frequent alcohol use and age indicated that those adolescents with weekly alcohol use at younger ages were most likely to exhibit aggressive behaviors and mood disorders. Although drug and alcohol use does not cause conduct disorder, it increases the risks associated with it. Drug intoxication itself can also aggravate the symptoms. Thus, all factors that increase the likelihood of regular substance use may promote and expand the disorder.

Violent Video Games and Violent Behavior. Longitudinal studies corroborate the contribution of media violence, including video gaming in middle-school children, with the expression of aggression in those adolescents. A review of the literature on the effect of violent video games on children and adolescents revealed that violent video game playing is related to aggressive affect, physiologic arousal, and aggressive behaviors. It stands to reason that the degree of exposure to violent games and the more restriction of activity would be related to a greater preoccupation with violent themes.

Psychological Factors. Poor emotion regulation among youth is associated with higher rates of aggression and conduct disorder. Emotion regulation is associated with social competence and can be observed even in children of preschool age—those children with greater degrees of emotion dysregulation exhibit higher levels of aggression. Poor modeling of impulse

control and the chronic lack of having their own needs met leads to a less developed sense of empathy.

Neurobiologic Factors. Neuroimaging studies utilizing MRI have used voxel-based morphometry methods to compare structural brain differences between children with conduct disorder compared to healthy controls. Studies have reported that children with conduct disorder had decreased gray matter in limbic brain structures and the bilateral anterior insula and left amygdala compared to healthy controls. A study investigated structural brain differences in children comorbid for oppositional defiant disorder or conduct disorder and ADHD compared to those with ADHD alone and healthy controls. Findings included decreased gray matter in ADHD and ADHD comorbid for oppositional defiant disorder or conduct disorder compared to controls in regions including bilateral temporal and occipital cortices and the left amygdala.

Neurotransmitter studies in children with conduct disorder suggest a low level of plasma dopamine β-hydroxylase, an enzyme that converts dopamine to norepinephrine, leading to a hypothesis of decreased noradrenergic functioning in conduct disorder. Other studies of conduct-disordered juvenile offenders have found high plasma serotonin levels in the blood. Evidence indicates that blood serotonin levels correlate inversely with levels of 5-HIAA in the CSF and that low 5-HIAA levels in CSF correlate with aggression and violence.

NEUROLOGIC FACTORS. An EEG study investigating resting frontal brain electrical activity, emotional intelligence, aggression, and rule-breaking in 10-year-old children found that aggressive children had significantly higher relative right frontal brain activity at rest compared with nonaggressive children. Frontal resting brain electrical activity likely reflects the ability to regulate emotion. Boys tended to show lower emotional intelligence than girls and more significant aggressive behavior than girls. No relationship, however, was found between emotional intelligence and pattern of frontal EEG activation.

Child Abuse and Maltreatment. Evidence shows that children chronically exposed to violence, physical or sexual abuse, and neglect,

particularly at a young age, are at high risk for demonstrating aggression. A study of female caregivers exposed to intimate partner violence revealed a strong association with offspring aggression and mood disturbance. Severely abused children and adolescents tend to be hypervigilant; in some cases, they misperceive benign situations as directly threatening and respond defensively with violence. Not all expressed aggressive behavior in adolescents is synonymous with conduct disorder; however, youth with a repetitive pattern of hypervigilance and violent responses are likely to violate the rights of others.

▲ 2.13 Anxiety Disorders of Infancy, Childhood, and Adolescence: Separation Anxiety Disorder, Generalized Anxiety Disorder, and Social Anxiety Disorder (Social Phobia)

Anxiety disorders are among the most common disorders in youth, affecting 10 to 20 percent of children and adolescents. Although observable anxiety behaviors mark normative development in infants, anxiety disorders in childhood predict a wide range of psychological difficulties in adolescence, including additional anxiety disorders, panic attacks, and depressive disorders. Fear is an expected response to a real or perceived threat; however, anxiety is the anticipation of future danger. The main characteristic of all the anxiety disorders is a recurrent emotional and physiologic arousal in response to excessive perceptions of perceived threat or danger. Anxiety disorders commonly found in youth include separation anxiety disorder, generalized anxiety disorder, social anxiety disorder, and selective mutism. We classify them by how the anxiety is experienced, the situations that trigger it, and the course that it tends to follow.

Separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder in children are often considered together in the evaluation process and differential diagnosis, and in developing treatment strategies,

because they are highly comorbid and have overlapping symptoms. A child with separation anxiety disorder, generalized anxiety disorder, or social anxiety disorder has a 60 percent chance of having at least one of the other two disorders as well. Of children with one of the above anxiety disorders, 30 percent have all three of them. Children and adolescents may also have additional comorbid anxiety disorders such as specific phobia or panic disorder. Separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder are distinguished from each other by the types of situations that elicit excessive anxiety and avoidance behaviors.

SEPARATION ANXIETY DISORDER

Separation anxiety is a universal human developmental phenomenon emerging in infants younger than 1 year of age and marking a child's awareness of separation from his or her mother or primary caregiver. Normative separation anxiety peaks between 9 and 18 months and will diminish by about 2½ years of age, enabling young children to develop a sense of comfort away from their parents in preschool. Separation anxiety or stranger anxiety most likely evolved as a human response that has survival value. The expression of transient separation anxiety is also typical in young children entering school for the first time. Approximately 15 percent of young children display intense and persistent fear, shyness, and social withdrawal when faced with unfamiliar settings and people. Young children with this pattern of significant behavioral inhibition are at higher risk for the development of separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder. Behaviorally inhibited children, as a group, exhibit characteristic physiologic traits, including higher than average resting heart rates, higher morning cortisol levels than average, and low heart rate variability. Separation anxiety disorder is developmentally inappropriate and excessive anxiety related to separation from the primary attachment figure. According to the DSM-5, separation anxiety disorder is a level of fear or anxiety regarding separation from their parents or primary caregiver which is beyond developmental expectations.

Furthermore, there may be a pervasive worry that harm will come to a parent upon separation, which leads to extreme distress and sometimes nightmares. The DSM-5 requires the presence of at least three symptoms

related to excessive worry about separation from a significant attachment figure for at least 4 weeks. The worries often take the form of refusal to go to school, fears, and distress on separation, repeated complaints of physical symptoms such as headaches and stomachaches when anticipating separation, and nightmares related to separation issues.

GENERALIZED ANXIETY DISORDER

Children with generalized anxiety disorder have significant distress in activities of daily life, often focused on the child's fears of incompetence in many areas, including school performance and in social settings. Also, children with generalized anxiety disorder, according to DSM-5, experience at least one of the following symptoms: restlessness, being easily fatigued, "mind going blank," irritability, muscle tension, or sleep disturbance. Children with generalized anxiety disorder tend to feel fearful in multiple settings and expect more negative outcomes when faced with academic or social challenges, compared with peers. Children and adolescents with generalized anxiety disorder may experience symptoms of autonomic hyperarousal such as tachycardia, shortness of breath, or dizziness, and are more likely than nonanxious youth to experience sweating, nausea, or diarrhea when they become anxious. Children and adolescents with generalized anxiety disorder tend to be overly concerned about potential natural disasters such as earthquakes or floods, and these worries can interfere with their daily activities. Finally, children and adolescents with generalized anxiety disorder are continuously worried about the quality of their performance in academics, sports, and other activities, and often seek excessive reassurance about their performance.

SOCIAL ANXIETY DISORDER (SOCIAL PHOBIA)

Children who experience intense discomfort and distress in social situations and are impaired by their fear of scrutiny or humiliation have the diagnosis of social anxiety disorder. Their distress may be expressed in the form of crying, tantrums, avoidance, freezing, or even becoming "mute" in these situations. According to DSM-5, this disorder is characterized by consistent anxiety and distress in almost all social situations. Any situation in which the child feels

exposed to possible scrutiny by others can provoke fear or anxiety, and the child will often try to avoid these feared social situations. Children must experience anxiety in the presence of peers, not only with adults, in order to receive the diagnosis. A child or adolescent with social anxiety disorder may exhibit the performance only type, which targets a specific type of performing, such as fear of public speaking. The performance-only type typically manifests in school or academic settings in which one must perform public presentations, such as in front of classmates in school.

Social anxiety disorder has significant implications for future accomplishments, since it is associated with lower levels of satisfaction in leisure activities, increased rates of school dropout, less productivity in the workplace as adults, and increased rates of remaining single. Despite the significant impairment caused by social anxiety disorder, up to half of the individuals with the disorder do not receive treatment.

DIAGNOSIS AND CLINICAL FEATURES

Separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder are highly related in children and adolescence because, in most children, overlapping symptoms, as well as comorbid disorders, emerge. Generalized anxiety disorder is the most common anxiety disorder among youth, more common in adolescents than in younger children; in almost one-third of these cases, a child with generalized anxiety disorder also exhibits separation anxiety disorder and social anxiety disorder.

Diagnostic criteria for separation anxiety disorder, according to the DSM-5, include three of the following symptoms for at least 4 weeks: persistent and excessive worry about losing, or possible harm befalling, significant attachment figures; persistent and excessive worry that an untoward event can lead to separation from a significant attachment figure; persistent reluctance or refusal to go to school or elsewhere because of fear of separation; persistent and excessive fear or reluctance to be alone or without major attachment figures at home or without significant adults in other settings; persistent reluctance or refusal to go to sleep without being near a major attachment figure or to sleep away from home; repeated nightmares involving the theme of separation; repeated complaints of physical symptoms, including headaches and stomachaches, when separation from significant

attachment figures is anticipated; and recurrent excessive distress when separation from home or significant attachment figures is anticipated or involved. Table 2-30 lists the criteria for separation anxiety disorder.

The following case history demonstrates separation anxiety disorder along with autonomic arousal symptoms.

Jake was a 9-year-old boy who was referred for outpatient evaluation by his family physician. He refused to sleep in his room alone at night and exhibited violent tantrums each morning in order to avoid going to school. Jake expressed recurrent fears that something terrible would happen to his mother. He worried that she would get into a car accident or that there would be a fire at home, and his mother would die. Developmental history revealed that Jake was anxious and irritable as an infant and toddler. He had trouble adjusting to babysitters in the preschool years. There was a history of panic disorder, agoraphobia in the mother, and major depression in his father. Jake became more concerned and territorial over his mother when his father left the family, and his mother became depressed. Jake always kept track of his mother's whereabouts and insisted that she stay at home.

Nighttime was a particularly difficult time at home. When Jake's mother tried to get Jake to remain in his room, Jake would whine and cry and insist that his mother lie in bed with him until he is asleep. He also expected his mother to sleep across the hall from his room throughout the evening. Jake's mother reported that each evening her son would get up and peek through the crack in her bedroom door, as frequently as every 10 minutes, to be sure that she was still there. Jake reported frequent nightmares that his mother died and that monsters prevented him from rescuing his mother, taking him away from his family forever.

During the daytime, Jake would shadow his mother around the house. Jake would agree to play a game with his sister in the lower level of the house only if his mother was nearby. When Jake's mother went upstairs, he would interrupt the game and follow her upstairs. He refused to sleep at a friend's house. Frequently, at home, as the evening progressed, Jake described a queasy sensation in his stomach mixed with feelings of sadness.

On school days, Jake usually complained of stomachaches and tried to stay home. Jake appeared distressed and panicky and would become violent when his mother attempted to drop him off at school. Once at school, he seemed calmer and less distressed, but frequently was seen in the nurse's office, complaining of nausea and seeking to be sent home. (Adapted from case material from Gail A. Bernstein, M.D. and Anne E. Layne, Ph.D.)

The essential feature of separation anxiety disorder is extreme anxiety precipitated by separation from parents, home, or other familiar surroundings. In contrast, in generalized anxiety disorder, fears are extended to adverse outcomes for all kinds of events, including academics, peer relationships, and family activities. In generalized anxiety disorder, a child or adolescent experiences at least one recurrent physiologic symptom, such as restlessness, poor concentration, irritability, or muscle tension. In social anxiety disorder, the child's fears peak during performance situations involving exposure to unfamiliar people or situations. Children and adolescents with social anxiety disorder have extreme concerns about being embarrassed, humiliated, or negatively judged. In each of the preceding anxiety disorders, the child's experience can approach terror or panic. The distress is higher than that normally expected for the child's developmental level and is not otherwise explainable. Morbid fears, preoccupations, and ruminations characterize separation anxiety disorder. Children with anxiety disorders overestimate the probability of danger and the likelihood of a negative outcome. Children with separation anxiety disorder and generalized anxiety disorder become overly fearful that someone close to them will be hurt or that something terrible will happen to them or their families, especially when they are away from their families. Many children with anxiety disorders are preoccupied with health and worry that their families or friends will become ill. Fears of getting lost, being kidnapped, and losing the ability to be in contact with their families are predominant among children with separation anxiety disorder.



Table 2-30
Separation Anxiety Disorder

	DSM-5	ICD-10
Diagnostic name	Separation Anxiety Disorder	Separation Anxiety Disorder of Childhood
Duration	Children: ≥4 wk Adults: ≥6 mo	
Symptoms	Inappropriate fear of separation from a loved one or attachment figure: • Distressed when separated or anticipating separation • Worries about losing them or harm coming to them • Worry about possible forced separation • Hesitant to leave person or home • Fears being alone, without the person • Needs to be near that person to sleep • Nightmares about separation • Physical symptoms of anxiety when separated or anticipating separation	• Fear of separation causes anxiety • Anxiety during early childhood • Severity is inappropriate to situations and stage of development
Required number of	≥3	

symptoms		
Psychosocial consequences of symptoms	Marked distress and/or psychosocial impairment	Impaired functioning
Exclusions (not result of):	Another mental disorder	
Comments		Note: classified as an Emotional disorder with onset specific to childhood

Adolescents with anxiety disorders may not directly express their worries; however, their behavioral patterns often reflect either separation anxiety or other anxiety if they exhibit discomfort about leaving home, engage in solitary activities because of fears about how they will perform in front of peers, or have distress when away from their families. Separation anxiety disorder in children is often manifested at the thought of travel or in the course of travel away from home. Children may refuse to go to camp, a new school, or even a friend's house. Frequently, a continuum exists between mild anticipatory anxiety before separation to pervasive anxiety after the separation has occurred. Premonitory signs include irritability, difficulty eating, whining, staying in a room alone, clinging to parents, and following a parent everywhere. Often, when a family moves, a child displays separation anxiety by intense clinging to the mother figure. Sometimes, geographical relocation anxiety causes feelings of acute homesickness or psychophysiological symptoms that break out when the child is away from home or is going to a new country. The child yearns to return home and becomes preoccupied with fantasies of how much better the old home was. Integration into the new life situation may become extremely difficult. Children with anxiety disorders may retreat from social or group activities and express feelings of loneliness because of their self-imposed isolation.

Sleep difficulties are frequent in children and adolescents with any anxiety disorder or in severe separation anxiety; a child or adolescent may require having someone remain with him or her until he or she falls asleep. An

anxious child may awaken and go to a parent's bed or even sleep at the parents' door to diminish anxiety. Nightmares and morbid fears may be expressions of anxiety.

Associated features of most anxiety disorders include fear of the dark and imaginary worries. Children may have the feeling that eyes are staring at them and monsters are reaching out for them in their bedrooms. Children with separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder often complain of somatic symptoms and may be more sensitive to changes in their bodies compared to youth without anxiety disorders. Children with separation anxiety disorder, generalized anxiety disorder, or social anxiety disorder are often more emotionally sensitive than peers and more easily brought to tears. Frequent somatic complaints accompanying anxiety disorders include gastrointestinal symptoms, nausea, vomiting, and stomachaches; unexplained pain in various parts of the body; sore throats; and flu-like symptoms. Older children and adolescents typically complain of somatic experiences classically reported by adults with anxiety, such as cardiovascular and respiratory symptoms—palpitations, dizziness, faintness, and feelings of strangulation. Physiologic signs of anxiety are a part of the diagnostic criteria for generalized anxiety disorder, but they are more often also experienced by children with separation anxiety and social anxiety disorder than the general population. The following case history demonstrates a young adolescent with generalized anxiety disorder.

Rachel was a 13-year-old girl referred for an evaluation by her pediatrician based on her chronic gastrointestinal complaints without any organic illness. In the interview, Rachel appeared withdrawn and meek but responsive to questions. She endorsed several worries that included concerns about her health, her parents' safety, her school performance, and her peer relationships. Rachel's most significant worries were related to her health and safety. Rachel's mother reported that Rachel had recently been very reluctant to play outside because she feared she would contract Lyme disease from a tick bite or West Nile virus from a mosquito bite. Rachel was also very distressed by news reports about catastrophic events locally and around the world (e.g., kidnapping, crime, terrorism). Her family and teachers described Rachel as overly

conscientious about her schoolwork and as often being concerned about adult matters (e.g., finances, parents' job security). Symptoms that accompanied Rachel's worries primarily involved stomach pain and problems with falling asleep. Rachel tended to be quite perseverative, repetitively verbalizing her worries even after receiving reassurance. Rachel admitted that she worried for hours each day and could not "turn off" her worried thoughts.

Rachel was the product of a healthy pregnancy and delivery. Her medical history was unremarkable, except for frequent gastrointestinal pain since kindergarten. Rachel was described as irritable and difficult to soothe as an infant. Her developmental milestones were within normal limits. She was described as very obedient and had no history of externalizing behavior problems. She was very concerned about her academic performance from an early age and earned A's with an occasional B. Rachel was somewhat shy in social situations but well-liked by her peers. Family history included depression in her maternal grandmother and a maternal history of generalized anxiety disorder, social anxiety, and separation anxiety disorder as a child. Rachel had two younger siblings who were high achievers and without notable problems. (Adapted from case material from Gail A. Bernstein, M.D. and Ann E. Layne, Ph.D.)

The next case history demonstrates an adolescent with multiple anxiety and depressive disorders.

Kate is a 15-year-old tenth grader who lives with her biologic parents and two sisters, ages 9 and 14 years. Kate is a very articulate teen who has always been a good student, although she never volunteers answers in school unless her teachers call on her. She gets along well with her sisters when at home, but ever since she entered high school in the ninth-grade year, she declines invitations to go to friends' homes, has turned down opportunities to go to parties, and has even stopped going on outings with her sisters to the neighborhood mall and the movies. Kate reports that she gets too nervous and blushes when she is with friends outside of the classroom at school because she cannot think of anything to say to them.

She reports that she is embarrassed to go shopping or to the movies with her sisters because they often run into neighborhood peers along the way and stop to chat, and this makes her feel “stupid,” because even though she is the oldest, she does not say anything and believes that her sisters’ friends will laugh at her shyness. Recently, one of her former best friends confronted her about why she had stopped “hanging out” with her friends. Kate had stopped eating lunch with her friends in school because she felt humiliated when they would talk about their weekend plans, and even when they invited her to join, she would just look the other way and ignore the conversation. Kate had become isolated, even in school, and admitted to her sister that she was lonely. Kate was brought for an evaluation after her younger sister commented to her mother that Kate spent all of her time alone whenever her sisters saw their friends and that she looked sad and stressed out whenever she was around peers. Kate was down, always in poor spirits, and had stopped interacting with her sisters. On rare occasions, Kate’s younger sister had invited Kate to parties or friend’s homes, but Kate had declined and burst into tears.

Kate was evaluated by a child psychiatrist who made the diagnoses of social anxiety disorder, generalized anxiety disorder, and major depression and recommended a combination of treatment options, including cognitive-behavioral therapy and a trial of fluoxetine. Kate and her family decided to try the medication first. Kate started 10 mg of fluoxetine, and over the next month was titrated to a dose of 20 mg. By the third week of the medication trial, Kate was noticeably less resistant to going out with her sisters to places where they were likely to encounter peers. Her sisters noticed that she did not seem as stressed and started to sit with peers at lunch in the school cafeteria occasionally. She stated that she did not feel as self-conscious as she used to in class and was willing to go to a friend’s house. She still declined to go to a birthday party of a peer that she did not know very well. Kate continued on the same medication, and within 2 months, she was significantly less anxious in social situations. She occasionally complained of a stomachache but tolerated the medication well. Her family was impressed when she requested they plan a birthday party for her 16th birthday and decided to invite 10 friends.

Pathology and Laboratory Examination

No specific laboratory measures help in the diagnosis of separation anxiety disorder, generalized anxiety disorder, or social anxiety disorder.

DIFFERENTIAL DIAGNOSIS

The presence of separation anxiety is a developmentally expected feature in a young child and often does not represent an impairing condition. Thus clinical judgment must be used in distinguishing normal anxiety from separation anxiety disorder in this age group. In older school-age children, a child experiencing more than normal distress is apparent when refusing school regularly. For children who resist school, it is essential to distinguish whether fear of separation, general worry about performance, or more specific fears of humiliation in front of peers or the teacher are driving the resistance. In many cases in which anxiety is the primary symptom, all three of the above-feared scenarios come into play. In generalized anxiety disorder, the focus of the anxiety is not mainly separation.



Table 2-31
Common Characteristics in Childhood Anxiety Disorders

Criteria	Separation Anxiety Disorder	Social Anxiety Disorder	Generalize Anxiety Di
Minimum duration to establish diagnosis	At least 4 wk	Persistent, typically at least 6 mo	At least 6 m
Precipitating stressors	Separation from home or attachment figures	Social situations with peers or other specific situations	Pressure fo type of performa activities scored, s performa
Peer relationships	Good when no	Tentative,	May appea

	separation is involved	overly inhibited	eager to peers sou for reass
Sleep	Reluctance or refusal to sleep away from home or not near attachment figure	May experience insomnia	Often difficu asleep
Psychophysiological symptoms	Stomachaches, headaches nausea, vomiting, palpitations, dizziness when anticipating separation	May exhibit blushing, inadequate eye contact, soft voice, or rigid posture	Stomachac nausea, l the throa shortnes breath, d palpitatio anticipati performir activity
Differential diagnosis	GAD, Soc AD, major depressive disorder, panic disorder with agoraphobia, PTSD, oppositional defiant disorder	GAD, SAD, major depressive disorder, dysthymic disorder, selective mutism, agoraphobia	SAD, Soc A attention- deficit/hy disorder, obsessiv compulsi disorder, depressiv disorder,

GAD, generalized anxiety disorder; Soc AD, social anxiety disorder; SAD, separation anxiety disorder; PTSD, posttraumatic stress disorder.

Adapted from Sidney Werkman, M.D.

When depressive disorders occur in children, we should evaluate for possible comorbidities, such as separation anxiety disorder, as well. We can

make a comorbid diagnosis of separation anxiety disorder and depressive disorder when the patient meets both criteria; the two diagnoses often coexist. Panic disorder with agoraphobia is uncommon before 18 years of age; the fear is of being incapacitated by a panic attack rather than of separation from parental figures. School refusal is a frequent symptom in separation anxiety disorder but is not pathognomonic of it. Children with other diagnoses, such as specific phobias, or social anxiety disorder, or fear of failure in school because of learning disorder, may also lead to school refusal. When school refusal occurs in an adolescent, the severity of the dysfunction is generally higher than when it emerges in a young child. Table 2-31 presents similar and distinguishing characteristics of childhood separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder.

COURSE AND PROGNOSIS

The course and the prognosis of separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder are varied and are related to the age of onset, the duration of the symptoms, and the development of comorbid anxiety and depressive disorders. Young children who can maintain attendance in school, after-school activities and peer relationships generally have a better prognosis than children or adolescents who refuse to attend school and withdraw from social activities. The large multisite randomized clinical trial Child/Adolescent Anxiety Multimodal Study (CAMS) provided acute treatment for children and adolescents with one or more anxiety disorders with sertraline medication alone, CBT alone, or both together. The investigators found that predictors of future remission included younger age of initiation of treatment, lower severity of anxiety, absence of a comorbid depressive or anxiety disorder, and the absence of social anxiety disorder as the primary anxiety disorder. A follow-up study of children and adolescents with mixed anxiety disorders over 3 years reported that up to 82 percent no longer met the criteria for the anxiety disorder at follow-up. Of the group followed, 96 percent of those with separation anxiety disorder were remitted at follow-up. Most children who recovered did so within the first year. Early age of onset and later age at diagnosis were factors in this study that predicted slower recovery. Close to one-third of the group studied, however, had developed another psychiatric disorder within the follow-up period, and

50 percent of these children developed another anxiety disorder. Studies have shown significant overlap between separation anxiety disorder and depressive disorders. In cases with multiple comorbidities, the prognosis is more guarded. Longitudinal data indicate that some children with severe school refusal continue to resist attending school into adolescence and remain impaired for many years.

TREATMENT

The treatment of child and adolescent separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder are often considered together, given the frequent comorbidity and overlapping symptomatology of these disorders. A multimodal comprehensive treatment approach usually includes psychotherapy, most often CBT, family education, family psychosocial intervention, and pharmacologic interventions, such as SSRIs. The best evidence-based treatments for childhood anxiety disorders include CBT and SSRIs. The comparative efficacy of CBT, SSRI medication, and their combination (CBT + SSRI) in the treatment of childhood anxiety disorders was investigated in the NIMH-funded CAMS. This double-blind, placebo-controlled, multi-site study included 488 children and adolescents with separation anxiety disorder, generalized anxiety disorder, or social anxiety disorder. The investigators treated them with either CBT alone, SSRI medication (sertraline) alone, both CBT and sertraline, or placebo. After an acute treatment phase of 12 weeks, those in the combined CBT + sertraline group had an 80.7 percent response rate of much or very much improved on the clinical global improvement (CGI) rating. Response rates for the CBT-only and sertraline-only groups were 59.7 and 54.9 percent, respectively. The placebo response was 23.7 percent. Over time, during an open follow-up, the combination of CBT plus sertraline continued to provide the most efficacy. All three treatments—CBT, sertraline, and their combination—were superior to placebo and thus effective treatments in childhood anxiety, but combined treatment was most likely to help children and adolescents with anxiety disorders. A trial of CBT may be applied first, if available, when a child can function sufficiently to engage in daily activities while obtaining this treatment. For a child with severe impairment, however, a combination of treatments is recommended. CBT is a first-line evidence-based treatment for

childhood anxiety disorders. A meta-analysis reviewed 16 RCTs of CBT for childhood anxiety disorders and found CBT to be consistently superior to a wait-list control group or a psychological placebo group. Exposure-based CBT has received the most empiric support among psychotherapeutic interventions for anxiety disorders in youth and is superior to wait-list control groups in reducing impairment and symptoms of anxiety.

Several psychosocial interventions have been designed specifically for anxiety disorders in young children. In a randomized clinical trial of CBT for 4- to 7-year-old children, the children received a manualized intervention called “Being Brave: A Program for Coping with Anxiety for Young Children and their Parents.” The intervention used a combination of parent-only sessions and child-and-parent sessions. Response rate, measured as much or very much improved on the Clinical Global Improvement Scale for Anxiety, was 69 percent among completers versus 32 percent of the wait-list controls. The treated children showed significantly better CGI improvement on social anxiety disorder, separation anxiety disorder, and specific phobia, but not on generalized anxiety disorder. This treatment, a developmentally modified parent–child CBT, shows promise in young children.

Coaching Approach behavior and Leading by Modeling (the CALM program) is an intervention aimed at treating anxiety disorders in children younger than 7 years of age, who are too young for traditional CBT. The CALM program draws on previous work with children aged 2 to 7 years through interventions that target a child’s undesired behavior by modifying parents’ behavior, called PCIT. The CALM program is a 12-session manual-based intervention that provides live, individualized coaching via a bug-in-the-ear receiver worn by the parent during sessions. It incorporates exposure tasks and promotes “brave” behavior with parent coaching. A pilot study using the CALM program with nine patients with a mean age of 5.4 years found that all treatment completers (seven patients and families) were global responders, and all but one showed functional improvement. Adapting the PCIT model for anxiety disorders in young children appears to be a promising approach to treating anxiety in early childhood.

A meta-analysis of RCTs of antidepressant agents for childhood anxiety provides evidence that multiple SSRIs, including fluvoxamine, fluoxetine, sertraline, and paroxetine, are efficacious in the treatment of childhood anxiety. Based on this evidence, SSRIs are the first choice of medication in

the treatment of anxiety disorders in children and adolescents.

A large, multisite investigation by the NIMH (Research Units in Pediatric Psychopharmacology [RUPP]) confirmed the safety and efficacy of fluvoxamine in the treatment of childhood separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder. This double-blind, placebo-controlled study of 128 children and adolescents revealed that 76 percent of children in the group treated with fluvoxamine showed significant improvement compared with 29 percent of those in the placebo group. Response to medication was noticeable after only 2 weeks of treatment. Fluvoxamine dosages ranged from 50 to 250 mg/day in children and up to 300 mg/day in adolescents. Children and adolescents with less comorbid depressive symptoms had the best response. Youths who responded continued fluvoxamine for 6 months, and almost all of them continued to be responders at the 6-month mark.

Several other randomized clinical trials have also supported the efficacy of SSRIs in the treatment of child and adolescent anxiety disorders. A randomized, controlled trial found fluoxetine, at a dose of 20 mg/day, to be safe and effective for children with these disorders, with minor side effects including gastrointestinal distress, headache, and drowsiness. Also, a randomized clinical trial for the treatment of generalized anxiety disorder in children lends support for the efficacy of sertraline. Finally, a large industry randomized clinical trial of paroxetine in the treatment of children with social anxiety disorder found that paroxetine was associated with a response in 78 percent of children treated. The dose of paroxetine was 10 to 50 mg/day.

The FDA has placed a “black box” warning on antidepressants, including all of the SSRI agents, used in the treatment of any childhood disorder, because of concerns about increased suicidality; however, no individual childhood anxiety study has found a statistically significant increase in suicidal thoughts or behaviors.

Currently, we do not recommend using tricyclic drugs due to their potentially serious cardiac adverse effects. Other agents, including buspirone and β-adrenergic receptor antagonists such as propranolol, have been used clinically in children with anxiety disorders, but currently, no data support their efficacy. Diphenhydramine may be used in the short term to control sleep disturbances in children with anxiety disorders. Open trials and one double-blind, placebo-controlled study suggested that alprazolam, a

benzodiazepine, may help to control anxiety symptoms in separation anxiety disorder. Clonazepam has been studied in open trials and may be useful in controlling symptoms of panic and other anxiety symptoms.

Although SSRIs and CBT alone and in combination have demonstrated efficacy in the treatment of anxiety disorders in youth, approximately 20 to 35 percent of children and adolescents with anxiety disorders do not appear to benefit. Several novel agents may be useful, some based on their effect on the NMDA system. For example, D-cycloserine (DCS), currently FDA approved in the treatment of pediatric tuberculosis, is a partial receptor agonist of the NMDA system and is hypothesized to augment the benefits of exposure treatment for phobias. Some evidence suggests that DCS may increase the speed of exposure interventions; however, there is no proof of long-term gains. Riluzole is an antiglutamatergic agent that decreases glutamatergic transmission by inhibiting glutamate release and inactivation of sodium channels in cortical neurons and blocking GABA reuptake. Due to its antiglutamatergic effects, riluzole may provide augmentation in the treatment of OCD and generalized anxiety disorder. Another agent, memantine, an NMDA receptor antagonist with FDA approval in the treatment of Alzheimer's disease, has been hypothesized to decrease anxiety due to its influence on the glutamatergic system. Published case reports have provided mixed results.

Although most childhood anxiety disorders wax and wane over time, school refusal associated with separation anxiety disorder is a psychiatric emergency. A comprehensive treatment plan involves the child, the parents, and the child's peers and school. Family interventions are critical in the management of separation anxiety disorder, especially in children who refuse to attend school, so that firm encouragement of school attendance is maintained while providing appropriate support. When a return to a full school day is overwhelming, the treatment team and school should help arrange a program so the child can progressively increase the time spent at school. Graded contact with an object of anxiety is a form of behavior modification useful for any type of separation anxiety. Some severe cases of school refusal require hospitalization. Cognitive-behavioral modalities include exposure to feared separations and cognitive strategies, such as coping self-statements aimed at increasing a sense of autonomy and mastery.

In summary, evidence-based treatments for anxiety disorders have focused on SSRIs and CBT. SSRIs are both safe and efficacious in the treatment of childhood anxiety disorders; however, in severe disorders, the evidence suggests that optimal treatment is to provide both CBT and SSRI antidepressant agents simultaneously.

EPIDEMIOLOGY

The prevalence of anxiety disorders has varied with the age group of the children surveyed and the diagnostic instruments used. The lifetime prevalence of any anxiety disorder in children and adolescents ranges from 10 to 27 percent. Anxiety disorders are common in preschoolers as well, and follow a similar epidemiologic profile as in older children. An epidemiologic survey using the PAPA found that 9.5 percent of preschoolers met criteria for any anxiety disorder, with 6.5 percent exhibiting generalized anxiety disorder, 2.4 percent meeting criteria for separation anxiety disorder, and 2.2 percent meeting criteria for social anxiety disorder. Separation anxiety disorder is estimated to be about 4 percent in children and young adolescents. Separation anxiety disorder is more common in young children than in adolescents and occurs equally in boys and girls. The onset may occur during preschool years, but is most common in children 7 to 8 years of age. The rate of generalized anxiety disorder in school-age children is estimated to be approximately 3 percent, the rate of social anxiety disorder is 1 percent, and the rate of simple phobias is 2.4 percent. In adolescents, lifetime prevalence for panic disorder is 0.6 percent; the prevalence of generalized anxiety disorder is 3.7 percent.

ETIOLOGY

Biopsychosocial Factors

Multiple investigations have found evidence for the influences of parental psychopathology and parenting styles on the emergence of anxiety disorders in childhood. Longitudinal studies have found that parental overprotection correlates with an increased risk of the development of anxiety disorders in children, and insecure parent–child attachment is associated with higher-than-expected rates of anxiety disorders in childhood. It is also well known that

maternal depression and anxiety have led to an increased risk of anxiety and depression in children. Psychosocial factors, in conjunction with a child's temperament, influence the degree of separation anxiety evoked in situations of brief separation and exposure to unfamiliar environments. The temperamental trait of shyness and withdrawal in unfamiliar situations is associated with a higher risk of developing separation anxiety disorder, generalized anxiety disorder, social anxiety disorder, or all three during childhood and adolescence.

External life stresses often coincide with the development of the disorder. The death of a relative, a child's illness, a change in a child's environment, or a move to a new neighborhood or school is frequent in the histories of children with separation anxiety disorder. In a vulnerable child, these changes probably intensify anxiety.

Neurophysiologic correlations exist with behavioral inhibition (extreme shyness); children with this constellation have a higher resting heart rate and an acceleration of heart rate with tasks requiring cognitive concentration. Additional physiologic correlates of behavioral inhibition include elevated salivary cortisol levels, elevated urinary catecholamine levels, and more significant papillary dilation during cognitive tasks.

Neuroimaging studies of adolescents with anxiety show increased activation of the amygdala compared to nonanxious adolescents when presented with anxiety-provoking stimuli. Furthermore, anxious adolescents maintain the hyperactivation of the amygdala over time, rather than showing an attenuation of the effect as in nonanxious adolescents. Structural studies of the amygdala in adolescents with anxiety have led to conflicting results, with some studies finding increased amygdala volumes, whereas other studies are finding decreased amygdala volumes.

Social Learning Factors

Fear, in response to a variety of unfamiliar or unexpected situations, may be unwittingly communicated from parents to children by direct modeling. If a parent is fearful, the child will probably have a phobic adaptation to new situations, especially to a school environment. There are much data to suggest that overprotective parenting promotes increased interpersonal sensitivity in healthy children and increases the risk of social anxiety disorder in children

with behavioral inhibition or other anxiety disorders such as separation anxiety disorder. Some parents appear to teach their children to be anxious by overprotecting them from expected dangers or by exaggerating the dangers. For example, a parent who cringes in a room during a lightning storm teaches a child to do the same. A parent who is afraid of mice or insects conveys the affect of fright to a child.

Conversely, a parent who becomes angry with a child when the child expresses fear of a given situation, for example, when exposed to animals, may promote a phobic concern in the child by exposing the child to the intensity of the anger expressed by the parent. When parents have anxiety disorders, this magnifies the social learning factors in the development of anxiety reactions. These factors may be pertinent in the development of separation anxiety disorder as well as in generalized anxiety disorder and social anxiety disorder. A recent study found no association between psychosocial hardships, such as ongoing family conflict, and behavioral inhibition among young children. It appears that temperamental predisposition to anxiety disorders emerges as a highly heritable constellation of traits and due to psychosocial stressors.

Genetic Factors

Genetic studies suggest that genes account for at least one-third of the variance in the development of anxiety disorders. Heritability for anxiety disorders in children and adolescents ranges from 36 to 65 percent, with the highest estimates found in younger children with anxiety disorders. Two heritable characteristics—behavioral inhibition (the tendency toward fear and withdrawal in new situations) and physiological hyperarousal—have both been found to impart significant risk factors for the future development of an anxiety disorder. However, although the temperamental constellation of behavioral inhibition, excessive shyness, the tendency to withdraw from unfamiliar situations, and the eventual emergence of anxiety disorders have a genetic contribution, one-third to two-thirds of young children with behavioral inhibition do not appear to go on to develop anxiety disorders.

Family studies have shown that the offspring of adults with anxiety disorders are at an increased risk of having an anxiety disorder themselves. Separation anxiety disorder and depression in children overlap, and the

presence of an anxiety disorder increases the risk of a future episode of a depressive disorder. The current consensus on the genetics of anxiety disorders suggests that what is inherited is a general predisposition toward anxiety, causing heightened levels of arousal, emotional reactivity, and increased negative affect, all of which increase the risk of developing separation anxiety disorder, generalized anxiety disorder, and social anxiety disorder.

2.14 Selective Mutism

Selective mutism, believed to be related to social anxiety disorder, although an independent disorder, is characterized in a child by persistent lack of speaking in one or more specific social situations, most typically, the school setting. A child with selective mutism may remain wholly silent or near-silent, in some cases only whispering in a school setting. Although selective mutism often begins before age 5 years, it may not be apparent until the child should be reading or speaking aloud in school. The current conceptualization of selective mutism highlights a convergence of underlying social anxiety, along with an increased likelihood of speech and language problems leading to the failure to speak in certain situations. Typically, children with the disorder are silent during stressful situations, whereas some may verbalize almost inaudibly single-syllable words. Despite an increased risk for delayed speech and language acquisition in children with selective mutism, children with this disorder are fully capable of speaking competently when not in a socially anxiety-producing situation. Some children with the disorder will communicate with eye contact or nonverbal gestures but not verbally when at school. Otherwise, children with selective mutism speak fluently at home and in many familiar settings. Selective mutism is believed to be related to social anxiety disorder because of its expression primarily in selective social situations.



Table 2-32
Selective Mutism

	DSM-5	ICD-10
Diagnostic name	Selective Mutism	Elective Mutism
Duration	≥1 mo (not 1st month of school)	
Symptoms	Persistent lack of speech	Persistent lack of speech • In certain social situations

	• In certain social situations • Occurs when expected to speak • Has ability to speak appropriately in other situations	• Has ability to speak appropriately in other situations • Related to specific emotions • For example: anxiety, social withdrawal, and sensitivity
Psychosocial consequences of symptoms	Presence of psychosocial impairment	
Exclusions (not result of):	Another mental disorder Communication disorder Lack of proficiency with the spoken language in a given situation	Pervasive developmental disorders Developmental disorders of speech and language Transient mutism as part of separation anxiety in young children
Comments		Note: classified under Disorders of social functioning with onset specific to childhood and adolescence

DIAGNOSIS AND CLINICAL FEATURES

The diagnosis of selective mutism is not difficult to make after it is clear that a child has adequate language skills in some environments but not in others. The mutism may have developed gradually or suddenly after a disturbing experience. The age of onset can range from 4 to 8 years. Mute periods are

most commonly manifested in school or outside the home; in rare cases, a child is mute at home but not in school. Children who exhibit selective mutism may also have symptoms of separation anxiety disorder, school refusal, and delayed language acquisition. Because social anxiety is almost always present in children with selective mutism, behavioral disturbances, such as temper tantrums and oppositional behaviors, may also occur in the home. Compared to children with other anxiety disorders, except social anxiety disorder, children with selective mutism tend to have less social competence and more social anxiety. Table 2-32 lists the criteria for selective mutism.

Janine is a 6-year-old Chinese American first-grade girl who lives with her biologic mother, father, and siblings. Janine's parents reported a 2-year history of not speaking at school, beginning in kindergarten, or to any children or adults outside of her family, despite speaking at home. At home, she reportedly is animated and quite talkative with her immediate family and a few young cousins as well. Although she speaks to adult relatives outside of her immediate family, her communication is often limited to one-word responses to their questions. By her parents' report, Janine also exhibits extreme social anxiety, to the point of "freezing" in certain situations when the attention is on her. At the time of her evaluation, Janine had not received prior treatment. Janine speaks fluent English as well as Mandarin, and, according to her parents, met all developmental milestones on time and appeared to have above-average intelligence. They also reported that Janine enjoys dancing, singing, and imaginative play with her sisters.

During the initial evaluation, Janine failed to make eye contact or respond verbally to the intake clinician. Janine's parents reported that this behavior is typical of her when in a new situation but that she communicates nonverbally and makes eye contact with most people once she "gets to know them." On request, Janine's parents provided a videotaped recording of Janine playing at home with her sisters. In the video, Janine was animated and was speaking spontaneously and fluently without apparent impairment. Janine received diagnoses of selective mutism and social anxiety disorder. The psychiatrist recommended CBT,

and Janine began to see a therapist.

The therapist instructed Janine and her mother to come up with lists of easy, medium, and most challenging “speaking” situations and lists of small, medium, and large rewards. These lists then became the basis for assignments for exposures and reinforcement for speaking tasks that gradually increased in difficulty. CBT sessions included time with Janine and her mother together to review past and future assignments and time with Janine and the therapist alone.

When treatment began, Janine did not communicate at all verbally or nonverbally with the therapist. The therapist gradually developed a rapport with Janine utilizing less stressful tasks such as whispering to her mother with the therapist in the corner, then nodding yes or no, pointing, whispering to a stuffed animal, whispering to her mother while facing the therapist, and eventually responding to the therapist directly. The therapist used animal puppets to enable Janine to “warm-up” without talking directly to the therapist. After three sessions, Janine began to speak to the therapist in a whisper. Janine received stickers for completing each speaking assignment, and, after filling up the sticker charts, she received rewards (a small toy or treat from the reward list).

The therapist gave Janine assignments that involved her teacher and classmates. These were implemented gradually and included waving to the teacher, playing an audiotape of her saying “hello” to the teacher, whispering “hello” to the teacher, speaking “hello” to the teacher in a regular voice, and so on. After 14 sessions, Janine could speak in complete sentences during class when called on. She also spoke to her teacher in front of several other students.

During the last few sessions, Janine’s mother took an increasingly active role in assigning and following up on speaking assignments. When Janine entered the second grade, it took only a few days for her to speak to her teacher and most peers in class. After completion of therapy, Janine’s mother continued to monitor Janine’s speaking behaviors and to promote speaking in new situations by encouraging (and rewarding) Janine’s gradual successes with novel people and situations. (Adapted from case material from Lindsey Bergman, Ph.D., and John Piacentini, Ph.D.)

Pathology and Laboratory Examination

No specific laboratory measures are useful in the diagnosis or treatment of selective mutism.

DIFFERENTIAL DIAGNOSIS

Differential diagnosis of children who are silent in social situations emphasizes ruling out communications disorder, autism spectrum disorder, and social anxiety disorder. Social anxiety disorder may be comorbid with selective mutism. Once confirming that the child is fully capable of speaking when comfortable, but not in stressful situations, we should consider an anxiety disorder. Shy children may exhibit a transient muteness in new, anxiety-provoking situations. These children often have histories of not speaking in the presence of strangers and of clinging to their mothers. Most children who are mute on entering school improve spontaneously and may be described as having transient adaptation shyness. We should distinguish between selective mutism and intellectual disabilities, pervasive developmental disorders, and expressive language disorder. In these disorders, the child may have an inability, rather than a refusal, to speak. In mutism secondary to conversion disorder, the mutism is pervasive. Children introduced into an environment which uses a different language may be reticent to begin using the new language. We should only diagnose selective mutism when children also refuse to converse in their native language and when they have gained communicative competence in the new language but refuse to speak it.

COURSE AND PROGNOSIS

Children with selective mutism are often excessively shy during the preschool years, but the onset of the full disorder is usually not evident until age 5 or 6 years. Many very young children with early symptoms of selective mutism in a transitional period when entering preschool have a spontaneous improvement over several months and never fulfill the criteria for the disorder. A typical pattern for a child with selective mutism is to speak almost exclusively at home with the nuclear family but not elsewhere, especially not at school. Consequently, a child with selective mutism may

have academic difficulties or even failure due to a lack of participation. Children with selective mutism are typically shy, anxious, and at increased risk for a depressive disorder. Many children with early-onset selective mutism remit with or without treatment. Recent data suggest that fluoxetine may influence the course of selective mutism, and treatment enhances recovery. Children in whom the disorder persists often have difficulty forming social relationships. Teasing and scapegoating by peers may cause them to refuse to go to school. Some children with any form of severe social anxiety are rigid, compulsive, negativistic, and prone to temper tantrums and oppositional and aggressive behavior at home. Other children with the disorder tolerate the feared situation by communicating with gestures, such as nodding, shaking the head, and saying “Uh-huh” or “No.” In one follow-up study, about one-half of children with selective mutism improved within 5 to 10 years. Children who do not improve by age 10 years appear to have a long-term course and a worse prognosis. As many as one-third of children with selective mutism, with or without treatment, may develop other psychiatric disorders, particularly other anxiety disorders and depression.

TREATMENT

The optimal treatment is a multimodal approach using psychoeducation for the family, CBT, and SSRIs as needed. Preschool children may also benefit from a therapeutic nursery. For school-age children, individual CBT is the first-line treatment. Family education and cooperation are beneficial. Published data on the successful treatment of children with selective mutism is scant. However, reasonable evidence indicates that children with social anxiety disorder respond to various SSRIs, and, currently, CBT treatments have also demonstrated efficacy in children with anxiety disorders.

A recent report of 21 children with selective mutism treated in an open trial with fluoxetine suggested that this medication may be useful for childhood selective mutism. Reports have confirmed the efficacy of fluoxetine in the treatment of adult social anxiety disorder and at least one double-blind, placebo-controlled study using fluoxetine with children with mutism. A large NIMH-funded study of anxiety disorders in children and adolescents called Research Units in Pediatric Psychopharmacology (RUPP), has shown distinct superiority of fluvoxamine over placebo in the treatment of a variety of

childhood anxiety disorders. Children with selective mutism may benefit similarly to those with social anxiety disorder, given the current belief that it is a subgroup of social anxiety disorder. SSRI medications that have been shown in randomized, placebo-controlled trials to have benefit in the treatment of children with social anxiety disorder include fluoxetine (20 to 60 mg/day), fluvoxamine (50 to 300 mg/day), sertraline (25 to 200 mg/day), and paroxetine (10 to 50 mg/day).

EPIDEMIOLOGY

The prevalence of selective mutism varies with age, with younger children at increased risk for the disorder. According to the DSM-5, the point prevalence of selective mutism using clinic or school samples ranges between 0.03 and 1 percent, depending on whether a clinical or community sample is studied. A sizeable epidemiologic survey in the United Kingdom reported a prevalence rate of selective mutism to be 0.69 percent in children 4 to 5 years of age. Another survey in the United Kingdom identified 0.06 percent of 7-year-olds as having selective mutism. Young children are more vulnerable to the disorder than older ones. Selective mutism appears to be more common in girls than in boys. Clinical reports suggest that many young children spontaneously “outgrow” this disorder as they age; the longitudinal course of the disorder remains to be studied.

ETIOLOGY

Genetic Contribution

Selective mutism may have many of the same etiologic factors leading to the emergence of social anxiety disorder. In contrast to other childhood anxiety disorders, however, children with selective mutism are at higher risk for delayed onset of speech or speech abnormalities that may be contributory. However, in addition to the speech and language factor, one survey found that 90 percent of children with selective mutism met diagnostic criteria for social anxiety disorder. These children showed high levels of social anxiety without notable psychopathology in other areas, according to parent and teacher ratings. Thus, selective mutism may not represent a distinct disorder but may be a subtype of social anxiety disorder. Maternal anxiety,

depression, and heightened dependence needs occur in families of children with selective mutism, similar to families with children who exhibit other anxiety disorders.

Parental Interactions

Maternal overprotection and anxiety disorders in parents may exacerbate interactions that unwittingly reinforce selective mutism behaviors. Children with selective mutism usually speak freely at home and only exhibit symptoms when under social pressure either in school or other social situations. Some children seem predisposed to selective mutism after early emotional or physical trauma; thus, some clinicians refer to the phenomenon as *traumatic mutism* rather than selective mutism.

Speech and Language Factors

Selective mutism is conceptualized as an anxiety-based refusal to speak; however, a higher-than-expected proportion of children with the disorder have a history of speech delay. An intriguing finding suggests that children with selective mutism are at higher risk for a disturbance in auditory processing, which may interfere with the efficient processing of incoming sounds. For the most part, however, speech and language problems in children with selective mutism are subtle and cannot account for the diagnosis.

▲ 2.15 Obsessive-Compulsive Disorder in Childhood and Adolescence

Data suggest that up to 25 percent of cases of OCD have their onset by 14 years of age. The overall clinical presentation of OCD in youth is similar to that in adults; however, compared to adults, children and adolescents with OCD more often do not consider their obsessional thoughts or repetitive behaviors to be unreasonable. In milder cases of OCD, a trial of CBT is the initial intervention. OCD in youth is often treated successfully with SSRIs or CBT alone or in combination. The results of a large-scale, randomized,

placebo-controlled study called the Pediatric OCD Treatment Study (POTS), demonstrated that the highest rates of remission in pediatric OCD are with a combination of both serotonergic agents and CBT treatment.

DIAGNOSIS AND CLINICAL FEATURES

Children and adolescents with obsessions or compulsions present for treatment due to the excessive time that they devote to their intrusive thoughts and repetitive rituals. Some children see their compulsive rituals as reasonable responses to their extreme fears and anxieties. Nevertheless, they are aware of their discomfort and inability to carry out usual daily activities promptly due to the compulsions, such as getting ready to leave their homes to go to school each morning.

The most commonly reported obsessions in children and adolescents include extreme fears of contamination—exposure to dirt, germs, or disease—followed by worries related to harm befalling themselves, family members, or fear of harming others due to losing control over aggressive impulses. Also commonly reported are obsessional needs for symmetry or exactness, hoarding, and excessive religious or moral concerns. Typical compulsive rituals among children and adolescents involve cleaning, checking, counting, repeating behaviors, or arranging items. Associated features in children and adolescents with OCD include avoidance, indecision, doubt, and slowness to complete tasks. In most cases of OCD among youth, obsessions and compulsions are present. According to the DSM-5, diagnosis of OCD is identical to that of adults, with the note that young children may not be able to articulate the aims of their compulsions in diminishing their anxiety. The DSM-5 has also added the following specifiers: with good, fair, poor, or absent insight; that is, the greater the belief in the OCD obsessions and compulsions, the weaker the insight. An additional specifier indicates whether the individual has a current or past history of a tic disorder. We discuss this disorder in Chapter 9, where Table 9-4 designates the DSM-5 diagnostic criteria for OCD.

Many children and adolescents who develop OCD have an insidious onset and may hide their symptoms as long as possible so that their rituals will not be challenged or disrupted. A minority of children, particularly males with early-onset, may have a rapid unfolding of multiple symptoms within a few

months. OCD is often comorbid with anxiety disorders, ADHD, and tic disorders, especially Tourette syndrome. Children with comorbid OCD and tic disorders are more likely to exhibit counting, arranging, or ordering compulsions and less likely to manifest excessive washing and cleaning compulsions. The high comorbidity of OCD, Tourette syndrome, and ADHD has led investigators to postulate a shared genetic vulnerability to all three of these disorders. It is crucial to search for comorbidity in children and adolescents with OCD so that the treatments can be optimal.

Jason, a 12-year-old boy in the sixth grade, was brought for evaluation by his parents, who expressed concerns over his repeated questions and anxiety regarding developing acquired immunodeficiency syndrome (AIDS). Jason was a high-functioning and well-adjusted boy who abruptly began to exhibit extremely disruptive behaviors related to his fears of AIDS approximately 2 to 3 months before the evaluation. Jason's new behaviors included relentless concerns about contracting illness, washing rituals, repeated expressions of uncertainty over his behavior, seeking reassurance, repeating rituals, and avoidance.

Jason repeatedly expressed his fear and belief that he was at risk for human immunodeficiency virus (HIV) through exposure to multiple strangers who were infected. For example, while riding in the car, if Jason saw a stranger from the window who appeared to him to be ill-kempt, he experienced a surge of extreme anxiety and obsessively agonized about whether the stranger could have AIDS and had exposed him to it. Despite his parents' reassurances about his safety and lack of exposure to illness, Jason insisted on vigorously washing for approximately 1 hour each time he reached home after being out. Jason continually expressed doubts about his behavior. He often asked his parents, "Did I use the s___ word? Did I use the f___ word?" Reassurance was only slightly calming. Jason, previously an excellent student, began to lose the ability to focus on schoolwork. While reading passages from assigned materials, Jason frequently experienced severe anxiety, wondering if he had missed a word or misunderstood the sentence, and proceeded to reread the material. Completing a page of written material began to take Jason 30 to 60 minutes. Over several

weeks, he was less and less able to complete assignments, following which he became very distressed over his deteriorating grades.

During Jason's evaluation, his family history suggested that Jason's older sister had experienced a period in which she, too, had similar but milder anxieties, with less interference in functioning, and she had never received any treatment for those symptoms.

At the intake interview, Jason presented as a preoccupied and sad boy who was cooperative with questioning. He did not volunteer much information, and he allowed his parents to recount the extent of his symptoms. Jason believed that his relentless concerns were reasonable and that he required repeated reassurance from his parents in order to continue his daily activities. Jason met full diagnostic criteria for OCD. Symptoms of depression were present but not sufficient for major depressive disorder.

He began CBT; however, Jason was so fearful of deviating from his rituals that he was unable to participate fully in his treatment, and he became despondent about his future. Jason refused to go to school due to his increasing distress associated with reading and his shame regarding his diminishing academic performance. Given his limited progress during the first 2 months of CBT, his psychiatrist added fluoxetine, titrated to 40 mg/day. Over 3 weeks, there was some improvement, and Jason was more amenable to cooperating with his CBT treatment. CBT and SSRI treatment continued over the next 3 months regularly. Over time, Jason finally began to show some flexibility with his rituals, and he was able to decrease the amount of time he spent with rituals. Once he had found some relief from his symptoms, Jason was able to focus more on his schoolwork and his family life. Follow-up over the next year was positive; Jason had maintained his gains from treatment, with only minimal interference from residual OCD symptoms. Jason's academic achievement improved, he was able to engage in activities with friends, and he spent almost no time preoccupied with obsessional thoughts of illness and cleansing rituals. (Adapted from a case courtesy of James T. McCracken, M.D.)

Pathology and Laboratory Examination

No specific laboratory measures are useful in the diagnosis of OCD.

Even when the onset of obsessions or compulsions appears to be associated with a recent infection with GABHS, antigens and antibodies to the bacteria do not indicate a causal relationship between GABHS and OCD.

DIFFERENTIAL DIAGNOSIS

Developmentally appropriate rituals in the play and behavior of young children should not be confused with OCD in that age group. Preschoolers often engage in ritualistic play and request a predictable routine such as bathing, reading stories, or selecting the same stuffed animal at bedtime to promote a sense of security and comfort. These routines allay developmentally normal fears and lead to reasonable completion of daily activities. On the other hand, extreme fears drive the obsessions or compulsions, and this significantly interferes with daily function because of the excessive time that they consume and the extreme distress that ensues when they are interrupted. The rituals of preschoolers generally become less rigid by the time they enter grade school, and school-age children do not typically experience a surge of anxiety when they encounter small changes in their routine.

Children and adolescents with generalized anxiety disorder, separation anxiety disorder, and social anxiety disorder experience intense worries that they repeatedly express; however, these are mundane compared to obsessions, which are often so extreme that they appear bizarre. A child with generalized anxiety disorder typically worries repeatedly about performance on academic examinations, whereas a child with OCD may experience repeated intrusive thoughts that he may harm someone he loves. The compulsions of OCD are not present in other anxiety disorders; however, children with autism spectrum disorders often display repetitive behaviors that may resemble OCD. In contrast with the rituals of OCD, children with autism spectrum disorder are not responding to anxiety, but are more often exhibiting stereotyped behaviors that are self-stimulating or self-comforting.

Children and adolescents with tic disorders such as Tourette syndrome may display complex repetitive, compulsive behaviors similar to the compulsions seen in OCD. Children and adolescents with tic disorders are at higher risk for the development of concurrent OCD.

Severe OCD symptoms may be difficult to distinguish from delusional symptoms, especially when the obsessions and compulsions are bizarre. In most adults, and often in youth with OCD, despite an inability to control their obsessions or resist completing compulsions, insight into their lack of reasonableness is preserved. That is, an individual's conviction in their beliefs often does not reach delusional intensity. When insight is present and underlying anxiety can be described, even in the face of significant dysfunction due to bizarre obsessions and compulsions, the diagnosis of OCD is suspect.

COURSE AND PROGNOSIS

OCD with onset in childhood and adolescence is most often a chronic, waxing, and waning disorder with variability in severity and outcome. Follow-up studies suggest that up to 40 to 50 percent of children and adolescents recover from OCD with minimal residual symptoms. A study of childhood OCD treatment with sertraline resulted in close to 50 percent of participants experiencing complete remission and partial remission in another 25 percent with a follow-up time of 1 year. Predictors of the best outcome were in those children and adolescents without comorbid disorders, including tic disorders and ADHD. A study of 142 children and adolescents with OCD followed over 9 years at the Maudsley Hospital in England found 41 percent to have a persistence of OCD, with 40 percent exhibiting an additional psychiatric diagnosis at follow-up. The main predictor for persistent OCD was the duration of illness at the time of the initial assessment. Approximately half of the follow-up group was still receiving treatment, and half believed that they needed continued treatment.

Neuropsychological functioning may also play a role in outcome and prognosis. A study of 63 youth with OCD who completed the Rey-Osterrieth Complex Figure (ROCF) along with specific subtests of the Wechsler Intelligence Scale for Children, Third Edition (WISC-III), found that 5-minute recall accuracy from the ROCF was positively correlated with response to treatment, particularly CBT. These findings imply that poorer performance on the ROCF and poor response to therapy may be in part due to executive functioning difficulties and that treatment may need to be modified to account for these obstacles.

Overall, the prognosis is hopeful for most children and adolescents with mild to moderate OCD. In about 10 percent of cases, OCD may represent a prodrome of a psychotic disorder in children and adolescents. In youth with subthreshold OCD symptoms, there is a high risk of developing the full OCD disorder within 2 years. Childhood OCD is responsive to available treatments, resulting in improvement, if not complete remission, in the majority of cases.

TREATMENT

CBT and SSRIs are both efficacious treatments for OCD in youth. CBT geared toward children of varying ages is based on the principle of developmentally appropriate exposure to the feared stimuli coupled with response prevention, leading to diminishing anxiety over time on exposure to feared situations. CBT manuals can ensure that the therapist can make developmentally appropriate interventions and that the child and parents receive comprehensive education.

Treatment guidelines for children and adolescents with mild to moderate OCD recommend a trial of CBT before initiating medication. However, the POTS, a multi-site National Institute of Health (NIH)–funded investigation of sertraline and CBT, each alone and in combination, for the treatment of childhood-onset OCD, revealed that the combination was superior to either treatment alone. Each treatment alone also provided encouraging levels of response. The mean daily dose of sertraline was 133 mg/day in the group administered the combination treatment, and 170 mg/day for the sertraline alone group. Improvement with the pharmacologic intervention of childhood OCD usually occurs within 8 to 12 weeks of treatment. Most children and adolescents who experienced remission with acute treatment using SSRIs were still responsive over a year. Among youth with OCD who obtain a partial response to a therapeutic trial of SSRI treatment, augmentation with a short-term OCD-specific CBT leads to a significantly greater response. Evidence shows that higher treatment expectations by patients and families link to better treatment response, greater compliance with home-based CBT assignments, less drop out of treatment, and reduced impairment.

In addition to individual CBT, both family and group CBT interventions are efficacious in the treatment of childhood OCD. Family CBT (FCBT)

intervention in the treatment of OCD in youth increases the response rates. A controlled comparison of family CBT and psychoeducation and relaxation (PRT) in 71 families of children with OCD showed that clinical remission rates in the FCBT group were significantly higher than those in the PRT group. The FCBT treatment reduced parent involvement and accommodation in their affected child's symptoms, which led to decreased symptomatology.

A randomized controlled study investigating web-camera delivered FCBT (W-CBT) compared to a waitlist condition assigned 31 families to one of the above conditions. Assessments were conducted immediately before and after treatment and at a 3-month follow-up for the W-CBT group. The W-CBT group was superior to the waitlist control group on all primary outcome measures, with large effect sizes. Eighty-one percent of the W-CBT group responded, compared to 13 percent of the waitlist group. The children maintained the gains at the 3-month follow-up assessment. The authors conclude that W-CBT may be efficacious in the treatment of OCD in youth and may be a promising tool for future dissemination.

Exposure and response prevention (ERP) was studied in a group format in youth with OCD in a community-based program. Group-based ERP was effective in reducing OCD symptom severity and depressive symptoms, but not anxiety symptoms, in a naturalistic treatment setting for children with OCD and comorbid anxiety or depressive features.

Multiple randomized clinical trials establish the efficacy of SSRIs for OCD in youth. A meta-analysis of 13 studies of SSRIs, including sertraline, fluvoxamine, fluoxetine, and paroxetine, has provided evidence of the efficacy of SSRIs with a moderate effect size. A randomized controlled clinical trial of citalopram versus fluoxetine in youth with OCD found that citalopram was as safe and effective as fluoxetine for the treatment of OCD in children and adolescents. There have been no apparent differences in the rate of response for the individual SSRIs.

Currently, three SSRIs: sertraline (at least 6 years), fluoxetine (at least 7 years), and fluvoxamine (at least 8 years), as well as clomipramine (at least 10 years), have received FDA approval for the treatment of OCD in youth. The black-box warning for antidepressants used in children for any disorder, including OCD, is applicable so that close monitoring for suicidal ideation or behavior is mandated when using these agents in children.

Typical side effects that emerge with the use of SSRIs include insomnia,

nausea, agitation, tremor, and fatigue. Dosage ranges for the various SSRIs found to have efficacy in randomized clinical trials are the following: fluoxetine (20 to 60 mg), sertraline (50 to 200 mg), fluvoxamine (up to 200 mg), and paroxetine (up to 50 mg).

Clomipramine was the first antidepressant studied in the treatment of OCD in childhood and the only tricyclic antidepressant that has FDA approval for the treatment of anxiety disorders in childhood. Clomipramine was efficacious in doses up to 200 mg, or 3 mg/kg, whichever is less. Nevertheless, clomipramine is not recommended as a first-line treatment due to its higher potential risks compared to other SSRIs, including the cardiovascular risk of hypotension and arrhythmia and seizure risk.

Pediatric patients with OCD who respond only partially to medications tend to have at least moderate to severe OCD symptoms, high ratings of global impairment, and significant comorbidity even after their partial response to an adequate trial of medication. Augmentation strategies with medications to enhance serotonergic effects, such as with atypical antipsychotics (e.g., risperidone), have demonstrated increased response when SSRIs yield a partial response. Aripiprazole augmentation in 39 adolescents with OCD who did not respond to two trials of monotherapy with SSRIs led to 59 percent of patients being rated as improved or very much improved. Patients who responded to aripiprazole were less impaired at baseline in functional impairment but not in the clinical severity of their OCD. Aripiprazole's final mean dose was 12.2 mg/day. This agent may be useful for pediatric OCD and warrants further controlled trials.

Given the lack of data on discontinuation, recommendations for maintaining medication such as stabilization, education about relapse risk, and tapering medication during the summer are likely in order to minimize academic compromise in case of relapse. For children and adolescents with more severe or multiple episodes of significant exacerbation of symptoms, treatment for more than a year is recommended. Overall, the efficacy of treatment for children and adolescents with OCD is high with choices of SSRIs and CBT.

EPIDEMIOLOGY

OCD is common among children and adolescents, with a point prevalence of

about 0.5 percent and a lifetime prevalence of 2 to 4 percent. The rate of OCD among youth rises exponentially with increasing age, with rates of 0.3 percent in children between the ages of 5 and 7 years, rising to rates between 0.6 and 1 percent among teens. According to the DSM-5, the prevalence of OCD in the United States is 1.2 percent, with a slightly higher rate in females. Rates of OCD among adolescents are higher than those for schizophrenia or bipolar disorder. Among young children with OCD, there appears to be a slight male predominance, which diminishes with age.

ETIOLOGY

Genetic Factors

Genetic factors contribute significantly to the development of OCD in early-onset illness. The rate of OCD among first-degree relatives of children and adolescents who develop OCD is 10 times greater than for the general population. Twin studies have shown that the concordance rates for OCD are higher for monozygotic twins (0.57) than for dizygotic twins (0.22); however, nongenetic factors play a role that may be equal to or greater than genetic contributions in some cases. OCD is a heterogeneous disorder that has been recognized for decades to run in families. Also, the presence of subclinical symptom constellations in family members appears to breed true. Genetic linkage studies have revealed evidence of susceptibility loci on chromosomes 1q, 3q, 6q, 7p, 9p, 10p, and 15q. The OCD collaborative genetics study found that the *Sapap3* gene was associated with grooming disorders and may be a promising candidate gene for OCD. There is evidence that the glutamate receptor–modulating genes may also be associated with and play a role in the emergence of OCD. Family studies have suggested a relationship between OCD and tic disorders such as Tourette syndrome. OCD and tic disorders likely share susceptibility factors, which may include both genetic and nongenetic factors.

Neuroimmunology

Immunologic contributions to the emergence of OCD are likely related to an inflammatory process in the basal ganglia associated with an immune response to a systemic infection that may trigger OCD and tics. A prototype

of this hypothesis has been the controversial association of OCD symptoms in a small subgroup of children and adolescents following documented exposure to or infection with *group A β-hemolytic streptococcus* (GABHS). Under this hypothesis, cases of infection-triggered OCD have been termed Pediatric Autoimmune Neuropsychiatric Disorders Associated with Streptococcus (PANDAS), and parallel an autoimmune process leading to a movement disorder much like Sydenham chorea following rheumatic fever. Some evidence from MRI studies has documented a proportional relationship between the size of the basal ganglia and the severity of OCD symptoms in a small sample. GABHS may be one of many physiologic stressors that can lead to an increase or emergence of OCD or tics; however, a prospective longitudinal study of youth with PANDAS followed over 2 years found no evidence of a temporal association between GABHS infections and OCD symptom exacerbations in children who met the criteria for PANDAS. The presentation of OCD in children and adolescents due to acute exposure to GABHS represents a minority of OCD cases in youth and remains controversial.

Neurochemistry

The evidence that SSRIs diminish symptoms of OCD, along with findings of altered sensitivity to the acute administration of 5-hydroxytryptamine (5-HT) agonists in individuals with OCD, supports the probability of serotonin's role in OCD. Also, the dopamine system is likely influential in OCD, especially in light of the frequent comorbidity of OCD with tic disorders in childhood. Clinical observations have indicated that obsessions and compulsions worsen with stimulants, as with patients who have comorbid ADHD. Dopamine antagonists, administered along with SSRIs, may augment the effectiveness of SSRIs in the treatment of OCD. Evidence suggests that multiple neurotransmitter systems may play a role in OCD.

Neuroimaging

Both CT and MRI of untreated children and adults with OCD have revealed smaller volumes of basal ganglia segments compared to healthy controls. A meta-analysis of voxel-based morphometry (VBM) to assess gray matter density compared 343 OCD patients with 318 healthy controls and found that

gray matter density in OCD patients was smaller in parietofrontal cortical regions (including the supramarginal gyrus, the dorsolateral prefrontal cortex, and the orbitofrontal cortex), but larger in the basal ganglia (the putamen) and anterior prefrontal cortex, compared to healthy controls. Increased gray matter volume in the basal ganglia of patients with OCD has been reported in other studies as well. These structural abnormalities in the prefrontal-basal ganglia are likely to be integrally involved in the pathophysiology of OCD. It is not clear whether the increases in gray matter in individuals with OCD occur before or after the symptoms emerge. In children, there is evidence of increased thalamic volume. Adult studies have provided evidence of hypermetabolism of frontal cortical-striatal-thalamocortical networks in untreated individuals with OCD. Of interest, imaging studies of before and after treatment have revealed that both medication and behavioral interventions lead to a reduction of orbitofrontal and caudate metabolic rates in children and adults with OCD.

2.16 Early-Onset Schizophrenia

Early-onset schizophrenia comprises childhood-onset and adolescent-onset schizophrenia. Childhood-onset schizophrenia is a rare and virulent form of schizophrenia now recognized as a progressive neurodevelopmental disorder. Childhood-onset is characterized by a more chronic course, with severe social and cognitive consequences and increased negative symptoms compared to adult-onset schizophrenia. Childhood-onset schizophrenia is an onset of psychotic symptoms before 13 years, believed to represent a subgroup of patients with schizophrenia with an increased heritable etiology, and evidence of widespread abnormalities in the development of brain structures including the cerebral cortex, white matter, hippocampus, and cerebellum. Children diagnosed with childhood-onset schizophrenia have higher than average rates of premorbid developmental abnormalities that appear to be nonspecific markers of abnormal brain development. Early-onset schizophrenia is an onset of disease before 18 years, including childhood-onset as well as adolescent-onset schizophrenia. Early-onset schizophrenia is associated with severe clinical course, poor psychosocial functioning, and increased severity

of brain abnormalities. Despite the more severe course, current evidence supports the efficacy of both psychosocial and pharmacologic interventions in the management of childhood-onset and, particularly, adolescent-onset schizophrenia.

Children with childhood-onset schizophrenia have more significant deficits in measures of IQ, memory, and tests of perceptuomotor skills compared with adolescent-onset schizophrenia. Increased impairment in childhood-onset schizophrenia of cognitive measures such as IQ, working memory, and perceptuomotor skills may be premorbid markers of illness rather than sequelae of the disorder. Although cognitive impairments are more considerable in younger patients with schizophrenia, the clinical presentation of schizophrenia remains remarkably similar across the ages, and the diagnosis of childhood-onset schizophrenia is continuous with that in adolescents and adults, with one exception: in childhood-onset schizophrenia, a failure to achieve expected social and academic functioning may replace a deterioration in functioning. Chapter 5 discusses the features of the disorder in detail.

DIAGNOSIS AND CLINICAL FEATURES

All of the symptoms included in adult-onset schizophrenia may be manifest in children and adolescents with the disorder. However, youth with schizophrenia are more likely to have a premorbid history of social rejection, poor peer relationships, clingy withdrawn behavior, and academic trouble than those with adult-onset schizophrenia. Some children with schizophrenia evaluated in middle childhood have early histories of delayed motor milestones and language acquisition similar to some symptoms of autism spectrum disorder.

The onset of schizophrenia in childhood is frequently insidious, starting with inappropriate affect or unusual behavior; it may take months or years for a child to meet all of the diagnostic criteria for schizophrenia.

Auditory hallucinations commonly occur in children with schizophrenia. The voices may reflect an ongoing critical commentary or command hallucinations may instruct children to harm or kill themselves or others. Hallucinatory voices may sound human or animal, or “bizarre,” for example, identified as “a computer in my head,” “Martians,” or the voice of someone

familiar, such as a relative. The childhood-onset schizophrenia project at the NIMH found high rates across all hallucination modalities. However, there were unexpectedly high rates of tactile, olfactory, and visual hallucinations among this study group of patients with childhood-onset schizophrenia. Visual hallucinations were associated with lower IQ and earlier age at onset of disease. Visual hallucinations are often frightening; affected children may "see" images of the devil, skeletons, scary faces, or space creatures. Transient phobic visual hallucinations occur in severely anxious or traumatized children who do not develop major psychotic disorders. Visual, tactile, and olfactory hallucinations may be a marker of more severe psychosis.

Delusions occur in up to half of children and adolescents with schizophrenia, in various forms, including persecutory, grandiose, and religious. Delusions increase in frequency with increased age. Blunted or inappropriate affect appears almost universally in children with schizophrenia. Children with schizophrenia may giggle inappropriately or cry without being able to explain why. Formal thought disorders, including loosening of associations and thought blocking, are common features among youth with schizophrenia. Illogic thinking and poverty of thought are also often present. Unlike adults with schizophrenia, children with schizophrenia do not have poverty of speech content, but they speak less than other children of the same intelligence and are ambiguous in the way they refer to persons, objects, and events. The communication deficits observable in children with schizophrenia include unpredictably changing the topic of conversation without introducing the new topic to the listener (loose associations). Children with schizophrenia also exhibit illogical thinking and speaking and tend to underuse self-initiated repair strategies to aid in their communication. When an utterance is unclear or vague, normal children attempt to clarify their communication with repetitions, revision, and more detail. Children with schizophrenia, on the other hand, fail to aid communication with revision, fillers, or starting over. These deficits are examples of negative symptoms in childhood schizophrenia.

Although core phenomena for schizophrenia seem to be universal across the age span, a child's developmental level significantly influences the presentation of the symptoms. Delusions of young children are less complex, therefore, than those of older children; for example, age-appropriate content, such as animal imagery and monsters, is likely to be a source of delusional

fear in young children. According to the DSM-5, a child with schizophrenia may experience deterioration of function, along with the emergence of psychotic symptoms, or the child may never achieve the expected level of functioning.

A 12-year-old sixth-grade boy named Ian, with a longstanding history of social isolation, academic problems, and temper outbursts, began to develop concerns that his parents might be poisoning his food. Over the next year, his symptoms progressed with increased suspiciousness and fearfulness, preoccupation with food, and beliefs that Satan was trying to communicate with him. Ian also appeared to be responding to auditory hallucinations that he believed were coming from the radio and television, which he found frightening and commanded him to harm his parents. Ian had also been informing his mother that their food had a strange smell and that he thought someone had poisoned it. At night, Ian would see frightening figures in his room. During this time, his parents also observed bizarre behaviors, including talking and yelling to himself, perseverating about devils and demons, and finally, assaulting family members because he thought they were evil. On one occasion, Ian scratched himself with a kitchen knife to “please God.” No predominant mood symptoms emerged, and there was no history of substance abuse found.

Developmentally, Ian was the product of a full-term pregnancy complicated by a difficult labor and forceps delivery. His milestones, both motor and speech, were delayed by about 6 months. However, his pediatrician reassured his parents that this was within the limits of healthy development. As a younger child, Ian tended to be quiet and socially awkward. His intellectual function was tested and was in the average range; however, academic achievement testing was consistently below grade level. Ian remained lonely and isolated, and he had great difficulty making friends.

Ian has had no medical problems, and his immunizations were up to date.

Ian’s family psychiatric history was significant for depression in a maternal aunt and a completed suicide in a maternal great-grandparent.

Ian was sent by ambulance to the hospital for the first time from school when he tried to jump off a balcony on the second story of his school in response to auditory hallucinations commanding him to kill himself. During his hospitalization, his parents reluctantly consented to a risperidone trial. The psychiatrist titrated Ian to 3 mg/day. His auditory hallucinations improved moderately after 2 weeks of treatment. However, he continued to be suspicious and mistrustful of his physicians and family.

Ian's family was perplexed as to what had caused Ian's severe symptoms, and the hospital treatment team met with his parents multiple times during his hospitalization to reassure them that they had not caused his illness and that their continued support might improve his chances of improvement. After discharge from the hospital 30 days later, Ian was placed in a special education program in a nonpublic school, and he was assigned a psychotherapist who met regularly with him individually and with his family. At the time of discharge from the hospital, Ian's symptoms had moderately improved, although he still had auditory hallucinations intermittently. Over the next 5 years after the onset of his illness, Ian had many exacerbations of his psychosis, and he was hospitalized nine times, including placement in a long-term residential program. Ian had received trials of olanzapine, quetiapine, and aripiprazole, each of which seemed to lead to improvement for a while, after which he was no longer responsive to the medications. Ian continued to receive individual cognitive-behavioral therapy and family therapy, and his family was very supportive. Even with these interventions, Ian's mental status continued to display tangential and disorganized thinking, paranoid delusions, loose associations, perseverative speech patterns, and a flat, at times, inappropriate affect. He had periods in which he resorted to pacing and muttering to himself, with no social interaction with others unless initiated by adults. Finally, Ian achieved significant improvement after being placed on clozapine therapy, although he remained mildly symptomatic. (Adapted from a case by Jon M. McClellan, M.D.)

Pathology and Laboratory Examinations

No specific laboratory tests are diagnostically specific for childhood-onset schizophrenia.

DIFFERENTIAL DIAGNOSIS

Making a diagnosis of childhood-onset schizophrenia is challenging. Very young children who report hallucinations, apparent thought disorders, language delays, and an inability to differentiate reality from fantasy may be manifesting phenomena better accounted for by other disorders such as PTSD, or sometimes developmental immaturity.

Nevertheless, the differential diagnosis of childhood-onset schizophrenia includes autism spectrum disorder, bipolar disorders, depressive disorders, psychotic disorders, multicomplex developmental syndromes, drug-induced psychosis, and psychosis caused by organic disease states. Children with childhood-onset schizophrenia have frequent comorbidities, including ADHD, oppositional defiant disorder, and major depression. Children with schizotypal personality disorder have some traits in common with children who meet diagnostic criteria for schizophrenia. Blunted affect, social isolation, eccentric thoughts, ideas of reference, and bizarre behavior occur in both disorders; however, in schizophrenia, overt psychotic symptoms, such as hallucinations, delusions, and incoherence, must be present at some point. Hallucinations alone, however, are not evidence of schizophrenia; patients must show either a deterioration of function or an inability to meet an expected developmental level to warrant the diagnosis of schizophrenia. Auditory and visual hallucinations can appear as self-limited events in nonpsychotic young children who are experiencing extreme stress or anxiety related to unstable home lives, abuse, or neglect or in children experiencing a significant loss.

Psychotic phenomena are common among children with major depressive disorder, in which both hallucinations and, less commonly, delusions may occur. The congruence of mood with psychotic features is most pronounced in depressed children, although children with schizophrenia may also seem sad. The hallucinations and delusions of schizophrenia are more likely to have a bizarre quality than those of children with depressive disorders. In children and adolescents with bipolar I disorder, it often is challenging to distinguish a first episode of mania with psychotic features from

schizophrenia if the child has no history of previous depressions. Grandiose delusions and hallucinations are typical of manic episodes, but clinicians often must follow the natural history of the disorder to confirm the presence of a mood disorder. Autism spectrum disorders share some features with schizophrenia, most notably difficulty with social relationships, an early history of delayed language acquisition, and ongoing communication deficits. However, hallucinations, delusions, and formal thought disorder are core features of schizophrenia and not the usual features of autism spectrum disorder. Autism spectrum disorder is usually diagnosed by 3 years of age, whereas schizophrenia with childhood-onset usually manifests after 5 years of age.

Among adolescents, alcohol and other substance abuse sometimes can result in a deterioration of function, psychotic symptoms, and paranoid delusions. Amphetamines, lysergic acid diethylamide (LSD), and phencyclidine (PCP) may lead to a psychotic state. Sudden, flagrant onset of paranoid psychosis may suggest substance-induced psychotic disorder. Medical conditions that can induce psychotic features include thyroid disease, systemic lupus erythematosus, and temporal lobe disease.

COURSE AND PROGNOSIS

Significant predictors of the course and outcome of childhood and early-onset schizophrenia include the child's premorbid level of functioning, the age of onset, IQ, response to psychosocial and pharmacologic interventions, degree of remission after the first psychotic episode, and degree of family support. Early age at onset and children with comorbid developmental delays, learning disorders, lower IQ, and premorbid behavioral disorders, such as ADHD and conduct disorder, are less treatment responsive and likely to have the most guarded prognoses. Predictors of a poorer course of childhood-onset schizophrenia include a family history of schizophrenia, young age and insidious onset, developmental delays and lower level of premorbid function, and chronicity or length of a first psychotic episode. Psychosocial and family stressors are known to influence the relapse rate in adults with schizophrenia, and high expression of negative emotion (EE) likely affects children with childhood-onset schizophrenia as well.

An essential factor in the outcome is the accuracy and stability of the

diagnosis of schizophrenia. One study reported that one-third of children who received an initial diagnosis of schizophrenia later have bipolar disorder in adolescence. Children and adolescents with bipolar I disorder may have a better long-term prognosis than those with schizophrenia. The NIMH-funded Treatment of Early-Onset Schizophrenia reported the outcome of neurocognitive functioning in 8- to 19-year-old youth with schizophrenia or schizoaffective disorders who participated in a randomized, double-blind clinical trial comparing molindone, olanzapine, and risperidone. The three medication groups yielded no group differences in neurocognitive functioning over a year; however, when data from the three groups were combined, a significant modest improvement was observed in several domains of neurocognitive functioning. The authors concluded that antipsychotic intervention in youth with early-onset schizophrenia spectrum disorders led to modest improvement in neurocognitive function.

TREATMENT

The treatment of childhood-onset schizophrenia requires a multimodal approach, including psychoeducation for families, pharmacologic interventions, psychotherapeutic interventions, social skills interventions, and appropriate educational placement. A recent RCT investigated the effectiveness of several psychosocial interventions on youth in an early prodromal stage characterized by changes in cognitive and social behavior. The interventions, termed integrated psychological interventions, specifically included CBT, group skills training, cognitive remediation therapy, multifamily psychoeducation, and supportive counseling on the prevention of psychosis. Of interest, the integrated psychological intervention was more effective than standard treatments in delaying the onset of psychosis over a 2-year follow-up period. These results sparked interest in the potential utility of psychosocial interventions to mediate psychosis and to alter the relapse rate and severity of illness over time. Children with childhood-onset schizophrenia may have less robust responses to antipsychotic medications than adolescents and adults. Family education and ongoing therapeutic family interventions are critical to maintaining the maximum level of support for the patient. Monitoring the most appropriate educational setting for a child with childhood-onset schizophrenia is essential, especially given the frequent

social skills deficits, attention deficits, and academic difficulties that often accompany childhood-onset schizophrenia.

Pharmacotherapy

Second-generation antipsychotics, serotonin-dopamine antagonists, are the current mainstay pharmacologic treatments for children and adolescents with schizophrenia. Current data include six randomized clinical trials in youth investigating the efficacy of second-generation antipsychotics for early-onset schizophrenia, with limited support for one agent over the others. Although clozapine, a serotonin receptor antagonist with some dopamine (D_2) antagonism, may be more effective in reducing positive and negative symptoms, it remains a choice of last resort in youth based on its side effects. To date, however, evidence from multisite randomized clinical trials supports some efficacy of risperidone, olanzapine, aripiprazole, and clozapine in the treatment of childhood- and adolescent-onset schizophrenia. Two randomized clinical trials using risperidone in adolescents with schizophrenia found risperidone at doses up to 3 mg/day to be superior to placebo. A multisite randomized 6-week controlled trial of olanzapine in adolescents with schizophrenia found that it was more efficacious than a placebo. An RCT of aripiprazole at two fixed doses found that it was superior to placebo in the treatment of positive symptoms of adolescent schizophrenia; however, more than 40 percent of subjects in the active medication group did not achieve remission. Finally, clozapine is more effective than haloperidol in improving both positive and negative symptoms in treatment-resistant schizophrenia in youth. More recently, a study compared clozapine to high doses of olanzapine and found that response rates were about twice as high for clozapine as olanzapine (66 percent vs. 33 percent) when the investigators defined response as 30 percent or better reduction in symptoms on the Brief Psychiatric Rating Scale and improvement on the Clinical Global Impression Scale. The Treatment of Early-Onset Schizophrenia Spectrum Disorders Study compared the efficacy of risperidone and olanzapine with those of molindone, a mid-potency first-generation antipsychotic. In this study, lacking a placebo group, each of these agents provided a similar therapeutic effect; however, fewer than half of the patients responded optimally. Despite the limited randomized controlled studies of second-generation

antipsychotics for the treatment of schizophrenia in youth, the FDA is progressively approving the use of these agents for pediatric schizophrenia and bipolar illness. In 2007, the FDA approved the use of risperidone and aripiprazole for the treatment of schizophrenia in 13- to 17-year-olds. The FDA approved the use of olanzapine and quetiapine in 2009 for the treatment of schizophrenia in 13- to 17-year-olds. Lurasidone and paliperidone were approved by the FDA in 2017 for the treatment of schizophrenia in adolescents.

A double-blind, randomized 8-week controlled trial compared the efficacy and safety of olanzapine to clozapine in childhood-onset schizophrenia. Children with childhood-onset schizophrenia who were resistant to at least two previous treatments with antipsychotics were randomized to treatment for 8 weeks with either olanzapine or clozapine followed by a 2-year open-label follow-up. Using the *Clinical Global Impression of Severity of Symptoms Scale and Schedule for the Assessment of Negative/Positive Symptoms,* clozapine was associated with a significant reduction in all outcome measures. In contrast, olanzapine showed improvement on some measures but not on all. The only statistically significant measure in which clozapine was superior to olanzapine was in alleviating negative symptoms compared with baseline. Clozapine was associated with more adverse events, such as lipid abnormalities and a seizure in one patient.

Several studies have provided evidence that risperidone, a benzisoxazole derivative, is as effective as the older high-potency first-generation antipsychotics, such as haloperidol, and causes less frequent severe side effects in the treatment of schizophrenia in older adolescents and adults. Published case reports and limited more extensive controlled studies have supported the efficacy of risperidone in the treatment of psychosis in children and adolescents. Risperidone can cause weight gain and dystonic reactions and other extrapyramidal adverse effects in children and adolescents. Olanzapine is generally well tolerated for extrapyramidal adverse effects compared with first-generation antipsychotics and risperidone, but it is associated with moderate sedation and significant weight gain.

Psychosocial Interventions

Psychosocial interventions aimed at family education and patient and family

support are critical components of the treatment plan for childhood-onset schizophrenia. Although there are not yet RCTs of psychosocial interventions in children and adolescents with schizophrenia, family therapy, psychoeducation, and social skills training have been shown to lead to improved clinical symptoms in young adults with a first episode of schizophrenia, and reviews of the adult literature support the benefit of CBT and cognitive remediation as adjunctive treatments to pharmacologic agents in adults. Psychotherapists who work with children with schizophrenia must take into account a child's developmental level in order to support the child's reality testing and be sensitive to the child's sense of self. Long-term supportive family interventions and cognitive-behavioral and remediation interventions, combined with pharmacotherapy, are likely to be the most effective approach to early-onset schizophrenia.

EPIDEMIOLOGY

Childhood-onset schizophrenia affects less than 1 in 40,000 children. For adolescents, this increases by at least 50-fold. Childhood-onset schizophrenia resembles the more severe forms of adult schizophrenia, although it has exceptionally high comorbidity rates. Comorbid disorders include ADHD, depressive disorders, anxiety disorders, speech and language disorders, and motor disturbances.

Boys seem to have a slight preponderance among children diagnosed with schizophrenia, with an estimated ratio of about 1.67 boys to 1 girl. Boys often become identified at a younger age than girls do. Schizophrenia rarely is diagnosed in children younger than 5 years of age. The prevalence of schizophrenia among the parents of children with schizophrenia is about 8 percent, which is about twice the prevalence in the parents of patients with adult-onset schizophrenia.

ETIOLOGY

Childhood-onset schizophrenia is a neurodevelopmental disorder in which complex interactions between genes and the environment result in abnormal early brain development. The consequences of the aberrant brain development in schizophrenia may not be fully evident until adolescence or

early adulthood; however, data support the hypothesis that white matter abnormalities and disturbances in myelination in childhood lead to abnormal connectivity between brain regions. The aberrant connectivity in various regions of the brain is likely a significant contributing factor in the psychotic symptoms and cognitive deficits in childhood-onset schizophrenia.

Genetic Factors

Estimates of heritability for childhood-onset schizophrenia have been as high as 80 percent. Children with childhood-onset schizophrenia have higher rates of schizophrenia among their relatives than do patients with adult-onset schizophrenia.

Endophenotype Markers for Childhood-Onset Schizophrenia. Currently, no reliable method can identify persons at the highest risk for schizophrenia in a given family. Neurodevelopmental abnormalities and higher-than-expected rates of neurologic soft signs and impairments in sustaining attention and in strategies for information processing appear among children at high risk. Increased rates of disturbed communication styles exist in family members of individuals with schizophrenia. Reports have documented higher-than-expected neuropsychological deficits in attention, working memory, and premorbid IQ among children who later develop schizophrenia and its spectrum disorders.

Magnetic Resonance Imaging (MRI) Studies

An NIMH prospective study of more than 100 patients with childhood-onset schizophrenia and their typically developing siblings has demonstrated progressive loss of gray matter, delayed and disrupted white matter growth, and a decline in cerebellar volume in those with childhood-onset schizophrenia. Although siblings of children with childhood-onset schizophrenia also showed some of these brain disruptions, the gray matter abnormalities were normalized over time in the siblings, indicating a protective mechanism in siblings that was not present in those children with childhood-onset schizophrenia. Furthermore, the hippocampal volume loss across the age span appears to be static among children with childhood-onset schizophrenia. An MRI NIMH study of more than 100 children with

childhood-onset schizophrenia and their typically developing siblings, studied for about two decades, documented that in childhood-onset schizophrenia, progressive brain gray matter loss occurs continuously over time. This gray matter shrinkage occurs with ventricular increases, with a pattern of loss originating in the parietal region and proceeding frontally to dorsolateral prefrontal and temporal cortices, including superior temporal gyri. Studies of childhood-onset schizophrenia at the NIMH provided evidence that early loss of parietal gray matter followed by frontal and parietal gray matter loss is more pronounced in childhood-onset schizophrenia than in schizophrenia with later onset. Other research utilized diffusion tensor images from children with childhood-onset schizophrenia versus controls and found increased diffusivities in the posterior corona radiata in children with childhood-onset schizophrenia, which implicated abnormal connectivity with the parietal lobes. These results contrasted with findings among subjects with later onset of schizophrenia in whom there were more abnormalities in the frontal lobes.

ATTENUATED PSYCHOSIS SYNDROME

Attenuated Psychosis Syndrome (APS) is a new diagnostic category included in DSM-5 as a condition for further study. It is a syndrome characterized by subthreshold psychotic symptoms, less severe than those found in psychotic disorders, but which are often present in prodromal psychotic states.

Debate and controversy among clinicians and researchers have surrounded the inclusion of APS in the DSM-5. Some believe that the identification and treatment of a prodromal syndrome of a psychotic disorder would either delay or diminish the severity of the future psychotic illness. However, others believe that identification of a prodromal syndrome, which may rarely, if ever, progress to a full psychotic illness, would lead to unnecessary exposure to antipsychotic agents with unpredictable and possibly harmful effects. There is an agreement, however, that patients with subthreshold prodromal psychotic symptoms are often impaired and need psychological and psychiatric intervention.

A recent meta-analysis reported that the rate of onset of psychotic disorders in those patients with prodromal psychotic symptoms was 18 percent at 6 months, 22 percent at 1 year, 29 percent at 2 years, and 36

percent at 3 years. In a follow-up study, of those with prodromal symptoms who went on to develop a threshold psychotic illness, 73 percent met the criteria for schizophrenia.

In children and adolescents, psychotic symptoms are not necessarily a hallmark of a threshold psychotic disorder compared to adults. For example, in 50 percent of children with major depressive episodes, psychotic symptoms were present. Also, epidemiologic studies have found that globally, auditory hallucinations occur in 9 percent to 21 percent of children and 8.4 percent of adolescents. Thus, in youth, the association between subthreshold psychotic symptoms and the emergence of future psychotic illness may not be a reliable predictor. Nevertheless, identification and follow up of youth with APS may provide an increased understanding of the longitudinal significance of these symptoms.

Diagnosis

Attenuated psychosis syndrome, according to DSM-5, is based on the presence of at least one of the following: delusions, hallucinations, or disorganized speech, which causes functional impairment. Although the symptoms may not have progressed to full psychotic severity, they must have been present at least once per week for 1 month and must have emerged or worsened in the past year. The symptoms must cause impairment and warrant clinical attention.

Attenuated delusions are described as either suspiciousness, persecutory or grandiose, resulting in a lack of trust in others, and a sense of danger. Attenuated delusions, in contrast to delusions of threshold illness, may lead to loosely organized beliefs about hostile intentions of others or danger; however, the delusions are not as fixed as they become in full-blown psychotic illness. Attenuated hallucinations include altered sensory perceptions such as the perception of murmurs, rumblings, or shadows that are disturbing, but which youth may doubt and challenge. Disorganized communication or speech may present as vague or confusing explanations or circumstantial or tangential communication. When severe but still in the attenuated range, thought blocking or loose associations may emerge; however, in contrast to psychotic illness, redirection is possible, and one can usually achieve a logical conversation. Although impairment is present in

APS, the individual retains awareness and insight into the mental changes that are occurring.

Treatment

A recent review of the literature on treatment trials with patients at ultra-high-risk for psychosis found that early intervention with both psychological interventions and pharmacologic agents can reduce symptoms and either delay or prevent the onset of a full psychotic illness. Other studies, however, found mixed results for early psychological or pharmacologic interventions to prevent the onset of psychotic illness. One study found that most patients who became frankly psychotic did so within a few months after joining the study, making it difficult to determine if these patients were already exhibiting early signs of the onset of schizophrenia when identified as prodromal.

A variety of treatment approaches exist, including treatment with risperidone, olanzapine, omega-3 polyunsaturated fatty acid (w-3PUFA), CBT, cognitive therapy (CT), and one using an integrated psychological intervention (IPI) including cognitive approaches, psychoeducation, and social skills intervention. A review of treatment effectiveness in APS found that receiving treatment was associated with a lower risk of psychotic illness at 1 year, 2 years, and 3 years. Given the limited data, however, it is not clear which interventions are most efficacious. Therefore, until additional treatment trials provide efficacy data, the safest choices for treatment of APS include psychological interventions rather than the use of antipsychotic agents. In summary, APS identifies a group of patients with psychotic-like phenomena that warrant interventions in order to improve their distress and functional levels. Further study is needed, however, to determine the relationship between APS and the development of schizophrenia and other psychotic illnesses.

Etiology

Genetic Factors. Family studies have demonstrated that genetic factors influence vulnerability for schizophrenia spectrum disorders and other psychotic disorders. To the extent that APS and schizophrenia are related,

genetic contributions are likely to be significant. Adoption and twin studies have confirmed that monozygotic twins have about a 50 percent concordance rate for schizophrenia compared to dizygotic twins who have a concordance rate of about 10 percent. Also, adopted children of parents with schizophrenia do not have higher rates of schizophrenia; but biologic children of schizophrenic parents do. However, genetic factors do not account fully for the emergence of schizophrenia spectrum disorders, since there is only a 50 percent concordance of exhibiting these disorders among monozygotic twins. Environmental factors also play an essential role.

Environmental Factors. Early environmental factors that increase the risk of developing schizophrenia include fetal malnutrition, hypoxia at birth, and possibly prenatal infections. Other environmental factors include trauma, stress, social adversity, and isolation. Finally, gene-environment interactions may influence an individual's sensitivity to adverse environmental events.

▲ 2.17 Adolescent Substance Use Disorders

Substance use is a public health concern among American youth. The most common substances used by adolescents in the United States are tobacco, alcohol, and marijuana. There are many other substances that adolescents may abuse. These include cocaine, heroin, inhalants, phencyclidine (PCP), lysergic acid diethylamide (LSD), dextromethorphan, anabolic steroids, and various club drugs, including 3,4-methylenedioxymethamphetamine (MDMA or Ecstasy), flunitrazepam, gamma-hydroxybutyrate (GHB), and ketamine.

Approximately 20 percent of 8th graders in the United States have tried illicit drugs, and about 30 percent of 10th through 12th graders have used an illicit substance. Alcohol remains the most common substance used and abused by adolescents. Binge drinking occurs in about 6 percent of adolescents, and teens with alcohol use disorders are at higher risk of problems with other substances as well.

The DSM-5 combines abuse and dependence under the diagnosis of substance use disorder. It also contains diagnoses for intoxication, withdrawal, and substance-induced disorders. These apply to youths as well

as adults. Some experts are concerned about the relevance of the criteria to adolescents, especially regarding tolerance and withdrawal. Some adolescents may develop tolerance to alcohol, for example, without any impairment. Withdrawal may have clinical significance. However, it does not correlate well with the level of severity.

Many risk and protective factors influence the age of onset and severity of substance use among adolescents. Psychosocial risk factors mediating the development of substance use disorders include parent modeling of substance use, family conflict, lack of parental supervision, peer relationships, and individual stressful life events. Protective factors that mitigate substance use among adolescents include variables such as stable family life, strong parent–child bond, consistent parental supervision, investment in academic achievement, and a peer group that models prosocial family and school behaviors. Interventions that diminish risk factors are likely to mitigate substance use.

Approximately one of five adolescents has used marijuana or hashish. Approximately one-third of adolescents have used cigarettes by age 17 years. Studies of alcohol use among adolescents in the United States have shown that by 13 years of age, one-third of boys and almost one-fourth of girls have tried alcohol. By 18 years of age, 92 percent of males and 73 percent of females reported trying alcohol, and 4 percent reported using alcohol daily. Of high school seniors, 41 percent reported using marijuana; 2 percent reported using the drug daily.

Drinking among adolescents follows adult demographic drinking patterns: The highest proportion of alcohol use occurs among adolescents in the northeast; whites are more likely to drink than are other groups; among whites, Roman Catholics are the least likely nondrinkers. The four most common causes of death in persons between the ages of 10 and 24 years are motor vehicle accidents (37 percent), homicide (14 percent), suicide (12 percent), and other injuries or accidents (12 percent). In pediatric trauma centers, more than one-third of adolescents need treatment for alcohol or drug use.

Studies considering alcohol and illicit drug use by adolescents as psychiatric disorders have demonstrated a higher prevalence of substance use, particularly alcoholism, among biologic children of alcoholics than among adopted youth. This finding is supported by family studies of genetic

contributions, by adoption studies, and by observing children of substance users reared outside the biologic home.

Numerous risk factors influence the emergence of adolescent substance abuse. These include parental belief in the harmlessness of substances, lack of anger control in families of substance abusers, lack of closeness and involvement of parents with children's activities, maternal passivity, academic difficulties, comorbid psychiatric disorders such as conduct disorder and depression, parental and peer substance use, impulsivity, and early onset of cigarette smoking. The higher the number of risk factors, the more likely it is that an adolescent will be a substance user.

DIAGNOSIS AND CLINICAL FEATURES

Chapter 4 discusses the substance-related disorders and their diagnoses, including substance use, substance intoxication, and substance withdrawal disorder.

We should make a diagnosis of alcohol or drug use in adolescents through a careful interview, observations, laboratory findings, and history provided by reliable sources. Many nonspecific signs may point to alcohol or drug use, and clinicians must be careful to corroborate hunches before jumping to conclusions. Substance use exists on a continuum with experimentation (the mildest use), regular use without visible impairment, abuse, and, finally, dependence. Changes in academic performance, nonspecific physical ailments, changes in relationships with family members, changes in one's peer group, unexplained phone calls, or changes in personal hygiene may indicate substance use in an adolescent. Many of these also occur with depression, prodromal psychotic disorders, and adjustment issues, for example, to a new school.

Substance use is related to a variety of behaviors, including early sexual experimentation, risky driving, destruction of property, stealing, "heavy metal" or electronic dance music (EDM), and, occasionally, preoccupation with cults or Satanism. Although none of these behaviors necessarily predicts substance use, at the extreme, many of these behaviors reflect alienation from the mainstream of developmentally expected social behavior. Adolescents with inadequate social skills may use a substance as a modality to join a peer group. In some cases, adolescents begin their substance use at home with

their parents, who also use substances to enhance their social interactions. Although no evidence indicates what determines a typical adolescent user of alcohol or drugs, many substance users seem to have underlying social skills deficits, academic difficulties, and less than optimal peer relationships.

Nicotine

Nicotine is one of the most addictive substances known; it involves cholinergic receptors and enhancing acetylcholine, serotonin, and β-endorphin release. Young teens who smoke cigarettes are also exposed to other drugs more frequently than nonsmoking peers.

Alcohol

Alcohol use in adolescents rarely results in the sequelae observed in adults with chronic use of alcohol, such as withdrawal seizures, Korsakoff syndrome, Wernicke aphasia, or cirrhosis of the liver. One report, however, has stated that adolescent exposure to alcohol may result in diminished hippocampal brain volume. Because the hippocampus is involved with attention, it is conceivable that adolescent alcohol use could result in compromised cognitive function, especially concerning attention.

Marijuana

The short-term effects of the active ingredient in marijuana, tetrahydrocannabinol (THC), include impairment in memory and learning, distorted perception, diminished problem-solving ability, loss of coordination, increased heart rate, anxiety, and panic attacks. Abrupt cessation of heavy marijuana use by adolescents can result in a withdrawal syndrome characterized by insomnia, irritability, restlessness, drug craving, depressed mood, and nervousness followed by anxiety, tremors, nausea, muscle twitches, increased sweating, myalgia, and general malaise. Typically, the withdrawal syndrome begins 24 hours after the last use, peaks at 2 to 4 days, and diminishes after 2 weeks. Marijuana use correlates with an increased risk of psychiatric disorders. Poor cognitive functioning has been associated with chronic marijuana use, although it is not clear whether marijuana impairs cognitive function. Deficits in verbal learning, memory, and attention occur in chronic marijuana users, and both acute and chronic

marijuana use is associated with changes in cerebral blood flow to specific brain regions, which can be detected by PET. Functional imaging studies suggest that there is less activity in brain regions involved with attention and memory in chronic marijuana users. A 15-year follow-up of 50,465 Swedish males in the military reported that participants who had used marijuana by 18 years of age were 2.4 times more likely to develop schizophrenia. Risks associated with chronic marijuana use include higher rates of motor vehicle accidents, impaired respiratory function, increased risk of cardiovascular disease, and potential increased risk for psychotic symptoms and disorders.

Cocaine

Cocaine can be sniffed or snorted, injected, or smoked. *Crack* is the term given to cocaine that has been changed to a free base for smoking. Cocaine's effects include constriction of peripheral blood vessels, dilated pupils, hyperthermia, increased heart rate, and hypertension. High doses or prolonged use of cocaine can induce paranoid thinking. There is an immediate risk of death secondary to cardiac arrest or from seizures followed by respiratory arrest. In contrast to stimulants used to treat ADHD, such as methylphenidate, cocaine quickly crosses the blood-brain barrier and moves off the dopamine transporter within 20 minutes; methylphenidate remains bound to dopamine for long periods.

Heroin

Heroin, a derivative of morphine, is produced from a poppy plant. Heroin usually appears as a white or brown powder that users can snort, but more commonly, they take it intravenously. Withdrawal symptoms include restlessness, muscle and bone pain, insomnia, diarrhea and vomiting, cold flashes with goosebumps, and kicking movements. Withdrawal occurs within a few hours after use; symptoms peak between 48 and 72 hours later and remit within about a week.

Club Drugs

Adolescents who frequent nightclubs, raves, bars, or music clubs also frequently use MDMA, GHB, flunitrazepam (Rohypnol), and ketamine. GHB, flunitrazepam (a benzodiazepine), and ketamine (an anesthetic) are

primarily depressants and can be added to drinks without detection because they are often colorless, tasteless, and odorless. Congress passed The Drug-Induced Rape Prevention and Punishment Act after finding that these drugs were used in date rapes. MDMA is a derivative of methamphetamine, a synthetic with both stimulant and hallucinogenic properties. MDMA can inhibit serotonin and dopamine reuptake. MDMA can result in dry mouth, increased heart rate, fatigue, muscle spasm, and hyperthermia.

Lysergic Acid Diethylamide

LSD is odorless, colorless, and has a slightly bitter taste. Higher doses of LSD can produce visual hallucinations and delusions and, in some cases, panic. The sensations experienced after the ingestion of LSD usually diminish after 12 hours. Flashbacks can occur up to 1 year after use. LSD can produce tolerance; that is, after multiple uses, one needs more to feel the same degree of intoxication.

COMORBIDITY

Rates of alcohol and marijuana use are reportedly higher in relatives of youth with depression and anxiety disorders. On the other hand, mood disorders are common among those with alcoholism. Evidence indicates another strong link between early antisocial behavior, conduct disorder, and substance abuse. Substance abuse is a form of behavioral deviance that, unsurprisingly, is associated with other forms of social and behavioral deviance. Early intervention with children who show early signs of social deviance and antisocial behavior may conceivably impede the processes that contribute to later substance abuse.

It is essential to know about all comorbid disorders, which may show differential responses to treatment. Surveys of adolescents with alcoholism show rates of 50 percent or higher for additional psychiatric disorders, especially mood disorders. A recent survey of adolescents who used alcohol found that more than 80 percent met the criteria for another disorder. The disorders most frequently present were depressive disorders, disruptive behavior disorders, and drug use disorders. These rates of comorbidity are even higher than those for adults. The diagnosis of alcohol use disorder was likely to follow, rather than precede, other disorders; that a large proportion

of adolescents with alcoholism have a previous childhood disorder may have both etiologic and treatment implications. In this survey, the onset of alcohol use disorders did not systematically precede other substance use disorders. In 50 percent of cases, alcohol use followed drug use. Alcohol use may be a gateway to drug use but is not in most cases. The presence of other psychiatric disorders was associated with an earlier onset of alcohol use disorder, but it did not seem to indicate a more protracted course of alcoholism.

TREATMENT

Interventions for substance use disorders in adolescents first require effective screening and identification of those teens in need of treatment. Once identifying a substance use disorder in a teen, we have a variety of treatment options.

Per the goals of the U.S. Substance Abuse Mental Health Services Administration (SAMHSA), a school-based alcohol and drug Screening, Brief Intervention, and Referral to Treatment (SBIRT) has been initiated in a study with 629 adolescents ages 14 to 17 years in 13 participating high schools in New Mexico. Initially, school-based health centers provided substance use screenings for all students seen in the clinic for any reason. Once identified, substance-using adolescents were offered either brief intervention by clinic staff (85.1 percent of those identified), whereas 14.9 percent received brief treatment or referral to treatment. The brief intervention was based on motivational interviewing to help the student to gain motivation for behavioral change, with referral for more intensive treatment if needed. Students who received the intervention, regardless of the severity of their substance use, reported decreases in self-reported drinking to intoxication at the 6-month follow-up.

The students reporting drug use self-reported using less at follow-up. Forty-two percent of the students reported alcohol use, and 37 percent reported intoxication. Eighty-five percent of study participants who reported drug use reported only marijuana use in the month before entering the study. The frequency of alcohol and marijuana as the most predominant substances in this age group is consistent with epidemiologic data. Overall, this school-based intervention had the advantage of being easily accessible to adolescents

and provided a graded option for treatment according to the severity of substance use. This study suggests that school-based programs for identifying and providing brief interventions for high school students is viable and merits further study.

Treatment of substance use disorders in adolescents can directly prevent the substance use behaviors, provide education for the patient and family, and address cognitive, emotional, and psychiatric factors that influence the substance use in a variety of settings such as a residential milieu, group, and individual psychosocial session.

One validated instrument used as a guide for clinicians in the treatment of adolescent substance use designates levels of care appropriate for the symptoms. This instrument, called the *Child and Adolescent Levels of Care Utilization Services* (CALOCUS), outlines six levels of care:

0: Basic services (prevention)
1: Recovery maintenance (relapse prevention)
2: Outpatient (once per week visits)
3: Intensive outpatient (2 or more visits per week)
4: Intensive integrated services (day treatment, partial hospitalization, wraparound services)
5: Nonsecure, 24-hour medically monitored service (group home, residential treatment facility)
6: Secure 24-hour medical management (inpatient psychiatric or highly programmed residential facility)

Treatment settings that serve adolescents with alcohol or drug use disorders include inpatient units, residential treatment facilities, halfway houses, group homes, partial hospital programs, and outpatient settings. Components of adolescent alcohol or drug use treatment include individual psychotherapy, drug-specific counseling, self-help groups (Alcoholics Anonymous [AA], Narcotics Anonymous [NA], Alateen, Al-Anon), substance abuse education and relapse prevention programs, and random urine drug testing. Family therapy and psychopharmacologic intervention are

also useful to add.

Before deciding on the most appropriate treatment setting for a particular adolescent, a screening process must take place in which structured and unstructured interviews help to determine the types of substances they are using as well as the quantities and frequencies. Determining coexisting psychiatric disorders is also critical. Rating scales can document pretreatment and posttreatment severity of abuse. The *Teen Addiction Severity Index* (T-ASI), the *Adolescent Drug and Alcohol Diagnostic Assessment* (ADAD), and the *Adolescent Problem Severity Index* (APSI) are several severity-oriented rating scales. The T-ASI is broken down into dimensions that include family function, school or employment status, psychiatric status, peer social relationships, and legal status.

After obtaining most of the information about substance use and the patient's overall psychiatric status, we must choose a treatment strategy, and an appropriate setting must be determined. Two very different approaches to the treatment of substance use disorders are the Minnesota model and the multidisciplinary professional model. The Minnesota model relies on the premise of AA; it is an intensive 12-step program with a counselor who functions as the primary therapist. The program uses self-help participation and group processes. Inherent in this treatment strategy is the need for adolescents to admit that substance use is problematic and that help is necessary. Furthermore, they must be willing to work toward altering their lifestyle to eradicate substance use.

The multidisciplinary professional model consists of a team of mental health professionals that usually is led by a physician. Following a case-management model, each member of the team has specific areas of treatment for which he or she is responsible. Interventions may include CBT, family therapy, and pharmacologic intervention. This approach usually is suited for adolescents with comorbid psychiatric diagnoses.

Cognitive-behavioral approaches to psychotherapy for adolescents with substance use generally require that adolescents be motivated to participate in treatment and refrain from further substance use. The therapy focuses on relapse prevention and maintaining abstinence.

Psychopharmacological interventions for adolescent alcohol and drug users are still in their early stages. The presence of mood disorders indicates the need for antidepressants, and generally, the SSRIs are the first line of

treatment. Occasionally, we might choose to substitute the illicit drug with another drug that is more amenable to the treatment situation; for example, using methadone instead of heroin. Adolescents are required to have documented attempts at detoxification and consent from an adult before they can enter such a treatment program.

Peter, a 16-year-old 11th grader, was admitted to substance abuse treatment for the second time, following a relapse and threats of suicide. Peter reported a longstanding history of ADHD, but he had been a good student and not had any difficulties until middle school. Peter reported an onset of substance use at age 13 years, a rapid progression in substance involvement since age 14 years, and current use of marijuana daily, drinking alcohol up to five times each week, and experimentation with a variety of substances, such as LSD and Ecstasy. After being discharged from the psychiatric hospital, Peter attended teen group sessions focusing on his substance use problems. Family sessions led to the realization that Peter's mother had been depressed for some time, and she entered into her own treatment. Peter was improving concerning his substance use; however, his depressive symptoms increased following 4 weeks of abstinence. His psychiatrist started him on fluoxetine. After titrating the medication to 30 mg, Perter remained on it for a month, at which time he showed improvement in mood and treatment compliance. Peter continued to attend teen AA meetings and outpatient therapy. The family conflict soon recurred, however, and Peter became noncompliant with outpatient treatment, medication, and meetings. He resumed old relationships with substance-using peers and relapsed into daily marijuana use and occasional alcohol use. (Courtesy of Oscar G. Bukstein, M.D.)

Efficacious treatments for cigarette smoking cessation include nicotine-containing gum, patches, or nasal spray or inhaler. Bupropion aids in diminishing cravings for nicotine and is beneficial in the treatment of smoking cessation.

Because comorbidity influences treatment outcome, it is essential to pay attention to other disorders, such as mood disorders, anxiety disorders, conduct disorder, or ADHD during the treatment of substance use disorders.

EPIDEMIOLOGY

Alcohol

The Centers for Disease Control and Prevention Youth Risk Behavior Survey found that 72.5 percent of high school students had tried at least one alcoholic drink, and 24.2 percent reported an episode of heavy drinking in the month preceding the survey. Findings from the Monitoring the Future Survey suggest that about 39 percent of adolescents have used alcohol before the eighth grade. Another survey found that drinking was a significant problem for 10 to 20 percent of adolescents. Seventy percent of eighth grade students reported drinking, and 54 percent reported drinking within the past year, 27 percent reported having gotten drunk at least once, and 13 percent reported binge drinking in the 2 weeks before the survey. By the 12th grade, 88 percent of high school students reported drinking, and 77 percent drank within the past year; 5 percent of 8th-grade students, 1.3 percent of 10th grade students, and 3.6 percent of 12th grade students reported daily alcohol use. In the age range of 13 to 17 years, in the United States, reports indicate there are 3 million problem drinkers and 300,000 adolescents with alcohol dependence. The gap between male and female alcohol consumers is narrowing.

Marijuana

For the last two decades, marijuana has been one of the most widely used drugs by young people in developed countries, and recently it has become highly used globally. The United Nations Office on Drugs and Crime estimated that 3.9 percent of people worldwide between ages 15 years and 64 use marijuana. Marijuana is the most commonly used illicit drug among high school students in the United States. About 10 percent of those who try marijuana become daily users and 20 to 30 percent become weekly users. Marijuana is a “gateway drug” because the strongest predictor of future cocaine use is frequent marijuana use during adolescence. Of 8th-, 10th-, and 12th-grade students, 10, 23, and 36 percent, respectively, report using marijuana, a slight decrease from the year preceding the survey. Of 8th-, 10th-, and 12th-grade students, 0.2, 0.8, and 2 percent, respectively, report

daily marijuana use. Prevalence rates for marijuana are highest among Native American males and females; these rates are nearly as high in white males and females and Mexican American males. The lowest annual rates occur in Latin American females, African American females, and Asian American males and females.

Cocaine

The annual cocaine use reported by high school seniors decreased by more than 30 percent between 1990 and 2000. Currently, about 0.5 percent of 8th-grade students, 1 percent of 10th-grade students, and 2 percent of 12th-grade students are estimated to have used cocaine. Crack cocaine use, however, is increasing in prevalence and is most common among those between the ages of 18 and 25.

Crystal Methamphetamine

Crystal methamphetamine, or “ice,” was at a relatively low level of use in adolescence about one decade ago of 0.5 percent, and has steadily increased to a recent rate of 1.5 percent among 12th graders.

Opioids

A survey of 7,374 high school seniors found that 12.9 percent reported nonmedical use of opioids. Of users, more than 37 percent reported intranasal administration of prescription opioids.

Lysergic Acid Diethylamide (LSD)

About 3 percent of 8th-grade students, 6 percent of 10th-grade students, and 9 percent of 12th-grade students used LSD. Of 12th-grade students, 0.1 percent report daily use. The current LSD rates are lower than the rates of LSD use during the past two decades.

3,4-Methylenedioxymethamphetamine (MDMA)

MDMA has increased in popularity over the last decade. In the United States, 5 percent of 10th graders and 8 percent of 12th graders use the drug. This is true even though its perceived danger has dramatically increased among

adolescents. There have been accidental adolescent deaths due to MDMA.

Gamma-Hydroxybutyrate (GHB)

Gamma-hydroxybutyrate, a club drug, has been found in surveys to have an annual prevalence rate of 1.1 percent for 8th graders, 1.0 percent for 10th graders, and a 1.6 percent rate of use for 12th graders.

Ketamine

Ketamine, another club drug, was found recently to have a rate of 1.3 percent annual prevalence for 8th graders, 2.1 percent for 10th graders, and 2.5 percent rate for 12th graders.

Flunitrazepam (Rohypnol)

Flunitrazepam (Rohypnol), a third club drug, has been found to have an annual prevalence rate of about 1 percent for all high school grades combined.

Anabolic Steroids

Despite reported knowledge of the risks of anabolic steroids among high school students, surveys over the last 5 years found rates of anabolic steroid use to be 1.6 percent among 8th graders and 2.1 percent among 10th graders. Up to 45 percent of 10th and 12th graders reported knowledge of the risks of anabolic steroids; however, over the last decade, it appears that high school seniors reported less disapproval of their use.

Inhalants

The use of inhalants in the form of glue, aerosols, and gasoline is relatively more common among younger than older adolescents. Among 8th-, 10th-, and 12th-grade students, 17.6, 15.7, and 17.6 percent, respectively, report using inhalants; 0.2 percent of 8th-grade students, 0.1 percent of 10th-grade students, and 0.2 percent of 12th-grade students report daily use of inhalants.

Multiple Substance Use

Among adolescents enrolled in substance abuse treatment programs, 96

percent are polydrug users; 97 percent of adolescents who abuse drugs also use alcohol.

ETIOLOGY

Genetic Factors

The concordance for alcoholism is reportedly higher among monozygotic than dizygotic twins. Studies of children of alcoholics reared away from their biologic homes have shown that these children have about a 25 percent chance of becoming alcoholics.

Psychosocial Factors

Among adolescents, substance use, particularly marijuana use, is strongly influenced by peers, and especially for those adolescents who report using marijuana for relaxation, the drug is used to escape from stress and as a social activity. There are data to suggest, however, that marijuana use is also associated with both social anxiety disorder and depressive symptoms. Among young adolescents who start using alcohol, tobacco, and marijuana at an early age, data suggest that they often come from families with low parental supervision. The risk of early initiation of substances is most significant for children below 11 years of age. Increased parental supervision during middle childhood years may diminish drug and alcohol sampling and ultimately diminish the risk of using marijuana, cocaine, or inhalants in the future.

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3 Neurocognitive Disorders

Advances in molecular biology diagnostic techniques and medication management have significantly improved the ability to recognize and treat cognitive disorders. Cognition includes memory, language, orientation, judgment, conducting interpersonal relationships, performing actions (praxis), and problem-solving. Cognitive disorders reflect disruption in one or more of these domains. Frequently, behavioral symptoms complicate the disorder. Cognitive disorders exemplify the complex interface among neurology, medicine, and psychiatry in that medical or neurologic conditions often lead to cognitive disorders that, in turn, are associated with behavioral symptoms. Of all psychiatric conditions, cognitive disorders best demonstrate how biologic insults result in behavioral symptomatology. The clinician must carefully assess the history and context of the presentation of these disorders before arriving at a diagnosis and treatment plan.

Cognitive disorders tend to defy Occam's razor, challenging clinicians and nosologists with multiplicity, comorbidity, and unclear boundaries. These are most concerning for elderly adults, the demographic group most at risk for cognitive disorders. Dementias of late-life are particularly problematic in this regard. Existing, although often unrecognized, dementia is a significant risk factor for superimposed delirium. Moreover, certain dementias, such as dementia with Lewy bodies or late stages of Alzheimer disease, may have chronic clinical presentations virtually indistinguishable from delirium except for temporal onset and the lack of an identifiable acute source.

Similarly, behavioral syndromes complicate the course of nearly all subjects developing progressive dementia. These syndromes include anxiety, depression, sleep problems, psychosis, and aggression. These symptoms can be as distressing and disabling as the primary cognitive disorder. Some of

these behavioral syndromes, such as psychosis, may themselves result from independent underlying biologies and may be additive with the primary neurodegenerative process.

The boundaries between types of dementia and between dementia and healthy aging can be similarly diffuse. Neuropathologic studies of both clinical and population samples have revealed a surprising truth. The most common neuropathologic presentation associated with dementia reveals mixtures of Alzheimer disease, vascular, and Lewy body pathologies. Pure syndromes are relatively less common, although often the dementia is ascribed to one of the coexisting pathologies. Strategies regarding how to understand or reconcile multiple pathologies in the clinic are needed, although they lag.

DEFINITION

The definitions of cognitive disorders follow. We will then describe each separately.

Delirium

Delirium describes a condition of short-term confusion and changes in cognition. There are four subcategories based on several causes: (1) general medical condition (e.g., infection), (2) substance-induced (e.g., cocaine, opioids, phencyclidine [PCP]), (3) multiple causes (e.g., head trauma and kidney disease), and (4) other or multiple etiologies (e.g., sleep deprivation, medications).

Dementia (Major Neurocognitive Disorder)

Dementia, also referred to as major neurocognitive disorder, in the fifth edition of DSM (DSM-5), is marked by severe impairment in memory, judgment, orientation, and cognition. The subcategories are (1) dementia of the Alzheimer type (Alzheimer Dementia), which usually occurs in persons older than 65 years of age and is manifested by progressive intellectual disorientation and dementia, delusions, or depression; (2) vascular dementia, caused by vessel thrombosis or hemorrhage; (3) human immunodeficiency virus (HIV) disease; (4) head trauma; (5) Pick disease or frontotemporal lobar

degeneration; (6) prion disease such as Creutzfeldt–Jakob disease (CJD), which is caused by a slow-growing transmittable virus; (7) substance-induced, caused by toxin or medication (e.g., gasoline fumes, atropine); (8) multiple etiologies; and (9) not specified (if the cause is unknown).

In DSM-5, mild neurocognitive disorder describes a less severe form of dementia.

Amnestic Disorder

Amnestic disorders are *major neurocognitive disorders caused by other medical conditions*. They are marked primarily by memory impairment in addition to other cognitive symptoms. The causes include (1) medical conditions (hypoxia), (2) toxins or medications (e.g., marijuana, diazepam), and (3) unknown causes.

CLINICAL EVALUATION

During history taking, the clinician seeks to elicit the development of the illness. Subtle cognitive disorders, fluctuating symptoms, and progressing disease processes may be tracked effectively. The clinician should obtain a detailed rendition of changes in the patient's daily routine involving such factors as self-care, job responsibilities, and work habits; meal preparation; shopping and personal support; interactions with friends; hobbies and sports; reading interests; religious, social, and recreational activities; and ability to maintain personal finances. Understanding the past life of each patient provides an invaluable source of baseline data regarding changes in function, such as attention and concentration, intellectual abilities, personality, motor skills, and mood and perception. The examiner explores what the patient considers essential to their lifestyle and how the clinical condition affects these. This helps to assess both the disease and the success of future therapies.

Table 3-1
Neuropsychiatric Mental Status Examination

A. General Description
1. General appearance, dress, sensory aids (glasses, hearing aid)

2. Level of consciousness and arousal
3. Attention to environment
4. Posture (standing and seated)
5. Gait
6. Movements of limbs, trunk, and face (spontaneous, resting, and after instruction)
7. General demeanor (including evidence of responses to internal stimuli)
8. Response to examiner (eye contact, cooperation, ability to focus on interview process)
9. Native or primary language

B. Language and Speech
1. Comprehension (words, sentences, simple and complex commands, and concepts)
2. Output (spontaneity, rate, fluency, melody or prosody, volume, coherence, vocabulary, paraphasic errors, complexity of usage)
3. Repetition
4. Other aspects
    a. Object naming
    b. Color naming
    c. Body part identification
    d. Ideomotor praxis to command

C. Thought
1. Form (coherence and connectedness)
2. Content
    a. Ideational (preoccupations, overvalued ideas, delusions)
    b. Perceptual (hallucinations)

D. Mood and Affect
1. Internal mood state (spontaneous and elicited; sense of humor)
2. Future outlook
3. Suicidal ideas and plans
4. Demonstrated emotional status (congruence with mood)

E. Insight and Judgment

1. Insight
    a. Self-appraisal and self-esteem
    b. Understanding of current circumstances
    c. Ability to describe personal, psychological, and physical status
2. Judgment
    a. Appraisal of major social relationships
    b. Understanding of personal roles and responsibilities

F. Cognition
1. Memory
    a. Spontaneous (as evidenced during interview)
    b. Tested (incidental, immediate repetition, delayed recall, cued recall, recognition; verbal, nonverbal; explicit, implicit)
2. Visuospatial skills
3. Constructional ability
4. Mathematics
5. Reading
6. Writing
7. Fine sensory function (stereognosis, graphesthesia, two-point discrimination)
8. Finger gnosis
9. Right–left orientation
10. “Executive functions”
11. Abstraction

Courtesy of Eric D. Caine, M.D., and Jeffrey M. Lyness, M.D.

Mental Status Examination

After taking a thorough history, the clinician’s primary tool is the assessment of the patient’s mental status. As with the physical examination, the mental status examination is a means of surveying functions and abilities to allow a definition of personal strengths and weaknesses. It is a repeatable, structured assessment of symptoms and signs that promotes effective communication among clinicians. It also establishes the basis for future comparison, essential

for documenting therapeutic effectiveness, and it allows comparisons between different patients, with a generalization of findings from one patient to another. Table 3-1 lists the components of a comprehensive neuropsychiatric mental status examination.

Cognition

When testing cognitive functions, the clinician should evaluate memory; visuospatial and constructional abilities; and reading, writing, and mathematical abilities. The assessment of abstraction ability is also practical. However, many things can affect proverb interpretation, such as culture and education, and any interpretation should happen in that context.

PATHOLOGY AND LABORATORY EXAMINATION

As with all medical tests, we should interpret psychiatric evaluations in the context of thorough clinical and laboratory assessments. Psychiatric and neuropsychiatric patients require a careful physical examination, particularly when issues exist that involve etiologically related or comorbid medical conditions. When consulting internists and other medical specialists, the clinician must ask specific questions to focus on the differential diagnostic process and use the consultation most effectively. In particular, most systemic medical or primary cerebral diseases that lead to psychopathological disturbances also manifest with a variety of peripheral or central abnormalities.

A screening laboratory evaluation is sought initially and may be followed by a variety of ancillary tests to increase the diagnostic specificity. Table 3-2 lists such procedures, some of which we also describe below.

Electroencephalography

Electroencephalography (EEG) is an easily accessible, noninvasive test of brain dysfunction that has a high sensitivity for many disorders but relatively low specificity. Beyond its recognized uses in epilepsy, EEG's highest utility is in detecting altered electrical rhythms associated with mild delirium, space-occupying lesions, and continuing complex partial seizures (in which the patient remains conscious, although behaviorally impaired). EEG is also

sensitive to metabolic and toxic states, often showing a diffuse slowing of brain activity. We discuss the EEG in more detail in Chapter 1.

Computed Tomography and Magnetic Resonance Imaging

Computed tomography (CT) and magnetic resonance imaging (MRI) have proved to be powerful neuropsychiatric research tools. Recent developments in MRI allow the direct measurement of structures such as the thalamus, basal ganglia, hippocampus, and amygdala, as well as temporal and apical areas of the brain and the structures of the posterior fossa. MRI has mostly replaced CT as the most utilitarian and cost-effective method of imaging in neuropsychiatry. CT is still the test of choice for patients with acute cerebral hemorrhages or hematomas, but these patients infrequently present in psychiatric settings. MRI better discriminates the interface between gray and white matter and is useful in detecting a variety of white matter lesions in the periventricular and subcortical regions. The pathophysiologic significance of such findings remains to be defined. White matter abnormalities occur in younger patients with multiple sclerosis or HIV infection and older patients with hypertension, vascular dementia, or dementia of the Alzheimer type. The prevalence of these abnormalities increases in healthy, aging individuals who have no defined disease process. In general, the highest utility of neuroimaging in the evaluation of patients with dementia arises from what it may exclude (tumors, vascular disease) rather than what it can demonstrate specifically.



Table 3-2
Screening Laboratory Tests

General Tests
Complete blood cell count
Erythrocyte sedimentation rate
Electrolytes
Glucose
Blood urea nitrogen and serum creatinine
Liver function tests
Serum calcium and phosphorus

Thyroid function tests
Serum protein
Levels of all drugs
Urinalysis
Pregnancy test for women of childbearing age
Electrocardiography

Ancillary Laboratory Tests

Blood

- Blood cultures
- Rapid plasma reagin test
- Human immunodeficiency virus (HIV) testing (enzyme-linked immunosorbent assay [ELISA] and Western blot)
- Serum heavy metals
- Serum copper
- Ceruloplasmin
- Serum B_{12}, red blood cell (RBC) folate levels

Urine

- Culture
- Toxicology
- Heavy metal screen

Electrography

- Electroencephalography
- Evoked potentials
- Polysomnography
- Nocturnal penile tumescence

Cerebrospinal fluid

- Glucose, protein
- Cell count
- Cultures (bacterial, viral, fungal)
- Cryptococcal antigen
- Venereal Disease Research Laboratory (VDRL) test

Radiography

- Computed tomography
- Magnetic resonance imaging

Positron emission tomography
Single photon emission computed tomography

Courtesy of Eric D. Caine, M.D., and Jeffrey M. Lyness, M.D.

Brain Biopsy

A brain needle biopsy can diagnose a variety of disorders: Alzheimer disease, autoimmune encephalopathy, and tumors. It is conducted stereotactically and indicated when no other investigative techniques such as MRI or lumbar puncture have been sufficient to make a diagnosis. The procedure is not without risk in that seizures may occur if scar tissue forms at the biopsy site.

Neuropsychologic Testing

Neuropsychologic testing provides a standardized, quantitative, reproducible evaluation of a patient's cognitive abilities. Such procedures may be useful for initial evaluation and periodic assessment. Tests are available that assess abilities across the broad array of cognitive domains, and many offer comparative normative groups or adjusted scores based on normative samples. The clinician seeking neuropsychologic consultation should understand enough about the strengths and weaknesses of selected procedures to benefit fully from the results obtained.

▲ 3.1 Delirium

Delirium is an acute decline in both the level of consciousness and cognition with particular impairment in attention. A life-threatening yet potentially reversible disorder of the central nervous system (CNS), delirium often involves perceptual disturbances, abnormal psychomotor activity, and sleep cycle impairment. Health care workers often underrecognize delirium. Part of the problem is that the syndrome has a variety of other names (Table 3-3).

The hallmark symptom of delirium is an impairment of consciousness, usually occurring in association with global impairments of cognitive functions. Abnormalities of mood, perception, and behavior are common psychiatric symptoms. Tremor, asterixis, nystagmus, incoordination, and

urinary incontinence are common neurologic symptoms. Classically, delirium has a sudden onset (hours or days), a brief and fluctuating course, and rapid improvement when we can identify the cause and eliminate it, but each of these characteristic features can vary in individual patients. Physicians must recognize delirium to identify and treat the underlying cause and to avert the development of delirium-related complications such as accidental injury because of the patient's clouded consciousness.



Table 3-3
Delirium by Other Names

Intensive care unit psychosis
Acute confusional state
Acute brain failure
Encephalitis
Encephalopathy
Toxic metabolic state
Central nervous system toxicity
Paraneoplastic limbic encephalitis
Sundowning
Cerebral insufficiency
Organic brain syndrome



Table 3-4
Delirium

	DSM-5	ICD-10
Name	Delirium	Delirium, not induced by alcohol and other psychoactive substances Delirium, not superimposed on dementia

		Delirium, superimposed on dementia Other delirium Delirium, unspecified
Symptoms	Impaired attention/awareness Acute onset, change from baseline. May fluctuate throughout the day Symptoms develop acutely, represent a change from baseline, and may fluctuate Impairment of other cognitive functions • Memory • Language • Orientation • Perception Evidence for physiologic cause of the impairment (from history, examination, laboratory or other data)	Nonspecific organic cognitive syndrome with disturbed consciousness, awareness, memory, motor function, sleep, and emotion, with variable duration and severity; it can be acute or subacute
Required number of symptoms	All the above categories	
Exclusions (not result of):	A neurocognitive disorder Coma or other cause of reduced arousal	
Symptom specifiers	Hyperactive Hypoactive	

	Mixed level of activity Based on cause: • Substance intoxication delirium, with or without use disorder (specify severity of use: mild/moderate/severe) • Substance withdrawal delirium, with or without use disorder (specify severity of use: mild/moderate/severe) • Medication-induced delirium • Delirium due to another medical condition • Delirium due to multiple etiologies	
Course Specifiers	**Acute:** hours to days **Persistent:** weeks to months	
Comments		If due to substance use, use “mental and behavioral disorders due to psychoactive substances”

DIAGNOSIS AND CLINICAL FEATURES

Table 3-4 compares the diagnostic approaches to delirium. Usually, the cause is one or more systemic or cerebral derangements that affect brain function.

A 70-year-old woman, Mrs. K, was brought to the emergency department by the police. The police had responded to complaints from neighbors that Mrs. K was wandering the neighborhood and was not taking care of herself. When the police found Mrs. K in her apartment, she was dirty, foul-smelling, and wearing nothing but a bra. Her apartment was also filthy with garbage and rotting food everywhere.

When interviewed, Mrs. K would not look at the interviewer and was confused and unresponsive to most of the questions asked. She knew her name and address but not the date. She was unable to describe the events that led to her admission.

The next day, the supervising psychiatrist attempted to interview Mrs. K. Her facial expression was still unresponsive, and she still did not know the month or the name of the hospital. She explained that the neighbors called the police because she was “sick” and that she did indeed feel sick and weak, with pains in her shoulder. She also reported not eating for three days. She denied ever being in a psychiatric hospital or hearing voices but acknowledged seeing a psychiatrist at one point because she had trouble sleeping. She said the doctor had prescribed medication, but she could not remember the name.

Table 3-5 lists the core features of delirium.

Associated clinical features are often present and may be prominent. The EEG usually shows diffuse slowing of background activity, although patients with delirium caused by alcohol or sedative-hypnotic withdrawal have low-voltage fast activity.

The primary neurotransmitter hypothesized to be involved in delirium is acetylcholine, and the major neuroanatomical area is the reticular formation. The reticular formation of the brainstem is the principal area regulating attention and arousal; the major pathway implicated in delirium is the dorsal tegmental pathway, which projects from the mesencephalic reticular formation to the tectum and thalamus. Several studies have reported that a variety of delirium-inducing factors result in decreased acetylcholine activity in the brain. One of the most common causes of delirium is toxicity from too many prescribed medications with anticholinergic activity. Researchers have

suggested other pathophysiologic mechanisms for delirium. In particular, the delirium associated with alcohol withdrawal correlates with hyperactivity of the locus ceruleus and its noradrenergic neurons. Serotonin and glutamate are also likely involved.



Table 3-5
Core Features of Delirium

- Altered consciousness (usually decreased level)
- Altered attention (diminished ability to focus, sustain, or shift attention)
- Other cognitive dysfunctions:
  - Disorientation (especially to time and space)
  - Impaired memory
- Typical course:
  - Relatively rapid onset (usually hours to days)
  - Brief duration (usually days to weeks)
  - Marked, unpredictable fluctuations in severity
  - Sometimes worse at night (sundowning), ranging from periods of lucidity to severe cognitive impairment and disorganization

PHYSICAL AND LABORATORY EXAMINATIONS

Delirium is usually diagnosed at the bedside and presents as a sudden onset of symptoms. A bedside mental status examination—such as the Mini-Mental State Examination—can be used to document the cognitive impairment and to provide a baseline from which to measure the patient's clinical course. The physical examination often reveals clues to the cause of the delirium (Table 3-6). The presence of a known physical illness or a history of head trauma or alcohol or other substance use disorder increases the likelihood of the diagnosis.

The laboratory workup of a patient with delirium should include standard tests and additional studies indicated by the clinical situation (Table 3-7). In delirium, the EEG characteristically shows a generalized slowing of activity and may be useful in differentiating delirium from depression or psychosis.

The EEG of a delirious patient sometimes shows focal areas of hyperactivity. In rare cases, it may be challenging to differentiate delirium related to epilepsy from delirium related to other causes.

DIFFERENTIAL DIAGNOSIS

Delirium versus Dementia

Several clinical features help distinguish delirium from dementia (Table 3-8). The major differential points between dementia and delirium are the time to development of the condition and the fluctuation in the level of attention in delirium compared with relatively consistent attention in dementia. The time to development of symptoms is usually short in delirium, and except for vascular dementia caused by stroke, it is usually gradual and insidious in dementia. Although both conditions include cognitive impairment, the changes in dementia are more stable over time and, for example, usually do not fluctuate over a day. A patient with dementia is usually alert; a patient with delirium has episodes of decreased consciousness. Occasionally, delirium occurs in a patient with dementia, a condition known as *beclouded dementia.* A dual diagnosis of delirium exists when there is a definite history of preexisting dementia.

Delirium versus Schizophrenia or Depression

We should differentiate delirium from schizophrenia and depressive disorder. Some patients with psychotic disorders, usually schizophrenia or manic episodes, can have periods of extremely disorganized behavior challenging to distinguish from delirium. In general, however, the hallucinations and delusions of patients with schizophrenia are more constant and better organized than those of patients with delirium. Patients with schizophrenia usually experience no change in their level of consciousness or their orientation. Patients with hypoactive symptoms of delirium may appear somewhat similar to severely depressed patients, but their confusion can help differentiate, as can an EEG. Other psychiatric diagnoses to consider in the differential diagnosis of delirium are brief psychotic disorder, schizophreniform disorder, and dissociative disorders. Patients with factitious disorders may attempt to simulate the symptoms of delirium but usually

reveal the factitious nature of their symptoms by inconsistencies on their mental status examinations, and an EEG can easily separate the two diagnoses.

COURSE AND PROGNOSIS

Although the onset of delirium is usually sudden, prodromal symptoms (e.g., restlessness and fearfulness) can occur in the days preceding the onset of florid symptoms. The symptoms of delirium usually persist as long as the causally relevant factors are present, although delirium generally lasts less than 1 week. After identification and removal of the causative factors, the symptoms of delirium usually recede over a 3- to 7-day period, although some symptoms may take up to 2 weeks to resolve completely. The older the patient and the longer the patient has been delirious, the longer the delirium takes to resolve. Recall of what transpired during a delirium, once it is over, is characteristically spotty; a patient may refer to the episode as a bad dream or a nightmare only vaguely remembered. As stated in the discussion on epidemiology, the occurrence of delirium is associated with a high mortality rate in the ensuing year, primarily because of the severe nature of the associated medical conditions that lead to delirium.

Whether delirium progresses to dementia has not been demonstrated in carefully controlled studies, although many clinicians believe that they have seen such a progression. A clinical observation that has been validated by some studies, however, is that depression or posttraumatic stress may follow a delirium.

TREATMENT

In treating delirium, the primary goal is to treat the underlying cause. When the underlying condition is anticholinergic toxicity, physostigmine salicylate, 1 to 2 mg intravenously or intramuscularly, may help. We may have to repeat the dose every 15 to 30 minutes. The other important goal of treatment is to provide physical, sensory, and environmental support. Physical support is necessary so that delirious patients do not get into situations in which they may have accidents. Patients with delirium should be neither sensory deprived nor overly stimulated by the environment. A familiar person staying

in the room, such as a friend or relative, may help. Familiar pictures and decorations; the presence of a clock or a calendar; and regular orientations to person, place, and time help make patients with delirium comfortable. Delirium can sometimes occur in older patients wearing eye patches after cataract surgery (“black-patch delirium”). Such patients can be helped by placing pinholes in the patches to let in some stimuli or by occasionally removing one patch at a time during recovery.

Pharmacotherapy

The three significant delirium symptoms that may require pharmacologic treatment are psychosis, agitation, and insomnia. A commonly used drug for psychosis is haloperidol. Depending on a patient’s age, weight, and physical condition, the initial dose may range from 2 to 5 mg intramuscularly, repeated in an hour if the patient remains agitated. As soon as the patient is calm, oral medication in liquid concentrate or tablet form should begin. Two daily oral doses should suffice, mostly at bedtime. The oral equivalent is about 1.5 times the parenteral dose. The effective total daily dose of haloperidol may range from 5 to 40 mg for most patients with delirium. Haloperidol may cause a prolonged QT interval. Clinicians should evaluate baseline and periodic electrocardiograms as well as monitor the cardiac status of the patient. Droperidol is a butyrophenone available as an alternative intravenous (IV) formulation, although careful monitoring of the electrocardiogram may be prudent with this treatment. The U.S. Food and Drug Administration (FDA) issued a Black Box Warning because of cases of patients with QT prolongation and torsades de pointes receiving droperidol. Because of its potential for serious proarrhythmic effects and death, it should be used only in patients who do not respond well to other treatments. Phenothiazines should be avoided in delirious patients because these drugs are associated with significant anticholinergic activity.



Table 3-6
Physical Examination of the Delirious Patient

Parameter	Finding	Clinical Implication
1. Pulse	Bradycardia	Hypothyroidism

		Stokes–Adams syndrome
		Increased intracranial pressure
	Tachycardia	Hyperthyroidism
		Infection
		Heart failure
2. Temperature	Fever	Sepsis
		Thyroid storm
		Vasculitis
3. Blood pressure	Hypotension	Shock
		Hypothyroidism
		Addison disease
	Hypertension	Encephalopathy
		Intracranial mass
4. Respiration	Tachypnea	Diabetes
		Pneumonia
		Cardiac failure
		Fever
		Acidosis (metabolic)
	Shallow	Alcohol or other substance intoxication
5. Carotid vessels	Bruits or decreased pulse	Transient cerebral ischemia
6. Scalp and face	Evidence of trauma	

7. Neck	Evidence of nuchal rigidity	Meningitis
		Subarachnoid hemorrhage
8. Eyes	Papilledema	Tumor
		Hypertensive encephalopathy
	Pupillary dilatation	Anxiety
		Autonomic overactivity (e.g., delirium tremens)
9. Mouth	Tongue or cheek lacerations	Evidence of generalized tonic–clonic seizures
10. Thyroid	Enlarged	Thyroid dysfunction
11. Heart	Arrhythmia	Inadequate cardiac output, possibility of emboli
	Cardiomegaly	Heart failure
		Hypertensive disease
12. Lungs	Congestion	Primary pulmonary failure
		Pulmonary edema
		Pneumonia
13. Breath	Alcohol	
	Ketones	Diabetes
14. Liver	Enlargement	Cirrhosis
		Liver failure

15. Nervous system		
a. Reflexes—muscle stretch	Asymmetry with Babinski signs	Mass lesion
		Cerebrovascular disease
		Preexisting dementia
	Snout	Frontal mass
		Bilateral posterior cerebral artery occlusion
b. Abducent nerve (sixth cranial nerve)	Weakness in lateral gaze	Increased intracranial pressure
c. Limb strength	Asymmetrical	Mass lesion
		Cerebrovascular disease
d. Autonomic	Hyperactivity	Anxiety
		Delirium

From Strub RL, Black FW. *Neurobehavioral Disorders: A Clinical Approach*. Philadelphia: FA Davis; 1981:121, with permission.



Table 3-7
Laboratory Workup of the Patient with Delirium

Standard studies
- Blood chemistries (including electrolytes, renal and hepatic indexes, and glucose)
- Complete blood count with white cell differential
- Thyroid function tests
- Serologic tests for syphilis
- Human immunodeficiency virus (HIV) antibody test
- Urinalysis
- Electrocardiogram

Electroencephalogram
Chest radiograph
Blood and urine drug screens
Additional tests when indicated
Blood, urine, and cerebrospinal fluid cultures
B_{12}, folic acid concentrations
Computed tomography or magnetic resonance imaging brain scan
Lumbar puncture and CSF examination

Agitation is a poorly defined symptom that can range from distressed anxiety to physical aggression. Because of the ambiguity of the term, agitation is challenging to study. However, some scales are available to aid research, such as the Richmond Agitation-Sedation Scale. Many medications have anecdotal support. However, studies on the pharmacologic treatment of agitation in delirium are disappointing. The antipsychotics are most commonly used. The approach is analogous to the above approach for psychosis in delirium, with adjustments for the severity and prolongation of the delirium. Sodium valproate given IV may be useful when antipsychotics fail. For severe agitation, dexmedetomidine, an α_2-adrenoreceptor agonist that is used as a sedative, can help sedate a patient, mainly a mechanically ventilated patient, without the deliriogenic effects of many other sedatives.

Second-generation antipsychotics, such as risperidone, clozapine, olanzapine, quetiapine, ziprasidone, and aripiprazole, may be considered for delirium management, but clinical trial experience with these agents for delirium is limited. Ziprasidone appears to have an activating effect and may not be appropriate in delirium management. Olanzapine is available for intramuscular (IM) use and as a rapidly disintegrating oral preparation. These routes of administration may be preferable for some patients with delirium who are poorly compliant with medications or who are too sedated to swallow medications safely.

Insomnia is difficult to treat. Benzodiazepines might be helpful; however, they risk worsening a patient’s confusion. Also, there is no conclusive evidence to support the use of benzodiazepines in non–alcohol-related delirium. If pain is the cause, a physician should not hesitate to prescribe

opioids for both their analgesic and sedative effects (Table 3-9). Melatonin is occasionally used but with mixed support and low-quality studies.

Treatment in Special Populations

Parkinson Disease. In Parkinson disease, antiparkinsonian agents can cause delirium. If coexistent dementia is present, delirium is twice as likely to develop in patients with Parkinson disease with dementia receiving antiparkinsonian agents than in those without dementia. Decreasing the dosage of the antiparkinsonian agent has to be weighed against a worsening of motor symptoms. If we cannot reduce the antiparkinsonian agents, or if the delirium persists after we do, an antipsychotic may help. However, this risks worsening the disorder. The FDA has approved pimavanserin for psychosis in Parkinson's. It appears to be an inverse agonist and antagonist activity of serotonin 5-HT_{2A} and possibly 5-HT_{2C}, with no appreciable affinity for dopamine. Clozapine may also help and has empirical support. Quetiapine has not been as rigorously studied as clozapine and may have parkinsonian side effects, but it is used in clinical practice to treat psychosis in Parkinson disease.



Table 3-8
Frequency of Clinical Features of Delirium Contrasted with Dementia

Symptom	Delirium	Dementia
Mental Status Examination		
Memory	Impaired, more short-term than long-term	Impaired, more long-term than short-term
Attention	Poor	Impaired although usually to a lesser degree
Orientation	Grossly disorganized	Varies

Sensorium	Prominent hallucinations (esp. visual or tactile)	Rare hallucinations
Thought content	May be paranoid	May be paranoid
Judgment, social skills and behavior	Grossly impaired	Initially relatively intact, worsens with time (depending on areas affected with frontal cortical damage causing more impairment)
Physical Examination		
Vital signs	Often abnormal	Dependent on cause
Neurologic examination	May be abnormal	Dependent on cause
Comorbid medical illnesses	Acute illnesses	Dependent on cause
Course		
Onset	Acute	Usually insidious
Short-term course	Fluctuates throughout day	Varies, but generally more stable
Long-term course	May improve (depending on underlying cause)	Usually chronic and progressive. May be gradual or stepwise depending on cause
Associated Features		
Comorbid	Acute illnesses	Dependent on cause

medical illnesses		
Most likely associated illnesses	Metabolic, toxic, or infectious illnesses	Neurodegenerative or vascular disorders

Terminally Ill Patients. When delirium occurs in the context of a terminal illness, issues about advanced directives and the existence of a health care proxy become more significant. This scenario emphasizes the importance of early development of advance directives for health care decision making while a person can communicate the wishes regarding the extent of aggressive diagnostic tests at life's end. The focus may change from an aggressive search for the etiology of the delirium to one of palliation, comfort, and assistance with dying.



Table 3-9
Pharmacologic Treatment of Delirium

Pharmacologic Agent	Dosage	Side Effects	Comments
Typical Antipsychotics			
Haloperidol	0.5–1 mg p.o. twice a day (may be given every 4–6 hr as needed, too)	Extrapyramidal side (EPS) effects Prolonged QTc	Most commonly used Can be given intramuscularly
Atypical Antipsychotics		All can prolong QTc duration	

Risperidone	0.5–1 mg a day	EPS concerns	Limited data in delirium
Olanzapine	5–10 mg a day	Metabolic syndrome	Higher mortality in dementia patients
Quetiapine	25–150 mg a day	More sedating	
Benzodiazepine			
Lorazepam	0.5–3 mg a day and as needed every 4 hr	Respiratory depression, paradoxical agitation	Best use in delirium secondary to alcohol or benzodiazepine withdrawal Can worsen other types of delirium

EPIDEMIOLOGY

Delirium is a common disorder, with most incidence and prevalence rates reported in elderly adults. In community studies, 1 percent of elderly persons age 55 years or older have delirium (13 percent in the age 85 years and older group in the community). Among elderly emergency department patients, 5 to 10 percent have delirium. At the time of admission to medical wards, between 15 and 21 percent of older patients meet the criteria for delirium-prevalent cases. Of patients free of delirium at the time of hospital admission, 5 to 30 percent reported subsequent incidences of delirium during hospitalization. Delirium occurs in 10 to 15 percent of general surgical patients, 30 percent of open-heart surgery patients, and more than 50 percent of patients treated for hip fractures. Delirium occurs in 70 to 87 percent of intensive care unit patients and up to 83 percent of all patients receiving end of life care. Sixty percent of patients in nursing homes or post-acute care

settings have delirium. An estimated 21 percent of patients with severe burns and 30 to 40 percent of patients with acquired immune deficiency syndrome (AIDS) have episodes of delirium during hospitalization. Delirium develops in 80 percent of terminally ill patients. The causes of postoperative delirium include the stress of surgery, postoperative pain, insomnia, pain medication, electrolyte imbalances, infection, fever, and blood loss. Table 3-10 shows the incidence and prevalence rates for delirium across settings.



Table 3-10
Delirium Incidence and Prevalence in Multiple Settings

Population	Prevalence Range (%)	Incidence Range (%)
General medical inpatients	10–30	3–16
Medical and surgical inpatients	5–15	10–55
General surgical inpatients	N/A	9–15 postoperatively
Critical care unit patients	16	16–83
Cardiac surgery inpatients	16–34	7–34
Orthopedic surgery patients	33	18–50
Emergency department	7–10	N/A
Terminally ill cancer patients	23–28	83
Institutionalized elderly	44	33

N/A, not available.

The risk for delirium could be conceptualized into two categories, predisposing and precipitating factors (Tables 3-11 and 3-12). Current approaches to delirium focus primarily on the precipitation factors and do

little to address the predisposing factors. Managing predisposing factors for delirium becomes essential in decreasing future episodes of delirium and the morbidity and mortality associated with it.



Table 3-11
Predisposing Factors for Delirium

Demographic characteristics
Age 65 years and older
Male sex
Cognitive status
Dementia
Cognitive impairment
History of delirium
Depression
Functional status
Functional dependence
Immobility
History of falls
Low level of activity
Sensory impairment
Hearing
Visual
Decreased oral intake
Dehydration
Malnutrition
Drugs
Treatment with psychoactive drugs
Treatment with drugs with anticholinergic properties
Alcohol abuse
Coexisting medical conditions
Severe medical diseases
Chronic renal or hepatic disease
Stroke
Neurologic disease

Metabolic derangements
Infection with human immunodeficiency virus
Fractures or trauma
Terminal diseases

Adapted from Inouye SK. Delirium in older persons. *N Engl J Med*. 2016;354(11):1157.



Table 3-12
Precipitating Factors for Delirium

Drugs
Sedative–hypnotics
Narcotics
Anticholinergic drugs
Treatment with multiple drugs
Alcohol or drug withdrawal
Primary neurologic diseases
Stroke, nondominant hemispheric
Intracranial bleeding
Meningitis or encephalitis
Intercurrent illnesses
Infections
Iatrogenic complications
Severe acute illness
Hypoxia
Shock
Anemia
Fever or hypothermia
Dehydration
Poor nutritional status
Low serum albumin levels
Metabolic derangements
Surgery
Orthopedic surgery
Cardiac surgery

Prolonged cardiopulmonary bypass
Noncardiac surgery
Environmental
Admission to intensive care unit
Use of physical restraints
Use of bladder catheter
Use of multiple procedures
Pain
Emotional stress
Prolonged sleep deprivation

Adapted from Inouye SK. Delirium in older persons. *N Engl J Med*. 2016;354(11):1157.

Advanced age is a significant risk factor for the development of delirium. Approximately 30 to 40 percent of hospitalized patients older than age 65 years have an episode of delirium, and another 10 to 15 percent of elderly persons exhibit delirium on admission to the hospital. Of nursing home residents older than age 75 years, 60 percent have repeated episodes of delirium. Male gender is also an independent risk factor for delirium.

Delirium is a poor prognostic sign. Rates of institutionalization are increased threefold for patients 65 years and older who exhibit delirium while in the hospital. The 3-month mortality rate of patients who have an episode of delirium is estimated to be 23 to 33 percent. The 1-year mortality rate for patients who have an episode of delirium may be as high as 50 percent. Elderly patients who experience delirium while hospitalized have a 21 to 75 percent mortality rate during that hospitalization. After discharge, up to 15 percent of these persons die within 1 month, and 25 percent die within 6 months.

ETIOLOGY

The significant causes of delirium are CNS disease (e.g., epilepsy), systemic disease (e.g., sepsis, cardiac failure), and either intoxication or withdrawal from pharmacologic or toxic agents (Table 3-13). When evaluating patients with delirium, clinicians should assume that any drug that a patient has taken could be a cause for delirium.

Table 3-13
Common Causes of Delirium

Central nervous system disorder	Seizure (postictal, nonconvulsive status, status)
	Migraine
	Head trauma, brain tumor, subarachnoid hemorrhage, subdural, epidural hematoma, abscess, intracerebral hemorrhage, cerebellar hemorrhage, nonhemorrhagic stroke, transient ischemia
Metabolic disorder	Electrolyte abnormalities
	Diabetes, hypoglycemia, hyperglycemia, or insulin resistance
Systemic illness	Infection (e.g., sepsis, malaria, erysipelas, viral, plague, Lyme disease, syphilis, or abscess)
	Trauma
	Change in fluid status (dehydration or volume overload)
	Nutritional deficiency
	Burns
	Uncontrolled pain
	Heat stroke
	High altitude (usually >5,000 m)
Medications	Pain medications (e.g., postoperative meperidine or morphine)

	Antibiotics, antivirals, and antifungals
	Steroids
	Anesthesia
	Cardiac medications
	Antihypertensives
	Antineoplastic agents
	Anticholinergic agents
	Neuroleptic malignant syndrome
Serotonin syndrome	
Over-the-counter preparations	Herbals, teas, and nutritional supplements
Botanicals	Jimsonweed, oleander, foxglove, hemlock, dieffenbachia, and *Amanita phalloides*
Cardiac	Cardiac failure, arrhythmia, myocardial infarction, cardiac assist device, cardiac surgery
Pulmonary	Chronic obstructive pulmonary disease, hypoxia, SIADH, acid–base disturbance
Endocrine	Adrenal crisis or adrenal failure, thyroid abnormality, parathyroid abnormality
Hematologic	Anemia, leukemia, blood dyscrasia, stem cell transplant
Renal	Renal failure, uremia, SIADH
Hepatic	Hepatitis, cirrhosis, hepatic failure
Neoplasm	Neoplasm (primary brain, metastases, paraneoplastic syndrome)

Drugs of abuse	Intoxication and withdrawal
Toxins	Intoxication and withdrawal
	Heavy metals and aluminum

SIADH, syndrome of inappropriate secretion of antidiuretic hormone.

▲ 3.2 Dementia (Major Neurocognitive Disorder)

Dementia refers to a disease process marked by progressive cognitive impairment in clear consciousness. Dementia does not refer to low intellectual functioning or mental retardation because these are developmental and static conditions, and the cognitive deficits in dementia represent a decline from a previous level of functioning. Dementia involves multiple cognitive domains, and cognitive deficits cause significant impairment in social and occupational functioning. There are several types of dementias based on etiology: Alzheimer disease, dementia of Lewy bodies, vascular dementia, frontotemporal dementia, traumatic brain injury (TBI), HIV, prion disease, Parkinson disease, and Huntington disease. Other medical and neurologic conditions can cause dementia, as can various substances.

The critical clinical points of dementia are the identification of the syndrome and the clinical workup of its cause. The disorder can be progressive or static, permanent, or reversible. We always assume an underlying cause, although, in rare cases, it is impossible to determine a specific cause. The potential reversibility of dementia is related to the underlying pathologic condition and the availability and application of effective treatment. Approximately 15 percent of people with dementia have reversible illnesses if we can initiate treatment before irreversible damage takes place.

DIAGNOSIS AND CLINICAL FEATURES

Table 3-14 compares the approaches to the general diagnosis of dementia. DSM-5 makes a distinction between major and minor cognitive disorders based upon levels of functioning, but the underlying etiology is similar.

Table 3-14
Approaches to the General Diagnosis of Dementia

	DSM-5	ICD-10
Major and Mild Neurocognitive Disorders		**Dementias**
Symptoms	Significant (**Major**) or moderate (**Minor**) decline in cognitive functioning: • Attention • Executive function • Learning/Memory • Language • Perceptual–motor • Social cognition (For each of the specific diagnoses, one should specify whether major or minor in the diagnosis [i.e., minor neurocognitive disorder with Lewy bodies])	Defined by the specific cause of Dementia
Required number of symptoms	≥1; see above criteria	
Exclusions (not result of):	• Delirium • Another mental disorder	
Psychosocial Impact	Some impairments in ability to independently perform activities of daily living	
Symptom Specifiers	See specifiers for the specific disorders With or without behavioral disturbance	See specifiers

		for the specific disorders
Severity Specifiers	Based on what is impaired and level of impairment and this refers to the current level: **Mild:** instrumental activities of daily life (e.g., cooking, paying bills) impaired **Moderate:** basic activities of daily life impaired (e.g., eating, dressing) **Severe:** impaired in all activities, fully dependent on others	

DSM-5 specifics a variety of types of dementia based on their cause. Tables 3-15 to 3-24 compare the diagnostic approaches to the major etiologies.

We make a diagnosis of dementia from the clinical examination, including a mental status examination, and on information from the patient's family, friends, and employers. Complaints of a personality change in a patient older than age 40 years suggest that we should consider a diagnosis of dementia.

Clinicians should note patients' complaints about intellectual impairment and forgetfulness as well as evidence of patients' evasion, denial, or rationalization aimed at concealing cognitive deficits. Excessive orderliness, social withdrawal, or a tendency to relate events in minute detail can be characteristic, and sudden outbursts of anger or sarcasm can occur. We should observe the patient's appearance and behavior. Lability of emotions, sloppy grooming, uninhibited remarks, silly jokes, or a dull, apathetic, or vacuous facial expression and manner suggest the presence of dementia, especially when coupled with memory impairment.

Memory impairment is typically an early and prominent feature in dementia, especially in dementias involving the cortex, such as dementia of the Alzheimer type. Early in the course of dementia, memory impairment is mild and usually most marked for recent events; people forget telephone numbers, conversations, and events of the day. As the course of dementia

progresses, memory impairment becomes severe, and the patient can only retain the earliest learned information (e.g., a person's place of birth).

Since memory is vital for orientation to person, place, and time, orientation can be progressively affected during a dementing illness. For example, patients with dementia may forget how to get back to their rooms after going to the bathroom. No matter how severe the disorientation seems, however, patients show no impairment in their level of consciousness.



Table 3-15
Neurocognitive Disorder due to Alzheimer Disease

DSM-5		ICD-10
Neurocognitive Disorder due to Alzheimer Disease		**Dementia in Alzheimer Disease**
Mild **Symptoms** • ↓ memory/learning • Progressive decline, no significant periods of stability • No evidence of mixed causes **Probable:** evidence of a genetic cause (family history/genetic testing) **Possible:** Above not met, all three	**Major** **Symptoms** • ↓ memory/learning + ≥1 other cognitive domain • Progressive decline, no significant periods of stability • **No evidence of mixed causes** **Probable:** evidence of a genetic cause (family history/genetic	**Dementia in Alzheimer disease with early onset** **Dementia in Alzheimer disease with late onset** **Dementia in Alzheimer disease, atypical or mixed type** **Dementia in Alzheimer disease, unspecified** **Early onset:** begins < age 65, rapid progression of functioning, involves higher levels of cognition **Late onset:** begins ≥ age 65, with slower progression,

symptoms present	testing) OR all three symptoms **Possible:** Some but not all the probable criteria met	preferentially affects memory Symptoms presenting with insidious onset, progressive decline over years, and include characteristic neuropathology and neurochemical features



Table 3-16
Frontotemporal Neurocognitive Disorder

DSM-5		**ICD-**
Frontotemporal Neurocognitive Disorder		**Dem** **Dise** **Brai**
Behavioral Variant **Affected:** social cognition/executive function **Spared:** learning/memory/perceptual–motor function **Associated Symptoms (≥3):** • Disinhibition • Apathy • ↓ sympathy/empathy • Perseveration/stereotypies • Hyperorality/dietary changes **Probable:** • evidence of a genetic	**Language Variant** **Affected:** language (word finding, naming, grammar, less spontaneous speech) **Spared:** learning/memory/perceptual–motor function **Probable:** • evidence of a genetic cause (family history/genetic testing) • Neuroimaging evidence of frontal/temporal pathology **Possible:** Some but not all the probable criteria met	Prog Onse **Affe** sk **Late** me fur

cause (family history/genetic testing)
- Neuroimaging evidence of frontal/temporal pathology

Possible:
- Some but not all the probable criteria met



Table 3-17
Neurocognitive Disorder with Lewy Bodies

DSM-5	ICD-10
Neurocognitive Disorder with Lewy Bodies	**Dementia in Other Specified Diseases Classified Elsewhere**
Core: • Fluctuating: • cognition • attention • alertness • Recurrent well-formed visual hallucinations • Parkinsonian symptoms (after cognitive decline) **Suggestive:** • REM sleep behavior disorder • Severe neuroleptic sensitivity **Probable:** ≥2 core features or 1 core/1 suggestive **Possible:** 1 core feature	Disorder is listed but not defined.

Table 3-18

Vascular Neurocognitive Disorder

DSM-5	ICD-10
Vascular Neurocognitive Disorder	**Vascular Dementia**
History + imaging/physical findings of cerebrovascular (CV) disease **Cognitive decline typical of vascular pathology (≥1):** • Cognitive decline after CV event • Cognitive areas mainly complex attention/frontal-executive function **Probable (≥1):** • Neuroimaging evidence of CV disease in brain • Neurocognitive symptoms are associated with the CV events • Clinical/genetic evidence of CV disease **Possible:** • No neuroimaging • No documented temporal relationship between CV events and cognitive changes	**Vascular dementia of acute onset: cognitive symptoms** develop after stroke(s) **Multi-infarct dementia:** progressive symptoms after repeated transient ischemic attacks **Subcortical Vascular dementia:** H/O hypertension, mainly white matter injury **Mixed cortical and subcortical vascular dementia** **Other vascular dementia** **Vascular dementia, unspecified**

Table 3-19
Neurocognitive Disorder due to Traumatic Brain Injury

DSM-5	ICD-10
Neurocognitive Disorder due to Traumatic Brain Injury	**Dementia in Other Specified Diseases Classified Elsewhere**
H/O traumatic brain injury, with associated noncognitive symptoms **Neurocognitive symptoms (≥1):** • Loss of consciousness • Amnesia following trauma • Confusion/disorientation • Neuroimaging evidence of injury/Neurologic symptoms	Disorder is listed but not defined.



Table 3-20
Neurocognitive Disorder due to HIV Infection

DSM-5	ICD-10
Neurocognitive Disorder due to HIV Infection	**Dementia in Human Immunodeficiency Virus (HIV)**
HIV diagnosis Associated cognitive symptoms	HIV diagnosis Associated cognitive symptoms



Table 3-21
Neurocognitive Disorder due to Prion Disease

DSM-5	ICD-10
Neurocognitive Disorder due to Prion Disease	**Dementia in Creutzfeldt-Jakob Disease**
Cognitive impairment: Subtle onset Rapid progression **Additional features:**	Significant neurologic symptoms Subacute course Progressive rapid decline

Motor symptoms, such as myoclonus/ataxia Biochemical marker on testing	



Table 3-22
Neurocognitive Disorder due to Parkinson Disease

DSM-5	ICD-10
Neurocognitive Disorder due to Parkinson Disease	**Dementia in Parkinson Disease**
Occurring in the context of Parkinson disease Subtle onset Gradual progression **Probable** (both of the following:) **Possible** (one of the following:) • No evidence of mixed etiology • Parkinson disease precedes cognitive symptoms	Occurring in the context of Parkinson's disease



Table 3-23
Neurocognitive Disorder due to Huntington Disease

DSM-5	ICD-10
Neurocognitive Disorder due to Huntington Disease	**Huntington Disease**
Occurring in the context of Huntington disease Subtle onset Gradual progression	Occurring in the context of Huntington disease Caused by autosomal dominant gene Slow progression Motor symptom (chorea)

Dementing processes that affect the cortex, primarily dementia of the Alzheimer type and vascular dementia, can affect patients' language abilities.

Psychiatric and Neurologic Changes

Personality. Changes in the personality of a person with dementia are especially disturbing for their families. Preexisting personality traits may accentuate during the development of dementia. Patients with dementia may also become introverted and seem to be less concerned than they previously were about the effects of their behavior on others. Persons with dementia who have paranoid delusions are generally hostile to family members and caretakers. Patients with frontal and temporal involvement are likely to have marked personality changes and may be irritable and explosive.

Hallucinations and Delusions. An estimated 20 to 30 percent of patients with dementia (primarily patients with dementia of the Alzheimer type) have hallucinations, and 30 to 40 percent have delusions, primarily of a paranoid or persecutory and unsystematized nature, although complex and sustained. Some patients also report well-systematized delusions. Physical aggression and other forms of violence are common in demented patients who also have psychotic symptoms.

Mood. In addition to psychosis and personality changes, depression and anxiety are significant symptoms in an estimated 40 to 50 percent of patients with dementia, although the full syndrome of depressive disorder may be present in only 10 to 20 percent. Patients with dementia also may exhibit pathologic laughter or crying—that is, extremes of emotions—with no apparent provocation.

Cognitive Change. In addition to the aphasias in patients with dementia, apraxias and agnosias are common. Other neurologic signs that can be associated with dementia are seizures, seen in approximately 10 percent of patients with dementia of the Alzheimer type and 20 percent of patients with vascular dementia, and atypical neurologic presentations, such as nondominant parietal lobe syndromes. Primitive reflexes, such as the grasp, snout, suck, tonic-foot, and palmomental reflexes, may be present on

neurologic examination, and myoclonic jerks are present in 5 to 10 percent of patients.



Table 3-24
Additional Diagnosis Types of Dementia in DSM-5

Neurocognitive disorder due to another medical condition: In the context of some other disorder not listed among the neurocognitive disorders
Neurocognitive disorder due to multiple etiologies
Substance/medication-induced neurocognitive disorder
Unspecified neurocognitive disorder

Patients with vascular dementia may have additional neurologic symptoms, such as headaches, dizziness, faintness, weakness, focal neurologic signs, and sleep disturbances, possibly attributable to the location of the cerebrovascular disease. Pseudobulbar palsy, dysarthria, and dysphagia are also more common in vascular dementia than in other dementing conditions.

Abstractions and the Catastrophic Reaction. Patients with dementia also exhibit a reduced ability to apply what Kurt Goldstein called the "abstract attitude." Patients have difficulty generalizing from a single instance, forming concepts, and grasping similarities and differences among concepts. Furthermore, the ability to solve problems, to reason logically, and to make sound judgments is compromised. Goldstein also described a catastrophic reaction marked by agitation secondary to the subjective awareness of intellectual deficits under stressful circumstances. Persons usually attempt to compensate for defects by using strategies to avoid demonstrating failures in intellectual performance; they may change the subject, make jokes, or otherwise divert the interviewer. Lack of judgment and poor impulse control appear commonly, particularly in dementias that primarily affect the frontal lobes. Examples of these impairments include coarse language, inappropriate jokes, neglect of personal appearance and hygiene, and a general disregard for the conventional rules of social conduct.

Sundowner Syndrome. Sundowner syndrome is characterized by

drowsiness, confusion, ataxia, and accidental falls. It occurs in older people who are overly sedated and in patients with dementia who react adversely to even a small dose of a psychoactive drug. The syndrome also occurs in demented patients when external stimuli, such as light and interpersonal orienting cues, are diminished.

Vascular Dementia

The general symptoms of vascular dementia are the same as those for dementia of the Alzheimer type, but the diagnosis of vascular dementia requires either clinical or laboratory evidence in support of a vascular cause of dementia. Vascular dementia is more likely to show a decremental, stepwise deterioration than is Alzheimer disease.

Substance-Induced Persisting Dementia

DSM lists substance-induced persisting dementia in two places, with the dementias, and with the substance-related disorders. The specific substances it cross-references are alcohol, inhalants, sedatives, hypnotics, or anxiolytics, and other or unknown substances.

Alcohol-Induced Persisting Dementia. For alcohol-induced persisting dementia, the patient must meet the criteria for dementia. Because amnesia can also occur in the context of Korsakoff psychosis, it is essential to distinguish between memory impairment accompanied by other cognitive deficits (i.e., dementia) and amnesia caused by thiamine deficiency. To complicate matters, however, evidence also suggests that other cognitive functions, such as attention and concentration, may also be impaired in Wernicke–Korsakoff syndrome. Also, alcohol abuse is frequently associated with mood changes, so poor concentration and other cognitive symptoms often observed in the context of major depression must also be ruled out. Prevalence rates differ considerably according to the population studied and the diagnostic criteria used, although alcohol-related dementia roughly accounts for 4 percent of dementias.

Pathology, Physical Findings, and Laboratory Examination

We should perform a comprehensive laboratory workup when evaluating a

patient with dementia. The purposes of the workup are to detect reversible causes of dementia and to provide the patient and family with a definitive diagnosis. The range of possible causes of dementia mandates selective use of laboratory tests. The evaluation should follow informed clinical suspicion based on the history and physical and mental status examination results. The continued improvements in brain imaging techniques, particularly MRI, have made the differentiation between dementia of the Alzheimer type and vascular dementia, in some cases, somewhat more straightforward than in the past. Single-photon emission computed tomography (SPECT) can be a useful adjunctive test to detect patterns of brain metabolism in various types of dementias.

A general physical examination is a routine component of the workup for dementia. It may reveal evidence of systemic disease causing brain dysfunction, such as an enlarged liver and hepatic encephalopathy, or it may demonstrate systemic disease related to particular CNS processes. The detection of Kaposi sarcoma, for example, should alert the clinician to the probable presence of AIDS and the associated possibility of AIDS dementia complex. Focal neurologic findings, such as asymmetrical hyperreflexia or weakness, are seen more often in vascular than in degenerative disease. Frontal lobe signs and primitive reflexes occur in many disorders and often point to more significant progression.

DIFFERENTIAL DIAGNOSIS

Dementia of the Alzheimer Type versus Vascular Dementia

Classically, vascular dementia has been distinguished from dementia of the Alzheimer type by the decremental deterioration that can accompany cerebrovascular disease over time. Although the discrete, stepwise deterioration may not be apparent in all cases, focal neurologic symptoms are more common in vascular dementia than in dementia of the Alzheimer type, as are the standard risk factors for cerebrovascular disease.

Vascular Dementia versus Transient Ischemic Attacks

Transient ischemic attacks (TIAs) are brief episodes of focal neurologic dysfunction lasting less than 24 hours (usually 5 to 15 minutes). Although a

variety of mechanisms may be responsible, the episodes are frequently the result of microembolization from a proximal intracranial arterial lesion that produces transient brain ischemia, and the episodes usually resolve without significant pathologic alteration of the parenchymal tissue. Approximately one-third of persons with untreated TIAs experience a brain infarction later; therefore, recognition of TIAs is an essential clinical strategy to prevent brain infarction.

Clinicians should distinguish episodes involving the vertebrobasilar system from those involving the carotid arterial system. In general, symptoms of vertebrobasilar disease reflect a transient functional disturbance in either the brainstem or the occipital lobe; carotid distribution symptoms reflect unilateral retinal or hemispheric abnormality. Anticoagulant therapy, antiplatelet agglutinating drugs such as aspirin, and extracranial and intracranial reconstructive vascular surgery are effective in reducing the risk of infarction in patients with TIAs.

Delirium

In general, delirium has a rapid onset, brief duration, cognitive impairment fluctuation during the day; nocturnal exacerbation of symptoms; marked disturbance of the sleep–wake cycle; and prominent disturbances in attention and perception (Table 3-8).

Depression

Some patients with depression have symptoms of cognitive impairment challenging to distinguish from symptoms of dementia. The clinical picture is sometimes referred to as *pseudodementia*, although the term *depression-related cognitive dysfunction* is preferable and more descriptive (Table 3-25). Patients with depression-related cognitive dysfunction generally have prominent depressive symptoms, more insight into their symptoms than do demented patients, and often a history of depressive episodes. Fortunately, memory impairment from depression usually responds to antidepressant medication.

Table 3-25
Major Clinical Features Differentiating Depression-Related

Cognitive Dysfunction from Dementia

Depression	Dementia
Clinical Course and History	
Family always aware of dysfunction and its severity	Family often unaware of dysfunction and its severity
Onset can be dated with some precision	Onset can be dated only within broad limits
Symptoms of short duration before medical help is sought	Symptoms usually of long duration before medical help is sought
Rapid progression of symptoms after onset	Slow progression of symptoms throughout course
History of previous psychiatric dysfunction common	History of previous psychiatric dysfunction unusual
Complaints and Clinical Behavior	
Patients usually complain much of cognitive loss	Patients usually complain little of cognitive loss
Patients' complaints of cognitive dysfunction usually detailed	Patients' complaints of cognitive dysfunction usually vague
Patients emphasize disability	Patients conceal disability
Patients highlight failures	Patients delight in accomplishments, however trivial
Patients make little effort to perform even simple tasks	Patients struggle to perform tasks
	Patients rely on notes, calendars, and so on to keep up

Patients usually communicate strong sense of distress	Patients often appear unconcerned
Affective change often pervasive	Affect labile and shallow
Loss of social skills often early and prominent	Social skills often retained
Behavior often incongruent with severity of cognitive dysfunction	Behavior usually compatible with severity of cognitive dysfunction
Nocturnal accentuation of dysfunction uncommon	Nocturnal accentuation of dysfunction common
Clinical Features Related to Memory, Cognitive, and Intellectual Dysfunctions	
Attention and concentration often well preserved	Attention and concentration usually faulty
“Don’t know” answers typical	Near-miss answers frequent
On tests of orientation, patients often give “don’t know” answers	On tests of orientation, patients often mistake unusual for usual
Memory loss for recent and remote events usually severe	Memory loss for recent events usually more severe than for remote events
Memory gaps for specific periods or events common	Memory gaps for specific periods unusual[a]
Marked variability in performance on tasks of similar difficulty	Consistently poor performance on tasks of similar difficulty

[a]Except when caused by delirium, trauma, seizures, and so on.

Reprinted with permission from Wells CE. Pseudodementia. *Am J Psychiatry*. 1979;136:898.

Malingering and Factitious Disorder

Persons who attempt to simulate memory loss, as in factitious disorder, do so erratically and inconsistently. In true dementia, one loses memory for time and place before memory for person, and recent memory before remote memory.

Schizophrenia

Although schizophrenia can be associated with some acquired intellectual impairment, its symptoms are much less severe than are the related symptoms of psychosis and thought disorder.

Normal Aging

Aging is not necessarily associated with any significant cognitive decline, but minor memory problems can occur as a normal part of aging. We sometimes call these normal occurrences *benign senescent forgetfulness, age-associated memory impairment,* or *normal benign age-related senescence.* They are distinguished from dementia by their minor severity and because they do not interfere significantly with a person's social or occupational behavior.

Other Disorders

Intellectual disability, which does not include memory impairment, occurs in childhood. Amnestic disorder is a circumscribed loss of memory and no deterioration. We should rule out a pituitary disorder, but they are a rare cause.

COURSE AND PROGNOSIS

The classic course of dementia is onset in the patient's 60s (although there may be subtle earlier signs), with gradual deterioration over 5 to 10 years, leading eventually to death. The age of onset and the rapidity of deterioration vary among different types of dementia and within individual diagnostic categories. The average survival expectation for patients with dementia of the Alzheimer type is approximately 8 years, with a range of 1 to 20 years. Data suggest that in persons with an early onset of dementia or with a family history of dementia, the disease is likely to have a rapid course. In a recent

study of 821 persons with Alzheimer disease, the median survival time was 3.5 years. After diagnosing dementia, patients must have a complete medical and neurologic workup because 10 to 15 percent of all patients with dementia have a potentially reversible condition if we initiate treatment before permanent brain damage occurs.

The most common course of dementia begins with several subtle signs that may, at first, be ignored by both the patient and the people closest to the patient. Gradual onset of symptoms is most commonly associated with dementia of the Alzheimer type, vascular dementia, endocrinopathies, brain tumors, and metabolic disorders. Conversely, the onset of dementia resulting from head trauma, cardiac arrest with cerebral hypoxia, or encephalitis can be sudden. Although the symptoms of the early phase of dementia are subtle, they become conspicuous as dementia progresses, and family members may then bring a patient to a physician's attention. People with dementia may be sensitive to the use of benzodiazepines or alcohol, which can precipitate agitated, aggressive, or psychotic behavior. In the terminal stages of dementia, patients become empty shells of their former selves—profoundly disoriented, incoherent, amnestic, and incontinent of urine and feces.

With psychosocial and pharmacologic treatment and possibly because of the self-healing properties of the brain, the symptoms of dementia may progress slowly for a time or may even recede somewhat. Symptom regression is certainly a possibility in reversible dementias (dementias caused by hypothyroidism, normal pressure hydrocephalus [NPH], and brain tumors) after initiating treatment. The course of dementia varies from a steady progression (commonly seen with dementia of the Alzheimer type) to an incrementally worsening dementia (commonly seen with vascular dementia) to stable dementia (as may be seen in dementia related to head trauma).

Psychosocial Determinants

The severity and course of dementia can be affected by psychosocial factors. The higher a person's premorbid intelligence and education, the better they can compensate for intellectual deficits. People who have a rapid onset of dementia use fewer defenses than do those who experience an insidious onset. Anxiety and depression can intensify and aggravate the symptoms.

TREATMENT

The first step in the treatment of dementia is the verification of the diagnosis. Accurate diagnosis is imperative because the progression may be halted or even reversed if one provides appropriate therapy. Preventive measures are essential, particularly in vascular dementia. Such measures might include changes in diet, exercise, and control of diabetes and hypertension. Pharmacologic agents might include antihypertensive, anticoagulant, or antiplatelet agents. Blood pressure control should aim for the higher end of the normal range because it may improve cognitive function in patients with vascular dementia. Blood pressure below the normal range can worsen cognitive function in patients with dementia. The choice of an antihypertensive agent can be significant in that β-adrenergic receptor antagonists are associated with exaggeration of cognitive impairment. Angiotensin-converting enzyme (ACE) inhibitors and diuretics have not been linked to exaggeration of cognitive impairment and may lower blood pressure without affecting cerebral blood flow, which presumably correlates with cognitive function. Surgical removal of carotid plaques may prevent subsequent vascular events in carefully selected patients. The general treatment approach to patients with dementia is to provide supportive medical care; emotional support for the patients and their families; and pharmacologic treatment for specific symptoms, including disruptive behavior.

Psychosocial Therapies

The deterioration of mental faculties has significant psychological meaning for patients with dementia. The experience of a sense of continuity over time depends on memory. Patients lose recent memory before remote memory in most cases of dementia, and many patients are profoundly distressed by clearly recalling how they used to function while observing their noticeable deterioration. At the most fundamental level, the self is a product of brain functioning. Patients' identities begin to fade as the illness progresses, and they can recall less and less of their past. Emotional reactions ranging from depression to severe anxiety to catastrophic terror can stem from the realization that the sense of self is disappearing.

Patients often benefit from supportive and educational psychotherapy in which the clinician clearly explains the nature and course of their illness.

They may also benefit from assistance in grieving and accepting the extent of their disability and from attention to self-esteem issues. We should try to maximize any areas of intact functioning by helping patients identify activities in which successful functioning is possible. A psychodynamic assessment of defective functions and cognitive limitations can also be useful. Clinicians can help patients find ways to deal with the defective functions, such as keeping calendars for orientation problems, making schedules to help structure activities, and taking notes for memory problems.

Psychodynamic interventions with family members of patients with dementia may be of great assistance. Those who take care of a patient struggle with feelings of guilt, grief, anger, and exhaustion as they watch a family member gradually deteriorate. A common problem that develops among caregivers involves their self-sacrifice in caring for a patient. They often suppress any developing resentment from this self-sacrifice because of the guilt feelings it produces. Clinicians can help caregivers understand the complex mixture of feelings associated with seeing a loved one decline and can provide understanding as well as permission to express these feelings. Clinicians must also be aware of the caregivers' tendencies to blame themselves or others for patients' illnesses and must appreciate the role that patients with dementia play in the lives of family members.

Pharmacotherapy

Clinicians may prescribe sedative-hypnotics for insomnia and anxiety, antidepressants for depression, and antipsychotic drugs for delusions and hallucinations, but they should be aware of the possible adverse drug effects that may be worse in older people (e.g., disinhibition, confusion, and oversedation). In general, we should avoid drugs with high anticholinergic activity.

Donepezil, rivastigmine, galantamine, and tacrine are cholinesterase inhibitors used to treat mild to moderate cognitive impairment in Alzheimer disease. They reduce the inactivation of the neurotransmitter acetylcholine and thus potentiate the cholinergic neurotransmitter, which in turn produces a modest improvement in memory and goal-directed thought. These drugs are most useful for persons with mild to moderate memory loss who have sufficient preservation of their basal forebrain cholinergic neurons to benefit

from augmentation of cholinergic neurotransmission.

Donepezil is well tolerated and widely used. We rarely use tacrine because of its potential for hepatotoxicity. Fewer clinical data are available for rivastigmine and galantamine, which appear more likely to cause gastrointestinal (GI) and neuropsychiatric adverse effects than does donepezil. None of these medications prevent the progressive neuronal degeneration of the disorder.

Memantine protects neurons from excessive amounts of glutamate, which may be neurotoxic. Sometimes we may combine the drug with donepezil. It appears to improve dementia sometimes.

Other Treatment Approaches. There are many drugs under development for cognitive-enhancing activity and this is a very active area of drug research. Novel strategies are focusing on the β-amyloid plaques and neurofibrillary tangles associated with Alzheimer and some other neurodegenerative disorders. In the case of amyloid plaques, attempts to reduce their formation, aggregation and to increase their clearance are underway, and several monoclonal and polyclonal antibody vaccines are being tested.

Estrogen replacement therapy may reduce the risk of cognitive decline in postmenopausal women. However the data is correlational and clinical trials to date are largely disappointing. A variety of other interventions have been proposed to improve cognitive outcomes in dementia, including cognitive stimulation, music therapy, reminiscence therapy, omega 3 fish oil, statins, ginkgo biloba, aromatherapy, or nonsteroidal anti-inflammatory drugs; however, all lack sufficient evidence to make any recommendation. Exercise improves cognition in healthy adults, and there is some evidence of benefit in dementia, particularly in the early stages.

EPIDEMIOLOGY

With the aging population, the prevalence of dementia is rising. The prevalence of moderate to severe dementia in different population groups is approximately 5 percent in the general population older than 65 years of age, 20 to 40 percent in the general population older than 85 years of age, 15 to 20 percent in outpatient general medical practices, and 50 percent in chronic care

facilities.

Of all patients with dementia, 50 to 60 percent have the most common type of dementia, dementia of the Alzheimer type (Alzheimer disease). Dementia of the Alzheimer type increases in prevalence with increasing age. For persons age 65 years, the prevalence for men is 0.6 percent and women, 0.8 percent. These rates increase with age, and at age 90, about one-fifth of people have the disorder. For all of these figures, 40 to 60 percent of cases are moderate to severe. The rates of prevalence (men to women) are 11 and 14 percent at age 85 years, 21 and 25 percent at age 90 years, and 36 and 41 percent at age 95 years. Patients with dementia of the Alzheimer type occupy more than 50 percent of nursing home beds. More than 2 million persons with dementia live at home. By 2050, current predictions suggest that there will be 14 million Americans with Alzheimer disease and, therefore, more than 18 million people with dementia.

The second most common type of dementia is vascular dementia, which is causally related to cerebrovascular diseases. Hypertension predisposes a person to the disease. Vascular dementias account for 15 to 30 percent of all dementia cases. Vascular dementia is most common in persons between the ages of 60 and 70 and is more common in men than in women. Approximately 10 to 15 percent of patients have coexisting vascular dementia and dementia of the Alzheimer type.

Other common causes of dementia, each representing 1 to 5 percent of all cases, include head trauma; alcohol-related dementias; and various movement disorder-related dementias, such as Huntington disease and Parkinson disease. Because dementia is a reasonably general syndrome, it has many causes, and clinicians must embark on a careful clinical workup of a patient with dementia to establish its cause.

ETIOLOGY

The most common causes of dementia in individuals older than 65 years of age are (1) Alzheimer disease, (2) vascular dementia, and (3) mixed vascular and Alzheimer disease. Other illnesses that account for approximately 10 percent include Lewy body dementia, Pick disease, frontotemporal dementias, NPH, alcoholic dementia, infectious dementia, such as HIV or syphilis, and Parkinson disease. Many types of dementias evaluated in

clinical settings can be attributable to reversible causes, such as metabolic abnormalities (e.g., hypothyroidism), nutritional deficiencies (e.g., vitamin B_{12} or folate deficiencies), or dementia syndrome caused by depression. See Table 3-26 for a review of possible etiologies of dementia.

Dementia of the Alzheimer Type

In 1907, Alois Alzheimer first described the condition that later assumed his name. He described a 51-year-old woman with a 4½-year course of progressive dementia. The final diagnosis of Alzheimer disease requires a neuropathologic examination of the brain; nevertheless, we can diagnose dementia of the Alzheimer type in the clinical setting after excluding other causes of dementia.

Genetic Factors. Although the cause of dementia of the Alzheimer type remains unknown, researchers have made impressive progress toward understanding the molecular basis of the amyloid deposits that are a hallmark of the disorder's neuropathology. Studies suggest as many as 40 percent of patients have a family history of dementia of the Alzheimer type. Thus, genetic factors are likely to play some part in the development of the disorder. Additional support for a genetic influence is the concordance rate for monozygotic twins, which is higher than the rate for dizygotic twins (43 percent vs. 8 percent, respectively). In several well-documented cases, the disorder has been transmitted in families through an autosomal dominant gene, although such transmission is rare. Alzheimer-type dementia has shown linkage to chromosomes 1, 14, and 21.



Table 3-26
Possible Etiologies of Dementia

Degenerative dementias
Alzheimer disease
Frontotemporal dementias (e.g., Pick disease)
Parkinson disease
Lewy body dementia
Idiopathic cerebral ferrocalcinosis (Fahr disease)

- Progressive supranuclear palsy

Miscellaneous

- Huntington disease
- Wilson disease
- Metachromatic leukodystrophy
- Neuroacanthocytosis

Psychiatric

- Pseudodementia of depression
- Cognitive decline in late-life schizophrenia

Physiologic

- Normal pressure hydrocephalus

Metabolic

- Vitamin deficiencies (e.g., vitamin B_{12}, folate)
- Endocrinopathies (e.g., hypothyroidism)
- Chronic metabolic disturbances (e.g., uremia)

Tumor

- Primary or metastatic (e.g., meningioma or metastatic breast or lung cancer)

Traumatic

- Dementia pugilistica, posttraumatic dementia
- Subdural hematoma

Infection

- Prion diseases (e.g., Creutzfeldt–Jakob disease, bovine spongiform encephalitis, Gerstmann–Sträussler syndrome)
- Acquired immune deficiency syndrome (AIDS)
- Syphilis

Cardiac, vascular, and anoxia

- Infarction (single or multiple or strategic lacunar)
- Binswanger disease (subcortical arteriosclerotic encephalopathy)
- Hemodynamic insufficiency (e.g., hypoperfusion or hypoxia)

Demyelinating diseases

- Multiple sclerosis

Drugs and toxins

- Alcohol

Heavy metals
Irradiation
Cognitive dysfunction due to medications (e.g., anticholinergics)
Carbon monoxide

AMYLOID PRECURSOR PROTEIN. The gene for amyloid precursor protein (APP) is on the long arm of chromosome 21. The process of differential splicing yields four forms of the APP. The β/A4 protein, the principal constituent of senile plaques, is a 42-amino acid peptide that is a breakdown product of APP. Down syndrome (trisomy 21) has three copies of the *APP* gene, and with a mutation at codon 717 in the *APP* gene, a pathologic process results in the excessive deposition of β/A4 protein. Whether the processing of abnormal APP is of primary causative significance in Alzheimer disease is unknown. However, many research groups are studying both the standard metabolic processing of APP and its processing in patients with dementia of the Alzheimer type in an attempt to answer this question.

MULTIPLE E4 GENES. One study implicated gene E4 in the origin of Alzheimer disease. People with one copy of the gene have Alzheimer disease three times more frequently than do those with no E4 gene, and people with two E4 genes have the disease eight times more frequently than do those with no E4 gene. Diagnostic testing for this gene is not contributory because it is found in persons without dementia and not found in all cases of dementia.

Neuropathology. The classic gross neuroanatomical observation of a brain from a patient with Alzheimer disease is diffuse atrophy with flattened cortical sulci and enlarged cerebral ventricles. The classic and pathognomonic microscopic findings are senile plaques, neurofibrillary tangles, neuronal loss (particularly in the cortex and the hippocampus), synaptic loss (perhaps as much as 50 percent in the cortex), and granulovascular degeneration of the neurons. Neurofibrillary tangles (Fig. 3-1) are composed of cytoskeletal elements, primarily phosphorylated tau protein, although other cytoskeletal proteins are also present. Neurofibrillary tangles are not unique to Alzheimer disease; they also occur in Down syndrome, dementia pugilistica (punch-drunk syndrome), Parkinson–

dementia complex of Guam, Hallervorden–Spatz disease, and the brains of healthy people as they age. Neurofibrillary tangles are common in the cortex, the hippocampus, the substantia nigra, and the locus ceruleus.

Senile plaques, also referred to as *amyloid plaques*, more strongly indicate Alzheimer disease, although they are also seen in Down syndrome and, to some extent, in healthy aging. Senile plaques are composed of a particular protein, β/A4, and astrocytes, dystrophic neuronal processes, and microglia. The number and the density of senile plaques present in postmortem brains correlate with the severity of the disease.

Neurotransmitters. The neurotransmitters usually implicated in the pathophysiologic condition of Alzheimer disease are acetylcholine and norepinephrine, both of which are likely hypoactive in Alzheimer disease. Several studies have reported data consistent with the hypothesis that specific degeneration of cholinergic neurons is present in the nucleus basalis of Meynert in persons with Alzheimer disease. Other data supporting a cholinergic deficit in Alzheimer disease demonstrate decreased acetylcholine and choline acetyltransferase concentrations in the brain. Choline acetyltransferase is the critical enzyme for the synthesis of acetylcholine, and a reduction in choline acetyltransferase concentration suggests a decrease in the number of cholinergic neurons present. Additional support for the cholinergic deficit hypothesis comes from the observation that cholinergic antagonists, such as scopolamine and atropine, impair cognitive abilities, whereas cholinergic agonists, such as physostigmine and arecoline, enhance cognitive abilities. The decrease in norepinephrine-containing neurons in the locus ceruleus found in some pathologic examinations of Alzheimer disease brains suggests decreased norepinephrine activity in the disease. Two other neurotransmitters implicated in the pathophysiologic condition of Alzheimer disease are the neuroactive peptides somatostatin and corticotropin; there are decreased concentrations of both in persons with Alzheimer disease.

Other Causes. Another theory to explain the development of Alzheimer disease is that an abnormality in the regulation of membrane phospholipid metabolism results in membranes that are less fluid—that is, more rigid—than usual. Several investigators are using molecular resonance spectroscopic imaging to assess this hypothesis directly in patients with dementia of the

Alzheimer type. Some have suggested that aluminum toxicity is a causative factor because of the high levels of aluminum found in the brains of some patients with Alzheimer disease, but this is no longer considered a significant etiologic factor. Excessive stimulation by the transmitter glutamate that may damage neurons is another theory of causation.

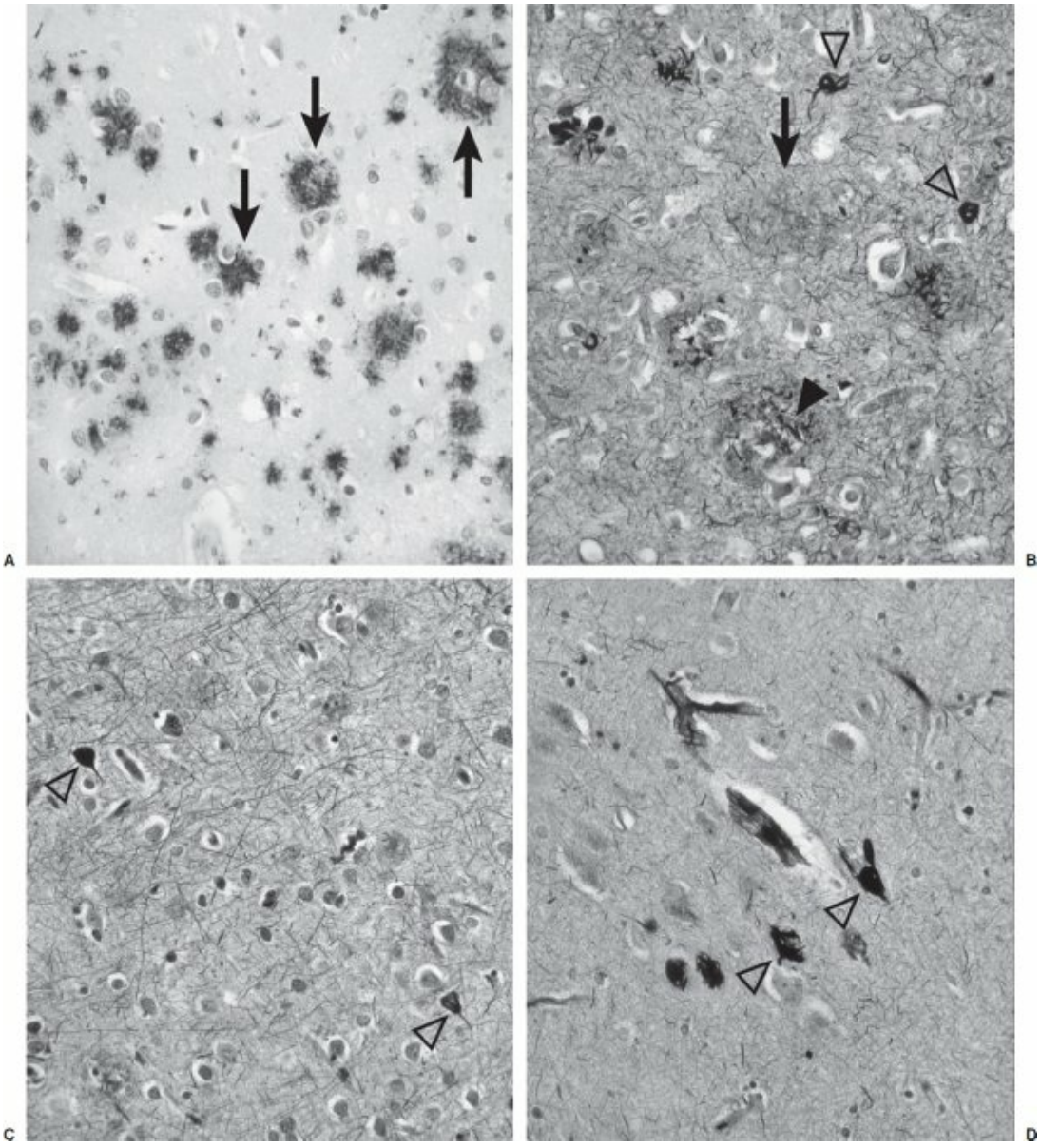


FIGURE 3-1
Photomicrographs of Alzheimer disease neuropathology. **A:** Deposition of insoluble fibrillar Aβ into plaques begins in the neocortex, labeled here using an antibody against Aβ and appearing as reddish-brown deposits (*arrows*). **B:** Bielchowsky stain of neocortex from an individual who died in advanced stages of Alzheimer disease (Braak stage VI). The Aβ plaques appear as dark brown in this preparation (*arrows*) and can be seen to be associated with dystrophic neuronal

processes (*arrowheads*) in which insoluble microtubule-associated protein τ (MAPT) aggregates appear as black deposits. This neurofibrillary pathology also appears extensively throughout the neuropil, and several neurofibrillary tangles can be seen (*open arrowheads*). **C:** Bielchowsky stain of neocortex from an individual who died in a less advanced disease stage (Braak stage IV). Although some neurofibrillary tangles are still evident (*open arrowheads*), the degree of neurofibrillary pathology in the neuropil is substantially diminished. **D:** Isolated neurofibrillary tangles (*open arrowheads*) in entorhinal cortex that can be seen in normal aging (Bielchowsky stain). Notice the lack of Aβ plaques and limited neuropil involvement. Aβ, Beta amyloid. (All images obtained at 200× magnification and provided courtesy of Dr. Ronald L. Hamilton, Department of Pathology, Division of Neuropathology, University of Pittsburgh School of Medicine.)

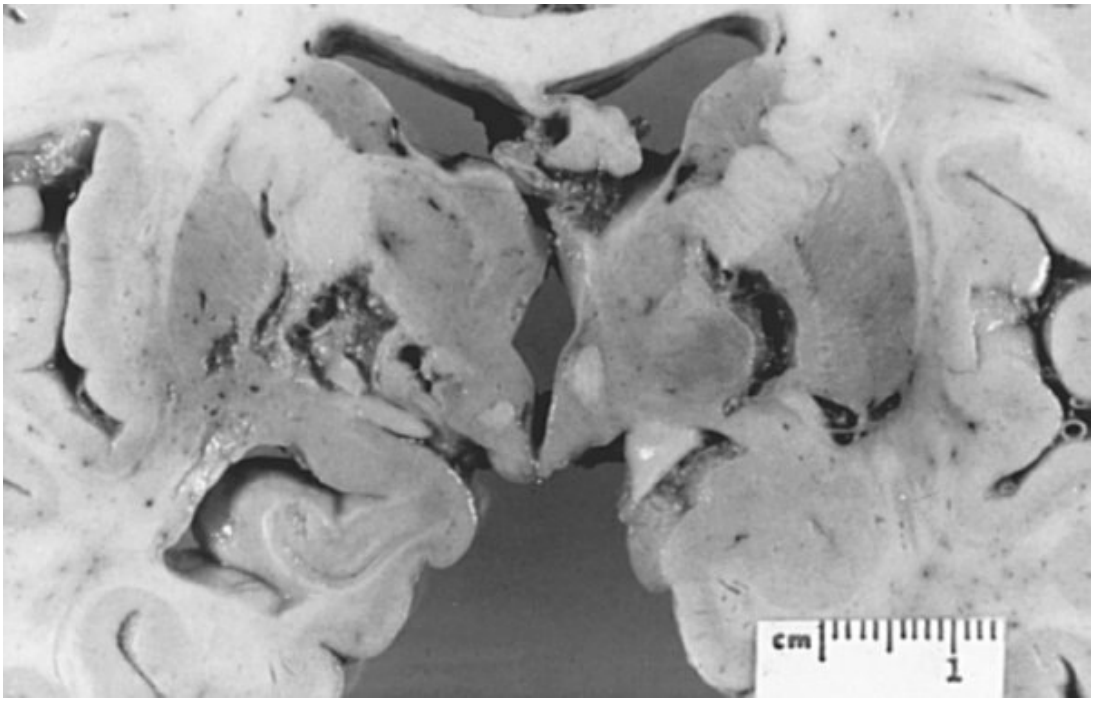


FIGURE 3-2
Gross appearance of the cerebral cortex on coronal section from a patient with vascular dementia. The multiple bilateral lacunar infarcts involve the thalamus, the internal capsule, and the globus pallidus. (Courtesy of Daniel P. Perl, M.D.)

Familial Multiple System Tauopathy with Presenile Dementia. A recently discovered type of dementia, familial multiple system tauopathy, shares some brain abnormalities found in people with Alzheimer disease. The gene that causes the disorder is likely on chromosome 17. The symptoms of the disorder include short-term memory problems and difficulty maintaining balance and walking. The onset of the disease is in the 40s and 50s. Persons with the disease live an average of 11 years after the onset of symptoms.

As in patients with Alzheimer disease, tau protein builds up in neurons and glial cells of persons with familial multiple system tauopathy. Eventually, the protein buildup kills brain cells. The disorder is not associated with the senile plaques seen with Alzheimer disease.

Mr. J, a 70-year-old retired businessman, was brought to psychiatric services on referral by the family physician. His wife claimed that Mr. J had become so forgetful that she was afraid to leave him alone, even at home. Mr. J retired at age 62 years after experiencing a decline in work performance during the previous 5 years. He also slowly gave up hobbies he once enjoyed (photography, reading, golf) and became increasingly quiet. However, his growing forgetfulness went unnoticed at home. Then one day, while walking in an area he knew well, he could not find his way home. From then on, his memory failure began to increase. He would forget appointments, misplace things, and lose his way around the neighborhood he resided in for 40 years. He failed to recognize people, even those he knew for many years. His wife had to start bathing and dressing him because he forgot how to do so himself.

On examination, Mr. J could not recall the time or place. He was only able to recall his name and place of birth. Mr. J seemed lost during the interview, only responding to questions with an occasional shrug of his shoulders. When asked to name objects or to recall words or numbers, Mr. J appeared tense and distressed. Mr. J had difficulty following instructions and was unable to dress or undress. His general medical condition was good. Laboratory examinations showed abnormalities on Mr. J's EEG and CT scans.

Vascular Dementia

The primary cause of vascular dementia, formerly referred to as *multi-infarct dementia*, is presumed to be multiple areas of cerebral vascular disease, resulting in a symptom pattern of dementia. Vascular dementia is most common in men, especially those with preexisting hypertension or other cardiovascular risk factors. The disorder affects primarily small- and medium-sized cerebral vessels, which undergo infarction and produce

multiple parenchymal lesions spread over wide areas of the brain (Fig. 3-2). The causes of the infarctions can include occlusion of the vessels by arteriosclerotic plaques or thromobemboli from distant origins (e.g., heart valves). An examination of a patient may reveal carotid bruits, funduscopic abnormalities, or enlarged cardiac chambers.

Binswanger Disease. Binswanger disease, also known as *subcortical arteriosclerotic encephalopathy*, is characterized by the presence of many small infarctions of the white matter that spare the cortical regions (Fig. 3-3). Although Binswanger disease was previously considered a rare condition, the advent of sophisticated and powerful imaging techniques, such as MRI, has revealed that the condition is more common than previously thought.



FIGURE 3-3
Binswanger disease. Cross-section demonstrating extensive subcortical white matter infarction, with sparing of the overlying gray matter. (Courtesy of Dushyant

Purohit, M.D., Neuropathology Division, Mount Sinai School of Medicine, New York, NY.)

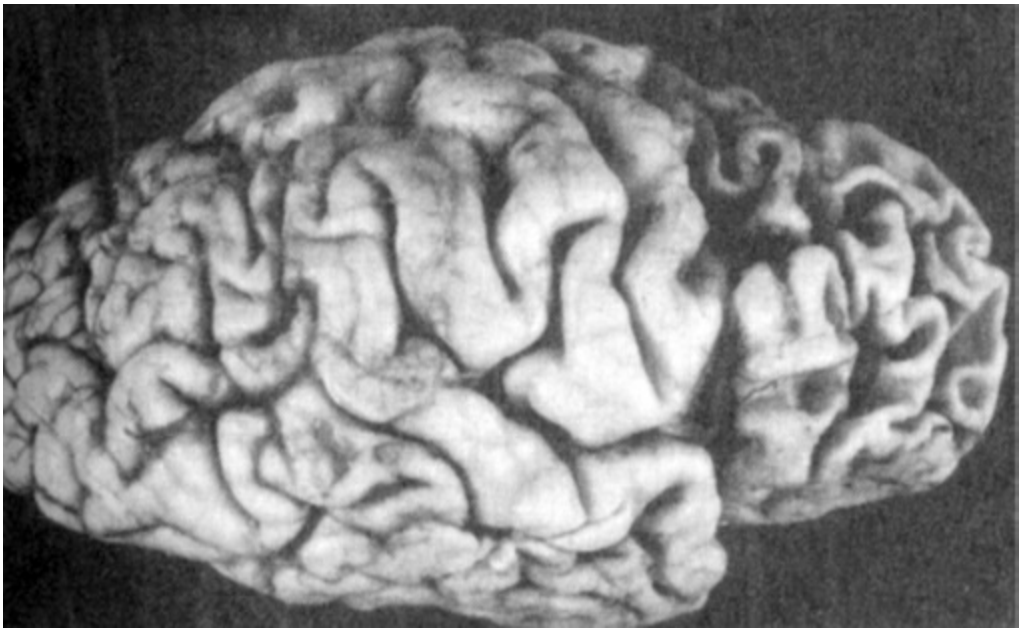


FIGURE 3-4
Pick disease gross pathology. This demonstrates the marked frontal and temporal atrophy seen in frontotemporal dementias, such as Pick disease. (Courtesy of Dushyant Purohit, M.D., Neuropathology Division, Mount Sinai School of Medicine, New York, NY.)

Frontotemporal Dementia (Pick Disease)

In contrast to the parietal–temporal distribution of pathologic findings in Alzheimer disease, Pick disease is a preponderance of atrophy in the frontotemporal regions. These regions also have neuronal loss, gliosis, and neuronal Pick bodies, which are masses of cytoskeletal elements. Pick bodies are seen in some postmortem specimens but are not necessary for the diagnosis. The cause of Pick disease is unknown, but the disease constitutes approximately 5 percent of all irreversible dementias. It is most common in men, especially those who have a first-degree relative with the condition. Pick disease is difficult to distinguish from dementia of the Alzheimer type. However, the early stages of Pick disease are more often characterized by personality and behavioral changes, with relative preservation of other cognitive functions, and it typically begins before 75 years of age. Familial cases may have an earlier onset, and some studies have shown that approximately half of the cases of Pick disease are familial (Fig. 3-4).

Features of Klüver–Bucy syndrome (e.g., hypersexuality, placidity, hyperorality) are much more common in Pick disease than in Alzheimer disease.

Table 3-27
Clinical Criteria for Dementia with Lewy Bodies (DLB)

The patient must have sufficient cognitive decline to interfere with social or occupational functioning. Of note, early in the illness, memory symptoms may not be as prominent as attention, frontosubcortical skills, and visuospatial ability. Probable DLB requires two or more core symptoms, whereas possible DLB only requires one core symptom.

Core features

- Fluctuating levels of attention and alertness
- Recurrent visual hallucinations
- Parkinsonian features (cogwheeling, bradykinesia, and resting tremor)

Supporting features

- Repeated falls
- Syncope
- Sensitivity to neuroleptics
- Systematized delusions
- Hallucinations in other modalities (e.g. auditory, tactile)

Adapted from McKeith LG, Galasko D, Kosaka K. Consensus guidelines for the clinical and pathologic diagnosis of dementia with Lewy bodies (DLB): Report of the consortium on DLB international workshop. 1996;47:1113–1124, with permission.

Lewy Body Disease

Lewy body disease is clinically similar to Alzheimer disease. However, it commonly presents with hallucinations, parkinsonian features, and extrapyramidal signs (Table 3-27). Lewy inclusion bodies are in the cerebral cortex (Fig. 3-5). The exact incidence is unknown. These patients often have Capgras syndrome (reduplicative paramnesia) as part of the clinical picture.

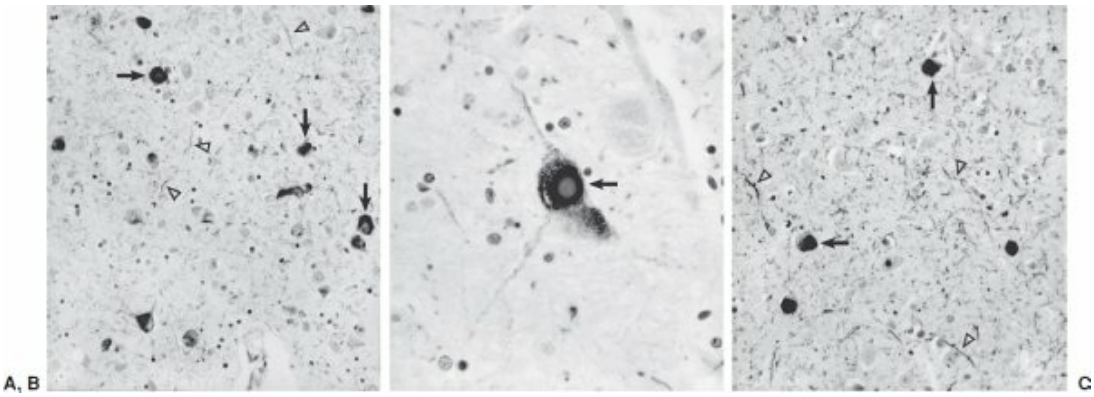


FIGURE 3-5
Photomicrographs of Lewy body pathology. **A:** Abnormal accumulation of α-synuclein aggregates demonstrated by immunocytochemistry in the amygdala of a subject with dementia. Lewy bodies appear as dense intracellular inclusions (*arrows*), but staining of neuronal processes can be seen throughout the neuropil (*arrowheads*). In individuals in whom Lewy body pathology occurs concurrently with Alzheimer disease, the amygdala is often the only region affected. **B:** Classic appearance of a Lewy body (*arrow*) in a large pigmented neuron of the substantia nigra. **C:** Lewy body pathology in the neocortex. Both Lewy bodies (*arrows*) and substantial labeling of neuronal processes in the neuropil (*arrowheads*) are evident. (Magnification for [A] and [B] 200×, for [C] 400×. All images provided courtesy of Dr. Ronald L. Hamilton, Department of Pathology, Division of Neuropathology, University of Pittsburgh School of Medicine.)

Table 3-28 Distinguishing Features of Subcortical and Cortical Dementias

Characteristic	Subcortical Dementia	Cortical Dementia	Recommended Tests
Language	No aphasia (anomia, if severe)	Aphasia early	FAS test Boston Naming test WAIS-R vocabulary test
Memory	Impaired recall	Recall and	Wechsler

	(retrieval) > recognition (encoding)	recognition impaired	memory scale; Symbol Digit Paired Associate Learning (Brandt)
Attention and immediate recall	Impaired	Impaired	WAIS-R digit span
Visuospatial skills	Impaired	Impaired	Picture arrangement, object assembly and block design; WAIS subtests
Calculation	Preserved until late	Involved early	Mini-Mental State
Frontal system abilities (executive function)	Disproportionately affected	Degree of impairment consistent with other involvement	Wisconsin Card Sorting Test; Odd Man Out test; Picture Absurdities
Speed of cognitive processing	Slowed early	Normal until late in disease	Trail making A and B: Paced Auditory Serial Addition Test (PASAT)
Personality	Apathetic, inert	Unconcerned	MMPI
Mood	Depressed	Euthymic	Beck and

			Hamilton depression scales
Speech	Dysarthric	Articulate until late	Verbal fluency (Rosen, 1980)
Posture	Bowed or extended	Upright	
Coordination	Impaired	Normal until late	
Motor speed and control	Slowed	Normal	Finger-tap; grooved pegboard
Adventitious movements	Chorea, tremor, tics, dystonia	Absent (Alzheimer dementia—some myoclonus)	
Abstraction	Impaired	Impaired	Category test (Halstead Battery)

From Pajeau AK, Román GC. HIV encephalopathy and dementia. In: J Biller, RG Kathol, eds. *The Psychiatric Clinics of North America: The Interface of Psychiatry and Neurolgy*. Vol. 15. Philadelphia, PA: WB Saunders; 1992:457.

Huntington Disease

Huntington disease is classically associated with the development of dementia. The dementia seen in this disease is the subcortical type of dementia, characterized by more motor abnormalities and fewer language abnormalities than in the cortical type of dementia (Table 3-28). The dementia of Huntington disease exhibits psychomotor slowing and difficulty with complex tasks, but memory, language, and insight remain relatively intact in the early and middle stages of the illness. As the disease progresses, however, the dementia becomes complete; the features distinguishing it from

dementia of the Alzheimer type are the high incidence of depression and psychosis in addition to the classic choreoathetoid movement disorder.

Parkinson Disease

As with Huntington disease, parkinsonism is a disease of the basal ganglia, commonly associated with dementia and depression. An estimated 20 to 30 percent of patients with Parkinson disease have dementia, and an additional 30 to 40 percent have measurable impairment in cognitive abilities. The slow movements of persons with Parkinson disease parallel the slow thinking of some affected patients, a feature that clinicians may refer to as *bradyphrenia*.

Mr. M, 77 years of age, came for a neurologic examination because he noticed his memory was slipping, and he was having difficulty concentrating, which interfered with his work. He complained of slowness and losing his train of thought. His wife stated that he was becoming withdrawn and was more reluctant to participate in activities he usually enjoyed. He denied symptoms of depression other than feeling mildly depressed about his disabilities. Two years prior, Mr. M developed an intermittent resting tremor in his right hand and a shuffling gait. Although a psychiatrist considered a diagnosis of Parkinson disease, the neurologist did not confirm it.

During an initial neurologic examination, Mr. M's spontaneous speech was hesitant and unclear (dysarthric). Cranial nerve examination was normal. Motor tone was increased slightly in the neck and all limbs. He performed alternating movements in his hands slowly. He had a slight intermittent tremor of his right arm at rest. Reflexes were symmetrical. Three weeks later, the patient had a neuropsychologic examination. On the examination, Mr. M had impairment of memory, naming, and constructional abilities.

HIV-Related Dementia

Encephalopathy in HIV infection is associated with dementia and is called *acquired immune deficiency syndrome* (AIDS) *dementia complex*, or *HIV dementia.* Patients infected with HIV experience dementia at an annual rate

of approximately 14 percent. An estimated 75 percent of patients with AIDS have involvement of the CNS at the time of autopsy. The development of dementia in people infected with HIV parallels the appearance of parenchymal abnormalities in MRI scans. *Cryptococcus* or *Treponema pallidum* can cause other infectious dementias.

Table 3-29
Criteria for Clinical Diagnosis of HIV Type 1-Associated Dementia Complex

Laboratory evidence for systemic human immunodeficiency virus (HIV) type 1 infection with confirmation by Western blot, polymerase chain reaction, or culture.
Acquired abnormality in at least *two* cognitive abilities for a period of at least 1 mo: attention and concentration, speed of processing information, abstraction and reasoning, visuospatial skills, memory and learning, and speech and language. The decline should be verified by reliable history and mental status examination. History should be obtained from an informant, and examination should be supplemented by neuropsychological testing.
Cognitive dysfunction causes impairment in social or occupational functioning. Impairment should not be attributable solely to severe systemic illness.
At least *one* of the following:
Acquired abnormality in motor function verified by clinical examination (e.g., slowed rapid movements, abnormal gait, incoordination, hyperreflexia, hypertonia, or weakness), neuropsychological tests (e.g., fine motor speed, manual dexterity, or perceptual motor skills), or both.
Decline in motivation or emotional control or a change in social behavior. This may be characterized by a change in personality with apathy, inertia, irritability, emotional lability, or a new onset of impaired judgment or disinhibition.
This does not exclusively occur in the context of a delirium.
Evidence of another etiology, including active central nervous system

opportunistic infection, malignancy, psychiatric disorders (e.g., major depression), or substance abuse, if present, is *not* the cause of the previously mentioned symptoms and signs.

Adapted from Working Group of the American Academy of Neurology AIDS Task Force. Nomenclature and research case definitions for neurologic manifestations of human immunodeficiency virus–type 1 (HIV-1) infection. *Neurology*. 1991;41:778–785, with permission.

The diagnosis of AIDS dementia complex is made by confirmation of HIV infection and exclusion of alternative pathology to explain cognitive impairment. The American Academy of Neurology AIDS Task Force developed research criteria for the clinical diagnosis of CNS disorders in adults and adolescents (Table 3-29). The AIDS Task Force criteria for AIDS dementia complex require laboratory evidence for systemic HIV, at least two cognitive deficits, and the presence of motor abnormalities or personality changes. Personality changes manifest with apathy, emotional lability, or behavioral disinhibition. The AIDS Task Force criteria also require the absence of clouding of consciousness or evidence of another etiology that could produce cognitive impairment. We should assess for cognitive, motor, and behavioral changes using physical, neurologic, and psychiatric examinations, in addition to neuropsychologic testing.

Head Trauma–Related Dementia

Dementia can be a sequela of head trauma. The so-called punch-drunk syndrome (dementia pugilistica) occurs in boxers after repeated head trauma over many years. The symptoms include emotional lability, dysarthria, and impulsivity. It also occurs in professional football players who developed dementia after repeated concussions over many years.

Mrs. S, 75 years of age, was brought to the emergency department after being found wandering her neighborhood in a confused and disoriented state. She was in good health until a few months before, when her husband had minor surgery, requiring 10-day hospitalization. About a month after her husband returned home, he and their two adult children, who do not reside with them, reported a noticeable change in Mrs. S's mental status. Mrs. S became hyperactive and appeared to have excessive

energy, was agitated and irritable, and had difficulty sleeping at night.

At the examination, Mrs. S was disoriented to time and place, agitated, and confused. Her husband revealed upon an interview that Mrs. S has for many years suffered from dizziness and lightheadedness upon standing and occasionally suffered from falls, none of which caused any significant damage. Not long before her confused symptoms began, Mrs. S had suffered a fall one night, and her husband found her the next morning lying next to the bed in a confused state. Because of her history of falls, neither Mr. S nor Mrs. S thought much of the incident. A CT scan revealed the presence of a subdural hematoma, which was surgically evacuated. Afterward, Mrs. S's confusion and disorientation cleared, and she returned to her normal state of functioning.

▲ 3.3 Major or Minor Neurocognitive Disorder due to Another Medical Condition (Amnestic Disorders)

The amnestic disorders cause impairment in memory as the primary sign and symptom, although other signs of cognitive decline may coexist. The authors of *Synopsis* believe amnestic disorder to be a clinically useful descriptive category of illness, but they are coded in DSM-5 as a neurocognitive disorder due to another medical condition with the specific medical condition noted.

The amnestic disorders are a broad category that results from a variety of diseases and conditions that have amnesia as the primary complaint. The syndrome is defined primarily by impairment in the ability to create new memories. Three different etiologies exist: amnestic disorder caused by a general medical condition (e.g., head trauma), substance-induced persisting amnestic disorder (e.g., caused by carbon monoxide poisoning or chronic alcohol consumption), and amnestic disorder not otherwise specified for cases in which the etiology is unclear.

DIAGNOSIS

The recognition of amnestic disorder occurs when there is impairment in the ability to learn new information or the inability to recall previously learned information, as a result of which there is significant impairment in social or occupational functioning and which is caused by a general medical condition (including physical trauma). Amnestic disorder may be transient, lasting for hours or days, or chronic, lasting weeks or months. We can make a diagnosis of substance-induced persisting amnestic disorder when evidence suggests that the symptoms are causatively related to the use of a substance. The DSM-5 refers clinicians to specific diagnoses within substance-related disorders: alcohol-induced disorder, sedative, hypnotic, or anxiolytic-induced disorder, and other (or unknown) substance-induced disorder.

CLINICAL FEATURES AND SUBTYPES

The central symptom of amnestic disorders is the development of a memory disorder characterized by an impairment in the ability to learn new information (anterograde amnesia) and an inability to recall previously remembered knowledge (retrograde amnesia). The symptom must result in significant problems for patients in their social or occupational functioning. The time in which a patient is amnestic can begin directly at the point of trauma or include a period before the trauma. The patient can also lose memory for the time during the physical insult (e.g., during a cerebrovascular event).

Short-term and recent memory is usually impaired. Patients cannot remember what they had for breakfast or lunch, the name of the hospital, or their doctors. In some patients, the amnesia is so profound that the patient cannot orient themselves to place and time, although the patient rarely loses orientation to person. Overlearned information or remote events usually remain intact, but there can be amnesia for even the less remote past (over the past decade). Immediate memory (tested, for example, by asking a patient to repeat six numbers) remains intact. With improvement, patients may experience a gradual shrinking of the time for which they lose memory, although some patients experience a gradual improvement in memory for the entire period.

The onset of symptoms can be sudden, as in trauma, cerebrovascular events, and neurotoxic chemical assaults, or gradual, as in nutritional

deficiency and cerebral tumors. The amnesia can be of short duration.

A variety of other symptoms can be associated with amnestic disorders. For patients with other cognitive impairments, a diagnosis of dementia or delirium is more appropriate than a diagnosis of an amnestic disorder. Both subtle and gross changes in personality can accompany the symptoms of memory impairment in amnestic disorders. Patients may be apathetic, lack initiative, have unprovoked episodes of agitation, or appear to be overly friendly or agreeable. Patients with amnestic disorders can also appear bewildered and confused and may attempt to cover their confusion with confabulatory answers to questions. Characteristically, patients with amnestic disorders do not have good insight into their neuropsychiatric conditions.

A local nursing home transferred a 73-year-old survivor of the Holocaust to the psychiatric unit. She was born in Germany to a middle-class family. Her internment in a concentration camp cut short her education. She immigrated to Israel after liberation from the concentration camp and later to the United States, where she married and raised a family. Premorbidly, she was described as a quiet, intelligent, and loving woman who spoke several languages. At 55 years of age, she had a significant carbon monoxide exposure when a gas line leaked while she and her husband slept. Her husband died of carbon monoxide poisoning, but the patient survived after a period of coma. After being stabilized, she displayed significant cognitive and behavioral problems. She had difficulty with learning new information and making appropriate plans. She retained the ability to perform activities of daily living but could not be relied on to pay bills, buy food, cook, or clean, despite appearing to have retained the intellectual ability to do these tasks. She moved to a nursing home after several difficult years at home and in the homes of relatives. In the nursing home, she was able to learn her way about the facility. She displayed little interest in scheduled group activities, hobbies, reading, or television. She had frequent behavioral problems. She repeatedly pressed staff to get her sweets and snacks and cursed them vociferously with racial epithets and disparaging comments on their weight and dress. On one occasion, she scratched the cars of several staff members with a key. Neuropsychological testing demonstrated severe

deficits in her delayed recall, intact performance on language and general knowledge measures, and moderate deficits in domains of executive function, such as concept formation and cognitive flexibility. She responded immediately to firmly set limits and rewards, but deficits in memory prevented long-term incorporation of these boundaries. Management involved the development of a behavioral plan and empirical trials of medications aimed at the amelioration of irritability.

Although one might suggest this is not a pure amnestic disorder, by far the most disabling symptom for this patient was her memory loss.

Cerebrovascular Diseases

Cerebrovascular diseases affecting the hippocampus involve the posterior cerebral and basilar arteries and their branches. Infarctions are rarely limited to the hippocampus; they often involve the occipital or parietal lobes. Thus, common accompanying symptoms of cerebrovascular diseases in this region are focal neurologic signs involving vision or sensory modalities. Cerebrovascular diseases affecting the bilateral medial thalamus, particularly the anterior portions, are often associated with symptoms of amnestic disorders. A few case studies report amnestic disorders from the rupture of an aneurysm of the anterior communicating artery, resulting in infarction of the basal forebrain region.

Multiple Sclerosis

The pathophysiologic process of multiple sclerosis involves the seemingly random formation of plaques within the brain parenchyma. When the plaques occur in the temporal lobe and the diencephalic regions, symptoms of memory impairment can occur. The most common cognitive complaints in patients with multiple sclerosis involve impaired memory, which occurs in 40 to 60 percent of patients. Characteristically, digit span memory is normal, but immediate recall and delayed recall of information are impaired. Memory impairment can affect both verbal and nonverbal material.

Korsakoff Syndrome

Korsakoff syndrome is an amnestic syndrome caused by thiamine deficiency,

most commonly associated with the poor nutritional habits of people with chronic alcohol use disorder. Other causes of poor nutrition (e.g., starvation), gastric carcinoma, hemodialysis, hyperemesis gravidarum, prolonged IV hyperalimentation, and gastric plication can also result in thiamine deficiency. Korsakoff syndrome is often associated with Wernicke encephalopathy, which is the associated syndrome of confusion, ataxia, and ophthalmoplegia. In patients with these thiamine deficiency–related symptoms, the neuropathologic findings include hyperplasia of the small blood vessels with occasional hemorrhages, hypertrophy of astrocytes, and subtle changes in neuronal axons. Although the delirium clears up within a month or so, the amnestic syndrome either accompanies or follows untreated Wernicke encephalopathy in approximately 85 percent of all cases.

Patients with Korsakoff syndrome typically demonstrate a change in personality as well, such that they display a lack of initiative, diminished spontaneity, and a lack of interest or concern. These changes appear frontal lobe–like, similar to the personality change ascribed to patients with frontal lobe lesions or degeneration. Indeed, such patients often demonstrate *executive function* deficits on neuropsychological tasks involving attention, planning, set-shifting, and inferential reasoning consistent with frontal pattern injuries. For this reason, Korsakoff syndrome is not a pure memory disorder, although it certainly is a useful paradigm of the more common clinical presentations for the amnestic syndrome.

The onset of Korsakoff syndrome can be gradual. Recent memory tends to be affected more than is remote memory, but this feature is variable. Confabulation, apathy, and passivity are often prominent symptoms in the syndrome. With treatment, patients may remain amnestic for up to 3 months and then gradually improve over the ensuing year. Administration of thiamine may prevent the development of additional amnestic symptoms, but the treatment seldom reverses severe amnestic symptoms when they are present. Approximately one-third to one-fourth of all patients recover completely, and approximately one-fourth of all patients have no improvement in their symptoms.

Alcoholic Blackouts

Some persons with severe alcohol abuse may exhibit the syndrome

commonly referred to as an alcoholic blackout. Characteristically, these persons awake in the morning with a conscious awareness of being unable to remember a period the night before during which they were intoxicated. Sometimes specific behaviors (hiding money in a secret place, and provoking fights) are associated with the blackouts.

Electroconvulsive Therapy

Electroconvulsive therapy treatments are usually associated with retrograde amnesia for several minutes before the treatment and anterograde amnesia after the treatment. The anterograde amnesia usually resolves within 5 hours. Mild memory deficits may remain for 1 to 2 months after a course of ECT treatments, but the symptoms are entirely resolved 6 to 9 months after treatment.

Head Injury

Head injuries (both closed and penetrating) can result in a wide range of neuropsychiatric symptoms, including dementia, depression, personality changes, and amnestic disorders. Amnestic disorders caused by head injuries are commonly associated with a period of retrograde amnesia leading up to the traumatic incident and amnesia for the traumatic incident itself. The severity of the brain injury correlates somewhat with the duration and severity of the amnestic syndrome. However, the best correlate of eventual improvement is the degree of clinical improvement in the amnesia during the first week after the patient regains consciousness.

Transient Global Amnesia

Transient global amnesia is the abrupt loss of the ability to recall recent events or to remember new information. The syndrome includes mild confusion and a lack of insight into the problem, a clear sensorium, and, occasionally, the inability to perform some well-learned complex tasks. Episodes last from 6 to 24 hours. Studies suggest that transient global amnesia occurs in 5 to 10 cases per 100,000 persons per year, although for patients older than age 50 years, the rate may be as high as 30 cases per 100,000 persons per year. The pathophysiology is unknown, but it likely involves ischemia of the temporal lobe and the diencephalic brain regions.

Several studies of patients with SPECT have shown decreased blood flow in the temporal and parietotemporal regions, particularly in the left hemisphere. Patients with transient global amnesia almost universally experience complete improvement, although one study found that approximately 20 percent of patients may have a recurrence of the episode, and another study found that approximately 7 percent of patients may have epilepsy. Patients with transient global amnesia differ from patients with TIAs in that fewer patients have diabetes, hypercholesterolemia, and hypertriglyceridemia, but more have hypertension and migrainous episodes.

PATHOLOGY AND LABORATORY EXAMINATION

Quantitative neuropsychologic testing can aid the diagnosis of amnestic disorder. Standardized tests also are available to assess recall of well-known historical events or public figures to characterize an individual's inability to remember previously learned information. Performance on such tests varies among individuals with amnestic disorder. Subtle deficits in other cognitive functions may occur in individuals with amnestic disorder. Memory deficits, however, constitute the predominant feature of the mental status examination and account largely for any functional deficits. No specific or diagnostic features are detectable on imaging studies such as MRI or CT. Damage of midtemporal lobe structures is typical, however, and may reflect enlargement of the third ventricle or temporal horns or in structural atrophy detected by MRI.

DIFFERENTIAL DIAGNOSIS

Table 3-30 lists the significant causes of amnestic disorders. To make the diagnosis, clinicians must obtain a patient's history, conduct a complete physical examination, and order all appropriate laboratory tests. Other diagnoses, however, can be confused with amnestic disorders.

Dementia and Delirium

Amnestic disorders can be distinguished from delirium because they occur in the absence of a disturbance of consciousness and are striking for the relative preservation of other cognitive domains.

Table 3-30
Major Causes of Amnestic Disorders

Thiamine deficiency (Korsakoff syndrome)
Hypoglycemia
Primary brain conditions
- Seizures
- Head trauma (closed and penetrating)
- Cerebral tumors (especially thalamic and temporal lobe)
- Cerebrovascular diseases (especially thalamic and temporal lobe)
- Surgical procedures on the brain
- Encephalitis due to herpes simplex
- Hypoxia (including nonfatal hanging attempts and carbon monoxide poisoning)
- Transient global amnesia
- Electroconvulsive therapy
- Multiple sclerosis

Substance-related causes
- Alcohol use disorders
- Neurotoxins
- Benzodiazepines (and other sedative-hypnotics)
- Many over-the-counter preparations (particularly anticholinergic or antihistaminic drugs)

Table 3-31
Comparison of Syndrome Characteristics in Dementia of the Alzheimer Type (DAT) and Amnestic Disorder

Characteristic	DAT	Amnestic Disorder
Onset	Insidious	Can be abrupt
Course	Progressive deterioration	Static or improvement
Anterograde	Impaired	Impaired

memory		
Retrograde memory	Impaired	Temporal gradient
Episodic memory	Impaired	Impaired
Semantic memory	Impaired	Intact
Language	Impaired	Intact
Praxis or function	Impaired	Intact

Table 3-31 outlines the critical distinctions between Dementia of the Alzheimer type (DAT) and amnestic disorders. Both disorders can have an insidious onset with slow progression, as in a Korsakoff psychosis in a chronic drinker. Amnestic disorders, however, can also develop precipitously, as in Wernicke encephalopathy, transient global amnesia, or anoxic insults. Although DAT progresses relentlessly, amnestic disorders tend to remain static or even improve after removing the offending cause. In terms of the actual memory deficits, the amnestic disorder and DAT still differ. DAT has an impact on retrieval in addition to encoding and consolidation. The deficits in DAT extend beyond memory to general knowledge (semantic memory), language, praxis, and general function. This is not the case for amnestic disorders. The dementias associated with Parkinson disease, AIDS, and other subcortical disorders demonstrate disproportionate impairment of retrieval, but relatively intact encoding and consolidation. The subcortical pattern dementias are also likely to display motor symptoms, such as bradykinesia, chorea, or tremor, that are not components of the amnestic disorders.

Normal Aging

Some minor impairment in memory may accompany healthy aging, but the requirement that the memory impairment causes significant impairment in social or occupational functioning should exclude normal aging from the diagnosis.

Dissociative Disorders

The dissociative disorders can sometimes be challenging to differentiate from amnestic disorders. Patients with dissociative disorders, however, are more likely to have lost their orientation to self and may have more selective memory deficits than do patients with amnestic disorders. For example, patients with dissociative disorders may not know their names or home addresses, but they are still able to learn new information and remember selected memories. Dissociative disorders are also often associated with emotionally stressful life events involving money, the legal system, or troubled relationships.

Factitious Disorders

Patients with factitious disorders who are mimicking an amnestic disorder often have inconsistent results on memory tests and have no evidence of an identifiable cause. These findings, coupled with evidence of primary or secondary gain for a patient, should suggest a factitious disorder.

COURSE AND PROGNOSIS

The course of an amnestic disorder depends on its etiology and treatment, particularly acute treatment. Generally, amnestic disorder has a static course. We see little improvement over time, but also no progression of the disorder. The exceptions are the acute amnesias, such as transient global amnesia, which resolves entirely over hours to days, and the amnestic disorder associated with head trauma, which improves steadily in the months after the trauma. Amnesia secondary to processes that destroy brain tissue, such as stroke, tumor, and infection, are irreversible, although it is, static once the disease process is halted.

TREATMENT

The primary approach to treating amnestic disorders is to treat the underlying cause. Although a patient is amnestic, supportive prompts about the date, the time, and the patient’s location can be helpful and can reduce the patient’s anxiety. After the resolution of the amnestic episode, psychotherapy of some type (cognitive, psychodynamic, or supportive) may help patients incorporate the amnestic experience into their lives.

Psychotherapy

Psychodynamic interventions may be of considerable value for patients who have amnestic disorders that result from insults to the brain. Understanding the course of recovery in such patients helps clinicians to be sensitive to the narcissistic injury inherent in damage to the CNS.

The first phase of recovery, in which patients are incapable of processing what happened because the ego defenses are overwhelmed, requires clinicians to serve as a supportive auxiliary ego who explains to a patient what is happening and provides missing ego functions. In the second phase of recovery, as the realization of the injury sets in, patients may become angry and feel victimized by the malevolent hand of fate. They may view others, including the clinician, as harmful or destructive, and clinicians must contain these projections without becoming punitive or retaliatory. Clinicians can build a therapeutic alliance with patients by explaining slowly and clearly what happened and by explaining the patient's internal experience. The third phase of recovery is integrative. As a patient accepts what has happened, a clinician can help the patient form a new identity by connecting current experiences of the self with past experiences. Grieving over the lost faculties may be an essential feature of the third phase.

Most patients who are amnestic because of brain injury engage in denial. Clinicians must respect and empathize with the patient's need to deny the reality of what has happened. Insensitive and blunt confrontations destroy any developing therapeutic alliance and can cause patients to feel attacked. In a sensitive approach, clinicians help patients accept their cognitive limitations by exposing them to these deficits bit by bit over time. When patients fully accept what has happened, they may need assistance in forgiving themselves and any others involved, so that they can get on with their lives. Clinicians must also be wary of being seduced into thinking that all of the patient's symptoms are directly related to the brain insult. An evaluation of preexisting personality disorders, such as borderline, antisocial, and narcissistic personality disorders, must be part of the overall assessment; many patients with personality disorders place themselves in situations that predispose them to injuries. These personality features may become a crucial part of the psychodynamic psychotherapy.

Centers for cognitive rehabilitation have rehabilitation-oriented therapeutic

milieu designed to promote recovery from brain injury, especially that from traumatic causes. Despite the high cost of extended care at these sites, which provide both long-term institutional and daytime services, there is limited data to define therapeutic effectiveness for the heterogeneous groups of patients who participate in such tasks as memory retaining.

EPIDEMIOLOGY

No adequate studies have reported on the incidence or prevalence of amnestic disorders. We are most likely to see them with alcohol use disorders and head injury. In general practice and hospital settings, the frequency of amnesia related to chronic alcohol abuse has decreased, and the frequency of amnesia related to head trauma has increased.

ETIOLOGY

The major neuroanatomical structures involved in memory and the development of an amnestic disorder are particular diencephalic structures such as the dorsomedial and midline nuclei of the thalamus and midtemporal lobe structures such as the hippocampus, the mamillary bodies, and the amygdala. Although amnesia is usually the result of bilateral damage to these structures, some cases of unilateral damage result in an amnestic disorder, and evidence indicates that the left hemisphere may be more critical than the right hemisphere in the development of memory disorders. Many studies of memory and amnesia in animals have suggested that other brain areas may also be involved in the symptoms accompanying amnesia. Frontal lobe involvement can result in such symptoms as confabulation and apathy, which occurs in patients with amnestic disorders.

Amnestic disorders have many potential causes (see Table 3-30). Thiamine deficiency, hypoglycemia, hypoxia (including carbon monoxide poisoning), and herpes simplex encephalitis all have a predilection for damaging the temporal lobes, particularly the hippocampi, and thus can be associated with the development of amnestic disorders. Similarly, when tumors, cerebrovascular diseases, surgical procedures, or multiple sclerosis plaques involve the diencephalic or temporal regions of the brain, the symptoms of an amnestic disorder may develop. General insults to the brain, such as seizures,

ECT, and head trauma, can also result in memory impairment. Transient global amnesia is presumed to be a cerebrovascular disorder involving transient impairment in blood flow through the vertebrobasilar arteries.

Many drugs are associated with the development of amnesia, and clinicians should review all drugs taken, including nonprescription drugs, in the diagnostic workup of a patient with amnesia. Benzodiazepines are the most commonly used prescription drugs associated with amnesia. All benzodiazepines can be associated with amnesia, especially if combined with alcohol. Triazolam has a reputation for causing amnesia. However, at standard doses, it is no more likely to cause memory loss than the other benzodiazepines. When using higher doses or combining it with alcohol, it can cause anterograde amnesia.

▲ 3.4 Neurocognitive and Other Psychiatric Disorders due to a General Medical Condition

Increasingly, scientific views of mental illness recognize that, whether caused by an identifiable anomaly (e.g., brain tumor), a neurotransmitter disturbance of unclear origin (e.g., schizophrenia), or a consequence of deranged upbringing or environment (e.g., personality disorder), all mental disorders ultimately share one common underlying theme: aberration in brain function. The treatments for those conditions, whether psychological or biologic, attempt to restore normal brain functioning.

The differential diagnosis for a mental syndrome in a patient should always include consideration of (1) any general medical condition that a patient may have and (2) any prescription, nonprescription, or illegal substances that a patient may be taking. Although some specific medical conditions are classically associated with mental syndromes, a much larger number of general medical conditions may occasionally cause mental syndromes.

The mental disorders caused by a general medical condition span the entire spectrum of diagnostic categories. Thus, one can have a cognitive disorder, mood disorder, sleep disorder, anxiety disorder, and psychotic disorder to

mention but a few that are caused or aggravated by a medical condition. In this section, neurocognitive disorders due to a general medical condition are described, including epilepsy, autoimmune disorders, and AIDS, of which psychiatrists should be aware.

SPECIFIC DISORDERS

Epilepsy

Epilepsy is the most common chronic neurologic disease in the general population and affects approximately 1 percent of the population in the United States. For psychiatrists, the significant concerns about epilepsy are consideration of an epileptic diagnosis in psychiatric patients, the psychosocial ramifications of a diagnosis of epilepsy for a patient, and the psychological and cognitive effects of commonly used anticonvulsant drugs. Concerning the first of these concerns, 30 to 50 percent of all persons with epilepsy have psychiatric difficulties sometime during their illness. The most common behavioral symptom of epilepsy is a change in personality. Psychosis and violence occur much less often than was previously believed.

Definitions. A seizure is a transient paroxysmal pathophysiologic disturbance of cerebral function caused by a spontaneous, excessive discharge of neurons. Patients are said to have epilepsy if they have a chronic condition characterized by recurrent seizures. The ictus, or ictal event, is the seizure itself. The nonictal periods include the preictal, postictal, and interictal periods. The ictal symptoms depend mainly on the brain origin for the seizure and how the activity spreads in the brain. The ictal event influences the interictal symptoms as do other neuropsychiatric and psychosocial factors, such as coexisting psychiatric or neurologic disorders, the presence of psychosocial stressors, and premorbid personality traits.

Classification. The two major categories of seizures are partial and generalized. Partial seizures involve epileptiform activity in localized brain regions. Generalized seizures involve the entire brain (Fig. 3-6). Table 3-32 outlines the classification system.

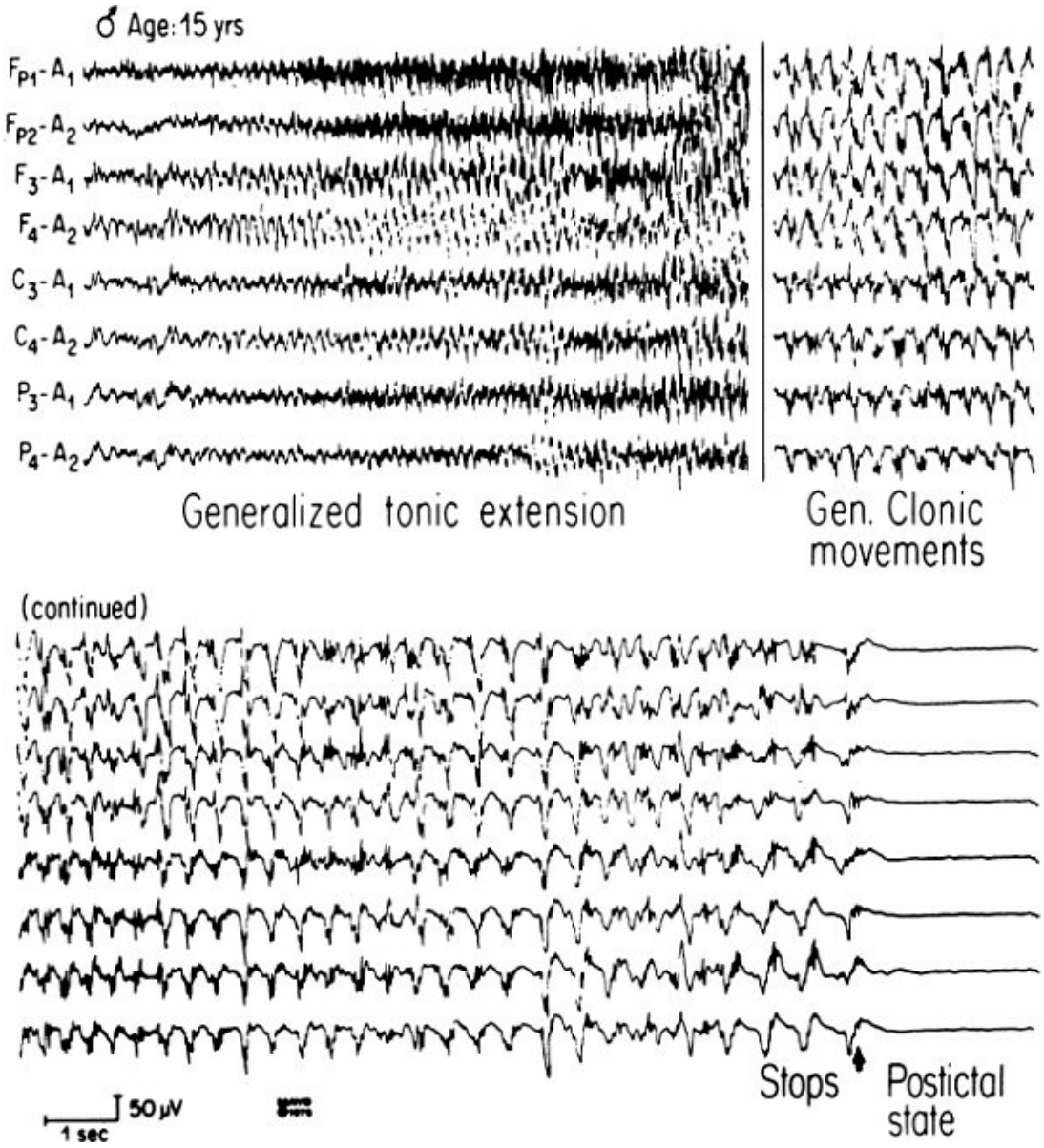


FIGURE 3-6
Electroencephalographic recording during generalized tonic–clonic seizure showing rhythmic sharp waves and muscles artifact during tonic phase, spike-and-wave discharges during clonic phase, and attenuation of activity during postictal state. (Courtesy of Barbara F. Westmoreland, M.D.)

GENERALIZED SEIZURES. Generalized tonic–clonic seizures exhibit the classic symptoms of loss of consciousness, generalized tonic–clonic movements of the limbs, tongue biting, and incontinence. Although the diagnosis of the ictal events of the seizure is relatively straightforward, the postictal state, characterized by a slow, gradual recovery of consciousness and cognition, occasionally presents a diagnostic dilemma for a psychiatrist in an emergency

department. The recovery period from a generalized tonic–clonic seizure ranges from a few minutes to many hours, and the clinical picture is that of a gradually clearing delirium. The most common psychiatric problems associated with generalized seizures involve helping patients adjust to a chronic neurologic disorder and assessing the cognitive or behavioral effects of anticonvulsant drugs.

Absence Seizure (Petit Mal). A complicated type of generalized seizure for a psychiatrist to diagnose is an absence or petit mal seizure. The epileptic nature of the episodes may go unrecognized because the characteristic motor or sensory manifestations of epilepsy may be absent or so slight that they do not arouse suspicion. Petit mal epilepsy usually begins in childhood between the ages of 5 and 7 years and ceases by puberty. Brief disruptions of consciousness, during which the patient suddenly loses contact with the environment, are characteristic of petit mal epilepsy, but the patient has no actual loss of consciousness and no convulsive movements during the episodes. The EEG produces a characteristic pattern of three-per-second spike-and-wave activity (Fig. 3-7). In rare instances, petit mal epilepsy begins in adulthood. Adult-onset petit mal epilepsy can present as sudden, recurrent psychotic episodes or deliriums that appear and disappear abruptly. The symptoms may include a history of falling or fainting spells.



Table 3-32
International Classification of Epileptic Seizures

I. Partial seizures (seizures beginning locally)
  A. Partial seizures with elementary symptoms (generally without impairment of consciousness)
    1. With motor symptoms
    2. With sensory symptoms
    3. With autonomic symptoms
    4. Compound forms
  B. Partial seizures with complex symptoms (generally with impairment of consciousness; temporal lobe or psychomotor seizures)

1. With impairment of consciousness only
2. With cognitive symptoms
3. With affective symptoms
4. With psychosensory symptoms
5. With psychosensory symptoms (automatisms)
6. Compound forms

C. Partial seizures secondarily generalized

II. Generalized seizures (bilaterally symmetrical and without local onset)

A. Absence (petit mal) seizures
B. Myoclonus
C. Infantile spasms
D. Clonic seizures
E. Tonic seizures
F. Tonic–clonic seizures (grand mal)
G. Atonic seizures
H. Akinetic seizures

III. Unilateral seizures

IV. Unclassified seizures (because of incomplete data)

Adapted from Gastaut H. Clinical and electroencephalographical classification of epileptic seizures. *Epilepsia*. 1970;11:102, with permission.

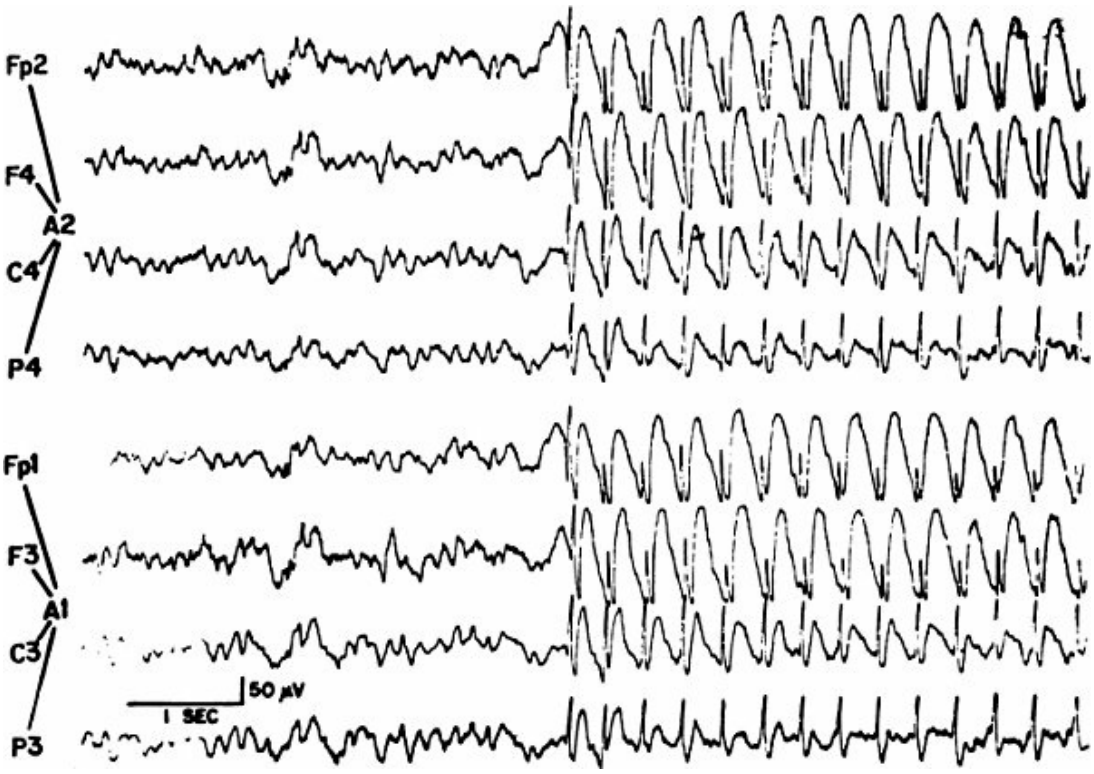


FIGURE 3-7
Petit mal epilepsy characterized by bilaterally synchronous, 3-Hz spike and slow-wave activity.

PARTIAL SEIZURES. Partial seizures can be simple (without alterations in consciousness) or complex (with an alteration in consciousness). Somewhat more than half of all patients with partial seizures have complex partial seizures. Other terms used for complex partial seizures are temporal lobe epilepsy, psychomotor seizures, and limbic epilepsy; these terms, however, are not accurate descriptions of the clinical situation. Complex partial epilepsy, the most common form of epilepsy in adults, affects approximately 3 of 1,000 persons. About 30 percent of patients with complex partial seizures have a major mental illness such as depression.

Symptoms

PREICTAL SYMPTOMS. Preictal events (auras) in complex partial epilepsy include autonomic sensations (e.g., fullness in the stomach, blushing, and changes in respiration); cognitive sensations (e.g., *déjá vu*, *jamais vu*, forced thinking, dreamy states); affective states (e.g., fear, panic, depression,

elation); and, classically, automatisms (e.g., lip-smacking, rubbing, chewing).

ICTAL SYMPTOMS. Brief, disorganized, and uninhibited behavior characterizes the ictal event. Although some defense attorneys may claim otherwise, rarely does a person exhibit organized, directed violent behavior during an epileptic episode. The cognitive symptoms include amnesia for the time during the seizure and a period of resolving delirium after the seizure. A seizure focus can be found on an EEG in 25 to 50 percent of all patients with complex partial epilepsy (Fig. 3-8). The use of sphenoidal or anterior temporal electrodes and sleep-deprived EEGs may increase the likelihood of finding an EEG abnormality. We might obtain multiple normal EEGs for a patient with complex partial epilepsy; therefore, we cannot use normal EEGs to exclude a diagnosis of complex partial epilepsy. The use of long-term EEG recordings (usually 24 to 72 hours) can help clinicians detect a seizure focus in some patients. Nasopharyngeal leads probably do not add much to the sensitivity of an EEG and cause discomfort for the patient.

INTERICTAL SYMPTOMS

Personality Disturbances. The most frequent psychiatric abnormalities reported in patients with epilepsy are personality disorders, and they are especially likely to occur in patients with epilepsy of temporal lobe origin. The most common features are religiosity, a heightened experience of emotions—a quality usually called *viscosity of personality—and changes in sexual behavior. The syndrome in its complete form is relatively rare, even in those with complex partial seizures of temporal lobe origin. Many patients are not affected by personality disturbances; others have a variety of disturbances that differ strikingly from the classic syndrome.*

A striking religiosity may be manifested not only by increased participation in overtly religious activities but also by unusual concern for moral and ethical issues, preoccupation with right and wrong, and heightened interest in global and philosophical concerns. The hyperreligious features can sometimes seem like the prodromal symptoms of schizophrenia and can result in a diagnostic problem in an adolescent or a young adult.

The symptom of viscosity of personality is usually most noticeable in a patient’s conversation, which is likely to be slow, serious, ponderous,

pedantic, overly replete with nonessential details, and often circumstantial. The listener may grow bored but be unable to find a courteous and successful way to disengage from the conversation. The speech tendencies are often mirrored in the patient's writing and result in a symptom known as *hypergraphia,* which some clinicians consider virtually pathognomonic for complex partial epilepsy.

Changes in sexual behavior can include hypersexuality, deviations in sexual interest, such as fetishism and transvestism, and, most commonly, hyposexuality. The hyposexuality is characterized both by a lack of interest in sexual matters and by reduced sexual arousal. Some patients with the onset of complex partial epilepsy before puberty may fail to reach an average level of sexual interest after puberty, although this characteristic may not disturb the patient. For patients with the onset of complex partial epilepsy after puberty, the change in sexual interest may be bothersome and worrisome.

Psychotic Symptoms. Interictal psychotic states are more common than ictal psychoses. Schizophrenia-like interictal episodes can occur in patients with epilepsy, particularly those with temporal lobe origins. An estimated 10 percent of all patients with complex partial epilepsy have psychotic symptoms. Risk factors for the symptoms include female gender, left-handedness, the onset of seizures during puberty, and a left-sided lesion.

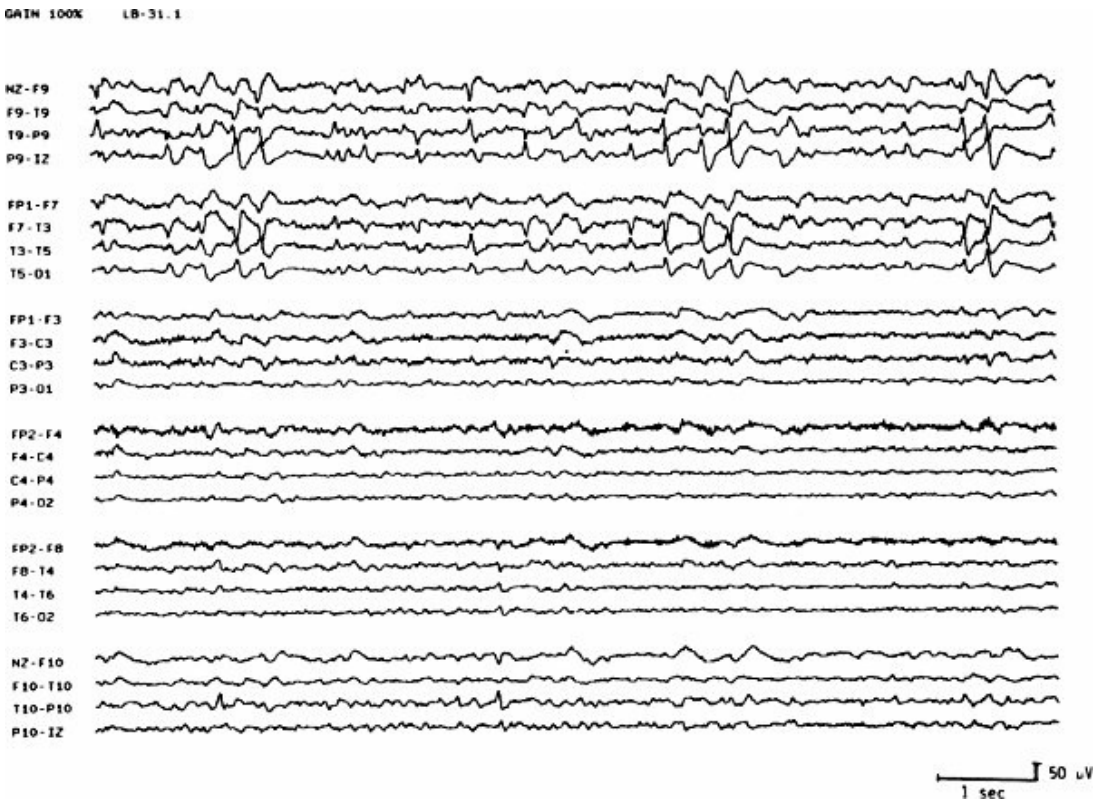


FIGURE 3-8
An interictal encephalograph in a patient with complex partial seizures reveals frequent left temporal spike discharges and rare, independent right temporal sharp-wave activity. (From Cascino GD. Complex partial seizures: Clinical features and differential diagnosis. *Psychiatr Clin North Am*. 1992;15:377, with permission.)

The onset of psychotic symptoms in epilepsy is variable. Classically, psychotic symptoms appear in patients who have had epilepsy for a long time, and the development of personality changes related to the epileptic brain activity precedes the onset of psychotic symptoms. The most characteristic symptoms of psychoses are hallucinations and paranoid delusions. Patients usually remain warm and appropriate in affect, in contrast to the abnormalities of affect commonly seen in patients with schizophrenia. The thought disorder symptoms in patients with psychotic epilepsy are most commonly those involving conceptualization and circumstantiality rather than the classic schizophrenic symptoms of blocking and looseness.

Violence. Episodic violence has been a problem in some patients with epilepsy, especially epilepsy of temporal and frontal lobe origin. Whether the violence is a manifestation of the seizure itself or is of interictal

psychopathological origin is uncertain. Most evidence points to the extreme rarity of violence as an ictal phenomenon. Only in rare cases should violence in the patient with epilepsy be attributed to the seizure itself.

Mood Disorder Symptoms. Mood disorder symptoms, such as depression and mania, are seen less often in epilepsy than are schizophrenia-like symptoms. The mood disorder symptoms that do occur tend to be episodic and appear most often when the epileptic foci affect the temporal lobe of the nondominant cerebral hemisphere. The importance of mood disorder symptoms may be attested to by the increased incidence of attempted suicide in people with epilepsy.

Diagnosis. A correct diagnosis of epilepsy can be particularly tricky when the ictal and interictal symptoms of epilepsy are severe manifestations of psychiatric symptoms in the absence of significant changes in consciousness and cognitive abilities. Psychiatrists, therefore, must maintain a high level of suspicion during the evaluation of a new patient and must consider the possibility of an epileptic disorder even in the absence of the classic signs and symptoms. Another differential diagnosis to consider is *psychogenic nonepileptic seizures* (previously called *pseudoseizures*), in which a patient has some conscious control over mimicking the symptoms of a seizure (Table 3-33).

For patients who have previously received a diagnosis of epilepsy, we should consider the appearance of new psychiatric symptoms as possibly representing an evolution in their epileptic symptoms. The appearance of psychotic symptoms, mood disorder symptoms, personality changes, or symptoms of anxiety (e.g., panic attacks) should cause a clinician to evaluate the control of the patient's epilepsy and to assess the patient for the presence of an independent mental disorder. In such circumstances, the clinician should evaluate the patient's compliance with the anticonvulsant drug regimen and should consider whether the psychiatric symptoms could be adverse effects from the antiepileptic drugs themselves. When psychiatric symptoms appear in a patient who has had epilepsy diagnosed or considered as a diagnosis in the past, the clinician should obtain results of one or more EEG examinations.

In patients who have not previously received a diagnosis of epilepsy, four

characteristics should cause a clinician to be suspicious of the possibility: the abrupt onset of psychosis in a person previously regarded as psychologically healthy, the abrupt onset of delirium without a recognized cause, a history of similar episodes with an abrupt onset and spontaneous recovery, and a history of previous unexplained falling or fainting spells.

Table 3-33
Differentiating Features of Psychogenic Nonepileptic Seizures (PNES) and Epileptic Seizures

Feature	Epileptic Seizures	PNES
Clinical Features		
Nocturnal seizure	Common	Uncommon
Stereotyped aura	Usually	None
Cyanotic skin changes during seizures	Common	None
Self-injury	Common	Rare
Incontinence	Common	Rare
Postictal confusion	Present	None
Body movements	Tonic or clonic or both	Nonstereotyped and asynchronous
Affected by suggestion	No	Yes
EEG Features		
Spike and waveforms	Present	Absent
Postictal slowing	Present	Absent
Interictal abnormalities	Variable	Variable

EEG, electroencephalogram.

From Stevenson JM, King JH. Neuropsychiatric aspects of epilepsy and epileptic seizures. In: Hales RE, Yodofsky SC, eds. *American Psychiatric Press Textbook of Neuropsychiatry*.

Washington, DC: American Psychiatric Press; 1987:220.

Treatment. First-line drugs for generalized tonic–clonic seizures are valproate and phenytoin. First-line drugs for partial seizures include carbamazepine, oxcarbazepine, and phenytoin. Ethosuximide and valproate are first-line drugs for absence (petit mal) seizures. Table 3-34 lists the drugs used for various types of seizures. Carbamazepine and valproic acid may help control the symptoms of irritability and outbursts of aggression, as do the typical antipsychotic drugs. Psychotherapy, family counseling, and group therapy may be useful in addressing the psychosocial issues associated with epilepsy. Also, clinicians should be aware that many antiepileptic drugs cause mild to moderate cognitive impairment, and we should consider an adjustment of the dosage or a change in medications if symptoms of cognitive impairment are a problem in a patient.

Brain Tumors

Brain tumors and cerebrovascular diseases can cause virtually any psychiatric symptom or syndrome, but cerebrovascular diseases, by the nature of their onset and symptom pattern, are rarely misdiagnosed as mental disorders. In general, tumors cause fewer psychopathological signs and symptoms than cerebrovascular diseases affecting a similar volume of brain tissue. The two critical approaches to the diagnosis of either condition are a comprehensive clinical history and a complete neurologic examination. The performance of the appropriate brain imaging technique is usually the final diagnostic procedure; the imaging should confirm the clinical diagnosis.

Clinical Features, Course, and Prognosis. Mental symptoms occur at some time during the illness in approximately 50 percent of patients with brain tumors. In approximately 80 percent of these patients with mental symptoms, the tumors are located in frontal or limbic brain regions rather than in parietal or temporal regions. Whereas meningiomas are likely to cause focal symptoms by compressing a limited region of the cortex, gliomas are likely to cause diffuse symptoms. Delirium is most often a component of rapidly growing, large, or metastatic tumors. If a patient’s history and a physical examination reveal bowel or bladder incontinence, a frontal lobe tumor should be suspected; if the history and examination reveal

abnormalities in memory and speech, we should suspect a temporal lobe tumor.

COGNITION. Impaired intellectual functioning often accompanies the presence of a brain tumor, regardless of its type or location.

LANGUAGE SKILLS. Disorders of language function may be severe, mainly if tumor growth is rapid. Defects of language function often obscure all other mental symptoms.



Table 3-34
Commonly Used Anticonvulsant Drugs

Drug	Use	Maintenance Dosage (mg/day)
Carbamazepine	Generalized tonic–clonic, partial	600–1,200
Clonazepam	Absence, atypical myoclonic	2–12
Ethosuximide	Absence	1,000–2,000
Gabapentin	Complex partial seizures (augmentation)	900–3,600
Lamotrigine	Complex partial seizures, generalized (augmentation)	300–500
Oxcarbazepine	Partial	600–2,400
Phenobarbital	Generalized tonic–clonic	100–200
Phenytoin	Generalized tonic–clonic, partial, status epilepticus	300–500
Primidone	Partial	750–1,000
Tiagabine	Generalized	32–56
Topiramate	Complex partial seizures (augmentation)	200–400

Valproate	Absence, myoclonic, generalized tonic–clonic, akinetic, partial seizures	750–1,000
Zonisamide	Generalized	400–600

MEMORY. Loss of memory is a frequent symptom of brain tumors. Patients with brain tumors exhibit an amnestic syndrome and retain no memory of events that occurred since the illness began. Events of the immediate past, even painful ones, are lost. Patients, however, retain old memories and are unaware of their loss of recent memory.

PERCEPTION. Prominent perceptual defects are often associated with behavioral disorders, mainly because patients must integrate tactile, auditory, and visual perceptions to function normally.

AWARENESS. Alterations of consciousness are common late symptoms of increased intracranial pressure caused by a brain tumor. Tumors arising in the upper part of the brainstem can produce a unique symptom called *akinetic mutism*, or *vigilant coma.* The patient is immobile and mute yet alert.

Colloid Cysts. Although they are not brain tumors, colloid cysts located in the third ventricle can exert physical pressure on structures within the diencephalon and produce such mental symptoms as depression, emotional lability, psychotic symptoms, and personality changes. The classic associated neurologic symptoms are position-dependent intermittent headaches.

Head Trauma

Head trauma can result in an array of mental symptoms and lead to a diagnosis of dementia due to head trauma or to mental disorder not otherwise specified due to a general medical condition (e.g., postconcussional disorder). The postconcussive syndrome remains controversial because it focuses on the full range of psychiatric symptoms, some serious, that can follow what seems to be minor head trauma.

Pathophysiology. Head trauma is a common clinical situation; an

estimated 2 million incidents involve head trauma each year. Head trauma most commonly occurs in people 15 to 25 years of age and has a male-to-female predominance of approximately 3 to 1. Gross estimates based on the severity of the head trauma suggest that virtually all patients with severe head trauma, more than half of patients with moderate head trauma, and about 10 percent of patients with mild head trauma have ongoing neuropsychiatric sequelae resulting from the head trauma. Head trauma can be divided grossly into penetrating head trauma (e.g., trauma produced by a bullet) and blunt trauma, in which there is no physical penetration of the skull. Blunt trauma is far more common than penetrating head trauma. Motor vehicle accidents account for more than half of all the incidents of blunt CNS trauma; falls, violence, and sports-related head trauma account for most of the remaining cases (Fig. 3-9).

Whereas brain injury from penetrating wounds localizes to the areas directly affected by the missile, brain injury from blunt trauma involves several mechanisms. During the actual head trauma, the head usually moves back and forth violently, so that the brain repeatedly hits against the skull as it and the skull mismatch their rapid deceleration and acceleration. This crashing results in focal contusions, and the stretching of the brain parenchyma produces diffuse axonal injury. Later developing processes, such as edema and hemorrhaging, can result in further damage to the brain.

Symptoms. The two significant clusters of symptoms related to head trauma are those of cognitive impairment and behavioral sequelae. After a period of posttraumatic amnesia, there is usually a 6- to 12-month period of recovery, after which the remaining symptoms are likely to be permanent. The most common cognitive problems are decreased speed in information processing, decreased attention, increased distractibility, deficits in problem-solving and in the ability to sustain an effort, and problems with memory and learning new information. A variety of language disabilities can also occur.



FIGURE 3-9
Severe contusion of the frontal poles has resulted in their atrophy and distortion. (Courtesy of Dr. H. M. Zimmerman.)

Behaviorally, the significant symptoms involve depression, increased

impulsivity, increased aggression, and changes in personality. Alcohol can exacerbate these symptoms, and it is often involved in the head trauma event itself. A debate has ensued about how preexisting character and personality traits affect the development of behavioral symptoms after head trauma. We do not have the critical studies needed to answer the question definitively, but the weight of opinion is leaning toward a biologically and neuroanatomically based association between the head trauma and the behavioral sequelae.

Treatment. The treatment of the cognitive and behavioral disorders in patients with head trauma is similar to the treatment approaches used in other patients with these symptoms. One difference is that patients with head trauma may be particularly susceptible to the side effects associated with psychiatric drugs; therefore, we should initiate treatment with these agents in lower dosages than usual, and they should be titrated upward more slowly than usual. We can use standard antidepressants to treat depression and either anticonvulsants or antipsychotics for aggression and impulsivity. Other approaches to the symptoms include lithium, calcium channel blockers, and β-adrenergic receptor antagonists.

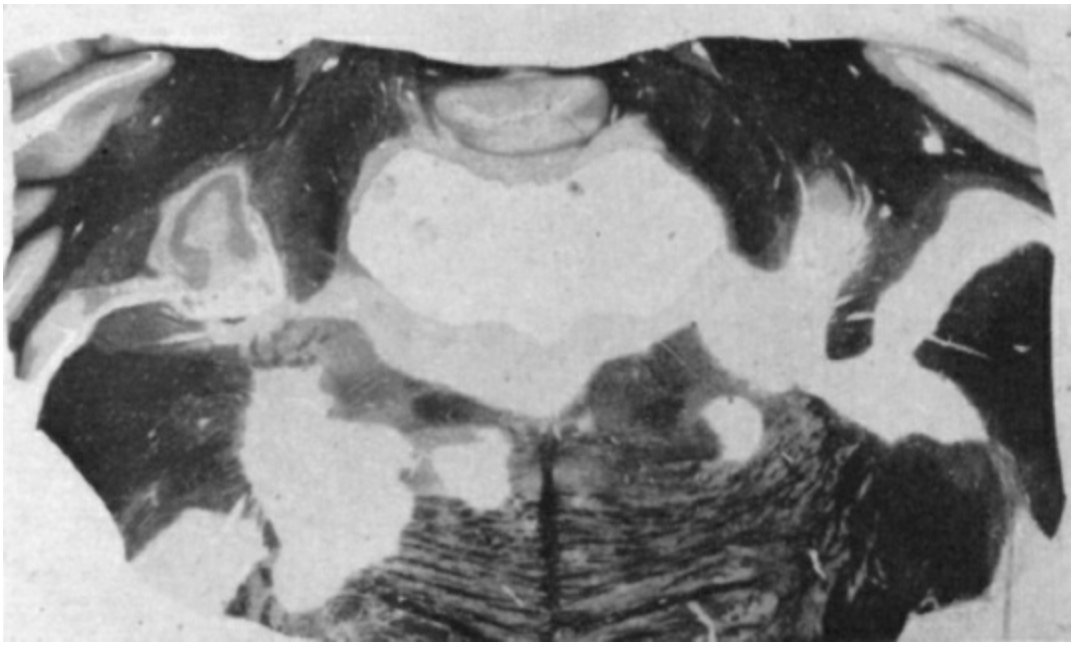


FIGURE 3-10
Multiple sclerosis. Irregular, seemingly punched out zones of demyelination are evident in this section through the level of the fourth ventricle. Myelin stain. 2.6×. (Courtesy of Dr. H. M. Zimmerman.)

Clinicians must support patients through individual or group psychotherapy and should support the primary caretakers through couples and family therapy. Patients with minor and moderate head trauma often rejoin their families and restart their jobs; therefore, all involved parties need help to adjust to any changes in the patient's personality and mental abilities.

Demyelinating Disorders

Multiple sclerosis (MS) is a major demyelinating disorder. Other demyelinating disorders include amyotrophic lateral sclerosis (ALS), metachromatic leukodystrophy, adrenoleukodystrophy, gangliosidoses, subacute sclerosing panencephalitis, and Kufs disease. All of these disorders can be associated with neurologic, cognitive, and behavioral symptoms.

Multiple Sclerosis. MS consists of multiple episodes of symptoms, pathophysiologically related to multifocal lesions in the white matter of the CNS (Fig. 3-10). The cause remains unknown, but studies have focused on slow viral infections and disturbances in the immune system. The estimated prevalence of MS in the Western Hemisphere is 50 per 100,000 people. The disease is much more frequent in cold and temperate climates than in the tropics and subtropics and more common in women than in men; it is predominantly a disease of young adults. In most patients, the onset occurs between the ages of 20 and 40 years.

The neuropsychiatric symptoms of MS include cognitive and behavioral types. Research reports have found that 30 to 50 percent of patients with MS have mild cognitive impairment (MCI) and that 20 to 30 percent of them have severe cognitive impairments. Although evidence indicates that patients with MS experience a decline in their general intelligence, memory is the most commonly affected cognitive function. The severity of the amnesia and the neurologic symptoms do not correlate well. Nor do they correlate with the duration of the illness.

The behavioral symptoms associated with MS are varied and can include euphoria, depression, and personality changes. Psychosis is a rare complication. Approximately 25 percent of persons with MS exhibit a euphoric mood. This mood is not hypomanic but somewhat more cheerful than their situation warrants and not necessarily in character with their

disposition before the onset of MS. Only 10 percent of patients with MS have a sustained and elevated mood, although it is still not genuinely hypomanic. Depression, however, is common; it affects 25 to 50 percent of patients with MS and results in a higher rate of suicide than is seen in the general population. Risk factors for suicide in patients with MS are male sex, the onset of MS before age 30 years, and a relatively recent diagnosis of the disorder. Personality changes are also frequent in patients with MS; they affect 20 to 40 percent of patients and often include irritability or apathy.

Amyotrophic Lateral Sclerosis. ALS is a progressive, noninherited disease of asymmetrical muscle atrophy. It begins in adult life and progresses over months or years to involve all the striated muscles except the cardiac and ocular muscles. In addition to muscle atrophy, patients have signs of pyramidal tract involvement. The illness is rare and occurs in approximately 1.6 persons per 100,000 annually. A few patients have concomitant dementia. The disease progresses rapidly, and death generally occurs within 4 years of onset.

Infectious Diseases

Herpes Simplex Encephalitis. Herpes simplex encephalitis, the most common type of focal encephalitis, most commonly affects the frontal and temporal lobes. The symptoms often include anosmia, olfactory and gustatory hallucinations, and personality changes and can also involve bizarre or psychotic behaviors. Complex partial epilepsy may also develop in patients with herpes simplex encephalitis. Although the mortality rate for the infection has decreased, many patients exhibit personality changes, symptoms of memory loss, and psychotic symptoms.

Rabies Encephalitis. The incubation period for rabies ranges from 10 days to 1 year, after which symptoms of restlessness, overactivity, and agitation can develop. Hydrophobia, present in up to 50 percent of patients, is characterized by an intense fear of drinking water. The fear develops from the severe laryngeal and diaphragmatic spasms that the patients experience when they drink water. When rabies encephalitis develops, the disease is fatal within days or weeks.

Neurosyphilis. Neurosyphilis (also known as general paresis) appears 10 to 15 years after the primary *Treponema* infection. Since the advent of penicillin, neurosyphilis has become a rare disorder, although AIDS is associated with reintroducing neurosyphilis into medical practice in some urban settings. Neurosyphilis generally affects the frontal lobes and results in personality changes, the development of poor judgment, irritability, and decreased care for self. Delusions of grandeur develop in 10 to 20 percent of affected patients. The disease progresses with the development of dementia and tremor until patients are paretic. The neurologic symptoms include Argyll Robertson pupils, which are small, irregular, and unequal and have light-near reflex dissociation, tremor, dysarthria, and hyperreflexia. Cerebrospinal fluid (CSF) examination shows lymphocytosis, increased protein, and a positive result on a Venereal Disease Research Laboratory (VDRL) test.

Chronic Meningitis. Chronic meningitis is becoming more common than earlier because of the immunocompromised condition of people with AIDS. The usual causative agents are *Mycobacterium tuberculosis, Cryptococcus* spp., and *Coccidioides* spp. The usual symptoms are headache, memory impairment, confusion, and fever.

Subacute Sclerosing Panencephalitis. Subacute sclerosing panencephalitis is a disease of childhood and early adolescence, with a 3-to-1 male-to-female ratio. The onset usually follows either an infection with measles or vaccination for measles. The initial symptoms may be behavioral change, temper tantrums, sleepiness, and hallucinations, but the classic symptoms of myoclonus, ataxia, seizures, and intellectual deterioration eventually develop. The disease progresses relentlessly to coma and death in 1 to 2 years.

Lyme Disease. Lyme disease is caused by infection with the spirochete *Borrelia burgdorferi* transmitted through the bite of the deer tick (*Ixodes scapularis*), which feeds on infected deer and mice. About 16,000 cases are reported annually in the United States.

A characteristic bull’s eye rash occurs at the site of the tick bite followed shortly after that by flulike symptoms. Impaired cognitive functioning and

mood changes are associated with the illness and may be the presenting complaint. These include memory lapses, difficulty concentrating, irritability, and depression.

No clear-cut diagnostic test is available. About 50 percent of patients become seropositive to *B. burgdorferi.* The prophylaxis vaccine is not always effective and is controversial. Treatment consists of a 14- to 21-day course of doxycycline, which results in a 90 percent cure rate. Specific psychiatric drugs can help treat the psychiatric sign or symptom (e.g., a benzodiazepine for anxiety). Left untreated, about 60 percent of persons develop a chronic condition. Such patients may get an erroneous diagnosis of a primary depression rather than one secondary to the medical condition. Support groups for patients with chronic Lyme disease are essential. Group members provide each other with emotional support that helps improve their quality of life.

Prion Disease. Prion disease is a group of related disorders caused by a transmissible infectious protein known as a *prion.* Included in this group are Creutzfeldt–Jakob disease (CJD), Gerstmann–Sträussler–Scheinker disorder (GSS), fatal familial insomnia (FFI), and kuru. A variant of CJD (vCJD), also called "mad cow disease," appeared in 1995 in the United Kingdom and is attributed to the transmission of bovine spongiform encephalopathy (BSE) from cattle to humans. Collectively, these disorders are also known as *subacute spongiform encephalopathy* because of shared neuropathologic changes that consist of (1) spongiform vacuolization, (2) neuronal loss, and (3) astrocyte proliferation in the cerebral cortex. Amyloid plaques may or may not be present.

ETIOLOGY. Prions are transmissible agents but differ from viruses in that they lack nucleic acid. Prions are mutated proteins generated from the human prion protein gene (PrP), located on the short arm of chromosome 20.

The PrP mutates into a disease-related isoform PrP-Super-C (PrP^{Sc}), which can replicate and is infectious. The neuropathologic changes that occur in prion disease are presumed to be caused by direct neurotoxic effects of PrP^{Sc}.

The specific prion disease that develops depends on the mutation of PrP that occurs. Mutations at PrP 178N/129V cause CJD, mutations at 178N/129M cause FFI, and mutations at 102L/129M cause GSS and kuru.

Other mutations of PrP occur, and research continues in this vital area of genomic identification. Some mutations are both fully penetrant and autosomal dominant and account for inherited forms of prion disease. For example, both GSS and FFI are inherited disorders, and about 10 percent of cases of CJD are also inherited. Prenatal testing for the abnormal PrP gene is available; whether or not we should do routine testing is an open question.

CREUTZFELDT–JAKOB DISEASE. First described in 1920, CJD is an invariably fatal, rapidly progressive disorder that occurs mainly in middle-aged or older adults. It manifests initially with fatigue, flu-like symptoms, and cognitive impairment. As the disease progresses, focal neurologic findings such as aphasia and apraxia occur. Psychiatric manifestations are protean and include emotional lability, anxiety, euphoria, depression, delusions, hallucinations, or marked personality changes. The disease progresses over months, leading to dementia, akinetic mutism, coma, and death.

The rates of CJD range from one to two cases per 1 million persons a year worldwide. The infectious agent self-replicates and can be transmitted to humans by inoculation with infected tissue and sometimes by ingestion of contaminated food. Iatrogenic transmission has been reported via transplantation of contaminated cornea or dura mater or to children via contaminated supplies of human growth hormone derived from infected persons. Neurosurgical transmission also occurs. Household contacts are not at higher risk for developing the disease than the general population unless there is direct inoculation.

Diagnosis requires a pathologic examination of the cortex, which reveals the classic triad of spongiform vacuolation, loss of neurons, and astrocyte cell proliferation. The cortex and basal ganglia are most affected. An immunoassay test for CJD in the CSF shows promise in supporting the diagnosis; however, this needs more testing. Although not specific for CJD, EEG abnormalities are present in nearly all patients, consisting of a slow and irregular background rhythm with periodic complex discharges. CT and MRI studies may reveal cortical atrophy later in the course of the disease. SPECT and positron emission tomography (PET) reveal heterogeneously decreased uptake throughout the cortex.

No known treatment exists for CJD. Death usually occurs within 6 months after diagnosis.

VARIANT CJD. In 1995, a vCJD appeared in the United Kingdom. The patients affected all died; they were young (younger than age 40 years), and none had risk factors of CJD. At autopsy, pathologists found a prion disease. The disease was transmitted between cattle and cattle to humans in the 1980s. BSE appears to have originated from sheep scrapie–contaminated feed given to cattle. Scrapie is a spongiform encephalopathy found in sheep and goats that does not cause human disease; however, it is transmissible to other animal species.

There have been more than 200 reported cases of vCJD to date, with the majority in the United Kingdom. Four cases have been reported in the United States. Clinicians must be alert to the diagnosis in young people with behavioral and psychiatric abnormalities in association with cerebellar signs such as ataxia or myoclonus. The psychiatric presentation of vCJD is not specific. Most patients have reported depression, withdrawal, anxiety, and sleep disturbance. Paranoid delusions have occurred. Neuropathologic changes are similar to those in CJD, with the addition of amyloid plaques.

Epidemiologic data gathering is ongoing. The incubation period for vCJD and the number of infected meat products required to cause infection is unknown. One patient had been a vegetarian for 5 years before contracting the disease. vCJD can be diagnosed antemortem by examining the tonsils with Western blot immunostains to detect PrP^{Sc} in lymphoid tissue. Diagnosis relies on the development of progressive neurodegenerative features in persons who have ingested contaminated meat or brains. No cure exists, and death usually occurs within 2 to 3 years after diagnosis. Prevention is dependent on careful monitoring of cattle for disease and feeding them grain instead of meat byproducts.

KURU. Kuru is an epidemic prion disease found in New Guinea that is caused by cannibalistic funeral rituals in which the practitioners eat the brains of the deceased. Women are more affected by the disorder than men, presumably because they participate in the ceremony to a greater extent. Death usually occurs within 2 years after symptoms develop. Neuropsychiatric signs and symptoms consist of ataxia, chorea, strabismus, delirium, and dementia. Pathologic changes are similar to those with other prion diseases: neuronal loss, spongiform lesions, and astrocytic proliferation. The cerebellum is most affected. Iatrogenic transmission of kuru has occurred when doctors

transplanted infected dura mater and corneas into normal recipients. Since the cessation of cannibalism in New Guinea, the incidence of the disease has decreased drastically.

GERSTMANN–STRäUSSLER–SCHEINKER DISEASE. First described in 1928, GSS is a neurodegenerative syndrome characterized by ataxia, chorea, and cognitive decline leading to dementia. The cause is a mutation in the PrP gene that is fully penetrant and autosomal dominant; thus, the disease is inherited, and affected families have been identified over several generations. Genetic testing can confirm the presence of abnormal genes before onset. Pathologic changes characteristic of prion disease are present: spongiform lesions, neuronal loss, and astrocyte proliferation. Amyloid plaques occur in the cerebellum. The onset of the disease occurs between 30 and 40 years of age. The disease is fatal within 5 years of onset.

FATAL FAMILIAL INSOMNIA. FFI is an inherited prion disease that primarily affects the thalamus. A syndrome of insomnia and autonomic nervous system dysfunction consisting of fever, sweating, labile blood pressure, and tachycardia occurs that is debilitating. Onset is in middle adulthood, and death usually occurs in 1 year. No treatment currently exists.

FUTURE DIRECTIONS. Determining how prions mutate to produce disease phenotypes and determining how they transmit between different mammalian species are essential areas of research. Public health measures to prevent transmission of animal disease to humans are ongoing and must be relentless, mainly because these disorders are invariably fatal within a few years of onset. Developing genetic interventions that prevent or repair damage to the normal prion gene offers the best hope of a cure. Psychiatrists must manage cases of persons who have the disease and those with hypochondriacal fears of having contracted the disease. In some patients, such fears can reach delusional proportions. Treatment is symptomatic and involves anxiolytics, antidepressants, and psychostimulants, depending on symptoms. Supportive psychotherapy may be of use in early stages to help patients and families cope with the illness.

Preventing unintentional human-to-human or animal-to-human transmission of prions remains the best way to limit the scope of these

diseases. Sporadic cases of CJD will still appear, however, because of the rare spontaneous mutation of the normal prion protein into the abnormal form. At present, little exists to offer patients with prion disease other than supportive treatment and emotional support.

Immune Disorders

The primary immune disorder in contemporary society is HIV and AIDS, but other immune disorders such as lupus erythematosus and autoimmune disorders that affect brain neurotransmitters (discussed below) can also present diagnostic and treatment challenges to mental health clinicians.

HIV Infection and AIDS. HIV is a retrovirus related to the human T-cell leukemia viruses (HTLV) and to retroviruses that infect animals, including nonhuman primates. At least two types of HIV exist, HIV-1 and HIV-2. HIV-1 is the causative agent for most HIV-related diseases; HIV-2, however, seems to be causing an increasing number of infections in Africa. Other types of HIV may exist, currently classified as HIV-O. HIV is present in the blood, semen, cervical and vaginal secretions, and, to a lesser extent, in saliva, tears, breast milk, and the CSF of those who are infected. HIV usually transmits through sexual intercourse or the transfer of contaminated blood from one person to another. Health providers should be aware of the guidelines for safe sexual practices and should advise their patients to practice safe sex (Table 3-35). The Centers for Disease Control and Prevention guidelines for the prevention of HIV from infected to uninfected persons is listed in Table 3-36.

After infection with HIV, AIDS develops in 8 to 11 years, although this time is gradually increasing because of early treatment. When a person is infected with HIV, the virus primarily targets T4 (helper) lymphocytes, so-called CD4+ lymphocytes, to which the virus binds because a glycoprotein (gp120) on the viral surface has a high affinity for the CD4 receptor on T4 lymphocytes. After binding, the virus can inject its ribonucleic acid (RNA) into the infected lymphocyte, where the RNA is transcribed into deoxyribonucleic acid (DNA) by the action of reverse transcriptase. The resultant DNA can then be incorporated into the host cell's genome and translated and eventually transcribed when the lymphocyte divides. After lymphocytes have produced viral proteins, the various components of the

virus assemble, and new mature viruses bud off from the host cell.

Table 3-35
AIDS Safe-Sex Guidelines

Safe Sexual Behaviors and Risk-Reduction Strategies for patients with HIV

- Adhering to ART and ongoing medical care, even if viral load is undetectable.
- Communicating HIV status with others.
- Correctly and consistently using condoms (to prevent STDs) and appropriate non–oil-based lubricants, even when negotiation of use occurs in the heat of the moment.
- Assessing relative risk of HIV transmission associated with various sexual activities (e.g., oral sex is less risky than receptive anal sex).
- Discussing how alcohol and/or drug use can impair judgment.
- Using PrEP for some HIV-negative partners, including women planning to become pregnant.
- Using PEP for emergencies for HIV-negative or unknown status partners (e.g., if a condom breaks or is not used and the patient is not virally suppressed).

Source: Centers for Disease Control and Prevention. Safer sexual behavior. https://www.cdc.gov/hiv/clinicians/treatment/safer-sex.html#discussion-topics

Table 3-36
Centers for Disease Control and Prevention Guidelines for the Prevention of HIV Transmission from Infected to Uninfected Persons

Infected persons should be counseled to prevent the further transmission of HIV by:

1. Informing prospective sex partners of their infection with HIV so they can take appropriate precautions. Abstention from sexual activity with another person is one option that would eliminate any

risk of sexually transmitted HIV infection.
2. Protecting a partner during any sexual activity by taking appropriate precautions to prevent that person's coming into contact with the infected person's blood, semen, urine, feces, saliva, cervical secretions, or vaginal secretions. Although the efficacy of using condoms to prevent infections with HIV is still under study, the consistent use of condoms should reduce the transmission of HIV by preventing exposure to semen and infected lymphocytes.
3. Informing previous sex partners and any persons with whom needles were shared of their potential exposure to HIV and encouraging them to seek counseling and testing.
4. For IV drug abusers, enrolling or continuing in programs to eliminate the abuse of IV substances. Needles, other apparatus, and drugs must never be shared.
5. Never sharing toothbrushes, razors, or other items that could become contaminated with blood.
6. Refraining from donating blood, plasma, body organs, other tissue, or semen.
7. Avoiding pregnancy until more is known about the risks of transmitting HIV from the mother to the fetus or newborn.
8. Cleaning and disinfecting surfaces on which blood or other body fluids have spilled in accordance with previous recommendations.
9. Informing physicians, dentists, and other appropriate health professionals of antibody status when seeking medical care, so that the patient can be appropriately evaluated.

HIV, human immunodeficiency virus; IV, intravenous.

From Centers for Disease Control (CDC). Additional recommendations to reduce sexual and drug abuse-related transmission of human T-lymphotropic virus type III/lymphadenopathy-associated virus. *MMWR Morb Mortal Wkly Rep*. 1986;35:152.

DIAGNOSIS

Serum Testing. Techniques are now widely available to detect the presence of anti-HIV antibodies in humans. The conventional test uses blood (time to result, 3 to 10 days), and the rapid test uses an oral swab (time to result, 20

minutes). Both tests are 99.9 percent sensitive and specific. Health care workers and their patients must understand that the presence of HIV antibodies indicates infection, not immunity to infection. Those who test positive have been exposed to the virus, have the virus within their bodies, have the potential to transmit the virus to another person, and will almost certainly eventually develop AIDS. Those who test negative either have not been exposed to HIV or were exposed but have not yet developed antibodies; the latter is possible if the exposure was less than a year prior. Seroconversion most commonly occurs 6 to 12 weeks after infection, although in rare cases, seroconversion can take 6 to 12 months.

COUNSELING. Although specific groups of persons who are at high risk for contracting HIV are a priority, anyone who wants a test should have it. We should try to ascertain the reason for the request, to detect unspoken concerns and motivations that may merit psychotherapeutic intervention.

Counseling should include a discussion of past practices that may have put the testee at risk for HIV infection and safe sexual practices. During posttest counseling, counselors should explain that a negative test finding implies that the person should espouse safe sexual behavior and avoid shared needles to remain free of infection. Those with positive results must receive counseling about safe practices and potential treatment options. They may need additional psychotherapeutic interventions if anxiety or depressive disorders develop after they discover that they are infected. A person may react to a positive HIV test finding with a syndrome similar to posttraumatic stress disorder. Adjustment disorder with anxiety or depressed mood may develop in as many as 25 percent of those informed of a positive HIV test result.

CONFIDENTIALITY. No one should be given an HIV test without previous knowledge and consent, although various jurisdictions and organizations, such as the military, now require HIV testing for all inhabitants or members. The results of an HIV test can be shared with other members of a medical team, although no one else should have the information except for exceptional circumstances. We should advise the patient against disclosing the result of HIV testing too readily to employers, friends, and family members; the information could result in discrimination in employment, housing, and insurance.

The major exception to restriction of disclosure is the need to notify potential and past sexual or IV substance use partners. If a treating physician knows that a patient who is HIV infected is putting another person at risk of becoming infected, the physician may try either to hospitalize the infected person involuntarily (to prevent danger to others) or to notify the potential victim. Clinicians should be aware of the laws about such issues, which vary among the states. These guidelines also apply to inpatient psychiatric wards when a patient with HIV infection may have had sex with other patients.

CLINICAL FEATURES

Nonneurologic Factors. About 30 percent of persons infected with HIV experience a flulike syndrome 3 to 6 weeks after becoming infected; most never notice any symptoms immediately or shortly after their infection. The flulike syndrome includes fever, myalgia, headaches, fatigue, GI symptoms, and sometimes a rash. The syndrome may include splenomegaly and lymphadenopathy.

The most common infection in persons affected with HIV who have AIDS is *Pneumocystis carinii* pneumonia, which is characterized by a chronic, nonproductive cough and dyspnea, sometimes sufficiently severe to result in hypoxemia and its resultant cognitive effects. For psychiatrists, the importance of these nonneurologic, nonpsychiatric complications lies in their biologic effects on patients' brain function (e.g., hypoxia in *P. carinii* pneumonia) and their psychological effects on patients' moods and anxiety states.

Neurologic Factors. An extensive array of disease processes can affect the brain of a patient infected with HIV (Table 3-37). The most important diseases for mental health workers to be aware of are *HIV mild neurocognitive disorder* and *HIV-associated dementia.*

Psychiatric Syndromes. HIV-associated dementia presents with the typical triad of symptoms seen in other subcortical dementias—memory and psychomotor speed impairments, depressive symptoms, and movement disorders. Patients may initially notice slight problems with reading, comprehension, memory, and mathematical skills, but these symptoms are

subtle and may be overlooked or discounted as fatigue and illness. The Modified HIV Dementia Scale is a useful bedside screen and can document disease progress with serial administration. The development of dementia in HIV-infected patients is generally a poor prognostic sign, and 50 to 75 percent of patients with dementia die within 6 months.

Table 3-37
Conditions Associated with Human Immunodeficiency Virus (HIV) Infection

Bacterial infections, multiple or recurrent[a]
Candidiasis of bronchi, trachea, or lungs
Candidiasis, esophageal
Cervical cancer, invasive[b]
Coccidioidomycosis, disseminated or extrapulmonary
Cryptococcosis, extrapulmonary
Cryptosporidiosis, chronic intestinal (>1 mo duration)
Cytomegalovirus disease (other than liver, spleen, or nodes)
Cytomegalovirus retinitis (with loss of vision)
Encephalopathy, HIV-related
Herpes simplex, chronic ulcers (>1 mo duration); or bronchitis, pulmonitis, or esophagitis
Histoplasmosis, disseminated or extrapulmonary
Isosporiasis, chronic intestinal (>1 mo duration)
Kaposi sarcoma
Lymphoid interstitial pneumonia or pulmonary lymphoid hyperplasia[a]
Lymphoma, Burkitt's (or equivalent term)
Lymphoma, immunoblastic (or equivalent term)
Lymphoma, primary, of brain
Mycobacterium avium complex or *Mycobacterium kansasii,* disseminated or extrapulmonary
Mycobacterium tuberculosis, any site (pulmonary[b] or extrapulmonary)
Mycobacterium, other species or unidentified species, disseminated

or extrapulmonary
Pneumocystis carinii pneumonia
Pneumonia, recurrent[b]
Progressive multifocal leukoencephalopathy
Salmonella septicemia, recurrent
Toxoplasmosis of brain
Wasting syndrome due to HIV

[a]Children younger than 13 years old.

[b]Added in the 1993 expansion of the AIDS surveillance case definition for adolescents and adults.

Adapted from 1993 revised classification system for HIV infection and expanded surveillance, case definition for AIDS among adolescents and adults. *MMWR Recomm Rep*. 1992:41.

HIV-associated neurocognitive disorder (also known as HIV encephalopathy) involves impaired cognitive functioning and reduced mental activity that interferes with work, domestic, and social functioning. No laboratory findings are specific to the disorder, and it occurs independently of depression and anxiety. Progression of HIV-associated dementia usually occurs but may be prevented by early treatment.

Delirium can result from the same causes that lead to dementia in patients with HIV. Clinicians have classified delirious states characterized by both increased and decreased activities. Delirium in patients infected with HIV is probably underdiagnosed, but it should always precipitate a medical workup of a patient infected with HIV to determine whether a new CNS-related process has begun.

Patients with HIV infection may have any of the anxiety disorders, but generalized anxiety disorder, posttraumatic stress disorder, and obsessive-compulsive disorder (OCD) are particularly common.

Adjustment disorder with anxiety or depressed mood occurs in 5 to 20 percent of HIV-infected patients. The incidence of adjustment disorder in HIV-infected patients is higher than usual in some special populations, such as military recruits and prison inmates.

Depression is a significant problem in HIV and AIDS. Approximately 4 to 40 percent of HIV-infected patients meet the criteria for depressive disorders. Major depression is a risk factor for HIV infection by its impact on behavior, intensification of substance abuse, exacerbation of self-destructive behaviors,

and promotion for poor partner choice in relationships. The pre-HIV infection prevalence of depressive disorders may be higher than usual in some groups who are at risk for contracting HIV. Depression can hinder effective treatment in infected persons. Patients with major depression are at increased risk of disease progression and death. HIV increases the risk of developing major depression through a variety of mechanisms, including direct injury to subcortical areas of the brain, chronic stress, worsening social isolation, and intense demoralization. Depression is higher in women than in men.

Mania can occur at any stage of HIV infection for individuals with preexisting bipolar disorder. AIDS mania is a type of mania that most commonly occurs in late-stage HIV infections and is associated with cognitive impairment. AIDS mania has a somewhat different clinical profile than bipolar mania. Patients tend to have cognitive slowing or dementia, and irritability is more characteristic than euphoria. AIDS mania is usually quite severe in its presentation and malignant in its course. It seems to be more chronic than episodic, has infrequent spontaneous remissions, and usually relapses with the cessation of treatment. One clinically significant presentation is the delusional belief that one has discovered the cure for HIV or has been cured, which may result in high-risk behaviors and the spread of the HIV infection.

Substance abuse is a primary vector for the spread of HIV. This is true not only for IV drug users and their partners but others impaired by substances. All addictions tend to increase impulsive and unsafe behaviors. Ongoing substance abuse has grave medical implications for HIV-infected patients. The accumulation of medical sequelae from chronic substance abuse can accelerate the process of immunocompromise and amplify the progressive burdens of the HIV infection itself. In addition to the direct physical effects caused by drugs, active substance use is highly associated with both nonadherence and reduced access to antiretroviral medication.

Suicidal ideation and suicide attempts may increase in patients with HIV infection and AIDS. The risk factors for suicide among persons infected with HIV include having friends who died from AIDS, recent notification of HIV seropositivity, relapses, complicated social issues relating to homosexuality, inadequate social and financial support, and the presence of dementia or delirium.

Psychotic symptoms are usually later-stage complications of HIV

infection. They require an immediate medical and neurologic evaluation and often require management with antipsychotic medications.

The worried well are people in high-risk groups who, although they tested negative and are disease-free, are anxious about contracting the virus. Repeated negative test results reassure some, but others cannot be reassured. Their worry well status can progress quickly to generalized anxiety disorder, panic attacks, OCD, and a somatic symptom and related disorder.

TREATMENT. Prevention is the primary approach to HIV infection. Primary prevention involves protecting persons from getting the disease; secondary prevention involves modification of the disease's course. All persons with any risk of HIV infection should be informed about safe-sex practices and about the necessity to avoid sharing contaminated hypodermic needles. The assessment of patients infected with HIV should include a complete sexual and substance use history, a psychiatric history, and an evaluation of the support systems available to them.

Pharmacotherapy. A growing list of agents that act at different points in viral replication has raised the hope that HIV might be permanently suppressed or eradicated from the body. These agents divide into five major drug classes. Reverse transcriptase inhibitors (RTIs) interfere with the critical step during the HIV life cycle known as reverse transcription. There are two types of RTIs: nucleoside/nucleotide RTIs (NRTIs), which are faulty DNA building blocks, and nonnucleoside RTIs (NNRTIs), which bind to RT, interfering with its ability to convert the HIV RNA into HIV DNA. Protease inhibitors interfere with the protease enzyme that HIV uses to produce infectious viral particles. Fusion or entry inhibitors interfere with the virus's ability to fuse with the cellular membrane, thereby blocking entry into the host cell. Integrase inhibitors block integrase, the enzyme HIV uses to integrate the genetic material of the virus into its target host cell. Multidrug combination products combine drugs from more than one class into a single product. The most common of this class of drugs is the highly active antiretroviral therapy (HAART). Table 3-38 lists the available agents in each of these categories.

Antiretroviral agents have many adverse effects. Of importance to psychiatrists is that protease inhibitors can increase levels of certain psychotropic drugs such as bupropion, meperidine, various benzodiazepines,

and selective serotonin reuptake inhibitors (SSRIs). We should take caution when prescribing psychiatric drugs to persons taking protease inhibitors.

Psychotherapy. Major psychodynamic themes for patients infected with HIV involved self-blame, self-esteem, and issues regarding death. The entire range of psychotherapeutic approaches may be appropriate for patients with HIV-related disorders. Both individual and group therapy can be useful. Individual therapy may be either short term or long term and may be supportive, cognitive, behavioral, or psychodynamic. Group therapy techniques can range from psychodynamic to completely supportive. It is crucial to include direct counseling regarding substance use and its potential adverse effects on the health of the patient. Specific treatments for particular substance-related disorders should be initiated if necessary for the total well-being of the patient.

Systemic Lupus Erythematosus. Systemic lupus erythematosus (SLE) is an autoimmune disease that involves inflammation of multiple organ systems. The officially accepted diagnosis of SLE requires a patient to have 4 of 11 criteria that have been defined by the American Rheumatism Association. Between 5 and 50 percent of patients with SLE have mental symptoms at the initial presentation, and approximately 50 percent eventually show neuropsychiatric manifestations. The primary symptoms are depression, insomnia, emotional lability, nervousness, and confusion. Treatment with steroids commonly induces further psychiatric complications, including mania and psychosis.

Autoimmune Disorders Affecting Brain Neurotransmitters. A group of autoimmune receptor-seeking disorders exists that causes encephalitis that mimics schizophrenia. Among those is anti-NMDA (N-methyl D-aspartate)-receptor encephalitis that causes dissociative symptoms, amnesia, and vivid hallucinations. The disorder occurs mostly in women; the memoir *Brain on Fire* vividly chronicled the writer's experience with it. There is no treatment, although intravenous immunoglobulins have proved useful. Recovery does occur, but some patients might require prolonged intensive care.

Table 3-38
Antiretroviral Agents

Generic Names	Trade Name	Usual Abbreviation
Reverse Transcriptase Inhibitors		
Nucleoside/nucleotide reverse transcriptase inhibitors		
Lamivudine and zidovudine	Combivir	
Emtricitabine	Emtriva	FTC
Lamivudine	Epivir	3TC
Abacavir and lamivudine	Epzicom	
Zidovudine, azidothymidine	Retrovir	ZDV or AZT
Abacavir, zidovudine, and lamivudine	Trizivir	
Tenofovir disoproxil fumarate and emtricitabine	Truvada	
Didanosine, dideoxyinosine	Videx	ddl
Enteric-coated didanosine	Videx EC	ddl EC
Tenofovir disoproxil fumarate	Viread	TDF
Stavudine	Zerit	d4t
Abacavir sulfate	Ziagen	ABC
Nonnucleoside Reverse Transcriptase Inhibitors		
Rilpivirine	Edurant	
Etravirine	Intelence	
Delavirdine	Rescriptor	DLV
Efavirenz	Sustiva	EFV
Nevirapine	Viramune	NVP

Protease Inhibitors		
Amprenavir	Agenerase	APV
Tipranavir	Aptivus	TPV
Indinavir	Crixivan	IDV
Saquinavir mesylate	Invirase	SQV
Lopinavir and ritonavir	Kaletra	LPV/RTV
Fosamprenavir calcium	Lexiva	FOS-APV
Ritonavir	Norvir	RTV
Darunavir	Prezista	
Atazanavir sulfate	Reyataz	ATV
Delfinavir mesylate	Viracept	NFV
Fusion/Entry Inhibitors		
Enfuvirtide	Fuzeon	T-20
Maraviroc	Selzentry	
Multi-Class Combination Products		
Efavirenz, emtricitabine, and tenofovir disoproxil fumarate	Atripla	
Emtricitabine, rilpivirine, and tenofovir disoproxil fumarate	Complera	

Endocrine Disorders

Thyroid Disorders. Hyperthyroidism includes symptoms of confusion, anxiety, and an agitated, depressive syndrome. Patients may also complain of being easily fatigued and of feeling generally weak. Insomnia, weight loss despite increased appetite, tremulousness, palpitations, and increased perspiration are also common symptoms. Serious psychiatric symptoms

include impairments in memory, orientation, and judgment, manic excitement, delusions, and hallucinations.

In 1949, Irvin Asher named hypothyroidism “myxedema madness.” In its most severe form, hypothyroidism includes symptoms of paranoia, depression, hypomania, and hallucinations. Slowed thinking and delirium can also be symptoms. The physical symptoms include weight gain, a deep voice, thin and dry hair, loss of the lateral eyebrow, facial puffiness, cold intolerance, and impaired hearing. Approximately 10 percent of all patients have residual neuropsychiatric symptoms after hormone replacement therapy.

Parathyroid Disorders. Dysfunction of the parathyroid gland results in the abnormal regulation of calcium metabolism. Excessive secretion of parathyroid hormone causes hypercalcemia, which can result in delirium, personality changes, and apathy in 50 to 60 percent of patients and cognitive impairments in approximately 25 percent of patients. Neuromuscular excitability, which depends on proper calcium ion concentration, is reduced, and muscle weakness may appear.

Hypocalcemia can occur with hypoparathyroidism and can result in neuropsychiatric symptoms of delirium and personality changes. If the calcium level decreases gradually, clinicians may see the psychiatric symptoms without the characteristic tetany of hypocalcemia. Other symptoms of hypocalcemia are cataract formation, seizures, extrapyramidal symptoms, and increased intracranial pressure.

Adrenal Disorders. Adrenal disorders disturb the normal secretion of hormones from the adrenal cortex and produce significant neurologic and psychological changes. Chronic adrenocortical insufficiency (Addison’s disease) is usually the result of adrenocortical atrophy or granulomatous invasion caused by tuberculous or fungal infection. Patients with this disorder exhibit mild mental symptoms, such as apathy, easy fatigability, irritability, and depression. Occasionally, confusion or psychotic reactions develop. Cortisone or one of its synthetic derivatives is effective in correcting such abnormalities.

Excessive quantities of cortisol produced endogenously by an adrenocortical tumor or hyperplasia (Cushing syndrome) lead to a secondary mood disorder, a syndrome of agitated depression, and often suicide.

Decreased concentration and memory deficits may also be present. Psychotic reactions, with schizophrenia-like symptoms, are seen in a few patients. The administration of high doses of exogenous corticosteroids typically leads to a secondary mood disorder similar to mania. Severe depression can follow the termination of steroid therapy.

Pituitary Disorders. Patients with total pituitary failure can exhibit psychiatric symptoms, particularly postpartum women who have hemorrhaged into the pituitary, a condition known as *Sheehan syndrome.* Patients have a combination of symptoms, especially of thyroid and adrenal disorders, and can show virtually any psychiatric symptom.

Metabolic Disorders

A common cause of organic brain dysfunction, metabolic encephalopathy can produce alterations in mental processes, behavior, and neurologic functions. We should consider the diagnosis whenever recent and rapid changes in behavior, thinking, and consciousness occur. The earliest signals are likely to be impairment of memory, particularly recent memory, and impairment of orientation. Some patients become agitated, anxious, and hyperactive; others become quiet, withdrawn, and inactive. As metabolic encephalopathies progress, confusion or delirium gives way to decreased responsiveness, stupor, and, eventually, death.

Hepatic Encephalopathy. Severe hepatic failure can result in hepatic encephalopathy, characterized by asterixis, hyperventilation, EEG abnormalities, and alterations in consciousness. The alterations in consciousness can range from apathy to drowsiness to coma. Associated psychiatric symptoms are changes in memory, general intellectual skills, and personality.

Uremic Encephalopathy. Renal failure is associated with alterations in memory, orientation, and consciousness. Restlessness, crawling sensations on the limbs, muscle twitching, and persistent hiccups are associated symptoms. In young people with brief episodes of uremia, the neuropsychiatric symptoms tend to be reversible; in older people with prolonged episodes of uremia, the neuropsychiatric symptoms can be irreversible.

Hypoglycemic Encephalopathy. Hypoglycemic encephalopathy can be caused either by excessive endogenous production of insulin or by excessive exogenous insulin administration. The premonitory symptoms, which do not occur in every patient, include nausea, sweating, tachycardia, and feelings of hunger, apprehension, and restlessness. As the disorder progresses, disorientation, confusion, and hallucinations, as well as other neurologic and medical symptoms, can develop. Stupor and coma can occur, and residual and persistent dementia can sometimes be a serious neuropsychiatric sequela of the disorder.

Diabetic Ketoacidosis. Diabetic ketoacidosis begins with feelings of weakness, easy fatigability, and listlessness and increasing polyuria and polydipsia. Headache and, sometimes, nausea and vomiting appear. Patients with diabetes mellitus have an increased likelihood of chronic dementia with general arteriosclerosis.

Acute Intermittent Porphyria. The porphyrias are disorders of heme biosynthesis that result in excessive accumulation of porphyrins. The triad of symptoms is acute, colicky abdominal pain, motor polyneuropathy, and psychosis. Acute intermittent porphyria is an autosomal dominant disorder that affects more women than men and has its onset between ages 20 and 50 years. The psychiatric symptoms include anxiety, insomnia, lability of mood, depression, and psychosis. Some studies have found that between 0.2 and 0.5 percent of chronic psychiatric patients may have undiagnosed porphyrias. Barbiturates precipitate or aggravate the attacks of acute porphyria, and we should never use them in this disorder (less a concern in modern treatments, however some headache treatments, e.g., Fioricet and Fiorinal, remain popular). We should also avoid those or similar medications in relatives or people with the disease.

Nutritional Disorders

Niacin Deficiency. Dietary insufficiency of niacin (nicotinic acid) and its precursor tryptophan are associated with pellagra, a globally occurring nutritional deficiency disease seen in association with alcohol abuse, vegetarian diets, and extreme poverty and starvation. The neuropsychiatric

symptoms of pellagra include apathy, irritability, insomnia, depression, and delirium; the medical symptoms include dermatitis, peripheral neuropathies, and diarrhea. Traditionally, some refer to the course of pellagra as the "five Ds": dermatitis, diarrhea, delirium, dementia, and death. The response to treatment with nicotinic acid is rapid, but dementia from prolonged illness may improve only slowly and incompletely.

Thiamine Deficiency. Thiamine (vitamin B_1) deficiency leads to beriberi, characterized chiefly by cardiovascular and neurologic changes, and to Wernicke–Korsakoff syndrome, which is most often associated with chronic alcohol abuse. Beriberi occurs primarily in Asia and areas of famine and poverty. The psychiatric symptoms include apathy, depression, irritability, nervousness, and poor concentration; severe memory disorders can develop with prolonged deficiencies.

Cobalamin Deficiency. Deficiencies in cobalamin (vitamin B_{12}) arise because of the failure of the gastric mucosal cells to secrete a specific substance, intrinsic factor, required for the normal absorption of vitamin B_{12} in the ileum. The deficiency state is characterized by the development of a chronic macrocytic megaloblastic anemia (pernicious anemia) and by neurologic manifestations resulting from degenerative changes in the peripheral nerves, the spinal cord, and the brain. Neurologic changes occur in approximately 80 percent of all patients. These changes are commonly associated with megaloblastic anemia, but they occasionally precede the onset of hematologic abnormalities.

Mental changes, such as apathy, depression, irritability, and moodiness, are frequent. In a few patients, encephalopathy and its associated delirium, delusions, hallucinations, dementia, and sometimes paranoid features are prominent and are sometimes called *megaloblastic madness.* The neurologic manifestations of vitamin B_{12} deficiency can be rapidly and entirely arrested by early and continued administration of parenteral vitamin therapy.

Toxins

Environmental toxins are becoming an increasingly severe threat to physical and mental health in contemporary society.

Mercury. Either inorganic or organic mercury can cause mercury poisoning. Inorganic mercury poisoning results in the “mad hatter” syndrome (previously seen in workers in the hat industry who softened mercury-containing felt by putting it in their mouths), with depression, irritability, and psychosis. Associated neurologic symptoms are headache, tremor, and weakness. Contaminated fish or grain can cause organic mercury poisoning and result in depression, irritability, and cognitive impairment. Associated symptoms are sensory neuropathies, cerebellar ataxia, dysarthria, paresthesias, and visual field defects. Mercury poisoning in pregnant women causes abnormal fetal development. No specific therapy is available, although chelation therapy with dimercaprol may help with acute poisoning.

Lead. Lead poisoning occurs when the amount of lead ingested exceeds the body’s ability to eliminate it. It takes several months for toxic symptoms to appear.

The signs and symptoms of lead poisoning depend on the level of lead in the blood. When lead reaches levels above 200 mg/L, symptoms of severe lead encephalopathy occur, with dizziness, clumsiness, ataxia, irritability, restlessness, headache, and insomnia. Later, an excited delirium occurs, with associated vomiting and visual disturbances, and progresses to convulsions, lethargy, and coma.

Treatment of lead encephalopathy should be instituted as rapidly as possible, even without laboratory confirmation, because of the high mortality rate. The treatment of choice to facilitate lead excretion is intravenous administration of calcium disodium edetate (calcium disodium versenate) daily for 5 days.

Manganese. Early manganese poisoning (sometimes called *manganese madness*) causes symptoms of headache, irritability, joint pains, and somnolence. An eventual picture appears of emotional lability, pathologic laughter, nightmares, hallucinations, and compulsive and impulsive acts associated with periods of confusion and aggressiveness. Lesions involving the basal ganglia and pyramidal system result in gait impairment, rigidity, monotonous or whispering speech, tremors of the extremities and tongue, masked facies (manganese mask), micrographia, dystonia, dysarthria, and loss of equilibrium. The psychological effects tend to clear 3 or 4 months

after the patient's removal from the site of exposure, but neurologic symptoms tend to remain stationary or to progress. No specific treatment exists for manganese poisoning, other than removal from the source of poisoning. The disorder occurs in persons working in refining ore, brick workers, and those making steel casings.

Arsenic. Chronic arsenic poisoning most commonly results from prolonged exposure to herbicides containing arsenic or from drinking water contaminated with arsenic. Silicon-based computer chip manufacturing also uses arsenic. Early signs of toxicity are skin pigmentation, GI complaints, renal and hepatic dysfunction, hair loss, and a characteristic garlic odor to the breath. Encephalopathy eventually occurs, with generalized sensory and motor loss. Chelation therapy with dimercaprol can help successfully treat arsenic poisoning.

▲ 3.5 Mild Cognitive Impairment

The past decade has seen the emergence of a new concept, *mild cognitive impairment* (MCI), which is defined as the presence of mild cognitive decline not warranting the diagnosis of dementia but with preserved basic activities of daily living.

DSM-5 classifies MCI as *mild neurocognitive disorder due to multiple etiologies or unspecified neurocognitive disorder*. It will most likely receive more attention in future revisions of the DSM.

DEFINITION

Although the term *mild cognitive impairment* has been in use for more than 25 years, it became a diagnostic category because of the need to fill the gap between cognitive changes associated with aging and cognitive impairment suggestive of dementia. The criteria proposed by the Mayo Clinic Alzheimer's Disease Research Center (MCADRC) are listed in Table 3-39. At this time, there are no international diagnostic criteria for MCI.

Table 3-39

Mild Cognitive Impairment Original Criteria

1. Memory complaint, preferably qualified by an informant
2. Memory impairment for age and education
3. Preserved general cognitive function
4. Intact activities of daily living
5. Not demented

Table 3-40
Terms Related to Mild Cognitive Impairment

Term	Author(s)	Year	Inclusion Criteria	Observations
Malignant senescent forgetfulness (MSF)	VA Kral	1962	Memory difficulties for recent events Lack of awareness regarding the memory deficit	Two-year follow-up showed a faster evolution of patients with MSF toward dementia
Age-associated memory impairment (AAMI)	NIMH (Crook, Bartus, and Ferris)	1986	Age-related memory disturbances leading to (1) subjective concern; (2) functional problem No underlying neurologic illness	Memory tests were validated on young populations, leading to high rates of AAMI in elderly adults
Age-associated cognitive decline (AACD)	International Psychogeriatric Association and World Health Organization (Levy)	1994	Cognitive deficits not meeting the criteria for dementia	Does not include prognosis regarding evolution to dementia Includes several kinds of cognitive decline (not exclusive memory decline)
Cognitively impaired no dementia (CIND)	Canadian Study of Health and Aging	1997	Age 65 yr and older	Includes static encephalopathies

Historical Perspective

The imprecise border between normal aging-related cognitive decline and dementia-related cognitive impairment has been described for several decades. Thus, in 1962, Kral introduced the terms *benign senescent forgetfulness* (forgetfulness for less essential facts and awareness of problems) and *malignant senescent forgetfulness* (memory problems for recent events and lack of awareness). In 1986, the National Institutes of Mental Health (NIMH) recommended the term *age-associated memory impairment* for age-related normal memory changes. In 1994, the International Psychogeriatrics Association presented the concept of *age-associated cognitive decline,* which described cognitive deficits, including but not limited to memory impairment, in the absence of dementia or other affecting cognitive conditions. *Cognitive impairment no dementia* was introduced in 1997 by the Canadian Study of Health and Aging to describe

the presence of nondemented cognitive impairment regardless of the underlying process (neurologic, psychiatric, medical). Several other classifications, including age-consistent memory impairment and late-life forgetfulness, are defined on the bases of performance on various cognitive tests (Table 3-40).

The exact place of MCI in the psychiatric nosology will be challenging. Based on the current definition of MCI, functional impairment is an exclusion criterion for MCI, but the same "functional impairment" is one of the standard criteria for defining psychiatric disorders. Further developments in finding biologic markers for MCI will probably contribute to a more solid conceptualization and, hopefully, treatment of patients with prodromal dementia.

CLINICAL PRESENTATION

The clinical picture of MCI is a function of the criteria used to define it. Memory impairment is necessary but has been challenging to quantify. One measure has been an objective loss of memory or other cognitive domain that is more than 1.5 standard deviations below the mean for individuals of similar age and education. Some have suggested subjective complaints of memory loss be a marker, but this runs the risk of many false-positive diagnoses.

Assessment

Neuropsychological Assessment. Most experts agree that earlier deficits occur in episodic (vs. semantic) memory. There is no consensus among experts concerning which memory tests and which cutoffs to use. There is a lack of norms, test scores do not have normal distributions, and multiple demographic characteristics influence test performance. Several experts have proposed that a scale such as the delayed recall task from the Consortium to Establish a Registry for Alzheimer Disease might be useful in detecting Alzheimer disease in the earliest stages. Brief mental status instruments (e.g., the Mini-Mental State Examination) are relatively insensitive for the detection of memory problems in MCI.

Biomarkers. In the past decade, researchers have studied several markers of progression from MCI to dementia of the Alzheimer type. Among these, apolipoprotein E4 (APOE4) allele carrier status has been one of the most prominent variables. For the amnestic MCI, APOE4 is a risk factor for a more rapid progression to Alzheimer disease. Several CSF markers are also possible predictors of disease progression: Pathologic low concentrations of $A\beta_{42}$ (the 42 amino acid form of β-amyloid), as well as pathologic high concentrations of total tau (t-tau) and phospho tau (p-tau), may differentiate early Alzheimer disease from normal aging. Locating alterations in the expression of proteins involved in the pathogenetic pathways of Alzheimer disease (proteomic approach) is another approach used to help early detection of Alzheimer disease. We can identify several proteins (cystatin C, β-2 microglobulin, and BEGF polypeptides) through new techniques, and currently, there are several proteins from both CSF and blood that correlate with Alzheimer disease pathology.

Genetics. Because MCI is likely a prodromal stage for several disorders (dementia of the Alzheimer type frontotemporal or vascular dementia), different genes are probably related to MCI. Four genes have a relationship with Alzheimer disease: the *APP* gene, presenilin-1 (*PSEN1*), presenilin-2 (*PSEN2*), and the apolipoprotein E (*APOE*) gene. Because the first three genes are involved in rare autosomal dominant forms of Alzheimer disease, screening for each of these mutations will have minimal value for the diagnosis of MCI in the general population. The *APOE* gene, a common genetic risk factor for early as well as for late-onset Alzheimer disease, has been studied more thoroughly in relationship to MCI, but the results have been inconsistent. Because the etiology of MCI is heterogeneous, a very large number of different genes likely underlie the pathology of MCI. We have not discovered many of these genes.

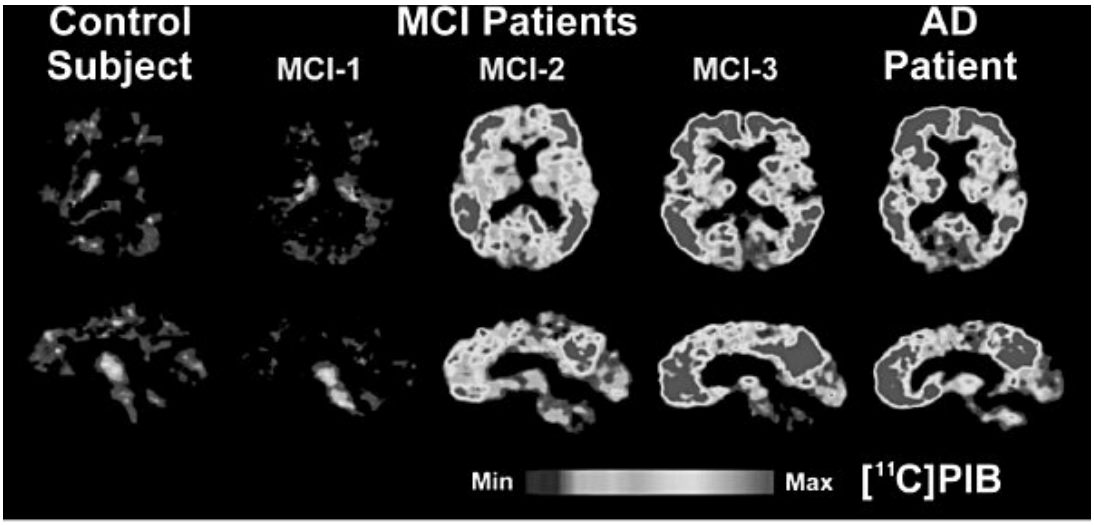


FIGURE 3-11
Positron emission tomography images obtained with the amyloid-imaging agent Pittsburgh Compound-B ([carbon-11]-PIB) in a normal individual with mild cognitive impairment (MCI; *center images*) and a patient with mild Alzheimer disease (AD) (*far right*). Some MCI patients have control-like levels of amyloid, some have Alzheimer disease–like levels of amyloid, and some have intermediate levels. (Courtesy of William E. Klunk, M.D., University of Pittsburgh, Department of Psychiatry, Pittsburgh, PA. All rights retained.)

Neuroimaging. Advances in neuroimaging studies aim to develop measures allowing the differentiation between MCI and healthy aging as well as within MCI among subjects who will convert to dementia of the Alzheimer type or will remain stable over time.

Structural studies of volumetric MCI showed early changes in the medial temporal structures, including neuronal atrophy, decreased synaptic density, and overall neuronal loss. Atrophy of the hippocampal volume and entorhinal cortex occur in MCI. Hippocampal formation atrophy predicts the rate of progression from MCI to dementia of the Alzheimer type. Three-dimensional modeling techniques have localized shape alteration and specific regions of atrophy within the hippocampus. Other methods, such as tensor-based morphometry, allow tracking brain changes in detail, quantifying tissue growth or atrophy throughout the brain and indicating the local rate at which one loses tissue. Other innovations in neuroimaging include MR relaxometry, imaging of iron deposition, diffusion tensor imaging, and high-field MRI scanning.

Perhaps the most promising development has been the advent of PET tracer compounds that visualize amyloid plaques and neurofibrillary tangles.

These new compounds—Pittsburgh Compound B (carbon-11-PIB) and fluorine-18-FDDNP—track pathology changes in the preclinical stages of Alzheimer disease. These specific tracers allow investigators to visualize the pathologic process and are also used to monitor progression from MCI to Alzheimer disease. However, the burden of β-amyloid plaques does not always correlate with the clinical stages, because some MCI subjects can present with a minimal burden similar to healthy control participants, but others have amyloid burden comparable to Alzheimer disease participants. A single biomarker will probably be insufficient to identify incipient Alzheimer disease. Thus, the combination of several markers further increases the accuracy of the prediction and will probably become the norm as described by recent studies (the combination of decreased parietal rCBF and CSF biomarkers as Aβ42, t-tau, and p-tau) (Fig. 3-11).

Diagnostic Differential

The Cognitive Continuum. The cognitive continuum describes the subtle pathway from age-related cognitive decline to MCI to dementia. Per this model, there is an overlap at both ends of MCI, which indicates that it can be quite challenging to identify the transition points (Fig. 3-12). In practice, differentiating MCI from age-related cognitive decline resides mainly on neuropsychological testing, showing a cognitive decline more severe for age and less education. The primary differentiation between MCI and dementia of the Alzheimer type resides in the lack of functional impairment in MCI.

COURSE AND PROGNOSIS

The typical rate at which MCI patients progress to dementia of the Alzheimer type is 10 to 15 percent per year and is associated with progressive loss of function. However, several studies have indicated that the diagnosis is not stable in both directions; patients can either convert to dementia of the Alzheimer type or revert to normal. This variability in course relates to the heterogeneous source of the subjects (clinical vs. community) as well as to the heterogeneous definition criteria used by different studies. Amnestic MCI occurs with increased morbidity compared with reference subjects.

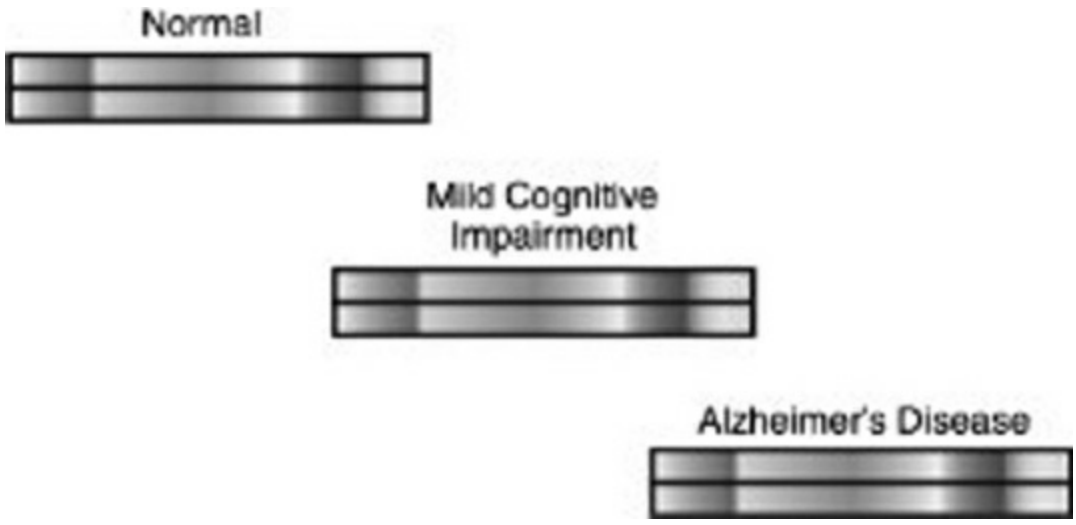


FIGURE 3-12
Cognitive continuum showing the overlap in the boundary between normal aging and the mild cognitive impairment and Alzheimer disease. (Reprinted with permission from Petersen RC, ed. *Mild Cognitive Impairment: Aging to Alzheimer's Disease*. New York: Oxford University Press, 2003.)

TREATMENT

There are no FDA-approved treatments for MCI at this time. MCI treatment involves adequate screening and diagnosis. Ideally, MCI treatment would also include improvement of memory loss, together with the prevention of further cognitive decline to dementia. Cognitive training programs may be mildly beneficial for compensating memory difficulties in MCI. Most likely, controlling vascular risk factors (high blood pressure, hypercholesterolemia, diabetes mellitus) helps to prevent the progression of MCI cases with underlying vascular pathology. Currently, sensitive tools (imaging techniques or biomarkers) are not available for MCI screening in the general population.

In the primary care setting, clinicians should maintain a high suspicion for subjective cognitive complaints and should corroborate these complaints with collateral information whenever possible. Also, identifying reversible causes of cognitive impairment (hypothyroidism, vitamin B_{12} deficiency, medication-induced cognitive impairment, depression) can further benefit some of the prodromal dementia MCI cases.

Currently, there is no evidence for the long-term efficacy of pharmacotherapies in reversing MCI. Several epidemiologic studies indicated a reduced risk of dementia in persons taking antihypertensive medications,

cholesterol-lowering drugs, antioxidants, and anti-inflammatory and estrogen therapy, but no randomized controlled trials verify these data. Concerning cognitive enhancers, most trials have ambiguous results (Table 3-41). Most of these studies had several problems, including (1) obtaining homogeneous samples and identifying potential beneficiaries of treatment; (2) treating a broader population, which led to large percentages of negative responses and problematic side effects; and (3) translation of the MCI construct into multiple cultures and languages and using dementia of the Alzheimer type diagnosis as the primary outcome, given the variability of this diagnosis in different countries.

Advances in MCI detection will be paramount for early detection and treatment of patients with Alzheimer disease; experts agree that disease-modifying treatments for Alzheimer disease will focus on cognitively intact individuals at increased risk. The field of identifying sensitive and specific biomarkers (biologic and neuroimaging markers) will probably witness exponential development in the coming years.

EPIDEMIOLOGY AND ETIOLOGY OF MCI

The recognition that Alzheimer disease pathology may exist in the brain long before the presence of clinical symptoms led to the focus on preclinical stages to characterize initial impairments that are associated with an increased risk of progression to Alzheimer disease.

The clinical expression of MCI is a result of the interaction among several risk factors and several protective factors. The most significant risk factors are related to the different types of neurodegeneration witnessed in dementias. There are different subtypes of MCI, most notably ones associated with amnesia. Other risk factors include the APOE4 allele status and cerebrovascular events in the form of either cerebrovascular accident or lacunar disease. The role of chronic exposure to high levels of cortisol, as seen in late-life depression, is also hypothesized to increase the risk for cognitive impairment through hippocampal volume reduction. The notion of "brain reserve" suggests that effects of brain size and neuron density may be protective against dementia despite the presence of neurodegeneration (a more significant number of neurons and synaptic connections and a bigger brain volume would protect against clinical manifestations of Alzheimer disease, despite the presence of neurodegeneration) (Fig. 3-13).

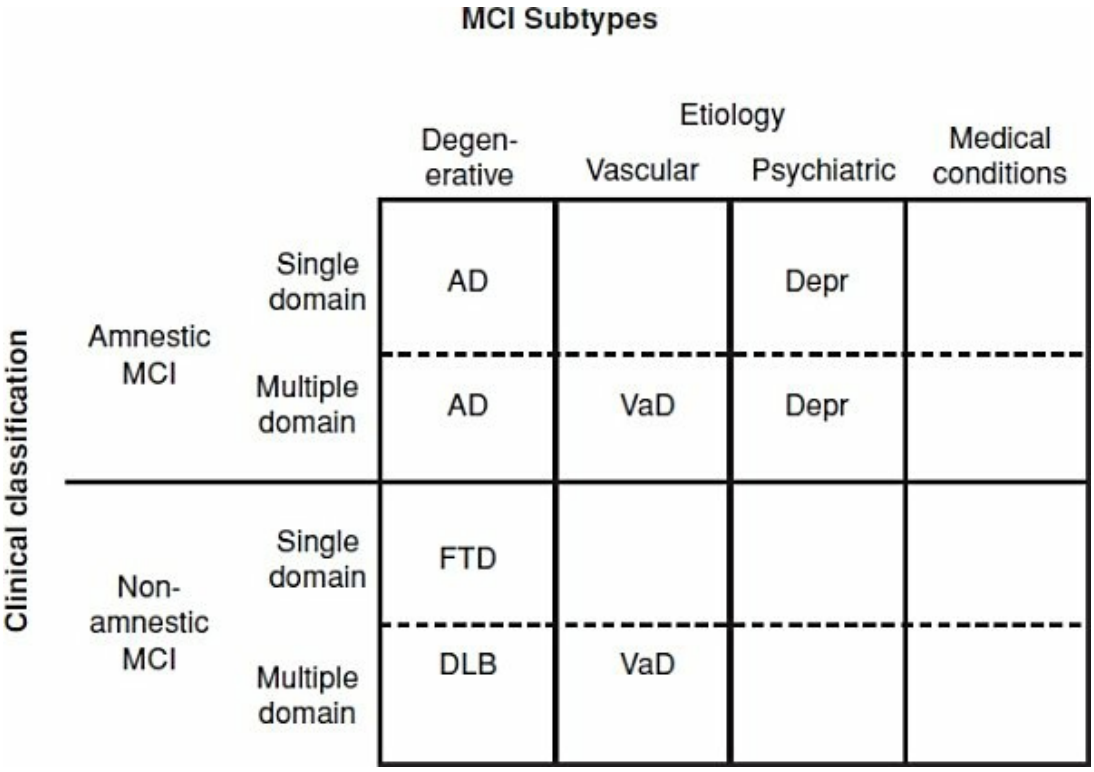


FIGURE 3-13
Outcome of clinical phenotypes of mild cognitive impairment (MCI) according to presumed etiology. AD, Alzheimer disease; Depr, depression; DLB, dementia with Lewy bodies; FRD, frontotemporal dementia; VaD, vascular dementia. (Adapted with permission from Petersen RC, ed. *Mild Cognitive Impairment: Aging to Alzheimer's Disease*. New York: Oxford University Press; 2003.)

Table 3-41
Treatment Trials for MCI

Study	N	Duration	Primary Outcome	Results	OBS	Sponsor
Donepezil + vitamin E (Thall et al., 1999)	769	3 yr	Conversion to AD	Partially positive (reduced risk of developing AD in the active arm group for the first 12 mo)	Amnestic MCI status and the presence of APOE4 allele–predictive of rate of progression to AD	ADCS
Donepezil (Salloway et al., 2004)	269	24 wk	ADAS-Cog total score; NYU PTIR	Negative	Positive results in secondary outcome measures (ADAS-Cog13)	Pfizer (The Donepezil "401" Study Group)
Donepezil (Doody et al., 2009)	821	48 wk	ADAS-Cog cognitive subscale; CDR-SB	Minimally positive with small decrease in ADAS-Cog score in donepezil group No significant change in CDR-SB	On secondary measures only global assessment differed significantly in favor of donepezil	Eisai and Pfizer
Rivastigmine (Feldman et al., 2007)	1,018	48 mo	Conversion to AD	Negative		Novartis
Galantamine (Winbald et al., 2008) Two studies	2,048		Progression of CDR score (from .5 to 1) Incident dementia	Negative	Attention assessed by DSST favored galantamine in both studies	Janssen-Cilag and Johnson & Johnson
Piribedil (Nagaraja et al., 2001)	60	12 wk	MMSE score	Favored piribedil	Mean MMSE change from baseline also favored intervention group	NIMH
Rofecoxib (Thall et al., 2005)	1,457	3–4 yr	Conversion to AD	Negative	Primary outcome favored placebo while secondary outcomes (ADS-cog, CDR) did not differentiate between rofecoxib and placebo	Merck
Nicotine patch (Newhouse et al., 2012)	74	6 mo	Attention on CCPT and global functioning on CGIC	Nicotine improved attention but not global functioning	Secondary outcome measures showed improvements in attention, memory, and psychomotor speed	
Piracetam	675	12 mo	Composite score extracted from eight tests	Negative		UCB Pharma
Ginkgo biloba (DeKosky et al., 2008) (Snitz et al., 2009)	3,069	6.1 yr	Incident dementia	Negative	No effect noted in normal cognitive either	NCCAM and NIH
		6.1 yr	Rates of change in MMSE, ADAS-Cog, and other cognitive domains	Negative		
B Vitamins (B_{12}, B_6, folic acid) (van Uffelen et al., 2008)	152	1 yr	Cognitive function at 6 and 12 mo	Negative	Moderate intensity walking improved memory in men and memory and attention in women	TNO-VU University Medical Center

MCI, mild cognitive impairment; AD, Alzheimer disease; ADCS, Alzheimer Disease Cooperative Study; CDR, Clinical Dementia Rating; NYU PTIR, New York University Paragraph Test Immediate Recall; DSST, Digit Symbol Substitution test; CCPT, Connors Continuous Performance Test; NCCAM, National Center for Complimentary Medicine.

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4 Substance Use and Addictive Disorders

4.1 General Features of the Substance-Related Disorders

This chapter covers substance use disorders with descriptions of the clinical phenomena associated with the following: alcohol; caffeine; cannabis; hallucinogens (including phencyclidine [PCP]); inhalants; opioids; sedatives, hypnotics, and anxiolytics; stimulants (including cocaine); tobacco; anabolic-androgenic steroids (AAS); and other substances, such as nitrous oxide and γ-hydroxybutyrate. We also discuss gambling disorder, as DSM-5 reclassified this as a non–substance-related addictive disorder in the most recent (fifth) edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5).

The substance use disorders share many features; however, the range of substances is broad in that there are differences in the pharmacology, intoxication, and associated behaviors that make for unique effects with many of the substances. For that reason, we begin with a general description of the substance use disorders in this section and discuss the features of each specific disorder in the following subsections of the chapter.

THE CLINICAL PRESENTATION

Behavioral, physical, and psychological dependence (see terminology below) are the hallmark of substance use disorders. While the direct effects of a

substance on behavioral, physical, and psychological function will vary depending on the particular substance used, the overall impact of the substance in causing impaired functioning has a similar pattern, regardless of substance. The symptoms typically fall within one of four categories, including pharmacologic symptoms, symptoms of impaired use, impairment in social domains, and use in risky or hazardous situations.

Terminology

Various terms have been used over the years to refer to drug abuse. For example, the term *dependence* has been and is used in one of two ways when discussing substance use disorders. In *behavioral dependence*, the emphasis is on substance-seeking activities and related evidence of pathologic use patterns, whereas *physical dependence* refers to the physical (physiologic) effects of multiple episodes of substance use. Psychological dependence, also referred to as habituation, is characterized by a continuous or intermittent craving (i.e., intense desire) for the substance to avoid a dysphoric state.

Somewhat related to dependence are the words addiction and addict. The word *addict* has acquired a pejorative connotation that ignores the concept of substance abuse as a medical disorder. The internet and other popular mediums have trivialized *Addiction*, as in the terms *TV addiction* and *chocolate addiction (or more commonly, "chocoholic");* however, the term still has value. There are common neurochemical and neuroanatomical substrates found among all addictions, whether it is to substances or gambling, sex, stealing, or eating. These various addictions may have similar effects on the activities of specific reward areas of the brain.

DIAGNOSIS

There are three major diagnostic categories in the DSM-5: (1) Substance Use Disorder; (2) Substance Intoxication; (3) Substance Withdrawal. Also, DSM includes a category of Substance-Induced Mental Disorder, to account for presentations of a psychiatric disorder that are likely due to a substance. This possibility is among the standard rule out criteria for most DSM-5 and ICD-10 mental disorders.

For both DSM-5 and ICD-10, the diagnoses are generic and do not differ among substances. Thus, we believe it is preferable to understand the overall

diagnostic criteria, with the understanding that we can then apply them to each of the substances discussed in later sections. In DSM-5, the disorders are substance-specific. Therefore, when diagnosing a disorder, the clinician should indicate the specific substance or drug used or that caused intoxication or withdrawal.

Substance Use Disorder

Substance use disorder is the diagnostic term for the prolonged use and abuse of a substance. When used, the specific substance is identified, for example, “alcohol use disorder” or “opioid use disorder.”

Table 4-1 lists the diagnostic approaches to substance use disorder in DSM-5 and ICD-10.

Substance Intoxication

Substance intoxication is the diagnosis used to describe a syndrome characterized by specific signs and symptoms resulting from recent ingestion or exposure to the substance. As with substance use disorder, in practice, the specific substance is identified, for example, “alcohol intoxication” or “opioid intoxication.” Table 4-2 lists the diagnostic approaches to substance intoxication in DSM-5 and ICD-10.

Substance Withdrawal

Substance withdrawal is the diagnosis used to describe a substance-specific syndrome that results from the abrupt cessation of heavy and prolonged use of a substance. As with substance use disorder, in practice, the specific substance is identified, for example, “alcohol withdrawal” or “opioid withdrawal.” Table 4-3 lists the diagnostic approaches to substance withdrawal in DSM-5 and ICD-10.



Table 4-1
Substance Use Disorder

	Substance Use Disorder	
Disorder	**DSM-5**	**ICD-10**

Diagnostic name	Alcohol use disorder Cannabis use disorder Hallucinogen use disorder Inhalant use disorder Opioid use disorder Sedative/hypnotic/anxiolytic use disorder Stimulant use disorder Tobacco use disorder	Mental and behavioral disorders due to • Use of alcohol • Use of opioids • Use of cannabinoids • Use of sedatives or hypnotics • Use of cocaine • Use of other stimulants, including caffeine • Use of hallucinogens • Use of tobacco • Use of volatile solvents Indicate if includes a dependence syndrome
Duration	Symptoms occur within 12 mo	
Symptoms	*Physiologic Symptoms:* 1. Tolerance 2. Withdrawal (does not apply to inhalants and hallucinogens) *Symptoms typical of addiction or obsessive use* 3. Craving to use	Tolerance and withdrawal states Adverse health effects of use: physiologic, cognitive or behavioral Craving or difficulty controlling

	4. Using more than intended 5. Difficulty stopping or reducing use 6. Spending significant time devoted to the substance (using, obtaining, recovering) 7. Use despite acknowledging health problems (physical, mental) because of use *Psychosocial sequelae of use* 8. Using despite social, occupational or other adverse consequences 9. Neglecting other responsibilities because of use 10. Neglecting other activities because of use 11. Risky or dangerous behaviors or situations because of use	substance use Use despite acknowledging adverse consequences of use Prioritizing the substance over other psychosocial obligations
Required number of symptoms	Two or more of the above	
Psychosocial consequences of symptoms	Marked impairment and/or distress	
Exclusions		***Psychotic state*** ***Nonalcoholic Korsakoff psychosis or syndrome***

Symptom specifiers	***In a controlled environment:*** (i.e., access to the substance is restricted) ***On maintenance therapy:*** (tobacco or opioids) ***For hallucinogens or inhalants, specify the specific substance***	***Associated specifiers:*** **Psychotic disorder** (following substance use) **Amnesic syndrome** (chronic memory loss because of use)
Course specifiers	**In early remission:** no symptoms for 3–12 mo (craving may still be present) **In sustained remission:** no symptoms for >12 mo (craving may still be present)	
Severity specifiers	Severity is measured by the number of symptoms present. See DSM-5 for ranges.	
Comments	Applies to the following substances: • Alcohol • Cannabis • Opioids • Sedatives, hypnotics, anxiolytics • Stimulants • Hallucinogens	Note: ICD additionally defines “harmful use” as pattern of use causing harm to health, both physical and/or mental. This classification is independent of

	• Inhalants • Tobacco	presence of a dependence syndrome.



Table 4-2
Substance Intoxication

	Substance Intoxication	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Alcohol intoxication Caffeine intoxication Cannabis intoxication Hallucinogen intoxication Inhalant intoxication Opioid intoxication Sedative/hypnotic/anxiolytic intoxication Stimulant intoxication	Mental and behavioral disorders due to • Use of alcohol • Use of opioids • Use of cannabinoids • Use of sedatives or hypnotics • Use of cocaine • Use of other stimulants, including caffeine • Use of hallucinogens • Use of tobacco • Use of volatile solvents Indicate if acute intoxication
Duration	Not specified	Not specified

Symptoms	Recent use of substance Mood or behavioral problems resulting from that use One or more substance-specific symptoms of intoxication (see individual substances for descriptions of the intoxication state)	Some mental change resulting from substance use: cognitive, psychological or behavioral Symptoms due to the substance resolve when stopped
Required number of symptoms	Specific to substance: Stimulants: ≥2 Caffeine: ≥5 or more symptoms Sedative/hypnotic/anxiolytics: ≥1 Opioids: ≥1 Inhalants: ≥2 Hallucinogens: ≥2 plus change in perception or alertness Cannabis: ≥2 Alcohol: ≥1	
Exclusions	Medical condition Other mental disorder Intoxication with another substance	Intoxication due to poisoning
Symptom specifiers	For cannabis, opioids, or stimulants: *with perceptual disturbance:* hallucinations but no loss or reality testing and not delirious	
Comments	Applies to the following substances: • Alcohol	

- Cannabis
- Opioids
- Sedatives, hypnotics, anxiolytics
- Stimulants
- Hallucinogens
- Inhalants
- Caffeine

COMORBIDITY

Comorbidity is the occurrence of two or more psychiatric disorders in a single patient at the same time. People seeking treatment for alcohol, cocaine, or opioid dependence have a high prevalence of additional psychiatric disorders; some studies have shown that up to 50 percent of addicts have a comorbid psychiatric disorder. Although opioid, cocaine, and alcohol abusers with current psychiatric problems are more likely to seek treatment, those who do not seek treatment are not necessarily free of comorbid psychiatric problems; such persons may have social supports that enable them to deny the impact that drug use is having on their lives. Two extensive epidemiologic studies have shown that even among representative samples of the population, those who meet the criteria for alcohol or drug abuse and dependence (excluding tobacco dependence) are also far more likely to meet the criteria for other psychiatric disorders.

In various studies, a range of 35 to 60 percent of patients with substance abuse or substance dependence also meet the diagnostic criteria for antisocial personality disorder. The range is even higher when investigators include persons who meet all the antisocial personality disorder diagnostic criteria, except the requirement that the symptoms started at an early age. That is, a high percentage of patients with substance abuse or substance dependence diagnoses have a pattern of antisocial behavior, whether it was present before the substance use started or developed during substance use. Patients with substance abuse or substance dependence diagnoses who have antisocial personality disorder are likely to use more illegal substances; to have more psychopathology; to be less satisfied with their lives; and to be more

impulsive, isolated, and depressed than patients with antisocial personality disorders alone.



Table 4-3
Substance Withdrawal

	Withdrawal	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Alcohol withdrawal Caffeine withdrawal Cannabis withdrawal Opioid withdrawal Sedative/hypnotic/anxiolytic withdrawal Stimulant withdrawal Tobacco withdrawal	Mental and behavioral disorders du to • Use of alcohol • Use of opioids • Use of cannabino • Use of sedatives hypnotics • Use of cocaine • Use of oth stimulants including caffeine • Use of hallucinog • Use of tobacco • Use of volatile solvents

		Indicate withdrawal state
Duration	Onset is specific to substance: sedatives/hypnotics/anxiolytics, stimulants: hours to days alcohol and cannabis: none listed opioids: minutes to days caffeine, tobacco: within 24 hr	
Symptoms	No or reduced use after prolonged or heavy use Substance-specific symptoms (see individual substances for typical symptoms of withdrawal)	• Group of symptoms occurring with cessation reduction substance following period of prolonged and/or regular us • Symptoms are time limited and substance specific
Required number of symptoms	Specific to substance: sedatives/hypnotics/anxiolytics: ≥2 alcohol: ≥2 stimulants: ≥2 plus dysphoric mood opioids: ≥3 caffeine: ≥3	

	cannabis: ≥3 tobacco: ≥4	
Psychosocial consequences of symptoms	Distress and/or impairment in functioning	
Exclusions	Medical condition Other mental disorder Intoxication with or withdrawal from another substance	
Symptom specifiers	For alcohol and sedative/hypnotic/anxiolytic withdrawal: *with perceptual disturbance:* hallucinations but no loss or reality testing and not delirious	Withdrawal sta with delirium
Comments	Applies to the following substances: • Alcohol • Cannabis • Opioids • Sedatives, hypnotics, anxiolytics • Stimulants • Tobacco • Caffeine	

Depression and Suicide

Depressive symptoms are common among persons diagnosed with substance abuse or substance dependence. About one-third to one-half of all those with opioid abuse or opioid dependence and about 40 percent of those with alcohol abuse or alcohol dependence meet the criteria for major depressive disorder sometime during their lives. Substance use is also a major precipitating factor for suicide. Persons who abuse substances are about 20

times more likely to die by suicide than the general population. About 15 percent of persons with alcohol abuse or alcohol dependence commit suicide. This frequency of suicide is second only to the frequency in patients with major depressive disorder.

TREATMENT APPROACH

Some persons who develop substance-related problems recover without formal treatment, especially as they age. For those patients with less severe disorders, such as nicotine addiction, relatively brief interventions are often as effective as more intensive treatments. Because these brief interventions do not change the environment, alter drug-induced brain changes, or provide new skills, a change in the patient's motivation (cognitive change) probably has the best impact on the drug-using behavior. For those individuals who do not respond or whose dependence is more severe, a variety of interventions described below appear to be effective.

It is useful to distinguish among specific procedures or techniques (e.g., individual therapy, family therapy, group therapy, relapse prevention, and pharmacotherapy) and treatment programs. Most programs use several specific procedures and involve a variety of professional disciplines as well as nonprofessionals who have special skills or personal experience with the specific substance problem. The best treatment programs combine specific procedures and disciplines with meeting the needs of the individual patient after a careful assessment.

There is no generally accepted classification system for either the specific procedures used in treatment or programs using various combinations of procedures. This lack of standardized terminology for categorizing procedures and programs presents a problem, even when we narrow the field of interest to treatment for a single substance, such as alcohol, tobacco, or cocaine. Except in carefully monitored research projects, even the definitions of specific procedures (e.g., individual counseling, group therapy, and methadone maintenance) tend to be so imprecise that usually just what transactions occur cannot be inferred. Nevertheless, for descriptive purposes, programs are often broadly grouped based on one or more of their salient characteristics: whether the program aims to merely control acute withdrawal and consequences of recent drug use (detoxification) or focuses on longer-

term behavioral change (rehabilitation); whether the program makes extensive use of pharmacologic interventions; and whether the program mainly uses individual therapy, Alcoholics Anonymous (AA) or other 12-step principles, or therapeutic community principles. For example, the National Institute on Drug Abuse suggested the following categories of treatment: long-term residential treatment, short-term residential treatment, outpatient treatment programs, individualized drug counseling, group counseling, and treatments for individuals involved in the criminal justice system.

Selecting a Treatment

Not all interventions apply to all types of substance use disorders, and some of the more coercive interventions used for illicit drugs do not apply to legal substances, such as tobacco. Addictive behaviors do not change abruptly but through a series of stages. Experts propose five stages in this gradual process: precontemplation, contemplation, preparation, action, and maintenance. For some types of addictions, the therapeutic alliance improves when the treatment approach is tailored to the patient's stage of readiness to change. Interventions for some drug use disorders may have a specific pharmacologic agent as a critical component; for example, disulfiram, naltrexone, or acamprosate for alcoholism; methadone, levomethadyl acetate, or buprenorphine for heroin addiction; and nicotine delivery devices, varenicline, or bupropion for tobacco dependence.

In general, brief interventions (e.g., a few weeks of detoxification, whether in or out of a hospital) used for persons who are severely dependent on illicit opioids have limited effect on outcome measured a few months later. Substantial reductions in illicit drug use, antisocial behaviors, and psychiatric distress among patients dependent on cocaine or heroin are much more likely following treatment lasting at least 3 months. Such a time-in-treatment effect occurs across very different modalities, from residential therapeutic communities to ambulatory methadone maintenance programs. Although some patients appear to benefit from a few days or weeks of treatment, a substantial percentage of users of illicit drugs drop out (or are dropped) from treatment before they have achieved significant benefits.

Treatment outcomes vary widely; this is partly due to the great variety of

patients that enter treatment as well as their circumstances. Programs based on similar philosophical principles and using what seem to be similar therapeutic procedures vary greatly in effectiveness, however. Some of the differences among programs that seem to be similar reflect the range and intensity of services offered. Programs with a professionally trained staff that provide more comprehensive services to patients with more severe psychiatric difficulties are more likely able to retain those patients in treatment and help them make positive changes. Differences in the skills of individual counselors and professionals can strongly affect outcomes.

Such generalizations concerning programs serving illicit drug users may not hold for programs dealing with those seeking treatment for alcohol, tobacco, or even cannabis problems uncomplicated by heavy use of illicit drugs. In such cases, relatively brief periods of individual or group counseling can produce long-lasting reductions in substance use. The outcomes usually considered in programs dealing with illicit drugs have typically included measures of social functioning, employment, and criminal activity, as well as decreased drug-using behavior.

Treatment of Comorbidity

Treatment of the severely mentally ill (primarily those with schizophrenia and schizoaffective disorders) who are also drug-dependent continues to pose problems for clinicians. Although some specialized facilities use both antipsychotic drugs and therapeutic community principles, for the most part, specialized addiction agencies have difficulty treating these patients. Generally, integrated treatment in which the same staff can treat both the psychiatric disorder and the addiction is more effective than either parallel treatment (mental health and a specialty addiction program providing care concurrently) or sequential treatment (treating either the addiction or the psychiatric disorder first and then dealing with the comorbid condition).

Services and Outcome

The extension of managed care into the public sector has produced a significant reduction in the use of hospital-based detoxification and the virtual disappearance of residential rehabilitation programs for alcoholics. Managed-care organizations, however, tend to assume that the relatively brief

courses of outpatient counseling that are effective with private-sector alcoholic patients are also useful with patients who are dependent on illicit drugs and who have minimal social supports. For the present, the trend is to provide the care that costs the least over the short term and to ignore studies showing that more services can produce better long-term outcomes.

Treatment is often a worthwhile social expenditure. For example, the treatment of antisocial illicit drug users in outpatient settings can decrease antisocial behavior and reduce rates of human immunodeficiency virus (HIV) seroconversion that more than offset the treatment cost. Treatment in a prison setting can decrease postrelease costs associated with drug use and rearrests. Despite such evidence, problems exist in maintaining public support for the treatment of substance dependence in both the public and private sectors. This lack of support suggests that these problems continue to be viewed, at least in part, as moral failings rather than as medical disorders.

The Epidemiology of Substance Use Disorders

The National Institute of Drug Abuse (NIDA) and other agencies, such as the National Survey of Drug Use and Health (NSDUH), conduct periodic surveys of the use of illicit drugs in the United States. As of 2017, more than 19 million persons older than the age of 12 years (about 7 percent of the total US population) had a substance-related disorder in the past year.

In general, more men than women abuse substances. Those who use substances earlier are more likely to develop a disorder. Among ethnic and racial groups in the United States, the highest lifetime rate is among American Indian or Alaska Natives; whites are more affected than blacks or African Americans. There is a sociodemographic effect as well. For example, those with some college education use more substances than those with less education, and the unemployed have higher rates than those with either part-time or full-time employment.

We will discuss the specific epidemiology of various substances in the appropriate section. Also, a comprehensive survey of drug use and trends in the United States is available at www.samhsa.gov.

ETIOLOGY

Substance use disorders are complicated psychiatric conditions, and, like

other psychiatric disorders, both biologic factors and environmental circumstances are etiologically significant.

The model of substance use disorders is the result of a process in which multiple interacting factors influence drug-using behavior and the loss of judgment for decisions about using a given drug. Although the actions of a given drug are critical in the process, we cannot assume that all drug-dependent people experience the effects of a particular drug similarly, or have the same set of motivating factors. Furthermore, different factors may be more or less important at different stages of the process. Thus, drug availability, social acceptability, and peer pressures may be the significant determinants of initial experimentation with a drug. However, other factors, such as personality and individual biology, probably are more important in how the effects of a given drug are perceived and the degree to which repeated drug use produces changes in the central nervous system (CNS). Still, other factors, including the particular actions of the drug, may represent primary determinants of whether drug use progresses to drug dependence. In contrast, others may still be valuable influences on the likelihood that drug use (1) leads to adverse effects or (2) to successful recovery from dependence.

We often hear the assertion that addiction is a “brain disease,” meaning that the critical processes that transform voluntary drug-using behavior to compulsive drug use are changes in the structure and neurochemistry of the brain of the drug user. Sufficient evidence now indicates that such changes in relevant parts of the brain do occur. The perplexing and unanswered question is whether these changes are both necessary and sufficient to account for the drug-using behavior. Many argue that they are not, that the capacity of drug-dependent individuals to modify their drug-using behavior in response to positive reinforcers or aversive contingencies indicates that the nature of addiction is more complex and requires the interaction of multiple factors.

The central element of drug dependence is the drug-using behavior itself. The decision to use a drug is influenced by immediate social and psychological situations as well as by the person’s more remote history. The use of the drug initiates a sequence of consequences that can be rewarding or aversive and which, through a process of learning, can result in a greater or lesser likelihood that the drug-using behavior will repeat. For some drugs, use also initiates the biologic processes associated with tolerance, physical

dependence, and sensitization. In turn, tolerance can reduce some of the adverse effects of the drug, permitting or requiring the use of larger doses, which then can accelerate or intensify the development of physical dependence. Above a certain threshold, the aversive qualities of a withdrawal syndrome provide a distinct recurrent motive for further drug use. Sensitization of motivational systems can increase the salience of drug-related stimuli.

The Neurobiology of Substance Use Disorders

Genetic Factors. Strong evidence from studies of twins, adoptees, and siblings brought up separately indicates that the cause of alcohol abuse has a genetic component. Many less conclusive data show that other types of substance abuse or substance dependence have a genetic pattern in their development. Researchers have used restriction fragment length polymorphism (RFLP) in the study of substance use disorders, and preliminary results show associations with genes that affect dopamine production.

Neurochemical Factors

RECEPTORS AND RECEPTOR SYSTEMS. Except for alcohol, researchers have identified particular neurotransmitters or neurotransmitter receptors involved with most substances of abuse. Some researchers base their studies on such hypotheses. The opioids, for example, act on opioid receptors. A person with too little endogenous opioid activity (e.g., low concentrations of endorphins) or with too much activity of an endogenous opioid antagonist may be at risk for developing an opioid use disorder. Even in a person with completely normal endogenous receptor function and neurotransmitter concentration, the long-term use of a particular substance of abuse may eventually modulate receptor systems in the brain so that the presence of the exogenous substance is needed to maintain homeostasis. Such a receptor-level process may be the mechanism for developing tolerance within the CNS. Demonstrating modulation of neurotransmitter release and neurotransmitter receptor function has proved difficult, however, and recent research focuses on the effects of substances on the second-messenger system and gene regulation.

Pathways and Neurotransmitters. The primary neurotransmitters possibly involved in developing substance use disorders are the opioid, catecholamine (mainly dopamine), and γ-aminobutyric acid (GABA) systems. The dopaminergic neurons in the ventral tegmental area (VTA) are particularly important. These neurons project to the cortical and limbic regions, especially the nucleus accumbens. This pathway is probably involved in the sensation of reward and may represent the primary mediator of the effects of such substances as amphetamine and cocaine. The locus ceruleus, the largest group of adrenergic neurons, probably mediates the effects of the opiates and the opioids. These pathways have collectively been called the *brain-reward circuitry.*

The Psychology of Substance Use Disorders

Psychodynamic Factors. The range of psychodynamic theories about substance abuse reflects the various popular theories during the last 100 years. According to classic theories, substance abuse is a masturbatory equivalent (some heroin users describe the initial "rush" as similar to a prolonged sexual orgasm), a defense against anxious impulses, or a manifestation of oral regression (i.e., dependency). New psychodynamic formulations relate substance use as a reflection of disturbed ego functions (i.e., the inability to deal with reality). As a form of self-medication, alcohol may control panic, opioids may diminish anger, and amphetamines may alleviate depression. Some addicts have great difficulty recognizing their inner emotional states, a condition called *alexithymia* (i.e., being unable to find words to describe their feelings).

Learning and Conditioning. Drug use, whether occasional or compulsive, can be viewed as behavior maintained by its consequences. Drugs can reinforce antecedent behaviors by terminating some noxious or aversive state such as pain, anxiety, or depression. In some social situations, drug use, apart from its pharmacologic effects, can be reinforced if it results in special status or the approval of friends. Each use of the drug evokes rapid positive reinforcement, either as a result of the rush (the drug-induced euphoria), alleviation of disturbed affects, alleviation of withdrawal symptoms, or any combination of these effects. Also, some drugs may

sensitize neural systems to the reinforcing effects of the drug. Eventually, the paraphernalia (needles, bottles, cigarette packs) and behaviors associated with substance use can become secondary reinforcers, as well as cues signaling the availability of the substance, and in their presence, craving or a desire to experience the effects increases.

Drug users respond to the drug-related stimuli with increased activity in limbic regions, including the amygdala and the anterior cingulate. A variety of drugs can have this effect, including cocaine, opioids, and cigarettes (nicotine). Of interest, cocaine-related stimuli can activate the same regions that are affected by sexual stimuli in both healthy controls and cocaine users.

In addition to the operant reinforcement of drug-using and drug-seeking behaviors, other learning mechanisms probably play a role in dependence and relapse. Opioid and alcohol withdrawal phenomena can be conditioned (in the Pavlovian or classical sense) to environmental or interoceptive stimuli. For a long time after withdrawal (from opioids, nicotine, or alcohol), the addict exposed to environmental stimuli previously linked with substance use or withdrawal may experience conditioned withdrawal, conditioned craving, or both. The feeling of craving may be independent of feelings of withdrawal. Conditions associated with the availability or use of a substance, such as watching someone else use heroin or light a cigarette or being offered some drug by a friend, bring about the most intense craving. Preexisting pathologies may superimpose on those learning and conditioning phenomena; however, the development of powerfully reinforced substance-seeking behavior does not require preexisting difficulties.

4.2 Alcohol-Related Disorders

Alcohol is a potent drug that causes both acute and chronic changes in almost all neurochemical systems. Thus alcohol abuse can produce severe, temporary psychological symptoms including depression, anxiety, and psychoses. Increasing levels of regular alcohol consumption can cause tolerance. Chronic use can cause such intense adaptation of the body that stopping drinking can precipitate withdrawal syndromes, including insomnia, autonomic nervous system hyperactivity, and anxiety. Therefore, in an

adequate evaluation of life problems and psychiatric symptoms in a patient, the clinician must consider the possibility that the clinical situation reflects the effects of alcohol.

DIAGNOSIS AND CLINICAL FEATURES

The DSM-5 and ICD-10 include several diagnoses related to alcohol, they generally follow the template for all substance use disorders (see Section 4.1 and Tables 4-1 through 4-3). A need for daily use of large amounts of alcohol for adequate functioning, a regular pattern of heavy drinking limited to weekends, and long periods of sobriety interspersed with binges of heavy alcohol intake lasting for weeks or months strongly suggest an alcohol use disorder. The drinking patterns are often associated with certain behaviors: the inability to cut down or stop drinking; repeated efforts to control or reduce excessive drinking by "going on the wagon" (periods of temporary abstinence) or by restricting drinking to certain times of the day; binges (e.g., remaining intoxicated throughout the day for at least 2 days); occasional consumption of large amounts of alcohol in a sitting (e.g., a fifth of spirits or the equivalent in wine or beer); amnestic periods for events occurring while intoxicated (blackouts); the continuation of drinking despite a severe physical disorder that the person knows the alcohol is exacerbating; and drinking nonbeverage alcohol, such as fuel and commercial products containing alcohol. Also, persons with alcohol use disorders show impaired social or occupational functioning because of alcohol use (e.g., violence while intoxicated, absence from work, job loss), legal difficulties (e.g., arrest for intoxicated behavior and traffic accidents while intoxicated), and arguments or difficulties with family members or friends about excessive alcohol consumption.

Mark, a 45-year-old divorced man, was examined in a hospital emergency room because he had been confused and unable to care for himself for the preceding 3 days. His brother, who brought him to the hospital, reported that the patient had consumed large quantities of beer and wine daily for more than 5 years. His home life and job were reasonably stable until his divorce 5 years prior. The brother indicated

that Mark's drinking pattern since the divorce had been approximately five beers and a fourth of wine a day. Mark often experienced blackouts from drinking and missed days of work frequently. As a result, Mark has lost several jobs in the past 5 years. Although he usually provides for himself marginally with small jobs, 3 days earlier, he ran out of money and alcohol and resorted to panhandling on the streets for cash to buy food. Mark had been poorly nourished, having one meal per day at best, and was relying on beer as his prime source of nourishment.

On examination, Mark alternates between apprehension and chatty, superficial warmth. He is keyed up and continually talks in a rambling and unfocused manner. His recognition of the physician varies; at times, he recognizes him, and other times he becomes confused and believes the doctor to be his other brother who lives in another state. On two occasions, he referred to the physician by said brother's name and asked when he arrived in town, evidently having lost track of the interview up to that point. He has a gross hand tremor at rest and is confused about the time. He believes he is in a parking lot rather than a hospital. Efforts at memory and calculation testing fail because Mark's attention shifts so rapidly.



Table 4-4
Levels of Impairment at Different Blood Alcohol Concentrations

Level (mg/dL)	Likely Impairment
20–30	Slowed motor performance and decreased thinking ability
30–80	Increases in motor and cognitive problems
80–200	Increases in incoordination and judgment errors Mood lability Deterioration in cognition
200–300	Nystagmus, marked slurring of speech, and alcoholic

	blackouts
>300	Impaired vital signs and possible death

Alcohol Intoxication

The DSM-5 diagnostic criteria for alcohol intoxication use the standard approach for all substance intoxications (see Section 4.1 and Table 4-3) and include evidence of recent ingestion of ethanol, maladaptive behavior, and at least one of several possible physiologic correlates of intoxication. As a conservative approach to identifying blood levels that are likely to have significant effects on driving abilities, the legal definition of intoxication in most states in the United States requires a blood concentration of 80- or 100-mg ethanol per deciliter of blood (mg/dL), which is the same as 0.08 to 0.10 g/dL. Table 4-4 lists the rough estimates of the levels of impairment likely to be seen at various blood alcohol concentrations for most people. Anyone who does not show significant levels of impairment in motor and mental performance at approximately 150 mg/dL probably has significant pharmacodynamic tolerance.

Alcohol Withdrawal

Alcohol withdrawal, even without delirium, can be severe; it can include seizures and autonomic hyperactivity. Conditions that may predispose to, or aggravate, withdrawal symptoms include fatigue, malnutrition, physical illness, and depression. The DSM-5 criteria for alcohol withdrawal require the cessation or reduction of alcohol use that was heavy and prolonged as well as the presence of specific physical or neuropsychiatric symptoms. The diagnosis also allows for the specification “with perceptual disturbances.”

The classic sign of alcohol withdrawal is tremulousness. However, the spectrum of symptoms can expand to include psychotic and perceptual symptoms (e.g., delusions and hallucinations), seizures, and the symptoms of delirium tremens (DTs), called alcohol delirium in DSM-5.

Table 4-5 outlines the progression of alcohol withdrawal from mild to life threatening symptoms. These are rough approximates, and there is great variation, for example, the syndrome of withdrawal sometimes skips the usual progression and, for example, goes directly to DTs.

Table 4-5
The Progression of Alcohol Withdrawal Symptoms

	Symptoms	Usual Time to Present
Mild	Tremulousness	6–8 hr
Moderate	Perceptual disturbances	8–12 hr
Severe	Seizures	12–24 hr
Life threatening	Delirium tremens	Within 72 hr

The tremor of alcohol withdrawal can be similar to either physiologic tremor, with a continuous tremor of high amplitude and of more than 8 Hz, or familial tremor, with bursts of tremor activity slower than 8 Hz. Other symptoms of withdrawal include general irritability and gastrointestinal (GI) symptoms (e.g., nausea and vomiting). Also standard are symptoms of autonomic hyperactivity, including anxiety, arousal, sweating, facial flushing, mydriasis, tachycardia, and mild hypertension. Patients experiencing alcohol withdrawal are generally alert but may startle easily.

Twenty-nine-year-old Mr. F had been a heavy drinker for 8 years. One evening after work, he started drinking with friends and drank throughout the evening. He fell asleep in the early morning hours and, upon awakening, had a strong desire to drink and decided not to attend work. He had several Bloody Mary's instead of food because the food did not appeal to him. He went to a local bar in the afternoon and consumed large quantities of beer. That evening he met with some friends and continued to drink.

This drinking pattern continued for the next week. At the beginning of the following week, Mr. F attempted to have a cup of coffee but found his hands too shaky to hold the cup. He eventually managed to pour himself some wine in a glass and drank as much as he could. His hands then became less shaky, but he now felt nauseous and began having dry heaves. He tried to drink repeatedly, but he could not keep the alcohol down. He felt very ill and anxious, so he contacted his physician, who

recommended he report to a hospital.

Upon evaluation, Mr. F was alert. He had a marked resting and intention tremor of the hands, and his tongue and eyelids were tremulous. He was oriented and had no memory impairment. When inquired about his drinking, Mr. F admitted to drinking several drinks each day for the past 8 years but claimed that his drinking never interfered with his work or his relations with colleagues or friends. He denied having any aftereffects from his drinking other than mild hangovers. He denied ever having a binge such as this before and denied ever needing to drink daily to function adequately. He admitted, however, that he has never tried to reduce or stop drinking.

Withdrawal Seizures. Seizures associated with alcohol withdrawal are stereotyped, generalized, and tonic–clonic in character. Patients often have more than one seizure 3 to 6 hours after the first seizure. Status epilepticus is relatively rare and occurs in less than 3 percent of patients. Seizure activity in patients with known alcohol abuse histories should still prompt clinicians to consider other causative factors, such as head injuries, CNS infections, CNS neoplasms, and other cerebrovascular diseases. Long-term severe alcohol abuse can result in hypoglycemia, hyponatremia, and hypomagnesemia—all of which can also be associated with seizures.

Delirium. We should always carefully monitor patients with recognized alcohol withdrawal symptoms to prevent progression to alcohol withdrawal delirium, the most severe form of the withdrawal syndrome, also known as DTs. Alcohol withdrawal delirium is a medical emergency that can result in significant morbidity and mortality. Patients with delirium are a danger to themselves and others. Because of the unpredictability of their behavior, patients with delirium may be assaultive or suicidal. They may act on hallucinations or delusional thoughts as if they were genuine dangers. Untreated, DTs have a mortality rate of 20 percent. This high rate of mortality is usually a result of an intercurrent medical illness such as pneumonia, renal disease, hepatic insufficiency, or heart failure. Although withdrawal seizures commonly precede the development of alcohol withdrawal delirium, delirium can also appear unheralded. The essential

feature of the syndrome is delirium occurring within 1 week after a person stops drinking or reduces the intake of alcohol. In addition to the symptoms of delirium, the features of alcohol intoxication delirium include a variety of symptoms, including autonomic symptoms, perceptual, and psychomotor. We list these in Table 4-6.



Table 4-6
Delirium Tremens Symptoms

Delirium
Confusion
Disorientation
Hallucinations (tactile common)
Delusions
Autonomic hyperactivity
Tachycardia
Diaphoresis
Fever
Hypertension
Anxiety
Insomnia
Fluctuating levels of psychomotor activity (ranging from lethargy to agitation)

About 5 percent of hospital patients with alcohol-related disorders have DTs. Because the syndrome usually develops on the third hospital day, a patient admitted for an unrelated condition may unexpectedly have an episode of delirium; this may be the first sign of a previously undiagnosed alcohol-related disorder. Episodes of DTs usually begin in a patient's 30s or 40s after 5 to 15 years of heavy drinking, typically of the binge type. Physical illness (e.g., hepatitis or pancreatitis) predisposes to the syndrome; a person in good physical health rarely has DTs during alcohol withdrawal.

R, a 40-year-old man, was admitted to the orthopedic department of a

general hospital after experiencing a fall down his stairs and breaking his leg. On the third day of his hospital stay, he became increasingly nervous and started to tremble. He was unable to sleep at night, talked incoherently, and was very anxious. Mr. R, when asked, denied an alcohol problem other than an occasional glass of wine.

When asked directly, his wife admitted that Mr. R drank large quantities of wine for over 4 years. During the previous year, his drinking would begin every evening when he came home from work and would not end until he fell asleep. On the evening of admission, the fall occurred before he was able to consume any alcohol.

During the few weeks before his admission, Mr. R had eaten very little. On several occasions, Mrs. R noticed that Mr. R was unable to recall even essential events from the previous day. He had a car accident 3 years prior but without significant injury. Mr. R had no other major health problems. His relationship with Mrs. R became problematic after he began drinking, and Mrs. R was seriously contemplating divorce. Mr. R had a tense relationship with his four children, and he often argued with them. Recently, the children have been trying to avoid Mr. R as much as possible.

On examination, Mr. R's speech was rambling and incoherent. He believed that he was still at work and that he had a job to finish. At times he thought the physicians and nurses were his coworkers. At times he picked at bugs that he could see on his bedsheets. He was disoriented in time and was startled easily by sounds from outside the room. He was sweating profusely and could not hold a glass without spilling some of the contents.

Alcohol-Induced Disorders

Alcohol-Induced Persisting Dementia. Alcohol-induced persisting dementia is a poorly studied, heterogeneous long-term cognitive problem that can develop in the course of alcoholism. Patients have decreased intellectual functioning, cognitive abilities, and memory. Recent memory difficulties are consistent with global cognitive impairment, an observation that helps to distinguish this from alcohol-induced persisting amnestic disorder. Brain

functioning tends to improve with abstinence, but perhaps half of all affected patients have long-term and even permanent disabilities in memory and thinking. Approximately 50 to 70 percent of these patients have increased brain ventricles and shrinkage of the cerebral sulci. However, these changes appear to be wholly or partially reversible after a year of abstinence.

Alcohol-Induced Persisting Amnestic Disorder. The essential feature of alcohol-induced persisting amnestic disorder is a disturbance in short-term memory caused by prolonged heavy use of alcohol. Because the disorder usually occurs in persons who have been drinking heavily for many years, it is rare in persons younger than age 35.

WERNICKE–KORSAKOFF SYNDROME. The classic names for alcohol-induced persisting amnestic disorder are Wernicke encephalopathy (a set of acute symptoms) and Korsakoff syndrome (a chronic condition). Whereas Wernicke encephalopathy is entirely reversible with treatment, only about 20 percent of patients with Korsakoff syndrome recover. The pathophysiologic connection between the two syndromes is thiamine deficiency, caused either by poor nutritional habits or by malabsorption problems. Thiamine is a cofactor for several critical enzymes and may also be involved in the conduction of the axon potential along the axon and synaptic transmission. The neuropathologic lesions are symmetrical and paraventricular, involving the mammillary bodies, the thalamus, the hypothalamus, the midbrain, pons, medulla, fornix, and cerebellum.

Wernicke encephalopathy, also called *alcoholic encephalopathy,* is an acute neurologic disorder characterized by ataxia (affecting the gait primarily), vestibular dysfunction, confusion, and a variety of ocular motility abnormalities, including horizontal nystagmus, lateral orbital palsy, and gaze palsy. These eye signs are usually bilateral but not necessarily symmetrical. Other eye signs may include a sluggish reaction to light and anisocoria (unequal pupillary size). Wernicke encephalopathy may clear spontaneously in a few days or weeks or may progress to Korsakoff syndrome.

Korsakoff syndrome is the chronic amnestic syndrome that can follow Wernicke encephalopathy. The cardinal features of Korsakoff syndrome are impaired mental syndrome (especially recent memory) and anterograde amnesia in an alert and responsive patient. The patient may or may not have

the symptom of confabulation.

Blackouts. Blackouts are similar to episodes of transient global amnesia in that they are discrete episodes of anterograde amnesia that occur in association with alcohol intoxication. The periods of amnesia can be particularly distressing when persons fear that they have unknowingly harmed someone or behaved imprudently while intoxicated. During a blackout, persons have relatively intact remote memory but experience a specific short-term memory deficit in which they are unable to recall events that happened in the previous 5 or 10 minutes. Because their other intellectual faculties are well preserved, they can perform complicated tasks and appear normal to casual observers. This deficit is directly caused by alcohol, which can block the consolidation of new memories into old memories, a process that likely involves the hippocampus and related temporal lobe structures.

Alcohol-Induced Psychotic Disorder. Approximately 3 percent of alcoholic persons experience auditory hallucinations or paranoid delusions in the context of heavy drinking or withdrawal. The most common auditory hallucinations are voices, but they are often unstructured. The voices are characteristically maligning, reproachful, or threatening, although some patients report that the voices are pleasant. The hallucinations usually last less than a week, but during that week, impaired reality testing is frequent. After the episode, most patients realize the hallucinatory nature of the symptoms.

Hallucinations after alcohol withdrawal are considered rare, and the syndrome is distinct from alcohol withdrawal delirium in that the patient has a clear sensorium. The hallucinations can occur at any age but usually, appear in persons abusing alcohol for a long time. Although the hallucinations usually resolve within a week, some linger; in these cases, clinicians must consider other psychotic disorders in the differential diagnosis.

Mr. G was a 40-year-old unemployed man living alone in a studio apartment and was brought to the hospital by the police. He contacted them, complaining that he heard voices of men on the street below his window talking about him and threatening to kill him. He stated that every time he looked out the window, the men had always disappeared.

Mr. G had a 15-year history of almost daily alcohol use. He was intoxicated each day and often experienced shakes upon awakening in the morning. On the previous day, he had only one glass of beer instead of his usual four because of gastrointestinal problems. He was fully alert and oriented.

Alcohol-Induced Mood Disorder. Heavy intake of alcohol over several days results in many of the symptoms observed in major depressive disorder, but the intense sadness markedly improves within several days to 1 month of abstinence. Eighty percent of people with alcoholism report having histories of intense depression, including 30 to 40 percent who were depressed for two or more weeks at a time. However, only 10 to 15 percent of alcoholic persons have ever had depression that meets the criteria for major depressive disorder when they have not been drinking heavily.

Even severe substance-induced depressions are likely to improve fairly rapidly with abstinence, without specific treatment for the depression. A logical approach for these substance-induced conditions is to teach the patient how to best view and deal with the temporary sadness through education and cognitive-behavioral treatment and to watch and wait at least 2 to 4 weeks before starting antidepressant medications.

A primary care doctor requested a consult for a 42-year-old woman with alcohol use disorder who complained of persisting severe depressive symptoms despite 5 days of abstinence. In the initial stage of the interview, she noted that she had “always been depressed” and felt that she “drank to cope with the depressive symptoms.” Her current complaint included a prominent sadness that had persisted for several weeks, difficulties concentrating, initial and terminal insomnia, and a feeling of hopelessness and guilt.

From a detailed discussion of the patient’s history, it became clear that there had been no major depressive episodes before her mid-20s when alcohol use disorder began. During 1 year of abstinence related to the birth and infancy of her son, her mood had significantly improved. The psychiatrist made a provisional diagnosis of an alcohol-induced mood disorder.

The psychiatrist offered the patient education, reassurance, and cognitive therapy to help her to deal with the depressive symptoms but did not prescribe antidepressant medications. The depressive symptoms remained at their original intensity for several additional days and then began to improve. After 3 weeks of abstinence, the patient was no longer significantly depressed, although she had some minor mood swings for several additional weeks. (Courtesy of Marc A. Shuckit, M.D.)

Alcohol-Induced Anxiety Disorder. Anxiety symptoms fulfilling the diagnostic criteria for alcohol-induced anxiety disorder are also common in the context of acute and protracted alcohol withdrawal. Almost 80 percent of alcoholic persons report panic attacks during at least one acute withdrawal episode; their complaints can, at times, be sufficiently intense for the clinician to consider diagnosing panic disorder. Similarly, during the first 4 weeks or so of abstinence, people with severe alcohol problems are likely to avoid some social situations for fear of being overwhelmed by anxiety (i.e., they have symptoms resembling social phobia); their problems can at times be severe enough to resemble agoraphobia. However, when patients only show symptoms of anxiety in the context of heavy drinking, or shortly after abstinence, then the symptoms are likely to diminish and disappear over time.

A primary care physician referred a 48-year-old woman for evaluation and treatment of her recent onset of panic attacks. These episodes occurred two to three times per week over the preceding 6 months, with each lasting typically between 10 and 20 minutes. Panic symptoms occurred regardless of levels of life stress and were not due to any medications or medical conditions. The workup included an evaluation of her laboratory test values, which revealed a carbohydrate-deficient transferrin (CDT) level of 28 U/L and a γ-glutamyltransferase value of 47. All other blood tests were within normal limits.

The atypical age of onset of the panic attacks, along with the blood results, encouraged the clinician to probe further regarding the pattern of alcohol-related life problems with both the patient and, separately, her spouse. This approach revealed a history of alcohol use disorder with an onset at approximately 35 years of age, with no evidence of panic

disorder before that date. During her periods of abstinence, the patient had no significant panic attacks after about 2 weeks. A working diagnosis of an alcohol use disorder, severe with an alcohol-induced anxiety disorder characterized by panic attacks, was made, and the patient was encouraged to abstain. The psychiatrist also recommended that the primary care doctor appropriately treat the patient for possible withdrawal symptoms. Over the subsequent 3 weeks after a taper of benzodiazepines used for the treatment of withdrawal, the panic symptoms diminished in intensity and subsequently disappeared. (Courtesy of Marc A. Schuckit, M.D.)

Alcohol-Induced Sexual Dysfunction. The formal diagnosis of symptoms of sexual dysfunction associated with alcohol intoxication is alcohol-induced sexual dysfunction, and we discuss this in the chapter on sexual dysfunctions (see Chapter 16).

Alcohol-Induced Sleep Disorder. We discuss alcohol-induced sleep disorders in the sleep disorders chapter (see Chapter 15).

Unspecified Alcohol-Related Disorder. The DSM-5 uses the diagnosis of unspecified alcohol-related disorder for alcohol-related disorders that do not meet the diagnostic criteria for any of the other diagnoses.



Table 4-7
Neurologic and Medical Complications of Alcohol Use

Alcohol intoxication
Acute intoxication
Pathologic intoxication (atypical, complicated, unusual)
Blackouts
Alcohol withdrawal syndromes
Tremulousness (the shakes or the jitters)
Alcoholic hallucinosis (horrors)
Withdrawal seizures (rum fits)
Delirium tremens (shakes)

- Nutritional diseases of the nervous system secondary to alcohol abuse
  - Wernicke–Korsakoff syndrome
  - Cerebellar degeneration
  - Peripheral neuropathy
  - Optic neuropathy (tobacco–alcohol amblyopia)
  - Pellagra
- Alcoholic diseases of uncertain pathogenesis
  - Central pontine myelinolysis
  - Marchiafava–Bignami disease
  - Fetal alcohol syndrome
  - Myopathy
  - Alcoholic dementia
  - Alcoholic cerebral atrophy
- Systemic diseases due to alcohol with secondary neurologic complications
  - Liver disease
  - Hepatic encephalopathy
  - Acquired (non-Wilsonian) chronic hepatocerebral degeneration
  - Gastrointestinal diseases
  - Malabsorption syndromes
  - Postgastrectomy syndromes
  - Possible pancreatic encephalopathy
- Cardiovascular diseases
  - Cardiomyopathy with potential cardiogenic emboli and cerebrovascular disease
  - Arrhythmias and abnormal blood pressure leading to cerebrovascular disease
  - Hematologic disorders
  - Anemia, leukopenia, thrombocytopenia (could possibly lead to hemorrhagic cerebrovascular disease)
- Infectious disease, especially meningitis (especially pneumococcal and meningococcal)
- Hypothermia and hyperthermia
- Hypotension and hypertension

Respiratory depression and associated hypoxia
Toxic encephalopathies, including alcohol and other substances
Electrolyte imbalances leading to acute confusional states and, rarely, local neurologic signs and symptoms
- Hypoglycemia
- Hyperglycemia
- Hyponatremia
- Hypercalcemia
- Hypomagnesemia
- Hypophosphatemia

Increased incidence of trauma
- Epidural, subdural, and intracerebral hematoma
- Spinal cord injury
- Posttraumatic seizure disorders
- Compressive neuropathies and brachial plexus injuries (Saturday night palsies)
- Posttraumatic symptomatic hydrocephalus (normal pressure hydrocephalus)
- Muscle crush injuries and compartmental syndromes

Other Alcohol-Related Neurologic Disorders. We are restricting our discussion to the significant neuropsychiatric syndromes associated with alcohol use. The complete list of neurologic syndromes is lengthy, Table 4-7 provides an example list of some neurologic and medical complications of alcohol use.

Alcoholic pellagra encephalopathy is one diagnosis of potential interest to psychiatrists presented with a patient who appears to have Wernicke–Korsakoff syndrome but who does not respond to thiamine treatment. We list the symptoms of alcoholic pellagra in Table 4-8.

Fetal Alcohol Syndrome. Data indicate that women who are pregnant or are breast-feeding should not drink alcohol. Fetal alcohol syndrome, the leading cause of intellectual disability in the United States, occurs when mothers who drink alcohol expose fetuses to alcohol in utero. The alcohol inhibits intrauterine growth and postnatal development. Microcephaly,

craniofacial malformations, and limb and heart defects are common in affected infants. Short adult stature and development of a range of adult maladaptive behavior are also associated with fetal alcohol syndrome.

Table 4-8
Possible Symptoms of Alcohol Pellagra

Confusion, ranging from mild to full delirium
Clouding of consciousness
Myoclonus
Oppositional hypertonias
Fatigue
Apathy
Irritability
Anorexia
Insomnia

Women with alcohol-related disorders have a 35 percent risk of having a child with defects. Although the precise mechanism of the damage to the fetus is unknown, the damage seems to result from exposure in utero to ethanol or its metabolites; alcohol may also cause hormone imbalances that increase the risk of abnormalities.

Laboratory Tests

The adverse effects of alcohol appear in standard laboratory tests, which can be useful diagnostic aids in identifying persons with alcohol-related disorders. Table 4-9 lists some of the more useful tests, all of which are related to the damaging effects of alcohol.

Table 4-9
Laboratory Test Associated with Heavy Drinking

State Markers of Heavy Drinking Useful in Screening for AUDs	
Test	**Relevant Range of Results**

γ-glutamyltransferase (GGT)	>35.0 U/L
Carbohydrate-deficient transferrin (CDT)	>3.0%
Mean corpuscular volume (MCV)	>91.0 μm^3
Serum glutamic oxaloacetic transaminase (aspartate aminotransferase)	>45.0 IU/L
Serum glutamic pyruvic transaminase (alanine aminotransferase)	>45.0 IU/L

COMORBIDITY

The psychiatric diagnoses most associated with alcohol-related disorders are other substance-related disorders, antisocial personality disorder, mood disorders, and anxiety disorders. Although the data are somewhat controversial, most suggest that persons with alcohol-related disorders have a markedly higher suicide rate than the general population.

Antisocial Personality Disorder

Researchers and clinicians have frequently reported a relation between antisocial personality disorder and alcohol-related disorders. Some studies suggest that antisocial personality disorder is particularly common in men with an alcohol-related disorder and can precede the development of an alcohol-related disorder. Other studies, however, suggest that antisocial personality disorder and alcohol-related disorders are entirely distinct entities that are not causally related.

Mood Disorders

About 30 to 40 percent of persons with an alcohol-related disorder meet the diagnostic criteria for major depressive disorder sometime during their lifetimes. Depression is more common in women than in men with these disorders. Several studies reported that depression is likely to occur in patients with alcohol-related disorders who have a high daily consumption of alcohol and a family history of alcohol abuse. Persons with alcohol-related disorders and major depressive disorder are at significant risk for attempting

suicide and are likely to have other substance-related disorder diagnoses. Some clinicians recommend antidepressant drug therapy for depressive symptoms that remain after 2 to 3 weeks of sobriety. Patients with bipolar disorder are at risk of developing an alcohol-related disorder; they may use alcohol to self-medicate their manic episodes.

Anxiety Disorders

Many persons use alcohol for its efficacy in alleviating anxiety. Although the comorbidity between alcohol-related disorders and mood disorders is widely recognized, perhaps 25 to 50 percent of all persons with alcohol-related disorders also meet the diagnostic criteria for an anxiety disorder. Phobias and panic disorder are particularly frequent comorbid diagnoses in these patients. Persons with anxiety may use alcohol to self-medicate symptoms of agoraphobia or social phobia. Still, an alcohol-related disorder is likely to precede the development of panic disorder or generalized anxiety disorder.

Suicide

Most estimates of the prevalence of suicide among persons with alcohol-related disorders range from 10 to 15 percent, although alcohol use itself may be involved in a much higher percentage of suicides. Some investigators have questioned whether the suicide rate among persons with alcohol-related disorders is as high as the numbers suggest. Table 4-10 lists factors associated with suicide among persons with alcohol-related disorders.

COURSE AND PROGNOSIS

Between 10 and 40 percent of alcoholic persons enter some formal treatment program. Several prognostic signs are favorable. Table 4-11 lists some of the most predictive factors, and the combination of these predicts at least a 60 percent chance for one or more years of abstinence. Few studies have documented the long-term course, but researchers agree that 1 year of abstinence is associated with a good chance for continued abstinence over an extended period. Alcoholic persons with severe drug problems (especially intravenous drug use or cocaine or amphetamine use disorders) and those who are homeless may have only a 10 to 15 percent chance of achieving 1

year of abstinence, however.

Table 4-10
Factors Associated with Suicide in Persons with Alcohol Use Disorder

Presence of a major depressive episode
Weak psychosocial support systems
A serious coexisting medical condition
Unemployment
Living alone

Accurately predicting whether any specific person will achieve or maintain abstinence is impossible, but the prognostic factors listed earlier are associated with an increased likelihood of abstinence. The factors reflecting life stability, however, probably explain only 20 percent or less of the course of alcohol use disorders. Many other factors, some that may be hard to measure, also affect the clinical course significantly. These somewhat intangible factors are likely to include motivational levels and the quality of the patient's social support system.

In general, alcoholic persons with preexisting independent major psychiatric disorders—such as antisocial personality disorder, schizophrenia, and bipolar disorder—are likely to run the course of their independent psychiatric illness. Clinicians should treat the patient with bipolar disorder who has alcoholism with appropriate pharmacotherapy and psychotherapy. The same can be said for an antisocial personality disorder, schizophrenia, or other psychiatric disorders. The goal is to minimize the symptoms of the independent psychiatric disorder in the hope that better life stability will be associated with a better prognosis for the patient's alcohol problems.

TREATMENT OF ALCOHOL USE DISORDERS

Three general steps are involved in treating the alcoholic person: intervention, detoxification, and rehabilitation. These approaches assume that we have already addressed the patient's medical functioning and addressed any psychiatric emergencies. Thus, for example, an alcoholic person with

symptoms of depression sufficiently severe to be suicidal requires inpatient hospitalization for at least several days until the suicidal ideation disappears. Similarly, a person presenting with cardiomyopathy, liver difficulties, or GI bleeding first needs adequate treatment of the medical emergency.

The patient with alcohol use disorder must then be brought face-to-face with the reality of the disorder (intervention), be detoxified if needed, and begin rehabilitation. The essentials of these three steps for an alcoholic person with independent psychiatric syndromes closely resemble the approaches used for the primary alcoholic person without independent psychiatric syndromes. In the former case, however, we should apply these treatments after stabilizing the psychiatric disorder as much as possible.



Table 4-11
Alcohol Use Disorders: Positive Prognostic Signs

No premorbid antisocial personality disorder
Good psychosocial functioning
Stable job
Stable family
No legal problems
Adherence to treatment

Intervention

The goal of intervention is to use the principles of motivational interviewing and brief interventions to help patients recognize the adverse consequences likely to occur if they do not stop drinking. A discussion of the presenting complaint (e.g., insomnia or depression) is a useful way to both show empathy and enhance motivation for change. As part of an intervention, we would emphasize how alcohol either created or contributed to these problems and reassure the patient that abstinence is possible. If the person does not respond to the first brief intervention or motivational interview, we can use the same nonjudgmental but persistent approach each time an alcohol-related problem arises. It is the persistence rather than exceptional interpersonal skills that usually gets results. A single intervention is rarely sufficient. Most alcoholic persons need a series of reminders of how alcohol contributed to

each developing crisis before they seriously consider abstinence as a long-term option.

JP, a 47-year-old physician, was confronted regarding his alcohol-related behaviors by his wife and 21-year-old daughter. They told him about his slurred speech on several recent occasions when the daughter called home, as well as a large number of wine bottles in the trash each week. JP's wife complained of the hours he spent alone in his study and his practice of staying up after she went to bed, retiring later with alcohol on his breath. She also related her concern about his consumption of about 10 or 12 drinks at a recent party. He also tended to isolate himself from the other guests. She then reminded him of his need to pack liquor when they go on trips where alcohol may not be readily available, and the tremor of his hands some mornings after being drunk the night before. The family shared their concern directly with JP at a time when he was not actively intoxicated, emphasizing specific times and events when his impairment with alcohol occurred. They had also made an appointment with the clinician at an alcohol and drug treatment program to facilitate the next step if the intervention was successful. (Adapted from Marc A. Schuckit, M.D.)

Detoxification

Most persons with alcohol use disorder have relatively mild symptoms when they stop drinking. If the patient is in relatively good health, is adequately nourished, and has a sound social support system, the depressant withdrawal syndrome usually resembles a mild case of the flu. Even intense withdrawal syndromes rarely approach the severity of symptoms described by some early textbooks in the field.

The essential first step in detoxification is a thorough physical examination. In the absence of a severe medical disorder or combined drug abuse, severe alcohol withdrawal is unlikely. The second step is to offer rest, adequate nutrition, and multiple vitamins, especially those containing thiamine.

Seizures. Seizures can occur suddenly and without other signs of withdrawal. About 1 percent of patients may have a single grand mal convulsion; the rare person has multiple fits, with the peak incidence on the second day of withdrawal.

Such patients require neurologic evaluation to rule out an independent seizure disorder. Assuming no other cause for the seizures is found, benzodiazepines are the treatment of choice for managing alcohol-related seizures. Anticonvulsants, although often used (particularly before the cause of the seizure is known), do not appear to offer additional benefit.

Mild or Moderate Withdrawal. Withdrawal develops because the brain has physically adapted to the presence of a brain depressant and cannot function adequately in the absence of the drug. Giving sufficient brain depressant on the first day to diminish symptoms and then weaning the patient off the drug over the next 5 days offers most patients optimal relief. It also minimizes the possibility that severe withdrawal will develop. Any CNS depressant (including benzodiazepines, barbiturates, or even alcohol) can work, but most clinicians choose a benzodiazepine for its relative safety. Many studies have found that benzodiazepines control most of the symptoms associated with alcohol withdrawal. We can give adequate treatment with either short-acting drugs (e.g., lorazepam), or long-acting substances (e.g., chlordiazepoxide and diazepam). We may choose to give benzodiazepines orally or parenterally; however, we should not give intramuscular (IM) diazepam or chlordiazepoxide as their IM absorption is erratic. We should titrate the dosage of the benzodiazepine, starting with a high dosage and lowering the dosage as the patient recovers. We should aim to give sufficient benzodiazepines to keep patients calm and sedated but not so sedated that we cannot arouse them for procedures and examinations.

Although benzodiazepines are the standard treatment for alcohol withdrawal, studies have shown that carbamazepine in daily doses of 800 mg is as effective as benzodiazepines and has the added benefit of minimal abuse liability. Carbamazepine use is gradually becoming common in the United States and Europe.

Table 4-12 gives an example of withdrawal treatment using chlordiazepoxide. When giving a long-acting agent, such as chlordiazepoxide, the clinician must avoid producing excessive sleepiness

through overmedication; if the patient is sleepy, it is best to omit the next scheduled dose. When taking a short-acting drug, such as lorazepam, the patient must not miss any dose because rapid changes in benzodiazepine concentrations in the blood can precipitate severe withdrawal.

A social model program of detoxification saves money by avoiding medications while using social supports. This less-expensive regimen can be helpful for mild or moderate withdrawal syndromes. Some clinicians also use β-adrenergic receptor antagonists (e.g., propranolol) or α-adrenergic receptor agonists (e.g., clonidine) to block the symptoms of sympathetic hyperactivity. However, these medications do not appear to be superior to benzodiazepines. Unlike the brain depressants, these other agents are not effective treatments for seizures or delirium.



Table 4-12
An Example Treatment for Alcohol Withdrawal

Day 1: chlordiazepoxide 25 mg by mouth 3–4 times daily, hold if sedated
1–2 additional doses if
The patient is jittery
Has increased tremor
Has autonomic dysfunction
Day 2: give the total dose used on day 1 minus 20% divided across 3–4 doses
Following days: continue a 20% decrease until no further medication is needed (4–5 days)

Severe Withdrawal. For the approximately 1 percent of alcoholic patients with extreme autonomic dysfunction, agitation, and confusion—that is, those with alcohol withdrawal delirium, or DTs—no optimal treatment has yet been developed. The best treatment for DTs is prevention. When that is not possible, the first step is to ask why such a severe and relatively uncommon withdrawal syndrome has occurred. The answer often relates to a severe concomitant medical problem that needs immediate treatment.

Once a delirium appears, the treatment remains similar to that for the standard withdrawal but with higher doses. Example regimens are 50 to 100

mg of chlordiazepoxide given every 4 hours orally, or lorazepam, given intravenously (IV) if oral medication is not possible. A high-calorie, high-carbohydrate diet supplemented by multivitamins is also essential.

Physically restraining patients with the DTs is risky; they may fight against the restraints to a dangerous level of exhaustion. When patients are disorderly and uncontrollable, it may be preferable to use a seclusion room. Adjunctive antipsychotic medications, such as haloperidol, are sometimes used to control severe agitation and hallucinations. However, antipsychotic medications may reduce the seizure threshold in patients. Dehydration, often exacerbated by diaphoresis and fever, can be corrected with fluids given by mouth or IV. Anorexia, vomiting, and diarrhea often occur during withdrawal. The emergence of focal neurologic symptoms, lateralizing seizures, increased intracranial pressure, or evidence of skull fractures or other indications of CNS pathology should prompt clinicians to examine a patient for neurologic disease.

Warm, supportive psychotherapy in the treatment of DTs is essential. Patients are often bewildered, frightened, and anxious because of their tumultuous symptoms, and skillful verbal support is imperative.

Protracted Withdrawal. Symptoms of anxiety, insomnia, and mild autonomic overactivity are likely to continue for 2 to 6 months after the acute withdrawal has disappeared. These protracted withdrawal symptoms may enhance the probability of relapse. Although there is little evidence that medications are helpful, it is possible that some of the medications used for the rehabilitation phase, especially acamprosate, may work by diminishing some of these symptoms. The clinician should warn the patient that some level of sleep problems or feelings of nervousness might remain after acute withdrawal and discuss cognitive and behavioral approaches that might be appropriate to help the patient feel comfortable.

Rehabilitation

For most patients, rehabilitation includes three major components, which are listed in Table 4-13. These components happen at a difficult time: the patient may still be experiencing withdrawal symptoms as well as life crises. Repetition will be necessary, with frequent reminders of how vital abstinence

is, and that it helps the patient develop new day-to-day support systems and coping styles.

Table 4-13
Three Components of Alcohol Rehabilitation

Continued efforts to increase and maintain high levels of motivation for abstinence
Work to help the patient readjust to a lifestyle free of alcohol
Relapse prevention

Table 4-14
Stages of Alcohol Rehabilitation Treatment

Early intensive period (2–4 wk)
Intervention (withdrawal treatment, preventing craving)
Optimizing physical and psychological functioning
Enhancing motivation
Reaching out to the family
Long term (3–6 mo)
Individual and group counseling
Judicious avoidance of psychotropic medications unless needed for independent disorders
Involvement in self-help groups such as AA

No single major life event, traumatic life period, or identifiable psychiatric disorder is known to be the sole cause of alcoholism. Regardless of any possible cause, after years of alcohol use, it develops a life of its own. Many alcoholic persons believe that the cause was depression, anxiety, life stress, or pain syndromes. However, usually, the reality is that alcohol contributed to the mood disorder, accident, or life stress, not vice versa.

The treatment approach remains the same regardless of setting, including both inpatient and outpatient treatments. The selection of the more intensive (and expensive) inpatient mode often depends on several factors. These include whether there are additional severe medical or psychiatric syndromes,

what other options are available, and whether the patient has previously failed outpatient treatment.

The treatment process in either setting involves an intensive early phase, which lasts 2 to 4 weeks, followed by longer-term care. Table 4-14 lists some of the approaches used in these phases.

Counseling

Counseling efforts in the first several months should focus on day-to-day life issues to help patients maintain a high level of motivation for abstinence and to enhance their functioning. Psychotherapy techniques that provoke anxiety or that require deep insights are not likely to be of benefit during the early months of recovery and, at least theoretically, may impair efforts to maintain abstinence. Thus, this discussion focuses on the efforts likely to characterize the first 3 to 6 months of care.

Both individual and group counseling is an option, and few data suggest that either approach is superior. The technique used is not likely to matter much and usually boils down to simple day-to-day counseling or almost any behavioral or psychotherapeutic approach focusing on the here and now. Treatment sessions should explore the consequences of drinking, the likely future course of alcohol-related life problems, and the marked improvement that results from abstinence. Whether in an inpatient or an outpatient setting, individual or group counseling is usually offered a minimum of three times a week for the first 2 to 4 weeks, followed by less intense efforts, perhaps once a week, for the subsequent 3 to 6 months.

Much time in counseling deals with how to build a lifestyle free of alcohol. Discussions cover the need for a sober peer group, a plan for social and recreational events without drinking, and approaches for reestablishing communication with family members and friends.

The third major component, relapse prevention, first identifies situations in which the risk for relapse is high. The counselor must help the patient develop modes of coping for when the craving for alcohol increases or when an event or emotional state makes a return to drinking likely. An essential part of relapse prevention is reminding the patient about the appropriate attitude toward slips. Short-term experiences with alcohol should not be an excuse for returning to regular drinking. Recovery is not an “all or none”

game. Instead, it is a process of trial and error; patients use slips as information to identify high-risk situations and to develop more appropriate coping techniques.

Most treatment efforts recognize the effects that alcoholism has on the significant persons in the patient's life, and an essential aspect of recovery involves helping family members and close friends understand alcoholism and realize that rehabilitation is an ongoing process that lasts for 6 to 12 or more months. Couples and family counseling and support groups for relatives and friends help the persons involved to rebuild relationships, to learn how to avoid protecting the patient from the consequences of any drinking in the future, and to be as supportive as possible of the alcoholic patient's recovery program.

Medication Treatment

Following detoxification, there is little reason to prescribe antidepressant or anxiety medications if the patient does not have an independent psychiatric disorder. Lingering levels of anxiety and insomnia as part of a reaction to life stresses and protracted abstinence should be treated with behavior modification approaches and reassurance. Medications for these symptoms are likely to lose their effectiveness much faster than the symptoms will disappear, and the patient may increase the dose, leading to subsequent problems. Similarly, sadness and mood swings can linger at low levels for several months. Controlled clinical trials, however, indicate no benefit in prescribing antidepressant medications or lithium to treat the average alcoholic person who has no independent or long-lasting psychiatric disorder. The mood disorder will clear before the medications can take effect, and patients who resume drinking while on the medications face significant potential dangers. With little or no evidence that the medications are effective, the dangers significantly outweigh any potential benefits from their routine use.

Data from double-blind trials support modest effects for two medications offered in the context of cognitive-behavioral therapy (CBT). The first involves the opioid antagonist naltrexone, which appears to decrease the craving for alcohol or blunt the rewarding effects of drinking. The COMBINE study (Combined Pharmacotherapies and Behavioral

Interventions) demonstrated that naltrexone improves clinical outcomes, and a Cochrane review of naltrexone concluded that naltrexone was effective in reducing heavy drinking days. The typical dose is 50 mg/day, and side effects include mild GI upset and lethargy.

The second medication, acamprosate, antagonizes neuronal overactivity related to the excitatory neurotransmitter glutamate, at least in part, by acting as an antagonist to N-methyl-D-aspartate (NMDA) receptors. Acamprosate may diminish mild anxiety, mood swings, and other sleep difficulties associated with the protracted withdrawal syndrome observed after the first 4 to 5 days of abstinence. The typical dose is about 2,000 mg divided into three doses per day and is associated with relatively mild side effects of GI problems such as diarrhea.

A third possible drug of interest is the alcohol-sensitizing agent disulfiram. Disulfiram is given in daily doses of 250 mg before discharging the patient from the intensive first phase of outpatient rehabilitation or inpatient care. The goal is to place the patient in a condition in which drinking alcohol precipitates an uncomfortable physical reaction, including nausea, vomiting, and a burning sensation in the face and stomach. Few data support disulfiram's effectiveness, most likely because patients stop it when they resume drinking. Many clinicians have stopped routinely prescribing the agent, partly in recognition of the dangers associated with the drug itself: mood swings, rare instances of psychosis, the possibility of increased peripheral neuropathies, the relatively rare occurrence of other significant neuropathies, and potentially fatal hepatitis. Moreover, we should not give disulfiram to patients with preexisting heart disease, cerebral thrombosis, diabetes, and some other conditions, because an alcohol reaction to the disulfiram could be fatal.

Table 4-15 gives a summary of medications used for alcohol dependence.

Two other medications, while not yet approved for the treatment of alcohol use disorders, are worth brief mention. Several studies using the anticonvulsant topiramate have reported improvement in drinking patterns. Additionally, some data suggest that the 5-HT_3 ondansetron might be beneficial in alcohol use disorder treatment. No evidence exists that antidepressant medications, such as selective serotonin reuptake inhibitors (SSRIs), lithium, or antipsychotic medications, are significantly effective in

the treatment of alcoholism.

Alcoholics Anonymous

Clinicians must recognize the potential importance of self-help groups such as AA. Members of AA have help available 24 hours a day, associate with a sober peer group, learn that it is possible to participate in social functions without drinking, and are given a model of recovery by observing the accomplishments of sober members of the group.

Learning about AA usually begins during inpatient or outpatient rehabilitation. The clinician can play a significant role in helping patients understand the differences between specific groups. Some groups are single gender, others are mixed. Certain groups may differ by socioeconomic level as well, or place great emphasis on religion. Such specialized groups may be a practical approach. Anyone can suffer from alcoholism, but a person may be more likely to stick with a group toward which they feel some affinity. Patients with coexisting psychiatric disorders may need some additional education about AA. The clinician should remind them that some members of AA may not understand their unique needs for medications and should arm the patients with ways of coping when group members inappropriately advise against medications. Although challenging to evaluate using double-blind controls, most studies indicate that participation in AA is associated with improved outcomes, as well as being cost-efficient.

Treatment of Alcohol-Related Disorders

Treatment of Alcohol-Induced Amnestic Disorders. In the early stages, Wernicke encephalopathy responds rapidly to large doses of parenteral thiamine, which is likely effective at preventing the progression to Korsakoff syndrome. The dosage of thiamine is usually initiated at 100 mg by mouth two to three times daily and continued for 1 to 2 weeks. In patients with alcohol-related disorders who are receiving IV administration of glucose solution, it is good practice to include 100 mg of thiamine in each liter of the glucose solution.

Table 4-15

Medications for Treating Alcohol Dependence

	Naltrexone	Acamprosate	Disulfiram
Action	Blocks opioid receptors, resulting in reduced craving and reduced reward in response to drinking	Not well understood. Likely an NMDA receptor antagonist and positive allosteric modulator of $GABA_a$ receptors	Inhibits intermedi metabolis alcohol, causing a buildup o acetaldeh and a rea of flushin sweating, nausea, a tachycard patient dr alcohol
Contraindications	Currently using opioids or in acute opioid withdrawal; anticipated need for opioid analgesics; acute hepatitis or liver failure	Severe renal impairment (CrCl ≤30 mL/min)	Concomitan of alcohol alcohol-containin preparati metronida coronary disease; s myocardi disease
Precautions	Other hepatic disease; renal impairment; history of	Moderate renal impairment (dose adjustment for CrCl between 30	High impuls likely to d while usir psychose (current o

	suicide attempts. If opioid analgesia is required, larger doses may be required, and respiratory depression may be deeper and more prolonged	and 50 mL/min); depression or suicidality	history); diabetes mellitus; epilepsy; hepatic dysfunctic hypothyro renal impairme rubber co dermatitis
Serious adverse reactions	Will precipitate severe withdrawal if patient is dependent on opioids; hepatotoxicity (uncommon at usual doses). Pregnancy Category C.	Anxiety; depression. Rare events include the following: suicide attempt, acute kidney failure, heart failure, mesenteric arterial occlusion, cardiomyopathy, deep thrombophlebitis, and shock. Pregnancy Category C.	Hepatitis; op neuritis; peripheral neuropath psychotic reactions. Pregnanc Category
Common side effects	Nausea; abdominal pain; constipation;	Diarrhea; flatulence; nausea; abdominal pain;	Metallic aftertaste dermatitis

	dizziness; headache; anxiety; fatigue	headache; back pain; infection; flu syndrome; chills; somnolence; decreased libido; amnesia; confusion	
Examples of drug interactions	Opioid analgesics (blocks action); yohimbine (use with naltrexone increases negative drug effects)	No clinically relevant interactions known	Amitriptyline anticoagu such as warfarin; diazepam isoniazid; metronida phenytoin theophylli warfarin; nonpresc drug-cont alcohol
Usual adult dosage	*Oral dose:* 50 mg daily *before prescribing:* evaluate for possible current opioid use; consider a urine toxicology screen for opioids, including synthetic	*Oral dose:* 666 mg (two 333-mg tablets) three times daily *or,* for patients with moderate renal impairment (CrCl 30–50 mL/min), reduce to 333 mg (one tablet) three times daily *before prescribing:* establish	*Oral dose:* 2 mg daily (125–500 *before prescribir* warn that patient sh not take disulfiram least 12 h drinking a that a disulfiram alcohol re

	opioids. Obtain liver function tests *Follow-up:* monitor liver function tests periodically	abstinence	can occu 2 wk afte last dose; (2) warn a alcohol in diet (e.g., sauces a vinegars) in medica and toilet *Follow-up* monitor li function t periodica

CrCl, creatinine clearance; GABA, γ-aminobutyric acid.

Treatment of Korsakoff syndrome is also thiamine given 100 mg by mouth two to three times daily; the treatment regimen should continue for 3 to 12 months. Few patients who progress to Korsakoff syndrome ever fully recover, although many have some improvement in their cognitive abilities with thiamine and nutritional support.

In a patient who appears to have Wernicke–Korsakoff syndrome but who does not respond to thiamine treatment, one should consider the diagnosis of alcoholic pellagra encephalopathy (see Other Alcohol-Related Neurologic Disorders). Such patients have a niacin (nicotinic acid) deficiency, and the specific treatment is 50 mg of niacin by mouth four times daily or 25 mg parenterally two to three times daily.

Treatment of Alcohol Withdrawal-Related Hallucinations. The treatment of alcohol withdrawal-related hallucinations is much like the treatment of DTs—benzodiazepines, adequate nutrition, and fluids, if necessary. If this regimen fails or for long-term cases, clinicians may add antipsychotics.

Treatment of Idiosyncratic Alcohol Intoxication. In treating idiosyncratic alcohol intoxication, clinicians must help protect patients from

harming themselves and others. Physical restraint may be necessary, but it is difficult because of the abrupt onset of the condition. Once restraining the patient, it may be helpful to inject an antipsychotic, such as haloperidol, to control assaultiveness. We should differentiate this condition from other causes of abrupt behavioral change, such as complex partial epilepsy. Some persons with the disorder reportedly showed temporal lobe spiking on an EEG after ingesting small amounts of alcohol.



Table 4-16
Alcohol Epidemiology

Condition	Population (%)
Ever had a drink	90
Current drinker	60–70
Temporary problems	40+
Alcohol use disorder (AUD) lifetime	Male: 15+
	Female: 10+
AUDs in psychiatric patients	30

The percentages listed here are approximations generated across studies.
From *Kaplan and Sadock's Comprehensive Textbook of Psychiatry*. 10th ed. 2017.

Epidemiology

Alcohol use disorder is among the most common psychiatric disorders observed in the Western world. Alcohol-related problems in the United States contribute to 88,000 deaths each year, with nearly 10,000 deaths as a result of alcohol-impaired driving. According to data from the World Health Organization (WHO), alcohol is the fifth leading risk factor for premature death and disability across the world. Table 4-16 lists some composite epidemiologic data on alcohol use.

Prevalence of Drinking. At some time during life, 90 percent of the population in the United States drinks, with most people beginning their alcohol intake in the early to middle teens. By the end of high school, 80

percent of students have consumed alcohol, and more than 60 percent have been intoxicated. At any time, two of three men are drinkers, with a ratio of persisting alcohol intake of approximately 1.3 men to 1.0 women, and the highest prevalence of drinking from the middle or late teens to the mid-20s.

Men and women with higher education and income are most likely to imbibe, and, among religious denominations, Jewish persons have the highest proportion who consume alcohol but among the lowest rates of alcohol use disorders. Other ethnicities, such as the Irish, have higher rates of severe alcohol problems, but they also have significantly higher rates of abstentions. Some estimates show that more than 60 percent of men and women in some Native American and Inuit tribes have at some time had an alcohol use disorder. In the United States, the average adult consumes 2.4 gallons of absolute alcohol a year, a considerable decrease from 5 gallons of absolute alcohol yearly intake in the 1700s.

Drinking alcohol-containing beverages is generally considered an acceptable habit in the United States. About 90 percent of all US residents have consumed alcohol at least once in their lives. About 56 percent of all US adults used alcohol in the past month. Alcohol is the third leading preventable cause of death in the United States, behind tobacco use and poor diet/physical inactivity. Excessive alcohol use is associated with 2.5 million years of potential life lost in the United States each year and causes 10 percent of all deaths among working adults. Although persons involved in automotive fatalities do not always meet the diagnostic criteria for an alcohol-related disorder, drunk drivers are involved in about 31 percent of all automotive fatalities. Alcohol use and alcohol-related disorders are associated with about 25 percent of all suicides.

PATHOLOGY

Effects of Alcohol

The term *alcohol* refers to a large group of organic molecules that have a hydroxyl group (–OH) attached to a saturated carbon atom. Ethyl alcohol, also called *ethanol,* is the standard form of drinkable alcohol, sometimes referred to as *beverage alcohol*. The chemical formula for ethanol is $CH_3–CH_2–OH$.

The characteristic tastes and flavors of alcohol-containing beverages result from their methods of production, which produce various congeners in the final product, including methanol, butanol, aldehydes, phenols, tannins, and trace amounts of various metals. Although the congeners may confer some differential psychoactive effects on the various alcohol-containing beverages, these differences are minimal compared with the effects of ethanol itself. A single drink typically contains about 12 g of ethanol, which is the content of 12 oz of beer (7.2 proof, 3.6 percent ethanol in the United States), one 4-oz glass of nonfortified wine, or 1 to 1.5 oz of an 80-proof (40 percent ethanol) liquor (e.g., whiskey or gin). In calculating patients' alcohol intake, however, clinicians should be aware that beers vary in their alcohol content, that beers are available in small and large cans and mugs, that glasses of wine range from 2 to 6 oz, and that mixed drinks at some bars and in most homes contain 2 to 3 oz of liquor. Nonetheless, using the moderate sizes of drinks, clinicians can estimate that a single drink increases the blood alcohol level of a 150-lb man by 15 to 20 mg/dL, which is about the concentration of alcohol that an average person can metabolize in 1 hour.

The possible beneficial effects of alcohol have been publicized, especially by the makers and the distributors of alcohol. Some epidemiologic data suggest that one or two glasses of red wine each day lower the incidence of cardiovascular disease; these findings, however, are highly controversial.

Absorption. The stomach absorbs about 10 percent of consumed alcohol, and the small intestine absorbs the rest. It takes about 30 to 90 minutes to reach peak blood concentration, depending on whether the alcohol was ingested on an empty stomach (which enhances absorption) or with food (which delays absorption). Also, rapid drinking reduces the time to peak concentration; slower drinking increases it. Absorption is most rapid, with beverages containing 15 to 30 percent alcohol (30 to 60 proof). There is some dispute about whether carbonation (e.g., in champagne and drinks mixed with seltzer) enhances the absorption of alcohol.

The body has protective devices against inundation by alcohol. For example, if the concentration of alcohol in the stomach becomes too high, mucus is secreted, and the pyloric valve closes. These actions slow the absorption and keep the alcohol from passing into the small intestine, where there are no significant restraints on absorption. Thus, a large amount of

alcohol can remain unabsorbed in the stomach for hours. Furthermore, pylorospasm often results in nausea and vomiting.

Once the bloodstream absorbs the alcohol, it distributes to all body tissues. Alcohol uniformly dissolves in the body's water, so tissues containing a high proportion of water receive a high concentration of alcohol. The intoxicating effects are more significant for a given blood alcohol concentration when the concentration is rising than when it is falling (the Mellanby effect). For this reason, the rate of absorption bears directly on the intoxication response.

Metabolism. Hepatic oxidation accounts for about 90 percent of alcohol metabolism. The remaining 10 percent is excreted unchanged by the kidneys and lungs. The rate of oxidation by the liver is constant and independent of the body's energy requirements. The body can metabolize about 15 mg/dL per hour, with a range of 10 to 34 mg/dL per hour. That is, the average person oxidizes three-fourths of an ounce of 40 percent (80 proof) alcohol in an hour. In persons with a history of excessive alcohol consumption, upregulation of the necessary enzymes results in rapid alcohol metabolism.

Two enzymes metabolize alcohol: alcohol dehydrogenase (ADH) and aldehyde dehydrogenase. ADH catalyzes the conversion of alcohol into acetaldehyde, which is a toxic compound; aldehyde dehydrogenase catalyzes the conversion of acetaldehyde into acetic acid. Disulfiram inhibits aldehyde dehydrogenase. Some studies have shown that women have a lower ADH blood content than men; this fact may account for woman's tendency to become more intoxicated than men after drinking the same amount of alcohol. The decreased function of alcohol-metabolizing enzymes in some Asian persons can also lead to more rapid intoxication and toxic symptoms.

Alcohol's Effects on the Brain

Biochemistry. Alcohol has prominent effects on almost all neurochemical systems, with different actions during intoxication versus withdrawal. Perhaps the most potent effects are on GABA complexes in the brain, especially on the GABA-A receptor ($GABA_A$), contributing to the sedating, sleep-inducing, anticonvulsant, and muscle-relaxing properties of alcohol. Ethanol also impacts NMDA receptors, with dampened stimulatory effects during intoxication and heightened activity during withdrawal. Alcohol

acutely increases dopamine and its metabolites, and chronic drinking changes dopamine receptor numbers and sensitivity, with related effects on intoxication and craving, in part through changes in the pleasure centers in the VTA of the brain. Alcohol also increases serotonin in the synapses and upregulates serotonin receptors, and acutely enhances the functioning of the opioid-related brain systems and impacts adenosine, acetylcholine, and cannabinoid 1 (CB1) receptors.

Behavioral Effects. Alcohol is a depressant, much like the barbiturates and the benzodiazepines, with which alcohol has some cross-tolerance and cross-dependence. At a level of 0.05 percent alcohol in the blood, thought, judgment, and restraint are loosened and sometimes disrupted. At a concentration of 0.1 percent, voluntary motor actions usually become perceptibly clumsy. Most states define legal intoxication as a 0.08 percent blood alcohol level. At 0.2 percent, the function of the entire motor area of the brain is measurably depressed, and the parts of the brain that control emotional behavior are also affected. At 0.3 percent, a person is commonly confused or may become stuporous; at 0.4 to 0.5 percent, the person falls into a coma. At higher levels, the primitive centers of the brain that control breathing and heart rate are affected, and death ensues secondary to direct respiratory depression or the aspiration of vomitus. Persons with long-term histories of alcohol use disorders, however, can tolerate much higher concentrations of alcohol than can alcohol-naïve persons; their alcohol tolerance may cause them to appear less intoxicated than they are.

Sleep Effects. Although alcohol consumed in the evening usually increases the ease of falling asleep (decreased sleep latency), alcohol also has adverse effects on sleep architecture. Specifically, alcohol use is associated with a decrease in rapid eye movement sleep (REM or dream sleep) and deep sleep (stage 4) and more sleep fragmentation, with more and longer episodes of awakening. Therefore, the idea that drinking alcohol improves a person’s sleep is a myth.

Other Physiologic Effects

LIVER. The most significant adverse effects of alcohol use are related to liver

damage. Alcohol use, even as short as week-long episodes of increased drinking, can result in an accumulation of fats and proteins, which produce the appearance of a fatty liver, sometimes found on physical examination as an enlarged liver. The association between fatty infiltration of the liver and severe liver damage remains unclear. Alcohol use, however, is associated with the development of alcoholic hepatitis and hepatic cirrhosis.

GASTROINTESTINAL SYSTEM. Long-term heavy drinking is associated with developing esophagitis, gastritis, achlorhydria, and gastric ulcers. The development of esophageal varices can accompany particularly heavy alcohol use; the rupture of the varices is a medical emergency often resulting in death by exsanguination. Disorders of the small intestine occasionally occur, and pancreatitis, pancreatic insufficiency, and pancreatic cancer are also associated with heavy alcohol use. Heavy alcohol intake can adversely affect the normal processes of food digestion and absorption. Alcohol use disorders also appear to inhibit the intestine's capacity to absorb various nutrients, such as vitamins and amino acids. This effect, coupled with the often poor dietary habits of those with alcohol-related disorders, can cause serious vitamin deficiencies, particularly of the B vitamins.

OTHER BODILY SYSTEMS. A significant intake of alcohol is associated with increased blood pressure, dysregulation of lipoprotein and triglyceride metabolism, and increased risk for myocardial infarction and cerebrovascular disease. Alcohol can affect the hearts of nonalcoholic persons who do not usually drink, increasing the resting cardiac output, the heart rate, and the myocardial oxygen consumption. Evidence indicates that alcohol intake can adversely affect the hematopoietic system and can increase the incidence of cancer, particularly head, neck, esophageal, stomach, hepatic, colonic, and lung cancer. Acute intoxication may also be associated with hypoglycemia, which, when unrecognized, may be responsible for some of the sudden deaths of persons who are intoxicated. Muscle weakness is another side effect of alcohol use disorders.

Alcohol intake also raises the blood concentration of estradiol in women. The increase in estradiol correlates with the blood alcohol level.

DRUG INTERACTIONS. The interaction between alcohol and other substances

can be dangerous, even fatal. Certain substances, such as alcohol and phenobarbital, are metabolized by the liver, and their prolonged use can lead to the acceleration of their metabolism. When persons with alcohol-related disorders are sober, this accelerated metabolism makes them unusually tolerant to many drugs such as sedatives and hypnotics; when they are intoxicated, however, these drugs compete with the alcohol for the same detoxification mechanisms, and potentially toxic concentrations of all involved substances can accumulate in the blood.

The effects of alcohol and other CNS depressants are usually synergistic. Sedatives, hypnotics, and drugs that relieve pain, motion sickness, head colds, and allergy symptoms must be used with caution by persons with alcohol-related disorders. Narcotics depress the sensory areas of the cerebral cortex and can produce pain relief, sedation, apathy, drowsiness, and sleep; high doses can result in respiratory failure and death. Increasing the dosages of sedative–hypnotic drugs, such as chloral hydrate and benzodiazepines, especially when combining them with alcohol, produces a range of effects from sedation to motor and intellectual impairment to stupor, coma, and death. Because sedatives and other psychotropic drugs can potentiate the effects of alcohol, it is important to warn patients about the dangers of combining CNS depressants and alcohol, particularly when they are driving or operating machinery.

ETIOLOGY

Many factors affect the decision to drink, the development of temporary alcohol-related difficulties, and the development of alcohol use disorders. The initiation of drinking probably depends mainly on social, religious, and personality characteristics, although genetic characteristics might also contribute. The reasons for beginning to drink may differ from the factors that later lead to an alcohol use disorder.

A similar interplay between genetic and environmental influences contributes to many medical and psychiatric conditions, and, thus, a review of these factors in alcohol use disorders offers information about complex genetic disorders overall. Dominant or recessive genes, although relevant, explain only rare conditions. Most disorders have some level of genetic predisposition that usually relates to a series of different genetically

influenced characteristics, each of which increases or decreases the risk for the disorder.

A series of genetic influences likely combine to explain approximately 60 percent of the proportion of risk for alcohol use disorders, with the environment responsible for the remaining proportion of the variance. The divisions offered in this section, therefore, are more heuristic than real, because it is the combination of a series of psychological, sociocultural, biologic, and other factors that are responsible for the development of severe, repetitive alcohol-related life problems.

Psychological Theories

A variety of theories assume that we use alcohol to reduce tension, increase feelings of power, and decrease the effects of psychological pain. Perhaps the most considerable interest has been paid to the observation that people with alcohol-related problems often report that alcohol decreases their feelings of nervousness and helps them cope with the day-to-day stresses of life. The psychological theories are built, in part, on the observation among nonalcoholic people that the intake of low doses of alcohol in a tense social setting or after a stressful day enhances feelings of well-being and improved ease of interactions. In high doses, especially at falling blood alcohol levels, however, most measures of muscle tension and psychological feelings of nervousness and tension are increased. Thus, the tension-reducing effects of this drug might have an impact most on light to moderate drinkers or add to the relief of withdrawal symptoms, but play a minor role in causing alcohol use disorders. The theories that focus on alcohol's potential to enhance one's esteem are challenging to evaluate definitively.

Psychodynamic Theories. Related to the disinhibiting or anxiety-lowering effects of lower doses of alcohol is the hypothesis that some people may use this drug to help them deal with self-punitive harsh superegos and unconscious stress. Also, classic psychoanalytical theory hypothesizes that at least some people with alcohol use disorders may have become fixated at the oral stage of development and use alcohol to relieve their frustrations by taking the substance by mouth. Hypotheses regarding arrested phases of psychosexual development, although heuristically useful, have had little

effect on the usual treatment approaches and are not the focus of extensive ongoing research.

Similarly, most studies have not been able to document an "addictive personality" present in most individuals with alcohol use disorders and associated with a propensity to lack control of the intake of a wide range of substances and foods. Although patients often have pathologic scores on personality during intoxication, withdrawal, and early recovery, many of these characteristics predate alcoholism, and most disappear with abstinence. Similarly, prospective studies of children of alcoholics without any co-occurring psychiatric disorders show a high risk for alcohol use disorders. As elsewhere described, one exception is the extreme impulsivity seen in the 15 to 20 percent of alcoholic men with an antisocial personality disorder, because they have high risks for criminality, violence, and multiple substance use disorders.

Behavioral Theories. Expectations about the rewarding effects of drinking, cognitive attitudes toward responsibility for one's behavior, and subsequent reinforcement after alcohol intake all contribute to the decision to drink again after the first experience with alcohol and to continue to imbibe despite problems. These issues are critical in efforts to modify drinking behaviors in the general population, and they contribute to some important aspects of rehabilitation of alcohol use disorders.

Sociocultural Theories. Sociocultural theories often extrapolate from social groups that have high and low rates of alcohol use disorders. Theorists hypothesize that ethnic groups, such as Jews, who introduce children to modest levels of drinking during religious rituals and eschew drunkenness, have low rates of alcohol use disorders. However, other groups have similar practices, such as French Catholics, but have higher rates of alcohol use disorders. Recent studies have found little evidence to suggest that introducing children to modest drinking in the home impacts the risks for heavy drinking and alcohol problems. So, it is likely that cultural attitudes toward drinking, drunkenness, and personal responsibility for consequences are essential contributors to the rates of alcohol-related problems in a society. In the final analysis, social and psychological theories are probably highly relevant because they outline factors that contribute to the onset of drinking,

the development of temporary alcohol-related life difficulties, and even alcohol use disorders. The problem is how to gather relatively definitive data to support or refute the theories.

Childhood History. Researchers have identified several factors in the childhood histories of persons with later alcohol-related disorders and in children at high risk for having an alcohol-related disorder because one or both of their parents are affected. In experimental studies, children at high risk for alcohol-related disorders possess, on average, a range of deficits on neurocognitive testing. These deficits include a low amplitude of the P300 wave on evoked potential testing and a variety of abnormalities on electroencephalography (EEG) recordings. Studies of high-risk offspring in their 20s have also shown a generally blunted effect of alcohol compared with that seen in persons whose parents lacked an alcohol-related disorder. These findings suggest that a heritable biologic brain function may predispose a person to an alcohol-related disorder. A childhood history of attention-deficit/hyperactivity disorder (ADHD), conduct disorder, or both, increases a child's risk for an alcohol-related disorder as an adult. Personality disorders, especially antisocial personality disorder, as noted earlier, also predispose a person to an alcohol-related disorder.



Table 4-17
Evidence that Alcohol Has a Genetic Influence

Close relatives have a 3–4 times increased risk for severe alcohol problems
The rate of alcohol problems increases with:
The number of relatives with AUDs
The severity of their illness
The closeness of their genetic relationship to the person studied
Higher concordance for severe AUDs identical twins than fraternal twins in most investigations, which estimate that genes explain 40–60% of the variance
Adoption studies show increased risk for children of patients with AUDs even when separated from their biologic parents since birth

and raised without any contact
The risk is not enhanced when the adoptive parents have AUD
Animals studies support the role of genes

Genetic Factors

Table 4-17 lists some lines of evidence that support the conclusion that genes have an essential influence on alcoholism. Twin studies suggest that genes explain 60 percent of the variance, with the remainder relating to nonshared, probably adult environmental influences. Animal studies support the importance of a variety of yet-to-be-identified genes in the free-choice use of alcohol, subsequent levels of intoxication, and some consequences.

4.3 Cannabis-Related Disorders

Cannabis is the most widely used illegal drug in the world. Over the past 40 years, cannabis use has become a standard part of youth culture in most developed societies, with first use now occurring in the mid-to-late teens. In the United States, there is a movement to legalize cannabis at the state level, with some states allowing the use of cannabis for medical purposes only and others for recreational use.

CLINICAL FEATURES

Most young people use cannabis to experience a “high,” characterized by feelings of mild euphoria, relaxation, and perceptual alterations. Cognitive changes include impaired short-term memory and attention that makes it easy for the user to become lost in pleasant reverie and have difficulty sustaining goal-directed mental activity. Motor skills, reaction time, motor coordination, and many forms of skilled psychomotor activity are impaired while the user is intoxicated.

DIAGNOSIS

The DSM-5 and ICD-10 include several diagnoses related to cannabis, and

they generally follow the template for all substance use disorders (see Section 4.1 and Tables 4-1 through 4-3). The DSM-5 includes the diagnoses of cannabis intoxication, cannabis intoxication delirium, cannabis withdrawal, cannabis use disorder, cannabis-induced psychotic disorder, cannabis-induced anxiety disorder, cannabis-induced sleep disorder, and unspecified cannabis-related disorder. The ICD-10 includes cannabis-related disorders under the general heading of "Mental and behavioral disorders due to psychoactive substance use."

Cannabis Intoxication

Cannabis intoxication commonly heightens users' sensitivities to external stimuli and subjectively slows the appreciation of time. In high doses, users may experience depersonalization and derealization. Cannabis use impairs motor skills, and this effect remains after the euphoriant effects have resolved. For 8 to 12 hours after using cannabis, users' impaired motor skills interfere with the operation of motor vehicles and other heavy machinery. Moreover, these effects are additive to those of alcohol.

Mr. M was an unemployed 20-year-old man who lived with his parents. He was brought to a hospital by some friends in a state of anxiety and agitation. He had been out at a restaurant, and after a couple of beers, he decided to have some cannabis. He had smoked cannabis on previous occasions; however, this time, he ate a lump of cannabis despite warnings from his friends. After about half an hour, Mr. M appeared tense and anxious and complained that everything was changing. He could see the faces of his friends increasing to about three times their original size. The room became distorted, and its proportions and colors kept altering. He felt that the other guests in the restaurant were talking about him and his friends menacingly, so he suddenly rushed outside because he felt that he was in danger. He became increasingly agitated and started running down the middle of the street, dodging in and out among the traffic. Eventually, his friends were able to catch him. They were unable to quiet his anxiety, however, and had a hard time persuading him to go with them to the hospital.

On examination, Mr. M appeared tense and apprehensive, looking

around the room as if he felt uneasy with the surroundings. He denied perceptual symptoms and did not believe that he was the subject of persecution. He was fully aware of his surroundings, but his attention was fleeting, and he did not always answer questions. There was no marked impairment of memory, and he was fully oriented.

Physical examination revealed conjunctival injection and an increased pulse rate of 120 beats/min, but otherwise, there were no abnormalities. The neurologic examination also revealed no abnormalities. In the course of a few hours, Mr. M quieted down and eventually left the hospital with his friends.

Cannabis Intoxication Delirium. Cannabis intoxication can markedly impair cognition and performance. Even modest doses of cannabis impair memory, reaction time, perception, motor coordination, and attention. High doses that also impair users' levels of consciousness have marked effects on cognitive measures.

Cannabis Withdrawal

Studies have shown that cessation of use in daily cannabis users results in withdrawal symptoms within 1 to 2 weeks of cessation. Table 4-18 lists typical cannabis withdrawal symptoms.



Table 4-18
Cannabis Withdrawal Symptoms

Irritability
Cannabis cravings
Nervousness
Anxiety
Insomnia
Disturbed or vivid dreaming
Decreased appetite
Weight loss
Depressed mood

Restlessness
Headache
Chills
Stomach pain
Sweating
Tremors

Cannabis Use Disorder

People who use cannabis daily for weeks to months are most likely to develop a cannabis use disorder. The risk of developing cannabis use disorder is around one in ten for anyone who uses cannabis. The earlier the age of first use, the more often cannabis has been used, and the longer it is used, the higher the risk of developing the disorder.

Cannabis-Induced Psychotic Disorder

Cannabis-induced psychotic disorder, in which there is a genuine psychotic process, is rare; transient paranoid ideation, however, is common.

Florid psychosis is somewhat commonplace in countries in which some persons have long-term access to high potency cannabis. The psychotic episodes are sometimes called “hemp insanity.” Cannabis use rarely causes the “bad-trip” experience we associate with hallucinogen intoxication. When cannabis-induced psychotic disorder does occur, the affected person is likely to have a preexisting personality disorder.

Cannabis-Induced Anxiety Disorder

Cannabis-induced anxiety disorder is common during acute intoxication, which in many persons, induces short-lived anxiety states often provoked by paranoid thoughts. In such circumstances, ill-defined and disorganized fears may induce a panic attack. Anxiety symptoms are the most common adverse effect of moderate cannabis use and correlate with the dose taken. Inexperienced users are much more likely to experience anxiety symptoms than are experienced users.

A friend gave a 35-year-old married man who was naïve to cannabis two

"joints." The man smoked the first of the two in the same manner that he usually smoked a cigarette (in about 3 to 5 minutes). Noting no significant effects, he proceeded immediately to smoke the second in the same amount of time. Within 30 minutes, he began to experience rapid heartbeat, dry mouth, mounting anxiety, and the delusional belief that his throat was closing up and that he was going to die. That belief induced further panic, and the patient was brought to the emergency department of a local hospital during the experience. There, the emergency team could not reassure him that he would not die. After sedating him with diazepam, some of his anxiety diminished. He eventually went to sleep, and on awakening after 5 hours, he was asymptomatic with full recall of previous events.

Unspecified Cannabis-Related Disorders

DSM-5 includes the category unspecified cannabis-related disorders for cannabis disorders that do not fit into the other diagnoses. Examples could be episodes of depressive or hypomanic symptoms, although these symptoms may suggest long-term cannabis use.

When either sleep disorder or sexual dysfunction symptoms are related to cannabis use, they almost always resolve within days or a week after cessation.

Flashbacks. There are case reports of persons who have experienced sensations related to cannabis intoxication after the short-term effects of the substance have disappeared. Whether flashbacks are related to cannabis use alone or the concomitant use of other substances (intended or tainted) remains a subject of debate.

Cognitive Impairment. Clinical and experimental evidence indicates that the long-term use of cannabis may produce subtle forms of cognitive impairment in the higher cognitive functions of memory, attention, and organization and the integration of complex information. This evidence suggests that the longer the period of heavy cannabis use, the more pronounced the cognitive impairment. Nonetheless, because the impairments in performance are subtle, it remains to be determined how significant they

are for everyday functioning. It also remains to be investigated whether these impairments resolve after an extended period of abstinence from cannabis.

Amotivational Syndrome. A controversial cannabis-related syndrome is *amotivational syndrome.* Whether the syndrome is related to cannabis use or reflects characterologic traits in a subgroup of persons regardless of cannabis use is under debate. Traditionally, a person's unwillingness to persist in a task—be it at school, at work, or in any setting that requires prolonged attention or tenacity—characterizes the syndrome, which is associated with long-term heavy use. People with the syndrome become apathetic and anergic, may gain weight, or appear slothful.

COMORBIDITY

Cannabis is often referred to as a "gateway drug," meaning that cannabis users are at high risk for other substance use disorders. Also, cannabis use may be comorbid with depression, anxiety, conduct disorder, and suicidality.

TREATMENT AND REHABILITATION

Treatment of cannabis use rests on the same principles as the treatment of other substances of abuse—abstinence and support. Abstinence involves direct interventions, such as hospitalization, or careful monitoring on an outpatient basis by the use of urine drug screens, which can detect cannabis for up to 4 weeks after use. We can use individual, family, and group psychotherapies to support the patient. Education should be a cornerstone of both abstinence and support programs. A patient who does not understand the intellectual reasons for addressing a substance use disorder has little motivation to stop. For some patients, an antianxiety drug may be useful for short-term relief of withdrawal symptoms. For other patients, cannabis use may be due to an underlying depressive disorder that may respond to specific antidepressant treatment.

Medical Use of Marijuana

People have used marijuana as a medicinal herb for centuries, and cannabis was listed in the US Pharmacopeia until the end of the 19th century as a

remedy for anxiety, depression, and GI disorders, among others.

Cannabis is currently a controlled substance, and the Drug Enforcement Agency (DEA) does not recognize any medical use for the substance. Despite this, patients and doctors have used the drug, sometimes illegally, to treat a variety of disorders, including nausea secondary to chemotherapy, HIV-associated weight loss, multiple sclerosis (MS) chronic pain, epilepsy, and glaucoma. We currently live in a time of differing views and conflicting laws regarding the use of cannabis. The US federal government continues to consider the drug to be illegal under any circumstances; the Supreme Court confirmed this in 2001. However, as of 2020, 15 states have legalized the recreational and medicinal use of marijuana, along with the District of Columbia, and several US territories. Many other states have some degree of medical exemption or have decriminalized the drug. Table 4-19 lists the status of cannabis laws in the United States and territories as of 2020, however this remains a rapidly changing field.

Dronabinol, a synthetic form of THC, has been approved by the US Food and Drug Administration (FDA) for the treatment of anorexia-associated weight loss in HIV and nausea and vomiting associated with chemotherapy; dronabinol is also under investigation for the treatment of obstructive sleep apnea. Nabilone, which is a synthetic cannabinoid, has been approved for the treatment of nausea and vomiting associated with chemotherapy. In 2013, cannabidiol was granted orphan drug status for the treatment of certain rare, intractable types of epilepsy in children. Additionally, nabiximols, an oral spray consisting of natural cannabis extracts, is currently being investigated for the treatment of cancer pain. A 2019 study found that nabiximols reduced cannabis use among individuals with cannabis use disorder when used along with concurrent behavioral therapy. Nabiximols is currently available by prescription in several countries outside of the United States for patients with neuropathic pain, MS, and other conditions.

EPIDEMIOLOGY

Cannabis is the most widely used illegal drug in the United States, with an estimated 24 million users aged 12 and older in 2016 (approximately 9 percent of the population). Of those individuals, roughly 4 million met criteria for a marijuana use disorder in the past year.

Prevalence and Recent Trends

The *Monitoring the Future* survey of adolescents in school indicates that 36 percent of twelfth graders used marijuana in the past year, which is slightly lower than the most recent peak in the 1990s (39 percent in 1997). Annual marijuana use by eighth and tenth graders is significantly lower than the corresponding rates in the 1990s; in 2018, the prevalence of marijuana use was 28 percent in tenth graders (compared to 35 percent in 1997) and 11 percent in eighth-graders (compared to 18 percent in 1997).

Demographic Correlates

Rates of cannabis use in the user's lifetime, the past year, and the past week are consistently higher among males than females, as are daily use and long-term daily use.



Table 4-19
The Legal Status of Cannabis as of 2020

Illegal
Alabama
Idaho
Kansas
Nebraska[a]
North Carolina[a]
South Carolina
Tennessee
Wyoming

Fully Legal
Alaska
California
Colorado
District of Columbia
Illinois
Maine

Massachusetts
Michigan
Nevada
Oregon
Vermont
Washington
Arizona[b]
Montana[b]
New Jersey[b]
South Dakota[b]

Medicinal Use

Arkansas
Connecticut[a]
Delaware[a]
Florida
Hawaii[a]
Louisiana
Maryland[a]
Minnesota[a]
Missouri[a]
New Hampshire[a]
New Mexico[a]
New York[a]
North Dakota[a]
Ohio[a]
Oklahoma
Pennsylvania
Rhode Island[a]
Utah
West Virginia
Mississippi[a]

CBD Oil Only

Georgia
Indiana
Iowa
Kentucky
Texas
Virginia[a]
Wisconsin

[a]Decliminalized.

[b]As of 2020 election, enactment is pending as of this date.

Information on the relationship between ethnicity and cannabis use is limited. Even in the extensive Monitoring the Future survey, samples from several years have to be combined to make reliable comparisons between the three largest ethnic groups. These show that African American students have lower rates of use in all grades than white or Hispanic students.

PATHOLOGY

Cannabis Preparations

The plant *Cannabis sativa* is the source for cannabis preparations. It has been used in China, India, and the Middle East for approximately 8,000 years, primarily for its fibers and secondarily for its medicinal properties. The plant occurs in male and female forms. The female plant contains the highest concentrations of cannabinoids. Delta-9-tetrahydrocannabinol (Δ9-THC) is primarily responsible for the psychoactive effects of cannabis. The most potent forms of cannabis come from the flowering tops of the plants or from the dried, black-brown, resinous exudate from the leaves, called *hashish* or *hash*. The cannabis plant is usually cut, dried, chopped, and rolled into cigarettes (commonly called “joints”). The common names for cannabis are marijuana, grass, pot, weed, tea, and Mary Jane, although there are many other names. The potency of marijuana preparations has increased in recent years because of improved agricultural techniques used in cultivation so that plants may contain up to 15 or 20 percent THC.

Physical Effects of Cannabis Use

Table 4-20 lists the most common physical effects of cannabis use. There is no documented case of death caused by cannabis intoxication alone; this reflects the substance's lack of effect on the respiratory rate. The most serious potential adverse effects of cannabis use are those caused by inhaling the same carcinogenic hydrocarbons present in conventional tobacco, and some data indicate that heavy cannabis users are at risk for chronic respiratory disease and lung cancer. The practice of smoking cannabis-containing cigarettes to their very ends, so-called "roaches," further increases the intake of tar (particulate matter).

Many reports indicate that long-term cannabis use is associated with other serious effects, also listed in Table 4-20. These reports, however, have not been conclusively replicated, and the association between these findings and cannabis use is uncertain.



Table 4-20
Physical Effects of Cannabis Use

Common
Dilation of the conjunctival blood vessels (red eyes)
Mild tachycardia
Increased appetite
Dry mouth
Orthostatic hypotension (higher doses)
Heavy chronic use (some of these are controversial)
Chronic respiratory disease
Lung cancer
Cerebral atrophy
Seizure susceptibility
Chromosomal damage
Birth defects
Impaired immune reactivity
Alterations in testosterone concentrations
Dysregulation of menstrual cycles

Neuropharmacology

As stated above, the principal component of cannabis is Δ9-THC; however, the cannabis plant contains more than 400 chemicals, of which about 60 are chemically related to Δ9-THC. In humans, Δ9-THC rapidly converts to 11-hydroxy-Δ9-THC, the metabolite that is active in the CNS.

A specific receptor for the cannabinols has been identified, cloned, and characterized. The cannabinoid receptor, a member of the G-protein–linked family of receptors, is linked to the inhibitory G protein (G_i), which inhibits adenylyl cyclase. The highest concentrations of the cannabinoid receptor are in the basal ganglia, hippocampus, and cerebellum, with lower concentrations in the cerebral cortex. The brainstem lacks this receptor, explaining the drug's minimal effects on respiratory and cardiac functions. Studies in animals have shown that cannabinoids affect monoamine and GABA neurons.

According to most studies, animals do not self-administer cannabinoids as they do most other substances of abuse. It is not clear whether the cannabinoids stimulate the so-called reward centers of the brain, such as the dopaminergic neurons of the VTA. Tolerance to cannabis does develop, however, as does psychological dependence. The evidence for physiologic dependence is not as strong. Withdrawal symptoms in humans are limited to modest increases in irritability, restlessness, insomnia, and anorexia and mild nausea; all these symptoms appear only when a person abruptly stops taking high doses of cannabis.

When smoking cannabis, the euphoric effects appear within minutes, peak in about 30 minutes, and last 2 to 4 hours. Some motor and cognitive effects last 5 to 12 hours. Oral cannabis is also frequent, usually as an additive in baked goods, such as brownies and cakes. It requires about two to three times as much oral cannabis to be as potent as smoked cannabis. Many variables affect the psychoactive properties of cannabis, including the potency of the cannabis used, the route of administration, the smoking technique, the effects of pyrolysis on the cannabinoid content, the dose, the setting, and the user's experience, expectations, and unique biologic vulnerability to the effects of cannabinoids.

4.4 Opioid-Related Disorders

The word *opioid* describes a class of psychoactive compounds, both naturally occurring and chemically synthesized, that are related to opiates: alkaloid compounds found as natural products in the opium poppy plant, *Papaver somniferumi*.

Many synthetic opioids have been manufactured, including meperidine, methadone, pentazocine, and propoxyphene.

Humans have used opioids for analgesic and other medicinal purposes for thousands of years, but they also have a long history of misuse for their psychoactive effects. Continued opioid misuse can result in abuse and dependence and cause disturbances in mood, behavior, and cognition that can mimic other psychiatric disorders. In developed countries, the opioid drug most frequently associated with abuse and dependence is heroin; however, there is growing public health concern about prescription opioids, which are widely available and have significant abuse liability. Opioid addiction affects the young and the old, the wealthy and the poor, and the professional and the unemployed. Over the last few decades, there have been significant advances in treatment and understanding of opioid use disorder. Opioid use disorder is often a chronic, relapsing disorder amenable to medical treatment and intervention. Table 4-21 lists various opioids that are used therapeutically in the United States, except for heroin.



Table 4-21
Opioids Available in the United States

Proprietary Name	Trade Name
Morphine	
Heroin (diacetylmorphine)	
Hydromorphone (dihydromorphinone)	Dilaudid
Oxymorphone (dihydrohydroxymorphinone)	Numorphan

Levorphanol	Levo-Dromoran
Methadone	Dolophine
Meperidine (pethidine)	Demerol, Pethadol
Fentanyl	Sublimaze
Codeine	
Hydrocodone (dihydrocodeinone)	Hycodan, Others
Drocode (dihydrocodeine)	Synalgos-Dc, Compal
Oxycodone (dihydrohydroxycodeinone)	Roxicodone, Oxycontin, Percodan, Percocet, Vicodin
Propoxyphene	Darvon, Others
Buprenorphine	Buprenex (parenteral), Sublocade (subcutaneous, extended-release)
Pentazocine	Talwin
Nalbuphine	Nubain
Butorphanol	Stadol

CLINICAL FEATURES

Opioids can be taken orally, snorted intranasally, and injected intravenously or subcutaneously. Opioids are subjectively addictive because of the euphoric high (the rush) that users experience, especially those who take the substances intravenously. The associated symptoms include a feeling of warmth, heaviness of the extremities, dry mouth, itchy face (especially the nose), and facial flushing. A period of sedation follows the initial euphoria, known in street parlance as “nodding off.” Opioid use can induce dysphoria, nausea, and vomiting in opioid-naive persons.

Diagnosis

The DSM-5 and ICD-10 include several diagnoses related to opioids, and they generally follow the template for all substance use disorders (see Section 4.1 and Tables 4-1 through 4-3). These include opioid use disorder, opioid intoxication, opioid withdrawal, opioid-induced sleep disorder, and opioid-induced sexual dysfunction. Opioid intoxication delirium can occur in hospitalized patients. Opioid-induced psychotic disorder, opioid-induced mood disorder, and opioid-induced anxiety disorder, by contrast, are uncommon with μ-agonist opioids but can occur with some mixed agonist–antagonist opioids acting at other receptors. The diagnosis of opioid-related disorder not elsewhere classified is for situations that do not meet the criteria for any of the other opioid-related disorders.

Opioid Use Disorder

Opioid use disorder is a pattern of maladaptive use of an opioid drug, leading to clinically significant impairment or distress and occurring within 12 months.

A surgeon referred a 42-year-old public relations executive to psychiatry after discovering the executive sneaking large quantities of a codeine-containing cough medicine into the hospital. The patient had been a heavy cigarette smoker for 20 years and had a chronic, hacking cough. He had come into the hospital for a hernia repair and found the pain from the incision unbearable when he coughed.

A back operation 5 years previously had led his doctors to prescribe codeine to help relieve the incisional pain at that time. Over the intervening 5 years, however, the patient had continued to use codeine-containing tablets and had increased his intake to 60 to 90 mg daily. He stated that he often “just took them by the handful—not to feel good, you understand, just to get by.” He spent considerable time and effort developing a circle of physicians and pharmacists to whom he would “make the rounds” at least three times a week to obtain new supplies of pills. He had tried several times to stop using codeine but had failed. During this period, he lost two jobs because of lax work habits, and his wife of 11 years divorced him.

Opioid Intoxication

Opioid intoxication includes maladaptive behavioral changes and specific physical symptoms of opioid use. In general, altered mood, psychomotor retardation, drowsiness, slurred speech, and impaired memory and attention in the presence of other indicators of recent opioid use strongly suggest a diagnosis of opioid intoxication.

Opioid Withdrawal

Opioid withdrawal develops after the cessation of or reduction in opioid use that has been heavy and prolonged. The withdrawal syndrome also occurs when administering an opioid partial agonist such as buprenorphine or an opioid antagonist such as naloxone or naltrexone to a patient who is regularly taking an opioid agonist. The withdrawal symptoms are usually the physiologic opposite of the intoxication symptoms. The withdrawal syndrome consists of severe muscle cramps and bone aches, profuse diarrhea, abdominal cramps, rhinorrhea, lacrimation, piloerection or gooseflesh (from which derives the term *cold turkey* for the abstinence syndrome), yawning, fever, pupillary dilation, hypertension, tachycardia, and temperature dysregulation, including hypothermia and hyperthermia. Persons with opioid dependence seldom die from opioid withdrawal unless they have a severe preexisting physical illness such as cardiac disease. Residual symptoms—such as insomnia, bradycardia, temperature dysregulation, and a craving for opioids—can persist for months after withdrawal. Associated features of opioid withdrawal include restlessness, irritability, depression, tremor, weakness, nausea, and vomiting. At any time during the abstinence syndrome, a single injection of morphine or heroin eliminates all the symptoms.

The general rule about the onset and duration of withdrawal symptoms is that substances with short durations of action tend to produce short, intense withdrawal syndromes, and substances with long durations of action produce prolonged but mild withdrawal syndromes. An exception to the rule, narcotic antagonist-precipitated withdrawal after long-acting opioid misuse can be severe. See Section 4.1, Table 4-3 for the diagnostic criteria for substance withdrawal.

When an opioid antagonist precipitates an abstinence syndrome, the

symptoms begin within seconds of such an intravenous injection and peak in about 1 hour. Opioid craving rarely occurs in the context of analgesic administration for pain from physical disorders or surgery. The full withdrawal syndrome, including an intense craving for opioids, usually occurs only secondary to abrupt cessation of use in persons with longstanding opioid misuse.

Morphine and Heroin. The morphine and heroin withdrawal syndrome begin 6 to 8 hours after the last dose, usually after a 1- to 2-week period of continuous use or after the administration of a narcotic antagonist. The withdrawal syndrome reaches its peak intensity during the second or third day and subsides during the next 7 to 10 days, but some symptoms may persist for 6 months or longer.

Methadone. Methadone withdrawal usually begins 1 to 3 days after the last dose and ends in 10 to 14 days.

Opioid Intoxication Delirium. Opioid intoxication delirium is most likely to happen when using opioids in high doses, or mixing them with other psychoactive compounds, or when a person with preexisting CNS disorder (e.g., brain damage or epilepsy) uses them.

Opioid-Induced Psychotic Disorder. The opioid-induced psychotic disorder can begin during opioid intoxication. Clinicians can specify whether hallucinations or delusions are the predominant symptoms.

Opioid-Induced Mood Disorder. An opioid-induced mood disorder can begin during opioid intoxication. Opioid-induced mood disorder symptoms can have a manic, depressed, or mixed nature, depending on a person's response to opioids. A person who comes to psychiatric attention with opioid-induced mood disorder usually has mixed symptoms, combining irritability, expansiveness, and depression.

Opioid-Induced Sleep Disorder and Opioid-Induced Sexual Dysfunction. Opioid-induced sleep disorder and opioid-induced sexual dysfunction are common, and both hypersomnia and insomnia are common complaints of patients who either have opioid use disorder or use opioids

regularly for therapeutic purposes. Common sexual dysfunctions for chronic opioid users include erectile dysfunction and orgasmic difficulties.

Unspecified Opioid-Related Disorder. The DSM-5 includes diagnoses for other opioid-related disorders with symptoms of delirium, abnormal mood, psychosis, abnormal sleep, and sexual dysfunction. Clinical situations that do not fit into these categories exemplify appropriate cases for the use of the DSM-5 diagnosis of an unspecified opioid-related disorder.

COMORBIDITY

Nearly 70 percent of men and 75 percent of women with opioid use disorder in their lifetimes have an additional psychiatric disorder. The most common comorbid disorders include mood disorders, antisocial personality disorder, anxiety disorders, and alcohol use disorder. Table 4-22 highlights the need to develop a broad-based treatment program that also addresses patients' associated psychiatric disorders.

In addition to the morbidity and mortality associated directly with the opioid-related disorders, the association between the transmission of both hepatitis and the HIV with intravenous opioid and opiate use is recognized as a leading national health concern.

TREATMENT AND REHABILITATION

Methadone is the current gold standard in the treatment of opioid use disorder with physiologic dependence. Methadone is used for detoxification and as a maintenance agent.

The opioid partial agonist buprenorphine binds to the receptor but activates it only partially. Due to buprenorphine's high affinity (binding strength) to the receptor and a low dissociation from the receptor, it often functions as an antagonist (blocker), preventing compounds with lower receptor affinity from activating the receptor. If a patient is actively using an agonist at the time they receive a dose of buprenorphine, the buprenorphine will abruptly displace the agonist from the receptor. Since the buprenorphine only has a partial effect at the receptor, it will cause a sudden drop in the level of agonist effect, resulting in a precipitated withdrawal. Buprenorphine is used to assist detoxification and as a maintenance agent.

Opioid antagonists (e.g., naloxone, nalmefene, and naltrexone) bind to the receptor but do not exert any activity. These medications function as antagonists—they compete for binding at the receptor with agonists and partial agonists, and if the dose of the antagonist is high enough, they may displace the agonist from the receptor and prevent the agonist from binding to the receptor and exerting any effect. Naloxone reverses an overdose; as it is a short-acting agent, it can be dosed very accurately without the risk of an unnecessarily prolonged effect. Naltrexone is a long-acting agent used following detoxification as a strategy to prevent recurrent physiologic dependence and relapse.

Overdose Treatment

The first task in overdose treatment is to ensure an adequate airway. Tracheopharyngeal secretions should be aspirated, and it may be necessary to insert an airway. The patient should be ventilated mechanically until the clinicians can give naloxone. Naloxone is administered intravenously at a slow rate—initially about 0.8 mg per 70 kg of body weight. Signs of improvement (increased respiratory rate and pupillary dilation) should occur promptly. In patients physiologically dependent on opioids, too much naloxone may produce signs of withdrawal. If no response to the initial dosage occurs, clinicians can repeat the naloxone administration after intervals of a few minutes. If there is no response after 4 to 5 mg, some think that nonopioid substances may be contributing to the CNS depression. The duration of action of naloxone is short compared with that of many opioids, such as methadone and levomethadyl acetate, and repeated administration may be required to prevent recurrence of opioid toxicity.



Table 4-22
Substance and Non–Substance-Related Psychiatric Disorders in Opioid Users

Mood, Anxiety, and Personality Disorders Among Individuals with Lifetime Opioid Use Disorders in NESARC by Gender				
(*N* = 355)	Men (%) (*N* = 223)	Women (%) (*N* = 578)	Total (%)	Odds Ratio (95% Confidence Interval)
Heroin/other opioids				
Abuse	75.6	67.5	73	0.67 (0.43, 1.04)
Dependence	24.4	32.5	27	1.49 (0.96, 2.31)
Cocaine				
Abuse	37.6	17	30.9	0.34 (0.20, 0.58)***
Dependence	17.4	20.5	18.4	1.22 (0.73, 2.03)
Marijuana				
Abuse	56.2	39.3	50.7	0.51 (0.33, 0.78)**
Dependence	19.4	19.9	19.6	1.03 (0.62, 1.72)
Alcohol				
Abuse	22.6	20.4	21.8	0.88 (0.55, 1.41)
Dependence	67.1	54.5	63	0.59 (0.40, 0.86)**
Sedatives				
Abuse	37.7	23.3	33	0.50 (0.30, 0.85)*
Dependence	9.1	12.9	10.3	1.48 (0.77, 2.87)
Amphetamine				
Abuse	35.9	28	33.3	0.70 (0.45, 1.08)
Dependence	13.3	18.5	15	1.47 (0.80, 2.72)
Hallucinogen				
Abuse	37.4	27.5	34.2	0.63 (0.40, 1.01)
Dependence	9	9.5	9.1	1.06 (0.51, 2.20)
Tranquilizers				
Abuse	37.7	22.7	32.9	0.49 (0.30, 0.78)**
Dependence	8.7	12.4	9.9	1.49 (0.80, 2.81)
Inhalants				
Abuse	13	6.7	10.9	0.48 (0.24, 0.95)*
Dependence	1.7	2.1	1.9	1.22 (0.30, 4.99)
Any other substance				
Abuse	23.1	21.6	22.6	0.92 (0.60, 1.41)
Dependence	72.4	62.3	69.1	0.63 (0.43, 0.91)*
Any mood or anxiety disorder	68.5	73.8	70.3	1.30 (0.84, 2.00)
Any mood disorder	54.1	71.2	59.7	2.10 (1.35, 3.26)**
Major depression	45.7	63.5	51.5	2.07 (1.38, 3.12)**
Dysthymia	15.3	25.3	18.6	1.88 (1.20, 2.93)**
Manic disorder	16.7	27.1	20.1	1.86 (1.17, 2.94)*
Hypomanic disorder	9	10.9	9.6	1.24 (0.61, 2.50)
Any anxiety disorder	33.6	50.1	39	1.98 (1.35, 2.92)***
Panic disorder w/o agoraphobia	13.2	18.3	14.9	1.46 (0.88, 2.45)
Panic disorder w/ agoraphobia	4.1	6.6	4.9	1.65 (0.63, 4.33)
Social phobia	9.3	19.8	12.7	2.42 (1.36, 4.30)**
Specific phobia	12.9	30.1	18.5	2.91 (1.69, 5.02)***
Generalized anxiety disorder	9.7	17.7	12.3	2.01 (1.13, 3.56)*
Any personality disorder	49.2	50.7	49.7	1.06 (0.74, 1.52)
Paranoid	12.9	22.2	15.9	1.93 (1.17, 3.21)*
Avoidant	9.5	12.4	10.4	1.35 (0.71, 2.57)
Dependent	3	2.3	2.8	0.77 (0.25, 2.37)
Obsessive compulsive	16	23	18.3	1.56 (0.95, 2.57)
Schizoid	11.5	12.8	11.9	1.13 (0.61, 2.11)
Histrionic	9.8	8.5	9.4	0.86 (0.42, 1.76)
Antisocial	34.1	22.4	30.3	0.56 (0.36, 0.87)*

Asterisks indicate significance level for *p*-values. **indicates $p \leq 0.05$; ***indicates $p \leq 0.01$; ****indicates $p \leq 0.001$.
Adapted from Grella CE, Karno MP, Warda US, Niv N, Moore AA. Gender and comorbidity among individuals with opioid use disorders in the NESARC study. *Addict Behav*. 2009;34(6–7):498–504.

Medically Supervised Withdrawal and Detoxification

Opioid Agents for Treating Opioid Withdrawal

METHADONE. Methadone is a synthetic narcotic (an opioid) that substitutes for heroin. When given to addicts to replace their typical substance of abuse, the drug suppresses withdrawal symptoms. The initial dose of methadone is usually 20 to 30 mg orally; if objective signs of withdrawal persist after the first dose, clinicians can repeat the dose after approximately 2 to 4 hours. As a general rule, initial stabilization does not require more than 40 mg during the first 24 hours, although it can be higher in the inpatient setting if clear signs of moderate or severe withdrawal are present after the first 6 to 10 hours of the observation period. After 24 to 48 hours of dose stabilization, the patient's methadone dose can be gradually reduced by 10 to 20 percent or by 5 mg/day. Most patients can complete inpatient detoxification within 1 week.

In the case of detoxification with methadone in an outpatient setting, the first-day dose should not exceed 40 mg. Over the next few days, clinicians may titrate the dose (generally by no more than 10-mg increments every 2 to 3 days) until the patient has no discernible withdrawal symptoms. In most cases, the dose of 40 to 60 mg/day is sufficient to prevent any opioid withdrawal; however, this dose may not be adequate to eliminate craving. If the patient continues to use opioids, the dose of methadone can increase. In that case, clinicians should postpone detoxification and the goal becomes stabilization—no withdrawal symptoms, craving, or opioid use. Generally, a period of dose stabilization, at a minimum of 4 weeks in duration, is necessary before a slow dose reduction schedule. In the outpatient setting, gradual dose reductions (e.g., 3 percent per week), have a higher likelihood of success, with even slower reductions once the total daily dose is down to 20 to 30 mg/day.

BUPRENORPHINE. As with methadone, buprenorphine is an opioid agonist used for treating opioid withdrawal. Physicians can dispense it on an outpatient basis, but they must demonstrate that they have received specialized training in its use. Buprenorphine is available in several formulations, including as monotherapy or combination therapy with buprenorphine plus naloxone. The monotherapy versions include a parenteral formulation and an extended-release subcutaneous injection. The combination product is preferred, as it carries a lower risk of diversion and misuse.

Buprenorphine can be used for detoxification in an inpatient setting under

direct monitoring. The first dose of buprenorphine begins when the patient experiences at least moderate severity of withdrawal, with several, preferably objective, signs present. The first dose of buprenorphine is generally 2 mg, and subsequent 2-mg doses are given every 2 to 4 hours until the successful elimination of withdrawal. Usually, the total daily dose needed to suppress withdrawal on the first day effectively is 8 to 12 mg. Buprenorphine taper is usually started on the second day, while the daily dose is decreased by 2 mg every day, with 1 mg decreases on the last 2 days of taper. The complete taper usually takes 5 to 7 days, and the long elimination half-life of buprenorphine (31 to 35 hours) helps to extend the effect after discontinuation.

Buprenorphine can also assist detoxification in the outpatient setting. The strategy is generally similar to the methadone-assisted detoxification. Ideally, the first day's dosing should occur under medical supervision to minimize the risk of buprenorphine-precipitated withdrawal. The physician usually instructs patients to arrive in withdrawal, which typically starts 12 hours after the last dose of a short-acting opioid or 24 hours after a long-acting opioid like methadone. The first dose of buprenorphine is usually 2 mg, and if there are no signs of precipitated withdrawal, subsequent 2-mg doses are given at the clinic every 2 hours, with additional doses given to take at home, until the elimination of withdrawal. If withdrawal worsens after the first buprenorphine dose, it is best to wait for 4 to 6 hours before giving the next buprenorphine dose while the precipitated withdrawal can be treated symptomatically with medications such as clonidine. Usually, the total daily dose needed on the first day is 8 to 12 mg. Clinicians can use subsequent daily dose increases of 2 to 4 mg to relieve opioid withdrawal and initiate opioid abstinence. Once establishing a period of stabilization, slow dose reductions can begin, usually in 2-mg dose increments and usually at weekly intervals. As with methadone, slower dosage reduction schedules are usually better tolerated and less likely to prompt relapse.



Table 4-23
Nonopioid Medications Used to Treat Withdrawal

Target Symptom	Medication Class	Example

Autonomic symptoms (e.g., sweating, restlessness, tremor, and rhinorrhea)	α_2-agonist	Clonidine
Anxiety and psychomotor agitation	Benzodiazepines (usually reserved for the inpatient setting)	Lorazepam
Insomnia	Hypnotics	Zolpidem, zopiclone
	Sedating antidepressants	Trazodone
	Atypical antipsychotics	Quetiapine
Musculoskeletal pain	Nonsteroidal anti-inflammatory drugs	Ibuprofen
	Antispasmodic agents	Cyclobenzaprine
GI distress and diarrhea	Antiemetic agents	Prochlorperazine
	Peripheral μ-opioid receptor agonists	Loperamide

Symptomatic Treatment of Opioid Withdrawal. It is also possible to treat an opioid withdrawal with nonopioid medications, primarily targeting several uncomfortable withdrawal symptoms. Table 4-23 lists some of these medications and their uses.

Medications for the Treatment of Opioid Use Disorder

For some individuals, opioid detoxification is not successful, and others may not feel ready for detoxification, although they may desire freedom from the daily struggle associated with opioid use and the associated psychosocial problems. In these individuals, opioid substitution therapy with either methadone or buprenorphine may be appropriate. For patients who can complete opioid detoxification, antagonist therapy can help them to maintain

abstinence.

Opioid Substitution Therapy. Opioid substitution therapy has several advantages. First, it frees persons with opioid use disorder from using injectable heroin or other illicit opioids and, thus, reduces the chance of spreading HIV or hepatitis through contaminated needles. Second, opioid agonists produce minimal euphoria and rarely cause drowsiness or depression when taken for a long time. Third, opioid agonists allow patients to engage in gainful employment instead of criminal activity. The major disadvantage of opioid agonist use is that patients remain dependent on a narcotic.

METHADONE. Meta-analytical studies show methadone's clear superiority over treatments without medication on improving treatment retention and reducing heroin or other illicit opioid use. The majority of patients treated in methadone programs also show decreased nonopioid drug use, criminal behavior, symptoms of depression, and increased gainful employment.

The effectiveness of opioid maintenance treatment depends on the dose and the quality of the additional services provided. The optimal dose is generally in the 80 to 120 mg/day range, with some patients requiring higher doses to achieve an optimal clinical response. Among the most common side effects of methadone are constipation, excessive sweating, decreased libido, and sexual dysfunction. In the United States, methadone can be dispensed only at an outpatient opioid treatment program (OTP) certified by SAMHSA and registered with the DEA or to a hospitalized patient in an emergency. SAMHSA-certified OTP facilities provide daily doses of methadone under direct supervision until the patient is stable enough to receive take-home doses.

BUPRENORPHINE/NALOXONE COMBINATION. Buprenorphine is a high-affinity partial μ-opioid receptor agonist, κ-receptor antagonist, and ORL-1 agonist. At lower doses of up to 16 mg, buprenorphine has a dose-dependent opiate-like effect, but at higher doses, there is progressively less increase in opiate-like effects. There is a ceiling to the μ-opioid agonist effects of buprenorphine, which limits respiratory depression and other adverse effects, increasing its safety profile. Buprenorphine has poor bioavailability and is usually administered sublingually as a tablet or as a film.

Naloxone is mixed with the buprenorphine in a four to one ratio; this is to decrease the risk of abuse. Because naloxone has poor sublingual bioavailability, it exerts minimal effect when taking the preparation as prescribed, but if it is dissolved and injected by an opioid-dependent person, it blocks some of the effects of buprenorphine and may precipitate withdrawal.

Treatment with buprenorphine is best suited for individuals who present with at least mild withdrawal symptoms. Unlike methadone, there is little concern about overdose during treatment initiation. However, combining buprenorphine with benzodiazepines or other sedative–hypnotics can produce excessive sedation, and there are case reports of fatalities associated with parenteral misuse of buprenorphine and a benzodiazepine. Common side effects of buprenorphine include constipation, nausea, headache, upset stomach, excessive sweating, somnolence, and decreased interest in sex. In the United States, to prescribe buprenorphine, physicians must complete a training course and receive a waiver granted by the DEA.

OPIOID ANTAGONISTS. Opioid antagonists block or antagonize the effects of opioids. Unlike methadone, they do not exert narcotic effects and do not cause dependence. Opioid antagonists include naloxone, which is used in the treatment of opioid overdose because it reverses the effects of narcotics, and naltrexone, the longest-acting (72 hours) antagonist. The theory for using an antagonist for opioid-related disorders is that blocking opioid agonist effects, particularly euphoria, discourages persons with physiologic dependence on opioids from substance-seeking behavior and, thus, deconditions this behavior. The major weakness of the antagonist treatment model is the lack of any mechanism that compels a person to continue to take the antagonist. Naltrexone can be taken orally or in a long-acting injectable formulation.

Psychotherapy

The entire range of psychotherapeutic modalities is appropriate for treating opioid-related disorders. Individual psychotherapy, behavioral therapy, CBT, family therapy, support groups (e.g., Narcotics Anonymous [NA]), and social skills training may all prove useful for specific patients. Patients with few social skills should receive social skills training. Most patients living with

family members should also receive family therapy.

Therapeutic Communities

Therapeutic communities are residences in which all members have a substance abuse problem.

Abstinence is the rule; to be admitted to such a community, a person must show a high level of motivation. The goals are to effect a complete change of lifestyle, including abstinence from substances; to develop personal honesty, responsibility, and useful social skills; and to eliminate antisocial attitudes and criminal behavior.

Education and Needle Exchange

Although the essential treatment of opioid use disorders is encouraging persons to abstain from opioids, education about the transmission of HIV must receive equal attention. It is essential to teach persons with opioid use disorders who use intravenous or subcutaneous routes of administration safe-sex practices. Free needle-exchange programs are often subject to intense political and societal pressures but, where allowed, should be made available to persons with opioid use disorders. Several studies have indicated that unsafe needle sharing is common when it is difficult to obtain enough clean needles and is also common in persons with legal difficulties, severe substance problems, and psychiatric symptoms.

Narcotic Anonymous

NA is a self-help group of abstinent drug addicts modeled on the 12-step principles of AA. Such groups now exist in most large cities and can provide useful group support. Although the scientific evidence supportive of the usefulness of such groups has been very limited, observational studies support its effectiveness, and therefore, patients receiving pharmacologic treatments should be encouraged to participate in the self-help groups.

Pregnant Women with Opioid Use Disorders

Illicit opioid use during pregnancy is associated with adverse effects for both the pregnant woman and the fetus but is not associated with any specific teratogenic effects. However, there is a higher risk of infections,

preeclampsia, miscarriage, premature rupture of membranes, and premature labor as well as stillbirths, premature infants, and infants with low birth weight and neonatal opioid withdrawal (also called the neonatal abstinence syndrome). Infants who survive may be infected with HIV and other diseases as a result of maternal high-risk behavior and are at risk of sudden infant death syndrome (SIDS). Most of these risks can be substantially reduced with methadone or buprenorphine maintenance treatment.

EPIDEMIOLOGY

The use and dependence rates derived from national surveys do not accurately reflect fluctuations in drug use among opioid-dependent and previously opioid-dependent populations. When the supply of illicit heroin increases in purity or decreases in price, use among that vulnerable population tends to increase, with subsequent increases in adverse consequences (emergency department visits) and requests for treatment. The number of current (past month) heroin users in the United States is about 475,000 in those ages 12 and up. That number grows to 3.3 million, or approximately 1.2 percent of the population, for current (past month) pain relievers. The majority (53 percent) of individuals who misuse pain relievers report obtaining them from friends or family. When considering all opioid use over the past year, the number is staggering: 11.8 million individuals 12 years old and older used opioids in the past year, representing approximately 4.4 percent of the population.

PATHOLOGY

Effects of Opioids

The physical effects of opioids include respiratory depression, pupillary constriction, smooth muscle contraction (including the ureters and the bile ducts), constipation, and changes in blood pressure, heart rate, and body temperature. The brainstem mediates the respiratory depressant effects.

Adverse Effects

The most common and most serious adverse effect associated with opioid-

related disorders is the potential transmission of hepatitis and HIV through the use of contaminated needles by more than one person. Persons can experience idiosyncratic allergic reactions to opioids, which result in anaphylactic shock, pulmonary edema, and death if they do not receive prompt and adequate treatment. Another serious adverse effect is an idiosyncratic drug interaction between meperidine and monoamine oxidase inhibitors (MAOIs), which can produce gross autonomic instability, severe behavioral agitation, coma, seizures, and death. Opioids and MAOIs should not be given together for this reason.

Opioid Overdose

Death from an overdose of an opioid is usually attributable to respiratory arrest from the respiratory depressant effect of the drug. The symptoms of overdose include marked unresponsiveness, coma, slow respiration, hypothermia, hypotension, and bradycardia. When presented with the clinical triad of coma, pinpoint pupils, and respiratory depression, clinicians should consider opioid overdose as a primary diagnosis. They can also inspect the patient's body for needle tracks in the arms, legs, ankles, groin, and even the dorsal vein of the penis.

Neuropharmacology

The primary effects of the opioid drugs are mediated via the opioid receptors, discovered in the early 1970s. The μ-opioid receptors are involved in the regulation and mediation of analgesia, respiratory depression, constipation, and drug dependence; the κ-opioid receptors, with analgesia, diuresis, and sedation; and the Δ-opioid receptors, with analgesia.

In 1975, the enkephalins, two endogenous pentapeptides with opioid-like actions, were identified. This discovery led to the identification of three classes of endogenous opioids within the brain. These include the endorphins, the dynorphins, and the enkephalins. Eric Simon was one of the discoverers, and he coined the term "endorphin," which was a contraction of "endogenous" and "morphine." Endorphins are involved in neural transmission and pain suppression. They are released naturally in the body when a person is physically hurt or severely stressed, and they may account for the absence of pain during acute injuries.

The endogenous opioids also have significant interactions with other neuronal systems, such as the dopaminergic and noradrenergic neurotransmitter systems. Several types of data indicate that the VTA dopaminergic neurons (which project to the cerebral cortex and the limbic system) mediate the rewarding properties of opioids.

Heroin, the most commonly abused opioid, is more lipid-soluble than morphine. It crosses the blood–brain barrier faster and has a more rapid and pleasurable onset than morphine. First introduced as a treatment for morphine addiction, heroin is more dependence-producing than morphine. Codeine, which occurs naturally as about 0.5 percent of the opiate alkaloids in opium, is absorbed quickly through the GI tract and is subsequently transformed into morphine in the body. Results of at least one study using positron emission tomography (PET) have suggested that one effect of all opioids is decreased cerebral blood flow (CBF) in selected brain regions in persons with opioid dependence. There is compelling evidence indicating that the endorphins are involved in other addictions, such as alcoholism, cocaine, and cannabinoid addiction. The opioid antagonist, naltrexone, has shown value in mitigating alcohol addiction.

Tolerance and Dependence. Tolerance to all actions of opioid drugs does not develop uniformly. Tolerance to some actions of opioids can be so high that a 100-fold increase in dose is required to produce the original effect. For example, terminally ill cancer patients may need 200 to 300 mg a day of morphine, whereas a dose of 60 mg can easily be fatal to an opioid-naive person. The symptoms of opioid withdrawal do not appear unless a person has been using opioids for a long time or when cessation is particularly abrupt, as occurs when a person takes an opioid antagonist. The long-term use of opioids results in changes in the number and sensitivity of opioid receptors, which mediate at least some of the effects of tolerance and withdrawal. Although long-term use is associated with increased sensitivity of dopaminergic, cholinergic, and serotonergic neurons, the effect of opioids on the noradrenergic neurons is probably the primary mediator of the symptoms of opioid withdrawal. Short-term use of opioids appears to decrease the activity of the noradrenergic neurons in the locus ceruleus; long-term use activates a compensatory homeostatic mechanism within the neurons, and opioid withdrawal results in rebound hyperactivity. This

hypothesis also explains why clonidine, an α_2-adrenergic receptor agonist that decreases the release of norepinephrine, is useful in the treatment of opioid withdrawal symptoms.

ETIOLOGY

Psychosocial Factors

Social attitudes, peer pressure, individual temperament, and drug availability all predispose people to problematic opioid use. Social stigma is less with prescription opioids than heroin, and prescription opioids are believed to be safer in comparison to heroin and are more widely available. As prescription opioid users develop some functional impairments, their opioid use often transitions to heroin and the IV route of administration due to lower cost and increasing tolerance for slower-acting agents. The presence of a significant number of opioid users in a geographic area can also create a subculture of opioid experimentation. In the United States, areas in which the prevalence of opioid use disorder is high also often have associated high crime rates, unemployment rates, and poorly performing school systems. While correlation does not equal causation, these factors can all be conceptualized as contributing to reduced resistance to opioid use and may worsen the prognosis and treatment efficacy. Most urban heroin users are children of single parents or divorced parents and are from families in which at least one other member has a substance-related disorder. Children from such settings are at high risk for opioid use disorder, especially if they also evidence behavioral problems in school or other signs of conduct disorder.

Biologic and Genetic Factors

Evidence now exists for common and drug-specific, genetically transmitted vulnerability factors that increase the likelihood of developing drug dependence. Individuals who abuse a substance from any category are more likely to abuse substances from other categories. Twin studies estimate that around 50 to 60 percent of the liability for heroin addiction is genetic.

Ongoing linkage and genome-wide association studies continue to investigate promising targets for understanding the genetics and genomics of opioid use disorder.

▲ 4.5 Sedative-, Hypnotic-, or Anxiolytic-Related Disorders

Anxiolytic or sedative–hypnotic drugs are on a continuum with their hypnotic or sleep-inducing effects. In addition to their psychiatric indications, clinicians also use these drugs as antiepileptics, muscle relaxants, anesthetics, and anesthetic adjuvants. Alcohol and all drugs of this class are cross-tolerant, and their effects are additive. Physical and psychological dependence develops to these drugs, and all are associated with withdrawal symptoms. In the practice of psychiatry and addiction medicine, the drug class that is most important clinically is the benzodiazepines.

The three major groups of drugs associated with this class of substance-related disorders are benzodiazepines, barbiturates, and barbiturate-like substances. We discuss each below.

Clinical Features

Specific Agents

BENZODIAZEPINES. Many benzodiazepines, differing primarily in their half-lives, are available in the United States. Examples of benzodiazepines are diazepam, flurazepam, oxazepam, and chlordiazepoxide. Benzodiazepines are used primarily as anxiolytics, hypnotics, antiepileptics, and anesthetics, as well as for alcohol withdrawal. After their introduction in the United States in the 1960s, benzodiazepines rapidly became the most prescribed drugs; about 15 percent of all persons in the United States have had a benzodiazepine prescribed by a physician. Increasing awareness of the risks for dependence on benzodiazepines and increased regulatory requirements, however, have decreased the number of benzodiazepine prescriptions. The DEA classifies all benzodiazepines as Schedule IV controlled substances.

Flunitrazepam, a benzodiazepine used in Mexico, South America, and Europe but not available in the United States, has become a drug of abuse. When taken with alcohol, it may lead to promiscuous sexual behavior and rape. It is illegal to bring flunitrazepam into the United States. Although

misused in the United States, it remains a standard anxiolytic in many countries.

Nonbenzodiazepine sedatives such as zolpidem, zaleplon, and eszopiclone —the so-called Z drugs—have clinical effects similar to the benzodiazepines and are also subject to misuse and dependence.

BARBITURATES. Barbiturates were popular before the introduction of benzodiazepines, but because of their high abuse potential, their use is much rarer today. Secobarbital was popular in the 1960s through the 1980s. It acquired many nicknames, including “reds,” “red devils,” “seggies,” and “downers,” among many others. Pentobarbital was also popular, and its nicknames included “yellow jackets,” “yellows,” and “nembies.” A secobarbital–amobarbital combination was also widely available, and was called “reds and blues,” “rainbows,” “double-trouble,” and “tooies.” Pentobarbital, secobarbital, and amobarbital are now under the same federal legal controls as morphine.

The first barbiturate, barbital, was introduced in the United States in 1903. Barbital and phenobarbital, introduced shortly after that, are long-acting drugs with half-lives of 12 to 24 hours. Amobarbital is an intermediate-acting barbiturate with a half-life of 6 to 12 hours. Pentobarbital and secobarbital are short-acting barbiturates with half-lives of 3 to 6 hours. Although barbiturates are useful and effective sedatives, they are highly lethal, with only ten times the standard dose producing coma and death.

BARBITURATE-LIKE SUBSTANCES. The most commonly abused barbiturate-like substance is methaqualone. Various manufacturers produced it in the United States until being discontinued in 1985. It is often used by young persons who believe that the substance heightens the pleasure of sexual activity. The street names for methaqualone include “mandrakes” (from the United Kingdom preparation Mandrax) and “soapers” (from the brand name Sopor). “Luding out” (from the brand name Quaalude) means getting high on methaqualone, often along with alcohol.

Other barbiturate-like substances include meprobamate, a carbamate derivative that has weak efficacy as an antianxiety agent but has muscle-relaxant effects, chloral hydrate, a hypnotic that is highly toxic to the GI system and when combined with alcohol, is known as a “Mickey Finn”; and

ethchlorvynol, a rapidly acting sedative agent with anticonvulsant and muscle-relaxant properties. All are subject to abuse.

Patterns of Abuse

Oral Use. Sedatives and hypnotics are usually taken orally, either occasionally to achieve a time-limited specific effect or regularly to obtain a constant, usually mild, intoxication state. The occasional use pattern is associated with young persons who take the substance to achieve specific effects—relaxation for an evening, intensification of sexual activities, and a short-lived period of mild euphoria. The user's personality and expectations about the substance's effects, as well as the setting, also affect the substance-induced experience. The regular use pattern is associated with middle-aged, middle-class persons who usually obtain the substance from a family physician as a prescription for insomnia or anxiety. Abusers of this type may have prescriptions from several physicians, and the pattern of abuse may go undetected until apparent signs of a substance use disorder are noticed by the person's family, coworkers, or physicians.

Intravenous Use. A severe form of misuse involves the intravenous use of this class of substances. The users are mainly young adults who are intimately involved with illegal substances. Intravenous barbiturate use is associated with a pleasant, warm, drowsy feeling, and users may be inclined to use barbiturates more than opioids because barbiturates are less costly. The physical dangers of injection include transmission of HIV, cellulitis, vascular complications from accidental injection into an artery, infections, and allergic reactions to contaminants. Intravenous use is associated with rapid and profound tolerance and dependence and severe withdrawal syndrome.

Overdose

Benzodiazepines. In contrast to the barbiturates and the barbiturate-like substances, the benzodiazepines have a large margin of safety when taken in overdoses, a feature that has contributed significantly to their rapid acceptance. The ratio of the lethal dose to effective dose is about 200 to 1 or higher, because of the minimal degree of respiratory depression associated

with the benzodiazepines. Table 4-24 lists the equivalent therapeutic doses of benzodiazepines. Even when users take grossly excessive amounts (more than 2 g) in suicide attempts, the symptoms include only drowsiness, lethargy, ataxia, some confusion, and mild depression of the user's vital signs. The most lethal overdoses occur when the user takes a benzodiazepine in combination with other sedatives, such as alcohol. In such cases, small doses of benzodiazepines can cause death. The availability of flumazenil, a specific benzodiazepine antagonist, has reduced the lethality of the benzodiazepines. Flumazenil can be used in emergency departments to reverse the effects of benzodiazepines.



Table 4-24
Approximate Therapeutic Equivalent Doses of Benzodiazepines

Generic Name	Trade Name	Dose (mg)
Alprazolam	Xanax	1
Chlordiazepoxide	Librium	25
Clonazepam	Klonopin	0.5–1.0
Clorazepate	Tranxene	15
Diazepam	Valium	10
Estazolam	Prosom	1
Flurazepam	Dalmane	30
Lorazepam	Ativan	2
Oxazepam	Serax	30
Temazepam	Restoril	20
Triazolam	Halcion	0.25
Quazepam	Doral	15
Zolpidem	Ambien	10

Zaleplon	Sonata	10

Barbiturates. Barbiturates are lethal when taken in an overdose because they induce respiratory depression. In addition to intentional suicide attempts, accidental or unintentional overdoses are common. Barbiturates in home medicine cabinets are a common cause of fatal drug overdoses in children. As with benzodiazepines, the lethal effects of the barbiturates are additive to those of other sedatives or hypnotics, including alcohol and benzodiazepines. Barbiturate overdose can induce coma, respiratory arrest, cardiovascular failure, and death.

The lethal dose varies with the route of administration and the degree of tolerance for the substance. For the most commonly abused barbiturates, the ratio of the lethal dose to effective dose ranges between 3:1 and 30:1. Dependent users often take an average daily dose of 1.5 g of a short-acting barbiturate, and some take as much as 2.5 g a day for months.

The lethal dose is not much higher for the long-term abuser than for the neophyte. Tolerance develops quickly, to the point at which withdrawal in a hospital becomes necessary to prevent accidental death from overdose.

Barbiturate-Like Substances. The barbiturate-like substances vary in their lethality and are usually intermediate between the relative safety of the benzodiazepines and the high lethality of the barbiturates. An overdose of methaqualone, for example, can result in restlessness, delirium, hypertonia, muscle spasms, convulsions, and, in very high doses, death. Unlike barbiturates, methaqualone rarely causes severe cardiovascular or respiratory depression, and most fatalities result from combining methaqualone with alcohol.

DIAGNOSIS

Sedative, Hypnotic, or Anxiolytic Use Disorder

The DSM-5 and ICD-10 include several diagnoses related to sedatives, and they generally follow the template for all substance use disorders (see Section 4.1 and Tables 4-1 through 4-3).

Sedative, Hypnotic, or Anxiolytic Intoxication

The intoxication syndromes induced by all these drugs are similar. Table 4-25 lists some of the common symptoms. Blood toxicology is the best way to confirm the diagnosis.

Benzodiazepines. Behavioral disinhibition is a typical result of benzodiazepine intoxication. It can potentially result in hostile or aggressive behavior in users, most commonly when they also take alcohol. Benzodiazepine intoxication is associated with less euphoria than is intoxication by other drugs in this class. This characteristic is the basis for the lower abuse and dependence potential of benzodiazepines than of barbiturates.

Barbiturates and Barbiturate-Like Substances. The clinical syndrome of barbiturate (and barbiturate-like substance) intoxication is indistinguishable from alcohol intoxication, at least in lower doses. Table 4-25 lists both symptoms common to all sedative–hypnotics as well as additional symptoms associated with barbiturates. Sluggishness usually resolves after a few hours, but, depending primarily on the half-life of the abused substance, the effects on judgment, mood, and motor skills may remain for 12 to 24 hours.



Table 4-25
Sedative–Hypnotic Intoxication Symptoms

Common symptoms
Incoordination
Dysarthria
Nystagmus
Impaired memory
Gait disturbance
Severe cases: stupor, coma, or death
Barbiturate intoxication (additional symptoms)
Sluggishness
Slow speech

Slow comprehension
Faulty judgment
Disinhibited sexual aggressive impulses
Narrowed range of attention
Emotional lability
Exaggerated basic personality traits
Hostility
Argumentativeness
Moroseness
Paranoid and suicidal ideation

Neurologic symptoms: nystagmus, diplopia, strabismus, ataxic gait, positive Romberg sign, hypotonia, and decreased superficial reflexes

Sedative, Hypnotic, or Anxiolytic Withdrawal

Benzodiazepines. The severity of the withdrawal syndrome associated with the benzodiazepines varies with the average dose and the duration of use, but a mild withdrawal syndrome can follow even short-term use of relatively low doses of benzodiazepines. A significant withdrawal syndrome is likely to occur at the cessation of dosages in the range of 40 mg a day for diazepam, for example, although 10 to 20 mg a day, taken for a month, can also result in a withdrawal syndrome when stopping it. The onset of withdrawal symptoms usually occurs 2 to 3 days after the cessation of use, but with long-acting drugs, such as diazepam, the latency before onset can be 5 or 6 days. The symptoms include anxiety, dysphoria, intolerance for bright lights and loud noises, nausea, sweating, muscle twitching, and sometimes seizures (generally at dosages of 50 mg a day or more of diazepam). Table 4-26 lists the signs and symptoms of benzodiazepine withdrawal.

Barbiturates and Barbiturate-Like Substances. The withdrawal syndrome for barbiturate and barbiturate-like substances range from mild symptoms (e.g., anxiety, weakness, sweating, and insomnia) to severe symptoms (e.g., seizures, delirium, cardiovascular collapse, and death).

Persons who have been abusing phenobarbital in the range of 400 mg a day may experience mild withdrawal symptoms; those who have been abusing the substance in the range of 800 mg a day can experience orthostatic hypotension, weakness, tremor, and severe anxiety. About 75 percent of these persons have withdrawal-related seizures. Users of dosages higher than 800 mg a day may experience anorexia, delirium, hallucinations, and repeated seizures.

Most symptoms appear in the first 3 days of abstinence, and seizures generally occur on the second or third day, when the symptoms are worst. If seizures do occur, they always precede the development of delirium. The symptoms rarely occur more than a week after stopping the substance. A psychotic disorder, if it develops, starts on the third to the eighth day. The various associated symptoms generally run their course within 2 to 3 days but can last as long as 2 weeks.



Table 4-26
Signs and Symptoms of the Benzodiazepine Discontinuation Syndrome

The following signs and symptoms may be seen when benzodiazepine therapy is discontinued; they reflect the return of the original anxiety symptoms (recurrence), worsening of the original anxiety symptoms (rebound), or emergence of new symptoms (true withdrawal):

- *Disturbances of mood and cognition*

anxiety, apprehension, dysphoria, pessimism, irritability, obsessive rumination, and paranoid ideation

- *Disturbances of sleep*

insomnia, altered sleep–wake cycle, and daytime drowsiness

- *Physical signs and symptoms*

tachycardia, elevated blood pressure, hyperreflexia, muscle tension, agitation/motor restlessness, tremor, myoclonus, muscle and joint pain, nausea, coryza, diaphoresis, ataxia, tinnitus, and grand mal seizures

- *Perceptual disturbances*

hyperacusis, depersonalization, blurred vision, illusions, and hallucinations

Other Sedative-, Hypnotic-, or Anxiolytic-Induced Disorders

Delirium. Delirium that is indistinguishable from DTs associated with alcohol withdrawal is seen more commonly with barbiturate withdrawal than with benzodiazepine withdrawal. Delirium associated with intoxication can be seen with either barbiturates or benzodiazepines if the dosages are sufficiently high.

Neurocognitive Disorders. The diagnosis of this disorder is complex because it is difficult to know whether a neurocognitive disorder is due to the substance use itself or associated features of substance use.

Psychotic Disorders. The psychotic symptoms of barbiturate withdrawal can be indistinguishable from those of alcohol-associated DTs. Agitation, delusions, and hallucinations are usually visual, but sometimes tactile or auditory features develop after about 1 week of abstinence. Psychotic symptoms associated with intoxication or withdrawal are more common with barbiturates than with benzodiazepines. DSM-5 diagnoses them as sedative, hypnotic, or anxiolytic withdrawal with perceptual disturbances when reality testing is intact (the individual is aware the drug is causing the psychotic symptoms). If reality testing is not intact (the individual believes the hallucinations are real), a diagnosis of substance/medication-induced psychotic disorder is more appropriate. Clinicians can further specify whether delusions or hallucinations are the predominant symptoms, including the type (e.g., auditory, visual, or tactile).

Other Disorders. Sedative, hypnotic, and anxiolytic use may also cause mood disorders, anxiety disorders, sleep disorders, and sexual dysfunctions.

Unspecified Sedative-, Hypnotic-, or Anxiolytic-Related Disorder. When none of the previously discussed diagnostic categories is appropriate for a person with sedative-, hypnotic-, or anxiolytic-related disorder, and he or she does not meet the diagnostic criteria for any general substance-related

disorder, the appropriate diagnosis is an unspecified sedative-, hypnotic-, or anxiolytic-related disorder.

TREATMENT AND REHABILITATION

Withdrawal

Benzodiazepines. Because some benzodiazepines are eliminated from the body slowly, symptoms of withdrawal can continue to develop for several weeks. To prevent seizures and other withdrawal symptoms, clinicians should gradually reduce the dosage. Table 4-27 lists guidelines for treating benzodiazepine withdrawal.

Barbiturates. To avoid sudden death during barbiturate withdrawal, clinicians must follow conservative clinical guidelines. Clinicians should not give barbiturates to a comatose or grossly intoxicated patient. A clinician should attempt to determine a patient's usual daily dose of barbiturates and then verify the dosage clinically. For example, a clinician can give a test dose of 200 mg of pentobarbital every hour until a mild intoxication occurs, but withdrawal symptoms are absent (Table 4-28). The clinician can then taper the total daily dose at a rate of about 10 percent of the total daily dose. Once the correct dosage is determined, the clinician can use a long-acting barbiturate for the detoxification period. During this process, the patient may begin to experience withdrawal symptoms, in which case the clinician should halve the daily decrement.

Table 4-27
Guidelines for Treatment of Benzodiazepine Withdrawal

1. Evaluate and treat concomitant medical and psychiatric conditions.
2. Obtain drug history and urine and blood samples for drug and ethanol assay.
3. Determine required dose of benzodiazepine or barbiturate for stabilization, guided by history, clinical presentation, drug–ethanol assay, and (in some cases) challenge dose.
4. Detoxification from supratherapeutic dosages:

a. Hospitalize if there are medical or psychiatric indications, poor social supports, or polysubstance dependence, or the patient is unreliable.
b. Some clinicians recommend switching to longer-acting benzodiazepine for withdrawal (e.g., diazepam, clonazepam); others recommend stabilizing on the drug that patient was taking or on phenobarbital.
c. After stabilization reduce dosage by 30% on the second or third day and evaluate the response, keeping in mind that symptoms that occur after decreases in benzodiazepines with short elimination half-lives (e.g., lorazepam) appear sooner than with those with longer elimination half-lives (e.g., diazepam).
d. Reduce dosage further by 10–25% every few days if tolerated.
e. Use adjunctive medications if necessary—carbamazepine, β-adrenergic receptor antagonists, valproate, clonidine, and sedative antidepressants have been used but their efficacy in the treatment of the benzodiazepine abstinence syndrome has not been established.

5. Detoxification from therapeutic dosages:
a. Initiate 10–25% dose reduction and evaluate response.
b. Dose, duration of therapy, and severity of anxiety influence the rate of taper and need for adjunctive medications.
c. Most patients taking therapeutic doses have uncomplicated discontinuation.

6. Psychological interventions may assist patients in detoxification from benzodiazepines and in the long-term management of anxiety.

Courtesy of Domenici A. Ciraulo, M.D., and Ofra Sarid-Segal, M.D.



Table 4-28
Pentobarbital Test Dose Procedure for Barbiturate Withdrawal

Symptoms After Test Dose of 200 mg Oral	Estimated 24-Hr Oral	Estimated 24-Hr Oral

Pentobarbital	Pentobarbital Dose (mg)	Phenobarbital Dose (mg)
Level I: asleep but arousable; withdrawal symptoms not likely	0	0
Level II: mild sedation; patient may have slurred speech, ataxia, nystagmus	500–600	150–200
Level III: patient is comfortable: no evidence of sedation; may have nystagmus	800	250
Level IV: no drug effect	1,000–1,200	300–600

In the withdrawal procedure, phenobarbital can substitute for the more commonly abused short-acting barbiturates. The effects of phenobarbital last longer, and because barbiturate blood levels fluctuate less, phenobarbital does not cause observable toxic signs or a severe overdose. An adequate dose is 30 mg of phenobarbital for every 100 mg of the short-acting substance. Clinicians should maintain the user for at least 2 days at that level before reducing the dose further. The regimen is analogous to the substitution of methadone for heroin.

After withdrawal is complete, the patient must overcome the desire to start retaking the substance. Although clinicians sometimes try to substitute a nonbarbiturate sedative or hypnotic for a barbiturate, this often replaces one substance dependence with another. If a user is to remain substance-free, follow-up treatment, usually with psychiatric help and community support, is vital. Otherwise, a patient will almost certainly return to barbiturates or a substance with similar hazards.

EPIDEMIOLOGY

According to the NSDUH, approximately 500,000 individuals aged 12 and older used sedatives in the past month. Persons older than 26 years have the

highest use. In 2016, more than 600,000 individuals met the criteria for a tranquilizer use disorder in the past year. Benzodiazepines are probably not abused as frequently as other substances to get “high,” or inducing a euphoric feeling. Instead, people use them to experience a general relaxed feeling.

PATHOLOGY

Neuropharmacology

The benzodiazepines, barbiturates, and barbiturate-like substances all have their primary effects on the $GABA_A$ receptor complex, which contains a chloride ion channel, a binding site for GABA, and a well-defined binding site for benzodiazepines. The barbiturates and barbiturate-like substances probably bind somewhere on the $GABA_A$ receptor complex. When one of these substances binds to the complex, the effect is to increase the affinity of the receptor for its endogenous neurotransmitter, GABA. GABA then increases the flow of chloride ions through the channel into the neuron. The influx of negatively charged chloride ions into the neuron is inhibitory and hyperpolarizes the neuron relative to the extracellular space.

Although all the substances in this class induce tolerance and physical dependence, we best understand the mechanisms for the benzodiazepines. Long-term benzodiazepine use attenuates the usual agonist effects on the receptor. Specifically, GABA stimulation of the $GABA_A$ receptors results in less chloride influx than was caused by GABA stimulation before the benzodiazepine administration. This downregulation of receptor response is not caused by a decrease in receptor number or by a decreased affinity of the receptor for GABA. The basis for the downregulation seems to be in the coupling between the GABA binding site and the activation of the chloride ion channel. This decreased efficiency in coupling may be regulated within the $GABA_A$ receptor complex itself or by other neuronal mechanisms.

Misuse Liability and Other Risks of Long-Term Use

Surveys of prescribing practices, patient-initiated dosage changes, and recreational or nonmedical benzodiazepine use along with reports from emergency departments, medical examiners, and law enforcement agencies

provide a complex picture of the public health implications of the benzodiazepines. The vast majority of medical and psychiatric patients use benzodiazepines appropriately, although rates of misuse by patients dependent on alcohol and other drugs may be higher than individuals with anxiety or insomnia without a history of substance use problems. Furthermore, long-term use is high among patients with psychiatric disorders and the elderly, with the latter group being sensitive to drug toxicity.

Also, long-term use of benzodiazepines may increase the risk of developing dementia. A recent meta-analysis found that this risk is higher for those taking benzodiazepines with a longer half-life and for those who have been taking benzodiazepines for a longer duration (more than 3 years), although other data suggest that there may not be an increased risk for dementia.

4.6 Stimulant-Related Disorders

CLINICAL FEATURES

Types of Stimulants

Amphetamines. The racemic amphetamine sulfate (Benzedrine) was first synthesized in 1887, and Smith, Kline, and French marketed it as a decongestant in the early 1930s. In 1930, benzedrine sulfate was introduced for the treatment of narcolepsy, postencephalitic parkinsonism, depression, and lethargy. In the 1970s, a variety of social and regulatory factors began to curb widespread amphetamine distribution. The current US FDA–approved indications for amphetamine are limited to ADHD; however, they are used for many symptoms, some of which we list in Table 4-29.

Table 4-29
Some Indications for Amphetamines

FDA approved
Attention-deficit/hyperactivity disorder

Narcolepsy

Off label uses
Obesity
Depression and dysthymia
Chronic fatigue syndrome
Acquired immunodeficiency syndrome (AIDS)
Cancer-related fatigue
End-of-life care (depressive symptoms)
Dementia
Multiple sclerosis
Fibromyalgia
Neurasthenia

PREPARATIONS. The main amphetamines currently available and used in the United States are dextroamphetamine, methamphetamine, a mixed dextroamphetamine-amphetamine salt, and the amphetamine-like compound methylphenidate. These drugs go by such street names as ice, crystal, crystal meth, and speed. As a general class, the amphetamines also called analeptics, sympathomimetics, stimulants, and psychostimulants. The typical amphetamines are used to increase performance and to induce a euphoric feeling, for example, by students studying for examinations, by long-distance truck drivers on trips, by business people with pressing deadlines, by athletes in competition, and by soldiers during wartime. Although not as addictive as cocaine, amphetamines are nonetheless addictive drugs.

Other amphetamine-like substances are ephedrine, pseudoephedrine, and phenylpropanolamine (PPA). These drugs, PPA in particular, can dangerously exacerbate hypertension, precipitate a toxic psychosis, cause an intestinal infarction, or result in death. The safety margin for PPA is particularly narrow, and three to four times the standard dose can result in life-threatening hypertension. In 2005, the FDA recalled medications containing PPA, and in 2006, they prohibited the sale of over-the-counter (OTC) medications containing ephedrine and regulated the sale of OTC medications containing pseudoephedrine, which was used illegally to make methamphetamine.

Amphetamine-type drugs with abuse potential also include phendimetrazine (which is Schedule II of the Controlled Substance Act [CSA]), and diethylpropion, benzphetamine, and phentermine (which are Schedules III or IV of the CSA). All of these drugs, most likely, are capable of causing all of the listed amphetamine-induced disorders. Modafinil, used in the treatment of narcolepsy, also has stimulant and euphorigenic effects in humans, but its toxicity and likelihood of producing amphetamine-induced disorders are unknown.

Methamphetamine is a potent form of amphetamine that abusers of the substance inhale, smoke, or inject intravenously. Its psychological effects last for hours and are particularly powerful. Unlike cocaine (see discussion later in this section), which is imported, methamphetamine is a synthetic drug that can be manufactured domestically in illicit laboratories.

Other agents called *substituted,* or *designer amphetamines* are discussed separately later in this section.

Cocaine. People have used cocaine in its raw form for more than 15 centuries. In the United States, cycles of widespread stimulant misuse and associated problems have occurred for more than 100 years. Cocaine and cocaine use disorders became a major public health issue in the 1980s when an epidemic of use spread throughout the country. Due to education and intervention, cocaine use has since declined. However, high rates of legal, psychiatric, medical, and social problems related to cocaine use still exist; thus, cocaine-related disorders remain a significant public health issue.

Cocaine is an alkaloid derived from the shrub *Erythroxylum coca,* which is indigenous to South America, where the leaves of the shrub are chewed by local inhabitants to obtain the stimulating effects. The cocaine alkaloid was first isolated in 1855 and first used as a local anesthetic in 1880. It continues to be used, especially for eye, nose, and throat surgery, for which its vasoconstrictive and analgesic effects are helpful. In 1884, Sigmund Freud made a study of cocaine's general pharmacologic effects, and, for some time, according to his biographers, he was addicted to the drug. In the 1880s and 1890s, cocaine was widely touted as a cure for many ills, and the 1899 *Merck Manual* listed it. It was the active ingredient in the beverage Coca-Cola until 1903. In 1914, as its addictive effects became known, cocaine was classified as a narcotic, alongside morphine and heroin.

METHODS OF USE. Because drug dealers often dilute cocaine powder with sugar or procaine, street cocaine varies greatly in purity. Amphetamines are occasionally combined with cocaine. The most common method of using cocaine is inhaling the finely chopped powder into the nose; a practice referred to as “snorting” or “tooting.” Other methods of ingesting cocaine are subcutaneous or intravenous injection and smoking (freebasing). Freebasing involves mixing street cocaine with chemically extracted pure cocaine alkaloid (the freebase) to get an increased effect. Smoking is also the method used to ingest crack cocaine. Inhaling is the least dangerous method of cocaine use; intravenous injection and smoking are the most dangerous. The most direct methods of ingestion are often associated with cerebrovascular diseases, cardiac abnormalities, and death. Although users can take cocaine orally, they rarely do this as it is the least effective route.

Crack. Crack, a freebase form of cocaine, is extremely potent. Dealers sell it in small, ready-to-smoke amounts, often called “rocks.” Crack cocaine is highly addictive; even one or two experiences with the drug can cause an intense craving for more. Users may resort to extremes of behavior to obtain the money to buy more crack. Reports from urban emergency departments have also associated extremes of violence with crack abuse.

Substituted Amphetamines. MDMA (3,4-methylenedioxymethamphetamine) is one of a series of substituted amphetamines that also includes MDEA, MDA (3,4-methylenedioxyamphetamine), DOB (2,5-dimethoxy-4-bromoamphetamine), PMA (paramethoxyamphetamine), and others. These drugs produce subjective effects resembling those of amphetamine and lysergic acid diethylamide (LSD), and in that sense, MDMA and similar analogs may represent a distinct category of drugs.

A methamphetamine derivative that came into use in the 1980s, MDMA was not technically subject to legal regulation at the time. Although a considered “designer drug,” meaning synthesized to evade legal regulation, in reality, it was synthesized and patented in 1914. At one time, it was legal and used as an aid for psychotherapy. However, the FDA never approved the drug. Its use raised questions of both safety and legality, because the related amphetamine derivatives MDA, DOB, and PMA had caused several overdose

deaths, and MDA was known to cause extensive destruction of serotonergic nerve terminals in the CNS. Using emergency scheduling authority, the DEA made MDMA a Schedule I drug under the CSA, along with LSD, heroin, and marijuana. Despite its illegal status, MDMA continues to be manufactured, distributed, and used in the United States, Europe, and Australia. Its use is frequent at extended dances ("raves") popular with adolescents and young adults.

Currently, no established clinical uses exist for MDMA, although before its regulation, there were several reports of its beneficial effects as an adjunct to psychotherapy.

After taking usual doses (100 to 150 mg), MDMA users experience an elevated mood and, according to various reports, increased self-confidence and sensory sensitivity, peaceful feelings coupled with insight, empathy, and closeness to persons, and decreased appetite. The effect on concentration varies with both increased and decreased concentration reported. Users also report dysphoric reactions, psychotomimetic effects, and even psychosis. Higher doses seem more likely to produce psychotomimetic effects. Sympathomimetic effects of tachycardia, palpitation, increased blood pressure, sweating, and bruxism are common. The subjective effects are prominent for about 4 to 8 hours, but that can vary widely depending on the dose and route of administration. The drug is usually taken orally but is also snorted and injected. Users report both tachyphylaxis and some tolerance.

Although not as toxic as MDA, MDMA can cause various toxicities as well as fatal overdoses. It does not appear to be neurotoxic when injected into the brains of animals, but humans and animals do metabolize it to MDA. In animals, MDMA produces selective, long-lasting damage to serotonergic nerve terminals. It is not clear whether the levels of the MDA metabolite reached in humans after the usual doses of MDMA suffice to produce lasting damage.

"Bath Salts". *Catha edulis*, or "Khat," is a plant indigenous to Eastern Africa and Southern Arabia. It releases a variety of psychoactive chemicals, including the stimulants cathinone and cathine. Khat is chewed like tobacco and held between the cheek and gums after chewing. The National Institute on Drug Abuse estimates that as many as 10 million people worldwide use Khat, including many in Eastern Africa and the Middle East, as well as an

unknown number of immigrants to the United States.

In the mid-2000s, synthetic cathinones began to appear, marketed as “legal high” chemicals. They were packaged under labels including “screen cleaner,” “jewelry cleaner,” and most notoriously “bath salts.” Produced as powders (generally marked “not for human consumption”), users can ingest them by oral, intranasal, inhaled (smoked) or intravenous routes. These compounds are substituted cathinones that are chemically modified to avoid regulatory control. There are a large number of specific compounds in the class, and a given package often contains a mix of agents, further increasing the risk of toxicity. Generally, substituted cathinones increase synaptic catecholamine levels by inhibiting the dopamine, serotonin, and norepinephrine reuptake transporters in a fashion similar to cocaine. Most also facilitate catecholamine release, similar to methamphetamine. Animal models indicate strong reinforcing properties of these compounds, suggesting that they are highly addictive.

“Club Drugs”. *Club drugs* is a generic term for drugs associated with dance clubs, bars, and all-night dance parties (raves). They include LSD, γ-hydroxybutyrate (GHB), ketamine, methamphetamine, MDMA (ecstasy), and Rohypnol or “roofies” (flunitrazepam). These represent various classes of drugs, with different chemical structures and a range of physical and subjective effects. GHB, ketamine, and Rohypnol are sometimes called *date rape drugs* because they produce disorienting and sedating effects, and often users cannot recall what occurred during all or part of an episode under the influence of the drug. These drugs may be given surreptitiously, such as slipped into a beverage, or a person might be convinced to take the drug, not understanding its effect.

Emergency department mentions of GHB, ketamine, and Rohypnol are relatively few. Of the club drugs, methamphetamine is the substance that accounts for the largest share of treatment admissions.

DIAGNOSIS

Stimulant Use Disorder

The DSM-5 and ICD-10 include several diagnoses related to stimulant use

disorder, and they generally follow the template for all substance use disorders (see Section 4.1 and Tables 4-1 through 4-3).

Amphetamine dependence can result in a rapid downward spiral of a person's abilities to cope with work- and family-related obligations and stresses. A person who abuses amphetamines requires increasingly high doses of amphetamine to obtain the usual high, and physical signs of chronic amphetamine use (e.g., decreased weight and paranoid ideas) almost always develop with continued use.

Mr. H, a 35-year-old married man, was admitted to a psychiatric hospital because he felt persecuted by gang members who were out to kill him. He could not explain why they wished to kill him, but he heard voices from people whom he suspected to be mob drug dealers, and they were discussing that they should kill him. He used methamphetamine for several years, so he had dealt with drug dealers before. He began using at age 27 at the persuasion of a friend to try it. After injecting 20 mg, he felt euphoric and powerful, and his sleepiness and fatigue disappeared. After a few tries, Mr. H found that he could not stop using it. He always thought about how he would obtain the drug and started increasing the dosage he used. During times that he could not get methamphetamine, he felt lethargic and sleepy and became irritable and dysphoric. Mr. H's wife learned of his drug use and attempted to persuade him to stop using it. He lost his job 2 months before his admission as he was repeatedly abusive to work colleagues because he felt that they were trying to harm him. With no income, Mr. H had to cut down his use of methamphetamine to only occasional usage. He finally decided to quit when his wife threatened to divorce him. Once he stopped using, he felt fatigued, seemed gloomy, and often sat in his favorite chair and did nothing. After a few weeks, Mr. H told his wife that he did not wish to leave the house because he had heard dealers on the street talking about him. He wanted all doors and windows locked, and he refused to eat in fear that the people were poisoning his food.

On examination, Mr. H seemed withdrawn, only giving short answers to questions. He was in clear consciousness and fully oriented and showed no marked impairment of cognitive functions. Physical and

neurologic testing showed no abnormalities except needle scars on his arms from methamphetamine injections. An EEG was normal.

Clinically and practically, clinicians may suspect cocaine use disorder in patients with unexplained changes in personality. Frequent changes associated with cocaine use are irritability, impaired ability to concentrate, compulsive behavior, severe insomnia, and weight loss. Colleagues at work and family members may notice a person's general and increasing inability to perform the expected tasks associated with work and family life. The patient may show new evidence of increased debt or inability to pay bills on time because of the vast sums used to buy cocaine. Individuals with cocaine use disorder often excuse themselves from work or social situations every 30 to 60 minutes to find a secluded place to inhale more cocaine. Because of the vasoconstricting effects of cocaine, users almost always develop nasal congestion, which they may attempt to self-medicate with decongestant sprays.

Mr. D, a 45-year-old married man, was referred by his therapist to a private outpatient substance use treatment program for evaluation and treatment of a possible cocaine problem. According to the therapist, Mr. D's wife expressed concern for a possible substance use problem on several occasions. A few days prior, Mr. D admitted to the therapist and his wife that he "occasionally" used cocaine for the past year. His wife insisted that he obtain treatment for his drug problem or else she would file for divorce. Mr. D reluctantly conceded to treatment but insisted that his cocaine use was not a problem and that he felt capable of stopping without entering a treatment program.

During the initial evaluation interview, Mr. D reported that he currently used cocaine, intranasally, 3 to 5 days a week and that this pattern has been continuing for a year and a half. On average, he consumes a total of 1 to 2 g of cocaine weekly. He mostly uses cocaine at work, in his office, or the bathroom. He usually started thinking about cocaine during his drive to work in the morning and once at work, was unable to avoid thinking about the cocaine in his desk drawer. Despite his attempts at distraction and postponing use, he usually takes his first line

of cocaine within an hour of arriving at work. On some days, he will take another two to three lines during the day, but, on days where he is frustrated and stressed, he may take a line or two every hour from morning until late afternoon. He rarely uses cocaine at home and never uses it in front of his wife or his three daughters. He occasionally takes a line or two during a weekday evening or weekends at home when everyone else is out of the house. He denies using alcohol or any other illicit drugs. He denies any history of alcohol or drug abuse and any history of emotional or marital problems.

Stimulant Intoxication

The diagnostic criteria for stimulant intoxication emphasize behavioral and physical signs and symptoms of stimulant use (Table 4-30). Persons use stimulants for their characteristic effects of elation, euphoria, heightened self-esteem, and perceived improvement on mental and physical tasks. With high doses, symptoms of intoxication include agitation, irritability, impaired judgment, impulsive and potentially dangerous sexual behavior, aggression, a generalized increase in psychomotor activity, and potentially, symptoms of mania. The significant associated physical symptoms are tachycardia, hypertension, and mydriasis.



Table 4-30
Signs and Symptoms of Stimulant Intoxication

- Mydriasis
- Psychomotor agitation or retardation
- Tachycardia or bradycardia
- Perspiration or chills
- Cardiac arrhythmias or chest pain
- Elevated or lowered blood pressure
- Dyskinesias
- Dystonias
- Weight loss
- Nausea or vomiting

- Muscular weakness
- Respiratory depression
- Confusion, seizures, or coma

Mrs. T, a 38-year-old married businesswoman, was admitted to the psychiatric service after a 3-month period in which she became increasingly mistrustful of others and suspicious of business associates. She took statements from others out of context, twisting their words and making inappropriately hostile and accusatory comments. On one occasion, Mrs. T physically attacked a coworker in a bar, accusing the coworker of having an affair with her husband and plotting with other coworkers to kill her.

One year previously, Mrs. T's physician prescribed her methylphenidate for narcolepsy due to daily irresistible sleep attacks and episodes of a sudden loss of muscle tone when she became emotionally excited. After taking the medication, Mrs. T became asymptomatic and was able to work effectively and have an active social life with family and friends.

In the 5 months before admission, Mrs. T had been using increasingly large doses of methylphenidate to maintain alertness late at night because of an increased amount of work that she could not complete during the day. She reported that during this time, she often could feel her heart racing and that she had trouble sitting still.

Mr. P, an 18-year-old man, was brought to a hospital emergency department via ambulance in the middle of the night. He was accompanied by a friend who called the ambulance as he felt that Mr. P was going to die. Mr. P was agitated and argumentative, his breathing was irregular and rapid, his pulse was rapid, and he had dilated pupils. His friend eventually admitted that they had used much cocaine that evening.

When his mother arrived at the hospital, Mr. P's condition had somewhat improved, although his loud singing created a commotion in

the emergency department. His mother states that Mr. P has some disciplinary problems; he is disobedient, resentful, and violently argumentative. He was arrested on a few occasions for shoplifting and for driving while intoxicated. His mother suspected that Mr. P was using drugs due to his behavior and because she heard him talk to his friends about drugs. However, she has no direct proof of his use.

Within 24 hours, Mr. P was well and willing to talk. He boastfully stated that he had been using alcohol and various drugs regularly since he was 13. It started with just alcohol and marijuana, but once he entered high school and became acquainted with older teenagers, he experimented with other drugs such as speed and cocaine. By the time he was 16, he was using combinations of alcohol, speed, marijuana, and cocaine. He settled on just cocaine after a year of mixing drugs.

Mr. P frequently skipped school, and, when he attended school, he was usually intoxicated. To support his habit, he acquired money in various schemes, such as borrowing money from friends that he had no intention of paying back or stealing car radios or stealing from his mother.

Despite his blatant admission of drug use, Mr. P denies having a problem. When asked about his ability to control his drug use, he defensively replied that, of course, he could, but as he does not see a problem, he has not tried.

Stimulant Withdrawal

After stimulant intoxication, a “crash” occurs with symptoms of anxiety, tremulousness, dysphoric mood, lethargy, fatigue, nightmares (accompanied by rebound REM sleep), headache, profuse sweating, muscle cramps, stomach cramps, and insatiable hunger. The withdrawal symptoms generally peak in 2 to 4 days and are resolved in 1 week. The most serious withdrawal symptom is depression, which can be particularly severe after the sustained use of high doses of stimulants and which can be associated with suicidal ideation or behavior. A person in the state of withdrawal can experience powerful and intense cravings for cocaine, mainly because taking cocaine can eliminate unpleasant withdrawal symptoms. Persons experiencing cocaine withdrawal often attempt to self-medicate with alcohol, sedatives, hypnotics, or antianxiety agents such as diazepam.

Stimulant Intoxication Delirium

Delirium associated with stimulant use generally results from high doses of a stimulant or sustained use, and so sleep deprivation affects the clinical presentation. The combination of stimulants with other substances and the use of stimulants by a person with preexisting brain damage can also cause delirium. It is not uncommon for university students who are using amphetamines to cram for examinations to exhibit this type of delirium.

Stimulant-Induced Psychotic Disorder

The hallmark of stimulant-induced psychotic disorder is the presence of paranoid delusions and hallucinations, which occurs in up to 50 percent of those who misuse stimulants. Auditory hallucinations are also common, but visual and tactile hallucinations are less common than paranoid delusions. Cocaine can also cause the sensation of bugs crawling beneath the skin (formication). The presence of these symptoms depends on the dose, duration of use, and the user's sensitivity to the substance. Cocaine-induced psychotic disorders are most familiar with intravenous use and crack users, and the psychotic symptoms are more common in men than in women. The treatment of choice for amphetamine-induced psychotic disorder is the short-term use of an antipsychotic medication such as haloperidol.

Mr. H is a 20-year-old college student who was functioning well until the weeks of his finals when he began taking large amounts of cocaine because he felt he was unprepared for his tests. He began having delusional beliefs that the police were following him at the request of his parents to spy on him. He also believed that his roommate would give reports to them about his study habits and social life. He was brought to the emergency department after he threatened to harm his roommate if he continued to report on him.

During his evaluation, Mr. P reported sleeplessness and auditory hallucinations that told him that his roommate was conspiring against him. He was very agitated and paced continuously. After admission to the hospital, Mr. P was given antipsychotics and sleeping medications and recovered in 3 days.

Stimulant-Induced Mood Disorder

Stimulant-induced mood disorder includes the diagnoses of stimulant-induced bipolar disorder and stimulant-induced depressive disorder, either of which can begin during either intoxication or withdrawal. In general, intoxication is associated with manic or mixed mood features, whereas withdrawal is associated with depressive mood features.

Stimulant-Induced Anxiety Disorder

The onset of stimulant-induced anxiety disorder can occur during intoxication or withdrawal. Stimulants can induce symptoms similar to those seen in panic disorder and phobic disorders, in particular.

Stimulant-Induced Obsessive-Compulsive Disorder

The onset of stimulant-induced obsessive-compulsive disorder (OCD) can occur during intoxication or withdrawal. After high doses of stimulants, some individuals develop time-limited stereotyped behaviors or rituals (i.e., picking at clothing, and arranging and rearranging items purposelessly) that share some features with the type of compulsions seen in OCD.

Stimulant-Induced Sexual Dysfunction

People sometimes use amphetamines as an antidote to the sexual side effects of serotonergic agents such as fluoxetine, but stimulants are often misused by persons to enhance sexual experiences. High doses and long-term use are associated with erectile disorder and other sexual dysfunctions.

Stimulant-Induced Sleep Disorder

Stimulant-induced sleep disorder can begin during either intoxication or withdrawal, and sleep dysfunction can vary depending on the onset. Stimulant intoxication can produce insomnia and sleep deprivation, whereas persons undergoing stimulant withdrawal can experience hypersomnolence and nightmares.

COMORBIDITY

Studies of the prevalence of non–substance-related psychiatric disorders

among individuals with stimulant use disorders have consistently found elevated levels of psychiatric disorders in this population. Results from the community-based Epidemiological Catchment Area (ECA) Study demonstrated elevated rates of non–substance-related mental disorders in those with cocaine use disorders. In this study, the estimated lifetime prevalence rate for mental disorders was 76 percent or 11 times that found in the general population; nearly 85 percent of those with cocaine use disorders also had co-occurring alcohol use disorders. Studies that have compared treatment-seeking individuals with cocaine use disorders to those who do not seek treatment have demonstrated higher rates of major depression and ADHD among the former group. Likewise, most studies report high levels of co-occurring other substance use disorders in individuals with amphetamine-type stimulant use disorders. Mood disorders, psychotic disorders, and anxiety disorders are also common. While psychotic symptoms are most commonly methamphetamine-induced, mood and anxiety disorders are often not substance-induced.

TREATMENT AND REHABILITATION

Detoxification and Early Treatment

The stimulant withdrawal syndrome is distinct from that of opioids, alcohol, or sedative–hypnotic agents as the physiologic effects are not severe enough to require inpatient or residential drug withdrawal. Thus, it is generally possible to engage in a therapeutic trial of outpatient withdrawal before deciding whether the patient will need a more intensive or controlled setting to stop the drug. Patients withdrawing from stimulants typically experience fatigue, dysphoria, disturbed sleep, and some craving; some may experience depression. No pharmacologic agents reliably reduce the intensity of withdrawal, but recovery over a week or two is generally uneventful. It may take longer, however, for sleep, mood, and cognitive function to recover fully.

Most stimulant users do not come to treatment voluntarily. Their experience with the substance is too positive, and the adverse effects are perceived as too minimal to warrant seeking treatment.

The major hurdle to overcome in the treatment of cocaine-related disorders

is the user's intense craving for the drug. Although animal studies have shown that cocaine is a potent inducer of self-administration, these studies have also shown that animals limit their use of cocaine when investigators add negative reinforcers to the cocaine intake. In humans, negative reinforcers may take the form of work and family-related problems brought on by cocaine use. Therefore, clinicians must take a broad treatment approach and include social, psychological, and perhaps biologic strategies in the treatment program.

Attaining abstinence from stimulants in patients may require complete or partial hospitalization to remove them from the social settings in which they had obtained or used stimulants. Frequent, unscheduled urine testing is almost always necessary to monitor patients' continued abstinence, especially in the first weeks and months of treatment. Relapse prevention therapy (RPT) relies on cognitive and behavioral techniques in addition to hospitalization and outpatient therapy to achieve the goal of abstinence.

Psychosocial Therapies

Psychological intervention usually involves individual, group, and family modalities. In individual therapy, therapists should focus on the dynamics leading to stimulant use, the perceived positive effects of the stimulant, and other ways to achieve these effects. Group therapy and support groups, such as NA, often focus on discussions with other persons who use stimulants and on sharing experiences and effective coping methods. Family therapy is often an essential component of the treatment strategy. Typical issues discussed in family therapy are the ways the patient's past behavior has harmed the family and the responses of family members to these behaviors. Therapy should also focus, however, on the future and on changes in the family's activities that may help the patient stay off the drug and direct energies in different directions. Clinicians can use this approach for outpatient treatment.

Network Therapy. Network therapy is a specialized type of combined individual and group therapy designed to ensure greater success in the office-based treatment of addicted patients. Network therapy uses both psychodynamic and cognitive-behavioral approaches to individual therapy while engaging the patient in a group support network. The group, composed

of the patient's family and peers, is used as a therapeutic network joining the patient and therapist at intervals in therapy sessions. The approach promotes group cohesiveness as a vehicle for engaging patients in this treatment. This network is managed by the therapist to provide cohesiveness and support and to promote compliance with treatment. Although network therapy has not received systematic, controlled evaluation, it is frequently applied in the psychiatric practice because it is one of the few manualized approaches designed for use by individual practitioners in an office setting.

Pharmacologic Adjuncts

Presently, no pharmacologic treatments produce decreases in stimulant use. However, investigators have explored several agents, and those are summarized below.

Amphetamines. Multiple medications have been investigated as potential treatment options for amphetamine-type stimulant disorders, though the results have been mostly disappointing. Most serotonergic medications studied have not demonstrated efficacy, though mirtazapine has shown some effectiveness in a small sample of men who have sex with men. The opioid-receptor antagonist naltrexone and the dopamine antidepressant bupropion have both shown promise in those with less severe amphetamine use disorders, but little effect on more dependent individuals. Other medications, including topiramate, aripiprazole, baclofen, gabapentin, and modafinil, have demonstrated little effectiveness in this population.

Cocaine. A variety of pharmacologic agents, most approved for other uses, have been, and are being tested clinically for the treatment of cocaine use disorder and relapse.

Cocaine users presumed to have preexisting ADHD or mood disorders have been treated with methylphenidate and lithium, respectively. Those drugs are of little or no benefit in patients without the disorders, and clinicians should adhere strictly to diagnostic criteria before using either of them in the treatment of cocaine use disorder. In patients with ADHD, slow-release forms of methylphenidate may be less likely to trigger cocaine craving, but the impact of such pharmacotherapy on cocaine use is not clear.

Investigators have explored many pharmacologic agents on the premise

that chronic cocaine use alters the function of multiple neurotransmitter systems, especially the dopaminergic and serotonergic transmitters regulating hedonic tone, and that cocaine induces a state of relative dopaminergic deficiency. Although the evidence for such alterations in dopaminergic function has been growing, it has been challenging to demonstrate that agents theoretically capable of modifying dopamine function can alter the course of treatment.

Tricyclic antidepressant drugs yielded some positive results when used early in treatment with minimally drug-dependent patients; however, they are of little or no use inducing abstinence in moderate or severe cases.

Also tried but not confirmed effective in controlled studies are other antidepressants, such as bupropion, MAOIs, SSRIs, antipsychotics, lithium, several different calcium channel inhibitors, and anticonvulsants. Several studies have found increased rates of abstinence with the use of topiramate for cocaine use disorders; however, there are also negative trials of topiramate in this population.

Disulfiram has demonstrated potential benefits in the pharmacotherapy of cocaine use disorder. Disulfiram has long been used in the treatment of alcohol use disorder because it inhibits an enzyme involved in the metabolism of ethanol. Though disulfiram was initially considered a potential treatment option for individuals with both alcohol and cocaine use disorders, studies demonstrated that disulfiram appears to have a direct effect on the metabolism of dopamine and cocaine itself; it acts as an inhibitor of dopamine β-hydroxylase, thereby slowing the breakdown of synaptic dopamine and increasing dopamine levels. Individuals who are co-administered cocaine and disulfiram describe more negative responses to cocaine, including anxiety, restlessness, and paranoia. These reactions may, in part, explain the effects of disulfiram in decreasing cocaine use. Issues affecting treatment response to disulfiram include challenges with treatment nonadherence, risks of inducing psychotic symptoms with combined cocaine and disulfiram use, and risks of serious medical consequences if a patient were to consume alcohol, given the frequent co-occurrence of alcohol and cocaine use disorders.

EPIDEMIOLOGY

Amphetamines

According to the 2017 National Survey on Drug Use and Health (NSDUH), 0.3 percent of persons 12 years or older were current users of methamphetamine, and 5.4 percent have a lifetime history of methamphetamine use. In 2016, 684,000 individuals older than 12 years met the criteria for a methamphetamine use disorder, representing 0.3 percent of the population.

Cocaine

Cocaine Use. In 2016, 1.9 million (0.7 percent) persons aged 12 years or older used cocaine in the past month. Persons aged 18 to 25 (1.6 percent) had a higher rate of past-month cocaine use than persons aged 26 or older (0.6 percent) and youths aged 12 to 17 (0.1 percent). The lifetime rate of cocaine use by persons aged 12 years and older is 14.9 percent.

Cocaine Use Disorder. In 2016, more than 850,000 (0.3 percent) persons aged 12 or older met the criteria for a cocaine use disorder. Persons aged 18 to 25 (0.6 percent) had the highest rate of meeting criteria for cocaine use disorder in the past year, followed by persons aged 26 or older (0.3 percent) and youths aged 12 to 17 (0.1 percent).

PATHOLOGY

As a result of actions in the CNS, stimulants can produce a sense of alertness, euphoria, and well-being. Users may experience decreased hunger and less need for sleep. Performance impaired by fatigue is usually improved. Some users believe that cocaine enhances sexual performance.

Adverse Effects

Stimulant use can cause medical complications both as a result of direct toxic effects and because of complications related to methods of preparation and administration. Stimulants are potent sympathomimetic drugs, causing physiologic effects, including vasoconstriction, increased heart rate, and elevated blood pressure through adrenergic stimulation. The combination of increased myocardial demand with constriction of coronary blood vessels can

precipitate angina pectoris and even myocardial infarction. Acute hypertension associated with stimulant use can also predispose to arterial hemorrhages, such as hemorrhagic stroke or aortic dissection. Vasoconstriction can involve damage to other end organs, including the brain, kidneys, or intestines. With intranasal use, decreased blood flow to the nasal mucosa frequently causes nasal symptoms (e.g., rhinorrhea, nosebleeds) and mucosal ulceration. A combination of vasoconstriction and bruxism associated with the use of methamphetamine causes markedly poor dentition, or "meth mouth." Grand mal seizures are frequent with stimulant overdose and may occur with routine use as well. Furthermore, individuals who misuse stimulants regularly may develop neuropsychological deficits, including problems with attention, learning, and memory. In many cases, these abnormalities persist during periods of abstinence.

Stimulant misuse can also precipitate behaviors that place an individual at increased risk of harm by violence or accidental injury. Unsafe sexual behaviors may lead to the transmission of HIV or other sexually transmitted diseases. The drugs can cause accidental injury, for example, driving while intoxicated. The associated illegal behaviors involved in obtaining cocaine or methamphetamine can also increase one's risk of violent injury.

Neuropharmacology

Amphetamines. All the amphetamines are rapidly absorbed orally and have a rapid onset of action, usually within 1 hour, when taken orally. The classic amphetamines are also taken intravenously and have an almost immediate effect by this route. Users can also inhale ("snort") nonprescribed amphetamines or designer amphetamines. Tolerance develops with both classic and designer amphetamines, although amphetamine users respond by taking more of the drug. Amphetamine is less addictive than cocaine, as evidenced by experiments on rats in which not all animals spontaneously self-administered low doses of amphetamine.

The classic amphetamines (i.e., dextroamphetamine, methamphetamine, and methylphenidate) produce their primary effects by causing the release of catecholamines, particularly dopamine, from presynaptic terminals. The effects are particularly potent for the dopaminergic neurons projecting from the VTA to the cerebral cortex and the limbic areas. Some neuroscientists call

this the *reward circuit pathway*, and its activation is probably the primary addicting mechanism for the amphetamines. The designer amphetamines cause the release of catecholamines (dopamine and norepinephrine) and serotonin, the neurotransmitter implicated as the significant neurochemical pathway for hallucinogens. Therefore, the clinical effects of designer amphetamines are a blend of the effects of classic amphetamines and those of hallucinogens.

Cocaine. In vitro animal studies using radiolabeled cocaine or cocaine analogs demonstrate high amounts of binding in dopamine-rich brain regions, including the caudate-putamen and the VTA. Moderate levels of cocaine binding also occur in the substantia nigra, amygdala, hypothalamus, and locus coeruleus. In humans, cocaine rapidly permeates the striatum, then quickly redistributes out of it with a half-life of about 20 minutes. The time course of self-reported "high" closely follows this pattern of rapid uptake and clearance.

Drug-related reinforcement is most strongly related to the dopaminergic projections from the VTA to the nucleus accumbens. Cocaine administration causes transient increases in extracellular dopamine levels in these regions by binding to the presynaptic dopamine transporter (DAT), thereby inhibiting dopamine reuptake. Subjective euphoria appears to be a function of both DAT occupancy (with a threshold of 50 percent occupancy to be detectable) and rapidity of onset, with the most robust responses occurring when dopaminergic neurotransmission is increasing dynamically.

ETIOLOGY

Genetic Factors

The most convincing evidence to date of a genetic influence on cocaine use disorder comes from studies of twins. Monozygotic twins have higher concordance rates for stimulant use disorder (cocaine, amphetamines, and amphetamine-like drugs) than dizygotic twins. The analyses indicate that genetic factors and unique (unshared) environmental factors contribute equally to the development of stimulant dependence.

Sociocultural Factors

Social, cultural, and economic factors are potent determinants of initial use, continuing use, and relapse. Excessive use is far more likely in countries where cocaine is readily available. Different economic opportunities may influence certain groups more than others to engage in selling illicit drugs, and selling is more likely to be carried out in familiar communities than in communities where the seller runs a high risk of arrest.

Learning and Conditioning

Learning and conditioning are also considered important in perpetuating cocaine use. Each inhalation or injection of cocaine yields a “rush” and a euphoric experience that reinforces the antecedent drug-taking behavior. Also, the environmental cues associated with substance use become associated with the euphoric state so that, long after a period of cessation, such cues (e.g., white powder and paraphernalia) can elicit memories of the euphoric state and reawaken craving for cocaine.

In cocaine abusers (but not in healthy controls), cocaine-related stimuli activate brain regions subserving episodic and working memory and produce EEG arousal (desynchronization). Increased metabolic activity in the limbic-related regions, such as the amygdala, parahippocampal gyrus, and dorsolateral prefrontal cortex, reportedly correlates with reports of craving for cocaine, but the degree of EEG arousal does not.

4.7 Tobacco-Related Disorders

Tobacco use disorder is among the most prevalent, deadly, and costly of substance use disorders. It is also one of the most ignored, particularly by psychiatrists. Despite recent research that shows commonalities between tobacco use disorder and other substance use disorders, tobacco use disorder differs from other substance use disorders in unique ways. Tobacco does not cause behavioral problems; therefore, few tobacco-dependent persons seek psychiatric treatment, nor are they referred. Tobacco is a legal drug, and most persons who stop tobacco use have done so without treatment. Thus a common but erroneous view is that, unlike alcohol and other illicit drugs, most smokers do not need treatment.

Several trends may change the reluctance of psychiatrists to play a role in treating tobacco use disorders: (1) the growing recognition that most psychiatric patients smoke and many die from the consequences of tobacco use disorders; (2) remaining smokers will be more and more likely to have psychiatric problems, which suggests that many need intensive treatments; and (3) the development of multiple pharmacologic agents to aid smokers in quitting.

CLINICAL FEATURES

Behaviorally, the stimulatory effects of nicotine produce improved attention, learning, reaction time, and problem-solving ability. Tobacco users also report that cigarette smoking lifts their mood, decreases tension, and lessens depressive feelings. Results of studies of the effects of nicotine on CBF suggest that short-term nicotine exposure increases CBF without changing cerebral oxygen metabolism, but long-term nicotine exposure decreases CBF. In contrast to its stimulatory CNS effects, nicotine acts as a skeletal muscle relaxant.

DIAGNOSIS

Tobacco Use Disorder

The DSM-5 includes a diagnosis for tobacco use disorder characterized by craving, persistent and recurrent use, tolerance, and withdrawal when stopping tobacco. Dependence on tobacco develops quickly, probably because nicotine activates the VTA dopaminergic system, the same system affected by cocaine and amphetamine. The development of dependence is enhanced by substantial social factors that encourage smoking in some settings and by the powerful effects of tobacco company advertising. Persons are likely to smoke if their parents or siblings smoke and serve as role models. Several recent studies have also suggested a genetic diathesis toward tobacco dependence. Most persons who smoke want to quit and have tried many times to quit but have been unsuccessful.

Tobacco Withdrawal

The DSM-5 does not have a diagnostic category for tobacco intoxication, but it does have a diagnostic category for nicotine withdrawal. Withdrawal symptoms can develop within 2 hours of smoking the last cigarette; they generally peak in the first 24 to 48 hours and can last for weeks or months. The common symptoms include an intense craving for tobacco, tension, irritability, difficulty concentrating, drowsiness and paradoxical trouble sleeping, decreased heart rate and blood pressure, increased appetite and weight gain, decreased motor performance, and increased muscle tension. A mild syndrome of tobacco withdrawal can appear when a smoker switches from regular to low-nicotine cigarettes.

TREATMENT

For those who already smoke, psychiatrists should advise them to quit smoking. For patients who are ready to stop smoking, it is best to set a “quit date.” Most clinicians and smokers prefer abrupt cessation, but because no good quality data indicate that abrupt cessation is better than gradual cessation, patient preference for gradual cessation should be respected. Brief advice should focus on the need for medication or group therapy, weight gain concerns, high-risk situations, making cigarettes unavailable, and so forth. Because relapse is often rapid, the first follow-up phone call or visit should be 2 to 3 days after the quit date. These strategies may double self-initiated quit rates. Table 4-31 presents a model, “The 5 A’s” for helping tobacco

cessation that uses a motivational interviewing approach.

Ms. H was a 45-year-old patient with schizophrenia who smoked 35 cigarettes per day. She began her cigarette use at approximately 20 years of age during the prodromal stages of her first psychotic break. During the first 20 years of treatment, no psychiatrist or physician advised her to stop smoking.

When the patient was 43 years of age, her primary physician recommended smoking cessation. Ms. H attempted to stop on her own but lasted only 48 hours, partly because her housemates and friends smoked. During a routine medication check, her psychiatrist recommended that she stop smoking, and Ms. H described her prior attempts. The psychiatrist and Ms. H discussed ways to avoid smokers and had the patient announce her intent to quit and request that her friends try not to smoke around her and to offer encouragement for her attempt to quit. The psychiatrist also noted that Ms. H became irritable, slightly depressed, and restless and that she had insomnia during prior cessation attempts, and thus recommended medications. Ms. H chose to use a nicotine patch plus nicotine gum as needed.

The psychiatrist had Ms. H call 2 days after she attempted to quit smoking. At this point, Ms. H stated that the patch and gum were helping. One week later, the patient returned after having relapsed back to smoking. The psychiatrist praised Ms. H for not smoking for 4 days. He suggested that Ms. H contact him again if she wished to try to stop again. Seven months later, during another medication check, the psychiatrist again asked Ms. H to consider cessation, but she was reluctant.

Two months later, Ms. H called and said she wished to try again. This time, the psychiatrist and Ms. H listed several activities that she could do to avoid being around friends who smoked, phoned Ms. H’s boyfriend to ask him to assist her in stopping, asked the nurses on the inpatient ward to call Ms. H to encourage her, plus enrolled Ms. H in a support group for the next 4 weeks. This time the psychiatrist prescribed the non-nicotine medication varenicline and followed her with 15-minute visits for each of the first 3 weeks. She had two “slips” but did not go back to smoking and remained an ex-smoker. (Adapted from John R. Hughes, M.D.)

Table 4-31
The 5 Major Steps to Intervention (The 5 A's)

1. **Ask:** Identify and document tobacco use status for every patient at every visit.
2. **Advise:** In a clear, strong, and personalized manner, urge every tobacco user to quit.
3. **Assess:** Is the tobacco user willing to make a quit attempt at this tin
4. **Assist:** For the patient willing to make a quit attempt, use counselin and pharmacotherapy to help him or her quit. (See https://www.ahrq.gov/prevention/guidelines/tobacco/clinicians/index. for materials to help).
5. **Arrange:** Schedule follow-up contact, in person or by telephone, preferably within the first week after the quit date.

Source: Five Major Steps to Intervention (The "5 A's"). Agency for Healthcare Research and Quality, Rockville, MD. Content last reviewed: December 2012. Accessed November 30, 2020. https://www.ahrq.gov/prevention/guidelines/tobacco/5steps.html

Psychosocial Therapies

Behavior therapy is the most widely accepted and well-proved psychological therapy for smoking. Skills training and relapse prevention identify high-risk situations and plan and practice behavioral or cognitive coping skills for those situations in which smoking occurs. Stimulus control involves eliminating cues for smoking in the environment. Aversive therapy has smokers smoke repeatedly and rapidly to the point of nausea, which associates smoking with unpleasant, rather than pleasant, sensations. Aversive therapy appears to be effective but requires a good therapeutic alliance and patient compliance.

Hypnosis. Some patients benefit from a series of hypnotic sessions. The hypnotist gives suggestions about the benefits of not smoking, which the patient assimilates into their cognitive framework. The clinician can also use posthypnotic suggestions that cause cigarettes to taste unpleasant or to produce nausea.

Psychopharmacological Therapies

Table 4-32 lists the pharmacotherapies typically used for tobacco use disorder.



Table 4-32
FDA-Approved Medications for Smoking Cessation

Product	Route of Administration	Availability	Usual Dose
Nicotine Replacement Medications			
NicoDerm, Habitrol, Generic versions	Patch	OTC	Varies by product and amount a person has smoked. See manufacturer guidelines for specific instructions
Nicorette, Commit, Generic versions	Oral (Gum, Lozenge)	OTC	
Nicotrol	Inhaled (Nasal Spray, Inhaler)	Prescription	
Non-Nicotine Medications			
Bupropion (marketed as Zyban, however brand discontinued by manufacturer, in US)	Pill	Prescription	Starting: 150 mg × 3 days After 3 days increase to 150 twice a day Begin 1 wk

			before quitting
Varenicline Tartrate (Chantix)	Pill	Prescription	Days 1–3: 0.5 mg once daily Days 4–7: 0.5 mg twice daily Day 8–End of treatment: 1 mg twice daily

Source: US Food and Drug Administration, www.fda.gov

Nicotine Replacement Therapies. All nicotine replacement therapies double cessation rates, presumably because they reduce nicotine withdrawal. Hospitals may use these therapies on the wards to reduce withdrawal. Replacement therapies use a short period of maintenance of 6 to 12 weeks, often followed by a gradual reduction period of another 6 to 12 weeks.

Nicotine polacrilex gum is an OTC product that releases nicotine via chewing and buccal absorption. A 2 mg variety for those who smoke fewer than 25 cigarettes a day and a 4 mg variety for those who smoke more than 25 cigarettes a day are available. Smokers are to use one to two pieces of gum per hour up to a maximum of 24 pieces per day after abrupt cessation. Venous blood concentrations from the gum are one-third to one-half between-cigarette levels. Acidic beverages (coffee, tea, soda, and juice) should not be used before, during, or after gum use because they decrease absorption. Compliance with the gum has often been a problem. Adverse effects are minor and include unpleasant taste and sore jaws. About 20 percent of those who quit take the gum for long periods, but 2 percent use it for longer than a year; long-term use does not appear to be harmful. The significant advantage of nicotine gum is its ability to provide relief in high-risk situations.

Nicotine lozenges deliver nicotine and are also available in 2- and 4-mg forms; they are useful, especially for patients who smoke a cigarette

immediately on awakening. Generally, 9 to 20 lozenges a day are used during the first 6 weeks, with a decrease in dosage after that. Lozenges offer the highest level of nicotine of all nicotine replacement products. Users must suck the lozenge until dissolved and not swallow it. Side effects include insomnia, nausea, heartburn, headache, and hiccups.

Nicotine patches also sold over the counter, are available in a 16-hour, no-taper preparation and a 24- or 16-hour tapering preparation. Patches are administered each morning and produce blood concentrations about half those of smoking. Compliance is high, and the only significant adverse effects are rashes and, with 24-hour wear, insomnia. Using gum and patches in high-risk situations increases quit rates by another 5 to 10 percent. After 6 to 12 weeks, patients should discontinue the patch.

Nicotine nasal spray, available only by prescription, produces nicotine concentrations in the blood that are more similar to those from smoking a cigarette, and it appears to be especially helpful for heavily dependent smokers. The spray, however, causes rhinitis, watering eyes, and coughing in more than 70 percent of patients.

The nicotine inhaler, also available only by prescription, was designed to deliver nicotine to the lungs, but the nicotine is absorbed in the upper throat. It delivers 4 mg per cartridge, and resultant nicotine levels are low. The primary asset of the inhaler is that it provides a behavioral substitute for smoking. The inhaler doubles quit rates. These devices require frequent puffing—about 20 minutes to extract 4 mg of nicotine—and have minor adverse effects.

Non-Nicotine Medications. Non-nicotine therapy may help smokers who object philosophically to the notion of replacement therapy and smokers who fail replacement therapy. Bupropion is an antidepressant medication that has both dopaminergic and adrenergic actions. Bupropion SR started at 150 mg/day for 3 days and increased to 150 mg twice a day for 6 to 12 weeks. Daily dosages of 300 mg double quit rates in smokers with and without a history of depression. In one study, a combined bupropion and nicotine patch had higher quit rates than either alone. Adverse effects include insomnia and nausea, but these are rarely significant. Contraindications to use of bupropion include a seizure disorder (the risk of seizures in those appropriately screened is less than 1 in 1,000), current/past bulimia or anorexia nervosa,

rennet/concurrent MAOI, or other bupropion use. Though bupropion at one point carried a black box warning regarding neuropsychiatric adverse events during smoking cessation, the FDA has since removed this warning. This medication can be started 1 to 2 weeks before a quit date and used for up to 6 months post quit. Of interest, another antidepressant, nortriptyline, appears to be useful for smoking cessation as well.

Varenicline is a partial agonist at the $\alpha_4\beta_2$ neuronal nicotinic acetylcholine receptor; it both relieves craving and withdrawal and, unlike other medications, reduces the reinforcing effects of nicotine by blocking dopaminergic stimulation responsible for smoking reinforcement/reward. While there were concerns about the FDA-issued black box warnings for neuropsychiatric adverse events (depressed mood, agitation, changes in behavior, suicidal ideation, and suicide), more recent research of varenicline in psychiatric patients has found that psychiatric symptoms do not worsen nor does the risk of suicide increase. The FDA has removed the black box warning associated with neuropsychiatric adverse events from varenicline. There is also a concern for potential cardiovascular adverse events in patients taking varenicline ("a small, increased risk of certain cardiovascular adverse events in people who have a cardiovascular disease"). Clinicians should counsel patients on these risks and monitor for symptoms related to mental status and cardiac status.

Combined Psychosocial and Pharmacologic Therapy

Several studies have shown that combining nicotine replacement and behavior therapy increases quit rates over either therapy alone.

EPIDEMIOLOGY

The *2018 Monitoring the Future Survey* reported that nicotine use via cigarettes is currently at a historic low. Thirty-day smoking rates among eighth, tenth and twelfth graders were 2.2 percent, 4.2 percent, and 7.6 percent, respectively. However, vaping, which involves the use of a battery-powered device to heat a liquid to or other substances until it aerosolizes chemicals for inhalation, increased significantly in 2018. Popular manufacturers of vaping liquids commonly sell them in flavors that are especially appealing to adolescents. Thirty-day nicotine vaping prevalence

rates in the eighth, tenth, and twelfth graders were 6 percent, 16 percent, and 21 percent, respectively, and were substantially higher than cigarette use. The increase in nicotine vaping was particularly high for tenth and twelfth graders (increasing by 9 and 11 percent for these grades).

The WHO estimates that there are 1 billion smokers worldwide, and more men than women smoke. Although the prevalence of smoking is decreasing overall, the number of persons smoking in the Eastern Mediterranean and African regions is increasing.

Tobacco is commonly smoked in cigarettes, as well as in cigars, snuff, chewing tobacco, and pipes. Additionally, there is an increasing prevalence of vaping and the use of e-cigarettes.

Currently, about 14 percent of Americans smoke. Dependence features appear to develop quickly. Classroom and other programs to prevent initiation are only mildly effective, but increased taxation does decrease initiation.

Nearly 70 percent of smokers indicate that they want to quit, and, as of 2015, more than 50 percent reported that they tried to quit in the past year. On a given attempt, only 30 percent remain abstinent for even 2 days, and only 5 to 10 percent stop permanently. Most smokers make 5 to 10 attempts, however, so eventually 50 percent of "ever smokers" quit. In the past, 90 percent of successful attempts to quit involved no treatment. With the advent of OTC and non-nicotine medications in 1998, about one-third of all attempts involved the use of medication.

In terms of the diagnosis of tobacco use disorder per se, about 20 percent of the population develop tobacco dependence at some point, making it one of the most prevalent psychiatric disorders. Approximately 85 percent of current daily smokers are tobacco-dependent. Tobacco withdrawal occurs in about 50 percent of smokers who try to quit.

According to the Centers for Disease Control and Prevention (CDC), regional differences exist in smoking throughout the United States. The 11 states with the highest prevalence of current smoking are Kentucky, West Virginia, Louisiana, Ohio, Indiana, Missouri, Oklahoma, Arkansas, Mississippi, Tennessee, and Alabama. Those states with the lowest prevalence are Utah and California.

Education

Level of education attainment correlated with tobacco use. Of adults who had not completed high school, 23 percent smoked cigarettes (and of those with a GED, 37 percent smoked), whereas only 7 percent of college graduates smoked.

Psychiatric Patients

Psychiatrists must be particularly concerned and knowledgeable about tobacco use disorders because of the high proportion of psychiatric patients who smoke. Approximately 50 percent of all psychiatric outpatients, 70 percent of outpatients with bipolar I disorder, almost 90 percent of outpatients with schizophrenia, and 70 percent of patients with substance use disorder smoke. Moreover, data indicate that patients with depressive disorders or anxiety disorders are less successful in their attempts to quit smoking than other persons; thus, a holistic health approach for these patients probably includes helping them address their smoking habits in addition to the primary mental disorder. The high percentage of patients with schizophrenia who smoke may be due to tobacco's ability to reduce their extraordinary sensitivity to outside sensory stimuli and to increase their concentration. In that sense, such patients are self-monitoring to relieve distress.

Death

Death is the primary adverse effect of cigarette smoking. Tobacco use is associated with more than 400,000 premature deaths each year in the United States, which is 20 percent of all deaths. Individuals who smoke tend to die 10 years earlier than nonsmokers. Cancer causes chronic obstructive pulmonary disease, other lung diseases, cancer, heart disease, stroke, and diabetes. The increased use of chewing tobacco and snuff (smokeless tobacco) is associated with the development of oropharyngeal cancer.

Researchers have found that tobacco smoke causes nearly 30 percent of cancer deaths in the United States, making tobacco the single most lethal carcinogen in the United States. Smoking (mainly cigarette smoking) causes cancer of the lung, upper respiratory tract, esophagus, bladder, and pancreas and probably of the stomach, liver, and kidney. Smokers are 15 to 30 times more likely than nonsmokers to develop lung cancer, and lung cancer has

surpassed breast cancer as the leading cause of cancer-related deaths in women. Despite these staggering statistics, smokers can dramatically lower their chances of developing smoke-related cancers by merely quitting.

PATHOLOGY

Neuropharmacology

The psychoactive component of tobacco is nicotine, which affects the CNS by acting as an agonist at the nicotinic subtype of acetylcholine receptors. About 25 percent of the nicotine inhaled during smoking reaches the bloodstream, through which nicotine reaches the brain within 15 seconds. The half-life of nicotine is about 2 hours. Nicotine likely produces its positive reinforcing and addictive properties by activating the dopaminergic pathway projecting from the VTA to the cerebral cortex and the limbic system. In addition to activating this dopamine reward system, nicotine causes an increase in the concentrations of circulating norepinephrine and epinephrine and an increase in the release of vasopressin, β-endorphin, adrenocorticotropic hormone (ACTH), and cortisol. These hormones contribute to the primary stimulatory effects of nicotine on the CNS.

Adverse Effects of Nicotine

Nicotine is a highly toxic alkaloid. Doses of 60 mg in an adult are fatal secondary to respiratory paralysis; doses of 0.5 mg are delivered by smoking an average cigarette. In low doses, the signs and symptoms of nicotine toxicity include nausea, vomiting, salivation, pallor (caused by peripheral vasoconstriction), weakness, abdominal pain (caused by increased peristalsis), diarrhea, dizziness, headache, increased blood pressure, tachycardia, tremor, and cold sweats. Toxicity is also associated with an inability to concentrate, confusion, and sensory disturbances. Nicotine is further associated with a decrease in the user’s amount of REM sleep. Tobacco use during pregnancy is associated with an increased incidence of low–birth-weight babies and an increased incidence of newborns with persistent pulmonary hypertension.

Health Benefits of Smoking Cessation

Smoking cessation has significant and immediate health benefits for persons of all ages and provides benefits for persons with and without smoking-related diseases. Former smokers live longer than those who continue to smoke. Smoking cessation decreases the risk of lung cancer and other cancers, myocardial infarction, cerebrovascular diseases, and chronic lung diseases. Women who stop smoking before pregnancy or during the first 3 to 4 months of pregnancy reduce their risk of having low–birth-weight infants to that of women who never smoked. The health benefits of smoking cessation substantially exceed any risks from the average 5-lb (2.3-kg) weight gain or any adverse psychological effects after quitting.

▲ 4.8 Caffeine-Related Disorders

Caffeine affects various neurobiologic and physiologic systems and produces significant psychological effects. Caffeine is not associated with any life-threatening illnesses, but its use can result in psychiatric symptoms and disorders. The habitual use of caffeine and its widely accepted integration into daily customs can lead to an underestimation of the role that caffeine may play in one's daily life and can make the recognition of caffeine-associated disorders particularly challenging. Hence, the clinician should be familiar with caffeine, its effects, and associated problems.

CLINICAL FEATURES AND DIAGNOSIS

Caffeine use is associated with five disorders: caffeine intoxication, caffeine withdrawal, caffeine use disorder, caffeine-induced anxiety disorder, and caffeine-induced sleep disorder. The diagnosis of caffeine intoxication or other caffeine-related disorders depends primarily on a comprehensive history of a patient's intake of caffeine-containing products. The history should cover whether a patient had experienced any symptoms of caffeine withdrawal during periods when caffeine consumption was either stopped or severely reduced. Table 4-33 lists some essential diagnoses to consider in the differential for caffeine-related disorders. The differential diagnosis should include the abuse of caffeine-containing OTC medications, anabolic steroids, and other stimulants, such as amphetamines and cocaine. A urine sample may

be needed to screen for these substances. The differential diagnosis should also include hyperthyroidism and pheochromocytoma.

Table 4-33
Differential Diagnosis for Caffeine-Related Disorders

Generalized anxiety disorder
Panic disorder with or without agoraphobia
Bipolar II disorder
Attention-deficit/hyperactivity disorder (ADHD)
Sleep disorders

Caffeine Intoxication

After the ingesting 50 to 100 mg of caffeine, the common symptoms can be pleasant (e.g., alertness) or only mildly annoying (diuresis). As the dose increases, the symptoms become more concerning; Table 4-34 lists these. Caffeine intoxication generally occurs above 250 mg. The annual incidence of intoxication is about 10 percent of the population. Above 10 g of caffeine —almost impossible to achieve with usual caffeinated beverages but possible with pills or other supplements—caffeine can be fatal.

Table 4-34
Symptoms and Signs of Caffeine Use

Mild to Moderate Use (50–100 mg)	Alertness	A mild sense of well-being	A sense of improved verbal and motor performance	Diuresis	Cardiac muscle stimulati

Table 4-35
Symptoms of Caffeine Withdrawal

Headache
Fatigue
Anxiety
Irritability
Mild depressive symptoms
Impaired psychomotor performance
Nausea
Vomiting
Craving for caffeine
Muscle pain and stiffness

Ms. B, a 30-year-old, went for consultation due to “anxiety attacks.” The attacks occurred mid- to late afternoon when Ms. B became restless, nervous, and quickly excited and sometimes was noticed to be flushed, sweating, and, according to coworkers, “talking a mile a minute.” In response to questioning, Ms. B admitted to consuming six to seven cups of coffee each day before the time the attacks usually occurred.

Caffeine Withdrawal

The appearance of withdrawal symptoms reflects the tolerance and physiologic dependence that develop with continued caffeine use. Several epidemiologic studies have reported symptoms of caffeine withdrawal in 50 to 75 percent of all caffeine users studied.

Table 4-35 lists the symptoms of caffeine withdrawal; the most common ones are headache and fatigue. The amount of caffeine and the abruptness of stopping it correlate with the number and severity of the withdrawal symptoms. Caffeine withdrawal symptoms begin about 12 to 24 hours after

the last dose, peak in 24 to 48 hours, and resolve within 1 week.

The induction of caffeine withdrawal can sometimes be iatrogenic. Physicians often ask their patients to discontinue caffeine intake before specific medical procedures, such as endoscopy, colonoscopy, and cardiac catheterization. Also, physicians often recommend that patients with anxiety symptoms, cardiac arrhythmias, esophagitis, hiatal hernias, fibrocystic disease of the breast, and insomnia stop caffeine intake. Some persons decide that it would be good for them to stop using caffeine-containing products. In all these situations, caffeine users should taper the use of caffeine-containing products over a 7- to 14-day period rather than stop abruptly.

Mr. F was a 43-year-old attorney whose wife brought him for a psychiatric consultation. Mr. F had been complaining of fatigue, loss of motivation, sleepiness, headache, nausea, and difficulty concentrating. His symptoms occurred mostly over the weekends. He withdrew from weekend social activities due to his symptoms, which worried Mrs. F because he seems fine during the week. Mr. F is in good health with no recent history of medical disorders.

Mr. F worked in hectic law practice, many times working 60-hour weeks, and barely sees his family during the week. At work, he is often anxious, restless, and always busy. He worries about his job so much that he has difficulty sleeping on weeknights. He denies any marital or family problems, apart from those related to his insisting on working on the weekend.

At work, Mr. F regularly consumes approximately four to five cups of coffee per day. He cut out coffee on the weekends because he felt that it might be contributing to his anxiety and sleeplessness.

Caffeine Use Disorder

Some people with problematic caffeine consumption might have a caffeine use disorder. DSM-5 includes this diagnosis in Section III, meant for conditions requiring further study. No studies have examined the course and prognosis for patients with a diagnosis of caffeine use disorder. Subjects with caffeine use disorder have reported continued use of caffeine despite repeated

efforts to discontinue their caffeine use.

Ms. G was a 35-year-old married homemaker with three children aged 8, 6, and 2. She took no prescription medications, took a multivitamin and vitamins C and E daily, did not smoke, and had no history of psychiatric problems. She drank moderate amounts of alcohol on the weekends, had smoked marijuana in college but had not used it since, and had no other history of illicit drug use.

She had started consuming caffeinated beverages while in college, and her current beverage of choice was caffeinated diet cola. Ms. G had her first soft drink early in the morning, shortly after getting out of bed, and she jokingly called it her "morning hit." She spaced out her bottles of soft drinks over the day, with her last bottle at dinnertime. She typically drank four to five 20-oz bottles of caffeinated diet cola each day.

She and her husband had argued about her caffeinated soft drink use in the past, and her husband had believed she should not drink caffeinated soft drinks while pregnant. However, she had continued to do so during each of her pregnancies. Despite a desire to stop drinking caffeinated soft drinks, she was unable to do so. She described having a strong desire to drink caffeinated soft drinks, and if she resisted this desire, she found that she could not think of anything else. She drank caffeinated soft drinks in her car, which had a manual transmission, and noted that she fumbled while shifting and holding the soft drink and spilled it in the car. She also noted that her teeth had become yellowed, and she suspected this was related to her tendency to swish the soft drink in her mouth before swallowing it. When asked to describe a time when she stopped using soft drinks, Ms. G reported that she ran out of it on the day one of her children was to have a birthday party, and she did not have time to leave her home to buy more. In the early afternoon of that day, a few hours before the scheduled start of the party, she felt extreme lethargy, a severe headache, irritability, and craving for a soft drink. She called her husband and told him she planned to cancel the party. She then went to the grocery store to buy soft drinks, and after drinking two bottles, she felt well enough to host the party.

Although initially expressing interest in decreasing or stopping her

caffeinated soft drink use, Ms. G did not attend scheduled follow-up appointments after her first evaluation. When finally contacted at home, she reported she had only sought help initially at her husband's request, and she had decided to try to cut down on her caffeine use on her own. (Courtesy of Eric Strain, M.D.)

Caffeine-Induced Disorders

Caffeine-Induced Anxiety Disorder. The anxiety related to caffeine use can resemble that of generalized anxiety disorder. Patients with the disorder may be perceived as "wired," overly talkative, and irritable; they may complain of not sleeping well and of having too much energy. Caffeine can induce and exacerbate panic attacks in persons with a panic disorder. Although there is no conclusive proof of a causative relation between caffeine and panic disorder, it is prudent to advise patients with panic disorder to avoid caffeine.

Mr. B was a 28-year-old male graduate student who was in good health and had no history of previous psychiatric evaluation or treatment. He took no medications, did not smoke or consume alcohol, and had no current or past history of illicit drug use.

His chief complaint was that he had begun feeling mounting "anxiety" when working in the laboratory where he was pursuing his graduate studies. His work had been progressing well, and he felt his relationship with his advisor was excellent and supportive, and he could not identify any problems with staff or peers that might explain his anxiety. He had been working long hours, but found the work exciting and had recently had his first paper accepted for publication.

Despite these successes, he reported feeling "crescendoing anxiety" as his day would progress. He noted that by the afternoon, he would be experiencing palpitations, bursts of his heart racing, tremors in his hands, and an overall feeling of "being on edge." He also noted nervous energy in the afternoons. These experiences were occurring daily and seemed confined to the laboratory (although he admitted he was in the laboratory

every day of the week).

During the history-taking, it became clear that Mr. B consumed large amounts of coffee. The staff made a large urn of caffeinated coffee each morning, and Mr. B routinely started with a large mug of coffee. Throughout the morning, he would consume three to four mugs of coffee (the equivalent of about six or eight 6-oz cups of coffee) and continued this for the afternoon. He occasionally had a single can of a caffeinated soft drink and used no other forms of caffeine regularly. Mr. B estimated that he drank a total of six to eight or more mugs of coffee per day (which was estimated to be at least 1,200 mg of caffeine per day). Once pointed out to him, he realized that this level of caffeine consumption was considerably higher than at any other time in his life. He admitted he liked the taste of coffee and felt a burst of energy in the morning when he drank coffee that helped him start his day.

Mr. B and his physician developed a plan to decrease his caffeine use by tapering off caffeine. Mr. B was successful in decreasing his caffeine use and had a reasonable resolution of his anxiety symptoms once his daily coffee use had markedly decreased. (Courtesy of Laura M. Juliano, Ph.D., and Roland R. Griffiths, Ph.D.)

Caffeine-Induced Sleep Disorder. Caffeine is associated with a delay in falling asleep, inability to remain asleep, and early morning awakening.

Caffeine-Related Disorder Not Elsewhere Classified. This category includes caffeine-related disorders that do not meet the criteria for caffeine use disorder, caffeine intoxication, caffeine withdrawal, caffeine-induced anxiety disorder, or caffeine-induced sleep disorder.

COMORBIDITY

Persons with caffeine-related disorders are more likely to have additional substance-related disorders than are those without diagnoses of caffeine-related disorders. About two-thirds of those who consume large amounts of caffeine daily also use sedative and hypnotic drugs. Significant caffeine consumption is associated with major depression, generalized anxiety disorder, panic disorder, antisocial personality disorder, and other substance

use disorders, including alcohol, cannabis, and cocaine.

Caffeine Use and Nonpsychiatric Illnesses

Despite a great deal of research, no one has conclusively demonstrated a significant health risk from routine caffeine use. Nonetheless, caffeine use is considered relatively contraindicated for various conditions; Table 4-36 lists these. Some of these are very controversial; for example, there may be a mild association between higher-daily caffeine use in women and delayed conception and slightly lower birth weight. Studies, however, have not found such associations, and effects, when found, are usually with relatively high daily dosages of caffeine (e.g., the equivalent of five cups of brewed coffee per day). For a woman who is considering pregnancy, especially if there is some difficulty in conceiving, it may be useful to counsel stopping caffeine use. Similarly, for a woman who becomes pregnant and has moderate- to high-daily caffeine consumption, a discussion about decreasing her daily caffeine use may be warranted.



Table 4-36
Possible Contraindications to Caffeine Use

Generalized anxiety disorder
Panic disorder
Primary insomnia
Gastroesophageal reflux
Cardiovascular disease (particularly hypertension and hypercholesterolemia)
Pregnancy
Infertility

TREATMENT

Analgesics, such as aspirin, usually control the headaches and muscle aches that may accompany caffeine withdrawal. Rarely do patients require benzodiazepines to relieve withdrawal symptoms. When using benzodiazepines, we should restrict them to small doses and short durations,

about 7 to 10 days at the longest.

The first step to reducing or eliminating caffeine is to have the patient determine their daily consumption of caffeine. A daily food diary can help determine an accurate dose. We may have to educate the patient about the many sources of caffeine in the diet. After several days of diet tracking, we can review the diary with the patient and determine the average-daily caffeine dose.

After obtaining an accurate accounting of all caffeine use, we can work with the patient to decide on a reasonable taper. A typical schedule would be a decrease of 10 percent every few days. Because most of the caffeine usually comes from beverages, gradually substituting decaffeinated beverages for caffeinated ones can be a simple strategy. The diary should be maintained, and the progress monitored. The tapering should be individualized for each patient so that the rate of caffeine decrease minimizes withdrawal symptoms. Although abrupt withdrawal is possible, most patients find the withdrawal too uncomfortable.

EPIDEMIOLOGY

Caffeine is the most widely consumed psychoactive substance in the world. In the United States, 87 percent of children and adults consume foods and beverages containing caffeine. Caffeine is found in more than 60 species of plants and belongs to the methylxanthine class of alkaloids, which also includes theobromine (found in chocolate) and theophylline (often used in the treatment of asthma).

Many drinks, foods, prescription medicines, and OTC medicines contain caffeine (Table 4-37). The mean caffeine intake among adults (over age 22) in the United States was estimated to be 300 mg/day, although, among coffee drinkers, the mean intake was 375 mg/day. A cup of coffee generally contains 100 to 150 mg of caffeine; tea contains about one-third as much. Many OTC medications contain one-third to one-half as much caffeine as a cup of coffee, and some migraine medications and OTC stimulants contain more caffeine than a cup of coffee. Cocoa, chocolate, and soft drinks contain significant amounts of caffeine, enough to cause some symptoms of caffeine intoxication in small children when they ingest a candy bar and a 12-oz cola drink.

Table 4-37
Caffeine Content of Common Foods and Medications

Product	Serving Size (Volume or Weight)	Typical Caffeine Content (mg)	Range (mg)
BEVERAGES			
Coffee			
Brewed/Drip	12 oz	200	107–420
Instant	12 oz	140	40–260
Espresso	1 oz	70	60–95
Decaffeinated	12 oz	8	0–20
Starbucks drip	12 oz	260	
Starbucks cappuccino	12 oz	75	
Starbucks espresso	1 oz	75	
Starbucks bottled Frappuccino	9.5 oz	90	
Starbucks decaffeinated	12 oz	20	
Tea			
Brewed	6 oz	40	30–90
Instant	6 oz	30	10–35
Canned or Bottled	12 oz	20	8–32
Soft drinks			
Typical caffeinated soft drink	12 oz	40	22–69

Mountain Dew/Diet Mt. Dew	12 oz	54	
Diet Coke	12 oz	47	
Sunkist/Diet Sunkist	12 oz	41	
Dr. Pepper/Diet Dr. Pepper	12 oz	41	
Pepsi	12 oz	38	
Diet Pepsi	12 oz	36	
Pepsi Max	12 oz	69	
Coke Classic	12 oz	35	
A & W Cream Soda	12 oz	29	
Barq’s Root Beer	12 oz	23	
A & W Diet Cream Soda	12 oz	22	
A & W Root Beer	12 oz	0	
7UP/Diet 7UP	12 oz	0	
Fanta Orange	12 oz	0	
Sprite/Diet Sprite	12 oz	0	
Canada Dry Ginger Ale	12 oz	0	
Cocoa/Hot Chocolate	6 oz	7	2–10
Chocolate Milk	6 oz	4	2–7
Caffeinated water			
Typical amount	16.9 oz	60	60–200
Water Joe	16.9 oz	60	

Buzzwater	16.9 oz	100 or 200	
Energy drinks			
Typical amount	varies	varies	50–500
AMP	16 oz	142	
Red Bull	8.4 oz	80	
Full Throttle	16 oz	160	
Monster	16 oz	160	
Rockstar	16 oz	160	
Spike Shooter	8.4 oz	300	
VPX Redline Energy	8 oz	316	
Wired-X-344	16 oz	344	
Rage inferno	24 oz	375	
Energy Shots			
Typical	varies	varies	
5-hour energy	2 oz	200	
5-hour energy extra strength	2 oz	230	
10-hour energy shot	1.93 oz	422	
FOODS			
Chocolate			
Hershey’s Chocolate Bar	1.55 oz	9	
Hershey’s Special Dark	1.45 oz	18	
Kit Kat Wafer Bar	1.5 oz	6	
Reese’s Peanut Butter	1.6 oz	4	

Cups			
Miscellaneous Foods			
Penguin Peppermints	1 mint	7	
Starbucks Classic Coffee Ice Cream	4 oz	30	
Dannon Coffee Yogurt	6 oz	30	
Jolt Caffeinated Gum	1 stick	33	
Powerbar Tangerine Powergel	41 g	50	
Stay-Alert Caffeinated Gum	1 stick	100	
PRESCRIPTION MEDICATIONS			
Headache/Migraine/Pain			
Fiorinal	2 capsules	80	
Fioricet/Esgic/Many others	2 tablets	80	
Cafergot	2 tablets	200	
Norgesic	2 tablets	60	
OVER-THE-COUNTER MEDICATIONS			
Stimulants			
Typical	1 tablet	100 or 200	100–200
Vivarin	1 tablet	200	
No-Doz/No-Doz Maximum	1 tablet	100 or 200	

Strength			
Ultra Pep-Back	1 tablet	200	
Analgesics			
Goody's Headache Powder	1 powder packet	32.5	
BC Fast Pain Relief	1 powder packet	33.3	
BC Arthritis Pain and Influenza	1 powder packet	38	
Anacin Advanced Headache	2 tablets	130	
Excedrin Extra Strength	2 tablets	130	
Menstrual Pain Relief/Diuretics			
Diurex Water Pills	2 tablets	100	
Midol Menstrual Complete	2 caplets	120	
Pamprin Max	2 caplets	130	
DIETARY SUPPLEMENTS/WEIGHT LOSS PRODUCTS			
Typical	1 or 2 tablets	varies	50–300
Dexatrim Max	1 caplet	50	
Metabolife Weight Management	2 tablets	101	
Metabolife Ultra	2 caplets	150	
Hydroxycut Weight Loss	2 caplets	200	

Formula		
Leptopril	2 capsules	220
Stacker 2	1 capsule	253
Stacker 3	1 capsule	254
Twinlab Ripped Fuel	2 capsules	220
Swarm Extreme Energizer	1 capsule	300
Xenadrine EFX	2 capsules	200

Caffeine values for all brand name products were obtained directly from product labels, the manufacturer's website, or customer service department.

Data from Juliano LM, Ferre S, Griffiths RR. The pharmacology of caffeine. In: Ries RK, Fiellin DA, Miller SC, Saitz R, eds. *ASAM Principles of Addiction Medicine*. 5th ed. Baltimore, MD: Lippincott Williams & Wilkins; 2014:180–200; McCusker RR, Fuehrlein B, Goldberger BA, Gold MS, Cone EJ. Caffeine content of decaffeinated coffee. *J Anal Toxicol*. 2006;30(8):611; McCusker RR, Goldberger BA, Cone EJ. Caffeine content of specialty coffees. *J Anal Toxicol*. 2003;27(7):520.

Special Populations

Cigarette smokers consume more caffeine than nonsmokers. This observation may reflect a shared genetic vulnerability to caffeine use and cigarette smoking. It may also be related to increased rates of caffeine elimination in cigarette smokers. Preclinical and clinical studies indicate that regular caffeine use can potentiate the reinforcing effects of nicotine.

Heavy use and clinical dependence on alcohol are associated with heavy use and clinical dependence on caffeine as well. Individuals with anxiety disorders tend to report lower levels of caffeine use, although one study showed that a more significant proportion of heavy caffeine consumers also use benzodiazepines. Several studies have also shown high-daily amounts of caffeine use in psychiatric inpatients. For example, several studies have found that such patients consume the equivalent of an average of five or more cups of brewed coffee each day. Finally, high-daily caffeine consumption occurs in prisoners.

PATHOLOGY AND ETIOLOGY

After exposure to caffeine, several factors influence continued consumption, such as the pharmacologic effects of caffeine, caffeine's reinforcing effects, genetic predispositions to caffeine use, and personal attributes of the consumer.

Neuropharmacology. Caffeine, a methylxanthine, is more potent than the other methylxanthine, theophylline. The half-life of caffeine in the human body is 3 to 10 hours, and the time of peak concentration is 30 to 60 minutes. Caffeine readily crosses the blood–brain barrier. Caffeine acts primarily as an antagonist of adenosine receptors. Adenosine receptors activate an inhibitory G protein (G_i), which then inhibits the formation of the second-messenger cyclic adenosine monophosphate (cAMP). Caffeine intake, therefore, results in an increase in intraneuronal cAMP concentrations in neurons with adenosine receptors.

Three cups of coffee are estimated to deliver enough caffeine to the brain caffeine to occupy about 50 percent of the adenosine receptors. Several experiments indicate that caffeine, especially at high doses or concentrations, can affect dopamine and noradrenergic neurons. Specifically, caffeine may enhance dopamine activity, a hypothesis that could explain clinical reports associating caffeine intake with an exacerbation of psychotic symptoms in patients with schizophrenia. Activation of noradrenergic neurons may be involved in the mediation of some symptoms of caffeine withdrawal.

Subjective Effects and Reinforcement. The subjective effect of single low to moderate doses of caffeine (i.e., 20 to 200 mg) is usually pleasurable. Studies have shown that these doses result in increased ratings on measures such as well-being, energy and concentration, and motivation to work. Also, these doses of caffeine produce decreases in ratings of feeling sleepy or tired. In the range of 300 to 800 mg (the equivalent of several cups of brewed coffee ingested at once), subjects tend to rate the effects of caffeine as more unpleasant, with feelings of anxiety and nervousness. Animal studies have generally found it difficult to demonstrate that caffeine functions as a reinforcer. However, well-controlled studies in humans have shown that people choose caffeine over placebo when given a choice under controlled experimental conditions. In habitual users, caffeine's ability to suppress withdrawal symptoms also reinforces the effects of caffeine. Thus, the profile

of caffeine's subjective effects and its ability to function as a reinforcer contribute to the regular use of caffeine.

Genetics and Caffeine Use. Some genetic predisposition may exist to continued coffee use after exposure to coffee. Some studies have shown higher concordance rates for monozygotic twins for total caffeine consumption, heavy use, caffeine tolerance, caffeine withdrawal, and caffeine intoxication, with heritabilities ranging between 35 and 77 percent. Multivariate structural equation modeling of caffeine use, cigarette smoking, and alcohol use suggest that a common genetic factor—polysubstance use—underlies the use of these three substances.

Effects on Cerebral Blood Flow. Most studies have found that caffeine results in global cerebral vasoconstriction, with a resultant decrease in CBF, although this effect may not occur in persons over 65 years of age. According to one study, tolerance does not develop to these vasoconstrictive effects, and the CBF shows a rebound increase after withdrawal from caffeine. Some clinicians believe that caffeine use can cause a similar constriction in the coronary arteries and produce angina in the absence of atherosclerosis.

4.9 Hallucinogen-Related Disorders

CLINICAL FEATURES

Hallucinogens, by definition, are intoxicants. Table 4-38 lists some of the psychological effects of hallucinogenic drugs. Hallucinations are usually visual, often of geometric forms and figures, but auditory and tactile hallucinations can occur. The sense of self is much changed, sometimes to the point of depersonalization, merging with the external world to the point of feeling separation of oneself from one's body.

Table 4-38
Psychological Effects of Hallucinogens

Heightened perceptions (richer colors, sharpened contours, richer

responses to music, smells or tastes)
Synesthesia
Changes in body image
Alterations in time and space
Intense, labile emotions
Suggestibility
Either sensitivity or detachment from others
Awareness of internal organs
The recovery of lost early memories
The release of unconscious material in symbolic form
Regression and the apparent reliving of past events, including birth
Introspective reflection and feelings of religious and philosophical insight
Depersonalization
Derealization
Anxiety, panic attacks
Flashbacks (hallucinogen-persisting perception disorder)
Mood disorders
Anxiety disorders
Psychosis
Delirium



Table 4-39
Overview of Representative Hallucinogens

Agent	Locale	Chemical Classification	Biologic Sources	Common Route	Typical Dose	Duration of Effects	Adverse Reactions
Lysergic acid diethylamide (LSD)	Globally distributed, semisynthetic	Indolealkylamine	Fungus in rye yields lysergic acid	Oral	100 µg	6–12 hr	Extensive, including pandemic 1965–1975
Mescaline	Southwestern US	Phenethylamine	Peyote cactus, *L. Williamsii*	Oral	200–400 mg or 4–6 cactus buttons	10–12 hr	Little or none verified
Methylenedioxyamphetamine (MDA)	US, synthetic	Phenethylamine	Synthetic	Oral	80–160 mg	8–12 hr	Documented
Methylenedioxymethamphetamine (MDMA)	US, synthetic	Phenethylamine	Synthetic	Oral	80–150 mg	4–6 hr	Documented
Psilocybin	Southern US, Mexico, south America	Phosphorylated hydroxylated DMT	Psilocybin mushrooms	Oral	4–6 mg or 5–10 g of dried mushroom	4–6 hr	Psychosis
Ibogaine	West central Africa	Indolealkylamine	Tabernanthe iboga	Eating powdered root	200–400 mg	8–48 hr	CNS excitation, death?
Ayahuasca	South American tropics	Harmine, other β-carbolines	Bark or leaves of *Banisteriopsis Caapi*	As a tea	300–400 mg	4–8 hr	None reported
Dimethyltryptamine	South America, synthetic	Substituted tryptamine	leaves of *Virola Calophylla*	As a snuff, iv	0.2 mg/kg iv	30 min	None reported
Morning glory	American tropics and warm zones	D-lysergic acid alkaloids	Seeds of *Ipomoea Violacea, Turbina Corymbosa*	Orally as infusion	7–13 seeds	3 hr	Toxic delirium
Nutmeg and mace	Warm zones of Europe, Africa, Asia	Myristicin and aromatic ethers	Fruit of *m. Fragrans*, commercial species	Orally or as a snuff	1 teaspoon, 5–15 g	Unknown	Similar to atropinism, with seizures, death
Yopo/Cohoba	Northern South America, Argentina	β-carbolines and tryptamines	Beans of Anadenanthera *Peregrina*	Smoked or as a snuff	Unknown	Unknown	Ataxia, hallucinations, seizures?
Bufotenin	Northern south America, Argentina	5-OH-dimethyl-tryptamine	Skin glands of toads; seeds of *A. Peregrina*	As a snuff or IV	Unknown	15 min	None reported
Phencyclidine (PCP)	US, synthetic	1-phenylcyclohexylpiperidine	Synthetic	Oral, smoked, as a snuff, IV	5–10 mg	4–6 hr	Psychotic
Ketamine	US, synthetic	(+/-)-2-(2-chlorophenyl)-2-(methylamino)-cyclohexanone	Synthetic	Oral, snorted, IV		1–2 hr	Psychotic

Adapted from Henry David Abraham, MD.

Specific Substances

Hallucinogens are natural and synthetic substances that are variously called *psychedelics* or *psychotomimetics.* The hallucinogens are classified as Schedule I controlled substances; the US FDA has decreed that they have no medical use and high abuse potential.

Table 4-39 lists some natural and synthetic hallucinogens. The most common naturally occurring hallucinogens are psilocybin, derived from certain mushrooms, and mescaline, derived from peyote cactus. The classic synthetic hallucinogen is LSD, synthesized in 1938 by Albert Hoffman, who accidentally ingested the drug and experienced the first LSD trip. Some researchers include some of the so-called designer amphetamines, such as 3,4-methylenedioxyamphetamine (MDMA), as hallucinogens, but we include them among the stimulants.

Phencyclidine and Ketamine. Phencyclidine (PCP; 1-1 [phenylcyclohexyl] piperidine), also known as *angel dust*, was developed as a novel anesthetic in the late 1950s. This drug and the closely related compound ketamine were called *dissociative anesthetics*, because subjects

taking them were awake but insensitive to, or dissociated from, the environment. Phencyclidine and ketamine exert their unique behavioral effects by blocking NMDA-type receptors for the excitatory neurotransmitter glutamate. They can cause a variety of symptoms, from anxiety to psychosis. Phencyclidine and ketamine are classified as Schedule II and Schedule III controlled substances, respectively. Although somewhat different from standard hallucinogens, both DSM-5 and we include among the hallucinogens given their psychological effects.

LSD. LSD is the prototype for a large class of hallucinogenic compounds with well-studied structure–activity relationships. LSD is a synthetic base derived from the lysergic acid nucleus from the ergot alkaloids. That family of compounds was discovered in rye fungus and was responsible for lethal outbreaks of St. Anthony's fire in the Middle Ages. The compounds are also present in morning glory seeds in low concentrations. Researchers have studied many homologs and analogs of LSD, but none more potent than LSD.

Table 4-40 lists the physiologic symptoms from LSD, which are typically few and mild. Usually, somatic symptoms appear first, then mood and perceptual changes, and, finally, psychological changes. The effects may overlap, and depending on the particular hallucinogen, the time of onset and offset varies. The intensity of LSD effects in a nontolerant user generally is proportional to dose, with 25 µg as an approximate threshold dose.

The significant difference between LSD, psilocybin, and mescaline is potency. A 100-µg dose of LSD is roughly equivalent to 10 to 15 mg of psilocybin, which is equivalent to 300 to 400 mg of mescaline. With mescaline, the onset of symptoms is slower and more nausea and vomiting occur, but in general, the perceptual effects are more similar than different.

Tolerance, particularly to the sensory and other psychological effects, is evident as soon as the second or third day of repeated LSD use. Four to 6 days free of LSD is necessary to lose significant tolerance. Tolerance is associated with frequent use of any of the hallucinogens.



Table 4-40
Physiologic Symptoms of LSD

Common:

Dilated pupils
Increased deep tendon motor reflexes
Muscle tension
Mild motor incoordination
Ataxia

Occasional:
Increased heart rate
Increased respiration
Hypertension
Nausea
Decreased appetite
Salivation
Sweating
Blurred vision

Cross-tolerance among these hallucinogens occurs, but not between amphetamine and LSD, despite the chemical similarity of amphetamine and mescaline. Previously distributed as tablets, liquid, powder, and gelatin squares, in recent years, LSD has been commonly distributed as "blotter acid." Sheets of paper are soaked with LSD, and dried and perforated into small squares. Manufacturers stamp popular designs on the paper. Each sheet contains as many as a few hundred squares; one square containing 30 to 75 µg of LSD is one chewed dose, more or less. Planned massive ingestion is uncommon but massive ingestion happens by accident.

The onset of action of LSD occurs within an hour, peaks in 2 to 4 hours, and lasts 8 to 12 hours. LSD is a sympathomimetic, and death caused by cardiac or cerebrovascular pathology related to hypertension or hyperthermia can occur. Some reports suggest that LSD can cause a syndrome similar to neuroleptic malignant syndrome. Death can also result from physical injury when LSD impairs a person's judgment: people may run into traffic or attempt to fly. The psychological effects are usually well tolerated, but when persons cannot recall experiences or appreciate that the experiences are substance-induced, they may fear the onset of insanity.

There is no clear evidence that LSD causes drastic personality changes or

chronic psychosis. Some heavy users of hallucinogens, however, may experience chronic anxiety or depression.

Many persons maintain that a single experience with LSD has given them an increased creative capacity, new psychological insights, relief from neurotic or psychosomatic symptoms, or a desirable change in personality. In the 1950s and 1960s, researchers were interested in the possible pharmacologic effects of LSD. However, results have been inconsistent. The principal value of these drugs for science has been for the basic neurosciences.

Phenethylamines. Phenethylamines are compounds with chemical structures similar to those of the neurotransmitters dopamine and norepinephrine. Mescaline (3,4,5-trimethoxyphenethylamine), a classic hallucinogen in every sense of the term, was the first hallucinogen and is described below.

Another series of phenethylamine analogs with hallucinogenic properties is the 3,4-methylenedioxyamphetamine (MDA)–related amphetamines. MDMA or ecstasy is the currently most popular and, to society, the most troublesome member of this large family of drugs. It is more a relatively mild stimulant than a hallucinogen. MDMA produces an altered state of consciousness with sensory changes and, most notable for some users, a feeling of enhanced personal interactions.

MESCALINE. Mescaline derives from the peyote cactus, which grows in the southwestern United States and northern Mexico. Mescaline human pharmacology was characterized in 1896 and synthesized 23 years later. Although we have recognized many psychoactive plants since antiquity, only mescaline's structure was understood, until we discovered LSD.

People usually take mescaline as peyote "buttons," picked from the small blue-green cacti *Lophophora williamsii* and *Lophophora diffusa*. The buttons are the dried, round, fleshy cacti tops. Mescaline is the active hallucinogenic alkaloid in the buttons. The use of peyote is legal for the Native American Church members in some states. Adverse reactions to peyote are rare during structured religious use. Peyote usually is not consumed casually because of its bitter taste and sometimes severe nausea and vomiting that precede the hallucinogenic effects.

The many structural variations of mescaline are relatively well characterized. One analog, 2,5-dimethoxy-4-methylamphetamine (DOM), also known as STP, an unusually potent amphetamine with hallucinogen properties, had a relatively brief period of illicit popularity in the 1960s. However, it appears to have disappeared from the illicit market.

Psilocybin and Its Analogs. An unusual collection of tryptamines has its origin in the world of fungi. The natural prototype is psilocybin itself. As many as 100 species of mushroom, mainly of the *Psilocybe* genus contain psilocybin or related homologs.

People usually ingest psilocybin as mushrooms. Many species of psilocybin-containing mushrooms exist worldwide. In the United States, large *Psilocybe cubensis* (gold caps) grow in Florida and Texas and are easily grown with cultivation kits advertised in drug-oriented magazines and on the internet. The tiny *Psilocybe semilanceata* (liberty cap) grows in lawns and pastures in the Pacific Northwest. Psilocybin remains active when the mushrooms are dried or cooked into foods.

Psilocybin mushrooms are used in religious activities by Mexican Indians. They are valued in Western society by users who prefer to ingest a mushroom rather than a synthetic chemical. Of course, one danger of eating wild mushrooms is misidentification and ingestion of a poisonous variety.

Although not originating from mushrooms, another analog of psilocybin is N,N-dimethyltryptamine (DMT). Many plants contain this substance, which is also found normally in human biofluids at very low concentrations. When DMT is taken parenterally or by sniffing, a brief, intense hallucinogenic episode can result. As with mescaline, DMT is one of the oldest, best documented, but least potent of the tryptamine hallucinogens. Researchers have evaluated synthesized homologs of DMT in humans, and have described the structure–activity relationships.

Phencyclidine. Phencyclidine and its related compounds are sold as a crystalline powder, paste, liquid, or drug-soaked paper (blotter). Most commonly, drug dealers add PCP to a cannabis- or parsley-containing cigarette. Experienced users report that the effects of 2 to 3 mg of smoked PCP occur in about 5 minutes and plateau in 30 minutes. The bioavailability of PCP is about 75 percent when taken by intravenous administration and

about 30 percent when smoked. The half-life of PCP in humans is about 20 hours, and the half-life of ketamine in humans is about 2 hours.



Table 4-41
Effects of PCP

Physical
Nystagmus
Hypertension
Hyperthermia
Less common:
- Head-rolling movements
- Stroking
- Grimacing
- Muscle rigidity on stimulation
- Vomiting

Psychological
Euphoria
Labile emotions
Bodily warmth
Tingling
Peaceful floating sensations
Depersonalization
Feelings of isolation and estrangement
Auditory and visual hallucinations
Alterations of body image
Distortions of space and time perception
Paranoia
Delusions
Intensified dependence
Confusion
Disorganization of thought
Irritability

Depression
Behavioral
Repetitive chanting speech
Aggressive behavior

The amount of PCP varies greatly from PCP-laced cigarette to cigarette; 1 g can make as few as four or as many as several dozen cigarettes. Less than 5 mg of PCP is considered a low dose, and doses above 10 mg are considered high. Dose variability makes it difficult to predict the effect, although smoking PCP is the easiest and most reliable way for users to titrate the dose.

Persons who have just taken PCP are frequently uncommunicative, appear to be oblivious, and report active fantasy production. They can have a range of side effects, such as those listed in Table 4-41.

The short-term effects last 3 to 6 hours and sometimes give way to a mild depression in which the user becomes irritable, somewhat paranoid, and occasionally belligerent or assaultive. The effects can last for several days. Users sometimes find that it takes 1 to 2 days to recover completely; laboratory tests show that PCP can remain in the patient's blood and urine for more than a week.

Ketamine. Ketamine is a dissociative anesthetic agent, derived initially from PCP, which is available for use in human and veterinary medicine. It has become a drug of abuse, with sources primarily from stolen supplies. It is available as a powder or in a solution for intranasal, oral, inhalational, or (rarely) intravenous use. Ketamine functions by working at the NMDA receptor and, as with PCP, can cause hallucinations and a dissociated state in which the patient has an altered sense of the body and reality and little concern for the environment.

Ketamine causes cardiovascular stimulation and no respiratory depression. On physical examination, the patient may be hypertensive and tachycardic, have increased salivation and bidirectional or rotary nystagmus, or both. The onset of action is within seconds when used intravenously, and analgesia lasting 40 minutes and dissociative effects lasting for hours may occur. Clinicians should monitor a patient's cardiovascular status and administer supportive care. There have been reports of dystonic reactions and

flashbacks, but a more common complication is a lack of concern for the environment or personal safety.

Ketamine has a briefer duration of effect than PCP. Peak ketamine levels occur approximately 20 minutes after intramuscular injection. After intranasal administration, the duration of effect is approximately 1 hour. Ketamine is N-demethylated by liver microsomal cytochrome P450 (CYP), especially CYP3A, into norketamine. Ketamine, norketamine, and dehydronorketamine are detectable in urine, with half-lives of 3, 4, and 7 hours, respectively. Urinary ketamine and norketamine levels vary widely from individual to individual and can range from 10 to 7,000 ng/mL after intoxication. As of yet, there are no formal studies of the relationship between serum ketamine levels and clinical symptoms. Often, Ketamine is combined with other drugs of abuse, especially cocaine. Ketamine may enhance cocaine metabolism. Recent years have seen a growing number of ketamine infusion clinics for the treatment of depression, along with the FDA approval of esketamine nasal spray for treatment-resistant depression.

Additional Hallucinogens

CANTHINONES. Canthinones are alkaloids similar to amphetamines naturally found in the khat plant and synthetically made and known as “bath salts.” They are CNS stimulants that cause a massive release of dopamine, and a single dose can last up to 8 hours. They produce profound toxic effects that can lead to seizures, strokes, or death. Hallucinations and delusions are common. They are swallowed, injected, or “snorted” to produce the desired euphoric effect.

IBOGAINE. Ibogaine is a complex alkaloid found in the African shrub *Tabernanthe iboga.* Ibogaine is a hallucinogen at the 400-mg dose range. The plant originates in Africa, where it is used in traditional ceremonies. It has not been a popular hallucinogen because of its unpleasant somatic effects when taken at hallucinogenic doses. However, psychiatrists may encounter patients exposed to ibogaine because of therapeutic claims.

AYAHUASCA. Ayahuasca, currently a popular subject of internet hallucinogen websites, initially referred to a decoction from one or more South American

plants. The substance contains the alkaloids harmaline and harmine. Both of those β-carboline alkaloids have hallucinogenic properties, but it also causes significant nausea. Amazon tribes discovered that adding leaves from plants containing substantial amounts of DMT enhanced the visual and sacramental impact of ayahuasca. Thus, when people use ayahuasca, it is often in combination with other drugs.

In recent years, the term *ayahuasca* has evolved to a less specific term to refer to any mixture of two things that are hallucinogenic when taken in combination. For example, harmine and harmaline are available as fine chemicals and, when taken along with many botanicals containing DMT, result in a mixture with hallucinogen properties, initially intense but usually of brief duration.

SALVIA DIVINORUM. American Indians in northern Oaxaca, Mexico, have used *Salvia divinorum* as a medicine and as a sacred sacrament, which is now widely discussed, advertised, and sold on the internet. Chewing the plant or smoking dried leaves can produce hallucinogen effects. Salvinorin-A, an active component in the plant, is parenterally potent, active at 250-µg doses when smoked, and of scientific and potential medical interest because it binds to the opioid κ-receptor.

DIAGNOSIS

Hallucinogen Use Disorder

Long-term hallucinogen use is uncommon. Some long-term users of PCP are said to be “crystallized,” a syndrome characterized by dulled thinking, decreased reflexes, loss of memory, loss of impulse control, depression, lethargy, and impaired concentration. Although psychological dependence occurs, it is rare, in part because each LSD experience is different and in part because there is no reliable euphoria.

B, a 16-year-old boy from divorced parents, was admitted to the psychiatric unit of a local hospital. During the previous night, he slashed his wrists, severing nerves and tendons in his left hand, and drifted in and out of consciousness during the night. He finally contacted the mother of

a friend who lived nearby in the morning who immediately brought him to the hospital.

B had a history of juvenile delinquency from the age of 13 when he began hanging out with some older boys at his junior high school. He and his friends shoplifted, stole, smoked marijuana, and took LSD. B's grades dropped, and he got in trouble at school on two occasions for getting into fights with other students.

On admission, B stated that he did not intend to commit suicide when he slashed his wrist. After some questioning, he revealed that he had been "dropping acid" with some friends, and after they left, he thought he heard the sirens of police cars approaching his home. He did not wish to get arrested, so he slashed his wrist and then lost consciousness. He denies feeling depressed, although he claims his life is pointless and felt it made no difference whether he lived or died.

Hallucinogen Intoxication

The hallmarks of hallucinogen intoxication are maladaptive behavioral and perceptual changes, along with particular physiologic signs (Table 4-42). The differential diagnosis for hallucinogen intoxication includes anticholinergic and amphetamine intoxication and alcohol withdrawal. The preferred treatment for hallucinogen intoxication is talking down the patient; during this process, guides can reassure patients that the symptoms are drug-induced, that they are not going crazy, and that the symptoms will resolve shortly. In the most severe cases, dopaminergic antagonists—for example, haloperidol—or benzodiazepines—for example, diazepam—can be used for a limited time. Hallucinogen intoxication usually lacks a withdrawal syndrome.



Table 4-42
Physiologic Changes from Hallucinogens

1. Pupillary dilation
2. Tachycardia
3. Sweating
4. Palpitations

5. Blurring of vision
6. Tremors
7. Incoordination

Short-term PCP intoxication can have potentially severe complications and should be considered a psychiatric emergency. Some patients may present within hours of ingesting PCP, but often 2 to 3 days elapse before they seek psychiatric help. Persons who lose consciousness are brought for help earlier than those who remain conscious. Most patients recover completely within a day or two, but some remain psychotic for as long as 2 weeks. Patients presenting in a coma often exhibit disorientation, hallucinations, confusion, and difficulty communicating on regaining consciousness. These symptoms may also occur in noncomatose patients, but their symptoms appear to be less severe. Behavioral disturbances sometimes are severe; they can include public masturbation, stripping off clothes, violence, urinary incontinence, crying, and inappropriate laughing. Patients frequently have amnesia for the entire period of the psychosis.

Police brought a 17-year-old male patient to the emergency room after finding him disoriented on the street. As the police attempted to question him, he became increasingly agitated; when they attempted to restrain him, he became assaultive. Attempts to question or to examine him in the emergency department evoked increased agitation.

Initially, it was impossible to determine vital signs or to draw blood. Based on the observation of horizontal, vertical, and rotatory nystagmus, clinicians made a preliminary diagnosis of PCP intoxication. Within a few minutes of being placed in a darkened examination room, his agitation markedly decreased. Blood pressure was 170/100; other vital signs were within normal limits. The patient agreed to take 20 mg of diazepam orally. Thirty minutes later, he was less agitated and could be interviewed, although he responded to questions in a fragmented fashion and was slightly dysarthric. He stated that he must have inadvertently taken a larger-than-usual dose of “dust,” which he reported having used once or twice a week for several years. He denied using any other substance and any history of mental disorder. He was confused and

disoriented to time and place. The qualitative toxicology screen revealed PCP and no other drugs. Results of neurologic examination were within normal limits, but he had brisk deep tendon reflexes. Some 90 minutes after arrival, his temperature, initially normal, was elevated to 38°C, his blood pressure had increased to 182/110, and he was poorly responsive to stimulation. He was admitted to a medical bed. His blood pressure and level of consciousness continued to fluctuate over the ensuing 18 hours. Results of hematologic and biochemical analyses of blood, as well as urinalyses, remained within normal limits. A history obtained from his family revealed that the patient had had multiple emergency room visits for complications from PCP use during the previous several years. He had completed a 30-day residential treatment program and had participated in several outpatient programs but had consistently relapsed. The doctors discharged the patient, with a referral to an outpatient treatment program, after his vital signs and level of consciousness had been within normal limits for 8 hours. At discharge, nystagmus and dysarthria were no longer present. (Courtesy of Daniel C. Javitt, M.D., Ph.D., and Stephen R. Zukin, M.D.)

Hallucinogen-Persisting Perception Disorder

Long after ingesting a hallucinogen, a person can experience a flashback of hallucinogenic symptoms. The DSM-5 calls this syndrome *hallucinogen-persisting perception disorder* and ICD-10 refers it as post hallucinogen perception disorder. Table 4-43 lists the comparative criteria for each disorder.

According to studies, from 15 to 80 percent of users of hallucinogens report having experienced flashbacks. They can be triggered by emotional stress, sensory deprivation (e.g., monotonous driving), or use a psychoactive substance, such as alcohol or marijuana.

Flashbacks are spontaneous, transitory recurrences of the substance-induced experience. Table 4-44 gives some examples of flashbacks. The episodes usually last a few seconds to a few minutes, but sometimes last longer. Most often, even in the presence of distinct perceptual disturbances, the person has insight into the pathologic nature of the disturbance. Suicidal behavior, major depressive disorder, and panic disorder are potential

complications.

When diagnosing this disorder several other similar syndromes should be considered in the differential. Migraines and seizures can cause flashback-like experiences. Posttraumatic stress disorder (PTSD) can also cause flashbacks. In addition, a variety of visual system abnormalities can cause perceptual problems that may resemble flashbacks.



Table 4-43
Flashbacks

	Flashbacks Due to Hallucinogen Use	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Hallucinogen-Persisting Perception Disorder	Post hallucinogen perception disorder
Symptoms	1. Reexperiencing, following cessation of use, of perceptual symptoms that were present during intoxication 2. Symptoms cause distress or functional impairment	• Substance-induced changed in cognition, personality, affect, behavior persisting beyond generally accepted period, with onset of symptoms directly related to use of substance • Flashbacks are episodic, short duration, and frequently identical or similar to perceptual distortions experienced during acute intoxication
Required number of symptoms	Both of the above	

Psychosocial consequences of symptoms	Significant distress and/or impairment	
Exclusions	**Result of:** • Medical condition • Another mental disorder	**Result of:** • Alcohol- or psychoactive substance-induced Korsakov syndrome • Alcohol- or psychoactive substance-induced psychotic state



Table 4-44
Examples of Flashbacks

Visual distortion
Geometric hallucinations
Hallucinations of sounds or voices
False perceptions of movement in peripheral fields
Flashes of color
Trails of images from moving objects
Positive afterimages and halos
Macropsia
Micropsia
Time expansion
Physical symptoms
Relived intense emotions

A 20-year-old undergraduate presented with a chief complaint of “seeing the air.” The visual disturbance consisted of perception of white pinpoint specks too numerous to count in both the central and peripheral visual fields. They were always present along with the perception of trails of moving objects left behind as they passed through the patient’s visual field. Attending a hockey game was difficult, as the brightly dressed

players left streaks of their images against the white of the ice for seconds at a time. The patient also described the false perception of movement in stable objects, usually in his peripheral visual fields, halos around objects, and positive and negative afterimages. Other symptoms included mild depression, daily bitemporal headache, and a loss of concentration in the last year.

The visual syndrome had gradually emerged over the last 3 months following experimentation with the hallucinogenic drug LSD-25 on three separate occasions. He feared he had sustained "brain damage" from the drug experience. He denied using any other agents. He had smoked marijuana twice a week for 7 months at age 17.

The patient had consulted two ophthalmologists, who found no visual abnormality. A neurologist's examination also proved negative. A therapeutic trial of an anticonvulsant medication resulted in a 50 percent improvement in the patient's visual symptoms and remission of his depression.

Hallucinogen Intoxication Delirium

Hallucinogen intoxication delirium is a relatively rare disorder beginning during intoxication in those who have ingested pure hallucinogens. An estimated 25 percent of all PCP-related emergency room patients may meet the criteria for hallucinogen intoxication delirium. Hallucinogens are often mixed with other substances, however, and those other substances or the subsequent drug–drug interactions can also cause delirium.

Hallucinogen-Induced Psychotic Disorders

If a patient using a hallucinogen has psychotic symptoms and inadequate reality testing, a diagnosis of a hallucinogen-induced psychotic disorder may be warranted. The course of this can be variable. Occasionally, a protracted psychotic episode is difficult to distinguish from a classic psychosis, such as from schizophrenia. Whether a chronic psychosis after drug ingestion is the result of drug ingestion, is unrelated, or is a combination of both the drug ingestion and predisposing factors is currently unanswerable. In some cases, however, the person had some predisposing disorder, such as schizoid

personality disorder.

A 22-year-old photography student presented to the hospital with an inappropriate mood and bizarre thinking. She had no prior psychiatric history. Nine days before admission, she ingested one or two psilocybin mushrooms. Following the immediate ingestion, the patient began to giggle. She then described euphoria, which progressed to auditory hallucinations and belief in the ability to broadcast her thoughts on the media. Two days later, she repeated the ingestion and continued to exhibit psychotic symptoms to the day of admission. When examined, she heard voices telling her she could be president and reported the sounds of “lambs crying.” She continued to giggle inappropriately, bizarrely turning her head from side to side ritualistically. She continued to describe euphoria, but with an intermittent sense of hopelessness in a context of thought blocking. Her self-description was “feeling lucky.” She received haloperidol, 10 mg twice a day, benztropine 1 mg three times a day, and lithium carbonate 300 mg twice a day. On this regimen, her psychosis improved after 5 days.

Hallucinogen-Induced Mood Disorder

Unlike cocaine-induced mood disorder and amphetamine-induced mood disorder, in which the symptoms are somewhat predictable, mood disorder symptoms accompanying hallucinogen abuse can vary. Abusers may experience manic-like symptoms with grandiose delusions or depression-like feelings and ideas or mixed symptoms. As with the hallucinogen-induced psychotic disorder symptoms, the symptoms of hallucinogen-induced mood disorder usually resolve after elimination from the body.

Hallucinogen-Induced Anxiety Disorder

Hallucinogen-induced anxiety disorder also varies in its symptom pattern, but few data about symptom patterns are available. Anecdotally, emergency department physicians who treat these patients frequently find that the syndrome resembles panic disorder. Anxiety is probably the most common symptom causing a PCP-intoxicated person to seek help in an emergency

room.

Unspecified Hallucinogen-Related Disorder

When a patient with a hallucinogen-related disorder does not meet the criteria for any of the standard hallucinogen-related disorders, this diagnosis may apply. An example would be a withdrawal syndrome associated with a hallucinogen, which has been occasionally reported, although DSM-5 does not include this diagnosis.

TREATMENT

Hallucinogen Intoxication

The most necessary treatment is reassurance and supportive care. Patients experiencing intense and unpleasant hallucinogen intoxication can be helped by a quiet environment, verbal reassurance, and the passage of time. More rapid relief of severe anxiety is likely after oral administration of 20 mg of diazepam or a parenteral benzodiazepine if oral use is not practical. Anxiety and other symptoms generally diminish within 20 minutes of medication administration, compared to hours with only psychological and environmental support; however, perceptual symptoms may persist. Patients may need gentle restraint if they present a danger to themselves or others, but clinicians should avoid restraints if possible. Antipsychotic medications, particularly if given at excessive doses, may worsen symptoms and are best avoided unless the diagnosis remains unclear and the behavior is unmanageable. The marketing of lower doses of LSD and a more sophisticated approach to the treatment of casualties by drug users themselves have combined to reduce the appearance of this once-common disorder in psychiatric treatment facilities.

Phencyclidine. Treatment of PCP intoxication aims to reduce systemic PCP levels and to address significant medical, behavioral, and psychiatric issues. For intoxication and PCP-induced psychotic disorder, although the resolution of current symptoms and signs is paramount, the long-term goal of treatment is to prevent relapse to PCP use. PCP levels can fluctuate over many hours or even days, especially after oral administration. A prolonged

period of clinical observation is, therefore, mandatory before concluding that no severe or life-threatening complications will ensue.

Trapping of ionized PCP in the stomach has led to the suggestion of continuous nasogastric suction as a treatment for PCP intoxication. This strategy, however, can be needlessly intrusive and can induce electrolyte imbalances. The preferred treatment is to administer activated charcoal, which binds PCP and diminishes its toxic effects in animal studies.

Trapping of ionized PCP in urine has led to the suggestion of urinary acidification as an aid to drug elimination. This strategy, however, may be ineffective and is potentially dangerous. The urine only excretes a small portion of PCP, metabolic acidosis itself carries significant risks, and acidic urine can increase the risk of renal failure secondary to rhabdomyolysis.

Hemodialysis and hemoperfusion are also ineffective, given the substantial volume of distribution of PCP.

No drug is known to function as a direct PCP antagonist. Any compound binding to the PCP receptor, located within the ion channel of the NMDA receptor, would block NMDA receptor-mediated ion fluxes, as PCP does. NMDA receptor mechanisms predict that pharmacologic strategies promoting NMDA receptor activation (e.g., administration of a glycine site agonist drug) would promote rapid dissociation of PCP from its binding sites. To date, there have been no clinical trials of NMDA agonists for PCP or ketamine intoxication in humans.

Because PCP disrupts sensory input, environmental stimuli can cause unpredictable, exaggerated, distorted, or violent reactions. A cornerstone of treatment, therefore, is to minimize sensory inputs. Patients should be evaluated and treated in a quiet and calm environment. Precautionary physical restraint is recommended by some authorities, with the risk of rhabdomyolysis from struggling against the restraints balanced by the avoidance of violent or disruptive behavior. Both antipsychotics and benzodiazepines are useful as sedatives, and neither is demonstrably superior. As PCP is an anticholinergic at high doses, drugs with anticholinergic properties (including some antipsychotics) should be avoided.

Hallucinogen-Persisting Perception Disorder. Treatment for hallucinogen-persisting perception disorder is palliative. The first step in the process is to identify the disorder; it is not uncommon for the patient to

consult several specialists before receiving a diagnosis. Pharmacologic approaches include long-acting benzodiazepines, such as clonazepam and, to a lesser extent, anticonvulsants, including valproic acid and carbamazepine. No drug is entirely adequate. Antipsychotics should only be used for psychotic symptoms because their side effects could exacerbate symptoms.

The second dimension of treatment is behavioral. We should instruct the patient to avoid gratuitous stimulation in the form of OTC drugs, caffeine, and alcohol, and avoidable physical and emotional stressors. Marijuana smoke is a particularly strong intensifier of the disorder, even when passively inhaled.

Finally, three comorbid conditions are associated with hallucinogen-persisting perception disorder: panic disorder, major depression, and alcohol use disorder. All these conditions require primary prevention and early intervention.

Hallucinogen-Induced Psychosis. Treatment of hallucinogen-induced psychosis does not differ from conventional treatment for other psychoses. In addition to antipsychotic medications, several agents are reportedly effective, including lithium carbonate, carbamazepine, and electroconvulsive therapy. Antidepressant drugs, benzodiazepines, and other anticonvulsants may have a role as well. Unlike schizophrenia, patients with this disorder lack negative symptoms—they have positive symptoms of psychosis but can still relate well to the psychiatrist. Also important are supportive, educational, and family therapies. The goals of treatment are the control of symptoms, minimal use of hospitals, daily work, the development and preservation of social relationships, and the management of comorbid illnesses such as alcohol use disorder.

EPIDEMIOLOGY

People have used hallucinogens for thousands of years, and drug-induced hallucinogenic states have been part of social and religious rituals. The discovery of LSD in 1943 increased the use and misuse of hallucinogens because it is easy and inexpensive to make such synthetic hallucinogens. They are also more potent than their botanical counterparts, and the development of synthetic hallucinogens is mostly responsible for the

psychiatric disorders associated with these drugs.

The incidence of hallucinogen use has had two notable periods of increase. Between 1965 and 1969, there was a 10-fold increase in the estimated annual number of initiates. This increase was driven primarily by the use of LSD. The second peak in first-time hallucinogen use occurred from around 1992 until 2000, fueled mainly by ecstasy (MDMA). Decreases in the initiation of both LSD and ecstasy were evident between then and 2013, coinciding with an overall drop in hallucinogen incidence from 1.6 to 1.1 million.

The NSDUH found that approximately 0.5 percent of persons age 12 years or older reported current use of hallucinogens. In 2016, the NSDUH added questions about the use of ketamine, DMT/AMT (α-methyltryptamine)/"Foxy" (5-methoxy-N,N-dimethyltryptamine [5-MeO-DMT]), and *Salvia divinorum*; for this reason, it is not possible to compare current rates of use with past rates. The lifetime prevalence of hallucinogen use disorder is about 0.6 to 1.7 percent. Approximately 5 percent of those with a lifetime history of hallucinogen use may develop hallucinogen dependence.

Hallucinogen use is most common among young (15 to 35 years of age) white men. About 60 percent of hallucinogen users are men. Persons 26 to 34 years of age show the highest use of hallucinogens, with 16 percent having used a hallucinogen at least once. Persons 18 to 25 years of age have the highest recent use of a hallucinogen. The ratio of whites to blacks who have used a hallucinogen is 2:1; the white to Hispanic ratio is about 1.5:1.

Cultural factors influence the use of hallucinogens; their use in the western United States is significantly higher than in the southern United States.

Hallucinogen use is associated with less morbidity and less mortality than many other substances of abuse. For example, one study found that only 1 percent of substance-related emergency department visits were related to hallucinogens, compared with 40 percent for cocaine-related problems. Of persons visiting the emergency department for hallucinogen-related reasons, however, more than 50 percent were younger than 20 years of age. Currently, there may be a resurgence in the popularity of hallucinogens among youths.

Phencyclidine

Phencyclidine and some related substances are relatively easy to synthesize in

illegal laboratories and relatively inexpensive to buy on the street. The variable quality of the laboratories, however, results in a range of potency and purity. PCP use varies most markedly with geography. Most users of PCP also use other substances, particularly alcohol, but also opiates, opioids, marijuana, amphetamines, and cocaine. Marijuana products often include PCP, with severe untoward effects on users. The actual rate of PCP use disorder is not known, but PCP is associated with 3 percent of substance abuse deaths and 32 percent of substance-related emergency department visits nationally.

PATHOLOGY

Neuropharmacology

LSD is a reasonable prototype for this group, although there is considerable variation. LSD probably acts on the serotonergic system—but exactly how, and in what direction, is not clear. Data at this time suggest that LSD acts as a partial agonist at postsynaptic serotonin receptors.

Most hallucinogens are well absorbed after oral ingestion, although some are inhaled, smoked, or injected. Tolerance to hallucinogens develops rapidly and is virtually complete after 3 or 4 days of continuous use. Tolerance also reverses quickly, usually in 4 to 7 days. Most people do not experience physical dependence or withdrawal, but they can develop psychological dependence, especially if they find the hallucinations pleasant or useful.

Phencyclidine and Ketamine. PCP and ketamine antagonize the NMDA subtype of glutamate receptors. PCP binds to a site within the NMDA-associated calcium channel and prevents the influx of calcium ions. PCP also activates the dopaminergic neurons of the VTA, which project to the cerebral cortex and the limbic system. Activation of these neurons is usually involved in mediating the reinforcing qualities of PCP.

Tolerance for the effects of PCP occurs in humans, although physical dependence generally does not occur. In animals that are administered more PCP per pound for longer times than most humans, PCP does induce physical dependence. Physical withdrawal in humans is rare, likely because the doses and durations are much lower. Although the physical dependence on PCP is

rare in humans, psychological dependence on both PCP and ketamine is common.

PCP usually comes from illicit laboratories, so impurities are likely. One such contaminant is 1-piperidenocyclohexane carbonitrite, which releases hydrogen cyanide in small quantities when ingested. Another contaminant is piperidine, which can be recognized by its strong, fishy odor.

4.10 Inhalant-Related Disorders

Inhalant drugs (also called *volatile substances* or *solvents*) are volatile hydrocarbons that vaporize to gaseous fumes at room temperature and are inhaled through the nose or mouth to enter the bloodstream via the transpulmonary route. These compounds are commonly found in many household products and are divided into four commercial classes: (1) solvents for glues and adhesives; (2) propellants (e.g., for aerosol paint sprays, hair sprays, and shaving cream); (3) thinners (e.g., for paint products and correction fluids); and (4) fuels (e.g., gasoline, propane). These drugs likely share some similar pharmacologic properties despite their chemical differences.

CLINICAL FEATURES

Persons, especially adolescents, like to inhale these products for their intoxicating effect. Inhalants are associated with several problems, including conduct disorder, mood disorders, suicidality, and physical abuse or neglect and sexual abuse. In some cases, early time-limited use of inhalants may signal a lifelong problem with externalizing behaviors and risk-taking propensity. A smaller subgroup uses inhalants chronically, and such use can cause multiple sequelae, including significant behavioral and organ pathology from the drugs' toxicity.

In small initial doses, inhalants can be disinhibiting and produce feelings of euphoria and excitement as well as pleasant floating sensations, the effects for which persons presumably use the drugs. High doses of inhalants can cause psychological symptoms of fearfulness, sensory illusions, auditory and visual hallucinations, and distortions of body size. The neurologic symptoms

can include slurred speech, decreased speed of talking, and ataxia. Long-term use can be associated with irritability, emotional lability, and impaired memory.

Tolerance for the inhalants does develop for some users; a withdrawal syndrome can accompany the cessation of inhalant use. The withdrawal syndrome does not occur frequently; when it does, it includes sleep disturbances, irritability, jitteriness, sweating, nausea, vomiting, tachycardia, and (sometimes) delusions and hallucinations.

DIAGNOSIS

The DSM-5 describes several disorders related to inhalants, including inhalant use disorder, inhalant intoxication, inhalant intoxication delirium, inhalant-induced neurocognitive disorder, inhalant-induced psychotic disorder, inhalant-induced mood and anxiety disorders, and other inhalant-induced disorders. The DSM-5 excludes anesthetic gases (e.g., nitrous oxide and ether) and short-acting vasodilators (e.g., amyl nitrite) from the inhalant-related disorders, these are classified within the generic category of “other (or unknown) substance-related disorders.” ICD-10 follows a similar approach.

Inhalant Use Disorder

Most persons probably use inhalants for a short time without developing dependence or abuse. Nonetheless, dependence and abuse of inhalants occur.

Inhalant Intoxication

The diagnostic criteria for inhalant intoxication specify the presence of maladaptive behavioral changes and at least two physical symptoms. People who become intoxicated on inhalants may have apathy, diminished social and occupational functioning, impaired judgment, and impulsive or aggressive behavior. They can also have nausea, anorexia, nystagmus, depressed reflexes, and diplopia. With high doses and long exposures, a user’s neurologic status can progress to stupor and unconsciousness, and a person may later be amnestic for the period of intoxication. Clinicians can sometimes identify a recent user of inhalants by rashes around the patient’s nose and mouth; unusual breath odors; the residue of the inhalant substances

on the patient's face, hands, or clothing; and irritation of the patient's eyes, throat, lungs, and nose. The disorder can be chronic, as in the following case.

> A 16-year-old single Hispanic female came to a university substance-treatment program for evaluation. The patient had been convicted for auto theft, using a weapon, and being out of control by her family. By age 15, she had regularly been using inhalants and drinking alcohol heavily. She had tried typewriter-erasing fluid, bleach, tile cleaner, hairspray, nail polish, glue, and gasoline, but preferred spray paint. She had sniffed paint many times each day for about 6 months at age 15, using a maximum of eight paint cans per day. The patient said, "It blacks out everything." Sometimes she had lost consciousness, and she believed that the paint had impaired her memory and made her "dumb." (Courtesy of Thomas J. Crowley, M.D.)

Inhalant Intoxication Delirium. The inhalants can directly cause delirium. Also, their interactions with other substances or the hypoxia caused by their use can cause delirium. If the delirium results in severe behavioral disturbances, short-term treatment with a dopamine receptor antagonist, such as haloperidol, may be necessary. Clinicians should avoid benzodiazepines because of the possibility of increasing the patient's respiratory depression.

Inhalant-Induced Neurocognitive Disorder. Inhalant-induced neurocognitive disorder, as with delirium, may result from the neurotoxic effects of the inhalants themselves, the neurotoxic effects of the metals (e.g., lead) commonly used in inhalants; or the impact of frequent and prolonged periods of hypoxia. Dementia caused by inhalants is likely to be irreversible in all but the mildest cases.

Inhalant-Induced Psychotic Disorder. Clinicians can specify hallucinations or delusions as predominant symptoms. Paranoid states are probably the most common psychotic syndromes during inhalant intoxication.

Inhalant-Induced Mood Disorder and Inhalant-Induced Anxiety

Disorder. Inhalant-induced mood disorder and inhalant-induced anxiety disorder allow the classification of inhalant-related disorders characterized by prominent mood and anxiety symptoms. Depressive disorders are the most common mood disorders associated with inhalant use, and panic disorders and generalized anxiety disorder are the most common anxiety disorders.

Other Inhalant-Induced Disorders. Other Inhalant-Induced Disorder is the recommended DSM-5 diagnosis for inhalant-related disorders that do not fit into one of the diagnostic categories discussed earlier.

TREATMENT

Inhalant intoxication, as with alcohol intoxication, usually requires no medical attention and resolves spontaneously. However, possible effects of the intoxication, such as coma, bronchospasm, laryngospasm, cardiac arrhythmias, trauma, or burns, need treatment. Otherwise, care primarily involves reassurance, quiet support, and attention to vital signs and level of consciousness. Sedative drugs, including benzodiazepines, are contraindicated because they worsen inhalant intoxication.

No established treatment exists for the cognitive and memory problems of inhalant-induced neurocognitive disorder. Severely deteriorated, inhalant-dependent, homeless adults could benefit from street outreach and extensive social service. Patients may require comprehensive support within their families or in foster or domiciliary care.

The course and treatment of inhalant-induced psychotic disorder are like those of inhalant intoxication. The disorder is brief, lasting a few hours to (at most) a very few weeks beyond the intoxication. Vigorous treatment of such life-threatening complications as respiratory or cardiac arrest is indicated, together with conservative management of the intoxication itself. Symptoms such as confusion, panic, and psychosis require special attention to patient safety. Severe agitation may require cautious control with haloperidol (5 mg intramuscularly per 70 kg body weight). Clinicians should avoid sedative drugs because they may aggravate psychosis. Inhalant-induced anxiety and mood disorders may precipitate suicidal ideation, and we should carefully evaluate patients for that possibility. Antianxiety medications and antidepressants are not useful in the acute phase of the disorder; they may be

of use in cases of coexisting anxiety or depressive illness.

Day Treatment and Residential Programs

Day treatment and residential programs have been used successfully, especially for adolescent abusers with combined substance use disorders and other psychiatric disorders. Treatment addresses the comorbid state, which, in most cases, is conduct disorder or, in some other cases, may be ADHD, major depressive disorder, persistent depressive disorder (dysthymia), and PTSD. Treatment for abuse or neglect, both very common in these patients, is also important. Both group and individual therapy that are behaviorally oriented are employed, with immediate rewards for progress toward objectively defined goals in treatment and consequences for lapses to previous behaviors. Patients attend on-site schools with special education teachers, together with planned recreational activities, and the programs provide birth control consultations. The patients' families, often very chaotic, are engaged in modifications of structural family therapy or multisystemic therapy, both of which have reasonable empirical support. Participation in 12-step programs is required. Treatment interventions are coordinated carefully with interventions by community social workers and probation officers. We can monitor progress with urine and breath samples analyzed for alcohol and other drugs at intake and frequently during treatment.

Treatment usually lasts 3 to 12 months. Table 4-45 lists some goals for treatment.

EPIDEMIOLOGY

Inhalant substances are readily available, legal, and inexpensive. These three factors contribute to the high use of inhalants among poor persons and young persons. Approximately 0.2 percent of persons age 12 and older in the United States are current users of inhalants. Adolescents (ages 12 to 17) are more likely to use inhalants than individuals in other age groups. According to the NSDUH, in 2016, only 0.2 percent of adults were using inhalants. In contrast, twice to three times as many adolescents were using inhalants. In adolescents, males and females use inhalants almost equally (with a slight increase in use for females), while in adults, inhalant use is rare in females.

Table 4-45
Goals of Treatment for Inhalant Disorder

Has practiced a plan to stay abstinent
Is showing fewer antisocial behaviors
Has a plan to continue any needed psychiatric treatment (e.g., treatment for comorbid depression)
Has a plan to live in a supportive, drug-free environment
Is interacting with the family in a more productive way
Is working or attending school
Is associating with drug-free, nondelinquent peers

Inhalant use among adolescents may be most common in those whose parents or older siblings use illegal substances, or among adolescents who have experienced childhood trauma or maltreatment. Inhalant use among adolescents is also associated with an increased likelihood of conduct disorder or antisocial personality disorder.

Pathology

Neuropharmacology. Inhalants most used by American adolescents are gasoline, glue (which usually contains toluene), spray paint, solvents, cleaning fluids, and assorted other aerosols. Sniffing vapor through the nose or huffing (taking deep breaths) through the mouth leads to transpulmonary absorption with very rapid drug access to the brain. Breathing through a solvent-soaked cloth, inhaling fumes from a glue-containing bag, huffing vapor sprayed into a plastic bag, or breathing vapor from a gasoline can are common. Approximately 15 to 20 breaths of 1 percent gasoline vapor produce several hours of intoxication. Inhaled toluene concentrations from a glue-containing bag may reach 10,000 ppm, and people may inhale fumes from several tubes each day. By comparison, one study of just 100 ppm of toluene showed that a 6-hour exposure produced a temporary neuropsychological performance decrement of approximately 10 percent.

Inhalants generally act as a CNS depressant. Tolerance for inhalants can develop, although withdrawal symptoms are usually relatively mild.

Inhalants are rapidly absorbed through the lungs and quickly delivered to the brain. The effects appear within 5 minutes and can last for 30 minutes to several hours, depending on the inhalant substance and the dose. The concentrations of many inhalant substances in the blood increase when combined with alcohol, perhaps because of the competition for hepatic enzymes.

Although about one-fifth of an inhalant substance is excreted unchanged by the lungs, the liver metabolizes the remainder. Inhalants are detectable in the blood for 4 to 10 hours after use, and blood samples should be taken in the emergency department when inhalant use is suspected.

Much like alcohol, inhalants have specific pharmacodynamic effects that are not well understood. Because their effects are generally similar and additive to the effects of other CNS depressants (e.g., ethanol, barbiturates, and benzodiazepines), some investigators have suggested that inhalants operate by enhancing the GABA system. Other investigators have suggested that inhalants work through membrane fluidization, which may also be a pharmacodynamic effect of ethanol.

Organ Pathology and Neurologic Effects. Inhalants are associated with many potentially dangerous adverse effects. The most serious of these is death, which can result from respiratory depression, cardiac arrhythmias, asphyxiation, aspiration of vomitus, or accident or injury (e.g., driving while intoxicated with inhalants).

Placing an inhalant-soaked rag and one’s head into a plastic bag, a standard procedure for inhalant users can cause coma and suffocation.

Chronic inhalant users may have numerous neurologic problems. Computed tomography (CT) and magnetic resonance imaging (MRI) reveal diffuse cerebral, cerebellar, and brainstem atrophy with white matter disease, a leukoencephalopathy. Single-photon emission CT (SPECT) of former solvent-abusing adolescents showed both increases and decreases in blood flow in different cerebral areas. Several studies of house painters and factory workers exposed to solvents for long periods also have found evidence of brain atrophy on CT scans, with decreased CBF.

Neurologic and behavioral signs and symptoms can include hearing loss, peripheral neuropathy, headache, paresthesias, cerebellar signs, persisting motor impairment, parkinsonism, apathy, poor concentration, memory loss,

visual–spatial dysfunction, impaired processing of linguistic material, and lead encephalopathy. White matter changes, or pontine atrophy on MRI, have been associated with worse intelligence quotient (IQ) test results. The combination of organic solvents with high concentrations of copper, zinc, and heavy metals can cause brain atrophy, temporal lobe epilepsy, decreased IQ, and a variety of EEG changes.

Other serious adverse effects associated with long-term inhalant use include irreversible hepatic disease or renal damage (tubular acidosis) and permanent muscle damage associated with rhabdomyolysis. Additional adverse effects include cardiovascular and pulmonary symptoms (e.g., chest pain and bronchospasm) as well as GI symptoms (e.g., pain, nausea, vomiting, and hematemesis). There are several clinical reports of *toluene embryopathy*, with signs such as those of fetal alcohol syndrome. These include low birth weight, microcephaly, shortened palpebral fissures, small face, low-set ears, and other dysmorphic signs. These babies reportedly develop slowly, show hyperactivity, and have cerebellar dysfunction. No convincing evidence indicates, however, that toluene, the best-studied inhalant, produces genetic damage in somatic cells.

4.11 Anabolic-Androgenic Steroid Use

The anabolic-androgenic steroids (AAS) are a family of hormones that includes testosterone, the natural male hormone, which, together with numerous synthetic analogs of testosterone, have been developed over the last 70 years (Table 4-46). These drugs exhibit various degrees of anabolic (muscle building) and androgenic (masculinizing) effects; none of these drugs display purely anabolic effects in the absence of androgenic effects. It is important not to confuse the AAS (testosterone-like hormones) with corticosteroids (cortisol-like hormones such as hydrocortisone and prednisone).



Table 4-46
Examples of Commonly Used Anabolic Steroids

Compounds Usually Administered Orally Department:
Fluoxymesterone (Halotestin, Android-F, Ultandren)
Methandienone (formerly called Methandrostenolone; Dianabol)
Methyltestosterone (Android, Testred, Virilon)
Mibolerone (Cheque Drops[a])
Oxandrolone (Anavar)
Oxymetholone (Anadrol, Hemogenin)
Mesterolone (Mestoranum, Proviron)
Stanozolol (Winstrol)

Compounds Usually Administered Intramuscularly:
Nandrolone decanoate (Deca-Durabolin)
Nandrolone phenpropionate (Durabolin)
Methenolone enanthate (Primobolan Depot)
Boldenone undecylenate (Equipoise[a])
Stanozolol (Winstrol-V[a])
Testosterone esters blends (Sustanon, Sten)
Testosterone cypionate
Testosterone enanthate (Delatestryl)
Testosterone propionate (Testoviron, Androlan)
Testosterone undecanoate (Andriol, Restandol)
Trenbolone acetate (Finajet, Finaplix[a])
Trenbolone hexahydrobencylcarbonate (Parabolan)

[a]Veterinary compound.

Note: Many of the brand names listed in this table are from other countries, but they are included because of the widespread illicit use of foreign steroid preparations in the United States.

Corticosteroids are hormones secreted by the adrenal gland, rather than by the testes. Corticosteroids have no muscle-building properties and, hence, little abuse potential; they treat many inflammatory conditions such as poison ivy or asthma. AAS, by contrast, have only limited legitimate medical applications, such as in the treatment of hypogonadal men and diseases associated with muscle wasting, such as acquired immunodeficiency syndrome (AIDS) and cancer. AAS, however, are widely used illicitly,

especially by boys and young men seeking to gain increased muscle mass and strength, either for athletic purposes or to improve personal appearance.

AAS do not have a diagnostic category in the DSM-5; instead, it is coded as one of the other or unknown substance-related disorders. ICD-10 includes the diagnosis in the category "Abuse of non–dependence-producing substances," which also includes antacids, herbal remedies, and vitamins.

CLINICAL FEATURES AND DIAGNOSIS

Steroids may initially induce euphoria and hyperactivity. After relatively short periods, however, their use can become associated with less desirable symptoms. Studies have reported that 2 to 15 percent of anabolic steroid abusers experience hypomanic or manic episodes, and a smaller percentage may have psychotic symptoms. Also disturbing is a correlation between steroid abuse and violence ("roid rage" in the parlance of users). Steroid abusers with no record of antisocial behavior or violence have committed murders and other violent crimes.

Steroids are addictive substances. When abusers stop taking steroids, they can become depressed, anxious, and concerned about the physical state of their bodies. Some male and female weight lifters may have muscle dysmorphia, a form of body dysmorphic disorder in which the individual feels that he or she is not sufficiently muscular and lean. There are some similarities between an athlete's and a patient with anorexia nervosa's self-perception; to an observer, both groups seem to distort the realistic assessment of the body.

Iatrogenic addiction is a consideration given the increasing number of geriatric patients who are receiving testosterone from their physicians in an attempt to increase libido and reverse some aspects of aging.

Mr. A is a 26-year-old single man. He is 69 in tall and presently weighs 204 lb, with a body fat of 11 percent. He reports that he began lifting weights at age 17, at which time he weighed 155 lb. About 2 years after beginning his weight lifting, he began taking AAS, which he obtained through a friend at his gymnasium. His first "cycle" (course) of AAS, lasting for 9 weeks, involved methandienone, 30 mg a day, orally, and

testosterone cypionate, 600 mg a week, intramuscularly. During these 9 weeks, he gained 20 lb of muscle mass. He was so pleased with these results that he took five further cycles of AAS over the next 6 years. During his most ambitious cycle, approximately 1 year ago, he used testosterone cypionate, 600 mg/wk; nandrolone decanoate, 400 mg a week; stanozolol, 12 mg a day; and oxandrolone, 10 mg a day.

During each of the cycles, Mr. A has noted euphoria, irritability, and grandiose feelings. These symptoms were most prominent during his most recent cycle when he felt "invincible." During this cycle, he also noted a decreased need for sleep, racing thoughts, and a tendency to spend excessive amounts of money. For example, he impulsively purchased a $2,700 stereo system when he could not afford to spend more than $500. He also became uncharacteristically irritable with his girlfriend, and on one occasion, put his fist through the side window of her car during an argument, an act inconsistent with his normally mild-mannered personality. After this cycle of AAS ended, he became mildly depressed for about 2 months.

Mr. A has used several drugs to lose weight in preparation for bodybuilding contests. These include ephedrine, amphetamine, triiodothyronine, and thyroxin. Recently, he has also begun to use the opioid agonist–antagonist nalbuphine intravenously (IV) to treat muscle aches from weight lifting. He also used oral opioids, such as controlled-release oxycodone, at least once a week. He uses oral opioids sometimes to treat muscle aches, but often to get high. He reports that the use of nalbuphine and other opioids is widespread among other AAS users of his acquaintance.

Mr. A exhibits characteristic features of muscle dysmorphia. He regularly checks his appearance, including his reflection in windows or even in the back of a spoon. He becomes anxious if he misses even one day of working out at the gym, and acknowledges that his preoccupation with weight lifting has cost him both social and occupational opportunities. Although he has a 48-in chest and 19-in biceps, he has frequently declined invitations to go to the beach or a swimming pool for fear that he would look too small when seen in a bathing suit. He is anxious because he has lost some weight since the end of his previous cycle of AAS and is eager to resume another cycle of AAS soon.

(Adapted from Harrison G. Pope, Jr., M.D., and Kirk J. Brower, M.D.)

Anabolic Steroid–Induced Mood Disorders

Table 4-47 lists some typical symptoms associated with anabolic steroid use. Although athletes using these drugs have long recognized that syndromes of anger and irritability could be associated with AAS use, the scientific literature did not recognize this syndrome until the late 1980s and 1990s. Since then, a series of observational field studies of athletes have suggested that some AAS users develop prominent hypomanic or even manic symptoms during AAS use.

A possible severe consequence of AAS-induced mood disorders may be violent or even homicidal behavior. Several published reports have anecdotally described individuals with no apparent history of psychiatric disorder, no criminal record, and no history of violence, who committed violent crimes, including murder, while under the influence of AAS. Although a causal link is difficult to establish in these cases, lawyers have presented evidence of AAS use in forensic settings as a possible mitigating factor in criminal behavior.



Table 4-47
Symptoms Associated with Anabolic Steroid Use

Euphoria
Hyperactivity
Irritability
Anger
Aggressiveness
Hypomania
Mania
Violent behavior (“Roid Rage”)
Homicidal behavior
Suicide
Psychosis
Withdrawal symptoms: somatization, depression

Depressive syndromes induced by AAS have occurred, and suicide is a risk. A brief and self-limited syndrome of depression occurs upon AAS withdrawal, probably as a result of the depression of the hypothalamic–pituitary–gonadal axis after exogenous AAS administration.

Anabolic Steroid–Induced Psychotic Disorder

Psychotic symptoms are rare in association with anabolic steroid use, but they occasionally occur, primarily in individuals who were using the equivalent of more than 1,000 mg of testosterone a week. Usually, these symptoms consist of grandiose or paranoid delusions, generally occurring in the context of a manic episode, although occasionally occurring in the absence of a frank manic syndrome. In most cases reported, psychotic symptoms have disappeared promptly (within a few weeks) after the discontinuation of the offending agent, although temporary treatment with antipsychotic agents was sometimes required.

Other Anabolic Steroid–Related Disorders

Symptoms of anxiety disorders, such as panic disorder and social phobia, can occur during AAS use. AAS use may serve as a "gateway to the use of opioid agonists or antagonists, such as nalbuphine, or to use of frank opioid agonists, such as heroin."

TREATMENT

Abstinence is the treatment goal of choice for patients manifesting AAS use disorders. To the extent that users of AAS abuse other addictive substances (including alcohol), traditional treatment approaches for substance-related disorders may be used. Nevertheless, AAS users may differ from other addicted patients in several ways that have implications for treatment. First, the euphorigenic and reinforcing effects of AAS may only become apparent after weeks or months of use in conjunction with intensive exercising. When compared with immediately and passively reinforcing drugs, such as cocaine, heroin, and alcohol, AAS use may entail more delayed gratification. Second, AAS users may manifest a more significant commitment to culturally endorsed values of physical fitness, success, victory, and goal directedness

than users of other illicit drugs. Finally, AAS users are often preoccupied with their physical attributes and may rely excessively on these attributes for self-esteem. Treatment, therefore, depends on a therapeutic alliance and a thorough and nonjudgmental understanding of the patient's values and motivations for using AAS.

Though there are no controlled trials of treatment for AAS-related disorders, following the generally recommended approaches for the related idiopathic syndrome may be appropriate. For example, muscle dysmorphia may improve with CBTs or the use of SSRIs.

AAS Withdrawal

Supportive therapy and monitoring are essential for treating AAS withdrawal because suicidal depressions can occur. Hospitalization may be required when suicidal ideation is severe. Patients should be educated about the possible course of withdrawal and reassured that symptoms are time-limited and manageable. Clinicians should reserve antidepressant agents for patients whose depressive symptomatology persists for several weeks after AAS discontinuation and who likely have a comorbid depressive disorder. SSRIs are the preferred agents because of their favorable adverse effect profile and their effectiveness in a reported case series of treated AAS users with major depressive disorder. Physical withdrawal symptoms are not life threatening and do not ordinarily require pharmacotherapy. Nonsteroidal anti-inflammatory drugs (NSAIDs) may be useful to treat musculoskeletal pain and headaches. AAS-induced hypogonadism, including loss of libido, erectile dysfunction, and possible major depression, may begin during AAS withdrawal and can sometimes be prolonged, requiring treatment by an endocrinologist. Patients with AAS-induced hypogonadism may benefit from physiologic exogenous doses of testosterone, gradually tapered over time, as well as clomiphene (to restore hypothalamic–pituitary–testicular [HPT] function) and HCG (to stimulate the testis to resume production of testosterone and spermatozoa).

EPIDEMIOLOGY

It is difficult to estimate the lifetime prevalence of illicit AAS use in the United States because most available data rely on anonymous surveys of

students, particularly students of high school age. There are two serious problems with the use of these data. First, the median age of onset of AAS use, both in the United States and other countries, is approximately 23 years of age—older than for any other major category of illicit drugs. Second, anonymous surveys are vulnerable to serious errors in the opposite direction as a result of false-positive responses to the survey questions about "steroids." When asking students a typical survey question, such as whether they have used "steroid pills or shots without a doctor's prescription," they may falsely respond in the affirmative when in fact they have taken only corticosteroids, or have purchased nutritional supplements OTC that they mistakenly believed were steroids. After allowing for these sources of error, the true prevalence of AAS in high school girls is close to 0 percent and in boys about 1 percent. However, by adulthood, perhaps 3 to 4 percent of American men have used AAS at some time in their lives, and approximately 1.8 percent of American women have used AAS. In general, males are more likely to take AAS than females, and athletes are more likely to take AAS than nonathletes.

The current high rates of steroid use among younger individuals appear to represent an essential shift in the epidemiology of steroid use. In the 1970s, most users were competition bodybuilders or other elite athletes. Since then, however, it appears that an increasing number of young men, and occasionally even young women may use these drugs purely to enhance personal appearance rather than for any athletic purpose.

PATHOLOGY

Pharmacology

All steroid drugs—including AAS, estrogens, and corticosteroids—are synthesized in vivo from cholesterol and resemble cholesterol in their chemical structure. Testosterone has a four-ring chemical structure containing 19 carbon atoms.

Average testosterone plasma concentrations for men range from 300 to 1,000 ng/dL. Generally, 200 mg of testosterone cypionate taken every 2 weeks restores physiologic testosterone concentrations in a hypogonadal male. A eugonadal male who initiates physiologic dosages of testosterone has

no net gain in testosterone concentrations because exogenously administered AAS shut down endogenous testosterone production via feedback inhibition of the hypothalamic–pituitary–gonadal axis. Consequently, illicit users take higher than therapeutic dosages to achieve supraphysiologic effects. The dose–response curve for anabolic effects may be logarithmic, which could explain why illicit users generally take 10 to 100 times the therapeutic dosages. Doses in this range usually require combinations of oral and injected AAS, which illicit AAS users often do. Transdermal testosterone, available by prescription for testosterone replacement therapy, may also be used.

Therapeutic Indications. The AAS are indicated primarily for testosterone deficiency (male hypogonadism), hereditary angioedema (a congenital skin disorder), and some rare forms of anemia caused by bone marrow or renal failure. Women may use AAS for metastatic breast cancer, osteoporosis, endometriosis, and adjunctive treatment of menopausal symptoms. However, these are not first-line treatments. In men, they have been used experimentally as a male contraceptive and for treating major depressive disorder and sexual disorders in eugonadal men. Recently, they have been used to treat wasting syndromes associated with AIDS. Controlled studies have also suggested that testosterone has antidepressant effects in some men infected with HIV with major depressive disorder and is also a supplementary (augmentation) treatment in some depressed men with low endogenous testosterone levels who are refractory to conventional antidepressants.

Adverse Reactions

The most common adverse medical effects of AAS involve the cardiovascular, hepatic, reproductive, and dermatologic systems.

The AAS produce an adverse cholesterol profile by increasing levels of low-density lipoprotein cholesterol and decreasing levels of high-density lipoprotein cholesterol. High-dose use of AAS can also activate hemostasis and increase blood pressure. Isolated case reports of myocardial infarction, cardiomyopathy, left ventricular hypertrophy, and stroke among users of AAS, including fatalities, have appeared.

Among the AAS-induced endocrine effects in men are testicular atrophy

and sterility, both usually reversible after discontinuing AAS, and gynecomastia, which may persist until surgical removal. In women, shrinkage of breast tissue, irregular menses (diminution or cessation), and masculinization (clitoral hypertrophy, hirsutism, and deepened voice) can occur. Masculinizing effects in women may be irreversible. Androgens taken during pregnancy could cause the masculinization of a female fetus. Dermatologic effects include acne and male pattern baldness. Abuse of AAS by children has led to concerns that AAS-induced premature closure of bony epiphyses could cause shortened stature. Other uncommon adverse effects include edema of the extremities caused by water retention, exacerbation of tic disorders, sleep apnea, and polycythemia.

ETIOLOGY

The primary reason for taking illicit AAS is to enhance either athletic performance or physical appearance. Taking AAS is reinforced because they can produce the athletic and physical effects that users desire, especially when combined with proper diet and training. Further reinforcement derives from winning competitions and from social admiration for physical appearance. AAS users also perceive that they can train more intensively for longer durations with less fatigue and with decreased recovery times between workouts.

Although the anabolic or muscle-building properties of AAS are important to those seeking to enhance athletic performance and physical appearance, psychoactive effects may also be valuable in the persistent and dependent use of AAS. Anecdotally, some AAS users report feelings of power, aggressiveness, and euphoria, which become associated with and can reinforce AAS taking.

▲ 4.12 Gambling Disorder

Although not related to a substance, gamblers share many features in common with other addictive disorders.

CLINICAL FEATURES

Pathologic gamblers often appear overconfident, somewhat abrasive, energetic, and free-spending. They often show apparent signs of personal stress, anxiety, and depression. They commonly have the attitude that money is both the cause of and the solution to all their problems. As their gambling increases, they may lie to obtain money and to continue gambling while hiding the extent of their gambling. They make no serious attempt to budget or save money. When without borrowing resources, they are likely to engage in antisocial behavior to obtain money for gambling. Their criminal behavior is typically nonviolent, such as forgery, embezzlement, or fraud, and they consciously intend to return or repay the money.

Complications include alienation from family members and acquaintances, the loss of life accomplishments, suicide attempts, and association with fringe and illegal groups. Arrest for nonviolent crimes may lead to imprisonment.

DIAGNOSIS

DSM-5 refers to this disorder as gambling disorder, whereas ICD-10 calls it pathologic gambling; Table 4-48 lists their comparative approaches to the disorder.

Previous editions of the DSM include pathologic gambling disorder in the impulse-control disorder category because of the patient's preoccupation or compulsion to gamble. However, the criteria for the disorder are more similar to substance-related disorders, given the compulsive behavior, tolerance, and withdrawal effects that develop. Substance use is often comorbid with gambling. Thus, in DSM-5, gambling disorder is included in the section on substance use and addictive disorders and is diagnosed as a non–substance-related disorder. ICD-10 continues to include it among the habit and impulse disorders, referring to it as pathologic gambling.

Gerry was a 35-year-old former auto dealership owner. Two of his uncles were compulsive gamblers, and his paternal grandfather had major depressive disorder. He played poker and had been a racecourse habitué since the age of 15 years. He had dropped out of college after a few months and became a car sales representative. Soon he was promoted to showroom manager and then went out on his own. By age 32 years, he

was a multimillionaire owner of a dealership chain, happily married with two children.

Gerry continued to gamble frequently. He was a successful weekend sports bettor, as well as a consistent winner at weekly gin rummy and poker games and occasional jaunts to Las Vegas and Atlantic City.

In the context of his wife giving birth to a stillborn child, Gerry started going to casinos more often, gradually increasing the size of bets at blackjack and craps. His sports wagers also escalated. His games at home gradually became boring—"there was zilch action." He began frequenting an illegal local poker parlor that featured high-stake action.

Over several years, Gerry slipped into a typical gambling spiral. He accumulated several million dollars in debts and lied to family and colleagues about his whereabouts. He raided his business and personal accounts, including his children's college funds, maxed out credit cards, and borrowed from loan sharks at exorbitant rates. He grew profoundly depressed and seriously thought of killing himself in a car crash so that his insurance would "take care of my family after I am gone."

Gerry's dire situation was unmasked when his Porsche was repossessed one Sunday morning. Initially, his wife threatened to divorce him. However, a wealthy relative intervened and bailed him out. He swore never to gamble again, entered Gamblers Anonymous, and within 2 months, resumed his frantic gambling.

Over the next decade, Gerry underwent four more episodes of recovery and relapse. His wife divorced him, he lost his dealerships, and he had to declare bankruptcy. Gerry finally enrolled in a pilot dual-diagnostic recovery program, where he was diagnosed with atypical bipolar disorder. His treatment included Gamblers Anonymous meetings, individual and family counseling, and pharmacotherapy with bupropion and lamotrigine.

Gerry eventually reconciled with his wife and family. He returned to selling cars, started living modestly, and continued to attend Gamblers Anonymous meetings regularly. However, he declared emphatically that he always considers himself one step away from becoming a "degenerate gambler" again. (Courtesy of Harvey Roy Greenberg, M.D.)

Table 4-48
Gambling Disorder

	Gambling Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic Name	Gambling Disorder	Pathologic Gambling
Duration	Symptoms present within a 12-month period	
Symptoms	• Needing higher stakes to achieve the same thrill • Becomes irritable when trying to stop • Unsuccessful attempts at reducing gambling • Preoccupation with thoughts about gambling • Gambling in response to distress • Gambling to recoup prior losses • Lies to hide gambling • Losses due to	Pattern of gambling that is dominant in the patient's life and resulting in adverse consequences in social, occupational, family, and other areas of life

	gambling: jobs, relationships, other opportunities • Borrows or steals money from others for gambling	
Required number of symptoms	Four or more of the above	
Psychosocial consequences of symptoms	Marked impairment and/or distress	
Exclusions	Manic episode	• Excessive gambling by manic patients • Gambling and betting NOS • Gambling in dissocial personality disorder
Course specifiers	**Episodic:** meets criteria more than once over time, separated by at least several months **Persistent:** continuous over several years **In early remission:** no symptoms for 3–12 mo **In sustained**	

	remission: no symptoms for >12 mo
Severity specifiers	Severity is measured by the number of symptoms present See DSM-5 for ranges

DIFFERENTIAL DIAGNOSIS

Social gambling is distinguished from pathologic gambling in that the former occurs with friends, on special occasions, and with predetermined acceptable and tolerable losses. Gambling that is symptomatic of a manic episode can usually be distinguished from pathologic gambling by the history of a marked mood change and the loss of judgment preceding the gambling.

Manic-like mood changes are frequent in pathologic gambling, but they always follow winning and are usually succeeded by depressive episodes because of subsequent losses. Persons with an antisocial personality disorder may have problems with gambling. When both disorders are present, clinicians should diagnose both.

COMORBIDITY

Significant comorbidity occurs between pathologic gambling and mood disorders (especially major depression and bipolarity) and other substance use and addictive disorders (notably, alcohol, stimulant, caffeine, and tobacco use disorders). Comorbidity also exists with ADHD (particularly in childhood), various personality disorders (notably, narcissistic, antisocial, and borderline personality disorders), and disruptive, impulse control, and conduct disorders. Although many pathologic gamblers have obsessive personality traits, full-blown OCD is uncommon in this group.

Table 4-49
The Four Phases of Pathologic Gambling

1. The winning phase, ending with a big win, equal to about a year's salary, which hooks patients. Women usually do not have a big win but use gambling as an escape from problems.
2. The progressive-loss phase, in which patients structure their lives around gambling and then move from being excellent gamblers to being stupid ones who take considerable risks, cash in securities, borrow money, miss work, and lose jobs.
3. The desperate phase, with patients frenziedly gambling with large amounts of money, not paying debts, becoming involved with loan sharks, writing bad checks, and possibly embezzling.
4. The hopeless stage of accepting that losses can never be made up, but the gambling continues because of the associated arousal or excitement. The disorder may take up to 15 yr to reach the last phase, but then, within a year or two, patients have totally deteriorated.

COURSE AND PROGNOSIS

Pathologic gambling usually begins in adolescence for men and late in life for women. The disorder waxes and wanes and tends to be chronic. Table 4-49 lists the four phases of pathologic gambling.

TREATMENT

Gamblers seldom come forward voluntarily to be treated. Legal difficulties, family pressures, or other psychiatric complaints bring gamblers to treatment. Gamblers Anonymous (GA) was founded in Los Angeles in 1957 and modeled on AA. It is accessible, at least in large cities, and is an effective treatment for gambling in some patients. GA is a method of inspirational group therapy that involves public confession, peer pressure, and the presence of reformed gamblers (as with sponsors in AA) available to help members resist the impulse to gamble. The dropout rate from GA is high, however. Hospitalization may sometimes help by removing patients from

their environments. Clinicians should not start insight-oriented psychotherapy until patients have been away from gambling for 3 months. At this point, patients who are pathologic gamblers may become excellent candidates for this form of psychotherapy. Family therapy is often valuable. CBT (e.g., relaxation techniques combined with visualization of gambling avoidance) has had some success.

Psychopharmacological treatment, once largely unsuccessful, now plays a significant role in the management of pathologic gamblers. Useful agents include antidepressants, notably SSRIs and bupropion; mood stabilizers, including sustained-release lithium and antiepileptics such as lamotrigine; atypical antipsychotics; and opioid agents such as naltrexone. In many patients, it is difficult to determine whether an antidepressant or mood stabilizer alleviates gambling cravings directly or via treatment of a comorbid condition, particularly depressive or bipolar disorders.

EPIDEMIOLOGY

Although we lack comprehensive worldwide statistics, available information indicates that 1 to 2 percent of the general population are problem gamblers. Problem gambling is more common in men and young adults than in women and older adults; however, escalation has been noted in the poor, notably poor minorities, adolescents, elderly retirees, and women. Pathologic gambling is associated with intimate partner violence—both as a perpetrator (for both men and women) and as a victim (typically women only). However, this violence is also due to comorbid psychiatric and substance use disorders.

ETIOLOGY

Psychosocial Factors

Several factors have been described as predisposing persons to develop the disorder: loss of a parent by death, separation, divorce, or desertion before a child is 15 years of age; inappropriate parental discipline (absence, inconsistency, or harshness); exposure to, and availability of, gambling activities for adolescents; a family emphasis on material and financial symbols; and a lack of family emphasis on saving, planning, and budgeting.

Psychoanalytic theory has focused on several core character difficulties.

Edmund Bergler suggested that compulsive gamblers have an unconscious desire to lose and gamble to relieve unconscious feelings of guilt, though these ideas are questionable. Another suggestion is that the gamblers are narcissists whose grandiose and omnipotent fantasies lead them to believe they can control events and even predict their outcome. Learning theorists view uncontrolled gambling as resulting from erroneous perceptions about control of impulses.

Biologic Factors

Many studies indicate that pathologic gamblers suffer from complex neurotransmitter dysregulation, similar to the abnormalities found in substance use disorders and patients with other behavioral/impulsive problems. Most major neurotransmitters have, at some point, been implicated as a significant factor in the dysfunction of cerebral sites and pathways mediating behavioral arousal, inhibition/disinhibition, reward/reinforcement, and psychophysiological stress.

The particular focus on the role of increased dopaminergic activity in pathologic gambling arose from the serendipitous discovery that approximately one-third of patients receiving dopamine agonists for Parkinson disease develop the illness without any previous history. The gambling excesses of these patients usually disappear upon withdrawal of the offending drug.

▲ 4.13 Other Disorders Related to the Addictive Disorders

INTERNET GAMING DISORDER

Also called *Internet Addiction,* such persons spend almost all their waking hours at their computer. Their patterns of use are repetitive and constant, and they are unable to resist intense urges to use the computer or to “surf” the internet. Internet addicts may gravitate to certain sites that meet specific needs (e.g., shopping, sex, and interactive games, among others). In DSM-5, there is a condition proposed for further study called “internet gaming

disorder," which refers to persons who continually use the internet to play games to the extent that it interferes with social relations and work performance. However, as mentioned earlier, the disorder need not be limited to games. Other activities may be involved. General population surveys show a wide range of reported prevalence, from 0 to almost 50 percent, with some of the highest percentages reported in South Asian countries.

Table 4-50
Symptoms of Internet Gaming Disorder

Activities associated with Gaming
Preoccupation w/ gaming
Withdrawal (discomfort when deprived of gaming)
Tolerance (spending more time to achieve same enjoyment from the game)
Compulsive Behavior
Lack of control around gaming
Social or Occupational Dysfunction
Decreased alternative activities
Negative effects on personal relationships, occupational or school performance due to gaming
Lying to loved ones or clinicians about the frequency of gaming
Relying on gaming to improve mood
Continued use despite recognize the harmful effects of compulsive gaming

Table 4-50 lists some of the common symptoms and effects of the disorder. They are organized so as to be consistent with the DSM-5 approach to substance use disorder (see Section 4.1).

Internet Use and Victims

The combination of anonymity, convenience, and escape (the ACE model) promotes the internet as a focus of psychopathology. It can be easy to conceal one's identity and even create alternate identities using various internet platforms. The resulting deception can take a malignant turn as sexual

predators deceive their victims with false identities only to exploit and harm them when they meet. These contacts are unregulated and difficult to detect except by monitoring and checking the computers used. There are frequent reports of minors having been lured into sometimes lethal situations by sexual predators. Occasionally there is a report of a couple that met to marry only to discover they had missed verifying crucial details, such as each other's sex.

Some people who make little use of the internet nonetheless become victims and enter treatment. The suicide of one teenager after reading untruths entered by a peer's malicious mother ("cyberbullying") has inspired laws to criminalize such behavior. Internet identity theft is also rampant. An underreported and growing problem, medical identity theft, is harder to detect and remedy, often requiring painstaking record correction.

MOBILE OR CELL PHONE COMPULSION

Some persons compulsively use mobile or smartphones. Often these individuals find themselves engaging in many of the same patterns of behavior as those with the proposed internet gaming disorder, though the focus is not specifically on the internet or gaming. These individuals may be preoccupied with checking for new messages (text or email), new social media posts, or other smartphone-related activities.

REPETITIVE SELF-MUTILATION

Persons who repeatedly cut themselves or do damage to their bodies may do so in a compulsive manner. Parasuicidal behavior is typical in, but not limited to, borderline personality disorder. Compulsive body piercing or tattooing may be a symptom of a paraphilia or a depressive equivalent.

In DSM-5, there is a proposed diagnosis called "non-suicidal self-injury" to refer to persons who repeatedly damage their bodies, who, however, do not wish to die, contrasted with those persons who harm themselves with real suicidal intent. Most individuals who engage in this behavior do so to obtain relief from dysphoric states or to resolve a conflict. Cutting the skin or inflicting bodily pain may release endorphins or raise dopamine levels in the brain, both of which contribute to a euthymic or elated mood, thus alleviating

depressed states of mind in those who practice self-mutilation.

COMPULSIVE SEXUAL BEHAVIOR

Some persons repeatedly seek out sexual gratification, often in perverse ways (e.g., exhibitionism). They are unable to control their behavior and may not experience feelings of guilt after an episode of acting-out behavior. Sometimes called *sexual addiction,* we discuss this condition in the chapter on sexual disorders (Chapter 16).

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5 Schizophrenia Spectrum and Other Psychotic Disorders

Although we tend to discuss schizophrenia as if it is a single disease, it likely comprises a group of disorders with heterogeneous etiologies, which includes patients whose clinical presentations, treatment response, and courses of illness vary. Signs and symptoms are variable and include changes in perception, emotion, cognition, thinking, and behavior. The expression of these symptoms varies across patients and over time, but the effect of the illness is always severe and is usually long-lasting. The disorder usually begins before age 25 years, persists throughout life, and affects persons of all social classes. Both patients and their families often suffer from inadequate care and social ostracism because of widespread ignorance about the disorder. Schizophrenia is one of the most common of the severe mental disorders, but its essential nature remains to be clarified. Thus, it is sometimes referred to as a syndrome, as a group of disorders, or, as in the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), the schizophrenia spectrum disorders. Clinicians should appreciate that the diagnosis of schizophrenia is based entirely on psychiatric history and mental status examination. There is no laboratory test for schizophrenia.

Because schizophrenia begins early in life, causes significant and long-lasting impairments, makes heavy demands for hospital care, and requires ongoing clinical care, rehabilitation, and support services, the financial cost of the illness in the United States likely exceeds that of all cancers combined. Patients with a diagnosis of schizophrenia account for 15 to 45 percent of homeless Americans, and about 5% of patients with schizophrenia per year are homeless. Worldwide, schizophrenia is one of the top 25 leading causes

of disability. This fact is striking, given its relatively low prevalence. However, this disorder affects not only individuals but families, caregivers, and societies overall. For this reason, indirect costs are enormous and often underestimated.

THE CLINICAL PRESENTATION

No clinical sign or symptom is pathognomonic for schizophrenia; every sign or symptom seen in schizophrenia occurs in other psychiatric and neurologic disorders. Therefore, a patient's history is essential for the diagnosis of schizophrenia; clinicians cannot diagnose schizophrenia only by results of a single mental status examination, as symptoms change with time. For example, a patient may have intermittent hallucinations and a varying ability to perform adequately in social situations, or significant symptoms of a mood disorder may come and go. The clinician should also take into account the patient's educational level, intellectual ability, and cultural and subcultural membership. An impaired ability to understand abstract concepts, for example, may reflect either the patient's education or intelligence. Religious organizations and cults may have customs that seem strange to outsiders but are typical to those within the cultural setting.

The appearance of a patient with schizophrenia can range from that of a completely disheveled, screaming, agitated person to an obsessively groomed, completely silent, and immobile person. Between these two extremes, patients may be talkative and may exhibit bizarre postures. Their behavior may become agitated or violent, apparently in an unprovoked manner, but usually in response to hallucinations. Patients with schizophrenia are often poorly groomed, fail to bathe, and dress too warmly for the prevailing temperatures. Other odd behaviors include tics, stereotypies, mannerisms, and, occasionally, echopraxia, in which patients imitate the posture or the behavior of the examiner.

Patient AB, a 32-year-old woman, began to lose weight and became careless about her work, which deteriorated in quality and quantity. She believed that other women at her place of employment were circulating slanderous stories concerning her and complained that a young man

employed in the same plant had put his arm around her and insulted her. Her family demanded that the charge be investigated, which showed not only that the charge was without foundation but also that the man in question had not spoken to her for months. One day she returned home from work, and as she entered the house, she laughed loudly, watched her sister-in-law suspiciously, refused to answer questions, and at the sight of her brother began to cry. She refused to go to the bathroom, saying that a man was looking in the windows at her. She ate no food, and the next day she declared that her sisters were “bad women,” that everyone was talking about her, and that someone had been having sexual relations with her, and although she could not see him, he was “always around.”

The patient was admitted to a public psychiatric hospital. As she entered the admitting office, she laughed loudly and repeatedly screamed in a loud tone, “She cannot stay here; she’s got to go home!” She grimaced and performed various stereotyped movements of her hands. When seen on the ward an hour later, she paid no attention to questions, although she talked to herself in a childish tone. She moved about constantly, walked on her toes in a dancing manner, pointed aimlessly about, and put out her tongue and sucked her lips in the manner of an infant. At times she moaned and cried like a child but shed no tears. As the months passed, she remained silly, childish, preoccupied, and inaccessible, grimacing, gesturing, pointing at objects in a stereotyped way, and usually chattering to herself in a peculiar high-pitched voice, with little of what she said being understood. Her condition continued to deteriorate, she remained unkempt, and she presented a picture of extreme introversion and regression, with no interest either in the activities of the institution or in her relatives who visited her. (Adapted from case of Arthur P. Noyes, M.D., and Lawrence C. Kolb, M.D.)

Catatonic stupor, often referred to simply as catatonia, is a condition in which patients seem completely lifeless and may exhibit such signs as muteness, negativism, and automatic obedience. Waxy flexibility, once a common sign in catatonia, has become rare, as has manneristic behavior. A person with a less extreme subtype of catatonia may show marked social withdrawal and egocentricity, a lack of spontaneous speech or movement,

and an absence of goal-directed behavior. Patients with catatonia may sit immobile and speechless in their chairs, respond to questions with only short answers, and only move when directed. Other overt behavior may include odd clumsiness or stiffness in body movements.

Localizing and nonlocalizing neurologic signs (also known as hard and soft signs, respectively) are more common in patients with schizophrenia than in other psychiatric patients. Nonlocalizing signs include dysdiadochokinesia, astereognosis, primitive reflexes, and diminished dexterity. The presence of neurologic signs and symptoms correlates with increased severity of illness, affective blunting, and a poor prognosis. Other abnormal neurologic signs include tics, stereotypies, grimacing, impaired fine motor skills, abnormal motor tone, and abnormal movements. Most patients are not aware of their abnormal involuntary movements.

In addition to the disorder of smooth ocular pursuit (saccadic movement), patients with schizophrenia have an elevated blink rate. The elevated blink rate may reflect hyperdopaminergic activity.

The inability of schizophrenia patients to perceive the prosody of speech or to inflect their speech is characteristic of disorders of the nondominant parietal lobe. Other parietal lobe–like symptoms in schizophrenia include the inability to carry out tasks (i.e., apraxia), right–left disorientation, and a lack of concern about the disorder.

Mood

Two common affective symptoms in schizophrenia are reduced emotional responsiveness, sometimes severe enough to warrant the label of anhedonia, and overly active and inappropriate emotions such as extremes of rage, happiness, and anxiety. A flat or blunted affect can be a symptom of the illness itself, of the parkinsonian adverse effects of antipsychotic medications, or depression, and differentiating these symptoms can be a clinical challenge. Overly emotional patients may describe exultant feelings of omnipotence, religious ecstasy, terror at the disintegration of their souls, or paralyzing anxiety about the destruction of the universe. Other feeling tones include perplexity, a sense of isolation, overwhelming ambivalence, and depression.

Thoughts

Psychotic disorders are, first and foremost, thought disorders and the disorder may affect either the process or content of their thought or both. A schizophrenia patient's thoughts may be challenging to understand and elicit. However, this is essential, as thought symptoms may represent the core symptoms of schizophrenia.

Disorders of thought concern the way we formulate ideas and languages. We sometimes call these *formal thought disorders*. The examiner infers a disorder from what and how the patient speaks, writes, or draws. The examiner may also assess the patient's thought process by observing their behavior, especially in carrying out discrete tasks (e.g., in occupational therapy).

When mild, thought disorders might present as stilted or vague. As it worsens, associations become looser. The patient may display circumstantiality, tangential thinking, perseverative thinking, neologisms, echolalia, verbigeration, word salad, and mutism.

Delusions

Delusions, the most obvious example of a disorder of thought content, are varied in schizophrenia and may assume persecutory, grandiose, religious, or somatic forms.

Patients may believe that an outside entity controls their thoughts or behavior or, conversely, that they extraordinarily control outside events (such as causing the sun to rise). Patients may have an intense and consuming preoccupation with esoteric, abstract, symbolic, psychological, or philosophic ideas. Patients may also worry about allegedly life-threatening but bizarre and implausible somatic conditions, such as the presence of aliens inside the patient's testicles affecting his ability to father children.

The phrase *loss of ego boundaries* describes the lack of a clear sense of where the patient's own body, mind, and influence end and where those of other animate and inanimate objects begin. For example, patients may think that other persons, the television, or the newspapers are referring to them (ideas of reference). Thought control, in which outside forces are controlling what the patient thinks or feels, is common, as is thought broadcasting, in which patients think others can read their minds or that they can broadcast

their thoughts through televisions.

Other signs of a loss of ego boundaries include the sense that the patient has physically fused with an external object (e.g., a tree or another person) or that the patient has disintegrated and fused with the entire universe (cosmic identity). With such a state of mind, some patients with schizophrenia doubt their gender or their sexual orientation. These symptoms should not be confused with gender identity problems.

Hallucinations

Any of the five senses may be affected by hallucinatory experiences in patients with schizophrenia. The most common hallucinations, however, are auditory, with voices that are often threatening, obscene, accusatory, or insulting. Two or more voices may converse among themselves, or a voice may comment on the patient's life or behavior.

A 48-year-old man, who had been diagnosed with schizophrenia while in the army at age 21 years, led an isolated and often frightened existence, living alone and supported by disability payments. Although he would confirm that he had chronic auditory hallucinations, he was never comfortable with discussing the content of these hallucinations, and a review of records showed this was a long-term pattern for the patient. Otherwise the patient had good rapport with his psychiatrist and was enthusiastic about the possibility of participating in a study of a novel antipsychotic agent. During the informed consent procedure, the patient asked about the possibility that the new medication might decrease his chronic auditory hallucinations. When it was acknowledged that any response was possible, including decreases in his hallucinations, the patient broke off the discussion abruptly and left the office. At a later visit, he reported that his most reliable pleasure in life was nightly discussions of gossip with hallucinations of voices he believed belonged to 17th-century French courtiers, and the chance that he might lose these conversations and the companionship they offered was too frightening for him to consider. (Adapted from Stephen Lewis, M.D., P. Rodrigo Escalona, M.D., and Samuel J. Keith, M.D.)

Visual hallucinations are common, but tactile, olfactory, and gustatory hallucinations are unusual; their presence should prompt the clinician to consider the possibility of an underlying medical or neurologic disorder that is causing the entire syndrome.

Psychotic disorders can affect other senses, as well. For example, *cenesthetic hallucinations* are unfounded sensations of altered states in bodily organs. Examples of cenesthetic hallucinations include a burning sensation in the brain, a pushing sensation in the blood vessels, and a cutting sensation in the bone marrow. Bodily distortions may also occur.

Cognition

Patients with schizophrenia are usually oriented to person, time, and place. The lack of such orientation should prompt clinicians to investigate the possibility of a medical or neurologic brain disorder. Some patients with schizophrenia may give incorrect or bizarre answers stemming from their delusions, such as “I am Christ; this is heaven, and it is AD 35.”

Memory, as tested in the mental status examination, is usually intact, but there can be minor cognitive deficiencies. It may not be possible, however, to get the patient to attend closely enough to the memory tests for the ability to be assessed adequately.

A significant development in the understanding of the psychopathology of schizophrenia is the appreciation of subtle cognitive impairment. In outpatients, cognitive impairment is a better predictor of the level of function than is the severity of psychotic symptoms. Patients with schizophrenia typically exhibit subtle cognitive dysfunction in the domains of attention, executive function, working memory, and episodic memory. Although a substantial percentage of patients have average intelligence quotients, every person who has schizophrenia may have cognitive dysfunction compared with what he or she would be able to do without the disorder. Although these impairments cannot function as diagnostic tools, they are strongly related to the functional outcome of the illness and, for that reason, have clinical value as prognostic variables, as well as for treatment planning.

The cognitive impairment seems already to be present when patients have their first episode and appear largely to remain stable throughout the early illness. There may be a small subgroup of patients who have true dementia in

late life that is not due to other cognitive disorders, such as Alzheimer disease. Cognitive impairments are also present in attenuated forms in nonpsychotic relatives of schizophrenia patients.

The cognitive impairments of schizophrenia have become the target of pharmacologic and psychosocial treatment trials. Hopefully, effective treatments for these impairments will become available soon.

Insight and Judgment and Reliability

Classically, patients with schizophrenia have poor insight into the nature and severity of their disorder. The so-called lack of insight is associated with poor compliance with treatment. When examining schizophrenia patients, clinicians should carefully define various aspects of insight, such as awareness of symptoms, trouble getting along with people, and the reasons for these problems. Such information can be clinically useful in tailoring a treatment strategy and theoretically useful in postulating what areas of the brain contribute to the observed lack of insight (e.g., the parietal lobes).

A patient with schizophrenia is no less reliable than any other psychiatric patient. The nature of the disorder, however, requires the examiner to verify relevant information through additional sources.

Safety Concerns

Patients with schizophrenia may be agitated and have little impulse control when ill. They may also have decreased social sensitivity and appear to be impulsive when, for example, they grab another patient’s cigarettes, change television channels abruptly, or throw food on the floor. Some impulsive behavior, including suicide and homicide attempts, may be in response to hallucinations commanding the patient to act.

Violent behavior (excluding homicide) is common among untreated schizophrenia patients, and the increased odds of a patient with schizophrenia committing acts of violence, compared to the general population, is 49% to 68%. Delusions of a persecutory nature, previous episodes of violence, and neurologic deficits are risk factors for violent or impulsive behavior. If a clinician feels fearful in the presence of a schizophrenic patient, they should take this as an internal clue that the patient may be on the verge of acting out violently. In such cases, the clinician should terminate the interview or

conduct it with an attendant at the ready.

Suicide. Suicide is the single leading cause of premature death among people with schizophrenia. The lifetime prevalence of suicidality in patients with schizophrenia is about 34.5 percent. Suicide attempts are made by 20 to 50 percent of the patients, with long-term rates of suicide estimated to be 10 to 13 percent. According to DSM-5, approximately 5 to 6 percent of schizophrenic patients die by suicide, but this is probably an underestimation. Often, suicide in schizophrenia seems to occur "out of the blue," without prior warnings or expressions of verbal intent. The most crucial factor is the presence of a major depressive episode. Epidemiologic studies indicate that up to 80 percent of schizophrenia patients may have a major depressive episode at some time in their lives. Some data suggest that those patients with the best prognosis (few negative symptoms, preservation of capacity to experience affects, better abstract thinking) can paradoxically also be at the highest risk for suicide. The profile of the patient at most significant risk is a young man who once had high expectations, declined from a higher level of functioning, realizes that his dreams are not likely to come true, and has lost faith in the effectiveness of treatment. Other possible contributors to the high rate of suicide include command hallucinations and drug abuse. Two-thirds or more of schizophrenic patients who commit suicide have seen an unsuspecting clinician within 72 hours of death. A large pharmacologic study suggests that clozapine may have particular efficacy in reducing suicidal ideation in schizophrenia patients with prior hospitalizations for suicidality. Adjunctive antidepressant medications may be effective for alleviating cooccurring major depression in schizophrenia.

Homicide. Despite the sensational attention that the news media provides when a patient with schizophrenia murders someone, the available data indicate that these patients are no more likely to commit homicide than is a member of the general population. When a patient with schizophrenia does commit homicide, it may be for unpredictable or bizarre reasons based on hallucinations or delusions. Possible predictors of homicidal activity are a history of previous violence, dangerous behavior while hospitalized, and hallucinations or delusions involving violence.

The Symptoms of Schizophrenia Can be Divided into Three Groupings

We can divide the symptoms of schizophrenia into three groups: positive, negative, and cognitive.

Positive Symptoms Are Abnormal Behaviors. Positive symptoms are symptoms that are present and usually observable. These are the symptoms associated with an acute psychotic episode and are primarily disorders of thought and presentation. They include hallucinations, delusions, and other bizarre behaviors. Table 5-1 lists examples of positive symptoms.

Negative Symptoms Are the Absence of Normal Behaviors. Negative symptoms are defined by their absence and sometimes also called deficit symptoms. They are commonly associated with the progression of the illness. These include the absence of affect, the absence of thought, the absence of motivation, the absence of pleasure, and the absence of attention. Table 5-2 lists examples of negative symptoms.

Cognitive Symptoms Are Impairments in Normal Cognitive Functions. The cognitive symptoms of schizophrenia may be subtle, particularly early in the disease process, but are very impairing and account for much of the disability associated with this disorder. They include impairments of attention, working memory, and executive functioning.



Table 5-1
Positive Symptoms

Hallucinations
Auditory hallucinations
Voices commenting
Voices conversing
Somatic or tactile hallucinations
Olfactory hallucinations
Visual hallucinations
Delusions

- Persecutory delusions
- Delusions of jealousy
- Delusions of guilt or sin
- Grandiose delusions
- Religious delusions
- Somatic delusions
- Delusions of reference
- Delusions of being controlled
- Delusions of mind reading
- Thought broadcasting
- Thought insertion
- Thought withdrawal

Bizarre behavior

- Clothing and behavior
- Social and sexual behavior
- Aggressive behavior
- Repetitive or stereotyped behavior

Positive formal thought disorder

- Derailment
- Tangentiality
- Incoherence
- Illogicality
- Circumstantiality
- Pressure of speech
- Distractible speech
- Clanging



Table 5-2
Negative Symptoms

Affective flattening or blunting

- Unchanging facial expressions

Decreased spontaneous movement
Paucity of expressive gesture
Poor eye contact
Affective nonresponsivity
Inappropriate affect
Lack of vocal inflections

Alogia
Poverty of speech
Poverty of content of speech
Blocking
Increased latency of response

Avolition—apathy
Grooming and hygiene
Impersistence at work or school
Physical anergia

Anhedonia—asociality
Recreational interests and activities
Sexual interest and activities
Intimacy and closeness
Relationships with friends

Attention
Social inattentiveness
Inattentiveness during testing

Presentation in Special Populations

The Disorder in Children and Adolescents. A small minority of patients manifest schizophrenia in childhood. Such children may at first present diagnostic problems, particularly with differentiation from mental retardation and autistic disorder. Recent studies have established that we should base our diagnosis of childhood schizophrenia on the same symptoms used for adult schizophrenia. Its onset is usually insidious, its course tends to

be chronic, and the prognosis is mostly unfavorable.

The Disorder in Older People. Late-onset schizophrenia is clinically indistinguishable from schizophrenia but has an onset after age 45 years. This condition tends to appear more frequently in women and tends to be characterized by a predominance of paranoid symptoms. The prognosis is favorable, and these patients usually do well on antipsychotic medication.

DIAGNOSIS

Schizophrenia

Table 5-3 compares the different approaches to diagnosing schizophrenia. The patient should have evidence of a psychotic disorder. However, the presence of hallucinations or delusions is not necessary for a diagnosis of schizophrenia; the patient's disorder is diagnosed as schizophrenia when the patient demonstrates any two of the several symptoms included under the broad category of "psychotic symptoms." These symptoms should persist for an extended time: 6 months for DSM-5 and 1 month for ICD-10. The DSM-5 diagnostic criteria include course specifiers (i.e., prognosis) that offer clinicians several options and describe actual clinical situations.

Catatonic Type. The catatonic type of schizophrenia, which was prevalent several decades ago, has become rare in Europe and North America. The classic feature of the catatonic type is a marked disturbance in motor function; this disturbance may involve stupor, negativism, rigidity, excitement, or posturing.



Table 5-3
Schizophrenia

	DSM-5	ICD-10
Diagnostic name	Schizophrenia	Schizophrenia
Duration	Symptoms present continuously for at least 6 mo	

Symptoms	Delusions Hallucinations Disorganization of speech Disorganization of behavior or catatonia Negative symptoms	Thought distortions Perceptual disorders Negative affect, often blunted Possible cognitive dysfunction Other possible symptoms: • Thought echo • Thought insertion or withdrawal • Thought broadcasting • Delusional perception • Delusions of control, influence, or passivity • Hallucinatory voices • Disordered/disorganized thinking • Negative symptoms
Required number of symptoms	≥2, including at least 1 of the first 3 listed	Defined by the first three listed, although the other symptoms are considered common
Psychosocial consequences of symptoms	Functional impairment	
Exclusions (not better explained by):	Substances Other medical conditions Other psychiatric conditions	Other neurologic diseases Schizoaffective disorder Epilepsy Psychoactive substances
Symptom Specifiers	**With catatonia,** defined as presence of three or more	**Paranoid schizophrenia**—primarily defined by delusions. Less or no disturbance of affect or

	of the following: • ↓ psychomotor activity/stupor • Catalepsy (holding a posture for an extended period) • Waxy flexibility (hold a position but movable to a new posture as if made of wax) • Mutism • Negativism • Posturing • Odd mannerisms • Stereotypic behaviors • Agitation • Grimacing • Echolalia (imitating another's speech) • Echopraxia (imitating another's movements)	volition **Hebephrenic schizophrenia—**negative affect with inappropriate mood, social isolation and unpredictable behavior **Catatonic schizophrenia—**psychomotor changes, such as posturing, odd mannerisms/affect, stupor vs. agitation **Undifferentiated Schizophrenia** **Residual schizophrenia—**chronic illness and cognitive changes resulting from a prolonged psychotic illness **Simple schizophrenia—**slow progressive development of changes in behavior and functioning, affective blunting without preceding psychotic symptoms **Other Schizophrenia** **Schizophrenia unspecified**
Course specifiers	**First episode, currently in**	

acute episode

First episode, currently in partial remission: currently less symptoms than needed for diagnosis

First episode, currently in full remission: 0 symptoms

Multiple episodes, currently in acute episode: ≥2 episodes

Multiple episodes, currently in partial remission

Multiple episodes, currently in full remission

Continuous

Unspecified

AC, age 32 years, was admitted to the hospital. On arrival, he was noted to be an asthenic, poorly nourished man with dilated pupils, hyperactive tendon reflexes, and a pulse rate of 120 beats/min. He showed many mannerisms, laid down on the floor, pulled at his foot, made undirected

violent striking movements, struck attendants, grimaced, assumed rigid and strange postures, refused to speak, and appeared to be having auditory hallucinations. When seen later in the day, he was found to be in a stuporous state. His face was without expression, he was mute and rigid, and he paid no attention to those about him or to their questions. His eyes were closed, and his eyelids could be separated only with effort. There was no response to pinpricks or other painful stimuli.

He gradually became accessible, and when asked concerning himself, he referred to his stuporous period as sleep and maintained that he had no recollection of any events occurring during it. He said, “I didn’t know anything. Everything seemed to be dark as far as my mind is concerned. Then I began to see a little light, like the shape of a star. Then my head got through the star gradually. I saw more and more light until I saw everything in a perfect form a few days ago.” He explained his mutism by saying that he had been afraid he would “say the wrong thing” and that he “didn’t know exactly what to talk about.” From his obviously inadequate emotional response and his statement that he was “a scientist and an inventor of the most extraordinary genius of the 20th century,” it was plain that he was still far from well. (Adapted from case of Arthur P. Noyes, M.D., Lawrence C. Kolb, M.D.)

Sometimes the patient shows a rapid alternation between extremes of excitement and stupor. Associated features include stereotypies, mannerisms, and waxy flexibility. Mutism is particularly common. During catatonic excitement, patients need careful supervision to prevent them from hurting themselves or others. Patients also often require medical care because of malnutrition, exhaustion, hyperpyrexia, or self-inflicted injury.

Subtypes from Previous Versions of DSM. Previous versions of the DSM described subtypes of schizophrenia based predominantly on the clinical features. These were: paranoid, disorganized, catatonic, undifferentiated, and residual subtype. DSM-5 no longer includes these as experts in the field frequently questioned their validity. ICD-10 does continue to include them. Although having some face validity for clinicians, they have only a weak relationship to biologic variables, poor long-term stability, and

poor predictive value.

Schizoaffective Disorder

Schizoaffective disorder has features of both schizophrenia and mood disorders. In current diagnostic systems, patients can receive the diagnosis of schizoaffective disorder if they fit into one of the following six categories: (1) patients with schizophrenia who have mood symptoms, (2) patients with a mood disorder who have symptoms of schizophrenia, (3) patients with both mood disorder and schizophrenia, (4) patients with a third psychosis unrelated to schizophrenia and mood disorder, (5) patients whose disorder is on a continuum between schizophrenia and mood disorder, and (6) patients with some combination of the above. Clinicians frequently use a preliminary diagnosis of schizoaffective disorder when they are uncertain of the diagnosis.

It remains unclear whether the disorder is a subtype of schizophrenia, a mood disorder, or the simultaneous expression of each. It is unlikely that this represents a simultaneous expression of both disorders, as it appears to be more common than would be expected for coincidental co-occurrence. Schizoaffective disorder may also be a third distinct type of psychosis, one that is unrelated to either schizophrenia or a mood disorder. The most likely possibility is that schizoaffective disorder is a heterogeneous group of disorders encompassing all of these possibilities. As will be discussed below, the genetic abnormalities in schizophrenia overlap with that for mood disorders, making an overlap between the disorders more likely.

In the DSM-5 criteria for schizoaffective disorder, the clinician must accurately diagnose the mood disorder, making sure it meets the criteria of either a manic or depressive episode but also determining the exact length of each episode (not always easy or even possible). Table 5-4 compares the diagnostic approaches to schizoaffective disorder.

The length of each episode is critical for two reasons. First, to meet the requirement that the psychotic symptoms must also occur independently of the mood symptoms, it is essential to know when the affective episode ends, and the psychosis continues. Second, the relative lengths of the mood and psychotic episodes should be roughly equal, which requires us to know the course of the episodes.

Mrs. P is a 47-year-old, divorced, unemployed woman who lived alone and who experienced chronic psychotic symptoms despite treatment with olanzapine 20 mg per day and citalopram 20 mg per day. She believed that she was getting messages from God and the police department to go on a mission to fight against drugs. She also believed that an organized crime group was trying to stop her in this pursuit. The onset of her illness began at age 20 years when she experienced the first of several depressive episodes. She also described periods when she felt more energetic and talkative, had a decreased need for sleep, and was more active, sometimes cleaning her house throughout the night. About 4 years after the onset of her symptoms, she began to hear "voices" that became stronger when she was depressed but were still present and disturbed her even when her mood was euthymic. About 10 years after her illness began, she developed the belief that policemen were everywhere and that the neighbors were spying on her. She was hospitalized voluntarily. Two years later, she had another depressive episode, and the auditory hallucinations told her she could not live in her apartment. She was tried on lithium, antidepressants, and antipsychotic medications but continued to be chronically symptomatic with mood symptoms as well as psychosis.

Mrs. P demonstrates a "classic" presentation of schizoaffective disorder in which clear depressive and hypomanic episodes are present in combination with continuous psychotic illness and first-rank symptoms. Her course is typical of many individuals with schizoaffective disorder.

Schizophreniform Disorder

The symptoms of schizophreniform are similar to those of schizophrenia. However, with schizophreniform disorder, the symptoms are short term, lasting at least 1 month but less than 6 months. Patients with schizophreniform disorder should then return to their baseline level of functioning.

Like schizoaffective disorder, schizophreniform disorder appears to be a heterogeneous disorder. Many patients seem to have a disorder that is similar to schizophrenia, while others may have a disorder more like a mood disorder.

Table 5-5 compares the diagnostic approaches to schizophreniform disorder.

The disorder is an acute psychotic disorder with a rapid onset. It lacks a long prodromal phase. Although many patients with schizophreniform disorder may experience functional impairment at the time of an episode, they are unlikely to report a progressive decline in social and occupational functioning. The initial symptom profile is the same as for schizophrenia in that two or more psychotic symptoms must be present.

By definition, patients with schizophreniform disorder have the symptoms for at least a month and return to their baseline state within 6 months. In some instances, the illness is episodic, with more than one episode occurring after long periods of full remission. If the combined duration of symptomatology exceeds 6 months, however, then schizophrenia should be considered.

Mr. C, a 28-year-old accountant, was brought to the emergency department by the police in handcuffs. He was disheveled and shouted and struggled with the police officers. It was apparent that he was hearing voices because he would respond to them with shouts such as, “Shut up! I told you I won’t do it!” However, when confronted about the voices, he denied hearing anything. Mr. C had a hypervigilant stare and jumped at the slightest noise. He stated that he must run away quickly because he knew he would be killed shortly otherwise.

Mr. C was functioning well until 2 months before hospitalization. He was an accountant at a prestigious company and had close friends and a live-in girlfriend. Most people who knew him would describe him as friendly, but he was occasionally quarrelsome.

When his girlfriend suddenly broke off the relationship and moved out of their apartment, Mr. C was distressed. However, he was convinced that he could win her back, so he began to “accidentally” run into her at her job or her new apartment with flowers and various gifts. When she strongly told him that she wanted nothing more to do with him and requested that he leave her alone, Mr. C was convinced that she wanted him dead. He became so preoccupied with this notion that his work began to suffer. Out of fear for his life, Mr. C took off from work frequently,

and when he did report to work, he was often tardy and did subpar work, making many errors. His supervisor confronted Mr. C about his behavior, threatening termination if it continued. Mr. C was embarrassed and resented his supervisor for the confrontation. He believed that his ex-girlfriend had hired the supervisor to kill him.

His beliefs were confirmed by a voice that would mock him. The voice told him time and again that he should quit his job, relocate to another city, and forget about his ex-girlfriend, but Mr. C refused, believing it would give them "more satisfaction than they deserved." He continued working, albeit cautiously, all the while fearing for his life.

Through it all, Mr. C believed himself to be the lone victim. He would awake abruptly at night from nightmares but would be able to fall right back to sleep. He had not lost any weight and had no other vegetative symptoms. His affect alternated between rage and terror. His mind was unusually alert and active, but he was not otherwise hyperactive, excessively energetic, or expansive. He did not display any formal thought disorder.

Mr. C was hospitalized and treated with antipsychotic medication. His symptoms remitted after several weeks of treatment, and he was well and able to return to work shortly after discharge.



Table 5-4
Schizoaffective Disorder

	DSM-5	ICD-10
Diagnostic name	Schizoaffective Disorder	Schizoaffective disorders
Duration	Mood symptoms present majority of time during the illness However, there is also a 2-wk period of psychotic symptoms without mood symptoms	

Symptoms	Meets criteria for a major depressive or manic episode Meets criteria for Schizophrenia	Symptoms of affective episode and schizophrenic symptoms
Required number of symptoms	See criteria for the individual disorders	
Psychosocial consequences of symptoms	Functional impairment	
Exclusions (not better explained by):	Substance use Another mental illness Another medical condition	Schizophrenia Depressive/manic episodes
Symptom Specifiers	**Bipolar Type:** manic episode **Depressive Type:** depressive episode **With catatonia:** See Table 5-3 for catatonia symptoms	**Schizoaffective disorder, manic type** **Schizoaffective disorder, depressive type** **Schizoaffective disorder, mixed type** **Schizoaffective disorder, unspecified** **Other schizoaffective disorder**
Course specifiers	**First episode, currently in acute episode** **First episode, currently in partial remission:**	

	currently less symptoms than needed for diagnosis **First episode, currently in full remission:** 0 symptoms **Multiple episodes, currently in acute episode:** ≥2 episodes **Multiple episodes, currently in partial remission** **Multiple episodes, currently in full remission** **Continuous** **Unspecified**



Table 5-5
Schizophreniform Disorder

	DSM-5	ICD-10
Diagnostic name	Schizophreniform Disorder	Acute and Transient psychotic disorder
Duration	≥1 mo, but <6 mo	<1 mo on average
Symptoms	Same as schizophrenia (see Table 5-3)	Symptoms of schizophrenia, including: Thought echo Thought insertion or withdrawal Thought broadcasting Delusional perception Delusions of control, influence, or passivity

		Hallucinatory voices Disordered/disorganized thinking Negative symptoms May or may not be associated with polymorphic (unstable, frequently changing) delusions, hallucinations and/or behavioral symptoms
Required number of symptoms	Same as schizophrenia (see Table 5-3)	
Exclusions (not better explained by):	Same as schizophrenia (see Table 5-3)	If symptoms persist, diagnosis should be changed to schizophrenia
Symptom Specifiers	**With catatonia:** See Table 5-3 for catatonia symptoms	
Course specifiers	**With good prognostic features:** ≥2 of following: Psychotic symptoms within 4 wk of initial behavioral changes Confusion Good premorbid function No negative	

	symptoms
Without good prognostic features	

Perhaps 60 to 80 percent of patients with this disorder will later develop schizophrenia. The remaining may have relapses of similar time-limited episodes of the disease, whereas some, although unfortunately only a few patients, will have only a single episode.

Brief Psychotic Disorder

Brief psychotic disorder is defined as a psychotic condition that involves the sudden onset of psychotic symptoms, which lasts 1 day or more but less than 1 month. Remission is full, and the individual returns to the premorbid level of functioning. Brief psychotic disorder is an acute and transient psychotic syndrome. The exact incidence and prevalence of brief psychotic disorder is not known, but it is generally considered uncommon. The disorder occurs more often among younger patients (20s and 30s) than among older patients. Brief psychotic disorder is more common in women than in men. Such epidemiologic patterns are sharply distinct from those of schizophrenia. Some clinicians indicate that the disorder may be seen most frequently in patients from low socioeconomic classes and in those who have experienced disasters or major cultural changes (e.g., immigrants). The age of onset in industrialized settings may be higher than in developing countries. Table 5-6 compares the diagnostic approaches to brief psychotic disorder.

The cause of the disorder is not known, however it is common in patients with personality disorders. Persons who have gone through major psychosocial stressors may be at greater risk for subsequent brief psychotic disorder.

The symptoms of brief psychotic disorder always include at least one major symptom of psychosis, such as hallucinations, delusions, and disorganized thoughts, usually with an abrupt onset, but do not always include the entire symptom pattern seen in schizophrenia. Some clinicians have observed that labile mood, confusion, and impaired attention may be more common at the onset of brief psychotic disorder than at the onset of

eventually chronic psychotic disorders. Characteristic symptoms in brief psychotic disorder include emotional volatility, strange or bizarre behavior, screaming or muteness, and impaired memory of recent events. Some of the symptoms suggest a diagnosis of delirium and warrant a medical workup, especially to rule out adverse reactions to drugs.



Table 5-6
Brief Psychotic Disorder

	DSM-5	ICD-10
Diagnostic name	Brief Psychotic Disorder	See Table 5-5 definition of Acute and Transient psychotic disorder—ICD 10 does not distinguish between these two disorders
Duration	≥1 day, <1 mo with return to baseline	
Symptoms	Same as for schizophrenia (see Table 5-3) except negative symptoms not included	
Required number of symptoms	1 of first 3 symptoms (delusions, hallucinations or disorganized speech) +/- behavioral symptoms	
Exclusions (not better explained	Culturally sanctioned response/behavior Another mental illness	

by):	Substance use Another medical condition
Symptom Specifiers	**With marked stressors** **Without marked stressors** **With catatonia:** See Table 5-3 for catatonia symptoms
Course specifiers	**With peripartum onset:** during pregnancy or ≤4 wk after delivery

A 20-year-old man was admitted to the psychiatric ward of a hospital shortly after starting military duty. During the first week after his arrival to the military base, he thought the other recruits looked at him in a strange way. He watched the people around him to see whether they were out “to get” him. He heard voices calling his name several times. He became increasingly suspicious and after another week had to be admitted for psychiatric evaluation. There he was guarded, scowling, skeptical, and depressed. He gave the impression of being very shy and inhibited. His psychotic symptoms disappeared rapidly when he was treated with an antipsychotic drug. However, he had difficulties in adjusting to hospital life. Transfer to a long-term medical hospital was considered, but after 3 months, a decision was made to discharge him to his home. He was subsequently judged unfit to return to military services.

The patient was the eldest of five siblings. His father was an intemperate drinker who became angry and brutal when drunk. The family was poor, and there were constant fights between the parents. As a child, the patient was inhibited and fearful and often ran into the woods

when troubled. He had academic difficulties.

When the patient got older, he preferred to spend time alone and disliked being with people. He occasionally took part in local parties. Although he was never a heavy drinker, he often got into fights when he had a drink or two.

The patient was reinterviewed by hospital personnel at 4 years, 7 years, and 23 years after his admission. He has had no recurrences of any psychotic symptoms and has been fully employed since 6 months after he left the hospital. He married, and at the last follow-up, he had two grown children.

After leaving the hospital, the patient worked for 2 years in a factory. For the past 20 years, he has managed a small business, and it has run well. He has been very happy at work and in his family life. He has made an effort to overcome his tendency toward isolation and has several friends.

The patient believes that his natural tendency is to be socially isolated and that his disorder was connected with the fact that in the military, he was forced to deal with other people. (Adapted from Laura J. Fochtmann, M.D., Ramin Mojtabai, M.D., Ph.D., M.P.H., and Evelyn J. Bromet, Ph.D.)

Delusional Disorder

The diagnosis of delusional disorder is made when a person exhibits one or more delusions of at least 1 month's duration that cannot be attributed to other psychiatric disorders. The delusions are often nonbizarre, meaning that the delusions are about situations that can occur in real life, such as being followed, infected, loved at a distance, and so on; that is, they usually have to do with phenomena that, although not real, are nonetheless possible. Several types of delusions may be present. Table 5-7 compares the diagnostic approaches to delusional disorder.



Table 5-7
Delusional Disorder

DSM-5	ICD-10

Diagnostic name	Delusional Disorder	Delusional Disorder
Duration	≥1 mo	
Symptoms	Delusions (see symptom specifiers for examples)	Delusions Persistent +/- hallucinations
Required number of symptoms	≥1	≥1 delusions
Psychosocial consequences of symptoms	No marked functional impairment	
Exclusions (not better explained by):	Schizophrenia Another medical condition Substance use Another mental illness	Personality disorder Psychosis Psychogenic reaction Schizophrenia
Symptom Specifiers	**Erotomanic type** **Grandiose type** **Jealous type** **Persecutory type** **Somatic type** **Mixed type** **Unspecified type** **With bizarre content:** if not related to reality or a life experience or not possible, include this specifier	
Course specifiers	**First episode, currently in acute episode**	

First episode, currently in partial remission: currently less symptoms than needed for diagnosis

First episode, currently in full remission: 0 symptoms

Multiple episodes, currently in acute episode: ≥2 episodes

Multiple episodes, currently in partial remission

Multiple episodes, currently in full remission

Continuous

Unspecified

Patients are usually well groomed and well dressed, without evidence of gross disintegration of personality or of daily activities, yet they may seem eccentric, odd, suspicious, or hostile. They are sometimes litigious and may make this inclination clear to the examiner. The most remarkable feature of patients with delusional disorder is that the mental status examination shows them to be quite normal except for a markedly abnormal delusional system. Patients may attempt to engage clinicians as allies in their delusions, but a clinician should not pretend to accept the delusion; this collusion further confounds reality and sets the stage for eventual distrust between the patient and the therapist.

Patients' moods are consistent with the content of their delusions. A patient with grandiose delusions is euphoric; one with persecutory delusions is suspicious. Whatever the nature of the delusional system, the examiner may sense some mild depressive qualities.

By definition, patients with delusional disorder do not have prominent or sustained hallucinations. A few delusional patients have other hallucinatory experiences—virtually always auditory rather than visual. They also have generally normal cognition apart from their delusion.

The disorders can be of several types, and in the diagnosis the clinician

should specify the type. The types include: persecutory (believing others are trying to harm them), jealous (believing that a lover or partner is unfaithful), erotomanic (believing that another person, often of higher status, is in love with them), somatic (believing the person has some physical disorder), and grandiose (a “delusion of grandeur”). There is also a mixed and unspecified type, for delusions that are a combination of types or not described by one of the categories. Of these types, persecutory and jealous are probably the most common.

Mrs. S, a 62-year-old-woman, was referred to a psychiatrist because of reports of being unable to sleep. She had previously worked full time taking care of children, and she played tennis almost every day and managed her household chores. However, she had now become preoccupied with the idea that her downstairs neighbor was doing a variety of things to harass her and wanted to get her to move away. At first, Mrs. S based her belief on certain looks that he gave her and damage done to her mailbox, but later she felt he might be leaving empty bottles of cleaning solutions in the basement so she would be overcome by fumes. As a result, the patient was fearful of falling asleep, convinced that she might be asphyxiated and unable to awaken in time to get help. She felt somewhat depressed and thought her appetite might be decreased from the stress of being harassed. However, she had not lost weight and still enjoyed playing tennis and going out with friends. At one point she considered moving to another apartment but then decided to fight back. The episode had gone on for 8 months when her daughter persuaded her to have a psychiatric assessment. In the interview, Mrs. S was pleasant and cooperative. Except for mild depressive symptoms and the specific delusion about being harassed by her neighbor, her mental status was normal.

Mrs. S had a past history of depression 30 years before, which followed the death of a close friend. She saw a counselor for several months and found this helpful, but she was not treated with medication. For the current episode, she agreed to take medications, although she believed her neighbor was more in need of treatment than she was. Her symptoms improved somewhat with risperidone 2 mg at bedtime and

clonazepam 0.5 mg every morning and at bedtime.

This patient presented with a single delusion regarding her neighbor that was within the realm of possibility (i.e., not bizarre). Other areas of her functioning were normal. Although mild depressive symptoms were present, she did not meet criteria for major depressive disorder. Her prior symptoms of depression appeared to be related to a normal bereavement reaction and had not required pharmacotherapy or hospitalization. Thus, her current presentation is one of delusional disorder, persecutory type and not major depressive disorder with psychotic features. In terms of treatment, the ability to create a working alliance with the patient, avoiding the discussion of the veracity of her delusion, and focusing on her anxiety, depression, and difficulty falling asleep enabled her psychiatrist to introduce the medications with beneficial results. (Courtesy of Laura J. Fochtmann, M.D., Ramin Mojtabai, M.D., Ph.D., M.P.H., and Evelyn J. Bromet, Ph.D.)

Mr. M was a 51-year-old married white man who lived with his wife in their own home and who worked full time driving a sanitation truck. Before his hospitalization, he became concerned that his wife was having an affair. He began to follow her, kept notes on his observations, and badgered her constantly about this, often waking her up in the middle of the night to make accusations. Shortly before admission, these arguments led to physical violence, and he was brought to the hospital by police. In addition to concerns about his wife's fidelity, Mr. M reported feelings of depression over his wife's "betrayal of [their] marriage vows," but he noted no changes in sleep, appetite, or work-related functioning. He was treated with a low dose of an antipsychotic medication and described being less concerned about his wife's behavior. After discharge, he remained on medications and was seen by a psychiatrist monthly, but 10 years later, he continued to believe that his wife was unfaithful. His wife noted that he sometimes became upset about the delusion but that he had not become aggressive or required readmission.

This patient experienced a fixed, encapsulated delusion of jealousy that did not interfere with his other activities and that showed a partial

response to antipsychotic medications. Although he initially reported feeling somewhat depressed over his wife's perceived infidelity, he did not have other symptoms suggestive of a major depressive episode. (Courtesy of Laura J. Fochtmann, M.D., Ramin Mojtabai, M.D., Ph.D., M.P.H., and Evelyn J. Bromet, Ph.D.)

Little is known about the epidemiology of delusional disorder as it is rare, and is, likely, a heterogeneous group of disorders that manifest as delusions. Despite many theories, the causes are not known. Many patients with the disorder can function well in society and never come to psychiatric attention. Psychiatrists may only meet them when they are evaluating them for another disorder, such as major depressive disorder. Other times, a physician may request a psychiatric consult for a patient they are evaluating for some other medical disorder and notice some odd responses. The disorder appears to be stable over time. Although reliable data are limited, patients with persecutory, somatic, and erotomanic delusions are thought to have a better prognosis than patients with grandiose and jealous delusions.

Other Psychotic Disorders

Patients may present with psychotic symptoms that are not easily described by one of the psychotic disorders. Some examples include patients with persistent auditory hallucination but no other symptoms, or delusions that present with significant mood symptoms. In some cases a clinician may consider the symptoms that seem psychotic to be so minor as not to deserve a diagnosis of a typical disorder; these can include very transient symptoms or presentations where the patient seems to have full insight into their delusions or hallucinations.

One pattern of psychosis that deserves mention is the presence of delusional symptoms in a partner of an individual with delusional disorder, commonly called "shared psychosis," or *folie á deux*. It is probably rare, but incidence and prevalence figures are lacking, and the literature consists almost entirely of case reports.

The disorder is characterized by the transfer of delusions from one person to another. Both persons are closely associated for a long time and typically live together in relative social isolation. In its most common form (and the

one recognized in DSM-5), the individual who first has the delusion (the primary case) is often chronically ill and typically is the influential member of a close relationship with a more suggestible person (the secondary case) who also develops the delusion. The person in the secondary case is frequently less intelligent, more gullible, more passive, or more lacking in self-esteem than the person in the primary case. If the pair separates, the secondary person may abandon the delusion, but this outcome is not seen uniformly. The occurrence of the delusion is attributed to the strong influence of the more dominant member. Old age, low intelligence, sensory impairment, cerebrovascular disease, and alcohol abuse are among the factors associated with this peculiar form of psychotic disorder. A genetic predisposition to idiopathic psychoses has also been suggested as a possible risk factor.

Other special forms have been reported, such as *folie simultanée*, in which two persons become psychotic simultaneously and share the same delusion. Occasionally, more than two individuals are involved (e.g., *folie á trois*, *quatre*, *cinq*; also *folie á famille*), but such cases are especially rare. The most common relationships in shared psychotic disorder are sister–sister, husband–wife, and mother–child, but other combinations have also been described. Almost all cases involve members of a single family.

A 52-year-old man was referred by the court for inpatient psychiatric examination, charged with disturbing the peace. He had been arrested for disrupting a trial, complaining of harassment by various judges. He had walked into a courtroom, marched to the bench, and begun to berate the probate judge. While in the hospital, he related a detailed account of conspiratorial goings-on in the local judiciary. A target of certain judges, he claimed he had been singled out for a variety of reasons for many years: he knew what was going on; he had kept records of wrongdoings; and he understood the significance of the whole matter. He refused to elaborate on the specific nature of the conspiracy. He had responded to it with frequent letters to newspapers, the local bar association, and even to a Congressional subcommittee. His mental state, apart from his story and a mildly depressed mood, was entirely normal.

A family interview revealed that his wife and several grown children

shared the belief in a judicial conspiracy directed against the patient. There was no change in delusional thinking in the patient or the family after 10 days of observation. The patient refused follow-up.

In this case, protection is provided by others who share the delusion and believe in the reasonableness of the response; such cases are uncommon, if not rare. (Courtesy of TC Manschreck, M.D.)

OBJECTIVE TESTS FOR THE DISORDER

Schizophrenia and the other psychotic disorders remain clinical diagnoses and no tests are sensitive or specific enough for diagnostic purposes. Most tests, such as serologic testing, are mainly used to rule out other causes of psychosis (such as syphilis or anti-NMDA receptor encephalitis). However, some tests are abnormal on average when comparing groups of patients with schizophrenia to patients without the disorder. Examples include computerized electroencephalogram (EEG), which show differences in event-related potentials.

Diagnostic and Rating Scales for Schizophrenia

Several diagnostic interviews and rating scales exist for schizophrenia. Some are useful for diagnosing the disorder, although they are mainly used in research settings.

Most scales measure the outcomes of various interventions, not the presence of a diagnosis. They focus on measuring clinical symptoms associated with the disorder. These include the Positive and Negative Syndrome Scale (PANSS), and the Brief Psychiatric Rating Scale (BPRS), both of which can track the significant symptoms of the disorder. We can assess for extrapyramidal symptoms with a variety of instruments, but the Simpson Angus Scale (SAS), the Abnormal Involuntary Movement Scale (AIMS) and Barnes Akathisia Rating Scale (BARS) are most common.

Psychological Testing

Patients with schizophrenia generally perform poorly on a wide range of neuropsychological tests. Vigilance, memory, and concept formation are most affected and consistent with pathologic involvement in the

frontotemporal cortex.

Objective measures of neuropsychological performance, such as the Halstead–Reitan battery and the Luria–Nebraska battery, often give abnormal findings, such as bilateral frontal and temporal lobe dysfunction, including impairments in attention, retention time, and problem-solving ability. Motor ability is also impaired, possibly related to brain asymmetry.

Intelligence Tests. When comparing groups of patients with schizophrenia with groups of psychiatric patients without schizophrenia or from the general population, schizophrenia patients tend to score lower on intelligence tests. Statistically, the evidence suggests that low intelligence is often present at the onset, and intelligence may continue to deteriorate with the progression of the disorder.

Projective and Personality Tests. Projective tests, such as the Rorschach test and the Thematic Apperception Test, may indicate bizarre ideation. Personality inventories, such as the Minnesota Multiphasic Personality Inventory, often give abnormal results in schizophrenia, but the contribution to diagnosis and treatment planning is minimal.

DIFFERENTIAL DIAGNOSIS FOR PSYCHOTIC DISORDERS

Secondary Psychotic Disorders

A wide range of nonpsychiatric medical conditions and a variety of substances can induce symptoms of psychosis and catatonia (Table 5-8). The most appropriate diagnosis for such symptoms is a psychotic disorder due to a general medical condition, catatonic disorder due to a general medical condition, or substance-induced psychotic disorder.

When evaluating a patient with psychotic symptoms, clinicians should follow the general guidelines for assessing nonpsychiatric conditions. First, clinicians should aggressively pursue an undiagnosed nonpsychiatric medical condition when a patient exhibits any unusual or rare symptoms or any variation in the level of consciousness. Second, clinicians should attempt to obtain a complete family history, including a history of medical, neurologic,

and psychiatric disorders. Third, clinicians should consider the possibility of a nonpsychiatric medical condition, even in patients with previous diagnoses of schizophrenia. A patient with schizophrenia is just as likely to have a brain tumor that produces psychotic symptoms, as is a patient without schizophrenia.



Table 5-8
Potential Medical Etiologies of Delusional Syndromes

Disease or Disorder Class	Examples
Neurodegenerative disorders	Alzheimer disease, Pick disease, Huntington disease, basal ganglia calcification, multiple sclerosis, metachromatic leukodystrophy
Other central nervous system disorders	Brain tumors, especially temporal lobe and deep hemispheric tumors; epilepsy, especially complex partial seizure disorder; head trauma (subdural hematoma); anoxic brain injury; fat embolism
Vascular disease	Atherosclerotic vascular disease, especially when associated with diffuse, temporoparietal, or subcortical lesions; hypertensive encephalopathy; subarachnoid hemorrhage, temporal arteritis
Infectious disease	HIIV/AIDS encephalitis lethargica, Creutzfeldt–Jakob disease, syphilis, malaria, acute viral encephalitis
Metabolic disorder	Hypercalcemia, hyponatremia, hypoglycemia, uremia, hepatic encephalopathy, porphyria
Endocrinopathies	Addison disease, Cushing syndrome, hyper-

	or hypothyroidism, panhypopituitarism
Vitamin deficiencies	Vitamin B_{12} deficiency, folate deficiency, thiamine deficiency, niacin deficiency
Medications	Adrenocorticotropic hormones, anabolic steroids, corticosteroids, cimetidine, antibiotics (cephalosporins, penicillin), disulfiram, anticholinergic agents
Substances	Amphetamines, cocaine, alcohol, cannabis, hallucinogens
Toxins	Mercury, arsenic, manganese, thallium

Mood Disorders

A patient with a major depressive episode may present with delusions and hallucinations, as can a patient with bipolar disorder. Delusions seen with psychotic depression are typically mood-congruent and involve themes such as guilt, self-depreciation, deserved punishment, and incurable illnesses. In mood disorders, psychotic symptoms resolve entirely with the resolution of depression. A depressive episode that is this severe may also result in loss of functioning, a decline in self-care, and social isolation, but these are secondary to the depressive symptoms and should not be confused with the negative symptoms of schizophrenia.

A full-blown manic episode often presents with delusions and sometimes hallucinations. Delusions in mania are most often mood-congruent and typically involve grandiose themes. The flight of ideas seen in mania may, at times, be confused with the thought disorder of schizophrenia. Special attention during the mental status examination of a patient with a flight of ideas is required to note whether the associative links between topics are conserved. However, the conversation is awkward for the observer to follow because of the patient's accelerated rate of thinking.

Personality Disorders

Various personality disorders may have some features of schizophrenia. Schizotypal, schizoid, and borderline personality disorders are personality

disorders with several overlapping symptoms. A severe obsessive-compulsive personality disorder may mask an underlying schizophrenic process. Personality disorders, unlike schizophrenia, have mild symptoms and a history of occurring throughout a patient's life; they also lack an identifiable date of onset.

Malingering and Factitious Disorders

For a patient who imitates the symptoms of schizophrenia but does not have the disorder, either malingering or factitious disorder may be an appropriate diagnosis. Although truly mimicking the symptoms of schizophrenia is difficult, especially in front of an experienced clinician, persons have faked schizophrenic symptoms and have been admitted into and treated at psychiatric hospitals. The condition of patients who are entirely in control of their symptom production may qualify for a diagnosis of malingering; such patients usually have some apparent financial or legal reason to want to be considered mentally ill. The condition of patients who are less in control of their falsification of psychotic symptoms may qualify for a diagnosis of factitious disorder. Some patients with schizophrenia, however, may falsely complain of an exacerbation of psychotic symptoms to obtain increased assistance benefits or to gain admission to a hospital.

COMORBIDITY

Substance Use Disorders

Comorbid substance use disorders in patients with schizophrenia are widespread, with a lifetime prevalence of 74%. Tobacco, alcohol, cannabis, and cocaine use disorders are most common, and almost half of the patients with schizophrenia will have a severe problem with drugs or alcohol during their lifetime.

The link between the two disorders is not clear, and experts in the field have proposed several explanations. The diathesis-stress model suggests that a biologically vulnerable person, when encountering external stress (of which a substance may be one), is more likely to develop schizophrenia. The self-medication hypothesis suggests that patients use substances to lessen their symptoms or side effects. The shared vulnerability model suggests that both

schizophrenia and substance use disorders share a common etiology and pathology. More research is required. However, the available evidence supports a shared vulnerability, in which a shared genetic risk or environmental insult leads to dysfunctions in specific key circuits, such as reward pathways. The result is the increased use of substances during adolescence, as well as the development of schizophrenia.

Complex Partial Epilepsy

Schizophrenia-like psychoses occur more frequently than expected in patients with complex partial seizures, especially seizures involving the temporal lobes. Factors associated with the development of psychosis in these patients include a left-sided seizure focus, medial temporal location of the lesion, and an early onset of seizures. Some psychotic symptoms are similar to symptoms of patients with complex partial epilepsy and may reflect the presence of a temporal lobe disorder when seen in patients with schizophrenia.

Obesity

Patients with schizophrenia appear to be more obese, with higher body mass indexes (BMIs) than age- and gender-matched cohorts in the general population. This obesity is due, at least in part, to the effect of many antipsychotic medications, as well as poor nutritional balance and decreased motor activity. This weight gain, in turn, contributes to an increased risk of cardiovascular morbidity and mortality, an increased risk of diabetes, and other obesity-related conditions such as hyperlipidemia and obstructive sleep apnea.

Diabetes Mellitus

Schizophrenia is associated with an increased risk of type II diabetes mellitus. This comorbidity is probably due, in part, to the association with obesity noted above, but there is also evidence that some antipsychotic medications cause diabetes through a direct mechanism.

Cardiovascular Disease

Many antipsychotic medications have direct effects on cardiac

electrophysiology. Also, obesity, increased rates of smoking, diabetes, and hyperlipidemia, and a sedentary lifestyle all independently increase the risk of cardiovascular morbidity and mortality.

HIV

Patients with schizophrenia appear to have a risk of HIV infection that is 1.5 to 2 times that of the general population. This association is likely due to increased risk behaviors, such as unprotected sex, multiple partners, and increased drug use.

Chronic Obstructive Pulmonary Disease

Rates of chronic obstructive pulmonary disease are increased in schizophrenia compared with the general population. The increased prevalence of smoking is a prominent contributor to this problem; it is not clear whether this is the only one.

Rheumatoid Arthritis

Patients with schizophrenia have a reduced risk of rheumatoid arthritis than is found in the general population. Researchers have replicated this inverse association several times, the significance of which is unknown. GWAS studies also indicate negative genetic correlations, suggesting that they may have shared pathogenesis but differential risks. The exact nature of this relationship remains a mystery and is an ongoing subject of investigation.

COURSE

Onset of Schizophrenia

Premorbid Signs and Symptoms. In theoretical formulations of the course of schizophrenia, premorbid signs and symptoms appear before the prodromal phase of the illness. The differentiation implies that premorbid signs and symptoms exist before the disease process evidences itself and that the prodromal signs and symptoms are parts of the evolving disorder. In the typical, but not invariable, premorbid history of schizophrenia, patients had schizoid or schizotypal personalities characterized as quiet, passive, and

introverted; as children, they had few friends. Preschizophrenic adolescents may have no close friends and no dates and may avoid team sports. They may enjoy watching movies and television, listening to music, or playing computer games to the exclusion of social activities. Some adolescent patients may show a sudden onset of obsessive-compulsive behavior as part of the prodromal picture.

A premorbid pattern of symptoms may be the first evidence of illness, although the importance of the symptoms is usually recognized only retrospectively. Nevertheless, although the family members often believe the disorder began with the first hospitalization, signs and symptoms have often been present for months or even years. The signs may have started with complaints about somatic symptoms, such as headache, back and muscle pain, weakness, and digestive problems. The initial diagnosis may be malingering, chronic fatigue syndrome, or somatic symptom disorder. Family and friends may eventually notice that the person has changed and is no longer functioning well in occupational, social, and personal activities. During this stage, a patient may begin to develop an interest in abstract ideas, philosophy, and the occult or religious questions (Fig. 5-1). Additional prodromal signs and symptoms can include markedly peculiar behavior, abnormal affect, unusual speech, bizarre ideas, and strange perceptual experiences.

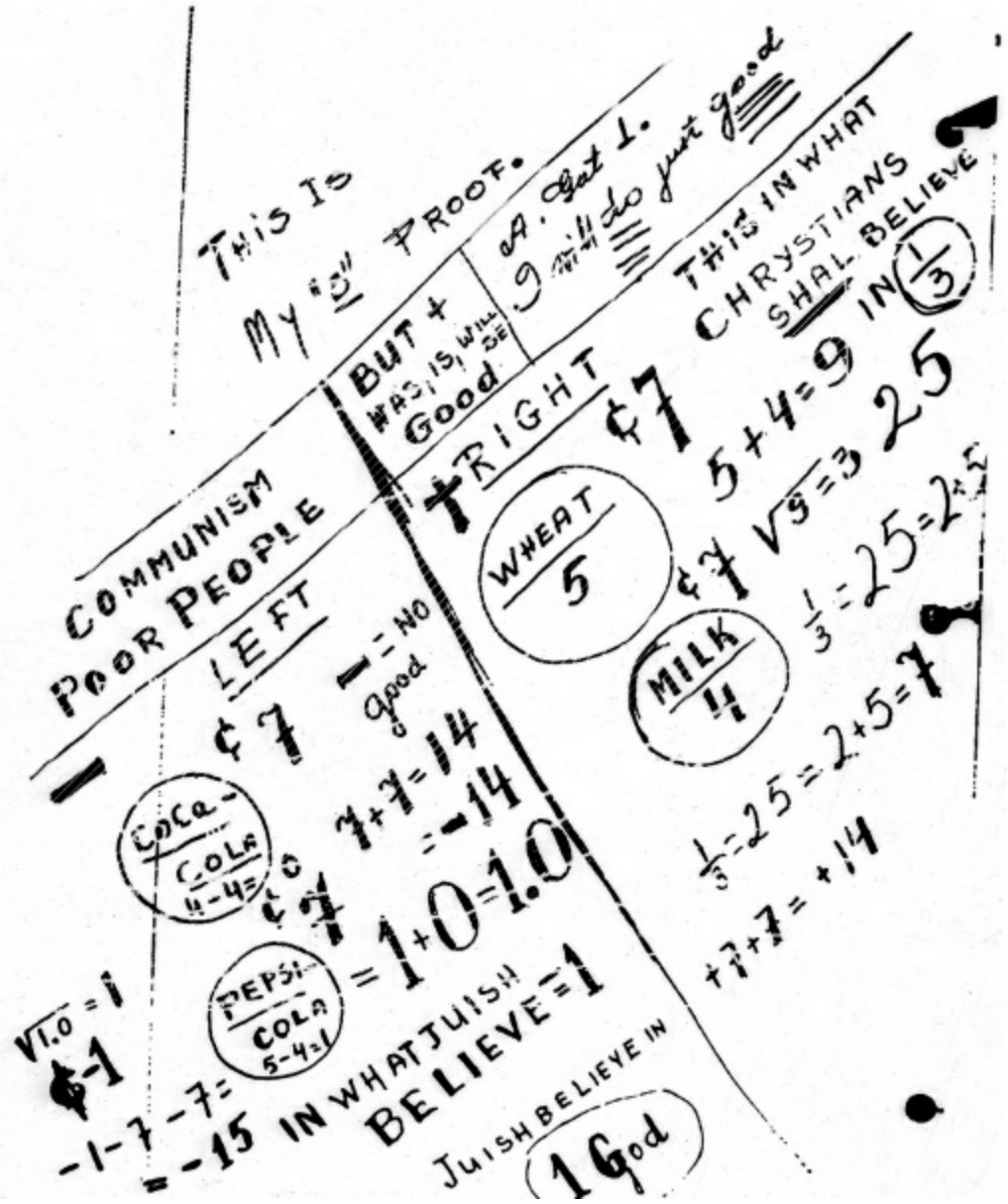


FIGURE 5-1
Schizophrenia patient schema. This illustrates his fragmented, abstract, and overly inclusive thinking and preoccupation with religious ideologies and mathematical proofs. (Courtesy of Heinz E. Lehmann.)

Emergence of Symptoms. Characteristically, the symptoms begin in adolescence and are followed by the development of prodromal symptoms in days to a few months. Social or environmental changes, such as going away to college, using a substance, or a relative's death, may precipitate the disturbing symptoms, and the prodromal syndrome may last a year or more before the onset of overt psychotic symptoms.

Duration. The classic course of schizophrenia is one of exacerbations and remissions. After the first psychotic episode, a patient gradually recovers and may then function relatively normally for a long time. Patients usually relapse, however, and the pattern of illness during the first 5 years after the diagnosis generally indicates the patient's course. Further deterioration in the patient's baseline functioning follows each relapse of psychosis. This failure to return to baseline functioning after each relapse was historically thought to be the primary distinction between schizophrenia and mood disorders. Sometimes a clinically observable postpsychotic depression follows a psychotic episode, and the schizophrenia patient's vulnerability to stress is usually lifelong. Positive symptoms tend to become less severe with time, but the socially debilitating negative or deficit symptoms may increase in severity.

An unmarried man, 27 years old, was brought to the mental hospital because he had on several occasions become violent toward his father. For a few weeks, he had hallucinations and heard voices. The voices eventually ceased, but he then adopted a strange way of life. He would sit up all night, sleep all day, and become very angry when his father tried to get him out of bed. He did not shave or wash for weeks, smoked continuously, ate very irregularly, and drank enormous quantities of tea.

In the hospital, he adjusted rapidly to the new environment and was found to be generally cooperative. He showed no marked abnormalities of mental state or behavior, except for his lack of concern for just about anything. He kept to himself as much as possible and conversed little with patients or staff. His personal hygiene had to be supervised by the nursing staff; otherwise, he would quickly become dirty and untidy.

Six years after his admission to the hospital, he is described as shiftless and careless, sullen and unreasonable. He lies on a couch all day long. Although many efforts have been made to get the patient to accept therapeutic work assignments, he refuses to consider any kind of regular occupation. In the summer, he wanders about the hospital grounds or lies under a tree. In the winter, he wanders through the tunnels connecting the various hospital buildings and is often seen stretched out for hours under the warm pipes that carry the steam through the tunnels. (Courtesy of

Heinz E. Lehmann, M.D.)

Although about one-third of all schizophrenia patients have some marginal or integrated social existence, most have lives characterized by aimlessness, inactivity, frequent hospitalizations, and, in urban settings, homelessness and poverty.

Other Disorders

Schizoaffective Disorder. The course of schizoaffective disorder is intermediate between schizophrenia and mood disorders, with a better course and prognosis than for schizophrenia and a worse one than for bipolar disorder or major depressive disorder.

Schizophreniform Disorder. Patients with schizophreniform disorder who do not develop schizophrenia have a better outcome than do patients with schizophrenia.

PROGNOSIS

Having a diagnosis of schizophrenia is associated with a reduction of life expectancy by as much as 20 percent. Persons with schizophrenia have a higher mortality rate from accidents and natural causes than the general population. Institution- or treatment-related variables do not explain the increased mortality rate. However, the higher rate may be related to the fact that the diagnosis and treatment of medical and surgical conditions in schizophrenia patients can be clinical challenges, and in some cases, clinical neglect may be a factor.

Reported remission rates range from 10 to 60 percent, and a reasonable estimate is that 20 to 30 percent of all schizophrenia patients can lead somewhat normal lives. About 20 to 30 percent of patients continue to experience moderate symptoms, and 40 to 60 percent of patients remain significantly impaired by their disorder for their entire lives. Patients with schizophrenia do much poorer than patients with mood disorders, although 20 to 25 percent of mood disorder patients are also severely disturbed at long-term follow-up.

Prognostic Indicators

Several studies have shown that over the 5- to 10-year period after the first psychiatric hospitalization for schizophrenia, only about 10 to 20 percent of patients can be described as having a good outcome. More than 50 percent of patients can be described as having a poor outcome, with repeated hospitalizations, exacerbations of symptoms, episodes of major mood disorders, and suicide attempts. Despite these glum figures, schizophrenia does not always run a deteriorating course, and several factors have been associated with a good prognosis (Table 5-9).

Negative symptoms are rare for schizophreniform disorder, but when they occur, they are considered a poor prognostic feature, and many patients with negative symptoms will later develop schizophrenia.

TREATMENT APPROACH

Although antipsychotic medications are the mainstay of the treatment for schizophrenia, research has found that psychosocial interventions, including psychotherapy, can augment clinical improvement. The complexity of schizophrenia usually renders any single therapeutic approach inadequate to deal with the multifaceted disorder. Psychosocial modalities should be integrated into the drug treatment regimen and should support it. Patients with schizophrenia benefit more from the combined use of antipsychotic drugs and psychosocial treatment than from either treatment used alone.



Table 5-9
Positive and Negative Prognostic Factors for Schizophrenia

Positive Prognostic Factors
Acute onset
Female sex
Living in a developed country
Poor Prognostic Factors
Insidious onset
Childhood or adolescent onset

Poor premorbid functioning
Cognitive impairment

Hospitalization

The development of effective antipsychotic drugs and changes in political and popular attitudes toward the treatment and the rights of persons who are mentally ill have dramatically changed the patterns of hospitalization for schizophrenia patients since the mid-1950s. Despite these changes, readmissions after the first hospitalization are frequent, perhaps 40 to 60 percent within 2 years. Patients with schizophrenia occupy about 50 percent of all psychiatric hospital beds and account for about 16 percent of all psychiatric patients who receive any treatment.

In current practice, hospitalization is indicated for diagnostic purposes, for stabilization of medications, for patients' safety because of suicidal or homicidal ideation, and for grossly disorganized or inappropriate behavior, including the inability to take care of basic needs such as food, clothing, and shelter. Establishing an active association between patients and community support systems is also a primary goal of hospitalization.

Short stays of 4 to 6 weeks are just as effective as long-term hospitalizations, and hospital settings with active behavioral approaches produce better results than do custodial institutions. Hospital treatment plans should orient toward practical issues of self-care, quality of life, employment, and social relationships. During hospitalization, clinicians should coordinate with aftercare facilities, including family homes, foster families, board-and-care homes, and halfway houses. Daycare centers and home visits by therapists or nurses can help patients remain out of the hospital for long periods and can improve the quality of their daily lives.

Pharmacotherapy

Patients usually first present for the treatment of acute psychotic symptoms, which require immediate attention. Treatment during the acute phase focuses on alleviating the most severe psychotic symptoms. This phase usually lasts from 4 to 8 weeks.

Antipsychotic medications are considered the mainstay of treatment, both in the acute and maintenance phases of the illness. Table 5-10 lists the

second-generation and commonly used first generation antipsychotics.

Most guidelines recommend starting with a second-generation antipsychotic. Although they are likely the most effective for patients with severe symptoms, and predominantly positive symptoms, they are useful across a broad range of symptoms and severity. There is little guidance for choosing a specific medication: some studies suggest they have individual differences in efficacy. However, the most significant difference between the drugs is in their side effects, and we should consider these carefully when choosing a treatment.

Agitation is a typical symptom in the acute phase. Clinicians have several options for managing agitation that results from psychosis. Antipsychotics and benzodiazepines can result in a relatively rapid calming of patients. With highly agitated patients, intramuscular administration of antipsychotics produces a more rapid effect. An advantage of an antipsychotic is that a single intramuscular injection of a first- or second-generation antipsychotic can often calm the patient without excessive sedation. Low-potency antipsychotics are often associated with sedation and postural hypotension, mainly when administering them intramuscularly. Intramuscular ziprasidone and olanzapine are similar to their oral counterparts in not causing substantial extrapyramidal side effects during acute treatment. This can be a significant advantage over haloperidol or fluphenazine, which can cause frightening dystonias or akathisia in some patients. A rapidly dissolving oral formulation of olanzapine may also be helpful as an alternative to an intramuscular injection.

Benzodiazepines are also useful for agitation during acute psychosis. Lorazepam has the advantage of reliable absorption when administering it either orally or intramuscularly. The use of benzodiazepines may also reduce the amount of antipsychotic that is needed to control psychotic patients.

Treating Side Effects. Patients frequently experience the side effects of an antipsychotic before they experience clinical improvement. Whereas a clinical response may take days or weeks after starting a drug, side effects may begin almost immediately. First-generation antipsychotics most commonly cause extrapyramidal side effects, and in the case of low potency, first-generation antipsychotics, sedation and postural hypotension, whereas second-generation antipsychotics cause weight gain and metabolic

derangements.

EXTRAPYRAMIDAL SIDE EFFECTS. Most first-generation antipsychotics cause extrapyramidal side effects, including parkinsonian symptoms, dystonias, and akathisia. Although less common in second-generation antipsychotics, extrapyramidal effects can still occur.

Clinicians have several alternatives for treating these side effects. These include reducing the dose of the antipsychotic, adding an antiparkinson medication, and changing the patient to a medication that is less likely to cause extrapyramidal side effects. The most effective antiparkinson medications are anticholinergic antiparkinson drugs. However, these medications have additional side effects, including dry mouth, constipation, blurred vision, and, often, memory loss. Also, these medications are often only partially effective, leaving patients with substantial amounts of lingering extrapyramidal side effects. Centrally acting β-blockers, such as propranolol, may help treat akathisia. Most patients respond to dosages between 30 and 90 mg per day.

If prescribing conventional antipsychotics, clinicians may consider prescribing prophylactic antiparkinson medications for patients who are likely to experience disturbing extrapyramidal side effects. These include patients who have a history of extrapyramidal side effect sensitivity and those treated with relatively high doses of high-potency drugs. Prophylactic antiparkinson medications may help when prescribing high-potency drugs for young men, who tend to have an increased vulnerability for developing dystonias. Again, these patients should be candidates for newer drugs.

Some individuals are highly sensitive to extrapyramidal side effects at the dose that is necessary to control their psychosis. For many of these patients, medication side effects may seem worse than the illness itself. These patients should be treated routinely with a medication less likely to cause extrapyramidal side effects. Risperidone may cause extrapyramidal side effects even at low doses—for example, 0.5 mg—but the severity and risk are increased at higher doses—for example, more than 6 mg. Olanzapine and ziprasidone are also associated with dose-related parkinsonism and akathisia.

TARDIVE DYSKINESIA. About 20 to 30 percent of patients on long-term treatment with a first-generation antipsychotic will exhibit symptoms of

tardive dyskinesia. About 3 to 5 percent of young patients receiving a first-generation drug develop tardive dyskinesia each year. The risk in elderly patients is much higher. Although severely disabling dyskinesia is uncommon, it can affect walking, breathing, eating, and talking when it occurs. Individuals who are more sensitive to acute extrapyramidal side effects appear to be more vulnerable to developing tardive dyskinesia. Patients with comorbid cognitive or mood disorders may also be more vulnerable to tardive dyskinesia than those with only schizophrenia.

Table 5-10
Pharmacologic, Formulation and Dosing Information for Second-Generation Antipsychotics and Selected First-Generation Antipsychotics

Antipsychotic	Principal Liver Enzyme Target	Protein Binding	Bio-Availability	Time to Peak Level	Half-Life	CPZ Dose Equivalent, mg[a]	Typical Starting Dose, mg[b]	Typical Dose Range, mg[b]	Maximum Approved Dose, mg[b]	Dose Strength, mg	Route of Administration/ Formulation
Second-Generation Antipsychotics											
Partial Dopamine D2 Agonists											
Aripiprazole	2D6 >3A4	>99%	87%	PO: 3–5 hr; IM short: 1–3 hr; IM long 5–7 hr	PO: 75 hr; IM long: 30–47 days	7.5	PO: 10–15; IM long (gluteal): 400 with 2 wk oral aripiprazole 10–20	PO: 10–30; IM long (gluteal): 400 every 4 wk	PO: 30; IM long: 400	Tablets: 2, 5, 10, 15, 20, 30; Disc: 10, 15; Liquid: 1 mg/mL (except: 30 mg = 25 mL); IM short: 9.75/1.3 mL; IM long qFwk (monohydrate): 300/1.5 mL, 400/1.9 mL IM long q4 or 6 (882 mg) wk (hastoxil): 441/1.6 mL, 662/2.4 mL, 882/3.2 mL	PO, disc, liquid; IM short, IM long
Brexpiprazole	2D6, 3A4	>99%	95%	4 hr	91 hr	N/A	0.5–1	2–4	4	0.25, 0.5, 1, 2, 3, 4	PO
Cariprazine	3A4 >2D6	91–97%	52% (1 mg)	3–6 hr	2–5 days (didesmethyl-cariprazine: 1–3 wk)	N/A	1.5	1.5–6	6	1.5, 3, 4.5, 6	PO
Dopamine D2—Serotonin 2A Antagonists											
Asenapine	1A2 >3A4	98	35% (≤2% if swallowed)	0.5–1.5 hr	First: 6 hr; terminal: 24 hr	7.5	5 bid	10–20	20	Tablets: 5, 10	Sublingual (avoid eating or drinking for 10 min after administration)
Clozapine	1A2 (30%) >2C19 (24%) >3A4 (22%) >2C9 (12%) >2D6 (6%)	97%	50–60%	1.5–2.5 hr	12 hr	50	12.5	50–600	900	Tablets: 25, 50, 100, 200; Disc: 12.5, 25, 100, 150, 200; Liquid: 50 mg/mL	PO, disc, liquid
Iloperidone	2D6 >3A4 >1A2	95%	96%	2–4 hr	18 hr	5	1 mg bid × 1 day; 2 mg × 1 day; then increase 2 mg/d daily to therapeutic dose	12–24	24	Tablets: 1, 2, 4, 6, 8, 10, 12	PO
Lurasidone	3A4	99%	9–19%	1–3 hr	18 hr	25	40–80	40–120	160	Tablets: 20, 40, 80, 120	PO (with ≥350 kcal meal)
Olanzapine	1A2, 2D6, 3A4	93%	60%	PO: 6 hr; IM short: 15–45 min; IM long: 7 days	PO: 30 hr; IM short: 30 hr; IM long: 30 days	5	PO: 5–10; IM long (gluteal): 210/2 wk or 405/4 wk (≈ oral 10), 300/2 wk (≈ oral 15 or 20)	PO: 10–20; IM long (gluteal): 150/2 wk or 300/4 wk (≈ oral 10), 210/2 wk or 405/2 wk (≈ oral 15), 300/2 wk (≈ oral 20)	20	Tablets: 2.5, 5, 7.5, 10, 15, 20; Disc: 5, 10, 15, 20; IM short: 10/2 mL; IM long q2 or 4 wk: 150/1.3 mL, 210/1.3 mL, 300/1.8 mL, 405/2.3 mL	PO, disc; IM short, IM long

Paliperidone	<10% first-pass hepatic clearance	74%	28%	PO: 24 hr; IM long: 13 days	PO: 23 hr; IM long: 25–49 days	3	PO: 6; IM long: 234 mg into deltoid day 1, then 156 mg day 8, then 39–234 into deltoid or gluteus every 4 wk	PO: 3–12; IM long (deltoid or gluteal): 117–234 every 4 wk	12	Tablets: 1.5, 3, 6, 9; IM long q4wk: 39 0.25 mL, 78/0.5 mL, 117/0.75 mL, 156/1 mL, 234/1.5 mL; IM long q12wk: 273 0.875 mL, 410/1.315 mL, 546/1.75 mL, 819/2.625 mL	PO (ER); IM long
Quetiapine	3A4	83%	100%	IR: 1.5 hr; XR: 6 hr	6–7 hr	75	IR: 25–100; XR: 200–300	IR: 150–750; XR: 400–800	800	Tablets: IR: 25, 50, 100, 200, 300, 400; XR: 50, 150, 200, 300, 400	PO (IR, XR)
Risperidone	2D6 >3A4	90%	70%	3 hr	3 hr	2	PO: 2; IM long (deltoid or gluteal): 25 with 3 wk oral risperidone	PO: 2–8; IM long (deltoid or gluteal): 12.5–50 into deltoid or gluteal muscle every 2 wk	16	Tablets: 0.25, 0.5, 1, 2, 3, 4; Diss: 0.5, 1, 2, 3, 4; Liquid: 1 mg/mL; IM long q2w: 12.5/1 mL, 25/2 mL, 37.5/2 mL, 50/2 mL	PO, diss, liquid; IM long
Ziprasidone	Aldehyde oxidase (2/1) >3A4 (1/1)	>99%	60%	PO: 6–8 hr; IM short: ≤60 min	PO: 7 hr; IM short: 2–5 hr	60	20–40 bid	80–160	160	Tablets: 20, 40, 60, 80 IM short: 20/1 mL	PO (with ≥500 kcal meal); IM short
first-Generation Antipsychotics (Dopamine D2 Antagonists)											
Chlorpromazine	2D6	>90	20%	2–4 hr	30 hr	100	PO: 25–100 mg; IM short: 25, followed by 25–50 mg as needed after 1–4 hr	PO: 200–800; IV/IM short: 400 mg every 4–6 hr	2,000	Tablets: 10, 25, 50, 100, 200; Spansule: 30, 75, 150; Syrup: 5 mL = 10 mg; Liquid: 25 mg/5 mL; IM short: 25/mL; Supp: 25, 100	PO, syrup; IM short; IV; rectal suppository
Fluphenazine	1A2	>90	<50%	PO: 2 hr; IM long: 8–10 hr	PO: 14–16 hr; IM long: 14 days	2	PO: 2.5–10 in 2–3 divided doses	PO: 1–5 in 2–3 divided doses; IM long (deltoid or gluteal): 12.5–25 mg every 2–4 wk	40	Tablets: 1, 2.5, 5, 10; Liquid: 2.5 or 5/mL; IM long: 25 mg/mL	PO; IM long
Haloperidol	3A4	92%	60%–70%	PO: 2–6 hr; IM short: 10–20 min; IM long: 6–7 days	PO: 18 hr; IM short: 10–20 hr; IM long: 3 wk	2	PO: 1–2; IM short: 2–5 mg every 4–8 hr as needed; IM long (deltoid or gluteal): 10–20 times the oral dose every 4 wk (≤100)	PO: 2–10; IM long (deltoid or gluteal): 10–15 times the oral dose every 4 wk (≤100)	PO: 60; IM long: 100	Tablets: 0.5, 1, 2, 5, 10, 20 Liquid: 2 or 10 mg/mL IM short: 5; IM long q4wk: 50 mg/mL, 100 mg/mL	PO; IM short, IM long
Perphenazine	2D6	>90%	40%	1–3 hr	9–12 hr	10	Inpatient: 8–16 divided in 2–3 doses; Outpatient: 4–8 divided in 3 doses	8–32	64	2, 4, 8, 16	PO

[a]Chlorpromazine dose equivalents (i.e., dose given in the table is equivalent to 100 mg of chlorpromazine).
[b]Doses need to be individualized based on efficacy and tolerability.
CPZ, chlorpromazine; Diss, dissolvable tablet; ER, extended release; IM, intramuscular; IR, immediate release; IV, intravenous; N/A, not available; q, every; PO, oral; Supp, suppository; XR, extended release.
Source: Package insert information for each medication.

The abnormal movements usually begin while the patient is receiving an antipsychotic or within 4 weeks of discontinuing an oral antipsychotic or 8 weeks after the withdrawal of a depot antipsychotic.

There is a lower risk of tardive dyskinesia with new-generation drugs. However, there still is some risk of tardive dyskinesia with newer drugs.

Recommendations for preventing and managing tardive dyskinesia include: (1) using the lowest effective dose of antipsychotics; (2) prescribing cautiously with children, elderly patients, and patients with mood disorders; (3) examining patients regularly for evidence of tardive dyskinesia; (4) considering alternatives to the current antipsychotic and considering dosage reduction when observing tardive dyskinesia; and (5) considering several options if the tardive dyskinesia worsens, including discontinuing the antipsychotic or switching to a different drug. Clozapine is effective in reducing severe tardive dyskinesia or tardive dystonia.

OTHER SIDE EFFECTS. Sedation and postural hypotension can be significant side effects for patients treated with low-potency first-generation drugs, such as perphenazine. These effects are often most severe during the initial dosing

of these medications. As a result, patients treated with these medications—or with clozapine—may require weeks to reach a therapeutic dose. Although most patients develop tolerance to sedation and postural hypotension, sedation may continue to be a problem. In these patients, daytime drowsiness may interfere with a patient's attempts to return to community life.

All antipsychotics elevate prolactin levels, which can result in galactorrhea and irregular menses. Long-term elevations in prolactin and the resultant suppression in gonadotropin-releasing hormone can cause suppression in gonadal hormones. These, in turn, may have effects on libido and sexual functioning. There is also concern that elevated prolactin may cause decreases in bone density and lead to osteoporosis. The concerns about hyperprolactinemia, sexual functioning, and bone density are based on experiences with prolactin elevations related to tumors and other causes. It is unclear if these risks are also associated with the lower elevations that occur with prolactin-elevating drugs.

HEALTH MONITORING IN PATIENTS RECEIVING ANTIPSYCHOTICS. Because of the effects of the second-generation antipsychotics on insulin metabolism, psychiatrists should monitor several health indicators, including BMI, fasting blood glucose, and lipid profiles. We should weigh patients and calculate their BMIs for every visit for at least 6 months after a medication change.

Clozapine. Clozapine is considered the most effective of antipsychotics, particularly in patients who have been unresponsive to other treatments. However, it is a challenging drug to administer, given its risk of both severe side effects (agranulocytosis in approximately 0.3 percent of those taking it, seizures in as high as 5 percent of those on doses above 600 mg, and, more rarely, myocarditis) and common ones (hypersalivation, sedation, tachycardia, weight gain, and postural hypotension). The concerns over agranulocytosis have resulted in a monitoring system in the United States in which patients who receive clozapine must be in a program of weekly blood monitoring for the first 6 months and biweekly monitoring for the next 6 months, and monthly after that.

These limitations have relegated clozapine to a later treatment option, and most guidelines recommend considering it after a patient has failed at least two other antipsychotic trials.

Duration and Prophylaxis

In the stable or maintenance phase, the illness is in a relative stage of remission. The goals during this phase are to prevent psychotic relapse and to assist patients in improving their level of functioning. Newer medications, with a substantially reduced risk of tardive dyskinesia, have diminished one of the major concerns about long-term treatment. During this phase, patients are usually in a relative state of remission with only minimal psychotic symptoms. Stable patients who are maintained on an antipsychotic have a much lower relapse rate than patients who have their medications discontinued. Data suggest that 16 to 23 percent of patients receiving treatment will experience a relapse within 1 year, and 53 to 72 percent will relapse without medications.

We should maintain patients experiencing a first episode of psychosis for at least a year. However, such patients still have a high chance of relapsing at least once over the following 5 years, and some experts believe that 1 year is inadequate. This is a particular concern when patients have achieved functional employment status or are involved in educational programs because they have a lot to lose if they experience another psychotic decompensation.

For patients experiencing two or more episodes, most experts recommend that we consider indefinite treatment.

Acute Treatment Failures

Addressing Noncompliance. Noncompliance with long-term antipsychotic treatment is very high. An estimated 40 to 50 percent of patients become noncompliant within 1 or 2 years. Long-acting injectable antipsychotics were developed to help improve adherence, particularly in patients likely to discontinue daily oral medications. Although there is some controversy in the literature, the weight of data suggests that long-term depot formulations do help adherence and decrease relapse, mainly when considered in real-world settings.

When beginning long-acting drugs, some oral supplementation is necessary while achieving peak plasma levels. Fluphenazine, haloperidol, risperidone, paliperidone, aripiprazole, and olanzapine have long-acting

injectable formulations.

There are several advantages to using long-acting injectable medication. Clinicians quickly know when noncompliance occurs and have some time to initiate appropriate interventions before the medication effect dissipates. Also, there is less day-to-day variability in blood levels, making it easier to establish a minimum effective dose. Finally, many patients prefer it to managing daily dosage schedules.

Selecting Second Treatment Options. When giving patients with acute schizophrenia an antipsychotic medication, approximately 60 percent will improve to the extent that they will achieve a complete remission or experience only mild symptoms. The remaining 40 percent of patients will improve but still demonstrate variable levels of positive symptoms that are resistant to the medications. One meta-analysis of available studies found that patients who did not show at least some response to treatment by the second week were less likely to benefit from the drug, and a treatment change may be the best option.

Rather than categorizing patients into responders and nonresponders, it is more accurate to consider the degree to which the illness improves with medication. Some resistant patients are so severely ill that they require chronic institutionalization. Others respond to an antipsychotic with substantial suppression of their psychotic symptoms but demonstrate persistent symptoms, such as hallucinations or delusions.

Before considering a patient a poor responder to a particular drug, it is vital to assure that they received an adequate trial of the medication. A 4- to 6-week trial on an adequate dose of an antipsychotic represents a reasonable trial for most patients. Patients who demonstrate even a mild amount of improvement during this period may continue to improve at a steady rate for 3 to 6 months. It may be helpful to confirm that the patient is receiving an adequate amount of the drug by monitoring the plasma concentration; that said, most of the accepted levels pertain to first-generation drugs, and there is less rationale for obtaining levels in patients who receive second-generation drugs. However, in any patient, a deficient plasma concentration may indicate that the patient has been noncompliant or, more commonly, only partially compliant. It may also suggest that the patient is a rapid metabolizer of the antipsychotic or is not absorbing the drug adequately. Under these conditions,

raising the dose may be helpful. If the level is relatively high, clinicians should consider whether side effects may be interfering with therapeutic response.

If the patient is responding poorly, one may increase the dose above the usual therapeutic level; however, higher than recommended doses usually do not improve response. Changing to another drug is preferable.

As discussed earlier, clozapine is useful for patients who respond poorly to other antipsychotics. Double-blind studies comparing clozapine with other antipsychotics indicated that clozapine had the most evident advantage over conventional drugs in patients with the most severe psychotic symptoms, as well as in those who had previously responded poorly to other antipsychotics.

Researchers and clinicians have tried several adjunctive medications, most with mixed success. These include lamotrigine, mirtazapine, donepezil, D-alanine, D-serine, estradiol, memantine, and allopurinol.

Other Somatic Treatments

Researchers have studied electroconvulsive therapy (ECT) in both acute and chronic schizophrenia. Studies in recent-onset patients indicate that ECT is about as effective as antipsychotic medications and more effective than psychotherapy. Other studies suggest that supplementing antipsychotic medications with ECT is more effective than antipsychotic medications alone. Antipsychotic medications should be administered during and after ECT treatment.

Some initial studies have suggested neuromodulation, using transcranial magnetic stimulation (TMS) or transcranial direct current stimulation (tDCS), may be useful for treating hallucinations or negative symptoms.

Although we no longer consider psychosurgery to be appropriate treatment, some research centers practice it on a limited experimental basis for severe, intractable cases.

Psychosocial Therapy

Psychotherapy is considered an essential part of treatment for schizophrenia, and patients who receive psychotherapy along with medications tend to have better adherence to therapy, less negative symptoms, and better overall functioning.

Currently, there is no convincing evidence to suggest that any single psychotherapeutic approach is preferable to another, although all tend to emphasize structured approaches rather than open-ended, exploratory methods. Psychosocial therapies include a variety of methods to increase social abilities, self-sufficiency, practical skills, and interpersonal communication in schizophrenia patients. The goal is to enable persons who are severely ill to develop social and vocational skills for independent living. Many sites use these approaches, including hospitals, outpatient clinics, mental health centers, day hospitals, and home or social clubs.

Cognitive Behavioral Therapy. Cognitive-behavioral therapy (CBT) has been used in schizophrenia patients to improve cognitive distortions, reduce distractibility, and correct errors in judgment. There are reports of ameliorating delusions and hallucinations in some patients using this method. Patients who might benefit generally have some insight into their illness, and clinicians usually use this approach after treating an acute psychotic episode. The approach generally incorporates cognitive restructuring, self-monitoring, and graded coping skills. Although at least one randomized controlled study showed a benefit for CBT as a sole treatment, most experts recommend it in combination with antipsychotic treatment.

Social Skills Training. Social skills training, also called behavioral skills therapy, can be directly supportive and useful for a patient. In addition to the psychotic symptoms seen in patients with schizophrenia, other noticeable symptoms involve the way the person relates to others, including poor eye contact, unusual delays in response, odd facial expressions, lack of spontaneity in social situations, and inaccurate perception or lack of perception of emotions in other people. Behavioral skills training addresses these behaviors through the use of videotapes of others and the patient, role-playing in therapy, and homework assignments for practicing the specific skills. Social skills training can reduce relapse rates and the need for hospitalization.

Group Therapy. Group psychotherapy for patients with schizophrenia can help improve social functioning and decrease negative symptoms. Group therapy for persons with schizophrenia generally focuses on real-life plans,

problems, and relationships. Groups may be behaviorally oriented, psychodynamically or insight-oriented, or supportive. Some investigators doubt that dynamic interpretation and insight therapy are valuable for typical patients with schizophrenia. However, group therapy is effective in reducing social isolation, increasing the sense of cohesiveness, and improving reality testing for patients with schizophrenia. Groups led in a supportive manner appear to be most helpful for schizophrenia patients.

Family-Oriented Therapies. Family therapy can be a significant adjunct to treatment and reduces relapse and rehospitalization rates compared to patients receiving standard care. Such therapy generally includes practical approaches such as psychoeducation and problem-solving training. Because patients with schizophrenia are often discharged in an only partially remitted state, a family to which a patient returns can often benefit from a brief but intensive (as often as daily) course of family therapy. The therapy should focus on the immediate situation and should include identifying and avoiding potentially troublesome situations. When problems do emerge with the patient in the family, the aim of the therapy should be to resolve the problem quickly.

In wanting to help, family members often encourage a relative with schizophrenia to resume regular activities too quickly, both from ignorance about the disorder and from denial of its severity. Without being overly discouraging, therapists must help both the family and the patient understand and learn about schizophrenia and must encourage discussion of the psychotic episode and the events leading up to it. Ignoring the psychotic episode, a common occurrence, often increases the shame associated with the event and does not exploit the freshness of the episode to understand it better. Psychotic symptoms often frighten family members, and talking openly with the psychiatrist and with the relative with schizophrenia often eases all parties. Therapists can direct later family therapy toward the long-range application of stress-reducing and coping strategies and the patient’s gradual reintegration into everyday life.

Therapists must control the emotional intensity of family sessions with patients with schizophrenia. The excessive expression of emotion during a session can damage a patient’s recovery process and undermine potentially successful future family therapy. Several studies have shown that family

therapy is especially useful in reducing relapses.

Case Management and Assertive Community Treatment. Because a variety of professionals with specialized skills, such as psychiatrists, social workers, and occupational therapists, among others, are involved in a treatment program, it is helpful to have one person aware of all the forces acting on the patient. The case manager helps coordinate this and ensures that the patient keeps appointments and complies with treatment plans. The case manager may also make home visits and even accompany the patient to work. The success of the program depends on the educational background, training, and competence of the individual case manager, which vary. Most important is maintaining a small caseload (less than 20 cases per manager), which can be challenging for overtaxed and underfunded systems.

The Assertive Community Treatment (ACT) program was initially developed by researchers in Madison, Wisconsin, in the 1970s, for the delivery of services for persons with chronic mental illness. Patients are assigned to one multidisciplinary team (e.g., case manager, psychiatrist, nurse, general physicians). The team has a fixed caseload of patients and delivers all services when and where needed by the patient, 24 hours a day, 7 days a week. ACT is a mobile and intensive intervention that provides treatment, rehabilitation, and support activities. These include home delivery of medications, monitoring of mental and physical health, in vivo social skills, and frequent contact with family members. There is a high staff-to-patient ratio (1:12). ACT programs can effectively decrease the risk of rehospitalization for persons with schizophrenia, but they are labor-intensive and expensive programs to administer.

Although the evidence is low in quality, there is some evidence to suggest that intensive case management can reduce hospitalization time and increase adherence to treatment. Intensive case management approaches seem to be most effective when they incorporate ACT.

Individual Psychotherapy. Studies of the effects of individual psychotherapy in the treatment of schizophrenia suggest that therapy is helpful and that the effects are additive to those of pharmacologic treatment. In psychotherapy with a schizophrenia patient, developing a therapeutic relationship that the patient experiences as safe is critical. The therapist’s

reliability, the emotional distance between the therapist and the patient, and the genuineness of the therapist as interpreted by the patient all affect the therapeutic experience. When considering psychotherapy for a patient with schizophrenia, we should think in decades, rather than sessions, months, or even years.

The best predictor of psychotherapy outcome is probably the strength of the therapeutic alliance. Schizophrenia patients who can form a positive therapeutic alliance are likely to remain in psychotherapy, to remain compliant with their medications, and to have good outcomes at 2-year follow-up evaluations.

The relationship between clinicians and patients differs from that encountered in the treatment of nonpsychotic patients. Establishing a relationship is often challenging. Persons with schizophrenia are desperately lonely, yet defend against closeness and trust; they are likely to become suspicious, anxious, or hostile or to regress when someone attempts to draw close.

Therapists should scrupulously respect a patient's distance and privacy and should demonstrate simple directness, patience, sincerity, and sensitivity to social conventions in preference to premature informality and the condescending use of first names. The patient is likely to perceive exaggerated warmth or professions of friendship as attempts at bribery, manipulation, or exploitation. In the context of a professional relationship, however, flexibility is essential in establishing a working alliance with the patient.

Vocational Therapy and Supported Employment. A variety of methods and settings can help patients regain old skills or develop new ones. These include sheltered workshops, job clubs, and part-time or transitional employment programs. Enabling patients to become gainfully employed is both a means toward and a sign of recovery. Many schizophrenia patients are capable of performing high-quality work despite their illness. Others may exhibit exceptional skill or even brilliance in a limited field as a result of some distinctive aspects of their disorder.

Supported employment focuses on helping patients to obtain competitive employment as opposed to sheltered workshops. It has strong data for helping patients to find and maintain jobs; it also is associated with less need for

treatment and improved self-esteem. It is less consistently associated with overall disease outcomes.

Art Therapy. Many schizophrenia patients benefit from art therapy, which provides them with an outlet for their constant bombardment of imagery. It helps them communicate with others and share their inner, often frightening world with others.

Cognitive Remediation. Cognitive remediation or cognitive training is a behavioral therapy that attempts to improve cognitive processes. It uses computer-generated exercises to influence neural networks to improve cognition, including working memory. This can then improve social functioning. One metanalysis determined that its effect size is in the medium range.

National Alliance on Mental Illness (NAMI). The NAMI and similar organizations offer support groups for family members and friends of patients who are mentally ill and for patients themselves. These organizations offer emotional and practical advice about obtaining care in the sometimes complex health care delivery system and are useful sources to which to refer family members. NAMI has also waged a campaign to destigmatize mental illness and to increase government awareness of the needs and rights of persons who are mentally ill and their families.

THE EPIDEMIOLOGY OF THE DISORDERS

Incidence and Prevalence

The worldwide lifetime prevalence of schizophrenia is about 0.7 percent. One meta-analysis of 101 prevalence rates published between 1990 and 2015 found that the mean global lifetime prevalence of psychotic disorders worldwide was 7.49 per 1,000 individuals. However, there was an extensive range across studies representing an approximately five-fold difference. These variations could largely be accounted for by the different methodologic approaches of different studies, including different populations and settings studied, different diagnostic criteria and diagnoses included, and differences in the study quality.

Table 5-11 lists some of the major epidemiologic studies of schizophrenia.

In the United States, about 0.05 percent of the total population is treated for schizophrenia in any single year. Perhaps only one-half of all patients with schizophrenia will obtain treatment, despite the severity of the disorder.

Sex

Schizophrenia is equally prevalent in men and women. The two sexes differ in the onset and course of illness. The onset of schizophrenia is earlier for men. Also, more than half of all male schizophrenia patients, but only one-third of all female schizophrenia patients, will be hospitalized for the disorder before age 25 years. Some studies have indicated that men are more likely to be impaired by negative symptoms than are women and that women are more likely to have better social functioning than are men before disease onset. In general, the outcome for female schizophrenia patients is better than that for male schizophrenia patients.



Table 5-11
Selected Prevalence Studies of Schizophrenia

Author	Country	Population	Method	Prevalence per 1,000 Population at Risk
Brugger (1931)	Germany	Area in Thuringia (n = 37,561); age 10+ yr	Census; interview of sample	2.4
Strömgren (1938); Bøjholm and Strömgren (1989)	Denmark	Island population (n = 50,000)	Census interviews; repeat census	3.9 → 3.3
Böök (1953); Böök et al. (1978)	Sweden	Genetic isolate (n = 9,000); age 15–50 yr	Census interviews; repeat census	9.5 → 17.0
Essen–Möller et al. (1956); Hagnell (1966)	Sweden	Community in southern Sweden	Census interviews; repeat census	6.7 → 4.5
Rin and Lin (1962); Lin et al. (1989)	Taiwan	Population sample	Census interviews; repeat census	2.1 → 1.4
Crocetti et al. (1971)	Croatia	Sample of 9,201 households	Census based on hospital records and interviews	5.9
Dube and Kumar (1972)	India	4 areas in Agra (n = 29,468)	Census based on hospital and clinic records	2.6
Rotstein (1977)	Russia	Population sample (n = 35,590)	Census based on hospital and clinic records	3.8
Keith et al. (1991)	USA	Aggregated data across 5 ECA sites	Sample survey; interviews	7.0 (point) 15.0 (lifetime)
Jeffreys et al. (1997)	UK	London health district (n = 112,127)	Census; interview of sample (n = 172)	5.1
Kebede et al. (1999)	Ethiopia	25 districts of Addis Ababa (n = 2,228,490)	Screening by self-report questionnaire, interviews of sample (n = 2,042)	7.0 (point) 9.0 (lifetime)
Jablensky et al. (2000)	Australia	4 urban areas (n = 1,084,978)	Census, screen for psychosis; interviews of sample (n = 980)	3.1–5.9 (point)[a] 3.9–6.9 (period, 1 yr)[b]
Waldo et al. (1999)	Micronesia	Island of Kosrae Genetic isolate	Screen of hospital records, interviews	6.8 (point)
Arajärvi et al. (2005)	Finland	Birth cohort (n = 14,817) Genetic isolate	Case register data; interviews of 55% of register cases	15.0 (lifetime) 19.0[c] (lifetime)
Wu et al. (2006)	USA (California)	Medicaid/Medicare health insurance data	20% random sample of insured subjects	5.1 (period, 1 yr)
Perälä et al. (2007)	Finland	National sample (n = 8,028)	Screen for psychosis, interviews of sample; register and case note data also used	10.0 (lifetime) 22.9[d] (lifetime)

[a]All psychoses.
[b]Schizophrenia and other nonaffective psychotic disorders.
[c]Schizophrenia spectrum disorders.
[d]Nonaffective psychotic disorders.
→ Changes in prevalence found in repeat surveys of the same population.

Age

In the United States, the peak ages of onset are between 10 and 25 years for men and 25 and 35 years for women. Unlike men, women display a bimodal age distribution, with a second peak occurring in middle age—approximately 3 to 10 percent of women with schizophrenia present with disease onset after age 40 years. About 90 percent of patients in treatment for schizophrenia are between 15 and 55 years old. The onset of schizophrenia before age 10 years or after age 60 years is scarce. Late-onset schizophrenia refers to the occurrence of the disorder after age 45 years.

Other Factors Affecting Epidemiology

Seasonality of Birth. Persons who develop schizophrenia are more likely to have been born in the winter or early spring. In the Northern Hemisphere, including the United States, persons with schizophrenia are more often born in the months from January to April. In the Southern Hemisphere, persons with schizophrenia are more often born in the months from July to September.

Maternal Factors Affecting the Fetus. Complications during delivery, maternal malnutrition during pregnancy, as well as other illnesses, appear to be risk factors for schizophrenia.

Early Life Experiences. Childhood trauma, social isolation, and other types of deprivation also seem to be risk factors.

Urban Upbringing. Several studies, mainly from the 1990s, have suggested that being born or living in a city is a risk for schizophrenia. This has been reported in various countries. The relationship appears to be more convincing for being raised in rather than being born in a city. The relationship may be "dose-related," meaning that the larger the city, the higher the risk.

Cannabis. There is an association between cannabis use and psychosis, and the use of cannabis may increase the risk of schizophrenia by as much as 40%, particularly in heavy users.

Cognitive Deficits May also be a Risk Factor. More recent research has suggested that cognitive deficits, such as poor verbal learning and memory and slower processing speed, can be a predictor of impending psychosis.

Other Psychotic Disorders

Schizoaffective disorder is less common than schizophrenia, possibly in the range of 0.5 to 0.8 percent, although the varying criteria used in different studies limits the findings. The sex differences are more typical of mood disorders in which there is a higher ratio of women than men having the depressed subtype but a similar ratio with the bipolar subtype. The bipolar

subtype may be more common in younger persons than in adults, who are more commonly depressed. As with schizophrenia, women tend to get it later than men.

We know little about the incidence, prevalence, and sex ratio of schizophreniform disorder. It appears to be about half as common as schizophrenia, more common in men, and most common in adolescents and young adults. The relatives of patients with schizophreniform disorder are more likely to have mood disorders than are the relatives of patients with schizophrenia. Many of those mood disorders include psychotic symptoms, however.

THE NEUROBIOLOGY OF THE DISORDER

Anatomic Findings

In the 19th century, neuropathologists failed to find a neuropathologic basis for schizophrenia, and thus they classified schizophrenia as a functional disorder. By the end of the 20th century, however, researchers had better tools and made significant strides in revealing a potential neuropathologic basis for schizophrenia.

Cerebral Ventricles. Computed tomography (CT) scans of patients with schizophrenia have consistently shown lateral and third ventricular enlargement and some reduction in cortical volume. Reduced volumes of cortical gray matter have been demonstrated during the earliest stages of the disease. Several investigators have attempted to determine whether the abnormalities detected by CT are progressive or static. Some studies have concluded that the lesions observed on CT scan are present at the onset of the illness and do not progress. Other studies, however, have concluded that the pathologic process visualized on CT scan continues to progress during the illness. Thus, whether an active pathologic process is continuing to evolve in schizophrenia patients is still uncertain.

Reduced Symmetry. There is a reduced symmetry in several brain areas in schizophrenia, including the temporal, frontal, and occipital lobes. This reduced symmetry is believed by some investigators to originate during fetal life and to be indicative of a disruption in brain lateralization during

neurodevelopment.

Limbic System. Because of its role in controlling emotions, researchers believe that the limbic system is involved in the pathophysiology of schizophrenia. Studies of postmortem brain samples from schizophrenia patients have shown a decrease in the size of the region, including the amygdala, the hippocampus, and the parahippocampal gyrus. This neuropathologic finding agrees with the observation made by magnetic resonance imaging studies of patients with schizophrenia. The hippocampus is not only smaller in size in schizophrenia but is also functionally abnormal, as indicated by disturbances in glutamate transmission. Brain tissue sections also show the disorganization of the neurons within the hippocampus in some patients with schizophrenia.

Prefrontal Cortex. There is considerable evidence from postmortem brain studies that support anatomical abnormalities in the prefrontal cortex in schizophrenia. Functional deficits in the prefrontal brain imaging region have also been demonstrated. Of interest, several symptoms of schizophrenia mimic those found in persons with prefrontal lobotomies or frontal lobe syndromes.

Thalamus. Some studies of the thalamus show evidence of volume shrinkage or neuronal loss in particular subnuclei. The medial dorsal nucleus of the thalamus, which has reciprocal connections with the prefrontal cortex, has been reported to contain a reduced number of neurons. The total number of neurons, oligodendrocytes, and astrocytes is reduced by 30 to 45 percent in schizophrenia patients. This putative finding does not appear to be due to the effects of antipsychotic drugs, because the volume of the thalamus is similar in size between patients with schizophrenia treated chronically with medication and neuroleptic-naive subjects.

Basal Ganglia and Cerebellum. The basal ganglia and cerebellum have been of theoretical interest in schizophrenia for at least two reasons. First, many patients with schizophrenia show odd movements, even in the absence of medication-induced movement disorders. The odd movements can include an awkward gait, facial grimacing, and stereotypies. Because the basal

ganglia and cerebellum are involved in the control of movement, disease in these areas is implicated in the pathophysiology of schizophrenia. Second, the movement disorders involving the basal ganglia (e.g., Huntington disease, Parkinson disease) are the ones most commonly associated with psychosis. Neuropathologic studies of the basal ganglia have produced variable and inconclusive reports about cell loss or the reduction of volume of the globus pallidus and the substantia nigra. Studies have also shown an increase in the number of D_2 receptors in the caudate, putamen, and the nucleus accumbens. The question remains, however, whether the increase is secondary to the patient having received antipsychotic medications. Some investigators have begun to study the serotonergic system in the basal ganglia; the clinical usefulness of antipsychotic drugs that are serotonin antagonists suggests a role for serotonin in psychotic disorders.

Physiologic Findings

Functional Imagining. PET studies have shown a variety of neurotransmitter abnormalities, including increased dopamine levels in the ventral striatum and reduced levels in the frontal cortex.

Studies using magnetic resonance spectroscopy have demonstrated increased glutamate levels, particularly in the prefrontal and medial temporal areas. Furthermore, concentrations of N-acetyl aspartate, a marker of neurons, were lower in the hippocampus and frontal lobes of patients with schizophrenia.

Electrophysiologic Findings. Electroencephalographic studies indicate that many schizophrenia patients have abnormal records, increased sensitivity to activation procedures (e.g., frequent spike activity after sleep deprivation), decreased alpha activity, increased theta and delta activity, possibly more epileptiform activity than usual, and possibly more left-sided abnormalities than usual. Schizophrenia patients also exhibit an inability to filter out irrelevant sounds and are extremely sensitive to background noise. The flooding of sound that results makes concentration difficult and may be a factor in the production of auditory hallucinations. This sound sensitivity may be associated with a genetic defect.

Evoked Potentials. A large number of abnormalities are seen in the evoked potential of patients with schizophrenia. The P300 is most studied; it is a large, positive evoked-potential wave that occurs about 300 milliseconds after a sensory stimulus is detected. The primary source of the P300 wave is likely in the limbic system structures of the medial temporal lobes. In patients with schizophrenia, the P300 is statistically smaller than in comparison groups. Abnormalities in the P300 wave are also common in children who, because they have affected parents, are at high risk for schizophrenia. Whether the characteristics of the P300 represent a state or a trait phenomenon remains controversial. Other evoked potentials reported to be abnormal in patients with schizophrenia are the N100 and the contingent negative variation. The N100 is a negative wave that occurs about 100 milliseconds after a stimulus, and the contingent negative variation is a slowly developing, negative-voltage shift following the presentation of a sensory stimulus that is a warning for an upcoming stimulus. The evoked-potential data likely indicates that, although patients with schizophrenia are unusually sensitive to a sensory stimulus (larger early evoked potentials), they compensate for the increased sensitivity by blunting the processing of information at higher cortical levels (indicated by smaller late evoked potentials).

Eye Movement Dysfunction. The inability to follow a moving visual target accurately is the defining basis for the disorders of smooth visual pursuit and disinhibition of saccadic eye movements seen in patients with schizophrenia. Eye movement dysfunction may be a trait marker for schizophrenia; it is independent of drug treatment and clinical state and also occurs in first-degree relatives of probands with schizophrenia. Various studies have reported abnormal eye movements in 50 to 85 percent of patients with schizophrenia compared with about 25 percent in psychiatric patients without schizophrenia and fewer than 10 percent in nonpsychiatrically ill control participants.

Prepulse Inhibition Deficits. Prepulse inhibition refers to a dysfunction of sensorimotor gating. Usually, a prepulse or weak stimulus, such as a noise, can decrease the startle reaction to a subsequent more substantial stimulus, such as a louder noise. As early as the 1970s, researchers found that patients

with schizophrenia lacked this normal inhibition, and this was later found in the relatives of schizophrenia patients as well. Dopamine regulates sensorimotor gating, which adds additional support for the importance of dopamine in schizophrenia.

Neurotransmitters and Receptors

Excessive dopamine release in patients with schizophrenia is associated with the severity of positive psychotic symptoms. Positron emission tomography studies of dopamine receptors have shown that patients with schizophrenia have increases in subcortical synaptic dopamine content and increased subcortical synthesis capacity. These appear localized to the area of the associative striatum. These abnormalities, particularly increased subcortical synaptic dopamine content, are associated with the positive symptoms of schizophrenia and positive treatment response. However, the dopamine abnormalities are not merely due to the symptoms, as these abnormalities precede the onset of illness. They also occur in individuals considered to be at high risk for schizophrenia.

In addition to dopamine, there are increased levels of glutamate and decreased levels of γ-aminobutyric acid (GABA). Some patients with schizophrenia show a loss of GABAergic neurons in the hippocampus.

Postmortem studies in schizophrenia have demonstrated decreased muscarinic and nicotinic receptors in the caudate-putamen, hippocampus, and selected regions of the prefrontal cortex. These receptors play a role in the regulation of neurotransmitter systems involved in cognition, which is impaired in schizophrenia.

As a result of these neurotransmitter alterations, there are alterations in a variety of receptors. For example, N-methyl-D-aspartate (NMDA) receptors appear to hypofunction as a result of glutamate and dopamine excesses.

Psychoneuroimmunology

Several immunologic abnormalities are associated with patients who have schizophrenia. The abnormalities include decreased T-cell interleukin-2 production, reduced number and responsiveness of peripheral lymphocytes, abnormal cellular and humoral reactivity to neurons, and the presence of brain-directed (antibrain) antibodies. The data can be interpreted variously as

representing the effects of a neurotoxic virus or an endogenous autoimmune disorder.

The observation further strengthens the hypothesis that several autoimmune diseases can cause psychosis, such as systemic lupus encephalitis. Similarly, some types of immune encephalitis, such as anti-NMDA receptor encephalitis, can cause a psychotic illness that resembles schizophrenia, at least early in the illness.

Psychoneuroendocrinology

Many reports describe neuroendocrine differences between groups of patients with schizophrenia and groups of control subjects. For example, the results of the dexamethasone-suppression test are abnormal in various subgroups of patients with schizophrenia, although the test lacks practical or predictive value.

Some data suggest decreased concentrations of luteinizing hormone or follicle-stimulating hormone, perhaps correlated with age of onset and length of illness. Two additional reported abnormalities may correlate with the presence of negative symptoms: a blunted release of prolactin and growth hormone on gonadotropin-releasing hormone or thyrotropin-releasing hormone stimulation and a blunted release of growth hormone on apomorphine stimulation.

Infections

The support for infections being operative in schizophrenia is mostly indirect. As already mentioned, persons with schizophrenia may be more likely to be born in the winter months, thus suggesting a season-specific risk factor such as a virus (there are other explanations, however, such as changes in diet).

Epidemiologic data show a high incidence of schizophrenia after prenatal exposure to influenza during several epidemics of the disease. Some studies show that the frequency of schizophrenia increases after exposure to influenza during the second trimester of pregnancy. Influenza is, of course, more prevalent during the winter than other seasons. Other data supporting a viral hypothesis are an increased number of physical anomalies at birth, an increased rate of pregnancy and birth complications, seasonality of birth consistent with a viral infection, geographical clusters of adult cases, and

seasonality of hospitalizations.

Viral theories stem from the fact that several specific viral theories have the power to explain the particular localization of pathology necessary to account for a range of manifestations in schizophrenia without overt febrile encephalitis.

Environmental Factors

In addition to infections, a variety of other factors, particularly in the prenatal period, can contribute a small but significant risk for schizophrenia. These include birth complications, trauma after delivery, nutritional deficiencies, and other factors that may negatively affect healthy development.

Our Understanding of the Genetics of Schizophrenia

Studies of Inheritance Patterns. At least as early as the 1930's, twin and family studies demonstrated that schizophrenia was an inherited disorder. The heritability of schizophrenia is estimated to be about 60 to 80 percent.

The likelihood of a person having schizophrenia correlates with the closeness of the relationship to an affected relative (e.g., first- or second-degree relative). In the case of monozygotic twins who have an identical genetic endowment, there is an approximately 50 percent concordance rate for schizophrenia. This rate is four to five times the concordance rate in dizygotic twins or the rate of occurrence found in other first-degree relatives (i.e., siblings, parents, or offspring). The role of genetic factors is further reflected by the drop-off in the occurrence of schizophrenia among second- and third-degree relatives, in whom one would hypothesize a decreased genetic loading. The finding of a higher rate of schizophrenia among the biologic relatives of an adopted-away person who develops schizophrenia, compared with the adoptive, nonbiologic relatives who raise the patient, provides further support to the genetic contribution to the etiology of schizophrenia.

Nevertheless, the monozygotic twin data demonstrate the fact that individuals who are genetically vulnerable to schizophrenia do not inevitably develop schizophrenia; other factors (e.g., environment) must be involved in determining a schizophrenia outcome. If a vulnerability–liability model of

schizophrenia is correct in its postulation of an environmental influence, then other biologic or psychosocial environmental factors may prevent or cause schizophrenia in genetically vulnerable individuals.

Some data indicate that the age of the father correlates with the development of schizophrenia. In studies of schizophrenia patients with no history of illness in either the maternal or paternal line, those born from fathers older than the age of 60 years were vulnerable to developing the disorder. Presumably, spermatogenesis in older men is subject to more significant epigenetic damage than in younger men.

Genetic Studies. The modes of genetic transmission in schizophrenia are unknown, but many genes are associated with schizophrenia. Newer studies using more direct methods, such as comparative genomic hybridization, small nucleotide polymorphism (SNP) chips, next generation sequencing (NGS), GWAS, and the Clustered Regularly Interspaced Short Palindromic Repeats-associated Nuclease 9 (CRISPR/Cas9) genomic editing have revolutionized the genetic research of schizophrenia, and the number of genes associated with schizophrenia continues to rise. Many of these are likely chance findings. However, at least some are plausible candidates for contributing to schizophrenia vulnerability.

Linkage and association genetic studies have provided strong evidence for several specific candidate genes. The best candidates are those involved in synaptic transmission, including those for various monoamine receptors and those involved in glutamate release and signaling.

Given the role dopamine plays in schizophrenia, it is no surprise that many genetic studies have concentrated on genes involved in dopamine regulation. For example, some studies implicate catecholamine O-methyltransferase (COMT) polymorphism, which is involved in dopamine metabolism (as well as other catecholamines).

Copy number variation (CNV) studies suggest that perhaps between 2 to 5 percent of schizophrenia may be due, in part, to genetic variants that are highly penetrant but very rare. These included genes that were involved in regulating synaptic function and neurodevelopment. An example of a rare variant that is associated with psychosis is the 22q11.2 microdeletion, which causes 22q11.2 deletion or velocardiofacial syndrome (also called DiGeorge syndrome), which occurs in about 1 in 4,000 live births.

However, most of the risk is explained by common alleles, numbering in the hundreds, each having a small effect on risk. GWAS studies have yielded several loci of interest, including the DRD genes which encode the dopamine receptors, as well as several other loci involved in functions thought essential to the pathogenesis of schizophrenia. These include genes involved with glutamate, calcium signaling, dendritic spine formation, and other aspects of neuronal and neurodevelopmental function. As will be described in greater detail below, subsequent research has focused on genes related to the immune system and involved in synaptic pruning.

Many implicated genes do not code for proteins such as those described above, but have a role in genetic processing, affecting transcription and various epigenetic factors.

The extensive and sometimes conflicting findings are likely related to the heterogeneity of the disorder, and researchers are working to develop endophenotypes that more closely map to the genetics of the disorder. For example, prepulse inhibition is a likely candidate for a schizophrenia endotype. Researchers have identified several other endophenotype traits. These have the benefit of being measurable and lend themselves to quantitative analysis.

THE PSYCHOLOGY OF THE DISORDER

Family Dynamics

In a classic early study of 4-year old British children, children rated as having a poor mother–child relationship had a six-fold increase in the risk for schizophrenia. However, this leaves open the question of which came first, the poor relationship or the child's disease-related inability to form close relationships. It is telling that those who were adopted away from their mothers were more likely to develop schizophrenia if raised in adverse circumstances. In another study, children of schizophrenic mothers raised in a kibbutz (a type of commune) were more likely to develop schizophrenia than children raised in a family home. Nevertheless, no well-controlled evidence indicates that a specific family pattern plays a causative role in the development of schizophrenia. Some patients with schizophrenia do come from dysfunctional families, just as do many nonpsychiatrically ill persons. It

is crucial, however, not to overlook pathologic family behavior that can significantly increase the emotional stress with which a vulnerable patient with schizophrenia must cope.

ETIOLOGY

Biologic Theories

The Dopamine Hypothesis. The simplest formulation of the dopamine hypothesis of schizophrenia posits that schizophrenia results from too much dopaminergic activity. The theory evolved from two observations. First, the efficacy and the potency of many antipsychotic drugs (i.e., the dopamine receptor antagonists [DRAs]) correlate with their ability to act as antagonists of the dopamine type 2 (D_2) receptor. Second, drugs that increase dopaminergic activity, notably cocaine and amphetamine, are psychotomimetic. There is also support from the fact that certain functions that are regulated by dopamine (e.g., prepulse inhibition) are abnormal in patients with schizophrenia.

The basic theory does not comment on whether dopaminergic hyperactivity results from too much dopamine, too many dopamine receptors, hypersensitivity of the receptors to dopamine, or a combination of these.

Which dopamine tracts in the brain are involved is also not specified. Historically, models of schizophrenia suggested that dysfunctions in the mesolimbic pathway were responsible for the positive symptoms of schizophrenia. This was due to a variety of observations, including that seizures and tumors in these regions produced schizophrenia-like symptoms. Also, amphetamines, which can induce psychosis, seemed to affect the nucleus accumbens, and antipsychotics injected in this area seemed to reverse this effect.

However, subsequent research suggests that the striatum (usually thought to be involved in motor function) plays an important role. This is suggested by functional neuroimaging studies, which found that the most significant differences in dopamine function found for schizophrenic patients were in the dorsal striatum and associated projections. This relation is also true for people at high risk for schizophrenia. This area has an integrative role, and

dysfunctions in this area could explain the associative deficits that occur in schizophrenia. Also, as this area has a vital role in habit formation, it may argue for a model of psychosis as a habitual or rigid type of thinking, in which a person has difficulty considering alternative explanations for an experience.

Serotonin. Current hypotheses posit serotonin excess as a cause of both positive and negative symptoms in schizophrenia. The robust serotonin antagonist activity of clozapine and other second-generation antipsychotics coupled with the effectiveness of clozapine to decrease positive symptoms in chronic patients has contributed to the validity of this proposition.

GABA. Some have implicated the inhibitory amino acid neurotransmitter γ-aminobutyric acid (GABA) in the pathophysiology of schizophrenia based on the finding that some patients with schizophrenia have a loss of GABAergic neurons in the hippocampus. GABA has a regulatory effect on dopamine activity, and the loss of inhibitory GABAergic neurons could lead to the hyperactivity of dopaminergic neurons.

Neuropeptides. Neuropeptides, such as substance P and neurotensin, are localized with the catecholamine and indoleamine neurotransmitters and influence the action of these neurotransmitters. Alteration in neuropeptide mechanisms could facilitate, inhibit, or otherwise alter the pattern of firing in these neuronal systems.

Glutamate. Glutamate is also of interest, as ingestion of phencyclidine, a glutamate antagonist, produces an acute syndrome similar to schizophrenia. The hypotheses proposed about glutamate include those of hyperactivity, hypoactivity, and glutamate-induced neurotoxicity.

Acetylcholine and Nicotine. As noted, some studies suggest deficits in muscarinic and nicotinic receptors. These receptors play a role in the regulation of neurotransmitter systems involved in cognition, which is impaired in schizophrenia.

Neural Circuits (The Disconnect Hypothesis). There has been a gradual evolution from conceptualizing schizophrenia as a disorder that

involves discrete areas of the brain to a perspective that views schizophrenia as a disorder of brain neural circuits. For example, as mentioned previously, the basal ganglia and cerebellum are reciprocally connected to the frontal lobes, and the abnormalities in frontal lobe function seen in some brain imaging studies may be due to disease in either area rather than in the frontal lobes themselves. It is possible that an early developmental lesion of the dopaminergic tracts to the prefrontal cortex results in the disturbance of the prefrontal and limbic system function and leads to the positive and negative symptoms and cognitive impairments observed in patients with schizophrenia.

Of particular interest in the context of neural circuit hypotheses linking the prefrontal cortex and limbic system are studies demonstrating a relationship between hippocampal morphologic abnormalities and disturbances in prefrontal cortex metabolism or function (or both). Data from functional and structural imaging studies in humans suggest that whereas dysfunction of the anterior cingulate basal ganglia thalamocortical circuit underlies the production of positive psychotic symptoms, dysfunction of the dorsolateral prefrontal circuit underlies the production of primary, enduring negative or deficit symptoms. There is a neural basis for cognitive functions that are impaired in patients with schizophrenia. The observation of the relationship among impaired working memory performance, disrupted prefrontal neuronal integrity, altered prefrontal, cingulate, and inferior parietal cortex, and altered hippocampal blood flow provides strong support for disruption of the normal working memory neural circuit in patients with schizophrenia. The involvement of this circuit, at least for auditory hallucinations, has been documented in several functional imaging studies that contrast hallucinating and nonhallucinating patients.

Viruses, Neurotoxicity, and Neuroinflammation. Most carefully conducted investigations that have searched for evidence of neurotoxic viral infections in schizophrenia have had negative results. However, as noted, indirect epidemiologic data support a viral role. Nonetheless, the inability to detect genetic evidence of viral infection reduces the significance of all circumstantial data. The possibility of autoimmune brain antibodies has some data to support it; the pathophysiologic process, if it exists, however, probably explains only a subset of the population with schizophrenia.

The neurotoxicity hypothesis suggests that psychosis can be toxic, increasing stress levels and associated cortisol, leading to brain changes. Also, the use of exogenous toxins such as cannabis and alcohol may further contribute. There is also some evidence that antipsychotic medications could contribute as well. As described below, however, the lack of evidence for neurodegeneration argues against this process.

Many studies have suggested that neuroinflammation plays a role in the pathogenesis of schizophrenia. Schizophrenia and certain autoimmune diseases have overlapping clinical, epidemiologic, and genetic features. Furthermore, at least a subset of patients with schizophrenia has laboratory findings indicating immune activation. These findings may be particularly prominent in patients with structural brain changes, although these results are inconsistent.

Several GWAS studies have found implicated genes that regulate the immune response, particularly the major histocompatibility complex (MHC) region and specific genes involved in synaptic pruning, lending more support for the neurodevelopmental hypothesis, discussed next.

Increasing Evidence Suggests that Schizophrenia Is a Neurodevelopmental Disorder. Schizophrenia is likely a disorder of neurodevelopment, in which the standard neural structure does not develop properly. Earlier researchers speculated that this would be the case based on the observation that many patients who developed schizophrenia in early adulthood had motor and cognitive impairments as younger persons. Also, as discussed above, obstetrical complications are a risk factor for schizophrenia. Furthermore, for many patients with schizophrenia, cognitive deficits do not deteriorate significantly after illness onset. Evidence for this includes the lack of gliosis found in the brains of schizophrenia patients, as well as no consistent evidence of neurodegeneration, which suggests that the significant deficits are not due to cell death but rather the lack of proper cell growth. Also, many of the genes implicated in schizophrenia have relatively higher expression before birth and are likely involved in early brain development. Some of the genes may even affect the placenta, making it more sensitive to environmental stress.

As noted, one of the strongest genetic associations found to date for schizophrenia has been with variations in the MHC. These genes, particularly

the C4 alleles, are involved in synaptic pruning during critical developmental periods, which occur during adolescence and adulthood, and may explain why symptoms of schizophrenia only become apparent during this period.

Schizophrenia Likely Involves Multiple Dysfunctions at Various Levels. We know from neurologic patients suffering various focal lesions that the lesions causing hallucinations are often in the networks associated with that sensory system; thus, visual hallucinations are associated with occipital lesions as well as other areas along the visual pathway, such as the striatum and thalamus. Similarly, auditory hallucinations can be caused by lesions in the auditory cortex, hippocampus, amygdala, or thalamus. However, most patients with these focal lesions retain insight and recognize their hallucinations for what they are. Lesions in the corticostriatal networks can cause loss of insight, and the dopamine-related dysfunctions in this network likely account for the delusional beliefs about the hallucinations and other disordered thoughts. Still, other circuits are involved in the affective reaction to these abnormal experiences. In this way, the symptoms of schizophrenia are probably due to multiple dysfunctions along several networks.

This approach to understanding schizophrenia helps account for why antipsychotics are only partly useful for treating schizophrenia. By normalizing the dopamine dysfunction, antipsychotics decrease the abnormally high dopamine signaling in the associative striatum, and reduce the psychotic symptoms associated with the disorder. However, they are acting only on this particular circuit, and therefore have little effect on the negative and cognitive symptoms of the disorder.

Psychosocial Theories

If schizophrenia is a disease of the brain, it is likely to parallel diseases of other organs (e.g., myocardial infarctions, diabetes) whose courses are affected by psychosocial stress. Thus, clinicians should consider both psychosocial and biologic factors affecting schizophrenia.

The disorder affects individual patients, each of whom has a unique psychological makeup. Although many psychodynamic theories about the pathogenesis of schizophrenia seem outdated, perceptive clinical

observations can help contemporary clinicians understand how the disease may affect a patient's psyche.

Psychoanalytic Theories. Sigmund Freud thought that schizophrenia resulted from developmental fixations early in life. These fixations produce defects in ego development, and he postulated that such defects contributed to the symptoms of schizophrenia. Margaret Mahler and Paul Federn concentrated on distortions in the mother–infant relationship, and Harry Stack Sullivan viewed schizophrenia as a disturbance in interpersonal relatedness. To Sullivan, schizophrenia is an adaptive method used to avoid panic, terror, and disintegration of the sense of self. The source of pathologic anxiety results from cumulative experiential traumas during development.

THE OBJECT RELATIONS DEFICIT MODEL OF SCHIZOPHRENIA. Object relationship theory posits that relations with other people are "introjected" and applied to new relationships. One can approach the challenge of new relationships by drawing on the internal models one collects from past relationships. In the case of schizophrenia, this normal process is disrupted, presumably due to errors in neurodevelopment that affect the filters that interpret information from the environment. Thus, the introjects one collects are distorted and incomplete, and interpreted as dangerous and threatening. As a result, the patient avoids relationships, thus having less chance for corrective experiences. Reality becomes terrifying, and the reaction is to create an alternate reality through psychotic thinking.

This theory has the benefit of accounting for both the positive and negative symptoms of schizophrenia. It also points at potential treatments. For example, the job of the therapist is to provide positive experiences that the patient can introject to help correct their distorted world view.

Regardless of the theoretical model, all psychodynamic approaches rely on the premise that psychotic symptoms have meaning in schizophrenia. Patients, for example, may become grandiose after an injury to their self-esteem. Similarly, all theories recognize that human relatedness may be terrifying for persons with schizophrenia. Although research on the efficacy of psychotherapy with schizophrenia shows mixed results, concerned persons who offer compassion and a sanctuary in the confusing world of schizophrenia must be a cornerstone of any overall treatment plan. Long-term

follow-up studies show that some patients who bury psychotic episodes do not benefit from exploratory psychotherapy, but those who can integrate the psychotic experience into their lives may benefit from some insight-oriented approaches. There is renewed interest in the use of long-term individual psychotherapy in the treatment of schizophrenia, especially when combined with medication.

Learning Theories. According to learning theorists, children who later have schizophrenia learn irrational reactions and ways of thinking by imitating parents who have significant emotional problems. In learning theory, the poor interpersonal relationships of persons with schizophrenia develop because of poor models for learning during childhood.

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6 Bipolar Disorders

The bipolar disorders include bipolar disorder type I, commonly referred to as manic depression, bipolar type II disorder, and cyclothymia. Historically, we considered these mood disorders to be on a continuum with depressive disorders—hence the concept of polarity, with depression at one end and mania the other. Since the 1980s and the introduction of DSM-III, we separated the bipolar disorders from depressive disorders mainly because the disorders have different epidemiologic characteristics, courses, and treatments. However, at various times experts in mood disorders have reconsidered this separation, noting that many patients with major depressive disorder have past episodes of at least some manic symptoms. Many authorities see considerable continuity between recurrent depressive and bipolar disorders. There continues to be widespread discussion and debate about the bipolar spectrum, which incorporates classic bipolar disorder, bipolar II, and recurrent depressions.

THE CLINICAL PRESENTATION

Manic Episodes

Manic patients are excited, talkative, sometimes amusing, and frequently hyperactive. Their speech is usually rapid and loud and challenging to interrupt. It is commonly referred to as *pressured*, appearing as if driven by some unknown urgency. Pressured speech is considered a hallmark of mania.

An elevated, expansive, or irritable mood is the hallmark of a manic episode. The elevated mood is euphoric and often infectious and can even cause a countertransferential denial of illness by an inexperienced clinician. Although uninvolved persons may not recognize the unusual nature of a

patient's mood, those who know the patient recognize it as abnormal. Alternatively, the mood may be irritable, especially when someone prevents a patient from some unrealistic plan. Patients often exhibit a change of predominant mood from euphoria early in the course of the illness to later irritability. They also have a low frustration tolerance, which can lead to feelings of anger and hostility. Manic patients may be emotionally labile, switching from laughter to irritability to depression in minutes or hours.

Manic patients describe rapid thoughts, as inferred from their speech. As the manic state increases, their speech contains puns, jokes, rhymes, or plays on words. They may seem clever, even brilliant. Manic patients are often easily distracted, and their cognitive functioning in the manic state is unrestrained, with an accelerated flow of ideas. At a still higher activity level, associations become loosened, the ability to concentrate fades, and flight of ideas, clanging, and neologisms appear. In acute manic excitement, speech can be incoherent and indistinguishable from that of a person with schizophrenia.

The manic patient's thought content includes themes of self-confidence and self-aggrandizement.

At times, manic patients are grossly psychotic and disorganized and require physical restraints and the intramuscular injection of sedating drugs.

A 37-year-old engineer had experienced three manic episodes for which he had been hospitalized; all three episodes were preceded by several weeks of moderate psychomotor retardation. Although he had responded to lithium each time, once outside the hospital, he had been reluctant to take it and eventually refused to do so. Now that he was "euthymic," after his third and most disruptive episode during which he had badly beaten his wife, he could more accurately explain how he felt when manic. He experienced mania as "God implanted in him," so he could serve as "testimony to man's communication with God." He elaborated as follows: "Ordinary mortals will never, never understand the supreme manic state which I'm privileged to experience every few years. It is so vivid, so intense, so compelling. When I feel that way, there can be no other explanation: To be manic is, ultimately, to be God. God himself must be supermanic: I can feel it when mania enters through my left brain

like laser beams, transforming my sluggish thoughts, recharging them, galvanizing them. My thoughts acquire such momentum, they rush out of my head, to disseminate knowledge about the true nature of mania to psychiatrists and all other ordinary mortals. That's why I will never accept lithium again—to do so is to obstruct the divinity in me." Although he was on the brink of divorce, he would not yield to his wife's plea to go back on lithium.

Delusions occur in 75 percent of all manic patients. Mood-congruent manic delusions are often concerned with great wealth, extraordinary abilities, or power. Bizarre and mood-incongruent delusions and hallucinations also appear in mania.

A 29-year-old female college graduate, mother of two children, and wife of a bank president, had experienced several manic and retarded depressive episodes that had responded to lithium carbonate. She was referred because she had developed the delusion that she had been involved in an international plot. Careful probing revealed that the delusion represented further elaboration, in a rather fantastic fashion, of a grandiose delusion that she had experienced during her last postpartum manic episode. She believed that she had played an important role in uncovering the plot, thereby becoming a national hero. Nobody knew about it, she contended, because the circumstances of the plot were top secret. She further believed that she had saved her country from the international scheme and suspected that she was singled out for persecution by the perpetrators of the plot. At one point, she had even entertained the idea that the plotters sent special radio communications to intercept and to interrupt her thoughts. As is typical in such cases, she was on a heavy dosage of a lithium–antipsychotic combination. The consultation was requested because the primary mood symptoms were under control, yet, she had not given up her grandiose delusion. She flippantly remarked, "I must be crazy to believe in my involvement in an international plot," but she could not help but believe in it. Over several months, seen typically in 60-minute sessions weekly, the patient had developed sufficient trust that the psychiatrist could gently challenge her

beliefs.

She was, in effect, told that her self-professed role in the international scheme was highly implausible and that someone with her superior education and high social standing could not entertain a belief, to use her own words, “as crazy as that.” She eventually broke into tears, saying that everyone in her family was so accomplished and famous that to keep up with them she had to be involved in something grand; in effect, the international scheme, she said, was her only claim to fame: “Nobody ever gives me credit for raising two kids, and throwing parties for my husband’s business colleagues: My mother is a dean, my older brother holds high political office; my sister is a medical researcher with five discoveries to her credit [all true], and who am I? Nothing. Now, do you understand why I need to be a national hero?” As she alternated, over subsequent months, between such momentary flashes of insight and delusional denial, antipsychotic medication was gradually discontinued. Maintained on lithium, she now only makes passing reference to the grand scheme. She was encouraged to pursue her career goal toward a master’s degree in library science. (Courtesy of HS Akiskal, M.D.)

Grossly, orientation and memory are intact, although some manic patients may be so euphoric that they answer questions testing orientation incorrectly. Emil Kraepelin called the symptom delirious mania.

We know less about the cognitive deficits associated with bipolar disorder than such chronic disorders like schizophrenia. However, there is some evidence to suggest they share some common deficits.

Impaired judgment is a hallmark of manic patients. They may break laws about credit cards, sexual activities, and finances and sometimes involve their families in financial ruin. Manic patients also have little insight into their disorder. Their disinhibition may result in other examples of poor judgment, such as making phone calls during inappropriate times of the day. Pathologic gambling, a tendency to disrobe in public places, wearing clothing and jewelry of bright colors in unusual or outlandish combinations, and inattention to small details (e.g., forgetting to hang up the telephone) are also symptomatic of the disorder. Patients act impulsively, and at the same time, with a sense of conviction and purpose. They are sometimes preoccupied

with religious, political, financial, sexual, or persecutory ideas that can evolve into complex delusional systems. Occasionally, manic patients become regressed and play with their urine and feces.

About 75 percent of all manic patients are assaultive or threatening at some time. Manic patients are at increased risk for suicide. However, the most significant risk seems to be when bipolar patients are depressed. Manic patients often drink alcohol excessively, perhaps in an attempt to self-medicate.

Depressive Episodes

The depressive episodes in bipolar disorder are similar to those described for depressive disorders. Many experts in the field feel that there are qualitative differences in the depressive episodes experienced by bipolar patients, and researchers have attempted to find reliable differences between bipolar I disorder depressive episodes and episodes of major depressive disorder, but the differences are elusive. Although the data are inconsistent and controversial, some clinicians report that the depressed patients who we later diagnose as having bipolar disorder often have hypersomnia, psychomotor retardation, psychotic symptoms, a history of postpartum episodes, a family history of bipolar I disorder, and a history of antidepressant-induced hypomania. Table 6-1 lists some differences between the depressive disorder seen in bipolar disorder and major depression.



Table 6-1
Differentiating Characteristics of Bipolar and Unipolar Depressions

	Bipolar	Unipolar
History of mania or hypomania (definitional)	Yes	No
Temperament and personality	Cyclothymic and extroverted	Dysthymic and introverted
Sex ratio	Equal	More women than men

Age of onset	Teens, 20s, and 30s	30s, 40s, and 50s
Postpartum episodes	More common	Less common
Onset of episode	Often abrupt	More insidious
Number of episodes	Numerous	Fewer
Duration of episode	3–6 mo	3–12 mo
Psychomotor activity	Retardation > agitation	Agitation > retardation
Sleep	Hypersomnia > insomnia	Insomnia > hypersomnia
Family History		
Bipolar disorder	Yes	±
Unipolar disorder	Yes	Yes
Alcoholism	Yes	Yes
Pharmacologic Response		
Most antidepressants	Induce hypomania–mania	±
Lithium carbonate	Prophylaxis	±

Bipolar Disorder in Children and Adolescents

It is easy to misdiagnose mania in adolescents as a conduct disorder, antisocial personality disorder, or schizophrenia. Symptoms of mania in adolescents may include psychosis, alcohol or other substance abuse, suicide attempts, academic problems, philosophical brooding, OCD symptoms, multiple somatic complaints, marked irritability resulting in fights, and other antisocial behaviors. Although we can see many of these symptoms in healthy adolescents, severe or persistent symptoms should cause clinicians to consider bipolar I disorder in the differential diagnosis.

DIAGNOSIS

Patients with both manic and depressive episodes or patients with manic episodes alone are said to have *bipolar disorder*. The terms unipolar mania and pure mania are sometimes used for patients who are bipolar but who do not have depressive episodes.

Three additional categories of mood disorders are hypomania, cyclothymia, and dysthymia. Hypomania is an episode of manic symptoms that do not meet the criteria for a manic episode. Cyclothymia and dysthymia are disorders that represent less severe forms of bipolar disorder and major depression, respectively.

Bipolar I Disorder

Bipolar I patients have at least one manic episode. It is what most people mean when they refer to bipolar as a disorder. Table 6-2 compares the different approaches to diagnosing bipolar I disorder.



Table 6-2
Bipolar I Disorder

Bipolar I Disorder		
	DSM-5	**ICD-10**
Name	Bipolar I Disorder	Bipolar Affective Disorder ***NOTE:** ICD-10 does not distinguish between Bipolar I and II, requiring only history of discrete episodes of mania, hypomania and/or depression, with episodes demarcated by switches in mood/affect.
Duration	• Manic episode: 1 wk+	

	• Hypomanic episode: 4 days+ • Major depressive episode: 2 wk+	
Symptoms	**Manic or hypomanic episodes** • Abnormally ↑ or irritable mood (required) • Grandiose thoughts • ↓ Need for sleep • Pressured speech • Racing and expansive thoughts • Distractibility • Hyperactivity • Impulsivity/high-risk activities **Depressive episodes** • Similar to that for major depressive disorder	• History of episodes of mania, hypomania and/or depression • Switches in mood/affect **Mania** • Abnormally ↑ or irritable mood (required) • ↑ Activity • ↑ Talkativeness • Flight of ideas/racing thoughts • Social disinhibition • ↓ Need for sleep • Grandiose thoughts • Distractibility • Impulsivity/recklessness • Hypersexuality **Hypomania** • Abnormally ↑ mood (required) • Psychomotor agitation • ↑ Talkativeness • Poor concentration/distractibility • ↓ Need for sleep • Hypersexuality • Impulsivity or ↑ spending • Over-familiarity **Depressive episode**

		• Depressed mood • Loss of interest or pleasure • Decreased energy *Additional symptoms:* • Low self-esteem • Excessive guilt or shame • Recurrent thoughts of death or suicide • Poor concentration • Psychomotor changes • Sleep disturbance • Change in appetite and/or weight
Required number of symptoms	At least 1 manic episode • Abnormally ↑ or irritable mood (required) ≥3 of the other symptoms (4 if irritable mood)	**Bipolar affective disorder, current episode hypomanic** • Hypomania • History of a prior affective episode (manic, hypomanic, depressed, mixed) **Bipolar affective disorder, current episode manic** • Mania • History of prior affective episode (manic, hypomanic, depressed, mixed) **Bipolar disorder, current episode mild/moderate/severe depression** • Depressive episode • History of prior affective

		episode (manic, hypomanic, depressed, mixed) **Bipolar disorder, current episode mixed** • Mixture or rapid alternation of hypomanic, manic and/or depressive symptoms • History of prior affective episode (manic, hypomanic, depressed, mixed)
Exclusions (not better explained by):	Drug abuse Medication effect Other medical condition Other psychiatric illness	Psychoactive substance use Another mental disorder
Psychosocial Impact	**Manic episode:** impaired functioning or needing hospitalization **Hypomanic episode:** No impairment or need for hospitalization **Depressive episode:** marked distress and/or psychosocial	

	impairment	
Symptom Specifiers	**With mixed features:** • Either depressive or manic/hypomanic episode • Additional symptoms of depressive or manic/hypomanic period (not full criteria) **With rapid cycling:** • ≥4 mood episodes in 1 yr • ≥2-mo period of partial/full remission between episodes **With melancholic features:** Similar to that for major depressive disorder **With atypical features:** Similar to that for major depressive disorder **With anxious distress:** ≥2 symptoms among the	**Current episode hypomanic** **Current episode manic without psychotic symptoms** **Current episode manic with psychotic symptoms** **mood congruent** **mood incongruent** **Current episode of depression** **Current episode mixed** **Currently in remission**—no symptoms. History of previous episodes.

	following: • Feeling tense • Restlessness • Difficulty with concentration due to worrying • ↑ fear without cause • Fear of loss of control **With mood-congruent psychotic features** **With mood-incongruent psychotic features** **With catatonia**	
Severity Specifiers	**Mild—**minimal symptoms, no to minimal impairment and/or distress **Moderate—**moderate symptomatology and impairment **Severe—**maximal symptoms, marked distress and impairment	**For current episode of depression** **Mild:** 2–3 symptoms **Moderate:** ≥4 symptoms, including ≥2 among loss of pleasure, depressed mood, and low energy **Severe:** symptoms are marked and distressing, suicidal thoughts are common **without psychotic symptoms** **with psychotic**

		symptoms **mood congruent** **mood incongruent**
Course Specifiers	**With peripartum onset:** • Episode occurs during pregnancy or within 4 wk after delivery **With seasonal pattern** • Pattern present for ≥2 yr **In partial remission** **In full remission**	

Patients with this disorder may have a single or recurrent episode. We consider manic episodes as distinct when they are separated by at least 2 months without significant symptoms of mania or hypomania. The episodes should not be due to another apparent cause, such as medication (including an antidepressant).

Bipolar II Disorder

Bipolar patients have hypomania rather than mania—manic-type symptoms that are not as severe or as impairing as full mania. Table 6-3 compares the different approaches to diagnosing bipolar II disorder.

It is easy to confuse other disorders, including dramatic but normal moods, with hypomania. For example, some patients with depression may be thrilled and very euphoric once emerging from a depressive episode. Many medications, including antidepressants, can induce hypomanic symptoms. The diagnostic approaches attempt to help distinguish hypomania from these other causes of a heightened mood.

Diagnostic Specifiers

Rapid Cycling. Some patients experience frequent manic episodes. When a patient has at least four such episodes in a year, we diagnose them with the rapid cycling subtype of bipolar I disorder. Patients with rapid cycling bipolar I disorder are likely to be female and to have had depressive and hypomanic episodes. No data indicate that rapid cycling has a familial pattern of inheritance; thus, an external factor such as stress or drug treatment may provoke rapid cycling.



Table 6-3
Bipolar II Disorder

Bipolar II Disorder		
	DSM-5	**ICD-10**
Name	Bipolar II Disorder	Bipolar Affective Disorder ***NOTE:** ICD-10 does not distinguish between Bipolar I and II, requiring only history of discrete episodes of mania, hypomania and/or depression, with episodes demarcated by switches in mood/affect.
Duration	See Table 6-2 for hypomanic and depressive episodes	See Table 6-2
Symptoms	**Hypomanic episodes** (see Table 6-2) **Depressive episodes** (see Table 6-2)	See Table 6-2

Required number of symptoms	≥1 hypomanic episode ≥1 depressive episode	
Exclusions (not better explained by):	Drug abuse Medication effect Other medical condition Another psychiatric illness Diagnose bipolar I if h/o manic episode	Psychoactive substance use Better explained by another mental disorder
Psychosocial Impact	• Hypomanic episode: NO marked impairment in functioning and NO hospitalization necessitated • Depressive episode: marked distress and/or impairment in psychosocial functioning	
Symptom Specifiers	**Current episode (see Table 6-2 for definitions)** **Depressed** **Hypomanic** **With anxious distress**	See Table 6-2

	With mixed features **With rapid cycling** **With melancholic features** **With atypical features** **With mood-congruent psychotic features** **With mood-incongruent psychotic features** **With catatonia**	
Severity Specifiers	**Mild—**minimal symptoms, no to minimal impairment and/or distress **Moderate—**moderate symptomatology and impairment **Severe—**maximal symptoms, marked distress and impairment (note, impairment occurs during depressive episode)	See Table 6-2
Course	**With peripartum**	

Specifiers	**onset—**see Table 6-2 **With seasonal pattern—see Table 6-2** **In partial remission** **In full remission**

With Seasonal Pattern. As with depressive disorders, mania can occur primarily during certain seasons. Some studies have found a higher prevalence of manic episodes in the spring and summer months. However, available research is most convincing for the seasonality of depressive episodes.

With Peripartum Onset. Mania occurring after pregnancy is a critical issue given the potential risk to the child.

Catatonia. Clinicians often do not associate catatonic symptoms with bipolar I disorder because of the marked contrast between the symptoms of stuporous catatonia and the classic symptoms of mania. They are associated with depressive episodes, however.

Other Bipolar Disorders

Cyclothymia. Cyclothymic disorder has also been appreciated clinically for some time as a less severe form of bipolar disorder. Patients with cyclothymic disorder have at least 2 years of frequently occurring hypomanic symptoms that cannot fit the diagnosis of a manic episode and of depressive symptoms that cannot fit the diagnosis of a major depressive episode. Table 6-4 compares the different approaches to diagnosing cyclothymia.



Table 6-4
Cyclothymic Disorder

Cyclothymic Disorder		
	DSM-5	**ICD-10**
Name	Cyclothymic Disorder	Cyclothymia
Duration	≥2 yr (≥1 for children) with depressive and hypomanic symptoms present ≥50% of the time	2-yr w/ periods of depression and elevated mood, without ever meeting criteria for a depressive episode or manic episode
Symptoms	Hypomanic episodes (see Table 6-2) Depressive episodes (see Table 6-2)	Unstable moods, many periods of depression or mild elation which are insufficient to be called hypomania or mild major depressive disorder
Required number of symptoms	Symptoms of hypomania and depression present, without ever meeting full criteria for a depressive episode or manic episode	Several episodes, symptoms are insufficient for a diagnosis of hypomania, major depression or another bipolar affective disorder
Exclusions (not better explained	• Substance use • Medication effect	Bipolar affective disorder (though history of bipolar

by):	• Other medical condition • Other mental illness (i.e., Bipolar I or II)	affective disorder may be present)
Psychosocial Impact	Marked distress or impairment in areas of functioning	
Symptom Specifiers	**With anxious distress:** including at least two symptoms among feeling tense, restlessness, difficulty with concentration due to worrying, excessive fear without identifiable cause, fear of loss of control	Affective personality disorder Cycloid personality Cyclothymic personality

Mr. B, a 25-year-old single man, came for evaluation due to irritability, insomnia, jumpiness, and excessive energy. He reported that such episodes lasted from a few days to a few weeks and alternated with longer periods of feeling hopeless, dejected, and worn out with thoughts of suicide. Mr. B reported having been this way for as long as he could remember. He had never been treated for his symptoms. He denied using drugs and said he had “only the occasional drink to relax.”

As a child, Mr. B went from one foster family to another and was an irresponsible and trouble-making child. He frequently ran away from home, was absent from school, and committed minor crimes. He ran away from his last foster family at the age of 16 years and drifted ever since, taking occasional odd jobs. When he became restless at one location or job, he quickly moved on to the next. He did not have close friends because he would form and end friendships quickly.

DIFFERENTIAL DIAGNOSIS

When a patient with bipolar I disorder has a depressive episode, the differential diagnosis is the same as that for a patient with major depressive disorder. However, when a patient is manic, there is a broad differential, including other mood disorders, psychiatric disorders, medical disorders, and substances. We list some of these in Table 6-5.

Psychotic Disorders

It can be tough to distinguish a manic episode from the acute psychosis of a patient experiencing a psychotic episode. Although challenging, a differential diagnosis is possible. Merriment, elation, and infectiousness of mood are much more common in manic episodes than in schizophrenia. The combination of a heightened mood, rapid speech, and hyperactivity weighs toward mania. Mania often begins rapidly and is a marked change from previous behavior. Family history is also helpful.

When evaluating patients with catatonia, clinicians should look carefully for a history of manic or depressive episodes and a family history of mood disorders. There is an unfortunate tendency to misdiagnose manic symptoms in persons from minority groups (mainly Black and Hispanic) as schizophrenic symptoms.

Personality Disorders

Hypomania can frequently be confused with the mood lability of personality disorders, particularly borderline personality disorder. Patients with a borderline personality disorder often have a severely disrupted life, similar to that of patients with bipolar II disorder, because of the multiple episodes of significant mood disorder symptoms.



Table 6-5
Differential Diagnosis of Mania

Medical
AIDS/HIV
Delirium
Hyperthyroidism
Postencephalitic syndrome

Substance induced
- Antidepressant-induced mania
- Steroid-induced mania
- Amphetamine-induced mania
- Cocaine-induced mania
- Phencyclidine-induced mania
- Alcohol intoxication
- L-Dopa–induced mania
- Bronchodilator-induced mania
- Decongestant-induced mania

Psychiatric
- Atypical psychosis
- Bipolar disorder
- Catatonic schizophrenia
- Schizoaffective disorder

A 19-year-old single woman presented with the chief complaint that “all men are bastards.” Since her early teens, with the onset of her menses, she had complained of extreme variability in her moods on a nearly daily basis; irritability with hostile outbursts was her main affect, although more-protracted hypersomnic depressions with multiple overdoses and wrist slashings had led to at least three hospitalizations. She also had migrainous headaches that, according to her mother, had motivated at least one of those overdoses. Despite her tempestuous and suicidal moods that led to these hospitalizations, she complained of “inner emptiness and a bottomless void.” She had used heroin, alcohol, and stimulants to overcome this troubling symptom. She also gave history of ice cream craving and frequent purging. She was talented in English and wrote much-acclaimed papers on the American confessional poet Anne Sexton. She said that she was mentally disturbed because of a series of stepfathers who had all forced “oral rape” on her when she was between 11 and 15 years of age. She subsequently gave herself sexually to any man that she met in bars, no longer knowing whether she was a “prostitute” or a “nice

little girl." On two occasions, she had inflicted cigarette burns inside her vagina "to feel something." She had also engaged in a "brief lesbian relationship" that ultimately left her "emptier" and guilt ridden; nonetheless, she now believed that she should burn in hell because she could not get rid of "obsessing" about the excitement of mutual cunnilingus with her much older female partner. The patient's mother, who owned an art gallery, had been married five times and gave a history of unmistakable hypomanic episodes; a maternal uncle had died from alcohol-induced cirrhosis. The patient's father, a prominent lawyer known for his "temper and wit," had committed suicide. The patient was given phenelzine, eventually raised to 75 mg/day, at which point the mother described her as "the sweet daughter she was before age 13." At her next premenstrual phase, the patient developed insomnia, ran away from home at night, started "dancing like a go-go girl, met an incredibly handsome man" of 45 years of age (a pornography shop owner), and had a clandestine marriage to him. After many dose adjustments, she is now maintained on a combination of lithium (900 mg/day) and divalproex 750 mg/day. The patient now attends college and has completed four semesters in art history. In addition to control of her irritable and suicidal moods, bulimic and migraine attacks have abated considerably. Her marriage has been annulled on the basis that she was not mentally competent at the time of the wedding. She is no longer promiscuous and now expresses fear of intimacy with men that she is attracted to. She is receiving individual psychotherapy for this problem.

Medical Conditions

In contrast to depressive symptoms, which are present in almost all psychiatric disorders, manic symptoms are more distinctive. However, a wide range of medical disorders and substances can cause manic symptoms. Antidepressant treatment can also be associated with the precipitation of mania in some patients.

COMORBIDITY

Whereas men more frequently present with substance use disorders, women

more frequently present with comorbid anxiety and eating disorders. In general, patients with bipolar disorder more frequently show comorbidity of substance use and anxiety disorders than do patients with unipolar major depression. In the Epidemiologic Catchment Area (ECA) study, the lifetime history of substance use disorders, panic disorder, and OCD was approximately twice as high among patients with bipolar I disorder (61 percent, 21 percent, and 21 percent, respectively) than in patients with unipolar major depression (27 percent, 10 percent, and 12 percent, respectively). Comorbid substance use disorders and anxiety disorders worsen the prognosis of the illness and markedly increase the risk of suicide.

Although cyclothymic disorder is sometimes diagnosed retrospectively in patients with bipolar I disorder, no identified personality traits are associated explicitly with bipolar I disorder.

COURSE

The natural history of bipolar I disorder is such that it is often useful to make a graph of a patient's disorder and to keep it up to date as the treatment progresses (Fig. 6-1).

Onset

About 5 to 10 percent of patients with an initial diagnosis of major depressive disorder have a manic episode 6 to 10 years after the first depressive episode. The mean age for this switch is 32 years, and it often occurs after two to four depressive episodes.

Bipolar I disorder most often starts with depression (75 percent of the time in women, 67 percent in men) and is a recurring disorder. Most patients experience both depressive and manic episodes, although 10 to 20 percent experience only manic episodes.

The incidence of bipolar I disorder in children and adolescents is about 1 percent, and the onset can be as early as age 8 years.

Manic symptoms are common in older persons, although the range of causes is broad and includes nonpsychiatric medical conditions, dementia, and delirium, as well as bipolar I disorder. The onset of true bipolar I disorder in older persons is relatively uncommon.

Duration

The manic episodes typically have a rapid onset (hours or days) but may evolve over a few weeks. An untreated manic episode lasts about 3 months; therefore, clinicians should not discontinue giving drugs before that time. Depressive episodes are generally similar to those for depressive disorders.

Of persons who have a single manic episode, 90 percent are likely to have another. As the disorder progresses, the time between episodes often decreases. After about five episodes, however, the interepisode interval often stabilizes at 6 to 9 months. Of persons with bipolar disorder, 5 to 15 percent have four or more episodes per year and are classified as rapid cyclers.

Bipolar II Disorder

The course and prognosis of bipolar II disorder indicate that the diagnosis is stable because there is a high likelihood that patients with bipolar II disorder will have the same diagnosis up to 5 years later. Bipolar II disorder is a chronic disease that warrants long-term treatment strategies.

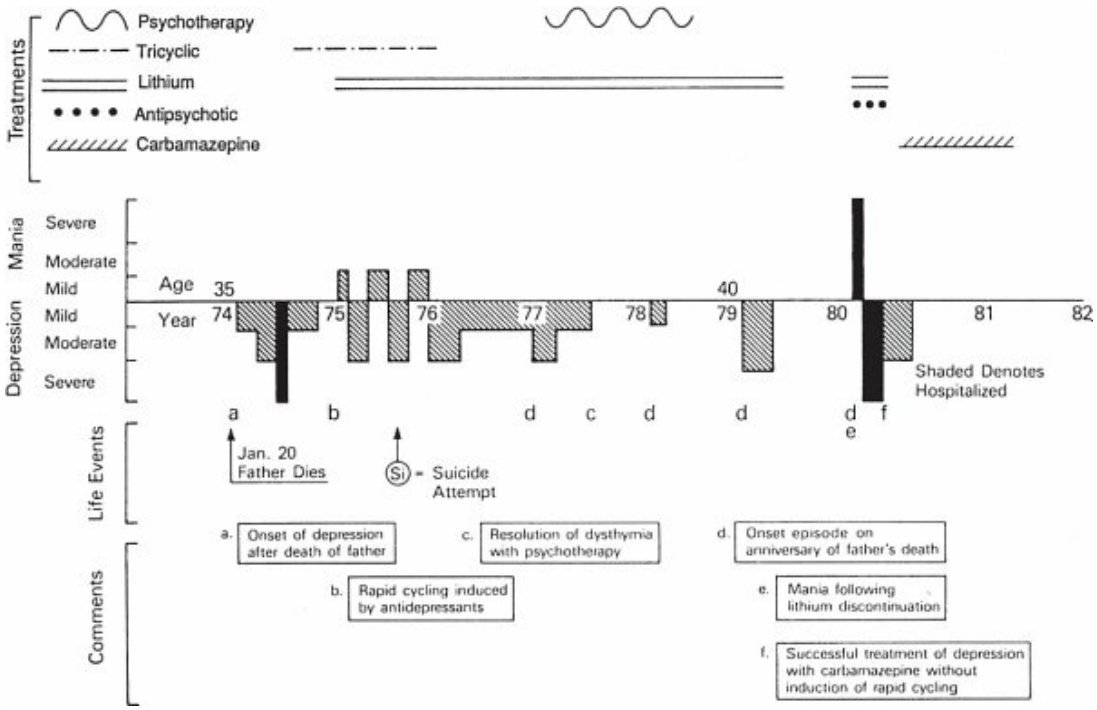


FIGURE 6-1
Graphing the course of a mood disorder. Prototype of a life chart. (Courtesy of Robert M. Post, M.D.)

PROGNOSIS

Patients with bipolar I disorder have a poorer prognosis than do patients with major depressive disorder. About 40 to 50 percent of patients with bipolar I disorder may have a second manic episode within 2 years of the first episode. Although lithium prophylaxis improves the course and prognosis of bipolar I disorder, probably only 50 to 60 percent of patients achieve significant control of their symptoms with lithium. About 7 percent of patients with bipolar I disorder do not have a recurrence of symptoms; 45 percent have more than one episode, and 40 percent have a chronic disorder. Patients may have from 2 to 30 manic episodes, although the mean number is about 9. About 40 percent of all patients have more than 10 episodes. On long-term follow-up, 15 percent of all patients with bipolar I disorder are well, 45 percent are well but have multiple relapses, 30 percent are in partial remission, and 10 percent are chronically ill. One-third of all patients with bipolar I disorder have chronic symptoms and evidence of significant social decline.

Prognostic Indicators

One 4-year follow-up study of patients with bipolar I disorder found that a premorbid poor occupational status, alcohol dependence, psychotic features, depressive features, interepisode depressive features, and male gender were all factors that contributed to a poor prognosis. A short duration of manic episodes, advanced age of onset, few suicidal thoughts, and few coexisting psychiatric or medical problems predict a better outcome.

TREATMENT APPROACH

Hospitalization

It is best to treat patients with severe mania in the hospital where aggressive dosing is possible, and it is possible to achieve an adequate response relatively quickly. Manic patients may test the limits of ward rules, shift responsibility for their acts onto others, or exploit the weaknesses of others, and they can create conflicts among staff members.

Choosing a Treatment

Medications are the treatments of choice for patients with bipolar disorders. However, psychotherapies can offer an essential adjunct to treatment.

Somatic Treatments

Pharmacotherapy

GENERAL CLINICAL GUIDELINES. We can divide the pharmacologic treatment of bipolar disorders into acute and maintenance phases. Bipolar treatment, however, also involves the formulation of different strategies for the patient who is experiencing mania or hypomania or depression. Lithium and its augmentation by antidepressants, antipsychotics, and benzodiazepines have been the principal approach to the illness. However, three anticonvulsant mood stabilizers—carbamazepine, valproate, and lamotrigine—are commonly used options, as well as a series of atypical antipsychotics. Often, it is necessary to try different medications before finding an optimal treatment. Furthermore, although one strives for monotherapy, in the case of bipolar disorder, polypharmacy is common.



Table 6-6
Recommendations for Pharmacologic Treatment of Acute Mania

First line	Monotherapy: lithium, divalproex, divalproex ER, olanzapine, risperidone, quetiapine, quetiapine XR, aripiprazole, ziprasidone, asenapine, paliperidone ER, cariprazine
	Adjunctive therapy with lithium or divalproex: risperidone, quetiapine, olanzapine, aripiprazole, asenapine
Second line	Monotherapy: carbamazepine, carbamazepine ER, ECT, haloperidol
	Combination therapy: lithium + divalproex
Third line	Monotherapy: chlorpromazine, clozapine, tamoxifen

	Combination therapy: lithium or divalproex + haloperidol, lithium + carbamazepine, adjunctive tamoxifen
Not recommended	Monotherapy: gabapentin, topiramate, lamotrigine, verapamil, tiagabine
	Combination therapy: risperidone + carbamazepine, olanzapine + carbamazepine

ECT, electroconvulsive therapy; XR or ER, extended release.
Modified from CANMAT/ISBD Guidelines.

Table 6-6 lists some recommended medications for treating acute mania.

Adherence to treatment, however, is often a problem because patients with mania frequently lack insight into their illness and refuse to take medication. Because impaired judgment, impulsivity, and aggressiveness combine to put the patient or others at risk, we must medicate patients when we need to protect themselves and others from harm.

INITIAL MEDICATION SELECTION. Figure 6-2 diagrams a strategy for selecting and preparing a bipolar patient for pharmacotherapy.

Acute Mania. The treatment of acute mania, or hypomania, usually is the most straightforward phase to treat. We can use agents alone or in combination to bring the patient down from a high.

Lithium Carbonate. Lithium carbonate is considered the prototypical mood stabilizer. However, because the onset of antimanic action with lithium can be slow, we often supplement it in the early phases of treatment by atypical antipsychotics, mood-stabilizing anticonvulsants, or high-potency benzodiazepines. Therapeutic lithium levels are between 0.6 and 1.2 mEq/L. The acute use of lithium has been limited in recent years by its unpredictable efficacy, problematic side effects, and the need for frequent laboratory tests. The introduction of newer drugs with more favorable side effects, lower toxicity, and less need for frequent laboratory testing has resulted in a decline in lithium use. For many patients, however, its clinical benefits can be remarkable.

Anticonvulsants. Valproate (valproic acid or divalproex sodium) has surpassed lithium in use for acute mania. Unlike lithium, valproate is only indicated for acute mania, although most experts agree it also has prophylactic effects. Normal dose levels of valproic acid are 750 to 2,500 mg/day, achieving blood levels between 50 and 120 µg/mL. Rapid oral loading with 15 to 20 mg/kg of divalproex sodium from day 1 of treatment has been well tolerated and associated with a rapid onset of response. Some laboratory testing is required during valproate treatment.

Carbamazepine has been used worldwide for decades as a first-line treatment for acute mania. The FDA approved it for acute mania in the United States in 2004. Typical doses of carbamazepine to treat acute mania range between 600 and 1,800 mg/day associated with blood levels of between 4 and 12 µg/mL. The keto congener of carbamazepine, oxcarbazepine, is better tolerated than carbamazepine, but the data for its efficacy are conflicting. A Cochrane review concluded there is insufficient evidence for this medication in acute mania.

Antipsychotics. The FDA approved many of the atypical antipsychotics for use in bipolar disorder. We list them in Table 6-7. Compared with older agents, such as haloperidol and chlorpromazine, atypical antipsychotics have a lesser liability for excitatory postsynaptic potential and tardive dyskinesia; many do not increase prolactin. However, many of them have the risk of weight gain with its associated medical problems. Some patients, however, require maintenance treatment with antipsychotic medication.

Acute Bipolar Depression

Lithium. There is limited evidence for lithium in bipolar depression. Early studies were promising, but later placebo-controlled studies did not confirm lithium's efficacy. More extensive studies that included lithium did suggest that lithium was at least as useful as other mood stabilizers for bipolar depression.

Anticonvulsants. The most promising anticonvulsant has been lamotrigine, which has several reasonable studies showing efficacy for bipolar depression. Its major limitation is that it must be titrated gradually to prevent a severe

skin rash. Evidence for valproate and other anticonvulsants is limited. Gabapentin and levetiracetam appear to be ineffective.

Antipsychotics. Several of the atypical antipsychotics have shown efficacy for bipolar depression. Quetiapine has the best evidence. It appears that quetiapine, in a modest dose (300 mg/day), is sufficient to improve symptoms. Olanzapine, lurasidone, and cariprazine also have positive studies, and the FDA approved lurasidone for this indication. Ziprasidone and aripiprazole do not appear to be effective.

Antidepressants. It remains controversial whether antidepressants are useful for the depressive phase of bipolar disorder. This is particularly true in patients with rapid cycling and mixed states. They seem to be less effective than for major depressive disorder and may induce cycling, mania, or hypomania. The risk of inducing mania seems highest for tricyclic antidepressants, monoamine oxidase inhibitors, and perhaps the serotonin norepinephrine reuptake inhibitors such as venlafaxine. Most experts agree that antidepressants are not appropriate as monotherapy for patients with bipolar disorder.

Whether we can use them as adjuncts is also controversial. Some antidepressants have some evidence of efficacy for adjunctive therapy, such as fluoxetine. On the whole, the available evidence suggests they may have some usefulness when combined with a mood stabilizer.

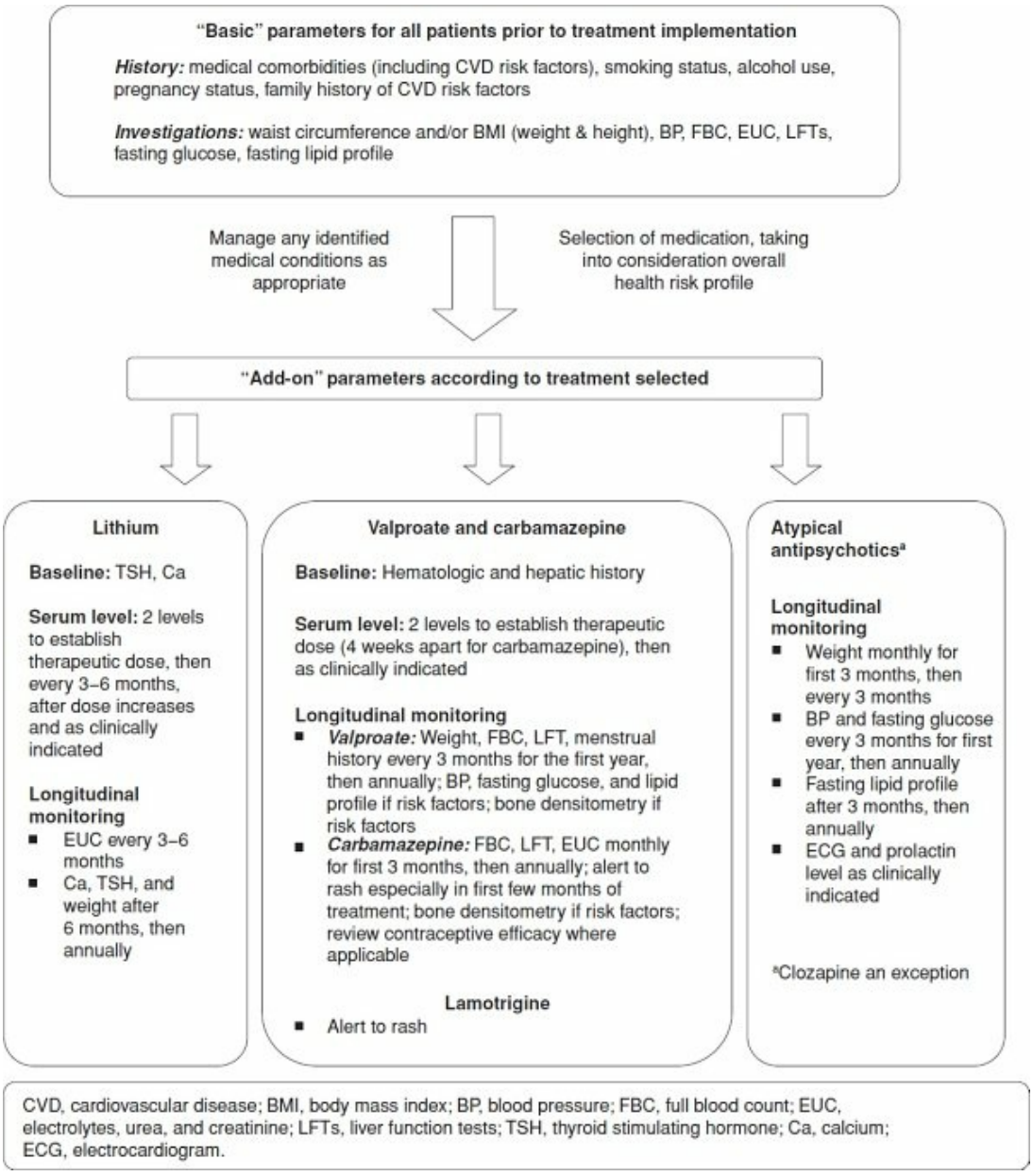


FIGURE 6-2
Recommendations for treatment monitoring in bipolar disorder. (Used with permission from ISBD safety monitoring guidelines.)

Electroconvulsive Therapy. Electroconvulsive therapy may also be useful for patients with bipolar depression who do not respond to lithium or other mood stabilizers and their adjuncts, particularly in cases in which strong suicidal tendency presents as a medical emergency.

Other Agents. Various clinicians and researchers have tried many other agents in an attempt to find more options for treatment. Most studies have not

been promising. Dopamine agonists, including modafinil and armodafinil, have some preliminary evidence. Omega-3-fatty acid adjunctive therapy has conflicting evidence. N-acetylcysteine (a glutathione precursor) has some preliminary evidence. Ketamine and other glutamatergic modulators may have a role in treatment-resistant patients.

Table 6-7
Atypical Antipsychotics for Bipolar Disorder: Efficacy Summary and Dose Ranges

		Efficacy					
				Prophylaxis			
Drug	Dose Range (mg)	Acute Mania	Acute Bipolar Depression	Mood Episodes	Mania	Depression	Comments
Olanzapine	5–20	Yes	Yes (see comments)	Yes	Yes	Yes	Improvement was seen mostly in sleep, appetite, and inner tension but not in core depressive symptoms Magnitude of benefit for depression less than for mania
Risperidone	1–6	Yes	No data	Yes	Yes	No	Prophylactic efficacy was demonstrated with Risperdal Consta but no studies with oral risperidone
Quetiapine	300–800	Yes	Yes	Yes	Yes	Yes	For bipolar depression, 300 mg/day is as effective as 600 mg/day (see text for guidance) Appears to have equal efficacy in preventing both mania and depression
Ziprasidone	80–160	Yes	No	Yes	Yes	No	Prophylaxis demonstrated for adjunctive therapy but no data for monotherapy
Aripiprazole	15–30	Yes	No	Yes	Yes	No	
Paliperidone	6–12	Yes	No data	Yes	Yes	No	Less effective than olanzapine in preventing mood episodes
Asenapine	10–20	Yes	No data	Yes	Yes	Yes	Numerically fewer patients in the asenapine group had relapse of manic and depressive episodes but the differences were not significant
Lurasidone	40–120	No data	Yes	Studies underway	Studies underway	Studies underway	Effective in depressed patients with mixed features
Cariprazine	3–12	Yes	Yes	No data	No data	No data	For bipolar depression, 1.5–3 mg/day is recommended, while for mania, up to 12 mg/day is appropriate

DURATION AND PROPHYLAXIS

Maintenance Treatment. Preventing recurrences of mood episodes is the greatest challenge facing clinicians. Not only must the chosen regimen achieve its primary goal—sustained euthymia—but the medications should not produce unwanted side effects that affect functioning. Sedation, cognitive impairment, tremor, weight gain, and rash are some side effects that lead to

treatment discontinuation.

Lithium, carbamazepine, and valproic acid, alone or in combination, are the most widely used agents for the long-term treatment of patients with bipolar disorder. For patients treated with long-term lithium, thyroid supplementation is often necessary to treat lithium-induced hypothyroidism.

Lamotrigine has prophylactic antidepressant and, potentially, mood-stabilizing properties. Lamotrigine appears to be superior at the acute and prophylactic treatment of the depressive phase of illness compared to the manic.

ACUTE TREATMENT FAILURES. Most patients will respond to treatment within 2 weeks. When they do not respond, we should consider using a different approach. When patients do not respond to initial treatment, it makes sense to try another first-line treatment as there are several from which to choose. Most patients will respond to one of these treatments, at least during their manic phase; depression may be more difficult.

SELECTING SECOND TREATMENT OPTIONS. In the rare case that first-line treatment fails, other options include haloperidol, carbamazepine, and combination treatment with lithium and valproate. If those fail, other antipsychotics could be considered.

Other Somatic Treatments. In addition to ECT (discussed above), transcranial magnetic stimulation and magnetic seizure therapy have limited, although promising, data.

Psychosocial Therapy

Psychotherapy can be a crucial adjunct for patients. The goals of this therapy include helping treatment adherence, promoting stability, and avoiding risk factors for the disorder. Cognitive-behavioral therapy, interpersonal and social rhythm therapy, and family focused therapy are all reasonable therapies to use. Table 6-8 summarizes psychotherapeutic treatments for bipolar disorder, including the presumed underlying mechanism and sample interventions.

Table 6-8
Efficacious Psychotherapeutic Treatments for Bipolar Disorder

Treatment	Conceptualization of Disorder Etiology	Sample Interventions
Cognitive-behavioral therapy	Biologic vulnerability interacting with stress Skills deficits limit ability to manage symptoms	Identify and challenge automatic thoughts that interfere with treatment adherence Engage in rewarding activities that provide increased routine and stability Practice communication skills with providers
Interpersonal and social rhythm therapy	Interpersonal vulnerabilities arising from early attachment and learned relationship patterns, plus disruption of social rhythms	Develop awareness of patterns in primary relationships and the therapeutic relationship Track and stabilize social rhythms Interpersonal skills training Communication analysis

Family-focused therapy	Biologic vulnerability exacerbated by negative expressed emotion in family environment	Education regarding the disorder, including precipitants, risk factors, and effective treatment Establish relapse prevention plan, agreed upon by all involved family members Communication skills training Problem-solving skills training

THE EPIDEMIOLOGY OF THE DISORDER(S)

Incidence and Prevalence

We show the US prevalence rates of different clinical forms of bipolar disorder in Figure 6-3. The annual incidence of bipolar illness is usually estimated to be less than 1 percent. Worldwide it is estimated to vary from 0.3 to 1.2 percent by country. However, it is difficult to estimate because it is easy to miss milder forms of bipolar disorder.

Sex

In contrast to major depressive disorder, bipolar I disorder has a roughly equal prevalence among men and women. Manic episodes are more common in men, and depressive episodes are more common in women. When manic episodes occur in women, they are more likely than men to present a mixed picture (e.g., mania and depression). Women also have a higher rate of having the rapid cycling subtype of bipolar I disorder.

Age

The onset of bipolar I disorder is earlier than that of major depressive disorder. The age of onset for bipolar I disorder ranges from childhood (as early as age 5 or 6 years) to 50 years or even older in rare cases, with a mean age of 30 years. The overall prevalence of bipolar disorder in adolescents is similar to adults. However, it increases with age; that is, the prevalence is about 2 percent in 13- and 14-year-olds, but it doubles by age 18.

Other Factors

Bipolar I disorder is more common in divorced and single persons than among married persons, but this difference may reflect the early onset and the resulting marital discord characteristic of the disorder.

There is a higher than average incidence of bipolar I disorder in upper socioeconomic groups. Bipolar I disorder is more common in persons who did not graduate from college than in college graduates. This may also reflect the relatively early age of onset for the disorder.

THE NEUROBIOLOGY OF THE DISORDER

Similar to the depressive disorders, the most consistent abnormality observed in bipolar disorders is an increased frequency of abnormal hyperintensities in subcortical regions, such as periventricular regions, the basal ganglia, and the thalamus. We find this more commonly in bipolar I disorder than in depressed adults. These hyperintensities likely reflect the deleterious neurodegenerative effects of recurrent episodes.

Our Understanding of the Genetics of Bipolar Disorder

Studies of Inheritance Patterns. A family history of bipolar disorder conveys a higher risk for mood disorders in general and, specifically, a much greater risk for bipolar disorder. Unipolar depression is typically the most common form of mood disorder in families of bipolar probands. One large study found a threefold increase in the rate of bipolar disorder and a twofold increase in unipolar disorder in the biologic relatives of bipolar probands. First-degree relatives have an approximately 10-fold risk of developing the disorder. Twin studies suggest that the heritability is between 0.7 and 0.8.

Genetic Studies. Linkage studies have historically implicated several regions, most notably chromosomes 18q and 22q. Several linkage studies have found evidence for the involvement of specific genes in clinical subtypes. For example, the linkage evidence on 18q mainly derived from bipolar II–bipolar II sibling pairs and families that had panic symptoms.

GWAS studies report an ever-growing list of associated loci. Current evidence suggests a pattern of polygenic risk—many susceptible loci, each with a small effect that contributes to the disorder. These genes do not appear to be specific to bipolar disorder but rather overlap with other severe psychiatric disorders, particularly schizophrenia. Studies have tended to focus on genes that could give clues to etiology, such as genes involved in neurodevelopment, or genes encoding for voltage-dependent calcium channels.

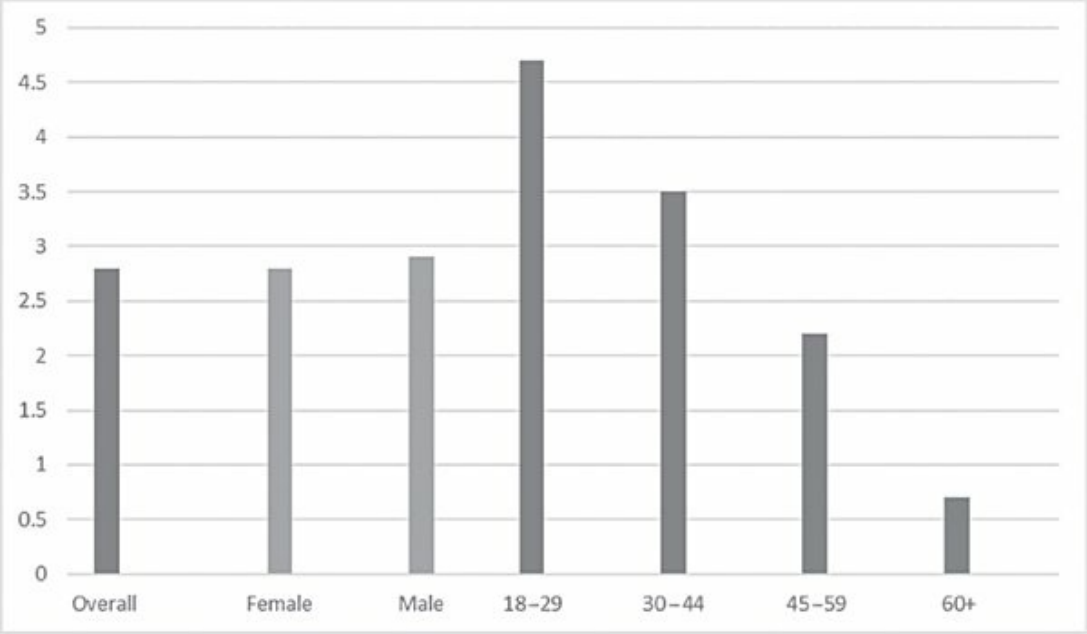


FIGURE 6-3
Past year prevalence of bipolar disorder in US Adults. (Data from the National Comorbidity Survey Replication [2001–2003].)

Perhaps the most important finding from genetic studies is that of the many genes found to be associated with bipolar disorder, most of them are also associated with schizophrenia.

Pathologic Imaging Findings

Structural Imaging. Individual studies have reported various areas of the brain with variations in gray matter volume, including the striatum, thalamus, amygdala, hippocampus, and pituitary—potentially meaningful findings as these are traditionally regions of interest in this disorder given their presumed functions. However, many of these are small studies with findings with conflicting results. At least one meta-analysis concluded that the evidence taken as a whole did not support morphologic differences in these areas.

There have been some consistent structural findings in other brain regions, including the inferior frontal gyrus, left insula, cerebellum, and left orbitofrontal gyrus, which are seen both in affected individuals and first-degree relatives. Although the significance of these findings is not well understood, many of these areas seem involved in emotional regulation, and the inferior frontal gyrus may have a particular role in executive function, particularly response inhibition. In several cases, the differences are of increased thickness in bipolar patients, suggesting that these brain changes may represent compensatory reactions to the disorder rather than etiologic risk factors.

Functional Imaging. There are fewer fMRI or PET studies than there are MRI studies; the majority examine the brain during the performance of various cognitive or emotional tasks and suggest either increased or decreased activations along various circuits. Consistent findings include abnormalities along a network that includes the superior and medial frontal cortex and insula, which showed increased activation compared with controls. As many of the areas are involved in executive functioning and working memory, this appears to be consistent with neuropsychological studies of the disorder. Other areas may show decreased activation, including the amygdala, basal ganglion, and limbic system. These findings may be evidence for downregulation of specific circuits, perhaps in response to the abnormal activity in the other circuits.

Inflammatory Markers

Some inflammatory markers, most notably interleukin-6, are elevated in bipolar patients. Markers of inflammation may be particularly evident during adolescence and become less evident in adulthood.

Other Neurochemical Findings. Some studies have found alterations in brain-derived neurotrophic factor, and in at least one study, this alteration may predict lithium responsiveness. Similarly, some studies of measures of oxidative stress in bipolar patients showed abnormalities that related to lithium response, however these findings are preliminary.

Assessment of the hypothalamic–pituitary–adrenal (HPA) axis suggests a dysregulated system, as evidenced by increased ACTH and cortisol levels in bipolar patients. However, these findings appear only to occur after individuals become symptomatic, suggesting that this is a result of the disorder.

THE PSYCHOLOGY OF THE DISORDER

Psychological studies have focused on neuropsychological findings, particularly of cognitive deficits associated with the disorder. Most commonly reported are deficits of executive functioning and verbal memory and learning. A variety of other related deficits, including deficits of cognitive flexibility, psychomotor speed, and attention, are also reported, although not consistently. Response inhibition is the executive function focused on the ability to prevent one's impulsive response to a situation when such a response would be inappropriate in the environmental context. This inhibition is often deficient in bipolar patients, as well as in first-degree relatives without bipolar disorder. It is particularly deficient in individuals who have bipolar disorder with psychotic features. Other executive function deficits include general impulsivity and risk-taking behavior. We see some of these same deficits with schizophrenia. However, they are less severe in schizophrenia and less likely to be recognized before the disorder appears. It may be that the cognitive deficits are indicative of a broader disorder that crosses traditional diagnostic lines and represents a spectrum of disorders defined by psychotic symptoms.

ETIOLOGY

Biologic Theories

As noted, several pathologic findings point at specific disrupted systems in the brain, particularly areas involved in the regulation of emotions and

executive functioning. What remains unclear are which findings are the results of the disorder and which are causative. For example, the significant disruptions seen in the HPA axis in patients with bipolar disorder seem to represent a neurobiologic scar that is seen only after symptoms occur.

As discussed above, converging evidence suggests that calcium signaling may be, at least in part, implicated in the disorder. The genetic abnormalities found in genes associated with voltage-gated calcium channels noted above, as well as the clinical observation that certain drugs have mechanisms of action involving calcium channels (e.g., antiepileptic drugs) lend support for such a theory. Furthermore, preclinical cellular studies show some evidence for increased intracellular calcium signaling in the neurons of bipolar patients. The evidence for calcium channel abnormalities in high risk but asymptomatic individuals, as well as first-degree relatives of patients with the disorder, suggests that this may be involved in the etiology of the disorder.

Neurodevelopment likely plays a role as well, like schizophrenia, and the fact that immunologic markers are more prevalent in bipolar individuals during adolescence than adulthood appears to support a developmental etiology for the disorder.

Psychosocial Theories

Most psychodynamic theories of mania view manic episodes as a defense against underlying depression.

Ms. G, a 42-year-old housewife and mother of a 4-year-old boy, developed symptoms of hypomania and later of frank mania without psychosis, when her only son was diagnosed with acute lymphocytic leukemia. A profoundly religious woman who had experienced 10 years of difficulty with conception, Ms. G was a devoted mother. She reported that she was usually rather down. Before her son's illness, she used to joke that she had become pregnant with him by divine intervention. When her son was diagnosed and subsequently hospitalized, he required painful medical tests and emergency chemotherapy, which made him very ill. The doctors regularly barraged Ms. G with bad news about his prognosis during the first few weeks of his illness.

Ms. G was ever present with her son at the hospital, never sleeping,

always caring for him, yet the pediatricians noted that as the child became more debilitated and the prognosis more grim, she seemed to bubble over with renewed cheerfulness, good humor, and high spirits. She could not seem to stop herself from cracking jokes to the hospital staff during her son's painful procedures, and as the jokes became louder and more inappropriate, the staff grew more concerned. During her subsequent psychiatric consultation (requested by the pediatric staff), Ms. G reported that her current "happiness and optimism" were justified by her sense of "oneness" with Mary, the mother of God. "We are together now, she and I, and she has become a part of me. We have a special relationship," she winked. Despite these statements, Ms. G was not psychotic and said that she was "speaking metaphorically, of course, only as a good Catholic would." Her mania resolved when her son achieved remission and was discharged from the hospital. (Courtesy of JC Markowitz, M.D. and BL Milrod, M.D.)

However, such approaches to understanding bipolar disorder have become less accepted as the biologic evidence has mounted. The role of the environment, particularly psychosocial stress, is likely crucial in the etiology. Karl Abraham, for example, believed that the manic episodes might reflect an inability to tolerate a developmental tragedy, such as the loss of a parent. The manic state may also result from a tyrannical superego, which produces excessive self-criticism that is then replaced by euphoric self-satisfaction. Bertram Lewin regarded the manic patient's ego as overwhelmed by pleasurable impulses, such as sex, or by feared impulses, such as aggression. Melanie Klein also viewed mania as a defensive reaction to depression, using manic defenses such as omnipotence, in which the person develops delusions of grandeur.

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7 Depressive Disorders

THE CLINICAL PRESENTATION

A depressed mood and a loss of interest or pleasure are the key symptoms of depression. Patients may say that they feel blue, hopeless, in the dumps, or worthless. For a patient, the depressed mood often has a distinct quality that differentiates it from the normal emotion of sadness or grief. Patients often describe the symptom of depression as one of agonizing emotional pain. Alternately they perceive it as a physical illness in which they feel exhausted and unmotivated. Others report feeling little, being unable to cry, and finding it difficult to experience any pleasure.

The classic presentation of a depressed patient is a person with a stooped posture, decreased movement, and a downward averted gaze. In practice, there is a considerable range of behaviors ranging from persons with no observable symptoms to the catatonically depressed patient. Among observable signs of depression, generalized psychomotor retardation is the most often described, in which patients show little spontaneous movement. At times it may be so severe as to be challenging to differentiate from catatonia.

Ms. A, a 34-year-old literature professor, presented to a mood clinic with the following complaint: “I am in a daze, confused, disoriented, staring. My thoughts do not flow, my mind is arrested.… I seem to lack any sense of direction, purpose.… I have such an inertia, I cannot assert myself. I cannot fight; I have no will.”

Psychomotor agitation may occur, including such behaviors as hand wringing and hair pulling. Many depressed patients have a decreased rate and volume of speech; they respond to questions with single words and exhibit delayed responses to questions.

When present, observable symptoms are helpful in the diagnosis; however, the lack of such symptoms does not imply that a patient has no disorder. It is not abnormal, for example, to encounter patients who can maintain a measure of social appropriateness, even smiling and laughing when with others, despite feeling internally miserable.

The most typical somatic symptoms of depression are called the *neurovegetative symptoms of depression* and typically include a variety of physical symptoms. Table 7-1 lists typical neurovegetative symptoms.

Almost all depressed patients (97 percent) complain about reduced energy; they have difficulty finishing tasks, are impaired at school and work, and have less motivation to undertake new projects. About 80 percent of patients complain of trouble sleeping, especially early morning awakening (i.e., terminal insomnia) and multiple awakenings at night, during which they ruminate about their problems. Many patients have decreased appetite and weight loss, but others experience increased appetite and weight gain and sleep longer than usual. These are sometimes called *reversed neurovegetative symptoms* or *atypical features*.

Depression is, by definition, a mood disorder, and disturbances of mood are at the core. Patients feel “bad” and may use words such as “sad,” “depressed,” “blue,” “down,” or similar words to describe this feeling. Here we use the term *dysphoria* to encompass these various depressive feelings as a way to avoid the confusion inherent in using the word “depressed” to mean both the diagnosis and the core symptom. Also, many patients are reluctant to use the word “depressed” as they may have difficulty coming to terms with their diagnosis. In such cases, other synonyms, including those above, may seem less threatening to the patient.

Some patients deny dysphoria altogether, but instead, describe feeling unable to enjoy things that are usually enjoyable to them. We call this *anhedonia* or a lack of pleasure. In addition to dysphoria and anhedonia, many depressed patients also report feeling anxious.

The critical point is that although sadness and even depression may be, in themselves, normal reactions, when encountered in the context of a

depressive disorder, patients do not experience them as normal. Patients will often feel genuinely ill and be able to distinguish their emotional state from "normal" feelings of sadness. Many will first interpret this as indicative of medical illness and present to their primary care doctor complaining of feeling "sick" rather than depressed.

Depressed patients customarily have negative views of the world and themselves. Their thought content often includes nondelusional ruminations about loss, guilt, suicide, and death. About 10 percent of all depressed patients have marked symptoms of a thought disorder, usually thought blocking and profound poverty of content.

A 42-year-old civil servant said that she was so paralyzed by depression that she felt that she had no personal initiative and volition left; she believed that some malignant force had taken over her actions and that it was commenting on every action that she was undertaking. The patient recovered fully with thymoleptic medication. There is no reason to believe that, in this patient, the feelings of somatic passability and running commentary indicated a schizophrenic process.

Depressed patients may complain of either delusions or hallucinations associated with their depressive episode: such patients may have a major depressive episode with psychotic features, often called psychotic depression. Delusions and hallucinations that are consistent with a depressed mood are said to be mood-congruent. Mood-congruent delusions in a depressed person include those of guilt, sinfulness, worthlessness, poverty, failure, persecution, and terminal somatic illnesses (such as cancer and a "rotting" brain). The content of mood-incongruent delusions or hallucinations is not consistent with a depressed mood. For example, a mood-incongruent delusion in a depressed person might involve grandiose themes of exaggerated power, knowledge, and worth.



Table 7-1
Neurovegetative Symptoms of Depression

Common:

Fatigue, low energy
Inattention
Insomnia, early morning awakening
Poor appetite, associated weight loss

Sometimes included:
Decreased libido and sexual performance
Menstrual irregularities
Worse depression in the AM

About two-thirds of all depressed patients contemplate suicide, and 10 to 15 percent commit suicide. Those recently hospitalized with a suicide attempt or suicidal ideation have a higher lifetime risk of successful suicide than those never hospitalized for suicidal ideation.

Depressed patients with psychotic features occasionally consider killing a person as a result of their delusional systems, but the most severely depressed patients often lack the motivation or the energy to act impulsively or violently. Patients with depressive disorders are at increased risk of suicide as they begin to improve and regain the energy needed to plan and carry out suicide (paradoxical suicide).

About 50 to 75 percent of all depressed patients have some measure of cognitive impairment. Cognitive symptoms include subjective reports of an inability to concentrate (84 percent of patients in one study) and impairments in thinking (67 percent of patients in another study).

Most depressed patients are oriented, although some may not have sufficient energy or interest to answer questions about these subjects during an interview. Memory can be challenging to disentangle from concentration difficulties. However, some patients appear to have genuine memory difficulties in addition to other cognitive deficits.

Some depressed patients sometimes seem unaware of their depression and do not complain of a mood disturbance even though they exhibit withdrawal from family, friends, and activities that previously interested them. Other depressed patients, particularly those with disordered thoughts, may be overly negative, hyperbolic in their symptomatic description, and hopeless about their future. It may be difficult to convince such patients that improvement is

possible.

We can assess judgment by reviewing patients' actions in the recent past and their behavior during the interview. Patients with particularly negative outlooks should be cautioned not to make important life decisions (e.g., around a relationship or job) until they are "thinking normally" again.

In interviews and conversations, depressed patients may emphasize the bad and minimize the good. They may be pessimistic about past treatment trials. A common clinical mistake is to unquestioningly believe a depressed patient who states that a previous trial of antidepressant medications did not work. Such statements may be false, and they require a confirmation from another source. Psychiatrists should not view patients' misinformation as an intentional fabrication; the admission of any hopeful information may be impossible for a person in a depressed state of mind.

Other patients, particularly those with poor insight, may have difficulty reporting symptoms or past episodes. Asking about changes in functioning rather than emotional states may be helpful ("have there been times that you felt unable to work, or care for your children?").

Presentation in Special Populations

Depression in Children and Adolescents. School phobia and excessive clinging to parents may be symptoms of depression in children. Poor academic performance, substance abuse, antisocial behavior, sexual promiscuity, truancy, and running away may be symptoms of depression in adolescents.

Depression in Older People. Depression is more common in older persons than it is in the general population. Various studies have reported prevalence rates ranging from 25 to almost 50 percent, although the percentage of these cases that are caused by major depressive disorder is uncertain. Several studies indicate that depression in older persons correlates with low socioeconomic status, the loss of a spouse, a concurrent physical illness, and social isolation. Other studies have indicated that depression in older persons is underdiagnosed and undertreated, perhaps particularly by general practitioners. The underrecognition of depression in older persons may occur because the disorder appears more often with somatic complaints

in older than in younger, age groups. Further, ageism may influence and cause clinicians to accept depressive symptoms as usual in older patients.

DIAGNOSIS

Depressive disorders can take many forms, depending on their severity and chronicity. The disorder that we most associate with "classic" depression is major depressive disorder, and this is the disorder most often referred to when someone reports that they suffer from depression. However, it is essential to understand the different varieties of depressive disorders, including those not included in some formal classifications, as all are significant sources of morbidity.

Major Depressive Disorder

The primary feature of major depressive disorder is the occurrence of at least one episode of major depression, which is significant depressive symptoms that last for a significant time. Table 7-2 compares the different approaches to diagnosing major depressive disorder.

With Psychotic Features. The presence of psychotic features in major depressive disorder reflects severe disease and is a poor prognostic indicator. We often categorize psychotic symptoms as either mood-congruent, that is, in harmony with the mood disorder ("I deserve punishment because I am so bad"), or mood-incongruent, not in harmony with the mood disorder. Patients with mood-incongruent psychotic symptoms may be more likely to have a comorbid primary psychotic disorder, such as schizoaffective disorder or schizophrenia.

With Melancholic Features. Melancholia is one of the oldest terms used in psychiatry, dating back to Hippocrates in the 4th century to describe the dark mood of depression. We still use it to refer to a depression characterized by severe anhedonia, early morning awakening, weight loss, and profound feelings of guilt (often over trivial events). It is not uncommon for patients who are melancholic to have suicidal ideation. Melancholia is associated with changes in the autonomic nervous system and endocrine functions. For that reason, we sometimes refer to melancholia as "endogenous depression" or

depression that arises in the absence of external life stressors or precipitants.

Table 7-2
Major Depressive Disorder

	DSM-5	ICD-10
Diagnostic name	Major Depressive Disorder	Major Depressive Episode
Duration	2 wk	
Symptoms	• Dysphoria or feeling depressed • Anhedonia • ↑ or ↓ weight or appetite • ↑ or ↓ sleep • ↑ or ↓ activity • ↓ energy • Depressing thoughts: worthlessness, guilt • ↓ concentration • Suicidal ideation/plan	• ↓ mood • ↓ energy • ↓ activity • ↓ capacity for enjoyment • ↓ interest • ↓ concentration • Fatigue after even minimal effort • Disturbed sleep/early morning awakening • Disturbed appetite/ ↓ weight • ↓ self esteem • ↓ self-confidence • Guilt or worthlessness • Mood unreactive to circumstances • Anhedonia • Worse symptoms in the AM • Psychomotor disturbance: agitation or retardation

		• ↓ libido
Required number of symptoms	5 (1 has to be one of the first two listed)	
Psychosocial consequences of symptoms	Distress or impaired functioning (social, occupational, or other significant areas)	Depends on severity
Exclusions (Not better explained by):	Medical illness Substance Other psychiatric disorder History of mania or hypomania	Adjustment disorder Conduct disorder Recurrent depressive disorder (which is considered a separate diagnosis)
Symptom specifiers	**With anxious distress** • 2+ symptoms of anxiety **With mixed features** • 3+ manic/hypomanic symptoms *during* the depressive episode (if occur independently, diagnose bipolar disorder) **With melancholic features** • Loss of pleasure or reactivity to pleasure • 3+ of the following • Severe depression/despair • Mood worse in AM	Depressive reaction Psychogenic depression Reactive depression

	• Early morning awakening • Psychomotor disturbance • Anorexia/weight loss • Guilt **With atypical features** • Mood reactivity • 2+ of following • Increased appetite/weight • Hyposomnia • Leaden paralysis • Rejection sensitivity **With mood-congruent psychotic features** **With mood-incongruent psychotic features** **With catatonia** • Must be present during most of depressive episode **With peripartum onset** **With seasonal pattern** • Usually occurs during a specific season	
Course specifiers	**With psychotic features** • Psychotic symptoms occurring only during the depressive	**Recurrent Depressive Disorder** (Coded as separate disorder): Repeated episodes of the above symptoms.

	episode • Mood-congruent • Mood-incongruent **In partial remission** • Full criteria no longer met **In full remission** • 0 symptoms for 2 mo	No mania.
Severity specifiers	**Mild:** minimal symptoms **Moderate:** between mild and severe **Severe:** # of symptoms and severity/dysfunction well beyond that required for diagnosis	**Mild** • 2–3 symptoms • Functions normal despite distress **Moderate** • 4+ symptoms • Difficulty with functioning **Severe** • Several symptoms marked and distressing • Loss of self-esteem/feels worthless and guilty • Suicidal ideation/acts • Somatic symptoms of depression **Severe with psychotic symptoms** • As above but with psychosis **Other** • Atypical depression

- Single episodes of "masked depression"

Unspecified

With Atypical Features. Patients with major depressive disorder with atypical features have specific, predictable characteristics. As described above, these tend to be the reverse of the neurovegetative symptoms. They may, for example, overeat or oversleep. Patients with atypical features have a younger age of onset and more severe psychomotor slowing. They also are more likely to have comorbid disorders, including anxiety disorders, substance use disorder, or somatic symptom disorder. They are easy to misdiagnose as having an anxiety disorder rather than a mood disorder. Patients with atypical features may also have a long-term course, a diagnosis of bipolar I disorder, or a seasonal pattern to their disorder.

Kevin, a 15-year-old adolescent, was referred to a sleep center to rule out narcolepsy. His main complaints were fatigue, boredom, and a need to sleep all the time. Although he had always started the day somewhat slowly, he now could not get out of bed to go to school. That alarmed his mother, prompting sleep consultation. Formerly a B student, he had been failing most of his courses in the 6 months before referral. Psychological counseling, predicated on the premise that his family's recent move from another city had led to Kevin's isolation, had not been beneficial. Extensive neurologic and general medical workup findings had also proven negative. He slept 12 to 15 hours per day but denied cataplexy, sleep paralysis, and hypnagogic hallucinations. During psychiatric interview, he denied being depressed but admitted that he had lost interest in everything except his dog. He had no drive, participated in no activities, and had gained 30 lb in 6 months. He believed that he was "brain damaged" and wondered whether it was worth living like that. The question of suicide disturbed him because it was contrary to his religious beliefs. These findings led to the prescription of an antidepressant. Not only did the antidepressant reverse the presenting complaints, but it also pushed him to the brink of a manic episode. (Courtesy of HS Akiskal, M.D.)

With Catatonic Features. As a symptom, catatonia can be present in several mental disorders, most commonly, schizophrenia and mood disorders. The hallmark symptoms of catatonia are stupor, blunted affect, extreme withdrawal, negativism, and marked psychomotor retardation. The presence of catatonic features in patients with mood disorders may have prognostic and treatment significance.

Postpartum Onset. We diagnose the postpartum subtype if the onset of symptoms is within 4 weeks postpartum. Postpartum mental disorders commonly include psychotic symptoms.

Seasonal Pattern. Patients with a seasonal pattern to their mood disorders tend to experience depressive episodes during a particular season, most commonly winter. The pattern has become known as seasonal affective disorder (SAD), although DSM-5 does not use this term. There is some controversy regarding whether this represents a subtype of major depressive disorder or a distinct entity. Either way, the presence of the disorder has implications for treatment, as patients with a seasonal pattern to their depression may preferentially respond to light therapy.

Dysthymic Disorder

Dysthymic disorder (also called dysthymia) is the presence of depressive symptoms that are less severe than those of major depressive disorder. Although less severe than a major depressive disorder, it is often more chronic. Table 7-3 compares the different approaches to diagnosing.

The most typical feature of dysthymia, also known as persistent depressive disorder, is the presence of a depressed mood that lasts most of the day and is present almost continuously. There are associated feelings of inadequacy, guilt, irritability, and anger; withdrawal from society; loss of interest; and inactivity and lack of productivity. The term dysthymia, which means “ill-humored,” was introduced in 1980. Before that time, we tended to classify patients having dysthymia as having “neurotic depression.”

Dysthymia is distinguished from major depressive disorder by the fact that patients complain that they have always been depressed. Thus, most cases are

of early-onset, beginning in childhood or adolescence, and, almost always, by a patient's 20s.

Table 7-3
Dysthymic Disorder

	DSM-5	ICD-10
Diagnostic name	Persistent Depressive Disorder	Dysthymia [F43.1]
Duration	2+ yr (1+ yr for children) ≤2-mo symptom free during illness	
Symptoms	Depressed mood most of the time ↓ appetite ↓ or ↑ sleep ↓ energy ↓ self-esteem ↓ concentration/decision-making ability Hopelessness	Chronically depressed mood that does not meet criteria for a depressive episode, though criteria may have been met in the past
Required number of symptoms	First symptom and 2+ of rest	
Psychosocial consequences of symptoms	Distress and functional impairment	
Exclusions	History of bipolar disorder Another mental illness Substance Another medical illness	Anxiety Depression Bereavement Schizophrenia
Symptom	**With pure dysthymic**	

specifiers	**syndrome:** criteria for depressive episode not met in the last 2 yr **With persistent major depressive episode:** has met diagnostic criteria for a depressive episode for entire 2-yr period **With intermittent major depressive episode, with current episode:** in current episode, 8 wk+ in last 2 yr with subdiagnostic symptoms episode **With intermittent major depressive episode, without current episode:** Not currently in episode, 1+ depressive episode in past 2 yr **With anxious distress** *≥2 of following:* • Feeling tense • Restlessness • ↓ concentration due to worrying • ↑ fear without apparent cause • Fear of loss of control **With mixed features**—depressive or manic/hypomanic

	episode, + with additional symptoms of other episode but subdiagnostic **With melancholic features:** see Table 7-2 **With atypical features:** see Table 7-2 **With psychotic features** • Mood-congruent • Mood-incongruent
Course specifiers	**With peripartum onset—** episode is during pregnancy or ≤4 wk after delivery
Severity specifiers	**Mild:** minimal symptoms **Moderate:** between mild and severe **Severe:** # of symptoms and severity/dysfunction well beyond that required for diagnosis

A 27-year-old male grade-school teacher presented with the chief complaint that life was a painful duty that had always lacked luster for him. He said that he felt “enveloped by a sense of gloom” that was nearly always with him. Although he was respected by his peers, he felt “like a grotesque failure, a self-concept I have had since childhood.” He stated that he merely performed his responsibilities as a teacher and that he had never derived any pleasure from anything he had done in life. He said that he had never had any romantic feelings; sexual activity, in which he

had engaged with two different women, had involved pleasureless orgasm. He said that he felt empty, going through life without any sense of direction, ambition, or passion, a realization that itself was tormenting. He had bought a pistol to put an end to what he called his "useless existence" but did not carry out suicide, believing that it would hurt his students and the small community in which he lived. (Courtesy of HS Akiskal, M.D.)

A late-onset subtype, much less prevalent and not well characterized clinically, has been identified among middle-aged and geriatric populations, mainly through epidemiologic studies in the community.

The family history of patients with dysthymia is typically replete with both depressive and bipolar disorders, which is one of the more robust findings supporting its link to primary mood disorder.

Other Diagnoses

Minor Depressive Disorder. Minor depressive disorder consists of episodes of depressive symptoms that are less severe than those seen in major depressive disorder. The difference between dysthymia and minor depressive disorder is primarily the episodic nature of the symptoms in the latter. Between episodes, patients with minor depressive disorder have a euthymic mood, but patients with dysthymia have virtually no euthymic periods. DSM-5 does not include this diagnosis. However, ICD-10 includes it as a mild type of depressive episode. DSM-5 would diagnose it as an "Other Specified Depressive Disorder."

Recurrent Brief Depressive Disorder. Recurrent brief depressive disorder consists of brief periods (less than 2 weeks) during which depressive episodes are present. Patients with the disorder would meet the diagnostic criteria for a major depressive disorder if their episodes lasted longer. Patients with recurrent brief depressive disorder differ from patients with dysthymia on two counts: they have an episodic disorder, and their symptoms are more severe. ICD-10 lists this as an "Other Recurrent Mood Disorder," whereas DSM-5 would diagnose it as an "Other Specified Depressive Disorder."

Double Depression. An estimated 40 percent of patients with major depressive disorder also meet the criteria for dysthymia, a combination often referred to as a double depression. Available data support the conclusion that patients with double depression have poorer prognoses than patients with major depressive disorder alone. The treatment of patients with double depression should be directed toward both disorders because the resolution of the symptoms of major depressive episode still leaves these patients with significant psychiatric impairment.

Objective Rating Scales for Depression

There is a growing literature suggesting that the use of objective scales can significantly improve the reliability of depressive diagnoses. They can also be useful for tracking the course of an episode. Several validated scales exist.

Clinician Administered Scales. The Hamilton Rating Scale for Depression (HAM-D) is a widely used depression scale. It contains up to 24 items, each rated on a 0 to 4 or 0 to 2 scale. The clinician evaluates the patient's answers to questions about feelings of guilt, thoughts of suicide, sleep habits, and other symptoms of depression, and derives the ratings from that interview. The usual scoring cutoffs are 10 to 13 for mild depression, 14 to 17 for mild-to-moderate depression, and >17 for moderate-to-severe depression.

Self-Administered Scales. The Zung Self-Rating Depression Scale is a 20-item report scale. A normal range is 34 or less; a depressed score is 50 or more. The scale provides a global index of the intensity of a patient's depressive symptoms, including the affective expression of depression.

The Raskin Depression Scale is a clinician-rated scale that measures the severity of a patient's depression, as reported by the patient and as observed by the physician, on a 5-point scale of three dimensions: verbal report, displayed behavior, and secondary symptoms. The scale has a range of 3 to 13; a normal score is 3, and a depressed score is 7 or more.

DIFFERENTIAL DIAGNOSIS

The symptoms of depressive disorders commonly overlap with other

syndromes and disorders. As a result, the differential is wide. Often a careful history and examination will make the differential clear. In other cases, one may have to observe the disorder over time before the diagnosis becomes clear.

General Medical Disorders

It is essential to consider whether a patient's depression is due to a general medical condition. Failure to obtain a good clinical history or to consider the context of a patient's current life situation can lead to diagnostic errors. Clinicians should have depressed adolescents tested for mononucleosis, and we should test patients who are markedly overweight or underweight for adrenal and thyroid dysfunctions. We should test patients with appropriate risk factors for HIV, and older patients for viral pneumonia and other medical conditions.

Table 7-4 lists some pharmacologic and medical conditions that can cause depression.



Table 7-4
Pharmacologic Factors and Physical Diseases Associated with Onset of Depression

Pharmacologic
Steroidal contraceptives
Reserpine, methyldopa
Anticholinesterase insecticides
Amphetamine or cocaine withdrawal
Alcohol or sedative–hypnotic withdrawal
Cimetidine, indomethacin
Phenothiazine antipsychotic drugs
Thallium, mercury
Cycloserine
Vincristine, vinblastine
Interferon

Endocrine–metabolic[a]
- Hypothyroidism and hyperthyroidism
- Hyperparathyroidism
- Hypopituitarism
- Addison disease
- Cushing syndrome
- Diabetes mellitus

Infectious
- General paresis (tertiary syphilis)
- Toxoplasmosis
- Influenza, viral pneumonia
- Viral hepatitis
- Infectious mononucleosis
- Acquired immune deficiency syndrome

Collagen
- Rheumatoid arthritis
- Lupus erythematosus

Nutritional
- Pellagra
- Pernicious anemia

Neurologic
- Multiple sclerosis
- Parkinson disease
- Head trauma
- Complex partial seizures
- Sleep apnea
- Cerebral tumors
- Cerebrovascular infarction (and disease)

Neoplastic

Abdominal malignancies
Disseminated carcinomatosis

[a]Cholesterol is not mentioned because low levels as a factor in depression have been inconsistently reported.

Many neurologic and medical disorders and pharmacologic agents can produce symptoms of depression. Patients with depressive disorders often first visit their general practitioners with somatic complaints. We can detect most medical causes of depressive disorders with a comprehensive medical history, a complete physical and neurologic examination, and routine blood and urine tests. The workup should include tests for thyroid and adrenal functions because disorders of both of these endocrine systems can appear as depressive disorders. In substance-induced mood disorder, a reasonable rule of thumb is that any drug a depressed patient is taking should be considered a potential factor in the mood disorder. Cardiac drugs, antihypertensives, sedatives, hypnotics, antipsychotics, antiepileptics, antiparkinsonian drugs, analgesics, antibacterials, and antineoplastics are all commonly associated with depressive symptoms.

Neurologic Conditions

The most common neurologic problems that manifest depressive symptoms are Parkinson disease, dementing illnesses, epilepsy, cerebrovascular diseases, and tumors. About 50 to 75 percent of all patients with Parkinson disease have marked symptoms of depressive disorder that do not correlate with the patient's physical disability, age, or duration of illness but do correlate with the presence of abnormalities found on neuropsychological tests. The motor symptoms of Parkinson disease can mask a depressive disorder as the motor symptoms are similar. Depressive symptoms often respond to antidepressant drugs or ECT. The interictal changes associated with temporal lobe epilepsy can mimic a depressive disorder, especially if the epileptic focus is on the right side. Depression is a frequent complicating feature of cerebrovascular diseases, particularly in the 2 years after the episode. Depression is more common in anterior brain lesions than in posterior brain lesions and, in both cases, often responds to antidepressant medications. Tumors of the diencephalic and temporal regions are

particularly likely to be associated with depressive disorder symptoms.

Dementia. Major depressive disorder can have a profound effect on concentration and even memory, and can occasionally be confused with a neurodegenerative illness such as Alzheimer disorder. At times the term “pseudodementia” has been used to describe this. However, that may be ill-advised as it implies that the cognitive symptoms are not genuine, which is not the case. Clinicians can usually differentiate cognitive symptoms of a major depressive disorder from the dementia of a disease, such as dementia of the Alzheimer type, on clinical grounds. The cognitive symptoms in major depressive disorder have a sudden onset, and other symptoms of the disorder, such as self-reproach, are also present. A diurnal variation in the cognitive problems, not seen in primary dementias, may occur. Whereas depressed patients with cognitive difficulties often do not try to answer questions, patients with dementia may confabulate. During an interview, depressed patients can sometimes be coached and encouraged into remembering, an ability that demented patients lack.

Other Mental Disorders

Depression can be a feature of virtually any mental disorder.

Other Mood Disorders. Clinicians must consider a range of diagnostic categories before arriving at a final diagnosis. We should determine whether a patient has had episodes of mania-like symptoms, which might indicate any one of the bipolar disorders. The depressive episode of a bipolar disorder may be identical to that of a major depressive disorder. However, there may be certain features that are more predictive of a bipolar disorder, and we list these in Table 7-5.



Table 7-5
Features of a Depressive Episode that Are More Predictive of Bipolar Disorder

Early age at onset
Psychotic depression before 25 yr of age

Postpartum depression, especially one with psychotic features
Rapid onset and offset of depressive episodes of short duration (less than 3 mo)
Recurrent depression (more than five episodes)
Depression with marked psychomotor retardation
Atypical features (reverse vegetative signs)
Seasonality
Bipolar family history
High-density three-generation pedigrees
Trait mood lability (cyclothymia)
Hyperthymic temperament
Hypomania associated with antidepressants
Repeated (at least three times) loss of efficacy of antidepressants after initial response
Depressive mixed state (with psychomotor excitement, irritable hostility, racing thoughts, and sexual arousal *during* major depression)

From Akiskal. Mood disorders. In: Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 10th ed. Philadelphia, PA: Wolters Kluwer; 2017.

If a patient's symptoms are solely depressive, the patient most likely has a depressive disorder. Even then, we must differentiate between the various depressive disorders discussed in this chapter. To do this, we must understand the severity and the course of the disorder in detail.

Other Mental Disorders. Substance-related disorders, psychotic disorders (Table 7-6), eating disorders, adjustment disorders, and somatoform disorders are all commonly associated with depressive symptoms and should be considered in the differential diagnosis of a patient with depressive symptoms.

Perhaps the most challenging differential is that between anxiety disorders with depression and depressive disorders with marked anxiety. Table 7-7 compares the unique features of depression with those of anxiety.

Table 7-6
Common Causes of Misdiagnosis of Mood Disorder as Schizophrenia

Reliance on cross-sectional rather than longitudinal picture
Incomplete interepisodic recovery equated with schizophrenic defect
Equation of bizarreness with schizophrenic thought disorder
Ascribing irritable and cantankerous mood to paranoid delusions
Mistaking depressive anhedonia and depersonalization for schizophrenic emotional blunting
Flight of ideas perceived as loose associations
Lack of familiarity with the phenomenologic approach in assessing affective delusions and hallucinations
Heavy weight given to incidental Schneiderian symptoms

Adapted from Akiskal HS, Puzantian VR. Psychotic forms of depression and mania. *Psychiatr Clin North Am*. 1979;2:419.

Table 7-7
Unique Cross-Sectional Profiles of Clinical Anxiety and Depression

Anxiety	Depression
Hypervigilance	Psychomotor retardation
Severe tension and panic	Severe sadness
Perceived danger	Perceived loss
Phobic avoidance	Loss of interest—anhedonia

Doubt and uncertainty	Hopelessness—suicidal
Insecurity	Self-deprecation
Performance anxiety	Loss of libido
	Early-morning awakening
	Weight loss

Reprinted with permission from Akiskal HS. Toward a clinical understanding of the relationship of anxiety and depressive disorders. In: Maser JP, Cloninger CR, eds. *Comorbidity of Mood and Anxiety Disorders*. Washington, DC: American Psychiatric Press; 1990.

Uncomplicated Bereavement. Uncomplicated bereavement is not a mental disorder even though about one-third of all bereaved spouses for a time meet the diagnostic criteria for major depressive disorder. Some patients with uncomplicated bereavement do develop a major depressive disorder, but we do not make this diagnosis unless the grief does not resolve. The severity and course of symptoms are significant differences. Table 7-8 lists features that would indicate that normal bereavement has progressed to a depressive disorder.

A 75-year-old widow was brought to treatment by her daughter because of severe insomnia and total loss of interest in daily routines after her husband's death 1 year before. She had been agitated for the first 2 to 3 months and thereafter "sank into total inactivity—not wanting to get out of bed, not wanting to do anything, not wanting to go out." According to her daughter, she was married at 21 years of age, had four children, and had been a housewife until her husband's death from a heart attack. Her past psychiatric history was negative; premorbid adjustment had been characterized by compulsive traits. During the interview, she was dressed in black; appeared moderately slowed; and sobbed intermittently, saying "I search everywhere for him …. I don't find him." When asked about life, she said, "Everything I see is black." Although she expressed no interest in food, she did not seem to have lost an appreciable amount of weight. The patient declined psychiatric care, stating that she "preferred to join her husband rather than get well." She was too religious to commit

suicide, but by refusing treatment, she felt that she would "pine away … find relief in death and reunion." (Courtesy of HS Akiskal, M.D.)

In cases of severe bereavement, regardless of diagnosis, some suggest that it would be clinically unwise to withhold antidepressants from many persons experiencing such intense mourning.

COMORBIDITY

Individuals with major depressive disorders are at an increased risk of having one or more additional comorbid disorders. The most frequent disorders are alcohol abuse or dependence, panic disorder, obsessive-compulsive disorder (OCD), and social anxiety disorder. Conversely, individuals with substance use disorders and anxiety disorders also have an elevated risk of lifetime or current comorbid depressive disorders.



Table 7-8
Signs that a Bereaved Person Has Progressed to a Depressive Disorder

Beliefs. Grieving persons and their relatives perceive bereavement as a normal reaction, whereas those with depressive disorder often view themselves as sick and may actually believe they are losing their minds.

Emotional reactivity. Unlike the melancholic person, the grieving person reacts to the environment and tends to show a range of positive effects.

Psychomotor activity. Marked psychomotor retardation is not observed in normal grief.

Guilt. Although bereaved persons often feel guilty about not having done certain things that they believe might have saved the life of the deceased loved one (guilt of omission), they typically do not experience guilt of commission.

Delusions. Delusions of worthlessness or sin and psychotic

experiences in general point toward mood disorder.

Suicidal ideation. Active suicidal ideation is rare in grief but common in major depressive disorder.

Behaviors. "Mummification" (i.e., keeping the belongings of the deceased person exactly as they were before his or her death) indicates serious psychopathology.

Anniversaries. Severe anniversary reactions should alert the clinician to the possibility of psychopathology.

From Akiskal. Mood disorders. In: Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 10th ed. Philadelphia, PA: Wolters Kluwer; 2017.

Anxiety

In anxiety disorders, DSM-5 notes the existence of mixed anxiety–depressive disorder. Significant symptoms of anxiety can and often do coexist with significant symptoms of depression. Whether patients who exhibit significant symptoms of both anxiety and depression are affected by two distinct disease processes or by a single disease process that produces both sets of symptoms are not yet resolved. Patients of both types may constitute a group of patients with mixed anxiety–depressive disorder.

Substance Use Disorder

Alcohol dependence frequently coexists with mood disorders. Both patients with major depressive disorder and those with bipolar I disorder are likely to meet the diagnostic criteria for an alcohol use disorder. The available data indicate that alcohol dependence is more strongly associated with a coexisting diagnosis of depression in women than in men. In contrast, the genetic and family data about men who have both a mood disorder and alcohol dependence indicate that they are likely to have two genetically distinct disease processes.

Substance-related disorders other than alcohol dependence are also commonly associated with mood disorders. The abuse of substances may be involved in precipitating an episode of illness or, conversely, may represent patients' attempts to treat their illnesses. Although manic patients seldom use sedatives to dampen their euphoria, depressed patients often use stimulants,

such as cocaine and amphetamines, to relieve their depression.

Patients with dysthymia commonly meet the diagnostic criteria for a substance-related disorder. This comorbidity can be logical; patients with dysthymia tend to develop coping methods for their chronically depressed state that involve substance abuse. Therefore, they are likely to use alcohol, stimulants such as cocaine, or marijuana, the choice perhaps depending primarily on a patient's social context. The presence of a comorbid diagnosis of substance abuse presents a diagnostic dilemma for clinicians; the long-term use of many substances can result in a symptom picture indistinguishable from that of dysthymia.

Medical Conditions

Depression commonly coexists with medical conditions, especially in older persons. When depression and medical conditions coexist, clinicians must try to determine whether the underlying medical condition is pathophysiologically related to the depression or whether any drugs that the patient is taking for the medical condition are causing the depression. Many studies indicate that the treatment of a coexisting major depressive disorder can improve the course of the underlying medical disorder, including cancer.

COURSE

Studies of the course and prognosis of mood disorders have generally concluded that mood disorders tend to have long courses and that patients tend to have relapses.

Major Depressive Disorder

Onset. About 50 percent of patients having their first episode of major depressive disorder exhibited significant depressive symptoms before the first identified episode. Therefore, early identification and treatment of early symptoms may prevent the development of a full depressive episode. Although symptoms may have been present, patients with major depressive disorder usually have not had a premorbid personality disorder. The first depressive episode occurs before age 40 years in about 50 percent of patients. Later onset is associated with the absence of a family history of mood

disorders, antisocial personality disorder, and alcohol abuse.

Duration. An untreated depressive episode lasts 6 to 13 months; most treated episodes last about 3 months. The withdrawal of antidepressants before 3 months has elapsed almost always results in the return of the symptoms. As the course of the disorder progresses, patients tend to have more frequent episodes that last longer. Over 20 years, the mean number of episodes is five or six.

Other Depressive Disorders

About 50 percent of patients with dysthymia experience an insidious onset of symptoms before age 25 years. Despite the early onset, patients often suffer from the symptoms for a decade before seeking psychiatric help and may consider early-onset dysthymia part of life. Patients with an early onset of symptoms are at risk for either major depressive disorder or bipolar I disorder in the course of their disorder. Studies of patients with the diagnosis of dysthymia indicate that about 20 percent progressed to major depressive disorder, 15 percent to bipolar II disorder, and fewer than 5 percent to bipolar I disorder.

The prognosis for patients with dysthymia varies. Antidepressive agents and specific types of psychotherapies (e.g., cognitive and behavior therapies) have positive effects on the course and prognosis of dysthymia. The available data about previously available treatments indicate that only 10 to 15 percent of patients are in remission 1 year after the initial diagnosis. About 25 percent of all patients with dysthymia never attain complete recovery. Overall, however, the prognosis is good with treatment.

PROGNOSIS

Major depressive disorder is not a benign disorder. It tends to be chronic, and patients tend to relapse. Patients hospitalized for the first episode of major depressive disorder have about a 50 percent chance of recovering in the first year. The percentage of patients recovering after repeated hospitalization decreases with time. Many unrecovered patients remain affected by a dysthymic disorder. About 25 percent of patients have a recurrence in the first 6 months after leaving the hospital. Another 30 to 50 percent recur in the

following 2 years. About 50 to 75 percent of all patients will have a recurrence within 5 years. The incidence of relapse is lower than these figures in patients who continue prophylactic psychopharmacological treatment and in patients who have had only one or two depressive episodes. In general, as a patient experiences more and more depressive episodes, the time between the episodes decreases, and the severity of each episode increases.

Prognostic Indicators

Many studies have focused on identifying both good and bad prognostic indicators in the course of major depressive disorder. Table 7-9 lists some of the positive and negative predictors.

TREATMENT APPROACH

The treatment of mood disorders is rewarding for psychiatrists. Specific treatments are available for depressive episodes, and data indicate that prophylactic treatment is also useful. We should be optimistic, as the prognosis for each episode is excellent. This fact tends to be welcome news for patients and families. Mood disorders are chronic, however, and the psychiatrist must educate the patient and the family about future treatment strategies.

Historically, patients with dysthymia either received no treatment or received long-term, insight-oriented psychotherapy. Contemporary data suggest that the treatments for dysthymia are similar to those for major depressive disorder.

There are several treatment goals. First, the patient's safety must be guaranteed. Second, a complete diagnostic evaluation of the patient is necessary. Third, we should initiate a treatment plan that addresses not only the immediate symptoms but also the patient's prospective well-being. Although current treatment emphasizes pharmacotherapy and psychotherapy addressed to the individual patient, stressful life events increase the chance of relapse. Thus, treatment should address the number and severity of stressors in patients' lives.

Table 7-10 lists the phases of treatment and the typical goals and activities for each phase.

Table 7-9
Prognostic Indicators for Depression

Positive Indicators
Clinical
Mild severity
No psychotic symptoms
Short hospital stay
No comorbid disease (medical/psychiatric)
No more than 1 hospitalization
Advanced age of onset
Psychosocial
Solid friendships during adolescence
Stable family functioning
Good social/occupational functioning in previous 5 yr
Negative Indicators
Comorbid dysthymic disorder and major depressive disorder
Substance use disorder
Anxiety symptoms
>1 previous episodes
Male

Hospitalization

The first and most critical decision a physician must make is whether to hospitalize a patient or attempt outpatient treatment. Definite indications for hospitalization are the risk of suicide or homicide, a patient's grossly reduced ability to get food and shelter, and the need for diagnostic procedures. A history of rapidly progressing symptoms and the rupture of a patient's natural support systems are also indications for hospitalization.

A physician may safely treat dysthymia and other forms of milder depression in the office, seeing the patient frequently. Clinical signs of impaired judgment, weight loss, or insomnia should be minimal. The patient's support system should be reliable, neither overinvolved nor

withdrawing from the patient. Any adverse changes in the patient's symptoms or behavior or the attitude of the patient's support system may suffice to warrant hospitalization.

Patients with mood disorders are often unwilling to enter a hospital voluntarily, and we may have to commit them involuntarily. These patients often cannot make decisions because of their slowed thinking, negative *Weltanschauung* (world view), and hopelessness.

Choosing a Treatment

Combined Treatments May Offer the Best Option. We often combine medication and psychotherapy. If physicians view mood disorders as fundamentally evolving from psychodynamic issues, their ambivalence about the use of drugs may result in a poor response, noncompliance, and probably inadequate dosages for too short a treatment period. Alternatively, if physicians ignore the psychosocial needs of a patient, the outcome of pharmacotherapy may be compromised. Several trials of a combination of pharmacotherapy and psychotherapy for chronically depressed outpatients have shown a higher response and higher remission rates for the combination than for either treatment used alone.

Although combination remains the general—and our—recommendation—we should point out that there are other views. Some have argued that most data suggest that single treatment, either psychopharmacological or psychotherapeutic, alone is sufficient for most people, and that combining treatments exposes patients to unnecessary costs and adverse effects.

Somatic Treatments

Pharmacotherapy. The efficacy of pharmacotherapy for major depressive disorder has been well established in more than 500 randomized controlled studies. In general, all available antidepressants show an at least modest effect compared with placebo. Efficacy is broadly similar across agents, at least when compared against placebo. However, tolerability and acceptability vary.

After establishing a diagnosis, we can formulate a pharmacologic treatment strategy. Accurate diagnosis is crucial because unipolar and bipolar spectrum

disorders require different treatment regimens.

The objective of pharmacologic treatment is symptom remission, not just symptom reduction. Patients with residual symptoms, as opposed to full remission, are more likely to experience a relapse or recurrence of mood episodes and to experience ongoing impairment of daily functioning.

The use of specific pharmacotherapy approximately doubles the chances that a depressed patient will recover in 1 month. All currently available antidepressants may take up to 3 to 4 weeks to exert significant therapeutic effects, although they may begin to show their effects earlier. The choice of antidepressants partly depends on the side effect profile that is least objectionable to a given patient's physical status, temperament, and lifestyle. Chapter 21 discusses the many classes of antidepressants. Although most antidepressants seem to have similar mechanisms of action and hence similar efficacy, there are some variations across classes. This variation gives us some diversity of choice. Although the first antidepressant drugs, the monoamine oxidase inhibitors (MAOIs) and tricyclic antidepressants (TCAs), are still in use, newer compounds have made the treatment of depression more "clinician and patient-friendly."



Table 7-10
Phases of Treatment

Treatment Phase	Duration	Goals	Activities
Acute and continuation	8–12 wk	Achieve symptomatic remission Monitor side effects Restore function	Establish therapeutic alliance Provide psychoeducation Select optimal antidepressant treatment(s) Supportive and measurement-based care Monitor progress

Maintenance	6–24 mo or longer	Return to full function and quality of life Prevention of recurrence	Continue psychoeducation Rehabilitate Manage comorbidities Monitor for recurrence

Modified from Lam RW, McIntosh D, Wang J, Enns MW, Kolivakis T, Michalak EE, Sareen J, Song WY, Kennedy SH, MacQueen GM, Milev RV, Parikh SV, Ravindran AV; CANMAT Depression Work Group. Canadian Network for Mood and Anxiety Treatments (CANMAT) 2016 clinical guidelines for the management of adults with major depressive disorder: section 1. Disease burden and principles of care. *Can J Psychiatry*. 2016;61(9):510–523.



Table 7-11
SSRI and SNRI Antidepressant Medications and Their Side Effect Profiles

		Side Effect Frequency	
Drug Class	**Recommended Dose Range (mg)**	**10–30%**	**>30%**
SSRI			
Citalopram	20–40	Nausea, dry mouth, sweating	None
Escitalopram	10–20	Male sexual dysfunction and nausea	None
Fluoxetine	20–60	Nausea, dry mouth, somnolence, nervousness, anxiety, insomnia, tremor, anorexia	None
Fluvoxamine	100–300	Dry mouth,	Nausea

		headaches, somnolence, agitation, insomnia, sweating, tremor, anorexia, dizziness, constipation	
Paroxetine	20–60	Nausea, diarrhea, dry mouth, headaches, somnolence, insomnia, sweating, asthenia, male sexual dysfunction, dizziness	None
Sertraline	50–200	Nausea, diarrhea, dry mouth, headaches, somnolence, insomnia, fatigue, tremor, male sexual dysfunction, dizziness	None
SNRI			
Venlafaxine	75–375	Headaches, somnolence, dry mouth, dizziness, nervousness, insomnia,	Nausea

		sweating, male sexual dysfunction	
Desvenlafaxine	50–100	Dry mouth, dizziness, nausea, sweating	None
Duloxetine	30–120	Nausea, dry mouth, constipation, insomnia, male sexual dysfunction	None
Levomilnacipran	20–80	Nausea, dry mouth, headaches, male sexual dysfunction	None
Other Second-Generation and Novel Antidepressants			
Agomelatine[a]	25–50	None	None
Bupropion	150–450	Insomnia, dry mouth, nausea	Headaches
Mirtazapine	15–60	Dry mouth, constipation, increased appetite, weight gain	Somnolence
Moclobemide[a]	300–600	None	None
Vilazodone	10–40	Diarrhea, nausea, headaches	None
Vortioxetine	10–20	Nausea	None

[a]Not routinely available in the United States.

Data from Kennedy SH, Lam RW, McIntyre RS, Tourjman SV, Bhat V, Blier P, Hasnain M, Jollant F, Levitt AJ, MacQueen GM, McInerney SJ, McIntosh D, Milev RV, Müller DJ, Parikh SV, Pearson NL, Ravindran AV, Uher R; CANMAT Depression Work Group. Canadian Network for Mood and Anxiety Treatments (CANMAT) 2016 clinical guidelines for the management of adults with major depressive disorder: section 3. Pharmacological treatments. *Can J Psychiatry*. 2016;61(9):540–560; Kennedy SH, Rizvi SJ. Chapter 246: SSRIs and related compounds. In: Stolerman I, ed. *Encyclopedia of Pharmacology*. New York: Springer; 2010.

Table 7-11 lists selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) by class as well as their side-effect profiles.

GENERAL CLINICAL GUIDELINES. The most common clinical mistake leading to an unsuccessful trial of an antidepressant drug is the use of too low a dosage for too short a time. Unless adverse events prevent it, the dosage of an antidepressant should be raised to the maximum recommended level and maintained at that level for at least 4 or 5 weeks before a drug trial is considered unsuccessful. Alternatively, if a patient is improving clinically on a low dosage of the drug, this dosage should not be raised unless clinical improvement stops before obtaining a maximal benefit. When a patient does not begin to respond to appropriate dosages of a drug after 2 or 3 weeks, clinicians may decide to obtain a plasma concentration of the drug if such a test is available. The test may indicate either noncompliance or a particularly unusual pharmacokinetic disposition of the drug and may thereby suggest an alternative dosage.

INITIAL MEDICATION SELECTION. The available antidepressants do not differ in overall efficacy, speed of response, or long-term effectiveness. Antidepressants, however, do differ in their pharmacology, drug–drug interactions, short- and long-term side effects, the likelihood of discontinuation symptoms, and ease of dose adjustment. Most often, we start with second- and third-generation antidepressants. Among them, the selective serotonin reuptake inhibitors (SSRIs) remain the most commonly used medications for depression.

Selection of the initial treatment depends on the chronicity of the condition, course of illness (a recurrent or chronic course increases the likelihood of subsequent depressive symptoms without treatment), family

history of illness and treatment response, symptom severity, concurrent general medical or other psychiatric conditions, prior treatment responses to other acute-phase treatments, potential drug–drug interactions, and patient preference.

In general, approximately 45 to 60 percent of all outpatients with uncomplicated (i.e., minimal psychiatric and general medical comorbidity), nonchronic, nonpsychotic major depressive disorder who begin treatment with medication respond (i.e., achieve at least a 50 percent reduction in baseline symptoms); however, only 35 to 50 percent achieve remission (i.e., the virtual absence of depressive symptoms).

DURATION AND PROPHYLAXIS. We should maintain antidepressant treatment for at least 6 months or the length of a previous episode, whichever is greater. When discontinuing antidepressant treatment, the drug dose should be tapered gradually over 1 to 2 weeks, depending on the half-life of the particular compound.

Prophylactic treatment with antidepressants is effective in reducing the number and severity of recurrences. One study concluded that when episodes are less than $2^1/_2$ years apart, we should recommend prophylactic treatment. Another factor suggesting prophylactic treatment is the seriousness of previous depressive episodes. Episodes that have involved significant suicidal ideation or impairment of psychosocial functioning may indicate that the risk of stopping treatment is too considerable.

Prevention of new mood episodes (i.e., recurrences) is the aim of the maintenance phase of treatment. Only patients with recurrent or chronic depressions are candidates for maintenance treatment. Several studies indicate that maintenance antidepressant medication appears to be safe and effective for the treatment of chronic depression.

TREATMENT OF SPECIFIC DEPRESSIVE DISORDERS. Clinical types of major depressive episodes may have varying responses to particular antidepressants or drugs other than antidepressants. Antidepressants with dual action on both serotonergic and noradrenergic receptors may have greater efficacy in melancholic depressions. We can treat patients with seasonal winter depression with light therapy.

Treatment of major depressive episodes with psychotic features may

require a combination of an antidepressant and an atypical antipsychotic. Several studies have also shown that ECT is useful for this indication—perhaps more effective than pharmacotherapy.

For those with atypical symptom features, strong evidence exists for the effectiveness of MAOIs. SSRIs and bupropion are also of use in atypical depression.

COMORBID DISORDERS. The simultaneous presence of another disorder can affect initial treatment selection. For example, the successful treatment of an OCD associated with depressive symptoms usually results in remission of the depression. Similarly, when panic disorder occurs with major depression, medications with demonstrated efficacy in both conditions are preferred (e.g., tricyclics and SSRIs). In general, the nonmood disorder dictates the choice of treatment in comorbid states.

Concurrent substance abuse raises the possibility of a substance-induced mood disorder, which must be evaluated by history or by requiring abstinence for several weeks. Abstinence often results in remission of depressive symptoms in substance-induced mood disorders. For those with continuing significant depressive symptoms, even with abstinence, an independent mood disorder is diagnosed and treated.

General medical conditions are established risk factors in the development of depression. The presence of a major depressive episode is associated with increased morbidity or mortality of many general medical conditions (e.g., cardiovascular disease, diabetes, cerebrovascular disease, and cancer).

THERAPEUTIC USE OF SIDE EFFECTS. Choosing more sedating antidepressants (such as mirtazapine or paroxetine) for more anxious, depressed patients or more activating agents (bupropion) for more psychomotor-retarded patients is not as helpful as one might think. For example, any short-term benefits with paroxetine or mirtazapine on symptoms of anxiety or insomnia may become liabilities over time. These drugs often continue to be sedating in the longer run, which can lead to patients prematurely discontinuing medication and increase the risk of relapse or recurrence. Some practitioners use adjunctive medications, such as hypnotics or anxiolytics, combined with antidepressants to provide more immediate symptom relief or to cover those side effects to which most patients ultimately adapt.

Understanding a patient's prior treatment history is essential because an earlier response typically predicts future responses. A documented failure on a properly conducted trial of a particular antidepressant class is grounds to choose an agent from an alternative class. The history of a first-degree relative responding to a particular drug is associated with a positive response to the same class of agents in the patient.

ACUTE TREATMENT FAILURES. Patients may not respond to medication, because (1) they cannot tolerate the side effects, even in the face of an excellent clinical response; (2) an idiosyncratic adverse event may occur; (3) the clinical response is not adequate; or (4) the wrong diagnosis has been made. Acute phase medication trials should last 4 to 6 weeks to allow for adequate time for meaningful symptom reduction. Most (but not all) patients who ultimately respond fully show at least a partial response by the fourth week, assuming an adequate dose. A "partial response" is defined as at least a 20 to 25 percent reduction in pretreatment depressive symptom severity. Patients who have not even a partial response in that time likely need a change of treatment. More extended periods—8 to 12 weeks or longer—are needed to define the ultimate degree of symptom reduction achievable with a medication. Approximately half of patients require a second medication treatment trial because the initial treatment is poorly tolerated or ineffective.

Selecting Second Treatment Options. When the initial treatment is unsuccessful, switching to an alternative treatment or augmenting the current treatment is a standard option. The choice between switching from the single initial treatment to a new single treatment (as opposed to adding a second treatment to the first one) rests on the patient's prior treatment history, the degree of benefit achieved with the initial treatment, and patient preference. As a rule, switching rather than augmenting is preferred after an initial medication failure. On the other hand, augmentation strategies are helpful with patients who have gained some benefit from the initial treatment but who have not achieved remission. A review of the different strategies suggested there is some evidence suggesting that both strategies do confer some benefit. However, the evidence is inadequate to suggest one strategy over another.

When switching from one monotherapy to another, the usual suggestion is

to pick a medication in a different class. For example, we might switch from an SSRI to an SNRI. However, when putting these assumptions to the test, it is difficult to find any advantage for any particular strategy. For example, in the landmark STAR*D study, which remains one of the most extensive studies of treatment strategies following initial failure, although medication switches were modestly helpful, both switches within and outside a class were equally effective.

Among augmentation options, several approaches have reasonable evidence. Several antipsychotics, most notably quetiapine and aripiprazole, are effective for augmentation. Lithium augmentation is also effective for augmenting both SSRIs and TCAs. There are also positive studies of thyroid hormone. However, this strategy is rarely used in clinical practice owing to the need for ongoing monitoring and potential adverse effects. Several other agents have been studied, including bupropion, buspirone, lamotrigine, methylphenidate, and pindolol; however, these have limited placebo-controlled data. One meta-analysis of the available data on switching strategies concluded that quetiapine and aripiprazole have the best evidence as augmentation agents. However, we should use these cautiously given their side effect profile.

NOVEL PHARMACOLOGIC AGENTS

Ketamine. The anesthetic agent ketamine is effective in treatment-resistant depression. It has a mechanism of action that inhibits the postsynaptic glutamate-binding protein N-methyl-D-aspartate (NMDA) receptor. Because abnormalities in glutamatergic signaling seem to have a role in major depressive disorder, this may account for its efficacy. Until recently, ketamine was only available intravenously, limiting its use as we had to monitor patients while they received an infusion of the drug over 30 minutes in a clinical setting. The most common side effects are dizziness, headache, and poor coordination, which are transitory. Dissociative symptoms, including hallucinations, may also occur. A positive response is usually seen within 24 hours, making this a genuinely novel substance in that it appears to act much more rapidly than standard antidepressants. However, the effect seems to be short-lived, and wears off after between 2 and 7 days, again limiting its use. There are little data on longer-term treatment as well as concern over associated adverse effects, particularly the psychogenic effect.

Also, the agent has an abuse potential, making it less attractive as a long-term option.

More recently, esketamine in a nasal spray formulation was approved by the FDA for treatment-resistant depression. Given the abuse risk, it is only available through a restricted distribution system. Most of the supportive evidence for the agent is from short-term use. However, one longer-term study suggested that continuation treatment could be effective for some patients.

Brexanolone. Brexanolone is an intravenous formulation of allopregnanolone, which was approved by the FDA in 2019 for the treatment of postpartum depression. It is a neuroactive steroid that also has a neuroactive effect, functioning as an allosteric modulator of $GABA_A$, *which may be central to the mechanism of postpartum depression. It is also only available in the United States through a restricted program that requires administrating the drug in a clinical setting over 60 hours. The most common adverse effects are sleepiness, dry mouth, loss of consciousness, and flushing. As with ketamine, brexanolone differs from standard antidepressants in its apparent rapid effect, reducing depressive symptoms as early as 24 hours after administration. In trials, the effect continued for at least 30 days. As of this writing, phase 3 trials for an oral version of the medication have promising results.*

Other Somatic Treatments

NEUROSTIMULATION

Vagal Nerve Stimulation. Experimental stimulation of the vagus nerve in several studies designed for the treatment of epilepsy found that patients showed improved mood. This observation led to the use of left vagal nerve stimulation (VNS) using an electronic device implanted in the skin, similar to a cardiac pacemaker. Preliminary studies have shown that many patients with chronic, recurrent major depressive disorder went into remission when treated with VNS. The mechanism of action of VNS to account for improvement is unknown. The vagus nerve connects to the enteric nervous system and, when stimulated, may cause the release of peptides that act as neurotransmitters.

Transcranial Magnetic Stimulation. Transcranial magnetic stimulation (TMS) shows promise as a treatment for depression. It involves the use of very short pulses of magnetic energy to stimulate nerve cells in the brain. The FDA has indicated this treatment for depression in adult patients who have failed to achieve satisfactory improvement from one prior antidepressant medication at or above the minimal effective dose and duration in the current episode.

Repetitive transcranial magnetic stimulation (rTMS) produces focal secondary electrical stimulation of targeted cortical regions. It is nonconvulsive, requires no anesthesia, has a safe side effect profile, and is not associated with cognitive side effects.

The patients do not require anesthesia or sedation and remain awake and alert. It is a 40-minute outpatient procedure that is prescribed by a psychiatrist and performed in a psychiatrist's office. The treatment is typically administered daily for 4 to 6 weeks. The most common adverse event related to treatment was scalp pain or discomfort.

TMS therapy is contraindicated in patients with implanted metallic devices or nonremovable metallic objects in or around the head.

PHOTOTHERAPY. Phototherapy (light therapy) was introduced in 1984 as a treatment for SAD (mood disorder with seasonal pattern). In this disorder, patients typically experience depression as the photoperiod of the day decreases with advancing winter. Women represent at least 75 percent of all patients with seasonal depression, and the mean age of presentation is 40 years. Patients rarely present older than the age of 55 years with SAD.

Phototherapy typically involves exposing the affected patient to bright light in the range of 1,500 to 10,000 lux or more, typically with a lightbox that sits on a table or desk. Patients sit in front of the box for approximately 1 to 2 hours before dawn each day. Some patients may also benefit from exposure after dusk. Alternatively, some manufacturers have developed light visors, with a light source built into the brim of the hat. These light visors allow mobility, but recent controlled studies have questioned the use of this type of light exposure. Trials have typically lasted 1 week, but longer treatment durations may be associated with a more significant response.

Phototherapy is usually well tolerated. Newer light sources tend to use lower light intensities and come equipped with filters; patients are instructed

not to look directly at the light source. As with any effective antidepressant, phototherapy, on rare occasions, has been implicated in switching some depressed patients into mania or hypomania.

In addition to seasonal depression, the other significant indication for phototherapy may be in sleep disorders. Phototherapy can decrease the irritability and diminished functioning associated with shift work. Sleep disorders in geriatric patients have reportedly improved with exposure to bright light during the day. Likewise, some evidence suggests that jet lag might respond to light therapy. Preliminary data indicate that phototherapy may benefit some patients with OCD that has a seasonal variation.

SLEEP DEPRIVATION. Sleep disturbances are common in depression. Depression can be associated with either hypersomnia or insomnia. Sleep deprivation may temporarily relieve depression in those who have unipolar depression. Approximately 60 percent of patients with depressive disorders exhibit significant but transient benefits from total sleep deprivation. The positive results usually reverse by the next night of sleep. Several strategies have been used in an attempt to achieve a more sustained response to sleep deprivation. One method used serial total sleep deprivation with a day or two of normal sleep in between. This method does not achieve a sustained antidepressant response because the depression tends to return during the "normal sleep" days. Another approach used phase delay in the time patients go to sleep each night, or partial sleep deprivation. In this method, patients may stay awake from 2 AM to 10 PM daily. Up to 50 percent of patients get same-day antidepressant effects from partial sleep deprivation, but this benefit also tends to wear off in time. In some reports, however, serial partial sleep deprivation has been used successfully to treat insomnia associated with depression. The third, and probably most useful strategy combines sleep deprivation with pharmacologic treatment of depression. Several studies have suggested that total and partial sleep deprivation followed by immediate treatment with an antidepressant or lithium sustains the antidepressant effects of sleep deprivation. Likewise, several reports have suggested that sleep deprivation accelerates the response to antidepressants. Sleep deprivation may also improve premenstrual dysphoria.

Table 7-12
Evidence-Based Psychotherapies for Major Depressive Disorder

Treatment	Conceptualization of Disorder Etiology	Sample Interventions
Behavioral therapy	Deficit of reinforcers, including pleasant activities and positive interpersonal contacts	Increase activity level Structured goal setting Interpersonal skills training
Cognitive-behavioral therapy	Interaction of beliefs with matching stressor	Identify and challenge automatic thoughts Engage in activities that provide evidence disproving dysfunctional beliefs Modify core beliefs by reviewing evidence
Interpersonal psychotherapy	Interpersonal vulnerabilities arising from early attachment and learned relationship patterns	Develop awareness of patterns in primary relationships and the therapeutic relationship Interpersonal skills training Communication analysis
Behavioral marital	Marital distress increases stress	Assertive communication

therapy	while impairing support resources	training Active listening exercises Problem-solving skills Increasing reinforcing behaviors toward spouse

Psychosocial Therapy

Three types of short-term psychotherapies—cognitive therapy, interpersonal therapy, and behavior therapy—have extensive studies for the treatment of major depressive disorder. In addition to these individual therapies, there are positive studies for behavioral marital therapy as well. Table 7-12 lists the evidence-based psychotherapies for major depressive disorder.

There are no generally accepted guidelines for choosing one therapy over another. The National Institute of Mental Health (NIMH) Treatment of Depression Collaborative Research Program did find some possible predictors of good response to specific treatments. Table 7-13 summarizes those findings.

Cognitive Therapy. Cognitive therapy, originally developed by Aaron Beck, focuses on the cognitive distortions postulated to be present in major depressive disorder. Such distortions include selective attention to the negative aspects of circumstances and unrealistically morbid inferences about consequences. For example, apathy and low energy result from a patient's expectation of failure in all areas. The goal of cognitive therapy is to alleviate depressive episodes and prevent their recurrence by helping patients identify and test negative cognitions; develop alternative, flexible, and positive ways of thinking; and rehearse new cognitive and behavioral responses.

In this example, the therapist takes a cognitive approach when a patient shows up late to an appointment.

Therapist: You mentioned that you were upset because you were late. Can you tell me what you were thinking when you realized you were late?
Patient: I figured you'd be angry.
Therapist: Okay, and when you thought I'd be angry, how did you feel?
Patient: Pretty nervous.
Therapist: So that is what we will focus on in here, how your thinking in different situations influences how you feel. In this case, you were late, thought "he'll be angry," and felt nervous. Our goal is to test out the beliefs, like "he'll be angry," and change them when they aren't healthy or accurate. The great thing is that you sort of implicitly tested out the belief that I'll be angry by showing up. Was I angry?
Patient: You didn't seem angry.



Table 7-13
Predictors of Response to Several Therapies for Depression

Treatment	Predictors
Interpersonal psychotherapy	Low social dysfunction High depression severity
Cognitive-behavioral therapy	Low cognitive dysfunction
Pharmacotherapy	Low cognitive dysfunction High depression severity High work dysfunction

Data from Sotsky SM, Glass DR, Shea MT, Pilkonis PA, Collins JF, Elkin I, Watkins JT, Imber SD, Leber WR, Moyer J, Oliveri ME. Patient predictors of response to psychotherapy and pharmacotherapy: findings in the NIMH Treatment of Depression Collaborative Research Program. *Am J Psychiatry*. 1991;148:997–1108.

Therapist: How anxious did you feel once we started talking?
Patient: Well, I'm feeling more comfortable.
Therapist: What thoughts do you have now about me being angry?

Patient: I don’t think you’re angry.
Therapist: So, this is an example of what we’re going to be doing in therapy—I’ll be helping you to identify thoughts that make you feel bad. We’re then going to work together to come up with ways to check them out and change them if they’re not true or accurate. You checked out whether I was angry by observing me, and you changed your thought, and, now, I get the impression that you feel better.

Studies have shown that cognitive therapy is useful in the treatment of major depressive disorder. Most studies found that cognitive therapy is equal in efficacy to pharmacotherapy and is associated with fewer adverse effects and better follow-up than pharmacotherapy. Some of the best-controlled studies have indicated that the combination of cognitive therapy and pharmacotherapy is more efficacious than either therapy alone, although other studies have not found that additive effect.

Interpersonal Therapy. Interpersonal therapy, developed by Gerald Klerman, focuses on one or two of a patient’s current interpersonal problems. This therapy has two assumptions. First, current interpersonal problems are likely to have their roots in early dysfunctional relationships. Second, current interpersonal problems are likely to be involved in precipitating or perpetuating the current depressive symptoms. Controlled trials have indicated that interpersonal therapy is effective for treatment of major depressive disorder and, not surprisingly, may be specifically helpful in addressing interpersonal problems. Some studies indicate that interpersonal therapy may be the most effective method for severe major depressive episodes when the treatment choice is psychotherapy alone.

The interpersonal therapy program usually consists of 12 to 16 weekly sessions and is characterized by an active therapeutic approach. Intrapsychic phenomena, such as defense mechanisms and internal conflicts, are not addressed. Discrete behaviors—such as lack of assertiveness, impaired social skills, and distorted thinking—may be addressed but only in the context of their meaning in, or their effect on, interpersonal relationships.

Behavior Therapy. Behavior therapy rests on the hypothesis that

maladaptive behavioral patterns result in a person's receiving little positive feedback and perhaps outright rejection from society. By addressing maladaptive behaviors in therapy, patients learn to function in the world in such a way that they receive positive reinforcement. Behavior therapy for major depressive disorder has not yet been the subject of many controlled studies. The limited data indicate that it is an effective treatment for major depressive disorder.

Psychoanalytically Oriented Therapy. Although not as well researched as those three therapies, many clinicians use psychoanalytically oriented psychotherapy as their primary method. What differentiates the short-term psychotherapy methods from the psychoanalytically oriented approach are the active and directive roles of the therapist, the directly recognizable goals, and the endpoints for short-term therapy.

The psychoanalytic approach to mood disorders uses psychoanalytic theories about depression. The goal of psychoanalytic psychotherapy is to effect a change in a patient's personality structure or character, not merely to alleviate symptoms. Improvements in interpersonal trust, capacity for intimacy, coping mechanisms, the capacity to grieve, and the ability to experience a wide range of emotions are some of the aims of psychoanalytic therapy. Treatment often requires the patient to experience periods of heightened anxiety and distress during therapy, which may continue for several years.

Accumulating evidence is encouraging about the efficacy of dynamic therapy. In a randomized, controlled trial comparing psychodynamic therapy with cognitive-behavioral therapy, the outcome of the depressed patients was the same in the two treatments.

Family Therapy. Family therapy is usually the first-line treatment for major depressive disorder, but increasing evidence indicates that helping a patient with a mood disorder to reduce and cope with stress can lessen the chance of a relapse. We should consider family therapy if the disorder jeopardizes a patient's marriage or family functioning or if the mood disorder is promoted or maintained by the family situation. Family therapy examines the role of the mood-disordered member in the overall psychological well-being of the whole family; it also examines the role of the entire family in the

maintenance of the patient's symptoms. Patients with mood disorders have a high rate of divorce, and about 50 percent of all spouses report that they would not have married or had children if they had known that the patient was going to develop a mood disorder.

THE EPIDEMIOLOGY OF DEPRESSION

Research on the epidemiology of depression, both in the United States and worldwide, has considerably broadened our understanding of the disorder. We have discovered that depression is much more common than once thought. It is also one of the more debilitating disorders known to humanity, as it often affects individuals during what should be their most productive years. Hence the depressive disorders are one of the most costly disorders in society.

Measuring the Incidence and Prevalence of Major Depressive Disorder Is Challenging

Measuring the true incidence of major depressive disorder is complicated and highly dependent on the methods used to collect the sample, definitions of the disorder, and the particular instruments used to measure depression. For example, studies that use self-assessment tend to report higher incidences than studies using clinician-rated instruments. The timing of a study likely affects the results as well. As depressive illness become less stigmatized, persons may be more willing to endorse symptoms of the disorder to the well-meaning strangers who knock at their door, call their phone, or email them and identify themselves as researchers.

Incidence and Prevalence

In a reasonably recent meta-analysis, which included 90 studies from 30 countries done from 1994 to 2014, with combined data on more than 1 million participants, the point prevalence for depression was 12.9 percent, the 1-year prevalence was 7.2 percent, and the lifetime prevalence was 10.8 percent. There was significant heterogeneity among the different studies.

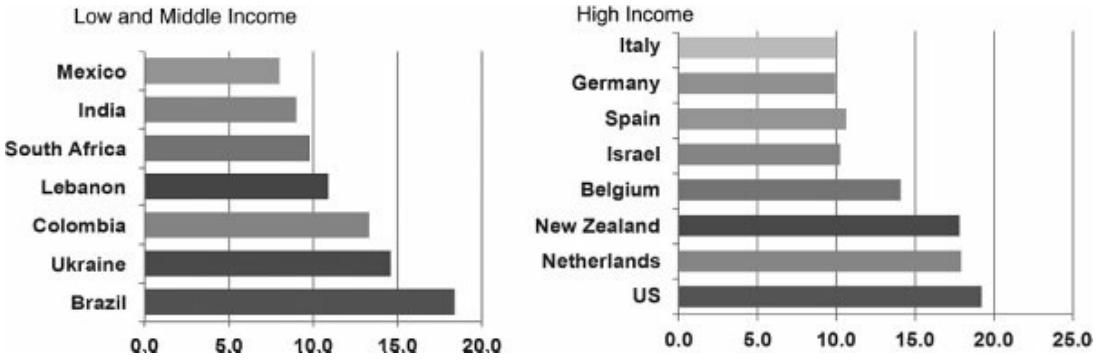


FIGURE 7-1
Lifetime prevalence of major depression across the world.

In the United States, the most recent epidemiologic data is from the National Survey on Drug Use and Health (NSDUH), sponsored by the Substance Abuse and Mental Health Services Administration (SAMHSA). Using a definition of a major depressive episode based on DSM-5, they found the overall 1-year prevalence of a major depressive episode to be 7.1 percent, hence very similar to the international data. Figure 7-1 summarizes some of the international prevalence data as it related to per capita income.

Sex

An almost universal observation, independent of country or culture, is that depression is more common in women than men. In the international meta-analysis, the aggregate prevalence for women was 14.4 percent versus 11.5 percent in men. In the United States, women were more likely to report a major depressive episode in the past year (8.7 percent) than men (5.3 percent).

There are various possible explanations, both biologic, psychological, and social, for this difference. A women's unequal status in many societies may make women more vulnerable to depression. Although this remains a valid hypothesis, it is notable that in societies where women have gained a better level of status, such as many Western societies, the ratio has not significantly changed. Although the explanation is undoubtedly complex and multifactorial, the weight of existing evidence suggests that biologic differences between men and women, such as variations in hormonal levels, do account for at least part of this difference.

Age

The mean age of onset for major depressive disorder is about 40 years, with 50 percent of all patients having an onset between the ages of 20 and 50 years. Major depressive disorder can also begin in childhood or old age. Some epidemiologic data suggest that the incidence of major depressive disorder may be increasing among younger persons. For example, in the SAMHSA study, major depressive disorder was almost twice as prevalent in adolescents as it was in adults, the most substantial difference being seen in adolescent females who reported a 1-year prevalence of major depression of 20 percent. The highest 1-year prevalence for major depressive disorder was in younger adults (adults ages 18 to 25).

Marital Status

Major depressive disorder occurs most often in persons without close interpersonal relationships and in those who are divorced or separated.

Socioeconomic and Cultural Factors

There is no proven correlation between socioeconomic status and major depressive disorder. Depression may be more common in rural areas than in urban areas. However, the most recent international meta-analysis described above did not find such a difference. It may be that technology and globalization have been able to narrow the differences between rural and urban settings.

Racial and ethnic differences may be particularly challenging to measure and may be highly dependent on method, approach, and locale. In the SAMSHA study, the highest prevalence was seen in White and Native American respondents.

THE NEUROBIOLOGY OF DEPRESSION

Since the time of Kraepelin, researchers have examined the brains, bodies, and behaviors of depressed patients in the search for clues of the underlying pathology of depression. This search is complicated by the fact that most experts on depression agree what we call major depressive disorder is not a single disease, but rather a collection of disorders with overlapping phenomenology but different etiologies and pathologies. Although this

assumption seems likely, we have yet to find definitive subtypes or distinct patterns that we can confidently group into separate disorders. This area of research is a rapidly changing field, and improvements in technology, including higher-resolution imaging, more efficient genotyping, and more powerful computing power are getting us closer to the search for valid "biotypes" of depression.

What follows here are examples of pathologic findings, most of which have been reasonably replicated by different investigators. Some are long known, others are recent. However, the growing collection of pathologic findings is helping us to understand what is happening when a patient says that they are "depressed."

Depression as a Disorder of Homeostasis

Depression and the Hypothalamic–Pituitary–Adrenal Axis. We have known for more than 50 years that, on average, depressed patients have an overactive hypothalamic–pituitary–adrenal (HPA) axis. Compared with controls, depressed patients have increased cortisol levels over 24 hours. We have documented elevated HPA activity in depression via excretion of urinary-free cortisol (UFC), 24-hour (or shorter time segments) intravenous (IV) collections of plasma cortisol levels, salivary cortisol levels, and tests of the integrity of feedback inhibition. Evidence of increased HPA activity is apparent in 20 to 40 percent of depressed outpatients and 40 to 60 percent of depressed inpatients.

This hypercortisolemia is due to increased corticotropin-releasing hormone (CRH) from the hypothalamus, along with decreased feedback inhibition. We can test for a disturbance of feedback inhibition using the dexamethasone test (DST). The DST was devised in the 1980s—patients were given dexamethasone—a synthetic steroid. This should create a negative feedback loop, and serum cortisol should decrease. However, depressed patients show an initial decrease but then "escape" from the suppression and return to abnormally high levels of cortisol. Although this is a replicable finding, it is not useful enough to be a diagnostic test, as it is neither sensitive (many patients diagnosed with depression have a normal DST) nor specific (many other conditions, such as Cushing syndrome, or simple environmental stress, can cause an abnormal DST).

Postmortem studies have shown increased neurons in the hypothalamus, which is likely what is driving the increased activity. This neuronal increase is likely in response to chronic stress.

Elevated HPA activity is a hallmark of the mammalian stress responses and one of the most unambiguous links between depression and the biology of chronic stress. Hypercortisolema in depression suggests one or more of the following central disturbances: decreased inhibitory serotonin tone, increased drive from norepinephrine, acetylcholine (ACh), or CRH; or decreased feedback inhibition from the hippocampus. Some studies in depressed humans indicate that a history of early trauma is associated with increased HPA activity accompanied by structural changes (i.e., atrophy or decreased volume) in the cerebral cortex.

DEPRESSION ALSO AFFECTS OTHER ENDOCRINE AND RELATED REGULATORY SYSTEMS

Thyroid Axis Activity. Approximately 5 to 10 percent of people evaluated for depression have previously undetected thyroid dysfunction, as reflected by an elevated basal thyroid-stimulating hormone (TSH) level or an increased TSH response to a 500-mg infusion of the hypothalamic neuropeptide thyroid-releasing hormone (TRH). Such abnormalities are often associated with elevated antithyroid antibody levels and, unless corrected with hormone replacement therapy, can compromise response to treatment. An even more significant subgroup of depressed patients (e.g., 20 to 30 percent) shows a blunted TSH response to TRH challenge. To date, the major therapeutic implication of a blunted TSH response is evidence of an increased risk of relapse despite preventive antidepressant therapy. Of note, unlike the dexamethasone-suppression test (DST), blunted TSH response to TRH does not usually normalize with effective treatment.

Growth Hormone. Growth hormone (GH) is secreted from the anterior pituitary after stimulation by NE and dopamine. Somatostatin, a hypothalamic neuropeptide, and CRH inhibit this secretion. Decreased CSF somatostatin levels occur in depression, as do increased levels in mania.

Prolactin. Prolactin is released from the pituitary by serotonin stimulation and inhibited by dopamine. Most studies have not found significant

abnormalities of basal or circadian prolactin secretion in depression, although some see a blunted prolactin response to various serotonin agonists. This response is uncommon among premenopausal women, suggesting that estrogen has a moderating effect.

Brain-Derived Neurotrophic Factor. Various growth factor proteins, of which brain-derived neurotrophic factor (or BDNF) is best understood, are responsible for the ongoing maintenance of neurons in the brain, and disruption of it causes a reduction in neuronal number and size. It is challenging to measure BDNF in the living subjects, but postmortem studies have suggested deficits in BDNF may correlate with psychopathology. For example, some studies have found that in postmortem examination of persons who died from suicide, the average BDNF in the prefrontal cortex (PFC) and hippocampus was lower than for control subjects. Indirect studies in live patients, looking at BDNF in plasma rather than the brain (the correlation between the two is not clear), found increases in serum BDNF in depressed patients who responded to antidepressants.

BDNF may provide clues to the relationship between chronic stress and depression as the activity of the gene coding for this growth factor decreases with chronic stress, as does neurogenesis.

Similarly, animal studies show that a variety of factors, including antidepressants, estrogen, lithium, and neurostimulation, all increase BDNF in the brain, suggesting a possible common link between known treatments for depression.

DEPRESSION ALSO DISRUPTS CIRCADIAN RHYTHM. It has also long been known that depression is associated with a premature loss of deep (slow-wave) sleep and an increase in nocturnal arousal. Four types of disturbance reflect the latter: (1) an increase in nocturnal awakenings, (2) a reduction in total sleep time, (3) increased phasic rapid eye movement (REM) sleep, and (4) increased core body temperature. The combination of increased REM drive and decreased slow-wave sleep results in a significant reduction in the first period of non-REM (NREM) sleep, a phenomenon referred to as *reduced REM latency.* Blunted secretion of GH may cause this reduction in latency, which usually increases with sleep onset. It is very persistent, often lasting beyond recovery from depression. Some patients manifesting a

characteristically abnormal sleep profile are known to be less responsive to psychotherapy and to have a higher risk of relapse or recurrence and may benefit preferentially from pharmacotherapy.

As with the HPA access, the association between sleep disturbances and depression is common, but it is neither sensitive nor specific enough for clinical use. About 40 percent of outpatients and 80 percent of inpatients show a typical pattern of reduced REM latency, increased REM density, and decreased sleep maintenance. False-negative findings are common in younger, hypersomnolent patients, who may experience an increase in slow-wave sleep during episodes of depression. Approximately 10 percent of otherwise healthy individuals have abnormal sleep profiles, as do many patients suffering from other medical or psychiatric disorders.

Neurotransmitter Deficits Are Involved. Much of the research into the neurobiology of depression has centered on the role played by neurotransmitters. This approach was understandable, given that most somatic treatments for depression appear to act by influencing neurotransmission. In the 1950s, researchers found that reserpine, an antihypertensive that acts by inhibiting the release of monoamine neurotransmitters, sometimes induced depression. During the same decade, others discovered the earliest antidepressants. Although the discoveries were serendipitous, it eventually became clear that they all interfered with the catabolism of monoamine neurotransmitters, and researchers presumed that this must be their mechanism of action. It was reasonable, then, to assume that depression was due to abnormally low levels of these monoamines. Early theories of depression explained the disorder as a deficit or imbalance in neurotransmitters: hence depressed patients were often told that they had a "chemical imbalance." Such a simplistic theory was, of course, overly optimistic, and subsequent neurochemical research did not find such simple relationships. However, neurotransmitters do play an essential role in the disorder, and subsequent findings have helped to clarify this role.

NOREPINEPHRINE. Perhaps the most compelling evidence for the central role of the noradrenergic system in depression is the well-replicated finding that antidepressant response is correlated with the downregulation or decreased sensitivity of β-adrenergic receptors. Also, presynaptic β_2-receptors appear to

have a role, as activation of these receptors results in a decrease in the amount of norepinephrine released. Presynaptic β_2-receptors are also located on serotonergic neurons and regulate the amount of serotonin released.

SEROTONIN. Most modern antidepressants act on serotonin, rather than norepinephrine, and as a result, serotonin has become the biogenic amine neurotransmitter most commonly associated with depression. The identification of multiple serotonin receptor subtypes has also increased the excitement within the research community about the development of even more specific treatments for depression. Besides treatment studies, other data indicate that serotonin is involved in the pathophysiology of depression. Depletion of serotonin may precipitate depression, and some patients with suicidal impulses have low cerebrospinal fluid (CSF) concentrations of serotonin metabolites and low concentrations of serotonin uptake sites on platelets.

DOPAMINE. Although norepinephrine and serotonin are the biogenic amines most often associated with the pathophysiology of depression, dopamine may also play a role. Dopamine activity may be reduced in depression and increased in mania. The discovery of new subtypes of the dopamine receptors and an increased understanding of the presynaptic and postsynaptic regulation of dopamine function have further enriched research into the relationship between dopamine and mood disorders. Drugs that reduce dopamine concentrations, including the reserpine mentioned above, as well as diseases that deplete dopamine such as Parkinson disease, all may cause depressive symptoms. In contrast, drugs that increase dopamine concentrations, such as tyrosine, amphetamine, and bupropion, reduce the symptoms of depression.

OTHER NEUROTRANSMITTER DISTURBANCES. Acetylcholine (ACh) is found in neurons that are distributed diffusely throughout the cerebral cortex. Cholinergic neurons have reciprocal or interactive relationships with all three monoamine systems. Abnormal levels of choline, which is a precursor to ACh, have been found at autopsy in the brains of some depressed patients. Cholinergic agonists can cause symptoms of lethargy, anergia, and psychomotor retardation. In an animal model of depression, strains of mice

that are super- or subsensitive to cholinergic agonists have been found susceptible or more resistant to animal models of depression. Cholinergic agonists can induce changes in HPA activity and sleep that mimic those associated with severe depression. Some patients with mood disorders in remission, as well as their never-ill first-degree relatives, have a trait-like increase in sensitivity to cholinergic agonists.

γ-Aminobutyric acid (GABA) has an inhibitory effect on ascending monoamine pathways, particularly the mesocortical and mesolimbic systems. Reductions of GABA occur in plasma, CSF, and brain GABA levels in depression. Animal studies have also found that chronic stress can reduce and eventually can deplete GABA levels. By contrast, antidepressants upregulate GABA receptors, and some GABAergic medications have weak antidepressant effects.

The amino acids glutamate and glycine are the primary excitatory and inhibitory neurotransmitters in the CNS. Glutamate and glycine bind to sites associated with the NMDA receptor and an excess of glutamatergic stimulation can cause neurotoxic effects. Importantly, a high concentration of NMDA receptors exists in the hippocampus. Glutamate, thus, may work in conjunction with hypercortisolemia to mediate the deleterious neurocognitive effects of severe recurrent depression. Emerging evidence suggests that drugs that antagonize NMDA receptors have antidepressant effects.

SECOND MESSENGERS AND INTRACELLULAR CASCADES. The binding of a neurotransmitter and a postsynaptic receptor triggers a cascade of membrane-bound and intracellular processes mediated by second messenger systems. Second messengers regulate the function of neuronal membrane ion channels. Increasing evidence also indicates that mood-stabilizing drugs act on these second messengers.

Immunologic Disturbance. Depressive disorders are associated with several immunologic abnormalities, including decreased lymphocyte proliferation in response to mitogens and other forms of impaired cellular immunity. These lymphocytes produce neuromodulators, such as corticotropin-releasing factor (CRF), and cytokines, peptides known as interleukins. There appears to be an association with clinical severity, hypercortisolism, and immune dysfunction, and the cytokine interleukin-1

may induce gene activity for glucocorticoid synthesis.

Findings from Brain Imaging

STRUCTURAL IMAGING. The most consistent abnormality observed in depressive disorders is the increased frequency of abnormal hyperintensities in subcortical regions, such as periventricular regions, the basal ganglia, and the thalamus. More common among elderly adults, these hyperintensities appear to reflect the deleterious neurodegenerative effects of recurrent affective episodes. Ventricular enlargement, cortical atrophy, and sulcal widening also have been reported in some studies. Some depressed patients also may have reduced hippocampal or caudate nucleus volumes, or both, suggesting more focal defects in relevant neurobehavioral systems. Diffuse and focal areas of atrophy are associated with increased illness severity, bipolarity, and increased cortisol levels.

Also, specific structures appear reduced in volume in patients with depression, for example, the hippocampus. This reduction may be due to neuronal loss due to the neurotoxic effects of the increased cortisol found in depressed patients. Although many areas of the brain are susceptible to this toxic effect, the hippocampus is particularly vulnerable, as it is rich in glutaminergic neurons. Other areas with reported brain loss include the PFC, cingulate gyrus, and cerebellum.

FUNCTIONAL IMAGINING. The most widely replicated positron emission tomography (PET) finding in depression is decreased anterior brain metabolism, which is generally more pronounced on the left side. From a different vantage point, depression may cause a relative increase in nondominant hemispheric activity. Furthermore, a reversal of hypofrontality occurs after shifts from depression into hypomania, such that higher left hemisphere reductions occur in depression compared with more significant right hemisphere reductions in mania. Other studies have observed more specific reductions of reduced cerebral blood flow or metabolism, or both, in the dopaminergically innervated tracts of the mesocortical and mesolimbic systems in depression. Evidence suggests that antidepressants at least partially normalize these changes.

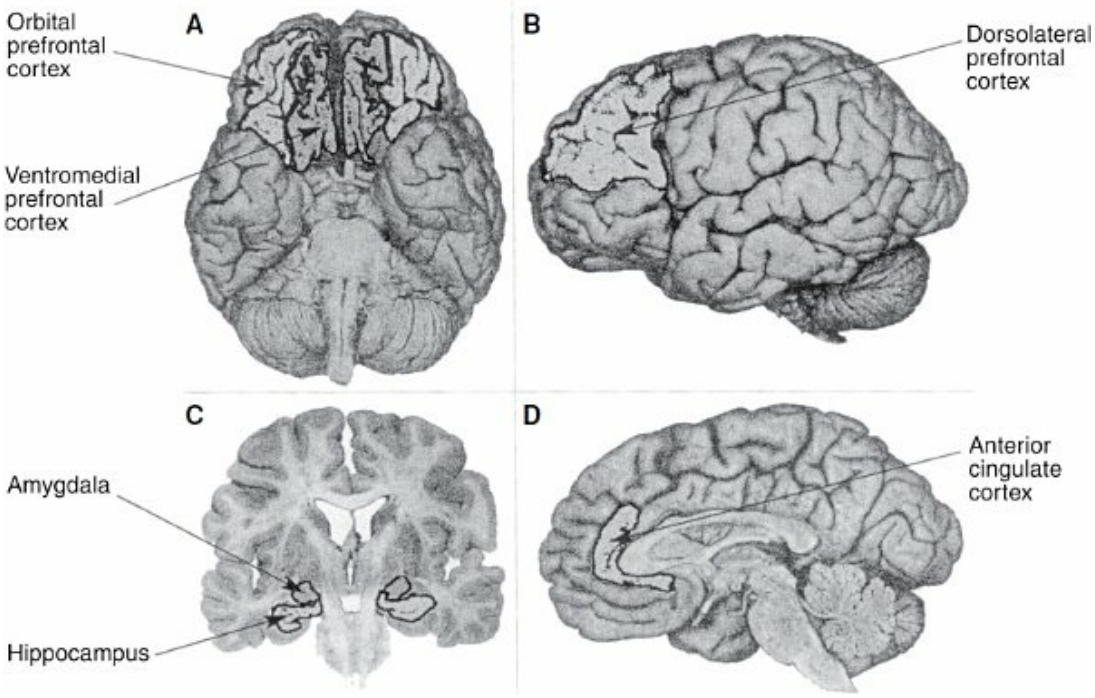


FIGURE 7-2
Key brain regions involved in affect and mood disorders. **A:** Orbital prefrontal cortex and the ventromedial prefrontal cortex. **B:** Dorsolateral prefrontal cortex. **C:** Hippocampus an¨d amygdala. **D:** Anterior cingulate cortex. (From Thase ME. Mood disorders neurobiology. In: Sadock BJ, Sadock VA, Ruiz P. (eds.). *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 10th ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2017.)

In addition to a global reduction of anterior cerebral metabolism, increased glucose metabolism occurs in several limbic regions, particularly among patients with relatively severe recurrent depression and a family history of mood disorder. During episodes of depression, increased glucose metabolism correlates with intrusive ruminations.

One meta-analysis found three consistent differences when comparing fMRI studies of depressed patients with healthy controls. They found activation of the pulvinar nucleus of the thalamus, more significant response to negative stimuli in the amygdala, insula and anterior cingulate, and lesser response in the dorsal striatum and dorsal lateral PFC.

Figure 7-2 illustrates some of the brain regions thought to be most critical to mood regulation.

Our Understanding of the Genetics of Depressive Disorders

Numerous family, adoption, and twin studies have long documented the heritability of mood disorders. Recently, however, the primary focus of genetic studies has been to identify specific susceptibility genes using molecular genetic methods.

Studies of Inheritance Patterns. Depression aggregates within families and meta-analytic studies suggest that first-degree relatives of persons with major depressive disorder have an increased odds ratio of 2.84 for developing the disease. Only limited adoption studies have been done, of varying quality and with mixed results, and most of what we know about the genetic contribution to the disorder comes from twin studies, which estimate the heritability to be 37 percent. It appears that this heritability is higher in women. The twin studies also concluded that shared environmental factors did not play a significant role in the familial aggregation of the disorder.

Genetic Studies. Genome-wide association studies have not found significant associations. Although there are various explanations for the lack of replicated findings, it likely relates to the fact that GWAS is powered to detect common variants, and that the genes relevant to depression risk are rare.

Candidate gene studies, which instead focus on specific genes of interest, have resulted in many reported findings. However, few of these are replicated. The most considerable focus has been on genes that might have a role in the proposed mechanism of depression, such as the gene involved in the encoding of serotonin (HTR1A), the serotonin transporter (5HTTP/SLC6A4), or the dopamine receptor (DRD4), and transporter (SLC6A3). Although some have replicated the most significant associations for these genes (most often for the 5HTTP/SLC6A4 gene), the effect size are small, and the sample sizes are smaller than those using GWAS, leading some experts to question the role of statistical error in the positive and negative findings. If some or all of the candidate gene studies are significant, the most likely interpretation of available findings is that much, or most of the genetic heritability for depression is the result of many genes each contributing a small effect, which only becomes meaningful when combined across thousands of genes. It is also possible, however, that there is some rare, still undetected genetic variant.

THE PSYCHOLOGY OF DEPRESSION

A great deal of psychological and sociologic research has also shed light on the pathology of depression. Perhaps the most meaningful contributions come from research on the role of stress and depression.

Stress and Depression

Some clinicians believe that life events play the primary or principal role in depression; others suggest that life events have only a limited role in the onset and timing of depression. The most compelling data indicate that the life event most often associated with the development of depression is losing a parent before age 11 years. The environmental stressor most often associated with the onset of an episode of depression is the loss of a spouse. Another risk factor is unemployment; persons out of work are three times more likely to report symptoms of an episode of major depression than those who are employed.

Personality

No single personality trait or type uniquely predisposes a person to depression; all humans, of whatever personality pattern, can and do become depressed under appropriate circumstances. Persons with certain personality disorders—OCPD, histrionic, and borderline—may be at higher risk for depression than persons with an antisocial or paranoid personality disorder. The latter can use projection and other externalizing defense mechanisms to protect themselves from their inner rage.

ETIOLOGY

As we can see, there is a great deal of data accumulating on various pathologies and dysregulations in the brain and body that are associated with depression. Many experts in the field have attempted to combine various lines of research into a coherent explanation of what causes us to suffer from a depressive disorder. Although there is no single unifying theory, several theories have emerged over the last century that attempt to account for the various clinical, psychological, and biologic findings in depression.

Biologic Theories

The Monoamine Hypothesis. As already noted, the fact that all known antidepressants acted on the monoamines (particularly catecholamines as noted above) coupled with the fact that drugs and diseases that deplete monoamines (e.g., reserpine and Parkinson disease) can cause depressive symptoms led to the supposition that depression is the result of some deficiency or dysregulation in brain monoamines.

Although the theory became, and to a point remains, prevalent, several lines of evidence suggested it was an inadequate explanation for the etiology of depression. Most notably, antidepressants take weeks to work, even though their presumed action of increasing monoamine levels happens quickly, often within hours after administration. Also, there is no convincing evidence of any neurochemical imbalance, for example, in studies of the CSF of depressed patients or postmortem brain tissue.

As noted in our discussion of the pathology of depression, there is good evidence to suggest that neurotransmitters do play an important role, and that receptor or second messenger relationships may explain some of the inconsistent findings. Some have attempted to modify the monoamine hypothesis, suggesting that it remains true, but is more complicated than initially thought. However, a consistent and coherent explanation of the role of neurotransmitters remains elusive, and we still do not know whether their relationship to depression is one of causation or a later effect of the disease.

Brain Dysfunction. Theories of brain dysregulation attempt to incorporate both findings of dysregulation with the increased and decreased responses to various negative stimuli observed in studies. Areas of interest include:

The amygdala. Part of the limbic system, the amygdala appears to be a crucial way station for processing novel stimuli of emotional significance and coordinating or organizing cortical responses.

The hippocampus. Adjacent to the amygdala, the hippocampus is most associated with learning and memory. Emotional or contextual learning appears to involve a direct connection between the hippocampus and the amygdala. Also, the hippocampus regulates the HPA axis by inhibiting activity.

The Prefrontal Cortex (PFC). We think of the PFC as the structure that

holds representations of goals and appropriate responses to obtain these goals. There are areas of specialization. For example, whereas left-sided activation of regions of the PFC is more involved in goal-directed or appetitive behaviors, regions of the right PFC are implicated in avoidance behaviors and inhibition of appetitive pursuits. Subregions in the PFC appear to localize representations of behaviors related to reward and punishment.

The anterior cingulate cortex (ACC). The ACC is involved in attention, motivation, and environmental exploration and appears to help integrate attentional and emotional inputs. The more rostral and ventral region has an affective subdivision that connects extensively with limbic regions of the brain. Activation of the ACC may facilitate control of emotional arousal, particularly when thwarting goal when attainment encountering novel problems.

There are various theories that attempt to integrate various pathologic findings into an understanding of the etiology of depression. Although there are many approaches and variations, many theories broadly fall into one of two categories: neurogenesis hypotheses and neuroplastic hypotheses.

Neurogenesis Hypotheses. Neurogenesis hypotheses suppose that the brain abnormalities leading to depression are the result of abnormalities in development such that there is a deficit in the number of newborn neurons in the brain.

There are several explanations for the cause of this deficit. Some versions attempt to link the role of stress, particularly chronic stress, in causing depression. Stress can cause increased activity in the HPA axis, which results in increased glucocorticoid production. Glucocorticoids are known to decrease neurogenesis. As neurogenesis continues in the adult hippocampus, this may be an area preferentially affected by stress. The hippocampus is then unable to adequately regulate the HPA axis, leading to continued elevated glucocorticoid levels in the brain and further inhibition of neurogenesis.

Neuroplasticity Hypotheses. Neuroplasticity hypotheses propose that the atrophy of already developed neurons causes depression. In this version, rather than a deficit of nerve growth, atrophy occurs in already mature neurons. As noted, chronic stress can increase glucocorticoid levels, which can cause atrophy. It can also decrease the expression of BDNF, which is

essential for the survival, growth, and differentiation of neurons in the brain; the result is atrophy. As this preferentially happens in the hippocampus, this can account for the finding of decreased hippocampal volume in depressed patients.

Psychosocial Theories

Life Events and Environmental Stress. A long-standing clinical observation is that stressful life events more often precede first, rather than subsequent, episodes of mood disorders. One theory proposed to explain this observation is that the stress accompanying the first episode results in long-lasting changes in the brain's biology. As a result, a person has a high risk of undergoing subsequent episodes of a mood disorder, even without an external stressor.

Ms. C, a 23-year-old woman, became acutely depressed when she was accepted to a prestigious graduate school. Ms. C had been working diligently toward this acceptance for the past 4 years. She reported being "briefly happy, for about 20 minutes" when she learned the good news but rapidly slipped into a hopeless state in which she recurrently pondered the pointlessness of her aspirations, cried constantly, and had to physically stop herself from taking a lethal overdose of her roommate's insulin. In treatment, she focused on her older brother, who had regularly insulted her throughout the course of her life, and how "he's not doing well." She found herself very worried about him. She mentioned that she was not used to being the "successful" one of the two of them. In connection with her depression, it emerged that Ms. C's brother had had a severe, life-threatening, and disfiguring pediatric illness that had required much family time and attention throughout their childhood. Ms. C had become "used to" his insulting manner toward her. In fact, it seemed that she required her brother's abuse of her in order not to feel overwhelmed by survivor guilt about being the "healthy, normal" child. "He might insult me, but I look up to him. I adore him. Any attention he pays to me is like a drug," she said. Ms. C's acceptance to graduate school had challenged her defensive and essential compensatory image of herself as

being less successful, or damaged, in comparison with her brother, thereby overwhelming her with guilt. Her depression remitted in psychodynamic psychotherapy as she better understood her identification with and fantasy submission to her brother. (Courtesy of J. C. Markowitz, M.D. and B. L. Milrod, M.D.)

Psychodynamic Theories of Depression. The psychodynamic understanding of depression defined by Sigmund Freud and expanded by Karl Abraham is known as the classic view of depression. That theory involves four key points: (1) disturbances in the infant–mother relationship during the oral phase (the first 10 to 18 months of life) predispose to subsequent vulnerability to depression; (2) depression can be linked to real or imagined object loss; (3) introjection of the departed objects is a defense mechanism invoked to deal with the distress connected with the object's loss; and (4) because the lost object is regarded with a mixture of love and hate, feelings of anger are directed inward at the self.

Ms. E, a 21-year-old college student, presented with major depression and panic disorder since early adolescence. She reported hating herself, crying constantly, and feeling profoundly hopeless in part because of the chronicity of her illness. Even at the time of presentation, she noted her sensitivity to her mother's moods. "My mother's just always depressed, and it makes me so miserable. I just don't know what to do," she said. "I always want something from her, I don't even know what, but I never get it. She always says the wrong thing, talks about how disturbed I am, stuff like that, makes me feel bad about myself." In one session, Ms. E poignantly described her childhood: "I spent a lot of time with my mother, but she was always too tired, she never wanted to do anything or play with me. I remember building a house with blankets over the coffee table and peeking out, spying on her. She was always depressed and negative, like a negative sink in the room, making it empty and sad. I could never get her to do anything." This patient experienced extreme guilt in her psychotherapy when she began to talk about her mother's depression. "I feel so bad," she sobbed. "It's like I'm saying bad things about her. And I love her so much, and I know she loves me. I feel it's so

disloyal of me." Her depression remitted in psychodynamic psychotherapy as she became more aware of and better able to tolerate her feelings of rage and disappointment with her mother. (Courtesy of J. C. Markowitz M.D. and B. L. Milrod, M.D.)

Cognitive Theory. According to cognitive theory, depression results from specific cognitive distortions present in persons susceptible to depression. These distortions, referred to as depressogenic schemata, are cognitive templates that perceive both internal and external data in ways that are altered by early experiences. Aaron Beck postulated a cognitive triad of depression that consists of (1) views about the self—a negative self-precept, (2) about the environment—a tendency to experience the world as hostile and demanding, and (3) about the future—the expectation of suffering and failure. Therapy consists of modifying these distortions. We summarize the elements of cognitive theory in Table 7-14.

Learned Helplessness. The learned helplessness theory of depression connects depressive phenomena to the experience of uncontrollable events. For example, when dogs in a laboratory are exposed to electrical shocks from which they cannot escape, they showed behaviors that differentiate them from dogs not exposed to such uncontrollable events. The dogs exposed to the shocks would not cross a barrier to stop the flow of electric shock when put in a new learning situation. They remained passive and did not move. According to the learned helplessness theory, the shocked dogs learned that outcomes were independent of responses, so they had both cognitive motivational deficits (i.e., they would not attempt to escape the shock) and emotional deficits (indicating decreased reactivity to the shock). In the reformulated view of learned helplessness as applied to human depression, internal causal explanations may produce a loss of self-esteem after adverse external events. Behaviorists who subscribe to this theory often stress that improvement of depression is contingent on the patient's learning a sense of control and mastery of the environment.



Table 7-14
Elements of Cognitive Theory

Element	Definition
Cognitive triad	Beliefs about oneself, the world, and the future
Schemas	Ways of organizing and interpreting experiences
Cognitive distortions	Persistent ways of thinking that are inaccurate and usually negatively biased
Arbitrary inference	Drawing a specific conclusion without sufficient evidence
Specific abstraction	Focus on a single detail while ignoring other, more important aspects of an experience
Overgeneralization	Forming conclusions based on too little and too narrow experience
Magnification and minimization	Over- or undervaluing the significance of a particular event
Personalization	Tendency to self-reference external events without basis
Absolutist, dichotomous thinking	Tendency to place experience into all-or-none categories

Courtesy of Robert M.A. Hirschfeld, M.D. and M. Tracie Shea, Ph.D.

Evolutionary Theory. Some theorists take an evolutionary perspective toward depression. From this perspective, depression is an adaptive response to perceived threats in the environment. From this perspective, depression is an adaptive response to perceived threats in the environment, and the tendency for depressed persons to withdraw from the environment in the face of possible threats could be protective. Beyond threats of bodily harm, social threats (e.g.,of exclusion or defeat) could also be considered threats in that such social threats can reduce one's perceived fitness, and hence, the desirability of a mate. The depressive response in which one decreases activity, withdraws from social situations, and approaches novel situations with a negative bias ("that person will think I am a loser") could be

interpreted as adaptive ways to reduce risk and avoid further social failures. There are many varieties of evolutionary theory, some of which stress the adaptive benefit of depression, others which see it as an outmoded response which may have had a benefit in ancient societies, but is counterproductive in our modern world. Other versions see the basic ability to vary mood as adaptive but suggest depression is a dysregulation of that normal function. These theories are difficult to test, as it is challenging to test the fitness of a particular trait that is very common in a population.

Integrative Approaches

It remains difficult, if not impossible, to incorporate all the various observations and research findings on depression into one grand unified theory. However, the various attempts by investigators and theoreticians to incorporate these findings have certain common features.

It seems clear that there exists some genetic vulnerability that puts persons at risk for depression. Such persons who then encounter stress, either external or internal, have epigenetic changes that differ from normal responses to stress; for example, they may be less able to transcribe certain growth factors in response to stress. This deficit results in subtle neuronal loss, and this loss is particularly apparent in some vulnerable areas (e.g., the hippocampus). The result is a disruption of the usual regulatory systems in the brain, causing changes in the expressions of transmitters, hormones, and other regulatory systems. This causes the physical changes that we associate with depression, and our behavioral response to this is the subjective feeling of depression.

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8

Anxiety Disorders

Everyone experiences anxiety. The experience of anxiety has two components: the awareness of the physiologic sensations (e.g., palpitations and sweating) and the awareness of being nervous or frightened. It is a diffuse, unpleasant, vague sense of apprehension, usually accompanied by autonomic symptoms. Although the physical and emotional symptoms of anxiety are similar to “fear,” it differs from fear in that it is not a response to an overt danger, but rather an impending one. Both fear and anxiety are normal adaptive responses to a potentially dangerous environment. They make us better prepared to survive a danger, either through fight, flight, or freezing, as the situation may warrant.

Much of psychology research and psychoanalytic thought is devoted to understanding what we would consider being normal anxiety. This chapter will concentrate, however, on pathologic anxiety: when it is inappropriately triggered and maladaptive, hence an anxiety disorder.

Anxiety disorders are the most prevalent psychiatric syndromes in the U.S. population. Nearly one-fifth of adults report a lifetime history of one of the major anxiety disorders, and 1 in 10 suffer from a current anxiety disorder. Anxiety disorders are associated with social impairment and, when occurring in childhood, can interfere with healthy development, having consequences for later social and occupational functioning. Worldwide, they are the sixth most significant contributor to nonfatal health loss globally, and they explain 10 percent of the disability-adjusted life years for all mental, neurologic, and substance use disorders, second only to major depression.

THE CLINICAL PRESENTATION

The anxious patient has a variety of symptoms, particularly cognitive and physical. Some of the symptoms, particularly the autonomic ones, are observable and quantifiable, making it easier to develop objective measures. It also makes research on these disorders somewhat more straightforward.

Anxiety can have a variety of physical manifestations. The most common is its effect on the autonomic system, including such symptoms as headache, perspiration, palpitations, tightness in the chest, and mild stomach discomfort. It can also produce motor symptoms, such as restlessness, indicated by an inability to sit or stand still for long. The particular constellation of symptoms present during anxiety tends to vary among persons.

The anxious individual may describe themselves as “nervous” or “frightened.” Their mood is frequently observable, particularly in patients who are experiencing a panic attack, as a classically fearful expression: eyes and mouth open, eyebrows raised. In patients with more chronic forms of anxiety, the expression may be more blunted and more akin to a depressed affect.

Thoughts may be more rapid, and in cases of severe anxiety, may become more disorganized. In extreme forms, a person may have great difficulty thinking clearly. For example, during a panic attack, a patient may ruminate or stammer.

In the setting of acute anxiety, the thoughts are focused on the perceived cause of the anxiety. Patients may catastrophize and overestimate the danger they are facing. In the case of a panic attack, somatic concerns of death from a cardiac or respiratory problem may be the primary focus of patients’ attention during panic attacks. Patients may believe that the palpitations and chest pain indicate that they are about to die. As many as 20 percent of such patients have syncopal episodes during a panic attack. The patients may present in emergency departments as young, physically healthy persons who nevertheless insist that they are about to die from a heart attack. As anxiety becomes chronic, thoughts may take the form of more negative thinking.

Although hallucinations are rare, patients with severe anxiety may have distortions in perception, not only of time and space but also of persons and the meanings of events.

In the proper dose, anxiety can help attention by increasing our alertness and focusing our attention. However, in excess, it can impair cognition.

Patients with severe anxiety can become confused. They can have difficulty focusing their attention, and as a result, may have trouble with recall.

Usually, anxiety does not reach a level where it significantly affects insight or judgment. However, patients can become selective in their interpretation of the environment, focusing on specific aspects and neglecting others to prove that they are justified in their reactions.

Although the patients do not tend to talk about suicidal ideation, they are at increased risk of committing suicide. Some studies have found that the lifetime risk of suicide in persons with panic disorder is higher than it is in persons with no mental disorder.

Presentation in Special Populations

Anxiety has many possible disorders, and different populations can present differently. For example, children or the elderly may present more with somatic symptoms than adults.

Certain cultural groups may have syndromes unique to their group, some of which represent culturally specific understandings of their body. Examples include the multiple anxiety syndromes related to the fear of “wind attacks” in Cambodians, or the syndrome *of Ataque de Nervios* (attack of nerves) in Puerto Rican and Dominican patients. In most cases, anxiety symptoms do not differ from those in other cultures. However, the emphasis may be on the specific ones relevant to beliefs about the underlying cause. In most cases, a full history and examination can elucidate the problem. It is often helpful to seek a cultural consultation for an understanding of the particular syndrome. In the past, there have been some broad overgeneralizations about different ethnic or cultural groups. However, these often reflect a bias and are usually not helpful.

DIAGNOSIS

Within this category, there are several disorders, including panic disorder (with and without agoraphobia), agoraphobia (without a history of panic disorder), specific phobia, social phobia, and generalized anxiety disorder.

Panic Disorder

Panic disorder is an acute intense attack of anxiety (a panic attack) accompanied by feelings of impending doom. The anxiety occurs during discrete periods of intense fear that can vary from several attacks during 1 day to only a few attacks during a year. Patients with panic disorder present with several comorbid conditions, most commonly agoraphobia, which refers to a fear of or anxiety regarding places from which escape might be difficult.

A panic attack is a sudden period of intense fear or apprehension that may last from minutes to hours. It is a symptom, and many disorders and situations besides panic disorder can cause panic attacks.

Mrs. K was a 35-year-old woman who initially presented for treatment at the medical emergency department at a large university-based medical center. She reported that while sitting at her desk at her job, she had suddenly experienced difficulty breathing, dizziness, tachycardia, shakiness, and a feeling of terror that she was going to die of a heart attack. A colleague drove her to the emergency department, where she received a full medical evaluation, including electrocardiography and routine blood work, which revealed no sign of cardiovascular, pulmonary, or other illness. She was subsequently referred for psychiatric evaluation, where she revealed that she had experienced two additional episodes over the past month, once when driving home from work and once when eating breakfast. However, she had not presented for medical treatment because the symptoms had resolved relatively quickly each time, and she worried that if she went to the hospital without ongoing symptoms, “people would think I’m crazy.” Mrs. K reluctantly took the phone number of a local psychiatrist but did not call until she experienced the fourth episode of a similar nature. (Courtesy of Erin B. McClure-Tone, Ph.D. and Daniel S. Pine, M.D.)

Table 8-1 compares the different approaches to diagnosing panic disorder.



Table 8-1
Panic Disorder

DSM-5	ICD-10

Diagnostic name	Panic Disorder	Panic disorder (episodic paroxysmal anxiety)
Duration	1 mo of worry after 1 panic attack Panic attacks occur during a discrete period	
Symptoms	Panic attacks: *Cardiopulmonary:* • Feeling short of breath • Palpitations • Chest discomfort *Gastrointestinal:* • Nausea or GI discomfort *Skin and systemic:* • Sweating • Chills or feeling flushed *Neurologic:* • Tremulousness • Dizziness • Numbness or tingling *Psychiatric:* • Derealization or depersonalization • Fear of losing control • Fear of dying Symptoms are abrupt and unpredictable Persistent symptoms between attacks, including: *Anticipatory anxiety:* fearing further attacks	Recurrent attacks of severe anxiety or panic Unpredictable *Physiologic symptoms:* • Chest pain • Palpitations • Difficulty breathing • Sweating • Dizziness *Additional symptoms* • Depersonalization • Derealization • Fear of losing control • Fear of dying

	Avoidance behaviors: trying to avoid real or imagined triggers to prevent another attack	
Required number of symptoms	*Panic attack:* 4+ symptoms ≥1 attack Avoidance behavior or anticipatory anxiety: 1+ month	
Exclusions (not result of):	Substance use Another medical condition Another mental disorder	Another mental illness
Comments	DSM-5 also includes a "Panic Attack Specifier" for discrete panic attacks without the additional criteria for a full panic disorder	

The diagnosis includes a variety of symptoms, including somatic, cognitive, and mood. Patients can feel the physical symptoms in many organs and systems, including cardiorespiratory, gastrointestinal, and otoneurologic. Panic attacks can occur as a part of another mental disorder, particularly the phobias and PTSD. They may or may not be associated with an identifiable situational stimulus and unexpected panic attacks are not uncommon. Some panic attacks do not fit easily into the distinction between unexpected and expected. We call these *situationally predisposed panic attacks*. They may or may not occur when encountering a specific trigger, or they may occur either immediately after exposure or after a considerable delay.

In the DSM-5, to be diagnosed with a panic disorder, a patient must have recurrent, unexpected attacks. This requirement is to help distinguish the disorder from the phobias and other possible causes of panic attacks.

Agoraphobia

Agoraphobia refers to a fear of or anxiety regarding places from which escape might be difficult. It can be the most disabling of the phobias because it can significantly interfere with a person's ability to function in work and social situations outside the home. Although agoraphobia often coexists with panic disorder, in that patients are afraid to leave the safety of home lest they have a panic attack in a public place, DSM-5 considers agoraphobia a separate condition that may or may not be comorbid with panic disorder. Often patients with agoraphobia alone still have some measure of anxiety, "panic-like" symptoms, although those symptoms may not reach the level of an actual panic attack.

Mrs. W was a 33-year-old married woman. She visited an anxiety clinic, reporting that she felt like she was having a heart attack whenever she left her home. Her disorder began 8 years earlier while attending a yoga class when she suddenly noticed a dramatic increase in her heartbeat, felt stabbing pains in her chest, and had difficulty breathing. She began sweating and trembling and felt dizzy. She immediately went to the emergency department, where they performed an electrocardiogram. No abnormalities were detected. Over the next few months, Mrs. W experienced similar attacks of 15 to 30 minutes' duration about four times per month. She often sought medical advice after each episode, and each time no physical abnormalities were detected. After experiencing a few of these attacks, Mrs. W became afraid of having an attack away from home and would not leave her home unless necessary, in which case she needed to have her cell phone or be accompanied by someone. Even so, she avoided crowded places such as malls, movie theaters, and banks, where rapid escape is sometimes blocked. Her symptoms and avoidance dominated her life, although she was aware that they were irrational and excessive. She experienced mild depression and restlessness and had difficulty sleeping.

Patients with agoraphobia rigidly avoid situations in which it would be challenging to obtain help. They prefer to be accompanied by a friend or a family member when leaving home, especially if their destination is crowded or closed-in. Severely affected patients may simply refuse to leave the house.

Patients may be terrified that they are going “crazy.”

Table 8-2 compares the different approaches to diagnosing agoraphobia. Several situations might cause anxiety, some involving confining spaces (public transportation, elevators, stores), open spaces (parks, shopping centers), or crowds. What they have in common is the fact that a person is away from home and safety and cannot quickly return to it.

Specific Phobia

The term *phobia* refers to excessive fear of a specific object, circumstance, or situation. A *specific phobia* is an intense, persisting fear of an object or situation, considered dangerous. The fear should be out of proportion to the actual threat. The diagnosis of specific phobia requires the development of intense anxiety, even to the point of panic, when exposed to the feared object. Persons with specific phobias may anticipate harm, such as being bitten by a dog, or may panic at the thought of losing control; for instance, if they fear elevators, they may also worry about fainting after the door closes.



Table 8-2
Agoraphobia

	DSM-5	ICD-10
Diagnostic name	Agoraphobia	Agoraphobia
Duration	≥6 mo	
Symptoms	Fear or anxiety from: Public transportation Open spaces Confined spaces Being in a line or in a crowd Being alone outside of home Avoidance of the situations, due to: Fear of having a panic attack while there Or, no access to a companion	Fear of: Leaving home Going to a public place Going to a crowded place Traveling alone

	to help withstand the situation Fear and avoidance is out of proportion to the potential threat	Avoidance of anxiety-provoking situations Frequently associated with panic disorders
Required number of symptoms	At least 1 of the above sources of fear	
Psychosocial consequences of symptoms	Marked distress or impairment	
Exclusions (not result of):	Another medical condition Another mental disorder	
Comments	This is for cases in which the above occurs without a panic disorder, despite the fear of having one	



Table 8-3
Specific Phobia

	DSM-5	ICD-10
Diagnostic name	Specific Phobia	Specific (Isolated) Phobias
Duration	Persistent and lasting ≥6 months	
Symptoms	Fear/anxiety about an object or situation Exposure to the object/situation	Phobias, restricted to highly specific situations, objects, or activities Exposure to the above causes panic

	causes immediate fear/anxiety Avoidance behaviors: of the object or situation The fear/anxiety is out of proportion to the likely threat	
Required number of symptoms	All the above	
Psychosocial consequences of symptoms	Marked distress or impairment	
Exclusions (not result of):	Another mental disorder	
Symptom specifiers	Animals Natural environment Blood–injection–injury Fear of blood Fear of injections and transfusions Fear of other medical care Fear of injury Situational Other	
Comments		Also contains categories for “Other phobic anxiety disorder” and an unspecified type

Mr. S was a successful lawyer who presented for treatment after his firm, to which he had previously been able to walk from home, moved to a new location that he could only reach by driving. Mr. S reported that he was “terrified” of driving, particularly on highways. Even the thought of getting into a car led him to worry that he would die in a fiery crash. His thoughts were associated with intense fear and numerous somatic symptoms, including a racing heart, nausea, and sweating. Although the thought of driving was terrifying in and of itself, Mr. S became nearly incapacitated when he drove on busy roads, often having to pull over to vomit. (Courtesy of Erin B. McClure-Tone, Ph.D. and Daniel S. Pine, M.D.)

Table 8-3 compares the different approaches to diagnosing specific phobia. There are many possible objects of the fear, including things (animals), environments (storms, dark rooms), situations (driving, flying, blood injections), and many other things that do not fit into these categories. What they have in common is the irrational fear that the object of one’s fear is harmful or dangerous.

In each case, the anxiety usually occurs immediately after exposure to the object or situation. The result is either avoidance or painful endurance. It should last for at least 6 months.

Social Anxiety Disorder

Social anxiety disorder (also referred to as *social phobia*) involves the fear of social situations, including situations that involve scrutiny or contact with strangers. The term social anxiety reflects the distinct differentiation of social anxiety disorder from a specific phobia, which is the intense and persistent fear of an object or situation. Persons with social anxiety disorder are fearful of embarrassing themselves in social situations (i.e., social gatherings, oral presentations, meeting new people). They may have specific fears about performing specific activities such as eating or speaking in front of others, or they may experience a vague, nonspecific fear of “embarrassing oneself.” In either case, the fear in social anxiety disorder is of the embarrassment that

may occur in the situation, not the situation itself.

Ms. B was a 29-year-old computer programmer who presented for treatment after she was offered a promotion to a managerial position at her firm. Although she wanted the raise and the increased responsibility that would come with the new job, which she had agreed to try on a probationary basis, Ms. B reported that she was reluctant to accept the position because it required frequent interactions with employees from other divisions of the company, as well as occasional public speaking. She stated that she had always felt nervous around new people, whom she worried would ridicule her for "saying stupid things" or committing a social faux pas. She also reported feeling "terrified" to speak before groups. These fears had not previously interfered with her social life and job performance. However, since starting her probationary job, Ms. B reported that they had become problematic. She noted that when she had to interact with others, her heart started racing, her mouth became dry, and she felt sweaty. At meetings, she had sudden thoughts that she would say something very foolish or commit a terrible social gaffe that would cause people to laugh. As a consequence, she had skipped several important meetings and left others early. (Courtesy of Erin B. McClure-Tone, Ph.D. and Daniel S. Pine, M.D.)

Table 8-4 compares the different approaches to diagnosing social anxiety disorder. Many people feel anxiety in social situations, and it is rare to find the person who feels no anxiety over speaking in public or attending a party where they will not know anyone. The key feature in social anxiety disorder is that this anxiety is significant to the point of being disabling, meaning that it is sufficient to cause clinically significant distress or impairment. It also occurs while one is scrutinized, and includes a fear of negative judgment. Furthermore, DSM-5 emphasizes that the fear should occur in situations in which the individual is under scrutiny and that the fear is of being "negatively evaluated" (embarrassed or rejected).



Table 8-4

Social Anxiety Disorder

	DSM-5	ICD-10
Diagnostic name	Social anxiety disorder	Social anxiety disorder
Duration	≤6 mo	
Symptoms	Fear of being judged/scrutinized in social situation Fear that others will notice the anxiety, causing additional judgment, embarrassment, or rejection Avoidance behaviors: of the feared social situations The fear/anxiety is out of proportion to the social risk	Fear of being judged/scrutinized by others Avoidance behaviors: of the feared social situations May be associated with low self-esteem May cause panic attacks
Required number of symptoms	All the above	
Psychosocial consequences of symptoms	Marked distress and/or impairment	
Exclusions (not result of):	Another mental illness Another medical illness	
Symptom specifiers	Performance only (performance anxiety)	

DSM-5 includes a specifier for performance anxiety (“performance only”), in which the fear is limited to public speaking or performance.

Generalized Anxiety Disorder

Generalized anxiety disorder is defined as excessive anxiety and worries about several events or activities most of the time for at least 6 months. The worry is difficult to control and is associated with somatic symptoms, such as muscle tension, irritability, difficulty sleeping, and restlessness. The worry usually involves a broad swath of everyday life, such as simple daily activities, timeliness, finances, or health. These are ordinary worries for many people. However, patients with a generalized anxiety disorder worry about them to the point where catastrophe seems possible, likely, and imminent. Another feature is that these concerns cannot be prioritized or put aside to deal with more pressing matters that may pop up. This inability to prioritize is a key feature that contributes to the pathologic effect that this disorder has on functioning.

Mr. G was a successful, married, 28-year-old teacher who presented for a psychiatric evaluation to treat mounting symptoms of worry and anxiety. Mr. G noted that for the preceding year, he had become more and more worried about his job performance. For example, although he had always been a respected and popular lecturer, he found himself worrying about his ability to engage students and convey the material effectively. Similarly, although he had always been financially secure, he increasingly worried that he was going to lose his wealth due to unexpected expenses. Mr. G noted frequent somatic symptoms that accompanied his worries. For example, he often felt tense and irritable while he worked and spent time with his family, and he had difficulty distracting himself from worries about the upcoming challenges for the next day. He reported feeling increasingly restless, especially at night, when his worries kept him from falling asleep. (Courtesy of Erin B. McClure-Tone, Ph.D. and Daniel S. Pine, M.D.)

Table 8-5 compares the different approaches to diagnosing generalized anxiety disorder. The distinction between generalized anxiety disorder and normal anxiety is that the worry is excessive, difficult to control, and causes impairment.

Objective Rating Scales for the Disorder

There are many scales for measuring anxiety. Some popular examples are the Beck Anxiety Inventory (BAI), Hospital Anxiety and Depression Scale (HADS), and the Generalized Anxiety Disorder scale (GAD-7), but there are many more. Many other broader scales include measures of anxiety. Some scales measure anxiety per se, while others help to identify a specific disorder. Scales may also differentiate between state and trait anxiety, which is situational anxiety versus anxiety that seems characteristic for a person and independent of the situation. An example for the latter is the State-Trait Anxiety Inventory.

DIFFERENTIAL DIAGNOSIS

Distinguishing Between the Anxiety Disorders

Anxiety disorders have overlapping symptoms. Sometimes it is difficult to distinguish between panic disorder, on the one hand, and specific and social phobias, on the other hand. Some patients who experience a single panic attack in a specific setting (e.g., an elevator) may go on to have long-lasting avoidance of the specific setting, regardless of whether they ever have another panic attack. These patients meet the diagnostic criteria for a specific phobia, and clinicians must use their judgment about what is the most appropriate diagnosis. In another example, a person who experiences one or more panic attacks may then fear to speak in public. Although the clinical picture is almost identical to the clinical picture in social anxiety disorder, we would not diagnose a social anxiety disorder because the avoidance of the public situation is based on fear of having a panic attack rather than on fear of the public speaking itself.



Table 8-5
Generalized Anxiety Disorder

	DSM-5	ICD-10
Diagnostic name	Generalized Anxiety Disorder	Generalized Anxiety Disorder

Duration	≤6 mo	
Symptoms	Excessive anxiety/worry Difficulty controlling/ managing worry *Anxiety characterized by:* • Restlessness • Fatigue • Poor concentration • Irritability • Muscle tension • Insomnia	Persistent anxiety *Anxiety characterized by:* • Shaking • Muscle tension • Sweating • Lightheadedness • Palpitations • GI symptoms
Required number of symptoms	First two criteria and 3+ of the specific symptoms	
Psychosocial consequences of symptoms	Marked distress and/or impairment	
Exclusions (not result of):	• Another mental disorder • Substance use • Another medical condition	Anxiety not associated with an object, event, or situation

Other Psychiatric Disorders

The differential diagnosis for agoraphobia includes all the psychiatric disorders that can cause anxiety or depression. A panic disorder is the most common cause of agoraphobia, and a separate diagnosis of agoraphobia is not necessary. Other disorders include major depressive disorder, schizophrenia, paranoid personality disorder, avoidant personality disorder, and dependent personality disorder.

Medical Disorders

Many medical disorders can cause anxiety or have symptoms that overlap with anxiety disorders.

Several medical conditions produce symptoms that are similar to panic disorder. Panic attacks are associated with a variety of endocrinologic disorders, including both hypo- and hyperthyroid states, hyperparathyroidism, and pheochromocytomas. Episodic hypoglycemia associated with insulinomas can also produce panic-like states, as can primary neuropathologic processes. These include seizure disorders, vestibular dysfunction, neoplasms, or the effects of both prescribed and illicit substances on the CNS. Finally, disorders of the cardiac and pulmonary systems, including arrhythmias, chronic obstructive pulmonary disease, and asthma, can produce autonomic symptoms and accompanying crescendo anxiety that can be difficult to distinguish from panic disorder. Clues of an underlying medical etiology to panic-like symptoms include the presence of atypical features during panic attacks, such as ataxia, alterations in consciousness, or bladder dyscontrol, onset of panic disorder relatively late in life, and physical signs or symptoms indicative of a medical disorder.

COMORBIDITY

Depressive symptoms are often present in panic disorder, and in some patients, a depressive disorder coexists with the panic disorder, and depression can complicate the symptom picture in anywhere from 40 to 80 percent of all patients, as estimated by various studies.

In addition to agoraphobia, other phobias and OCD can coexist with panic disorder.

Alcohol and other substance use disorders occur in about 20 to 40 percent of all patients, and OCD may also develop.

COURSE

Panic attacks, by definition, have a sudden onset and relatively short duration. The first panic attack is often completely spontaneous, although panic attacks occasionally follow excitement, physical exertion, sexual activity, or moderate emotional trauma.

The attack often begins with 10 minutes of rapidly increasing symptoms. The significant mental symptoms are extreme fear and a sense of impending death and doom. Patients usually cannot name the source of their fear; they may feel confused and have trouble concentrating. The physical signs often include tachycardia, palpitations, dyspnea, and sweating. Patients often try to leave whatever situation they are in to seek help. The attack generally lasts 20 to 30 minutes and rarely more than an hour.

Panic disorder usually has its onset in late adolescence or early adulthood, although onset during childhood, early adolescence, and midlife do occur. Some data implicate increased psychosocial stressors with the onset of panic disorder, although in most cases, there are no identifiable psychosocial stressors.

Panic disorder is a chronic disorder, although its course is variable, both among patients and within a single patient. The available long-term follow-up studies of panic disorder are difficult to interpret because they have not controlled for the effects of treatment. Nevertheless, about 30 to 40 percent of patients seem to be symptom-free at long-term follow-up, about 50 percent have symptoms that are sufficiently mild not to affect their lives significantly, and about 10 to 20 percent continue to have significant symptoms.

After the first one or two panic attacks, patients may be relatively unconcerned about their condition; with repeated attacks, however, the symptoms may become a significant concern. Between attacks, patients may have anticipatory anxiety about having another attack. Patients may attempt to keep the panic attacks secret and thereby cause their families and friends concern about unexplained changes in behavior. The frequency and severity of the attacks can fluctuate. Panic attacks can occur several times in a day or less than once a month. Excessive intake of caffeine or nicotine can exacerbate the symptoms.

When agoraphobia is part of a panic disorder, improving panic symptoms often also improves agoraphobia. For rapid and complete reduction of agoraphobia, behavior therapy is useful. Agoraphobia without a history of panic disorder is often incapacitating and chronic, and depressive disorders and alcohol use disorder often complicate its course.

The presence of comorbid disorders, particularly alcohol and substance use disorders, complicates the course of anxiety disorders. Patients with good premorbid functioning and symptoms of brief duration tend to have a

favorable prognosis.

Most of the other anxiety disorders also have long-term courses with multiple relapses. Overall these are all chronic disorders. Generalized anxiety disorder commonly has multiple relapses, although some may occur long after the initial episode, sometimes giving a false sense of security. Clinicians should regularly monitor the symptoms.

Most of these disorders also have an increased risk of suicide, and clinicians should monitor this as well.

TREATMENT APPROACH

With treatment, most patients exhibit dramatic improvement in their anxiety symptoms. Treatments include pharmacologic, psychologic, and combined treatments for all anxiety disorders. Meta-analyses generally suggest that pharmacologic treatment has the largest effect size of the various options. However, depending on the anxiety disorder, there are many pharmacologic, psychotherapeutic, and combined options available.

Hospitalization

Patients rarely require hospitalization unless they need a diagnostic workup, for example, to rule out a medical cause. Also, we may hospitalize patients with comorbid disorders such as substance use or those who are suicidal.

Pharmacotherapy

Among the medication options, selective serotonin reuptake inhibitors are the first-line agents for most anxiety disorders, including panic disorder, generalized anxiety disorder, and social anxiety disorder. Some non-SSRIs are also useful, for example, venlafaxine for panic disorder, generalized anxiety disorder, and social anxiety disorder. Tricyclic antidepressants are useful for panic disorder as well, although less popular due to their side effects. Many clinicians consider mirtazapine to be useful for anxiety disorders, owing to its sedative effect. However, there are few studies of its use for anxiety disorder.

Benzodiazepines remain one of the most popular medications used for anxiety disorder, perhaps the most popular. Most treatment guidelines

suggest that they be mostly limited to short-term use, either as an adjunct to SSRIs during the initial treatment phase or for acute use during exacerbations of the anxiety. Generally, most guidelines recommend that we should only consider long-term benzodiazepines for patients who do not respond or cannot tolerate the SSRIs. The main concerns are the potential for dependence, as well as the cognitive and other side effects. Tolerance to the anxiolytic effects does not seem to develop.

Antipsychotics and anticonvulsants are not recommended as initial therapy but may have some role for treatment-resistant patients. Among them, quetiapine is popular and may be useful as a second-line treatment for generalized anxiety disorder.

Buspirone is an azapirone, and effective for the treatment of generalized anxiety disorders. It is given in three divided doses during the day. The time to effect is similar to that for antidepressants, several weeks, and may take as long as several months. It may be useful as an adjunct to antidepressants for other anxiety disorders; however, most of the evidence is anecdotal.

β-Blockers, such as propranolol, are sometimes used for anxiety disorders, particularly social anxiety disorder. However, the available evidence does not support this use. Anecdotally, many consider it useful for social anxiety, particularly performance anxiety. The presumed mechanism is the drug's ability to block many of the physiologic symptoms of anxiety. However, there is little data to support this, and at least one study showed no effect. One study concluded it might be useful for performance anxiety—specifically musical performances—however, it also had significant side effects that could impair performance.

Antihistamines, such as hydroxyzine, are also used, particularly as alternatives to benzodiazepines for acute treatment. It has some evidence for its use in generalized anxiety disorder. We know little about its long-term effect.

Table 8-6 lists the common classes of drugs used for anxiety disorders and the evidence supporting their use.

Treatment Approach. Table 8-7 gives some pointers for using medications to treat patients with anxiety.

For most anxiety disorders, a conservative approach is, to begin with, an SSRI. Benzodiazepines are most useful when a patient requires rapid control

of severe anxiety symptoms. In those cases, a short-term benzodiazepine, such as lorazepam or alprazolam, is helpful in the short term. Concurrently, an SSRI should be started and slowly increased.

For panic disorder, SSRIs or the serotonin–norepinephrine reuptake inhibitor (SNRI) venlafaxine are first-line options. Tricyclic antidepressants or MAOIs are effective but less preferably given their side effects. Other agents, such as mirtazapine, are other second-line options. As mentioned, the benzodiazepines are effective options but mainly used for short-term use or exacerbations. Long-term antidepressant therapy is useful for preventing relapse, with studies suggesting that the beneficial effects last for up to 1 to 3 years. When appropriate, maintenance therapy should be discontinued very slowly.

For generalized anxiety disorder, the approach is similar, with SSRIs and SNRIs being the first-line treatments. Reasonable alternatives if those are not effective include agomelatine, pregabalin, buspirone, and quetiapine.

For social anxiety disorder, the approach is similar to those already described, with SSRIs and SNRIs being the first-line choice. Pregabalin and clonazepam also have strong evidence, although the same caveats for other benzodiazepines apply to clonazepam. Phenelzine is effective as well but is rarely used because of the side effects. Tricyclic antidepressants, buspirone, and quetiapine are not recommended owing to their side effects. As noted above, β-blockers may be useful for performance anxiety. However, their use does not appear to generalize to other types of social anxiety.



Table 8-6
Evidence-Based Recommendations for Monotherapy of Anxiety Disorders

	Anxiety Disorders		
Classes and Agents	**GAD**	**PD**	**SAD**
SSRIs	First line	First line	First line
SNRIs	First line	First line	First line
TCAs	Second line	Second line	Not

			recommended
MAOIs	Insufficient evidence	Second line	Second line
RIMA-Moclobemide	Insufficient evidence	Insufficient evidence	Second Line
Other Psychotropics—Agomelatine, Buspirone, Mirtazapine	Second line	Second line	Second line
Benzodiazepines	Second line	Second line	Second line
Atypical Antipsychotics—Quetiapine	Second line	Insufficient evidence	Not recommended
Anticonvulsant—Pregabalin	Second line	Insufficient evidence	Second line

Note: Beta-adrenergic blocking agents are recommended only for performance anxiety.
RIMA, reversible monoamine oxidase inhibitor; GAD, generalized anxiety disorder; PD, panic disorder; SAD, social anxiety disorder.

In the case of specific phobia, psychotherapies, particularly behavioral therapies, are the first-line choice. SSRIs may be helpful, although there are few studies of SSRIs for this disorder.

For agoraphobia, the main goal of pharmacotherapy is to treat the panic attacks that are usually comorbid. Early studies did not support the use of medications for “pure” agoraphobia; however, there has been little research since.

Psychosocial Therapies

There is strong support for a variety of psychosocial therapies for anxiety disorders, including cognitive-behavioral therapy (CBT), behavioral therapies, and interpersonal therapy. Studies of CBT suggest it has a substantial effect on generalized anxiety disorder, panic disorder, and social anxiety disorder, although this evidence is tempered by concerns about

treatment bias in the studies. Some treatment guidelines consider individual CBT to be a first-line treatment for social anxiety disorder. Group psychotherapy, mainly using CBT techniques, is also useful for social anxiety disorder. In vivo exposure therapy is the treatment of choice for specific phobia.

Cognitive Therapy. The two major foci of cognitive therapy for panic disorder are instruction about a patient's false beliefs and information about panic attacks. The instruction about false beliefs centers on the patient's tendency to misinterpret mild bodily sensations as indicating impending panic attacks, doom, or death. The information about panic attacks includes explanations that when panic attacks occur, they are time-limited and not life-threatening.



Table 8-7
Pharmacotherapy of Anxiety Disorders: Key Pointers

- SSRIs are the first-line option.
- Start low and go slow.
- Routine increase to higher doses not recommended, but a subgroup might benefit.
- Concomitant benzodiazepines as initial therapy in the short term may be useful.
- 8–12 weeks of pharmacotherapy at optimum doses may be needed to assess efficacy.
- Good evidence for the benefit of maintenance treatment at least up to 6 months.

Behavioral Therapies. In behavior therapy, the underlying assumption is that change can occur without the development of psychologic insight into underlying causes. Techniques include positive and negative reinforcement, systematic desensitization, flooding, implosion, graded exposure, response prevention, thought stopping, relaxation techniques, panic control therapy, self-monitoring, and hypnosis. For patients with specific phobias, the approach usually involves gradually increasing the patient's exposure to the

feared stimulus (e.g., climbing to increasingly higher floors in a building for a patient with a fear of heights) while practicing relaxation techniques until the patient masters each successive step.

Interpersonal Psychotherapy. There is good evidence for interpersonal psychotherapy, particularly interpersonal skills training for social anxiety disorder. The assumption is that such patients have interpersonal deficits that contribute to their anxiety and lack the skills to interact with others effectively. The result is that patients experience more of the "punishments" and less of the rewards from social interaction than most other people.

Virtual Therapy. Computer programs exist that allow us to treat a variety of anxiety disorders, including agoraphobia, specific phobia, and social anxiety disorder. As an example, patients with agoraphobia may experience a virtual environment in which they are in a crowded space (e.g., a supermarket). As they identify with the avatars in repeated computer sessions, they can learn to master their anxiety until they are ready to try the exposure in real life. This approach is particularly useful for situations that are not easily reproduced inside or near a clinician's office (i.e., flying).

Supportive Psychotherapy. Supportive psychotherapy involves the use of psychodynamic concepts and a therapeutic alliance to promote adaptive coping. Adaptive defenses are encouraged and strengthened, and maladaptive ones are discouraged. The therapist assists in reality testing and may offer behavioral advice. Although lacking empirical support, clinicians frequently incorporate some degree of supportive psychotherapy as an adjunct to medication.

Insight-Oriented Psychotherapy. In insight-oriented psychotherapy, the goal is to increase the patient's development of insight into psychologic conflicts that, if unresolved, can manifest as symptomatic behavior. For a time, it was the classic treatment for many types of anxiety, with the assumption that the anxiety represented an underlying psychodynamic conflict. Some studies suggest that psychodynamic psychotherapy can help decrease anxiety symptoms, and may have a lasting effect. However, the studies tend to have methodologic problems, and we lack the large,

comparative studies needed to determine efficacy.

THE EPIDEMIOLOGY OF THE DISORDER(S)

Incidence and Prevalence

Anxiety disorders make up one of the most common groups of psychiatric disorders. The National Comorbidity Study reported that one of four persons met the diagnostic criteria for at least one anxiety disorder and that there is a 12-month prevalence rate of 17.7 percent.

Most epidemiologic studies of panic disorder suggest a 12-month prevalence between 0.2 and 1.1 percent. The lowest rate reported rate was 0.1 percent (Nigeria), and the highest was 6.9 percent (Italy), with a median of 2.3 percent.

Most epidemiologic studies of generalized anxiety disorder suggest a 12-month prevalence between about 2.1 and 3.1 percent in the United States. The ranges vary widely, with ranges from 0 percent (Nigeria) to 2.6 percent (Germany).

The 12-month prevalence of social anxiety disorder varies widely across regions. The United States has the highest reported rates (6.8 percent), and China the lowest (0.2 percent). The median is about 4 percent to 5 percent. It may be that these differences involve differing cultural concepts of social fears or even difficulties in translating this concept to different cultures.

The 12-month prevalence of agoraphobia is mostly consistent across regions, with a range from 0 (China) to 0.8 percent (U.S.). An exception is South Africa, with a reported rate of 4.8 percent.

The rates of specific phobias vary widely, with a range of 1.9 percent (China) to 8.7 percent (U.S.).

Table 8-8 compares the 12-month and lifetime prevalence of anxiety disorders across different countries.

Sex

Women have higher rates of almost all anxiety disorders than do men. The difference is twofold for most disorders. The exception is social anxiety disorder, where the ratio is about equal. This difference is true across all ages but is most evident during early and mid-adulthood.

Age

The anxiety disorders have one of the earliest onsets of all psychiatric disorders. Most of the disorders begin in childhood or adolescence; the median age is 12.

In general, phobic disorders are the most stable of the disorders over time. Panic and generalized anxiety disorder are much like major depressive disorder in that they tend to have exacerbations and remissions over the life span.

Sociocultural and Ethnic Variables

Anxiety disorders appear to be common in persons of lower socioeconomic status and educational level. However, these relationships are very complex.

Some studies have reported higher rates of anxiety disorders in African Americans and lower rates in Hispanics.

THE NEUROBIOLOGY OF THE DISORDER

Genetics

Like most psychiatric disorders, anxiety disorders are complex disorders. Studying anxiety disorders has an advantage over many, as fear and anxiety are common to animals and are easily observed, at least compared to other mental phenomena. Thus, animal models have much informed the research on genetics and other aspects of the disorder. Here, we will, however, concentrate on human studies.

Studies of Inheritance Patterns. A relatively large number of family studies have supported the familial aggregation of the major anxiety disorders, including panic disorder, generalized anxiety disorder, phobias, and agoraphobia. For panic disorder, the familial risk was highest for early-onset panic disorder. In the case of social anxiety disorder, it was strongest for the generalized subtype.

Several large twin studies have demonstrated higher inheritance for monozygotic than dizygotic twins, suggesting a genetic component. The genetic contribution appears to be about 30 percent or more, perhaps as high as 60 percent for the phobias.

Genetic Studies. Among candidate gene studies, linkage studies have had inconsistent results, which is not a surprise, given the assumption that this is a complex disorder. Thus we are unlikely to find highly penetrant genes. Association studies usually rely on candidate genes, usually chosen for their presumed relevance to the disorder or because of prior studies suggesting an association. Most studied are the genes for the neurotransmitter systems and the stress response. Again, the results have been inconsistent.

Genome-wide studies (GWAS) do not require the same a priori assumptions or degree of penetrance. A limitation is the large sample sizes needed, and this limits many of the genetic studies in psychiatry. The most studied has been panic disorder. One study found support for immune-related involvement for the disorder. Another study found two associated single nucleotide polymorphisms (SNPs), a finding later replicated in another study. The genes are involved in mRNA expression in the frontal cortex. Although intriguing, the results remain preliminary. A large research collaboration, the Psychiatric Genetics Consortium, has found several possible novel risk genes; again, these results are tentative.

To date, there is little data on gene–environment interactions, given the substantial sample sizes needed. However, with converging evidence suggesting a role for both cumulative and specific life events, this is an essential area for investigation. There is also limited but compelling evidence for the role of epigenetics in anxiety disorders. However, this area is just emerging.

Neuroimaging Studies

Preclinical data on fear circuitry has guided the search for neuroanatomical associations with anxiety. Thus, the amygdala, and the other components of the frontoamygdala connections (the perirhinal cortex, ventrolateral prefrontal cortex (vlPFC), and anterior insula) have been the most studied. Another focus is the hippocampus, which plays a critical role in fear learning and extinction. Other areas include the posterior and lateral orbitofrontal cortex (OFC), anterior insula, and vlPFC, which show increased neurophysiologic activity in patients with phobias.

Table 8-8

Lifetime and 12-Month Prevalence Rates of DSM-IV Anxiety Disorders in International Community Studies of Adults

	GAD		Panic Disorder		Separation Anxiety		Agoraphobia		Social Phobia		Specific Phobia		All Anxiety	
Study	12 mo	LT	12 mo	LT	12 mo	LT	12 mo	LT	12 mo	LT	12 mo	LT	12 mo	LT
U.S. (NESARC)[a]	2.1	4.1	2.1	5.1	—	—	0.1	0.2	2.8	5.0	7.1	9.4	11.1[h]	17.2[h]
Iran[b]	—	1.3	—	1.3	—	—	—	0.7	—	0.8	—	—	—	8.4
Australia (NSMHWB)[c]	2.6	—	1.1	—	—	—	0.5	—	1.3	—	—	—	5.6	—
Germany (GHS-MHS)[d]	1.5	—	2.3	—	—	—	12.6[i]	—	12.6[i]	—	12.6[i]	—	14.5	—
Italy, Sesto Fiorentino[e]	—	6.9	—	6.9	—	—	—	0.4	—	3.7	—	1.5	—	16.9
Korea (KECA-R)[f]	0.8	1.6	0.1	1.6	—	—	0.2	0.2	0.4	0.5	3.4	3.8	5.3	6.9
WHO WMH Surveys[g]														
U.S.	3.1	5.7	2.7	4.7	1.9	9.2	0.8	1.4	6.8	12.1	8.7	12.5	18.1	28.8
Mexico	—	0.9	—	1.0	0.9	4.5	—	1.0	—	2.9	—	7.0	—	14.3
Belgium	—	1.9	—	16	0.1	1.4	—	—	—	2.0	—	6.8	—	13.1
Italy	0.5	1.9	0.6	16	0.0	1.5	0.4	1.2	1.0	2.1	2.7	5.7	5.1	11.1
Ukraine	1.2	1.9	1.3	1.9	—	—	0.2	0.3	1.5	2.6	—	—	6.1	3.8
Lebanon	1.3	—	0.2	—	1.9	6.9	0.3	—	1.1	—	8.2	—	11.2	—
Nigeria	0.0	0.1	0.1	0.2	0.0	0.2	0.2	0.4	0.3	0.3	3.5	5.4	4.1	5.7
New Zealand	—	6.0	—	2.7	—	—	—	1.2	—	9.4	—	10.8	—	24.9
Japan	1.2	—	0.5	—	—	—	0.3	—	0.8	—	2.7	—	4.8	—
China	0.8	—	0.2	0.4	0.2	1.3	0.0	0.0	0.2	0.5	1.9	2.6	2.7	4.8
South Africa	1.4	—	0.8	—	—	—	4.8	—	1.9	—	—	—	8.1	—

[a]From Hasin DS, Grant BF. The National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) Waves 1 and 2: Review and summary of findings. *Soc Psychiatry Psychiatr Epidemiol*. 2015;50(11):1609–1640.
[b]Mohammadi MR, Davidian H, Noorbala AA, Malekafzali H, Naghavi HR, Pouretemad HR, Yazdi SAB, Rahgozar M, Alaghebandrad J, Amini H, Razzaghi EM, Mesgarpour B, Soori H, Mohammadi M, Ghanizadeh A. An epidemiological survey of psychiatric disorders in Iran. *Clin Pract Epidemol Ment Health*. 2005;26;1:16.
[c]Oakley Browne MA, Wells JE, Scott KM, McGee MA; New Zealand Mental Health Survey Research Team. Lifetime prevalence and projected lifetime risk of DSM-IV disorders in Te Rau Hinengaro: the New Zealand Mental Health Survey. *Aust N Z J Psychiatry*. 2006;40(10):865–874.
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[g]The WHO World Mental Health Survey Consortium. Prevalence, severity, and unmet need for treatment of mental disorders in the World Health Organization World Mental Health Surveys. *JAMA*. 2004;291:2581–2590.
[h]Not including PTSD.
[i]Any phobia, including social phobia, specific phobia, and agoraphobia.

Neurochemical systems are also a focus, including central noradrenergic systems, central serotonergic and dopaminergic, and GABA systems.

Studies of fear conditioning in healthy humans are similar to those of animals, confirming that the fear circuitry is conserved across species. The amygdala, ventromedial PFC (vmPFC), and hippocampus have been implicated in both PET and fMRI studies.

Studies of panic disorder suggest that there are abnormalities both when patients are at rest and during an acute panic attack. In the resting state,

hippocampal and parahippocampal areas are implicated, whereas during panic, the associations are in the insular and striatal regions, with reduced activity in other regions such as the PFC. MRI studies also suggest abnormalities of the gray matter volume in the parahippocampal and temporal regions. Also interesting are MRI studies of brain lactate, which suggest an exaggerated response to hypocapnia, suggesting a suffocation response. Receptor-binding studies show abnormalities in GABA and serotonergic, particularly 5-HT_{1A}, binding.

Studies in specific phobias suggest activation of anterior paralimbic regions and sensory association cortices related to the particular phobia (be it visual, auditory, or other senses). These suggest hypersensitivity to specific threat-related cues. Also, the amygdala is again implicated.

In social anxiety disorder, functional neuroimaging studies show an exaggerated response to social stimuli, particularly in the medial temporal lobe structures. Most commonly, investigators find amygdala hyperresponsivity to social threats.

Unlike the other disorders, generalized anxiety disorder does not show clear findings of amygdala hyperactivity. The core features are more typical of emotional dysregulation, with disruptions of the functional connections between the anterior cingulate and the amygdala, as well as the uncinate fasciculus. This may suggest weaker frontoamygdala connectivity and fear overgeneralization. Preliminary evidence suggests that effective treatments target these areas, including CBT and pharmacotherapies.

Psychological Studies. Behavioral scientists have helped to elucidate the psychology of fear and anxiety through both animal and human studies. A full discussion of the psychological research is well beyond the scope of this chapter, but most relevant to anxiety disorders are the work of Pavlov and later scientists on the nature of conditioning. In the classic studies, the scientist exposes an animal to an neutral stimulus, such as an auditory tone, and then presents the tone while introducing an aversive stimulus, such as an electric shock. With the repeated pairing of the two stimuli, the neutral stimulus becomes associated with the aversive one, to the point where it alone can elicit the same response even when removing the aversive stimulus. The result is that the animal has been conditioned or learned to fear the neutral stimuli. In behavioral science, the neutral stimulus is called the

conditioned stimulus and the aversive one the unconditioned stimulus. If the scientist presents the conditioned stimulus without the unconditioned one and repeats this often enough, the animal will no longer associate the two stimuli; this is called extinction. How long extinction learning takes will depend on the particular stimuli and the individual animal. It also can depend on context, and a different setting can change the response. Behavioral studies suggest that the animal does not forget the conditioning. Instead, it is now competing with a new memory. Under appropriate circumstances, the pairing, and associated fear response, can be reinstated.

As humans are animals, they experience conditioning and extinction as well. The relevance to anxiety disorders, particularly phobias, should be obvious. Research in conditioning has gone well beyond this simplified description to identify and elaborate the many subtleties of conditioned learning.

ETIOLOGY

Biologic Theories

Neurochemicals and the Fight-or-Flight Reaction. When we perceive stress, our bodies activate certain neurotransmitters and neuropeptides. The result is the "fight-or-flight" reaction. This response is adaptive in specific contexts. However, overgeneralization of their reaction can be disabling and underlies many of the anxiety disorders.

Chronic activation can cause alterations in a variety of systems. Some evidence suggests that early-life stressors can cause this, making us predisposed to anxiety disorders later in life. Why some of us are more susceptible than others, under similar circumstances, is an area of active investigation.

Several neurochemicals and systems are thought to be most relevant to the mechanism between anxiety disorders, including the monoamines, the hypothalamic–pituitary–adrenal (HPA) axis, corticotropin-releasing hormone (CRH) and other chemicals. Broadly put, patients with anxiety disorders have symptoms that suggest an exaggerated noradrenergic system output, along with increased autonomic and sympathetic activation. Table 8-9 summarizes the major neurochemical systems thought to be relevant to anxiety disorders.

Table 8-9
Neurochemical Systems Involved in Anxiety Disorders and Related Treatment Approaches

Neurochemical	Brain Regions	Association with Anxiety Disorders	Treatm Approa (Curre Future
Noradrenergic system	Locus coeruleus, amygdala, hippocampus, hypothalamus, prefrontal cortex	Unrestrained, excessive system activation	SNRIs ther anxi (alo SSR Propra perf anxi
Hypothalamic–pituitary–adrenal (HPA) axis	Hippocampus, amygdala, hypothalamus, prefrontal cortex	Dysregulated HPA axis function (excessive cortisol release, abnormal feedback) in some studies	Cortiso adm und SAD pho Mifepri stud and
Corticotropin-releasing hormone (CRH)	Prefrontal and cingulate cortices, amygdala, hippocampus, hypothalamus, bed nucleus of stria terminalis, nucleus accumbens,	Persistently increased CRH concentration	To date rece anta faile dem effic clini

	periaqueductal gray matter, locus coeruleus, dorsal raphe nuclei		
Neurosteroids	Hippocampus, amygdala, cortex	Abnormal peripheral levels of neurosteroids in PD; possibly also in GAD and SAD (inconsistent findings)	Paroxe incr peri of allo in o PD not stud Synthe neu und dev
Arginine vasopressin (AVP)	Paraventricular nucleus of hypothalamus, septum, hippocampus, cortex	Single nucleotide polymorphism in AVP V1b receptor gene linked to PD	AVP V1 anta (SS faile dem effic in a New AV rece anta curr stud
Dopaminergic system	Amygdala, nucleus accumbens,	Excessive mesocortical dopamine	Buprop NDR prim

	prefrontal cortex	release, persistently high levels of dopamine in prefrontal cortex	dep som adju anxi
Serotonergic system	Dorsal raphe nuclei, amygdala, hippocampus, prefrontal cortex	Low activity of postsynaptic 5-HT1A receptors in PD and SAD	SSRIs first- for a disc
γ-amino butyric acid (GABA)	Substantia nigra, globus pallidus, hypothalamus, periaqueductal gray matter, hippocampus, amygdala, anterior cingulate	Reduced GABA-A and benzodiazepine binding in PD Reduced GABA levels in PD, possible imbalance between tonic GABAergic inhibition and glutamate-mediated excitation	Tiagab (sel reup inhil viga (inh GAE tran are trea Tiagab findi Topirar volta sodi chai pote GAE findi Compo sele spe rece

			subt deve **Note:** C and alth stru rela do r GAB
Glutamate	Amygdala, hippocampus, frontal and cingulate cortices	Possible imbalance between tonic GABAergic inhibition and glutamate-mediated excitation in PD	Efficacy adju exp ther acro and Glycine inhi stud Metabo rece mod und NMDA anta pote anxi prel effic riluz
Neuropeptide Y (NPY)	Amygdala, hippocampus, brainstem, nucleus accumbens,	Low NPY levels in PTSD, less well studied in anxiety disorders	Intrana adm may anxi inve

	locus coeruleus, hypothalamus		Y1 and rece pote for t
Galanin	Prefrontal cortex, amygdala, hippocampus, locus coeruleus	Very few studies in patients with anxiety disorders Galanin gene polymorphism associated with PD, only in women	Potenti mod futu no k stud
Cholecystokinin (CCK)	Cerebral cortex, hippocampus, amygdala, caudate, putamen, thalamus, hypothalamus	Lower CSF levels of CCK in PD	To date rece anta faile dem effic or P
Oxytocin	Hypothalamus, ventral tegmental area, amygdala	Oxytocin gene receptor polymorphism associated with increased risk for anxiety in individuals with early life stress	Intrana adm ben SAD stud ben
Endocannabinoid system	Prefrontal cortex, hypothalamus, amygdala, hippocampus	Dysregulation of endocannabinoid signaling	Cannal redu in pa SAD pub und

	Studies inhil pote trea way (no clas anxi nee anxi

GAD, generalized anxiety disorder; PD, panic disorder; SAD, social anxiety disorder; NMDA, *N*-methyl-D-aspartate; NDRI, norepinephrine–dopamine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor; SNRI, serotonin–norepinephrine reuptake inhibitor; DCS, D-cycloserine; FAAH, fatty acid amide hydrolase.

Preclinical Studies of Fear Learning. As we mentioned above, the ubiquity of fear and anxiety in many animals makes it relatively easy to study through animal models. Using conditioned models of fear, researchers can study the anatomical correlates of fear anxiety. It is daunting to summarize the many aspects of what we know about the etiology of fear and anxiety. In its broadest sense, one can imagine a stimulus, collected through afferent systems. That stimulus is then processed and evaluated to assess whether it is averse. This assessment involves including previous experiences and environmental contexts associated with the stimulus. What results is a fear or anxiety response, which employs a range of behavioral, endocrine, and autonomic responses to react to that stimulus.

Many areas of the brain and connections between those areas are involved in this response. For example, the amygdala plays a vital role. The amygdala has many subnuclei, called the amygdaloid complex, each with a unique function. The lateral nucleus of the amygdala (LA), for example, is the primary interface for visual, auditory, and somatic sensory information received from the thalamus and cortex. The connections between the thalamus and this nucleus appear to be central to fear conditioning related to these particular stimuli.

Once entering the L.A., it sends the neural representation of the stimulus to various other amygdaloid nuclei. It is then modulated by an array of systems,

including ones that supply context from memory, or about different homeostatic states in the body. Some of the most extensive projections are to the basal (BA) nuclei, which are involved in forming the long-lasting traces for fear conditioning. After much processing, including from other cortical regions, the output nuclei of the amygdala organize the behavioral responses that reflect the sum of the activity of the many nuclei. The central nucleus of the amygdala then outputs this to various motor, autonomic, and neuroendocrine systems involved in expressing fear. This includes the hypothalamus, midbrain, and medulla. For example, the hypothalamus can then activate the CRH to start the associated stress response. The hippocampus, among other structures, also has a critical role in fear learning and extinction and helps in the development of the emotional responses associated with fear.

This discussion is just a brief introduction into what we know and theorize about the neurobiology of anxiety disorders. Our *Comprehensive Textbook of Psychiatry* has a much expanded discussion as well as references to many of the relevant articles. Figure 8-1 summarizes the stages of the fear response, as well as some of the brain regions involved in these stages.

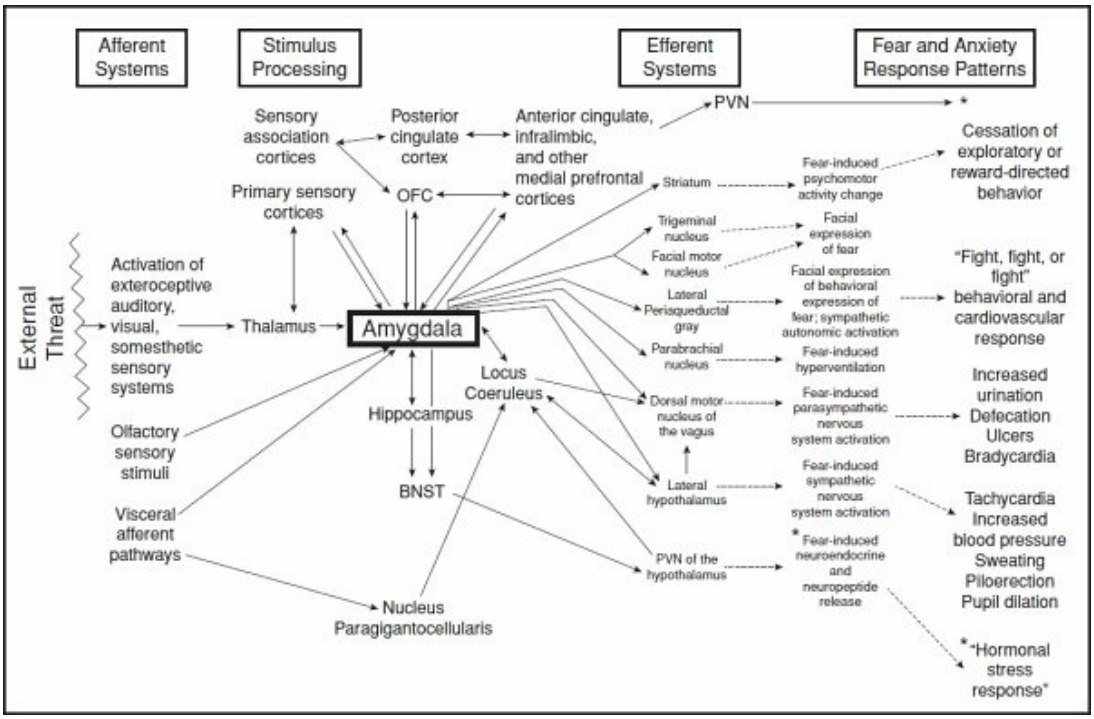


FIGURE 8-1
Neuroanatomical circuitry underlying fear and anxiety, organized by systems that evaluate the salience of potentially threatening stimuli (to the left) and systems

that organize and mediate the expression of fear and anxiety behaviors (to the right). BNST, bed nucleus of the stria terminalis; OFC, orbitofrontal cortex; PVN, paraventricular nucleus. (From Charney DS, Drevets WC. The neurobiological basis of anxiety disorders. In: Davis K, Charney DS, Coyle J, Nemeroff CB, eds. *Psychopharmacology. The Fifth Generation of Progress*. New York: Lippincott Williams & Wilkins; 2002:901.)

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9 Obsessive-Compulsive and Related Disorders

This chapter includes disorders characterized by repetitive, intrusive thoughts and/or repeated mental acts or behaviors. These disorders include obsessive-compulsive disorder (OCD), body dysmorphic disorder (BDD), hoarding disorder (HD), hair-pulling disorder (trichotillomania), and excoriation (skin-picking) disorder, as well as some related disorders.

THE CLINICAL PRESENTATION

Obsessive-Compulsive Disorder

OCD is a diverse group of symptoms that include intrusive thoughts, rituals, preoccupations, and compulsions. Obsessions are intrusive and unwanted repetitive thoughts, urges, or impulses that often lead to a marked increase in anxiety or distress. Compulsions are repetitive behaviors or mental acts that are done in response to obsessions, or in a rigid, rule-bound way. These recurrent obsessions or compulsions cause severe distress to the person. The obsessions or compulsions are time-consuming and interfere significantly with the person's routine, occupational functioning, usual social activities, or relationships. A patient with OCD may have only obsessions or only compulsions, but in most cases, both obsessions and compulsions are present.

Obsessions and compulsions are the essential features of OCD. An idea or an impulse intrudes itself insistently and persistently into a person's conscious awareness. Typical obsessions associated with OCD include thoughts about contamination ("My hands are dirty") or doubts ("I forgot to turn off the stove").

The typical patterns of OCD symptoms cluster within several limited dimensions, including contamination/cleansing, forbidden thoughts/checking, symmetry/ordering, and hoarding symptoms (see below). Most patients with OCD typically have good insight into the fact that their OCD beliefs are not real. However, some patients have poor insight, and a small number lack any insight—their obsessions are delusional. Such patients may be misdiagnosed with a psychotic disorder and receive inappropriate treatment.

The presentation of obsessions and compulsions is heterogeneous in adults (Table 9-1) and children and adolescents (Table 9-2). The symptoms of an individual patient can overlap and change with time, but OCD has four major symptom patterns, as outlined below.

Contamination/Cleansing. The most common pattern is an obsession of contamination, followed by washing or accompanied by compulsive avoidance of the presumably contaminated object. The feared object is often hard to avoid (e.g., feces, urine, dust, or germs). Patients may rub the skin off their hands by excessive handwashing or may be unable to leave their homes because of fear of germs. Although anxiety is the most common emotional response to the feared object, obsessive shame and disgust are also common. Patients with contamination obsessions usually believe that the contamination is spread from object to object or person to person by the slightest contact.

Pathologic Doubt/Checking. The second most common pattern is an obsession of doubt, followed by compulsions of checking. The obsession often implies some danger of violence (e.g., forgetting to turn off the stove or not locking a door). The checking may involve multiple trips back into the house to check the stove, for example. These patients have obsessional self-doubt and always feel guilty about having forgotten or committed something.

Intrusive/Forbidden Thoughts. In the third most common pattern, there are intrusive obsessional thoughts without a compulsion. Patients usually experience thoughts of a sexual or aggressive act that is reprehensible to the patient. Patients obsessed with thoughts of aggressive or sexual acts may report themselves to the police or confess to a priest. Suicidal ideation may also be obsessive (and unlikely to be acted on), but we should assess for actual risk.

Symmetry/Ordering. The fourth most common pattern is the need for symmetry or precision, which can lead to a compulsion of slowness. Patients can take hours to eat a meal or shave their faces.

Other Symptom Patterns. Religious obsessions and compulsive hoarding are common in patients with OCD. Compulsive hairpulling and nail-biting are behavioral patterns related to OCD. Masturbation may also be compulsive.

Ms. K was referred for psychiatric evaluation by her general practitioner. In the interview, Ms. K described a long history of checking rituals that had caused her to lose several jobs and had damaged numerous relationships. She reported, for example, that she often had the thought that she had not locked the door to the car. It was difficult for her to leave her car until she had repeatedly checked that it was secure. She had broken several car door handles with the vigor of her checking and had been up to an hour late to work because she spent too much time checking her car door.

Similarly, she had recurrent thoughts that she had left the door to her apartment unlocked. She returned several times to check her door before she left for work. She reported that checking doors decreased her anxiety about security. Ms. K reported that she had occasionally tried to leave her car or apartment without checking the door (e.g., when she was already late for work). However, she found that she became so worried that someone would steal her car or break into her apartment that she had difficulty going anywhere. Ms. K reported that her obsessions about security had become so extreme over the past 3 months that she had lost her job due to recurrent tardiness. She recognized the irrational nature of her obsessive concerns but could not bring herself to ignore them. (Courtesy of Erin B. McClure-Tone, Ph.D., and Daniel S. Pine, M.D.)



Table 9-1
Obsessive-Compulsive Symptoms in Adults

Variable	%
Obsessions (*N* = 200)	
Contamination	45
Pathologic doubt	42
Somatic	36
Need for symmetry	31
Aggressive	28
Sexual	26
Other	13
Multiple obsessions	60
Compulsions (*N* = 200)	
Checking	63
Washing	50
Counting	36
Need to ask or confess	31
Symmetry and precision	28
Hoarding	18
Multiple comparisons	48
Course of illness (*N* = 100)[a]	
Type	
Continuous	85
Deteriorative	10
Episodic	2
Not present	71

Present	29

[a]Age at onset: men, 17.5 ± 6.8 years; women, 20.8 ± 8.5 years.

From Rasmussen SA, Eisen JL. The epidemiology and differential diagnosis of obsessive compulsive disorder. *J Clin Psychiatry*. 1992;(53 Suppl):4–10, with permission.

Body Dysmorphic Disorder

Persons with BDD have persistent preoccupations about one or more perceived defects or flaws in one's appearance. The defects or flaws appear slight or are not observable to others. Their concerns about their appearance result in a range of mental acts or behaviors, including comparing themselves with others, checking in the mirror, or camouflaging their perceived flaws.

The most common concerns (Table 9-3) involve the face and head, particularly those involving specific parts (e.g., the skin, nose shape and size, hair), although symptoms may focus on any area of the body. Sometimes the concern is vague and confusing to understand, such as extreme concern over a "scrunchy" chin. Throughout their illness, patients worry about five to seven different body areas. More than a quarter of patients are concerned with the symmetry of their appearance.

Commonly associated symptoms include ideas or delusions of reference, frequent mirror checking or avoidance of reflective surfaces, and attempts to hide the presumed deformity. The ideas or delusions of reference are usually about others noticing the alleged flaw. The effects on a person's life can be significant, and it is essential to assess for avoidance due to BDD symptoms, which can range from minor social avoidance to being housebound. Patients with BDD have less insight into the truth of their beliefs than those with OCD. Only a quarter of BDD patients have reasonable insight, and around one-third have absent insight.

Table 9-2
Reported Obsessions and Compulsions for 70 Consecutive Child and Adolescent Patients

Major Presenting Symptom	No. (%) Reporting Symptom at Initial Interview[a]

Obsession	
Concern or disgust with bodily wastes or secretions (urine, stool, saliva), dirt, germs, environmental toxins	30 (43)
Fear something terrible may happen (fire, death or illness of loved one, self, or others)	18 (24)
Concern or need for symmetry, order, or exactness	12 (17)
Scrupulosity (excessive praying or religious concerns out of keeping with patient's background)	9 (13)
Lucky and unlucky numbers	6 (8)
Forbidden or perverse sexual thoughts, images, or impulses	3 (4)
Intrusive nonsense sounds, words, or music	1 (1)
Compulsion	
Excessive or ritualized handwashing, showering, bathing, toothbrushing, or grooming	60 (85)
Repeating rituals (e.g., going in and out of door, up and down from chair)	36 (51)
Checking doors, locks, stove, appliances, car brakes	32 (46)
Cleaning and other rituals to remove contact with contaminants	16 (23)
Touching	14 (20)
Ordering and arranging	12 (17)

Measures to prevent harm to self or others (e.g., hanging clothes a certain way)	11 (16)
Counting	13 (18)
Hoarding and collecting	8 (11)
Miscellaneous rituals (e.g., licking, spitting, special dress pattern)	18 (26)

[a]Multiple symptoms recorded, so total exceeds 70.

From Rapoport JL. The neurobiology of obsessive-compulsive disorder. *JAMA*. 1988;260:2889, with permission.

Ms. R, a 28-year-old single woman, presented with the complaint that she is "ugly" and that she feels others are laughing at her, despite family telling her she was attractive. She first became preoccupied with her appearance when she was 13. She became obsessed with her "facial defects," feeling, for example, that her nose was fat, or her eyes were too far apart. Up until this point, Ms. R was confident, a good student, and socially active. However, her fixation on her face caused her to withdraw socially and have difficulty concentrating in school, which in turn hurt her grades.

Ms. R dropped out of high school and went for her GED due to her preoccupation. She began to frequently pick at "blemishes" and hairs on her face. She frequently checked herself in mirrors and other reflective surfaces (e.g., spoons, windows), and found herself thinking about her defects almost daily. Her family and others could not convince her that there was nothing wrong with her appearance.



Table 9-3
Location of Imagined Defects in 30 Patients with Body Dysmorphic Disorder[a]

Location	*N*	%
Hair[b]	19	63

Nose	15	50
Skin[c]	15	50
Eyes	8	27
Head, face[d]	6	20
Overall body build, bone structure	6	20
Lips	5	17
Chin	5	17
Stomach, waist	5	17
Teeth	4	13
Legs, knees	4	13
Breasts, pectoral muscles	3	10
Ugly face (general)	3	10
Ears	2	7
Cheeks	2	7
Buttocks	2	7
Penis	2	7
Arms, wrists	2	7
Neck	1	3
Forehead	1	3
Facial muscles	1	3
Shoulders	1	3
Hips	1	3

[a]Total is greater than 100% because most patients had “defects” in more than one location.
[b]Involved head hair in 15 cases, beard growth in 2 cases, and other body hair in 3 cases.
[c]Involved acne in 7 cases, facial lines in 3 cases, and other skin concerns in 7 cases.

[d]Involved concerns with shape in 5 cases and size in 1 case.

From Phillips KA, McElroy SL, Keck PE Jr., Pope HG, Hudson JL. Body dysmorphic disorder: 30 cases of imagined ugliness. *Am J Psychiatry*. 1993;150:303, with permission.

Hoarding Disorder

Persons with HD have persistent and profound difficulty discarding or parting with their possessions. The disorder leads to significant congestion and clutter and substantive distress or impairment, differentiating it from regular collecting. Compulsive hoarding is a common and often disabling phenomenon associated with impairment in such functions as eating, sleeping, and grooming. Hoarding may result in health problems and poor sanitation, particularly when hoarding of animals is involved. It may also lead to injuries from fire or falling. As a rule, people with the disorder acquire things of little or no value and cannot throw them away.

What drives the behavior appears to be the fear of losing items that the patient believes will be needed later and a distorted belief about or an emotional attachment to possessions. Most hoarders do not perceive their behavior to be a problem. Many perceive their behavior to be reasonable and part of their identity. Most hoarding patients accumulate possessions passively rather than intentionally, thus clutter accumulates gradually. Commonly hoarded items include newspapers, mail, magazines, old clothes, bags, books, lists, and notes. Hoarding poses risks to not only the patient but also to those around them. Clutter accumulated from hoarding has caused deaths from fire or other accidents. It can also attract pest infestations that can pose a health risk both to the patient and residents around them. Many sufferers have been evicted from their homes or threatened with eviction as a result of their hoarding. In severe cases, hoarding can interfere with work, social interaction, and necessary activities such as eating or sleeping.

The pathologic nature of hoarding comes from the inability to organize possessions and keep them organized. Many patients hoard rather than make decisions about discarding items. Patients with HD also overemphasize the importance of recalling information and possessions. For example, a hoarder will keep old newspapers and magazines because they believe that important information may be lost. Patients may believe that forgetting the information will lead to severe consequences and prefer to keep their possessions within sight so as not to forget them.

Ms. T, a 55-year-old single woman, presented to a therapist accompanied by her adult son, who expressed concern about Ms. T's inability to "throw things away." He reported that Ms. T's home was extremely cluttered with "needless things." Whenever he attempted to help her "organize things," however, Ms. T would become agitated and argumentative. Ms. T confirmed her son's complaint and reported having this difficulty for as long as she could remember, but never really viewed it as a problem.

Over the past 5 years, Ms. T's home had become increasingly cluttered to the point that it became more and more challenging to move around within it. She was able to keep the kitchen and bathroom relatively clutter-free. However, boxes and bags filled with papers, magazines, clothes, and miscellaneous gifts and trinkets filled the rest of her home. Her living room was the most affected. Her son could no longer visit his mother as so uncomfortable at her home. Ms. T admits that his avoidance of her home has been a significant source of depression for her. Ms. T had enjoyed entertaining her family and friends but had not done so in years as she felt that her home was no longer suitable. She had made a few attempts to clean out her home but was unable to discard most items. When asked why she was keeping them, she replied, "I may need them later."

Hair-Pulling Disorder (Trichotillomania)

Hair-pulling disorder is a chronic disorder characterized by repetitive hairpulling, leading to variable hair loss that may be visible to others. It is also known as *trichotillomania*, from the Greek *trich* (hair) and *tillein* (to pull or pluck), a term coined by a French dermatologist Francois Hallopeau in 1889. Although once considered rare, it appears to be relatively common. It resembles both obsessive-compulsive and impulse control disorder: increased tension before the hairpulling leads to the behavior and then subsequent relief or satisfaction.

Hair-pulling disorder, or trichotillomania, is characterized by pulling out one's hair. The pulling is not for cosmetic reasons, but instead, individuals often describe an irresistible urge to pull out their hair. This hairpulling

results in noticeable hair loss and is associated with repeated attempts to decrease or stop hairpulling. Patients with hair-pulling disorder may experience an increasing sense of tension before engaging in the behavior and achieve a sense of release or gratification from pulling out their hair. All areas of the body may be affected, most commonly the scalp (Fig. 9-1). Other areas involved are eyebrows, eyelashes, and beard; trunk, armpits, and pubic areas are less commonly involved.

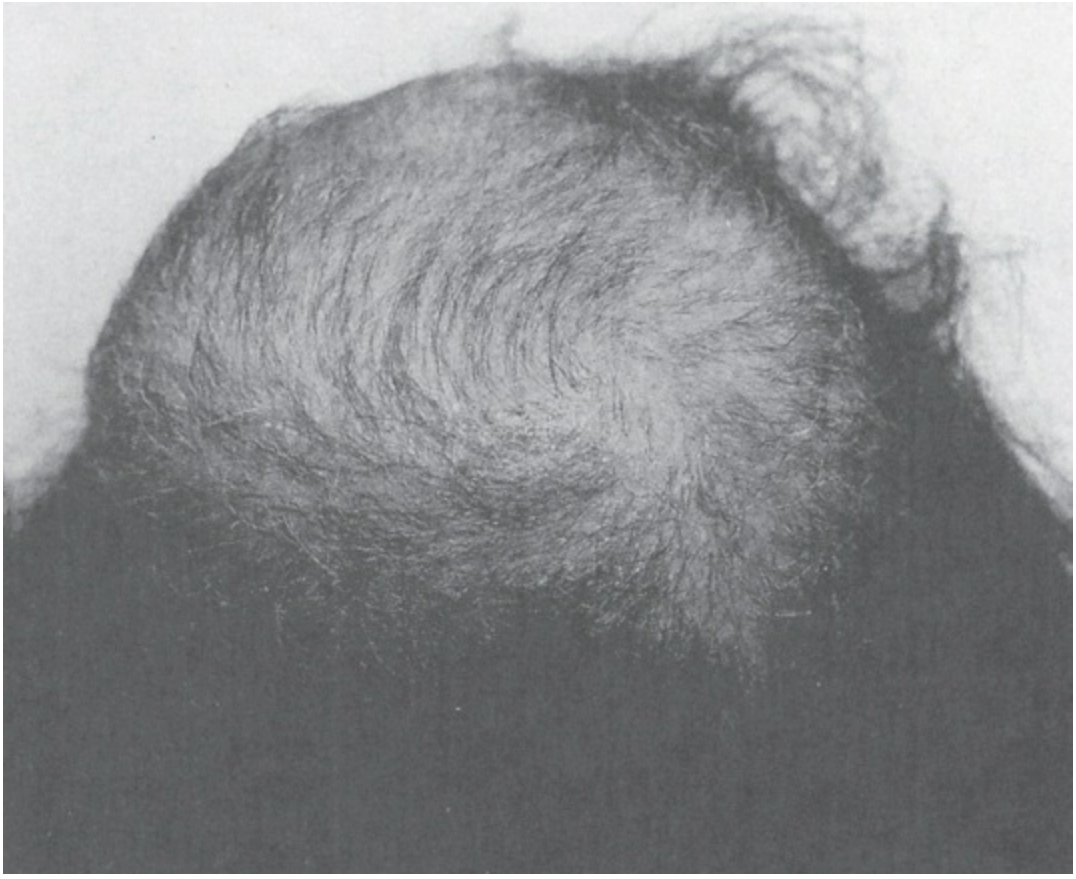


FIGURE 9-1
Hair-pulling disorder (trichotillomania). Note the typical findings of an area of incomplete alopecia involving the frontal and vertex scalp. (From Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.)

There are (at least) two types of hairpulling. *Focused pulling* is the use of an intentional act to control unpleasant personal experiences, such as an urge, bodily sensation (e.g., itching or burning), or thought. In contrast, *automatic pulling* occurs outside the person's awareness and, most often, during

sedentary activities. Most patients have a combination of these types.

The hair loss usually appears as short, broken strands near long, healthy hairs in the affected areas. No abnormalities of the skin or scalp are present. The hairpulling is usually painful, although pruritus and tingling may occur in the affected area. *Trichophagy,* mouthing of the hair, may follow the hair plucking. Complications of trichophagy include trichobezoars, malnutrition, and intestinal obstruction. Patients usually deny the behavior and often try to hide the resultant alopecia. The patients may also self-mutilate in other ways, such as headbanging, nail-biting, scratching, gnawing, or excoriation.

Ms. C, a 27-year-old single woman, came to a local clinic complaining of persistent hairpulling. She first started at age 11 when she began to pick the hairs at the nape of her neck. She would persistently pick at the hair until there was almost none left. Fortunately, her hair was long, so no one noticed the lack of hair at the back of her neck. Over the years, her hair picking progressed until she began picking hair from her entire head, leaving noticeable small bald patches. She strategically hid the bald patches by brushing over the remainder of her hair or with carefully placed scarves and hats. Despite her habit, Ms. C's daily functioning was average. She earned good grades in school and was a year away from getting her master's degree.

Ms. C's habit was constant, occurring every day, often without her noticing it. She could only be reading an assignment for school, and eventually, her hand would find its way into her hair to find hair to pull. Soon she would notice a small pile of hairs in her book or on her lap, indicating that she had been pulling her hair out for a while. Whenever she tried to stop herself from pulling her hair, she would become increasingly nervous and anxious until she resumed the hairpulling. Her hair-pulling sessions lasted anywhere from 10 minutes to an hour.

Excoriation (Skin-Picking) Disorder

People with excoriation disorder repeatedly pick their skin, sufficiently to cause lesions. Individuals with skin-pulling attempt to decrease or stop skin-picking; however, they are unable to do so. While some picking of scabs is

likely universal, in the case of excoriation disorder, the behavior causes clinically significant distress or impairment. The historical literature describes cases of “neurotic excoriation” or “psychogenic excoriation,” however, it was only with *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) that the disorder became a separate diagnosis.

Skin-picking may be from any area of the body but is most commonly from the face, followed by the hands, fingers, arms, and legs (Fig. 9-2). Patients may pick from multiple sites. In severe cases, skin-picking can result in physical disfigurement and medical consequences that require medical or surgical interventions (e.g., skin grafts or radiosurgery).

Patients may experience tension before picking and relief or gratification after picking. Patients may pick to relieve stress, tension, and other negative feelings. Despite the feeling of relief felt from picking, patients often feel guilty or embarrassed at their behavior. Up to 87 percent of patients report feeling embarrassed by the picking and 58 percent report avoiding social situations. Many patients use bandages, makeup, or clothing to hide their picking. Of skin-picking patients, 15 percent report suicidal ideation due to their behavior, and about 12 percent have attempted suicide.

Ms. J, a 22-year-old single woman, presented to a psychiatrist at the urging of her dermatologist because of compulsively picking her facial skin. She picked at it every day up to three times a day in sessions lasting from 20 minutes to over an hour. She had massive scarring and lesions on her face. She went to a physician 6 months prior when one of the lesions had become infected.

Ms. J began picking her face at age 11 at the onset of puberty. At first, she only picked at acne that formed on her face, but as the urge to pick became greater, she started picking at clear patches of skin as well. Due to the scaring and lesions, Ms. J became increasingly withdrawn and avoided all social engagements. She reported feeling great tension before picking, and only felt relief after she began picking.



FIGURE 9-2
Skin-picking disorder. Multiple erythematous and pigmented maculae and crusted erosions on chin. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009, with permission.)

Obsessive-Compulsive or Related Disorder due to Another Medical Condition

As with most psychiatric disorders, when OCD (or another disorder in this category) is due to some other medical condition, we use a different diagnosis. We include a discussion of medical causes of obsessive and compulsive symptoms in our section on differential diagnosis.

Substance-Induced Obsessive-Compulsive or Related Disorder

Sometimes the symptoms described in this chapter may be due to a substance, including drugs, medications, and alcohol (either during use or during withdrawal), and is not part of a delirium. In such cases, we would use this separate diagnosis.

Other Specified Obsessive-Compulsive or Related Disorder

This category is for patients who have symptoms characteristic of obsessive-compulsive and related disorder but do not meet the full criteria for any specific obsessive-compulsive or related disorder. This diagnosis is appropriate under three situations: (1) an atypical presentation, (2) another specific syndrome not listed in DSM-5 (such as Olfactory Reference Syndrome, described next), and (3) the information presented is insufficient to make a full diagnosis of an obsessive-compulsive or related disorder.

Olfactory Reference Syndrome. People with olfactory reference syndrome have a false belief that they have a foul body odor that is not perceived by others. The preoccupation leads to repetitive behaviors such as washing the body or changing clothes. Patients with the disorder may have good, fair, poor, or absent insight into the behavior. The syndrome is predominant in males and single persons. The mean age of onset is 25 years of age. The belief of a subjective sense of smell that does not exist externally may rise to the level of a somatic delusion, in which a diagnosis of a delusional disorder may be more appropriate. The syndrome has been frequently described in the psychiatric literature, usually classified as a delusion of perception. Whether or not it deserves a unique diagnosis remains controversial.

In assessing a patient with olfactory reference syndrome, it is essential to exclude other medical causes. Some patients with temporal lobe epilepsy may complain of smelling foul odors. Local irritations of the hippocampus from pituitary tumors may also cause olfactory sensations. Patients with inflammation of the frontal, ethmoidal, or sphenoidal sinuses may also have a subjective sense of offensive odors. DSM-5 includes olfactory reference syndrome in the “other specified” designation for the obsessive-compulsive and related disorder of DSM-5.

DIAGNOSIS

Obsessive-Compulsive Disorder

Table 9-4 lists the diagnostic approaches to OCD in DSM-5 and ICD-10. We can usually diagnose OCD with a thorough psychiatric history and

examination. Patients often attempt to neutralize obsessions with compulsions. These compulsions are usually not credibly connected to the thoughts, or they are excessive. When assessing the negative impact of OCD symptoms, it is important also to address avoidance of triggers of obsessions and compulsions.

Body Dysmorphic Disorder

Table 9-5 lists the diagnostic approaches to BDD in DSM-5 and ICD-10. The DSM-5 diagnostic criteria for BDD stipulate preoccupation with a perceived defect in appearance or overemphasis of a slight defect. It also requires that at some point during the disorder, the patient has compulsive behaviors or mental acts related to the preoccupation. These compulsions may include mirror checking, excessive grooming, or comparing their appearance to that of others. The preoccupation causes patients significant emotional distress or markedly impairs their ability to function in important areas.

Hoarding Disorder

Table 9-6 lists the diagnostic approaches to HD in DSM-5 and ICD-10. We diagnose HD using the significant features of the disorder: patients acquire large amounts of useless or valueless possessions, they cannot throw them away, and this leads to significant clutter in living areas. If their space is not cluttered, it is because others have stepped in to clean it. Finally, it should cause significant distress or functional impairment. The fifth edition of the DSM-5 includes diagnostic specifiers that relate to insight, which may be rated poor, fair, or good. Some patients are completely unaware of the full extent of the problem and resistant to treatment. At times, delusional beliefs about hoarded items are present.

Hair-Pulling Disorder (Trichotillomania)

Table 9-7 lists the diagnostic approaches to hair-pulling disorder in DSM-5 and ICD-10. It emphasizes the repetitive and compulsive nature of the behavior as well as the fact that it is not motivated by cosmetic reasons.

Excoriation (Skin-Picking) Disorder

Table 9-8 lists the diagnostic approaches to excoriation (skin-picking)

disorder in DSM-5 and ICD-10. DSM-5 diagnostic criteria for skin-picking disorder require recurrent skin-picking resulting in skin lesions and repeated attempts to decrease or stop picking. The skin-picking must cause clinically relevant distress or impairment in functioning. The skin-picking behavior cannot be attributed to another medical or mental condition and cannot be a result of a substance use disorder (e.g., cocaine or methamphetamine use). A thorough physical examination is crucial before a psychiatric diagnosis.

DIFFERENTIAL DIAGNOSIS

Medical Conditions

Several primary medical disorders can produce syndromes bearing a striking resemblance to OCD.

OCD-like symptoms have been reported in children following group A beta-hemolytic streptococcal infection and have been called *pediatric autoimmune neuropsychiatric disorders associated with streptococcus* (PANDAS). They may result from an autoimmune process that leads to inflammation of the basal ganglia that disrupts cortical–striatal–thalamic axis functioning.

The current conceptualization of OCD as a disorder of the basal ganglia derives from the phenomenologic similarity between idiopathic OCD and OCD-like disorders that are associated with basal ganglia diseases, such as Sydenham chorea and Huntington disease. It is essential to search for the neurologic signs of these disorders when assessing a patient with presumed OCD. Also, OCD often develops before age 30 years, and new-onset OCD in an older individual should raise questions about potential neurologic contributions to the disorder.



Table 9-4
Obsessive-Compulsive Disorder

	Obsessive-Compulsive Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Obsessive-Compulsive Disorder	Obsessive-Compulsive Disorder

Symptoms	1. Obsessions (Intrusive thoughts that accompany behaviors and that individuals may attempt to ignore or counteract through action) 2. Compulsions (repetitive actions, commonly driven by an obsessive thought, performed to relieve anxiety or fear)	• Recurrent obsessional thoughts and/or compulsive acts • Obsessions: ideas, images, or impulses occurring recurrently • Compulsions: repeated stereotyped behaviors meant to neutralize a thought/anxiety or prevent a perceived adverse event • Attempts at resisting thoughts and behaviors are frequently unsuccessful and cause worsening anxiety
Required number of symptoms	Obsessive thoughts, compulsions, or both	Presence of obsessions, compulsions, or both
Psychosocial consequences of symptoms	Symptoms and behaviors consume at least 1 hr/day or cause significant distress and impairment	

Exclusions (not result of):	Medical illness Substance Other psychiatric disorder	Obsessive-compulsive personality disorder
Symptom specifiers	**Tic related** • current or prior tic disorder **With good or fair insight** • understanding that beliefs surrounding actions may not be true **With poor insight** • feeling that beliefs surrounding actions are likely true **With absent insight and/or delusional beliefs** • strong conviction that beliefs surrounding actions are true	• Predominantly obsessional thoughts or ruminations • Predominantly compulsive acts (obsessional rituals) • Mixed obsessional thoughts and acts
Severity specifiers	With or without insight (per above)	

Many medical disorders can cause alopecia. A biopsy may be necessary to distinguish hair-pulling disorder from alopecia areata and tinea capitis.

Many dermatologic conditions can cause skin-picking. Conditions associated with itching (such as scabies) can cause picking, as well as other medical conditions (such as in Prader–Willi syndrome).

Tourette Disorder. OCD is closely related to Tourette disorder, as the two conditions frequently co-occur, both in individuals over time and within families. About 90 percent of persons with Tourette disorder have compulsive symptoms, and as many as two-thirds meet the diagnostic criteria for OCD.

Classically, patients with Tourette disorder have recurrent vocal and motor tics that bear only a slight resemblance to OCD. The premonitory urges that precede tics often strikingly resemble obsessions, however, and many of the more complicated motor tics are very similar to compulsions.

Other Psychiatric Conditions

Many other psychiatric disorders include obsessive-compulsive symptoms, and we must also rule out these conditions when diagnosing OCD. OCD exhibits a superficial resemblance to obsessive-compulsive personality disorder, which is associated with an obsessive concern for details, perfectionism, and other similar personality traits. However, only OCD is associated with genuine obsessions and compulsions. Worries in generalized anxiety disorder are about real-life concerns, and they tend to be less irrational and ego-dystonic than in OCD. Similarly, the ruminations of depression and preoccupations of mania are typically mood-congruent and ego-syntonic; in neither case are symptoms neutralized by compulsions.

Psychotic symptoms often lead to obsessive thoughts and compulsive behaviors, bodily preoccupations, or hoarding behaviors that can be difficult to distinguish from the obsessive-compulsive and related disorder. Patients with psychotic disorders, however, cannot acknowledge the unreasonableness of their behavior. Also, psychotic patients typically have a host of other features that are not characteristic of OCD.

In some patients with BDD, their preoccupations have a delusional intensity. In such cases, delusional disorder, somatic type can be made, generally in addition to BDD, to capture the true nature of the patient's presentation.



Table 9-5
Body Dysmorphic Disorder

Body Dysmorphic Disorder

Disorder	DSM-5	ICD-10
Diagnostic name	Body Dysmorphic Disorder	Body Dysmorphic Disorder
Symptoms	1. Preoccupation with physical appearance and subjective flaws 2. Presence of repetitive behaviors at some point during the disorder • Checking • Grooming • Skin-picking • Recurrent thoughts/perseveration on comparing physical appearance to that of others	Excessive preoccupation with image or appearance
Required number of symptoms	Both of the above, causing significant impairment and/or distress and excluding other causes of symptoms	
Psychosocial consequences of symptoms	Symptoms and behaviors cause significant distress and impairment	Significant distress or impairment in areas of functioning
Exclusions (not result of):	Medical illness Substance Eating disorder Other psychiatric disorder	

Symptom specifiers	**With muscle dysmorphia** • preoccupation with perceived low muscle mass or small body build **With good or fair insight** • understanding that beliefs may not be true **With poor insight** • feeling that beliefs are likely true **With absent insight and/or delusional beliefs** • strong conviction that beliefs are true
Severity specifiers	With or without insight (per above)

Similarly, major depression is often associated with obsessive thoughts that, at times, border on actual obsessions. However, in the case of major depression, the obsessive symptoms are only found during a depressive episode.

Individuals with an avoidant personality disorder or social phobia may worry about being embarrassed by imagined or real defects in appearance. However, this concern is usually not prominent, persistent, distressing, or impairing.

Other psychiatric disorders may cause hoarding symptoms, including autism spectrum disorders (where collections may reflect a specific interest) and psychotic disorders. We should only diagnose HD if the hoarding symptoms are independent of these other disorders.

Neurodevelopmental and neurodegenerative conditions may be associated with hoarding. For example, in Prader–Willi Syndrome, individuals are characterized by hoarding of food, obesity, and skin-picking. Alzheimer disease is another example. Damage to specific brain regions (anterior ventromedial prefrontal and cingulate cortices) may cause excessive and

indiscriminate hoarding activity.

Patients with anorexia nervosa have a bodily preoccupation, but the focus is on weight. In gender dysphoria, it is a discomfort with, or a sense of wrongness about, one's primary and secondary sex characteristics. In major depressive disorder, there may be mood-congruent cognitions involving appearance that occur exclusively during a major depressive episode.

Olfactory reference syndrome may result in other compulsive or repetitive behaviors such as washing or changing one's clothes. Individuals with BDD have obsessional preoccupations about their appearance and may have associated compulsive behaviors (e.g., mirror checking). We should only include multiple diagnoses when the specific focus of concerns of the obsessions extends beyond the usual focus of that disorder.

Some patients with OCD may hoard as a response to an unwanted obsession. This situation is most familiar with obsessions around symmetry, incompleteness, or fears of harm or contamination (e.g., keeping an object in case discarding that object leads to harm). In contrast, in HD, there is no associated obsession; instead, the individual wishes to retain saved items and is distressed by the thought of discarding them. In some cases, both OCD and HD may be comorbid. Skin-picking may also occur in response to obsessions about contamination. Skin-picking is also common in BDD, but in these individuals, it is a response to concerns about body appearance, and a second diagnosis is not necessary.

Patients with factitious dermatitis or *dermatitis artefacta* pick their skin as a type of self-inflicted injury and use more elaborate methods than simple excoriation to self-induce skin lesions. It is seen in 0.3 percent of dermatology patients and has a female:male ratio of 8:1. It can present at any age but occurs most frequently in adolescents and young adults. It may include a variety of skin lesions, such as blisters, ulcers, erythema, edema, purpura, and sinuses. The morphology of factitious dermatitis lesions is often bizarre and linear, with clear-cut, angulated, or geometric edges. The presence of entirely healthy, unaffected skin adjacent to the horrific-looking lesions is a clue to the diagnosis of factitious dermatitis (Fig. 9-3). Also, the patient's description of the history of the lesions is usually vague and lacks detail about the appearance and evolution of the lesions.

Table 9-6
Hoarding Disorder

Disorder	Hoarding Disorder	
	DSM-5	**ICD-10**
Diagnostic name	Hoarding Disorder	Hoarding Disorder
Symptoms	1. Excessive attachment to possessions 2. Distress related to being separated from items 3. Excessive accumulation of possessions 4. Accumulation results in significant distress, impairment, and/or safety/health compromise	• Difficulty parting with objects due to perceived need to save possessions • Distress and functional impairment associated with being separated from possessions
Required number of symptoms	All of the above	
Psychosocial consequences of symptoms	Symptoms and behaviors cause significant distress and impairment	Symptoms and behaviors cause significant distress and impairment
Exclusions (not result of):	Medical illness Neurocognitive disorder Other psychiatric disorder	
Symptom specifiers	**With excessive acquisition** • hoarding	

	accompanied by excessive acquisition of additional possessions **With good or fair insight** • understanding that actions are problematic **With poor insight** • belief that actions may not be problematic **With absent insight and/or delusional beliefs** • strong conviction that actions are not problematic
Severity specifiers	**With or without insight (per above)**

Normal Behaviors

Certain normal behaviors can become more prominent in specific individuals without being a disorder. We live in a society concerned with appearance; some persons are more concerned than others. Some of us are more careful about avoiding contamination or germs and may wash or avoid touching potentially infected persons or items. Living during a global pandemic naturally blurs the line between contamination phobia and understandable caution. We expect this overlap between pathologic and prudent to continue long after the pandemic. Many of us occasionally remove stray hair, often for cosmetic reasons. We may bite at a fingernail or pick at a scab. Collecting is an enjoyable and sometimes lucrative hobby for many people.

Table 9-7
Hair-Pulling Disorder (Trichotillomania)

	Trichotillomania	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Trichotillomania (Hair-Pulling Disorder)	Trichotillomania
Symptoms	1. Recurrent hairpulling causing hair loss 2. Hairpulling despite repeated attempts to change/end behaviors 3. Actions cause distress and/or impaired functioning	• Presence of noticeable hair loss due to recurrent hairpulling • Hair-pulling behavior preceded by increased anxiety that is relieved by hairpulling
Required number of symptoms	All of the above	
Psychosocial consequences of symptoms	Symptoms and behaviors cause significant distress and impairment	
Exclusions (not result of):	Medical illness Other psychiatric disorder	Pre-existing inflammation of skin Response to delusion or hallucination Stereotyped movement disorder

Table 9-8
Excoriation (Skin-Picking)

	Excoriation Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Excoriation (Skin-Picking) Disorder	Excoriation (Skin-Picking) Disorder
Symptoms	1. Recurrent skin-picking causing lesions to the skin 2. Ongoing skin-picking despite repeated attempts to change/end behaviors 3. Actions cause distress and/or impaired functioning	See criteria for obsessive-compulsive disorders
Required number of symptoms	All of the above	
Psychosocial consequences of symptoms	Symptoms and behaviors cause significant distress and impairment	
Exclusions (not result of):	Other psychiatric disorder	Factitious dermatitis

In each case, the behavior is desired, usually restricted (e.g., a collector may only collect certain toys) and often enjoyable. These disorders, however, represent impulsive behaviors that cause distress and impairment.

COMORBIDITY

Obsessive-Compulsive Disorder

Persons with OCD are commonly affected by other mental disorders. The lifetime prevalence for major depressive disorder in persons with OCD is

about 67 percent and for social phobia about 25 percent. Suicide is a risk for all patients with OCD. Other common comorbid psychiatric diagnoses in patients with OCD include alcohol use disorders, generalized anxiety disorder, specific phobia, panic disorder, eating disorders, and personality disorders. OCD exhibits a superficial resemblance to obsessive-compulsive personality disorder, which is associated with an obsessive concern for details, perfectionism, and other similar personality traits. The incidence of Tourette disorder in patients with OCD is 5 to 7 percent, and 20 to 30 percent of patients with OCD have a history of tics.

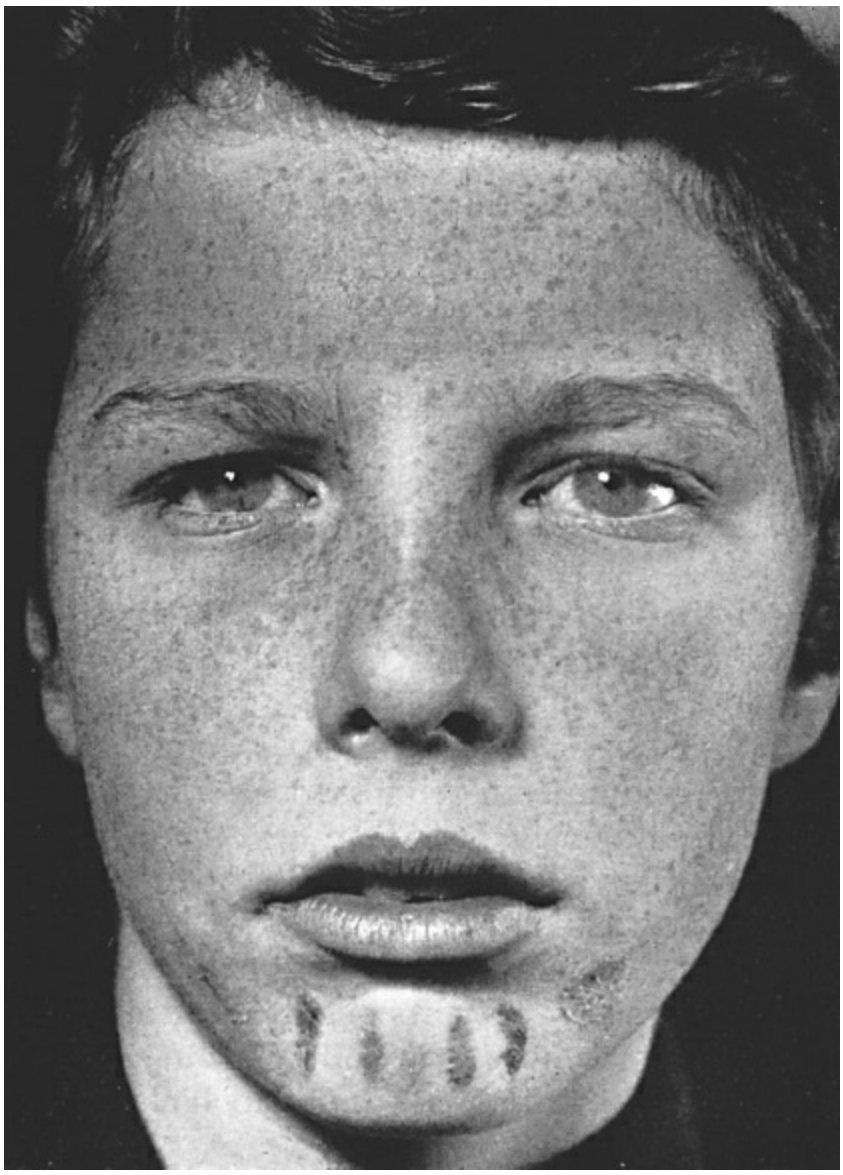


FIGURE 9-3
Typical self-produced lesions with scabbing. (From Douthwaite AH, ed. *French's*

Index of Differential Diagnosis. 7th ed. Baltimore, MD: Williams & Wilkins; 1954, with permission.)

Body Dysmorphic Disorder

Of the various obsessive-compulsive and related disorders, it appears that BDD is the one most associated with comorbid depression with a lifetime rate of 75 percent, as well as with suicidality, with 80 percent having experienced suicidal ideation. About one-third of BDD patients have a lifetime history of OCD, and around 30 percent of BDD patients experience panic attacks triggered by appearance concerns. BDD is also associated with high levels of rejection sensitivity and low self-esteem. Many patients with BDD use substances to self-medicate symptoms of social anxiety or emotional pain.

Hoarding Disorder

Comorbid OCD is common in patients with HD. Other prevalent comorbid disorders include generalized anxiety disorder and major depressive disorder.

Hair-Pulling Disorder

Comorbid conditions include other body-focused repetitive behavior disorders, with excoriation disorder the most common. OCD is also more prevalent in hair-pulling disorder than in the general population. More than half of treatment-seeking individuals with hair-pulling disorder have a comorbid psychiatric disorder, with mood and anxiety disorders the most common. Also, hair-pulling disorder can result in medical sequelae (e.g., trichobezoars) and decreased quality of life.

Excoriation Disorder

Comorbid conditions in excoriation disorder include other body-focused repetitive behavior disorders, with hair-pulling disorder the most common. OCD and BDD are also more prevalent in excoriation disorder than in the general population. Other common comorbid psychiatric disorders in excoriation disorder include mood and anxiety disorders.

COURSE AND PROGNOSIS

Obsessive-Compulsive Disorder

The course of OCD is generally chronic, and in the absence of intervention, symptoms may persist for decades. Nevertheless, recent data emphasize that long-term outcomes can be positive. Thus, patients should be encouraged to remain hopeful and to try a range of different evidence-based treatments.

Body Dysmorphic Disorder

The course of BDD is generally chronic, and in the absence of intervention, symptoms may persist for decades. Earlier age of onset and more severe symptoms at intake may predict a worse course.

Hoarding Disorder

HD has a somewhat later course than most obsessive-compulsive and related disorders. In the absence of intervention, its course appears to be often chronic and progressive. Hoarding symptoms often begin in childhood and adolescence, with diagnostic criteria met only in the 30s, and worsening of symptoms in each subsequent decade.

Hair-Pulling Disorder

There are relatively little data on the long-term course of hair-pulling disorder. In the absence of intervention, its course is likely often chronic. However, some data indicate that in a proportion of cases, remission does occur.

Excoriation Disorder

Those data which exist on the long-term course of excoriation disorder suggest that, without intervention, this is often chronic, albeit with fluctuations in severity over time. As is the case in other obsessive-compulsive and related disorders, individuals often report that they have not sought medical attention because they are embarrassed by their symptoms, feel that they should stop the symptoms by themselves, or because they are unaware that the symptoms may be part of a known condition or that there are efficacious treatments.

TREATMENT

Obsessive-Compulsive Disorder

Pharmacotherapy. Effective medications include clomipramine and selective serotonin reuptake inhibitors (SSRIs, which, along with clomipramine and other serotonergic medications, are sometimes called serotonin reuptake inhibitor [SRIs]). Head-to-head comparisons of clomipramine and SSRIs show equal efficacy but better tolerability of the SSRIs. In general, clinicians must use higher doses to achieve an optimal effect. Examples are doses of 80 mg of fluoxetine or 200 mg of sertraline. Some patients respond to even higher doses. Exceeding the recommended doses is not, however, advisable for clomipramine or citalopram, due to safety concerns. There is also some evidence that response to SRIs is slower in OCD than in depression; clinicians are therefore encouraged to prescribe an SRI for at least 12 weeks before deciding to switch agents or augment with another psychotropic. A general rule of thumb is that we should continue to use the dose that was required to obtain a response for at least 1 to 2 years. When discontinuing medication, there is a real risk of relapse, so this should be done very gradually, with small adjustments made every few months.

Patients who fail to respond to one SRI may respond to another.

The most extensive evidence base is for antipsychotics in adult OCD. Among those, risperidone and aripiprazole are best supported as adjuncts. Some studies have indicated that these are particularly useful in patients with tics, but not all data are consistent. When a response to these agents occurs, it typically happens relatively quickly (e.g., 4 weeks). When using dopamine receptor blockers, it is crucial to monitor adverse events.

Memantine, riluzole, ketamine, lamotrigine, and *N*-acetylcysteine (NAC) are other promising augmentation agents.

Psychotherapy. There is a long history of understanding OCD symptoms from a psychodynamic perspective, but there is no rigorous evidence that a psychodynamic approach is useful in the treatment of OCD. Instead, empirical research on the psychotherapy of OCD has emerged from behavioral and cognitive perspectives. Exposure and response prevention

(ERP) has proven to be a particularly effective component of treatment, with more recent work also focusing on the potential role for cognitive and mindfulness approaches. The essential principle of ERP is to minimize avoidance of distressing cues, and instead to encourage in vivo exposure to feared situations and imaginal exposure to feared consequences. After such exposures, it is important not to engage in the usual compulsive response. The therapist and patient then develop a hierarchy of gradual exposure to more distressing cues and follow it. As in the case of pharmacotherapy, it is essential to ensure that the dose and duration of ERP are sufficient.

Other cognitive approaches to OCD focus on thought distortions (e.g., inflated responsibility). The therapist and patient collaborate to test such distortions and replace them with more adaptive thinking patterns. More recently, mindfulness techniques have supplemented these approaches. Family sessions may also be helpful, particularly with younger patients, to better understand OCD and the principles of ERP.

Combined Treatment. Clinical guidelines indicate that either pharmacotherapy or psychotherapy is a reasonable first-line intervention for the management of OCD. In younger patients, it seems reasonable to begin cognitive-behavioral therapy (CBT) first. In patients with more severe symptoms or comorbid depression, it seems reasonable to begin pharmacotherapy first. There is some evidence that the combination of CBT and pharmacotherapy is particularly useful.

Other Somatic Therapies. Targeted neurosurgical lesions to cortico-striatal-thalamic-cortical (CSTC) tracts have been shown efficacious in a proportion of highly treatment-refractory patients. Approaches have included anterior cingulotomy and capsulotomy, among other techniques. The more recent focus on neurosurgical work has been on deep brain stimulation, with implantation of electrodes to stimulate specific brain regions. Transcranial magnetic stimulation has also been studied for OCD, although data are not yet sufficiently strong to motivate for routine clinical use. Electroconvulsive therapy does not seem to be useful for OCD.

Body Dysmorphic Disorder

SSRIs are efficacious in the treatment of BDD, including delusional BDD.

There is some evidence that, as in the case of OCD, doses higher than those ordinarily used in the treatment of depression are beneficial in BDD, as are longer durations. Again, the duration of treatment should be longer, with gradual tapering of dosage and close monitoring of emergent symptoms.

Patients who do not respond to one SRI may respond to another. There is anecdotal evidence that buspirone augmentation of SRIs is useful in partial responders. There is little controlled data on using dopamine receptor blockers as augmenting agents; what exists is not encouraging. However, clinicians have reported anecdotal evidence for using aripiprazole as an augmenting agent. Also, irreversible MAOIs have some positive, although experiential, reports.

Relation to Plastic Surgery. Few data exist about the number of patients seeking plastic surgery who have BDD. One study found that only 2 percent of the patients in a plastic surgery clinic had the diagnosis, but DSM-5 reports the figure to be 7 to 8 percent. The overall percentage may be much higher, however. Surgical requests are varied: removal of facial sags, jowls, wrinkles, or puffiness; rhinoplasty; breast reduction or enhancement; and penile enlargement. Men who request penile enlargements and women who request cosmetic surgery of the labia of the vagina or the lips of the mouth may be suffering from this disorder. Commonly associated with the belief about appearance is an unrealistic expectation of how much surgery will correct the defect. As reality sets in, the person realizes they have not solved their problems by merely altering a perceived cosmetic defect. These patients, hopefully, will explore their feelings of inadequacy using psychotherapy. The patients may take out their unfulfilled expectations and anger by taking legal action against their plastic surgeons or becoming depressed.

Hoarding Disorder

HD is challenging to treat. Although it shows similarities to OCD, effective treatments for OCD have shown little benefit for patients with HD. In one study, only 18 percent of patients responded to medication and CBT. The challenges posed by hoarding patients to typical CBT treatment include poor insight into the behavior and low motivation and resistance to treatment.

The most effective treatment for the disorder is a cognitive-behavioral

model that includes training in decision making and categorizing; exposure and habituation to discarding; and cognitive restructuring. Treatment includes both office and in-home sessions. The role of the therapist in this model is to assist in the development of decision-making skills, to provide feedback about normal saving behavior, and to identify and challenge the patient's erroneous beliefs about possessions. The goal in treatment is to get rid of a significant amount of possessions and make the living space livable. Also, treatment can provide the patients with the skills to maintain a positive balance between the number of possessions and livable space.

Evidence for pharmacologic treatment relies on limited and uncontrolled data that support the use of SSRIs or venlafaxine.

Hair-Pulling Disorder

Early data indicated that hair-pulling disorder, like OCD and BDD, responded more robustly to clomipramine than to desipramine. Unfortunately, subsequent trials of SSRIs and venlafaxine in hair-pulling disorder have not consistently demonstrated efficacy. The most significant positive clinical trial to date is with NAC (1200 to 2400 mg/day), a nutraceutical that acts on the glutamatergic system. However, a trial of NAC in pediatric hair-pulling disorder was negative. There are small controlled trials indicating that dopamine receptor blockers may also be useful in these conditions. Given their favorable side-effect profile, it may be reasonable to use NAC as a first-line agent in adults with hair-pulling disorder, to use SSRIs when patients have comorbid conditions for which these agents are efficacious, and to consider low-dose dopamine receptor blockers in more treatment-refractory cases.

Habit reversal training (HRT) is a set of cognitive-behavioral techniques that have been shown efficacious in the treatment of childhood and adult hair-pulling disorder. Components of HRT include awareness training, competing for response training, and social support. Also, therapists use stimulus control (SC) to change the patient's environment so that hairpulling is more effortful or less reinforcing. Controlled trials of augmentation of HRT/SC with acceptance and commitment therapy, dialectical behavior therapy, or cognitive therapy, have also suggested efficacy in hair-pulling disorder.

Excoriation Disorder

Excoriation disorder is challenging to treat, and there are few data on effective treatments. Most patients do not actively seek treatment due to embarrassment or because they believe their condition is untreatable. There is support for the use of SSRIs. Studies comparing fluoxetine against placebo have shown fluoxetine to be superior in reducing skin-picking.

Other agents have limited data. Lamotrigine has demonstrated inconsistent results. Glutamatergic agents, such as the nutraceutical NAC, have some positive, although anecdotal reports, as do dopamine receptor blockers. Nonpharmacologic treatments include HRT and brief CBT.

EPIDEMIOLOGY

Obsessive-Compulsive Disorder

The rates of OCD are relatively consistent, with a lifetime prevalence in the general population estimated at 2 to 3 percent. Community studies indicate a higher prevalence of OCD in females, but in clinical contexts, there are roughly equal numbers of male and female patients. The mean age of onset is about 19 years, with early-onset more common in males. The onset of the disorder typically occurs in adolescence or childhood, and onset after age 30 is rare.

Body Dysmorphic Disorder

Point prevalence estimates in the United States and elsewhere have ranged from 1.7 to 2.4 percent, indicating that this is one of the more common obsessive-compulsive and related disorders. Furthermore, clinical studies have found high rates of BDD in general adult psychiatric inpatients, in cosmetic surgery clinics, and dermatology clinics.

The mean age of onset is in adolescence. As in the case of OCD, community studies indicate a higher prevalence of females with BDD, while clinical reports indicate a roughly even male:female ratio.

Hoarding Disorder

Most prevalence studies of HD have not used DSM-5 diagnostic criteria.

Available data indicate that HD is remarkably prevalent in Western countries, with one study that employed DSM-5 criteria yielding a point prevalence of 1.5 percent. The male:female ratio is estimated to be about 1. In contrast, in clinics, females are more common than males, perhaps indicating greater insight. Unlike the picture seen in most obsessive-compulsive and related disorders, the prevalence of hoarding increases with age, and in clinical settings, the average age of first treatment is around 50.

Hair-Pulling Disorder

Studies of select populations, such as college students, indicate that the point prevalence of trichotillomania is around 0.5 to 2 percent. Age of onset is typically at menarche, and female:male ratio is about 4:1 in adults. Pediatric samples include an equal distribution between males and females.

Excoriation Disorder

Community-based prevalence studies of excoriation disorder have found a point prevalence of 1.4 to 5.4 percent. Age of onset is somewhat more variable than hair-pulling disorder, although the mean age is 12 years, typically coinciding with the onset of puberty. Although many more females than males have excoriation disorder, the ratio is more even than in the case of hair-pulling disorder.

ETIOLOGY

Obsessive-Compulsive Disorder

OCD is now primarily viewed as a neuropsychiatric disorder that is mediated by specific neurocircuits. From a cognitive-affective neuroscience perspective, OCD has several specific neuropsychological impairments, including cognitive inflexibility and alterations in the habit system. Early clinical work established that OCD could be precipitated by damage to striatal circuitry, as was seen in the global influenza epidemic in the early 20th century. More recent advances have delineated the role of neuroanatomy, neurocircuitry, neurotransmitters, and other molecules involved in the pathogenesis of OCD.

Cognitive and neuropsychological studies of OCD have found several

alterations, including a range of impairments in executive function, and with evidence for cognitive inflexibility, motor impulsivity, and excessive habit formation. Such work is consistent with the hypothesis that OCD represents impaired control of automated, habitual behaviors.

Several brain regions, including the anterior cingulate, orbitofrontal cortex, and striatum, have consistently been implicated in OCD. There is, however, evidence that many other regions have an important role (Fig. 9-4). One hypothesis has been that cortico–striatal–thalamic–cortical (CSTC) circuitry plays a particularly important role in mediating the cognitive-affective impairments seen in OCD, with activation or inhibition of different components of this circuitry driving the compulsive and impulsive features of obsessive-compulsive and related disorders. Early imaging work was seminal in showing that both pharmacotherapy and CBT were able to normalize functional alterations in CSTC circuitry in OCD.

Also, data indicate that a range of neurotransmitter systems contribute to OCD, including serotonergic, dopaminergic, glutamatergic, and GABAergic systems. A serotonin hypothesis of OCD has been influential; a range of evidence points to the involvement of this system in OCD, and patients respond to treatment with SRIs but not to treatment with noradrenergic reuptake inhibitors. At the same time, there is relatively little evidence that serotonin plays a causal role in OCD. Furthermore, a range of evidence points to the involvement of the dopaminergic system in OCD, and, notably, augmentation of serotonin reuptake blockers with dopamine receptor blockers as a first-line intervention for treatment-refractory OCD. Most recently, there has been some interest in other systems, including the glutamatergic system, with evidence that some glutamatergic agents may be useful in this condition.

Genetic findings may increasingly influence future work on the cognitive-affective neuroscience of OCD. Twin and family studies indicate that genetic susceptibility plays a role in OCD, and more so in childhood-onset OCD.

Body Dysmorphic Disorder

There is some evidence that specific cognitive-affective impairments characterize BDD. For example, neuropsychological studies have suggested deficits in executive functioning and visual processing. Brain imaging and

neurogenetic studies have provided some insights into the potential mechanisms that underlie such deficits, and into potential overlaps with other obsessive-compulsive and related disorders. For example, CSTC circuits may be involved, as well as circuits involved in visual processing. Family and twin data provide preliminary evidence for genetic susceptibility, as well as a relationship to OCD. However, the etiology of BDD is still far from clear.

Hoarding Disorder

Patients with HD may demonstrate impairment in various neuropsychological domains, including spatial planning, working memory, response inhibition, and set-shifting. Structural and functional brain imaging studies have indicated that the relevant neurocircuitry overlaps only partially with that involved in OCD. Research on patients with neurologic disorders has suggested that ventromedial prefrontal/anterior cingulate cortices and medial temporal regions are involved in hoarding. Also, animal studies have pointed to a possible role for the dopaminergic system in hoarding, and twin studies indicate genetic susceptibility for developing pathologic hoarding.

Hair-Pulling Disorder

From a cognitive-affective neuroscience perspective, neuropsychological investigations of hair-pulling disorder have demonstrated some evidence of deficits in working memory and visual–spatial learning. Structural and functional brain imaging studies have suggested some involvement of CSTC circuits relevant to habit learning, although data is sparse. There is also evidence of the involvement of brain regions associated with reward processing and affect regulation. Family and twin studies provide evidence for genetic susceptibility, as well as a relationship to OCD and other body-focused repetitive behaviors. Several candidate genes may be associated with hair-pulling disorder, but the evidence is preliminary.

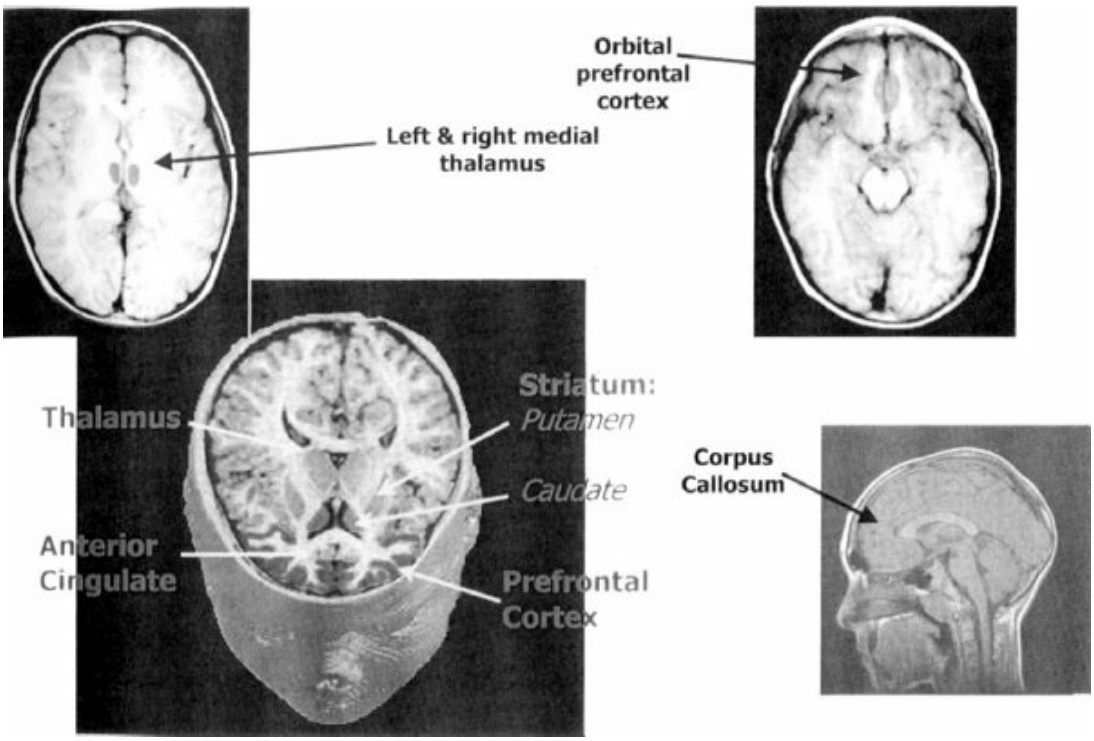


FIGURE 9-4
Brain regions implicated in the pathophysiology of obsessive-compulsive disorder. (From Rosenberg DR, MacMillan SN, Moore GJ. Brain anatomy and chemistry may predict treatment response in paediatric obsessive-compulsive disorder. *Int J Neuropsychopharmacol*. 2001;4(2):179–190, with permission.)

Excoriation Disorder

From a cognitive-affective neuroscience perspective, neuropsychological investigations of excoriation disorder have demonstrated some overlap with OCD regarding motor impulsivity. Brain imaging studies have similarly suggested some involvement of CSTC circuitry, although the data is sparse, and other regions also appear to be implicated. There is also some evidence that suggests that the dopaminergic system plays a role in excoriation disorder. Family and twin studies provide evidence for genetic susceptibility, as well as a relationship to OCD and other body-focused repetitive behaviors. As with hair-pulling disorder, preliminary evidence has identified several candidate genes.

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10 Trauma- and Stressor-Related Disorders

This chapter includes disorders characterized by exposure to significant stress or trauma. These disorders include posttraumatic stress disorder (PTSD), acute stress disorder, and adjustment disorders.

THE CLINICAL PRESENTATION

Persons suffering from one of these disorders develop emotional and behavioral symptoms in response to a significant stressor or traumatic event.

Posttraumatic Stress Disorder and Acute Stress Disorder

Persons who have PTSD or acute stress disorder have increased stress and anxiety following exposure to a traumatic or stressful event. Traumatic or stressful events may include experiencing a violent accident or crime, military combat, assault, being kidnapped, being involved in a natural disaster, being diagnosed with a life-threatening illness, or experiencing systematic physical or sexual abuse. A PTSD patient typically relives the trauma or tries to avoid reminders of it. They experience negative thoughts and moods about the event and feel hyperaroused or hyperactive. The patient may relive the trauma in their dreams or experience “flashbacks” or waking thoughts about the ordeal.

The stressors causing both acute stress disorder and PTSD are sufficiently overwhelming to affect almost everyone. They can arise from experiences in war, torture, natural catastrophes, assault, rape, and serious accidents.

Physicians have appreciated the link between acute mental syndromes and

traumatic events for more than 200 years. After the American Civil War, physicians documented trauma-related syndromes, and early psychoanalytic writers, including Sigmund Freud, noted a relationship between neurosis and trauma. In World Wars I and II, these disorders were given various names, including “battle fatigue,” “shell shock,” and “soldier’s heart.” Moreover, increasing documentation of mental reactions to the Holocaust, to a series of natural disasters, and assault contributed to the growing recognition of a close relation between trauma and psychopathology.

Mrs. M sought treatment for symptoms that she developed in the wake of an assault that had occurred about 6 weeks before her psychiatric evaluation. While leaving work late one evening, Mrs. M was attacked in a parking lot next to the hospital in which she worked. She was raped and badly beaten but was able to escape and call for help. On referral, Mrs. M reported frequent intrusive thoughts about the assault, including nightmares about the event and recurrent intrusive visions of her assailant. She now took the bus to work to avoid the scene of the attack and changed her work hours to avoid leaving the building after dark. Also, she reported that she had difficulty interacting with men, particularly those who resembled her attacker, and that she consequently avoided such interactions whenever possible. Mrs. M described increased irritability, difficulty staying asleep at night, poor concentration, and an increased focus on her environment, particularly after dark. (Courtesy of Erin B. McClure-Tone, Ph.D., and Daniel S. Pine, M.D.)

Clinical Features. Individuals with PTSD relive distressing instances of the traumatic event, with vivid emotional proximity and high, imperative intensity. They organize their lives trying to contain and mitigate the persistent effects of the traumatic experience. For those traumatized in a war zone, they often feel as if the war never ended. Victims of rape, assault, or torture describe difficulties engaging and trusting other humans. Constantly reliving the trauma in the present, PTSD patients’ lives become a series of effortful attempts to avoid reminders of the traumatic event. They scan the environment for threat signals, which they fearfully expect, and remain on guard, tense, restless, and exhausted. The persistence of symptoms despite

the termination of the threat, combined with an inability to regain a sense of safety, are core features of PTSD. Another core feature is the involuntary, uncontrollable, and intense nature of symptoms.

Adjustment Disorders

People with adjustment disorders have an emotional response to a stressful event. It is one of the few diagnostic entities that directly links an external stressor to the development of symptoms. Typically, the stressor involves financial issues, a medical illness, or a relationship problem. The symptom complex that develops may involve anxious or depressive affect or may present with a disturbance of conduct. The symptoms must begin within 3 months of the stressor. The *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) includes a variety of subtypes of the disorder, including adjustment disorder with depressed mood, mixed anxiety and depressed mood, disturbance of conduct, mixed disturbance of emotions and conduct, and unspecified type.

Clinical Features. Patients with adjustment disorders develop intense emotional or behavioral symptoms in response to one or more external stressors. The intensity of the symptoms is subjectively considered beyond what one would expect in the given situation and impairs the patient's ability to function.

Other Specified or Unspecified Trauma- or Stressor-Related Disorder

As with other psychiatric diagnoses, the DSM-5 includes separate categories to account for persons whose symptoms do not meet the full criteria for another trauma- or stressor-related disorder.

DIAGNOSIS

Posttraumatic Stress Disorder and Acute Stress Disorder

The diagnosis includes several categories of symptoms, including symptoms of intrusion, avoidance, negative mood or cognitions, and hyperarousal. These symptoms cause significant functional impairment and are present for

more than a month.

We outline the diagnostic approach to posttraumatic stress disorder in Table 10-1.

As with most disorders, substance use or a medical condition should not better explain the symptoms.

With acute stress disorder, the primary feature differentiating this disorder from PTSD is the time course, with the symptoms of acute stress disorder occurring 3 days to 1 month following a traumatic event. Acute stress disorder can have any of the symptoms of PTSD; however, they do not have to have all the domains. A person who experiences a minimum of nine symptoms from any of these domains within 3 days to 1 month of a traumatic event meets the criteria for acute stress disorder.

We outline the diagnostic approach to acute stress disorder in Table 10-2.

Three weeks after a train derailment, a 42-year-old budget analyst presented to the mental health clinic. He noted that he was embarrassed to seek care, as he was previously a firefighter, but felt he needed “some reassurance that what I’m experiencing is normal.” He reported that, since the wreck, he had been feeling nervous and on edge. He experienced some difficulty focusing his attention at work, and he had occasional intrusive recollections of “the way the ground just shook; the tremendous ‘bang’ and then the screaming when the train rolled over.” He noted that he had spoken with five business colleagues who were also on the train, and three acknowledged similar symptoms. However, they said that they were improving. He was more concerned about the frequency of tearful episodes, sometimes brought on by hearing the name of a severely injured friend, but, at other times, occurring “for no particular reason.” Also, he noted that, when he evacuated the train, rescue workers gave him explicit directions about where to report, and, although the man complied, he now felt extremely guilty about not returning to the train to assist in the rescue of others. He reported a modest decrease in appetite and denied weight loss but noted that he had stopped jogging during his lunch break. He had difficulty initiating sleep, so he had begun consuming a “glass or two” of wine before bed to help with this. He did not feel rested on awakening. He denied suicidal

ideation or any psychotic symptoms. His sister had taken an antidepressant several years ago, but he did not desire medication. He feared that side effects could further diminish his ability to function at the workplace and could cause him to gain weight. (Courtesy of D. M. Benedek, M.D., R. J. Ursano, M.D., and H. C. Holloway, M.D.)

Adjustment Disorders

Clinicians tend to use the diagnostic category of adjustment disorders liberally. Although this disorder must follow a stressor, the symptoms do not necessarily begin immediately. Up to 3 months may elapse between a stressor and the development of symptoms. Symptoms do not always subside as soon as the stressor ceases; if the stressor continues, the disorder may be chronic. The disorder can occur at any age, and its symptoms vary considerably, with depressive, anxious, and mixed features most common in adults. Physical symptoms, which are most common in children and the elderly, can occur in any age group. Manifestations may also include impulsivity, violence, excessive alcohol consumption, and suspiciousness.

The clinical presentations of adjustment disorder can vary widely.

We outline the diagnostic approach to adjustment disorder in Table 10-3.

Adjustment Disorder with Depressed Mood. In adjustment disorder with depressed mood, symptoms may manifest as depression, hyposomnia, low self-esteem, or suicidal ideation. We should distinguish this disorder from a major depressive disorder and uncomplicated bereavement.

Adjustment Disorder with Anxiety. Symptoms such as generalized anxiety, increased motor activity, and situational anxiety are present in adjustment disorder with anxiety, which we should differentiate from anxiety disorders.

Adjustment Disorder with Mixed Anxiety and Depressed Mood. In adjustment disorder with mixed anxiety and depressed mood, patients exhibit features of both anxiety and depression that do not meet the criteria for an already established anxiety disorder or depressive disorder.

A 48-year-old married woman, in good health, with no previous psychiatric difficulties, presented to the emergency room, reporting that she had overdosed on a handful of antihistamines shortly before she arrived. She described her problems as having started 2 months earlier, soon after her husband unexpectedly requested a divorce. She felt betrayed after having devoted much of her 20-year marriage to being a wife, mother, and homemaker. She was sad and tearful at times, and she occasionally had difficulty sleeping. Otherwise, she had no vegetative symptoms and enjoyed time with family and friends. She felt desperate and suicidal after she realized that "he no longer loves me." After crisis intervention in the emergency setting, she responded well to individual psychotherapy for over 3 months. She occasionally required benzodiazepines for anxiety during the period of treatment. By the time of discharge, she had returned to her baseline function. She came to terms with the possibility of life after divorce and was exploring her best options under the circumstances. (Courtesy of Jeffrey W. Katzman, M.D., and C. M. A. Geppert, M.D., Ph.D., M.P.H.)

Adjustment Disorder with Disturbance of Conduct. In adjustment disorder with disturbance of conduct, symptoms include impulsivity, lack of insight, and violent behavior. We should differentiate this category from conduct disorder and antisocial personality disorder.



Table 10-1
Posttraumatic Stress Disorder Diagnoses in DSM-5 and ICD-10

	Posttraumatic Stress Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Posttraumatic Stress Disorder	Posttraumatic Stress Dis
Duration	≥1 mo	Symptoms arise in weeks months following trau

		event
Symptoms	1. History of exposure to (directly experiencing, repeated exposure witnessing in person, learning of occurrence in close acquaintance) actual threatened death, severe injury, or sexual trauma 2. Intrusive symptoms • Involuntary intrusive memories • In children <6 yr, may see reenactment of event through play • Recurrent nightmares/dreams of the event • In children <6 yr, frightening dreams without identifiable content may be present • Dissociative responses or reliving of prior experience (i.e., flashbacks)	1. Symptoms arise as a or prolonged respons single or recurrent str experience that is life threatening or catastr severity 2. Traumatic event wou distress in most indivi 3. Symptoms may inclu • Flashbacks • Dreams/nightmares • Emotional blunting a subjective numbne • Social withdrawal/detachr • Anhedonia • Avoidance • Hyperarousal/hyper • Sleep disturbance 4. May or may not be as with co-occurring anx and/or depression

- In children <6 yr, may see reenactment of event through play
- Psychological distress related to exposure to stimuli that are reminders of prior trauma
- Presence of physiologic response to stimuli that are reminders of prior trauma

3. Pattern of avoidance of stimuli associated with prior experience of trauma
   - Avoidance of memories related to trauma
   - Avoidance of external reminders of trauma
4. Negative mood or cognitions related to trauma
   - Impairment in memories related to event
   - Negative perceptions of self and others
   - Cognitive

	distortions related to event • Excessive guilt, anger or fear • Diminished interest and social withdrawal • Subjective detachment from others • Difficulty experiencing positive feelings in response to previously pleasurable stimuli 5. Altered level of arousal • Irritability and/or anger • Risk taking • Hypervigilance • Increased startle response • Difficulties with concentration • Sleep disturbances	
Required number of symptoms	In addition to history of exposure to trauma, must have • at least one symptom of intrusion • at least one	Any of the above

	symptom of avoidance • at least two symptoms of negative mood/cognition • at least two symptoms of arousal alterations
Psychosocial consequences of symptoms	Marked distress and impairment in functioning
Exclusions	• Exposure through media, electronics, movie, or photo • Related to substance use • Related to another medical condition
Symptom specifiers	**With dissociative symptoms:** • **Depersonalization:** perception of feeling outside one's own body • **Derealization:** perception of surrounding environment being unreal or distorted
Course specifiers	**With delayed expression:** • All diagnostic

	criteria not met until 6 mo or more after initial traumatic event	
Severity specifiers		



Table 10-2
Acute Stress Disorder Diagnoses in DSM-5 and ICD-10

	Acute Stress Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Acute Stress Disorder	Acute Stress Reaction
Duration	3 days to <1 mo following exposure to trauma	Symptoms arise transiently in response to emotional or physical stress and usually resolve within hours or days
Symptoms	1. History of exposure to (directly experiencing, repeated exposure witnessing in person, learning of occurrence in close acquaintance) actual threatened death, severe injury, or sexual trauma 2. Presence of symptoms in the following categories	1. Cognitive “daze” 2. Narrowed attention 3. Difficulties with comprehension 4. Disorientation 5. Autonomic symptoms of panic 6. May be followed by further withdrawal from the situation or possible agitation/excitability

	(same as PTSD categories) • Intrusion • Negative mood/cognition • Dissociation • Avoidance • Arousal	7. May have partial or complete amnesia related to the event
Required number of symptoms	Nine symptoms from among the above five categories	
Psychosocial consequences of symptoms	Marked distress and impairment in functioning	
Exclusions (not result of):	Substance Other medical condition Brief psychotic disorder or other psychiatric illness	



Table 10-3
Adjustment Disorder Diagnoses in DSM-5 and ICD-10

	Adjustment Disorders	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Adjustment Disorder	Adjustment Disorders
Duration	Occurring within 3 mo of an acute stressor and resolving within 6 mo of resolution of stressor	
Symptoms	1. Emotional or	1. Subjective

	behavioral changes 2. Marked distress/impairment that is felt to be out of proportion to the stressor itself	distress/emotional disturbance 2. Arises during period of adaptation to a change or stress 3. Symptoms of depressed mood or anxiety may arise, and conduct disturbance may be noted in adolescents
Required number of symptoms	All of the above	
Psychosocial consequences of symptoms	Marked distress/functional impairment	Often result in impairment in social functioning
Exclusions (not result of):	Exacerbation of existing mental illness Normal bereavement Other psychiatric illness	
Symptom specifiers	**With depressed mood:** • Predominant low mood, hopelessness **With anxiety:** • Predominant anxiety, worry **With mixed anxiety and depressed mood:**	

- Both low mood and anxiety present

With mixed disturbance of emotions and conduct:

- Mood and conduct disturbance present

Unspecified

Adjustment Disorder with Mixed Disturbance of Emotions and Conduct. A combination of disturbances of emotions and conduct sometimes occurs. Examples include excessive alcohol ingestion, suspiciousness, hostility, defrauding behavior, and homicidal ideation. Clinicians are encouraged to try to make one or the other diagnosis in the interest of clarity.

Adjustment Disorder Unspecified. Adjustment disorder unspecified is a residual category for atypical maladaptive reactions to stress. Examples include inappropriate responses to the diagnosis of physical illness, such as massive denial, severe noncompliance with treatment, and social withdrawal, without significant depressed or anxious mood.

Other Specified or Unspecified Trauma- or Stressor-Related Disorder

DSM-5 uses the categories of “other specified trauma- or stressor-related disorder” or “unspecified trauma- or stressor-related disorder” uses for patients who develop emotional or behavioral symptoms in response to an identifiable stressor, but do not meet the full criteria for any of the above disorders. The primary difference between these two categories is that “other specified trauma- or stressor-related disorder” is used when it is essential to convey the reason that the patient does not meet the full criteria for one of the

other disorders.

DIFFERENTIAL DIAGNOSIS FOR THE TRAUMA- AND STRESSOR-RELATED DISORDERS

Generally, the time requirements and diagnostic criteria help to differentiate PTSD, acute stress disorder, and other psychiatric disorders. With adjustment disorder, one of the challenges is that there are no clear criteria that define the stressors that are required to make this diagnosis. PTSD and acute stress disorder have the nature of the stressor better characterized and include a defined constellation of affective, cognitive, and autonomic symptoms. In contrast, the stressor in adjustment disorder can be of any severity, with a wide range of possible symptoms. When the response to an extreme stressor does not meet the acute stress or posttraumatic disorder threshold, the adjustment disorder diagnosis would be appropriate.

We list some common disorders that can also cause posttraumatic symptomatology in Table 10-4.

Posttraumatic Stress Disorder and Acute Stress Disorder

Because patients often exhibit complex reactions to trauma, the clinician must be careful to exclude other syndromes as well when evaluating patients presenting in the wake of trauma. It is particularly important to recognize potentially treatable medical contributors to posttraumatic symptomatology, especially head injury during the trauma. A careful history and examination can help detect medical contributors. Epilepsy, alcohol use disorders, and other substance-related disorders can also either cause or exacerbate the symptoms. Acute intoxication or withdrawal from some substances may also present a clinical picture that is difficult to distinguish from the disorder until the effects of the substance have worn off.



Table 10-4
Differential Diagnosis of Posttraumatic Stress Disorder

Medical causes	Traumatic brain injury Epilepsy Alcohol use disorder

	Substance-related disorders Acute substance or alcohol withdrawal
Psychiatric diagnoses	Panic disorder Generalized anxiety disorder Major depressive disorder Dissociative identity disorder Borderline personality disorder Factitious disorders Acute stress disorder

Symptoms of PTSD can be difficult to distinguish from both panic disorder and generalized anxiety disorder because all three syndromes are associated with prominent anxiety and autonomic arousal. Keys to correctly diagnosing PTSD include a careful review of the time course relating the symptoms to a traumatic event. PTSD is also associated with intrusion symptoms and avoidance of anything that reminds the person of the trauma, something not usually present in panic or generalized anxiety disorder. Major depression is also a frequent concomitant of PTSD. Although the two syndromes are not generally difficult to distinguish phenomenologically, it is essential to note the presence of comorbid depression because this can influence the treatment of PTSD. We also must differentiate PTSD from several related disorders that can look similar, including borderline personality disorder, dissociative disorders, and factitious disorders. Borderline personality disorder can be difficult to distinguish from PTSD. The two disorders can coexist or even be causally related. Patients with dissociative disorders do not usually have the same degree of avoidance and arousal that PTSD patients do. The time course can distinguish PTSD and acute stress disorder.

Adjustment Disorders

Although uncomplicated bereavement often produces temporarily impaired social and occupational functioning, the person’s dysfunction remains within the expectable bounds of a reaction to the loss of a loved one and, thus, is not considered adjustment disorder. We discuss uncomplicated bereavement among the depressive disorders in Chapter 7.

We must differentiate adjustment disorder from many other disorders, and we list some of these in Table 10-5. Patients with an adjustment disorder are impaired in social or occupational functioning and show symptoms beyond the usual and expectable reaction to the stressor. Because no absolute criteria help to distinguish an adjustment disorder from another condition, clinical judgment is necessary. Some patients may meet the criteria for both an adjustment disorder and a personality disorder. If the adjustment disorder follows a physical illness, the clinician must make sure that the symptoms are not a continuation or another manifestation of the illness or its treatment.

COMORBIDITY

Posttraumatic Stress Disorder

Comorbidity rates are high among patients with PTSD, with about two-thirds having at least two other disorders. Common comorbid conditions include depressive disorders, substance-related disorders, anxiety disorders, and bipolar disorders. Comorbid disorders make persons more vulnerable to develop PTSD.



Table 10-5
Differential Diagnosis of Adjustment Disorders

Major depressive disorder
Brief psychotic disorder
Generalized anxiety disorder
Somatic symptom disorder
Substance-related disorder
Conduct disorder
Antisocial personality disorder
Posttraumatic stress disorder

Adjustment Disorders

Most psychiatric disorders can co-occur with adjustment disorder. However, it is critical to ensure that another psychiatric disorder does not best explain

the person’s response to the stressor.

COURSE AND PROGNOSIS

Posttraumatic Stress Disorder

Immediately following exposure to a traumatic event, many people experience symptoms such as shock, nightmares, and intrusive thoughts about the event. However, we cannot make a diagnosis of PTSD unless enough symptoms persist for at least 1 month. Symptoms can fluctuate over time and may be most intense during periods of stress. Untreated, about 30 percent of patients recover completely, 40 percent continue to have mild symptoms, 20 percent continue to have moderate symptoms, and 10 percent remain unchanged or become worse. After 1 year, about 50 percent of patients will recover.

We list some predictors of good prognosis in Table 10-6.

In general, the very young and the very old have more difficulty with traumatic events than do those in midlife. For example, about 80 percent of young children who sustain a burn injury show symptoms of PTSD 1 or 2 years after the initial injury; only 30 percent of adults who suffer such an injury have symptoms of PTSD after 1 year. Presumably, young children do not yet have adequate coping mechanisms to deal with the physical and emotional insults of the trauma. Likewise, older persons are likely to have more rigid coping mechanisms than younger adults and to be less able to muster a flexible approach to dealing with the effects of trauma. Furthermore, physical disabilities typical of late life can exacerbate the symptoms. Examples include impairments of the nervous system and the cardiovascular system, such as reduced cerebral blood flow, failing vision, palpitations, and arrhythmias. Pre-existing psychiatric disability also increases the effects of stressors. PTSD that is comorbid with other disorders is often more severe and perhaps more chronic and may be challenging to treat. The availability of social supports may also influence the development, severity, and duration of PTSD. In general, patients who have a good network of social support are less likely to have the disorder and to experience it in its severe forms and are more likely to recover faster.

Adjustment Disorders

With appropriate treatment, the overall prognosis of an adjustment disorder is generally favorable. Most patients return to their previous level of functioning within 3 months. Some persons (particularly adolescents) who receive a diagnosis of an adjustment disorder later have mood disorders or substance-related disorders. Adolescents usually require a longer time to recover than adults.



Table 10-6
Positive Prognostic Factors for PTSD

Rapid onset of the symptoms
Short duration of the symptoms (less than 6 mo)
Good premorbid functioning
Strong social supports
Absence of other psychiatric, medical, or substance-related disorders or other risk factors

Research has disclosed a risk for suicide not previously fully appreciated. Some studies suggest that a high proportion of patients with adjustment disorders have had past suicide attempts and recent suicidality. This data is difficult to interpret, given the extensive and sometimes inconsistent use of this diagnosis; however, it is critical to evaluate for suicidality in patients with this disorder.

A 16-year-old high school senior's first relationship ended when his girlfriend rejected him. In the weeks after the end of the relationship, he began to exhibit dysphoric mood accompanied by anxiety and psychomotor agitation. He had received counseling in junior high school when his parents divorced, and he began using alcohol and marijuana. His school suspended him during his freshman year for fighting. A month after the breakup, he began to tell his parents that life was no longer worth living without his former girlfriend. Two months later, his parents came home from work and found him hanging in the garage with a note

stating he could not go on alone. (Courtesy of J. W. Katzman, M.D., and C. M. A. Geppert, M.D., Ph.D., M.P.H.)

TREATMENT APPROACH

Posttraumatic Stress Disorder

We list some of the most common treatments for PTSD and related disorders in Table 10-7.

Pharmacotherapy. Currently, selective serotonin reuptake inhibitors (SSRIs), particularly sertraline and paroxetine, have the most robust evidence for efficacy. There are promising initial findings for the noradrenergic and serotonergic reuptake inhibitor venlafaxine, the atypical antipsychotic risperidone, and the anticonvulsant topiramate. We lack evidence for the effectiveness of benzodiazepines, despite their continued use in practice; one meta-analysis went so far as to consider benzodiazepines relatively contraindicated in this population. Finally, the alpha 1-adrenergic antagonist prazosin and the atypical antipsychotics show some efficacy in treatment-resistant PTSD, and prazosin shows effectiveness for treating nightmares. Insomnia is prevalent and disabling in chronic PTSD, with low-dose trazodone preferable to benzodiazepines. The orexin antagonist suvorexant is promising for managing trauma-related insomnia.



Table 10-7
Treatments for PTSD

Pharmacologic interventions	Strong evidence SSRIs (sertraline, paroxetine) Promising initial evidence SNRI (venlafaxine) Atypical antipsychotic (risperidone) Anticonvulsant (topiramate) Limited evidence Alpha 1-adrenergic antagonist (prazosin)

	Lack of evidence Benzodiazepines Symptom-specific Prazosin (nightmares) Orexin (insomnia) Trazodone (insomnia)
Psychotherapeutic interventions	Trauma-focused cognitive-behavioral therapy (TFCBT), Prolonged exposure (PE) therapy Cognitive processing therapy (CPT) Eye movement desensitization and reprocessing therapy (EMDR) Present centered therapy (PCT) Cognitive processing therapy (CPT) Psychodynamic psychotherapy Group and family therapy

Psychotherapy. Several therapies are effective for PTSD and related disorders; we list these in Table 10-7. Trauma-focused cognitive-behavioral therapy (TFCBT) includes prolonged exposure (PE) therapy, which focuses on reexperiencing the traumatic event through repeatedly engaging with the memories (imaginal exposure) and everyday reminders (in vivo exposure) rather than avoiding triggers. Eye movement desensitization and reprocessing therapy (EMDR) involves repeatedly recalling distressing images while receiving sensory inputs. Present centered therapy (PCT) focuses on the current relationship and work challenges rather than the trauma, and cognitive processing therapy (CPT) emphasizes correcting faulty attributions including posttraumatic overgeneralizations of the world as dangerous, uncontrollable, and unpredictable. Psychodynamic psychotherapy may also be useful in the treatment of many patients with PTSD. In some cases, reconstruction of the traumatic events with associated abreaction and catharsis may be therapeutic. Still, psychotherapy must be tailored to the individual, because reexperiencing the trauma overwhelms some patients.

In addition to individual therapy techniques, group therapy and family

therapy are effective for PTSD. The advantages of group therapy include the sharing of traumatic experiences and support from other group members. Group therapy has been particularly successful with Vietnam veterans and survivors of catastrophic disasters such as earthquakes. Family therapy often helps sustain a marriage through periods of exacerbated symptoms. Hospitalization may be necessary when symptoms are particularly severe or when the risk of suicide or other violence exists.

Adjustment Disorders

Psychotherapy. Psychotherapy remains the treatment of choice for adjustment disorders. Group therapy can be particularly useful for patients who have had similar stresses—for example, a group of retired persons or patients having renal dialysis. Individual psychotherapy interventions include supportive psychological approaches, cognitive-behavioral, problem-solving techniques, and psychodynamic interventions. After successful therapy, patients sometimes emerge from an adjustment disorder stronger than in the premorbid period, although no pathology was evident during that period. Because there is an explicit stressor, some clinicians mistakenly deemphasize psychotherapy and believe that the disorder will remit spontaneously. This viewpoint, however, ignores the fact that many persons exposed to the same stressor experience different symptoms, and in adjustment disorders, the response is pathologic. Psychotherapy can help persons adapt to stressors that are not reversible or time-limited and can serve as a preventive intervention if the stressor does remit.

Crisis Intervention. Crisis intervention and case management are short-term treatments aimed at helping persons with adjustment disorders resolve their situations quickly by supportive techniques, suggestion, reassurance, environmental modification, and even hospitalization, if necessary. The frequency and length of visits for crisis support vary according to patients' needs; daily sessions may be required, sometimes two or three times each day. Flexibility is essential in this approach.

Pharmacotherapy. There is limited evidence for the efficacy of pharmacologic interventions in persons with adjustment disorder. Still, it may

be reasonable to use medication to treat specific symptoms for a brief time. The judicious use of medications can help patients with adjustment disorders, but we should only use them for short periods. Depending on the type of adjustment disorder, a patient may respond to an antianxiety agent or an antidepressant. It is often helpful and important to treat insomnia and severe anxiety with pharmacologic interventions in the short run. It is not as clear whether it helps to treat the depressive symptoms using medication. Clinicians have found some SSRIs to be useful for treating some subthreshold depressive symptoms and may benefit certain subtypes of adjustment disorders. Pharmacologic intervention in this population should mainly augment psychosocial strategies and not serve as the primary modality.

EPIDEMIOLOGY

Posttraumatic Stress Disorder

About 6.8 percent of the general population has a lifetime prevalence of PTSD, with a current past year prevalence estimated at 3.5 percent. The lifetime prevalence rate is 9.7 percent in women and 3.6 percent in men. According to the National Vietnam Veterans Readjustment Study (NVVRS), 30.9 percent of men developed full-blown PTSD after having served in the war, and 26.9 percent of women developed the disorder. A 25-year follow-up of the NVVRS, the National Vietnam Veteran Longitudinal Study (NVVLS), was conducted to assess the course of PTSD 40 years after the war. This study found that the current prevalence of war zone PTSD was 4.5 percent, and the lifetime prevalence of war zone PTSD was 17 percent.

Although PTSD can appear at any age, it is most prevalent in young adults because they tend to be more exposed to precipitating situations. Children can also have the disorder. The disorder is most likely to occur in those who are single, divorced, widowed, socially withdrawn, or of low socioeconomic level, but anyone can be affected, and no one is immune. The most important risk factors, however, for this disorder are the severity, duration, and proximity of a person’s exposure to the actual trauma. A familial pattern seems to exist for this disorder, and first-degree biologic relatives of persons with a history of depression have an increased risk for developing PTSD

following a traumatic event.

Adjustment Disorders

Few studies have examined the prevalence of adjustment disorder in community samples. One study that did assess for the presence of adjustment disorder was the European Outcome of Depression International Network (ODIN) study, which found that less than 1 percent of subjects had adjustment disorder. Another study examined the prevalence of adjustment disorder in a primary care setting and found that 2.94 percent of the sample met the criteria for adjustment disorder. There is a higher prevalence of adjustment disorder in the medical setting—especially patients with cancer and those receiving palliative care. For example, one study found that 11 percent of individuals with mixed cancer diagnoses had an adjustment disorder.

ETIOLOGY

Posttraumatic Stress Disorder

Risk Factors for PTSD. Even when faced with overwhelming trauma, most persons do not experience PTSD symptoms. The National Comorbidity Study found that 60 percent of males and 50 percent of females had experienced some significant trauma. In contrast, the reported lifetime prevalence of PTSD, as mentioned earlier, was much lower. Similarly, events that may appear mundane or less than catastrophic to most persons can produce PTSD in some. Evidence indicates a dose–response relationship between the degree of trauma and the likelihood of symptoms.

A meta-analysis of risk factors for PTSD in the community found that female gender, age at trauma, race, lower education, childhood abuse, higher severity of trauma exposure, lack of social support, and additional life stress increased risk for PTSD. A second meta-analysis identified seven predictors: prior trauma, prior psychological adjustment, family history of psychopathology, more significant perceived life threat during the trauma, lower posttrauma social support, more considerable emotional distress during exposure, and higher dissociation during exposure. For combat-related PTSD,

identified risk factors include lower education, nonofficer ranks, army service, combat specialization, higher numbers of deployments, longer cumulative length of deployments, more adverse life events, prior trauma exposure, and previous psychological problems. Various aspects of the trauma period also constituted risk factors, including higher levels of combat exposure, discharging a weapon, witnessing someone being wounded or killed, severe trauma, and deployment-related stressors. Another study reported that killing enemy combatants, prisoners of war, and civilians in the war zone increased the risk of PTSD.

Table 10-8 summarizes vulnerability factors that appear to play etiologic roles in the disorder.

Genetics and Risk for PTSD. Genes account for about 30 percent of the variance in the risk for PTSD. A study of Vietnam veteran twin pairs reported that monozygotic twins of veterans with combat-related PTSD had more significant mood disorder symptoms than monozygotic co-twins of combat controls or dizygotic co-twins of veterans with PTSD. The findings suggest a shared genetic vulnerability to PTSD and mood disorders. Other studies have indicated that PTSD symptoms are moderately heritable, with the remaining vulnerability accounted for by unique environmental experiences. Several common genetic variants are associated with PTSD, including polymorphisms in FKBP5, PACAP1, COMT, DRD2, GABA alpha-2 receptor, G-protein signaling 2 (RSG2), an SNP in an intergenic region of the fourth chromosome, and an estrogen response element on ADCYAP R1. Also, the s/s genotype of the serotonin transporter gene may interact with childhood adversity to increase PTSD risk.



Table 10-8
Predisposing Vulnerability Factors in Posttraumatic Stress Disorder

Presence of childhood trauma
Borderline, paranoid, dependent, or antisocial personality disorder traits
Inadequate family or peer support system
Being female

Genetic vulnerability to psychiatric illness
Recent stressful life changes
Perception of an external locus of control (natural cause) rather than an internal one (human cause)
Recent excessive alcohol intake

Adjustment Disorders

By definition, a stressor precipitates an adjustment disorder. The severity of the stressor or stressors does not always predict the severity of the disorder; the stressor severity is a complex function of degree, quantity, duration, reversibility, environment, and personal context. For example, the loss of a parent is different for a child 10 years of age than for a person 40 years of age. Personality organization and cultural or group norms and values also contribute to the disproportionate responses to stressors.

Stressors may be single, such as the death of a loved one, divorce, losing a job, or medical illness. Alternatively, they can be a combination of any of these. Stressors may be recurrent, such as seasonal business difficulties, or continuous, such as chronic illness or poverty. A discordant intrafamilial relationship can produce an adjustment disorder that affects the entire family system, or the disorder may be limited to a patient who was perhaps the victim of a crime or who has a physical illness. Sometimes, adjustment disorders occur in a group or community setting, and the stressors affect several persons, as in a natural disaster or racial, social, or religious persecution. Specific developmental stages, such as beginning school, leaving home, getting married, becoming a parent, failing to achieve occupational goals, having the last child leave home, and retiring, are often associated with adjustment disorders.

Psychodynamic Factors. Pivotal to understanding adjustment disorders is an understanding of three factors: the nature of the stressor, the conscious and unconscious meanings of the stressor, and the patient's pre-existing vulnerability. A concurrent personality disorder or organic impairment may make a person vulnerable to adjustment disorders. Vulnerability is also associated with the loss of a parent during infancy or growing up in a dysfunctional family. Actual or perceived support from crucial relationships

can affect behavioral and emotional responses to stressors.

Clinicians must undertake a detailed exploration of a patient's experience of the stressor. Some patients commonly place all the blame on a particular event when a less apparent event may have had more significant psychological meaning for the patient. Current events may reawaken past traumas or disappointments from childhood, so patients should be encouraged to think about how the current situation relates to similar past events.

Throughout early development, each child develops a unique set of defense mechanisms to deal with stressful events. Because of higher amounts of trauma or greater constitutional vulnerability, some children have less mature defensive constellations than other children. This disadvantage may cause them as adults to react with substantially impaired functioning when facing a loss, divorce, or a financial setback; those who have developed mature defense mechanisms are less vulnerable and bounce back more quickly from the stressor. Resilience is also crucially determined by the nature of children's early relationships with their parents. Studies of trauma repeatedly indicate that supportive, nurturing relationships prevent traumatic incidents from causing permanent psychological damage.

Psychodynamic clinicians must consider the relationship between a stressor and the human developmental life cycle. When adolescents leave home for college, for example, they are at high developmental risk for reacting with a temporary symptomatic picture. Similarly, if the young person who leaves home is the last child in the family, the parents may be particularly vulnerable to a reaction of adjustment disorder. Moreover, middle-aged persons who are confronting their mortality may be especially sensitive to the effects of loss or death.

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11

Dissociative Disorders

In psychiatry, *dissociation* is an unconscious defense mechanism involving the segregation of any group of mental or behavioral processes from the rest of the person's psychic activity. Dissociative disorders involve this mechanism so that there is a disruption in one or more mental functions, such as memory, identity, perception, consciousness, or motor behavior. The disturbance may be sudden or gradual, transient or chronic, and psychological trauma is often the cause.

DIAGNOSTIC AND CLINICAL FEATURES

Dissociative Amnesia

The main feature of dissociative amnesia is the inability to recall important personal information, usually related to a significant trauma or stressor, that is too extensive to be explained by ordinary forgetfulness. The disorder cannot result from the direct physiologic effects of a substance or a neurologic or other general medical condition.

Table 11-1 lists the different types of dissociative amnesia.

A 45-year-old, divorced, left-handed, male bus dispatcher was seen in psychiatric consultation on a medical unit. He had been admitted with an episode of chest discomfort, light-headedness, and left-arm weakness. He had a history of hypertension and had a medical admission in the past year for ischemic chest pain, although he had not suffered a myocardial infarction. Psychiatric consultation was called, because the patient complained of memory loss for the previous 12 years, behaving and

responding to the environment as if it were 12 years previously (e.g., he did not recognize his 8-year-old son, insisted that he was unmarried, and denied recollection of current events, such as the name of the current president). Physical and laboratory findings were unchanged from the patient's usual baseline. Brain computed tomography (CT) scan was normal.

On mental status examination, the patient displayed intact intellectual function but insisted that the date was 12 years earlier, denying recall of his entire subsequent personal history and of current events for the past 12 years. He was perplexed by the contradiction between his memory and current circumstances. The patient described a family history of brutal beatings and physical discipline. He was a decorated combat veteran, although he described amnestic episodes for some of his combat experiences. In the military, he had been a champion golden glove boxer noted for his powerful left hand.

He was educated about his disorder and given the suggestion that his memory could return as he could tolerate it, perhaps overnight during sleep or perhaps over a longer time. If this strategy was unsuccessful, hypnosis or an amobarbital interview was proposed. (Adapted from a case of Richard J. Loewenstein, M.D., and Frank W. Putnam, M.D.)

Classic Presentation. The classic disorder is an overt, florid, dramatic clinical disturbance that quickly presents for medical attention. A history of extreme acute trauma is typical. It also commonly develops, however, in the context of profound intrapsychic conflict or emotional stress. Patients may present with physical symptoms, alterations in consciousness, depersonalization, derealization, trance states, spontaneous age regression, and even ongoing anterograde dissociative amnesia. There is a significant risk of depression and suicidal ideation. No single personality profile or antecedent history is consistently reported in these patients, although a prior personal or family history of somatoform or dissociative symptoms can predispose individuals to develop acute amnesia during traumatic circumstances. Many of these patients have histories of prior adult or childhood abuse or trauma. In wartime cases, as in other forms of combat-related posttraumatic disorders, the most crucial variable in the development

of dissociative symptoms, however, appears to be the intensity of combat.

Table 11-2 presents the mental status evaluation of dissociative amnesia.

Nonclassic Presentation. These patients frequently come to treatment for a variety of symptoms, such as depression or mood swings, substance abuse, sleep disturbances, somatoform symptoms, anxiety and panic, suicidal or self-mutilating impulses and acts, violent outbursts, eating problems, and interpersonal problems. Self-mutilation and violent behavior in these patients may also occur. Amnesia may also occur for flashbacks or behavioral reexperiencing episodes related to trauma.

Dissociative Fugue. DSM-5 treats dissociative fugue as a subtype of dissociative amnesia, whereas it is a separate diagnosis in ICD-10. It can occur in patients with both dissociative amnesia and dissociative identity disorder.

Dissociative fugue is as sudden, unexpected travel away from home or one's customary place of daily activities. Also, the person cannot recall some or all of one's past. Along with the amnesia, the person is confused about their identity or may even assume a new identity. The disturbance is not due to the direct physiologic effects of a substance or a general medical condition.

Dissociative fugues have been described to last from minutes to months. Some patients report multiple fugues. In some extreme cases of posttraumatic stress disorder (PTSD), a person may awaken from a nightmare in a fugue state. Children or adolescents may be more limited than adults in their ability to travel. Thus, fugues in this population may be brief and involve only short distances.

A teenage girl was continually sexually abused by her alcoholic father and another family friend. She was threatened with perpetration of sexual abuse on her younger siblings if she told anyone about the abuse. The girl became suicidal but felt that she had to stay alive to protect her siblings. She ran away from home after being raped by her father and several of his friends. She traveled to a part of the city where she had lived previously with the idea of finding her grandmother with whom she had lived before the abuse began. She traveled by public transportation and

walked the streets, apparently without attracting attention. After approximately 8 hours, she was stopped by the police in a curfew check. When questioned, she could not recall recent events or give her current address, insisting that she lived with her grandmother. On initial psychiatric examination, she was aware of her identity, but she believed that it was 2 years earlier, giving her age as 2 years younger and insisting that none of the events of recent years had occurred. (Courtesy of Richard J. Loewenstein, M.D., and Frank W. Putnam, M.D.)



Table 11-1
Types of Dissociative Amnesia

Localized amnesia: Inability to recall events related to a circumscribed period of time

Selective amnesia: Ability to remember some, but not all, of the events occurring during a circumscribed period of time

Generalized amnesia: Failure to recall one's entire life

Continuous amnesia: Failure to recall successive events as they occur

Systematized amnesia: Failure to remember a category of information, such as all memories relating to one's family or to a particular person



Table 11-2
Mental Status Examination Questions for Dissociative Amnesia

If answers are positive, ask the patient to describe the event. Make sure to specify that the symptom does not occur during an episode of intoxication.

1. Do you ever have blackouts? Blank spells? Memory lapses?
2. Do you lose time? Have gaps in your experience of time?
3. Have you ever traveled a considerable distance without recollection of how you did this or where you went exactly?

4. Do people tell you of things you have said and done that you do not recall?
5. Do you find objects in your possession (such as clothes, personal items, groceries in your grocery cart, books, tools, equipment, jewelry, vehicles, weapons, and so on) that you do not remember acquiring? Out-of-character items? Items that a child might have? Toys? Stuffed animals?
6. Have you ever been told or found evidence that you have talents and abilities that you did not know that you had? For example, musical, artistic, mechanical, literary, athletic, or other talents? Do your tastes seem to fluctuate a lot? For example, food preference, personal habits, taste in music or clothes, and so forth.
7. Do you have gaps in your memory of your life? Are you missing parts of your memory for your life history? Are you missing memories of some important events in your life? For example, weddings, birthdays, graduations, pregnancies, birth of children, and so on.
8. Do you lose track of or tune out conversations or therapy sessions as they are occurring? Do you find that, while you are listening to someone talk, you did not hear all or part of what was just said?
9. What is the longest period of time that you have lost? Minutes? Hours? Days? Weeks? Months? Years? Describe.

Adapted from Loewenstein RJ. An office mental status examination for chronic complex dissociative symptoms and multiple personality disorder. *Psychiatr Clin North Am*. 1991;14(3):567–604.

After the termination of a fugue, the patient may experience perplexity, confusion, trance-like behaviors, depersonalization, derealization, and conversion symptoms, in addition to amnesia. Some patients may terminate a fugue with an episode of generalized dissociative amnesia.

As the patient with dissociative fugue begins to become less dissociated, he or she may display mood disorder symptoms, intense suicidal ideation, and PTSD or anxiety disorder symptoms. In classic cases, the person has an alter identity under whose auspices the patient lives for a period. Many of these latter cases are better classified as dissociative identity disorder or, if

using DSM-5, as other specified dissociative disorder with features of dissociative identity disorder.

Differential Diagnosis

Table 11-3 lists the differential diagnosis of dissociative amnesia.

Ordinary Forgetfulness and Nonpathologic Amnesia. Ordinary forgetfulness is benign and unrelated to stressful events. In dissociative amnesia, memory loss is more extensive than in nonpathologic amnesia. There are other types of nonpathologic amnesia as well, including infantile and childhood amnesia, amnesia for sleep and dreaming, and hypnotic amnesia.

Dementia, Delirium, and Amnestic Disorders due to Medical Conditions. In patients with dementia, delirium, and amnestic disorders due to medical conditions, memory loss is part of a more extensive set of cognitive, language, attentional, behavioral, and memory problems. Loss of memory for personal identity is usually not found without evidence of a marked disturbance in many domains of cognitive function. Causes of organic amnestic disorders include Korsakoff psychosis, cerebral vascular accident (CVA), postoperative amnesia, postinfectious amnesia, anoxic amnesia, and transient global amnesia (see below). Electroconvulsive therapy (ECT) may also cause marked temporary amnesia, as well as persistent memory problems in some cases. Here, however, memory loss for autobiographical experience is unrelated to traumatic or overwhelming experiences and seems to involve many different types of personal experiences, most commonly those occurring just before or during the ECT treatments.

Posttraumatic Amnesia. In posttraumatic amnesia caused by brain injury, we usually find a history of a clear-cut physical trauma, a period of unconsciousness or amnesia, or both. Also, there is objective clinical evidence of brain injury.

Seizure Disorders. In most seizure cases, the clinical presentation differs significantly from that of dissociative amnesia, with clear-cut ictal events and

sequelae. In complex–partial seizures, patients may wander or show semipurposeful behavior, or both, during seizures or in postictal states, for which subsequent amnesia occurs. Rarely, patients with recurrent, complex–partial seizures present with ongoing bizarre behavior, memory problems, irritability, or violence, leading to a differential diagnostic puzzle. Seizure patients in an epileptic fugue often exhibit abnormal behavior, however, including confusion, perseveration, and abnormal or repetitive movements. Patients with pseudoepileptic seizures may also have dissociative symptoms, such as amnesia and an antecedent history of psychological trauma. In some of these cases, the diagnosis can be clarified only by telemetry or ambulatory electroencephalographic (EEG) monitoring.



Table 11-3
Differential Diagnosis of Dissociative Amnesia

- Ordinary forgetfulness
  - Age-related cognitive decline
- Nonpathologic forms of amnesia
  - Infantile and childhood amnesia
  - Amnesia for sleep and dreaming
  - Hypnotic amnesia
- Dementia
- Delirium
- Amnestic disorders
- Neurologic disorders with discrete memory loss episodes
  - Posttraumatic amnesia
  - Transient global
- Acute stress disorder
- Posttraumatic stress disorder
- Somatization disorder
- Psychotic episode
  - Lack of memory for psychotic episode when returns to nonpsychotic state
- Mood disorder episode
  - Lack of memory for aspects of episode of mania when depressed and vice versa or when euthymic
- Factitious disorder
- Malingering
- Psychophysiologic symptoms or disorders
  - Asthma and breathing problems
  - Perimenstrual disorders
  - Irritable bowel syndrome
  - Gastroesophageal reflux disease
  - Somatic memory
- Affective symptoms
  - Depressed mood, dysphoria, or

amnesia	anhedonia
Amnesia related to seizure disorders	Brief mood swings or mood lability
Substance-related amnesia	Suicidal thoughts and attempts or self-mutilation
Alcohol	Guilt and survivor guilt
Sedative–hypnotics	Helpless and hopeless feelings
Anticholinergic agents	Obsessive-compulsive symptoms
Steroids	Ruminations about trauma
Marijuana	Obsessive counting, singing
Narcotic analgesics	Arranging
Psychedelics	Washing
Phencyclidine	Checking
Methyldopa (Aldomet)	
Pentazocine (Talwin)	
Hypoglycemic agents	
β-Blockers	
Lithium carbonate	
Many others	
Other dissociative disorders	
Dissociative fugue	
Dissociative identity disorder	
Dissociative disorder not otherwise specified	

Substance-Related Amnesia. A variety of substances and intoxicants can cause amnesia. Table 11-3 lists some common offending agents.

Transient Global Amnesia. Transient global amnesia can be mistaken for dissociative amnesia, especially because stressful life events may precede either disorder. In transient global amnesia, however, there is the sudden onset of complete anterograde amnesia and learning abilities; pronounced

retrograde amnesia; preservation of memory for personal identity; anxious awareness of memory loss with repeated, often perseverative, questioning; overall normal behavior; lack of gross neurologic abnormalities in most cases; and rapid return of baseline cognitive function, with a persistent short retrograde amnesia. The patient usually is older than 50 years of age and shows risk factors for cerebrovascular disease, although epilepsy and migraine are also possible causes.

Dissociative Identity Disorder. Patients with dissociative identity disorder can present with acute forms of amnesia and fugue episodes. These patients, however, have symptoms beyond that found in dissociative amnesia. Concerning amnesia, most patients with dissociative identity disorder, and those with an unspecified dissociative disorder with dissociative identity disorder features, report multiple forms of complex amnesia, including recurrent blackouts, fugues, unexplained possessions, and fluctuations in skills, habits, and knowledge.

Acute Stress Disorder and Posttraumatic Stress Disorder. Many trauma spectrum disorders, including acute stress disorder and PTSD, are associated with dissociative amnesia. When diagnoses of acute stress disorder or PTSD are appropriate, we should add the co-occurring diagnosis of dissociative amnesia if the amnesia persists beyond the immediate trauma.

Malingering and Factitious Amnesia. No absolute way exists to differentiate dissociative amnesia from factitious or malingered amnesia. Malingerers may continue their deception even during hypnotically or barbiturate-facilitated interviews. A patient who presents to psychiatric attention seeking to recover repressed memories as a chief complaint most likely has a factitious disorder or has been subject to suggestive influences. Most of these individuals do not describe bona fide amnesia when carefully questioned but are often insistent that they must have been abused in childhood to explain their unhappiness or dysfunction. Malingering of dissociative fugue can occur in individuals who are attempting to flee a situation involving legal, financial, or personal difficulties, as well as in soldiers who are attempting to avoid combat or unpleasant military duties. No test, battery of tests, or set of procedures exist that invariably distinguish

actual dissociative symptoms from malingering. Many malingerers confess spontaneously or when confronted. In the forensic context, when a patient claims to have a fugue, the examiner should always carefully consider the diagnosis of malingering.

DEPERSONALIZATION/DEREALIZATION DISORDER

Depersonalization is the persistent or recurrent feeling of detachment or estrangement from one's self. The individual may report feeling like an automaton or watching himself or herself in a movie. Derealization is somewhat related and refers to feelings of unreality or of being detached from one's environment. The patient may describe his or her perception of the outside world as lacking lucidity and emotional coloring, as though dreaming or dead.



Table 11-4
Depersonalization/Derealization Disorder

	DSM-5	ICD-10
Name	**Depersonalization/Derealization Disorder**	**Depersonalizati Syndrome**
Symptoms	Depersonalization: • Feeling detached from oneself • Feeling as if observing oneself • Sense of unreality • Sense of distorted time • Sense of distorted perceptions Derealization • Sense of unreality • Feeling detached from one's surroundings	Perceived chang activity, such t body, or surro Unreal Remote Automatized Loss of emotio Feeling estrang
# Symptoms needed	Either depersonalization or derealization as above	Any of above Intact reality testi

	Intact reality testing	
Exclusions (not result of):	Substance use Another medical condition Another mental disorder	Schizophrenic Depressive disor Phobic disorders Obsessive-compu
Psychosocial impact	Marked distress and/or impairment	

Table 11-4 compares the diagnostic approaches to depersonalization/derealization disorder.

Several distinct components comprise the experience of depersonalization, including a sense of (1) bodily changes, (2) duality of self as observer and actor, (3) feeling cut off from others, and (4) feeling cut off from one's own emotions. Patients experiencing depersonalization often have great difficulty expressing what they are feeling. Trying to express their subjective suffering with banal phrases, such as "I feel dead," "Nothing seems real," or "I'm standing outside of myself," depersonalized patients may not adequately convey to the examiner the distress they experience. While complaining bitterly about how this is ruining their life, they may nonetheless appear remarkably without distress.

Ms. R was a 27-year-old, unmarried, graduate student with a master's degree in biology. She complained about intermittent episodes of "standing back," usually associated with anxiety-provoking social situations. When asked about a recent episode, she described presenting in a seminar course. "All of a sudden, I was talking, but it didn't feel like it was me talking. It was very disconcerting. I had this feeling, 'Who's doing the talking?' I felt like I was just watching someone else talk. Listening to words come out of my mouth, but I wasn't saying them. It wasn't me. It went on for a while. I was calm, even sort of peaceful. It was as if I was very far away. In the back of the room somewhere—just watching myself. But the person talking didn't even seem like me really. It was like I was watching someone else." The feeling lasted the rest of that day and persisted into the next, during which time it gradually

dissipated. She thought that she remembered having similar experiences during high school but was confident that they occurred at least once a year during college and graduate school.

As a child, Ms. R reported frequent intense anxiety from overhearing or witnessing the frequent violent arguments and periodic physical fights between her parents. In addition, the family was subject to many unpredictable dislocations and moves owing to the patient's father's intermittent difficulties with finances and employment. The patient's anxieties did not abate when the parents divorced when she was a late adolescent. Her father moved away and had little further contact with her. Her relationship with her mother became increasingly angry, critical, and contentious. She was unsure if she experienced depersonalization during childhood while listening to her parents' fights. (Adapted from a case of Richard J. Loewenstein, M.D., and Frank W. Putnam, M.D.)

Differential Diagnosis

The variety of conditions associated with depersonalization complicates the differential diagnosis of depersonalization disorder. Depersonalization can result from a medical condition or neurologic condition, intoxication or withdrawal from illicit drugs, as a side effect of medications, or can be associated with panic attacks, phobias, PTSD, or acute stress disorder, major depression, schizophrenia, illness anxiety disorder, or another dissociative disorder. A thorough medical and neurologic evaluation is essential, including standard laboratory studies, an EEG, and any indicated drug screens. Drug-related depersonalization is typically transient. However, persistent depersonalization can follow an episode of intoxication with a variety of substances, including marijuana, cocaine, and other psychostimulants. A range of neurologic conditions, including seizure disorders, brain tumors, postconcussion syndrome, metabolic abnormalities, migraine, vertigo, and Ménière disease, have been reported as causes. Depersonalization caused by organic conditions tends to be primarily sensory without the elaborated descriptions and personalized meanings common to psychiatric etiologies.

DISSOCIATIVE IDENTITY DISORDER

Dissociative identity disorder, previously called multiple personality disorder, has been the most extensively researched of all the dissociative disorders. It is the presence of two or more distinct identities or personality states. The identities or personality states, sometimes called *alters, self-states, alter identities,* or *parts,* among other terms, differ from one another in that each presents as having its pattern of perceiving, relating to, and thinking about the environment and self, in short, its personality. It is the paradigmatic dissociative psychopathology in that the symptoms of all the other dissociative disorders occur in patients with dissociative identity disorder: amnesia, often with fugue, depersonalization, derealization, and similar symptoms.

Diagnostic Features

The critical feature in diagnosing this disorder is the presence of two or more distinct personality states. Also, individuals with this disorder often report losing time beyond what one would expect from ordinary forgetfulness. There are many other diverse signs and symptoms, making diagnosis difficult. Table 11-5 describes the many other associated symptoms commonly found in patients with dissociative identity disorder, and Table 11-6 compares the different diagnostic approaches to dissociative identity disorder.

Mental Status

A careful and detailed mental status examination is essential in making the diagnosis. It is easy to mistake patients with this disorder as suffering from schizophrenia, borderline personality disorder, or outright malingering. Table 11-7 lists the questions clinicians should ask to make the proper diagnosis.

Memory and Amnesia Symptoms

Dissociative disturbances of memory are manifest in several fundamental ways and are frequently observable in clinical settings. As part of the general mental status examination, clinicians should routinely inquire about experiences of losing time, blackout spells, and significant gaps in the

continuity of recall for personal information. Dissociative time loss experiences are too extensive to be explained by normal forgetting and typically have sharply demarcated onsets and offsets.

Patients with dissociative identity disorder often report significant gaps in autobiographical memory, especially for childhood events. Dissociative gaps in the autobiographical recall are usually sharply demarcated and do not fit the usual decline in autobiographical recall for younger ages.



Table 11-5
Associated Symptoms Commonly Found in Dissociative Identity Disorder

Posttraumatic stress disorder symptoms
Intrusive symptoms
Hyperarousal
Avoidance and numbing symptoms
Somatic symptoms
Conversion and pseudoneurologic symptoms
Seizure-like episodes
Pain symptoms
Headache, abdominal, musculoskeletal, pelvic pain
Psychophysiologic symptoms or disorders
Asthma and breathing problems
Perimenstrual disorders
Irritable bowel syndrome
Gastroesophageal reflux disease
Somatic memory
Affective symptoms
Depressed mood, dysphoria, or anhedonia
Brief mood swings or mood lability
Suicidal thoughts and attempts of self-mutilation
Helpless and hopeless feelings
Obsessive-compulsive symptoms
Ruminations about trauma
Obsessive counting, singing

Arranging
Washing
Checking

Ms. A, a 33-year-old married woman employed as a librarian in a school for disturbed children, presented to psychiatric attention after discovering her 5-year-old daughter “playing doctor” with several neighborhood children. Although this event was of little consequence, the patient began to become fearful that her daughter would be molested. The patient was seen by her internist and was treated with antianxiety agents and antidepressants, but with little improvement. She sought psychiatric consultation from several clinicians but repeated, appropriate trials of antidepressants, antianxiety agents, and supportive psychotherapy resulted in limited improvement. After the death of her father from complications of an alcohol use disorder, the patient became more symptomatic. He was estranged from the family since the patient was approximately 12 years of age, owing to his drinking and associated antisocial behavior.

Psychiatric hospitalization was precipitated by the patient’s arrest for disorderly conduct in a nearby city. She was found in a hotel, in revealing clothing, engaged in an altercation with a man. She denied knowledge of how she had come to the hotel, although the man insisted that she had come there under a different name for a voluntary sexual encounter.

On psychiatric examination, the patient described dense amnesia for the first 12 years of her life, with the feeling that her “life started at 12 years old.” She reported that, for as long as she could remember, she had an imaginary companion, an elderly black woman, who advised her and kept her company. She reported hearing other voices in her head: several women and children, as well as her father’s voice repeatedly speaking to her in a derogatory way. She reported that since age 12, she had episodes of amnesia: for work, for her marriage, for the birth of her children, and her sex life with her husband. She reported perplexing changes in skills; for example, people told her that she played the piano well but had no conscious awareness that she could do so. Her husband reported that she had always been “forgetful” of conversations and family activities. He

also noted that, at times, she would speak like a child; at times, she would adopt a southern accent; and, at other times, she would be angry and provocative. She frequently had little recall of these episodes.

When questioned about her early life, the patient appeared to enter a trance and stated, “I just don’t want to be locked in the closet” in a childlike voice. Inquiry about this produced rapid shifts in a state between alter identities who differed in manifested age, facial expression, voice tone, and knowledge of the patient’s history. One spoke in an angry, expletive-filled manner and appeared irritable and preoccupied with sexuality. She discussed the episode with the man in the hotel and stated that it was she who had arranged it. Gradually, the alters described a history of family chaos, brutality, and neglect during the first 12 years of the patient’s life, until her mother, who also struggled with alcohol use disorder, achieved sobriety and fled her husband, taking her children with her. The patient, in the alter identities, described episodes of physical abuse, sexual abuse, and emotional torment by the father, her siblings, and her mother.

After an assessment of family members, the patient’s mother also met diagnostic criteria for dissociative identity disorder, as did her older sister, who also had been molested. A brother met diagnostic criteria for PTSD, major depression, and alcohol use disorder. (Adapted from a case of Richard J. Loewenstein, M.D., and Frank W. Putnam, M.D.)



Table 11-6
Dissociative Identity Disorder

	DSM-5	ICD-10
Name	**Dissociative Identity Disorder**	**Other Dissociative (Conversion) Disorders**
Symptoms	Sense of one’s identity is distorted Sense of having	

	at least two personality states, each with unique Sense of self-control Affect Behavior Memory Perception Sensory motor function Memory lapses about past events
# Symptoms needed	All the above
Exclusions (not result of):	Acceptable cultural beliefs Normal forgetting Substance use Another medical condition Fantasy play in children
Psychosocial impact	Marked distress and/or impairment

Comments	Listed among the other disorders, along with Ganser syndrome and psychogenic confusion/twilight state. Symptoms are not detailed.



Table 11-7
Mental Status Examination Questions for Dissociative Identity Disorder Process Symptoms

If answers are positive, ask the patient to describe the event. Make sure to specify that the symptom does not occur during an episode of intoxication.

1. Do you act so differently in one situation compared to another situation that you feel almost like you were two different people?
2. Do you feel that there is more than one of you? More than one part of you? Side of you? Do they seem to be in conflict or in a struggle?
3. Does that part (those parts) of you have its (their) own independent way(s) of thinking, perceiving, and relating to the world and the self? Have its (their) own memories, thoughts, and feelings?
4. Does more than one of these entities take control of your behavior?
5. Do you ever have thoughts or feelings, or both, that come from inside you (outside you) that you cannot explain? That do not feel like thoughts or feelings that you would have? That seem like thoughts or feelings that are not under your control (passive influence)?
6. Have you ever felt that your body was engaged in behavior that did not seem to be under your control? For example, saying things, going places, buying things, writing things, drawing or creating things, hurting yourself or others, and so forth? That your body does not seem to belong to you?
7. Do you ever feel that you have to struggle against another part of

you that seems to want to do or to say something that you do not wish to do or to say?

8. Do you ever feel that there is a force (pressure, part) inside you that tries to stop you from doing or saying something?
9. Do you ever hear voices, sounds, or conversations in your mind? That seem to be discussing you? Commenting on what you do? Telling you to do or not do certain things? To hurt yourself or others? That seem to be warning you or trying to protect you? That try to comfort, support, or soothe you? That provide important information about things to you? That argue or say things that have nothing to do with you? That have names? Men? Women? Children?
10. I would like to talk with that part (side, aspect, facet) of you (of the mind) that is called the "angry one" (the Little Girl, Janie, that went to Atlantic City last weekend and spent lots of money, etc.). Can that part come forward now, please?
11. Do you frequently have the experience of feeling like you are outside yourself? Inside yourself? Beside yourself, watching yourself as if you were another person?
12. Do you ever feel disconnected from yourself or your body as if you (your body) were not real?
13. Do you frequently experience the world around you as unreal? As if you are in a fog or daze? As if it were painted? Two-dimensional?
14. Do you ever look in the mirror and not recognize who you see? See someone else there?

Adapted from Loewenstein RJ. An office mental status examination for chronic complex dissociative symptoms and multiple personality disorder. *Psychiatr Clin North Am*. 1991;14:567–604, with permission.

Dissociative Alterations in Identity

Clinically, dissociative alterations in identity may first manifest with odd first-person plural or third-person singular or plural self-references. Also, patients may refer to themselves using their first names or make depersonalized self-references, such as "the body," when describing themselves and others. Patients often describe a profound sense of

concretized internal division or personified internal conflicts between parts of themselves. In some instances, these parts may have proper names, or the patient may refer to them by their predominate affect or function, for example, “the angry one” or “the wife.” Patients may suddenly change how they refer to others, for example, “the son” instead of “my son.”

Other Associated Symptoms

Most patients with dissociative identity disorder meet the criteria for a mood disorder, usually one of the depression spectrum disorders. Frequent, rapid mood swings are common but are usually part of posttraumatic and dissociative phenomena, not a cyclic mood disorder. Considerable overlap may exist between PTSD symptoms of anxiety, disturbed sleep, and dysphoria, and mood disorder symptoms.

Obsessive-compulsive personality traits are common in dissociative identity disorder, and intercurrent obsessive-compulsive disorder (OCD) symptoms occur in patients with dissociative identity disorder, with a subgroup manifesting severe OCD symptoms. OCD symptoms commonly have a posttraumatic quality: repeatedly checking to be sure that no one can enter the house or the bedroom, compulsive washing to relieve a feeling of being dirty because of abuse, and repetitive counting or singing in one’s head to distract from anxiety over being abused, for example.

Child and Adolescent Presentations

Children and adolescents manifest the same core dissociative symptoms and secondary clinical phenomena as adults. Age-related differences in autonomy and lifestyle, however, may significantly influence the clinical expression of dissociative symptoms in youth. Younger children have a less linear and less continuous sense of time and often are not able to self-identify dissociative discontinuities in their behavior. Often additional informants, such as teachers and relatives, are available to help document dissociative behaviors.

Several normal childhood phenomena, such as imaginary companionship and elaborated daydreams, must be carefully differentiated from pathologic dissociation in younger children. The clinical presentation may be that of an elaborated or autonomous imaginary companionship, with the imaginary companions taking control of the child’s behavior, often experienced through

passive influence experiences or auditory pseudohallucinations, or both, that command the child to behave in specific ways.

Differential Diagnosis

Table 11-8 lists the most common differentials for dissociative identity disorder.

Factitious, Imitative, and Malingered Dissociative Identity Disorder. Indicators of falsified or *imitative dissociative identity disorder* include those typical of other factitious or malingering presentations. These include symptom exaggeration, lies, use of symptoms to excuse antisocial behavior (e.g., amnesia only for bad behavior), amplification of symptoms when under observation, refusal to allow collateral contacts, legal problems, and pseudologia fantastica. Patients with genuine dissociative identity disorder are usually confused, conflicted, ashamed, and distressed by their symptoms and trauma history. Those with the nongenuine disorder frequently show little dysphoria about their disorder.



Table 11-8
Differential Diagnosis of Dissociative Identity Disorder

Comorbidity versus differential diagnosis
Affective disorders
Psychotic disorders
Anxiety disorders
Posttraumatic stress disorder
Personality disorders
Neurocognitive disorders
Neurologic and seizure disorders
Somatic symptom disorders
Factitious disorders
Malingering
Other dissociative disorders
Deep-trance phenomena

OTHER SPECIFIED OR UNSPECIFIED DISSOCIATIVE DISORDER

The categories of specified or unspecified dissociative disorders cover all the conditions characterized by a primary dissociative response that does not meet diagnostic criteria for one of the other DSM-5 dissociative disorders. Specified dissociative disorders include chronic and recurrent syndromes of mixed dissociative symptoms; identity disturbance due to prolonged and intense coercive persuasion (e.g., brainwashing); acute dissociative reactions to stressful events; and dissociative trance.

Dissociative Trance

Dissociative trance is manifest by a temporary, marked alteration in the state of consciousness or by loss of the customary sense of personal identity without the replacement by an alternate sense of identity. A variant of this, possession trance, involves single or episodic alternations in the state of consciousness, characterized by the exchange of the person's customary identity with a new identity usually attributed to a spirit, divine power, deity, or another person. In this possessed state, the individual exhibits stereotypical and culturally determined behaviors or experiences an entity controlling them. There must be partial or full amnesia for the event. The trance or possession state must not be a commonly accepted part of cultural or religious practice and must cause significant distress or functional impairment in one or more of the usual domains. Finally, the dissociative trance state must not occur exclusively during a psychotic disorder and is not the result of any substance use or general medical condition.

Brainwashing

DSM-5 describes this dissociative disorder as "identity disturbance due to prolonged and intense coercive persuasion." Brainwashing occurs mostly in the setting of political reform, as has been described at length with the Cultural Revolution in communist China, war imprisonment, torture of political dissidents, terrorist hostages, and, more familiarly, in Western culture, totalitarian cult indoctrination. It implies that under conditions of adequate stress and duress, those in power can make individuals comply with

their demands, thereby undergoing significant changes in their personality, beliefs, and behaviors. Persons subjected to such conditions can undergo considerable harm, including loss of health and life, and they typically manifest a variety of posttraumatic and dissociative symptoms.

Some have likened the first stage in coercive processes to the artificial creation of an identity crisis, with the emergence of a new pseudoidentity that manifests characteristics of a dissociative state. Under circumstances of extreme and malignant dependency, overwhelming vulnerability, and danger to one's existence, individuals develop a state characterized by an extreme idealization of their captors, with ensuing identification with the aggressor and externalization of their superego, regressive adaptation known as *traumatic infantilism,* paralysis of will, and a state of frozen fright. The coercive techniques include isolation of the subject, degradation, control of overall communications and essential daily functions, induction of fear and confusion, peer pressure, assignment of repetitive and monotonous routines, the unpredictability of environmental supplies, renunciation of past relationships and values, and various deprivations. Even though physical or sexual abuse, torture, and extreme sensory deprivation and physical neglect can be part of this process, they are not required to define a coercive process. As a result, victims manifest extensive posttraumatic and dissociative symptomatology, including drastic alteration of their identity, values, and beliefs; reduction of cognitive flexibility with regression to simplistic perceptions of good versus evil and dominance versus submission; numbing of experience and blunting of affect; trance-like states and diminished environmental responsiveness; and, in some cases, more severe dissociative symptoms such as amnesia, depersonalization, and shifts in identity.

Ganser Syndrome

Ganser syndrome is a poorly understood condition characterized by the giving of approximate answers (*paralogia*), together with a clouding of consciousness, and, often, hallucinations and other dissociative, somatoform, or conversion symptoms.

The symptom of *passing over* (*vorbeigehen*) the correct answer for a related but incorrect one is the hallmark of Ganser syndrome. The approximate answers often just miss the mark but bear an obvious relation to

the question, indicating that the person must have understood. When asked how old she was, a 25-year-old woman answered, “I’m not five.” If asked to do simple calculations (e.g., 2 + 2 = 5); for general information (the capital of the United States is New York); to identify simple objects (a pencil is a key); or to name colors (green is gray), the patient with Ganser syndrome gives erroneous but comprehensible answers. They may even answer incorrectly to rhetorical questions (“how many legs are on a three-legged stool?” “Four?”)

A clouding of consciousness also occurs, usually manifest by disorientation, amnesias, loss of personal information, and some impairment of reality testing. Visual and auditory hallucinations occur in roughly one-half of the cases. Neurologic examination may reveal what Ganser called *hysterical stigmata*, for example, nonneurologic analgesia or shifting hyperalgesia. It must include other dissociative symptoms, such as amnesias, conversion symptoms, or trance-like behaviors.

Differential Diagnosis. The examiner should conduct a thorough neurologic and medical evaluation, given the reported frequent history of organic brain syndromes, seizures, head trauma, and psychosis in Ganser syndrome. Differential diagnoses include organic dementia, depressive pseudodementia, the confabulation of Korsakoff syndrome, organic dysphasia, and reactive psychoses. Patients with dissociative identity disorder occasionally may also exhibit Ganser-like symptoms.

COMORBIDITY

Common comorbidities found in dissociative disorders include depressive disorders, adjustment disorder, anxiety disorders, trauma and stress-related disorders, eating disorders, OCD, somatic symptom disorders, and conversion disorder. Individuals with dissociative disorders may also meet the criteria for personality disorders, with reports of co-occurring avoidant, borderline, dependent, and obsessive-compulsive personality disorders predominating.

COURSE AND PROGNOSIS

Dissociative Amnesia

We know little about the clinical course of dissociative amnesia. After removing the person to safety from traumatic or overwhelming circumstances, the acute dissociative amnesia frequently spontaneously resolves. At the other extreme, some patients do develop chronic forms of generalized, continuous, or severe localized amnesia and are profoundly disabled and require high levels of social support, such as nursing home placement or intensive family caretaking. Clinicians should try to restore patients' lost memories to consciousness as soon as possible; otherwise, the repressed memory may form a nucleus in the unconscious mind around which future amnestic episodes may develop.

In those who develop a dissociative fugue, most fugue states are relatively brief, lasting from hours to days. Most individuals appear to recover, although refractory dissociative amnesia may persist in rare cases. Some studies have described recurrent fugues in most individuals presenting with an episode of dissociative fugue.

Depersonalization/Derealization Disorder

Depersonalization after traumatic experiences or intoxication commonly remits spontaneously after removal from the traumatic circumstances or ending of the episode of intoxication. Depersonalization accompanying mood, psychotic, or other anxiety disorders commonly remit with the definitive treatment of these conditions.

Depersonalization disorder may have an episodic, relapsing and remitting, or chronic course. Many patients with chronic depersonalization may have a severe and chronic functional impairment. The mean age of onset is usually late adolescence or early adulthood.

Dissociative Identity Disorder

We know little about the natural history of untreated dissociative identity disorder. Some individuals with untreated dissociative identity disorder may continue in abusive relationships or violent subcultures, or both, that may result in the traumatization of their children, with the potential for additional family transmission of the disorder. Many authorities believe that some percentage of patients with undiagnosed or untreated dissociative identity disorder die by suicide or because of their risk-taking behaviors.

Prognosis is poorer in patients with comorbid medical disorders, psychotic disorders (*not* dissociative identity disorder pseudopsychosis), and severe medical illnesses. Refractory substance use and eating disorders also suggest a poorer prognosis. Other factors that usually indicate a poorer outcome include significant antisocial personality features, current criminal activity, ongoing perpetration of abuse, and current victimization, with the refusal to leave abusive relationships. Repeated adult traumas with recurrent episodes of acute stress disorder may severely complicate the clinical course.

TREATMENT

Dissociative Amnesia

Psychotherapy

PHASE-ORIENTED TREATMENT. Phase-oriented treatment is the current standard of care for the treatment of dissociative amnesia, although there are no systematic studies with large cohorts of dissociative amnesia patients. This phasic treatment follows the three-stage phasic treatment model developed for the treatment of complex PTSD and dissociative identity disorder, which is discussed extensively in the section on treatment of dissociative identity disorder, below. When applied to dissociative amnesia treatment, memory recall is a central issue, because the loss of memory for personal identity and gaps in current autobiographical memory are acutely disabling symptoms that require relatively rapid intervention.

COGNITIVE THERAPY. Cognitive therapy may have specific benefits for individuals with trauma disorders. Identifying the specific cognitive distortions about the trauma may provide an entrée into autobiographical memory for which the patient experiences amnesia. As the patient becomes able to correct cognitive distortions, particularly about the meaning of prior trauma, a more detailed recall of traumatic events may occur.

HYPNOSIS. Hypnotic interventions can help to contain, modulate, and titrate the intensity of symptoms; to facilitate controlled recall of dissociated memories; to provide support and ego strengthening for the patient; and, finally, to promote working through and integration of dissociated material.

Also, the patient can learn self-hypnosis to apply containment and calming techniques in his or her everyday life. Successful use of containment techniques, whether hypnotically facilitated or not, also increases the patient's sense that he or she can more effectively be in control of alternations between intrusive symptoms and amnesia.

GROUP PSYCHOTHERAPY. Time-limited and longer-term group psychotherapies can help combat veterans with PTSD and survivors of childhood abuse. During group sessions, patients may recover memories for which they have had amnesia. Supportive interventions by the group members or the group therapist, or both, may facilitate integration and mastery of the dissociated material.

Somatic Therapies. No known pharmacotherapy exists for dissociative amnesia other than pharmacologically facilitated interviews. The interviews have used a variety of sedatives, including sodium amobarbital, thiopental, oral benzodiazepines, and amphetamines.

Pharmacologically facilitated interviews using intravenous amobarbital or diazepam are used primarily in working with acute amnesias and conversion reactions, among other indications, in general hospital medical and psychiatric services. This procedure is also occasionally useful in refractory cases of chronic dissociative amnesia when patients are unresponsive to other interventions. The material uncovered in a pharmacologically facilitated interview needs to be processed by the patient in his or her usual conscious state. The Joint Commission now considers pharmacologically facilitated interviews to be conscious sedation, requiring the presence of an anesthesiologist.

Depersonalization/Derealization Disorder

Clinicians working with patients with depersonalization/derealization disorder often find them to be a singularly clinically refractory group. However, benzodiazepines, serotonin reuptake inhibitors, and stimulants appear to be partially efficacious for some patients. There is absolutely no empirical evidence for the efficacy of typical or atypical antipsychotic medications in depersonalization/derealization disorder, and indeed these medications may increase a feeling of emotional deadness and lack of

emotional response to oneself or the world. Also, opioid antagonists such as naltrexone and cognitive enhancers have had some benefit in the clinical treatment of specific depersonalization/derealization disorder patients. Early data suggest the possible efficacy of repetitive transcranial magnetic stimulation (rTMS) treatment.

Many different types of psychotherapy may help depersonalization disorder: psychodynamic, cognitive, cognitive-behavioral, hypnotherapeutic, and supportive. However, the response is not robust for many patients. Stress management strategies, distraction techniques, reduction of sensory stimulation, relaxation training, and physical exercise may be somewhat helpful in some patients.

Dissociative Identity Disorder

Psychotherapy. The current model of treatment of the dissociative identity disorder patient follows a phasic model that is the current standard of care for complex posttraumatic disorders. The phases include: (1) development of safety and symptom stabilization; (2) (optionally) focused, in-depth attention to traumatic material; and (3) integration or reintegration, in which the individual moves away from a life adaptation based on chronic traumatization and victimization. These phases are relatively heuristic, and aspects of each may be part of the others.

STAGE 1 PHASIC TRAUMA TREATMENT: STABILIZATION AND SAFETY. Stabilization of the individual with dissociative identity disorder is vital to the successful negotiation of all aspects of treatment. Stabilization focuses on the safety and management of core dissociative identity disorder and comorbid symptoms. The majority of patients with dissociative identity disorder come to treatment because of overwhelming PTSD and dissociative symptoms and because of some form of self-destructive behavior or violence against others.

Before establishing safety, the patient in reality or symbolically lives in a world of ongoing trauma, whether perpetrated on the patient by others or by the patient on him/herself due to chronic forms of repeated self-destruction and high-risk behavior. The clinician must logically prioritize basic life and health over other interventions. The clinician may need to resort to many different levels and types of interventions to protect the patient, including

hospitalization, specialized substance use or eating disorders programs, police assistance and shelter for victims of intimate partner violence, social service interventions to protect minor children, and assistance with housing, and access to medical care.

STAGE 2 PHASIC TRAUMA TREATMENT: WORK ON TRAUMATIC MEMORIES. For patients who stabilize and form a reasonable working alliance in treatment, longer-term treatment goals involve the detailed, affectively intense, psychotherapeutic processing of life experiences, and the transformation of the meaning of these experiences for the individual. Authorities emphasize that, in most cases, intensive, detailed psychotherapeutic work with traumatic memories should only start after the patient has demonstrated the ability to use symptom management skills independently, after the self-state system can work together in a reasonably cooperative way, and after a stable therapeutic relationship has been established.

The patient should be able to give informed consent and have a realistic understanding of the potential risks and benefits of intensive focus on traumatic material. Potential risks may include temporary, acute worsening of PTSD, affective, somatoform, and self-destructive symptoms, and short-term interference with daily activities. Long-term benefits may include significant amelioration of dissociative and PTSD symptoms, decreases in subjective self-division, the fusion of identities, and freeing of psychological energy for daily life. The patient must be able to understand that the goal is the integration of dissociated thoughts, feelings, recollections, and perceptions, not the exhumation or expulsion of memories per se.

The patient should not be in the midst of an acute life crisis or significant life change. Also, we should first stabilize comorbid medical and psychiatric disorders. The patient should have the ego strength and psychosocial resources to withstand the rigors of the process, and there must be adequate resources, such as support by significant others, and, potentially, support for additional sessions.

Experienced clinicians attempt to carefully structure affectively intense sessions focused on traumatic material, with attention to affect modulation, restabilization of the patient before concluding the session, and reasonable availability to assist the patient supportively between sessions. It may require many sessions to fully explicate the cognitive and emotional meaning of

traumatic events, so that they may become part of the patient's repertoire of nondissociated, ordinary memories for life experience.

STAGE 3 PHASIC TRAUMA TREATMENT: FUSION, INTEGRATION, RESOLUTION, AND RECOVERY. Throughout treatment, we may observe significant unification of dissociated mental processes. Self-states lose distinctness and decrease compartmentalization of thoughts, memories, and affects. The patient develops a more unified sense of self. Transference is modified, consistent with these changes. Amnesia and switching become less apparent than before. The fusion of states results in the psychological merging of two or more entities at a point in time, with a subjective experience of loss of all separateness. The term *integration* is sometimes used synonymously with *fusion* but is a more general term, describing the process of undoing all forms of dissociative division during treatment. Some patients seem to proceed to a complete fusion of all states. They may shift in self-representation, and develop a consistent and continuous sense of self. Many patients never attain a complete fusion of their personalities but achieve a therapeutic *resolution:* improved communication, collaboration, and cooperation among self-states leading to relative stability and adequate function.

HYPNOSIS. Hypnotherapeutic interventions can often alleviate self-destructive impulses or reduce symptoms, such as flashbacks, dissociative hallucinations, and passive-influence experiences. Teaching the patient self-hypnosis may help with crises outside of sessions. Hypnosis can be useful for accessing specific alter personality states, and their sequestered affects and memories. Hypnosis can create relaxed mental states in which the subject can examine adverse life events without overwhelming anxiety. Clinicians using hypnosis should obtain specialized training in trauma populations. Clinicians should be aware of current controversies over the impact of hypnosis on accurate reporting of recollections and should use appropriate informed consent for its use.

GROUP THERAPY. In therapy groups, including general psychiatric patients, the emergence of alter personalities can be disruptive to the group process by eliciting excess fascination or by frightening other patients. Therapy groups composed only of patients with dissociative identity disorder seem to be more

successful, although the groups must be carefully structured, must provide firm limits, and should generally focus only on here-and-now issues of coping and adaptation.

FAMILY THERAPY. Family or couples therapy is often crucial for long-term stabilization and to address pathologic family and marital processes that are common in patients with dissociative identity disorder and their family members. Education of family and concerned others about dissociative identity disorder and its treatment may help family members cope more effectively with dissociative identity disorder and PTSD symptoms in their loved ones. Group interventions for education and support of family members have also been found helpful. Sex therapy may be a critical part of couples' treatment, because patients with dissociative identity disorder may become intensely phobic of intimate contact for periods, and spouses may have little idea how to deal with this helpfully.

SELF-HELP GROUPS. Patients with dissociative identity disorder usually have a negative outcome to self-help groups or 12-step groups for incest survivors. A variety of problematic issues occur in these settings, including the intensification of PTSD symptoms because of discussion of trauma material without clinical safeguards, exploitation of the patient with dissociative identity disorder by predatory group members, contamination of the patient's recall by group discussions of trauma, and a feeling of alienation even from these other reputed sufferers of trauma and dissociation.

EXPRESSIVE AND OCCUPATIONAL THERAPIES. Expressive and occupational therapies, such as art and movement therapy, have proved particularly helpful in the treatment of patients with dissociative identity disorder. Art therapy may help with the containment and structuring of severe dissociative identity disorder and PTSD symptoms, as well as to permit these patients safer expression of thoughts, feelings, mental images, and conflicts that they have difficulty verbalizing. Movement therapy may facilitate the normalization of body sense and body image for these severely traumatized patients. Occupational therapy may help the patient with focused, structured activities that can be completed successfully and may help with grounding and symptom management.

EYE MOVEMENT DESENSITIZATION AND REPROCESSING. Eye movement desensitization and reprocessing (EMDR) is a treatment for PTSD. No systematic studies have been done in dissociative identity disorder patients using EMDR. Case reports suggest that some dissociative identity disorder patients may be destabilized by EMDR procedures, especially those with acutely increased PTSD and dissociative symptoms. Some authorities believe that EMDR can be a helpful adjunct for later phases of treatment in well-stabilized dissociative identity disorder outpatients. The International Society for the Study of Trauma and Dissociation dissociative identity disorder treatment guidelines suggest that EMDR only be used in this patient population by clinicians who have taken advanced EMDR training, are knowledgeable and skilled in phasic trauma treatment for dissociative disorders and have received supervision in the use of EMDR in dissociative identity disorder.

DIALECTICAL BEHAVIOR THERAPY. Dialectical behavior therapy (DBT) is a treatment developed by the psychologist Marsha Linehan for borderline personality disorder. Its concepts and techniques may be helpful to structure aspects of stabilization throughout dissociative identity disorder treatment since many patients with borderline personality disorder report significant trauma histories and fit the complex trauma construct. Helpfully for the patient with dissociative identity disorder, DBT focuses on the primacy of safety, therapy-interfering behaviors, affect regulation strategies, distress tolerance, and radical acceptance.

However, DBT may also have notable limitations for the patient with a dissociative identity disorder. It does not focus on stabilization and management of dissociative symptoms such as depersonalization/derealization and dissociative amnesia, flashbacks, and unconscious reenactments, traumatic transference, and related trauma-based symptoms.

Psychopharmacologic Interventions. Guidelines for the use of medications with dissociative patients emphasize the need to identify specific treatment-responsive symptoms rather than attempting to treat the dissociation per se. Medications may help attenuate symptoms to assist the patient in stabilizing during treatment. We should advise patients that the

medication response is likely to be partial, devising the best “shock-absorber” system for the patient at a given time. In general, success is more likely if the target symptoms are present across most or all personality states, rather than confined to one or a few. Because rapid symptom oscillations are characteristic in this population, it is not advisable to chase each symptom shift with a medication change.

The most robust medication treatment effect in this patient population is the response of nightmares to the α_1-adrenergic antagonist prazosin. If the patient tolerates the blood pressure effects, dosages of prazosin as high as 25 mg daily, in single or twice-daily doses, can be useful. Daytime dosing may also be effective for control of flashbacks and intrusive symptoms. Doxazosin, a longer-acting α_1-adrenergic antagonist, may also help with nightmares and intrusive symptoms in PTSD.

Antidepressant medications can reduce depression and stabilize mood. A variety of PTSD symptoms, especially intrusive and hyperarousal symptoms, are partially medication responsive. Clinicians report some success with SSRI, tricyclic, and monoamine oxidase (MAO) antidepressants, β-blockers, clonidine, anticonvulsants, and benzodiazepines in reducing intrusive symptoms, hyperarousal, and anxiety in patients with a dissociative identity disorder. Patients with obsessive-compulsive symptoms may respond to serotonergic antidepressants. Emerging data suggest that naltrexone may help ameliorate recurrent self-injurious behaviors in a subset of traumatized patients.

The atypical neuroleptics, such as risperidone, quetiapine, ziprasidone, aripiprazole, and olanzapine, may be more effective and better tolerated than typical neuroleptics for overwhelming anxiety and intrusive PTSD symptoms in patients with dissociative identity disorder. However, weight gain and metabolic syndrome may be a particularly problematic side effect for these nonpsychotic patients. Occasionally, an extremely disorganized, overwhelmed, chronically ill patient with dissociative identity disorder, who has not responded to trials of other neuroleptics, responds favorably to a trial of clozapine.

Electroconvulsive Therapy. For some patients, ECT helps to ameliorate refractory mood disorders and does not worsen dissociative memory

problems. Clinical experience in tertiary care settings for severely ill patients with dissociative identity disorder suggests that a clinical picture of major depression with persistent, refractory melancholic features across all alter states may predict a positive response to ECT. This response is usually only partial, however, as is typical for most successful somatic treatments in the dissociative identity disorder population.

Table 11-9 lists target symptoms and somatic treatments for dissociative identity disorder.

Dissociative Trance

There are no systematic studies of dissociative trance disorder in the absence of pathologic possession. In general, treatment should focus on the patient understanding the nature of trance states and their relationship to more ordinary phenomena such as daydreaming. Assessment of hypnotizability may also allow the clinician to produce the same symptoms the patient is experiencing with a hypnotic induction. This may lead to an increased sense of control and may be enough to assist the patient in recognizing autohypnotic states and having strategies to manage these. In more chronic cases, supportive–expressive psychotherapy, like that for conversion disorder, may be most successful.



Table 11-9
Medications for Associated Symptoms in Dissociative Identity Disorder

Medications and somatic treatments for posttraumatic stress disorder (PTSD), affective disorders, anxiety disorders, and obsessive-compulsive disorder (OCD)
Selective serotonin reuptake inhibitors (no preferred agent, except for OCD symptoms)
Fluvoxamine (for OCD presentations)
Clomipramine (for OCD presentations)
Tricyclic antidepressants
Monoamine oxidase inhibitors (if patient can reliably maintain diet safely)

- Electroconvulsive therapy (for refractory depression with persistent melancholic features across all dissociative identity disorder alters)
- Mood stabilizers (more useful for PTSD and anxiety than mood swings)
- Divalproex
- Lamotrigine
- Oral or intramuscular benzodiazepines

Medications for sleep problems

- Low-dose trazodone
- Low-dose mirtazapine
- Low-dose tricyclic antidepressants
- Low-dose neuroleptics
- Benzodiazepines (often less helpful for sleep problems in this population)
- Zolpidem
- Anticholinergic agents (diphenhydramine, hydroxyzine)

Medications for self-injury, addictions

- Naltrexone

Brainwashing

There are no empirical treatment studies of individuals subjected to extreme coercion applied in the service of indoctrination into a belief system or an attempt at identity alteration. The basic principles of phased trauma treatment would seem to provide an organizing framework for their care, as for victims of other forms of torture. Family and social interventions may be necessary because of the duress and disruption accompanying the precipitating events and the social effects of the profound changes in the patient's attitudes, behaviors, and beliefs, as well as sequestration from everyday events during the period of captivity.

Ganser Syndrome

No systematic treatment studies have been conducted, given the rarity of this condition. In most case reports, the patient has been hospitalized and provided with a protective and supportive environment. In some instances,

low doses of antipsychotic medications may be beneficial. Confrontation or interpretations of the patient's approximate answers are not productive, but the exploration of possible stressors may be helpful. Hypnosis and amobarbital narcosynthesis can also help patients reveal the underlying stressors that preceded the development of the syndrome, with the concomitant cessation of the Ganser symptoms.

Usually, a relatively rapid return to normal function occurs within days, although some cases may take a month or more to resolve. The individual is typically amnesic for the period of the syndrome.

EPIDEMIOLOGY

Dissociative Amnesia

Dissociative amnesia is in a range of approximately 2 to 6 percent of the general population. There is no known difference in incidence between men and women. Cases generally begin to be reported in late adolescence and adulthood. Dissociative amnesia can be especially challenging to assess in preadolescent children because of their more limited ability to describe the subjective experience.

Depersonalization/Derealization Disorder

Transient experiences of depersonalization and derealization are prevalent in normal and clinical populations. They are the third most commonly reported psychiatric symptoms, after depression and anxiety. Transient symptoms are common in the general population, with a lifetime prevalence of 26 to 74 percent, and 31 to 66 percent at the time of a traumatic event. Symptoms can occur with the use of illicit drugs, especially marijuana, hallucinogens, ketamine, and 3,4-methylenedioxymethamphetamine (MDMA or Ecstasy). They have been described after certain types of meditation, deep hypnosis, extended mirror or crystal gazing, and sensory deprivation experiences. They are also common after mild to moderate head injury, wherein little or no loss of consciousness occurs, but they are significantly less likely if unconsciousness lasts for more than 30 minutes. Depersonalization/derealization symptoms also occur in patients with epilepsy, both as a manifestation of the aura as well as the seizure itself,

particularly in complex–partial seizure patients. The prevalence of depersonalization/derealization disorder is approximately 2.5 percent, with no age or gender differences.

Dissociative Identity Disorder

Few systematic epidemiologic data exist for dissociative identity disorder. Based on community samples, the prevalence in the general population is approximately 1 to 1.5 percent. Developmental studies indicate that the ratio of female to male dissociative identity disorder cases steadily increases from 1 to 1 in early childhood to approximately 8 to 1 by late adolescence in North American samples.

Ganser Syndrome

The overall frequency of Ganser syndrome has declined with time. Men outnumber women by approximately 2 to 1, and cases have been reported in a variety of cultures. Three of Ganser's first four cases were convicts, leading some authors to consider it to be a disorder of penal populations and, thus, an indicator of potential malingering.

ETIOLOGY

Dissociative Amnesia

In many cases of acute dissociative amnesia, the psychosocial environment out of which the amnesia develops is massively conflictual, with the patient experiencing intolerable emotions of shame, guilt, despair, rage, and desperation. These usually result from conflicts over unacceptable urges or impulses, such as intense sexual, suicidal, or violent compulsions. Traumatic experiences such as physical or sexual abuse can induce the disorder. In some cases, the trauma is caused by a betrayal by a trusted, needed other (betrayal trauma). This betrayal may influence how the event is processed and remembered.

Depersonalization/Derealization Disorder

Psychodynamic. Traditional psychodynamic formulations have

emphasized the disintegration of the ego or have viewed depersonalization as an affective response in defense of the ego. These explanations stress the role of overwhelming, painful experiences or conflictual impulses as triggering events.

Traumatic Stress. A substantial proportion—typically one-third to one-half—of patients in clinical depersonalization case series have histories of significant trauma. Several studies of accident victims find as many as 60 percent of those with a life-threatening experience report at least transient depersonalization during the event or immediately after that. Military training studies find that stress and fatigue can evoke symptoms of depersonalization and derealization and affect performance.

Neurobiologic Theories. There are several neurobiologic theories related to depersonalization and derealization symptoms. These include animal models of *submissive immobility* when an organism determines it is unable to defend itself from a predator, like the concept of "playing dead"; disturbances in inferior bilateral frontal, medial prefrontal, cingulate, insula, and temporoparietal cortex; and neurochemical dysregulation, with various studies implicating the NMDA glutamate receptor, neuropeptide Y, norepinephrine, serotonin, and the opioid system.

Dissociative Identity Disorder

Severe experiences of early childhood trauma are the likely cause of dissociative identity disorder. From 85 to 97 percent of dissociative identify disorder cases reported severe childhood trauma or maltreatment. Physical and sexual abuse are common sources of childhood trauma.

Ganser Syndrome

Some case reports identify precipitating stressors, such as personal conflicts and financial reverses, whereas others note organic brain syndromes, head injuries, seizures, and medical or psychiatric illness. Psychodynamic explanations are common in the older literature; in some cases, there are clear medical or neurologic causes. The physiologic insults from such a disorder may act as acute stressors, precipitating the syndrome in vulnerable

individuals. Some patients have reported significant histories of childhood maltreatment and adversity.

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12

Somatic Symptom and Related Disorders

Everyone experiences somatic symptoms, and most can cope with them effectively. However, some people's lives are overwhelmed by their somatic concerns. Sometimes the somatic concerns stem from well-established major medical illnesses; sometimes, the origins of the concerns are never quite clear. What is common in both situations are the pervasive and overwhelming thoughts and behaviors centered on these sensations. DSM-5 incorporated this perspective because the older perspective (only counting or cataloging symptoms labeled as medically unexplained) embodied in the previous versions of the disorder was unreliable and such perspective often put the doctor and the patient at odds over the question of the legitimacy and "reality" of the patient's symptoms and personal suffering. ICD-10 continues the older approach. However, the planned revision will be more in line with DSM-5.

THE CLINICAL PRESENTATION

Persons suffering from one of these disorders have one or more somatic symptoms that become all-consuming or lead to notable impairment in their day-to-day lives. These symptoms are no longer required to be medically unexplained, as such a distinction in itself is unreliable, and because psychiatrists commonly treat patients with medically established diagnoses who are disproportionately troubled by or preoccupied with their physical symptoms.

Somatic Symptom Disorder

Patients with somatic symptom disorder believe that they have some severe yet undetected disease, and evidence to the contrary does not persuade them otherwise. They may maintain a belief that they have a particular disease or, over time, may transfer their belief to another disease. They are fixated on one or more somatic symptoms that they are convinced are evidence of illness. For some individuals, their convictions persist despite negative laboratory results, the benign course of the alleged disease over time, and appropriate reassurances from physicians. Others may have a genuine medical condition about which they develop excessive and unreasonable anxiety, and this is also a manifestation of somatic symptom disorder. Patients with somatic symptom disorder often experience symptoms of depression and anxiety, in addition to their somatic symptoms.

Illness Anxiety Disorder

Patients with illness anxiety disorder, like those with somatic symptom disorder, believe they have a serious but undiagnosed disease despite evidence to the contrary. They may maintain a belief that they have a particular disease or, as time progresses, they may transfer their belief to another disease. Their convictions persist despite negative laboratory results, the benign course of the alleged disease over time, and appropriate reassurances from physicians. Their preoccupation with illness interferes with their interaction with family, friends, and coworkers. They are often addicted to internet searches about their feared illness, inferring the worst from information (or misinformation) they find there. Unlike somatic symptom disorder, however, these individuals do not have significant physical symptoms. Sometimes people with this disorder develop a fear of going to medical appointments, while other times, they seek excessive reassurance about their health from medical providers.

Conversion Disorder (Functional Neurologic Symptom Disorder)

Persons with conversion disorder (also called functional neurologic symptom disorder) present with what appears to be a neurologic condition. The symptoms may be motor or sensory but are incompatible with known neurologic conditions. Often the illness is preceded by conflicts or other stressors and may seem to be associated with apparent psychological factors.

Individuals with conversion disorder do not intentionally produce these symptoms or deficits. Conversion motor symptoms mimic syndromes such as paralysis, ataxia, dysphagia, or seizure disorder (nonepileptic seizures [NESs]), and the sensory symptoms mimic neurologic deficits such as blindness, deafness, or anesthesia. There can also be disturbances of consciousness (e.g., amnesia, fainting spells).

Psychological Factors Affecting Other Medical Conditions

Patients with this disorder have physical disorders caused by or adversely affected by emotional or psychological factors. A medical condition must always be present to make the diagnosis. Common clinical examples include denial and refusal of treatment for an acute condition (such as myocardial infarct or abdominal emergencies) by individuals with certain personality styles (e.g., domineering or controlling), the exacerbation of asthma or irritable bowel attacks by anxiety, and the manipulation of insulin by an individual with diabetes, or diuretics in the case of hypertensive patients, in efforts to lose weight.

Factitious Disorder

Patients with factitious disorder feign, misrepresent, simulate, cause, induce, or aggravate illness to receive medical attention, regardless of whether or not they are ill. Thus, they may inflict painful, deforming, or even life-threatening injuries on themselves, their children, or other dependents. The primary motivation is not the avoidance of duties, financial gain, or anything concrete. The motivation is simply to receive medical care and to partake in the medical system.

Factitious disorders can lead to significant morbidity or even mortality. Therefore, even the patients falsify their presenting complaints, health professionals must take the medical and psychiatric needs of these patients seriously, as their self-induced symptoms can result in significant harm or even death. Historically this disorder was called “Munchausen syndrome,” a reference to the Baron Munchausen, legendary for his outrageously exaggerated stories of his military career.

Other Specified and Unspecified Somatic Symptom and Related

Disorders

Patients with other specified somatic symptom and related disorders present with somatic symptoms that do not meet the threshold for another disorder. For example, they may present with symptoms consistent with illness anxiety disorder, except that the symptoms do not meet the duration criterion; in this case, the diagnosis would be brief illness anxiety disorder.

When there is not enough information to make a specific diagnosis, then clinicians should use the unspecified somatic symptom and related disorder diagnosis.

DIAGNOSIS

Somatic Symptom Disorder

According to the DSM-5, individuals with somatic symptom disorder present with one or more somatic complaints that result in significant angst or functional impairment. Also, they must be anxious about their symptoms or be preoccupied with them. The analogous diagnosis in ICD-10, as well in the previous version of the DSM is somatization disorder. The biggest difference between the two concepts is whether there needs to be evidence that there is no underlying medical cause for the disorder—ICD-10 does, and DSM-IV did require this whereas DSM-5 does not. Table 12-1 compares the approaches to this diagnosis.

Mr. K, a white man in his mid-50s, consulted a general medicine clinic complaining of gastrointestinal problems. He had a long list of physical symptoms and concerns, mostly related to the gastrointestinal system. These included abdominal pain, left lower quadrant cramps, bloating, persistent sense of fullness in stomach hours after eating, intolerance to foods, constipation, decrease in physical stamina, heart palpitations, and feelings that “skin is getting yellow” and “not getting enough oxygen.” A review of systems disclosed disturbances from virtually every organ system, including tired eyes with blurred vision, sore throat and “lump” in the throat, heart palpitations, irregular heartbeat, dizziness, trouble breathing, and general weakness.

The patient reported that symptoms started more than 20 years ago. Over this time, psychiatrists, general practitioners, and other medical specialists evaluated him, including surgeons. He used the internet regularly and traveled extensively in search of expert evaluations, seeking new procedures and diagnostic assessments. He had undergone repeated colonoscopies, sigmoidoscopies, and computed tomographic (CT) scans, magnetic resonance imaging (MRI) studies, and ultrasound examinations of the abdomen that had revealed Barrett esophagus but no other pathology. He was on disability and had been unable to work for more than 2 years due to his condition.

About 3 years before his visit to the medical clinic, his abdominal complaints and his fixed belief that he had an intestinal obstruction led to an exploratory surgical intervention for the first time, apparently with negative findings. However, according to the patient, the surgery "got things even worse," and since then, he had undergone at least five more operations. During these surgeries, he has undergone subtotal colectomies and ileostomies due to possible "adhesions" to rule out "mechanical" obstruction. However, available records from some of the surgeries do not disclose any specific pathology other than "intractable constipation." Pathologic specimens were also inconclusive.

A complete physical and neurologic examination showed a well-developed, well-nourished male who was afebrile. A complete physical and neurologic examination was normal except for examination of the abdomen, which revealed multiple abdominal scars. Right ileostomy was present, with soft stool in the bag and active bowel sounds. There was no point tenderness and no abdominal distension. During the examination, the patient kept pointing to an area of "hardness" in the left lower quadrant that he thought was a "tight muscle strangling his bowels." However, the examination did not disclose any palpable mass. Skin and extremities were all within normal limits, and all joints had a full range of motion and no swelling. Musculature was well developed. Neurologic examination was within normal limits. His primary care physician scheduled brief monthly visits by, during which the doctor performed brief physicals, reassured the patient, and allowed the patient to talk about "stressors." The physician avoided invasive tests or diagnostic procedures, did not prescribe any medications and avoided telling the

patient that the symptoms were mental or “all in his head.” The primary care physician then referred the patient back to psychiatry.

The psychiatrist confirmed a long list of physical symptoms that started in his 20s, most of which remained medically unexplained. The psychiatric examination revealed some anxiety symptoms, including apprehension, tension, uneasiness, and somatic components such as blushing and palpitations that seemed particularly prominent in social situations. Possible symptoms of depression included mild dysphoria, low energy, and sleep disturbance, all of which the patient blamed on his “medical” problems. The mental status examination showed that Mr. K’s mood was rather somber and pessimistic, although he denied feeling sad or depressed. Affect was irritable. He was somatically focused and had little if any, psychological insight. The examination revealed the presence of a few life stressors (unemployment, financial problems, and family issues) that the patient quickly discounted as unimportant. The psychiatrist diagnosed somatic symptom disorder, severe.

Although the patient continued to deny having any psychiatric problems or any need for psychiatric intervention or treatment, he agreed to a few regular visits to continue to assess his situation. He refused to have anyone from his family involved in this process. Efforts to engage the patient with formal therapy such as cognitive-behavioral therapy (CBT) or a medication trial were all futile, so he was seen only for “supportive psychotherapy,” with the hope of developing rapport and preventing additional iatrogenic complications.

During the follow-up period, the patient was operated on at least one more time and continued to complain of abdominal bloating and constipation and to rely on laxatives. The patient continued to believe he had an obstruction of his intestines; this bordered on the delusional. However, he continued to refuse pharmacologic treatment. The only medication he accepted was a low-dose benzodiazepine for anxiety. He continued to monitor his intestinal function 24 hr/day and to seek evaluation by prominent specialists, traveling to high-profile specialty centers far from home in search of solutions. (Courtesy of J. I. Escobar, M.D.)

Table 12-1
Somatic Symptom Disorder (DSM-5) and Somatization Disorder (ICD-10)

	DSM-5	ICD-10
Name	**Somatic Symptom Disorder**	**Somatization Disorder**
Duration	≥6 mo	≥2 yr
Symptoms	• Preoccupied with symptoms and their potential seriousness. • Anxious about symptoms and health • Time and energy devoted to the symptoms	• Recurring symptoms • Symptoms frequently change • Prolonged involvement with the medical system • High use of healthcare services
# Symptoms needed	≥1	
Exclusion (not the result of):		Malingering Medical cause for symptoms on workup
Psychosocial impact	Distress or impairment	Social, interpersonal, and familial disruption
Symptom specifiers	With predominant pain	
Severity specifiers	***Mild:*** 1 symptom ***Moderate:*** ≥2 symptoms ***Severe:*** ≥2 symptoms, with	Undifferentiated somatoform disorder: <2 yr, less severe symptoms Somatoform autonomic

	multiple symptomatic complaints or 1 that is severe	dysfunction: autonomic arousal symptoms or other less specific complaints Persistent somatoform pain disorder (pain is the predominant symptom)
Course specifiers	***Persistent:*** >6 mo of severe symptoms and impairment	

Although DSM-5 specifies that the symptoms must be present for at least 6 months, transient manifestations can occur after significant stresses, most commonly the death or severe illness of someone important to the patient. It can also occur after a severe illness that resolved but left the patient shaken by the experience. Such states that last fewer than 6 months are diagnosed as "Other Specified Somatic Symptom and Related Disorders" in DSM-5. Transient somatic symptom disorder responses to external stress generally remit with a resolution of the stress, but they can become chronic if friends or healthcare professionals reinforce the concerns.

Illness Anxiety Disorder

The primary DSM-5 diagnostic criteria for illness anxiety disorder are that patients be preoccupied with the false belief that they have or will develop a severe disease in the presence of few if any physical signs or symptoms. With illness anxiety disorder, patients have relatively minor somatic symptoms and instead focus on concerns that they will get sick or have an undiagnosed illness; on the other hand, with somatic symptom disorder, there are significant health concerns along with substantial somatic symptoms. The analogous disorder in ICD-10 and the earlier DSM-IV is hypochondriasis. Table 12-2 compares the two disorders.

Conversion Disorder (Functional Neurologic Symptom Disorder)

A conversion reaction is a rather acute and temporary loss or alteration in

motor or sensory function that requires substantial discordance between the symptoms displayed and any neurologic condition, such that it would be impossible for the patient's presentation to be consistent with a neurologic disease. Because the onset frequently coincides with psychological issues (conflict), early theorists speculated that such issues were "converted" to neurologic symptoms. The problem is that stress is omnipresent in life, and many patients present with conversion without an obvious stressor. The classic syndromes represent neurologic syndromes such as paralysis, seizures, or blindness. In DSM-5, we can indicate the type of symptom the patient is experiencing. Table 12-3 compares the approaches to diagnosing this disorder in DSM-5 and ICD-10. Table 12-4 lists common symptoms of conversion disorder. Table 12-5 lists examples of significant physical examination findings often found in disorder.



Table 12-2
Illness Anxiety Disorder (DSM-5) and Hypochondriasis (ICD-10)

	DSM-5	ICD-10
Name	**Illness Anxiety Disorder**	**Hypochondriacal Disorder**
Duration	≥6 mo	
Symptoms	• Preoccupied with having or getting a serious illness without good reason • Anxious about the belief • Excess health-related behaviors related to the belief ("care-seeking"), maladaptive behaviors, or avoidance behaviors ("care-avoiding")	Preoccupied with having a serious medical illness Anxiety Somatic complaints Concerns about appearance Distress or depression associated

		with the above
# Symptoms needed	All the above	
Exclusion (not the result of):	Another mental disorder	
Symptom specifiers	Care-seeking type Care-avoidant type	
Comments		Body dysmorphic disorder is included with this diagnosis as a subtype



Table 12-3
Conversion Disorder

	DSM-5	ICD-10
Name	**Conversion Disorder (Functional Neurologic Symptom Disorder)**	**Dissociative (Conversion) Disorders**
Symptoms	• ≥ Voluntary motor or sensory symptoms • No correlation between any medical or neurologic illness and the symptoms on examination	
Exclusion (not the result of):	Another mental illness Another medical condition	
Psychosocial impact	Marked distress and/or functional impairment	

Symptom specifiers	**With weakness or paralysis** **With abnormal movement** **With swallowing symptoms** **With speech symptom** **With attacks or seizures** **With anesthesia or sensory loss** **With special sensory symptoms** (i.e., vision, olfaction, hearing) **With mixed symptoms** **Can also specify with or without psychological stressor**	**Dissociative stupor** **Dissociative motor disorder** (limb, speech, or gait) **Dissociative convulsions** **Dissociative anesthesia and sensory loss** **Mixed dissociative (conversion) disorders**
Course specifiers	**Acute episode** (<6 mo) **Persistent** (≥6 mo)	

Mr. J is a 28-year-old single man who works in a factory. He was brought to an emergency department by his father, complaining that he had lost his vision while sitting in the back seat on the way home from a family gathering. He had been playing volleyball at the gathering but had sustained no significant injury except for the volleyball hitting him in the head a few times. He was initially reluctant to play volleyball because of his mediocre athleticism and was a last-minute addition to the team. He recalls having some problems with seeing during the game, but his vision did not become ablated until he was in the car on the way home. By the time he got to the emergency department, his vision was improving, although he still complained of blurriness and mild diplopia. The physician could attenuate the double vision by having the patient focus on different items at different distances.

On examination, Mr. J was fully cooperative, somewhat uncertain

about why this would have occurred, and rather nonchalant. Pupillary, oculomotor, and general sensorimotor examinations were routine. After being cleared medically, the physician referred the patient to a mental health center for further evaluation.

At the mental health center, the patient recounts the same story as he did in the emergency department. His father continued to accompany him. He said his vision improved when his father stopped the car, and they talked about the day's events. He spoke with his father about how he had felt embarrassed and somewhat conflicted about playing volleyball but felt pressure to play. Further history from the patient and his father revealed that this young man had been shy as an adolescent, particularly around athletic participation. He had never had another episode of visual loss. He did recount feeling anxious and sometimes not feeling well in his body during athletic activities.

Discussion with the patient at the mental health center focused on the potential role of psychological and social factors in acute vision loss. The patient was somewhat perplexed by this but was also amenable to discussion. He stated that he recognized that he began seeing and feeling better when his father pulled off to the side of the road and discussed things with him. Doctors stated that they did not know the cause of the vision loss and that it would likely not return. The patient and his father were satisfied with the medical and psychiatric evaluation and agreed to return for care if there were any further symptoms. The patient was appointed a follow-up time at the outpatient psychiatric clinic. (Courtesy of Michael A. Hollifield, M.D.)



Table 12-4
Common Symptoms of Conversion Disorder

Motor Symptoms	Sensory Deficits
Involuntary movements	Anesthesia, especially of extremities
Tics	Midline anesthesia
Blepharospasm	Blindness

Torticollis	Tunnel vision
Opisthotonos	Deafness
Seizures	**Visceral Symptoms**
Abnormal gait	Psychogenic vomiting
Falling	Pseudocyesis
Astasia–abasia	Globus hystericus
Paralysis	Swooning or syncope
Weakness	Urinary retention
Aphonia	Diarrhea

(Courtesy of Frederick G. Guggenheim, M.D.)



Table 12-5
Distinctive Physical Examination Findings in Conversion Disorder

Condition	Test	Conversion Findings
Anesthesia	Map dermatomes	Sensory loss does not conform to recognized pattern of distribution
Hemianesthesia	Check midline	Strict half-body split
Astasia–abasia	Walking, dancing	With suggestion, those who cannot walk may still be able to dance; alteration of sensory and motor findings with suggestion
Paralysis, paresis	Drop paralyzed hand onto face	Hand falls next to face, not on it
	Hoover test	Pressure noted in examiner's

		hand under paralyzed leg when attempting straight leg raising
	Check motor strength	Give-away weakness
Coma	Examiner attempts to open eyes	Resists opening; gaze preference is away from doctor
	Ocular cephalic maneuver	Eyes stare straight ahead, do not move from side to side
Aphonia	Request a cough	Essentially normal coughing sound indicates cords are closing
Intractable sneezing	Observe	Short nasal grunts with little or no sneezing on inspiratory phase; little or no aerosolization of secretions: minimal facial expression; eyes open; stops when asleep; abates when alone
Syncope	Head-up tilt test	Magnitude of changes in vital signs and venous pooling do not explain continuing symptoms
Tunnel vision	Visual fields	Changing pattern on multiple examinations
Profound monocular blindness	Swinging flashlight sign (Marcus Gunn)	Absence of relative afferent pupillary defect
	Binocular visual	Sufficient vision in “bad eye”

	fields	precludes plotting normal physiologic blind spot in good eye
Severe bilateral blindness	"Wiggle your fingers, I'm just testing coordination"	Patient may begin to mimic new movements before realizing the slip
	Sudden flash of bright light	Patient flinches
	"Look at your hand"	Patient does not look there
	"Touch your index fingers"	Even blind patients can do this by proprioception

(Courtesy of Frederick G. Guggenheim, M.D.)

Sensory Symptoms. In conversion disorder, anesthesia and paresthesia are common, especially of the extremities. All sensory modalities can be involved, and the distribution of the disturbance is usually inconsistent with either central or peripheral neurologic disease. Thus, clinicians may see the characteristic stocking-and-glove anesthesia of the hands or feet or the hemianesthesia of the body beginning precisely along the midline.

Conversion disorder symptoms may involve the organs of special sense and can produce deafness, blindness, and tunnel vision. These symptoms can be unilateral or bilateral, but neurologic evaluation reveals intact sensory pathways. In conversion disorder blindness, for example, patients walk around without collisions or self-injury, their pupils react to light, and their cortical-evoked potentials are normal.

Motor Symptoms. The motor symptoms of conversion disorder include abnormal movements, gait disturbance, weakness, and paralysis. Gross rhythmical tremors, choreiform movements, tics, and jerks may be present. The movements generally worsen when calling attention to them. One gait disturbance seen in conversion disorder is *astasia–abasia*, which is a wildly

ataxic, staggering gait accompanied by gross, irregular, jerky truncal movements and thrashing and waving arm movements. Patients with the symptoms rarely fall; if they do, they are generally not injured.

Other common motor disturbances are paralysis and paresis involving one, two, or all four limbs, although the distribution of the involved muscles does not conform to the neural pathways. Reflexes remain normal; the patients have no fasciculations or muscle atrophy (except after long-standing conversion paralysis); electromyography findings are normal.

Seizure Symptoms. NESs are another symptom of conversion disorder. Clinicians may find it challenging to differentiate NESs from an actual seizure by clinical observation alone. Some patients with NESs also have a coexisting epileptic disorder, which may complicate the clinical picture. Tongue-biting, urinary incontinence, and injuries after falling can occur in NESs, although these symptoms are generally not present. Patients with NESs retain pupillary and gag reflexes after their seizure-like activity, and patients have no postseizure increase in prolactin concentrations.

Other Associated Features. Several psychological symptoms are also associated with conversion disorder.

PRIMARY GAIN. Patients achieve primary gain by keeping internal conflicts outside their awareness. Symptoms have symbolic value; they represent an unconscious psychological conflict.

SECONDARY GAIN. Patients accrue tangible advantages and benefits as a result of being sick; for example, being excused from obligations and difficult life situations, receiving support and assistance that might not otherwise be forthcoming, and controlling others' behavior.

LA BELLE INDIFFÉRENCE. *La belle indifférence* ("beautiful indifference") is a patient's inappropriately cavalier attitude toward severe symptoms; that is, the patient seems to be unconcerned about what appears to be a significant impairment. That bland indifference may also occur in some seriously ill medical patients who develop a stoic attitude. The presence or absence of *la belle indifférence* is not pathognomonic of conversion disorder, but it is often

associated with the condition.

IDENTIFICATION. Patients with conversion disorder may unconsciously model their symptoms on those of someone important to them. For example, a parent or a person who has recently died may serve as a model for conversion disorder. During pathologic grief reaction, bereaved persons commonly have symptoms of the deceased.

Psychological Factors Affecting Other Medical Conditions

According to DSM-5, individuals with this condition have a medical illness that is significantly impacted by psychological influences or patterns of behavior. We should only make the diagnosis when the effect of the psychological issue on the medical condition is unambiguous and leads to documentable effects on the course and outcome of the medical condition. Psychological or behavioral factors include psychological distress, patterns of interpersonal interaction, coping styles, and maladaptive health behaviors such as denial of symptoms or poor adherence to medical recommendations.

The reverse situation, the psychiatric or psychological consequences of having a medical condition, is more appropriately classified as an adjustment disorder. Other ambiguous situations where psychological and physical symptoms coexist (the so-called "comorbidities") present particular challenges. For example, many somatic, anxiety, and mood disorders are associated with chronic medical conditions (diabetes, hypertension, hypercholesterolemia) or "functional" syndromes (irritable bowel, migraine headaches, and several others).

Mr. A, a 55-year-old man, was hospitalized in the intensive care unit following a cardiac arrest. He had been experiencing severe substernal chest pain but "ignored it." He became diaphoretic and reported reluctantly to a nearby emergency room where he had a cardiac arrest while awaiting evaluation. He was successfully resuscitated and transferred to the ICU. He emphatically rejected having had a heart attack, took his EKG leads off, and was preparing to leave the unit against medical advice. The psychiatric consultant obtained a history that the patient's father and his brother had died of coronary disease. At the

time of the initial assessment, the patient was irritable and somewhat anxious. He agreed to stay in the ICU overnight after his wife insisted that he stay, and the psychiatrist explained the tests further. The psychiatrist prescribed a low dose of a benzodiazepine, and, on the next day, he was agreeable to remain in the hospital for further treatment.

Factitious Disorder

Factitious disorder is the faking of physical or psychological signs and symptoms. Symptoms can be imposed on self or imposed on another. Table 12-6 compares criteria for the disorder and Table 12-7 list some clues that should trigger suspicion of this disorder.

Factitious Disorder Imposed on Self. We diagnose this condition when an individual feigns having a medical or psychiatric illness in order to achieve the sick role. Patients may simply misrepresent their symptoms, may cause injury to themselves, or may use other dishonest methods in order to appear as if they are sick.

Some patients show psychiatric symptoms judged to be fake. This determination can be difficult and is often made only after a prolonged investigation. The feigned symptoms frequently include depression, hallucinations, dissociative and conversion symptoms, and bizarre behavior. Because the patient's condition does not improve after routine therapeutic measures, he or she may receive large doses of psychoactive drugs and may undergo electroconvulsive therapy. Patients may appear depressed and may explain their depression by offering a false history of the recent death of a significant friend or relative. Elements of the history that may suggest factitious bereavement include a violent or bloody death, a death under dramatic circumstances, and the dead person being a child or a young adult. Other patients may describe either recent and remote memory loss or both auditory and visual hallucinations. Some patients may use psychoactive substances to produce symptoms, such as stimulants to produce restlessness or insomnia, or hallucinogens to produce distortions of reality. Combinations of psychoactive substances can produce very unusual presentations. Other psychological symptoms include pseudologia fantastica and impostorship.

Table 12-6
Factitious Disorder

	DSM-5	ICD-10
Name	**Factitious Disorder**	**Intentional Production or Feigning of Symptoms or Disabilities, Either Physical or Psychological (Factitious Disorder)**
Symptoms	**Factitious disorder imposed on self:** • Fabricates their symptoms and clinical findings • Claims to be ill • Purposely misleads health professionals • No apparent external reward or motivation (outside of the illness role) **Factitious disorder imposed on another:** • As above except the object of the feigned illness or injury is another person.	Feigns symptoms No apparent external reward or motivation Aim is to take on the sick role
Exclusion (not the result of):	Another mental disorder	Factitious dermatitis Malingering
Course	**Single episode**	

specifiers	**Recurrent episodes** (2 or more discrete events)
Comments	*Note: In factitious disorder imposed on another, the perpetrator receives the diagnosis, not the victim.*



Table 12-7
Clues That Should Trigger Suspicion of Factitious Disorder

1. The patient has sought treatment at various different hospitals or clinics
2. The patient is an inconsistent, selective, or misleading informant; he or she resists allowing the treatment team access to outside sources of information
3. The course of the illness is atypical and does not follow the natural history of the presumed disease
4. A remarkable number of tests, consultations, and medical and surgical treatments have been done to little or no avail
5. The magnitude of symptoms consistently exceeds objective pathology or symptoms have proved to be exaggerated by the patient
6. Some findings are discovered to have been self-induced or at least worsened through self-manipulation
7. The patient might eagerly agree to or request invasive medical procedures or surgery
8. Physical evidence of a factitious cause might be discovered during the course of treatment
9. The patient predicts deteriorations or there are exacerbations shortly before their scheduled discharge

10. A diagnosis of factitious disorder has been explicitly considered by at least one healthcare professional
11. The patient is noncompliant with diagnostic or treatment recommendations or is disruptive on the unit
12. Evidence from laboratory or other tests disputes information provided by the patient
13. The patient has a history of work in the healthcare field
14. The patient engages in gratuitous, self-aggrandizing lying
15. The patient has been prescribed (or obtained) opiate drugs when not indicated
16. While seeking medical or surgical intervention, the patient opposes psychiatric assessment

In *pseudologia fantastica,* the patient mixes limited factual material with extensive and colorful fantasies. The listener's interest pleases the patient and, thus, reinforces the symptom. In addition to distortions of the history, patients often give false and conflicting accounts about other areas of their lives (e.g., they may claim the death of a parent, to play on the sympathy of others). Imposture is commonly related to lying in these cases. Many patients assume the identity of a prestigious person. They may, for example, report being war heroes and attribute their surgical scars to wounds received during battle or in other dramatic and dangerous exploits. Similarly, they may say that they have ties to accomplished or renowned figures.

Other patients may feign physical symptoms suggesting a disorder involving any organ system. They are familiar with the diagnoses of most disorders that usually require hospital admission or medication and can give excellent histories capable of deceiving even experienced clinicians. Clinical presentations are myriad and include hematoma, hemoptysis, abdominal pain, fever, hypoglycemia, lupus-like syndromes, nausea, vomiting, dizziness, and seizures. For example, the patient may contaminate urine with blood or feces or take anticoagulants to simulate bleeding disorders, or insulin to produce hypoglycemia. Such patients often insist on surgery and claim adhesions from previous surgical procedures. They may acquire a "gridiron" or washboard-like abdomen from multiple procedures. Complaints of pain, especially that simulating renal colic, are common, with the patients wanting

narcotics. In about half the reported cases, these patients demand treatment with specific medications, usually analgesics. Once in the hospital, they continue to be demanding and challenging. After each test result is negative, they may accuse doctors of incompetence, threaten litigation, and become generally abusive. Some may sign out abruptly shortly before they believe the healthcare team is going to confront them with their factitious behavior. They then go to another hospital in the same or another city and begin the cycle again. Specific predisposing factors are actual physical disorders during childhood, leading to extensive medical treatment. Other factors may include a grudge against the medical profession, employment in the healthcare industry, or a significant relationship with a physician in the past.

Table 12-8 provides a comprehensive overview of a variety of signs and symptoms that may be faked and mistaken for genuine illness. The table also includes the means of simulation and possible methods of detection.

Factitious Disorder Imposed on Another (formerly Factitious Disorder by Proxy). In this diagnosis, a person intentionally produces physical signs or symptoms in another person who is under the first person's care. One apparent purpose of the behavior is for the caretaker to assume the sick role indirectly; another is to be relieved of the caretaking role by having the child hospitalized. The most common cause of factitious disorder imposed on another involves a mother who deceives medical personnel into believing that her child is ill. The deception may involve a false medical history, contamination of laboratory samples, alteration of records, or induction of injury and illness in the child. Table 12-9 lists some clues that should trigger suspicion of this disorder.

BC, a 1-month-old girl, was admitted for the evaluation of fever. The medical team requested a psychiatric consult because of inconsistencies in the mother's reporting of medical information despite her presentation as a knowledgeable and caring mother who worked as an emergency medical technician. BC's mother reported her own diagnosis of ovarian cancer when she was 3 months pregnant with BC. She reported undergoing a hysterectomy during her cesarean section, and that she had been getting radiation therapy at a local hospital since BC's birth. The

pediatrician called the local hospital with the mother's permission and learned that she had a corpus luteum cyst removed at 3 months' gestation and mild hydronephrosis but no cancer or hysterectomy. BC's mother, when confronted with this, replied that she might need a kidney transplant for the hydronephrosis.

On further exploration, the medical team discovered that the mother had brought her children to multiple emergency rooms, giving inaccurate histories that prompted excessive testing. At one visit, she told clinicians that her 2-year-old son had lupus and hypergammaglobulinemia, and at another visit, that he had asthma and seizures. She also pursued a minor cosmetic surgical procedure for him against his pediatrician's recommendation.

Clinicians suspected that BC's mother intentionally fabricated symptoms, such as by warming BC's thermometer, and that she did not actively induce symptoms in her children. She was faithful in keeping medical appointments, and her children appeared healthy and well cared for, despite her factitious behavior. The mother denied a psychiatric history but agreed to let the clinicians contact the local psychiatric hospital. That hospital revealed her history of depression, anorexia, panic disorder, and a suicide attempt resulting in psychiatric hospitalization. After the hospitalization, she received psychotherapy and medication, which she stopped a few months before this presentation. During BC's admission for fever, her mother agreed to resume psychiatric treatment. The team made a referral to social services, and the pediatrician scheduled regular follow-up visits for the children.

Table 12-8
Methods of Factitious Symptom Production, Suggestive Signs, and Confirmatory Tests by Systems

Symptom	Method of Factitious Symptom Manufacture	Signs Suggestive Factitiousness
Infectious Disease		
Fever	Injecting infectious material	Higher than 41°C

	into vein/intravenous line	
	Ingesting thyroid hormone	Not accompanied b other vital sign abnormalities
	Drinking hot fluids	Does not follow diu pattern
	Manipulating thermometers with lightbulb or heating pad	No diaphoresis with defervescence
	Substituting another thermometer	
	Warm wax/wet cotton in ears	Measuring fresh uri specimen tempe
Bacteremia	Injection of contaminated substance	Polymicrobial bacte
		Absence of urologic/biliary/G obstruction
		Stool flora/pet flora
HIV/AIDS	False history/reports	Normal CD4 count/undetecta viral load/Ab neg
Gastrointestinal		
Diarrhea	Laxative abuse (magnesium, castor oil, phenolphthalein)	Metabolic alkalosis

		Acute hyperchlorem metabolic acidos with normal anio
		Decreased serum bicarbonate with metabolic acidos
		High daily stool volu
		Melanosis coli on sigmoidoscopy
		Cathartic colon on b enema
		Low urine potassiur concentration
		High fecal fluid pota concentration
		Fecal fluid osmolali <290 mOsm/kg
	Adding water to stool sample	High stool osmolar >125 mOsmol/kg
Vomiting	Ipecac/cathartic abuse	Hypokalemia
		High urine potassiu concentration
		Low urine chloride concentration
		Metabolic alkalosis increased serum bicarbonate

Pancreatitis	Spitting into urine sample (salivary amylase)	Hyperamylasuria w normal serum amylase
GI bleeding	Injection of blood from transfusions	Nasogastric tube sh blood despite nc endoscopy
	Ingestion of NSAIDs/salicylates	Single-stripe sign o colonoscopy
Obstruction	Ingestion of loperamide	
Renal		
Diuresis	Diuretic abuse	High urine potassiu concentration
Bartter syndrome	Loop diuretic abuse	Hypokalemia, urine chloride low or variable
	Self-induced vomiting	
Metabolic		
Hypervitaminosis A	Vitamin A abuse	Increased gamma-glutamyltransfer Increased bilirubin
Hypokalemia	Laxative abuse	See above
Hyperkalemia	Injection of urine into blood sample	Potassium level incompatible wit
Hypernatremia	Salt load	Fractional excretion sodium high
		Gastric aspirate sal concentration >2 mmol/L

Urologic		
Hematuria	Adding blood from another wound/meat	Lack of red blood cell/hemoglobin
	Traumatizing urethra	Lack of distorted re blood cells
	Inserting foreign bodies into the bladder	
	Ingestion of anticoagulants	
	Addition of coloring to urine	
Proteinuria	Injection of egg protein in bladder	Large day-to-day variations in urin protein concentr
		Serum albumin concentration re in normal range
		Lack of other signs nephrotic syndro
Bacteriuria	Injecting bacteria into bladder or urine specimens	
Urinary calculi	Adding pepper grains to urine	
	Inserting stones into bladder	
	Submitting quartz, feldspar as samples	

Hematologic		
Anemia	Self-bloodletting	Decreased serum i ferritin, iron-bind capacity
		Decreased bone m iron concentratio
		No evidence of blee
Sickle cell crisis		Serum protein electrophoresis normal
Pancytopenia	Ingestion of chemotherapy	
Hemorrhage/purpura	Ingestion of anticoagulant (rodenticide/Coumadin)	Prolonged prothrom time (PT)
	Heparin injection	Prolonged partial thromboplastin t (PTT) with norm
	Ingestion of quinidine	Purpura with thrombocytopen
Endocrine		
Hypoglycemia	Injection of insulin	Serum insulin

	Ingestion of oral hypoglycemic (sulfonylurea) Ingestion of tolbutamide Manipulation of testing strips	concentration >1 mU/L
Hyperglycemia	Insufficient administration of insulin	
Hyperthyroidism	Ingestion of thyroid hormone	Goiter/eye findings absent Increased thyroxine liothyronine leve Low TSH in thyroid
Cushing syndrome	Injecting/ingesting glucocorticoid Adding glucocorticoid to urine	Plasma/urinary cort increased Plasma ACTH low/undetectable Normal corticosterone/h cortisol

Pheochromocytoma	Injection of epinephrine, isoproterenol	Normal chromogran level
	Ingestion of stimulant	Lack of metabolites epinephrine (metanephrines)
	Injection of epinephrine into urine sample	Lack of norepineph increase after glucagon
		Normal response to clonidine suppre test
Hyperaldosteronism	Ingestion of black licorice/glycyrrhizic acid	Treatment-resistant hypokalemia
		Metabolic alkalosis
		Hypernatremia
Cardiovascular		
Arrhythmia	Rearranging of EKG leads Ingestion of digitalis, beta blockers, calcium channel blockers	Unusual pattern on
Hypertension	Valsalva maneuver	Office BPs normal, BPs high
	Ingestion of stimulants	
Infarction	Beating self with towel to raise creatine kinase	EKGs don’t change
		Enzymes do not ris appropriately

Dermatologic		
Cheilitis granulomatosa	Self-inoculation with polyvinylpyrrolidone (PVP)	
Dermatitis artefacta	Self-mutilation	Distribution of lesio reachable areas
Erythematous lesion	Applying alcohol to lesion	
Nonhealing wounds	Injection of air/contaminants/foreign bodies	Heal when casted Resistant to treatm Subcutaneous emphysema
Obstetrical		
Ectopic pregnancy	Injection of hCG	Negative urine b-hC Widely varying b-hC levels
Vaginal discharge	Intravaginal insertion	Inconsistent pH Vaginal wall shows abrasion/denies intercourse
Neurologic		
Movement disorders	Ingestion of neuroleptics to induce parkinsonism	MRI/EEG/EMG inconsistent Hoover sign/tremor entrainable/distr Prolactin elevated w antipsychotics
Multiple sclerosis		Examination incons
Pseudoseizures		History inconsistent

		Semiology inconsis
		Never observed sei by others
Pulmonary		
Asthma	Tape interfering with pulse oximeters	Intractable hypoxia
	Inhalation of irritant (talc)	Intractable status asthmaticus
	Ingestion of allergens	
Cystic fibrosis	Manipulation of sweat test	Normal x-rays
		Normal examinatior of clubbing
Hemoptysis	Ingestion of anticoagulants	Normal bronchoscopy/la blood
	Cough abrasion of lung	
	Self-induced trauma to respiratory tract	
Respiratory failure		Incongruous arteria blood gas
		Marked hypocapnia
Rheumatologic		
Arthritis	Insertion of metal fragments in joints	
Lupus	Feigned history/borderline positive ANA	Repeat serum autoimmune tes

		negative Normal complemen active flare
Vasculitis	Injection of contaminants to create purpura	
Ophthalmologic		
Conjunctivitis	Instillation of foreign substance	Purulent discharge purposely left or and eyelashes Severity of discharg greater than sev of redness Severity of swelling than severity of redness Mainly inferior conjunctival involvement Cornea uninvolved
Anisocoria	Instillation of atropine eye drops	

Table 12-9
Clues Triggering Suspicion for Factitious Disorder Imposed on Another Person

1. Diagnosis does not match the objective findings
2. Signs or symptoms are bizarre
3. Caregiver or suspected offender does not express relief or pleasure when told that dependent is improving or that dependent does not have a particular illness

4. Inconsistent histories of symptoms from different observers
5. Caregiver insists on invasive or painful procedures or hospitalizations
6. Caregiver's behavior does not match expressed distress or report of symptoms (e.g., unusually calm)
7. Signs and symptoms begin only in the presence of one caregiver
8. Sibling or another dependent has or had an unusual or unexplained illness or death
9. Sensitivity to multiple environmental substances or medicines
10. Failure of the dependent's illness to respond to its normal treatments or unusual intolerance to those treatments
11. Caregiver publicly solicits sympathy or donations or benefits because of the dependent's rare illness
12. Extensive unusual illness history in the caregiver or caregiver's family; caregiver's history of somatization disorders
13. Caregiver seeks other medical opinions when told the dependent does not have illness
14. Caregiver perseverates about borderline abnormal results of no clinical relevance despite repeated reassurance, or refutes the validity of normal results

Other Specified or Unspecified Somatic Symptom and Related Disorders

This DSM-5 category describes conditions characterized by one or more unexplained physical symptoms, which are below the threshold for a diagnosis of somatic symptom or a related disorder. The symptoms are not better explained solely by another medical, psychiatric, or substance use disorder, and they cause clinically significant distress or impairment. Disorders in this category generally have too few symptoms are too brief to meet the criteria for a full disorder.

EVALUATION OF SOMATIC SYMPTOM AND RELATED DISORDERS

Patients with severe and protracted somatic syndromes approach medical

encounters with a mixture of unrealistic expectations, pessimism, and distrust of the medical profession. The patients base these attitudes on their previous experiences with physicians, in which the physicians showed a lack of interest or disbelief in the patient's complaints and suffering. Building a trusting alliance must begin with respect for the patient's symptoms and an acknowledgment of their validity. Active, receptive listening, tolerance for repetition, and a "neutral" approach (avoiding being dismissive, confrontational, or overly reassuring) are essential skills.

Many of these patients often bring "thick" charts that include descriptions of many clinical encounters and multiple tests and procedures that are redundant and ordered without a clear rationale. The prospect of reviewing these medical records is challenging and often leads to a negative attitude from the physician. The physician should keep an open mind, despite forewarnings in the medical records, and perform an independent assessment of the patient. Avoid an emphasis on psychological questioning and interpretations at this stage. The patient may perceive premature reassurance as disinterest or dismissiveness.

Many patients with somatic symptom and related disorders do not readily acknowledge or recognize "emotional" issues and will feel more comfortable dealing with questions related to their physical symptoms than questions related to psychological issues. Taking a history of multiple somatic complaints can be carried over into subsequent appointments. A thorough physical examination, including neurologic examination, should follow history-taking at the initial visit, and the clinician should perform briefer physical assessments at subsequent visits. An inclusive drug history, including prescription and nonprescription remedies, should also be completed. As history-taking moves along, attitudes, beliefs, and attributions should become more evident and patterns of interaction and illness behavior discernible.

In the case of conversion disorder, there should be careful attention to any history of trauma, sexual abuse, physical abuse, and family history of conversion symptoms. Additionally, a neurologic workup of suspected conversion disorder should include routine laboratory studies, electroencephalograms, and other special studies (such as MRI) to rule out organic causes.

In the case of factitious disorder, the psychiatric examination should

emphasize securing information from any available friends, relatives, or other informants, because interviews with reliable outside sources often reveal the false nature of the patient's illness. Although time-consuming and tedious, corroborating all the facts presented by the patient about previous hospitalizations and medical care is essential. Psychiatric evaluation is requested on a consultation basis in about 50 percent of cases, usually after a simulated illness is suspected. Under these circumstances, it is necessary to avoid pointed or accusatory questioning that may provoke truculence, evasion, or flight from the hospital. A danger may exist in provoking frank psychosis when using vigorous confrontation. In some instances, the feigned illness serves an adaptive function and is a desperate attempt to ward off further disintegration.

DIFFERENTIAL DIAGNOSIS

Somatic Symptom Disorder

We should differentiate somatic symptom disorder from nonpsychiatric medical conditions, especially disorders that show symptoms that are not necessarily easily diagnosed. Such diseases include acquired immunodeficiency syndrome (AIDS), endocrinopathies, myasthenia gravis, multiple sclerosis, degenerative diseases of the nervous system, systemic lupus erythematosus, and occult neoplastic disorders.

Somatic symptom disorder is differentiated from illness anxiety disorder by the emphasis on illness anxiety disorder on fear of having a disease rather than a concern about many symptoms. Patients with illness anxiety disorder usually complain about fewer symptoms than patients with somatic symptom disorder; they are primarily concerned about being sick.

Conversion disorder is acute and generally transient and usually involves a symptom rather than a particular disease. The presence or absence of *la belle indifférence* is an unreliable feature with which to differentiate the two conditions.

Patients with body dysmorphic disorder wish to appear normal, but believe that others notice that they are not, whereas those with somatic symptom disorder seek out attention for their presumed diseases.

Somatic symptom disorder can also occur in patients with depressive

disorders and anxiety disorders. Patients with panic disorder may initially complain that they are affected by a disease (e.g., heart trouble), but careful questioning during the medical history usually uncovers the classic symptoms of a panic attack. Delusional beliefs occur in schizophrenia and other psychotic disorders but can be differentiated from somatic symptom disorder by their delusional intensity and by the presence of other psychotic symptoms. Also, schizophrenic patients' somatic delusions tend to be bizarre, idiosyncratic, and out of keeping with their cultural milieus, as illustrated in the case below.

> A 52-year-old man complained, "my guts are rotting away." Even after an extensive medical workup, the physician could not reassure the man that he was not ill.

Somatic symptom disorder is distinguished from factitious disorder and from malingering in that patients with somatic symptom disorder experience and do not simulate the symptoms they report.

Illness Anxiety Disorder

Again, we must differentiate illness anxiety disorder from other medical conditions. Too often, doctors dismiss these patients as "chronic complainers" and do not perform a careful medical examination. Patients with an illness anxiety disorder are differentiated from those with somatic symptom disorder by the emphasis on illness anxiety disorder on fear of having a disease versus the emphasis on somatic symptom disorder on concern about many symptoms. However, both may exist to varying degrees in each disorder. Patients with illness anxiety disorder usually complain about fewer symptoms than patients with somatic symptom disorder. Conversion disorder is differentiated from illness anxiety disorder by the fact that it is acute, generally transient, and usually involves a symptom rather than a particular disease. The fear of illness can also occur in patients with depressive and anxiety disorders. If a patient meets the full diagnostic criteria for both illness anxiety disorder and another major mental disorder, such as major depressive disorder or generalized anxiety disorder, the patient should

receive both diagnoses. Illness anxiety disorder can be differentiated from obsessive-compulsive disorder by the singularity of their beliefs and by the absence of compulsive behavioral traits, but there is often an obsessive quality to the patient's fear. Delusional beliefs can be differentiated from illness anxiety disorder by their delusional intensity and the presence of other psychotic symptoms.

Conversion Disorder (Functional Neurologic Symptom Disorder)

One of the significant problems in diagnosing conversion disorder is the difficulty in definitively ruling out a medical disorder. Concomitant nonpsychiatric medical disorders are common in hospitalized patients with conversion disorder, and evidence of a current or previous neurologic disorder or a systemic disease affecting the brain occurs in 18 to 64 percent of such patients. An estimated 25 to 50 percent of patients classified as having conversion disorder eventually receive diagnoses of neurologic or nonpsychiatric medical disorders that could have caused their earlier symptoms. Thus, a thorough medical and neurologic workup is essential in all cases. If the suggestion resolves the symptom, such as with hypnosis, or parenteral amobarbital or lorazepam, the symptoms are probably the result of conversion disorder.

Neurologic disorders (e.g., dementia and other degenerative diseases), brain tumors, and basal ganglia disease are part of the differential diagnosis. For example, weakness may be confused with myasthenia gravis, polymyositis, acquired myopathies, or multiple sclerosis. Optic neuritis can resemble conversion disorder blindness. Other diseases that can cause confusing symptoms are Guillain–Barré syndrome, Creutzfeldt–Jakob disease, periodic paralysis, and early neurologic manifestations of AIDS. Conversion disorder symptoms occur in schizophrenia, depressive disorders, and anxiety disorders, but these other disorders are associated with their distinct symptoms that eventually make differential diagnosis possible.

The demarcation between conversion disorder and somatic symptom disorder is not that clear, and conversion symptoms may form part of the constellation of symptoms seen in somatic symptom disorder. One differentiating feature is that conversion disorder requires that the presenting symptoms be inconsistent with neurologic conditions, whereas somatic

symptom disorder does not require this inconsistency.

In both malingering and factitious disorder, the symptoms are under conscious, voluntary control. A malingerer's history is usually more inconsistent and contradictory than that of a patient with conversion disorder, and a malingerer's fraudulent behavior is goal-directed.

Psychological Factors Affecting Other Medical Conditions

Differential diagnosis can be complicated and should include other somatic symptom disorders (somatic symptom disorder, conversion disorder, factitious disorder) as well as personality disorders (e.g., borderline personality disorder).

Factitious Disorder

Any disorder in which physical signs and symptoms are prominent are part of the differential diagnosis, and the physician must explore the possibility of authentic or concomitant physical illness. Additionally, a history of many surgeries in patients with factitious disorder may predispose such patients to complications or actual diseases, necessitating even further surgery. Factitious disorder is on a continuum between somatic symptom and related disorders and malingering, the goal being to assume the sick role.

A factitious disorder is differentiated from conversion disorder by the voluntary production of factitious symptoms, the extreme course of multiple hospitalizations, and the seeming willingness of patients with a factitious disorder to undergo an extraordinary number of mutilating procedures. Patients with conversion disorder are not usually conversant with medical terminology and hospital routines, and their symptoms have a direct temporal relation or symbolic reference to specific emotional conflicts.

Illness anxiety disorder differs from factitious disorder in the lack of voluntary production of symptoms.

Because of their pathologic lying, lack of close relationships with others, hostile and manipulative manner, and associated substance use and criminal history, clinicians may think that patients with factitious disorder have an antisocial personality disorder. Antisocial persons, however, do not usually volunteer for invasive procedures or resort to a way of life marked by repeated or long-term hospitalization. Because of attention-seeking and an

occasional flair for the dramatic, they may also be diagnosed with a histrionic personality disorder. However, not all such patients have a dramatic flair, and many are withdrawn and bland. Consideration of the patient's chaotic lifestyle, history of disturbed interpersonal relationships, identity crisis, substance abuse, self-damaging acts, and manipulative tactics may lead to the diagnosis of borderline personality disorder.

We should distinguish factitious disorders from malingering. Malingerers have a manifest, recognizable environmental goal in producing signs and symptoms. They may seek hospitalization to secure financial compensation, evade the police, avoid work, or merely obtain free bed and board for the night, but they always have some apparent end for their behavior. Moreover, these patients can usually stop producing their signs and symptoms when the symptoms are no longer useful to them or when the risk becomes too high.

COMORBIDITY

Somatic Symptom Disorder

Individuals with somatic symptom disorders often have co-occurring anxiety disorders and depressive disorders, as well as comorbid medical illnesses. It is possible to have a diagnosed medical condition and receive a co-occurring diagnosis of somatic symptom disorder when the individual experiences more considerable distress and anxiety about the illness than would be expected.

Illness Anxiety Disorder

Given that illness anxiety disorder is new to DSM-5, there is limited data on comorbidity. However, a 2017 study demonstrated comorbidity of illness anxiety disorder and both depressive and anxiety disorders, especially generalized anxiety disorder and panic disorder.

Conversion Disorder (Functional Neurologic Symptom Disorder)

Medical and, especially, neurologic disorders, frequently occur among patients with conversion disorders. What one typically sees in these comorbid neurologic or medical conditions is an elaboration of symptoms stemming from the original lesion.

Depressive disorders, anxiety disorders, and somatic symptom disorder

often occur alongside conversion disorder. It is rarer with schizophrenia, however, it does occur. Personality disorders also frequently accompany conversion disorder, especially histrionic, dependent, and antisocial personality disorders, or individuals with passive–aggressive personality styling. Conversion disorders can occur, however, in persons with no predisposing medical, neurologic, or psychiatric disorder.

Psychological Factors Affecting Other Medical Conditions

All individuals with this diagnosis have at least one other medical condition that is comorbid, as per the DSM-5 diagnostic criteria.

Factitious Disorder

Many persons diagnosed with factitious disorder have comorbid psychiatric diagnoses (e.g., mood disorders, personality disorders, or substance-related disorders).

COURSE AND PROGNOSIS

Somatic Symptom Disorder

The course of the disorder is usually episodic; the episodes last from months to years with equally long quiescent periods. There may be an apparent association between exacerbations of somatic symptoms and psychosocial stressors. Although no well-conducted extensive outcome studies exist, an estimated one-third to one-half of all patients with somatic symptom disorder eventually improve significantly. A good prognosis is associated with high socioeconomic status, treatment-responsive anxiety or depression, sudden onset of symptoms, the absence of a personality disorder, the absence of childhood adversity, and the absence of a related nonpsychiatric medical condition. Most children with the disorder will recover by late adolescence or early adulthood.

Illness Anxiety Disorder

Because the disorder is a new diagnosis, there are no reliable data about the prognosis. One may extrapolate from the course of somatic symptom

disorder, which is usually episodic; the episodes last from months to years and have equally long quiescent periods in between. A good prognosis is associated with high socioeconomic status, treatment-responsive anxiety or depression, sudden onset of symptoms, the absence of a personality disorder, and the absence of a related nonpsychiatric medical condition.

Conversion Disorder (Functional Neurologic Symptom Disorder)

The onset of conversion disorder is usually acute, but a crescendo of symptomatology may also occur. Symptoms or deficits are usually of short duration, and approximately 95 percent of acute cases remit spontaneously, usually within 2 weeks in hospitalized patients. If symptoms have been present for 6 months or longer, the prognosis for symptom resolution is less than 50 percent and diminishes further, the longer that conversion is present. Recurrence occurs in one-fifth to one-fourth of people within 1 year of the first episode. Thus, one episode is a predictor for future episodes. Acute onset, presence of clearly identifiable stressors at the time of onset, a short interval between onset and the institution of treatment, and above-average intelligence are all good prognostic indicators. Paralysis, aphonia, and blindness are associated with a good prognosis, whereas tremor and seizures are poor prognostic factors.

Psychological Factors Affecting Other Medical Conditions

This condition can occur at any age. Little has been reported about course and prognosis.

Factitious Disorder

Factitious disorders typically begin in early adulthood, although they can appear during childhood or adolescence. The onset of the disorder or of discrete episodes of seeking treatment may follow a real illness, loss, rejection, or abandonment. Usually, the patient or a close relative had a hospitalization in childhood or early adolescence for genuine physical illness. After that, a long pattern of successive hospitalizations begins insidiously and evolves. As the disorder progresses, the patient becomes knowledgeable about medicine and hospitals.

Factitious disorders are incapacitating to the patient and often produce

severe trauma or untoward reactions related to treatment. A course of repeated or long-term hospitalization is incompatible with meaningful vocational work and sustaining interpersonal relationships. The prognosis, in most cases, is poor.

TREATMENT OF SOMATIC SYMPTOM AND RELATED DISORDERS

Treatment Approach

Reviews of controlled studies have shown a good efficacy of psychotherapy, particularly of the cognitive-behavioral therapy (CBT) type, and some evidence for the efficacy of pharmacotherapy in the treatment of several somatic syndromes. Kroenke and colleagues found 34 randomized controlled trials on somatoform disorders, which included more than 1,000 patients. Of these, 13 were CBT trials, 5 were antidepressant trials, and 16 involved other modalities. Overall, CBT showed the highest efficacy, with virtually all reported studies showing a significant effect of CBT on symptoms compared to no treatment. Antidepressants seem to work in painful syndromes, and the data favored older antidepressants (tricyclics) over newer ones (serotonin reuptake inhibitors [SSRIs]). Other interventions, including the use of a psychiatric consultation letter, brief psychodynamic psychotherapy, exercise, and more recently, biofeedback, have also shown beneficial effects on some of these symptoms. A more recent article reviewed 15 randomized controlled trials of the efficacy of CBT in somatoform disorders and found that CBT improved somatic, anxiety, and depressive symptoms and improved social functioning.

However, patients with somatic symptom and related disorders sometimes are not open to psychological explanations of their symptoms or psychological or psychiatric treatment. More specific treatment guidelines are listed for each diagnosis, below.

Somatic Symptom Disorder

Therapeutic goals should be modest at first and limited to small, attainable gains such as a decrease in medical visits, a commitment to a single primary care physician, and the avoidance of unnecessary tests and procedures. The

physician should prepare for a long-term commitment to the patient. Physicians should limit specialist consultations unless there is evidence of comorbid physical conditions. Some of the elements of CBT, such as diary keeping, should be incorporated into the primary care physician's treatment strategy. Recommendations for exercise, yoga, relaxation, meditation, and massage may be useful and are generally better accepted by patients than psychological treatments.

The use of a brief "consultation letter" intended for primary care physicians has demonstrated effectiveness in improving patients' functional capacity and decreasing their utilization of health services. This consultation letter provides physicians with "do's and don'ts" regarding their encounters with patients with multiple medically unexplained physical symptoms and briefly instructs them on using of a few fundamental management techniques. The letter should urge the physicians to see these patients during regularly scheduled appointments, perform brief physical examinations focusing on the area of discomfort at each visit, avoid unnecessary diagnostic procedures, invasive treatments, and hospitalizations, avoid using statements such as "symptoms are all in your head," and briefly allow/encourage patients to talk about "stressors."

A body of research has demonstrated the efficacy of CBT for the treatment of somatic symptom disorder and related syndromes. CBT-type interventions appear to help patients by modifying thoughts and behaviors associated with somatization. In the typical CBT program, the therapist systematically introduces patients to several behavioral techniques, which include relaxation training and graded increases in activities. From a cognitive perspective, CBT helps patients to identify thoughts that contribute to increased stress, inactivity, and health concern. Often, patients with this condition tend to think catastrophically about their physical symptoms. Such thoughts lead them to conclude that they are sick and must limit physical activity, creating a cycle that perpetuates the somatic process. Reports indicate that brief psychodynamic therapy of unexplained somatic symptoms may be useful, although most of the recent evidence in terms of randomized controlled trials involves CBT.

In general, we should avoid medications when treating patients with somatic symptom disorder except in the presence of clearly delineated anxious, depressive, or psychotic symptoms. Despite the development of new

pharmacologic agents such as dual (norepinephrine and serotonin) reuptake inhibitors and limited data on gabapentin, there is no evidence that a purely pharmacologic approach targeting the treatment of somatic symptom disorder will be sufficient.

Illness Anxiety Disorder

Various group and individual therapies, including CBT and psychodynamic therapy, have been proposed over the years to manage patients with health anxiety. Several controlled trials have demonstrated efficacy for CBT interventions, making CBT the prototype, first-line treatment for illness anxiety disorder. Other types of therapy, including mindfulness training, exposure therapy, and acceptance and commitment therapy, have also demonstrated some efficacy.

Limited data exist to guide pharmacologic treatment of illness anxiety disorder. There have been a few randomized, placebo-controlled trials that demonstrated benefit with SSRIs. Most reports have been anecdotal, single-blind studies. A recent randomized controlled trial of 195 patients with hypochondriasis demonstrated the efficacy and safety of fluoxetine in this patient population. Interestingly, the addition of CBT to fluoxetine conferred a slight added benefit.

Conversion Disorder (Functional Neurologic Symptom Disorder)

Many conversion syndromes have an acute, benign course and may remit spontaneously with understanding and support. Early intervention can forestall potential chronicity and the progression into a well-entrenched somatization disorder. Once chronicity has developed, intensive treatment may use all treatment modalities, including hospitalization, individual or group therapy, insight-oriented therapies, behavioral techniques, hypnosis, sodium amytal interview, physical therapy, biofeedback, relaxation training, and medication (primarily for comorbid anxiety, depression, or other somatoform disorders). Psychological interpretations or explanations do not work well early in the process, but reassuring patients that critical tests are normal and that symptoms will eventually improve may be helpful. Any implication to the patient that he or she is malingering is very counterproductive. Behavioral interventions should focus on improving self-

esteem, the capacity for emotional expression and assertiveness, and the ability to communicate comfortably with others. With chronic conversion, muscle contractures can occur, and physical therapy is necessary. Even in the absence of such contractures, however, many conversion patients find that physical therapy can be helpful for muscular symptoms or balance problems. The process of slowly progressive exercises and activity can help restore functioning.

Psychological Factors Affecting Other Medical Conditions

Treatment frequently involves communication with the patient's primary medical team as well as family. The psychoeducational intervention clarifies the role that emotional and behavioral factors play in aggravating the underlying medical condition. Medication to treat another underlying psychiatric disorder may be necessary.

Factitious Disorder

No specific psychiatric therapy has been effective in treating factitious disorders. It is a clinical paradox that patients with the disorders simulate severe illness and seek and submit to unnecessary treatment while they deny to themselves and others their actual illness and thus avoid possible treatment for it. Ultimately, the patients elude meaningful therapy by abruptly leaving the hospital or failing to keep follow-up appointments.

Treatment, thus, is best focused on management rather than on cure. Table 12-10 lists the guidelines for treating and managing factitious disorder. The three primary goals in the treatment and management of factitious disorders are (1) to reduce the risk of morbidity and mortality, (2) to address the underlying emotional needs or psychiatric diagnosis underlying factitious illness behavior, and (3) to be mindful of legal and ethical issues. Perhaps the single most crucial factor in successful management is a physician's early recognition of the disorder. In this way, physicians can forestall a multitude of painful and potentially dangerous diagnostic procedures for these patients. There must be a good liaison between psychiatrists and the medical or surgical staff. Although a few cases of individual psychotherapy exist in the literature, no consensus exists about the best approach. In general, working in concert with the patient's primary care physician is more effective than

working with the patient in isolation.

Table 12-10
Guidelines for Management and Treatment of Factitious Disorder

Keep in mind that active pursuit of a prompt diagnosis can minimize the risk of morbidity and mortality.
Minimize harm. Avoid unnecessary tests and procedures, especially if they are invasive. Treat according to clinical judgment, keeping in mind that subjective complaints may be deceptive.
Arrange regular interdisciplinary meetings to reduce conflict and splitting among staff. Manage staff countertransference.
Steer the patient toward psychiatric treatment in an empathic, nonconfrontational, face-saving manner. Avoid aggressive direct confrontation.
Treat underlying psychiatric disturbances. In psychotherapy, address coping strategies and emotional conflicts.
Appoint a primary care provider as a gatekeeper for all medical and psychiatric treatment.
Consider involving risk management and bioethicists from an early point.
Consider appointing a guardian for medical and psychiatric decisions.
As a behavioral disincentive, consider prosecution for fraud.

Table 12-11
Pediatric Factitious Disorder Imposed on Another—Basic Principles of Management

Make sure the child is safe.

Make sure the child's future safety is also assured.

Allow treatment to occur in the least restrictive setting possible.

A pediatrician should serve as "gatekeeper" for medical care utilization.

All other physicians should coordinate care with the gatekeeper.

Child-protective services should be informed whenever a child is harmed.

Family psychotherapy and/or individual psychotherapy should be instituted for the perpetrating parent and the child.

Health insurance companies, school officials, and other nonmedical sources should be asked to report possible medical abuse to the physician gatekeeper. Permission of a parent or of child-protective services must first be obtained.

The possibility should be considered of admitting the child to an inpatient or partial hospital setting to facilitate diagnostic monitoring of symptoms and to institute a treatment plan.

The child may require placement in another family. The perpetrating parent may need to be removed from the child through criminal prosecution and incarceration.

The personal reactions of physicians and staff members are of great significance in treating and establishing a working alliance with these patients, who invariably evoke feelings of futility, bewilderment, betrayal, hostility, and even contempt. In essence, staff members must abandon a fundamental element of their relationship with patients—accepting the truthfulness of the patients' statements. One appropriate psychiatric intervention is to suggest to the staff ways of remaining aware that even though the patient's illness is factitious, the patient is ill. Physicians should try not to feel resentment when patients humiliate their diagnostic prowess, and they should avoid any unmasking ceremony that sets up the patients as adversaries and precipitates their flight from the hospital. The staff should not

perform unnecessary procedures or discharge patients abruptly, both of which are manifestations of anger.

In cases of factitious disorder imposed on another, physicians have obtained a legal intervention in several instances, particularly with children. The senselessness of the disorder and the denial of false action by parents are obstacles to successful court action and often make conclusive proof unobtainable. In such cases, the treatment team should notify the child welfare and make arrangements for ongoing monitoring of the children's health (see Table 12-11 for interventions for pediatric factitious disorder imposed on another).

EPIDEMIOLOGY

Somatic Symptom Disorder

The prevalence of somatic symptom disorder is not yet known. Data do exist, however, for somatization disorder as defined by DSM-IV. Studies in the United States, Puerto Rico, Germany, and Italy found lifetime prevalence rates of "full" somatization disorder ranging from 0.1 percent in the United States to 0.8 percent in Germany. In contrast, lifetime rates of "abridged" somatization disorder ranged from 5.6 percent in Germany to 19 percent in Puerto Rico. Much higher prevalence rates have been reported for more broadly defined somatic symptom clusters.

Illness Anxiety Disorder

The prevalence of this disorder is unknown, aside from using data that relate to hypochondriasis, which gives a prevalence of 4 to 6 percent in a general medical clinic population. In other surveys, up to 10 percent of persons in the general population worry about becoming sick and incapacitated as a result.

Conversion Disorder (Functional Neurologic Symptom Disorder)

The epidemiologic information on conversion disorder is limited. Estimates vary broadly: Less than 1 percent in the general population, 5 to 14 percent among general hospital medical/surgical referrals to psychiatry consultation services, and 5 to 25 percent in treated psychiatric outpatients. The disorder appears to be more frequent in females and can occur in children as young as

7 or 8 years old. It is rare after the age of 35 years.

Psychological Factors Affecting Other Medical Conditions

The prevalence of this condition is unknown. In mental health systems, it is most frequently diagnosed in consultation–liaison psychiatry settings.

Factitious Disorder

No comprehensive epidemiologic data on factitious disorder exist. However, it is estimated to comprise approximately 1 percent of the healthcare-seeking population. Factitious disorder imposed on another account for less than 0.04 percent of reported child abuse in the United States each year.

ETIOLOGY

Somatic Symptom Disorder

Early family studies suggested an association between hysteria/Briquet syndrome in females and antisocial personality in their male first-degree relatives. Also, studies of adopted-away children highlighted the presence of alcoholism and violence in biologic fathers of women with somatization disorder. However, since developing the new diagnostic criteria, there have been no major epidemiologic studies.

Some have postulated that persons with this disorder augment and amplify their somatic sensations; they have low thresholds for and low tolerance of physical discomfort. For example, what persons usually perceive as abdominal pressure, persons with somatic symptom disorder experience as abdominal pain. They may focus on bodily sensations, misinterpret them, and become alarmed by them because of a faulty cognitive scheme.

Others have proposed a social learning model. They see the symptoms of this disorder as a request for admission to the sick role made by a person facing seemingly insurmountable and unsolvable problems. The sick role offers an escape that allows a patient to avoid noxious obligations, to postpone unwelcome challenges, and to evade usual duties and obligations.

Illness Anxiety Disorder

The etiology is unknown. The social learning model described for somatic symptom disorder may apply to this disorder as well, with a similar dynamic to what we described above.

Conversion Disorder (Functional Neurologic Symptom Disorder)

Biologic Factors. The neuropsychologic conceptualization speaks of an inherent deficit in certain brain functions, especially those in the dominant hemisphere that may interfere with verbal associations. Also, functional MRI studies have shown differences in brain activation between patients with conversion disorder and controls.

Psychological Factors The behavioral theory attributes conversion disorder to faulty childhood learning, with the nonadaptive behavioral responses used for secondary gain and control of interpersonal relationships. The psychoanalytic theory, on the other hand, describes symptoms as compromise formations with primary gain of conflict resolution through a partial expression of the conflict without conscious awareness of its significance. Some have suggested a strong relationship between childhood traumatization by sexual or physical abuse and a later propensity for conversion disorder. Other studies, however, do not confirm such an association.

Factitious Disorder

Etiology is generally not clear, except that a common denominator is that these patients tend to be avid medical service seekers. The underlying motivations for the behaviors are likely unconscious. Two factors may underlie most cases of factitious disorder: (1) an affinity for the medical system, and (2) poor, maladaptive coping skills. In the case of factitious disorder imposed on another, psychodynamic theories predominately view the disorder as an objectification of the child to serve the parent's psychological needs.

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13 Feeding and Eating Disorders

This chapter deals with several feeding and eating disorders, including anorexia nervosa, bulimia nervosa, binge-eating disorder, and other specified feeding and eating disorders. There are different feeding and eating disorders that are typically associated with childhood and adolescence (e.g., pica, rumination disorder, and avoidant/restrictive food intake disorder); Neurodevelopmental Disorders and Other Childhood Disorders (Chapter 2) covers those disorders.

CLINICAL FEATURES

Anorexia Nervosa

The term anorexia nervosa comes from the Greek term for “loss of appetite” and a Latin word implying nervous origin. Anorexia nervosa is a syndrome characterized by three essential criteria, one behavioral, one psychopathological, and the last, physiologic. The first is self-induced starvation, to a significant degree (behavioral). The second is a relentless drive for thinness or a morbid fear of fatness (psychopathological). The third criterion is the presence of medical signs and symptoms resulting from starvation (physiologic). Anorexia nervosa is often, but not always, associated with disturbances of body image, the perception that one is distressingly large despite evident medical starvation. The distortion of body image is disturbing when present; it is, however, not pathognomonic, invariable, or required for diagnosis. Two subtypes of anorexia nervosa exist: restricting and binge/purge. The theme in all anorexia nervosa subtypes is the highly disproportionate emphasis placed on thinness as a vital source of self-esteem, with weight, and to a lesser degree, shape, becoming the overriding

and consuming daylong preoccupation of thoughts, mood, and behaviors.

Approximately half of anorexic persons will lose weight by drastically reducing their total food intake. The other half of these patients will not only diet but will also regularly engage in binge eating, followed by purging behaviors. Some patients routinely purge after eating small amounts of food. Anorexia nervosa is much more prevalent in females than in males and usually has its onset in adolescence. Hypotheses of an underlying psychological disturbance in young women with the disorder include conflicts surrounding the transition from girlhood to womanhood. Some have also suggested that psychological issues related to feelings of helplessness and difficulty establishing autonomy also contribute to the development of the disorder. Bulimic symptoms can occur as a separate disorder or as part of anorexia nervosa. Persons with either disorder are excessively preoccupied with weight, food, and body shape. The outcome of anorexia nervosa varies from spontaneous recovery to a waxing and waning course to death.

Bulimia Nervosa

People with bulimia nervosa have episodes of binge eating combined with inappropriate ways of stopping weight gain. Physical discomfort—for example, abdominal pain or nausea—terminates the binge eating, which is often followed by feelings of guilt, depression, or self-disgust. Unlike patients with anorexia nervosa, those with bulimia nervosa typically maintain average body weight.

The term bulimia nervosa derives from the terms for “ox-hunger” in Greek and “nervous involvement” in Latin. For some patients, bulimia nervosa may represent a failed attempt at anorexia nervosa, sharing the goal of becoming very thin, but occurring in an individual less able to sustain prolonged semistarvation or severe hunger as consistently as classic restricting anorexia nervosa patients. For others, eating binges represent “breakthrough eating” episodes of giving in to hunger pangs generated by efforts to limit eating to maintain a socially desirable level of thinness. Still, others use binge eating as a means to self-medicate during times of emotional distress. Regardless of the reason, eating binges provoke panic as individuals feel that their eating has been out of control. The unwanted binges lead to subsequent attempts to avoid the feared weight gain by a variety of compensatory behaviors, such as

purging or excessive exercise.

Binge-Eating Disorder

Individuals with binge-eating disorder engage in recurrent binge eating during which they eat an abnormally large amount of food over a short time. Unlike bulimia nervosa, patients with binge-eating disorder do not compensate in any way after a binge episode (e.g., vomiting, laxative use). Binge episodes often occur in private, generally include foods of dense caloric content, and, during the binge, the person feels he or she cannot control his or her eating.

Other Specified Feeding or Eating Disorders

This diagnostic category also includes eating conditions that may cause significant distress but do not meet the full criteria for a classified eating disorder. Conditions included in this category include night-eating syndrome, purging disorder, and subthreshold forms of anorexia nervosa, bulimia nervosa, and binge-eating disorder.

Night-Eating Syndrome. Night-eating syndrome is characterized by the consumption of large amounts of food after the evening meal. Individuals generally have little appetite during the day and suffer from insomnia.

Purging Disorder. Purging disorder is characterized by recurrent purging behavior after consuming a small amount of food in persons of average weight who have a distorted view of their weight or body image. Purging behavior includes self-induced vomiting, laxative abuse, enemas, and diuretics. This behavior should not be associated with anorexia nervosa. Purging disorder is differentiated from bulimia nervosa because purging behavior occurs after eating small quantities of food or drink and does not occur as a result of a binge episode.

DIAGNOSIS AND CLINICAL FEATURES

Anorexia Nervosa

The onset of anorexia nervosa usually occurs between the ages of 10 and 30

years. We summarize the diagnostic criteria in DSM-5 and ICD-10 for the disorder in Table 13-1.

Intense fear of gaining weight and becoming obese is present in all patients with the disorder and undoubtedly contributes to their lack of interest in and even resistance to therapy. Most aberrant behavior directed toward losing weight occurs in secret. Patients with anorexia nervosa usually refuse to eat with their families or in public places. They lose weight by drastically reducing their total food intake, with a disproportionate decrease in high-carbohydrate and fatty foods.

The term anorexia is a misnomer because the loss of appetite is usually rare until late in the disorder. Evidence that patients are frequently thinking about food is their passion for collecting recipes and for preparing elaborate meals for others. Some patients cannot continuously control their voluntary restriction of food intake, and so have eating binges. These binges usually occur secretly, often at night, and are frequently followed by self-induced vomiting. Patients abuse laxatives and even diuretics to lose weight, and ritualistic exercising, extensive cycling, walking, jogging, and running are everyday activities.

Patients with the disorder show peculiar behavior surrounding food. They hide food all over the house and frequently carry large quantities of candies in their pockets and purses. While eating meals, they try to dispose of food in their napkins or hide it in their pockets. They cut their meat into tiny pieces and spend a great deal of time rearranging these pieces on their plates. If someone confronts the patient about the behavior, they often deny that it is unusual or flatly refuse to discuss it.

Obsessive-compulsive behavior, depression, and anxiety are other psychiatric symptoms of anorexia nervosa most frequently noted clinically. Patients tend to be rigid and perfectionist, and somatic complaints, especially epigastric discomfort, are usual. Compulsive stealing, usually of candies and laxatives but occasionally of clothes and other items, may occur.

Patients with the disorder frequently have poor sexual adjustment. Many adolescent patients with anorexia nervosa have delayed psychosocial sexual development; in adults, a markedly decreased interest in sex often accompanies the onset of the disorder. A minority of anorexic patients have a premorbid history of promiscuity, substance abuse, or both, but during the disorder show a decreased interest in sex.

Patients usually come to medical attention when their weight loss becomes apparent. As the weight loss grows profound, physical signs such as hypothermia (as low as 35°C), dependent edema, bradycardia, hypotension, and lanugo (the appearance of neonatal-like hair) appear, and patients show a variety of metabolic changes. These and other medical complications are listed in Table 13-2.



Table 13-1
Anorexia Nervosa Diagnoses

	Anorexia Nervosa	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Anorexia nervosa	Anorexia nervosa
Symptom	• Food restriction leading to abnormally low weight • Fear of gaining weight and behaviors to prevent weight gain • Distorted body image • Misperceiving weight or shape • Self-evaluation based on thinness, lack of recognition of the seriousness of one's weight	• Deliberate weight loss induced by the patient • Fear of being fat or flabby • Associated physiologic disturbances due to low weight • Specific behaviors: • Restricted diet • Excess exercise • Binging behavior • Use of appetite suppressants and diuretics
Required number of	All	

symptom		
Exclusions		Loss of appetite (physical or psychological)
Symptom specifiers	Restricting type: Primary behaviors are dieting, fasting, and exercise for last 3 mo Binge eating type: Binging and purging behavior for last 3 mo. Can include use of laxatives or other substances to aid purging	
Course specifiers	Partial remission: Symptoms present but no longer an abnormally low weight Full remission: No symptoms for a sustained period	
Severity specifiers	Severity is measured by BMI level See DSM-5 for ranges	
Comments		If lacking core symptoms, the diagnosis of “atypical anorexia nervosa” is made



Table 13-2
Medical Complications of the Eating Disorders

Disorder and System	Consequence

Affected	
Anorexia Nervosa	
Vital signs	Bradycardia, hypotension with marked orthostatic changes, hypothermia, poikilothermia
General	Muscle atrophy, loss of body fat
Central nervous system	Generalized brain atrophy with enlarged ventricles, decreased cortical mass, seizures, abnormal electroencephalogram
Cardiovascular	Peripheral (starvation) edema, decreased cardiac diameter, narrowed left ventricular wall, decreased response to exercise demand, superior mesenteric artery syndrome
Renal	Prerenal azotemia
Hematologic	Anemia of starvation, leukopenia, hypocellular bone marrow
Gastrointestinal	Delayed gastric emptying, gastric dilatation, decreased intestinal lipase and lactase
Metabolic	Hypercholesterolemia, nonsymptomatic hypoglycemia, elevated liver enzymes, decreased bone mineral density
Endocrine	Low luteinizing hormone, low follicle-stimulating hormone, low estrogen or testosterone, low/normal thyroxine, low triiodothyronine, increased reverse triiodothyronine, elevated cortisol, elevated growth hormone, partial diabetes insipidus, increased prolactin
Bulimia Nervosa and Binge Eating and Purging Type Anorexia Nervosa	
Metabolic	Hypokalemic alkalosis or acidosis, hypochloremia, dehydration

Renal	Prerenal azotemia, acute and chronic renal failure
Cardiovascular	Arrhythmias, myocardial toxicity from emetine (ipecac)
Dental	Lingual surface enamel loss, multiple caries
Gastrointestinal	Swollen parotid glands, elevated serum amylase levels, gastric distention, irritable bowel syndrome, melanosis coli from laxative abuse
Musculoskeletal	Cramps, tetany

Subtypes. Anorexia nervosa has two clinical subtypes: food restricting and purging. In the food-restricting category, present in approximately 50 percent of cases, food intake is highly restricted (usually with attempts to consume fewer than 300 to 500 calories per day and no fat grams) and the patient may be relentlessly and compulsively overactive, with overuse athletic injuries. In the purging subtype, patients alternate attempts at rigorous dieting with intermittent binge or purge episodes. Purging represents a secondary compensation for the unwanted calories, most often accomplished by self-induced vomiting, frequently by laxative abuse, less frequently by diuretics, and occasionally with emetics. Sometimes, repetitive purging occurs without prior binge eating, after ingesting only relatively few calories. Both types may be socially isolated and have depressive disorder symptoms and diminished sexual interest. Overexercising and perfectionistic traits are also common in both types.

Those who practice binge eating and purging share many features with persons who have bulimia nervosa without anorexia nervosa. Those who binge eat and purge tend to have families in which some members are obese, and they have histories of heavier body weights before the disorder than do persons with the restricting type. Binge eating–purging persons are likely to be associated with substance abuse, impulse control disorders, and personality disorders. Persons with restricting anorexia nervosa often have obsessive-compulsive traits concerning food and other matters. Some persons with anorexia nervosa may purge but not binge.

Patients with anorexia nervosa are often secretive, deny their symptoms,

and resist treatment. In almost all cases, relatives or intimate acquaintances must confirm a patient's history. The mental status examination usually shows a patient who is alert and knowledgeable on the subject of nutrition and who is preoccupied with food and weight.

A young woman who weighed 10 percent above the average weight but was otherwise healthy, functioning well, and working hard as a university student joined a track team. She started training for hours a day, more than her teammates, began to perceive herself as fat, and thought that her performance would be enhanced if she lost weight. She started to diet and reduced her weight to 87 percent of the "ideal weight" for her age according to standard tables. At her point of maximum weight loss, her performance declined, and she pushed herself even harder in her training regimen. She started to feel apathetic and morbidly afraid of becoming fat. Her food intake became restricted, and she stopped eating anything containing fat. Her menstrual periods became skimpy and infrequent but did not cease. (Courtesy of Arnold E. Andersen, M.D. and Joel Yager, M.D.)

Bulimia Nervosa

People with bulimia nervosa have binge episodes with compensatory behaviors. Like persons with anorexia nervosa, they fear becoming fat; however, they are not severely thin. We summarize the diagnostic criteria in DSM-5 and ICD-10 for the disorder in Table 13-3.



Table 13-3
Bulimia Nervosa Diagnoses

	Bulimia Nervosa	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Bulimia nervosa	Bulimia nervosa
Duration	Occurs at least once a week for	

	3 mo	
Symptom	1. Repeated binge-eating episodes • Occurring within 2 hr period • Loss of control during the episode 2. Compensatory behaviors to prevent weight gain, such as • Vomiting • Laxatives • Diuretics, other substances • Excessive exercise • Fasting 3. Self-evaluation based on weight and shape	• Binging • Preoccupation with weight control • Preoccupation with body shape and weight • Associated physiologic disturbances due to repeated vomiting • Specific compensatory behaviors: • Vomiting • Use of purgatives
Required number of symptom	All 3	Not specified
Exclusions	Does not occur only during episodes of anorexia nervosa	
Course specifiers	Partial remission: Full symptoms no longer met for a sustained time Full remission: No symptoms for a sustained period	
Severity specifiers	Severity is measured by the average number of compensatory behaviors in	

a weeks' time
See DSM-5 for ranges

When making a diagnosis of bulimia nervosa, clinicians should explore the possibility that the patient has experienced a brief or prolonged prior bout of anorexia nervosa, which is present in approximately half of those with bulimia nervosa. The binging behavior commonly precedes vomiting by about 1 year.

Vomiting is frequent and is often induced by sticking a finger down the throat. Some patients can vomit at will. Vomiting decreases the abdominal pain and the feeling of being bloated and allows patients to continue eating without fear of gaining weight. The acid content of vomitus can damage tooth enamel, a not uncommon finding in patients with the disorder. Depression, sometimes called postbinge anguish, often follows the episode. During binges, patients eat food that is sweet, high in calories, and generally soft or smooth textured, such as cakes and pastry. Some patients prefer bulky foods without regard to taste. The food is eaten secretly and rapidly and is sometimes not even chewed.

Jean was a 25-year-old woman admitted to an inpatient unit after a superficial suicide attempt. During her initial evaluation, she revealed that she had struggled with bulimia nervosa since her early teens. She said that she has always tended "to be on the heavy side" and would respond by skipping meals during the day, only to binge on "junk food" in the evening. Her binge eating occurred almost daily, and at least two or three times a week, Jean would force herself to vomit afterward. She also used daily laxatives and ran for at least an hour a day. Despite these behaviors, she could not lose weight, which distressed her to the point that she cut her wrists, although Jean admitted she had really "only scratched myself." She drank socially and did not use illicit drugs. On examination, she appeared somewhat overweight with a BMI of 29. Her dentition showed signs of enamel erosion, and she had calluses on her fingers from her induced vomiting. There were superficial scratches on her wrist from when she cut herself. She had normal electrolytes on

laboratory evaluation.

Jean had a high school education and had been married for 2 years, and she described her marriage as “rocky” as her husband frequently traveled on business. She said she rarely saw her parents, who were “always criticizing me.” She did not work and said she was bored most of the time. On several occasions, while her husband was away, she would meet up with former boyfriends and sometimes slept with them. She felt that her guilt over these affairs contributed to her depression and suicidal ideation.

During her hospital stay, she had nutritional counseling and ate regularly scheduled meals, for which she had to be observed. After meals, she could use the bathroom but was instructed not to flush the toilet. This continued for 2 days until a staff member found a plastic bag of partially digested food in a trash can, after which the staff would observe her for an hour after meals. Initially, she would use a hospital courtyard for jogging, and when that was restricted, the evening staff observed her doing calisthenics in her room. Several staff members also saw her flirting with the male patients, and her therapist reminded her of the unit rules regarding patient relationships. Her treatment consisted of a behavioral schedule and frequent observation while her therapy focused on her difficulties with self-esteem and interpersonal interactions. Her psychiatrist initiated fluoxetine, 20 mg. Her purging behaviors decreased over the next 2 weeks to the point where she was considered appropriate for outpatient treatment.

Most patients with bulimia nervosa are within their normal weight range, but some may be underweight or overweight. These patients are concerned about their body image and their appearance, worried about how others see them, and concerned about their sexual attractiveness. Most are sexually active, compared with anorexia nervosa patients, who are not interested in sex. They may have a history of pica and struggles during meals.



Table 13-4
Binge-Eating Disorder

	Binge-Eating Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Binge-eating disorder	Bulimia nervosa, nonpurging type
Duration	Occurs at least once a week for 3 mo	
Symptom	• Repeated binge-eating episodes • Occurring within 2 hr period • Loss of control during the episode • Binge-associated behaviors or reactions • Eating quicker than normal • Eating until uncomfortably full • Eating despite not being hungry • Eating alone to avoid embarrassment • Feeling guilty or disgusted after eating	Similar to ICD-10 bulimia nervosa minus the specific compensatory behaviors
Required number of symptom	Binge behaviors plus at least three of the associated behaviors or reactions	
Psychosocial consequences of symptoms	Marked distress	

Exclusions	No compensatory behavior as in bulimia nervosa
Course specifiers	Partial remission: Occurrences are less frequent (<1 episode/wk) Full remission: No symptoms for a sustained period
Severity specifiers	Severity is measured by the average number of compensatory behaviors in a weeks' time See DSM-5 for ranges

Binge-Eating Disorder

Individuals with binge-eating disorder engage in frequent binges, often independent of feeling hungry. We summarize the diagnostic criteria in DSM-5 and ICD-10 for the disorder in Table 13-4.

Other Specified Feeding and Eating Disorder

Night-Eating Syndrome. As implied by the name, night-eating syndrome includes recurrent episodes of hyperphagia or night eating. It may be associated with insomnia and a lack of desire for food in the morning.

Patients with night-eating syndrome usually consume a large portion of their daily calorie intake after the evening meal. They may also wake up during the night and eat upon awakening. Some patients believe that they can only sleep if they eat. Depressed mood is common among these patients, especially during the evening and night hours.

PATHOLOGY AND LABORATORY EXAMINATION

Anorexia Nervosa

A patient with anorexia nervosa must have a thorough general physical and neurologic examination. If the patient is vomiting, a hypokalemic alkalosis may be present. Because most patients are dehydrated, serum electrolyte levels must be determined initially and periodically. Hospitalization may be necessary to deal with medical complications.

A complete blood count often reveals leukopenia with a relative lymphocytosis in emaciated patients with anorexia nervosa. If binge eating and purging are present, serum electrolyte determination reveals hypokalemic alkalosis. Fasting serum glucose concentrations are often low during the emaciated phase, and serum salivary amylase concentrations may increase if the patient is vomiting.

Electrocardiographic (ECG) changes, such as T-wave flattening or inversion, ST segment depression, and lengthening of the QT interval, have been noted in the emaciated stage of anorexia nervosa. The ST segment and T-wave changes are usually secondary to electrolyte disturbances; emaciated patients have hypotension and bradycardia. ECG changes may also result from potassium loss, which can lead to death. Young girls may have a high serum cholesterol level. All these values revert to normal with nutritional rehabilitation and cessation of purging behaviors. Endocrine changes related to being underweight may occur. These can include amenorrhea, mild hypothyroidism, and hypersecretion of corticotrophin-releasing hormone, which revert to normal with weight gain (see Table 13-2).

Bulimia Nervosa

Bulimia nervosa can result in electrolyte abnormalities and various degrees of starvation, although it may not be as apparent as in low-weight patients with anorexia nervosa. Thus, even normal-weight patients with bulimia nervosa should have laboratory studies of electrolytes and metabolism. In general, thyroid function remains intact in bulimia nervosa, but patients may show nonsuppression on a dexamethasone-suppression test. Dehydration and electrolyte disturbances are likely to occur in patients with bulimia nervosa who purge regularly. These patients commonly exhibit hypomagnesemia and hyperamylasemia. Although not a core diagnostic feature, many patients with bulimia nervosa have menstrual disturbances. Hypotension and bradycardia

occur in some patients.

DIFFERENTIAL DIAGNOSIS

Anorexia Nervosa

The differential diagnosis of anorexia nervosa is complicated, as patients often deny symptoms, are secretive about their eating habits, and resist seeking treatment. Thus, it may be difficult to identify the mechanism of weight loss and the patient's associated ruminative thoughts about distortions of body image.

We should differentiate the eating disorders from one another. Attention to specific criteria, including whether the patient is of average weight, is essential. However, the two conditions can coexist.

We must ascertain that a patient does not have a medical illness that can account for the weight loss (e.g., a brain tumor or cancer). Weight loss, peculiar eating behaviors, and vomiting can occur in several mental disorders. Depressive disorders and anorexia nervosa have several features in common, such as depressed feelings, crying spells, sleep disturbance, obsessive ruminations, and occasional suicidal thoughts. The two disorders, however, have several distinguishing features. In general, a patient with a depressive disorder has decreased appetite, whereas a patient with anorexia nervosa claims to have a normal appetite and to feel hungry; only in the severe stages of anorexia nervosa do patients have decreased appetite.

In contrast to depressive agitation, the hyperactivity seen in anorexia nervosa is planned and ritualistic. The preoccupation with recipes, the caloric content of foods, and the preparation of gourmet feasts is typical of patients with anorexia nervosa but is absent in patients with a depressive disorder. In depressive disorders, patients have no intense fear of obesity or disturbance of body image.

Weight fluctuations, vomiting, and peculiar food handling may occur in somatization disorder. On rare occasions, a patient may have both a somatization disorder and anorexia nervosa, in which case we should diagnose both. In general, patients with somatization disorder have less weight loss than patients with anorexia nervosa and do not fear weight gain. Amenorrhea for 3 months or longer is unusual in somatization disorder.

Patients with schizophrenia that include delusions about food rarely are concerned with caloric content. More likely, they believe someone poisoned the food. Patients with schizophrenia are rarely preoccupied with a fear of becoming obese and do not have the hyperactivity seen with anorexia nervosa. Patients with schizophrenia have bizarre eating habits but not the entire syndrome of anorexia nervosa.

There are rare conditions of unknown etiology in which a hyperactive vagus nerve causes changes in eating patterns that are associated with weight loss, sometimes severe. In such cases, we may see bradycardia, hypotension, and other parasympathomimetic signs and symptoms. Because the vagus nerve relates to the enteric nervous system, eating may be associated with gastric distress such as nausea or bloating. Patients do not generally lose their appetite. Treatment is symptomatic. Anticholinergic drugs can reverse hypotension and bradycardia, which may be life-threatening.

Bulimia Nervosa

We cannot make the diagnosis of bulimia nervosa if the binge-eating and purging behaviors occur exclusively during episodes of anorexia nervosa. In such cases, the diagnosis is anorexia nervosa, binge eating–purging type. Additionally, we should distinguish bulimia nervosa from binge-eating disorder, which typically includes binge-eating behaviors but no compensatory or purging behaviors.

We must ascertain that patients have no neurologic disease, such as epileptic-equivalent seizures, central nervous system tumors, Klüver–Bucy syndrome, or Kleine–Levin syndrome. Klüver–Bucy syndrome includes visual agnosia, compulsive licking and biting, an examination of objects by the mouth, inability to ignore any stimulus, placidity, altered sexual behavior (hypersexuality), and altered dietary habits, especially hyperphagia. The syndrome is exceedingly rare and is unlikely to cause a problem in the differential diagnosis. Kleine–Levin syndrome consists of periodic hypersomnia lasting for 2 to 3 weeks and hyperphagia. As in bulimia nervosa, the onset is usually during adolescence, but the syndrome is more common in men than in women.

Some patients with bulimia nervosa have multiple comorbid impulsive behaviors, including substance abuse, and lack of ability to control

themselves in such diverse areas as money management (resulting in impulse buying and compulsive shopping) and sexual relationships (often resulting in a short, passionate attachment and promiscuity). They exhibit self-mutilation, chaotic emotions, and chaotic sleeping patterns. They usually meet the criteria for borderline personality disorder and other mixed personality disorders and, not infrequently, bipolar II disorder. When bulimia nervosa co-occurs with one of these disorders, we should note both diagnoses.

Binge-Eating Disorder

Binge-eating disorder and bulimia nervosa share the same core feature of recurrent binge eating. Binge-eating disorder is distinct from bulimia nervosa, however, in that binge-eating disorder patients do not report repetitive compensatory behaviors such as vomiting, laxative abuse, or excessive dieting. Binge-eating disorder is distinct from anorexia nervosa in that patients do not exhibit an extreme drive for thinness and are of average weight or are obese.

Other Specified Feeding and Eating Disorder

Night-Eating Syndrome. Night-eating syndrome is common among patients with other eating disorders, particularly bulimia nervosa and binge-eating disorder. Although we can find night eating in patients with bulimia nervosa and binge-eating disorder, it is the characteristic sign of night-eating disorder. Also, the amount of food consumed during eating episodes is usually lower in night-eating disorder than in bulimia nervosa and binge-eating disorder. Unlike other eating disorders, patients with night eating syndrome are not overly concerned about body image and weight. Patients with night-eating syndrome are also at higher risk for obesity and metabolic syndrome.

Recurrent episodes of involuntary eating characterize a sleep-related eating disorder during the night. These episodes can lead to severe consequences such as the ingesting of nonedible foods or substances, dangerous behaviors while searching for or preparing food, and sleep-related injury. The eating episodes usually occur after the patient has gone to sleep and may occur while the patient is unconscious or asleep. The sleep-related eating disorder

also has high comorbidity with sleepwalking, restless legs syndrome, and obstructive sleep apnea, conditions that are rarely found among night-eating syndrome patients. Episodes of sleep-related eating disorder have been reported after the use of certain medications, including zolpidem, triazolam, olanzapine, and risperidone.

COMORBIDITY

Anorexia Nervosa

Table 13-5 lists comorbid psychiatric conditions associated with anorexia nervosa. Overall, anorexia nervosa is associated with depression in 50 percent of cases, social phobia in 22 percent of cases, and obsessive-compulsive disorder (OCD) in 35 percent of cases.

The suicide rate is higher in persons with the binge eating–purging type of anorexia nervosa than in those with the restricting type.

Binge eating–purging persons are likely to be associated with substance abuse, impulse control disorders, and personality disorders. Persons with restricting anorexia nervosa often have obsessive-compulsive traits concerning food and other matters. Some persons with anorexia nervosa may purge but not binge.

Bulimia Nervosa

Bulimia nervosa occurs in individuals with high rates of mood disorders and impulse control disorders. Bulimia nervosa also co-occurs with substance use disorders, particularly alcohol. Patients with bulimia nervosa also have increased rates of anxiety disorders, bipolar I disorder, dissociative disorders, and histories of sexual abuse. Patients with bulimia nervosa who have concurrent seasonal affective disorder and patterns of atypical depression (with overeating and oversleeping in low-light months) may manifest seasonal worsening of both bulimia nervosa and depressive features. Individuals with bulimia nervosa who purge may be at risk for certain medical complications such as hypokalemia from vomiting or laxative abuse and hypochloremic alkalosis. Those who vomit repeatedly are at risk for gastric and esophageal tears, although these complications are rare.

Binge-Eating Disorder

Binge-eating disorder is associated with mood, anxiety, and substance use disorders. Also, nearly half of individuals with binge-eating disorder are obese and are at risk for medical complications associated with obesity. Patients with binge-eating disorder are also more likely to have an unstable weight history with frequent episodes of weight cycling (the gaining or losing of more than 10 kg). The disorder may be associated with insomnia, early menarche, neck or shoulder and lower back pain, chronic muscle pain, and metabolic disorders.

Table 13-5
Lifetime Comorbidities of Various Psychiatric Disorders with Eating Disorders

	No Eating Disorder	Anorexia Nervosa (%)		Bulimia Nervosa (%)		Binge-Eating Disorder (%)	
Diagnosis	%	%	AOR[a]	%	AOR[a]	%	AOR[a]
MDD	10	8.7	0.7	31.0	3.2[b]	35.4	3.9[b]
Dysthymia	1.7	1.1	0.6	6.7	3.6[b]	5.8	3.1[b]
Bipolar disorder type 1 or 2	2.8	2.1	0.7	18.5	7.3[b]	9.0	3[b]
Any anxiety disorder	12.8	10.9	0.7	49.9	5.7[b]	45.3	4.6[b]
Agoraphobia[c]	2.2	3.5	1.8	7.5	2.7[b]	7.1	2.6[b]
GAD	0.9	0.0	n/a[d]	4.4	4.0	2.0	1.8
Social phobia	5	9.2	1.8	20.3	3.9[b]	26.3	5.9[b]
Specific phobia	14.2	20.5	1.5	36.7	3.1[b]	32.1	2.6[b]
Panic disorder[c]	2.1	0.9	0.4	11.1	5.2[b]	8.7	4[b]
PTSD	3.4	8.7	2.0	26.5	7.6[b]	13.2	3[b]
Seasonal affective disorder	6.5	11.1	1.8	22.1	3.5[b]	16.8	2.6[b]
Any substance use or dependence	24.5	23.9	0.9	66.2	5[b]	65.2	5[b]
Alcohol use disorder	6.1	9.1	1.6	14.3	3.1[b]	13.9	2.4[b]
Substance use disorder	8.3	13.0	1.7	19.3	2.8[b]	22.5	3.2[b]
Any drug use	10.8	13.0	1.3	20.1	2.2[b]	26.8	3.1
ADHD[e]	8.6	2.3	0.2	20.0	3.6[b]	12.6	2.1[b]
Oppositional defiant disorder[e]	6.9	30.4	5.1[b]	24.4	4[b]	32.8	6.2[b]
Conduct disorder[e]	10.4	5.7	0.5	29.0	3.5[b]	28.5	3.6[b]

AOR, adjusted odds ratio.
[a]Adjusted for age, sex, and race/ethnicity.
[b]$p < 0.05$.
[c]Agoraphobia is assessed without panic disorder, while panic disorder is assessed with or without agoraphobia.
[d]No adolescents had both AN and GAD.
[e]Disorders are assessed using both parent and child report ($n = 6,483$).
Adapted from Hudson JI, Hiripi E, Pope HG Jr, Kessler RC. The prevalence and correlates of eating disorders in the National Comorbidity Survey Replication. *Biol Psychiatry*. 2007;61(3):348–358.

COURSE AND PROGNOSIS

Anorexia Nervosa

The precise course of the illness varies substantially, although specific patterns have emerged in the literature. Follow-up studies of patients with anorexia nervosa reveal that, at the time of assessment, approximately 30 to 50 percent have achieved full recovery, and 10 to 20 percent remain chronically ill. The remainder improve but continue to struggle with certain disordered behaviors. The chronically ill group often requires multiple hospitalizations. Of note, anorexia nervosa has a mortality rate as high as that associated with any psychiatric illness. Compared to the general population, individuals with the illness are up to six times more likely to die. The majority of deaths are attributable to medical complications of low weight and malnourishment, but a smaller, yet significant, proportion of deaths (approximately 1 in 5) are due to suicide.

Although additional research is needed to identify predictors of outcome in anorexia nervosa, studies have found that adolescents with a shorter duration of illness tend to have a better prognosis, emphasizing the importance of early detection and intervention. Studies examining inpatients with anorexia nervosa just before and after discharge have also identified specific predictors of posttreatment outcomes. Individuals who achieve full weight-restoration on an inpatient unit and maintain their weight in the first month after discharge are, perhaps unsurprisingly, more likely to remain at a healthy weight up to a year after treatment. Lower BMI at discharge and weight loss in the first month after treatment, on the other hand, predict more unsatisfactory long-term outcomes. Additionally, individuals who demonstrate an ability to consume a diet that is high in variety and energy density (i.e., a greater concentration of kcal/g) before discharge seem to do better after treatment.

Bulimia Nervosa

Bulimia nervosa is characterized by higher rates of partial and full recovery compared with anorexia nervosa. Those receiving treatment fare much better than those who are untreated. Untreated patients tend to remain chronic or may show small, but generally unimpressive, degrees of improvement with time. Research has yet to identify any clear predictors of outcome in bulimia nervosa. Studies have examined a variety of potential prognostic factors,

including duration of illness, age of onset, illness severity, comorbid diagnoses, and personality characteristics, with mixed results. In individuals who receive treatment for bulimia nervosa, studies have found that rapid symptom reduction predicts better treatment outcomes.

Binge-Eating Disorder

We know little about the course of binge-eating disorder. Severe obesity is a long-term effect in over 3 percent of patients with the disorder. One prospective study of women in the community with binge-eating disorder suggested that by 5 years of follow-up, fewer than one-fifth of the sample still had clinically significant eating disorder symptoms.

Other Specified Feeding and Eating Disorder

Night-Eating Syndrome. The age of onset for night-eating syndrome ranges from the late teens to the late 20s and has a long-lasting course with periods of remission with treatment. Patients who experience poor sleep quality are more likely to develop diabetes, obesity, hypertension, and cardiovascular disease.

TREATMENT APPROACH

Treatment of Anorexia Nervosa

Given the complicated psychological and medical implications of anorexia nervosa, a comprehensive treatment plan, including hospitalization when necessary and both individual and family therapy, is recommended. It is important to consider behavioral, interpersonal, and cognitive approaches. In many cases, medication may also help.

Hospitalization. The first consideration in the treatment of anorexia nervosa is to restore patients' nutritional state; dehydration, starvation, and electrolyte imbalances can seriously compromise health and, in some cases, lead to death. When deciding whether to hospitalize a patient, we should consider the patient's medical condition and the amount of structure needed to ensure cooperation. In general, patients with anorexia nervosa who are 20

percent below the normal weight for their height require inpatient programs, and patients who are 30 percent below their expected weight require psychiatric hospitalization for 2 to 6 months.

Inpatient psychiatric programs for patients with anorexia nervosa generally use a combination of a behavioral management approach, individual psychotherapy, family education and therapy, and, in some cases, psychotropic medications. Staff members should maintain a firm yet supportive approach to patients, often through a combination of positive reinforcers (praise) and negative reinforcers (restriction of exercise). The program must have some flexibility for individualizing treatment to meet patients' needs and cognitive abilities. Patients must become willing participants for treatment to succeed in the long run.

Most patients are uninterested in psychiatric treatment and even resist it; they are brought to a doctor's office unwillingly by agonizing relatives or friends. The patients rarely accept the recommendation of hospitalization without arguing and criticizing the proposed program. Emphasizing the benefits, such as relief of insomnia and depressive signs and symptoms, may help persuade the patients to admit themselves willingly to the hospital. Relatives' support and confidence in the physicians and treatment team are essential for reinforcing the firm recommendations made on the unit. We should warn patients' families that the patients will resist admission and, for the first several weeks of treatment, will make many dramatic pleas for their families' support to obtain release from the hospital program. We should consider compulsory admission or commitment only when the risk of death from the complications of malnutrition is likely. On rare occasions, patients prove that the doctor's statements about the probable failure of outpatient treatment are wrong. They may gain a specified amount of weight by the time of each outpatient visit, but such behavior is uncommon, and a period of inpatient care is usually necessary.

HOSPITAL MANAGEMENT. The following considerations apply to the general management of patients with anorexia nervosa during a hospitalized treatment program. Patients should be weighed daily, early in the morning after emptying the bladder. The staff should record daily fluid intake and urine output. If vomiting is occurring, hospital staff members must monitor serum electrolyte levels regularly and watch for the development of

hypokalemia. Because patients often regurgitate food after meals, the staff may be able to control vomiting by making the bathroom inaccessible for at least 2 hours after meals or by having an attendant in the bathroom to prevent the opportunity for vomiting. Constipation in these patients is relieved when they begin to eat normally. We may occasionally use stool softeners, but never laxatives. If diarrhea occurs, it usually means that patients are surreptitiously taking laxatives. Because of the risk of refeeding syndrome when patients immediately start eating an enormous number of calories, the hospital staff should start patients on a low-caloric intake initially (e.g., 1,000 to 1,400 kcals/day), and increase slowly by approximately 400 kcals every few days. It is wise to give these calories in six equal feedings throughout the day so that patients need not eat a large amount of food at one sitting. Giving patients a liquid food supplement may be advisable, because they may be less apprehensive about gaining weight slowly with formula than by eating food. After discharging patients from the hospital, clinicians usually find it necessary to continue outpatient supervision of the problems identified in the patients and their families.

Psychotherapy

FAMILY-BASED THERAPY. Family-based therapy (FBT) is an effective treatment for anorexia nervosa, particularly in patients under the age of 18. FBT, also known as the Maudsley method, generally consists of three phases of treatment. In phase one, treatment focuses on the restoration of the patient's physical health, with decisions about what or when the patient will eat made by the parents. Once the patient has begun to gain weight and shown improvement in symptoms of anorexia nervosa, FBT moves on to phase two. In this phase, the patient gradually begins to take responsibility for decisions about eating. In phase three, the focus shifts to the patient's growth and development.

COGNITIVE-BEHAVIORAL THERAPY. Cognitive and behavioral therapy principles can be applied in both inpatient and outpatient settings and have been found effective for inducing weight gain. Monitoring is an essential component of cognitive-behavioral therapy (CBT). Therapists teach patients to monitor their food intake, their feelings and emotions, their binging and purging behaviors, and their problems in interpersonal relationships. They also teach

the patients cognitive restructuring to identify automatic thoughts and to challenge their core beliefs. Problem-solving is a specific method whereby patients learn how to think through and devise strategies to cope with their food-related and interpersonal problems. These techniques can help address a patient's vulnerability to rely on anorectic behavior as a means of coping.

DYNAMIC PSYCHOTHERAPY. We sometimes use dynamic expressive-supportive psychotherapy in the treatment of patients with anorexia nervosa, but their resistance may make the process difficult and painstaking. Because patients view their symptoms as constituting the core of their specialness, therapists must avoid excessive investment in trying to change their eating behavior. We should gear the opening phase of the psychotherapy process toward building a therapeutic alliance. Patients may experience early interpretations as though someone else was telling them what they feel and thereby minimizing and invalidating their own experiences. Therapists who empathize with patients' points of view and take an active interest in what their patients think and feel, however, convey to patients that their autonomy is respected. Above all, psychotherapists must be flexible, persistent, and durable in the face of patients' tendencies to defeat any efforts to help them.

Pharmacotherapy. Pharmacologic studies have not yet identified any medication that yields a definitive improvement of the core symptoms of anorexia nervosa. Some reports support the use of atypical antipsychotics, particularly olanzapine, for weight gain, although larger studies or meta-analyses have not supported this. When using atypical antipsychotics, the metabolic and cardiac risks associated with these medications, particularly in a population already at risk for cardiac complications, require close monitoring. Antidepressants, including selective serotonin reuptake inhibitors (SSRIs) and tricyclic antidepressants (TCAs), have been tried by patients with anorexia nervosa with variable results, though typically antidepressants are not helpful while patients are in an undernourished state. In patients with anorexia nervosa and coexisting depressive disorders, we should treat the depressive condition. Concern exists about the use of tricyclic drugs in low-weight, depressed patients with anorexia nervosa, who may be vulnerable to hypotension, cardiac arrhythmia, and dehydration. In some patients, the depression improves with weight gain and normalized nutritional status.

Treatment of Bulimia Nervosa

Most patients with uncomplicated bulimia nervosa do not require hospitalization. In general, patients with bulimia nervosa are not as secretive about their symptoms as patients with anorexia nervosa. Therefore, outpatient treatment is usually not difficult, but psychotherapy is frequently stormy and prolonged. In some cases—when eating binges are out of control, outpatient treatment does not work, or a patient exhibits such additional psychiatric symptoms as suicidality and substance abuse—hospitalization may become necessary. Also, electrolyte and metabolic disturbances resulting from severe purging may necessitate hospitalization.

Psychotherapy

COGNITIVE-BEHAVIORAL THERAPY. CBT should be considered the benchmark, first-line treatment for bulimia nervosa. The data supporting the efficacy of CBT are based on strict adherence to rigorously implemented, highly detailed, manual-guided treatments that include about 18 to 20 sessions over 5 to 6 months. CBT implements several cognitive and behavioral procedures to (1) interrupt the self-maintaining behavioral cycle of binging and dieting and (2) alter the individual's dysfunctional cognitions; beliefs about food, weight, body image; and overall self-concept.

OTHER MODALITIES. Given its effectiveness in anorexia nervosa, clinicians have also used FBT for bulimia nervosa. Additionally, controlled trials have shown that a variety of novel ways of administering and facilitating CBT are effective for bulimia nervosa. These include "stepped-care" programs and internet-based platforms, computer-facilitated programs, email-enhanced programs, and administration of CBT via telemedicine to remote areas. Finally, emerging evidence suggests that dialectical behavior therapy may be effective.

Pharmacotherapy. Antidepressant medications help treat bulimia nervosa, particularly the SSRI fluoxetine. Fluoxetine can reduce binge eating and purging, independent of the presence of a mood disorder. Dosages of fluoxetine that are effective in decreasing binge eating, however, may be higher (60 to 80 mg a day) than those used for depressive disorders. Other

antidepressants that may be helpful include other SSRIs (although concerns about prolonged QT intervals with higher doses limit the utility of citalopram in this population), TCAs (particularly amitriptyline and desipramine), trazodone, and monoamine oxidase inhibitors (MAOIs). Bupropion is contraindicated due to an increased risk of seizure in this population. In general, most of the antidepressants other than fluoxetine have been effective at dosages usually given in the treatment of depressive disorders. Medication is helpful in patients with comorbid depressive disorders and bulimia nervosa. Topiramate may have some efficacy in reducing binge episodes in bulimia nervosa, as may lisdexamfetamine. Evidence indicates that CBT and medications (particularly fluoxetine) are the most effective combination.

Treatment of Binge-Eating Disorder

Psychotherapy. CBT is the most effective psychological treatment for binge-eating disorder and should be considered a first-line treatment. CBT can lead to decreases in binge eating and associated problems (e.g., depression); however, studies have not shown marked weight loss as a result of CBT. CBT combined with psychopharmacological treatments such as SSRIs shows better results than CBT alone. Exercise has also shown a reduction in binge eating when combined with CBT. Interpersonal psychotherapy (IPT) is also effective in the treatment of binge-eating disorder; however, therapy focuses more on the interpersonal problems that contribute to the disorder rather than disturbances in eating behavior. There is also some evidence for the use of dialectical behavior therapy for binge-eating disorder.

Psychopharmacotherapy. Symptoms of binge eating may benefit from medication treatment, with strong evidence supporting the use of lisdexamfetamine for both weight loss and reduction of binge episodes. Antidepressant medications have demonstrated improvement in binge eating, but typically do not result in sustained weight loss; these include fluoxetine, fluvoxamine, citalopram, escitalopram, sertraline, duloxetine, and bupropion. The anticonvulsants topiramate and zonisamide may improve binge-eating disorder, particularly with moderate weight loss. Topiramate may also reduce binge episodes.

Most, but not all, studies show that medication added to CBT is more effective than medication alone. For example, studies indicate that CBT did better than fluvoxamine or desipramine as a monotherapy for binge-eating disorder; however, when CBT was combined with these agents, more improvement was seen in terms of weight loss compared with CBT alone.

Treating of Other Specified Feeding and Eating Disorder

Night-Eating Syndrome. Various studies have shown positive results in patients treated with SSRIs who showed improvement in nighttime awakenings, nocturnal eating, and post-evening caloric intake. Weight loss and a reduction in nocturnal eating are associated with an addition of topiramate to medication regimens.

In patients with comorbid major depression and night-eating syndrome, bright light therapy has shown to decrease depressed mood. CBT has also been helpful.

EPIDEMIOLOGY

Anorexia Nervosa

Although a substantial proportion of those diagnosed with subthreshold anorexia nervosa in DSM-IV now have a formal diagnosis in DSM-5, combined rates of subthreshold and threshold anorexia nervosa have remained relatively stable since the 1970s. Various studies have, however, documented a significant increase in incidence rates (the number of new cases in the population over a set time) in the high-risk group of 15- to 19-year-old females in recent years. It remains unclear if this increase signifies an earlier age of illness onset or more rapid recognition and intervention. Across all age groups and genders, a review of recent epidemiologic studies found that lifetime prevalence of DSM-5 anorexia nervosa is between 2.4 and 4.3 percent, approximately double the rate of cases diagnosed using DSM-IV criteria. Among girls and women, point prevalence estimates of DSM-5 anorexia nervosa is between 0.6 and 0.7 percent.

Although epidemiologic studies indicate that it may occur as many as ten times more frequently in females, anorexia nervosa does affect males. Recent

research suggests that the illness, which continues to be perceived by many as a female disorder, may be even more common in men and boys than previously believed. It may go unrecognized in men for a variety of reasons. In particular, men may be reluctant to seek treatment out of shame, and clinicians may be less likely to recognize the syndrome in male versus female patients. Historically, many consider anorexia nervosa to be a disease of white, wealthy women; however, it impacts individuals of all racial and socioeconomic backgrounds.

Bulimia Nervosa

As with anorexia nervosa, rates of bulimia nervosa have increased with changes to diagnostic criteria. However, before the recent modifications appearing in DSM-5, studies suggested that the incidence of the illness had decreased in recent years. Using DSM-5 criteria, the lifetime prevalence of bulimia nervosa in women is approximately 2 percent, and point prevalence is close to 0.6 percent. The average age of onset also seems to have decreased, although this finding may be an artifact of earlier detection. Like anorexia nervosa, the disorder is more common in women than in men.

Binge-Eating Disorder

Binge-eating disorder is the most common eating disorder and the least gender-divided of the three. In the United States, the lifetime prevalence of binge-eating disorder is about 3.6 percent for women and 2.1 percent for men. Rates of binge-eating disorder are particularly high in obese and overweight individuals.

Other Specified Feeding and Eating Disorder

Night-Eating Syndrome. Night-eating syndrome occurs in approximately 2 percent of the general population; however, it has a higher prevalence among patients with insomnia, obesity (10 to 15 percent), eating disorders, and other psychiatric disorders. The disorder usually begins in early adulthood.

ETIOLOGY

Biologic, social, and psychological factors are important to the cause of feeding and eating disorders. However, the precise mechanisms of causation remain elusive. Current theories posit that individuals with eating disorders, in particular, possess a set of predisposing traits (e.g., biologic, genetic, or personality vulnerability) which, when triggered by a precipitating event (e.g., the stress of puberty, the decision to go on a diet), result in illness. Once the illness sets in, several factors work to maintain it (e.g., the social rewards of weight loss, the effects of the starvation state in anorexia nervosa). A discussion of the current understanding of the etiology of feeding and eating disorders is detailed below.

Anorexia Nervosa

Genetic and Biologic Factors. Although we have not identified a specific gene in anorexia nervosa, several lines of evidence suggest that genetic vulnerability plays an important role in the development of the illness. Family studies, in particular, have provided useful information about genetic susceptibility, demonstrating that individuals with a family history of anorexia nervosa are much more likely than those with no family history to receive a diagnosis during their lifetime. The risk of developing the illness is up to 11 times greater in individuals with a first-degree relative who has experienced the illness. A relative with a history of a different eating disorder also increases this risk. Twin studies have demonstrated that concordance rates of anorexia nervosa are substantially higher in monozygotic twins as compared to dizygotic twins, suggesting that something more than the family environment is at play in increasing the odds of illness development. Studies of the serotonin transporter offer preliminary evidence that this gene may interact with environmental stressors to play a role in the development of anorexia, further emphasizing that a convergence of elements is necessary for illness development. Finally, once the illness sets in, the biologic and psychological changes to the body that occur in the starvation state, including depression and obsessionality, may help to maintain the illness.

Developmental Factors. Anorexia nervosa tends to develop during adolescence, suggesting that factors particular to this period may put individuals at risk for the syndrome. Adolescence represents a time of

increased biologic, psychological, and social change. The experience of going through puberty and experiencing changes to body shape or weight may serve as a major stressor for some, triggering or worsening body dissatisfaction and low self-esteem. Additionally, several major social and psychological transitions occur throughout adolescence, including identity and role formation, increasing independence from parents, and the initiation of romantic relationships. These stressors and others may work to catalyze the eating disorder.

Psychological Factors. Several psychological factors appear to confer added risk for developing anorexia nervosa. Certain personality traits, including high levels of perfectionism, self-discipline, harm avoidance, and self-criticism, are common in individuals with the illness. Individuals with the restricting subtype, in particular, exhibit low impulsivity and are much more likely to delay rewards than individuals without the illness. Cognitive inflexibility is usually prominent, as well. In a subset of individuals, mood and anxiety disorders or symptoms precede the development of the anorexia nervosa. OCD and obsessive-compulsive personality traits also appear to serve as vulnerability factors.

Environmental and Social Factors. Environmental and social factors, small and large, from experiences in the family or school setting to cultural ideals, may play a role in the development of anorexia nervosa. The influence of family functioning style as a potential predisposing factor remains controversial. No specific family functioning style appears to be either a necessary or sufficient requirement for developing an eating disorder. As in most psychiatric disorders, various family dysfunctional styles appear to act as nonspecific vulnerability factors and also hamper recovery.

Activities that emphasize weight may increase the probability of developing anorexia nervosa or another eating disorder. Ballet, gymnastics, modeling, and weight-restricted sports like wrestling or light-weight rowing may lead to preoccupation with body form and unhealthy attempts to control weight. As many of these activities likely select for individuals who are higher in perfectionism, another risk factor for anorexia nervosa, they may be even more likely to contribute to illness onset.

Although it is a common misperception that Western media causes eating

disorders, the cultural ideal of thinness, which is certainly perpetuated by the media, may fuel the overvaluation of shape and weight that marks both anorexia nervosa and bulimia nervosa. Indeed, studies in Fiji, where Western media was not available until the 1990s, showed a significant increase in cases of eating disorders following the introduction of Western television shows to the country. Nonetheless, classic anorexia nervosa remains rare, and research has firmly established that the media alone does not cause eating disorders.

Bulimia Nervosa

Genetic and Biologic Factors. As in anorexia nervosa, genetics likely play a role in the development of bulimia nervosa, as evidenced by twin and family studies. Individuals with a family history of bulimia nervosa, mood disorder, substance use disorder, or obesity are at higher risk for developing the syndrome.

Neurobiologic disturbances are also present in individuals with bulimia nervosa and may increase the probability of binge eating. In particular, individuals with the disorder tend to display delayed gastric emptying, increased stomach capacity, and reduced secretion of cholecystokinin (CCK), a peptide hormone released by the small intestine that helps to signal satiety during food consumption. Together, disruptions to these neurobiologic processes may put individuals at elevated risk for overeating.

Developmental Factors. The same psychological and social stressors of puberty and adolescence that put individuals at risk for anorexia nervosa also confer risk for bulimia nervosa.

Psychological Factors. Unlike individuals with anorexia nervosa, those with bulimia nervosa tend to exhibit high levels of novelty seeking and impulsivity. They also tend to display elevated levels of harm avoidance, negative emotionality, and stress reactivity. Afflicted individuals are more likely than others to experience a comorbid substance use disorder and engage in self-harm, leading to speculation that a subset of individuals with bulimia nervosa may have a propensity toward impulsivity across a range of problematic behaviors. Mood and anxiety disturbances appear to serve as risk

factors for bulimia nervosa, as well.

Just as the starvation state works to maintain anorexia nervosa, the cycle of binge eating and purging often becomes self-sustaining in bulimia nervosa. Individuals with bulimia nervosa tend to restrict their food intake outside of binge episodes, which puts them at increased risk for episodes of overeating. In turn, after experiencing a binge episode and the resulting compensatory behaviors, individuals often renew their commitment to restrictive eating in an attempt to avoid future overeating episodes. Thus, this cycle tends to become self-perpetuating and works to maintain the disorder.

Environmental and Social Factors. Individuals with bulimia nervosa are likely affected by similar environmental and social stressors as those with anorexia nervosa.

Binge-Eating Disorder

We know even less about the etiology of binge-eating disorder than of anorexia or bulimia nervosa. Research thus far has suggested that a history of childhood obesity, mood disorder, and negative family dynamics may put individuals at higher risk for developing the illness. In some individuals, dietary restriction appears to play a role in precipitating episodes of binge eating, as in bulimia nervosa. Many individuals with binge-eating disorder, however, seem to experience binge eating outside of the context of dietary restriction. For these individuals, intense emotions may provoke binge episodes.

Other Specified Feeding and Eating Disorder

Night-Eating Syndrome. We also know little about the cause of night-eating disorder. However, researchers have studied the hormones melatonin, leptin, ghrelin, and cortisol as they relate to this disorder. Night-eating syndrome also appears to run in families.

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14 Elimination Disorders

The developmental milestones of mastering control over bowel and bladder function are complex processes that involve motor and sensory functions, coordinated through frontal lobe activities, and regulated by neurons in the pons and midbrain area. Mastery of bowel and bladder function occurs over several months for the typical toddler. Infants generally void small volumes of urine approximately every hour, commonly stimulated by feeding, and may have incomplete emptying of the bladder. As the infant matures to be a toddler, bladder capacity increases, and between 1 and 3 years of age, cortical inhibitory pathways develop that allow the child to have voluntary control over reflexes that control the bladder muscles. The ability to have muscular control over the bowel occurs even before bladder control for most toddlers, and the assessment of fecal soiling includes determining whether the clinical presentation occurs with or without chronic constipation and overflow soiling. The typical sequence of developing control over bowel and bladder functions is the development of nocturnal fecal continence, diurnal fecal continence, diurnal bladder control, and nocturnal bladder control. Bowel and bladder control develops gradually over time. Toilet training is affected by many factors, such as a child's intellectual capacity and social maturity, cultural determinants, and the psychological interactions between child and parents. The ability to control bowel and bladder functions depends on the maturation of neurobiologic systems, so that children with developmental delays may also display delayed continence of bowel and bladder. When children exhibit incontinence of urine or feces regularly, it is troubling to the children and families, and often misunderstood as voluntary misbehavior.

Encopresis (repeated passage of feces into inappropriate places) and enuresis (repeated urination into bed or clothes) are the two elimination

disorders described in DSM-5. ICD-10 contains the diagnoses of nonorganic encopresis and enuresis, which they list in the category of other behavioral and emotional disorders with onset usually occurring in childhood and adolescence. We should not make these diagnoses until after age 4 years, for encopresis, and after age 5 years for enuresis, the ages at which a typically developing child masters these skills. Healthy development encompasses a range of time in which a given child can devote the attention, motivation, and physiologic skills to exhibit competency in elimination processes.

We discuss each disorder separately in this chapter.

ENCOPRESIS

Clinical and Diagnostic Features

According to both DSM-5 and ICD-10, encopresis occurs when a child passes feces into inappropriate places regularly. Table 14-1 compares the diagnostic approaches for encopresis. Encopresis may be present in children who have bowel control and intentionally deposit feces in their clothes or other places for a variety of emotional reasons. Anecdotal reports have suggested that occasionally encopresis is attributable to an expression of anger or rage in a child whose parents have been punitive or of hostility at a parent. In a case such as this, once a child develops this inappropriate repetitive behavior eliciting negative attention, it is difficult to break the cycle of continuous negative attention. In other children, sporadic episodes of encopresis can occur during times of stress—for example, proximal to the birth of a new sibling—but in such cases, the behavior is usually transient and does not fulfill the diagnostic criteria for the disorder.

Encopresis can also be present on an involuntary basis in the absence of physiologic abnormalities. In these cases, a child may not exhibit adequate control over the sphincter muscles, either because the child is absorbed in another activity or because he or she is unaware of the process. The feces may be of normal, near-normal, or liquid consistency. Some involuntary soiling occurs from chronic retaining of stool, which may result in a liquid overflow. In rare cases, the involuntary overflow of stool results from psychological causes of diarrhea or anxiety disorder symptoms. DSM-5 includes specifiers to indicate whether or not the disorder includes

constipation and overflow incontinence.

Studies have indicated that children with encopresis who do not have gastrointestinal illnesses have high rates of abnormal anal sphincter contractions. This finding is particularly prevalent among children with encopresis with constipation and overflow incontinence who have difficulty relaxing their anal sphincter muscles when trying to defecate. Children with constipation, who have difficulties with sphincter relaxation, are not likely to respond well to laxatives in the treatment of their encopresis. Children with encopresis without abnormal sphincter tone are likely to improve over a short period.

Jack was a 7-year-old boy with daily encopresis, enuresis, and history of hoarding behaviors, along with hiding the feces around the house. He lived with his adoptive parents, having been removed from his biologic parents at the age of 3 years because of neglect and physical abuse. He was reported to be cocaine-addicted at birth but was otherwise healthy. Jack's biologic mother was a known methamphetamine and alcohol user, and his father had spent time in jail for drug dealing. Jack had always been enuretic at night, and until this year, he had a history of daytime enuresis as well. Jack had a short attention span, was highly impulsive, and had great difficulty staying in his seat at school and remaining on task. He had reading difficulties and was in a contained special education classroom because of his disruptive behavior as well as his academic difficulties. Despite experiencing physical abuse, he has not experienced flashbacks or other symptoms that would indicate the presence of posttraumatic stress disorder. Jack was treated for attention-deficit/hyperactivity disorder (ADHD) with a good response to methylphenidate.



Table 14-1
Encopresis

	DSM-5	ICD-10
Name	**Encopresis**	**Nonorganic**

		Encopresis
Duration	≥1/mo for ≥3 mo	
Symptoms	Eliminating feces in clothes or floor Can be voluntary or unintentional	Eliminating feces in inappropriate places Can be voluntary or unintentional
Exclusion	Age <4 y Substance use Another medical condition (not including ones that are causing constipation)	Another medical condition
Severity specifiers	With constipation and overflow incontinence Without constipation and overflow incontinence	

Jack's adoptive family sought help at a university hospital's outpatient program that had expertise in the behavioral treatments of many psychiatric disorders, including encopresis. The treatment program combined the use of regular laxatives and a bowel training method with cognitive-behavioral therapy for Jack and his family. Jack was started on a regimen of daily polyethylene glycol (PEG) solution and was seen by a pediatrician who was able to perform a manual disimpaction under sedation. Following that, Jack was continued on daily PEG solution combined with therapy. He learned to empty his bowel while sitting on the toilet for 10 minutes after each meal, whether or not he felt like he had to go. He soon was eager to stay on this regular bathroom schedule and felt proud when he was able to have a bowel movement in the toilet. Over 3 months, Jack improved, and at 6 months, he was almost entirely better. (Courtesy of Edwin J. Mikkelsen, M.D. and Caroly Pataki, M.D.)

Pathology and Laboratory Examination

Although no specific test indicates a diagnosis of encopresis, clinicians must rule out medical illnesses, such as Hirschsprung disease, before making a diagnosis. The physician should perform an abdominal examination to help determine whether fecal retention is responsible for encopresis with constipation and overflow incontinence, and an abdominal x-ray can help determine the degree of constipation present. Tests to determine whether sphincter tone is abnormal are generally not conducted in uncomplicated cases of encopresis.

Differential Diagnosis

In encopresis with constipation and overflow incontinence, constipation can begin as early as the child's first year and can peak between the second and fourth years. Soiling usually begins by age 4. There are frequent liquid stools and hard fecal masses in the colon and the rectum on abdominal palpation and rectal examination. Complications include impaction, megacolon, and anal fissures.

The chief differential medical problem is aganglionic megacolon or Hirschsprung disease, in which a patient may have an empty rectum and no desire to defecate, but may still have an overflow of feces. The disorder occurs in 1 in 5,000 children; signs appear shortly after birth. Faulty nutrition is rarely the cause of encopresis with constipation and overflow incontinence. Other conditions are rare as well, including structural disease of the anus, rectum, and colon, adverse drug effects, or nongastrointestinal medical (endocrine or neurologic) disorders.

Course and Prognosis

The outcome of encopresis depends on the etiology, the chronicity of the symptoms, and coexisting behavioral problems. In some cases, encopresis is self-limiting, and it rarely continues beyond middle adolescence. Encopresis in children who have associated physiologic factors, such as poor gastric motility and an inability to relax the anal sphincter muscles, is more challenging to treat than that in those with constipation but normal sphincter tone.

Encopresis is a particularly objectionable disorder to family members, who

may assume that the behavior is due to “laziness,” and family tensions are often high. Peers are intolerant of the developmentally inappropriate behavior and typically taunt and reject a child with encopresis. Many affected children have abysmally low self-esteem and feel constant social rejection. Psychologically, a child may appear blunted toward the symptoms or less frequently may be entrenched in a pattern of encopresis as a mode of expressing anger. The outcome of encopresis is influenced by a family’s willingness and ability to participate in treatment without being overly punitive and by the child’s ability and motivation to engage in treatment.

Treatment

A typical treatment plan for a child with encopresis includes daily oral administration of laxatives such as PEG at 1 g/kg per day, and often a surgical disimpaction under general anesthesia before administering maintenance laxatives. Also help is a cognitive-behavioral intervention to help the child begin regular attempts to have bowel movements in the toilet, and to diminish anxiety related to bowel movement. By the time a child presents for treatment, considerable family discord and distress are common. We must reduce family tensions about the symptom and establish a nonpunitive atmosphere. Similarly, we should help to reduce the child’s embarrassment at school. A treatment plan may include arranging for many changes of underwear with a minimum of embarrassment. Education of the family and correction of misperceptions that a family may have about soiling must occur before treatment. Laxatives are not necessary for children who are not constipated and do have reasonable bowel control, but regular timed intervals on the toilet may be useful with these children as well.

A report confirms the success of an interactive parent–child family guidance intervention for young children with encopresis based on psychological and behavioral interventions for children younger than age 9 years.

Supportive psychotherapy and relaxation techniques may be useful in treating the anxieties and other sequelae of children with encopresis, such as low self-esteem and social isolation. Family interventions can be helpful for children who have bowel control but who continue to deposit their feces in inappropriate locations. An optimal outcome occurs when a child achieves a

feeling of control over his or her bowel function.

Epidemiology

Encopresis affects about 3 percent of 4-year-olds and 1.6 percent of 10-year-old children. Incidence rates for encopretic behavior decrease drastically with increasing age. Between the ages of 10 and 12 years, it affects about 0.75 percent of typically developing children. Globally, the community prevalence of encopresis ranges from 0.8 to 7.8 percent. In Western cultures, bowel control is established in more than 95 percent of children by their fourth birthday and 99 percent by the fifth birthday. Encopresis is virtually absent in youth with normal intellectual function by the age of 16 years. Males are from three to six times more likely to have encopresis than females. A significant relationship exists between encopresis and enuresis.

Etiology

Ninety percent of chronic childhood encopresis is likely functional. Children with this disorder typically withhold feces by contracting their gluteal muscles, holding their legs together, and tightening their external anal sphincter. In some cases, this is an entrenched behavioral response to previously painful bowel movements due to hard stool, which leads to fear of defecation and withholding behaviors. Encopresis involves an often-complicated interplay between physiologic and psychological factors leading to an avoidance of defecation. However, when children chronically hold in bowel movements, the result is often fecal impaction and eventual overflow soiling. This pattern occurs in more than 75 percent of children with encopretic behavior. This standard set of circumstances in most children with encopresis supports a behavioral intervention with a focus on ameliorating constipation while increasing appropriate toileting behavior. Inadequate training or the lack of appropriate toilet training may delay a child’s attainment of continence.

Evidence indicates that some encopretic children have lifelong inefficient and ineffective sphincter control. Other children may soil involuntarily, either because of an inability to control the sphincter adequately or because of excessive fluid caused by a retentive overflow.

In about 5 to 10 percent of cases, medical conditions cause fecal

incontinence, including abnormal innervation of the anorectal region, ultrashort-segment Hirschsprung disease, neuronal intestinal dysplasia, or spinal cord damage.

One study found encopresis to occur with significantly higher frequency among children with known sexual abuse and other psychiatric disorders, compared to samples of healthy children. Encopresis, however, is not a specific indicator of sexual abuse.

It is evident that once a child has developed a pattern of withholding bowel movements and attempts to defecate have become painful, the child's fear and resistance to changing the pattern are high. Battles with parents who insist that their untreated child attempts to defecate may aggravate the condition and cause secondary behavioral difficulties. Untreated children with encopresis, however, frequently end up being socially ostracized and rejected. The social consequences of soiling can lead to the development of emotional problems. On the other hand, children with encopresis who clearly can control their bowel function adequately but chronically deposit feces of relatively normal consistency in abnormal places are likely to have pre-existing neurodevelopmental problems. Occasionally, a child has a specific fear of using the toilet, leading to a phobia.

Encopresis, in some cases, can be considered secondary, that is, emerging after a period of healthy bowel habits in conjunction with a disruptive life event, such as the birth of a sibling or a move to a new home. When encopresis manifests after a long period of fecal continence, it may reflect a regressive developmental behavior based on a severe stressor, such as parental separation, loss of a best friend, or unexpected academic failure.

Megacolon. Most children with encopresis retain feces and become constipated, either voluntarily or secondary to painful defecation. In some cases, a subclinical pre-existing anorectal dysfunction exists that contributes to constipation. In either case, the resulting chronic rectal distention from large, hard fecal masses can cause loss of tone in the rectal wall and desensitization to pressure. Thus, children in this situation become even less aware of the need to defecate, and overflow encopresis occurs, usually with relatively small amounts of liquid or soft stool leaking out.

ENURESIS

Clinical and Diagnostic Features

Enuresis is the repeated voiding of urine into a child's clothes or bed; the voiding may be involuntary or intentional. The child must exhibit a developmental or chronologic age of at least 5 years. Children with enuresis are at higher risk for ADHD compared with the general population. They are also more likely to have comorbid encopresis. DSM-5 and the ICD-10 break down the disorder into three types: nocturnal only, diurnal only, and nocturnal and diurnal (Table 14-2).

Pathology and Laboratory Examination

No single laboratory finding is pathognomonic of enuresis, but clinicians must rule out organic factors, such as the presence of urinary tract infections, which may predispose a child to enuresis. Structural obstructive abnormalities may be present in up to 3 percent of children with apparent enuresis. Usually, clinicians defer sophisticated radiographic studies in uncomplicated cases of enuresis with no signs of repeated infections or other medical problems.



Table 14-2
Enuresis

	DSM-5	ICD-10
Name	**Enuresis**	**Nonorganic Enuresis**
Duration	≥2/wk for ≥3 mo	
Symptoms	Urinating into bed or clothes Can be voluntary or unintentional	Involuntary voiding of urine Can be nighttime or daytime
Exclusions (not result of):	Age <5 y Substance use Another	Another medical condition (i.e., structural abnormality, epilepsy, neurologic disorder)

	medical condition
Symptom specifiers	Nocturnal only Diurnal only Nocturnal and Diurnal

Differential Diagnosis

Medical causes of bladder dysfunction must be investigated and ruled out. Urinary tract infections, obstructions, or anatomical conditions are found most often in children who experience both nocturnal and diurnal enuresis combined with urinary frequency and urgency. We should consider various causes of genitourinary pathology. Depending on the signs and symptoms, these can include obstructive uropathy, spina bifida occulta, and cystitis—there are many other causes of polyuria and enuresis, such as diabetes mellitus and diabetes insipidus. Also, we should consider disturbances of consciousness and sleep, such as seizures, intoxication, and sleepwalking disorder, during which a child urinates; and adverse effects from medications.

Course and Prognosis

Enuresis is often self-limited, and a child with enuresis may have a spontaneous remission. Most children who master the task of control over their bladder gain self-esteem and improved social confidence when they become continent. About 80 percent of affected children have never achieved a year-long period of dryness. Enuresis after at least 1 dry year usually begins between the ages of 5 and 8 years; if it occurs much later, especially during adulthood, we should investigate for other causes. Some evidence indicates that the late onset of enuresis in children is more frequently associated with a concomitant psychiatric difficulty than is enuresis without at least 1 dry year. Relapses occur in children with enuresis who are becoming dry spontaneously and in those in treatment. The significant emotional and social difficulties of these children usually include poor self-image, decreased self-esteem, social embarrassment and restriction, and intrafamilial conflict. The course of children with enuresis may be influenced by whether they receive

appropriate evaluation and treatment for common comorbid disorders such as ADHD.

Treatment

A relatively high rate of spontaneous remission of enuresis occurs over time in childhood; however, in many cases, interventions are necessary because enuresis is causing functional impairment. The first step in any treatment plan is to review appropriate toilet training. If the parents have not attempted toilet training with the child, we should guide the parents and the patient in this undertaking. It is helpful to keep records, to determine a baseline, and follow the child's progress; it may also be therapeutic in itself. A star chart may be particularly helpful. Other useful techniques include restricting fluids before bed and night lifting to toilet train the child. Intervention with alarm therapy, which is triggered by wet underwear, has been a mainstay of treatment for enuresis. Alarm therapy works by alerting a child to respond when voiding begins during sleep. The alarm is a battery-operated device that can be attached to a child's underwear or a mat. The alarm is triggered as soon as voiding begins by emitting a loud noise that awakens the child. The success of this method depends on the child's ability to awaken promptly and respond to the alarm by getting up and voiding in the toilet. A child who can respond optimally is at least 6 or 7 years old.

Another simple intervention for children with enuresis and bowel dysfunction is to assess whether chronic constipation is contributing to urinary dysfunction and to consider increasing dietary fiber to diminish constipation.

Behavioral Therapy. Classical conditioning with the bell (or buzzer) and pad (alarm) apparatus is generally the most effective treatment for enuresis, with dryness resulting in more than 50 percent of cases. Bladder training—encouragement or reward for delaying micturition for increasing times during waking hours—has also been used. Although sometimes useful, this method is decidedly inferior to the bell and pad.

Pharmacotherapy. We can consider medication when enuresis is causing social, family or functional impairment, and when dietary and behavioral interventions are not sufficient. When the problem interferes significantly

with a child's functioning, we can consider several medications, although the problem often recurs when we discontinue the medications.

Desmopressin, an antidiuretic compound that is available as an intranasal spray, has shown success in reducing enuresis. The reduction of enuresis has varied from 10 to 90 percent with the use of desmopressin. In most studies, enuresis recurred shortly after discontinuation of this medication. Adverse effects that can occur with desmopressin include headache, nasal congestion, epistaxis, and stomachache. The most severe adverse effect reported with the use of desmopressin to treat enuresis was a hyponatremic seizure experienced by a child.

Antidepressants, particularly imipramine, were the first medications used for enuresis, and are still used at times. In this case, we are taking advantage of the anticholinergic side effects of the medication. However, it does have a considerable side effect profile. Reboxetine (not available in the United States), a norepinephrine reuptake inhibitor that has a noncardiotoxic side effect profile, may be a safer alternative to the classically used imipramine in the treatment of childhood enuresis.

Psychotherapy. Psychotherapy may be useful in dealing with coexisting psychiatric problems and the emotional and family difficulties that arise secondary to chronic enuresis. However, the behavioral therapy approaches described above are the treatments of choice.

Epidemiology

The prevalence of enuresis ranges from 5 to 10 percent in 5-year-olds. It is about 1.5 to 5 percent in 9- to 10-year-olds, and about 1 percent in adolescents 15 years and older. The prevalence of enuresis decreases with increasing age. Enuretic behavior is considered developmentally appropriate among young toddlers, precluding diagnoses of enuresis; however, enuretic behavior occurs in 82 percent of 2-year-olds, 49 percent of 3-year-olds, and 26 percent of 4-year-olds regularly.

Although most children with enuresis do not have a comorbid psychiatric disorder, children with enuresis are at higher risk for the development of another psychiatric disorder.

Nocturnal enuresis is about 50 percent more common in boys and accounts

for about 80 percent of children with enuresis. Diurnal enuresis is also seen more often in boys who often delay voiding until it is too late. A spontaneous resolution of nocturnal enuresis is about 15 percent per year. Nocturnal enuresis consists of an average volume of voided urine, whereas when a child voids small volumes of urine at night, other medical causes may be present.

Etiology

Enuresis involves complex neurobiologic systems that include contributions from cerebral and spinal cord centers, motor and sensory functions, and autonomic and voluntary nervous systems. Neurons regulate urination in the pons and midbrain regions. Bladder detrusor muscle contraction occurs whenever bladder capacity is reached, which can lead to enuresis in a sleeping child. Therefore, excessive volumes of urine produced at night may lead to enuresis at night in children without any physiologic abnormalities. Nighttime enuresis often occurs in the absence of a specific neurogenic cause. Daytime enuresis may develop based on behavioral habits developed over time.

Daytime enuresis may occur in the absence of neurologic abnormalities resulting from habitual, voluntary tightening of the external sphincter during urges to urinate. The pattern may become set in a young child who may start with a normal or overactive detrusor muscle in the bladder, but with repeated attempts to prevent leaking or urination when there is an urge to void. Over time, the sensation of the urge to urinate diminishes, and the bladder does not empty regularly, leading to enuresis at night when the bladder is relaxed and can empty without resistance. This immature pattern of urinating can account for some cases of enuresis, especially when the pattern has been in place since early childhood. Most children are not enuretic by intention or even with awareness until after they are wet. Physiologic factors often play a role in the development of enuresis, and behavioral patterns are likely to maintain the maladaptive urination. Healthy bladder control, which is acquired gradually, is influenced by neuromuscular and cognitive development, socioemotional factors, toilet training, and genetic factors. Difficulties in one or more of these areas can delay urinary continence.

Genetic factors play a role in the expression of enuresis, given that the emergence of enuresis is significantly higher in first-degree relatives. A

longitudinal study of child development found that children with enuresis were about twice as likely to have concomitant developmental delays as those who did not have enuresis. About 75 percent of children with enuresis have a first-degree relative who has or has had enuresis. A child's risk for enuresis is more than seven times higher if the father was enuretic. The concordance rate is higher in monozygotic twins than in dizygotic twins. There is a strong genetic component, and much can be accounted for by tolerance for enuresis in some families and by other psychosocial factors.

Studies indicate that children with enuresis with a normal anatomical bladder capacity report urge to void with less urine in the bladder than children without enuresis. Other studies report that nocturnal enuresis occurs when the bladder is full because of lower than expected levels of nighttime antidiuretic hormone. This low level of the hormone could lead to higher-than-usual urine output. Enuresis does not appear to be related to a specific stage of sleep or time of night; instead, bedwetting appears randomly. In most cases, the quality of sleep is average. Little evidence indicates that children with enuresis sleep more soundly than other children.

Psychosocial stressors appear to precipitate enuresis in a subgroup of children with the disorder. In young children, the disorder may relate to the birth of a sibling, hospitalization, the start of school, separation of a family due to divorce, or a move to a new environment.

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15 Sleep–Wake Disorders

To understand sleep and its disorders, it helps to begin with sleep's three essential characteristics: (1) Sleep is a process required for proper brain function. Failure to sleep impairs thought processes, mood regulation, and a host of normal physiologic functions. (2) Sleep is not a single process; there are several distinct types of sleep. These different types of sleep differ both qualitatively and quantitatively. Each type of sleep has unique characteristics, functional importance, and regulatory mechanisms. Selectively depriving one particular type of sleep produces compensatory rebound when an individual is allowed to sleep ad lib. (3) Sleep is not a passive process; during sleep, there is a high degree of brain activation and metabolism.

The science of sleep is a fascinating ever-growing topic, that is, perhaps unfortunately, beyond the repertoire of the average psychiatrist, and we refer the interested reader to the Comprehensive Textbook's discussion of the basic science of sleep, discussed about the neural sciences.

Several basic mechanisms regulate sleep, and when these systems go awry, sleep disorders occur. Sleep disorders are both dangerous and expensive. Hypersomnolence is a serious, potentially life-threatening condition affecting not only the sleepy individual but also their family, coworkers, and society in general. Sleep-related motor vehicle accidents represent a significant public safety concern, with some states enacting criminal statutes to deter sleepy driving. Investigations link many major industrial catastrophes to sleepiness. Research shows that sleep-disordered breathing contributes to hypertension, heart failure, and stroke. Sleep disorders' direct cost per annum in the United States is about $16 billion, with indirect costs ranging upward to more than $100 billion.

THE CLINICAL PRESENTATION

This chapter's disorders all describe disturbances in healthy sleep. Among them, however, there is considerable variation.

It is helpful to understand the stages of sleep and electrophysiologic criteria. Table 15-1 and Figure 15-1 illustrate some of the features of normal sleep.

Insomnia Disorder

Persons with insomnia primarily have difficulty falling asleep, difficulty staying asleep, or trouble waking early with an inability to fall back to sleep, sufficient to impair their functioning. In children, this may manifest as resistance to caregiver designation of bedtime or difficulty sleeping without some sort of intervention from the caregiver.

Hypersomnolence Disorder

Hypersomnolence broadly refers to excessive sleepiness and time sleeping. During the day, people are drowsy and have reduced attention. Excessive sleepiness can be a serious, debilitating, potentially life-threatening condition. It affects not only the patient but their family, colleagues, and the public. Not due to disrupted sleep or circadian problems, hypersomnolence disorder likely results from some fundamental neurologic sleep regulation dysfunction.

Narcolepsy

People with narcolepsy have an overwhelming desire to sleep and may suddenly fall asleep, even if it is not appropriate to do so. Also, they may experience cataplexy, or the sudden loss of muscle tone, usually with continuing full or partial consciousness. Laughter, or other strong emotions, commonly precipitate the cataplexy. Cataplexy can range from transient weakness in the knees to total paralysis while the patient is fully conscious. Episodes may last from several seconds to minutes. Usually, the patient is unable to speak and may fall to the floor. Sleep paralysis and hypnagogic (or hypnopompic) hallucinations may occur.

Sleep-Related Breathing Disorders

People who present with sleep-related breathing disorders experience an interruption of normal respiration that impacts their sleep, often resulting in CNS arousal and, consequently, daytime sleepiness. Sleep-related breathing disorders include conditions involving large airway obstruction during sleep, breathing cessation resulting from central respiratory mechanisms, and hypoventilation without breathing cessation. Those affected may experience sleep-related breathing cessation (sleep apnea), in which the patient stops breathing for 10 seconds or more during sleep. They may instead have reduced breathing or *hypopnea.* Airway obstruction is the usual case of these impairments, however central (brainstem) pathology can also be the cause. When asleep, breathing cessations or reductions typically provoke CNS arousal, significant oxyhemoglobin desaturation, or both. Sleep apnea can *obstructive, central,* or *mixed,* depending on the cause.

The clinical presentation of the sleep-related breathing disorders varies depending on the mechanism of the disorder. Table 15-2 lists some of these symptoms.

Circadian Rhythm Sleep–Wake Disorders

Circadian rhythm sleep disorders include a wide range of conditions involving a misalignment between desired and actual sleep periods. This collection of sleep disorders shares the same primary underlying etiology—a desynchrony between an individual's internal circadian biologic clock and the desired or conventional sleep–wake cycle. The circadian pacemaker is in the suprachiasmatic nucleus (SCN). SCN firing oscillates with an almost sinusoidal pattern, the period of which is 24 hours, and the output correlates with the daily fluctuations in core body temperature. Mismatched circadian clock and desired schedules can arise from improper phase relationships between the two, travel across time zones, or dysfunctions in the basic biologic rhythm. Under normal circumstances, the internal circadian pacemaker is reset each day by bright light, social cues, stimulants, and activity. In cases in which these factors fail to reentrain the circadian rhythm, the circadian sleep disorders occur.

Table 15-1



Stages of Sleep—Electrophysiologic Criteria

	Electroencephalogram	Electrooculogram	Electrom
Wakefulness	Low-voltage, mixed frequency activity; α (8–13 cps) activity with eyes closed	Eye movements and eye blinks	High toni and vo mover
Non–rapid eye movement sleep			
Stage 1	Low-voltage, mixed frequency activity; θ (3–7 cps) activity, vertex sharp waves	Slow eye movements	Tonic act slightl decrea from wakef
Stage 2	Low-voltage, mixed frequency background with sleep spindles (12–14 cps bursts) and K complexes (negative sharp wave followed by positive slow wave)	None	Low tonic
Stage 3	High-amplitude (≥75 μV) slow waves (≤2 cps) occupying 20–50% of epoch	None	Low tonic
Stage 4	High-amplitude slow waves occupy >50% of epoch	None	Low tonic

REM sleep	Low-voltage, mixed frequency activity; saw-tooth waves, θ activity, and slow α activity	REMs	Tonic ato phasic twitch

cps, cycles per second; REM, rapid eye movement.

Criteria from Rechtschaffen A, Kales A. *A Manual of Standardized Terminology, Techniques, and Scoring System for Sleep Stages of Human Subjects*. Los Angeles, CA: Brain Information Service/UCLA Brain Research Institute; 1968, with permission.

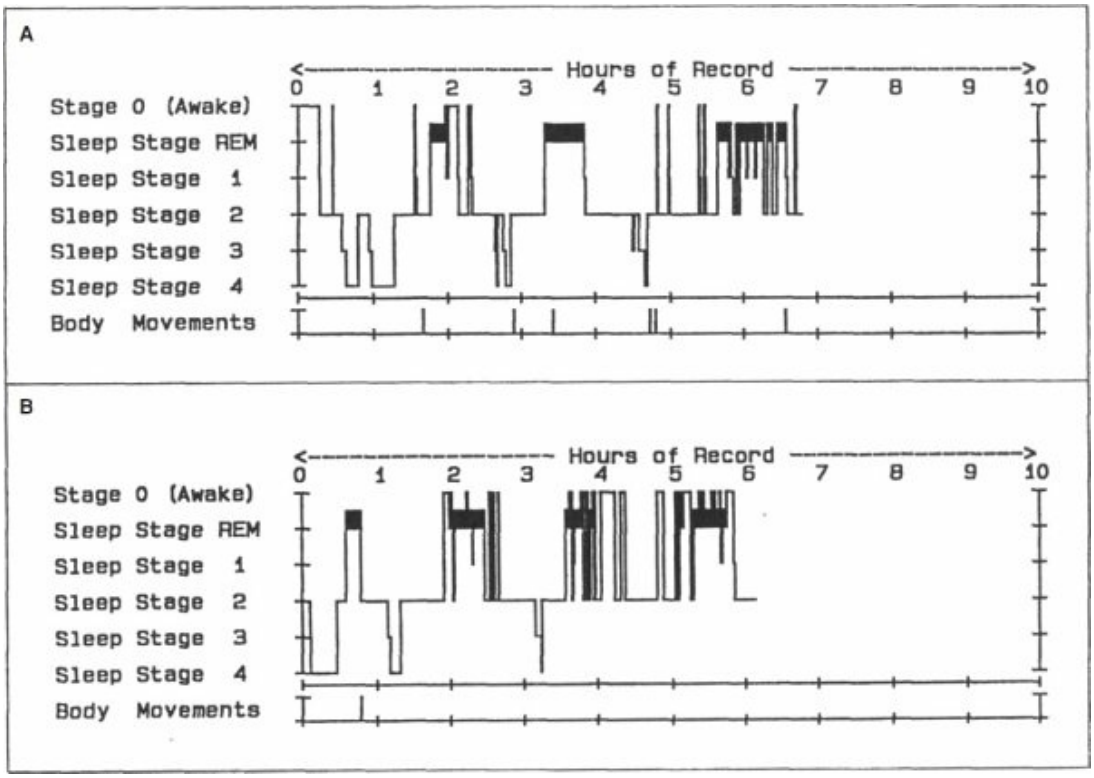


FIGURE 15-1

Sleep histograms showing normal sleep and insomnia. Sleep stage histograms comparing normal sleep (**Panel A**) with that found in a patient with major depressive disorder (**Panel B**). Difficulty maintaining sleep and early morning awakenings are common complaints in patients with depression. **Panel B** illustrates the electrophysiologic correlates of these complaints beginning, in this case, after approximately 2 hours of sleep. Sleep continuity becomes disrupted as morning approaches. Also present is a markedly reduced latency to REM sleep. This sleep feature is characteristic of this patient population and is thought by some to reflect cholinergic–aminergic imbalance.



Table 15-2

Sleep Apnea Symptoms Based on Cause

Obstructive Sleep Apnea
Excessive sleepiness
Snoring
Obesity
Restless sleep
Nocturnal awakenings with choking or gasping for breath
Morning dry mouth
Morning headaches
Heavy nocturnal sweating
Central Sleep Apnea
Breathing cessations unrelated to airway flow limitations
Insomnia
Daytime sleepiness
Morning headaches

Parasomnias and Sleep-Related Movement Disorders

Parasomnias are disorders of partial arousal. In general, the parasomnias are a diverse collection of sleep disorders characterized by physiologic or behavioral phenomena that occur during or are potentiated by sleep. One conceptual framework views many parasomnias as overlaps or intrusions of one sleep–wake state into another. Usually, we divide sleep into three basic states: wakefulness, non–rapid eye movement (NREM) sleep, and REM. Each has a unique neurologic organization. During wakefulness, both the body and brain are active. In NREM sleep, both the body and brain are much less active. REM sleep, however, pairs an atonic body with an active brain (capable of creating elaborate dream fantasies). Regional cerebral blood flow, magnetic resonance imaging (MRI), and other imaging studies confirm increased brain activation during REM sleep. It certainly appears that in some parasomnias, there are state boundary violations. For example, arousal disorders (sleepwalking and sleep terrors) involve momentary or partial wakeful behaviors suddenly occurring in NREM sleep. Similarly, isolated sleep paralysis is the persistence of REM sleep atonia into the wakefulness

transition, whereas REM sleep behavior disorder (RBD) is the failure of the mechanism creating paralytic atonia such that individuals act out their dreams.

There are many sleep-related movement disorders. These disorders typically involve relatively simple bodily movements that impact sleep. Table 15-3 lists examples of these disorders.

DIAGNOSIS

Three different nosologies provide classification systems for sleep disorders: (1) the Diagnostic and Statistical Manual of Mental Disorders, fifth edition (DSM-5), (2) the International Classification of Sleep Disorders, third edition (ICSD-3), and (3) the International Classification of Diseases, 10th edition (ICD-10). Most general psychiatrists use the DSM-5 approach. However, Sleep Medicine specialists often prefer the ICSD-3. We have included a listing of ICSD sleep–wake disorders for reference (Table 15-4).



Table 15-3
Sleep-Related Movement Disorders

Restless legs syndrome
Periodic limb movement disorder
Sleep-related leg cramps
Sleep-related bruxism
Sleep-related rhythmic movement disorder
Benign sleep myoclonus of infancy
Propriospinal myoclonus at sleep onset
Sleep-related movement disorders due to:
Medical conditions
Medications
Substances

Insomnia Disorder

The primary diagnostic criteria descriptively relate to how insomnia affects sleep. To be diagnosed with insomnia, the person must have difficulty falling

asleep, staying asleep, or waking early with difficulty getting back to sleep. While there are no formal criteria for diagnostic subtypes of insomnia, the ICSD-3 does describe each subtype briefly. We describe the insomnia subtypes here to highlight possible insomnia nuances and complications. We present a comparison of the different approaches to diagnosing insomnia in Table 15-5.

Psychophysiologic Insomnia. Psychophysiologic insomnia (PPI) involves conditioned arousal with the thought of sleeping. Objects related to sleep (e.g., the bed, the bedroom) likewise have become conditioned stimuli that evoke insomnia. Daytime adaptation is usually good; however, there can be extreme tiredness, and the affected person can become desperate. PPI often occurs in combination with stress and anxiety disorders, delayed sleep phase syndrome, and hypnotic use and withdrawal. Table 15-6 lists some typical characteristics of PPI.

DESCRIPTION

Ms. W is a 41-year-old, divorced white female presenting with a two- and one-half year complaint of sleeplessness. She has some difficulty falling asleep (30- to 45-minute sleep-onset latency) and awakens every hour or two after sleep onset. These awakenings may last 15 minutes to several hours, and she estimates approximately 4.5 hours of sleep on an average night. She rarely takes daytime naps notwithstanding feeling tired and edgy. The patient describes her sleep problem with the following words, “It seems like I never get into a deep sleep. I have never been a heavy sleeper, but now the slightest noise wakes me up. Sometimes I have a hard time getting my mind to shut down.” She views the bedroom as an unpleasant place of sleeplessness and states, “I tried staying at a friend’s house where it is quiet, but then I couldn’t sleep because of the silence.”

At times, Ms. W is unsure whether she is asleep or awake. She has a history of clock watching (to time her wakefulness) but stopped doing this when she realized it was contributing to the problem. Reportedly the insomnia is unrelated to seasonal changes, menstrual cycle, or time-zone translocation. Her basic sleep hygiene is good. Appetite and libido are unchanged. She denies mood disturbance, except that she is quite

frustrated and concerned about sleeplessness and its effect on her work. Her work involves sitting at a microscope for 6 hours out of a 9-hour working day and meticulously documenting her findings. Her final output has not suffered, but she must now "double check" for accuracy.

She describes herself as a worrier and a Type-A personality. The patient does not know how to relax. For example, on vacation she continually worries about things that can go wrong. She will not even begin to unwind until she has arrived at the destination, checked in, and unpacked. Even then, she is unable to relax.

Medical history is unremarkable except for tonsillectomy (age 16 years), migraine headaches (current), and diet-controlled hypercholesterolemia. She takes naproxen as needed for headache. She does not currently drink caffeinated beverages, smoke tobacco, or drink alcoholic beverages. She does not use recreational drugs.

The problem with insomnia began after relocation to a new city and place of employment. She attributes her insomnia to the noisy neighborhood in which she now lives. She first sought treatment 18 months ago. Her family practice physician diagnosed depression, and she was started on fluoxetine that made her "climb the walls." Antihistamines were tried next with similar results. She was then switched to low-dose trazodone (for sleep) and developed nausea. After these medical interventions, she sought medical care elsewhere. Zolpidem 5 mg was prescribed, but it made her feel drugged and upon discontinuation she had withdrawal effects. Another family practice physician diagnosed "nonspecific anxiety disorder" and began buspirone; an experience she describes as "having an alien try to climb out of my skin." Buspirone was discontinued. Paroxetine was tried for 8 weeks with no effect. Finally, a psychiatrist was consulted, who diagnosed adult attention-deficit disorder (without hyperactivity) and suggested treatment with methylphenidate. At this point, the patient was convinced that a stimulant would not help her insomnia and demanded referral to a sleep disorders center.

DISCUSSION

Ms. W's symptoms fall into the broad category of an insomnia, and the symptoms began after having moved from one city to another. Environmental sleep disorder (noise) and adjustment sleep disorder (new

job, city, and apartment) are likely initial diagnoses. However, a more chronic, endogenous problem has become operative. What is it? Ms. W is a "worrier" and meticulous, but she doesn't reach diagnostic criteria for personality or anxiety disorders. Dyssomnia associated with mood disorder should be considered in any patient with sleep maintenance problems and early morning awakening insomnia. However, this patient does not have other significant signs of depression. Unfortunately, many patients are misdiagnosed with depression or "masked depression" on the sole basis of an insomnia complaint and unsuccessfully treated with antidepressant medication. Ms. W's job demands long hours with focused concentration. Her job performance has been superior for many years notwithstanding insomnia. Thus, a diagnosis of attention-deficit disorder is unlikely. Idiopathic insomnia implies a childhood complaint, which Ms. W denies.

The likely working diagnosis is PPI. There may be some sleep state misperception (sometimes unclear of whether she is awake or asleep), but this cannot adequately account for the constellation of symptoms. An initial treatment plan should include further documentation of the sleep pattern using a sleep log. Behavioral treatments will likely benefit this patient. Medications with sedative effects are sometimes useful during initial treatment of PPI. However, thus far in this patient they have done more harm than good. She is likely to be a challenging patient to treat.



Table 15-4
Outline of the International Classification of Sleep Disorders, Third Edition

I. Insomnia
- A. Chronic Insomnia Disorder
- B. Short-Term Insomnia Disorder
- C. Other Insomnia Disorder
- D. Isolated Symptoms and Normal Variants
  - 1. Excessive Time in Bed
  - 2. Short Sleeper

II. Sleep-Related Breathing Disorders

A. Obstructive Sleep Apnea Disorders
1. Obstructive Sleep Apnea, Adult
2. Obstructive Sleep Apnea Disorder, Pediatric

B. Central Sleep Apnea Syndromes
1. Central Sleep Apnea with Cheyne–Stokes Breathing
2. Central Apnea due to a Medical Disorder without Cheyne–Stokes Breathing
3. Central Sleep Apnea due to High-Altitude Periodic Breathing
4. Central Sleep Apnea due to a Medication or Substance
5. Primary Central Sleep Apnea
6. Primary Central Sleep Apnea of Infancy
7. Primary Central Sleep Apnea of Prematurity
8. Treatment-Emergent Central Sleep Apnea

C. Sleep-Related Hypoventilation Disorders
1. Obesity Hypoventilation Syndrome
2. Congenital Central Alveolar Hypoventilation Syndrome
3. Late-Onset Central Hypoventilation and Hypothalamic Dysfunction
4. Idiopathic Central Alveolar Hypoventilation
5. Sleep-Related Hypoventilation due to a Medication or Substance
6. Sleep-Related Hypoventilation due to a Medical Disorder

D. Sleep-Related Hypoxemia Disorder
1. Sleep-Related Hypoxemia

E. Isolated Symptoms and Normal Variants
1. Snoring
2. Catathrenia

III. Central Disorders of Hypersomnolence
A. Narcolepsy type 1
B. Narcolepsy type 2
C. Idiopathic Hypersomnia
D. Kleine–Levin Syndrome
E. Hypersomnia due to a Medical Disorder
F. Hypersomnia due to a Medication or Substance
G. Hypersomnia Associated with a Psychiatric Disorder
H. Insufficient Sleep Syndrome

I. Isolated Symptoms and Normal Variants
1. Long Sleeper

IV. Circadian Rhythm Sleep–Wake Disorders
A. Delayed Sleep–Wake Phase Disorder
B. Advanced Sleep–Wake Phase Disorder
C. Irregular Sleep–Wake Rhythm Disorder
D. Non–24-h Sleep–Wake Rhythm Disorder
E. Shift Work Disorder
F. Jet Lag Disorder
G. Circadian Sleep–Wake Disorder Not Otherwise Specified

V. Parasomnias
A. NREM-Related Parasomnias
1. Confusional Arousals
2. Sleepwalking
3. Sleep Terrors
4. Sleep-Related Eating Disorder

B. REM-Related Parasomnias
1. REM Sleep Behavior Disorder
2. Recurrent Isolated Sleep Paralysis
3. Nightmare Disorder

C. Other Parasomnias
1. Exploding Head Syndrome
2. Sleep-Related Hallucinations
3. Sleep Enuresis
4. Parasomnia due to Medical Disorder
5. Parasomnia due to a Medication or Substance
6. Parasomnia, Unspecified
7. Isolated Symptoms and Normal Variants

a. Sleep Talking

VI. Sleep-Related Movement Disorders
A. Restless Legs Syndrome
B. Periodic Limb Movement Disorder
C. Sleep-Related Leg Cramps
D. Sleep-Related Bruxism
E. Sleep-Related Rhythmic Movement Disorder
F. Benign Sleep Myoclonus of Infancy

G. Propriospinal Myoclonus at Sleep Onset
H. Sleep-Related Movement Disorder due to Medical Disorder
I. Sleep-Related Movement Disorder due to a Medication or Substance
J. Sleep-Related Movement Disorder, Unspecified
K. Isolated Symptoms and Normal Variants
1. Excessive Fragmentary Myoclonus
2. Hypnagogic Foot Tremor and Alternating Leg Muscle Activation
3. Sleep Starts (Hypnic Jerks)

VII. Other Sleep Disorders



Table 15-5
Insomnia Disorder

	DSM-5	ICSD-3	IC
Name	Insomnia disorder	Insomnia disorder	D
Duration	Symptoms occur at least 3 nights/wk, lasting at least 3 mo	Occurs at least 3×/wk	
Symptoms	Dissatisfaction with quality of sleep due to either: • Difficulty falling asleep • Difficulty	1. Problem with sleep initiation or maintenance 2. Symptoms occurring despite adequate opportunities for sleep 3. Symptoms resulting in daytime consequences due to poor sleep	D

	maintaining sleep • Early morning awakenings		
# Symptoms needed	Any of the above	All the above three symptoms	
Exclusion (not the result of):	Medical disorder A medication or substance Another sleep–wake disorder Insufficient opportunity for sleep		N N S S
Psychosocial impact	Causes significant distress and/or impairment in functioning	Required presence of significant daytime impairment/consequences due to impaired sleep	
Symptom specifiers	• **With non-sleep disorder mental comorbidity** (including substance use disorder) • **With other medical comorbidity** • **With other sleep disorder**		

Severity specifiers		
Course specifiers	**Episodic** (lasting 1–3 mo) **Persistent** (lasting ≥3 mo) **Recurrent** (two or more episodes occurring within 1 y)	**Short-term insomnia disorder:** <3 mo **Chronic insomnia disorder:** ≥3 mo
Comments		This is often occurring in the setting of an acute stressor

Idiopathic Insomnia. Idiopathic insomnia characterizes patients with a lifelong inability to obtain adequate sleep. The insomnia predates any psychiatric condition, and other etiologies must be ruled out or treated, including PPI, environmental sleep disturbances, and poor sleep hygiene.

Paradoxical Insomnia. Paradoxical insomnia, at its core, involves a dissociation between sleep and its usual attendant unconsciousness. In paradoxical insomnia, a person thinks he or she is awake and having insomnia even though the brain electrophysiologic activity pattern is consistent with the correlates of healthy sleep. We should consider this disorder when a patient complains of difficulty initiating or maintaining sleep without any objective evidence of sleep disruption. Paradoxical insomnia can occur in individuals who are free from psychopathology; however, it may represent a somatic delusion or hypochondriasis. Some patients with paradoxical insomnia have obsessional features regarding bodily functions.



Table 15-6
Some Characteristics of Psychophysiologic Insomnia

1. Excessive worry about not being able to sleep

2. Trying too hard to sleep
3. Rumination—inability to clear one's mind while trying to sleep
4. Increased muscle tension when getting into bed
5. Other somatic manifestations of anxiety
6. Ability to fall asleep when not trying (e.g., while watching television)
7. Sleeping better away from the person's own bedroom

Inadequate Sleep Hygiene. Inadequate sleep hygiene refers to insomnia produced by behaviors that are not conducive to good sleep. Many behaviors can interfere with sleep. Some of these behaviors increase arousal, for example, consuming caffeine or nicotine at night or engaging in excessive emotional or physical stimulation within a few hours of bedtime. Other behaviors interfere with sleep architecture, including daytime naps and a significant variation of the daily sleep–wake schedule.

Behavioral Insomnia of Childhood. Children with this subtype of insomnia depend on specific stimulation, objects, or setting for initiating or returning to sleep. Without the presence of, for example, a stuffed animal or a parent, the child has trouble falling asleep. Alternatively, without adequate limit-setting by the caregiver, bedtime stalling ("Dad, I'm thirsty—can I get some water?") or bedtime refusal ("I'm not tired! I don't want to go to sleep!") can ensue.

Insomnia Comorbid with Mental Disorder. This type of insomnia is the most common. Sleep disorder centers report that 35 percent of patients seen with insomnia have a mental disorder. Of these, major depressive disorder (MDD) is the most common disorder. Other common disorders include bipolar disorder, schizophrenia, and generalized anxiety disorder.

Insomnia Comorbid with Medical Condition. Insomnia accompanies many medical and neurologic conditions. Given pain's potential for disturbing sleep, all medical conditions producing pain can (and usually do) disturb sleep. Unfortunately, a synergy exists between pain and sleep, such that poor sleep lowers the pain threshold. This vicious cycle can present a difficult treatment challenge. However, reducing pain can also improve sleep,

and improving sleep can reduce pain. In other medical conditions, sleep disturbance appears to be secondary. For example, patients with sleep-related gastroesophageal reflux disease (GERD) often have insomnia. Treating the reflux results in sleep improvement but insomnia treatments seldom, if ever, relieve nocturnal GERD. Patients with chronic obstructive pulmonary disease (COPD) commonly suffer from both sleep-onset and sleep-maintenance insomnia. Neurodegenerative disorders are also frequently associated with sleep disorders.

Insomnia due to Drug or Substance. Many prescription drugs, even when taken properly, can disturb sleep. We list some common examples in Table 15-7.

Alcohol and hypnotic use initially promote sleep onset because of their sedating properties. A problem occurs when sleep quality is adversely affected, tolerance develops after chronic use, or withdrawal begins. Alcohol may relax a tense person and thereby decrease latency to sleep; however, sleep later in the night will usually be fragmented by arousals. As tolerance develops to the alcohol, higher amounts or more frequent dosing are needed to sustain the effects. Furthermore, during withdrawal or after tolerance develops, insomnia may rebound to a level more severe than the initial disturbance.



Table 15-7
Medications and Drugs Causing Insomnia

Antiparkinsonian drugs
Decongestants (e.g., pseudoephedrine)
Anorectics
Stimulants
Antiepileptic medications
SSRI antidepressants
Caffeine
Alcohol (rebound insomnia)
Sedatives (rebound insomnia)

Caffeine (the active ingredient in coffee) and theobromine (the active ingredient in chocolate) are methylxanthines and act as psychostimulants in the central nervous system (CNS). Psychostimulants increase sleep latency, reduce sleep efficiency, and decrease total sleep time. Caffeine's half-life is 3 to 7 hours and may interfere with sleep when consumed in large quantities throughout the day or even in smaller portions closer to bedtime. Some individuals are hypersensitive to methylxanthines, and any coffee or chocolate can trigger difficulty falling asleep or awakening after a couple of hours of sleep with difficulty getting back to sleep.

Finally, abuse of illicit substances, particularly stimulants (such as cocaine and amphetamines), interfere with sleep onset and sleep maintenance. Unlike with alcohol, discontinuation of these substances will cause hypersomnolence.

Hypersomnolence Disorder

DSM-5 includes hypersomnolence disorder as a discreet diagnosis, whereas ICSD and ICD consider it more broadly. ICSD refers to it as "idiopathic hypersomnia" and questions whether it is a single disorder or rather a group of disorders with different underlying causes. Table 15-8 compares the different approaches to diagnosing hypersomnolence.

In general, we should consider this disorder when a patient complains of frequently feeling sleepy despite getting adequate sleep. They may nap during the day. Despite getting adequate sleep, they do not feel refreshed when waking.

Kleine–Levin Syndrome. Kleine–Levin syndrome is a relatively rare condition consisting of recurrent periods of prolonged sleep (from which they are arousable) with intervening periods of healthy sleep and alert waking. During the episodes of hypersomnia, wakeful periods are usually marked by withdrawal from social contacts and return to bed at the first opportunity. Kleine–Levin syndrome is the best-recognized recurrent hypersomnia though it is uncommon. It predominantly afflicts males in early adolescence; however, it can also affect females and older people. With few exceptions, the first attack occurs between the ages of 10 and 21 years. However, there are some reports of onset in the fourth and fifth decades of life. In its classic

form, the recurrent episodes include extreme sleepiness (18- to 20-hour sleep periods), voracious eating, hypersexuality, and disinhibition (e.g., aggression). Episodes typically last for a few days up to several weeks and appear once to 10 times per year. A monosymptomatic hypersomnolent form can occur. The disorder is usually sporadic but familial cases are reported.

Narcolepsy

Table 15-9 lists the criteria for narcolepsy. People with narcolepsy have irresistible sleep episodes, lapses into sleep, or nap frequently.

The discovery that narcolepsy is strongly associated with a hypocretin (orexin) deficit radically changed diagnostic practice. Cataplexy had been considered a core feature of the disorder, but we now know that there are variants that occur without cataplexy.

Sleep-Related Breathing Disorders

DSM-5 includes three disorders under the category of sleep-related breathing disorders: obstructive sleep apnea-hypopnea, central sleep apnea, and sleep-related hypoventilation. Table 15-10 compares the diagnostic approaches to sleep-related breathing disorders.



Table 15-8
Hypersomnolence Disorder

	DSM-5	ICS-3	ICD-10
Name	Hypersomnolence disorder	Idiopathic hypersomnia	Nonorg hype
Duration	≥3×/wk for at least 3 mo		
Symptoms	Multiple episodes of sleeping within the same day Main sleep lasts >9 h, but is nonrestorative	Subjective sleepiness MSLT shows mean sleep latency ≤8 min <2 sleep-onset rapid	Excess slee Sleep a Difficult becc arou

	Difficult to become fully aroused when woken	eye movement periods on the MSLT and overnight polysomnogram	when
# Symptoms needed	At least one of the above	All the above	
Exclusion (not the result of):	Another sleep disorder Substance use Another mental or medical disorder	Cataplexy Hypocretin-1 deficiency Sleep deprivation Other causes of hypersomnolence	Medical Narcole
Psychosocial impact	Causes significant distress and/or impairment in functioning		
Symptom specifiers	With mental disorder With medical condition With another sleep disorder		
Severity specifiers	Mild: occurs 1–2 days a week Moderate: occurs 3–4 days a week Severe: occurs 5–7 days a week		
Course specifiers	Acute: <1 mo Subacute: 1–3 mo Persistent: >3 mo		

Comments	It is unclear whether this is a single disorder or multiple with different underlying mechanisms	This is u asso with othe diso

Obstructive Sleep Apnea-Hypopnea. Obstructive sleep apnea/hypopnea, often called obstructive sleep apnea (OSA), occurs when the airway partially or fully collapses during sleep. Decreased oxygen saturation and increased respiratory effort leads to arousals and sleep fragmentation. Predisposing factors for OSA include being male, reaching middle age, being obese, and having micrognathia (undersized lower jaw), retrognathia (posteriorly positioned lower jaw), nasopharyngeal abnormalities, hypothyroidism, and acromegaly. Patients with OSA may also have hypertension, erectile failure in men, depression, heart failure, nocturia, polycythemia, and memory impairment as a result of obstructive sleep apnea-hypopnea. Obstructive apnea and hypopnea episodes can occur in any stage of sleep but are more typical during REM sleep, NREM stage 1, and NREM stage 2 sleep.

Mr. J is a 28-year-old single African American male with an approximately 10-year history of fatigue and sleepiness in the daytime. He began to recognize the daytime sleepiness as a problem in his freshman year of college when he would fall asleep in class or in the dormitory. He admits that his sleep–wake schedule was disrupted during college due to taking long naps and then having to stay up until 1:00 or 2:00 am to complete his studies. His grades and social life suffered, and he describes himself as depressed, isolated, and hopeless about his planned future as a certified public accountant.

As a child, Mr. J says he slept “normally.” In high school he felt best with 10 hours of sleep per night and was able to function well in the daytime. Mr. J denies abuse of alcohol or drugs. He does not use tobacco and drinks about 8 to 10 cups of coffee per day. Family history is

negative for known sleep or psychiatric disorders. Physical examination findings are noncontributory except for body mass index of 29. Routine labs were normal, including TSH.

Mr. J's excessive sleepiness has continued to the present day, notwithstanding some improved sleep hygiene. Improvements include more consistency in bedtime, trying not to nap, and a torturous month-long trial without caffeine. He remains dysphoric and discouraged about his future, blaming his chronic sleepiness as the continuing impediment to his life plans. "I'm just tired of being tired," he says.

Currently, his bedtime is between 10:00 and 10:30 pm; his wake-up alarm is set for 6:30 am. He oversleeps at least once a week on workdays and sleeps from 10:30 pm until 10:00 am on weekends to "catch up." He has difficulty awakening and feels unrefreshed or mildly refreshed. By drinking six to eight cups of coffee in the morning, he is usually able to avoid dozing during the morning. Luckily, he works independently and can schedule client appointments during this relatively alert time. After lunch he routinely falls asleep at the computer while working. He sleeps for 20 to 60 minutes and is usually awakened by his secretary. He then drinks another two cups of coffee and continues with his work. Unexpected napping can also occur later in the afternoon or evening, and he has "nodded off" while driving. He sleeps alone; however, he has been told that he snores loudly. He does not awaken gasping or choking. He denies hypnagogic hallucinations and sleep paralysis but thinks he may feel weak after the rare occasions when he participates in a heated argument.

DISCUSSION

Mr. J has one of the hypersomnias. Most consistent with his history are obstructive sleep apnea syndrome, idiopathic hypersomnia, sleep deprivation in a long sleeper, dyssomnia associated with mood disorder, and narcolepsy. The ancillary symptoms of narcolepsy are absent, with the possible exception of cataplexy. When cataplexy occurs clearly, the diagnosis of narcolepsy is strongly indicated. However, Mr. J's possible infrequent weakness during heated arguments is equivocal for cataplexy. His persistent desire for a 10-hour sleep period would be unusual for a patient with narcolepsy.

A long sleeper or an individual with idiopathic hypersomnia requires prolonged sleep periods and may awaken groggy as does Mr. J. The main differentiating feature is that whenever consistently given a chance to have a full nightly sleep period (usually 10 to 12 hours), the long sleeper does not experience excessive daytime sleepiness. Furthermore, there may be associated autonomic nervous system dysfunction or polysomnographic evidence of elevated slow-wave sleep percentage in patients with idiopathic hypersomnia.

Sleepiness associated with a low-grade chronic depression may be difficult to differentiate from other causes of hypersomnolence. Polysomnography, psychiatric interview, and psychometric testing can be helpful. Mr. J relates his dysphoria to sleepiness and not vice versa; nonetheless, a dyssomnia associated with mood disorder should be considered.

Obstructive sleep apnea syndrome is a strong possibly. Mr. J is overweight (BMI = 29) and snores loudly. Many patients are unaware of gasping or choking for breath. Often, family members witness cessation of breathing during sleep and urge patients to seek treatment. However, Mr. J lives and sleeps alone.

Polysomnography is recommended in patients suspected of obstructive sleep apnea syndrome, narcolepsy, or idiopathic hypersomnia. These disorders usually require lifelong treatment and have significant morbidity and mortality if untreated.



Table 15-9
Narcolepsy

	DSM-5	ICSD-3	ICD-10
Name	Narcolepsy	Narcolepsy	Narcolepsy cataplexy
Duration	≥3×/wk for at least 3 mo		
Symptoms	Excessive daytime sleepiness with • Cataplexy	1. Excessive sleepiness 2. Cataplexy	Episodes of daytime sleepines

	• CSF hypocretin deficiency • REM sleep latency • ≤15 min on polysomnography • OR mean sleep latency ≤8 min on multiple sleep latency testing with ≥ sleep-onset REM sleep periods	and/or hypocretin 1 deficiency 3. Mean sleep latency of ≤8 min and two sleep-onset rapid eye movement periods (SOREMP) within 15 min of sleep onset	occurring despite adequate sleep, an often associate with cata and uncontro lapses in consciou
# Symptoms needed	Excessive daytime sleepiness is required, plus at least one other symptom as above.	**Type 1:** All of the above criteria met **Type 2:** Cataplexy and/or hypocretin deficiency not present	
Exclusion (not the result of):	Substance use disorder Another sleep–wake disorder	Sleep deprivation Another sleep-related disorder	Nightmares Nonorganic disorders Sleep terrors Sleepwalkin
Symptom specifiers	**Narcolepsy without cataplexy but with**	**Types 1 and 2** (see	

	hypocretin deficiency **Narcolepsy with cataplexy but without hypocretin deficiency** **Autosomal dominant cerebellar ataxia, deafness, and narcolepsy** (due to genetic mutation and characterized by late onset in 30s and 40s) **Autosomal dominant narcolepsy, obesity, and type 2 diabetes** (associated with mutation in myelin oligodendrocyte glycoprotein gene) **Narcolepsy secondary to another medical condition**	above)
Severity specifiers	**Mild:** <1/wk, 1–2 naps daily mildly, or intermittently disturbed sleep. **Moderate:** cataplexy 1/day or close to that. Disturbed sleep, need for multiple naps. **Severe:** near constant.	

	Multiple attacks/day. Disturbed sleep. Resistant to medications.

Central Sleep Apnea. Central sleep apnea is a diminished effort to breathe. There are many possible causes, DSM defines three of them: idiopathic CSA, Cheyne–Stokes breathing, and CSA comorbid with opioid use. Other causes include heart failure, high altitude, brainstem lesions, metabolic conditions, specific drugs or substances (CNS depressants), congenital abnormalities, and positive airway pressure (PAP) treatment. The critical characteristic linking the various CSA syndromes is the diminished breathing is not caused by airway obstruction. ICSD-3 includes several subtypes of CSA in addition to those included in DSM-5, which we discuss below, along with the DSM-5 subtypes.



Table 15-10
Sleep-Related Breathing Disorders

	DSM-5	ICSD-3
Name	Breathing-related sleep disorders: • **Obstructive sleep apnea-hypopnea** • **Central sleep apnea** • **Sleep-related hypoventilation**	Sleep-related breathing disorders: • **Central sleep apnea syndromes** • Central sleep apnea with Cheyne–Stokes breathing • Central sleep apnea due to a medical disorder without Cheyne–Stokes breathing • Central sleep apnea due to

		high-altitude periodic breathing • Primary central sleep apnea • **Obstructive sleep apnea disorders** • **Sleep-related hypoventilation disorders** • Obesity hypoventilation syndrome • Congenital central alveolar hypoventilation syndrome • Late-onset central hypoventilation with hypothalamic dysfunction • Idiopathic central alveolar hypoventilation
Symptoms	**<u>Obstructive Sleep Apnea-Hypopnea:</u>** Evidence on polysomnography of ≤5 obstructive apneas or hypopneas/hour of sleep with either: • Nocturnal breathing disturbance (i.e., snoring, breathing pauses) • Daytime sleepiness,	**<u>Obstructive Sleep Apnea-Hypopnea:</u>** 15+ events/hour of sleep on polysomnogram OR 5 obstructive events/hour on polysomnography w/ daytime symptoms of fatigue or sleepiness **<u>Central Sleep Apnea:</u>** Above symptoms but they persist despite

	fatigue Evidence on polysomnography of 15+ obstructive apneas and/or hypopneas/hour of sleep **<u>Central Sleep Apnea:</u>** 5+ central apneas/hour of sleep on polysomnography **<u>Sleep-Related Hypoventilation:</u>** Decreased respiration causing increased CO_2 levels, as correlated by polysomnography	positive airway pressure relieving an obstruction **<u>Sleep-Related Hypoventilation:</u>** ↑ alveolar CO_2 (direct measurement or by end-tidal CO_2 measurement) *For obesity hypoventilation syndrome:* Daytime ↑ $PaCO_2$ (>45 mm Hg) in a patient with BMI of >30 kg/m^2
# Symptoms needed	**Obstructive Sleep Apnea-Hypopnea:** One of the above criteria must be met **Central Sleep Apnea:** All criteria met **Sleep-related hypoventilation:** All criteria met	**Obstructive Sleep Apnea-Hypopnea:** One of the above criteria must be met **Central Sleep Apnea:** All criteria met **Sleep-related hypoventilation:** All criteria met
Exclusion (not the result of):	Another sleep disorder Use of a substance or medication	Another sleep disorder Use of a substance/medication
Symptom specifiers	<u>Obstructive Sleep Apnea-Hypopnea:</u> n/a <u>Central Sleep Apnea:</u>	

	• **Idiopathic central sleep apnea** • **Cheyne–Stokes breathing** • **Central sleep apnea comorbid with opioid use** <u>Sleep-Related Hypoventilation:</u> • **Idiopathic hypoventilation** • **Congenital central alveolar hypoventilation** • **Comorbid sleep-related hypoventilation**
Severity specifiers	<u>Obstructive Sleep Apnea-Hypopnea:</u> **Mild:** apnea hypopnea index is <15 **Moderate:** apnea hypopnea index 15–30 **Severe:** apnea hypopnea index >30 <u>Central Sleep Apnea:</u> Severity based on frequency of breathing disturbance as well as severity of oxygen desaturation/sleep fragmentation <u>Sleep-Related Hypoventilation:</u> Severity based on degree

	of hypoxemia/hypercarbia, in addition to presence of end-organ impairment. Blood gas abnormality during wakefulness is an indicator of more severe illness.

IDIOPATHIC CSA. Patients with idiopathic CSA typically have low normal arterial carbon dioxide tension ($PaCO_2$) while awake and have a high ventilatory response to CO_2. They present with daytime sleepiness, insomnia, or awakening with shortness of breath. Respiratory cessations during sleep are independent of ventilatory effort. Polysomnography reveals five or more central apneas per hour of sleep.

CHEYNE–STOKES BREATHING. Cheyne–Stokes breathing is a unique breathing pattern consisting of prolonged hyperpnea during which tidal volume gradually waxes and wanes in a crescendo–decrescendo fashion. The hyperpnea alternates with apnea and hypopnea episodes that are associated with reduced ventilatory effort. This pattern is most common in older men with congestive heart failure or stroke. As with idiopathic CSA, the patient presents with daytime sleepiness, insomnia, and awakening short of breath.

CSA COMORBID WITH OPIOID USE. This syndrome is the third subtype of CSA in DSM-5, specified if opioid use disorder is present. There is an association with chronic use of long-acting opioid medications and impairment of neuromuscular respiratory control leading to CSA.

CSA DUE TO HIGH-ALTITUDE PERIODIC BREATHING. Central apnea at sleep onset is universal at elevations above 7,600 m but can occur at 1,500 m in some individuals (especially with a rapid ascent). This subtype is no longer included in DSM-5 but may still have clinical significance. Periods of central apnea alternate with periods of hyperpnea in a 12- to 34-second cycle. This

response is an extension of normal respiratory control at sleep onset where medullary pH receptors raise their set-point and require lower pH to respond. At high altitudes, hyperventilation causes a hypocapnic alkalosis that reduces ventilation during sleep. Sleep architecture may suffer, with increased duration in stages 1 and 2 and less slow-wave sleep. REM sleep may not be affected.

CSA DUE TO MEDICAL DISORDER WITHOUT CHEYNE–STOKES BREATHING. This form of CSA is usually caused by a brainstem lesion associated with a wide range of variable etiologies. Cardiac and renal disorders can also cause central apnea. Diagnostic criteria require a polysomnographically verified rate of 5 or more central apneas and hypopneas per hour of sleep.

CSA DUE TO DRUG OR SUBSTANCE USE. A variety of drugs or drug combinations can provoke central apnea episodes, most notably long-acting opiates. However, other substances or medications can also alter neuromuscular control leading to CSA. Diagnostic criteria are a central apnea index (number of episodes per hour) of 5 or more per hour.

PRIMARY SLEEP APNEA OF INFANCY. This form of CSA involves prolonged apneas or hypopneas with concomitant hypoxemia, bradycardia, or both. This condition afflicts preterm neonates, presumably because their brainstems are not fully developed. The condition may be exacerbated by other medical problems that further compromise the infant's physiologic and developmental status.

Sleep-Related Hypoventilation. DSM-5 includes three types of sleep-related hypoventilation disorders: (1) idiopathic hypoventilation, (2) congenital central alveolar hypoventilation, and (3) comorbid sleep-related hypoventilation (which results from a medical disorder, such as a cervical spinal cord injury, COPD, or a neuromuscular disorder, for example). Polysomnography must demonstrate either episodes with elevated CO_2 levels and decreased respirations or, if not monitoring the CO_2, episodes of persistently low oxyhemoglobin levels that are not provoked by apnea or hypopnea.

Circadian Rhythm Sleep–Wake Disorders

DSM-5 lists six types of circadian rhythm sleep–wake disorders: delayed sleep phase type, advanced sleep phase type, irregular sleep–wake type, non–24-hour sleep–wake type, shift work type, and unspecified type. DSM-5 does not include jet lag type, but ICSD does. Table 15-11 compares the diagnostic approaches to the circadian rhythm sleep–wake disorders.



Table 15-11
Circadian Rhythm Sleep Disorders

	DSM-5	ICSD-3	ICD-10
Name	Circadian rhythm sleep-wake disorders	Circadian rhythm sleep–wake disorders: • **Shift work disorder** • **Jet lag disorder** • **Delayed sleep–wake phase disorder** • **Advanced sleep–wake phase disorder** • **Irregular sleep–wake rhythm disorder** • **Non–24-h sleep–wake rhythm disorder**	Disorders of sleep–wak schedule **Delayed slee phase syndrome** **Irregular slee wake patt**

Duration	**Episodic** (lasting 1–3 mo) **Persistent** (lasting ≥3 mo) **Recurrent** (two or more episodes occurring within 1 y)	Symptoms ≥3 mo (exception of jet lag disorder)	
Symptoms	• Disturbed sleep due to altered circadian rhythm • Excessive daytime sleepiness and/or insomnia	Chronic pattern of sleep–wake rhythm disruption due to abnormality in circadian timing/schedule Presence of insomnia and/or excessive sleepiness	Persistent sle disturbanc (i.e., insom or excessiv daytime sleepiness due to alte circadian rhythm and environme demands Disruption of normal 24- circadian cycle
# Symptoms needed	All the above	All the above	One of the ab
Exclusion (not the result of):	Substance use Another medical condition		Nightmares Nonorganic sleep disorders Sleep terrors Sleepwalking

Psychosocial impact	Marked distress and/or impairment in functioning	Distress or impairment in functioning
Symptom specifiers	**Delayed sleep phase type** (sleep onset and awakening times are later) Specify if **familial** Specify if **overlapping with non–24-h sleep–wake type** **Advanced sleep phase type** (sleep onset and awakening times are earlier) Specify if **familial** **Irregular sleep–wake type** (disorganized cycle) **Non–24-h sleep–wake type** does not follow 24-h	

	pattern, with daily drift of progressively later sleep onset/wake time) **Shift work type** **Unspecified type**	
Course specifiers	**Episodic** (lasting 1–3 mo) **Persistent** (lasting ≥3 mo) **Recurrent** (two or more episodes occurring within 1 y)	
Comments		Circadian rhythm disorders can be the result of underlying medical, psychiatric, or neurologic disorders

Delayed Sleep Phase Type. The delayed sleep phase circadian disorder occurs when the biologic clock runs slower than 24 hours or is shifted later than the desired schedule. This produces a phase delay in the sleepiness–alertness cycle. Individuals with delayed sleep phase are more alert in the evening and early nighttime, stay up later and are more tired in the morning. These individuals are sometimes called *night owls.*

Advanced Sleep Phase Type. Advanced sleep phase occurs when the circadian rhythm cycle shifts forward. Therefore, the sleepiness cycle moves earlier. Individuals with advanced sleep phase are drowsy in the evening, want to retire to bed earlier, awaken earlier, and are more alert in the early morning. Individuals with this pattern of advanced sleep phase are sometimes called *early birds* or *larks.*

Irregular Sleep–Wake Type. The irregular sleep–wake pattern occurs when the circadian sleep–wake rhythm is absent or pathologically diminished. The sleep–wake pattern is temporally disorganized, and the timing of sleep and wakefulness is unpredictable. Individuals with this condition have a healthy amount of sleep during 24 hours; however, it is fragmented into three or more episodes that occur irregularly. There are symptoms of insomnia at night and excessive sleepiness during the day. Long daytime naps and inappropriate nocturnal wakefulness occur. This usually impairs activities of daily life.

Non–24-Hour Sleep–Wake Type. When the circadian sleep–wake pacemaker has a cycle length greater or less than 24 hours and is not reset each morning, a person may develop this type of circadian rhythm disorder. Under normal circumstances, resynchronization of the circadian rhythm occurs daily in response to the light–dark cycle. Problems occur incrementally when internal and environmental clocks become more and more out of phase. If the circadian clock's period is longer than 24 hours and does not reset each day, the patient experiences progressively worsening sleep-onset insomnia and daytime sleepiness. Sleep problems peak when circadian and environmental clocks are 12 hours out of phase and then begin to lessen, as the sleep phase continues to advance. Eventually, the clocks align, and the sleep–wake cycle is normal for a few days. Then, unfortunately, the problem begins again.

For this reason, non–24-hour sleep–wake disorder has been called *periodic insomnia* and *periodic excessive sleepiness.* Traumatic brain injury (TBI) can cause this, as can blindness.

Shift Work Type. Many service industries require 24-hour operation (e.g., transportation, health care). Similarly, as cultures became more capital

intensive, mining and manufacturing became around-the-clock enterprises. The number of individuals doing shift work has been increasing steadily for decades. Shift workers commonly have insomnia, excessive sleepiness, or both. Some individuals require only a short time to adjust to a shift change, whereas others have great difficulty. Frequent shift rotation adds to the problem.

Furthermore, to meet social demands, shift workers often adopt an unshifted sleep–wake schedule on weekends and holidays. Even those individuals who try to stay shifted usually retain an unshifted circadian rhythm. The result can be severe insomnia when attempting to sleep and excessive sleepiness when attempting to remain awake. The result is profound sleep deprivation as circadian rhythm continues to conflict with the sleep–wake schedule. The natural low point in the normal sleep–wake rhythm occurs at approximately 3:00 to 5:00 AM—the time frame during which transportation and industrial accidents commonly occur as a direct consequence of sleepiness.

Jet Lag Type. Removed from DSM-5, the other classifications still recognize jet lag. With the advent of high-speed air travel, induced desynchrony between circadian and environmental clocks became possible. Thus, the term *jet lag* came into use. When an individual rapidly travels across many time zones, they either induce a circadian phase advance or a phase delay, depending on the direction of travel. Typically, traveling across one or two time zones will not cause a significant problem. However, it can be challenging to adjust to overseas travel. Individuals who frequently travel for business can find themselves quite impaired at the time they need to make crucial decisions.

Furthermore, “night owls” will experience greater difficulty adjusting to eastward travel because resynchronization requires phase advance. Similarly, “larks” theoretically will have more difficulty with westward travel. The number of time zones crossed is a critical factor. Usually, healthy individuals can quickly adapt to one or two time zone changes per day; therefore, natural adjustment to an 8-hour translocation may take 4 or more days.

Unspecified Type. During illnesses that keep patients bedridden during hospitalizations, and in some forms of dementia, individuals often sleep

erratically. The resulting chaotic sleep–wake pattern adversely affects the circadian rhythm. Medications can also exacerbate the sleep–wake cycle breakdown. Sleep in patients in the intensive care unit is disturbed by noise, light, and therapeutic and monitoring procedures. The resulting disorganized sleep–wake pattern can produce a significant sleep disorder. Also, abuse of recreational street drugs (e.g., methamphetamine, ecstasy) is associated with individuals remaining awake overnight or continuously for several days at a time. These episodes of prolonged wakefulness ultimately produce periods of profound hypersomnia.

Parasomnias and Sleep-Related Movement Disorders

DSM 5 includes only three of the 10 specific parasomnias referenced in the ICSD-3. Also, DSM-5 includes restless legs syndrome (RLS), which ICSD-3 classifies as one of the seven specific sleep-related movement disorders. We will discuss all of these disorders together.

NREM-Related Parasomnias. Table 15-12 compares the diagnostic approaches to the NREM-related parasomnias.

SLEEPWALKING. Sleepwalking, in its classic form, as the name implies, is a condition in which an individual arises from bed and ambulates without fully awakening. It is sometimes called somnambulism, and individuals can engage in a variety of complex behaviors while unconscious. Usually, sleepwalking occurs during slow-wave sleep, and characteristically begins toward the end of the first or second slow-wave sleep episodes. Sleep deprivation and interruption of slow-wave sleep appear to exacerbate, or even provoke, sleepwalking in susceptible individuals. Sleepwalking episodes may range from sitting up and attempting to walk to conducting an involved sequence of semipurposeful actions. The sleepwalker can often successfully interact with the environment (e.g., avoiding tripping over objects). However, the sleepwalker may interact with the environment inappropriately, which sometimes results in injury (e.g., stepping out of an upstairs window or walking into the roadway). There are cases in which sleepwalkers have committed acts of violence. An individual who is sleepwalking is difficult to awaken. Once awake, the sleepwalker will usually appear confused. It is best to gently attempt to lead sleepwalkers back to bed rather than to attempt to

awaken them by grabbing, shaking, or shouting. In their confused state, sleepwalkers may think they are being attacked and may react violently to defend themselves. Sleepwalking in adults is rare, has a familial pattern, and may occur as a primary parasomnia or secondary to another sleep disorder (e.g., sleep apnea). By contrast, sleepwalking is very common in children and has a peak prevalence between the ages of 4 and 8 years. After adolescence, it usually disappears spontaneously. Nightly to weekly sleepwalking episodes associated with physical injury to the patient and others are considered severe. There are "specialized" forms of sleepwalking, most notably sleep-related eating behavior.



Table 15-12
NREM Sleep Disorders

	DSM-5	ICSD-3	ICD-10
Name	NREM sleep arousal disorders	NREM-related parasomnias	Nonorganic disorders sleep–wa schedule Sleepwalk (somna Sleep terr (night te Nightmare
Symptoms	Recurrent episodes of incomplete awakening from sleep, accompanied by either • Sleepwalking • Sleep terrors Little dream recall, if any	Recurrent episodes of incomplete awakening associated with abnormal behaviors or experiences (confused arousal, sleepwalking,	Sleepwalkin altered st consciou which sle wakefuln combined episodes associate low level awarenes no recall

	Amnesia for episodes	sleep terrors, sleep-related eating disorders) Absent or inappropriate responsiveness during episodes Limited or no cognition of dream report Partial or complete amnesia for the event	on awake Sleep terrors nighttime episodes and pani associate motor mo and voca in additio levels of autonom associate little to no of the eve Nightmares: associate fear and emotion, presence recall of c content c awakenir associate vocalizat body mov
# Symptoms needed	All the above criteria met		
Exclusion (not the result of):	Effects of substance or medication Other mental or medical disorders		
Psychosocial	Marked		

impact	impairment and/or distress	
Symptom specifiers	Sleep terror type Sleepwalking type • With sleep-related eating • With sleep-related sexual behavior	DSM include nightmar disorder separate

DESCRIPTION

Ms. R is a 20-year-old white woman who was referred with symptoms of talking, mumbling, and crying out during sleep. At least twice per week she screams in her sleep. She is bothered with excessive sleepiness and falling asleep inappropriately, such as during a conversation. When inactive, she is tired and sleepy, even after a full 8-hour night of sleep. However, she has energy when motivated and leads a vigorous life. Once, she awakened outside of her apartment and her roommate had to let her back in because she had locked herself out. She does not recall the sleepwalking episode or other nocturnal wanderings but sometimes remembers yelling. From the history, crying seems to occur in light sleep but she rarely recalls any sleep-related thoughts or dreams. However, there is a history of occasional nightmares and bruxism. The patient uses an oral appliance to protect her teeth. Leg-kicking and mild snoring without gasping or choking are noted. The patient also complains of leg-kicking during sleep. Her sleep–wake schedule is irregular, and she averages between 5 and 7 hours of sleep per night. She occasionally awakens with a headache in the morning.

Previous health history includes a hospitalization for febrile convulsions during infancy, ophthalmologic surgery for strabismus during childhood, and tonsillectomy as a teenager. Health is otherwise excellent. The patient does not smoke tobacco or drink alcohol.

DISCUSSION

By history Ms. R has one or more of the parasomnias. Sleep talking alone does not require a sleep study, but this patient has nocturnal wanderings. Polysomnography with clinical EEG is indicated to rule out unrecognized nocturnal seizure disorder or other organic factors inducing sleepwalking. Sleepwalking is common and not necessarily considered abnormal in young children; however, in the adult it is rare and merits careful evaluation. Ms. R's excessive daytime sleepiness is likely due to insufficient sleep (5 to 7 hours per night) and possibly parasomnia-related disruption. Interestingly, many parasomnias are exacerbated by sleep deprivation, as is nocturnal seizure disorder.

Sleep studies were performed using comprehensive, attended, laboratory polysomnography. Prior to the overnight study, a clinical EEG was performed. The clinical EEG study did not reveal any significant abnormal EEG activity during baseline, photic stimulation, or hyperventilation. An extended EEG montage was used during the sleep study. Overall sleep quality was within the normal range. Sleep efficiency was 96 percent and latency to sleep was 1 minute. REM sleep percentage was elevated (31 percent) and latency to REM sleep was less than normal (57 minutes). Slow-wave sleep was normal in percentage, but EEG delta activity was very high amplitude. The overall macroarchitectural sleep pattern suggested rebound from sleep deprivation.

By contrast, sleep microarchitecture contained many abnormal features. The clinicians observed high amplitude paroxysmal EEG bursts. Excessively prolonged sleep spindles were noted, and rhythmic K complexes were observed. There was one arousal out of slow-wave sleep with rhythmic EEG discharges alternating with sharp waves. Sharps and spikes occurred several times; however, the focus was difficult to localize (possibly right temporal lobe). There were frequent body movements and full body jerks, most of which occurred during NREM sleep. There were episodes of moaning during slow-wave sleep and laughing during stage 2 sleep that was followed by high amplitude theta bursts and REM sleep. The clinicians observed frequent movements and arousals from REM sleep, but no REM-related spikes or sharp waves. Seizure-like EEG activity was noted during the night and occurred predominantly during slow-wave sleep. However, the patient did not attempt to sleepwalk.

Sharp wave and spike activity increased during the final 45 minutes of the sleep study.

The patient did not have any sleep-related breathing impairment and SaO_2 nadir was 90 percent. She had no periodic limb movements during sleep, and polygraphic features associated with RLS were absent.

SLEEP TERRORS. Sleep terrors involve sudden arousal with intense fearfulness. They usually begin with a piercing scream or cry and are accompanied by behavioral manifestations of intense anxiety bordering on panic. Autonomic and behavioral correlates of fear are typically present. Typically, an individual experiencing a sleep terror sits up in bed, is unresponsive to stimuli, and, if awakened, is confused or disoriented. Vocalizations may occur, but they usually are incoherent.

Notwithstanding the intensity of these events, amnesia for the episodes usually occurs. Like sleepwalking, these episodes usually arise from slow-wave sleep. Fever and CNS depressant withdrawal potentiate sleep terror episodes. Unlike nightmares, in which an elaborate dream sequence unfolds, sleep terrors may be devoid of images or contain only fragments of very brief but frighteningly vivid and sometimes static images. It is sometimes called *pavor nocturnus*, incubus, or night terror. As with other slow-wave sleep parasomnias, sleep deprivation can provoke or exacerbate sleep terrors. Psychopathology is seldom associated with sleep terrors in children; however, a history of a traumatic experience or frank psychiatric problems is often comorbid in adults with this disorder. Severity ranges from less than once per month to almost nightly occurrence (with injury to the patient or others).

CONFUSIONAL AROUSALS. ICSD-3 defines confusional arousals as a milder form of NREM sleep parasomnias. It is common in young children. The child will typically partially awaken from sleep and sit up. The child is confused, but lies back down and resumes sleep. Confusional arousals, sleepwalking, and sleep terrors likely lie on a continuum.

SLEEP-RELATED EATING DISORDER. This diagnosis is subsumed under the DSM-5 classification NREM sleep arousal disorders and specified as

“sleepwalking with sleep-related eating.” In this disorder, eating may become obsessional, and the patient may eat several small meals during a night. The individual may be unaware of the activity, and weight gain can become a problem. During the recurrent sleep-related eating episodes, the individual may consume unusual foods or food combinations or may injure themselves or create a hazard while preparing food (e.g., starting a fire on the stove). They may also experience health consequences from these episodes.

REM-Related Parasomnias. Table 15-13 compares the diagnostic approaches to the REM-related parasomnias.

NIGHTMARE DISORDER. Nightmares are frightening or terrifying dreams. Sometimes called *dream anxiety attacks*, they produce sympathetic activation and ultimately awaken the dreamer. Nightmares occur in REM sleep and usually evolve from a long, complicated dream that becomes increasingly frightening. Having aroused to wakefulness, the person typically remembers the dream (in contrast to sleep terrors). Some nightmares are recurrent, and reportedly when they occur in association with posttraumatic stress disorder, they may be recollections of actual events. Common in children ages 3 to 6 years, nightmares are rare in adults. Frequent and distressing nightmares are sometimes responsible for insomnia because the individual is afraid to sleep. Most people with nightmares are free from psychiatric conditions. Traumatic events are known to induce nightmares, sometimes immediately, but at other times delayed. The nightmares can persist for many years. Several medications can provoke nightmares, including L-DOPA and β-adrenergic blockers. Withdrawal from REM suppressant medications may induce nightmares, as well. Finally, drug or alcohol abuse is associated with nightmares.



Table 15-13
REM Sleep Disorders

	DSM-5	ICSD-3	ICD-10
Name	REM sleep behavior disorder	REM-related parasomnias	REM s beh

			disc
Symptoms	Repeated arousal during sleep Vocalization or complex motor behavior Occurs during REM sleep Not confused when awakened Either: REM sleep without atonia on polysomnography OR history consistent with REM sleep behaviors with established diagnosis of a synucleinopathy (i.e., Parkinson disease)	Sleep-related vocalizations or complex motor behaviors Occurs during REM sleep as documented by polysomnography (or presumed based on history) REM sleep without atonia on polysomnography	Recurr epis of s ofte viol beh duri REM May ca harı self othe Difficul awa Associ with ena
# Symptoms needed	All the above	All the above	
Exclusion (not the result of):	Medication Substance use disorder Another medical condition A mental disorder	Another sleep disorder Medical or neurologic disorder Mental disorder Medication Substance use	

			disorder Absence of epileptiform activity during REM sleep
Psychosocial impact	Marked distress and/or impairment in functioning		



Table 15-14
Nightmare Disorder

	DSM-5	ICSD-3	ICD-10
Name	Nightmare disorder	Nightmare disorder (classified under REM-related parasomnia; see Table 15-13)	Nonorganic disorders of the sleep–wake schedule Sleep terrors Nightmares
Symptoms	Pattern of long, dysphoric, threatening, vivid dreams Dreams are recalled On awakening, quickly becomes awake and alert	Recurrent, highly dysphoric or anxiety-laden nightmares Dreams are remembered upon awakening	Sleep terrors: Nighttime episodes of terror and panic associated with motor movement and vocalization High levels of autonomic tone

			Little to no recall of the event <u>Nightmares:</u> Dreams associated with fear and negative emotion Intact recall of dream content
# Symptoms needed	All of the above		
Exclusion (not the result of):	Effects of a substance or medication Coexisting mental or medical disorder		Not associated with vocalization and body movement
Psychosocial impact	Marked distress and/or impairment in areas of functioning	Marked distress and/or impairment in areas of functioning	
Symptom specifiers	• **During sleep onset** • **With associated nonsleep**		

	disorder (including substance use disorder) • **With associated other medical condition** • **With associated other sleep disorder**
Severity specifiers	**Mild** (less than one episode per week) **Moderate** (one or more episodes per week, but less than nightly) **Severe** (nightly episodes)
Course specifiers	**Acute** (duration ≤1 mo) **Subacute** (1–6 mo)

Persistent (≥6 mo)	

Frequently occurring nightmares often produce a “fear of sleeping” type of insomnia. In turn, insomnia may provoke sleep deprivation, which is known to exacerbate nightmares; hence, a vicious cycle.

ICSD includes it among the REM-related sleep disorders whereas DSM-5 classifies this as a separate disorder. ICD considers this a “nonorganic disorder” of the sleep–wake schedule, along with sleep terrors. Table 15-14 compares the diagnostic approaches.

REM SLEEP BEHAVIOR DISORDER. RBD involves a failure of the patient to have atonia (sleep paralysis) during the REM stage sleep. The result is that the patient enacts his or her dreams. Under normal circumstances, the REM-related hypopolarization of α and γ motor neurons immobilizes the dreamer. Without this paralysis or with intermittent atonia, a dreamer could punch, kick, leap, while dreaming. The activity correlates with dream imagery.

Unlike sleepwalking, the individual seems unaware of the actual environment but instead is acting on the dream sensorium. Thus, a sleepwalker may calmly go to a bedroom window, open it, and step out. By contrast, a person with RBD would more likely dive through the window, thinking it is a dream-visualized lake. Patients and bed partners frequently sustain an injury, which is sometimes severe (e.g., lacerations, fractures). A wide variety of drugs and comorbid conditions can precipitate or worsen RBD.

RECURRENT ISOLATED SLEEP PARALYSIS. Sleep paralysis is, as the name implies, an inability to make voluntary movements during sleep. It becomes a parasomnia when it occurs at sleep onset or on awakening, a time when the individual is partially conscious and aware of the surroundings. This inability to move can be extremely distressing, especially when coupled with the feeling that there is an intruder in the house or when hypnagogic hallucinations are occurring. Sleep paralysis is part of the tetrad of symptoms associated with narcolepsy; however, it can occur (with or without hypnagogia) in individuals that have neither cataplexy nor excessive daytime

sleepiness. Although it is sometimes frightening, sleep paralysis is a feature of normal REM sleep, briefly intruding into wakefulness. The paralysis may last from one to several minutes.

Interestingly, the occurrence of sleep paralysis with hypnagogia may account for a variety of experiences in which the sleeper is confronted or attacked by some sort of "creature." The standard description is that the paralyzed sleeper feels a "presence" nearby, and the creature talks, attacks, or sits on the sleeper's chest and then vanishes. Whether it is called an incubus, "Old Hag," a vampire, ghost oppression (*kanashibari* in Japanese), witch riding, or an alien encounter, all these experiences have elements that are similar to descriptions of sleep paralysis. Irregular sleep, sleep deprivation, psychological stress, and shift work increase the likelihood of sleep paralysis.

Other Parasomnias

EXPLODING HEAD SYNDROME. Individuals with the parasomnia "hear" a loud imagined noise or a sense of an explosion. They experience it as inside their head as they are falling asleep or awakening. The experience can occur just once or recurrently. There is no pain associated with the noise, but the individual may be concerned that they are having a stroke or that something is very wrong. Even a single episode can trigger severe insomnia. There are no known neurologic consequences to this syndrome.

SLEEP-RELATED HALLUCINATIONS. This parasomnia typically involves visual images occurring at sleep onset (hypnagogic) or on awakening (hypnopompic) from sleep. Sometimes challenging to differentiate from dreams, they are common in patients with narcolepsy. Complex hallucinations are rare and usually happen with abrupt awakening and without remembering dreaming. Images tend to be vivid and immobile and persist for several minutes. They usually appear when turning on a light. The images can be frightening.

SLEEP ENURESIS. Sleep enuresis is a disorder in which the individual urinates during sleep while in bed. It is commonly called *bed-wetting* and has primary and secondary forms. In children, primary sleep enuresis is the continuance of bed-wetting since infancy. Secondary enuresis refers to relapse after toilet training was complete, and there was a period during which the child

remained dry. Usually, after toilet training, bed-wetting spontaneously resolves before age 6 years.

Parental primary enuresis increases the likelihood that the children will also have enuresis. A single recessive gene is suspected. Secondary enuresis in children may occur with the birth of a sibling and represent a "cry for attention." Secondary enuresis can also be associated with nocturnal seizures, sleep deprivation, and urologic anomalies. In adults with sleep-disordered breathing, sleep enuresis may occur. In most cases, embarrassment, shame, and guilt are the most substantial consequences. Nonetheless, if we do not address sleep enuresis, it may leave psychosocial scars.

PARASOMNIA DUE TO DRUG OR SUBSTANCE USE AND PARASOMNIA DUE TO MEDICAL CONDITIONS. Many drugs and substances can trigger parasomnias, particularly those agents that lighten sleep; however, alcohol is notorious for producing sleepwalking (even in individuals who have taken sleeping pills). Table 15-15 lists some medications and drugs that can worsen parasomnias.

Seizure disorder should always be on the top of a differential diagnosis list for most parasomnias. The American Academy of Sleep Medicine practice guidelines concerning the indications for polysomnography includes using sleep testing to rule out seizures when diagnosing sleep terror, sleepwalking, RBD, nightmares, and other parasomnias. Sleep-related breathing disorders are also known to trigger sleepwalking, enuresis, sleep terror, confusional arousal, and nightmares. RBD is associated with a variety of neurologic conditions, including Parkinson disease, dementia, progressive supranuclear palsy, Shy–Drager syndrome (a movement disorder with autonomic arousal symptoms), narcolepsy, and others.

Restless Legs Syndrome. RLS is an uncomfortable, subjective sensation of the limbs, usually the legs, sometimes described as a "creepy-crawly" feeling, and the irresistible urge to move the legs when at rest or while trying to fall asleep. Patients often report the sensation of ants walking on the skin and crawling feelings in their legs. It tends to be worse at night, and moving the legs or walking helps to alleviate the discomfort. Thus, as the individual is lying in bed and relaxing, he or she is disturbed by these sensations. Then he or she moves the legs and again tries to fall asleep. This cycle sometimes continues for hours and results in profound insomnia. Table 15-16 compares